

## TECHNICAL MANUAL

# AVIATION UNIT AND INTERMEDIATE MAINTENANCE FOR ARMY MODELS UH-60A, UH-60L, EH-60A, HH-60A AND HH-60L HELICOPTERS

## CHAPTER 4 TROUBLESHOOTING PROCEDURES

\*This manual together with TM 1-1520-237-23-1, TM 1-1520-237-23-2, TM 1-1520-237-23-3, TM 1-1520-237-23-5, TM 1-1520-237-23-6, TM 1-1520-237-23-7, TM 1-1520-237-23-8, TM 1-1520-237-23-9, TM 1-1520-237-23-10, TM 1-1520-237-23-11, TM 1-1520-237-23-12, TM 1-1520-237-23-13, TM 1-1520-237-23-14, TM 1-1520-237-23-15, TM 1-1520-237-23-16, TM 1-1520-237-23-17, TM 1-1520-237-23-18, TM 1-1520-237-23-19 and TM 1-1520-237-23-20, dated 17 April 2006, supersedes TM 1-1520-237-23-1, TM 1-1520-237-23-2, TM 1-1520-237-23-3, TM 1-1520-237-23-4, TM 1-1520-237-23-5, TM 1-1520-237-23-6, TM 1-1520-237-23-7, TM 1-1520-237-23-8, TM 1-1520-237-23-9, TM 1-1520-237-23-10-1, TM 1-1520-237-23-10-2, TM 1-1520-237-23-10-3, TM 1-1520-237-23-11, dated 1 May 2003 including all changes.

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## HEADQUARTERS, DEPARTMENT OF THE ARMY

### 17 April 2006



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## WARNING SUMMARY

Personnel performing operations, procedures, and practices which are included or implied in this technical manual shall observe the following warnings. To disregard these warnings and precautionary information can cause serious injury, death, or an aborted mission.

### WARNING

**FIRST AID** - refer to FM 4-25-11.

**AC POWER** - before applying ac power to the helicopter, make sure that the area around the stabilator is clear of personnel and equipment. Should the stabilator be in any position other than trailing edge fully down, it may reposition automatically to fully down when ac power is applied.

**CARBON MONOXIDE** - during cold weather operations when preheating is necessary, all personnel shall become acquainted with the various types of heaters and where they are to be used. Be careful of accumulations of carbon monoxide and damage to heat-sensitive equipment.

**CONSUMABLE MATERIALS** - observe all cautions and warnings on the containers when using consumables. When applicable wear necessary protective gear during handling and use. If a consumable is flammable or explosive, MAKE SURE consumable and its vapors are kept away from heat, spark, and flame. MAKE SURE helicopter is properly grounded and firefighting equipment is readily available for use.

**SAFETY PINS** - make sure all safety pins are installed in ejector racks prior to performing any maintenance. To disregard this warning can cause serious injury, death, or an aborted mission.

**ELECTRICAL GROUNDING OF THE HELICOPTER** - army regulations require the grounding of the helicopter when doing all fueling and defueling operations. Do not operate helicopter electrical switches, except those essential for servicing during fueling and defueling. Do not smoke or use flame during fueling and defueling operations.

**HIGH VOLTAGE** - there are dangerous voltages in the helicopter. Use extreme care when working with equipment having these voltages.

**HYDRAULIC SYSTEMS** - there are dangerous high hydraulic system pressures in the helicopter. Use extreme care when working on systems having this pressure.

**USE OF FIRE EXTINGUISHER** - when fire extinguishers are used in a confined area, ventilate immediately. Serious injury or death could result if the area is not ventilated or the user does not use a self-contained breathing device.

**MAIN ROTOR PYLON SLIDING COVER** - when opening and closing main pylon sliding cover, keep hands away from sharp edges near track, mating end, and especially ventilation blower inlet hole.

**TECHNICAL ACETONE** - technical Acetone is flammable and toxic to eyes, skin, and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well-ventilated areas (or use approved respirator). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

**ALKALINE AQUEOUS CLEANER** - alkaline cleaner is harmful to skin and eyes. Use adequate ventilation. Operators should wear protective gloves, aprons, and goggles. Upon contact with cleaning compound, wash affected area for at least 20 minutes with clean water. Seek medical attention.

**CLEANING COMPOUND SOLVENT** - cleaning Compound Solvent is combustible and toxic to eyes, skin, and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well-ventilated areas (or use approved respirator). Keep away from open flames or other sources of ignition.

**ELECTRON** - use electron in well ventilated area only. Avoid prolonged breathing of vapors. Avoid bodily contact. The use of chemical gloves and chemical splash goggles are required. Do not use near heat, spark or flame. This solvent is reactive with acids and oxidizers; do not mix or cross-apply with other chemicals. Organic vapor respirator with dust and mist filter is recommended when solvent is spray applied. Keep containers closed between applications.

- Provide mechanical ventilation if used in confined spaces. Coordinate the use of this material with your supporting industrial hygiene and safety offices. Ensure you read and understand the material safety data sheet (MSDS) for electron prior to use.

**ISOPROPYL ALCOHOL** - isopropyl alcohol is a volatile and flammable liquid which must be kept away from flame and other ignition sources. It must be used only in well ventilated area. Personnel must wear protective gloves and eye protection. Excessive inhalation of vapors or contact with skin must be avoided. If contact occurs with eyes or skin, flush area thoroughly with water. If ingestion occurs, induce vomiting and call a doctor. Remove to fresh air if overexposed to vapor.

**DS-108** - use in well ventilated area. Avoid breathing vapors. Organic vapor respirator with dust and mist filter is recommended. May be harmful by inhalation, ingestion, or skin absorption. Avoid bodily contact. Chemical safety glasses/goggles and chemical resistant gloves are required. Combustible, keep away from heat, spark, or flame. Vapors are heavier than air and may travel over a distance to an ignition source and then flash back. Do not mix or cross apply with other chemicals.

**LOCKWIRE** - in critical applications use lockwire instead of safety cable which is specifically prohibited from use in critical applications. The use of safety cable is restricted in these specific applications with a warning.

**PURPOSE OF A WARNING** - alerts personal to operation, procedure, practice, etc., which if not strictly observed, could result in personal injury or loss of life.

CHANGE  
NO. 2

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WASHINGTON, D.C., 15 December 2007

**TECHNICAL MANUAL**

**AVIATION UNIT AND INTERMEDIATE MAINTENANCE  
FOR  
ARMY MODELS  
UH-60A, UH-60L, EH-60A, HH-60A AND HH-60L HELICOPTERS**

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6. Replace the following work packages with their revised version.

Work Package Number

WP 0145 00  
WP 0152 00  
WP 0156 00

**Change  
NO. 2**

By Order of the Secretary of the Army:

Official:

A handwritten signature in black ink, reading "Joyce E. Morrow". The signature is written in a cursive, flowing style.

JOYCE E. MORROW  
*Administrative Assistant to the  
Secretary of the Army*  
0733118

GEORGE W. CASEY, JR.  
*General, United States Army  
Chief of Staff*

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## TECHNICAL MANUAL

# AVIATION UNIT AND INTERMEDIATE MAINTENANCE FOR ARMY MODELS UH-60A, UH-60L, EH-60A, HH-60A, AND HH-60L HELICOPTERS

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i and ii  
Chapter 4 Title and Blank  
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i and ii  
Chapter 4 Title and Blank  
Cover

6. Replace the following work packages with their revised version.

### Work Package Number

WP 0141 00  
WP 0144 00  
WP 0145 00  
WP 0146 00  
WP 0148 00

Work Package Number

WP 0149 00  
WP 0151 00  
WP 0153 00  
WP 0155 00  
WP 0157 00  
WP 0160 00  
WP 0162 00  
WP 0163 00  
WP 0167 00  
WP 0168 00  
WP 0170 00

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Work Package Number

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WP 0170 02  
WP 0170 03  
WP 0170 04

By Order of the Secretary of the Army:

PETER J. SCHOOMAKER  
*General, United States  
Army  
Chief of Staff*

Official:



JOYCE E. MORROW  
*Administrative Assistant to the  
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| Change   | 1 | 21 March 2007    |
| Change   | 2 | 15 December 2007 |

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WASHINGTON, D.C., 17 April 2006

## AVIATION UNIT AND INTERMEDIATE MAINTENANCE FOR

### Army Models UH-60A, UH-60L, EH-60A, HH-60A, and HH-60L Helicopters

#### REPORTING ERRORS AND RECOMMENDING IMPROVEMENTS.

You can help improve this manual. If you find any mistakes or if you know of a way to improve these procedures, please let us know. Mail your letter or DA Form 2028 (Recommended Changes to Publications and Blank Forms) located in the back of this manual directly to: Commander, U.S. Army Aviation and Missile Command, ATTN: AMSAM-MMC-MA-NP, Redstone Arsenal, AL 35898-5000. A reply will be furnished to you. You may also provide DA Form 2028 information to AMCOM via e-mail, fax, or the World Wide Web. Our fax number is: DSN 788-6546 or Commercial 256-842-6546. Our e-mail address is: [2028@redstone.army.mil](mailto:2028@redstone.army.mil). Instructions for sending an electronic 2028 may be found at the back of this manual immediately preceding the hard copy 2028. For the World Wide Web use: <https://amcom2028.redstone.army.mil>.

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**CHAPTER 4**  
**TROUBLESHOOTING PROCEDURES**  
**FOR**  
**ARMY MODELS UH-60A, UH-60L, EH-60A, HH-60A,**  
**AND HH-60L**  
**HELICOPTERS**





## UNIT LEVEL

## UTILITY SYSTEM

## FIRE DETECTION SYSTEM

This WP supersedes WP 0141 00, dated 17 April 2006.

## INITIAL SETUP:

**Test Equipment**

Multimeter, AN/PSM-45A

WP 0790 00

WP 0791 00

WP 0800 00

**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01

WP 0824 00

Electrical Repairer Toolkit, SC 5180-99-B06

WP 0825 00

WP 0925 00

**Personnel Required**

Aircraft Electrician MOS 15F (1)

WP 1205 00

UH-60 Helicopter Repairer MOS 15T (1)

WP 1206 00

WP 1207 00

**References**

TM 1-1520-237-10

WP 1685 00

WP 0090 00

WP 1747 00

WP 0100 00

WP 0104 00

WP 0785 00

**Equipment Condition**

Battery Power Available, TM 1-1520-237-10

External Electrical Power Available WP 1685 00  
or APU Operational, TM 1-1520-237-10

## SPECIAL INSTRUCTIONS

## NOTE

See Figure 1 for component location diagram and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

The fire detection system operational/troubleshooting procedure is subdivided as follows:

- Quick Check
- Detailed Check

**QUICK CHECK OPERATIONAL/TROUBLESHOOTING PROCEDURE****PROCEDURE****Step****NOTE**

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- The photosensitive material in the fire detectors is sensitive to red light. Sunlight filtered through smoke or haze, or at sunrise or sunset, may contain greater amounts of red light. This could trigger the fire detectors and cause a false fire indication. For this reason, fire detectors must be shielded from sunlight.

1. Make sure the following circuit breakers are pushed in:

**CIRCUIT BREAKER****CIRCUIT BREAKER PANEL**

FIRE DET NO. 1 ENG

Upper console

FIRE DET NO. 2 ENG

Upper console

APU FIRE DET

Lower console

2. Turn on battery power (TM 1-1520-237-10).
3. Turn FIRE DET TEST switch to position 1.

**Indication/Condition**

- a. #1 ENG EMER OFF T-handle lights shall be on.
- b. #2 ENG EMER OFF T-handle lights shall be on.
- c. APU T-handle lights shall be on.

**Corrective Action**

1. If Indication/Conditions are not as specified, do the following:
  - a. Turn off battery power.
  - b. Do DETAILED CHECK, in this work package.

**PROCEDURE****Step**

4. Turn FIRE DET TEST switch to position 2.

**Indication/Condition**

- a. #1 ENG EMER OFF T-handle lights shall be on.
- b. #2 ENG EMER OFF T-handle lights shall be on.



QUICK CHECK OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

- c. APU T-handle lights shall be off.

Corrective Action

- 1. If Indication/Conditions are not as specified, do the following:
  - a. Turn off battery power.
  - b. Do DETAILED CHECK, in this work package.

Step

- 5. Turn FIRE DET TEST switch to OPER.
- 6. Turn off battery power.

Indication/Condition

Power shall be off.

Corrective Action

None Required

DETAILED CHECK OPERATIONAL/TROUBLESHOOTING PROCEDURE  
PROCEDURE

Step

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- The photosensitive material in the fire detectors is sensitive to red light. Sunlight filtered through smoke or haze, or at sunrise or sunset, may contain greater amounts of red light. This could trigger the fire detectors and cause a false fire indication. For this reason, fire detectors must be shielded from sunlight.

- 1. Make sure the following circuit breakers are pushed in:

| CIRCUIT BREAKER                                                                                                                                                                                                                                          | CIRCUIT BREAKER PANEL |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| FIRE DET NO. 1 ENG                                                                                                                                                                                                                                       | Upper console         |
| FIRE DET NO. 2 ENG                                                                                                                                                                                                                                       | Upper console         |
| APU FIRE DET                                                                                                                                                                                                                                             | Lower console         |
| 2. Turn on electrical power. <b>HH-60A HH-60L</b> Turn pilot's or copilot's multifunction display (MFD) switch ON and press T6 switch (illuminates all MFD legends for 10 seconds, then only the active caution and advisory legends will be displayed). |                       |
| 3. Make sure upper console FIRE DET TEST switch is at OPER.                                                                                                                                                                                              |                       |

**DETAILED CHECK OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Indication/Condition**

- a. #1 ENG EMER OFF T-handle lights shall be off.
- b. #2 ENG EMER OFF T-handle lights shall be off.
- c. APU T-handle lights shall be off.

**Corrective Action**

- 1. If Indication/Condition a. is not as specified, go to #1 ENG EMER OFF T-HANDLE LIGHTS ARE ON, in this work package.
- 2. If Indication/Condition b. is not as specified, go to #2 ENG EMER OFF T-HANDLE LIGHTS ARE ON, in this work package.
- 3. If Indication/Condition c. is not as specified, check for less than 9.0 vdc between P285-1 and P285-10.
  - a. If voltage is less than 9.0 vdc, replace APU fire detector control amplifier (WP 1205 00).
  - b. If voltage is greater than 9.0 vdc, replace APU fire detector (WP 1206 00).

**PROCEDURE****Step**

- 4. Pull out the following circuit breakers:

**CIRCUIT BREAKER****CIRCUIT BREAKER PANEL**

FIRE DET NO. 2 ENG

Upper console

APU FIRE DET

Lower console

- 5. Turn FIRE DET TEST switch to position 1.

**Indication/Condition**

- a. #1 ENG EMER OFF T-handle lights and pilot's and copilot's master warning FIRE capsules shall go on.
- b. #2 ENG EMER OFF and APU T-handle lights shall be off.

**Corrective Action**

- 1. If Indication/Condition a. is not as specified, go to #1 ENG EMER OFF T-HANDLE LIGHTS AND MASTER WARNING FIRE CAPSULES ARE OFF, in this work package.
- 2. If both #1 ENG EMER OFF T-handle lights or both master warning FIRE capsules are off, replace left relay panel (WP 0825 00).
- 3. If one master warning FIRE capsule is off, go to FIRE CAPSULE ON ONE MASTER WARNING PANEL DOES NOT GO ON DURING FIRE DETECTOR TEST, in this work package.
- 4. If Indication/Condition b. is not as specified, replace left relay panel (WP 0825 00).

**PROCEDURE****Step**

- 6. Turn FIRE DET TEST switch to position 2.

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**DETAILED CHECK OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Indication/Condition**

#1 ENG EMER OFF T-handle lights, and pilot's and copilot's master warning FIRE capsules shall be on.

**Corrective Action**

If Indication/Condition is not as specified, go to #1 ENG EMER OFF T-HANDLE LIGHTS AND MASTER WARNING FIRE CAPSULES ARE OFF, in this work package.

**Step**

7. Pull out FIRE DET NO. 1 ENG circuit breaker on upper console.
8. Push in FIRE DET NO. 2 ENG circuit breaker on upper console.
9. Turn FIRE DET TEST switch to position 1.

**Indication/Condition**

- a. #2 ENG EMER OFF T-handle lights and pilot's and copilot's master warning FIRE capsules shall go on.
- b. #1 ENG EMER OFF T-handle lights shall be off.

**Corrective Action**

1. If Indication/Condition a. is not as specified, go to #2 ENG EMER OFF T-HANDLE LIGHTS AND MASTER WARNING FIRE CAPSULES ARE OFF, in this work package.
2. If both #2 ENG EMER OFF T-handle lights or both master warning panel FIRE capsules are off, replace left relay panel (WP 0825 00).
3. If Indication/Condition b. is not as specified, replace left relay panel (WP 0825 00).

**Step**

10. Turn FIRE DET TEST switch to position 2.

**Indication/Condition**

#2 ENG EMER OFF T-handle lights and pilot's and copilot's master warning FIRE capsules shall be on.

**Corrective Action**

If Indication/Condition is not as specified, go to #2 ENG EMER OFF T-HANDLE LIGHTS ARE OFF AND MASTER WARNING FIRE CAPSULES ARE OFF, in this work package.

**Step**

11. Pull out FIRE DET NO. 2 ENG circuit breaker on upper console.
12. Push in APU FIRE DET circuit breaker on lower console.
13. Turn FIRE DET TEST switch to position 1.

**Indication/Condition**

APU T-handle lights and pilot's and copilot's master warning FIRE capsules shall go on.

**Corrective Action**

1. If Indication/Condition is not as specified, go to APU T-HANDLE LIGHTS AND MASTER WARNING FIRE CAPSULES DO NOT GO ON, in this work package.

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**DETAILED CHECK OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued

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2. If both APU T-handle lights or both master warning FIRE capsules are off, replace left relay panel (WP 0825 00).

**Step**

14. Turn FIRE DET TEST switch to OPER.

**NOTE**

Flashlight shall be positioned between 6 to 8 inches from the fire detector. The flashlight beam shall be aimed directly at the fire detector.

15. Shine a flashlight with a red filter on APU fire detector.

**Indication/Condition**

APU T-handle lights shall go on.

**Corrective Action**

1. If Indication/Condition is not as specified, check for 28 vdc between P414-A and P414-B.
  - a. If voltage is as specified, replace APU fire detector (WP 1206 00).
  - b. If voltage is not as specified, repair/replace wiring (WP 1747 00).

**Step****NOTE**

Flashlight shall be positioned between 6 to 8 inches from the fire detector. The flashlight beam shall be aimed directly at the fire detector.

16. Shine a flashlight with other than red or blue filter on APU fire detector.

**Indication/Condition**

APU T-handle lights shall not go on.

**Corrective Action**

If Indication/Condition is not as specified, replace APU fire detector (WP 1206 00).

**Step**

17. Push in FIRE DET NO. 1 ENG circuit breaker on upper console.

**NOTE**

Flashlight shall be positioned between 6 to 8 inches from the fire detector. The flashlight beam shall be aimed directly at the fire detector.

18. Shine a flashlight with a red filter on No. 1 engine firewall fire detector.

**Indication/Condition**

#1 ENG EMER OFF T-handle lights shall go on.

**Corrective Action**

1. If Indication/Condition is not as specified, check for 28 vdc between P413-A and P413-B.
  - a. If voltage is as specified, replace No. 1 engine firewall fire detector (WP 1206 00).

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**DETAILED CHECK OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**

- b. If voltage is not as specified, repair/replace wiring (WP 1747 00).

**Step****NOTE**

Flashlight shall be positioned between 6 to 8 inches from the fire detector. The flashlight beam shall be aimed directly at the fire detector.

- 19. Shine a flashlight with other than red or blue filter on No. 1 engine firewall fire detector.

**Indication/Condition**

#1 ENG EMER OFF T-handle lights shall not go on.

**Corrective Action**

If Indication/Condition is not as specified, replace No. 1 engine fire detector (WP 1206 00).

**Step****NOTE**

Flashlight shall be positioned between 6 to 8 inches from the fire detector. The flashlight beam shall be aimed directly at the fire detector.

- 20. Shine a flashlight with a red filter on No. 1 engine deck fire detector.

**Indication/Condition**

#1 ENG EMER OFF T-handle lights shall go on.

**Corrective Action**

- 1. If Indication/Condition is not as specified, check for 28 vdc between P286-A and P286-B.
  - a. If voltage is as specified, replace No. 1 engine deck fire detector (WP 1206 00).
  - b. If voltage is not as specified, repair/replace wiring (WP 1747 00).

**Step****NOTE**

Flashlight shall be positioned between 6 to 8 inches from the fire detector. The flashlight beam shall be aimed directly at the fire detector.

- 21. Shine a flashlight with other than red or blue filter on No. 1 engine deck fire detector.

**Indication/Condition**

#1 ENG EMER OFF T-handle lights shall not go on.

**Corrective Action**

If Indication/Condition is not as specified, replace No. 1 engine deck fire detector (WP 1206 00).

**Step**

- 22. Push in FIRE DET NO. 2 ENG circuit breaker on upper console.

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**DETAILED CHECK OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued**NOTE**

Flashlight shall be positioned between 6 to 8 inches from the fire detector. The flashlight beam shall be aimed directly at the fire detector.

23. Shine a flashlight with a red filter on No. 2 engine firewall fire detector.

**Indication/Condition**

#2 ENG EMER OFF T-handle lights shall go on.

**Corrective Action**

1. If Indication/Condition is not as specified, check for 28 vdc between P415-A and P415-B.
  - a. If voltage is as specified, replace No. 2 engine firewall fire detector (WP 1206 00).
  - b. If voltage is not as specified, repair/replace wiring (WP 1747 00).

**Step****NOTE**

Flashlight shall be positioned between 6 to 8 inches from the fire detector. The flashlight beam shall be aimed directly at the fire detector.

24. Shine a flashlight with other than red or blue filter on No. 2 engine firewall fire detector.

**Indication/Condition**

#2 ENG EMER OFF T-handle lights shall not go on.

**Corrective Action**

If Indication/Condition is not as specified, replace No. 2 engine firewall fire detector (WP 1206 00).

**Step****NOTE**

Flashlight shall be positioned between 6 to 8 inches from the fire detector. The flashlight beam shall be aimed directly at the fire detector.

25. Shine a flashlight with a red filter on No. 2 engine deck fire detector.

**Indication/Condition**

#2 ENG EMER OFF T-handle lights shall go on.

**Corrective Action**

1. If Indication/Condition is not as specified, check for 28 vdc between P287-A and P287-B.
  - a. If voltage is as specified, replace No. 2 engine deck fire detector (WP 1206 00).
  - b. If voltage is not as specified, repair/replace wiring (WP 1747 00).

---

**DETAILED CHECK OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued**Step****NOTE**

Flashlight shall be positioned between 6 to 8 inches from the fire detector. The flashlight beam shall be aimed directly at the fire detector.

26. Shine a flashlight with other than red or blue filter on No. 2 engine deck fire detector.

**Indication/Condition**

#2 ENG EMER OFF T-handle lights shall not go on.

**Corrective Action**

If Indication/Condition is not as specified, replace No. 2 engine deck fire detector (WP 1206 00).

**Step**

27. Turn FIRE DET TEST switch to position 1.

**Indication/Condition**

#1 ENG EMER OFF, #2 ENG EMER OFF, and APU T-handles lights shall go on.

**Corrective Action**

If Indication/Condition is not as specified, repeat operational/troubleshooting procedure.

**Step**

28. Rotate INST LT PILOT FLT counterclockwise to OFF, then rotate 1/4 turn clockwise.
29. Place caution/advisory panel **HH-60A HH-60L** instrument panel INDICATOR LTS **■** BRT/DIM-TEST switch to BRT/DIM.

**Indication/Condition**

#1 ENG EMER OFF, #2 ENG EMER OFF, and APU T-handles lights shall go on dimly.

**Corrective Action**

1. If all T-handles remain bright, go to T-HANDLES REMAIN BRIGHT, in this work package.
2. If any T-handle remains bright, replace left relay panel (WP 0825 00).

**Step**

30. Turn FIRE DET TEST switch to OPER.
31. **HH-60A HH-60L** Turn MFD switch OFF. **■**
32. Turn off electrical power.

**Indication/Condition**

Power shall be off.

**Corrective Action**

None Required

**#1 ENG EMER OFF T-HANDLE LIGHTS ARE ON****SYMPTOM**

#1 ENG EMER OFF T-handle lights are on.

**MALFUNCTION**

#1 ENG EMER OFF T-handle lights are on.

**CORRECTIVE ACTION****1. Check for less than 9.0 vdc between P283-1 and P283-10.**

- a. If voltage is as specified, go to **2**.
- b. If voltage is not as specified, replace No. 1 engine firewall detector (WP 1206 00). Go to **3**.

**2. Check for less than 9.0 vdc between P283-2 and P283-10.**

- a. If voltage is as specified, replace No. 1 engine fire detector control amplifier (WP 1205 00). Go to **3**.
- b. If voltage is not as specified, replace No. 1 engine deck fire detector (WP 1206 00). Go to **3**.

**3. Procedure completed.****#2 ENG EMER OFF T-HANDLE LIGHTS ARE ON****SYMPTOM**

#2 ENG EMER OFF T-handle lights are on.

**MALFUNCTION**

#2 ENG EMER OFF T-handle lights are on.

**CORRECTIVE ACTION****1. Check for less than 9.0 vdc between P284-1 and P284-10.**

- a. If voltage is as specified, go to **2**.
- b. If voltage is not as specified, replace No. 2 engine firewall detector (WP 1206 00). Go to **3**.

**2. Check for less than 9.0 vdc between P284-2 and P284-10.**

- a. If voltage is as specified, replace No. 2 engine fire detector control amplifier (WP 1205 00). Go to **3**.
- b. If voltage is not as specified, replace No. 2 engine deck fire detector (WP 1206 00). Go to **3**.

**3. Procedure completed.**



**#1 ENG EMER OFF T-HANDLE LIGHTS AND MASTER WARNING FIRE CAPSULES ARE OFF****SYMPTOM**

#1 ENG EMER OFF T-handle lights and master warning FIRE capsules are off.

**MALFUNCTION**

#1 ENG EMER OFF T-handle lights and master warning FIRE capsules are off.

**CORRECTIVE ACTION**

- 1. Set FIRE DET TEST switch S15 to position 2, observe No. 1 engine T-handle warning light lamps and master warning panel FIRE capsules shall go on.**
  - a. If No. 1 engine T-handle warning light lamps and master warning panel FIRE capsules go on, go to **2**.
  - b. If No. 1 engine T-handle warning light lamps and master warning panel FIRE capsules do not go on, go to **6**.
- 2. Set FIRE DET TEST switch S15 to position 1, check continuity between terminals BC1 and B2 of FIRE DET TEST switch S15.**
  - a. If continuity is present, go to **3**.
  - b. If continuity is not present, replace FIRE DET TEST switch S15 (WP 1207 00). Go to **15**.
- 3. Check for 15 vdc between P283-1 (+) and P283-10 (gnd).**
  - a. If voltage is as specified, replace No. 1 engine fire detector control amplifier (WP 1205 00). Go to **15**.
  - b. If voltage is not as specified, go to **4**.
- 4. Check for 15 vdc between P413-E and ground.**
  - a. If voltage is as specified, go to **5**.
  - b. If voltage is not as specified, repair/replace wiring between terminal BC1 of FIRE DET TEST switch S15 and P413-E (WP 1747 00). Go to **15**.
- 5. Check continuity between P413-C and P283-1.**
  - a. If continuity is present, replace No. 1 engine firewall fire detector (WP 1206 00). Go to **15**.
  - b. If continuity is not present, repair/replace wiring between P413-C and P283-1 (WP 1747 00). Go to **15**.
- 6. Check for 28 vdc between P283-11 (+) and P283-10 (gnd).**
  - a. If voltage is as specified, go to **7**.
  - b. If voltage is not as specified, go to **13**.
- 7. Set FIRE DET TEST switch S15 to position 1, check for 15 vdc between P283-1 (+) and P283-10 (gnd).**
  - a. If voltage is as specified, go to **8**.
  - b. If voltage is not as specified, go to **10**.
- 8. Install an insulated jumper wire between P283-11 and P283-9, observe No. 1 engine T-handle warning light lamps and master warning panel FIRE capsules shall go on.**

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**#1 ENG EMER OFF T-HANDLE LIGHTS AND MASTER WARNING FIRE CAPSULES ARE OFF - Continued**

- a. If No. 1 engine T-handle warning light lamps and master warning panel FIRE capsules go on, replace No. 1 engine fire detector control amplifier (WP 1205 00). Go to **15**.
- b. If No. 1 engine T-handle warning light lamps and master warning panel FIRE capsules do not go on, go to **9**.
- 9. Check continuity between P283-9 and P902-M.**
  - a. If continuity is present, replace left relay panel (WP 0825 00). Go to **15**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **15**.
- 10. Check for 15 vdc between terminal B2 of FIRE DET TEST switch S15 and ground.**
  - a. If voltage is as specified, replace FIRE DET TEST switch S15 (WP 1207 00). Go to **15**.
  - b. If voltage is not as specified, go to **11**.
- 11. Check for 28 vdc between P902-P and ground.**
  - a. If voltage is as specified, go to **12**.
  - b. If voltage is not as specified, go to **14**.
- 12. Check for 9,800 to 10,200 ohms between J902-F and J902-P.**
  - a. If correct reading is present, repair/replace wiring between P902-F and terminal B2 of FIRE DET TEST switch S15 (WP 1747 00). Go to **15**.
  - b. If correct reading is not present, replace left relay panel (WP 0825 00). Go to **15**.
- 13. Check continuity between P283-10 and ground.**
  - a. If continuity is present, go to **14**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **15**.
- 14. Check for 28 vdc between terminal 1 of FIRE DET NO. 1 ENG circuit breaker and ground.**
  - a. If voltage is as specified, repair/replace wiring between terminal 1 of FIRE DET NO. 1 ENG circuit breaker and P902-P (WP 1747 00). Go to **15**.
  - b. If voltage is not as specified, replace FIRE DET NO. 1 ENG circuit breaker (WP 0824 00). Go to **15**.
- 15. Procedure completed.**

**FIRE CAPSULE ON ONE MASTER WARNING PANEL DOES NOT GO ON DURING FIRE DETECTOR TEST SYMPTOM**

FIRE capsule on one master warning panel does not go on during fire detector test.

**MALFUNCTION**

FIRE capsule on one master warning panel does not go on during fire detector test.

**CORRECTIVE ACTION**

- 1. Check for burnt-out lamp.**
  - a. If lamp is good, go to **2**.
  - b. If lamp is not good, replace lamp (WP 0791 00). Go to **7**.

**FIRE CAPSULE ON ONE MASTER WARNING PANEL DOES NOT GO ON DURING FIRE DETECTOR TEST**

- Continued

**2. Check continuity between J243-G and J902-H.**

- a. If continuity is present, go to **3**.
- b. If continuity is not present, replace left relay panel (WP 0825 00). Go to **7**.

**3. Check helicopter configuration.**

- a. **EMEP** , go to **4**.
- b. **W/O EMEP** , go to **6**.

**4. EMEP Remove and check pin filtered adapters P130 and P131 (WP 0925 00).**

- a. If pin filtered adapters are good, go to **5**.
- b. If pin filtered adapters are not good, replace pin filtered adapters as required (WP 0925 00). Go to **7**.

**5. EMEP Install pin filtered adapters (WP 0925 00). Go to 6.****6. Check continuity between P130-4 and P131-4.**

- a. If continuity is present, replace master warning panel (WP 0790 00). Go to **7**.
- b. If trouble remains, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to **7**.
- c. If continuity is not present, replace wiring, as required, between (WP 1747 00), then go to **7**:
  - P130-4 and P243-G (pilot's master warning panel)
  - P131-4 and P902-H (copilot's master warning panel)

**7. Procedure completed.****#1 ENG EMER OFF T-HANDLE LIGHTS AND MASTER WARNING FIRE CAPSULES ARE OFF****SYMPTOM**

#1 ENG EMER OFF t-handle lights and master warning FIRE capsules are off.

**MALFUNCTION**

#1 ENG EMER OFF t-handle lights and master warning FIRE capsules are off.

**CORRECTIVE ACTION****1. Set FIRE DET TEST switch S15 to position 2 and check continuity between terminals BC2 and B8 of FIRE DET TEST switch S15.**

- a. If continuity is present, go to **2**.
- b. If continuity is not present, replace FIRE DET TEST switch S15 (WP 1207 00). Go to **6**.

**2. Check for 15 vdc between P286-E and ground.**

- a. If voltage is as specified, go to **3**.
- b. If voltage is not as specified, go to **5**.

**3. Check continuity between P286-C and P283-2.**

- a. If continuity is present, go to **4**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **6**.

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**#1 ENG EMER OFF T-HANDLE LIGHTS AND MASTER WARNING FIRE CAPSULES ARE OFF - Continued****4. Check for 15 vdc between P283-2 and ground.**

- a. If voltage is as specified, replace No. 1 engine fire detector control amplifier (WP 1205 00). Go to **6**.
- b. If voltage is not as specified, replace No. 1 engine deck flame detector (WP 1206 00). Go to **6**.

**5. Check continuity between terminals B2 and B8 of FIRE DET TEST switch S15.**

- a. If continuity is present, repair/replace wiring between terminal BC2 of FIRE DET TEST switch S15 and P286-E (WP 1747 00). Go to **6**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **6**.

**6. Procedure completed.****#2 ENG EMER OFF T-HANDLE LIGHTS AND MASTER WARNING FIRE CAPSULES ARE OFF****SYMPTOM**

#2 ENG EMER OFF T-handle lights and master warning FIRE capsules are off.

**MALFUNCTION**

#2 ENG EMER OFF T-handle lights and master warning FIRE capsules are off.

**CORRECTIVE ACTION****1. Set FIRE DET TEST switch S15 to position 2, observe No. 2 engine T-handle warning light lamps and master warning panel FIRE capsules shall go on.**

- a. If No. 2 engine T-handle warning light lamps and master warning panel FIRE capsules go on, go to **2**.
- b. If No. 2 engine T-handle warning light lamps and master warning panel FIRE capsules do not go on, go to **6**.

**2. Set FIRE DET TEST switch S15 to position 1, check continuity between terminals AC1 and A2 of FIRE DET TEST switch S15.**

- a. If continuity is present, go to **3**.
- b. If continuity is not present, replace FIRE DET TEST switch S15 (WP 1207 00). Go to **15**.

**3. Check for 15 vdc between P284-1 (+) and P284-10 (gnd).**

- a. If voltage is as specified, replace No. 2 engine fire detector control amplifier (WP 1205 00). Go to **15**.
- b. If voltage is not as specified, go to **4**.

**4. Check for 15 vdc between P415-E and ground.**

- a. If voltage is as specified, go to **5**.
- b. If voltage is not as specified, repair/replace wiring between terminal AC1 of FIRE DET TEST switch S15 and P415-E (WP 1747 00). Go to **15**.

**5. Check continuity between P415-C and P284-1.**

- a. If continuity is present, replace No. 2 engine firewall fire detector (WP 1206 00). Go to **15**.

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**#2 ENG EMER OFF T-HANDLE LIGHTS AND MASTER WARNING FIRE CAPSULES ARE OFF - Continued**

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- b. If continuity is not present, repair/replace wiring between P415-C and P284-1 (WP 1747 00). Go to **15**.
- 6. Check for 28 vdc between P284-11 (+) and P284-10 (gnd).**
  - a. If voltage is as specified, go to **7**.
  - b. If voltage is not as specified, go to **13**.
- 7. Set FIRE DET TEST switch S15 to position 1, check for 15 vdc between P284-1 (+) and P284-10 (gnd).**
  - a. If voltage is as specified, go to **8**.
  - b. If voltage is not as specified, go to **10**.
- 8. Install an insulated jumper wire between P284-11 and P284-9, observe No. 2 engine T-handle warning light lamps and master warning panel FIRE capsules shall go on.**
  - a. If No. 2 engine T-handle warning light lamps and master warning panel FIRE capsules go on, replace No. 2 engine fire detector control amplifier (WP 1205 00). Go to **15**.
  - b. If No. 2 engine T-handle warning light lamps and master warning panel FIRE capsules do not go on, go to **9**.
- 9. Check continuity between P284-9 and P243-E.**
  - a. If continuity is present, replace left relay panel (WP 0825 00). Go to **15**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **15**.
- 10. Check for 15 vdc between terminal A2 of FIRE DET TEST switch S15 and ground.**
  - a. If voltage is as specified, replace FIRE DET TEST switch S15 (WP 1207 00). Go to **15**.
  - b. If voltage is not as specified, go to **11**.
- 11. Check for 28 vdc between P902-R and ground.**
  - a. If voltage is as specified, go to **12**.
  - b. If voltage is not as specified, go to **14**.
- 12. Check for 9,800 to 10,200 ohms between J902-E and J902-R.**
  - a. If correct reading is present, repair/replace wiring between P902-E and terminal A2 of FIRE DET TEST switch S15 (WP 1747 00). Go to **15**.
  - b. If correct reading is not present, replace left relay panel (WP 0825 00). Go to **15**.
- 13. Check continuity between P284-10 and ground.**
  - a. If continuity is present, go to **14**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **15**.
- 14. Check for 28 vdc between terminal 1 of FIRE DET NO. 2 ENG circuit breaker and ground.**
  - a. If voltage is as specified, repair/replace wiring between terminal 1 of FIRE DET NO. 2 ENG circuit breaker and P902-R (WP 1747 00). Go to **15**.
  - b. If voltage is not as specified, replace FIRE DET NO. 2 ENG circuit breaker (WP 0824 00). Go to **15**.

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**#2 ENG EMER OFF T-HANDLE LIGHTS AND MASTER WARNING FIRE CAPSULES ARE OFF - Continued****15. Procedure completed.****#2 ENG EMER OFF T-HANDLE LIGHTS ARE OFF AND MASTER WARNING FIRE CAPSULES ARE OFF****SYMPTOM**

#2 ENG EMER OFF T-handle lights are off and master warning FIRE capsules are off.

**MALFUNCTION**

#2 ENG EMER OFF T-handle lights are off and master warning FIRE capsules are off.

**CORRECTIVE ACTION**

- 1. Set FIRE DET TEST switch S15 to position 2 and check continuity between terminals AC2 and A8 of FIRE DET TEST switch S15.**
  - a. If continuity is present, go to **2**.
  - b. If continuity is not present, replace FIRE DET TEST switch S15 (WP 1207 00). Go to **6**.
- 2. Check for 15 vdc between P287-E and ground.**
  - a. If voltage is as specified, go to **3**.
  - b. If voltage is not as specified, go to **5**.
- 3. Check continuity between P287-C and P284-2.**
  - a. If continuity is present, go to **4**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **6**.
- 4. Check for 15 vdc between P284-2 and ground.**
  - a. If voltage is as specified, replace No. 2 engine fire detector control amplifier (WP 1205 00). Go to **6**.
  - b. If voltage is not as specified, replace No. 2 engine deck flame detector (WP 1206 00). Go to **6**.
- 5. Check continuity between terminals A2 and A8 of FIRE DET TEST switch S15.**
  - a. If continuity is present, repair/replace wiring between terminal AC2 of FIRE DET TEST switch S15 and P287-E (WP 1747 00). Go to **6**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **6**.
- 6. Procedure completed.**

**APU T-HANDLE LIGHTS AND MASTER WARNING FIRE CAPSULES DO NOT GO ON****SYMPTOM**

APU T-handle lights and master warning FIRE capsules do not go on.

**MALFUNCTION**

APU T-handle lights and master warning FIRE capsules do not go on.

**CORRECTIVE ACTION**

- 1. Set FIRE DET TEST switch S15 to position 1, check for 15 vdc between terminal CC1 of FIRE DET TEST switch S15 and ground.**

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APU T-HANDLE LIGHTS AND MASTER WARNING FIRE CAPSULES DO NOT GO ON - Continued

- a. If voltage is as specified, go to **2**.
  - b. If voltage is not as specified, go to **7**.
- 2. Check for 28 vdc between P285-11 (+) and P285-10 (gnd).**
  - a. If voltage is as specified, go to **3**.
  - b. If voltage is not as specified, go to **6**.
- 3. Check for 15 vdc between P285-1 (+) and P285-10 (gnd).**
  - a. If voltage is as specified, go to **4**.
  - b. If voltage is not as specified, go to **10**.
- 4. Install an insulated jumper wire between P285-11 and P285-9, observe APU T-handle warning light lamps and master warning panel FIRE capsules shall go on.**
  - a. If APU T-handle warning light lamps and master warning panel FIRE capsules go on, replace APU fire detector control amplifier (WP 1205 00). Go to **13**.
  - b. If APU T-handle warning light lamps and master warning panel FIRE capsules do not go on, go to **5**.
- 5. Check continuity between P285-9 and P902-L.**
  - a. If continuity is present, replace left relay panel (WP 0825 00). Go to **13**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **13**.
- 6. Check continuity between P285-10 and ground.**
  - a. If continuity is present, go to **12**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **13**.
- 7. Check for 15 vdc between terminal C2 of FIRE DET TEST switch S15 and ground.**
  - a. If voltage is as specified, replace FIRE DET TEST switch S15 (WP 1207 00). Go to **13**.
  - b. If voltage is not as specified, go to **8**.
- 8. Check for 28 vdc between P902-N and ground.**
  - a. If voltage is as specified, go to **9**.
  - b. If voltage is not as specified, go to **12**.
- 9. Check for 9,800 to 10,200 ohms between J902-G and J902-N.**
  - a. If correct reading is present, repair/replace wiring between P902-G and terminal C2 of FIRE DET TEST switch S15 (WP 1747 00). Go to **13**.
  - b. If correct reading is not present, replace left relay panel (WP 0825 00). Go to **13**.
- 10. Check for 15 vdc between P414-E and ground.**
  - a. If voltage is as specified, go to **11**.
  - b. If voltage is not as specified, repair/replace wiring between terminal CC1 of FIRE DET TEST switch S15 and P414-E (WP 1747 00). Go to **13**.
- 11. Check continuity between P414-C and P285-1.**
  - a. If continuity is present, replace APU fire detector (WP 1206 00). Go to **13**.

---

**APU T-HANDLE LIGHTS AND MASTER WARNING FIRE CAPSULES DO NOT GO ON - Continued**


---

- b. If continuity is not present, repair/replace wiring between P414-C and P285-1 (WP 1747 00). Go to **13**.

**12. Check for 28 vdc between terminal 1 of APU FIRE DET circuit breaker and ground.**

- a. If voltage is as specified, repair/replace wiring between terminal 1 of APU FIRE DET circuit breaker and P902-N (WP 1747 00). Go to **13**.
- b. If voltage is not as specified, replace APU FIRE DET circuit breaker (WP 0824 00). Go to **13**.

**13. Procedure completed.**

**T-HANDLES REMAIN BRIGHT**

**SYMPTOM**

T-handles remain bright.

**MALFUNCTION**

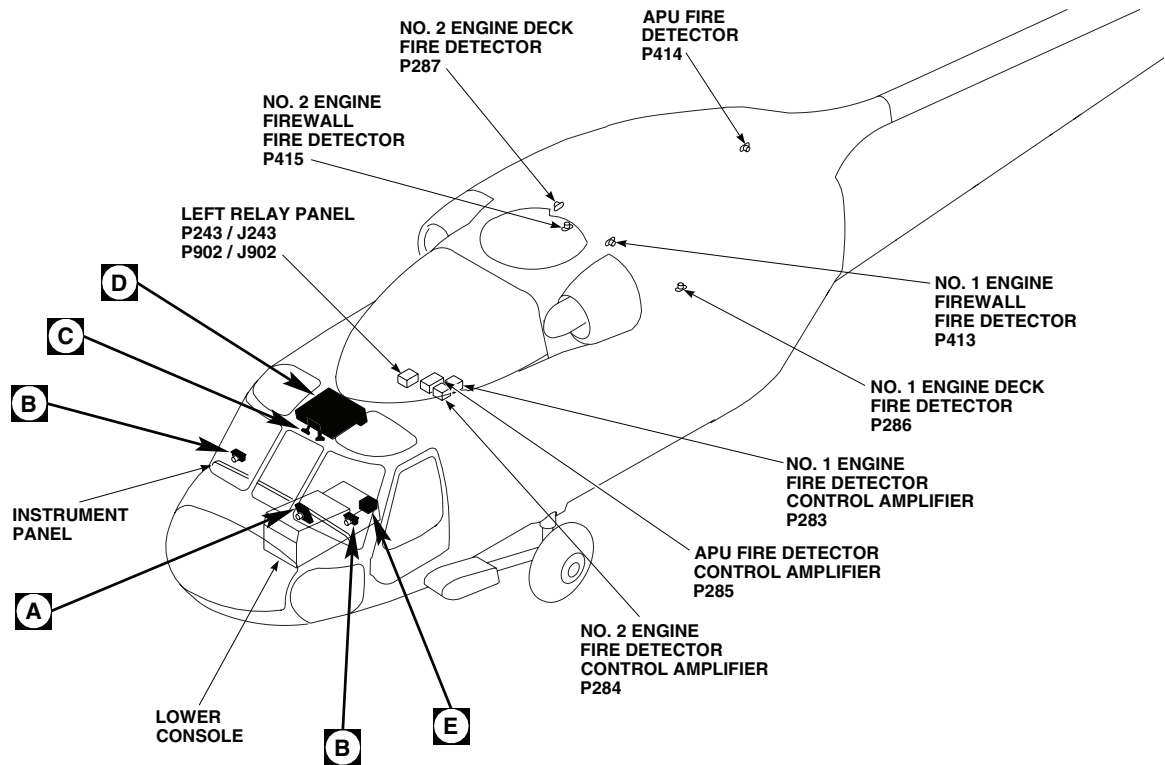
T-handles remain bright.

**CORRECTIVE ACTION**

- 1. Do caution/advisory capsules HH-60A HH-60L or MFD legends go dim?**
  - a. If caution/advisory capsules HH-60A HH-60L or MFD legends are dim, go to **2**.
  - b. If caution/advisory capsules HH-60A HH-60L or MFD legends are not dim, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to **7**.
- 2. Do instrument panel and lower console lights go dim?**
  - a. If lights are dim, replace left relay panel (WP 0825 00). Go to **7**.
  - b. If lights are not dim, go to **3**.
- 3. Check helicopter configuration.**
  - a. **EMEP** , go to **4**.
  - b. **W/O EMEP** , go to **6**.
- 4. EMEP Remove and check pin filtered adapter P118 (WP 0925 00).**
  - a. If pin filtered adapter is good, go to **5**.
  - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **7**.
- 5. EMEP Install pin filtered adapter (WP 0925 00). Go to 6.**
- 6. Check continuity between P902-g HH-60A HH-60L and P118-T or P1000-D .**
  - a. If continuity is present, replace left relay panel (WP 0825 00).
  - b. If trouble remains, replace caution/advisory panel BRT/DIM-TEST switch (WP 0785 00) or HH-60A HH-60L instrument panel INDICATOR LTS BRT/DIM-TEST switch (WP 0800 00) . Go to **7**.
  - c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **7**.
- 7. Procedure completed.**



## LOCATION



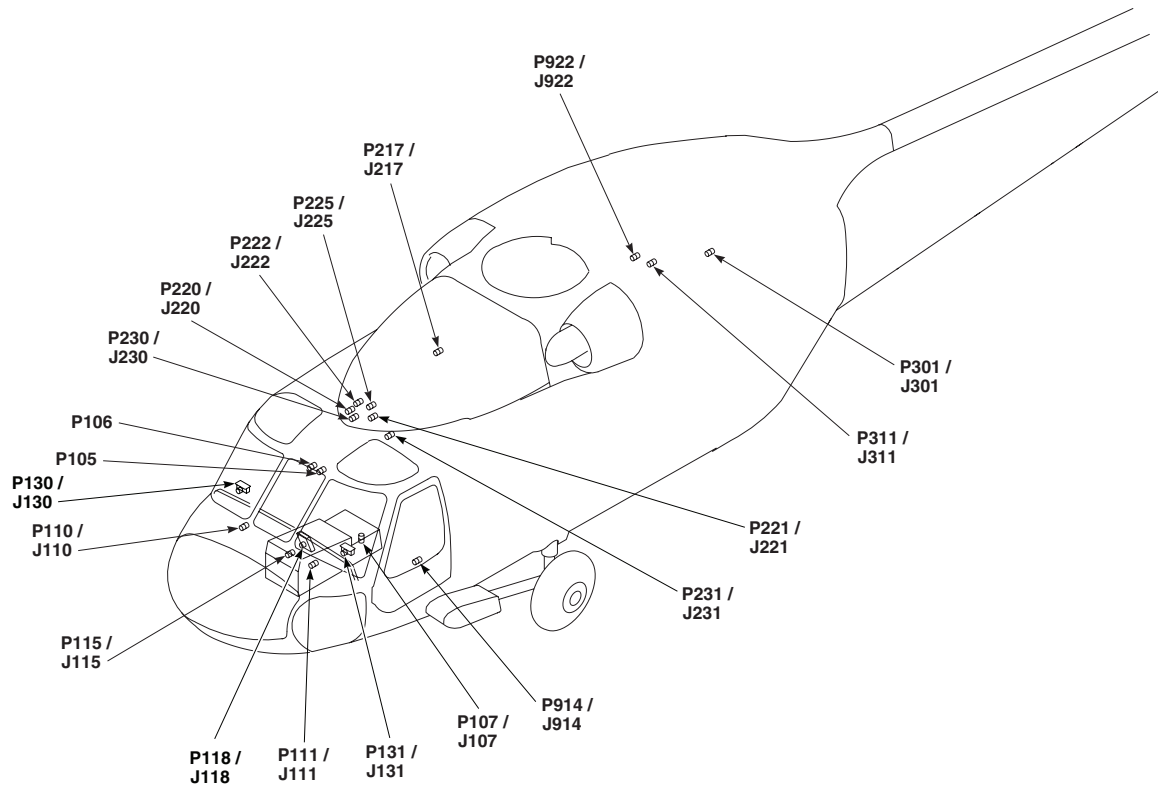
| TERMINAL BOARD/<br>DISCONNECT PLUG/<br>RECEPTACLE | LOCATION/<br>CONNECTION POINT |
|---------------------------------------------------|-------------------------------|
| P105                                              | NO. 1 ENGINE QUADRANT         |
| P106                                              | NO. 2 ENGINE QUADRANT         |
| P107 / J107                                       | COCKPIT, BL 9 LH, STA 236     |
| P110 / J110                                       | COCKPIT, BL 7.5 RH, STA 197   |
| P111 / J111                                       | COCKPIT, BL 7.5 LH, STA 197   |
| P115 / J115                                       | COCKPIT, BL 6 RH, STA 210     |

| TERMINAL BOARD/<br>DISCONNECT PLUG/<br>RECEPTACLE | LOCATION/<br>CONNECTION POINT            |
|---------------------------------------------------|------------------------------------------|
| P117 / J117                                       | CAUTION / ADVISORY PANEL                 |
| P118 / J118                                       | CAUTION / ADVISORY PANEL                 |
| P130 / J130                                       | PILOT'S MASTER WARNING PANEL             |
| P131 / J131                                       | COPILOT'S MASTER WARNING PANEL           |
| P217 / J217                                       | MAIN ROTOR PYLON DECK BL 0, STA 284      |
| P220 / J220                                       | MAIN ROTOR PYLON DECK BL 3 RH, STA 243   |
| P221 / J221                                       | MAIN ROTOR PYLON DECK BL 3 LH, STA 245.5 |
| P222 / J222                                       | MAIN ROTOR PYLON DECK BL 3 RH, STA 244   |
| P225 / J225                                       | MAIN ROTOR PYLON DECK BL 3 LH, STA 244   |
| P230 / J230                                       | CABIN CEILING, BL 8 RH, STA 247          |
| P231 / J231                                       | CABIN CEILING, BL 9 LH, STA 247          |

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Figure 1. Fire Detection System Location Diagram. (Sheet 1 of 4)

## LOCATION - Continued



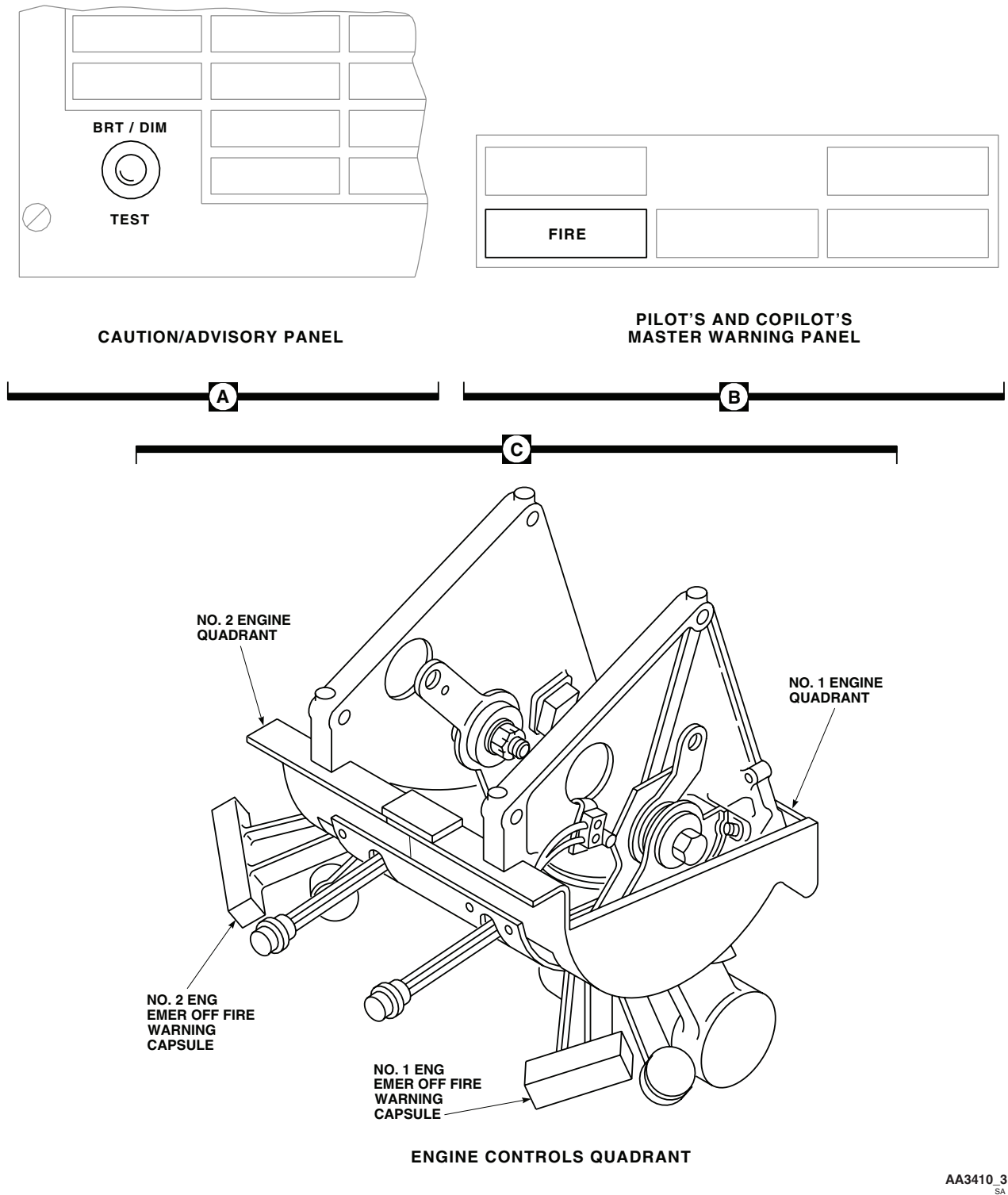
| TERMINAL BOARD/<br>DISCONNECT PLUG/<br>RECEPTACLE | LOCATION/<br>CONNECTION POINT                      |
|---------------------------------------------------|----------------------------------------------------|
| P243 / J243                                       | LEFT RELAY PANEL                                   |
| P283                                              | NO. 1 ENGINE FIRE<br>DETECTOR CONTROL<br>AMPLIFIER |
| P284                                              | NO. 2 ENGINE FIRE<br>DETECTOR CONTROL<br>AMPLIFIER |
| P285                                              | APU FIRE DETECTOR<br>CONTROL AMPLIFIER             |
| P286                                              | NO. 1 ENGINE DECK FIRE<br>DETECTOR                 |
| P287                                              | NO. 2 ENGINE DECK FIRE<br>DETECTOR                 |

| TERMINAL BOARD/<br>DISCONNECT PLUG/<br>RECEPTACLE | LOCATION/<br>CONNECTION POINT           |
|---------------------------------------------------|-----------------------------------------|
| P301 / J301                                       | TRANSITION DECK<br>BL 12.88 LH, STA 415 |
| P311 / J311                                       | CABIN CEILING, BL 16.5 LH,<br>STA 383   |
| P413                                              | NO. 1 ENGINE FIREWALL<br>FIRE DETECTOR  |
| P414                                              | APU FIRE DETECTOR                       |
| P415                                              | NO. 2 ENGINE FIREWALL<br>FIRE DETECTOR  |
| P902 / J902                                       | LEFT RELAY PANEL                        |
| P914 / J914                                       | COCKPIT, BL 24 LH,<br>STA 247           |
| P922 / J922                                       | CABIN CEILING, BL 10.7 LH,<br>STA 380   |

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Figure 1. Fire Detection System Location Diagram. (Sheet 2 of 4)

LOCATION - Continued



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Figure 1. Fire Detection System Location Diagram. (Sheet 3 of 4)

LOCATION - Continued

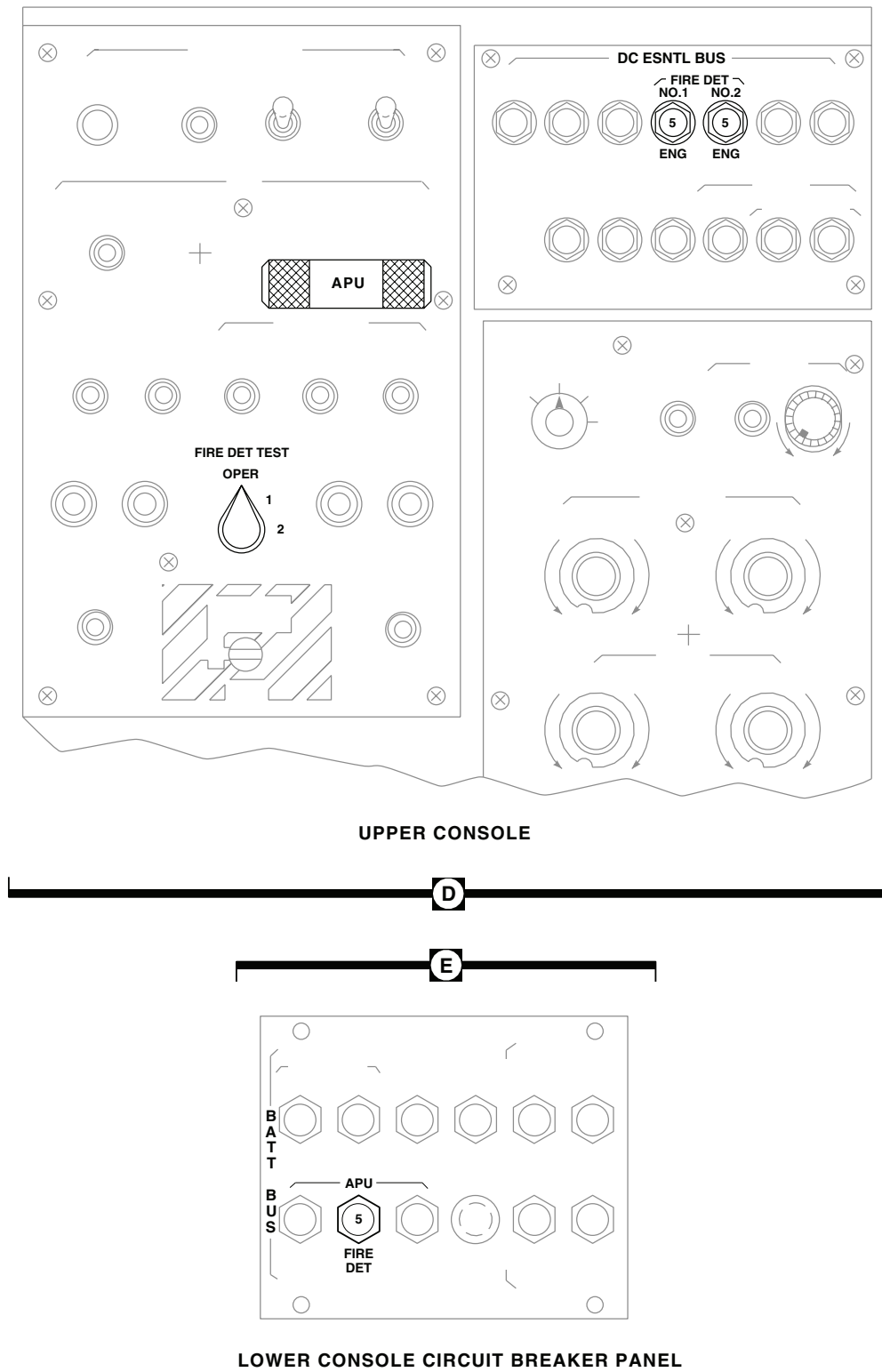


Figure 1. Fire Detection System Location Diagram. (Sheet 4 of 4)

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## SCHEMATICS

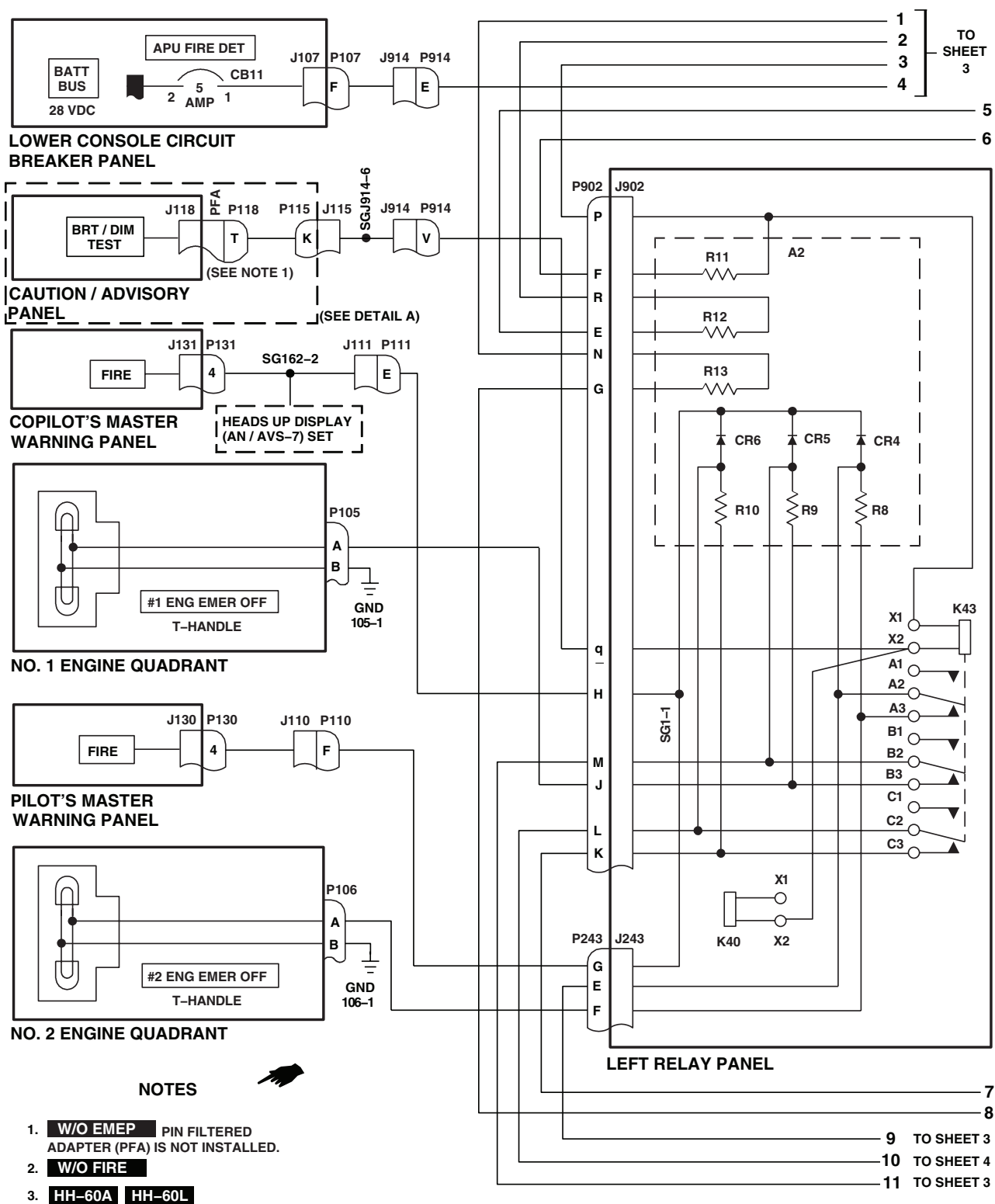
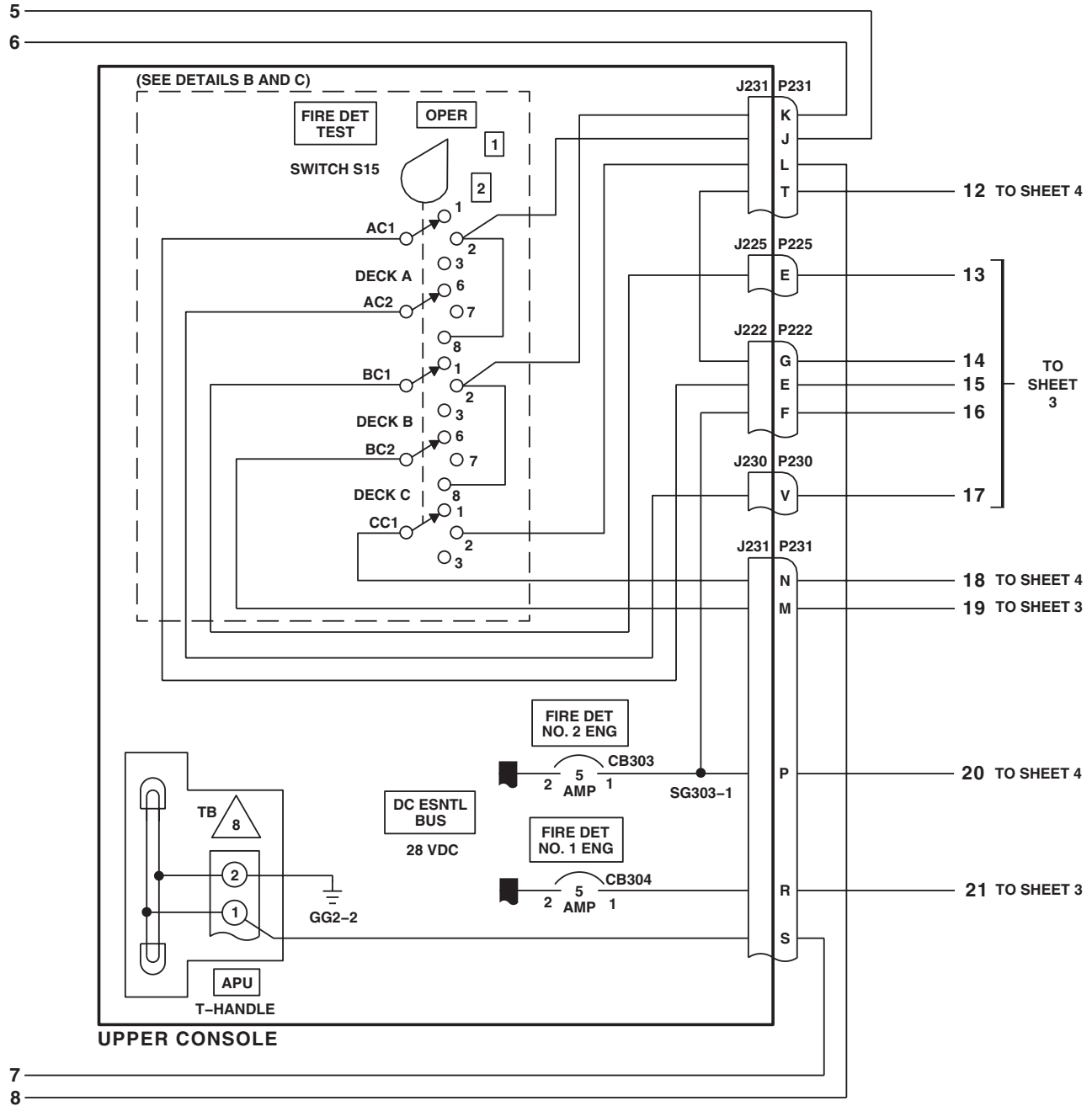


Figure 2. Fire Detection System Schematic Diagram. (Sheet 1 of 5)

SCHEMATICS - Continued



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Figure 2. Fire Detection System Schematic Diagram. (Sheet 2 of 5)

SCHEMATICS - Continued

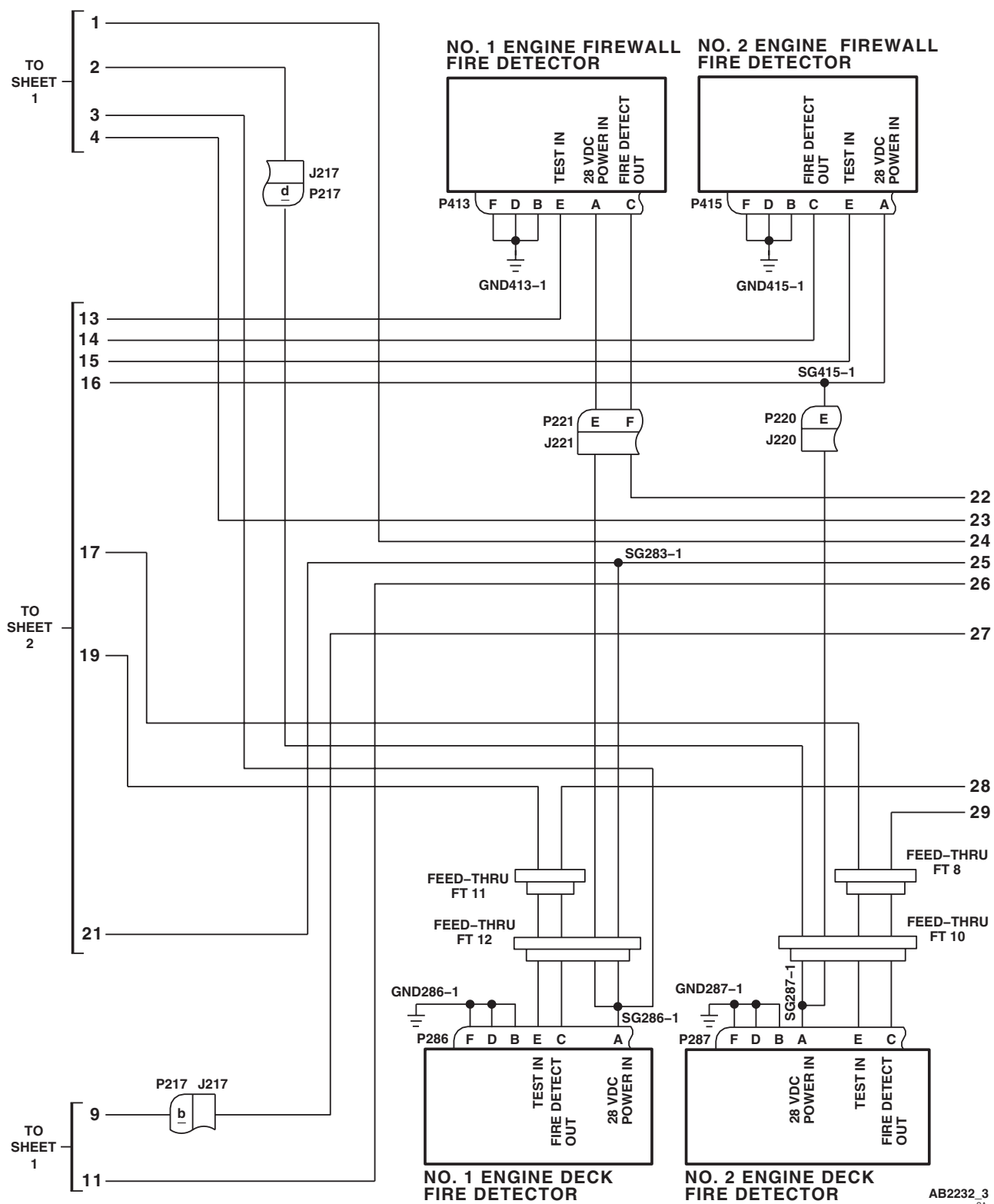
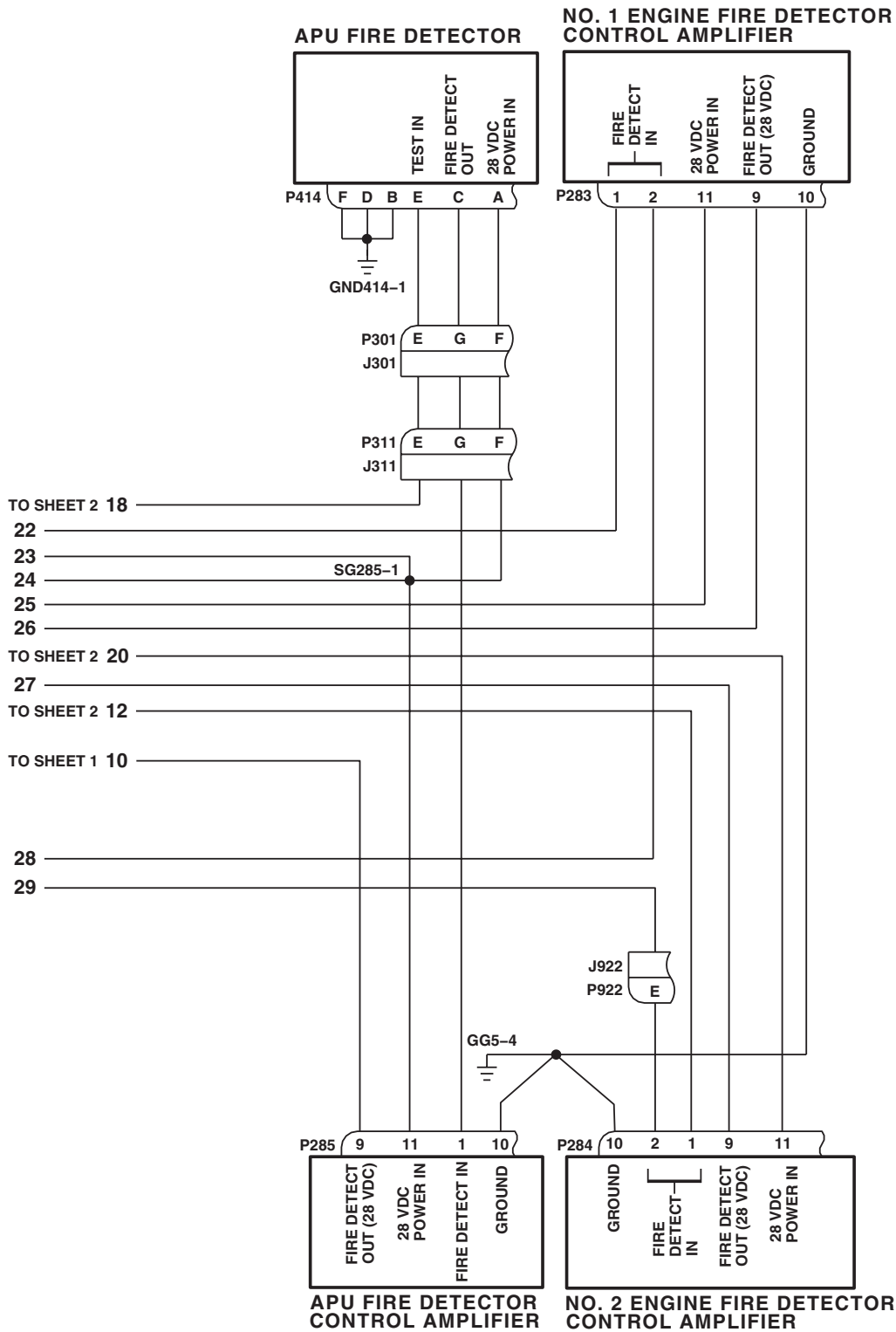


Figure 2. Fire Detection System Schematic Diagram. (Sheet 3 of 5)

SCHEMATICS - Continued

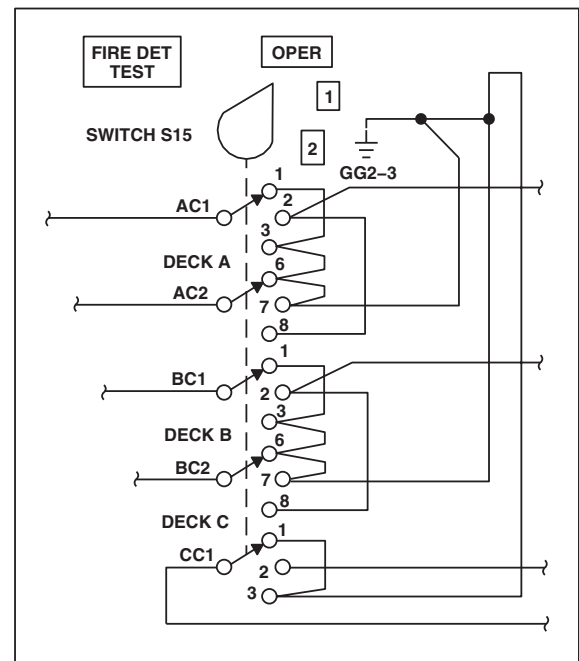
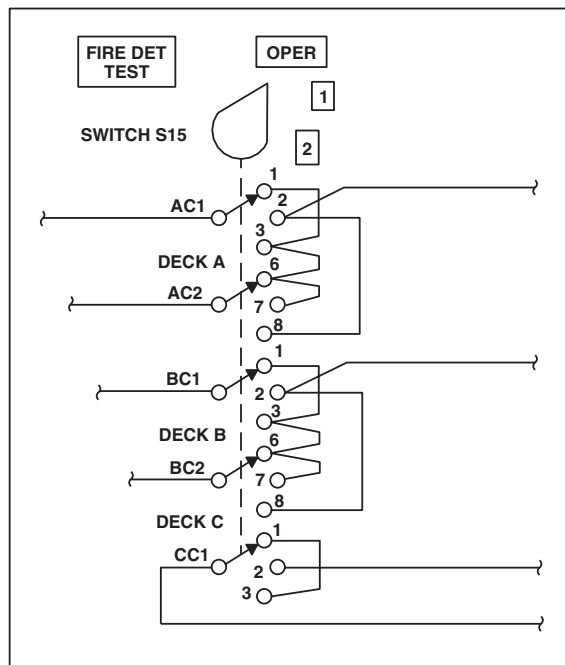
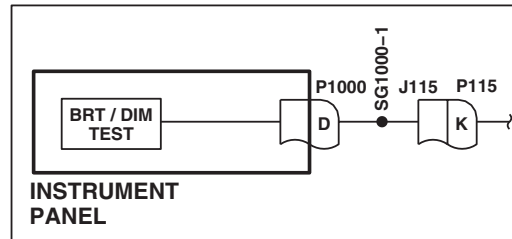


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Figure 2. Fire Detection System Schematic Diagram. (Sheet 4 of 5)



SCHEMATICS - Continued



AB2232\_5  
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Figure 2. Fire Detection System Schematic Diagram. (Sheet 5 of 5)

END OF WORK PACKAGE



## UNIT LEVEL

## UTILITY SYSTEM

## FIRE EXTINGUISHING SYSTEM

## INITIAL SETUP:

## Test Equipment

Multimeter, AN/PSM-45A

WP 0824 00

WP 0825 00

WP 0857 00

## Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01

WP 1195 00

Electrical Repairer Toolkit, SC 5180-99-B06

WP 1200 00

WP 1201 00

## Personnel Required

Aircraft Electrician MOS 15F (1)

WP 1203 00

UH-60 Helicopter Repairer MOS 15T (1)

WP 1400 00

WP 1685 00

WP 1747 00

## References

WP 0104 00

WP 0501 00

## Equipment Condition

External Electrical Power Available, WP 1685 00

## SPECIAL INSTRUCTIONS

## NOTE

See Figure 1 for component location diagram and **FIRE** Figure 2  **W/O FIRE** Figure 3  for schematic diagrams as an aid in operational/troubleshooting.

The fire extinguishing system operational/troubleshooting procedure is subdivided as listed below. An operational/troubleshooting procedure for the directional control valve is in DIRECTIONAL CONTROL CHECK VALVE, in this work package.

- No. 1 Main Fire Extinguishing Bottle
- No. 2 Main Fire Extinguishing Bottle
- No. 1 Reserve Fire Extinguishing Bottle
- No. 2 Reserve Fire Extinguishing Bottle
- Directional Control Valve Check
- Shutdown

**SETUP OPERATIONAL/TROUBLESHOOTING PROCEDURE****PROCEDURE****Step****CAUTION**

To prevent possible firing of a squib, never place multimeter leads on fire extinguisher bottle terminals.

**NOTE**

- When connecting multimeter leads to wiring terminals, connect positive lead to smaller terminal and negative lead to larger terminal. Always see Figure 1, Figure 2, and Figure 3 for proper connections.
- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Turn off external electrical power.
2. Disconnect battery by removing cover and disconnecting both electrical connectors (WP 0857 00).

**NOTE**

Make sure wiring terminals are inside rubber boots, and do not allow them to contact airframe.

3. Disconnect all wire terminals from main and reserve fire extinguishing containers (WP 1195 00).
4. Make sure FIRE EXTGH circuit breaker on lower console circuit breaker panel is pushed in.

**Indication/Condition**

FIRE EXTGH circuit breaker shall be pushed in.

**Corrective Action**

None Required

## NO. 1 MAIN FIRE EXTINGUISHING BOTTLE OPERATIONAL/TROUBLESHOOTING PROCEDURE

### PROCEDURE

#### Step

#### CAUTION

To prevent possible firing of a squib, never place multimeter leads on fire extinguisher bottle terminals.

#### NOTE

- When connecting multimeter leads to wiring terminals, connect positive lead to smaller terminal and negative lead to larger terminal. Always see Figure 1, Figure 2, and Figure 3 for proper connections.
- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Do SETUP, in this work package, if not already done.
2. **FIRE** Connect multimeter leads to wires marked W3163A16 and W3162A16N. ■
3. **W/O FIRE** Connect multimeter leads to wires marked W3163A20 and W3162A20N. ■
4. Connect battery (WP 0857 00).
5. Turn on external electrical power.
6. Pull engine controls quadrant #1 ENG EMER OFF lever back.
7. Place upper console FIRE EXTGH switch to MAIN, then release to OFF.

#### Indication/Condition

Multimeter shall indicate 28 vdc when switch is placed to MAIN.

#### Corrective Action

If result is not as specified, go to MULTIMETER CONNECTED TO NO.1 MAIN FIRE BOTTLE WIRING HAS NO VOLTAGE INDICATION WHEN #1 ENG EMER OFF T-HANDLE IS USED AND FIRE EXTGH SWITCH IS PLACED TO MAIN, in this work package.

#### Step

8. Push #1 ENG EMER OFF lever forward.

#### NOTE

- When performing operational check, move engine T-handle back or pull APU T-handle down as indicated by order, then place FIRE EXTGH switch to position indicated.
  - After completing each step, return all T-handles to the OFF position.
9. Perform a No. 1 Main Fire Extinguishing Bottle Priority check by performing the operational checks in NO. 1 MAIN FIRE EXTINGUISHING BOTTLE, in this work package.

**NO. 1 MAIN FIRE EXTINGUISHING BOTTLE OPERATIONAL/TROUBLESHOOTING PROCEDURE -**  
Continued

**Table 1. No. 1 Main Fire Extinguishing Bottle.**

| <b>FIRE BOTTLE TESTED</b> | <b>STEP NUMBER</b> | <b>CHECK FIRST</b> | <b>CHECK SECOND</b> | <b>CHECK THIRD</b> | <b>FIRE EXTINGUISHER SWITCH POSITION</b> |
|---------------------------|--------------------|--------------------|---------------------|--------------------|------------------------------------------|
| #1 Main                   | 1                  | #2 ENG*            | #1 ENG              | -                  | MAIN                                     |
| #1 Main                   | 2                  | #1 ENG             | #2 ENG*             | -                  | MAIN                                     |
| #1 Main                   | 3                  | #1 ENG             | APU                 | -                  | MAIN                                     |
| #1 Main                   | 4                  | #2 ENG             | APU                 | -                  | MAIN                                     |
| #1 Main                   | 5                  | APU                | #2 ENG*             | #1 ENG             | MAIN                                     |
| #1 Main                   | 6                  | #1 ENG             | #2 ENG              | APU                | MAIN                                     |
| #1 Main                   | 7                  | APU                | #1 ENG              | -                  | MAIN                                     |
| #1 Main                   | 8                  | APU                | #2 ENG*             | -                  | MAIN                                     |

\*Pull engine T-handle back, then push forward.

**Indication/Condition**

Multimeter shall indicate 28 vdc.

**Corrective Action**

If results are not as specified, replace right relay panel (WP 0825 00).

**Step**

10. Pull upper console APU T-handle down. Visually check directional control valve.

**Indication/Condition**

Sound of directional control valve being energized shall be heard and position indicator shall turn.

**Corrective Action**

If result is not as specified, go to DIRECTIONAL CONTROL VALVE DOES NOT OPERATE WHEN NO. 1 MAIN IS USED FOR APU, in this work package.

**Step**

11. Place FIRE EXTGH switch to MAIN, then release to OFF.

**Indication/Condition**

Multimeter shall indicate 28 vdc when switch is placed to MAIN.

**Corrective Action**

If result is not as specified, go to MULTIMETER CONNECTED TO NO. 1 MAIN FIRE BOTTLE WIRING HAS NO VOLTAGE INDICATION WHEN APU T-HANDLE IS USED AND FIRE EXTGH SWITCH IS PLACED TO MAIN, in this work package.

---

**NO. 1 MAIN FIRE EXTINGUISHING BOTTLE OPERATIONAL/TROUBLESHOOTING PROCEDURE -**  
Continued**Step**

12. Push APU T-handle up while assistant is observing directional control valve.

**Indication/Condition**

Sound of directional control valve deenergizing shall be heard and position indicator shall turn.

**Corrective Action**

If result is not as specified, replace directional control valve (WP 1200 00).

**Step**

13. Turn off external electrical power.
14. Disconnect battery (WP 0857 00).
15. Disconnect P243 and P902 from left relay panel.
16. Install an insulated jumper wire between P243-Z and P902-g (simulates actuation of impact switch).
17. Connect battery (WP 0857 00).

**Indication/Condition**

Meter shall indicate 28 vdc.

**Corrective Action**

If result is not as specified, go to MULTIMETER HAS NO VOLTAGE INDICATION WHEN ACTUATION OF IMPACT SWITCH IS SIMULATED, in this work package.

**Step**

18. Disconnect battery (WP 0857 00).
19. Remove jumper wire from P243 and P902. Reconnect to left relay panel.
20. Disconnect multimeter leads from No. 1 main wire terminals.
21. If no other check is to be done, go to SHUTDOWN, in this work package.

**Indication/Condition**

Shutdown shall be performed.

**Corrective Action**

None Required

## **NO. 2 MAIN FIRE EXTINGUISHING BOTTLE OPERATIONAL/TROUBLESHOOTING PROCEDURE**

### **PROCEDURE**

#### **Step**

#### **CAUTION**

To prevent possible firing of a squib, never place multimeter leads on fire extinguisher bottle terminals.

#### **NOTE**

- When connecting multimeter leads to wiring terminals, connect positive lead to smaller terminal and negative lead to larger terminal. Always see Figure 1, Figure 2, and Figure 3 for proper connections.
- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Do SETUP, in this work package, if not already done.
2. Connect multimeter leads to wires labeled W3164A20 and W3165A20N.
3. Connect battery (WP 0857 00).
4. Turn on external electrical power.
5. Pull engine controls quadrant #2 ENG EMER OFF lever back.
6. Place FIRE EXTGH switch to MAIN, then release to OFF.

#### **Indication/Condition**

Multimeter shall indicate 28 vdc when switch is placed to MAIN.

#### **Corrective Action**

If result is not as specified, go to MULTIMETER CONNECTED TO MAIN FIRE BOTTLE WIRING HAS NO VOLTAGE INDICATION WHEN #2 ENG EMER OFF T-HANDLE IS USED AND FIRE EXTGH SWITCH IS PLACED TO MAIN, in this work package.

#### **Step**

7. Push #2 ENG EMER OFF lever forward (OFF position).

#### **NOTE**

- When performing operational check, move engine T-handle back or pull APU T-handle down as indicated by order, then place FIRE EXTGH switch to position indicated.
  - After completing each step, return all T-handles to the OFF position.
8. Perform a No. 2 Main Fire Extinguishing Bottle Priority check by performing the operational checks in NO. 2 MAIN FIRE EXTINGUISHING BOTTLE, in this work package.



**NO. 2 MAIN FIRE EXTINGUISHING BOTTLE OPERATIONAL/TROUBLESHOOTING PROCEDURE -**  
 Continued

**Table 2. No. 2 Main Fire Extinguishing Bottle.**

| <b>FIRE BOTTLE TESTED</b>                     | <b>STEP NUMBER</b> | <b>CHECK FIRST</b> | <b>CHECK SECOND</b> | <b>CHECK THIRD</b> | <b>FIRE EXTINGUISHER SWITCH POSITION</b> |
|-----------------------------------------------|--------------------|--------------------|---------------------|--------------------|------------------------------------------|
| #2 Main                                       | 1                  | #1 ENG             | #2 ENG              | -                  | MAIN                                     |
| #2 Main                                       | 2                  | APU                | #2 ENG              | -                  | MAIN                                     |
| #2 Main                                       | 3                  | APU                | #2 ENG              | #1 ENG*            | MAIN                                     |
| #2 Main                                       | 4                  | APU                | #1 ENG              | #2 ENG             | MAIN                                     |
| #2 Main                                       | 5                  | #1 ENG             | #2 ENG              | APU**              | MAIN                                     |
| *Pull engine T-handle back, then push forward |                    |                    |                     |                    |                                          |
| **Pull APU T-handle down, then push up        |                    |                    |                     |                    |                                          |

**Indication/Condition**

Multimeter shall indicate 28 vdc.

**Corrective Action**

If results are not as specified, replace right relay panel (WP 0825 00).

**Step**

9. Disconnect multimeter leads from No. 2 main wire terminals.
10. If no other check is to be done, go to SHUTDOWN, in this work package.

**Indication/Condition**

Shutdown shall be performed.

**Corrective Action**

None Required

## **NO. 1 RESERVE FIRE EXTINGUISHING BOTTLE OPERATIONAL/TROUBLESHOOTING PROCEDURE**

### **PROCEDURE**

#### **Step**

#### **CAUTION**

To prevent possible firing of a squib, never place multimeter leads on fire extinguisher bottle terminals.

#### **NOTE**

- When connecting multimeter leads to wiring terminals, connect positive lead to smaller terminal and negative lead to larger terminal. Always see Figure 1, Figure 2, and Figure 3 for proper connections.
- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Do SETUP, in this work package, if not already done.
2. Connect multimeter leads to wires labeled W3167A20 and W3166A20N.
3. Connect battery (WP 0857 00).
4. Turn on external electrical power.
5. Pull engine controls quadrant #1 ENG EMER OFF lever back.
6. Place FIRE EXTGH switch to RESERVE, then release to OFF.

#### **Indication/Condition**

Multimeter shall indicate 28 vdc when switch is placed to RESERVE.

#### **Corrective Action**

If result is not as specified, go to MULTIMETER CONNECTED TO NO. 1 RESERVE FIRE BOTTLE WIRING HAS NO VOLTAGE INDICATION WHEN #1 ENG EMER T-HANDLE IS USED AND FIRE EXTGH SWITCH IS PLACED TO RESERVE, in this work package.

#### **Step**

7. Push #1 ENG EMER OFF lever forward.

#### **NOTE**

- When performing operational check, move engine T-handle back or pull APU T-handle down as indicated by order, then place FIRE EXTGH switch to position indicated.
  - After completing each step, return all T-handles to the OFF position.
8. Perform a No. 1 Reserve Fire Extinguishing Bottle Priority check by performing the operational checks in NO. 1 RESERVE FIRE EXTINGUISHING BOTTLE, in this work package.

**NO. 1 RESERVE FIRE EXTINGUISHING BOTTLE OPERATIONAL/TROUBLESHOOTING PROCEDURE -**  
 Continued

**Table 3. No. 1 Reserve Fire Extinguishing Bottle.**

| <b>FIRE BOTTLE TESTED</b> | <b>STEP NUMBER</b> | <b>CHECK FIRST</b> | <b>CHECK SECOND</b> | <b>CHECK THIRD</b> | <b>FIRE EXTINGUISHER SWITCH POSITION</b> |
|---------------------------|--------------------|--------------------|---------------------|--------------------|------------------------------------------|
| #1 Reserve                | 1                  | #2 ENG*            | #1 ENG              | -                  | RESERVE                                  |
| #1 Reserve                | 2                  | APU                | #1 ENG              | -                  | RESERVE                                  |
| #1 Reserve                | 3                  | APU                | #1 ENG*             | -                  | RESERVE                                  |
| #1 Reserve                | 4                  | #1 ENG             | APU                 | -                  | RESERVE                                  |
| #1 Reserve                | 5                  | #2 ENG             | APU                 | -                  | RESERVE                                  |
| #1 Reserve                | 6                  | APU                | #2 ENG*             | #1 ENG             | RESERVE                                  |
| #1 Reserve                | 7                  | #1 ENG             | #2 ENG              | APU                | RESERVE                                  |

\*Pull engine T-handle back, then push forward.

**Indication/Condition**

Multimeter shall indicate 28 vdc.

**Corrective Action**

If results are not as specified, replace right relay panel (WP 0825 00).

**Step**

9. Pull APU T-handle down, and place FIRE EXTGH switch to RESERVE, then release to OFF.

**Indication/Condition**

Multimeter shall indicate 28 vdc when switch is placed to RESERVE.

**Corrective Action**

If result is not as specified, go to MULTIMETER CONNECTED TO NO. 1 RESERVE FIRE BOTTLE WIRING HAS NO VOLTAGE INDICATION WHEN APU T-HANDLE IS USED AND FIRE EXTGH SWITCH IS PLACED TO RESERVE, in this work package.

**Step**

10. Push APU T-handle up. Visually check directional control valve.

**Indication/Condition**

Sound of directional control valve deenergizing shall be heard and position indicator shall turn.

**Corrective Action**

If result is not as specified, replace directional control valve (WP 1200 00).

**Step**

11. Disconnect multimeter leads from No. 1 reserve wire terminals.

---

**NO. 1 RESERVE FIRE EXTINGUISHING BOTTLE OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**

12. If no other check is to be done, go to SHUTDOWN, in this work package.

**Indication/Condition**

Shutdown shall be performed.

**Corrective Action**

None Required

**NO. 2 RESERVE FIRE EXTINGUISHING BOTTLE OPERATIONAL/TROUBLESHOOTING PROCEDURE PROCEDURE****Step****CAUTION**

To prevent possible firing of a squib, never place multimeter leads on fire extinguisher bottle terminals.

**NOTE**

- When connecting multimeter leads to wiring terminals, connect positive lead to smaller terminal and negative lead to larger terminal. Always see Figure 1, Figure 2, and Figure 3 for proper connections.
- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Do SETUP, in this work package, if not already done.
2. **FIRE** Connect multimeter leads to wires marked W3168A16 and W3169A16N. ■
3. **W/O FIRE** Connect multimeter leads to wires marked W3168A20 and W3169A20N. ■
4. Connect battery (WP 0857 00).
5. Turn on external electrical power.
6. Pull engine controls quadrant #2 ENG EMER OFF lever back.
7. Place FIRE EXTGH switch to RESERVE, then release to OFF.

**Indication/Condition**

Multimeter shall indicate 28 vdc when switch is placed to RESERVE.

**Corrective Action**

If result is not as specified, go to MULTIMETER CONNECTED TO NO. 2 RESERVE FIRE BOTTLE WIRING HAS NO VOLTAGE INDICATION WHEN #2 ENG EMER OFF T-HANDLE IS USED AND FIRE EXTGH SWITCH IS PLACED TO RESERVE, in this work package.

**Step**

8. Push #2 ENG EMER OFF lever forward.

## NO. 2 RESERVE FIRE EXTINGUISHING BOTTLE OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

### NOTE

- When performing operational check, move engine T-handle back or pull APU T-handle down as indicated by order, then place FIRE EXTGH switch to position indicated.
  - After completing each step, return all T-handles to the OFF position.
9. Perform a No. 2 Reserve Fire Extinguishing Bottle Priority check by performing the operational checks in NO. 2 RESERVE FIRE EXTINGUISHING BOTTLE, in this work package.

**Table 4. No. 2 Reserve Fire Extinguishing Bottle.**

| FIRE BOTTLE TESTED | STEP NUMBER | CHECK FIRST | CHECK SECOND | CHECK THIRD | FIRE EXTINGUISHER SWITCH POSITION |
|--------------------|-------------|-------------|--------------|-------------|-----------------------------------|
| #2 Reserve         | 1           | #1 ENG      | #2 ENG       | -           | RESERVE                           |
| #2 Reserve         | 2           | APU         | #2 ENG       | -           | RESERVE                           |
| #2 Reserve         | 3           | APU         | #1 ENG       | #2 ENG      | RESERVE                           |

### Indication/Condition

Multimeter shall indicate 28 vdc.

### Corrective Action

If results are not as specified, replace right relay panel (WP 0825 00).

### Step

10. Turn off external electrical power.
11. Disconnect battery (WP 0857 00).
12. Disconnect P243 and P902 from left hand relay panel.
13. Install an insulated jumper wire across P243-Z and P902-g.
14. Connect battery (WP 0857 00).

### Indication/Condition

Multimeter shall indicate 28 vdc.

### Corrective Action

If result is not as specified, go to MULTIMETER HAS NO VOLTAGE INDICATION WHEN ACTUATION OF IMPACT SWITCH IS SIMULATED, in this work package.

### Step

15. Disconnect battery (WP 0857 00).
16. Remove jumper wires from P243 and P902.
17. Connect P243 and P902 to left relay panel.
18. Disconnect multimeter leads from No. 2 reserve wire terminals.

---

**NO. 2 RESERVE FIRE EXTINGUISHING BOTTLE OPERATIONAL/TROUBLESHOOTING PROCEDURE -**  
Continued

19. If no other check is to be done, go to SHUTDOWN, in this work package.

**Indication/Condition**

Shutdown shall be performed.

**Corrective Action**

None Required

**DIRECTIONAL CONTROL VALVE CHECK OPERATIONAL/TROUBLESHOOTING PROCEDURE**  
**PROCEDURE****Step****CAUTION**

To prevent possible firing of a squib, never place multimeter leads on fire extinguisher bottle terminals.

**NOTE**

- When connecting multimeter leads to wiring terminals, connect positive lead to smaller terminal and negative lead to larger terminal. Always see Figure 1, Figure 2, and Figure 3 for proper connections.
- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Do SETUP, in this work package, if not already done.
2. Connect battery (WP 0857 00).
3. Turn on external electrical power.
4. On upper console, pull down APU T-Handle while listening to directional control valve.

**Indication/Condition**

Sound of directional control valve being energized shall be heard and position indicator shall turn from neutral position to energized position.

**Corrective Action**

If result is not as specified, go to DIRECTIONAL CONTROL VALVE DOES NOT OPERATE WHEN APU T-HANDLE IS PULLED DOWN, in this work package.

**Step**

5. On upper console, push in APU T-Handle while listening to directional control valve.

**Indication/Condition**

Sound of directional control valve deenergizing shall be heard and position indicator shall pivot back to neutral position.

**Corrective Action**

If result is not as specified, go to DIRECTIONAL CONTROL VALVE DOES NOT OPERATE WHEN APU T-HANDLE IS PULLED DOWN, in this work package.

---

**DIRECTIONAL CONTROL VALVE CHECK OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Step**

6. If no other check is to be done, go to SHUTDOWN, in this work package.

**Indication/Condition**

Shutdown shall be performed.

**Corrective Action**

None Required

**SHUTDOWN OPERATIONAL/TROUBLESHOOTING PROCEDURE****PROCEDURE****Step****WARNING**

- With APU T-handle pushed up and #1 and #2 ENG EMER OFF levers pushed forward, check for no voltage on terminals before connecting wires to fire extinguisher containers. Failure to do so could result in injury to personnel.
- To prevent accidental firing of an explosive squib, observe all safety precautions regarding the handling and storage of explosives.

1. Turn off and disconnect external electrical power.
2. Disconnect battery (WP 0857 00).
3. Connect helicopter wiring to fire extinguishing containers (WP 1195 00).
4. Connect battery (WP 0857 00).

**Indication/Condition**

Battery shall be connected.

**Corrective Action**

None Required

**MULTIMETER CONNECTED TO NO. 1 MAIN FIRE BOTTLE WIRING HAS NO VOLTAGE INDICATION WHEN #1 ENG EMER OFF HANDLE IS USED AND FIRE EXTGH SWITCH IS PLACED TO MAIN****SYMPTOM**

Multimeter connected to No. 1 main fire bottle wiring has no voltage indication when #1 ENG EMER OFF handle is used and FIRE EXTGH switch is placed to MAIN.

**MALFUNCTION**

Multimeter connected to No. 1 main fire bottle wiring has no voltage indication when #1 ENG EMER OFF handle is used and FIRE EXTGH switch is placed to MAIN.

**CORRECTIVE ACTION**

1. Make sure #1 ENG EMER OFF handle is pulled back and FIRE EXTGH switch is placed to MAIN. Go to 2.

---

**MULTIMETER CONNECTED TO NO. 1 MAIN FIRE BOTTLE WIRING HAS NO VOLTAGE INDICATION WHEN #1 ENG EMER OFF HANDLE IS USED AND FIRE EXTGH SWITCH IS PLACED TO MAIN - Continued**

2. **Check for 28 vdc between terminal of wire removed from squib's power terminal stud and ground.**
  - a. If voltage is as specified, go to **3**.
  - b. If voltage is not as specified, go to **5**.
3. **Check continuity between:**
  - **J313-B and J900-M**
  - **J313-B and J900-N**
  - a. If continuity is present, go to **4**.
  - b. If continuity is not present, replace right relay panel (WP 0825 00). Go to **16**.
4. **Check continuity between P900-M and ground.**
  - a. If continuity is present, repair/replace wiring between P313-B and terminal of ground wire removed from squib's ground terminal (WP 1747 00). Go to **16**.
  - b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **16**.
5. **Check continuity between wire terminal 1 of the No. 1 main squib and P313-C.**
  - a. If continuity is present, go to **6**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **16**.
6. **Check continuity between #1 ENG EMER OFF T-handle switch terminals C and NC.**
  - a. If continuity is present, go to **7**.
  - b. If continuity is not present, replace #1 ENG EMER OFF T-handle (WP 0501 00). Go to **16**.
7. **Check for 28 vdc between #1 ENG EMER OFF T-handle switch terminal C and ground.**
  - a. If voltage is as specified, go to **8**.
  - b. If voltage is not as specified, go to **15**.
8. **Check for 28 vdc between FIRE EXTGH switch terminal 4 and ground.**
  - a. If voltage is as specified, go to **9**.
  - b. If voltage is not as specified, go to **13**.
9. **Check for 28 vdc between P900-U and ground.**
  - a. If voltage is as specified, go to **10**.
  - b. If voltage is not as specified, repair/replace wiring between P900-U and of FIRE EXTGH switch terminal 4 (WP 1747 00). Go to **16**.
10. **Check helicopter configuration.**
  - a. **WINTER** , go to **11**.
  - b. **W/O WINTER** , go to **12**.
11. **WINTER Check for 28 vdc between P244B-P and ground.**
  - a. If voltage is as specified, replace right relay panel (WP 0825 00). Go to **16**.
  - b. If voltage is not as specified, replace winterization kit (WP 1400 00). Go to **16**.



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**MULTIMETER CONNECTED TO NO. 1 MAIN FIRE BOTTLE WIRING HAS NO VOLTAGE INDICATION WHEN #1 ENG EMER OFF HANDLE IS USED AND FIRE EXTGH SWITCH IS PLACED TO MAIN - Continued**

- c. If trouble remains, go to **12**.
- 12. Check for 28 vdc between P244-P and ground.**
  - a. If voltage is as specified, replace right relay panel (WP 0825 00). Go to **16**.
  - b. If voltage is not as specified, repair/replace wiring between #1 ENG EMER OFF T-handle switch terminal NC and P244-P (WP 1747 00). Go to **16**.
- 13. Check for 28 vdc between of FIRE EXTGH circuit breaker terminal 1 (pilot's circuit breaker panel) and ground.**
  - a. If voltage is as specified, go to **14**.
  - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **16**.
- 14. Check continuity between FIRE EXTGH circuit breaker terminal 1 (pilot's circuit breaker panel) and FIRE EXTGH switch terminal 5.**
  - a. If continuity is present, replace switch (WP 1207 00). Go to **16**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **16**.
- 15. Check for 28 vdc between FIRE EXTGH circuit breaker terminal 1 (lower console circuit breaker panel) and ground.**
  - a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and terminal C of #1 ENG EMER OFF T-handle (WP 1747 00). Go to **16**.
  - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **16**.
- 16. Procedure completed.**

**DIRECTIONAL CONTROL VALVE DOES NOT OPERATE WHEN NO. 1 MAIN IS USED FOR APU**

**SYMPTOM**

Directional control valve does not operate when NO. 1 MAIN is used for APU.

**MALFUNCTION**

Directional control valve does not operate when NO. 1 MAIN is used for APU.

**CORRECTIVE ACTION**

- 1. Pull APU T-handle down and place FIRE EXTGH switch to MAIN. Go to 2.**
- 2. Check for 28 vdc between directional control valve terminals 1 and 2.**
  - a. If voltage is as specified, replace directional control valve (WP 1200 00). Go to **7**.
  - b. If voltage is not as specified, go to **3**.
- 3. Check continuity between directional control valve terminal 2 and ground.**
  - a. If continuity is present, go to **4**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **7**.
- 4. Check helicopter configuration.**
  - a. **WINTER** , go to **5**.
  - b. **W/O WINTER** , go to **6**.
- 5. **WINTER** Check for 28 vdc between J244A-N and ground.**

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**DIRECTIONAL CONTROL VALVE DOES NOT OPERATE WHEN NO. 1 MAIN IS USED FOR APU - Continued**

- a. If voltage is as specified, replace winterization kit (WP 1400 00). Go to **7**.
  - b. If trouble remains, repair/replace wiring between P244-N and directional control valve terminal 1 (WP 1747 00). Go to **7**.
  - c. If voltage is not as specified, go to MULTIMETER CONNECTED TO NO. 1 MAIN FIRE BOTTLE WIRING HAS NO VOLTAGE INDICATION WHEN APU T-HANDLE IS USED AND FIRE EXTGH SWITCH IS PLACED TO MAIN, in this work package.
- 6. W/O WINTER Check for 28 vdc between J244-N and ground.**
- a. If voltage is as specified, repair/replace wiring between P244-N and directional control valveterminal 1 (WP 1747 00). Go to **7**.
  - b. If voltage is not as specified, go to MULTIMETER CONNECTED TO NO. 1 MAIN FIRE BOTTLE WIRING HAS NO VOLTAGE INDICATION WHEN APU T-HANDLE IS USED AND FIRE EXTGH SWITCH IS PLACED TO MAIN, in this work package.
- 7. Procedure completed.**

**MULTIMETER CONNECTED TO NO. 1 MAIN FIRE BOTTLE WIRING HAS NO VOLTAGE INDICATION WHEN APU T-HANDLE IS USED AND FIRE EXTGH SWITCH IS PLACED TO MAIN****SYMPTOM**

Multimeter connected to No. 1 main fire bottle wiring has no voltage indication when APU T-handle is used and FIRE EXTGH switch is placed to MAIN.

**MALFUNCTION**

Multimeter connected to No. 1 main fire bottle wiring has no voltage indication when APU T-handle is used and FIRE EXTGH switch is placed to MAIN.

**CORRECTIVE ACTION**

- 1. Make sure APU T-handle is pulled down and FIRE EXTGH switch is placed to MAIN. Go to 2.**
- 2. Check for 28 vdc between P900-X and ground.**
  - a. If voltage is as specified, go to **3**.
  - b. If voltage is not as specified, go to **5**.
- 3. Check for 28 vdc between P900-T and ground.**
  - a. If voltage is as specified, replace right relay panel (WP 0825 00). Go to **7**.
  - b. If voltage is not as specified, go to **4**.
- 4. Check for 28 vdc between FIRE EXTGH switch terminal 1 and ground.**
  - a. If voltage is as specified, repair/replace wiring between switch terminal 1 and P900-T (WP 1747 00). Go to **7**.
  - b. If voltage is not as specified, replace switch (WP 1203 00). Go to **7**.
- 5. Check for 28 vdc between APU T-handle terminal NO and ground.**
  - a. If voltage is as specified, repair/replace wiring between APU T-handle terminal NO and P900-X (WP 1747 00). Go to **7**.
  - b. If voltage is not as specified, go to **6**.

---

**MULTIMETER CONNECTED TO NO. 1 MAIN FIRE BOTTLE WIRING HAS NO VOLTAGE INDICATION WHEN APU T-HANDLE IS USED AND FIRE EXTGH SWITCH IS PLACED TO MAIN - Continued****6. Check for 28 vdc between APU T-handle terminal C and ground.**

- a. If voltage is as specified, replace T-handle (WP 1201 00). Go to **7**.
- b. If voltage is not as specified, repair/replace wiring between T-handle terminal C and FIRE EXTGH switch terminal 2 (WP 1747 00). Go to **7**.

**7. Procedure completed.****MULTIMETER HAS NO VOLTAGE INDICATION WHEN ACTUATION OF IMPACT SWITCH IS SIMULATED****SYMPTOM**

Multimeter has no voltage indication when actuation of impact switch is simulated.

**MALFUNCTION**

Multimeter has no voltage indication when actuation of impact switch is simulated.

**CORRECTIVE ACTION****1. Place an insulated jumper wire between P902-g and P243-Z. Go to 2.****2. Check for 28 vdc between P900-S and ground.**

- a. If voltage is as specified, replace right relay panel (WP 0825 00). Go to **6**.
- b. If voltage is not as specified, go to **3**.

**3. Remove jumper wire from P902 and P243. Go to 4.****4. Check continuity between P900-S and P243-Z.**

- a. If continuity is present, go to **5**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **6**.

**5. Check for 28 vdc between P902-g and ground.**

- a. If voltage is as specified, replace left relay panel (WP 0825 00). Go to **6**.
- b. If voltage is not as specified, repair/replace wiring between P902-g and FIRE EXTGH circuit breaker terminal 1 (lower console circuit breaker panel) (WP 1747 00). Go to **6**.

**6. Procedure completed.****MULTIMETER CONNECTED TO NO. 2 MAIN FIRE BOTTLE WIRING HAS NO VOLTAGE INDICATION WHEN #2 ENG EMER OFF T-HANDLE IS USED AND FIRE EXTGH SWITCH IS PLACED TO MAIN****SYMPTOM**

Multimeter connected to No. 2 main fire bottle wiring has no voltage indication when #2 ENG EMER OFF T-handle is used and FIRE EXTGH switch is placed to MAIN.

**MALFUNCTION**

Multimeter connected to No. 2 main fire bottle wiring has no voltage indication when #2 ENG EMER OFF t-handle is used and FIRE EXTGH switch is placed to MAIN.

**CORRECTIVE ACTION**

- 1. Make sure #2 ENG EMER OFF T-handle is pulled back and FIRE EXTGH switch is placed to MAIN. Go to 2.**

---

**MULTIMETER CONNECTED TO NO. 2 MAIN FIRE BOTTLE WIRING HAS NO VOLTAGE INDICATION WHEN #2 ENG EMER OFF T-HANDLE IS USED AND FIRE EXTGH SWITCH IS PLACED TO MAIN -**

Continued

- 2. Check for 28 vdc between terminal of wire removed from squib's power terminal stud and ground.**
  - a. If voltage is as specified, go to **3**.
  - b. If voltage is not as specified, go to **4**.
- 3. Check continuity between P313-E and terminal of ground wire removed from squib's ground terminal.**
  - a. If continuity is present, replace right relay panel (WP 0825 00). Go to **8**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **8**.
- 4. Check continuity between wire terminal from squib and P313-D.**
  - a. If continuity is present, go to **5**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **8**.
- 5. Check continuity between #2 ENG EMER OFF T-handle terminals C and NC.**
  - a. If continuity is present, go to **6**.
  - b. If continuity is not present, replace T-handle (WP 0501 00). Go to **8**.
- 6. Check for 28 vdc between #2 ENG EMER OFF T-handle terminal C and ground.**
  - a. If voltage is as specified, go to **7**.
  - b. If voltage is not as specified, repair/replace wiring between FIRE EXTGH switch terminal 5 and T-handle terminal C (WP 1747 00). Go to **8**.
- 7. Check for 28 vdc between P900-P and ground.**
  - a. If voltage is as specified, replace right relay panel (WP 0825 00). Go to **8**.
  - b. If voltage is not as specified, repair/replace wiring between #2 ENG EMER OFF T-handle terminal NC and P900-P (WP 1747 00). Go to **8**.
- 8. Procedure completed.**

**MULTIMETER CONNECTED TO NO. 1 RESERVE FIRE BOTTLE WIRING HAS NO VOLTAGE INDICATION WHEN #1 ENG EMER OFF T-HANDLE IS USED AND FIRE EXTGH SWITCH IS PLACED TO RESERVE****SYMPTOM**

Multimeter connected to No. 1 reserve fire bottle wiring has no voltage indication when #1 ENG EMER OFF T-handle is used and FIRE EXTGH switch is placed to RESERVE.

**MALFUNCTION**

Multimeter connected to No. 1 reserve fire bottle wiring has no voltage indication when #1 ENG EMER OFF T-handle is used and FIRE EXTGH switch is placed to RESERVE.

**CORRECTIVE ACTION**

- 1. Make sure #1 ENG EMER OFF T-handle is pulled back and FIRE EXTGH switch is placed to RESERVE. Go to 2.**
- 2. Check for 28 vdc between terminal of wire removed from squib's power terminal stud and ground.**
  - a. If voltage is as specified, go to **3**.

**MULTIMETER CONNECTED TO NO. 1 RESERVE FIRE BOTTLE WIRING HAS NO VOLTAGE INDICATION WHEN #1 ENG EMER OFF T-HANDLE IS USED AND FIRE EXTGH SWITCH IS PLACED TO RESERVE -**  
Continued

- b. If voltage is not as specified, go to **4**.
- 3. Check continuity between P313-F and terminal of ground wire removed from squib's ground terminal.**
  - a. If continuity is present, replace right relay panel (WP 0825 00). Go to **8**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **8**.
- 4. Check continuity between squib's wire terminal and P313-G.**
  - a. If continuity is present, go to **5**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **8**.
- 5. Check for 28 vdc between FIRE EXTGH switch terminal 3 and ground.**
  - a. If voltage is as specified, go to **6**.
  - b. If voltage is not as specified, go to **7**.
- 6. Check for 28 vdc between P900-W and ground.**
  - a. If voltage is as specified, replace right relay panel (WP 0825 00). Go to **8**.
  - b. If voltage is not as specified, repair/replace wiring between P900-W and switch (WP 1747 00). Go to **8**.
- 7. Check for 28 vdc between FIRE EXTGH switch terminal 2 and ground.**
  - a. If voltage is as specified, replace switch (WP 1207 00). Go to **8**.
  - b. If voltage is not as specified, repair/replace wiring between switch terminal 2 and FIRE EXTGH circuit breaker terminal 1 (lower console circuit breaker panel) (WP 1747 00). Go to **8**.
- 8. Procedure completed.**

**MULTIMETER CONNECTED TO NO. 1 RESERVE FIRE BOTTLE WIRING HAS NO VOLTAGE INDICATION WHEN APU T-HANDLE IS USED AND FIRE EXTGH SWITCH IS PLACED TO RESERVE**

#### **SYMPTOM**

Multimeter connected to No. 1 reserve fire bottle wiring has no voltage indication when APU T-handle is used and FIRE EXTGH switch is placed to RESERVE.

#### **MALFUNCTION**

Multimeter connected to No. 1 reserve fire bottle wiring has no voltage indication when APU T-handle is used and FIRE EXTGH switch is placed to RESERVE.

#### **CORRECTIVE ACTION**

- 1. Make sure APU T-handle is pulled down and FIRE EXTGH switch is placed to RESERVE. Go to 2.**
- 2. Check for 28 vdc between P900-V and ground.**
  - a. If voltage is as specified, replace right relay panel (WP 0825 00). Go to **4**.
  - b. If voltage is not as specified, go to **3**.
- 3. Check for 28 vdc between FIRE EXTGH switch terminal 6 and ground.**
  - a. If voltage is as specified, repair/replace wiring between switch terminal 6 and P900-V (WP 1747 00). Go to **4**.

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**MULTIMETER CONNECTED TO NO. 1 RESERVE FIRE BOTTLE WIRING HAS NO VOLTAGE INDICATION WHEN APU T-HANDLE IS USED AND FIRE EXTGH SWITCH IS PLACED TO RESERVE - Continued**

- b. If voltage is not as specified, replace switch (WP 1207 00). Go to **4**.

**4. Procedure completed.****MULTIMETER CONNECTED TO NO. 2 RESERVE FIRE BOTTLE WIRING HAS NO VOLTAGE INDICATION WHEN #2 ENG EMER OFF T-HANDLE IS USED AND FIRE EXTGH SWITCH IS PLACED TO RESERVE****SYMPTOM**

Multimeter connected to No. 2 reserve fire bottle wiring has no voltage indication when #2 ENG EMER OFF T-handle is used and FIRE EXTGH switch is placed to RESERVE.

**MALFUNCTION**

Multimeter connected to No. 2 reserve fire bottle wiring has no voltage indication when #2 ENG EMER OFF T-handle is used and FIRE EXTGH switch is placed to RESERVE.

**CORRECTIVE ACTION**

- 1. Make sure #2 ENG EMER OFF T-handle is pulled back and FIRE EXTGH switch placed to RESERVE. Go to 2.**
- 2. Check for 28 vdc between terminal of wire removed from squib's power terminal stud and ground.**
  - a. If voltage is as specified, go to **3**.
  - b. If voltage is not as specified, go to **6**.
- 3. Check helicopter configuration.**
  - a. **FIRE** , go to **4**.
  - b. **W/O FIRE** , go to **5**.
- 4. FIRE Check continuity between P313-A and terminal of ground wire removed from squib's ground terminal.**
  - a. If continuity is present, replace right relay panel (WP 0825 00). Go to **7**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **7**.
- 5. W/O FIRE Check continuity between P313-J and terminal ground wire removed from squib's ground terminal.**
  - a. If continuity is present, replace right relay panel (WP 0825 00). Go to **7**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **7**.
- 6. Check continuity between squib's wire terminal and P313-H.**
  - a. If continuity is present, replace right relay panel (WP 0825 00). Go to **7**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **7**.
- 7. Procedure completed.**

**DIRECTIONAL CONTROL VALVE DOES NOT OPERATE WHEN APU T-HANDLE IS PULLED DOWN****SYMPTOM**

Directional control valve does not operate when APU T-handle is pulled down.

**MALFUNCTION**

Directional control valve does not operate when APU T-handle is pulled down.

**CORRECTIVE ACTION**

- 1. Check for 28 vdc between directional control valve terminals 1 and 2.**
  - a. If voltage is as specified, replace directional control valve (WP 1200 00). Go to **11**.
  - b. If voltage is not as specified, go to **2**.
- 2. Check continuity between directional control valve terminal 2 and ground.**
  - a. If continuity is present, go to **3**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **11**.
- 3. Check helicopter configuration.**
  - a. **WINTER** , go to **4**.
  - b. **W/O WINTER** , go to **5**.
- 4. WINTER Check for 28 vdc between J244A-N and ground.**
  - a. If voltage is as specified, replace winterization kit (WP 1400 00). Go to **11**.
  - b. If trouble remains, repair/replace wiring between P244-N and directional control valve terminal 1 (WP 1747 00). Go to **11**.
  - c. If voltage is not as specified, go to **6**.
- 5. W/O WINTER Check for 28 vdc between J244-N and ground.**
  - a. If voltage is as specified, repair/replace wiring between P244-N and directional control valve terminal 1 (WP 1747 00). Go to **11**.
  - b. If voltage is not as specified, go to **6**.
- 6. Check for 28 vdc between P900-X and ground.**
  - a. If voltage is as specified, go to **7**.
  - b. If voltage is not as specified, go to **9**.
- 7. Check for 28 vdc between P900-T and ground.**
  - a. If voltage is as specified, replace right relay panel (WP 0825 00). Go to **11**.
  - b. If voltage is not as specified, go to **8**.
- 8. Check for 28 vdc between FIRE EXTGH switch terminal 1 and ground.**
  - a. If voltage is as specified, repair/replace wiring between switch terminal 1 and P900-T (WP 1747 00). Go to **11**.
  - b. If voltage is not as specified, replace switch (WP 1203 00). Go to **11**.
- 9. Check for 28 vdc between APU T-handle terminal NO and ground.**

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**DIRECTIONAL CONTROL VALVE DOES NOT OPERATE WHEN APU T-HANDLE IS PULLED DOWN -**  
Continued

- a. If voltage is as specified, repair/replace wiring between APU T-handle terminal NO and P900-X (WP 1747 00). Go to **11**.
- b. If voltage is not as specified, go to **10**.

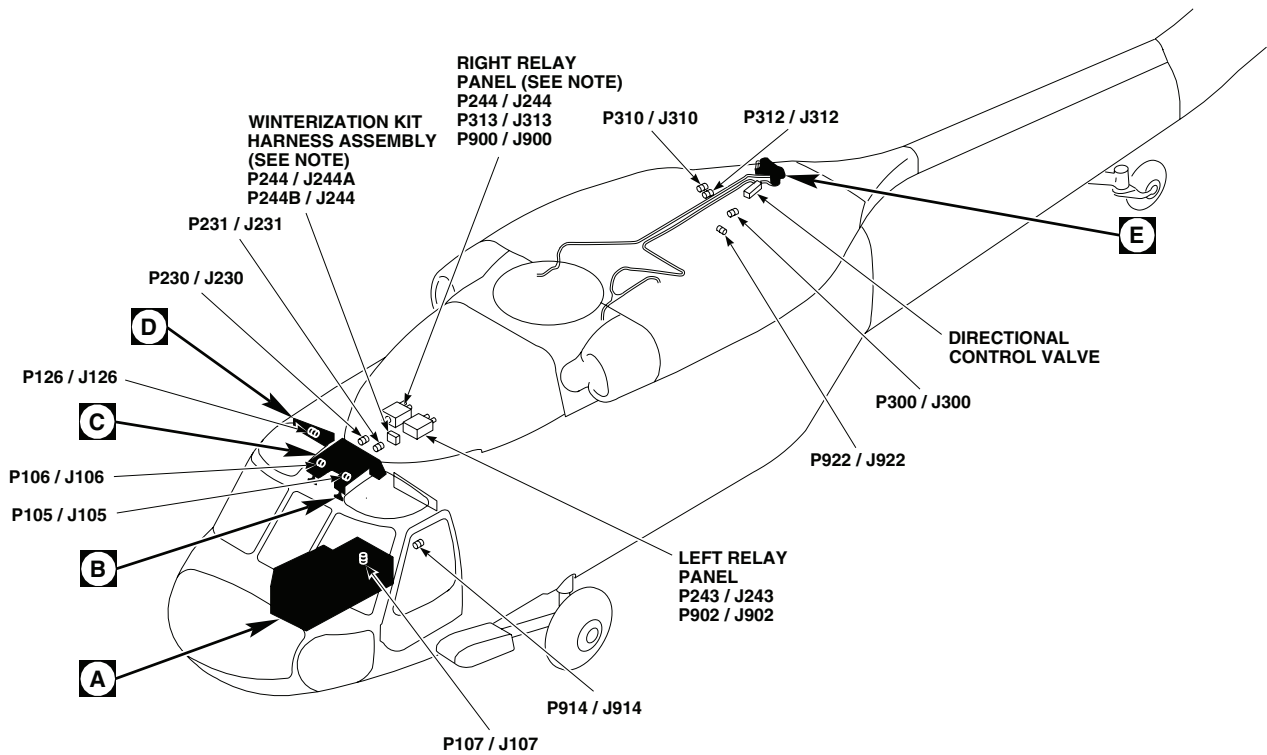
**10. Check for 28 vdc between APU T-handle terminal C and ground.**

- a. If voltage is as specified, replace T-handle (WP 1201 00). Go to **11**.
- b. If voltage is not as specified, repair/replace wiring between T-handle terminal C and FIRE EX-TGH switch terminal 2 (WP 1747 00). Go to **11**.

**11. Procedure completed.**



LOCATION



NOTE

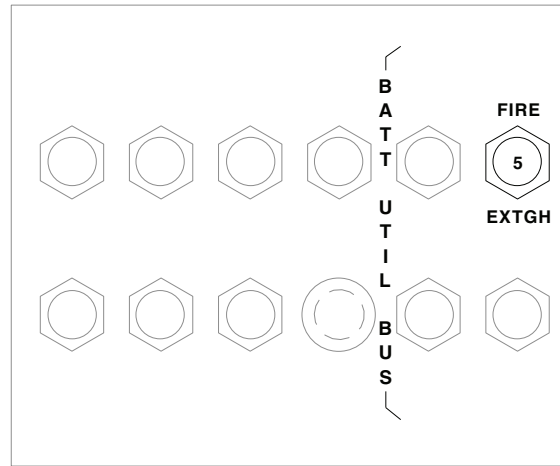
WINTER

| TERMINAL BOARD /<br>DISCONNECT PLUG /<br>RECEPTACLE | LOCATION /<br>CONNECTION POINT                      |
|-----------------------------------------------------|-----------------------------------------------------|
| P105 / J105                                         | NO. 1 ENGINE CONTROL<br>QUADRANT                    |
| P106 / J106                                         | NO. 2 ENGINE CONTROL<br>QUADRANT                    |
| P107 / J107                                         | BATT BUS PANEL                                      |
| P126 / J126                                         | COCKPIT CEILING<br>BL 28 RH, STA 243                |
| P230 / J230                                         | CABIN CEILING<br>BL 8 RH, STA 247                   |
| P231 / J231                                         | CABIN CEILING<br>BL 9 LH, STA 247                   |
| P243 / J243                                         | LEFT RELAY PANEL<br>ASSEMBLY                        |
| P244 / J244                                         | RH RELAY PANEL<br>ASSEMBLY (SEE NOTE)               |
| P244 / J244A<br>P244B / J244                        | WINTERIZATION KIT<br>HARNESS ASSEMBLY<br>(SEE NOTE) |

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Figure 1. Fire Extinguishing System Location Diagram. (Sheet 1 of 4)

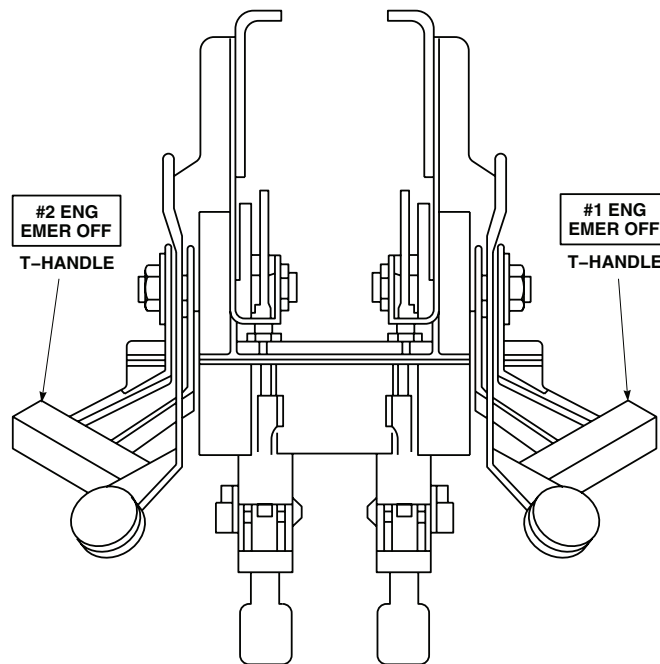
LOCATION - Continued



LOWER CONSOLE  
CIRCUIT BREAKER PANEL



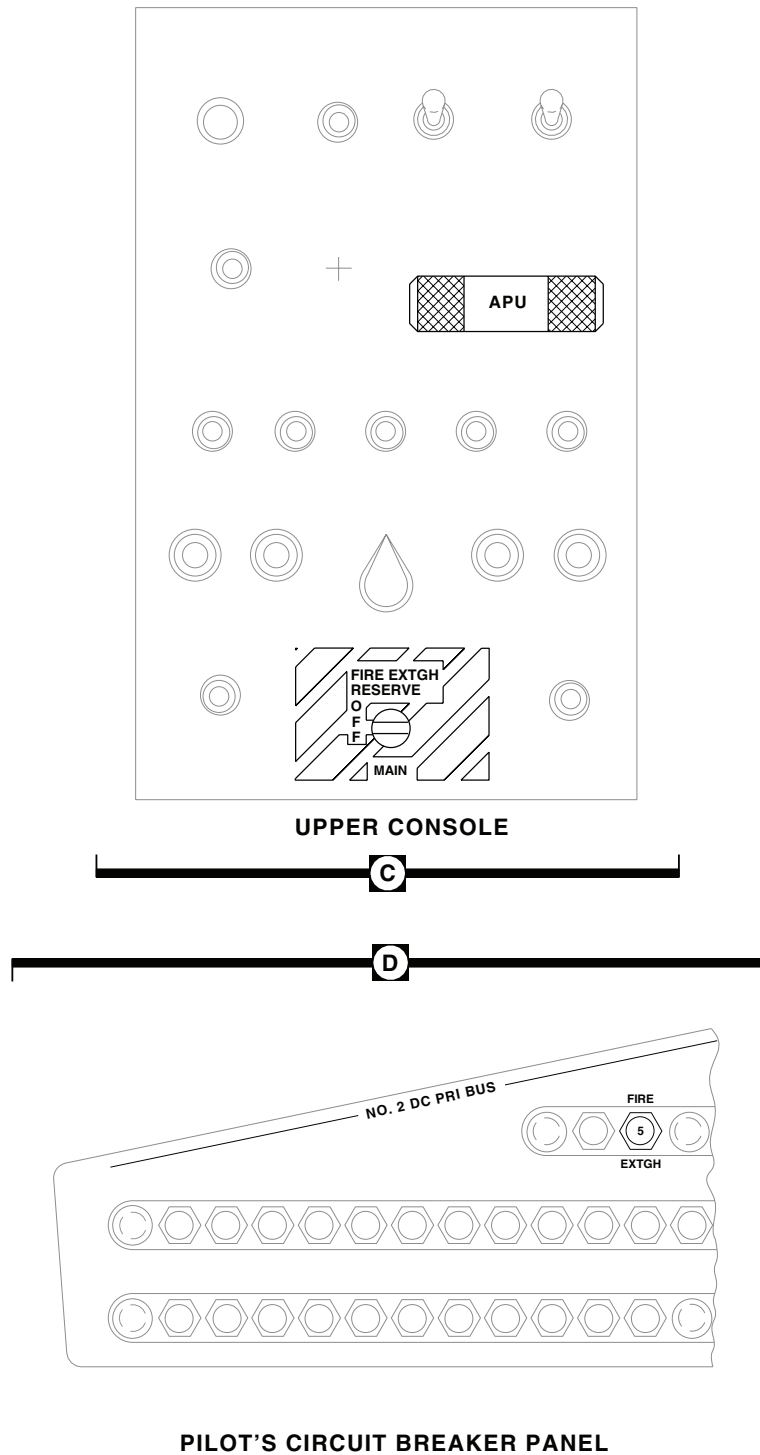
| TERMINAL BOARD/<br>DISCONNECT PLUG/<br>RECEPTACLE | LOCATION/<br>CONNECTION POINT         |
|---------------------------------------------------|---------------------------------------|
| P300 / J300                                       | TRANSITION DECK<br>BL 15 RH, STA 415  |
| P310 / J310                                       | CABIN CEILING<br>BL 17 RH, STA 387    |
| P312 / J312                                       | CABIN CEILING<br>BL 16.50 LH, STA 383 |
| P313 / J313                                       | RIGHT RELAY PANEL<br>ASSEMBLY         |
| P900 / J900                                       | RIGHT RELAY PANEL<br>ASSEMBLY         |
| P902 / J902                                       | LEFT RELAY PANEL<br>ASSEMBLY          |
| P914 / J914                                       | COCKPIT<br>BL 24 LH, STA 247          |
| P922 / J922                                       | CABIN CEILING<br>BL 10.7 LH, STA 380  |



AA2794\_2A  
SA

Figure 1. Fire Extinguishing System Location Diagram. (Sheet 2 of 4)

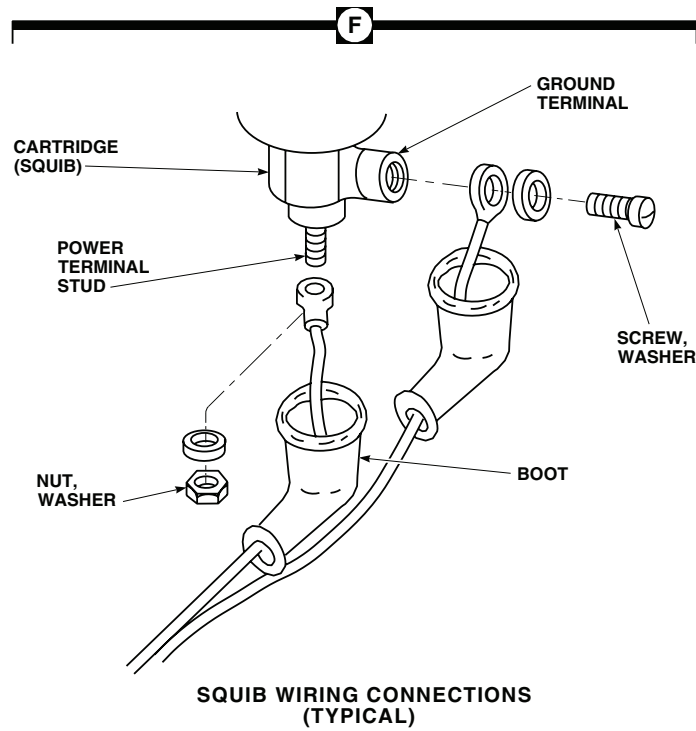
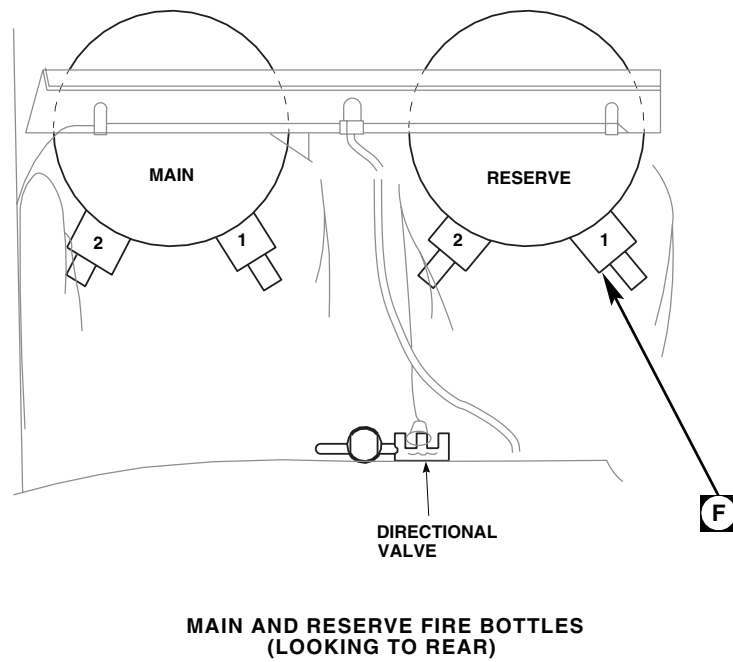
LOCATION - Continued



AA2794\_3A  
SA

Figure 1. Fire Extinguishing System Location Diagram. (Sheet 3 of 4)

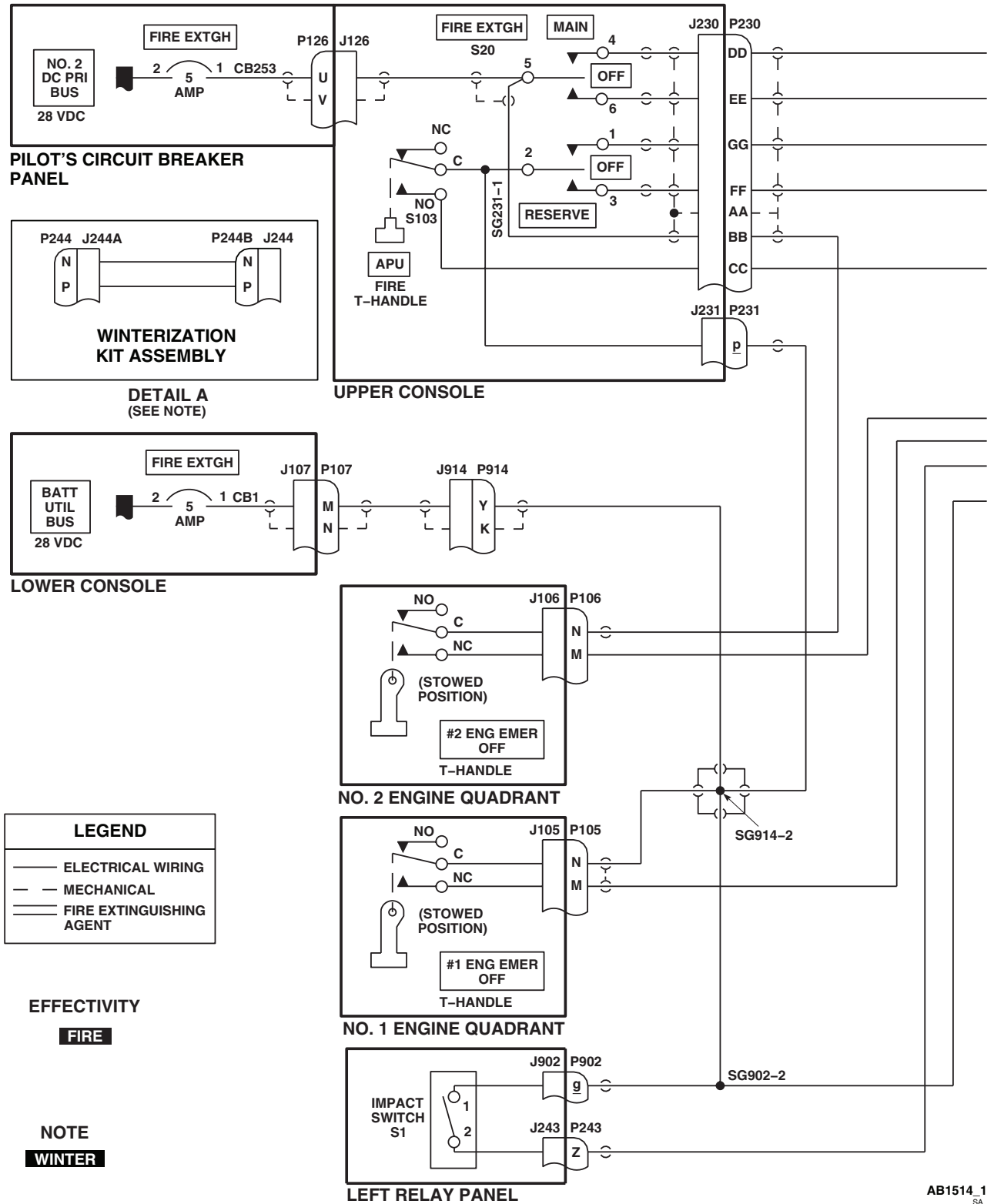
LOCATION - Continued



AA2794\_4  
SA

Figure 1. Fire Extinguishing System Location Diagram. (Sheet 4 of 4)

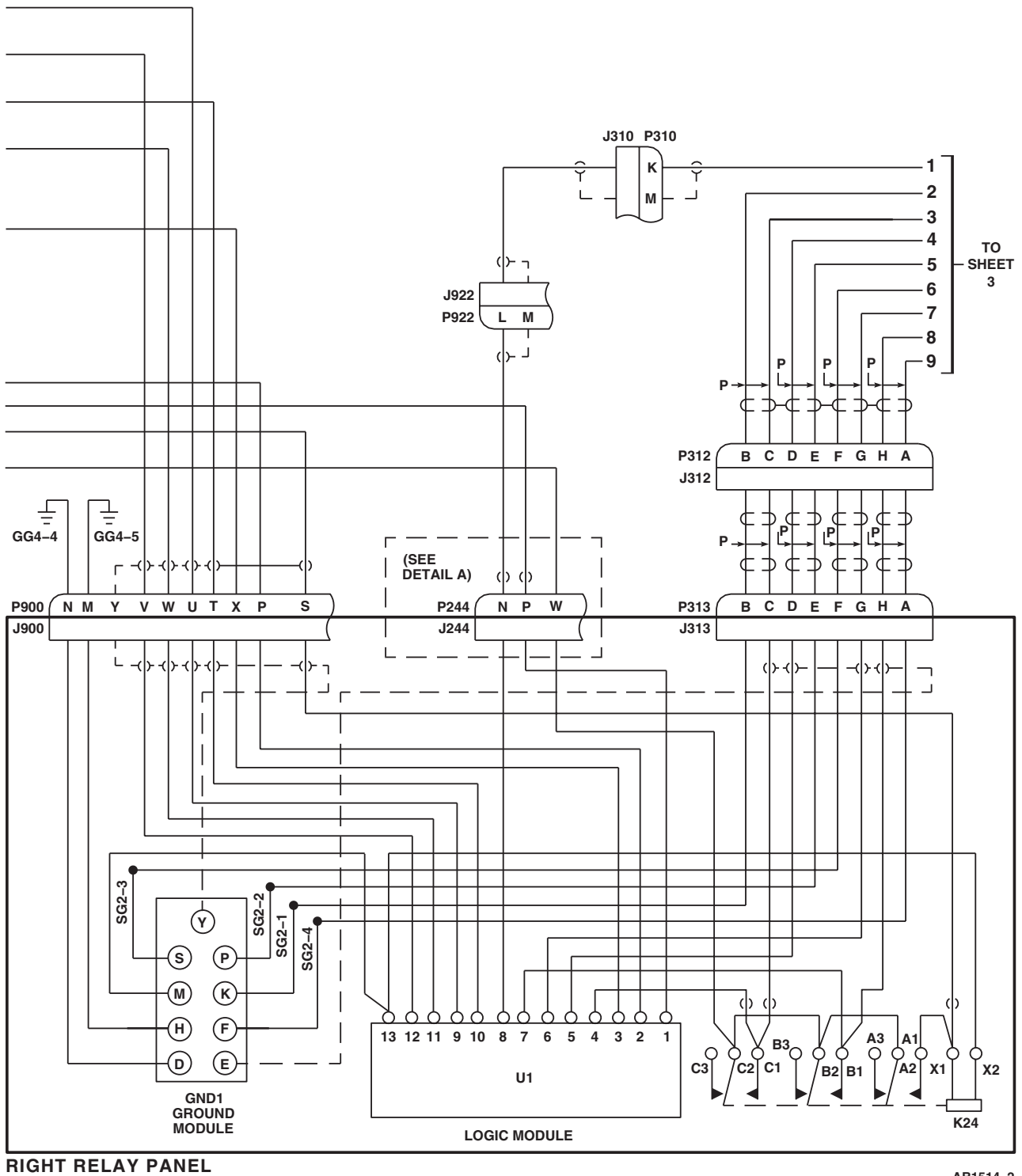
SCHEMATICS



AB1514\_1  
SA

Figure 2. Fire Extinguishing System Schematic Diagram. (Sheet 1 of 4)

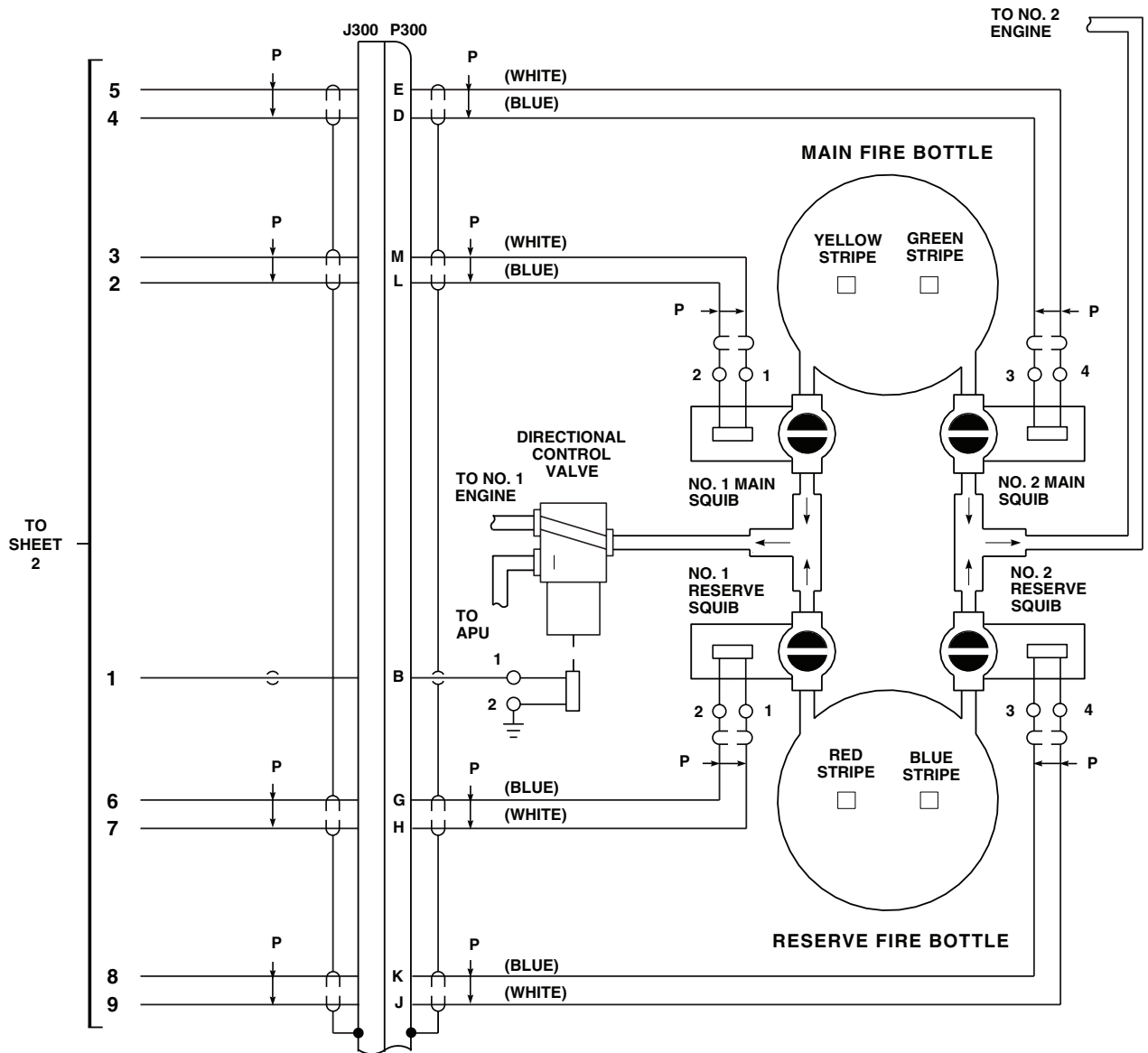
SCHEMATICS - Continued



AB1514.2  
SA

Figure 2. Fire Extinguishing System Schematic Diagram. (Sheet 2 of 4)

SCHEMATICS - Continued



AB1514\_3  
SA

Figure 2. Fire Extinguishing System Schematic Diagram. (Sheet 3 of 4)

## SCHEMATICS - Continued

| FIRE BOTTLE SQUIB<br>WIRING CONNECTION CHART |                                |                       |                              |                           |
|----------------------------------------------|--------------------------------|-----------------------|------------------------------|---------------------------|
| FIRE<br>BOTTLE<br>SQUIB                      | WIRE CODE<br>(PRINTED ON WIRE) | WIRE<br>COLOR<br>CODE | CONNECTION<br>POINT TO SQUIB | FIRE BOTTLE<br>COLOR CODE |
| MAIN<br>(NO. 1)                              | W3163A16<br>W3162A16N          | WHITE<br>BLUE         | #6 STUD<br>#8 SCREW          | YELLOW STRIPE             |
| MAIN<br>(NO. 2)                              | W3164A20<br>W3165A20N          | BLUE<br>WHITE         | #6 STUD<br>#8 SCREW          | GREEN STRIPE              |
| RESERVE<br>(NO. 1)                           | W3167A20<br>W3166A20N          | WHITE<br>BLUE         | #6 STUD<br>#8 SCREW          | RED STRIPE                |
| RESERVE<br>(NO. 2)                           | W3168A16<br>W3169A16N          | BLUE<br>WHITE         | #6 STUD<br>#8 SCREW          | BLUE STRIPE               |

Figure 2. Fire Extinguishing System Schematic Diagram. (Sheet 4 of 4)



## SCHEMATICS - Continued

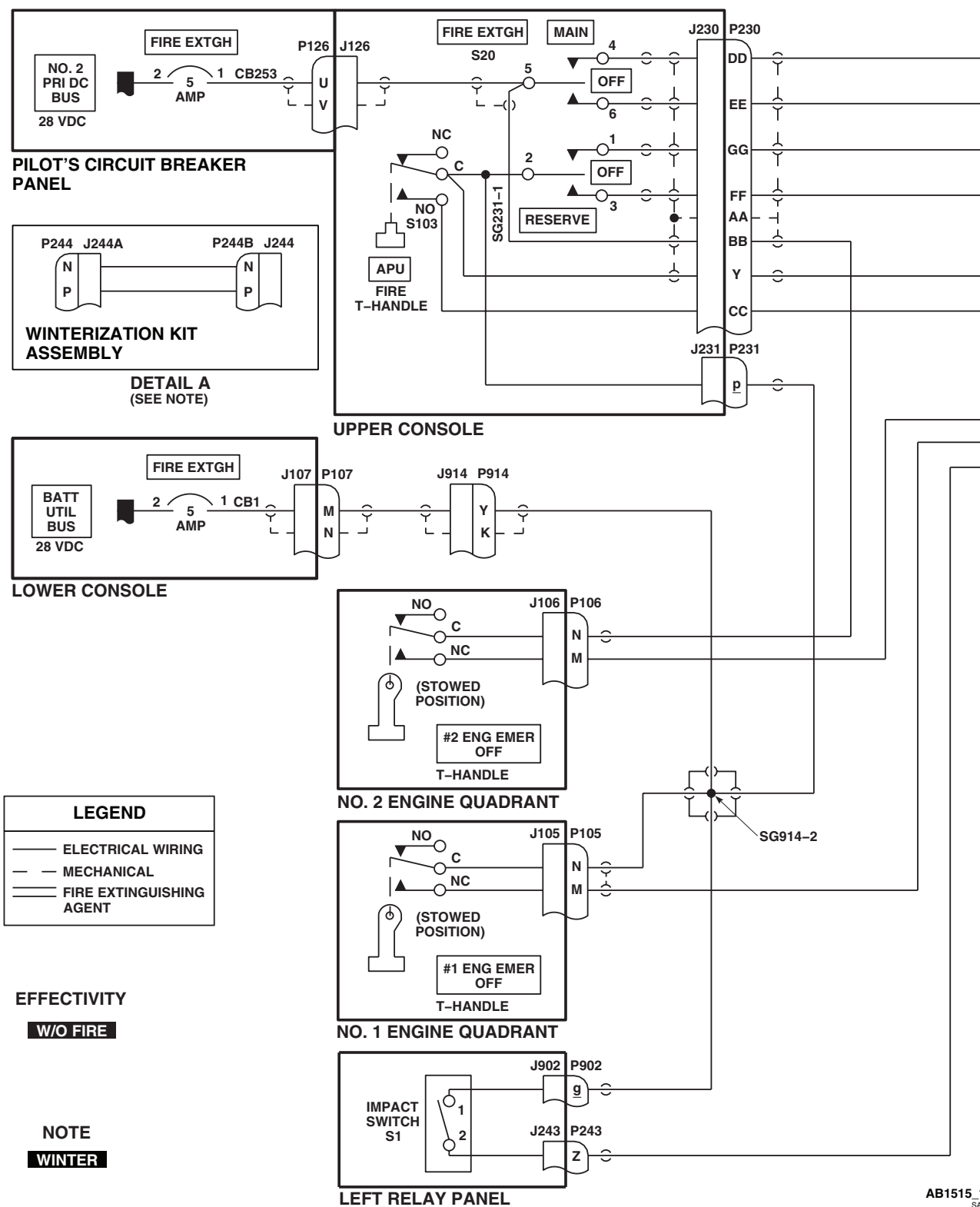


Figure 3. Fire Extinguishing System Schematic Diagram. (Sheet 1 of 4)

SCHEMATICS - Continued

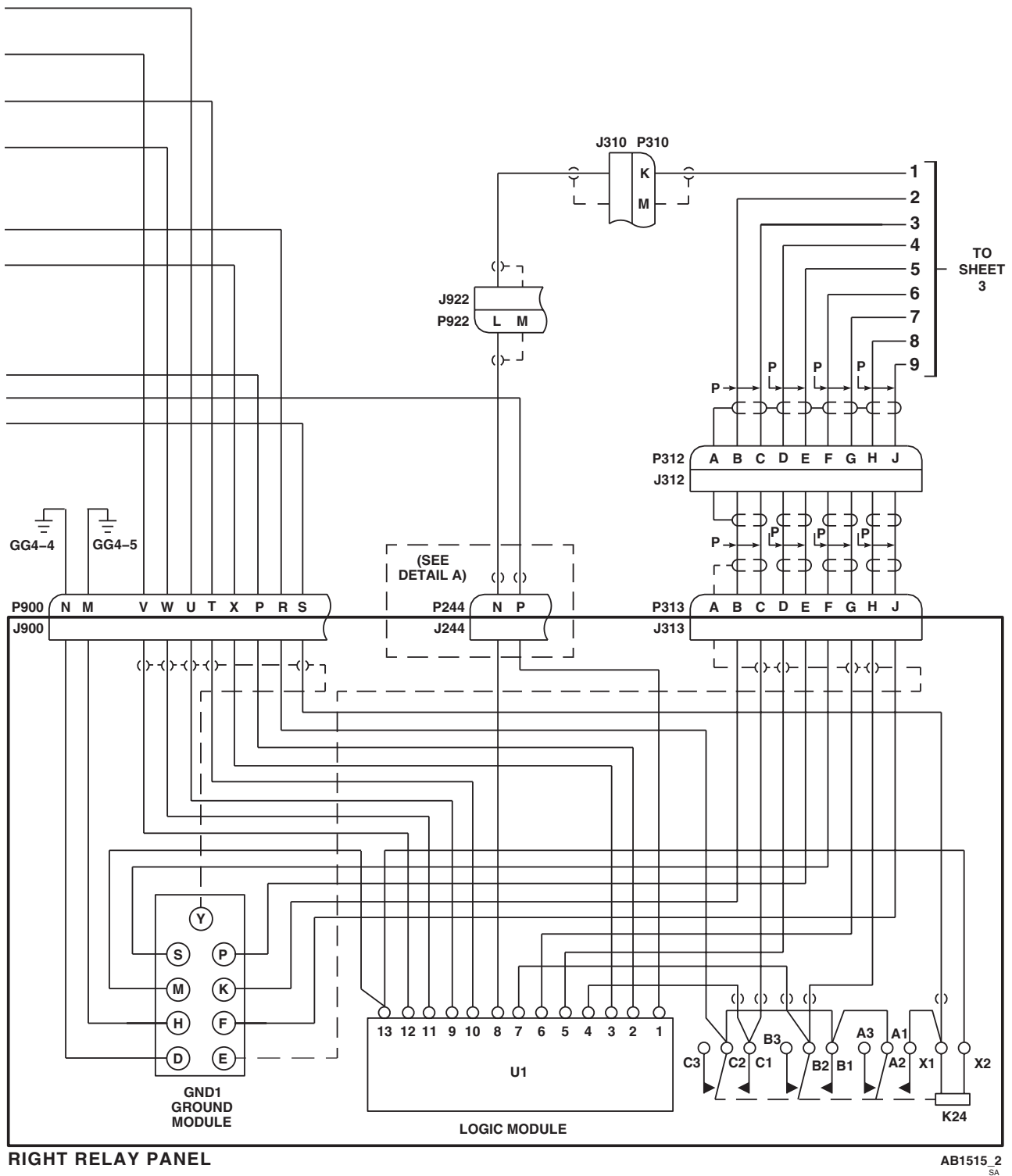
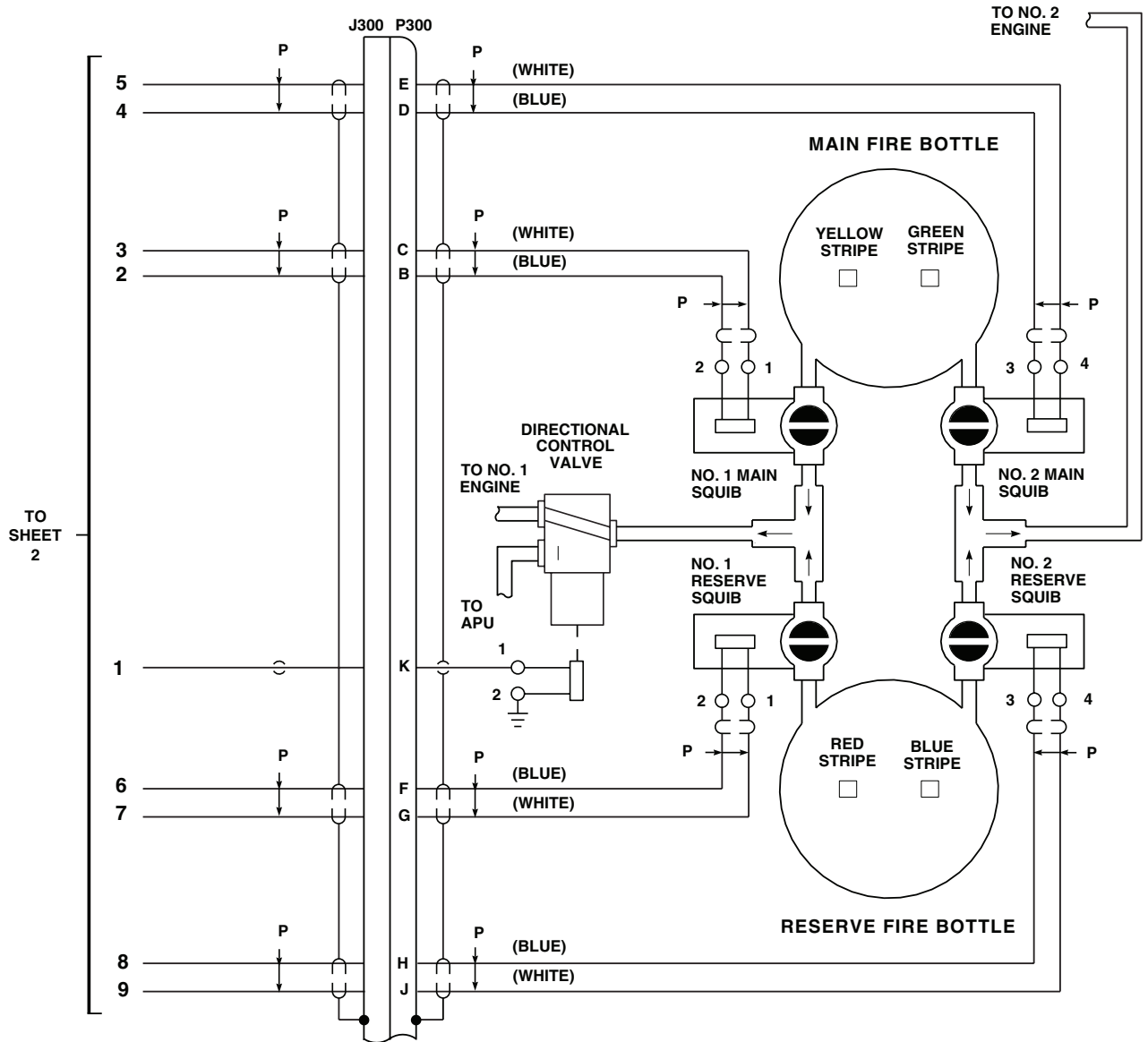


Figure 3. Fire Extinguishing System Schematic Diagram. (Sheet 2 of 4)

SCHEMATICS - Continued



AB1515\_3  
SA

Figure 3. Fire Extinguishing System Schematic Diagram. (Sheet 3 of 4)

## SCHEMATICS - Continued

| FIRE BOTTLE SQUIB<br>WIRING CONNECTION CHART |                                |                       |                              |                           |
|----------------------------------------------|--------------------------------|-----------------------|------------------------------|---------------------------|
| FIRE<br>BOTTLE<br>SQUIB                      | WIRE CODE<br>(PRINTED ON WIRE) | WIRE<br>COLOR<br>CODE | CONNECTION<br>POINT TO SQUIB | FIRE BOTTLE<br>COLOR CODE |
| MAIN<br>(NO. 1)                              | W3163A20<br>W3162A20N          | WHITE<br>BLUE         | #6 STUD<br>#8 SCREW          | YELLOW STRIPE             |
| MAIN<br>(NO. 2)                              | W3164A20<br>W3165A20N          | BLUE<br>WHITE         | #6 STUD<br>#8 SCREW          | GREEN STRIPE              |
| RESERVE<br>(NO. 1)                           | W3167A20<br>W3166A20N          | WHITE<br>BLUE         | #6 STUD<br>#8 SCREW          | RED STRIPE                |
| RESERVE<br>(NO. 2)                           | W3168A20<br>W3169A20N          | BLUE<br>WHITE         | #6 STUD<br>#8 SCREW          | BLUE STRIPE               |

AB1515\_4  
SA

Figure 3. Fire Extinguishing System Schematic Diagram. (Sheet 4 of 4)

END OF WORK PACKAGE

## UNIT LEVEL

## UTILITY SYSTEM

## WINDSHIELD WIPER SYSTEM

## INITIAL SETUP:

## Test Equipment

Multimeter, AN/PSM-45A

WP 0824 00

WP 0828 00

WP 1208 00

## Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01

WP 1209 00

Electrical Repairer Toolkit, SC 5180-99-B06

WP 1210 00

WP 1211 00

## Personnel Required

Aircraft Electrician MOS 15F (1)

WP 1213 00

UH-60 Helicopter Repairer MOS 15T (1)

WP 1685 00

WP 1747 00

## References

TM 1-1520-237-10

WP 0104 00

## Equipment Condition

External Electrical Power Available, WP 1685 00  
or APU Operational, TM 1-1520-237-10

## WINDSHIELD WIPER SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE

## PROCEDURE

## Step

## CAUTION

To prevent scratching windshield, make sure windshield is kept wet during operational/troubleshooting procedure, or install cotter pin in holdoff pin hole.

## NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- See Figure 1 for component location diagram and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

1. Make sure WSHLD WIPER circuit breaker on copilot's circuit breaker panel is pushed in.
2. Turn on electrical power.

## NOTE

Windshield wiper controls consist of a rotary WINDSHIELD WIPER switch with PARK, OFF, LOW, and HI positions.

3. Turn WINDSHIELD WIPER switch to LOW, and then to HI.

## Indication/Condition

Both pilot's and copilot's wipers shall operate in both LOW and HI.

---

**WINDSHIELD WIPER SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued**Corrective Action**

1. If Indication/Condition is not as specified, go to WIPERS DO NOT OPERATE, in this work package.
2. If only one wiper operates, go to ONLY ONE WIPER OPERATES, in this work package.
3. If both wipers do not operate in LOW only, go to WIPERS DO NOT OPERATE IN LOW, in this work package.
4. If both wipers do not operate in HI only, go to WIPERS DO NOT OPERATE IN HI, in this work package.

**Step**

4. Turn WINDSHIELD WIPER switch to LOW.

**Indication/Condition**

Observe that both wipers shall produce the proper sweep and effectively wipe windshield.

**Corrective Action**

1. If wipers do not produce proper sweep, adjust converter (WP 1208 00).
2. If wipers do not wipe effectively, check wiper arm for 4 to 5 pounds of tension.
  - a. If tension is correct, replace wiper blades (WP 1211 00).
  - b. If tension is not correct, adjust wiper arm tension (WP 1213 00).

**Step**

5. Turn WINDSHIELD WIPER switch to OFF, and then to PARK.

**Indication/Condition**

Both wipers shall return to the normal park position (wiper blades parallel to inboard edge of windshield).

**Corrective Action**

If Indication/Condition is not as specified, go to WIPERS DO NOT RETURN TO PARK, in this work package.

**Step**

6. Turn off electrical power.
7. If necessary, remove cotter pins from holdoff pin hole.

**Indication/Condition**

Cotter pins shall be removed from holdoff pin hole.

**Corrective Action**

None Required

**WIPERS DO NOT OPERATE****SYMPTOM**

Wipers do not operate.

**MALFUNCTION**

Wipers do not operate.

**CORRECTIVE ACTION**

1. **Check for 115 vac between WINDSHIELD WIPER circuit breaker terminals and WINDSHIELD WIPER switch terminals:**
  - **A2 and 11**
  - **B2 and 21**
  - **C2 and 31**
  - a. If voltage is as specified, go to **2**.
  - b. If voltage is not as specified, go to **5**.
2. **Check flex shaft for kinks.**
  - a. If kinks are present, replace flex shaft (WP 1210 00). Go to **6**.
  - b. If kinks are not present, go to **3**.
3. **Check converter for binds (disconnect flex shaft and turn converter drive shaft).**
  - a. If binds are present, replace converter (WP 1208 00). Go to **6**.
  - b. If binds are not present, go to **4**.
4. **Make sure shaft couplings are installed.**
  - a. If shaft couplings are installed, replace motor (WP 1209 00). Go to **6**.
  - b. If shaft couplings are not installed, install couplings (WP 1210 00). Go to **6**.
5. **Check for 115 vac between WINDSHIELD WIPER circuit breaker:**
  - **Terminal A2 and ground**
  - **Terminal B2 and ground**
  - **Terminal C2 and ground**
  - a. If voltage is as specified, repair/replace wiring between circuit breaker and switch as required (WP 1747 00). Go to **6**.
  - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **6**.
6. **Procedure completed.**

**ONLY ONE WIPER OPERATES****SYMPTOM**

Only one wiper operates.

**MALFUNCTION**

Only one wiper operates.

---

**ONLY ONE WIPER OPERATES - Continued****CORRECTIVE ACTION****1. Check for disconnected or broken flex shaft.**

- a. If disconnected or broken flex shaft is present, connect or replace shaft (WP 1210 00). Go to **3.**
- b. If disconnected or broken flex shaft is not present, go to **2.**

**2. Check for broken shaft or missing couplings.**

- a. If broken shaft or missing couplings are present, replace shaft or couplings as required (WP 1210 00). Go to **3.**
- b. If broken shaft or missing couplings are not present, replace converter (WP 1208 00). Go to **3.**

**3. Procedure completed.****WIPERS DO NOT OPERATE IN LOW****SYMPTOM**

Wipers do not operate in low.

**MALFUNCTION**

Wipers do not operate in low.

**CORRECTIVE ACTION****1. Turn WINDSHIELD WIPER switch to LOW. Go to 2.****2. Check for 115 vac between:**

- P108-C and ground
- P108-E and ground
- P108-G and ground

- a. If voltage is as specified, replace motor (WP 1209 00). Go to **4.**
- b. If voltage is not as specified, go to **3.**

**3. Check continuity between:**

- P108-C and terminal 32 (WINDSHIELD WIPER switch)
- P108-E and terminal 22 (WINDSHIELD WIPER switch)
- P108-G and terminal 12 (WINDSHIELD WIPER switch)

- a. If continuity is present, replace switch (WP 0828 00). Go to **4.**
- b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **4.**

**4. Procedure completed.**



**WIPERS DO NOT OPERATE IN HI****SYMPTOM**

Wipers do not operate in HI.

**MALFUNCTION**

Wipers do not operate in HI.

**CORRECTIVE ACTION**

1. Turn WINDSHIELD WIPER switch to HI. Go to 2.
2. Check for 115 vac between:
  - P108-B and ground
  - P108-D and ground
  - P108-F and ground
  - a. If voltage is as specified, replace motor (WP 1209 00). Go to 4.
  - b. If voltage is not as specified, go to 3.
3. Check continuity between:
  - P108-B and terminal 13 (WINDSHIELD WIPER switch)
  - P108-D and terminal 33 (WINDSHIELD WIPER switch)
  - P108-F and terminal 23 (WINDSHIELD WIPER switch)
  - a. If continuity is present, replace switch (WP 0828 00). Go to 4.
  - b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to 4.
4. Procedure completed.

**WIPERS DO NOT RETURN TO PARK****SYMPTOM**

Wipers do not return to park.

**MALFUNCTION**

Wipers do not return to park.

**CORRECTIVE ACTION**

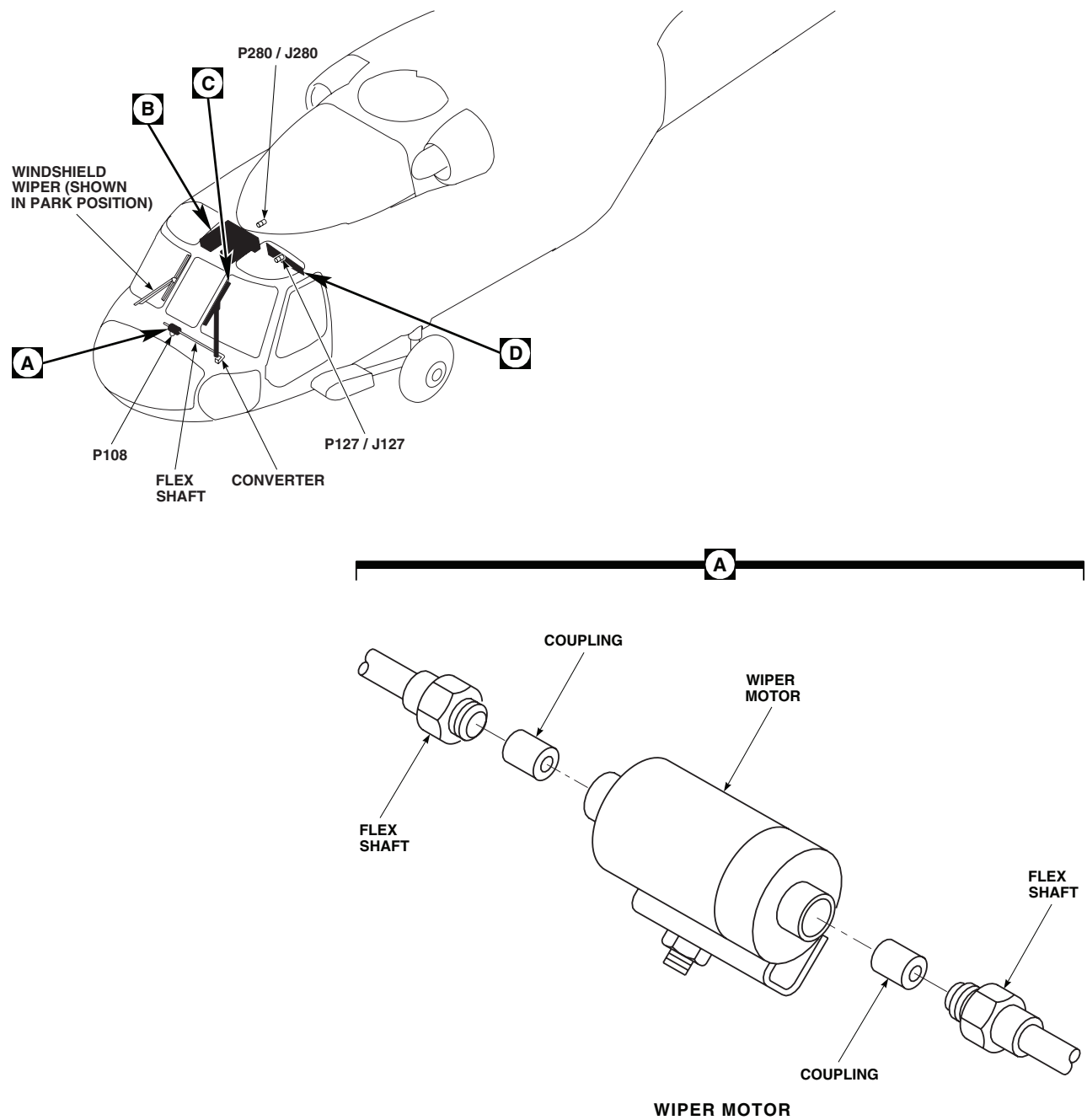
1. Turn WINDSHIELD WIPER switch to PARK. Go to 2.
2. Check for 115 vac between:
  - P108-A and ground
  - P108-H and ground
  - a. If voltage is as specified, replace motor (WP 1209 00). Go to 4.
  - b. If voltage is not as specified, go to 3.
3. Check continuity between:

---

**WIPERS DO NOT RETURN TO PARK - Continued**

- **P108-A and terminal 28 (WINDSHIELD WIPER switch)**
  - **P108-H and terminal 18 (WINDSHIELD WIPER switch)**
    - a. If continuity is present, replace switch (WP 0828 00). Go to **4.**
    - b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **4.**
- 4. Procedure completed.**

LOCATION



| TERMINAL BOARD/<br>DISCONNECT PLUG/<br>RECEPTACLE | LOCATION/<br>CONNECTION POINT        |
|---------------------------------------------------|--------------------------------------|
| P108                                              | WINDSHIELD WIPER<br>MOTOR            |
| P127 / J127                                       | COCKPIT CEILING<br>BL 28 LH, STA 247 |
| P280 / J280                                       | CABIN CEILING, BL 6 LH,<br>STA 247   |

AA3412\_1  
SA

Figure 1. Windshield Wiper System Location Diagram. (Sheet 1 of 2)

LOCATION - Continued

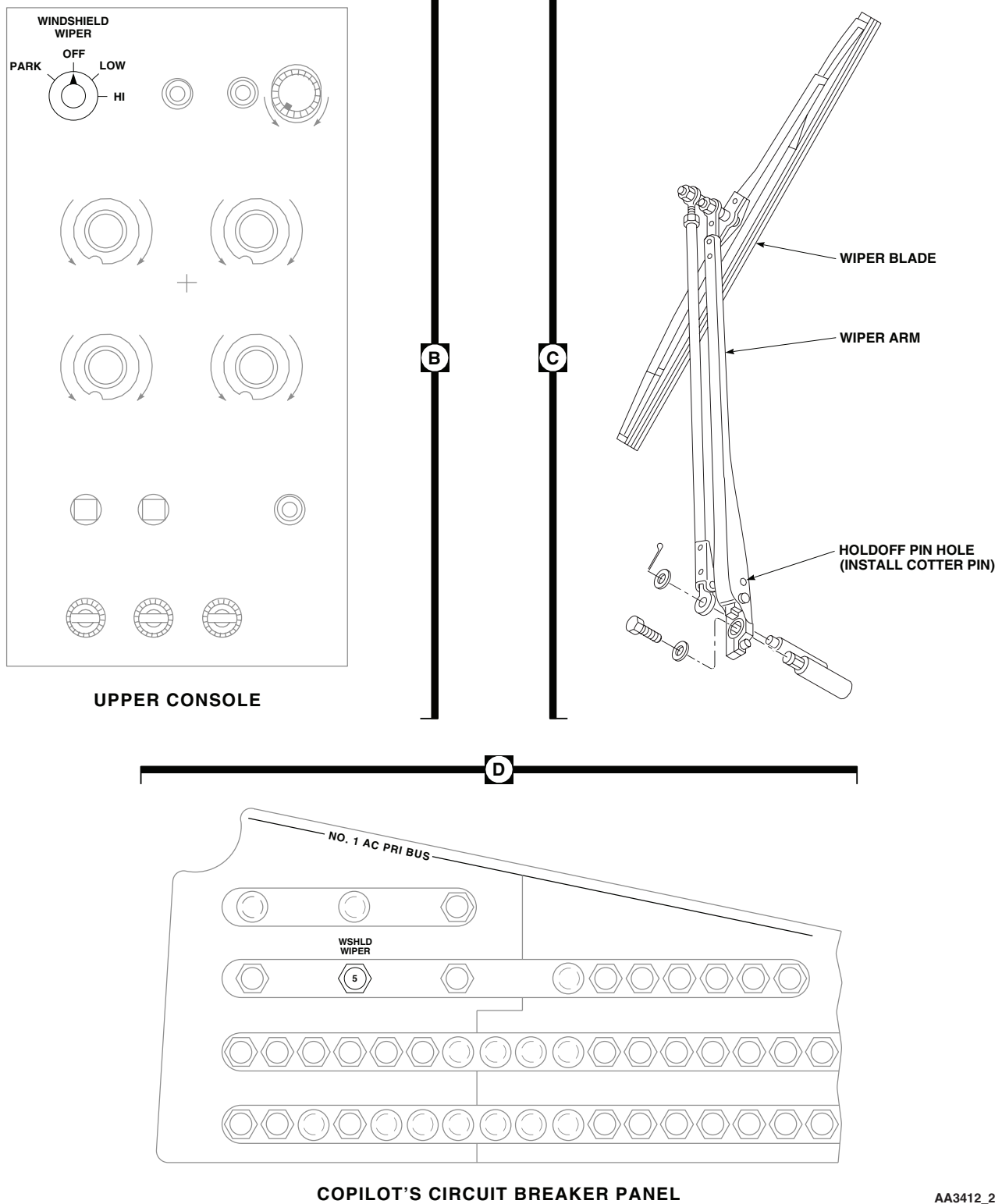
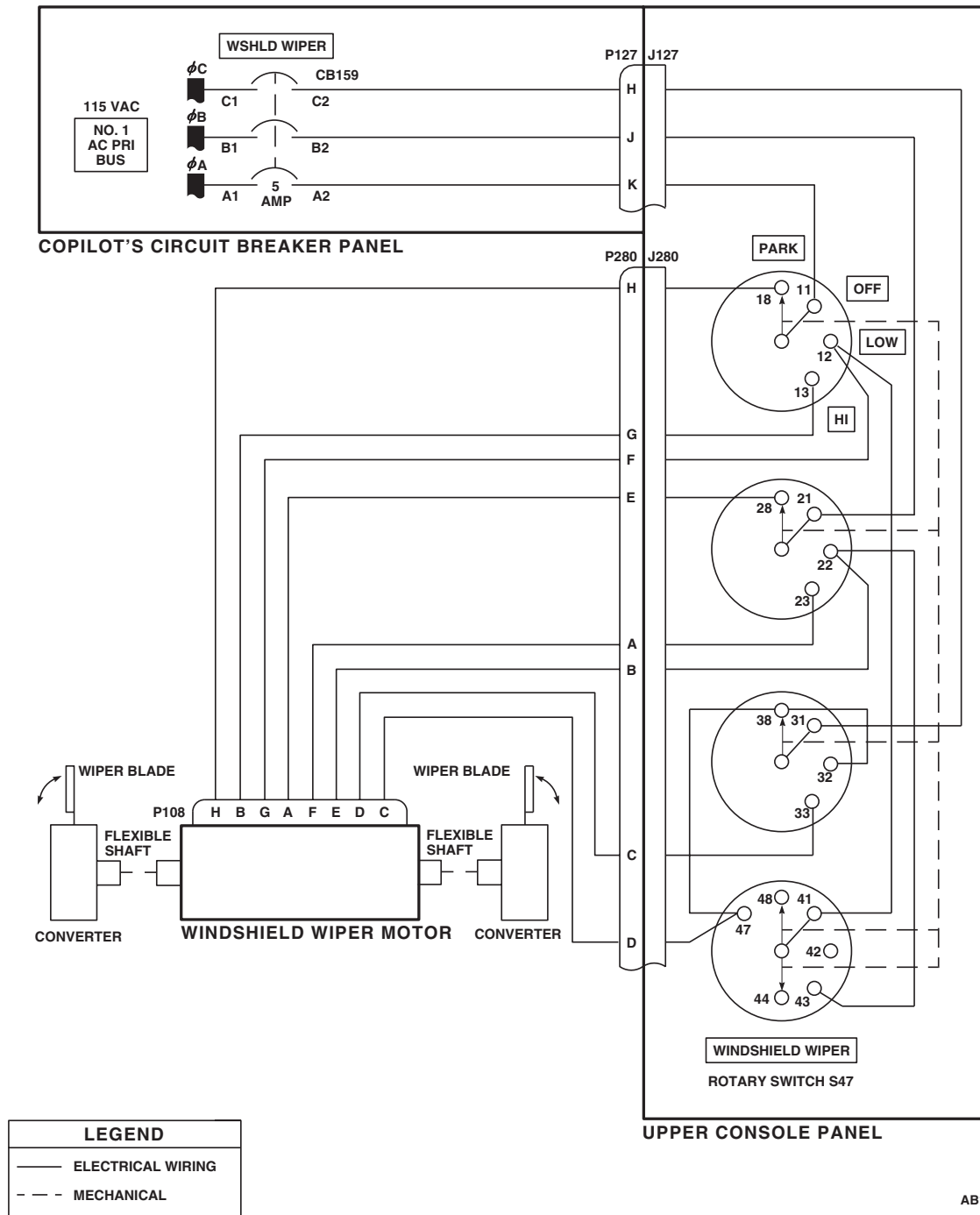


Figure 1. Windshield Wiper System Location Diagram. (Sheet 2 of 2)

AA3412\_2  
SA

SCHEMATICS



AB1458  
SA

Figure 2. Windshield Wiper System Schematic Diagram.

END OF WORK PACKAGE



## UNIT LEVEL

## UTILITY SYSTEM

## WINDSHIELD ANTI-ICE SYSTEM

This WP supersedes WP 0144 00, dated 17 April 2006.

## INITIAL SETUP:

## Test Equipment

Multimeter, AN/PSM-45A

WP 0226 00

WP 0227 00

WP 0824 00

## Tools and Special Tools

Electrical Repairer Toolkit, SC 5180-99-B06

Temperature Probe, 0790

Thermometer, 389-3L

WP 1214 00

WP 1215 00

WP 1400 00

WP 1685 00

WP 1747 00

## Personnel Required

Aircraft Electrician MOS 15F (1)

## Equipment Condition

Ambient Temperature Below 80°F (26.7°C)

External Electrical Power Available, WP 1685 00

or APU Operational, TM 1-1520-237-10

## References

TM 1-1520-237-10

WP 0104 00

WP 0120 00

## WINDSHIELD ANTI-ICE SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE

## PROCEDURE

## Step

## NOTE

- Windshield anti-ice system cannot be checked out in ambient temperatures of 80°F (26.7°C) or higher. System also may not go on if windshield is in direct sunlight.
- The windshield anti-ice system is inoperative and cannot be checked when all of these conditions exist: Both primary ac generators are off, the APU generator is on, and the backup hydraulic pump is running. These conditions are indicated by lighting of the **EH-60A UH-60A UH-60L** APU GEN ON and BACK-UP PUMP ON caution/advisory panel capsules **HH-60A HH-60L** APU GEN ON and BACK-UP PUMP ON multifunction display legends . When these capsules or legends are on, do not do the operational/troubleshooting procedure.
- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- **EH-60A** When any WINDSHIELD ANTI-ICE switch is set to ON, the environmental control system is disabled.
- See Figure 1 for component location diagram, **HCW** Figure 2 and **W/O HCW** Figure 3 for schematic diagrams, and Figure 4 for temperature/resistance chart as an aid in operational/troubleshooting.

1. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.

---

**WINDSHIELD ANTI-ICE SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**

2. Turn on electrical power.
3. Place upper console WINDSHIELD ANTI-ICE COPILOT switch to ON.

**Indication/Condition**

Heat shall be felt on copilot's windshield in about five minutes.

**Corrective Action**

If Indication/Condition is not as specified, go to COPILOT'S WINDSHIELD DOES NOT HEAT, in this work package.

**Step**

4. Connect temperature probe to thermometer.
5. Place probe in contact with copilot's windshield in area of either sensor and check windshield temperature.

**Indication/Condition**

Windshield temperature shall be between 75°F (23.9°C) and 115°F (46.1°C).

**Corrective Action**

If Indication/Condition is not as specified, go to COPILOT'S WINDSHIELD TEMPERATURE IS NOT BETWEEN 75°F (23.9°C) AND 115°F (46.1°C), in this work package.

**Step**

6. Place upper console WINDSHIELD ANTI-ICE COPILOT switch to OFF.
7. Place upper console WINDSHIELD ANTI-ICE PILOT switch to ON.

**Indication/Condition**

Heat shall be felt on pilot's windshield in about five minutes.

**Corrective Action**

If Indication/Condition is not as specified, go to PILOT'S WINDSHIELD DOES NOT HEAT, in this work package.

**Step**

8. Connect temperature probe to thermometer.
9. Place probe in contact with pilot's windshield in area of either sensor and check windshield temperature.

**Indication/Condition**

Windshield temperature shall be between 75°F (23.9°C) and 115°F (46.1°C).

**Corrective Action**

If Indication/Condition is not as specified, go to PILOT'S WINDSHIELD TEMPERATURE IS NOT BETWEEN 75°F (23.9°C) AND 115°F (46.1°C), in this work package.

**Step**

10. Place upper console WINDSHIELD ANTI-ICE PILOT switch to OFF.



---

**WINDSHIELD ANTI-ICE SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****NOTE**

**HCW** Steps 11. through 14. apply. ■

11. Place upper console WINDSHIELD ANTI-ICE CTR switch to ON.

**Indication/Condition**

Heat shall be felt on center windshield in about five minutes.

**Corrective Action**

If Indication/Condition is not as specified go to CENTER WINDSHIELD DOES NOT HEAT, in this work package.

**Step**

12. Connect temperature probe to thermometer.
13. Place probe in contact with center windshield in area of either sensor and check windshield temperature.

**Indication/Condition**

Windshield temperature shall be between 75°F (23.9°C) and 115°F (46.1°C).

**Corrective Action**

If Indication/Condition is not as specified, go to CENTER WINDSHIELD TEMPERATURE IS NOT BETWEEN 75°F (23.9°C) AND 115°F (46.1°C), in this work package.

**Step**

14. Place upper console WINDSHIELD ANTI-ICE CTR switch to OFF.
15. Turn off electrical power.

**Indication/Condition**

Power shall be off.

**Corrective Action**

None Required

**COPILOT'S WINDSHIELD DOES NOT HEAT****SYMPTOM**

Copilot's windshield does not heat.

**MALFUNCTION**

Copilot's windshield does not heat.

**CORRECTIVE ACTION**

1. **Check that resistance between P275-A and P275-E is within limits shown in Figure 4 for windshield sensor temperature.**
  - a. If resistance is within limits, go to **2**.
  - b. If resistance is not within limits, go to COPILOT'S WINDSHIELD TEMPERATURE IS NOT BETWEEN 75°F (23.9°C) AND 115°F (46.1°C), step 2., in this work package.
2. **Check for 28 vdc between P275-C and ground.**

---

COPILOT'S WINDSHIELD DOES NOT HEAT - Continued

- a. If voltage is as specified, go to **3**.
  - b. If voltage is not as specified, go to **7**.
3. **Check for 115 vac between:**
  - **P274-B (phase A) and ground**
  - **P274-C (phase B) and ground**
  - **P274-E (phase C) and ground**
  - a. If voltage is as specified, go to **4**.
  - b. If voltage is not as specified, go to **6**.
4. **Check continuity between:**
  - **P274-A and terminal 2 (copilot's windshield terminal block)**
  - **P274-G and terminal 1 (copilot's windshield terminal block)**
  - **P274-F and terminal 6 (copilot's windshield terminal block)**
  - a. If continuity is present, go to **5**.
  - b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **14**.
5. **Check continuity between P274-D and ground.**
  - a. If continuity is present, replace copilot's anti-ice controller (WP 1214 00). Go to **14**.
  - b. If continuity is not as present, repair/replace wiring (WP 1747 00). Go to **14**.
6. **Check for 115 vac between COPILOT WSHLD ANTI-ICE circuit breaker:**
  - **Terminal A2 and ground**
  - **Terminal B2 and ground**
  - **Terminal C2 and ground**
  - a. If voltage is as specified, repair/replace wiring between (WP 1747 00) then go to **14**:
    - **Terminal A2 and P274-B**
    - **Terminal B2 and P274-B**
    - **Terminal C2 and P274-B**
  - b. If voltage is not as specified, replace COPILOT WSHLD ANTI-ICE circuit breaker (WP 0824 00). Go to **14**.
7. **Check helicopter configuration.**
  - a. **WINTER** , go to **8**.
  - b. **W/O WINTER** , go to **9**.
8. **WINTER** **Check for 28 vdc between P244B-G and ground.**
  - a. If voltage is as specified, go to **10**.
  - b. If voltage is not as specified, replace **WINTER** (WP 1400 00). Go to **14**.

---

**COPILOT'S WINDSHIELD DOES NOT HEAT - Continued**

- c. If trouble remains, go to **9**.
- 9. **Check for 28 vdc between P244-G and ground.**
  - a. If voltage is as specified, go to **13**.
  - b. If voltage is not as specified, go to **11**.
- 10. **Check continuity between P244-F and P275-C.**
  - a. If continuity is present, troubleshoot right relay panel (WP 0120 00). Go to **14**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 11. **Check for 28 vdc between WINDSHIELD ANTI-ICE COPILOT switch terminal 3 and ground.**
  - a. If voltage is as specified, repair/replace wiring between switch terminal 3 and P244-G (WP 1747 00). Go to **14**.
  - b. If voltage is not as specified, go to **12**.
- 12. **Check for 28 vdc between WINDSHIELD ANTI-ICE COPILOT switch terminal 2 and ground.**
  - a. If voltage is as specified, replace switch (WP 1215 00). Go to **14**.
  - b. If voltage is not as specified, go to **13**.
- 13. **Check continuity between COPILOT WSHLD ANTI-ICE circuit breaker terminal 1 and WINDSHIELD ANTI-ICE COPILOT switch terminal 2.**
  - a. If continuity is present, replace circuit breaker (WP 0824 00). Go to **14**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 14. **Procedure completed.**

**COPILOT'S WINDSHIELD TEMPERATURE IS NOT BETWEEN 75°F (23.9°C) AND 115°F (46.1°C)****SYMPTOM**

Copilot's windshield temperature is not between 75°F (23.9°C) and 115°F (46.1°C).

**MALFUNCTION**

Copilot's windshield temperature is not between 75°F (23.9°C) and 115°F (46.1°C).

**CORRECTIVE ACTION**

- 1. **Check that resistance between P275-A and P275-E is within limits shown in Figure 4 for windshield sensor temperature.**
  - a. If resistance is within limits, replace copilot's anti-ice controller (WP 1214 00). Go to **4**.
  - b. If resistance is not within limits, go to **2**.
- 2. **Check that resistance between terminals 7 and 8 of copilot's windshield terminal block is within limits shown in Figure 4 for windshield sensor temperature.**
  - a. If resistance is within limits, repair/replace wiring between copilot's windshield terminal block terminal 8 and P275-A, and terminal 7 and P275-E (WP 1747 00). Go to **4**.
  - b. If resistance is not within limits, go to **3**.

---

**COPILOT'S WINDSHIELD TEMPERATURE IS NOT BETWEEN 75°F (23.9°C) AND 115°F (46.1°C) - Continued**

3. **Check continuity between copilot's windshield terminal block:**
  - **TB1-3 and TB1-8**
  - **TB1-4 and TB1-7**
  - a. If continuity is present, replace copilot's windshield (WP 0226 00). Go to **4**.
  - b. If continuity is not present, repair/replace terminal block wiring (WP 1747 00). Go to **4**.
4. **Procedure completed.**

**PILOT'S WINDSHIELD DOES NOT HEAT****SYMPTOM**

Pilot's windshield does not heat.

**MALFUNCTION**

Pilot's windshield does not heat.

**CORRECTIVE ACTION**

1. **Check that resistance between P271-A and P271-E is within limits shown in Figure 4 for windshield sensor temperature.**
  - a. If resistance is within limits, go to **2**.
  - b. If resistance is not within limits, go to PILOT'S WINDSHIELD TEMPERATURE IS NOT BETWEEN 75°F (23.9°C) AND 115°F (46.1°C), step 2., in this work package.
2. **Check for 28 vdc between P271-C and ground.**
  - a. If voltage is as specified, go to **3**.
  - b. If voltage is not as specified, go to **7**.
3. **Check for 115 vac between:**
  - **P270-B (phase A) and ground**
  - **P270-C (phase B) and ground**
  - **P270-E (phase C) and ground**
  - a. If voltage is as specified, go to **4**.
  - b. If voltage is not as specified, go to **6**.
4. **Check continuity between:**
  - **P270-A and terminal 2 (pilot's windshield terminal block)**
  - **P270-F and terminal 6 (pilot's windshield terminal block)**
  - **P270-G and terminal 1 (pilot's windshield terminal block)**
  - a. If continuity is present, go to **5**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **12**.
5. **Check continuity between P270-D and ground.**
  - a. If continuity is present, replace pilot's anti-ice controller (WP 1214 00). Go to **12**.

---

**PILOT'S WINDSHIELD DOES NOT HEAT - Continued**

- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **12**.
- 6. **Check for 115 vac between PILOT WSHLD ANTI-ICE circuit breaker:**
  - **Terminal A2 and ground**
  - **Terminal B2 and ground**
  - **Terminal C2 and ground**
- a. If voltage is as specified, repair/replace wiring between (WP 1747 00) then go to **12**:
  - Terminal A2 and P270-B
  - Terminal B2 and P270-C
  - Terminal C2 and P270-E
- b. If voltage is not as specified, replace PILOT WSHLD ANTI-ICE circuit breaker (WP 0824 00). Go to **12**.
- 7. **Check for 28 vdc between P900-F and ground.**
  - a. If voltage is as specified, go to **8**.
  - b. If voltage is not as specified, go to **9**.
- 8. **Check continuity between P900-E and P271-C.**
  - a. If continuity is present, troubleshoot right relay panel (WP 0120 00). Go to **12**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **12**.
- 9. **Check for 28 vdc between WINDSHIELD ANTI-ICE PILOT switch terminal 3 and ground.**
  - a. If voltage is as specified, repair/replace wiring between switch terminal 3 and P900-F (WP 1747 00). Go to **12**.
  - b. If voltage is not as specified, go to **10**.
- 10. **Check for 28 vdc between WINDSHIELD ANTI-ICE PILOT switch terminal 2 and ground.**
  - a. If voltage is as specified, replace switch (WP 1215 00). Go to **12**.
  - b. If voltage is not as specified, go to **11**.
- 11. **Check continuity between PILOT WSHLD ANTI-ICE circuit breaker terminal 1 and WINDSHIELD ANTI-ICE PILOT switch terminal 2.**
  - a. If continuity is present, replace circuit breaker (WP 0824 00). Go to **12**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **12**.
- 12. **Procedure completed.**

**PILOT'S WINDSHIELD TEMPERATURE IS NOT BETWEEN 75°F (23.9°C) AND 115°F (46.1°C)****SYMPTOM**

Pilot's windshield temperature is not between 75°F (23.9°C) and 115°F (46.1°C).

**MALFUNCTION**

Pilot's windshield temperature is not between 75°F (23.9°C) and 115°F (46.1°C).

**CORRECTIVE ACTION**

1. **Check that resistance between P271-A and P271-E is within limits shown in Figure 4 for windshield sensor temperature.**
  - a. If resistance is within limits, replace pilot's anti-ice controller (WP 1214 00). Go to **4**.
  - b. If resistance is not within limits, go to **2**.
2. **Check that resistance between terminals 7 and 8 of pilot's windshield terminal block is within limits shown in Figure 4 for windshield sensor temperature.**
  - a. If resistance is within limits, repair/replace wiring between pilot's windshield terminal block terminal 8 and P271-A, terminal 7 and P271-E (WP 1747 00). Go to **4**.
  - b. If resistance is not within limits, go to **3**.
3. **Check continuity between pilot's windshield terminal block:**
  - **3 and 8**
  - **4 and 7**
  - a. If continuity is present, replace pilot's windshield (WP 0226 00). Go to **4**.
  - b. If continuity is not present, repair/replace terminal block wiring (WP 1747 00). Go to **4**.
4. **Procedure completed.**

**CENTER WINDSHIELD DOES NOT HEAT****SYMPTOM**

Center windshield does not heat.

**MALFUNCTION**

Center windshield does not heat.

**CORRECTIVE ACTION**

1. **Check that resistance between P384-A and P384-E is within limits shown in Figure 4 for windshield sensor temperature.**
  - a. If resistance is within limits, go to **2**.
  - b. If resistance is not within limits, go to CENTER WINDSHIELD TEMPERATURE IS NOT BETWEEN 75°F (23.9°C) AND 115°F (46.1°C), step 2., in this work package.
2. **Check for 28 vdc between P384-C and ground.**
  - a. If voltage is as specified, go to **3**.
  - b. If voltage is not as specified, go to **7**.

---

**CENTER WINDSHIELD DOES NOT HEAT - Continued**

3. **Check for 115 vac between:**
  - **P383-B and ground**
  - **P383-C and ground**
  - **P383-E and ground**
  - a. If voltage is as specified, go to **4**.
  - b. If voltage is not as specified, go to **6**.
4. **Check continuity between:**
  - **P383-A and terminal 2 (center windshield terminal block)**
  - **P383-F and terminal 6 (center windshield terminal block)**
  - **P383-G and terminal 1 (center windshield terminal block)**
  - a. If continuity is present, go to **5**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **12**.
5. **Check continuity between P383-D and ground.**
  - a. If continuity is present, replace anti-ice controller (WP 1214 00). Go to **12**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **12**.
6. **Check for 115 vac between CTR WSHLD ANTI-ICE circuit breaker:**
  - **Terminal A2 and ground**
  - **Terminal B2 and ground**
  - **Terminal C2 and ground**
  - a. If voltage is as specified, repair/replace wiring between CTR WSHLD ANTI-ICE circuit breaker (WP 1747 00) then go to **12**:
    - **Terminal A2 and P383-B**
    - **Terminal B2 and P383-C**
    - **Terminal C2 and P383-E**
  - b. If voltage is not as specified, replace CTR WSHLD ANTI-ICE circuit breaker (WP 0824 00). Go to **12**.
7. **Check for 28 vdc between P900-D and ground.**
  - a. If voltage is as specified, go to **8**.
  - b. If voltage is not as specified, go to **9**.
8. **Check continuity between P900-C and P384-C.**
  - a. If continuity is present, troubleshoot right relay panel (WP 0120 00). Go to **12**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **12**.
9. **Check for 28 vdc between WINDSHIELD ANTI-ICE CTR switch terminal 3 and ground.**

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**CENTER WINDSHIELD DOES NOT HEAT - Continued**

- a. If voltage is as specified, repair/replace wiring between switch terminal 3 and P900-D (WP 1747 00). Go to **12**.
  - b. If voltage is not as specified, go to **10**.
10. **Check for 28 vdc between WINDSHIELD ANTI-ICE CTR switch terminal 2 and ground.**
  - a. If voltage is as specified, replace switch (WP 1215 00). Go to **12**.
  - b. If voltage is not as specified, go to **11**.
11. **Check continuity between WINDSHIELD ANTI-ICE CTR switch terminal 2 and WINDSHIELD ANTI-ICE CTR circuit breaker terminal 1.**
  - a. If continuity is present, replace circuit breaker (WP 0824 00). Go to **12**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **12**.
12. **Procedure completed.**

**CENTER WINDSHIELD TEMPERATURE IS NOT BETWEEN 75°F (23.9°C) AND 115°F (46.1°C)****SYMPTOM**

Center windshield temperature is not between 75°F (23.9°C) and 115°F (46.1°C).

**MALFUNCTION**

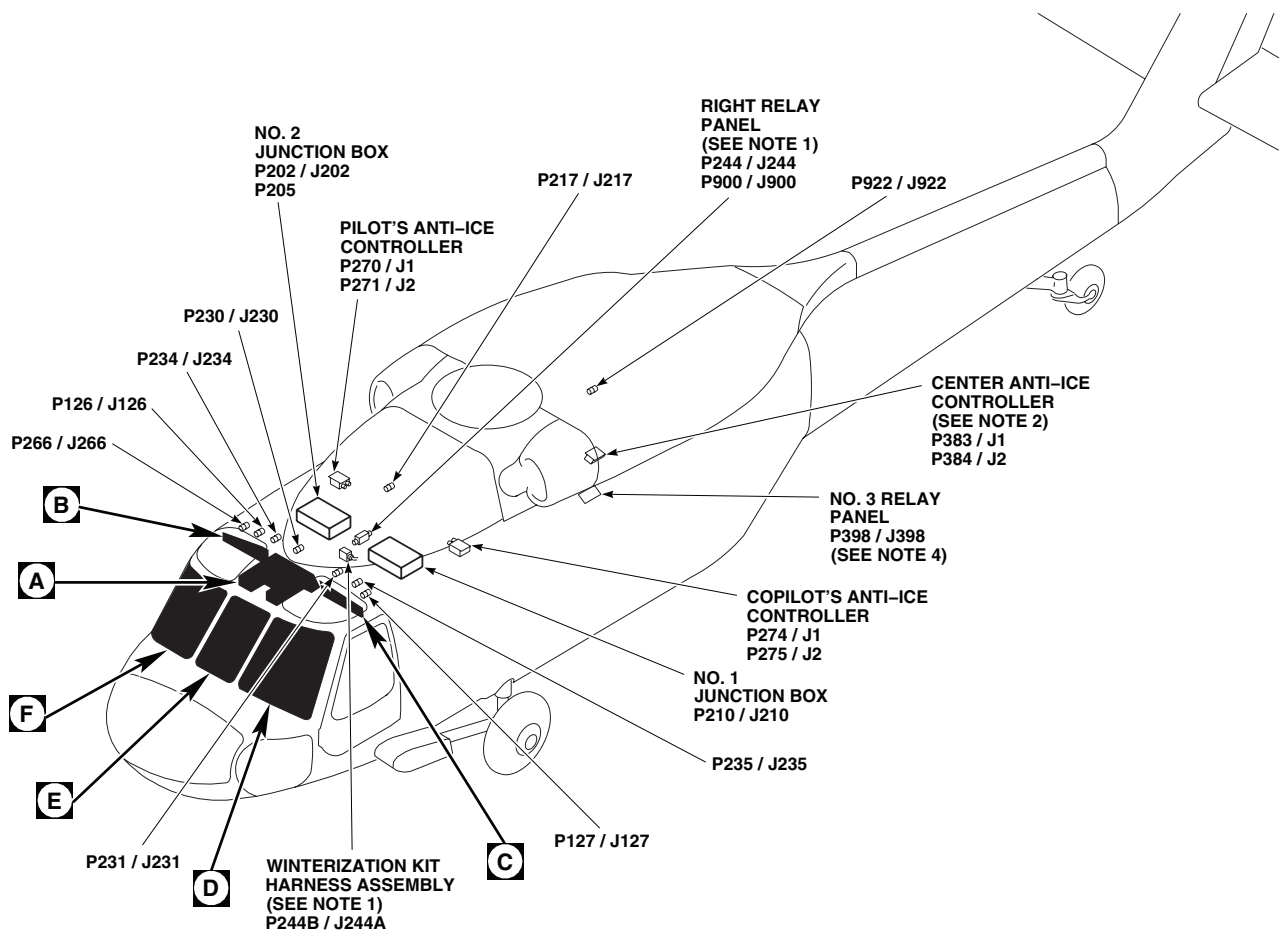
Center windshield temperature is not between 75°F (23.9°C) and 115°F (46.1°C).

**CORRECTIVE ACTION**

1. **Check that resistance between P384-A and P384-E is within limits shown in Figure 4 for windshield sensor temperature.**
  - a. If resistance is within limits, replace center anti-ice controller (WP 1214 00). Go to **4**.
  - b. If resistance is not within limits, go to **2**.
2. **Check that resistance between terminals 7 and 8 of center windshield terminal block is within limits shown in Figure 4 for windshield sensor temperature.**
  - a. If resistance is within limits, repair/replace wiring between center windshield terminal block terminal 7 and P384-E, and terminal 8 and P384-A (WP 1747 00). Go to **4**.
  - b. If resistance is not within limits, go to **3**.
3. **Check continuity between center windshield terminal block:**
  - **3 and 8**
  - **4 and 7**
  - a. If continuity is present, replace center windshield (WP 0227 00). Go to **4**.
  - b. If continuity is not present, repair/replace terminal block wiring (WP 1747 00). Go to **4**.
4. **Procedure completed.**



LOCATION



NOTES

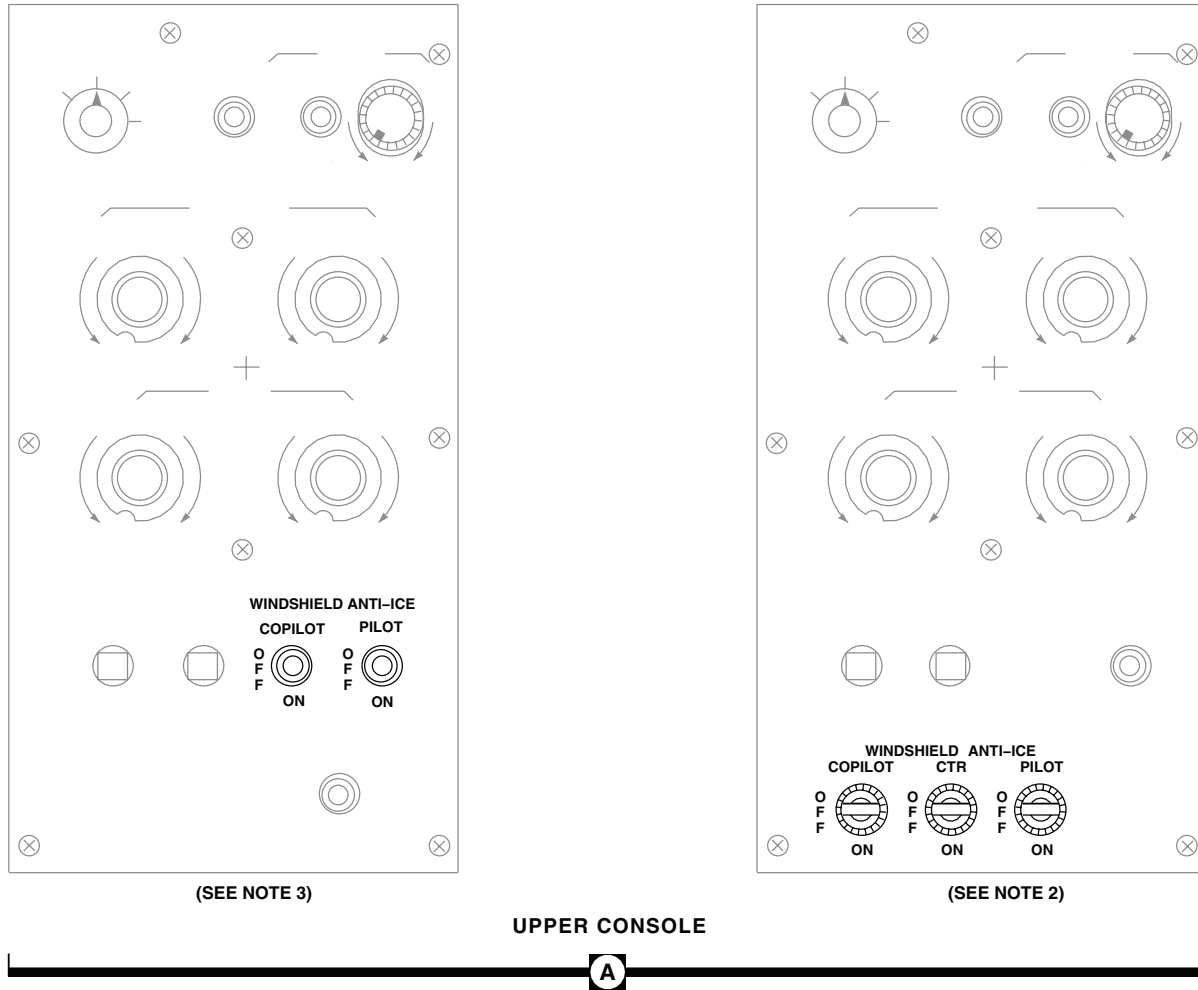
- 1. **WINTER**
- 2. **HCW**
- 3. **W/O HCW**
- 4. **EH-60A**

| TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE | LOCATION / CONNECTION POINT                               |
|-----------------------------------------------|-----------------------------------------------------------|
| P126 / J126                                   | COCKPIT CEILING, BL 28 RH, STA 243                        |
| P127 / J127                                   | COCKPIT CEILING, BL 28 LH, STA 243                        |
| P202 / J202                                   | NO. 2 JUNCTION BOX                                        |
| P205                                          | NO. 2 JUNCTION BOX                                        |
| P210 / J210                                   | NO. 1 JUNCTION BOX                                        |
| P217 / J217                                   | MAIN ROTOR PYLON DECK, BL 0, STA 284                      |
| P230 / J230                                   | CABIN CEILING, BL 8 RH, STA 247                           |
| P231 / J231                                   | CABIN CEILING, BL 9 LH, STA 247                           |
| P234 / J234                                   | BEHIND PILOT'S CIRCUIT BREAKER PANEL, BL 20 RH, STA 247   |
| P235 / J235                                   | BEHIND COPILOT'S CIRCUIT BREAKER PANEL, BL 20 LH, STA 247 |
| P244 / J244<br>P244B / J244A                  | RIGHT RELAY PANEL (SEE NOTE 1)                            |

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Figure 1. Windshield Anti-ice System Location Diagram. (Sheet 1 of 4)

LOCATION - Continued



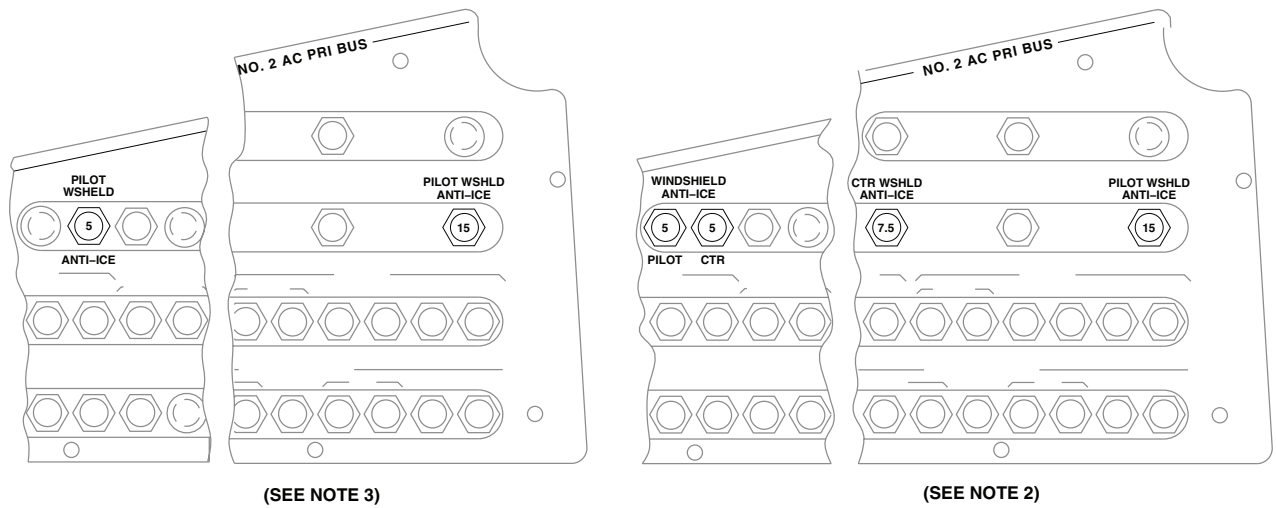
| TERMINAL BOARD/<br>DISCONNECT PLUG/<br>RECEPTACLE | LOCATION/<br>CONNECTION POINT                        |
|---------------------------------------------------|------------------------------------------------------|
| P266 / J266                                       | BEHIND PILOT'S CIRCUIT<br>BREAKER PANEL,<br>BL 25 RH |
| P270 / J1                                         | PILOT'S ANTI-ICE<br>CONTROLLER                       |
| P271 / J2                                         | PILOT'S ANTI-ICE<br>CONTROLLER                       |
| P274 / J1                                         | COPILOT'S ANTI-ICE<br>CONTROLLER                     |
| P275 / J2                                         | COPILOT'S ANTI-ICE<br>CONTROLLER                     |

| TERMINAL BOARD/<br>DISCONNECT PLUG/<br>RECEPTACLE | LOCATION/<br>CONNECTION POINT         |
|---------------------------------------------------|---------------------------------------|
| P383 / J1<br>(SEE NOTE 2)                         | CENTER ANTI-ICE<br>CONTROLLER         |
| P384 / J2<br>(SEE NOTE 2)                         | CENTER ANTI-ICE<br>CONTROLLER         |
| P398 / J398 (SEE NOTE 4)                          | NO. 3 RELAY PANEL                     |
| P900 / J900                                       | RIGHT RELAY PANEL                     |
| P922 / J922                                       | CABIN CEILING,<br>BL 10.7 LH, STA 380 |

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Figure 1. Windshield Anti-ice System Location Diagram. (Sheet 2 of 4)

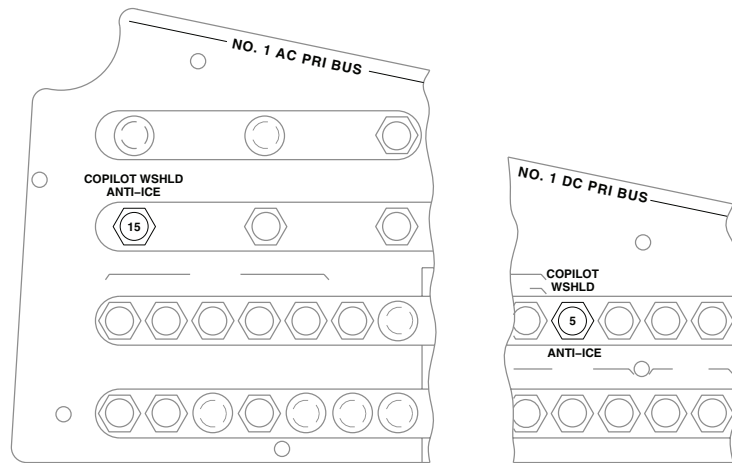
LOCATION - Continued



PILOT'S CIRCUIT BREAKER PANEL

**B**

**C**

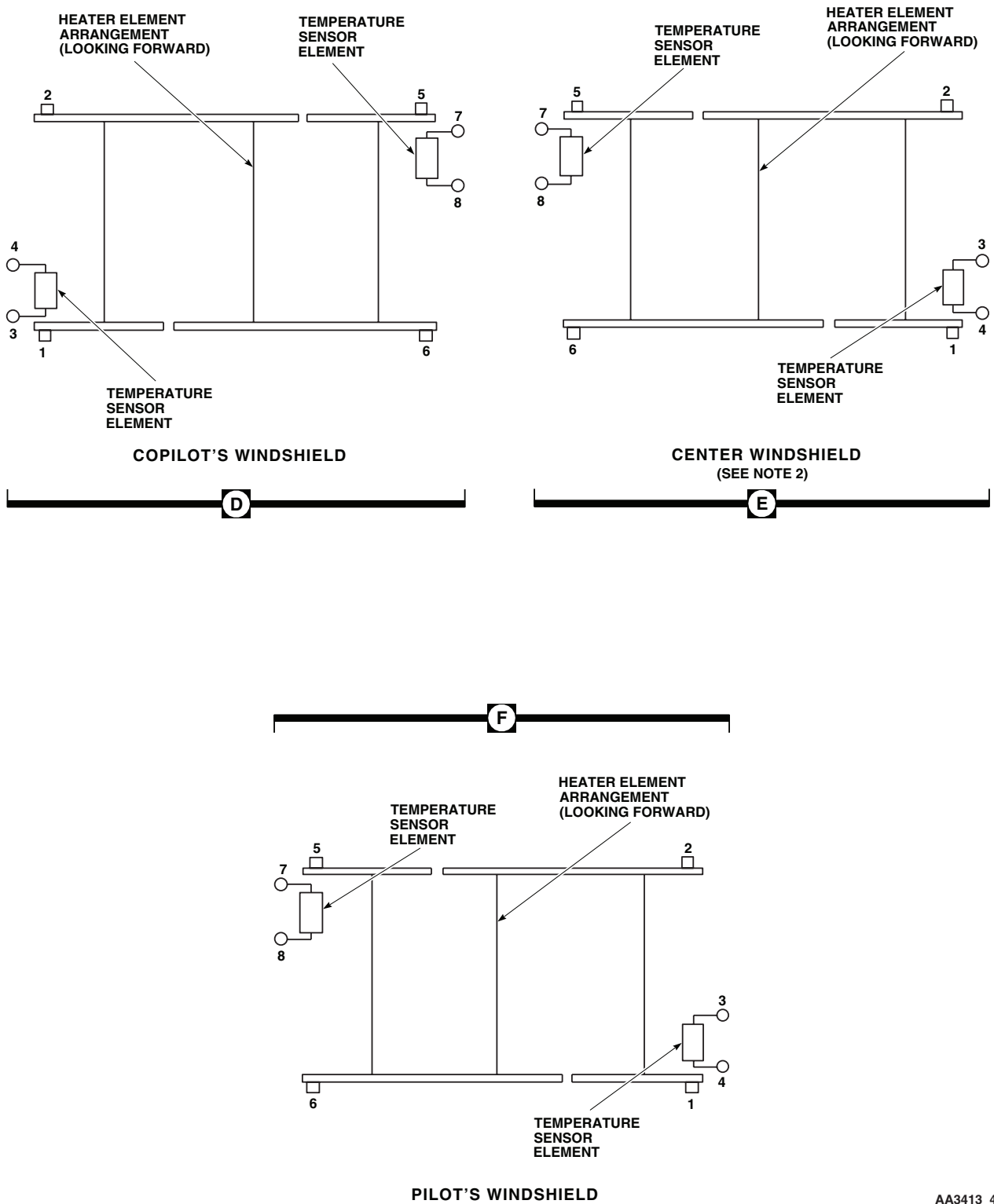


COPLOT'S CIRCUIT BREAKER PANEL

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Figure 1. Windshield Anti-ice System Location Diagram. (Sheet 3 of 4)

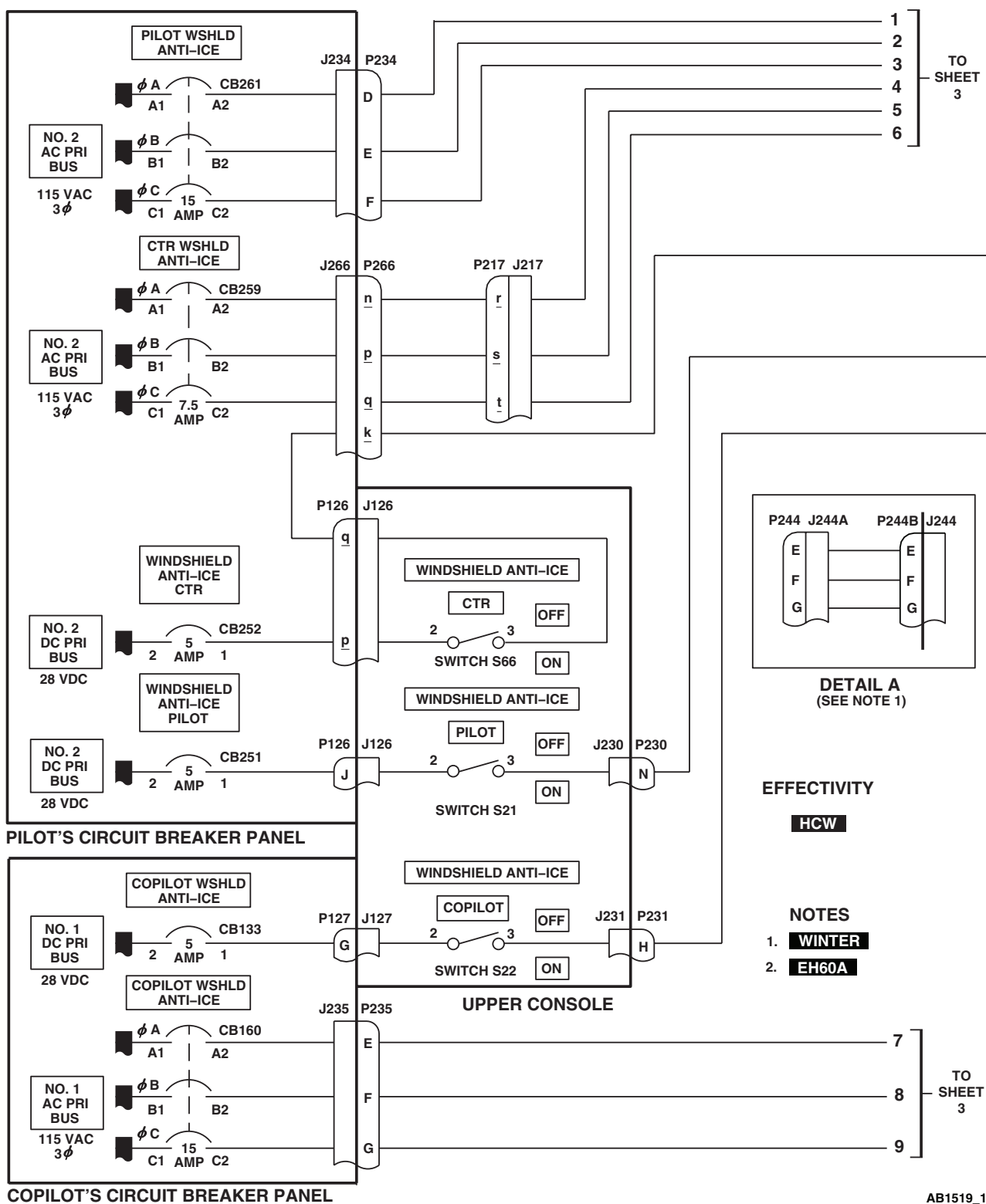
LOCATION - Continued



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Figure 1. Windshield Anti-ice System Location Diagram. (Sheet 4 of 4)

## SCHEMATICS

Figure 2. Windshield Anti-ice System Schematic Diagram **HCW**. (Sheet 1 of 4)

SCHEMATICS - Continued

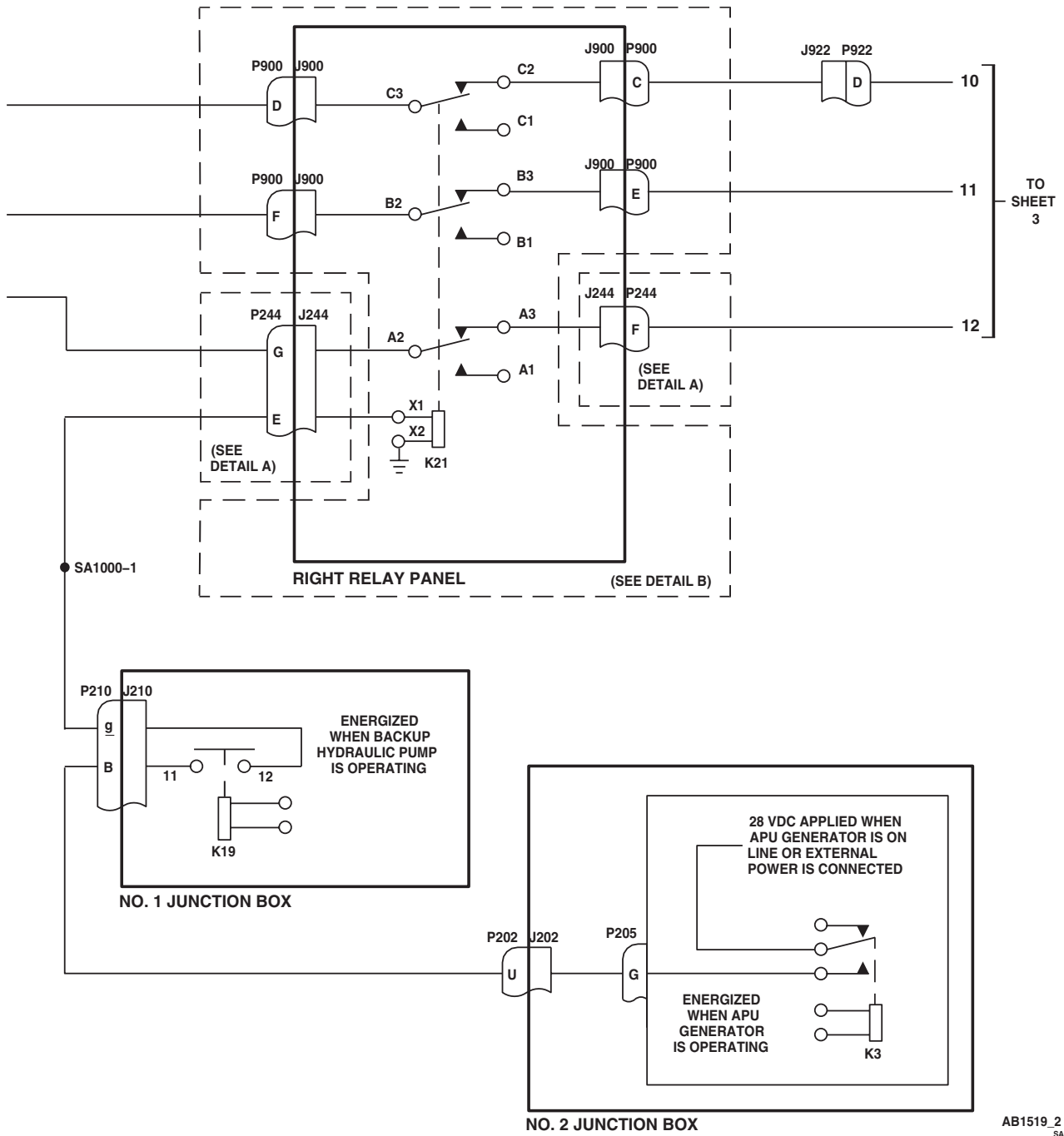
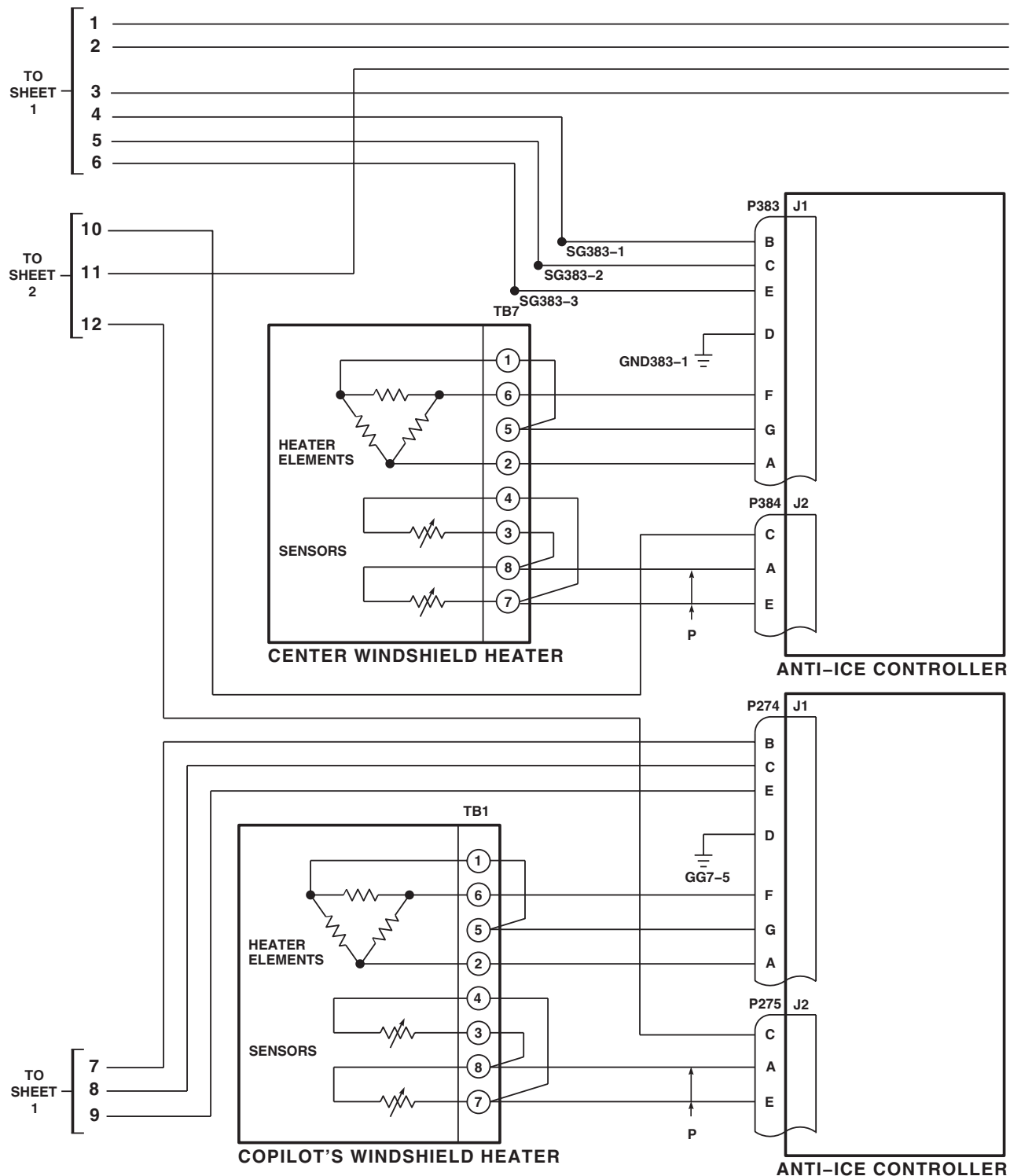
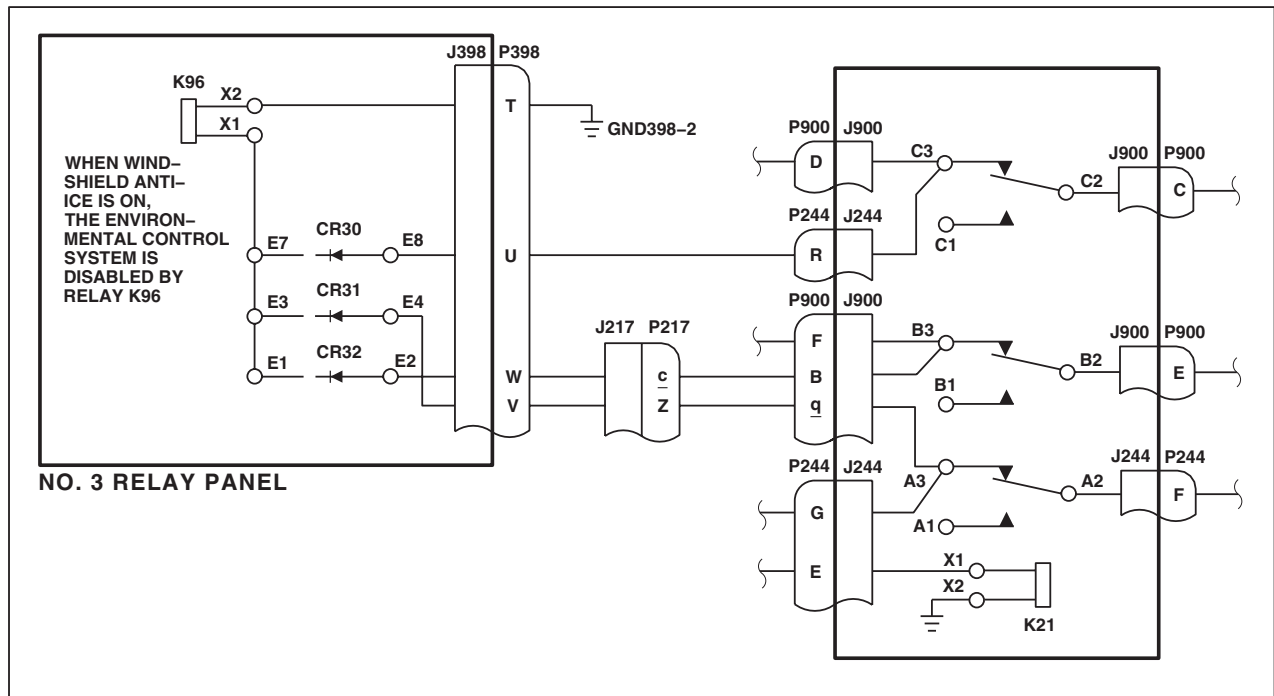
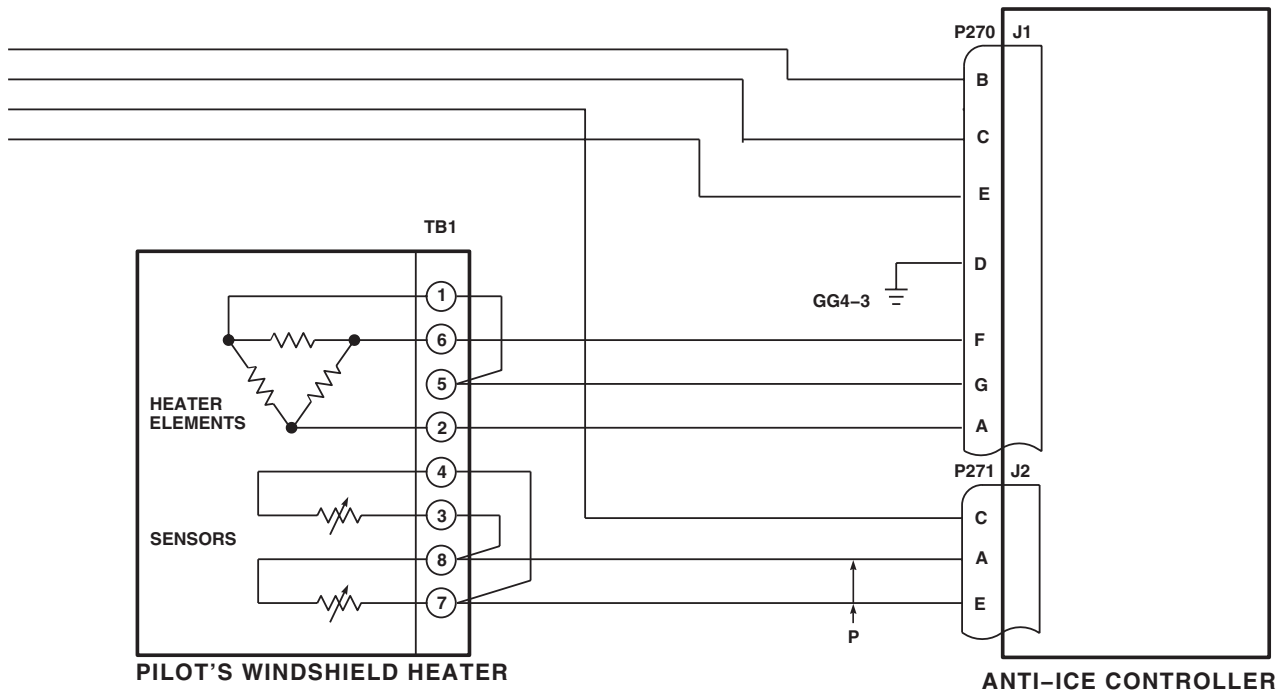


Figure 2. Windshield Anti-ice System Schematic Diagram **HCW** . (Sheet 2 of 4)

## SCHEMATICS - Continued

AB1519\_3  
SAFigure 2. Windshield Anti-ice System Schematic Diagram **HCW**. (Sheet 3 of 4)

SCHEMATICS - Continued

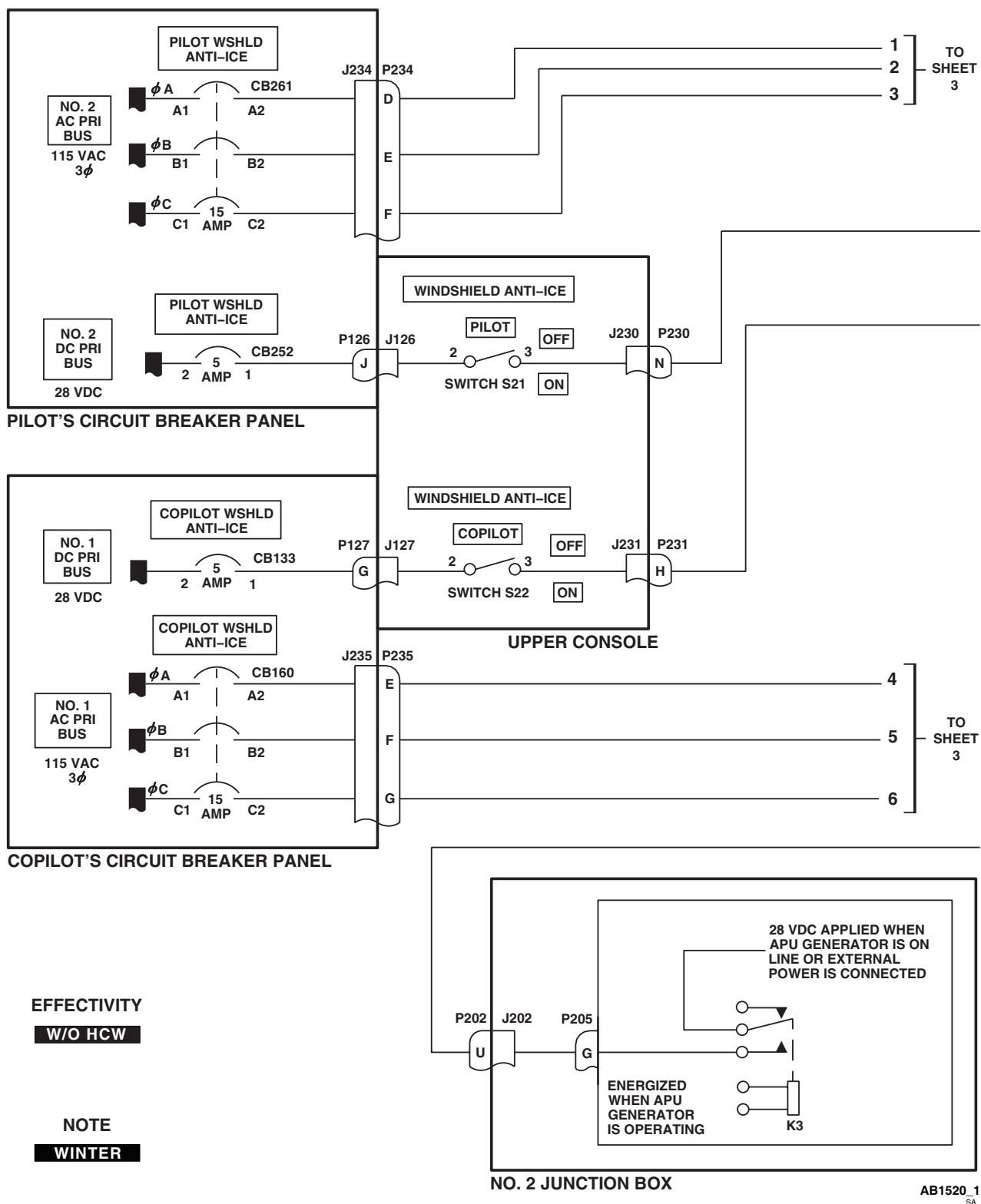


AB1519 4  
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Figure 2. Windshield Anti-ice System Schematic Diagram **HCW** . (Sheet 4 of 4)



## SCHEMATICS - Continued

Figure 3. Windshield Anti-ice System Schematic Diagram **W/O HCW** . (Sheet 1 of 3)

SCHEMATICS - Continued

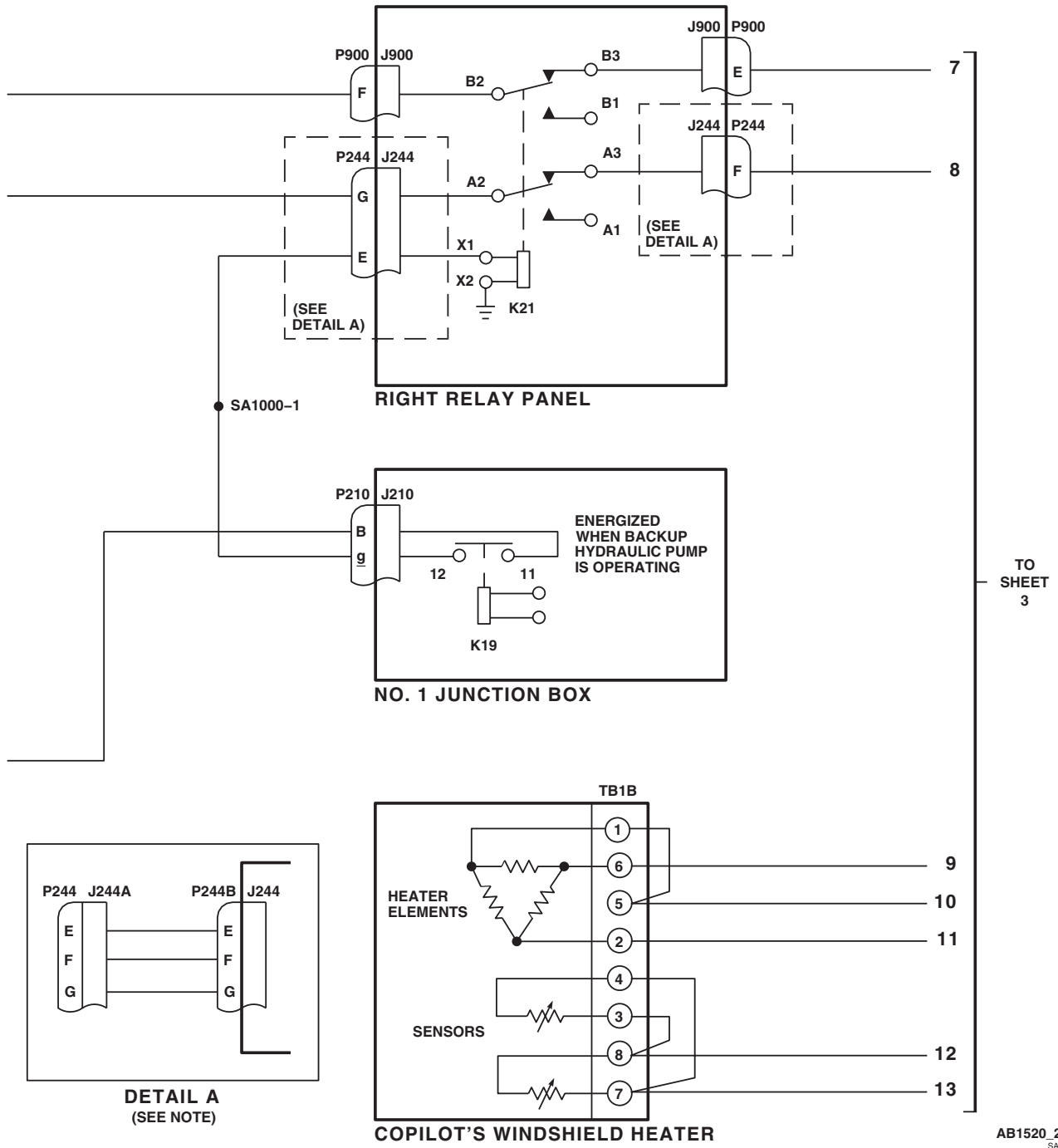
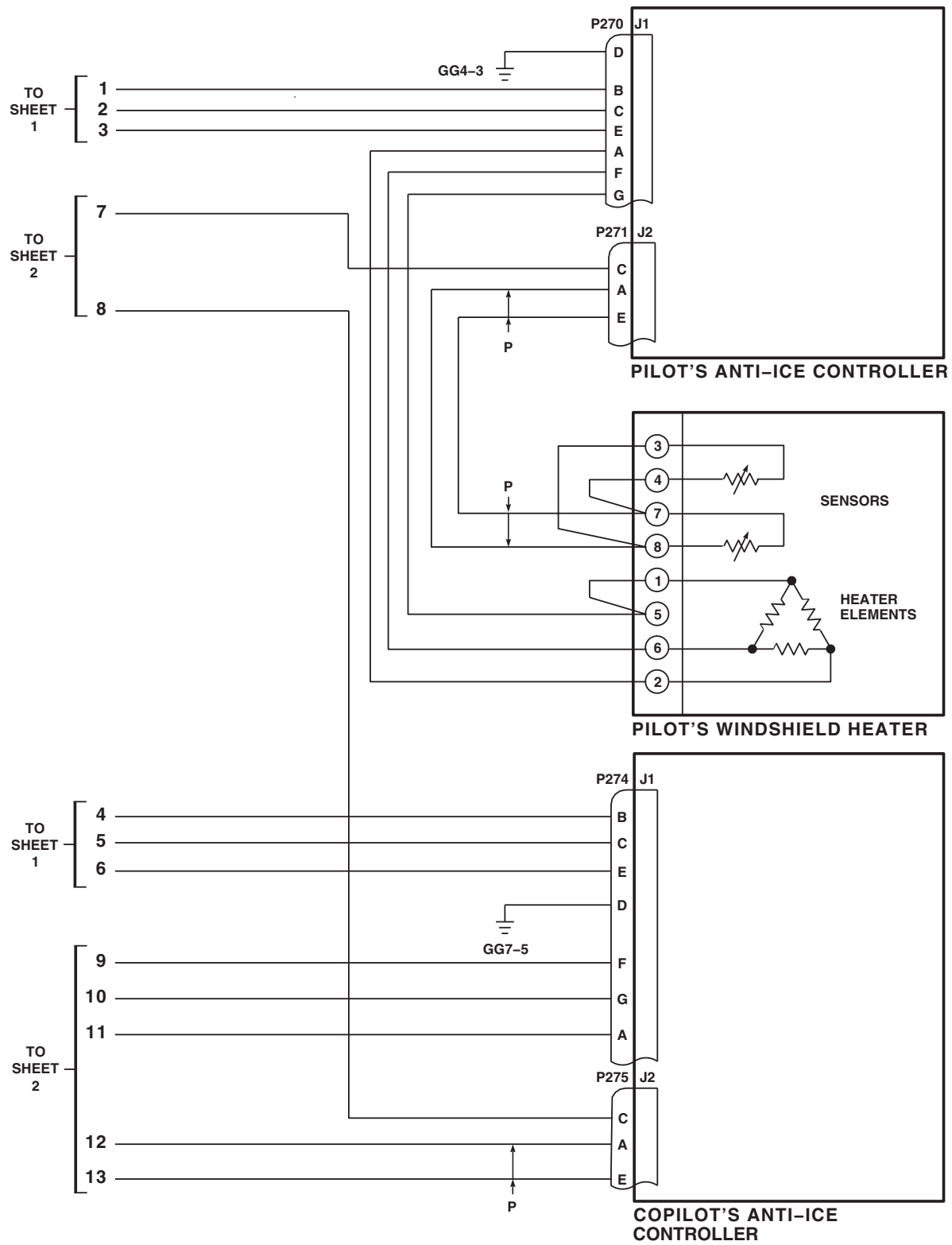
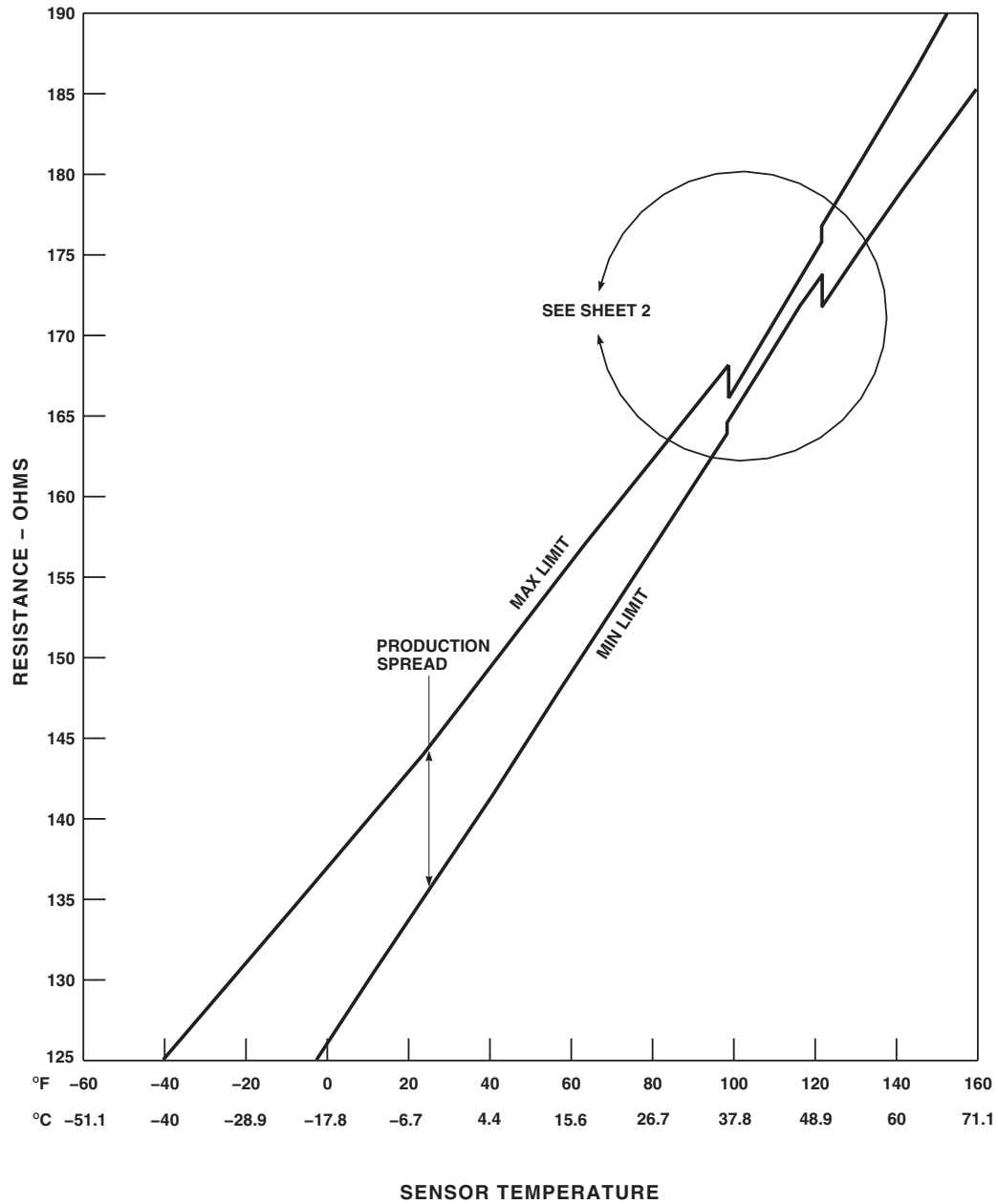


Figure 3. Windshield Anti-ice System Schematic Diagram **W/O HCW** . (Sheet 2 of 3)

## SCHEMATICS - Continued

AB1520\_3  
SAFigure 3. Windshield Anti-ice System Schematic Diagram **W/O HCW** . (Sheet 3 of 3)

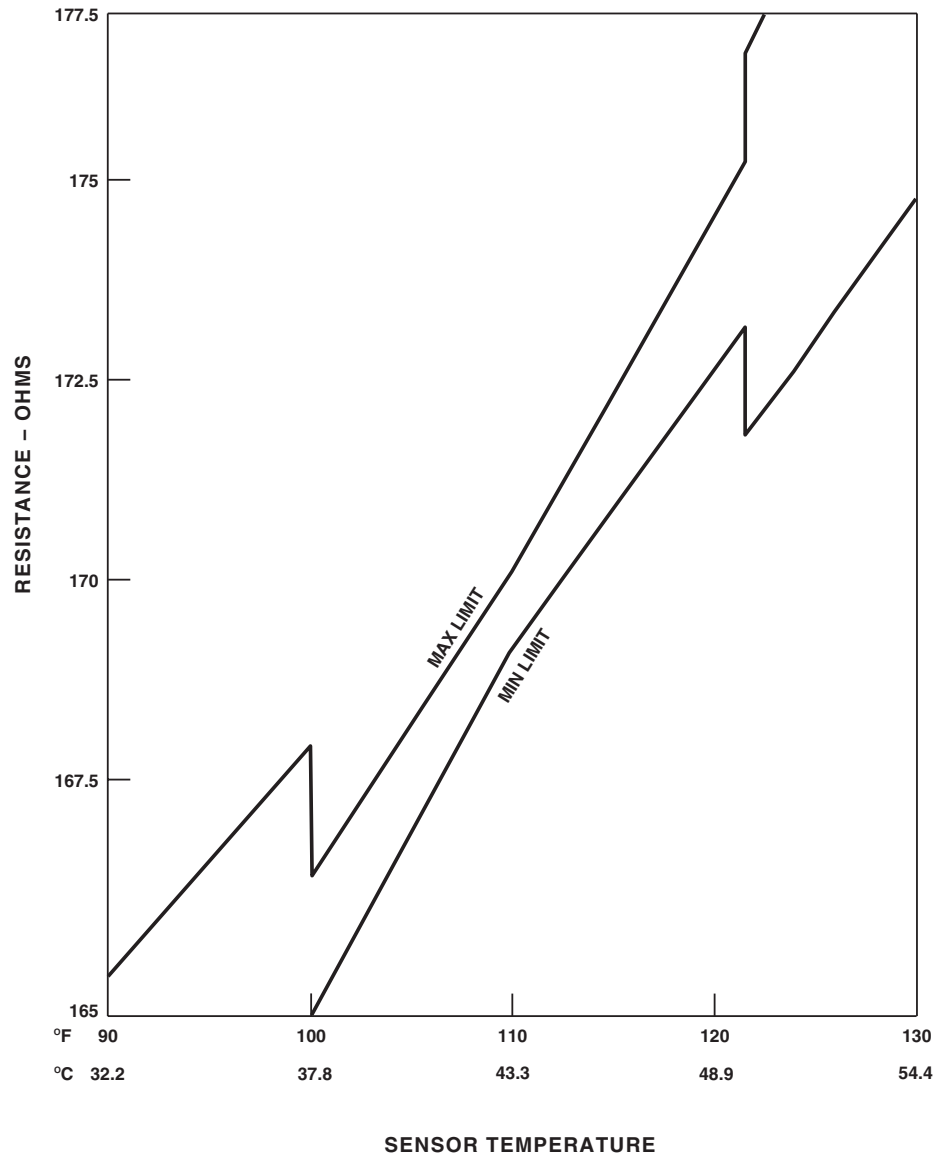
SCHEMATICS - Continued



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Figure 4. Windshield Anti-ice System Temperature/Resistance Chart. (Sheet 1 of 2)

SCHEMATICS - Continued



AA6438\_2  
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Figure 4. Windshield Anti-ice System Temperature/Resistance Chart. (Sheet 2 of 2)

END OF WORK PACKAGE



## UNIT LEVEL

## UTILITY SYSTEM

## BLADE DEICING SYSTEM

This WP supersedes WP 0145 00, dated 21 March 2007.

## INITIAL SETUP:

**Test Equipment**

Clamp-On Ammeter, 749  
Multimeter, AN/PSM-45A

WP 0849 00

WP 0925 00

WP 1220 00

WP 1221 00

**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01  
Electrical Repairer Toolkit, SC 5180-99-B06  
Locally-Made Blade Deice Test Block Figure 148,  
WP 1805 00  
Locally-Made Blade Deice Test Harness Figure 147,  
WP 1805 00

WP 1223 00

WP 1224 00

WP 1226 00

WP 1228 00

WP 1230 00

WP 1231 00

WP 1232 00

**Personnel Required**

Aircraft Electrician MOS 15F (Two)  
UH-60 Helicopter Repairer MOS 15T (1)

WP 1233 00

WP 1234 00

WP 1235 00

**References**

TM 1-1520-237-10

WP 1236 00

WP 1237 00

WP 1238 00

WP 1239 00

WP 1240 00

WP 1241 00

WP 1243 00

WP 1244 00

WP 1246 00

WP 1248 00

WP 1400 00

WP 1685 00

WP 1747 00

WP 1805 00

WP 0090 00

WP 0100 00

WP 0101 00

WP 0102 00

WP 0104 00

WP 0122 00

WP 0555 00

WP 0576 00

WP 0585 00

WP 0785 00

WP 0799 00

WP 0824 00

WP 0825 00

WP 0834 00

WP 0835 00

WP 0840 00

WP 0845 00

**Equipment Condition**

External Electrical Power Available, WP 1685 00  
and No. 1 and No. 2 Generators Operational,  
TM 1-1520-237-10  
Rotor at 100%, TM 1-1520-237-10

## SPECIAL INSTRUCTIONS

### NOTE

See Figure 1 for component location diagram, **EH-60A** Figure 2  and **HH-60A HH-60L UH-60A UH-60L** Figure 3  for schematic diagrams, and Figure 4 for temperature/resistance chart as an aid in operational/troubleshooting.

If main rotor blade erosion protection kit is installed, do not perform blade deicing system operational/troubleshooting procedure.

The blade deicing system operational/troubleshooting procedure shall be performed no more than one hour after passing a minimum of one ampere through all 12 legs of the blade heater. This can be done by performing the blade deice preflight test (TM 1-1520-237-MTF).

If system was recently installed, do steps 1. through 67. to check operation of blade deicing system. To check blade deicing system for unscheduled maintenance, do steps 1. through 63.

## BLADE DEICING SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE

### PROCEDURE

#### Step

### CAUTION

To prevent damage to the blades, if PWR MAIN RTR or PWR TAIL RTR capsules go on for 15 seconds or more, immediately remove electrical power from helicopter.

### NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. **EMEP** Remove and check pin filtered adapters from P134, P324, and P325. Replace if necessary (WP 0925 00). 
2. **EMEP** Install pin filtered adapters (WP 0925 00). 
3. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
4. Place blade deice control panel POWER switch to OFF and MODE switch to AUTO.
5. Place blade de-ice test panel test function switch to NORM.
6. Make sure upper console BACKUP HYD PUMP switch is at OFF.

### NOTE

Unless otherwise specified, ignore the PWR MAIN RTR and/or PWR TAIL RTR lamps if either goes on for less than 10 seconds.

7. Turn on external electrical power. **HH-60A HH-60L** Turn pilot's or copilot's multifunction display (MFD) switch ON and press T6 switch (illuminates all MFD legends for 10 seconds, then only the active caution and advisory legends will be displayed). 



---

**BLADE DEICING SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**

8. Place caution/advisory panel BRT/DIM-TEST switch to TEST and hold or **HH-60A HH-60L** press T6 switch on MFD. ■

**Indication/Condition**

All caution/advisory capsules or MFD legends shall go on.

**Corrective Action**

If Indication/Condition is not as specified, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00).

**Step**

9. Release caution/advisory panel BRT/DIM-TEST switch.

**Indication/Condition**

Caution/advisory panel #1 GEN, #2 GEN, or EXT PWR CONNECTED (if external power is connected) capsules or legends shall remain on.

**Corrective Action**

If Indication/Condition is not as specified, troubleshoot AC electrical system (WP 0101 00).

**Step**

10. **UH-60A UH-60L HH-60A HH-60L** Pull out ICE DET (DC) circuit breaker on mission readiness circuit breaker panel. ■ **EH-60A** Pull out ICE DET (DC) circuit breaker on copilot's circuit breaker panel. ■

**Indication/Condition**

Icing rate meter shall indicate zero and FAIL flag shall appear.

**Corrective Action**

If Indication/Condition is not as specified, replace icing rate meter (WP 1221 00).

**Step**

11. Push in ICE DET (DC) circuit breaker.

**Indication/Condition**

- a. Icing rate meter FAIL flag shall disappear from view.
- b. Icing rate meter needle shall deflect and return to zero.

**Corrective Action**

1. If Indication/Conditions a. and b. are not as specified, go to ICING RATE METER FAIL FLAG IN VIEW AND METER NEEDLE DID NOT DEFLECT AND RETURN TO ZERO, in this work package.
2. If either Indication/Condition is not as specified, replace icing rate meter (WP 1221 00).

---

**BLADE DEICING SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Step****NOTE**

- To avoid the possibility of obtaining a false indication, wait at least 20 seconds after power is turned on before pressing the icing rate meter PRESS TO TEST pushbutton.
- Allow at least 75 seconds between each PRESS TO TEST pushbutton actuation to allow test cycle to complete.

12. Press and release icing rate meter PRESS TO TEST pushbutton and simultaneously start timer (instrument panel clock). **RECORD RESULT.**

**Indication/Condition**

- a. Icing rate meter needle shall move to half scale (1.0 grams per cubic centimeter LWC) with a tolerance of 1/8 inch on indicator scale.
- b. In 15 to 25 seconds after pressing PRESS TO TEST pushbutton, ICE DETECTED capsule or legend shall go on.
- c. In 40 to 60 seconds after pressing PRESS TO TEST button, icing rate meter needle shall return to below zero, and ICE DETECTED capsule or **HH-60A HH-60L** MFD ICE DETECTED legend  shall go off.
- d. Within 75 seconds after pressing PRESS TO TEST button, icing rate meter needle shall move to zero to complete test.

**Corrective Action**

1. If Indication/Condition a. is not as specified, replace icing rate meter (WP 1221 00).
2. If Indication/Condition b. is not as specified, go to ICE DETECTED CAPSULE OR LEGEND DOES NOT GO ON, in this work package.
3. If Indication/Condition c. is not as specified, replace icing rate meter (WP 1221 00).
4. If Indication/Condition d. is not as specified, replace icing rate meter (WP 1221 00).

**Step****CAUTION**

To prevent damage to main rotor blades, remove locally-made blade de-ice test block (test block) (WP 1805 00) from ice detector when icing rate needle moves. When removing test block, be careful not to hit and damage sensing probe.

**NOTE**

- Icing rate meter needle may hold value for 50 to 70 seconds after test block is removed and the ICE DETECTED capsule or **HH-60A HH-60L** MFD ICE DETECTED legend  may go on.
- The greater the pressure on the test block, the greater the icing rate meter needle indication.

13. Carefully place test block against upper half of ice detector probe.

**Indication/Condition**

Icing rate meter needle shall move to some value between zero and full scale.

---

**BLADE DEICING SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Corrective Action**

1. If Indication/Condition is not as specified, go to ICING RATE METER NEEDLE DOES NOT MOVE, in this work package.
2. If the ICE DETECTED capsule or **HH-60A HH-60L** MFD ICE DETECTED legend **■** came on in step 13., wait up to 2 minutes after block is removed for ICE DETECTED capsule or **HH-60A HH-60L** MFD ICE DETECTED legend **■** to go off before continuing. If capsule does not go off as specified, replace icing rate meter (WP 1221 00).

**Step**

14. Reset instrument panel clock.
15. **EMEP** Remove and check pin filtered adapter from P117. Replace if necessary (WP 0925 00). **■**
16. **EMEP** Install pin filtered adapter on P117 (WP 0925 00). **■**
17. Make sure the following caution/advisory capsules or **HH-60A HH-60L** MFD legends **■** are off:
  - #1 CONV
  - #2 CONV
  - MR DE-ICE FAIL
  - TR DE-ICE FAIL
  - MR DE-ICE FAULT
  - ICE DETECTED

**Indication/Condition**

The specified capsules or legends shall be off.

**Corrective Action**

1. If #1 CONV or #2 CONV capsule or legend is on, troubleshoot DC electrical system (WP 0102 00).
2. If M/R DE-ICE FAIL, T/R DE-ICE FAIL or M/R DE-ICE FAULT capsule or legend is on, go to WITH BLADE DEICE CONTROL PANEL POWER SWITCH OFF, TR DE-ICE FAIL, MR DE-ICE FAIL, OR MR DE-ICE FAULT CAPSULE OR LEGEND IS ON, in this work package.
3. If ICE DETECTED capsule or legend is on, check for 28 vdc between P117-63 or **HH-60A HH-60L** P1019R-49 or P1020R-49 **■** and ground.
  - a. If voltage is as specified, replace icing rate meter (WP 1221 00).
    - (1) If trouble remains, replace ice detector (WP 1244 00).
  - b. If voltage is not as specified, troubleshoot caution/advisory warning system (WP 0090 00) or **HH-60A HH-60L** MFD/caution/advisory system (WP 0100 00) **■**.

**Step**

18. Make sure blade de-ice test panel PWR MAIN RTR and PWR TAIL RTR lamps are off.

**Indication/Condition**

The specified lamps shall be off.

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**BLADE DEICING SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued**Corrective Action**

1. If both PWR MAIN RTR and PWR TAIL RTR lamps are on, check for 28 vdc between APU/GENERATOR INTERLOCK RELAY terminals K64-X1 and K64-X2.
  - a. If voltage is as specified, replace blade deice control panel (WP 1223 00).
  - b. If voltage is not as specified, replace K64 (WP 1231 00).
2. If PWR MAIN RTR lamp is on, go to PWR MAIN RTR LAMP IS ON, in this work package.
3. If PWR TAIL RTR lamp is on, go to PWR TAIL RTR LAMP IS ON, in this work package.

**Step**

19. Momentarily press blade de-ice test panel PWR MAIN RTR lamp.

**Indication/Condition**

PWR MAIN RTR lamp shall go on, then off.

**Corrective Action**

If Indication/Condition is not as specified, go to PWR TAIL RTR LAMP DOES NOT GO ON WHEN PRESS-TO-TEST, in this work package.

**Step**

20. Momentarily press blade de-ice test panel PWR TAIL RTR lamp.

**Indication/Condition**

PWR TAIL RTR lamp shall go on, then off.

**Corrective Action**

1. If Indication/Condition is not as specified, unscrew lamp cap and check lamp.
  - a. If lamp is good, replace blade de-ice test panel (WP 1240 00).
  - b. If lamp is not good, replace lamp (WP 1241 00).

**Step**

21. Momentarily press blade deice control panel TEST IN PROGRESS lamp.

**Indication/Condition**

TEST IN PROGRESS lamp shall go on, then off.

**Corrective Action**

If Indication/Condition is not as specified, go to TEST IN PROGRESS LAMP DOES NOT CYCLE ON AND OFF, in this work package.

---

**BLADE DEICING SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Step****CAUTION**

To prevent damage to the blades, do not test blade deice system more than once in 30 minutes when rotor head is not turning.

**NOTE**

Backup pump may run to recharge APU hydraulic start accumulator. Wait until BACKUP PUMP ON capsule or **HH-60A HH-60L** MFD BACKUP PUMP ON legend  goes off before starting step 22. If backup pump goes on while doing steps 22. through 35., and APU generator is only power source for the test, the test will automatically terminate. This will be indicated by the M/R DE-ICE FAIL caution capsule or **HH-60A HH-60L** MFD M/R DE-ICE FAIL legend  and the BACKUP PUMP ON advisory capsule or **HH-60A HH-60L** MFD BACKUP PUMP ON legend  going on. If this occurs, place blade deice control panel POWER switch to OFF, wait until backup pump goes off, and then continue operational/troubleshooting procedure starting with step 22.

22. Place blade deice control panel POWER switch to TEST, and simultaneously start instrument panel clock.

**RECORD RESULT.****Indication/Condition**

- a. TEST IN PROGRESS lamp shall go on and stay on for 105 to 135 seconds.
- b. MR DE-ICE FAIL, T/R DE-ICE FAIL, and M/R DE-ICE FAULT capsules or legends shall remain off for full 105 to 135 seconds of test.
- c. PWR MAIN RTR lamp shall go on for up to 11 seconds at end of test.
- d. PWR TAIL RTR lamp shall go on for up to 4 seconds at end of test.

**Corrective Action**

1. If Indication/Condition a. is not as specified, go to TEST IN PROGRESS LAMP DOES NOT FUNCTION AS SPECIFIED, in this work package.
2. If M/R DEICE FAIL and M/R DEICE FAULT capsules or legends go on, go to MR DEICE FAIL AND MR DEICE FAULT CAPSULES OR LEGENDS GO ON, in this work package.
3. If M/R DE-ICE FAIL capsule or legend goes on, go to MR DE-ICE FAIL CAPSULE OR LEGEND GOES ON, in this work package.
4. If T/R DE-ICE FAIL capsule or legend goes on, go to TR DE-ICE FAIL CAPSULE OR LEGEND GOES ON, in this work package.
5. If M/R DE-ICE FAIL and T/R DE-ICE FAIL capsules or legends both go on, go to MR DE-ICE FAIL AND TR DE-ICE FAIL CAPSULES OR LEGENDS ARE BOTH ON, in this work package.
6. If M/R DE-ICE FAULT capsule or legend goes on, go to MR DE-ICE FAULT CAPSULE OR LEGEND GOES ON, in this work package.
7. If PWR MAIN RTR lamp remains on, go to PWR TAIL RTR LAMP IS ON, in this work package.
8. If PWR TAIL RTR lamp remains on, go to PWR TAIL RTR LAMP DOES NOT CYCLE ON AND OFF AND/OR PWR TAIL RTR LAMP DOES NOT GO ON WHEN PRESS-TO-TEST, in this work package.
9. If Indication/Condition c. or d. is not as specified, go to PWR MAIN RTR AND PWR TAIL RTR LAMPS DO NOT GO ON, in this work package.

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**BLADE DEICING SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Step**

23. Reset instrument panel clock.

**WARNING**

Droop stop hinge pins and cams become very hot within 30 to 60 seconds after blade deice control panel POWER switch is placed to TEST. To prevent injury, only momentarily touch droop stop cams.

24. Touch each droop stop cam.

**Indication/Condition**

Droop stop cams shall be warm to touch.

**Corrective Action**

If Indication/Condition is not as specified, go to DROOP STOP CAMS ARE NOT WARM WHEN POWER SWITCH IS IN TEST, in this work package.

**Step**

25. Place blade deice control panel POWER switch to OFF.  
26. Place blade de-ice test panel test function switch to SYNC 1.  
27. Place blade deice control panel POWER switch to TEST.

**Indication/Condition****NOTE**

MR DE-ICE FAULT capsule or **HH-60A HH-60L** legends  may go on.

MR DE-ICE FAIL capsule or legend shall go on.

**Corrective Action**

1. If Indication/Condition is not as specified, check for 115 vac between P134-L and P134-T.
  - a. If voltage is as specified, replace blade de-ice test panel (WP 1240 00).
  - b. If voltage is not as specified, repair/replace wiring between P134-L and P906-W (WP 1747 00).

**Step**

28. Place blade deice control panel POWER switch to OFF.

**Indication/Condition**

MR DE-ICE FAIL capsule or **HH-60A HH-60L** legends  shall go off.

**Corrective Action**

If Indication/Condition is not as specified, replace rotor blade deice controller (WP 1226 00).

**Step**

29. Place blade de-ice test panel test function switch to SYNC 2.  
30. Place blade deice control panel POWER switch to TEST.

---

**BLADE DEICING SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Indication/Condition****NOTE**

MR DE-ICE FAULT capsule or **HH-60A HH-60L** legends  may go on.

MR DE-ICE FAIL capsule or **HH-60A HH-60L** legends  shall go on.

**Corrective Action**

1. If Indication/Condition is not as specified with test function switch in SYNC 2, check for open between P134-N and P134-P.
  - a. If open is present, replace rotor blade deice controller (WP 1226 00).
  - b. If open is not present, replace blade de-ice test panel (WP 1240 00).

**Step**

31. Place blade deice control panel POWER switch to OFF.

**Indication/Condition**

MR DE-ICE FAIL capsule shall go off.

**Corrective Action**

If Indication/Condition is not as specified, replace rotor blade deice controller (WP 1226 00).

**Step**

32. Place blade de-ice test panel test function switch to OAT.
33. Place blade deice control panel POWER switch to TEST.

**Indication/Condition**

MR DE-ICE FAIL and T/R DE-ICE FAIL capsules or **HH-60A HH-60L** legends  shall go on.

**Corrective Action**

1. If M/R DE-ICE FAIL capsule or **HH-60A HH-60L** legends  goes on but T/R DE-ICE FAIL capsule does not, replace rotor blade deice controller (WP 1226 00).
2. If T/R DE-ICE FAIL capsule or **HH-60A HH-60L** legends  goes on but M/R DE-ICE FAIL capsule does not, replace rotor blade deice controller (WP 1226 00).
3. If neither capsule goes on with test function switch in OAT position, check continuity between J134-X and J134-W.
  - a. If continuity is present, replace rotor blade deice controller (WP 1226 00).
  - b. If continuity is not present, replace blade de-ice test panel (WP 1240 00).

**Step**

34. Place blade deice control panel POWER switch to OFF.

**Indication/Condition**

MR DE-ICE FAIL and T/R DE-ICE FAIL capsules or **HH-60A HH-60L** legends  shall go off.

**BLADE DEICING SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Corrective Action**

If Indication/Condition is not as specified, replace rotor blade deice controller (WP 1226 00).

**Step**

35. Place blade de-ice test panel test function switch to NORM.
36. Start APU (TM 1-1520-237-10).
37. With qualified personnel at controls start engines and adjust rotor speed to 100% (TM 1-1520-237-10).
38. Place upper console GENERATORS - NO. 1 and NO. 2 switches to ON.

**Indication/Condition**

#1 GEN and #2 GEN capsules shall be off.

**Corrective Action**

If Indication/Condition is not as specified, troubleshoot AC electrical system (WP 0101 00 or WP 0122 00).

**Step****CAUTION**

To prevent rotor blades from overheating in ambient temperatures above 70°F (21°C), operate rotor at 100% for 5 minutes before doing deice EOT check. Do not do deice EOT check if FAT is above 100°F (38°C).

39. Place blade de-ice test panel test function switch to EOT.
40. Place blade deice control panel MODE switch to MANUAL-M.

**NOTE**

- PWR MAIN RTR lamp may flash at start of test.
- MR DE-ICE FAULT capsule or **HH-60A HH-60L** legends   may go on.

41. Place blade deice control panel POWER switch to ON, and simultaneously start instrument panel clock. **RECORD RESULT.**

**Indication/Condition**

- a. TR DE-ICE FAIL capsule or **HH-60A HH-60L** legends   shall go on 18 to 26 seconds after POWER switch is placed ON.
- b. MR DE-ICE FAIL capsule or **HH-60A HH-60L** legends   shall go on 50 to 70 seconds after POWER switch is placed ON.

**Corrective Action**

1. If T/R DE-ICE FAIL capsule or legend goes on, but M/R DE-ICE FAIL capsule or legend does not, check continuity between P325-W and P134-S.
  - a. If continuity is present, replace blade de-ice test panel (WP 1240 00).
    - (1) If trouble remains, replace rotor blade deice controller (WP 1226 00).
  - b. If continuity is not present, repair/replace wiring (WP 1747 00).



**BLADE DEICING SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**

2. If M/R DE-ICE FAIL capsule or legend goes on but T/R DE-ICE FAIL capsule or legend does not, check continuity between P325-V and P134-U.
3. If neither capsule or legend goes on, check continuity between P134-R and ground.
  - a. If continuity is present, replace blade de-ice test panel (WP 1240 00).
    - (1) If trouble remains, replace rotor blade deice controller (WP 1226 00).
  - b. If continuity is not present, repair/replace wiring (WP 1747 00).

**Step**

42. Place blade deice control panel POWER switch to OFF.
43. Place blade de-ice test panel test function switch to NORM.
44. Place blade deice control panel POWER switch to ON, and wait 30 seconds.

**Indication/Condition**

MR DE-ICE FAIL, T/R DE-ICE FAIL, and M/R DE-ICE FAULT capsules or **HH-60A HH-60L** legends  shall remain off.

**Corrective Action**

1. If M/R DE-ICE FAIL capsule or legends goes on, check for 115 vac between MN RTR BUS CONT terminals L1, L2, L3 and ground.
  - a. If voltage is as specified, replace rotor blade deice controller (WP 1226 00).
  - b. If voltage is not as specified, replace K2 NO. 2 GEN CNTOR (WP 0835 00).
2. If T/R DE-ICE FAIL capsule or legend goes on, check for 115 vac between TAIL ROTOR DE-ICE CONTACTOR terminals A1, B1, C1 and ground.
  - a. If voltage is as specified, replace rotor blade deice controller (WP 1226 00).
  - b. If voltage is not as specified, replace K1 NO. 1 GEN CNTOR (WP 0834 00).
3. If M/R DE-ICE FAIL and T/R DE-ICE FAIL capsules or legends go on, go to MR DE-ICE FAIL AND TR DE-ICE FAIL CAPSULES OR LEGENDS ARE ON, in this work package.
4. If M/R DE-ICE FAULT capsule or legend go on, replace NO. 2 generator contactor K2 (WP 0835 00).

**Step**

45. Place blade deice control panel POWER switch to OFF.
46. Place upper console GENERATORS - NO. 1 switch to OFF and APU switch to ON.
47. Place blade deice control panel POWER switch to ON and wait 30 seconds.

**Indication/Condition**

MR DE-ICE FAIL, T/R DE-ICE FAIL, and M/R DE-ICE FAULT capsules or **HH-60A HH-60L** legends  shall remain off.

**Corrective Action**

If Indication/Condition is not as specified, go to MR DE-ICE FAIL, TR DE-ICE FAIL AND/OR MR DE-ICE FAULT CAPSULES OR LEGENDS GO ON, in this work package.

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**BLADE DEICING SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Step**

48. Place blade deice control panel POWER switch to OFF.
49. Place upper console GENERATORS - NO. 1 switch to ON, and NO. 2 switch to OFF.
50. Place blade deice control panel POWER switch to ON, and wait 30 seconds.

**Indication/Condition**

MR DE-ICE FAIL, T/R DE-ICE FAIL, and M/R DE-ICE FAULT capsules or **HH-60A HH-60L** legends  shall remain off.

**Corrective Action**

If Indication/Condition is not as specified, go to MR DE-ICE FAIL, TR DE-ICE FAIL AND/OR MR DE-ICE FAULT CAPSULES OR LEGENDS GO ON, in this work package.

**Step**

51. Place blade deice control panel POWER switch to OFF.
52. Place upper console GENERATORS - NO. 2 switch to ON, and APU switch to OFF.
53. Place blade deice control panel POWER switch to ON and wait 30 seconds.

**Indication/Condition**

MR DE-ICE FAIL, T/R DE-ICE FAIL and M/R DE-ICE FAULT capsules or **HH-60A HH-60L** legends  shall remain off.

**Corrective Action**

If Indication/Condition is not as specified, go to MR DE-ICE FAIL, TR DE-ICE FAIL AND MR DE-ICE FAULT CAPSULES OR LEGENDS GO ON, in this work package.

**Step**

54. Place blade deice control panel POWER switch to OFF.
55. Place upper console GENERATORS - NO. 2 switch to OFF.

**CAUTION**

To prevent damage to the blades, place blade deice control panel POWER switch to OFF if Indication/Condition of step 56. is not obtained within 30 seconds.

56. Place blade deice control panel POWER switch to ON.

**Indication/Condition**

MR DE-ICE FAIL and M/R DE-ICE FAULT capsules or **HH-60A HH-60L** legends  shall go on immediately, and T/R DE-ICE FAIL capsule or **HH-60A HH-60L** legends  shall go on after 30 seconds.

**Corrective Action**

1. If none of the capsules or legends go on, go to NONE OF THE CAPSULES OR LEGENDS GO ON, in this work package.
2. If only M/R capsules or legends, or T/R capsule or legends go on, replace rotor blade deice controller (WP 1226 00).

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**BLADE DEICING SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Step**

57. Place blade deice control panel POWER switch to OFF.
58. Place upper console GENERATORS - APU switch to ON and NO. 1 switch to OFF.
59. Place blade deice control panel POWER switch to ON.

**Indication/Condition**

MR DE-ICE FAIL and M/R DE-ICE FAULT capsules or **HH-60A HH-60L** legends  shall go on immediately, and T/R DE-ICE FAIL capsule shall go on after 20 seconds.

**Corrective Action**

If Indication/Condition is not as specified, go to WITH BLADE DEICE POWER SWITCH ON, MR DE-ICE FAIL, TR DE-ICE FAIL, OR BOTH CAPSULES/LEGENDS GO ON WITH MODE SWITCH PLACED TO T, L, OR M, in this work package.

**Step**

60. Place blade deice control panel MODE switch to AUTO.
61. Place blade deice control panel POWER switch to OFF.
62. Shut down engines, rotor, and APU (TM 1-1520-237-10).
63. **HH-60A HH-60L** Turn MFD switch OFF. 
64. Turn off electrical power.

**NOTE**

To obtain accurate resistance values of the rotor blade heater mat elements, a minimum of one ampere voltage current must be passed through blade heater mat elements for a minimum of 15 seconds or the blade deice operational/troubleshooting procedure must be done.

65. Select maximum resistance range on multimeter.
66. Within one hour of operational/troubleshooting procedure, check resistance between the following pins on J841, J842, J843, and J844:
  - 1 and 2
  - 1 and 3
  - 2 and 3
  - 4 and 5
  - 4 and 6
  - 5 and 6
  - 7 and 8
  - 7 and 9
  - 8 and 9
  - 10 and 11

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**BLADE DEICING SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**

- 10 and 12
- 11 and 12

**Indication/Condition**

Resistance shall be between 7 and 10 ohms.

**Corrective Action**

If Indication/Condition is not as specified, replace affected main rotor blade (WP 0576 00).

**Step**

67. Within one hour of operational/troubleshooting procedure, check resistance between the following pins on J611, J612, J613, and J614:

- 1 and 2
- 3 and 4
- 5 and 6

**Indication/Condition**

Resistance shall be between 6 and 8 ohms.

**Corrective Action**

If Indication/Condition is not as specified, replace affected tail rotor blade (WP 0585 00).

**ICING RATE METER FAIL FLAG IN VIEW AND METER NEEDLE DID NOT DEFLECT AND RETURN TO ZERO****SYMPTOM**

Icing rate meter FAIL flag in view and meter needle did not deflect and return to zero.

**MALFUNCTION**

Icing rate meter FAIL flag in view and meter needle did not deflect and return to zero.

**CORRECTIVE ACTION****1. Check for 28 vdc between P132-F and P132-G.**

- a. If voltage is as specified, go to **2**.
- b. If voltage is not as specified, go to **6**.

**2. Check for 115 vac between P132-S and P132-G.**

- a. If voltage is as specified, go to **3**.
- b. If voltage is not as specified, go to **5**.

**3. Check continuity between:**

- P132-A and P326-1
- P132-B and P326-2
- P132-C and P326-3
- P132-D and P326-4

**ICING RATE METER FAIL FLAG IN VIEW AND METER NEEDLE DID NOT DEFLECT AND RETURN TO ZERO - Continued**

- **P132-E and P326-5**
- **P132-T and P326-6**
- a. If continuity is present, go to **4**.
- b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **8**.
- 4. Check continuity between:**
  - **P326-8 and ground**
  - **P326-9 and ground**
  - a. If continuity is present, replace ice detector (WP 1244 00). Go to **8**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **8**.
  - c. If trouble remains, replace icing rate meter (WP 1221 00). Go to **8**.
- 5. Check for 115 vac between ICE DET circuit breaker terminal 1 and ground.**
  - a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P132-S (WP 1747 00). Go to **8**.
  - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **8**.
- 6. Check continuity between P132-G and ground.**
  - a. If continuity is present, go to **7**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **8**.
- 7. Check for 28 vdc between ICE DET circuit breaker terminal 1 and ground.**
  - a. If voltage is as specified, repair/replace wiring between P132-F and ICE DET circuit breaker terminal 1 (WP 1747 00). Go to **8**.
  - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **8**.
- 8. Procedure completed.**

**ICE DETECTED CAPSULE OR LEGEND DOES NOT GO ON****SYMPTOM**

ICE DETECTED capsule or legend does not go on.

**MALFUNCTION**

ICE DETECTED capsule or legend does not go on.

**CORRECTIVE ACTION**

- 1. Check helicopter configuration.**
  - a. **EMEP** , go to **2**.
  - b. **W/O EMEP** , go to **4**.
- 2. EMEP Remove and check pin filtered adapter P117 or HH-60A HH-60L P1019R or P1020R (WP 0925 00).**
  - a. If pin filtered adapter is good, go to **3**.
  - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **11**.

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**ICE DETECTED CAPSULE OR LEGEND DOES NOT GO ON** - Continued

3. **EMEP** Install pin filtered adapter (WP 0925 00). Go to 4.
4. Check for 28 vdc between P117-63 or **HH-60A HH-60L** P1019R-49 or P1020R-49 and ground.
  - a. If voltage is as specified, replace caution/advisory panel (WP 0785 00) or MFD (WP 0799 00). Go to 11.
  - b. If voltage is not as specified, go to 5.
5. Check continuity between P117-63 or **HH-60A HH-60L** P1019R-49 or P1020R-49 and P132-J.
  - a. If continuity is present, go to 6.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to 11.
6. Check for 28 vdc between P132-H and P132-G.
  - a. If voltage is as specified, replace icing rate meter (WP 1221 00). Go to 11.
  - b. If voltage is not as specified, go to 7.
7. Check continuity between P132-H and P133-T.
  - a. If continuity is present, go to 8.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to 11.
8. Check for 28 vdc between P133-L and ground.
  - a. If voltage is as specified, replace blade deice control panel (WP 1223 00). Go to 11.
  - b. If voltage is not as specified, go to 9.
9. Check for 28 vdc between J906-N and ground.
  - a. If voltage is as specified, repair/replace wiring between P906-N and P133-L (WP 1747 00). Go to 11.
  - b. If voltage is not as specified, go to 10.
10. Check for 28 vdc between DE-ICE CNTRLR circuit breaker terminal 1 and ground.
  - a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and J906-N (WP 1747 00). Go to 11.
  - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to 11.
11. Procedure completed.

**ICING RATE METER NEEDLE DOES NOT MOVE****SYMPTOM**

Icing rate meter needle does not move.

**MALFUNCTION**

Icing rate meter needle does not move.

**CORRECTIVE ACTION**

1. Check continuity between:
  - P132-A and P326-1

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ICING RATE METER NEEDLE DOES NOT MOVE - Continued

- P132-B and P326-2
- P132-C and P326-3
- P132-D and P326-4
- P132-E and P326-5
- P132-T and P326-6

- a. If continuity is present, go to **2**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **5**.

**2. Check helicopter configuration.**

- a. **EMEP** , go to **3**.
- b. **W/O EMEP** , go to **4**.

**3. Check for corresponding resistances between:**

- J326-1 and J326-2 Greater than 1000 ohms
- J326-3 and J326-4 Greater than 1000 ohms
- J326-5 and J326-8 4.9 to 5.9 ohms
- J326-6 and J326-9 47 to 61 ohms

- a. If corresponding resistances are as specified, replace icing rate meter (WP 1221 00). Go to **5**.
- b. If corresponding resistances are not as specified, replace ice detector (WP 1244 00). Go to **5**.

**4. Check for corresponding resistances between:**

- J326-1 and J326-2 15 to 21 ohms
- J326-3 and J326-4 15 to 21 ohms
- J326-5 and J326-8 4.9 to 5.9 ohms
- J326-6 and J326-9 47 to 61 ohms

- a. If corresponding resistances are as specified, replace icing rate meter (WP 1221 00). Go to **5**.
- b. If corresponding resistances are not as specified, replace ice detector (WP 1244 00). Go to **5**.

**5. Procedure completed.**

**WITH BLADE DEICE CONTROL PANEL POWER SWITCH OFF, T/R DE-ICE FAIL, M/R DE-ICE FAIL, OR M/R DE-ICE FAULT CAPSULE OR LEGEND IS ON****SYMPTOM**

With blade deice control panel power switch OFF, T/R DE-ICE FAIL, M/R DE-ICE FAIL, or M/R DE-ICE FAULT capsule or legend is on.

**MALFUNCTION**

With blade deice control panel power switch OFF, T/R DE-ICE FAIL, M/R DE-ICE FAIL, or M/R DE-ICE FAULT capsule or legend is on.

**CORRECTIVE ACTION**

1. With power applied, pull out DE-ICE CNTRLR circuit breaker. Go to 2.
2. Does caution/advisory capsule or MFD legend go off?
  - a. If capsule or legend is off, replace deice controller (WP 1226 00). Go to 7.
  - b. If capsule or legend is not off, go to 3.
3. Check helicopter configuration.
  - a. **EMEP** , go to 4.
  - b. **W/O EMEP** , go to 6.
4. **EMEP** Remove and check pin filtered adapter P117 or **HH-60A HH-60L** P1019R or P1020R (WP 0925 00).
  - a. If pin filtered adapter is good, go to 5.
  - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to 7.
5. **EMEP** Install pin filtered adapter (WP 0925 00). Go to 6.
6. Check for 28 vdc between:
  - P117-15 and ground
  - P117-31 and ground
  - P117-47 and ground
  - **HH-60A HH-60L** P1019R-27 or P1020R-27 and ground
  - **HH-60A HH-60L** P1019R-41 or P1020R-41 and ground
  - **HH-60A HH-60L** P1019R-82 or P1020R-82 and ground
  - a. If voltage is as specified, troubleshoot wiring for short circuit to 28 vdc (WP 1747 00). Go to 7.
  - b. If voltage is not as specified, replace caution/advisory panel (WP 0785 00) or MFD (WP 0799 00). Go to 7.
7. Procedure completed.



**PWR MAIN RTR LAMP IS ON****SYMPTOM**

PWR MAIN RTR lamp is on.

**MALFUNCTION**

PWR MAIN RTR lamp is on.

**CORRECTIVE ACTION**

1. Check helicopter configuration.
  - a. **EMEP** , go to 2.
  - b. **W/O EMEP** , go to 4.
2. **EMEP** Remove and check pin filtered adapter P134 (WP 0925 00).
  - a. If pin filtered adapter is good, go to 3.
  - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to 6.
3. **EMEP** Install pin filtered adapter (WP 0925 00). Go to 4.
4. Check for 13 vac between:
  - P134-D and ground
  - P134-E and ground
  - P134-F and ground
  - a. If voltage is as specified, go to 5.
  - b. If voltage is not as specified, replace blade deice test panel (WP 1240 00). Go to 6.
5. Check for 28 vdc between K60 MAIN ROTOR CONTACTOR terminals X1 and X2.
  - a. If voltage is as specified, replace K64 APU/Generator interlock relay (WP 1231 00). Go to 6.
  - b. If voltage is not as specified, replace contactor (WP 1233 00). Go to 6.
6. Procedure completed.

**PWR TAIL RTR LAMP IS ON****SYMPTOM**

PWR TAIL RTR lamp is on.

**MALFUNCTION**

PWR TAIL RTR lamp is on.

**CORRECTIVE ACTION**

1. Check helicopter configuration.
  - a. **EMEP** , go to 2.
  - b. **W/O EMEP** , go to 4.
2. **EMEP** Remove and check pin filtered adapter P134 (WP 0925 00).
  - a. If pin filtered adapter is good, go to 3.

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**PWR TAIL RTR LAMP IS ON** - Continued

- b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **6**.
3. **EMEP** Install pin filtered adapter (WP 0925 00). Go to **4**.
4. Check for 115 vac between:
  - P134-A and ground
  - P134-B and ground
  - P134-C and ground
  - a. If voltage is as specified, go to **5**.
  - b. If voltage is not as specified, replace blade deice test panel (WP 1240 00). Go to **6**.
5. Check for 28 vdc between K61 TAIL ROTOR CONTACTOR terminals X1 and X2.
  - a. If voltage is as specified, replace K64 APU/Generator interlock relay (WP 1231 00). Go to **6**.
  - b. If voltage is not as specified, replace contactor (WP 1230 00). Go to **6**.
6. Procedure completed.

**PWR TAIL RTR LAMP DOES NOT CYCLE ON AND OFF AND/OR PWR TAIL RTR LAMP DOES NOT GO ON WHEN PRESS-TO-TEST****SYMPTOM**

PWR TAIL RTR lamp does not cycle on and off and/or PWR TAIL RTR lamp does not go on when press-to-test.

**MALFUNCTION**

PWR TAIL RTR lamp does not cycle on and off and/or PWR TAIL RTR lamp does not go on when press-to-test.

**CORRECTIVE ACTION**

1. Momentarily press PWR TAIL RTR lamp. Go to **2**.
2. Did PWR TAIL RTR lamp go on, then off?
  - a. If lamp did go on, then off, go to **3**.
  - b. If lamp did not go, then off, go to **9**.
3. Check helicopter configuration.
  - a. **EMEP** , go to **4**.
  - b. **W/O EMEP** , go to **6**.
4. **EMEP** Remove and check pin filtered adapter P134 (WP 0925 00).
  - a. If pin filtered adapter is good, go to **5**.
  - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **10**.
5. **EMEP** Install pin filtered adapter (WP 0925 00). Go to **6**.
6. Check for 28 vdc between P134-H and P134-T on blade de-ice test panel.
  - a. If voltage is as specified, replace blade de-ice test panel (WP 1240 00). Go to **10**.
  - b. If voltage is not as specified, go to **7**.
7. Check continuity between P134-T and ground.

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**PWR TAIL RTR LAMP DOES NOT CYCLE ON AND OFF AND/OR PWR TAIL RTR LAMP DOES NOT GO ON WHEN PRESS-TO-TEST - Continued**

- a. If continuity is present, go to **8**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **10**.
- 8. Check continuity between P134-H and P902-h.**
  - a. If continuity is present, replace left relay panel (WP 0825 00). Go to **10**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **10**.
- 9. Unscrew PWR MAIN RTR lamp cap and check for burnt-out lamp.**
  - a. If lamp is good, replace blade de-ice test panel (WP 1240 00). Go to **10**.
  - b. If lamp is not good, replace lamp (WP 1241 00). Go to **10**.
- 10. Procedure completed.**

**TEST IN PROGRESS LAMP DOES NOT CYCLE ON AND OFF****SYMPTOM**

TEST IN PROGRESS lamp does not cycle on and off.

**MALFUNCTION**

TEST IN PROGRESS lamp does not cycle on and off.

**CORRECTIVE ACTION**

- 1. Unscrew lamp cap and check for burnt-out lamp.**
  - a. If lamp is good, go to **2**.
  - b. If lamp is not good, replace lamp (WP 1224 00). Go to **5**.
- 2. Check for 28 vdc between P133-K and P133-D.**
  - a. If voltage is as specified, replace BLADE DEICE control panel (WP 1223 00). Go to **5**.
  - b. If voltage is not as specified, go to **3**.
- 3. Check continuity between P133-D and ground.**
  - a. If continuity is present, go to **4**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **5**.
- 4. Check continuity between P133-K and P902-h.**
  - a. If continuity is present, replace left relay panel (WP 0825 00). Go to **5**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **5**.
- 5. Procedure completed.**

**TEST IN PROGRESS LAMP DOES NOT FUNCTION AS SPECIFIED****SYMPTOM**

TEST IN PROGRESS LAMP does not function as specified.

**MALFUNCTION**

TEST IN PROGRESS LAMP does not function as specified.

**CORRECTIVE ACTION**

- 1. Check helicopter configuration.**
  - a. **EMEP** , go to **2**.
  - b. **W/O EMEP** , go to **4**.
- 2. EMEP Remove and check pin filtered adapter P325 (WP 0925 00).**
  - a. If pin filtered adapter is good, go to **3**.
  - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **12**.
- 3. EMEP Install pin filtered adapter (WP 0925 00). Go to 4.**
- 4. Place POWER switch on blade deice control panel to TEST. Go to 5.**
- 5. Check for 28 vdc between P325-Y and P325-H.**
  - a. If voltage is as specified, go to **6**.
  - b. If voltage is not as specified, go to **10**.
- 6. Check continuity between P133-N and P325-T.**
  - a. If continuity is present, go to **7**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **12**.
- 7. Check for 15 vdc between P133-N and ground.**
  - a. If voltage is as specified, go to **8**.
  - b. If voltage is not as specified, replace blade deice controller (WP 1226 00). Go to **12**.
- 8. Check for 28 vdc between P133-K and ground.**
  - a. If voltage is as specified, replace blade deice controller (WP 1226 00). Go to **12**.
  - b. If voltage is not as specified, go to **9**.
- 9. Check continuity between P133-K and P902-h.**
  - a. If continuity is present, replace left relay panel (WP 0825 00). Go to **12**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **12**.
- 10. Check continuity between P325-H and P133-U.**
  - a. If continuity is present, go to **11**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **12**.
- 11. Check for 28 vdc between P133-R and P133-D.**
  - a. If voltage is as specified, replace BLADE DEICE control panel (WP 1223 00). Go to **12**.

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**TEST IN PROGRESS LAMP DOES NOT FUNCTION AS SPECIFIED - Continued**

- b. If voltage is not as specified, repair/replace wiring between P133-R and J121-A (WP 1747 00). Go to **12**.

**12. Procedure completed.****MR DEICE FAIL AND M/R DEICE FAULT CAPSULES OR LEGENDS GO ON****SYMPTOM**

MR DEICE FAIL and M/R DEICE FAULT capsules or legends go on.

**MALFUNCTION**

MR DEICE FAIL and M/R DEICE FAULT capsules or legends go on.

**CORRECTIVE ACTION****1. Disconnect P841 and P1001. Go to 2.****2. Check continuity between:**

- P841-1 and connector shell
- P841-2 and connector shell
- P841-3 and connector shell
- P841-4 and connector shell
- P841-5 and connector shell
- P841-6 and connector shell
- P841-7 and connector shell
- P841-8 and connector shell
- P841-9 and connector shell
- P841-10 and connector shell
- P841-11 and connector shell
- P841-12 and connector shell

a. If continuity is present, replace harness (WP 1220 00). Go to **25**.

b. If continuity is not present, go to **3**.

**3. Check for less than 1 ohm resistance between:**

- P841-1 and P1001-1
- P841-2 and P1001-2
- P841-3 and P1001-3
- P841-5 and P1001-4
- P841-6 and P1001-5
- P841-8 and P1001-6
- P841-9 and P1001-7

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**MR DEICE FAIL AND M/R DEICE FAULT CAPSULES OR LEGENDS GO ON - Continued**

- **P841-11 and P1001-8**
  - **P841-12 and P1001-9**
  - a. If resistance is as specified, go to **4**.
  - b. If resistance is not as specified, replace harness (WP 1220 00). Go to **25**.
- 4. Check continuity between:**
- **P841-1 and P841-4**
  - **P841-4 and P841-7**
  - **P841-7 and P841-10**
  - a. If continuity is present, reconnect P841. Go to **5**.
  - b. If continuity is not present, replace harness (WP 1220 00). Go to **25**.
- 5. Disconnect P845. Go to 6.**
- 6. Check for less than 1 ohm resistance between:**
- **P845-1 and P1001-10**
  - **P845-2 and P1001-11**
  - a. If resistance is as specified, reconnect P845 and P1001. Go to **7**.
  - b. If resistance is not as specified, replace harness (WP 1220 00). Go to **25**.
  - c. If trouble still remains, replace droop stop heater pin (WP 0555 00). Go to **25**.
- 7. Disconnect P842 and P1002. Go to 8.**
- 8. Check continuity between:**
- **P842-1 and connector shell**
  - **P842-2 and connector shell**
  - **P842-3 and connector shell**
  - **P842-4 and connector shell**
  - **P842-5 and connector shell**
  - **P842-6 and connector shell**
  - **P842-7 and connector shell**
  - **P842-8 and connector shell**
  - **P842-9 and connector shell**
  - **P842-10 and connector shell**
  - **P842-11 and connector shell**
  - **P842-12 and connector shell**
  - a. If continuity is present, replace harness (WP 1220 00). Go to **25**.
  - b. If continuity is not present, go to **9**.

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MR DEICE FAIL AND M/R DEICE FAULT CAPSULES OR LEGENDS GO ON - Continued**9. Check for less than 1 ohm resistance between:**

- P842-1 and P1002-1
- P842-2 and P1002-2
- P842-3 and P1002-3
- P842-5 and P1002-4
- P842-6 and P1002-5
- P842-8 and P1002-6
- P842-9 and P1002-7
- P842-11 and P1002-8
- P842-12 and P1002-9

a. If resistance is as specified, go to **10**.

b. If resistance is not as specified, replace harness (WP 1220 00). Go to **25**.

**10. Check continuity between:**

- P842-1 and P842-4
- P842-4 and P842-7
- P842-7 and P842-10

a. If continuity is present, reconnect P842. Go to **11**.

b. If continuity is not present, replace harness (WP 1220 00). Go to **25**.

**11. Disconnect P846. Go to 12.****12. Check for less than 1 ohm resistance between:**

- P846-1 and P1002-10
- P846-2 and P1002-11

a. If resistance is as specified, reconnect P846 and P1002 then go to **13**.

b. If resistance is not as specified, replace harness (WP 1220 00). Go to **25**.

c. If trouble remains, replace droop stop heater pin (WP 0555 00). Go to **25**.

**13. Disconnect P843 and P1003. Go to 14.****14. Check continuity between:**

- P843-1 and connector shell
- P843-2 and connector shell
- P843-3 and connector shell
- P843-4 and connector shell
- P843-5 and connector shell
- P843-6 and connector shell

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**MR DEICE FAIL AND M/R DEICE FAULT CAPSULES OR LEGENDS GO ON - Continued**

- **P843-7 and connector shell**
  - **P843-8 and connector shell**
  - **P843-9 and connector shell**
  - **P843-10 and connector shell**
  - **P843-11 and connector shell**
  - **P843-12 and connector shell**
- a. If continuity is present, replace harness (WP 1220 00). Go to **25**.
- b. If continuity is not present, go to **15**.
- 15. Check for less than 1 ohm resistance between:**
- **P843-1 and P1003-1**
  - **P843-2 and P1003-2**
  - **P843-3 and P1003-3**
  - **P843-5 and P1003-4**
  - **P843-6 and P1003-5**
  - **P843-8 and P1003-6**
  - **P843-9 and P1003-7**
  - **P843-11 and P1003-8**
  - **P843-12 and P1003-9**
- a. If resistance is as specified, go to **16**.
- b. If resistance is not as specified, replace harness (WP 1220 00). Go to **25**.
- 16. Check continuity between:**
- **P843-1 and P843-4**
  - **P843-4 and P843-7**
  - **P843-7 and P843-10**
- a. If continuity is present, reconnect P843. Go to **17**.
- b. If continuity is not present, replace harness (WP 1220 00). Go to **25**.
- 17. Disconnect P847. Go to 18.**
- 18. Check for less than 1 ohm resistance between:**
- **P847-1 and P1003-10**
  - **P847-2 and P1003-11**
- a. If resistance is as specified, reconnect P847 and P1003. Go to **19**.
- b. If resistance is not as specified, replace harness (WP 1220 00). Go to **25**.
- c. If trouble remains, replace droop stop heater pin (WP 0555 00). Go to **25**.



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**MR DEICE FAIL AND M/R DEICE FAULT CAPSULES OR LEGENDS GO ON - Continued**

**19. Disconnect P844 and P1004. Go to 20.**

**20. Check continuity between:**

- **P844-1 and connector shell**
- **P844-2 and connector shell**
- **P844-3 and connector shell**
- **P844-4 and connector shell**
- **P844-5 and connector shell**
- **P844-6 and connector shell**
- **P844-7 and connector shell**
- **P844-8 and connector shell**
- **P844-9 and connector shell**
- **P844-10 and connector shell**
- **P844-11 and connector shell**
- **P844-12 and connector shell**

a. If continuity is present, replace harness (WP 1220 00). Go to **25**.

b. If continuity is not present, go to **21**.

**21. Check for less than 1 ohm resistance between:**

- **P844-1 and P1004-1**
- **P844-2 and P1004-2**
- **P844-3 and P1004-3**
- **P844-5 and P1004-4**
- **P844-6 and P1004-5**
- **P844-8 and P1004-6**
- **P844-9 and P1004-7**
- **P844-11 and P1004-8**
- **P844-12 and P1004-9**

a. If resistance is as specified, go to **22**.

b. If resistance is not as specified, replace harness (WP 1220 00). Go to **25**.

**22. Check continuity between:**

- **P844-1 and P844-4**
- **P844-4 and P844-7**
- **P844-7 and P844-10**

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**MR DEICE FAIL AND M/R DEICE FAULT CAPSULES OR LEGENDS GO ON - Continued**

- a. If continuity is present, reconnect P844. Go to **23**.
  - b. If continuity is not present, replace harness (WP 1220 00). Go to **25**.
- 23. Disconnect P848. Go to 24.**
- 24. Check for less than 1 ohm resistance between:**
- **P848-1 and P1004-10**
  - **P848-2 and P1004-11**
- a. If resistance is as specified, replace droop stop heater pin (WP 0555 00). Go to **25**.
  - b. If resistance is not as specified, replace harness (WP 1220 00). Go to **25**.
- 25. Procedure completed.**

**MR DE-ICE FAIL CAPSULE OR LEGEND GOES ON****SYMPTOM**

MR DE-ICE FAIL capsule or legend goes on.

**MALFUNCTION**

MR DE-ICE FAIL capsule or legend goes on.

**CORRECTIVE ACTION**

- 1. Does capsule or legend go on immediately after placing POWER switch to TEST?**
  - a. If capsule or legend is immediately on, go to **2**.
  - b. If capsule or legend is not on immediately, go to **9**.
- 2. Check helicopter configuration.**
  - a. **EMEP** , go to **3**.
  - b. **W/O EMEP** , go to **5**.
- 3. **EMEP** Remove and check pin filtered adapter P134 (WP 0925 00).**
  - a. If pin filtered adapter is good, go to **4**.
  - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **87**.
- 4. **EMEP** Install pin filtered adapter (WP 0925 00). Go to 5.**
- 5. Check for resistance proportional to outside air temperature between P134-Y and P134-Z.**
  - a. If resistance is present, go to **6**.
  - b. If resistance is not present, go to **8**.

**NOTE**

Performance of the following resistance check must be completed within one hour of performing operating/troubleshooting procedure steps 1. through 64. to achieve valid resistance measurements.

- 6. Check for resistance of 7 to 10 ohms between:**
  - **J841-1 and J841-2**

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MR DE-ICE FAIL CAPSULE OR LEGEND GOES ON - Continued

- J841-1 and J841-3
- J841-2 and J841-3
- J841-4 and J841-5
- J841-4 and J841-6
- J841-5 and J841-6
- J841-7 and J841-8
- J841-7 and J841-9
- J841-8 and J841-9
- J841-10 and J841-11
- J841-10 and J841-12
- J841-11 and J841-12
- J842-1 and J842-2
- J842-1 and J842-3
- J842-2 and J842-3
- J842-4 and J842-5
- J842-4 and J842-6
- J842-5 and J842-6
- J842-7 and J842-8
- J842-7 and J842-9
- J842-8 and J842-9
- J842-10 and J842-11
- J842-10 and J842-12
- J842-11 and J842-12
- J843-1 and J843-2
- J843-1 and J843-3
- J843-2 and J843-3
- J843-4 and J843-5
- J843-4 and J843-6
- J843-5 and J843-6
- J843-7 and J843-8
- J843-7 and J843-9

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**MR DE-ICE FAIL CAPSULE OR LEGEND GOES ON - Continued**

- J843-8 and J843-9
- J843-10 and J843-11
- J843-10 and J843-12
- J843-11 and J843-12
- J844-1 and J844-2
- J844-1 and J844-3
- J844-2 and J844-3
- J844-4 and J844-5
- J844-4 and J844-6
- J844-5 and J844-6
- J844-7 and J844-8
- J844-7 and J844-9
- J844-8 and J844-9
- J844-10 and J844-11
- J844-10 and J843-12
- J844-11 and J844-12

- a. If resistance is as specified, go to **7**.
- b. If resistance is not as specified, replace malfunctioning main rotor blade (WP 0576 00). Go to **87**.

**7. Disconnect P325 from deice controller. Does capsule or legend go off?**

- a. If capsule or legend is off, replace rotor blade deice controller (WP 1226 00). Go to **87**.
- b. If capsule or legend is not off, replace blade deice test panel (WP 1240 00). Go to **87**.

**8. Check continuity between:**

- P122-A and P134-Y
- P122-B and P134-Z

- a. If continuity is present, replace oat sensor (WP 1243 00). Go to **87**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **87**.

**9. Does PWR MAIN RTR lamp flash and M/R DE-ICE FAIL capsule or legend go on in 6 to 80 seconds or greater than 120 seconds after POWER switch is placed to TEST?**

- a. If PWR MAIN RTR lamp flashes and M/R DE-ICE FAIL capsule or legend is on, replace rotor blade deice controller (WP 1226 00). Go to **87**.
- b. If PWR MAIN RTR lamp does not flash and M/R DE-ICE FAIL capsule or legend is not on, go to **10**.

**10. Place POWER switch to OFF. Go to 11.**

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**MR DE-ICE FAIL CAPSULE OR LEGEND GOES ON - Continued**

11. Turn off electrical power. Go to 12.
12. Disconnect wire H6373A20 from TB3-3. Go to 13.
13. Connect locally-made test harness P1 (WP 1805 00) to DC utility receptacle and P2 to AC utility receptacle. Go to 14.
14. Connect DC alligator clip to TB3-3 and AC alligator clip to TB3-5. Go to 15.
15. Connect the negative lead of a multimeter, set on the 50 vdc scale, to TB3-1 and the multimeter positive lead to ground. Go to 16.
16. Place test harness switch to OFF. Go to 17.
17. Disconnect P841, P842, P843, and P844 from rotor blades. Go to 18.
18. Turn on electrical power. Go to 19.
19. Place test harness switch alternately to ON and OFF. Go to 20.
20. Multimeter shall indicate alternately between 0 and -30 vdc. Go to 21.
21. After a maximum of 8 on-off pulses of test harness switch, does multimeter indicate between -7 and -3 volts?
  - a. If multimeter is as specified, go to 22.
  - b. If multimeter is not as specified, go to 42.
22. Place test harness switch alternately to ON and OFF seven times. Go to 23.
23. Voltage at TB3-1 shall be 0 vdc. Go to 24.
24. Turn off electrical power. Go to 25.
25. Disconnect test harness. Go to 26.
26. Connect wire H6373A20 to TB3-3. Go to 27.
27. Connect P841, P842, P843, and P844 to blades. Go to 28.
28. Turn on electrical power. Go to 29.
29. Place POWER switch to TEST. Go to 30.
30. Check that multimeter at TB3-1 indicates alternately 0 and -30 vdc.
  - a. If multimeter indicates alternately 0 vdc and -30 vdc, go to 31.
  - b. If multimeter does not indicate alternately 0 vdc and -30 vdc, go to 52.
31. Place POWER switch to OFF. Go to 32.
32. Turn off electrical power. Go to 33.
33. Place clamp-on ammeter around  $\varnothing$ A wire at TB3-2. Go to 34.
34. Turn on electrical power. Go to 35.
35. Monitor ammeter as POWER switch is placed to TEST. Go to 36.
36. Check that current pulses are indicated on ammeter.
  - a. If pulses are indicated, replace rotor blade deice controller (WP 1226 00). Go to 87.
  - b. If pulses are not indicated, go to 37.
37. Place POWER switch to OFF, and then to TEST. Go to 38.

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**MR DE-ICE FAIL CAPSULE OR LEGEND GOES ON - Continued**

**38. Check for 115 vac between:**

- TB3-2 and ground
- TB3-4 and ground

- a. If voltage is as specified, go to **39**.
- b. If voltage is not as specified, go to **44**.

**39. Place POWER switch to OFF. Go to 40.**

**40. Turn off electrical power. Go to 41.**

**41. Check for resistance of 7 to 10 ohms between:**

- J841-1 and J841-2
- J841-1 and J841-3
- J841-2 and J841-3
- J841-4 and J841-5
- J841-4 and J841-6
- J841-5 and J841-6
- J841-7 and J841-8
- J841-7 and J841-9
- J841-8 and J841-9
- J841-10 and J841-11
- J841-10 and J841-12
- J841-11 and J841-12
- J842-1 and J842-2
- J842-1 and J842-3
- J842-2 and J842-3
- J842-4 and J842-5
- J842-4 and J842-6
- J842-5 and J842-6
- J842-7 and J842-8
- J842-7 and J842-9
- J842-8 and J842-9
- J842-10 and J842-11
- J842-10 and J842-12
- J842-11 and J842-12

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**MR DE-ICE FAIL CAPSULE OR LEGEND GOES ON - Continued**

- J843-1 and J843-2
- J843-1 and J843-3
- J843-2 and J843-3
- J843-4 and J843-5
- J843-4 and J843-6
- J843-5 and J843-6
- J843-7 and J843-8
- J843-7 and J843-9
- J843-8 and J843-9
- J843-10 and J843-11
- J843-11 and J843-12
- J844-1 and J844-2
- J844-1 and J844-3
- J844-2 and J844-3
- J844-4 and J844-5
- J844-4 and J844-6
- J844-5 and J844-6
- J844-7 and J844-8
- J844-7 and J844-9
- J844-8 and J844-9
- J844-10 and J844-11
- J844-10 and J843-12
- J844-11 and J844-12

a. If resistance is as specified, go to **42**.

b. If resistance is not as specified, replace malfunctioning main rotor blade (WP 0576 00). Go to **87**.

**42. Remove distributor and slipring from helicopter (WP 1220 00). Go to 43.**

**43. Check continuity between the following for slipring input wire and slipring output terminal strip respectively:**

- S1 and S1
- S2 and S2
- S3 and S3
- øA and øA

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**MR DE-ICE FAIL CAPSULE OR LEGEND GOES ON - Continued**

- **øB and øB**
  - **øC and øC**
  - a. If continuity is present, replace main rotor blade deice distributor (WP 1220 00). Go to **87**.
  - b. If continuity is not present, replace slipring brush blocks (WP 1220 00). Go to **87**.
  - c. If trouble remains, repair/replace wiring (WP 1747 00). Go to **87**.
  - d. If trouble still remains, replace slipring assembly (WP 1220 00). Go to **87**.
- 44. Place POWER switch to OFF, and then to TEST. Go to 45.**
- 45. Check for 115 vac between K60 contactor:**
- **Terminal A1 and ground**
  - **Terminal B1 and ground**
  - a. If voltage is as specified, replace current transformer (WP 1237 00). Go to **87**.
  - b. If voltage is not as specified, go to **46**.
- 46. Place POWER switch to OFF, and then to TEST. Go to 47.**
- 47. Check for 115 vac between K60 contactor:**
- **Terminal A2 and ground**
  - **Terminal B2 and ground**
  - a. If voltage is as specified, replace K60 contactor (WP 1233 00). Go to **87**.
  - b. If voltage is not as specified, go to **48**.
- 48. Place POWER switch to OFF, and then to TEST. Go to 49.**
- 49. Check for 115 vac between K62 contactor:**
- **Terminal T1 and ground**
  - **Terminal T2 and ground**
  - a. If voltage is as specified, replace current limiter either CL10 or CL11, or both (WP 1228 00). Go to **87**.
  - b. If voltage is not as specified, go to **50**.
- 50. Place POWER switch to OFF, and then to TEST. Go to 51.**
- 51. Check for 115 vac between K62 contactor:**
- **Terminal L1 and ground**
  - **Terminal L2 and ground**
  - a. If voltage is as specified, replace contactor (WP 1234 00). Go to **87**.
  - b. If voltage is not as specified, replace current limiter CL7 and CL8 (WP 1228 00). Go to **87**.
- 52. Place POWER switch to OFF, and then to TEST. Go to 53.**
- 53. Check for 115 vac between:**
- **TB3-2 and ground**



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MR DE-ICE FAIL CAPSULE OR LEGEND GOES ON - Continued

- TB3-4 and ground
  - TB3-5 and ground
  - a. If voltage is as specified, go to **54**.
  - b. If voltage is not as specified, go to **60**.
- 54. Check helicopter configuration.**
- a. **EMEP** , go to **55**.
  - b. **W/O EMEP** , go to **57**.
- 55. EMEP Remove and check pin filtered adapter P325 (WP 0925 00).**
- a. If pin filtered adapter is good, go to **56**.
  - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **87**.
- 56. EMEP Install pin filtered adapter (WP 0925 00). Go to 57.**
- 57. Check continuity between P325-a and TB3-3.**
- a. If continuity is present, replace rotor blade deice controller (WP 1226 00). Go to **87**.
  - b. If continuity is not present, go to **58**.
- 58. Check continuity between P325-J and P134-P.**
- a. If continuity is present, go to **59**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **87**.
- 59. Check continuity between P134-N and TB3-1.**
- a. If continuity is present, replace blade deice test panel (WP 1240 00). Go to **87**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **87**.
- 60. Place POWER switch to OFF, and then to TEST. Go to 61.**
- 61. Check for 115 vac between K60 MAIN ROTOR CONTACTOR:**
- Terminal A1 and ground
  - Terminal B1 and ground
  - Terminal C1 and ground
  - a. If voltage is as specified, replace current transformer (WP 1237 00). Go to **87**.
  - b. If voltage is not as specified, go to **62**.
- 62. Place POWER switch to OFF, and then to TEST. Go to 63.**
- 63. Check for 115 vac between K60 MAIN ROTOR CONTACTOR:**
- Terminal A2 and ground
  - Terminal B2 and ground
  - Terminal C2 and ground
  - a. If voltage is as specified, go to **64**.
  - b. If voltage is not as specified, go to **82**.

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**MR DE-ICE FAIL CAPSULE OR LEGEND GOES ON - Continued**

- 64. Check for 28 vdc between K60 MAIN ROTOR CONTACTOR terminals X1 and X2.**
  - a. If voltage is as specified, replace contactor (WP 1233 00). Go to **87**.
  - b. If voltage is not as specified, go to **65**.
- 65. Check continuity between K60 MAIN ROTOR CONTACTOR terminal X2A and ground.**
  - a. If continuity is present, go to **66**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **87**.
- 66. Check continuity between K64 APU/GENERATOR INTERLOCK relay terminal A1 and K60 MAIN ROTOR CONTACTOR terminal X1.**
  - a. If continuity is present, go to **67**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **87**.
- 67. Check for 28 vdc between K64 APU/GENERATOR INTERLOCK relay terminal A2 and ground.**
  - a. If voltage is as specified, go to **68**.
  - b. If voltage is not as specified, go to **77**.
- 68. Check for 28 vdc between K64 APU/GENERATOR INTERLOCK relay terminals X1 and X2.**
  - a. If voltage is as specified, replace relay (WP 1231 00). Go to **87**.
  - b. If voltage is not as specified, go to **69**.
- 69. Check continuity between K64 APU/GENERATOR INTERLOCK relay terminal X2 and ground.**
  - a. If continuity is present, go to **70**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **87**.
- 70. Check for 28 vdc between K65 BACK-UP HYDRAULIC PUMP INTERLOCK relay terminal A3 and ground.**
  - a. If voltage is as specified, repair/replace wiring (WP 1747 00). Go to **87**.
  - b. If voltage is not as specified, go to **71**.
- 71. Check for 28 vdc between K65 BACK-UP HYDRAULIC PUMP INTERLOCK relay terminal A2 and ground.**
  - a. If voltage is as specified, go to **72**.
  - b. If voltage is not as specified, go to **75**.
- 72. Check for 28 vdc between K65 BACK-UP HYDRAULIC PUMP INTERLOCK relay terminals X1 and X2.**
  - a. If voltage is as specified, go to **73**.
  - b. If voltage is not as specified, replace relay (WP 1232 00). Go to **87**.
- 73. Check continuity between K65 BACK-UP HYDRAULIC PUMP INTERLOCK relay terminal X1 and J329-a.**
  - a. If continuity is present, go to **74**.

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MR DE-ICE FAIL CAPSULE OR LEGEND GOES ON - Continued

- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **87**.
- 74. Check for 28 vdc between P329-a and ground.**
- a. If voltage is as specified, troubleshoot AC electrical system (WP 0101 00 or WP 0122 00). Go to **87**.
- b. If voltage is not as specified, replace main rotor de-ice junction box (WP 1238 00). Go to **87**.
- 75. Check continuity between K65 BACK-UP HYDRAULIC PUMP INTERLOCK relay terminal A2 and J329-Z.**
- a. If continuity is present, go to **76**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **87**.
- 76. Check continuity between P133-U and P329-Z.**
- a. If continuity is present, replace blade de-ice control panel (WP 1223 00). Go to **87**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **87**.
- 77. Check continuity between K64 APU/GENERATOR INTERLOCK relay terminal A2 and J329-M.**
- a. If continuity is present, repair/replace wiring (WP 1747 00). Go to **87**.
- b. If continuity is not present, go to **78**.
- 78. Check helicopter configuration.**
- a. **EMEP** , go to **79**.
- b. **W/O EMEP** , go to **81**.
- 79. EMEP Remove and check pin filtered adapter from P325 (WP 0925 00).**
- a. If pin filtered adapter is good, go to **80**.
- b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **87**.
- 80. EMEP Install pin filtered adapter (WP 0925 00). Go to 81.**
- 81. Check continuity between P329-M and P325-h.**
- a. If continuity is present, replace rotor blade deice controller (WP 1226 00). Go to **87**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **87**.
- 82. Place POWER switch to OFF, and then ON. Go to 83.**
- 83. Check for 115 vac between K62 MN RTR BUS CONT:**
- **Terminal T1 and ground**
  - **Terminal T2 and ground**
  - **Terminal T3 and ground**
- a. If voltage is as specified, replace current limiters CL10, CL11, and CL12 (WP 1228 00). Go to **87**.
- b. If trouble remains, repair/replace wiring between MN RTR BUS CONT terminals T1, T2, T3, and K60 main rotor contactor terminals A2, B2, and C2 (WP 1747 00). Go to **87**.
- c. If voltage is not as specified, go to **84**.

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**MR DE-ICE FAIL CAPSULE OR LEGEND GOES ON - Continued****84. Check for 115 vac between K62 MN RTR BUS CONT:**

- **Terminal L1 and ground**
- **Terminal L2 and ground**
- **Terminal L3 and ground**

- a. If voltage is as specified, go to **85**.
- b. If voltage is not as specified, replace current limiters CL7, CL8, and CL9 (WP 1228 00). Go to **87**.

**85. Check for 28 vdc between K62 MN RTR BUS CONT terminals X1 and X2.**

- a. If voltage is as specified, replace contactor (WP 1234 00). Go to **87**.
- b. If voltage is not as specified, go to **86**.

**86. Check continuity between K62 MN RTR BUS CONT terminal X2 and ground.**

- a. If continuity is present, repair/replace wiring between J333-A and contactor terminal X1 (WP 1747 00). Go to **87**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **87**.

**87. Procedure completed.****TR DE-ICE FAIL CAPSULE OR LEGEND GOES ON****SYMPTOM**

TR DE-ICE FAIL capsule or legend goes on.

**MALFUNCTION**

TR DE-ICE FAIL capsule or legend goes on.

**CORRECTIVE ACTION****1. Does T/R DE-ICE FAIL capsule or legend go on immediately after placing POWER switch to TEST?**

- a. If capsule or legend goes on, go to **2**.
- b. If capsule or legend does not go on, go to **7**.

**2. Check helicopter configuration.**

- a. **EMEP** , go to **3**.
- b. **W/O EMEP** , go to **5**.

**3. **EMEP** Remove and check pin filtered adapter P325 (WP 0925 00).**

- a. If pin filtered adapter is good, go to **4**.
- b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **67**.

**4. **EMEP** Install pin filtered adapter (WP 0925 00). Go to 5.****5. Disconnect P325 from deice controller. Does T/R DE-ICE FAIL capsule or legend go off?**

- a. If capsule is off, replace rotor blade deice controller (WP 1226 00). Go to **6**.

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TR DE-ICE FAIL CAPSULE OR LEGEND GOES ON - Continued

- b. If capsule is not off, replace blade de-ice test panel (WP 1240 00). Go to **6**.
- 6. Connect P325 to deice controller. Go to 67.**
- 7. Does PWR-TAIL RTR lamp flash and T/R DE-ICE FAIL capsule or legend go on in less than 75 seconds or more than 120 seconds?**
  - a. If lamp flashes and capsule or legend is on in less than 75 seconds or more than 120 seconds, replace blade de-ice test panel (WP 1240 00). Go to **67**.
  - b. If lamp does not flash and capsule or legend is not on in less than 75 seconds or more than 120 seconds, go to **8**.
- 8. Turn off electrical power. Go to 9.**
- 9. Place 20 amp clamp-on ammeter around wire H6389A14 connected to K61 TAIL ROTOR CONTACTOR terminal A2. Go to 10.**
- 10. Turn on electrical power. Go to 11.**
- 11. Place POWER switch to TEST. Go to 12.**
- 12. Check that between 75 to 105 seconds after POWER switch is placed to TEST, a current pulse occurs.**
  - a. If pulse is present, replace blade deice controller (WP 1226 00). Go to **67**.
  - b. If pulse is not present, go to **13**.
- 13. Place POWER switch to OFF. Go to 14.**
- 14. Turn off electrical power. Go to 15.**
- 15. Disconnect P611, P612, P613, and P614. Go to 16.**
- 16. Turn on electrical power. Go to 17.**
- 17. Place POWER switch to OFF, and then TEST. Go to 18.**
- 18. Check for 115 vac between K61 TAIL ROTOR CONTACTOR:**
  - Terminal A2 and ground
  - Terminal B2 and ground
  - Terminal C2 and ground
  - a. If voltage is as specified, go to **19**.
  - b. If voltage is not as specified, go to **34**.
- 19. Check helicopter configuration.**
  - a. **EMEP** , go to **20**.
  - b. **W/O EMEP** , go to **22**.
- 20. **EMEP** Remove and check pin filtered adapter P324 (WP 0925 00).**
  - a. If pin filtered adapter is good, go to **21**.
  - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **67**.
- 21. **EMEP** Install pin filtered adapter (WP 0925 00). Go to 22.**
- 22. Check continuity between K61 TAIL ROTOR CONTACTOR:**

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**TR DE-ICE FAIL CAPSULE OR LEGEND GOES ON - Continued**

- **Terminal A2 and P324-A**
  - **Terminal B2 and P324-B**
  - **Terminal C2 and P324-C**
- a. If continuity is present, go to **23**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **67**.
- 23. Disconnect P134. Go to 24.**
- 24. Check for resistance of 16 to 22 ohms between:**
- **P324-D and P324-E**
  - **P324-D and P324-H**
  - **P324-E and P324-H**
- a. If resistance is as specified, replace blade deice controller (WP 1226 00). Go to **67**.
  - b. If resistance is not as specified, go to **25**.
- 25. Check continuity between:**
- **P324-D and P327-1**
  - **P324-E and P327-2**
  - **P324-H and P327-3**
- a. If continuity is present, go to **26**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **67**.
- 26. Connect P134. Go to 27.**
- 27. Check for resistance of 6 to 9 ohms between:**
- **J611-1 and J611-2**
  - **J611-3 and J611-4**
  - **J611-5 and J611-6**
  - **J612-1 and J612-2**
  - **J612-3 and J612-4**
  - **J612-5 and J612-6**
  - **J613-1 and J613-2**
  - **J613-3 and J613-4**
  - **J613-5 and J613-6**
  - **J614-1 and J614-2**
  - **J614-3 and J614-4**
  - **J614-5 and J614-6**
- a. If resistance is as specified, go to **28**.

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**TR DE-ICE FAIL CAPSULE OR LEGEND GOES ON - Continued**

- b. If resistance is not as specified, replace malfunctioning tail rotor blade (WP 0585 00). Go to **67**.

**28. Check continuity between tail rotor brush assembly connector pin 1 and P614-2.**

- a. If continuity is present, go to **29**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **67**.

**29. Check continuity between P611-3 and P614-2.**

- a. If continuity is present, go to **30**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **67**.
- c. If T/Rouble remains, replace tail rotor slipring rotor (WP 1246 00). Go to **67**.

**30. Check continuity between tail rotor brush assembly connector pin 2 and P611-1.**

- a. If continuity is present, go to **31**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **67**.

**31. Check continuity between P611-1 and P613-6.**

- a. If continuity is present, go to **32**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **67**.
- c. If T/Rouble remains, replace tail rotor slipring rotor (WP 1246 00). Go to **67**.

**32. Check continuity between tail rotor brush assembly connector pin 3 and P614-4.**

- a. If continuity is present, go to **33**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **67**.
- c. If T/Rouble remains, replace tail rotor slipring rotor (WP 1246 00). Go to **67**.

**33. Check continuity between P614-4 and P614-5.**

- a. If continuity is present, go to **34**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **67**.

**34. Check continuity between the following:**

- **P611-2 and P612-1**
- **P611-4 and P612-3**
- **P611-5 and P614-6**
- **P611-6 and P612-5**
- **P612-2 and P613-1**
- **P612-4 and P613-3**
- **P612-6 and P613-5**
- **P613-2 and P614-1**
- **P613-4 and P614-3**

- a. If continuity is present, go to **35**.

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**TR DE-ICE FAIL CAPSULE OR LEGEND GOES ON - Continued**

- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **67**.
- c. If T/Rouble remains, replace tail rotor slipring brush assembly (WP 1248 00). Go to **67**.
- 35. Place POWER switch to OFF, and then to TEST. Go to 36.**
- 36. Check for 115 vac between K61 TAIL ROTOR CONTACTOR:**
  - **Terminal A1 and ground**
  - **Terminal B1 and ground**
  - **Terminal C1 and ground**
- a. If voltage is as specified, go to **37**.
- b. If voltage is not as specified, go to **48**.
- 37. Check for 28 vdc between K61 TAIL ROTOR CONTACTOR terminals X1 and X2.**
  - a. If voltage is as specified, replace contactor (WP 1230 00). Go to **67**.
  - b. If voltage is not as specified, go to **38**.
- 38. Check continuity between K61-X2 and ground.**
  - a. If continuity is present, go to **39**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **67**.
- 39. Check continuity between P329-V and K61-X1.**
  - a. If continuity is present, go to **40**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **67**.
- 40. Check continuity between K64 APU/GENERATOR INTERLOCK relay terminal B1 and J329-V.**
  - a. If continuity is present, go to **41**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **67**.
- 41. Place POWER switch to OFF, then to TEST. Go to 42.**
- 42. Check for 28 vdc between K64 APU/GENERATOR INTERLOCK relay terminal B2 and ground.**
  - a. If voltage is as specified, replace relay (WP 1231 00). Go to **67**.
  - b. If voltage is not as specified, go to **43**.
- 43. Check continuity between K64 APU/GENERATOR INTERLOCK relay terminal B2 and J329-U.**
  - a. If continuity is present, go to **44**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **67**.
- 44. Check helicopter configuration.**
  - a. **EMEP** , go to **45**.
  - b. **W/O EMEP** , go to **47**.
- 45. **EMEP** Remove and check pin filtered adapter P325 (WP 0925 00).**



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TR DE-ICE FAIL CAPSULE OR LEGEND GOES ON - Continued

- a. If pin filtered adapter is good, go to **46**.
- b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **67**.
- 46. EMEP Install pin filtered adapter (WP 0925 00). Go to 47.**
- 47. Check continuity between P329-U and P325-j.**
  - a. If continuity is present, replace rotor blade deice controller (WP 1226 00). Go to **67**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **67**.
- 48. Check for 115 vac between K63 TAIL ROTOR DE-ICE CONTACTOR:**
  - Terminal A2 and ground
  - Terminal B2 and ground
  - Terminal C2 and ground
  - a. If voltage is as specified, replace current limiters CL13, CL14, and CL15 (WP 1228 00). Go to **67**.
  - b. If voltage is not as specified, go to **49**.
- 49. Check for 115 vac between K63 TAIL ROTOR DE-ICE CONTACTOR:**
  - Terminal A1 and ground
  - Terminal B1 and ground
  - Terminal C1 and ground
  - a. If voltage is as specified, go to **50**.
  - b. If voltage is not as specified, go to **52**.
- 50. Check for 28 vdc between K63 TAIL ROTOR DE-ICE CONTACTOR terminals X1 and X2.**
  - a. If voltage is as specified, replace contactor (WP 1235 00). Go to **67**.
  - b. If voltage is not as specified, go to **51**.
- 51. Check continuity between K63 TAIL ROTOR DE-ICE CONTACTOR terminal X2 and ground.**
  - a. If continuity is present, repair/replace wiring between J333-A and contactor terminal X1 (WP 1747 00). Go to **67**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **67**.
- 52. Check for 115 vac between DE-ICE PWR TAIL ROTOR circuit breaker:**
  - Terminal A2 and ground
  - Terminal B2 and ground
  - Terminal C2 and ground
  - a. If voltage is as specified, repair/replace wiring between (WP 1747 00), then go to **67**:
    - Terminal A1 and K61-C1
    - Terminal B1 and K61-B1

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TR DE-ICE FAIL CAPSULE OR LEGEND GOES ON - Continued

- Terminal C1 and K61-A1
- b. If voltage is not as specified, go to **53**.
- 53. Check helicopter configuration.**
  - a. **UH-60A UH-60L** , go to **54**.
  - b. **EH-60A** , go to **55**.
- 54. UH-60A UH-60L HH-60A HH-60L** Replace DE-ICE PWR TAIL ROTOR circuit breaker (WP 0824 00). Go to **67**.
- 55. Check for 115 vac between DE-ICE PWR TAIL ROTOR circuit breaker:**
  - Terminal A2 and ground
  - Terminal B2 and ground
  - Terminal C2 and ground
  - a. If voltage is as specified, go to **56**.
  - b. If voltage is not as specified, go to **66**.
- 56. EH-60A** Disconnect wire connected to DE-ICE PWR TAIL ROTOR circuit breaker terminal **5A**. Go to **57**.
- 57. EH-60A** Check wire for continuity to ground.
  - a. If continuity is present, go to **58**.
  - b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **67**.
- 58. EH-60A** Disconnect wire connected to DE-ICE PWR TAIL ROTOR circuit breaker terminal **3**. Go to **59**.
- 59. EH-60A** Check wire for continuity to ground.
  - a. If continuity is present, go to **60**.
  - b. If continuity is not present, go to **62**.
- 60. EH-60A** Pull T/R DE-ICE PWR circuit breaker. Go to **61**.
- 61. EH-60A** Check continuity between wire from terminal **3** and ground.
  - a. If continuity is present, T/Roubleshoot using vibration manual. Go to **67**.
  - b. If continuity is not present, replace circuit breaker (WP 0824 00). Go to **67**.
- 62. EH-60A** Leave wire disconnected from DE-ICE PWR TAIL ROTOR circuit breaker terminal **3**. Go to **63**.
- 63. EH-60A** Check continuity between T/R DE-ICE PWR circuit breaker terminal **1** and ground.
  - a. If continuity is present, repair/replace wiring between circuit breaker terminal **1** and wire connected to terminal **3** of DE-ICE PWR TAIL ROTOR circuit breaker (WP 1747 00). Go to **67**.
  - b. If continuity is not present, go to **64**.
- 64. EH-60A** Pull T/R DE-ICE PWR circuit breaker. Go to **65**.
- 65. EH-60A** Check continuity between T/R DE-ICE PWR circuit breaker terminal **2** and ground.

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**TR DE-ICE FAIL CAPSULE OR LEGEND GOES ON - Continued**

- a. If continuity is present, replace circuit breaker (WP 0824 00). Go to **67**.
- b. If continuity is not present, repair/replace wiring between circuit breaker terminal 2 and ground (WP 1747 00). Go to **67**.

**66. Check for 115 vac between K1 NO. 1 GEN CNTOR:**

- **Terminal A2 and ground**
  - **Terminal B2 and ground**
  - **Terminal C2 and ground**
- a. If voltage is as specified, repair/replace wiring between contactor and DE-ICE PWR TAIL ROTOR circuit breaker (WP 1747 00). Go to **67**.
  - b. If voltage is not as specified, T/Roubleshoot AC electrical system (WP 0101 00 or WP 0122 00). Go to **67**.

**67. Procedure completed.****MR DE-ICE FAIL AND T/R DE-ICE FAIL CAPSULES OR LEGENDS ARE BOTH ON****SYMPTOM**

MR DE-ICE FAIL and T/R DE-ICE FAIL capsules or legends are both on.

**MALFUNCTION**

MR DE-ICE FAIL and T/R DE-ICE FAIL capsules or legends are both on.

**CORRECTIVE ACTION**

- 1. Place POWER switch to OFF, and then to TEST. Go to 2.**
- 2. Check for 28 vdc between P333-A and ground.**
  - a. If voltage is as specified, repair/replace wiring between J333-A and MN RTR BUS CONT terminal X1 and TAIL ROTOR DE-ICE contactor (WP 1747 00). Go to **19**.
  - b. If voltage is not as specified, go to **3**.
- 3. Check continuity between P333-A and P329-X.**
  - a. If continuity is present, go to **4**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **19**.
- 4. Place POWER switch to OFF, and then to TEST. Go to 5.**
- 5. Check for 28 vdc between K64 APU/GENERATOR INTERLOCK relay terminal C1 and ground.**
  - a. If voltage is as specified, repair/replace wiring between relay terminal C1 and J329-X (WP 1747 00). Go to **19**.
  - b. If voltage is not as specified, go to **6**.
- 6. Check for 28 vdc between K64 APU/GENERATOR INTERLOCK relay terminal C2 and ground.**
  - a. If voltage is as specified, go to **7**.
  - b. If voltage is not as specified, go to **12**.

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**MR DE-ICE FAIL AND T/R DE-ICE FAIL CAPSULES OR LEGENDS ARE BOTH ON - Continued**

- 7. Check for 28 vdc between K64 APU/GENERATOR INTERLOCK relay terminals X1 and X2.**
  - a. If voltage is as specified, replace relay (WP 1231 00). Go to **19**.
  - b. If voltage is not as specified, go to **8**.
- 8. Check continuity between K64 APU/GENERATOR INTERLOCK relay terminal X2 and ground.**
  - a. If continuity is present, go to **9**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **19**.
- 9. Check for 28 vdc between main rotor blade deice junction box terminal E4 and ground.**
  - a. If voltage is as specified, replace diode CR15 (WP 1239 00). Go to **19**.
  - b. If voltage is not as specified, go to **10**.
- 10. Check continuity between main rotor blade deice junction box terminal E4 and J329-Z.**
  - a. If continuity is present, repair/replace wiring between P329-Z and P133-U (WP 1747 00). Go to **19**.
  - b. If continuity is not present, go to **11**.
- 11. Check continuity between main rotor blade deice junction box terminal E4 and BACK-UP PUMP INTERLOCK relay terminal A3.**
  - a. If continuity is present, replace relay (WP 1232 00). Go to **19**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **19**.
- 12. Check for 28 vdc between P329-W and ground.**
  - a. If voltage is as specified, repair/replace wiring between APU/Generator interlock relay terminal C2 and J329-W (WP 1747 00). Go to **19**.
  - b. If voltage is not as specified, go to **13**.
- 13. Check continuity between P329-W and P212-H.**
  - a. If continuity is present, go to **14**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **19**.
- 14. Check for 28 vdc between P212-J and ground.**
  - a. If voltage is as specified, replace No. 1 generator line contactor (WP 0834 00). Go to **19**.
  - b. If voltage is not as specified, go to **15**.
- 15. Check continuity between P212-J and P204-J.**
  - a. If continuity is present, go to **16**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **19**.
- 16. Check for 28 vdc between P204-H and ground.**
  - a. If voltage is as specified, replace No. 2 generator line contactor (WP 0835 00). Go to **19**.
  - b. If voltage is not as specified, go to **17**.

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**MR DE-ICE FAIL AND T/R DE-ICE FAIL CAPSULES OR LEGENDS ARE BOTH ON - Continued****17. Check for 28 vdc between J107-K and ground.**

- a. If voltage is as specified, repair/replace wiring between P107-K and P204-H (WP 1747 00). Go to **19**.
- b. If voltage is not as specified, go to **18**.

**18. Check continuity between J107-K and APU GEN CONTR circuit breaker terminal 1.**

- a. If continuity is present, replace circuit breaker (WP 0824 00). Go to **19**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **19**.

**19. Procedure completed.****MR DE-ICE FAULT CAPSULE OR LEGEND GOES ON****SYMPTOM**

MR DE-ICE FAULT capsule or legend goes on.

**MALFUNCTION**

MR DE-ICE FAULT capsule or legend goes on.

**CORRECTIVE ACTION****1. Does M/R DE-ICE FAULT capsule or legend go on immediately after POWER switch placed to TEST?**

- a. If capsule or legend goes on, replace rotor blade deice controller (WP 1226 00). Go to **18**.
- b. If capsule or legend does not go on, go to **2**.

**2. Place POWER switch to OFF, and then to TEST. Go to 3.****3. Check for 115 vac between:**

- TB3-2 and ground
- TB3-4 and ground

- a. If voltage is as specified, go to **4**.
- b. If voltage is not as specified, go to **11**.

**4. Check continuity between:**

- T14-T1 and P325-K
- T14-T2 and P325-L
- T14-T3 and P325-N

- a. If continuity is present, go to **5**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **18**.

**5. Check for resistance of 85 to 95 ohms between:**

- T14-T1 and T14-T4
- T14-T2 and T14-T5
- T14-T3 and T14-T6

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**MR DE-ICE FAULT CAPSULE OR LEGEND GOES ON** - Continued

- a. If resistance is as specified, go to **6**.
- b. If resistance is not as specified, replace T/Ransformer T14 (WP 1237 00). Go to **18**.

**6. Place POWER switch to OFF. Go to 7.**

**7. Turn off electrical power. Go to 8.**

**NOTE**

Performance of the following resistance check must be completed within one hour of performing operating/troubleshooting procedure steps 1. through 64. to achieve valid resistance measurements.

**8. Check for resistance of 7 to 10 ohms between:**

- J841-1 and J841-2
- J841-1 and J841-3
- J841-2 and J841-3
- J841-4 and J841-5
- J841-4 and J841-6
- J841-5 and J841-6
- J841-7 and J841-8
- J841-7 and J841-9
- J841-8 and J841-9
- J841-10 and J841-11
- J841-10 and J841-12
- J841-11 and J841-12
- J842-1 and J842-2
- J842-1 and J842-3
- J842-2 and J842-3
- J842-4 and J842-5
- J842-4 and J842-6
- J842-5 and J842-6
- J842-7 and J842-8
- J842-7 and J842-9
- J842-8 and J842-9
- J842-10 and J842-11
- J842-10 and J842-12
- J842-11 and J842-12

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**MR DE-ICE FAULT CAPSULE OR LEGEND GOES ON - Continued**

- J843-1 and J843-2
  - J843-1 and J843-3
  - J843-4 and J843-5
  - J843-2 and J843-3
  - J843-4 and J843-6
  - J843-5 and J843-6
  - J843-7 and J843-8
  - J843-7 and J843-9
  - J843-8 and J843-9
  - J843-10 and J843-11
  - J843-10 and J843-12
  - J843-11 and J843-12
  - J844-1 and J844-2
  - J844-1 and J844-3
  - J844-2 and J844-3
  - J844-4 and J844-5
  - J844-4 and J844-6
  - J844-5 and J844-6
  - J844-7 and J844-8
  - J844-7 and J844-9
  - J844-8 and J844-9
  - J844-10 and J844-11
  - J844-10 and J843-12
  - J844-11 and J844-12
- a. If resistance is as specified, go to **9**.
- b. If resistance is not as specified, replace malfunctioning main rotor blade (WP 0576 00). Go to **18**.
- 9. Remove distributor and slipring from helicopter (WP 1220 00). Go to 10.**
- 10. Check continuity between the following for slipring input wire and slipring output terminal strip respectively:**
- S1 and S1
  - S2 and S2
  - S3 and S3

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**MR DE-ICE FAULT CAPSULE OR LEGEND GOES ON - Continued**

- **øA and øA**
  - **øB and øB**
  - **øC and øC**
- a. If continuity is present, replace main rotor blade deice distributor (WP 1220 00). Go to **18**.
  - b. If continuity is not present, replace slipring brush blocks (WP 1220 00). Go to **18**.
  - c. If T/Rouble remains, repair/replace wiring (WP 1747 00). Go to **18**.
  - d. If T/Rouble still remains, replace slipring (WP 1220 00). Go to **18**.
- 11. Place POWER switch to OFF, and then to TEST. Go to 12.**
- 12. Check for 115 vac between main rotor contactor:**
- **Terminal A1 and ground**
  - **Terminal B1 and ground**
- a. If voltage is as specified, replace T/Ransformer (WP 1237 00). Go to **18**.
  - b. If voltage is not as specified, go to **13**.
- 13. Place POWER switch to OFF, and then to TEST. Go to 14.**
- 14. Check for 115 vac between K60 MAIN ROTOR CONTACTOR:**
- **Terminal A2 and ground**
  - **Terminal B2 and ground**
- a. If voltage is as specified, replace contactor (WP 1233 00). Go to **18**.
  - b. If voltage is not as specified, go to **15**.
- 15. Place POWER switch to OFF, and then to TEST. Go to 16.**
- 16. Check for 115 vac between K62 MN RTR BUS CONT:**
- **Terminal T1 and ground**
  - **Terminal T2 and ground**
- a. If voltage is as specified, repair/replace wiring between MN RTR BUS CONT terminals T1 and T2 and main rotor contactor terminals A2 and B2 (WP 1747 00). Go to **18**.
  - b. If voltage is not as specified, go to **17**.
- 17. Check for 115 vac between K62 MN RTR BUS CONT:**
- **Terminal L1 and ground**
  - **Terminal L2 and ground**
- a. If voltage is as specified, replace contactor (WP 1234 00). Go to **18**.
  - b. If voltage is not as specified, replace current limiter CL7 or CL8 as required (WP 1228 00). Go to **18**.
- 18. Procedure completed.**



**PWR MAIN RTR AND PWR TAIL RTR LAMPS DO NOT GO ON****SYMPTOM**

PWR MAIN RTR and PWR TAIL RTR lamps do not go on.

**MALFUNCTION**

PWR MAIN RTR and PWR TAIL RTR lamps do not go on.

**CORRECTIVE ACTION**

1. **Disconnect P324 and P325. Go to 2.**
2. **Check helicopter configuration.**
  - a. **EMEP** , go to 3.
  - b. **W/O EMEP** , go to 5.
3. **EMEP Remove and check pin filtered adapter P134 (WP 0925 00).**
  - a. If pin filtered adapter is good, go to 4.
  - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to 9.
4. **EMEP Install pin filtered adapter (WP 0925 00). Go to 5.**
5. **Check for 28 vdc between P134-M and ground.**
  - a. If voltage is as specified, replace blade de-ice test panel (WP 1240 00). Go to 9.
  - b. If voltage is not as specified, go to 6.
6. **Check continuity between P133-T and P134-M.**
  - a. If continuity is present, go to 7.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to 9.
7. **Check continuity between J133-L and J133-T.**
  - a. If continuity is present, go to 8.
  - b. If continuity is not present, replace blade de-ice control panel (WP 1240 00). Go to 9.
8. **Check for 28 vdc between DE-ICE CONTRLR circuit breaker terminal 1 and ground.**
  - a. If voltage is as specified, repair/replace wiring between terminal 1 and P133-L (WP 1747 00). Go to 9.
  - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to 9.
9. **Procedure completed.**

**DROOP STOP CAMS ARE NOT WARM WHEN POWER SWITCH IS IN TEST****SYMPTOM**

Droop stop cams are not warm when POWER switch is in TEST.

**MALFUNCTION**

Droop stop cams are not warm when POWER switch is in TEST.

**CORRECTIVE ACTION**

1. **Check helicopter configuration.**

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**DROOP STOP CAMS ARE NOT WARM WHEN POWER SWITCH IS IN TEST - Continued**

- a. **EMEP** , go to **2**.
- b. **W/O EMEP** , go to **5**.
- 2. EMEP Is more than one droop stop heater cool?**
  - a. If more than one droop stop heater cool, go to **4**.
  - b. If there is no more than one droop stop heater cool, go to **3**.
- 3. EMEP Check for 200 vac between pins 1 and 2 of affected droop stop heater.**
  - a. If voltage is as specified, replace affected droop stop heater pin assembly (WP 0555 00). Go to **7**.
  - b. If voltage is not as specified, go to **4**.
- 4. Check continuity between:**
  - **P845-1 and P1001-10**
  - **P845-2 and P1001-11**
  - **P846-1 and P1002-10**
  - **P846-2 and P1002-11**
  - **P847-1 and P1003-10**
  - **P847-2 and P1003-11**
  - **P848-1 and P1004-10**
  - **P848-2 and P1004-11**
  - a. If continuity is present, replace main rotor blade deice distributor (WP 1220 00). Go to **7**.
  - b. If continuity is not present, replace harness as required (WP 1747 00). Go to **7**.
- 5. W/O EMEP Is more than one droop stop heater cool?**
  - a. If more than one droop stop heater cool, replace main rotor blade deice distributor (WP 1220 00). Go to **7**.
  - b. If there is not more than one droop stop heater cool, go to **6**.
- 6. W/O EMEP Check for 200 vac between pins 1 and 2 of affected droop stop heater.**
  - a. If voltage is as specified, replace affected droop stop heater pin assembly (WP 0555 00). Go to **7**.
  - b. If voltage is not as specified, replace main rotor blade deice distributor (WP 1220 00). Go to **7**.
- 7. Procedure completed.**

**MR DE-ICE FAIL AND T/R DE-ICE FAIL CAPSULES OR LEGENDS ARE ON****SYMPTOM**

MR DE-ICE FAIL and T/R DE-ICE FAIL capsules or legends are on.

**MALFUNCTION**

MR DE-ICE FAIL and T/R DE-ICE FAIL capsules or legends are on.

**CORRECTIVE ACTION**

- 1. With ON-OFF-TEST switch at ON, check for 28 vdc between P329-Y and ground.**
  - a. If voltage is as specified, go to **2**.
  - b. If voltage is not as specified, go to **11**.
- 2. Check for 28 vdc between main rotor blade deice junction box terminal E6 and ground.**
  - a. If voltage is as specified, go to **3**.
  - b. If voltage is not as specified, repair/replace wiring between terminal E6 and J329-Y (WP 1747 00). Go to **12**.
- 3. Check helicopter configuration.**
  - a. **EMEP** , go to **4**.
  - b. **W/O EMEP** , go to **6**.
- 4. EMEP Remove and check pin filtered adapter P325 (WP 0925 00).**
  - a. If pin filtered adapter is good, go to **5**.
  - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **12**.
- 5. EMEP Install pin filtered adapter (WP 0925 00). Go to 6.**
- 6. With GENERATORS NO. 1 and NO. 2 switches ON and rotor at 100%, check for 28 vdc between P325-P and ground.**
  - a. If voltage is as specified, replace rotor blade deice controller (WP 1226 00). Go to **12**.
  - b. If voltage is not as specified, go to **7**.
- 7. Check continuity between P205-F and P325-P.**
  - a. If continuity is present, go to **8**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **12**.
- 8. Check for 28 vdc between P205-E and ground.**
  - a. If voltage is as specified, replace APU/EXTERNAL POWER contactor (WP 0840 00). Go to **12**.
  - b. If voltage is not as specified, go to **9**.
- 9. Check continuity between P205-E and P206-L.**
  - a. If continuity is present, go to **10**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **12**.
- 10. Check for 28 vdc between P206-M and ground.**

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**MR DE-ICE FAIL AND T/R DE-ICE FAIL CAPSULES OR LEGENDS ARE ON** - Continued

- a. If voltage is as specified, replace AC bus tie contactor (WP 0845 00). Go to **12**.
- b. If voltage is not as specified, repair/replace wiring between P206-M and DE-ICE CNTRLR circuit breaker (WP 1747 00). Go to **12**.

**11. Check continuity between P329-Y and P133-J.**

- a. If continuity is present, replace blade deice control panel (WP 1223 00). Go to **12**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **12**.

**12. Procedure completed.****MR DE-ICE FAIL, T/R DE-ICE FAIL AND/OR M/R DE-ICE FAULT CAPSULES OR LEGENDS GO ON****SYMPTOM**

MR DE-ICE FAIL, T/R DE-ICE FAIL and/or M/R DE-ICE FAULT capsules or legends go on.

**MALFUNCTION**

MR DE-ICE FAIL, T/R DE-ICE FAIL and/or M/R DE-ICE FAULT capsules or legends go on.

**CORRECTIVE ACTION**

- 1. Did M/R DE-ICE FAIL, T/R DE-ICE FAIL and M/R DE-ICE FAULT capsules or legends go on?**
  - a. If capsules or legends are on, go to **2**.
  - b. If capsules or legends are not on, go to **21**.
- 2. Check wiring to TB5-1, TB5-3, and TB5-5.**
  - a. If wiring is good, go to **3**.
  - b. If wiring is not good, repair/replace wiring (WP 1747 00). Go to **24**.
- 3. Check helicopter configuration.**
  - a. **EMEP** , go to **4**.
  - b. **W/O EMEP** , go to **6**.
- 4. EMEP Remove and check pin filtered adapter P325 (WP 0925 00).**
  - a. If pin filtered adapter is good, go to **5**.
  - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **24**.
- 5. EMEP Install pin filtered adapter (WP 0925 00). Go to 6.**
- 6. With GENERATORS NO. 1 switch at OFF and GENERATORS NO. 2 and APU switches at ON, check for 28 vdc between P325-P and ground.**
  - a. If voltage is as specified, go to **7**.
  - b. If voltage is not as specified, go to **8**.
- 7. Check wiring between T10 and T3.**
  - a. If wiring is good, replace current T/Ransformer T10 (WP 1236 00). Go to **24**.
  - b. If T/Rouble remains, replace T/Ransformer T3 (WP 0840 00). Go to **24**.
  - c. If wiring is not good, repair/replace wiring (WP 1747 00). Go to **24**.

MR DE-ICE FAIL, T/R DE-ICE FAIL AND/OR M/R DE-ICE FAULT CAPSULES OR LEGENDS GO ON -  
Continued

8. **Check helicopter configuration.**
  - a. **EH-60A** , go to 9.
  - b. **UH-60A UH-60L HH-60A HH-60L** , go to 16.
9. **EH-60A Check continuity between P325-P and No. 1 junction box CR20-E7.**
  - a. If continuity is present, go to 10.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to 24.
10. **EH-60A Check for 28 vdc between No. 1 junction box CR20-E8 and ground.**
  - a. If voltage is as specified, replace CR20 in No. 1 junction box (WP 0849 00). Go to 24.
  - b. If voltage is not as specified, go to 11.
11. **EH-60A Check continuity between No. 1 junction box CR20-E8 and P205-F.**
  - a. If continuity is present, go to 12.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to 24.
12. **EH-60A Check for 28 vdc between P205-E and ground.**
  - a. If voltage is as specified, replace APU/EXTERNAL POWER contactor (WP 0840 00). Go to 24.
  - b. If voltage is not as specified, go to 13.
13. **EH-60A Check continuity between P205-E and P206-L.**
  - a. If continuity is present, go to 14.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to 24.
14. **EH-60A Check for 28 vdc between P206-M and ground.**
  - a. If voltage is as specified, replace AC bus tie contactor (WP 0845 00). Go to 24.
  - b. If voltage is not as specified, go to 15.
15. **EH-60A Check for 28 vdc between SEC MON BUS CONTR circuit breaker terminal 1 and ground.**
  - a. If voltage is as specified, repair/replace wiring between P206-M and circuit breaker (WP 1747 00). Go to 24.
  - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to 24.
16. **UH-60A UH-60L HH-60A HH-60L Check continuity between P325-P and P205-F.**
  - a. If continuity is present, go to 17.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to 24.
17. **UH-60A UH-60L HH-60A HH-60L Check for 28 vdc between P205-E and ground.**
  - a. If voltage is as specified, replace APU/EXTERNAL POWER contactor (WP 0840 00). Go to 24.
  - b. If voltage is not as specified, go to 18.
18. **UH-60A UH-60L HH-60A HH-60L Check continuity between P205-E and P206-L.**
  - a. If continuity is present, go to 19.

**MR DE-ICE FAIL, T/R DE-ICE FAIL AND/OR M/R DE-ICE FAULT CAPSULES OR LEGENDS GO ON -**  
Continued

- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **24**.
- 19. **UH-60A UH-60L HH-60A HH-60L Check for 28 vdc between P206-M and ground.**
  - a. If voltage is as specified, replace AC bus tie contactor (WP 0845 00). Go to **24**.
  - b. If voltage is not as specified, go to **20**.
- 20. **UH-60A UH-60L HH-60A HH-60L Check for 28 vdc between DE-ICE CNTRLR circuit breaker terminal 1 and ground.**
  - a. If voltage is as specified, repair/replace wiring between P206-M and terminal 1 (WP 1747 00). Go to **24**.
  - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **24**.
- 21. **Did T/R DE-ICE FAIL capsule or legend go on?**
  - a. If capsule or legend is on, go to **22**.
  - b. If capsule or legend is not on, go to **23**.
- 22. **Check wiring to K63 TAIL ROTOR DE-ICE CONTACTOR.**
  - a. If wiring is good, replace contactor (WP 1235 00). Go to **24**.
  - b. If wiring is not good, repair/replace wiring (WP 1747 00). Go to **24**.
- 23. **Check wiring to K62 MN RTR BUS CONT.**
  - a. If wiring is good, replace contactor (WP 1234 00). Go to **24**.
  - b. If wiring is not good, repair/replace wiring (WP 1747 00). Go to **24**.
- 24. **Procedure completed.**

**MR DE-ICE FAIL, T/R DE-ICE FAIL AND M/R DE-ICE FAULT CAPSULES OR LEGENDS GO ON**

**SYMPTOM**

MR DE-ICE FAIL, T/R DE-ICE FAIL and M/R DE-ICE FAULT capsules or legends go on.

**MALFUNCTION**

MR DE-ICE FAIL, T/R DE-ICE FAIL and M/R DE-ICE FAULT capsules or legends go on.

**CORRECTIVE ACTION**

- 1. **Check continuity between P325-P and P329-L.**
  - a. If continuity is present, go to **2**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **13**.
- 2. **With rotor at 100% and GENERATORS NO. 1 and NO. 2 switches at ON, check for 28 vdc between main rotor blade deice junction box terminal E1 and ground.**
  - a. If voltage is as specified, replace diode CR11 (WP 1239 00). Go to **13**.
  - b. If voltage is not as specified, go to **3**.
- 3. **Check continuity between P204-K and E1 in main rotor blade de-ice junction box.**
  - a. If continuity is present, go to **4**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **13**.

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MR DE-ICE FAIL, T/R DE-ICE FAIL AND M/R DE-ICE FAULT CAPSULES OR LEGENDS GO ON - Continued

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4. With rotor at 100% and GENERATORS NO. 1 and NO. 2 switches at ON, check for 28 vdc between P204-L and ground.
  - a. If voltage is as specified, replace No. 2 generator line contactor (WP 0835 00). Go to 13.
  - b. If voltage is not as specified, go to 5.
5. Check continuity between P204-L and P212-L.
  - a. If continuity is present, go to 6.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to 13.
6. Check for 28 vdc between P212-K and ground.
  - a. If voltage is as specified, replace No. 1 generator line contactor (WP 0834 00). Go to 13.
  - b. If voltage is not as specified, go to 7.
7. Check helicopter configuration.
  - a. **WINTER** , go to 8.
  - b. **W/O WINTER** , go to 9.
8. **WINTER** Check continuity between J244B-L and P212-K.
  - a. If continuity is present, go to 9.
  - b. If continuity is not present, replace **WINTER** (WP 1400 00). Go to 13.
  - c. If trouble remains, go to 9.
9. Check continuity between P244-L and P212-K.
  - a. If continuity is present, go to 10.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to 13.
10. Check helicopter configuration.
  - a. **WINTER** , go to 11.
  - b. **W/O WINTER** , go to 12.
11. **WINTER** Check for 28 vdc between J244B-V and ground.
  - a. If voltage is as specified, replace right relay panel (WP 0825 00). Go to 13.
  - b. If voltage is not as specified, replace **WINTER** (WP 1400 00). Go to 13.
  - c. If trouble remains, go to 12.
12. Check for 28 vdc between P244-V and ground.
  - a. If voltage is as specified, replace right relay panel (WP 0825 00). Go to 13.
  - b. If voltage is not as specified, repair/replace wiring between P244-V and DE-ICE CNTRLR circuit breaker terminal 1 (WP 1747 00). Go to 13.
13. Procedure completed.

**NONE OF THE CAPSULES OR LEGENDS GO ON****SYMPTOM**

None of the capsules or legends go on.

**MALFUNCTION**

None of the capsules or legends go on.

**CORRECTIVE ACTION**

1. **Check helicopter configuration.**
  - a. **EMEP** , go to 2.
  - b. **W/O EMEP** , go to 4.
2. **EMEP Remove and check pin filtered adapter P325 (WP 0925 00).**
  - a. If pin filtered adapter is good, go to 3.
  - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to 7.
3. **EMEP Install pin filtered adapter (WP 0925 00). Go to 4.**
4. **With GENERATORS NO. 1, GENERATORS NO. 2, and APU switches at OFF, check for 28 vdc between P325-P and ground.**
  - a. If voltage is as specified, go to 5.
  - b. If voltage is not as specified, replace rotor blade deice controller (WP 1226 00). Go to 7.
5. **Disconnect P205 from APU/External power contactor. Go to 6.**
6. **Check for 28 vdc between P325-P and ground.**
  - a. If voltage is as specified, replace No. 2 generator line contactor (WP 0835 00). Go to 7.
  - b. If voltage is not as specified, replace APU/EXTERNAL POWER contactor (WP 0840 00). Go to 7.
7. **Procedure completed.**

**WITH BLADE DEICE POWER SWITCH ON, M/R DE-ICE FAIL, T/R DE-ICE FAIL, OR BOTH CAPSULES/LEGENDS GO ON WITH MODE SWITCH PLACED TO T, L, OR M****SYMPTOM**

With blade deice power switch ON, M/R DE-ICE FAIL, T/R DE-ICE FAIL, or both capsules/legends go on with mode switch placed to T, L, or M.

**MALFUNCTION**

With blade deice power switch ON, M/R DE-ICE FAIL, T/R DE-ICE FAIL, or both capsules/legends go on with mode switch placed to T, L, or M.

**CORRECTIVE ACTION**

1. **Does system work in AUTO mode?**
  - a. If system works in AUTO, go to 2.
  - b. If system does not work in AUTO, do operational/troubleshooting procedure steps 25. through 38.



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WITH BLADE DEICE POWER SWITCH ON, M/R DE-ICE FAIL, T/R DE-ICE FAIL, OR BOTH CAPSULES/  
LEGENDS GO ON WITH MODE SWITCH PLACED TO T, L, OR M - Continued

**2. Check helicopter configuration.**

- a. **EMEP** , go to **3**.
- b. **W/O EMEP** , go to **5**.

**3. EMEP Remove and check pin filtered adapter P325 (WP 0925 00).**

- a. If pin filtered adapter is good, go to **4**.
- b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **6**.

**4. EMEP Install pin filtered adapter (WP 0925 00). Go to 5.**

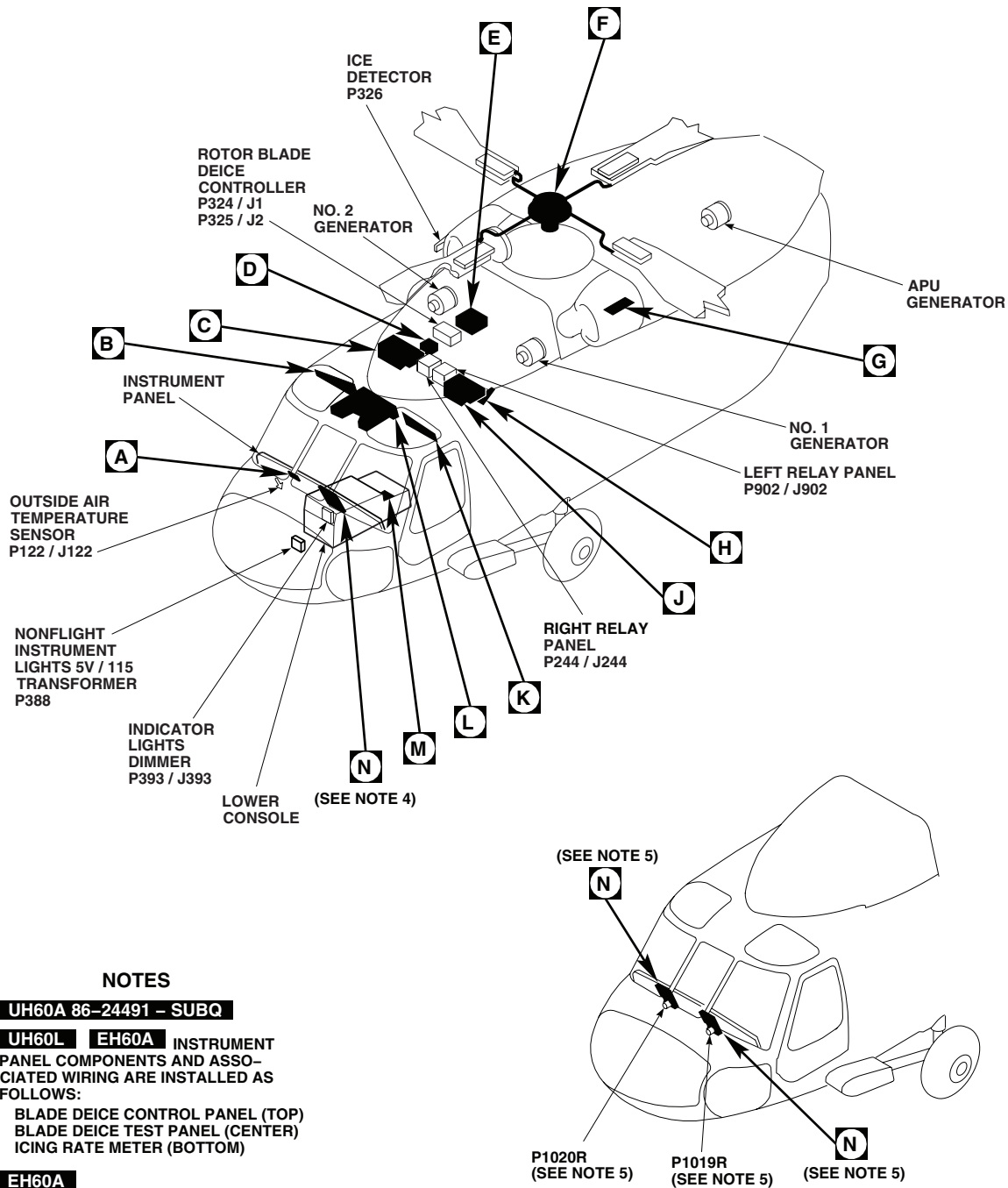
**5. Check continuity between:**

- **P133-G and P325-f**
- **P133-H and P325-e**
- **P133-S and P325-g**

- a. If continuity is present, replace blade de-ice test panel (WP 1240 00). Go to **6**.
- b. If trouble remains, replace rotor blade deice controller (WP 1226 00). Go to **6**.
- c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **6**.

**6. Procedure completed.**

## LOCATION



## NOTES

1. UH60A 86-24491 - SUBQ

UH60L EH60A INSTRUMENT PANEL COMPONENTS AND ASSOCIATED WIRING ARE INSTALLED AS FOLLOWS:

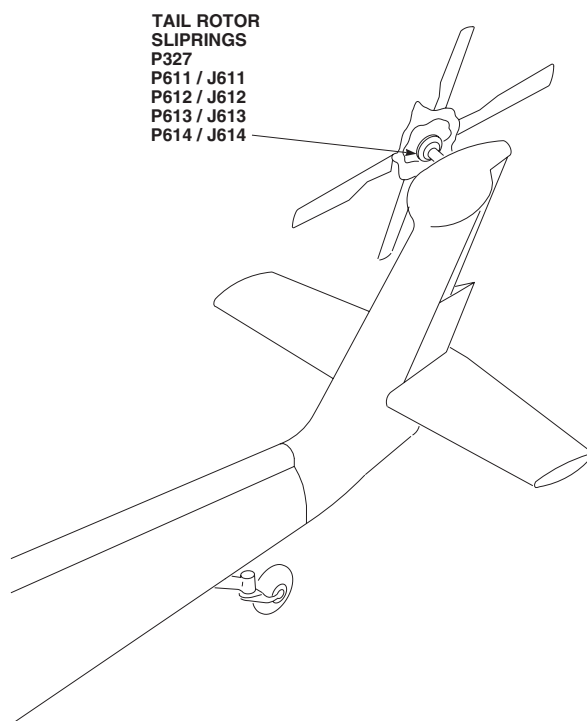
BLADE DEICE CONTROL PANEL (TOP)  
BLADE DEICE TEST PANEL (CENTER)  
ICING RATE METER (BOTTOM)

2. EH60A
3. UH60A UH60L HH60A  
HH60L
4. UH60A UH60L EH60A
5. HH60A HH60L

AA3425\_1B  
SA

Figure 1. Blade Deicing System Location Diagram. (Sheet 1 of 10)

## LOCATION - Continued



| TERMINAL BOARD /<br>DISCONNECT PLUG /<br>RECEPTACLE | LOCATION /<br>CONNECTION POINT                                 |
|-----------------------------------------------------|----------------------------------------------------------------|
| P107 / J107                                         | COCKPIT, BL 9 LH, STA 236                                      |
| P111 / J111                                         | COCKPIT, BL 7.5 LH,<br>STA 197                                 |
| P113 / J113                                         | COCKPIT, BL 6 LH, STA 197                                      |
| P114 / J114                                         | COCKPIT, BL 8 RH, STA 200                                      |
| P115 / J115                                         | COCKPIT, BL 6 RH, STA 210                                      |
| P117 / J117                                         | CAUTION / ADVISORY<br>PANEL (SEE NOTE 4)                       |
| P118 / J118                                         | CAUTION / ADVISORY<br>PANEL (SEE NOTE 4)                       |
| P121 / J121                                         | COCKPIT, BL 10 RH,<br>STA 915                                  |
| P122 / J122                                         | (OAT) OUTSIDE AIR-<br>TEMPERATURE SENSOR                       |
| P127 / J127                                         | COCKPIT CEILING, BL 28 LH,<br>STA 243                          |
| P132 (SEE NOTE 1)                                   | ICING RATE METER                                               |
| P133 / J133 (SEE NOTE 1)                            | BLADE DEICE CONTROL<br>PANEL                                   |
| P134 / J1 (SEE NOTE 1)                              | BLADE DEICE TEST PANEL                                         |
| P202 / J202                                         | NO. 2 JUNCTION BOX                                             |
| P203 / J203                                         | NO. 2 JUNCTION BOX                                             |
| P204 / J204                                         | NO. 2 JUNCTION BOX                                             |
| P205                                                | NO. 2 JUNCTION BOX                                             |
| P206                                                | NO. 2 JUNCTION BOX                                             |
| P210 / J210                                         | NO. 1 JUNCTION BOX                                             |
| P212 / J212                                         | NO. 1 JUNCTION BOX                                             |
| P217 / J217                                         | MAIN ROTOR PYLON DECK<br>BL 0, STA 284                         |
| P219 / J219                                         | CABIN FLOOR, BL 25 RH,<br>STA 247                              |
| P224 / J224                                         | COCKPIT CEILING, BL 3 RH,<br>STA 247                           |
| P230 / J230                                         | CABIN CEILING, BL 8 RH,<br>STA 245                             |
| P231 / J231                                         | UPPER COCKPIT LH, AFT<br>OUTBD BL 6.05, STA 247                |
| P237 / J237                                         | BEHIND COPILOT'S<br>CIRCUIT BREAKER PANEL<br>BL 25 LH          |
| P244 / J244                                         | RIGHT RELAY PANEL                                              |
| P249 / J249                                         | BEHIND COPILOT'S<br>CIRCUIT BREAKER PANEL<br>BL 23 LH, STA 247 |
| P266 / J266                                         | BEHIND PILOT'S CIRCUIT<br>BREAKER PANEL, BL 25 RH              |
| P280 / J280                                         | CABIN CEILING, BL 6 LH,<br>STA 247                             |
| P324 / J1                                           | ROTOR BLADE DEICE<br>CONTROLLER                                |
| P325 / J2                                           | ROTOR BLADE DEICE<br>CONTROLLER                                |

AA3425\_2A  
SA

Figure 1. Blade Deicing System Location Diagram. (Sheet 2 of 10)

## LOCATION - Continued

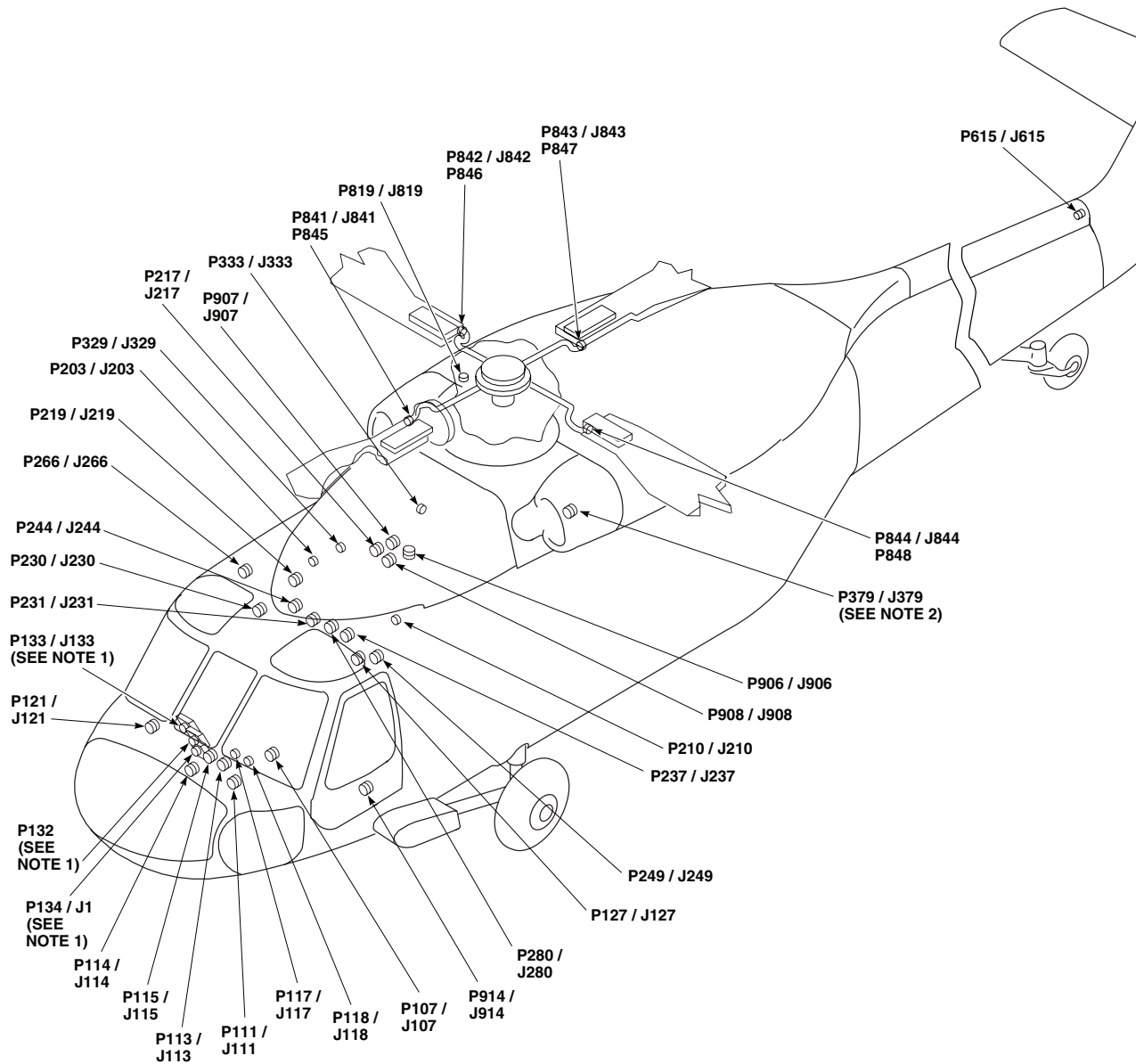
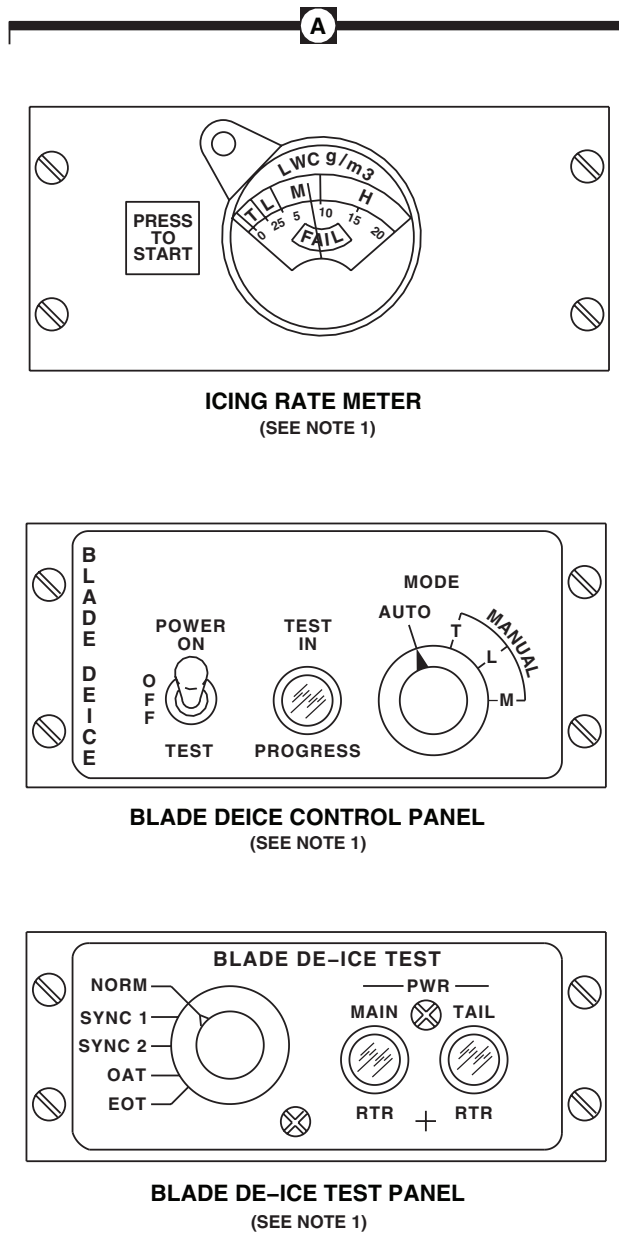
AA3425\_3  
SA

Figure 1. Blade Deicing System Location Diagram. (Sheet 3 of 10)

## LOCATION - Continued

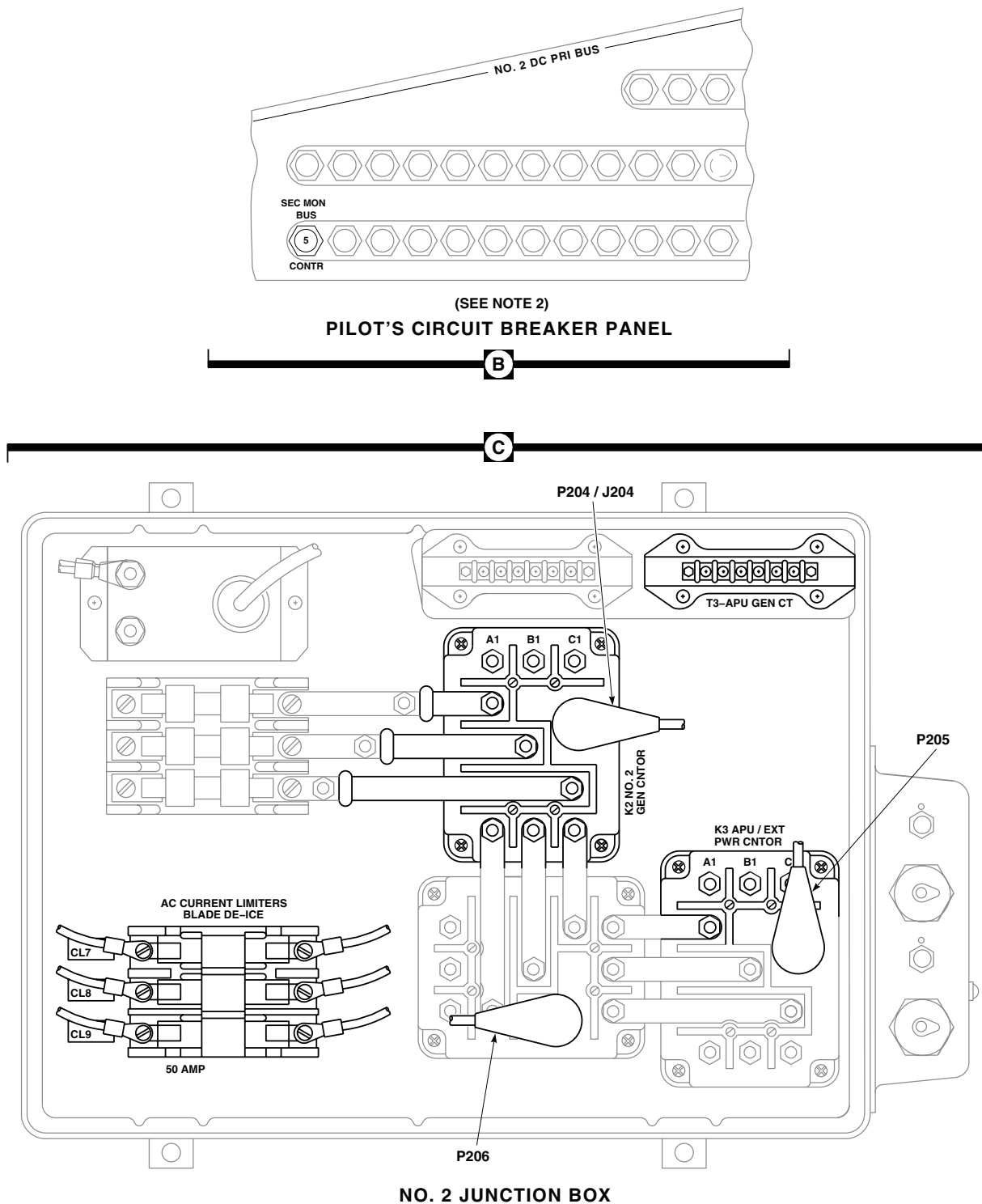


| TERMINAL BOARD /<br>DISCONNECT PLUG /<br>RECEPTACLE | LOCATION /<br>CONNECTION POINT                           |
|-----------------------------------------------------|----------------------------------------------------------|
| P326                                                | ICE DETECTOR                                             |
| P327                                                | TAIL ROTOR SLIPRINGS                                     |
| P329 / J329                                         | MAIN ROTOR BLADE DEICE<br>JUNCTION BOX                   |
| P332 / J332                                         | AUXILIARY AC JUNCTION<br>BOX                             |
| P333 / J333                                         | AUXILIARY AC JUNCTION<br>BOX                             |
| P379 / J379 (SEE NOTE 2)                            | BL 3.50 RH, STA 379                                      |
| P388                                                | NONFLIGHT INSTRUMENT<br>LIGHTS 5V / 115V<br>TRANSFORMERS |
| P393 / J393                                         | INDICATOR LIGHTS<br>DIMMER                               |
| P611 / J611                                         | TAIL ROTOR BLADE                                         |
| P612 / J612                                         | TAIL ROTOR BLADE                                         |
| P613 / J613                                         | TAIL ROTOR BLADE                                         |
| P614 / J614                                         | TAIL ROTOR BLADE                                         |
| P615 / J615                                         | TAIL CONE, BL 3 RH,<br>STA 651                           |
| P819 / J819                                         | MAIN ROTOR PYLON DECK<br>BL 23 RH, STA 352               |
| P841 / J841                                         | NO. 1 MAIN ROTOR BLADE                                   |
| P842 / J842                                         | NO. 2 MAIN ROTOR BLADE                                   |
| P843 / J843                                         | NO. 3 MAIN ROTOR BLADE                                   |
| P844 / J844                                         | NO. 4 MAIN ROTOR BLADE                                   |
| P845                                                | DROOP STOP HEATER NO. 1                                  |
| P846                                                | DROOP STOP HEATER NO. 2                                  |
| P847                                                | DROOP STOP HEATER NO. 3                                  |
| P848                                                | DROOP STOP HEATER NO. 4                                  |
| P902 / J902                                         | LEFT RELAY PANEL                                         |
| P906 / J906                                         | CABIN CEILING, BL 0,<br>STA 284                          |
| P907 / J907                                         | CABIN CEILING, BL 0,<br>STA 288                          |
| P908 / J908                                         | CABIN CEILING, BL 0,<br>STA 279                          |
| P914 / J914                                         | COCKPIT, BL 24 LH,<br>STA 247                            |
| P1001 / J1001                                       | MAIN ROTOR BLADE<br>DEICE DISTRIBUTOR                    |
| P1002 / J1002                                       | MAIN ROTOR BLADE<br>DEICE DISTRIBUTOR                    |
| P1003 / J1003                                       | MAIN ROTOR BLADE<br>DEICE DISTRIBUTOR                    |
| P1004 / J1004                                       | MAIN ROTOR BLADE<br>DEICE DISTRIBUTOR                    |
| P1019R / J3                                         | COPILOT'S MULTIFUNCTION<br>DISPLAY (SEE NOTE 5)          |
| P1020R / J3                                         | PILOT'S MULTIFUNCTION<br>DISPLAY (SEE NOTE 5)            |

AA3425\_4A  
SA

Figure 1. Blade Deicing System Location Diagram. (Sheet 4 of 10)

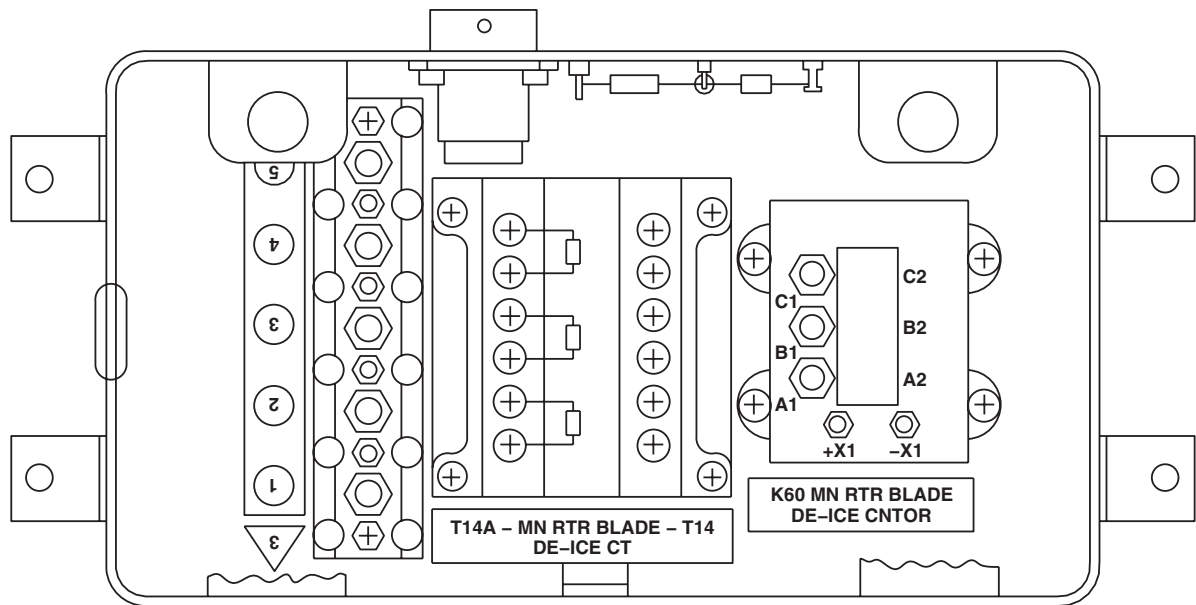
LOCATION - Continued



AA3425\_5A  
SA

Figure 1. Blade Deicing System Location Diagram. (Sheet 5 of 10)

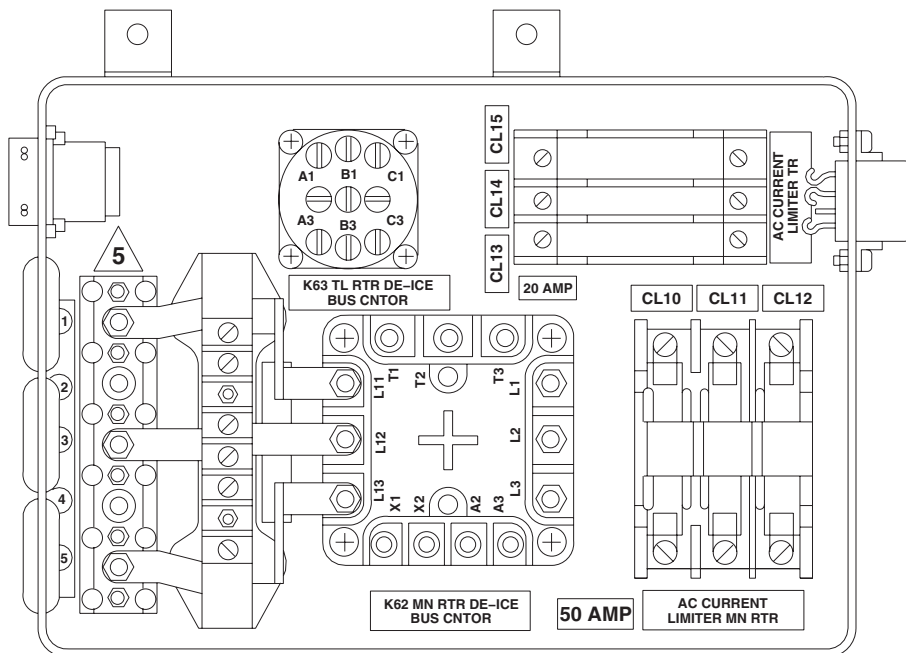
LOCATION - Continued



MAIN ROTOR DEICE JUNCTION BOX

D

E

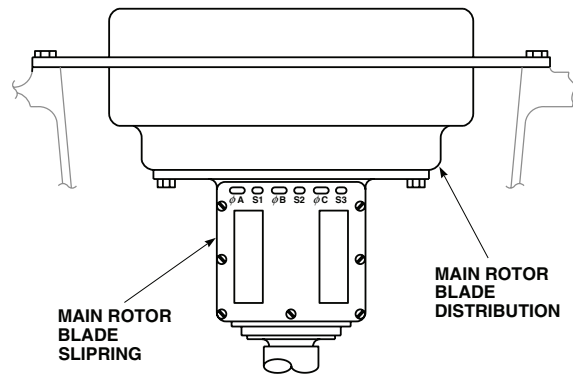


AUXILIARY AC JUNCTION BOX

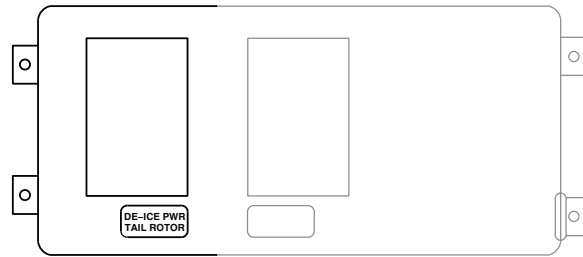
AA3425\_6  
SA

Figure 1. Blade Deicing System Location Diagram. (Sheet 6 of 10)

LOCATION - Continued



MAIN ROTOR DISTRIBUTION  
AND SLIPRINGS

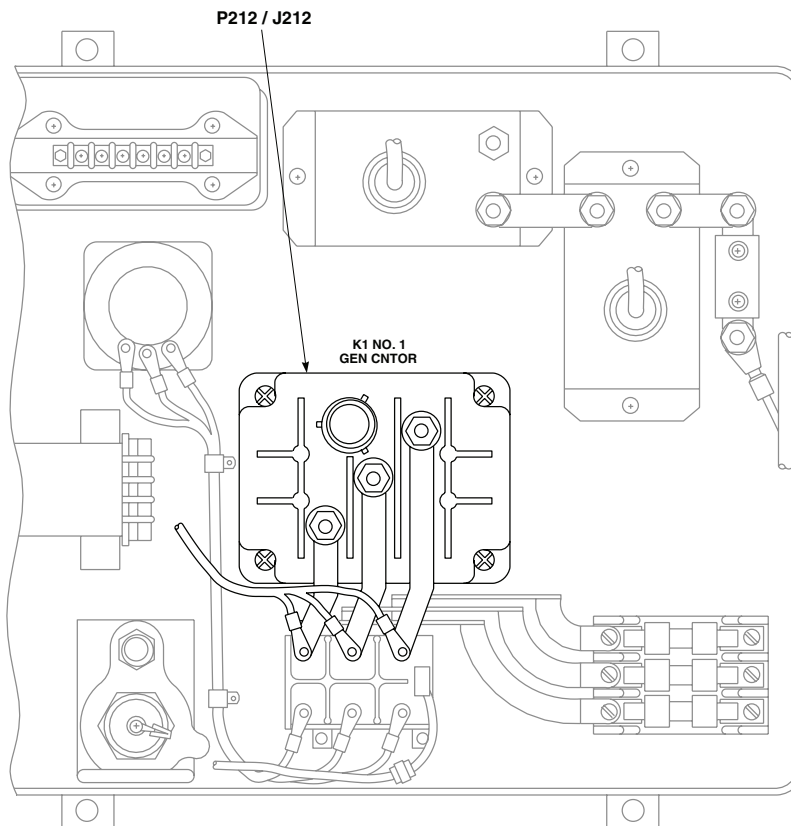


REMOTE CONTROL CIRCUIT BREAKER BOX  
(SEE NOTE 2)

F

G

H



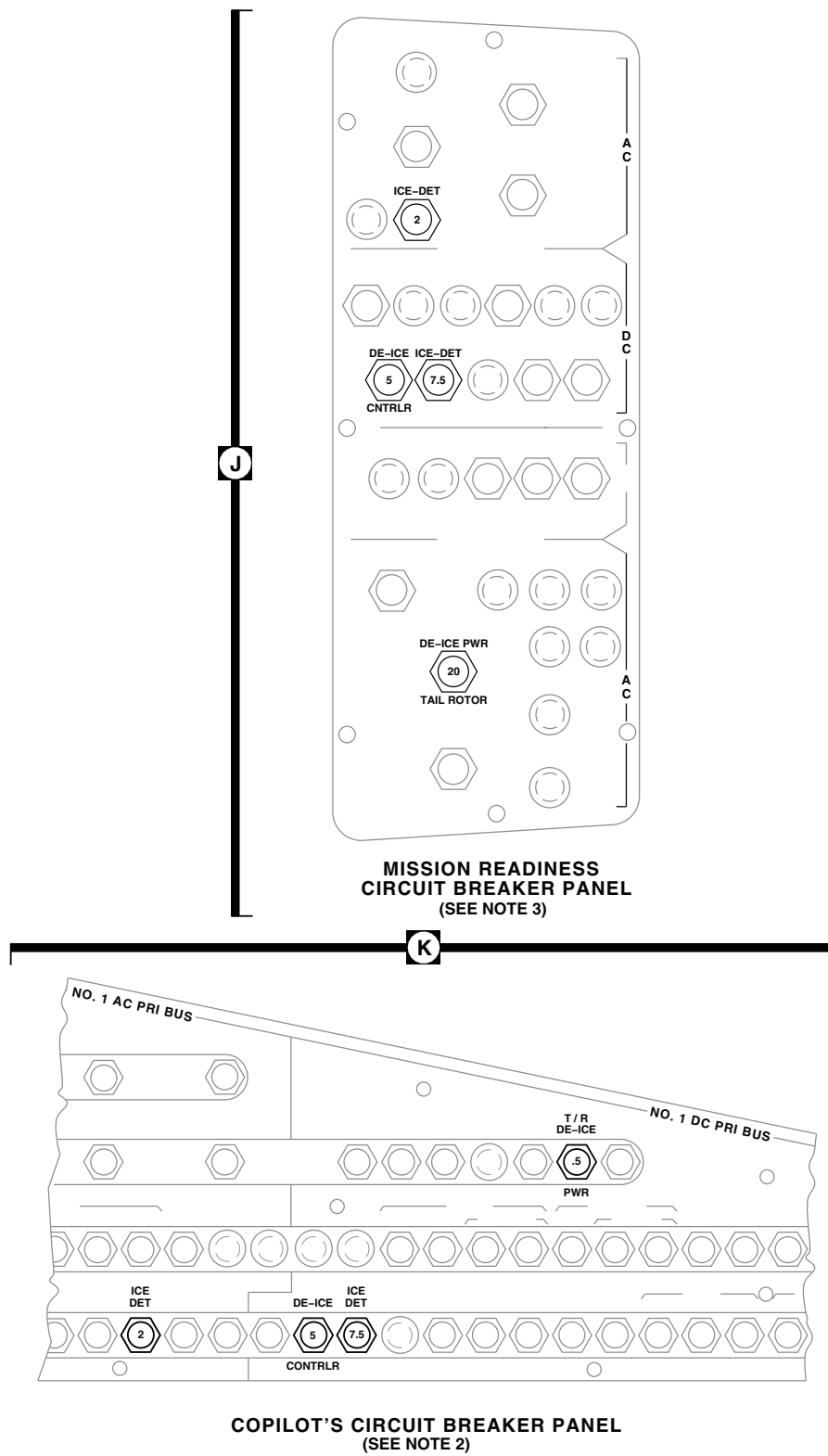
NO. 1 JUNCTION BOX

AA3425\_7  
SA

Figure 1. Blade Deicing System Location Diagram. (Sheet 7 of 10)



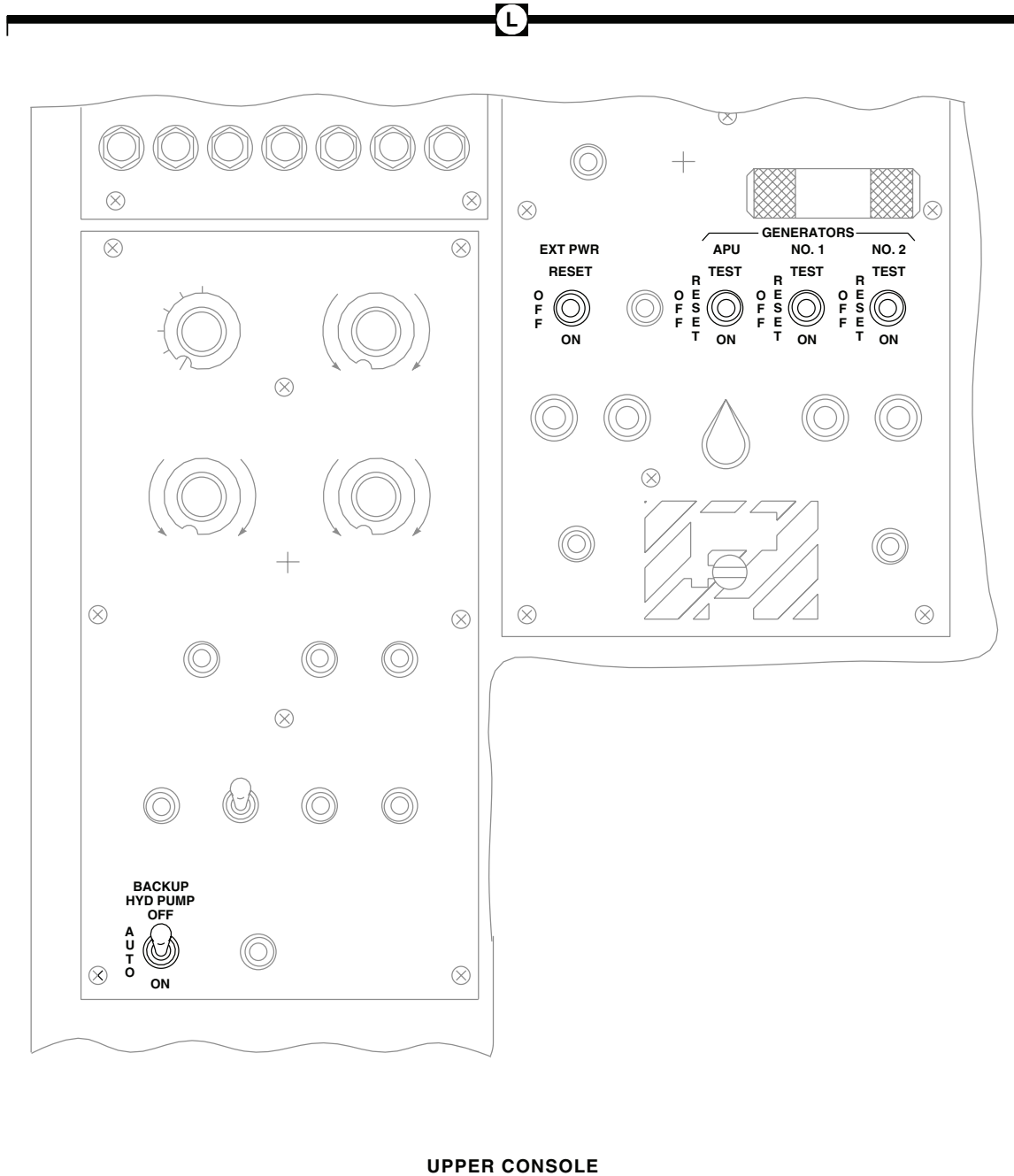
LOCATION - Continued



AA3425\_8  
SA

Figure 1. Blade Deicing System Location Diagram. (Sheet 8 of 10)

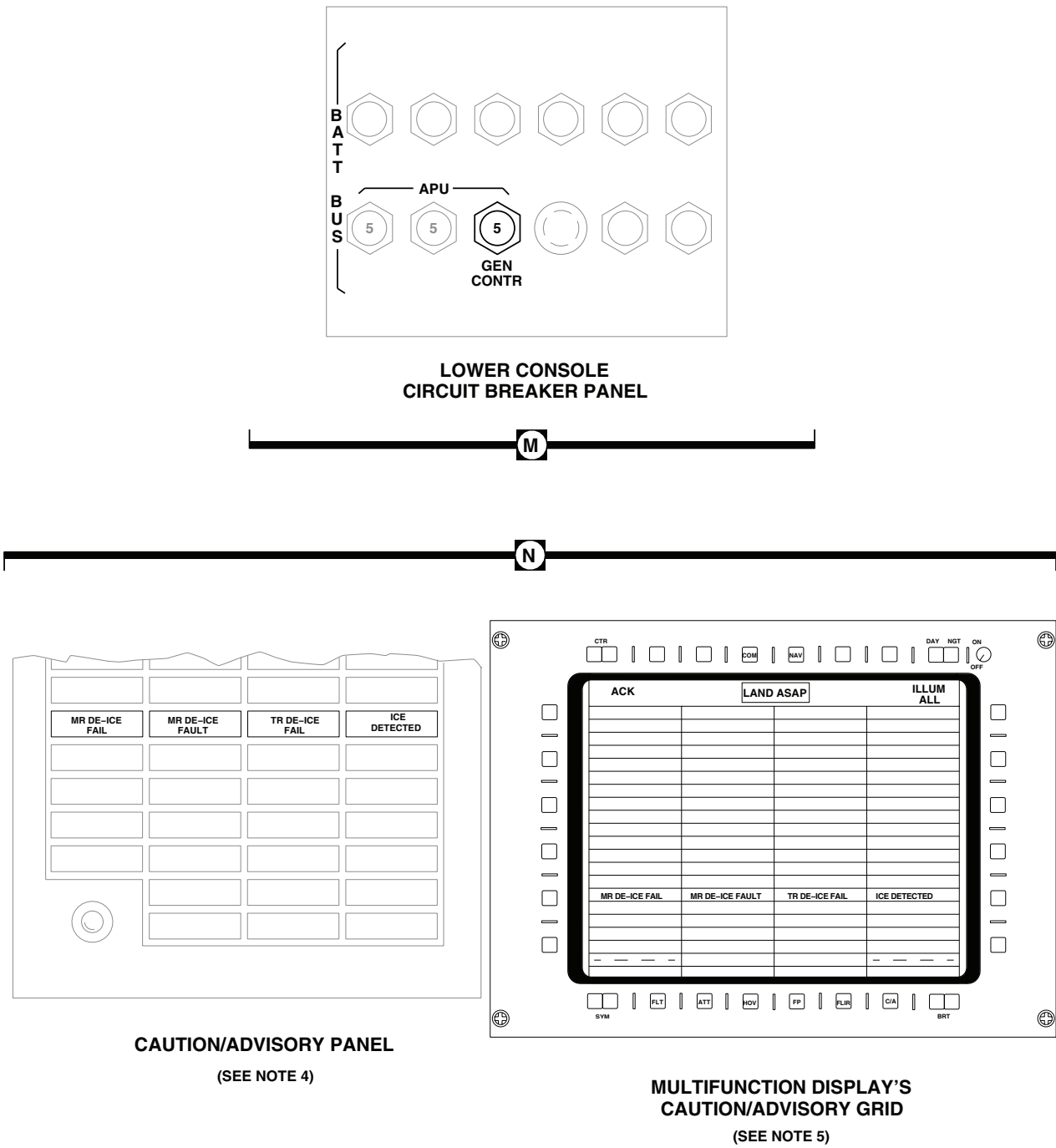
LOCATION - Continued



AA3425 9  
SA

Figure 1. Blade Deicing System Location Diagram. (Sheet 9 of 10)

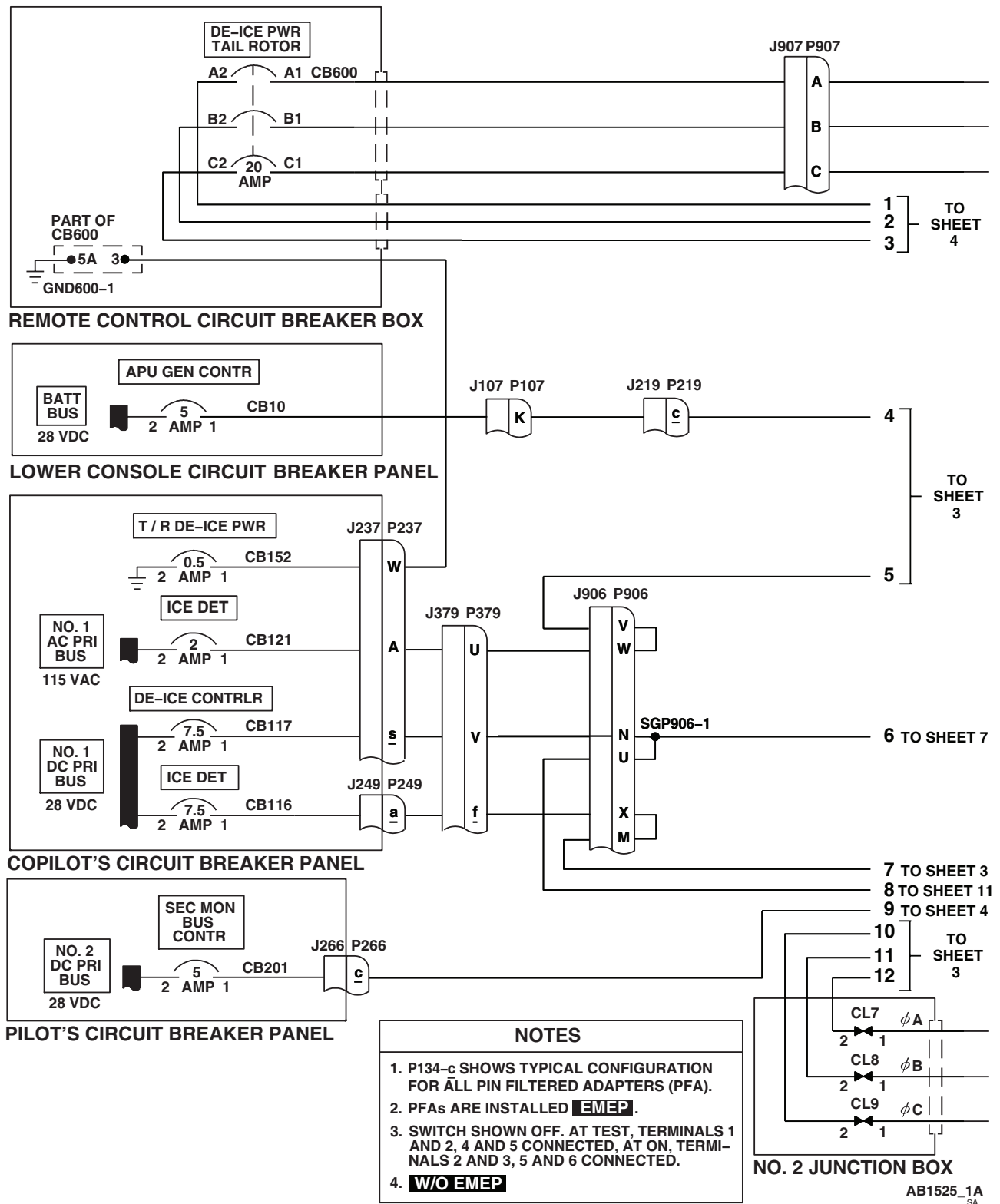
LOCATION - Continued



AA3425\_10B  
SA

Figure 1. Blade Deicing System Location Diagram. (Sheet 10 of 10)

## SCHEMATICS

Figure 2. Blade Deicing System Schematic Diagram **EH-60A** . (Sheet 1 of 19)

SCHEMATICS - Continued

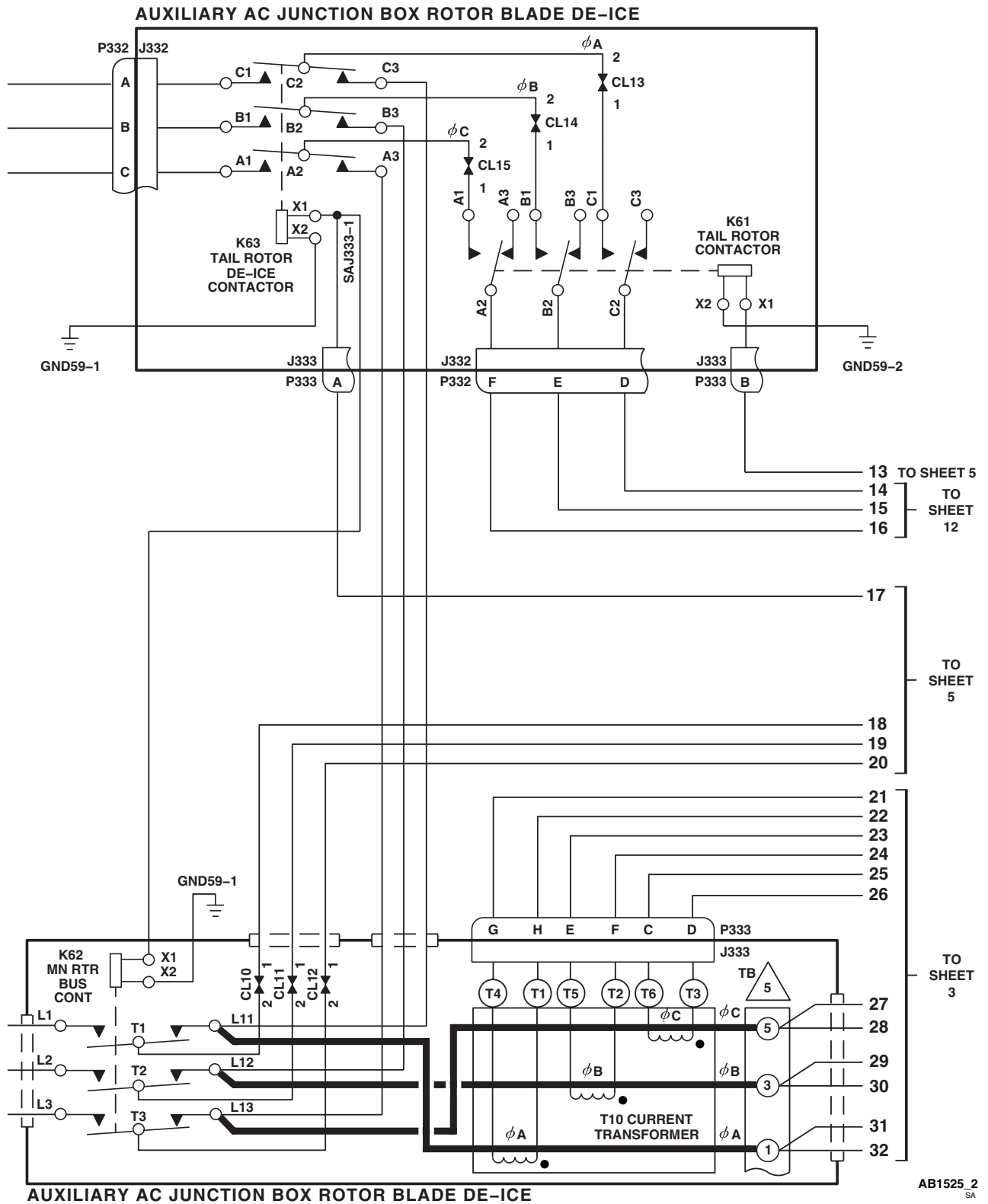


Figure 2. Blade Deicing System Schematic Diagram **EH-60A** . (Sheet 2 of 19)

SCHEMATICS - Continued

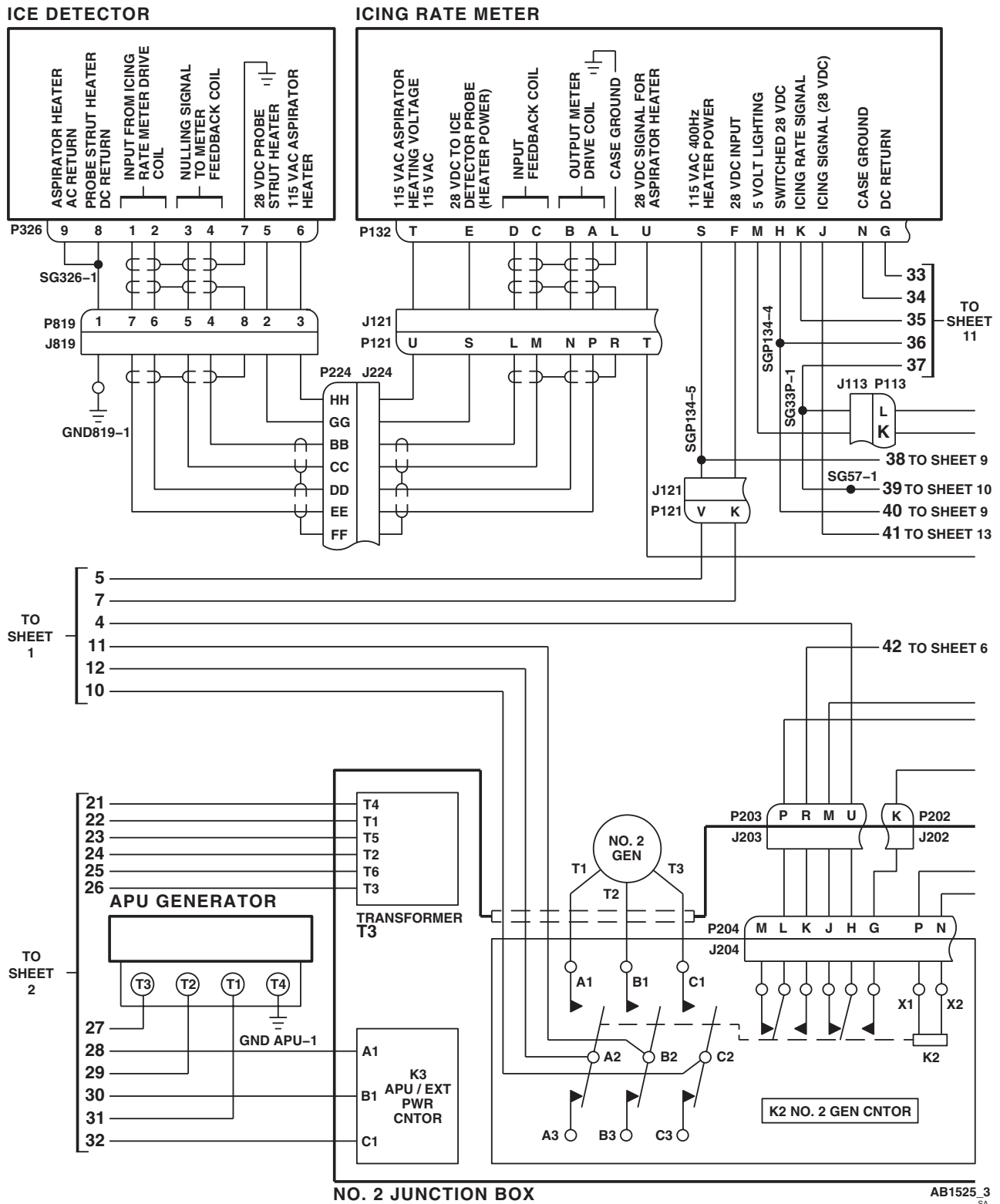
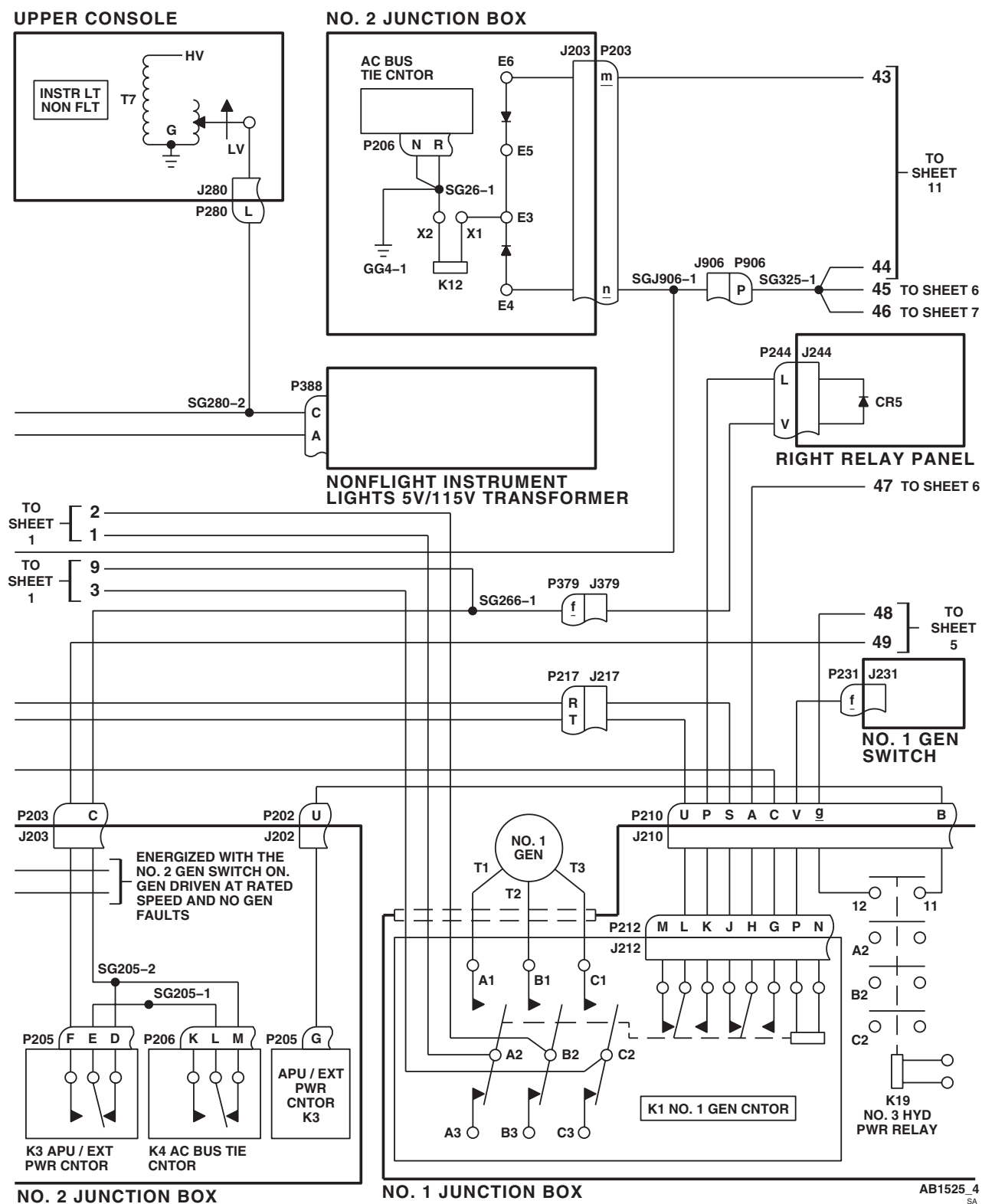
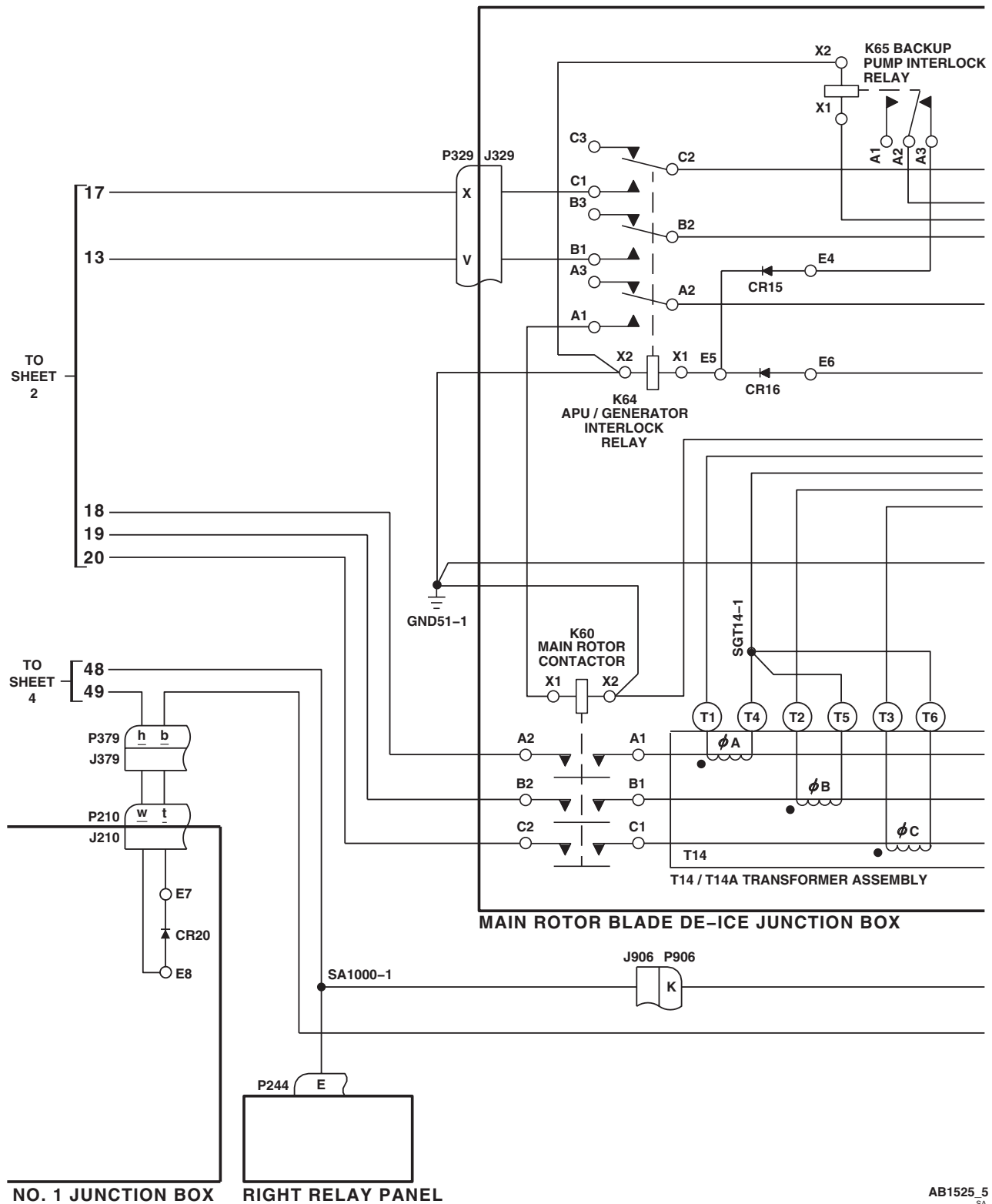


Figure 2. Blade Deicing System Schematic Diagram **EH-60A** . (Sheet 3 of 19)

## SCHEMATICS - Continued

Figure 2. Blade Deicing System Schematic Diagram **EH-60A** . (Sheet 4 of 19)

SCHEMATICS - Continued

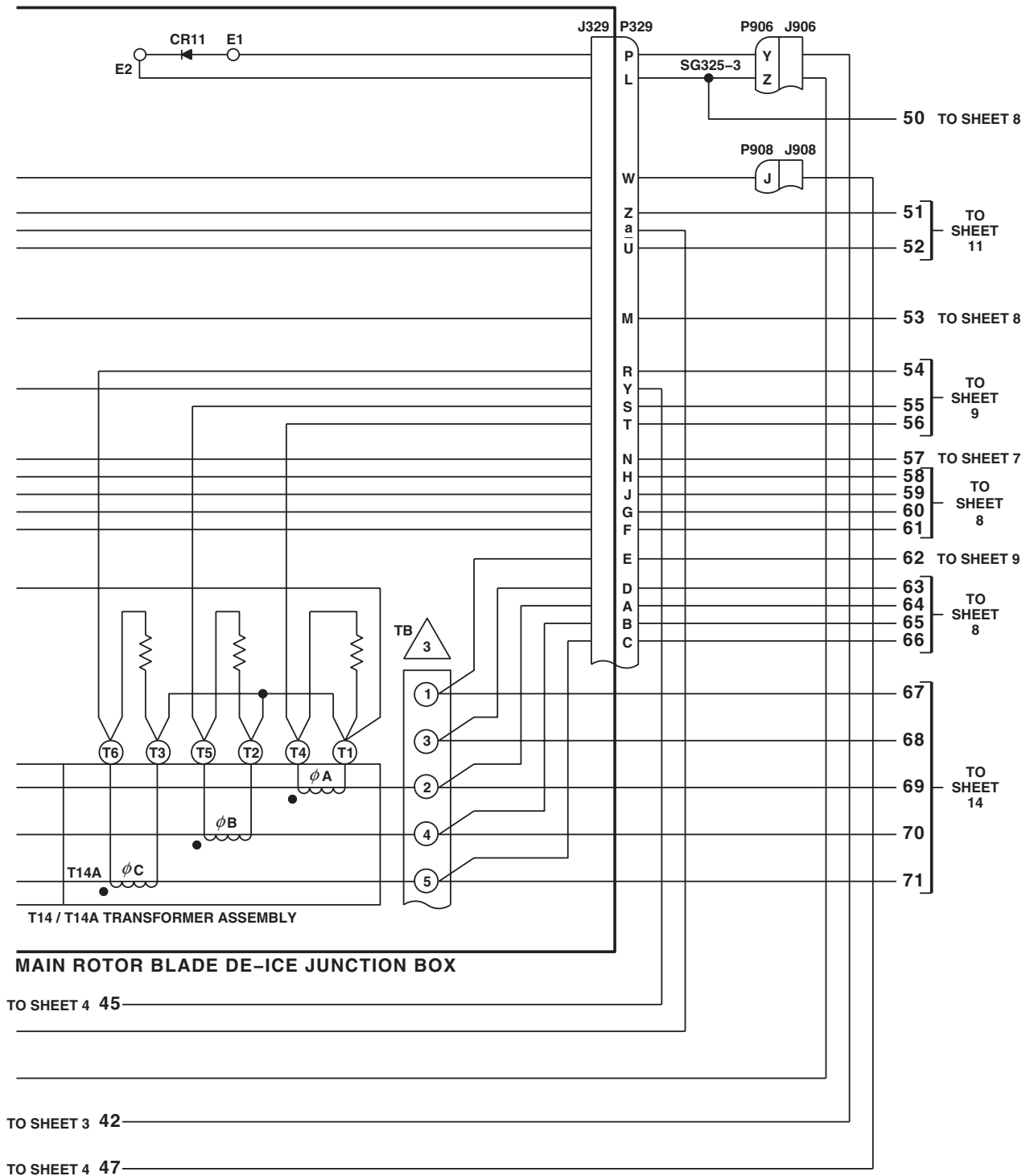


AB1525 5  
SA

Figure 2. Blade Deicing System Schematic Diagram **EH-60A** . (Sheet 5 of 19)



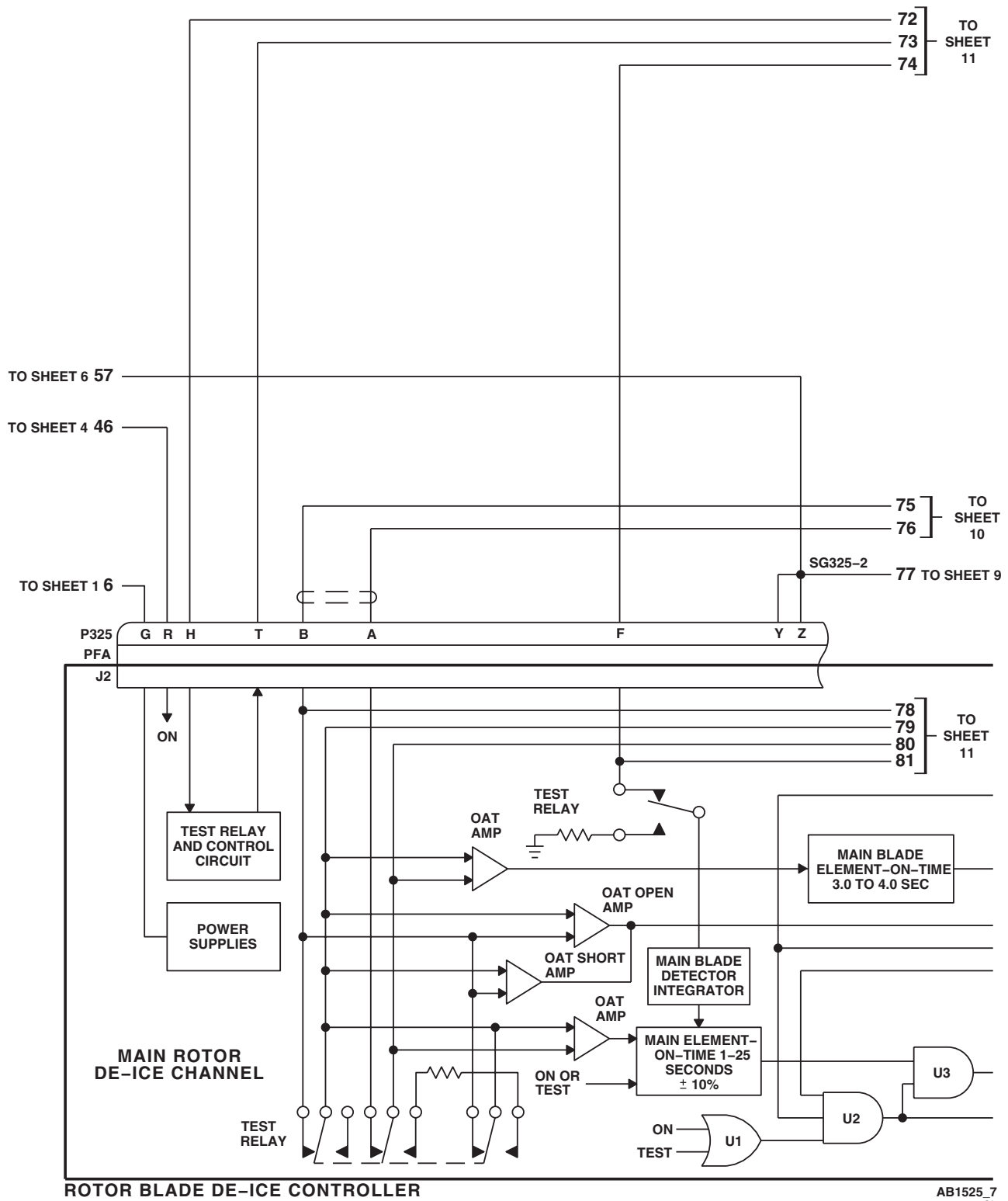
SCHEMATICS - Continued



AB1525\_6A  
SA

Figure 2. Blade Deicing System Schematic Diagram **EH-60A** . (Sheet 6 of 19)

### SCHEMATICS - Continued



SCHEMATICS - Continued

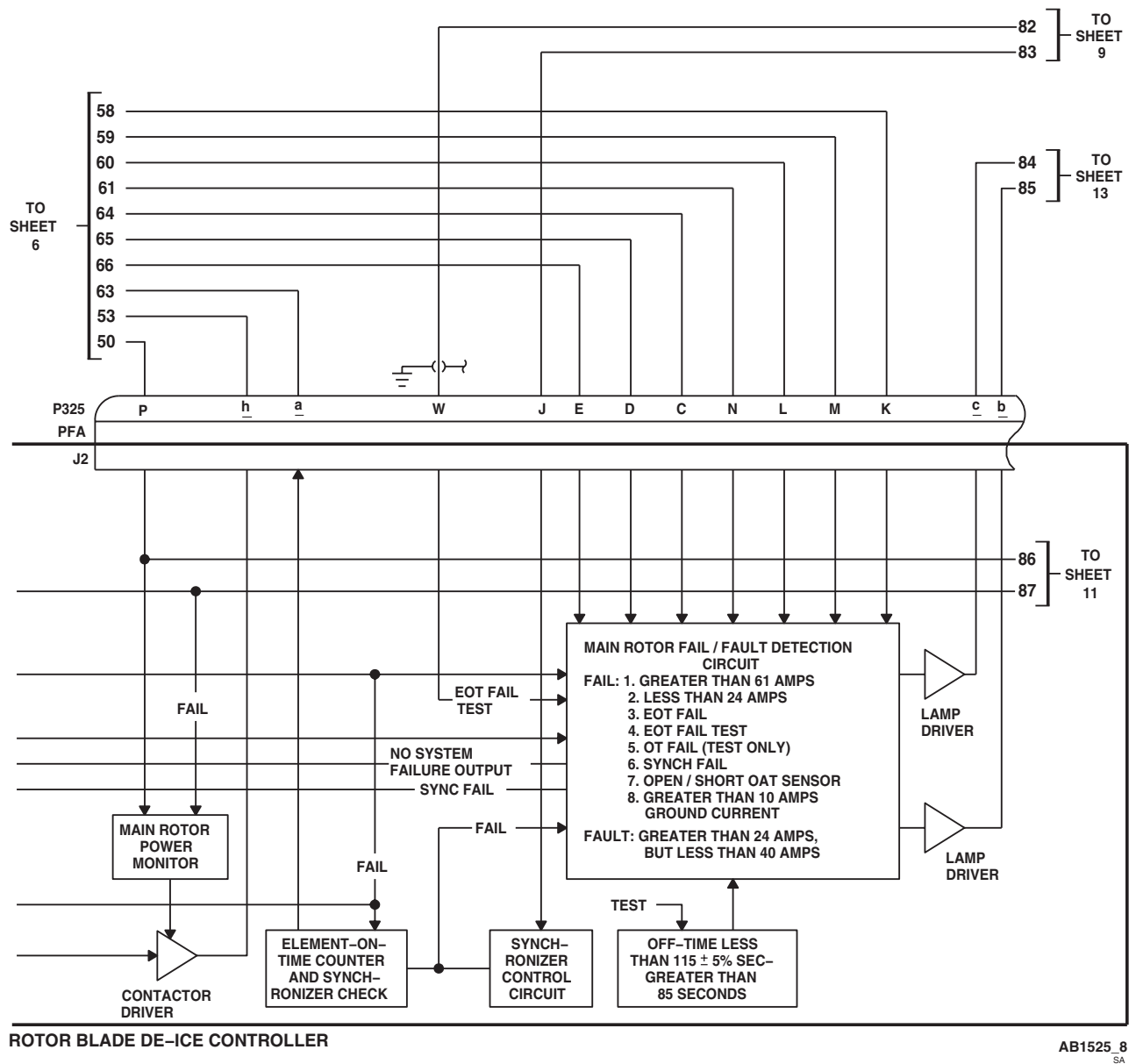


Figure 2. Blade Deicing System Schematic Diagram **EH-60A** . (Sheet 8 of 19)

SCHEMATICS - Continued

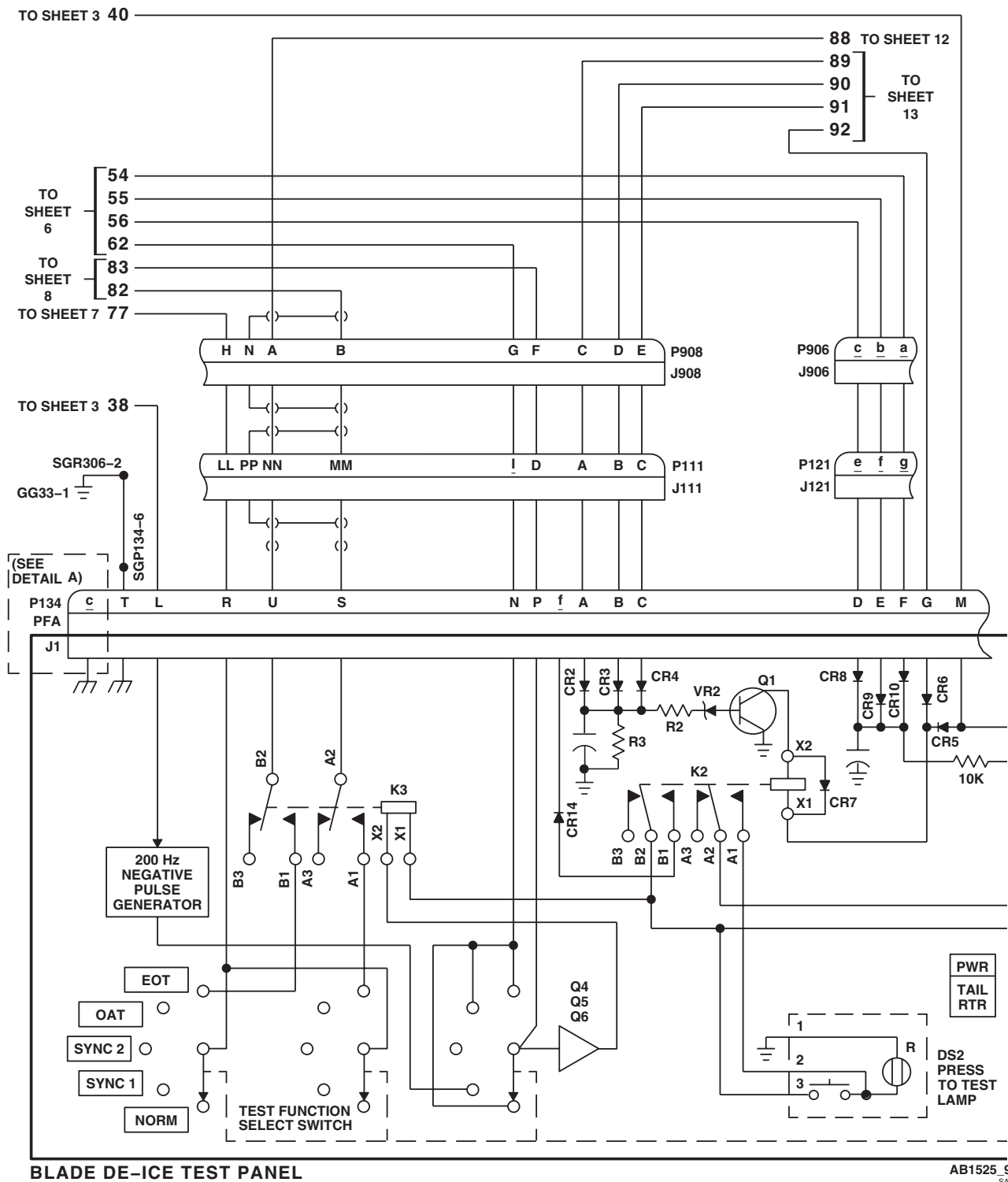


Figure 2. Blade Deicing System Schematic Diagram **EH-60A** . (Sheet 9 of 19)

SCHEMATICS - Continued

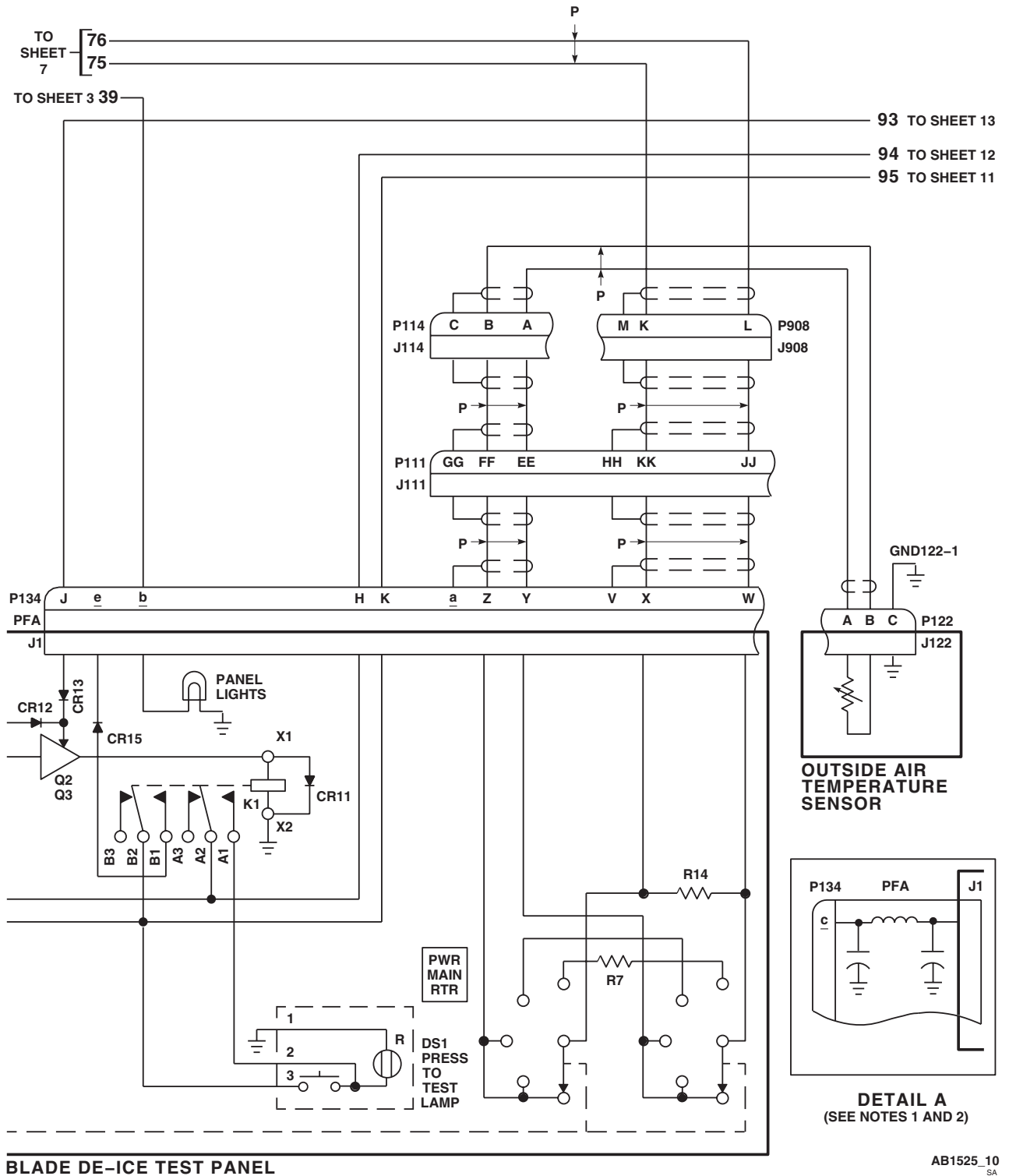
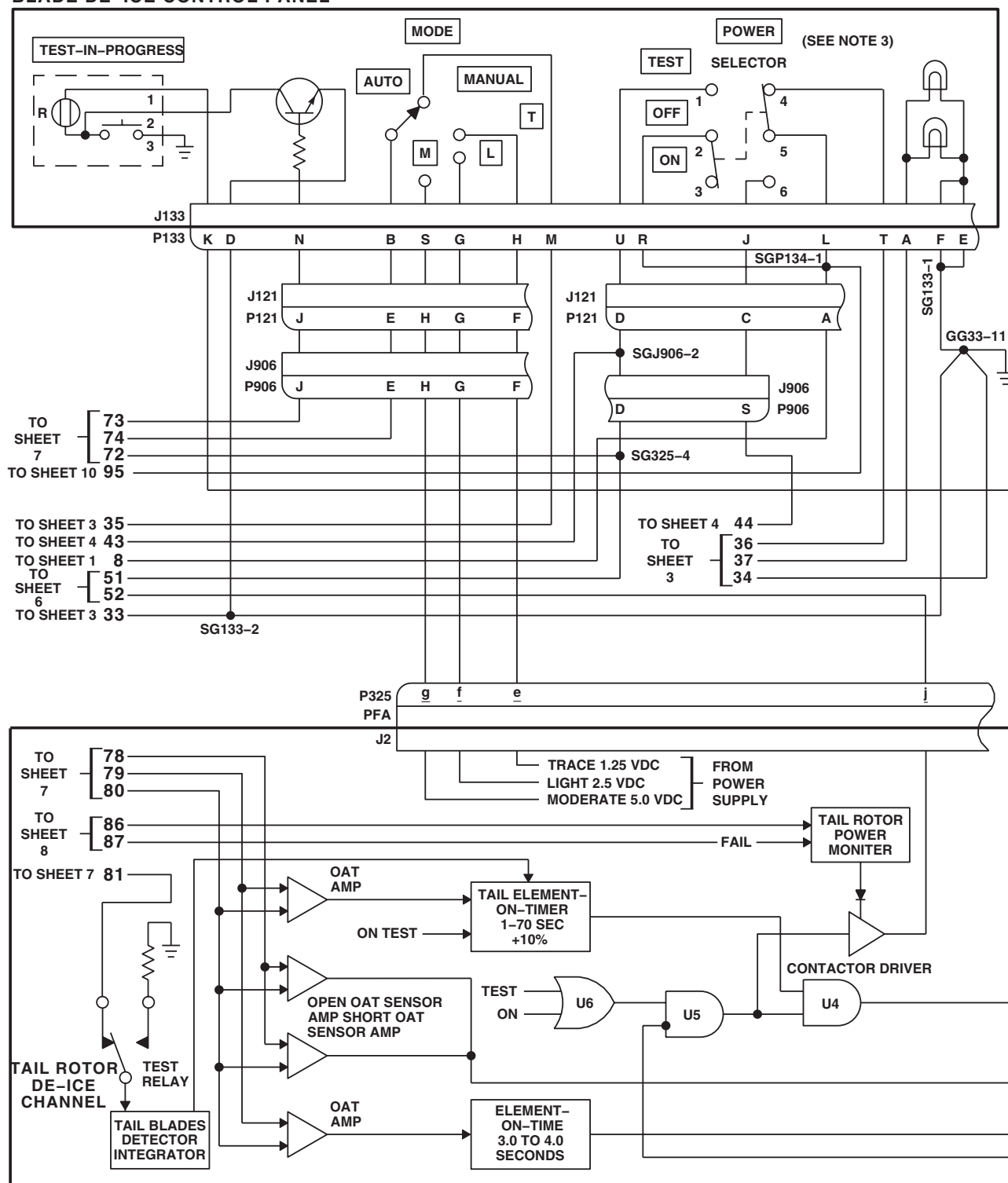


Figure 2. Blade Deicing System Schematic Diagram **EH-60A** . (Sheet 10 of 19)

## SCHEMATICS - Continued

## BLADE DE-ICE CONTROL PANEL

AB1525 11  
SAFigure 2. Blade Deicing System Schematic Diagram **EH-60A** . (Sheet 11 of 19)

SCHEMATICS - Continued

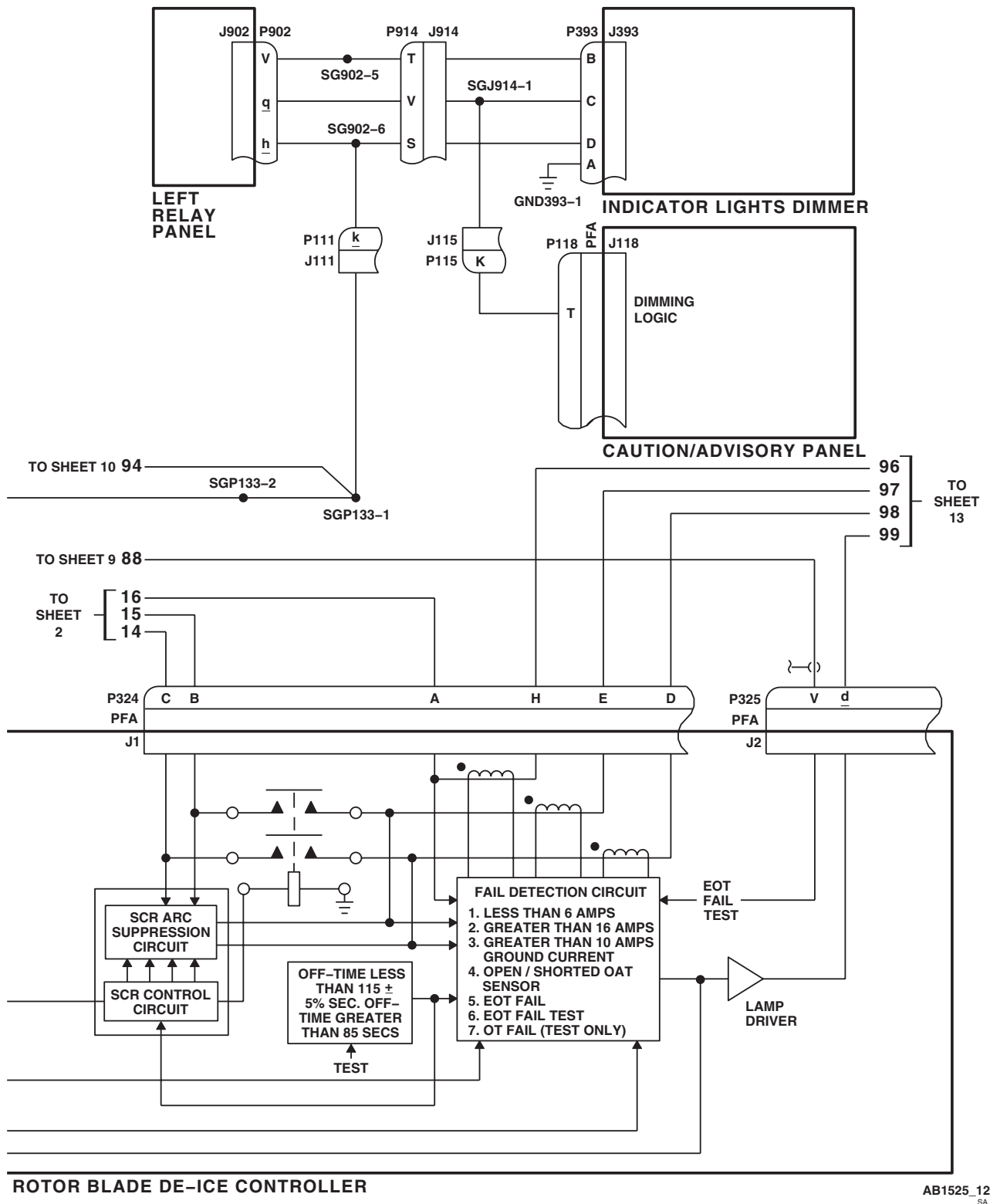
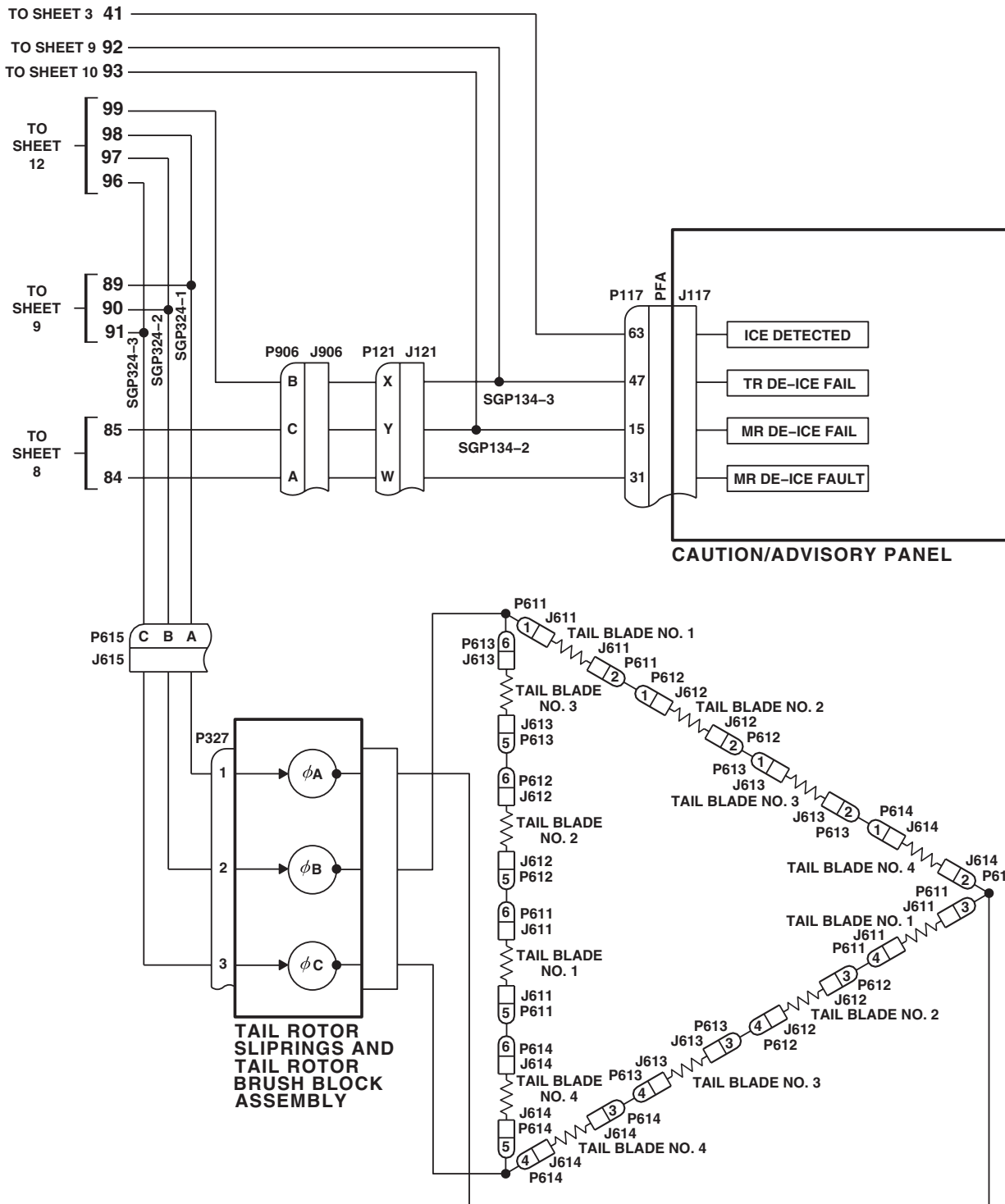


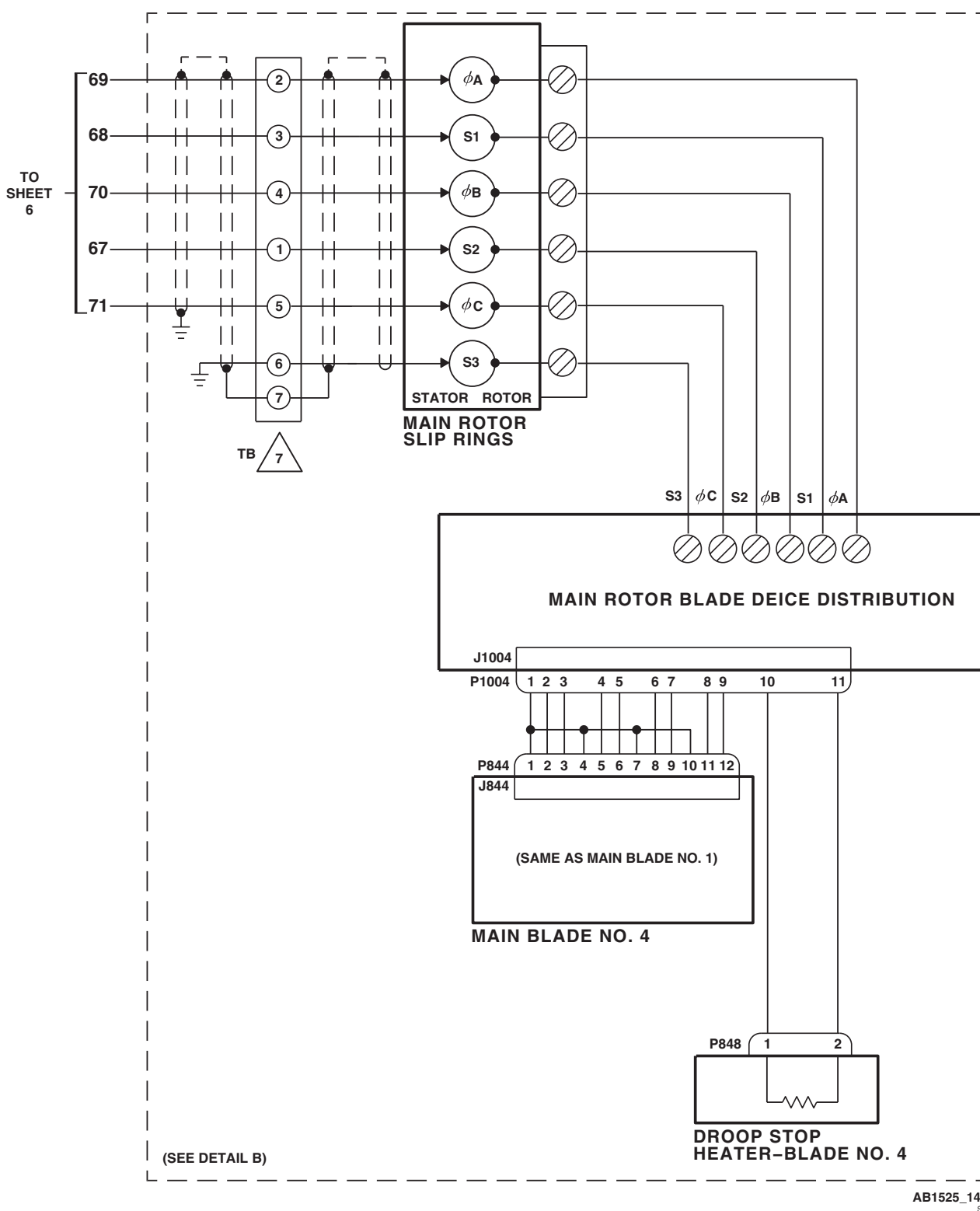
Figure 2. Blade Deicing System Schematic Diagram **EH-60A** . (Sheet 12 of 19)

## SCHEMATICS - Continued

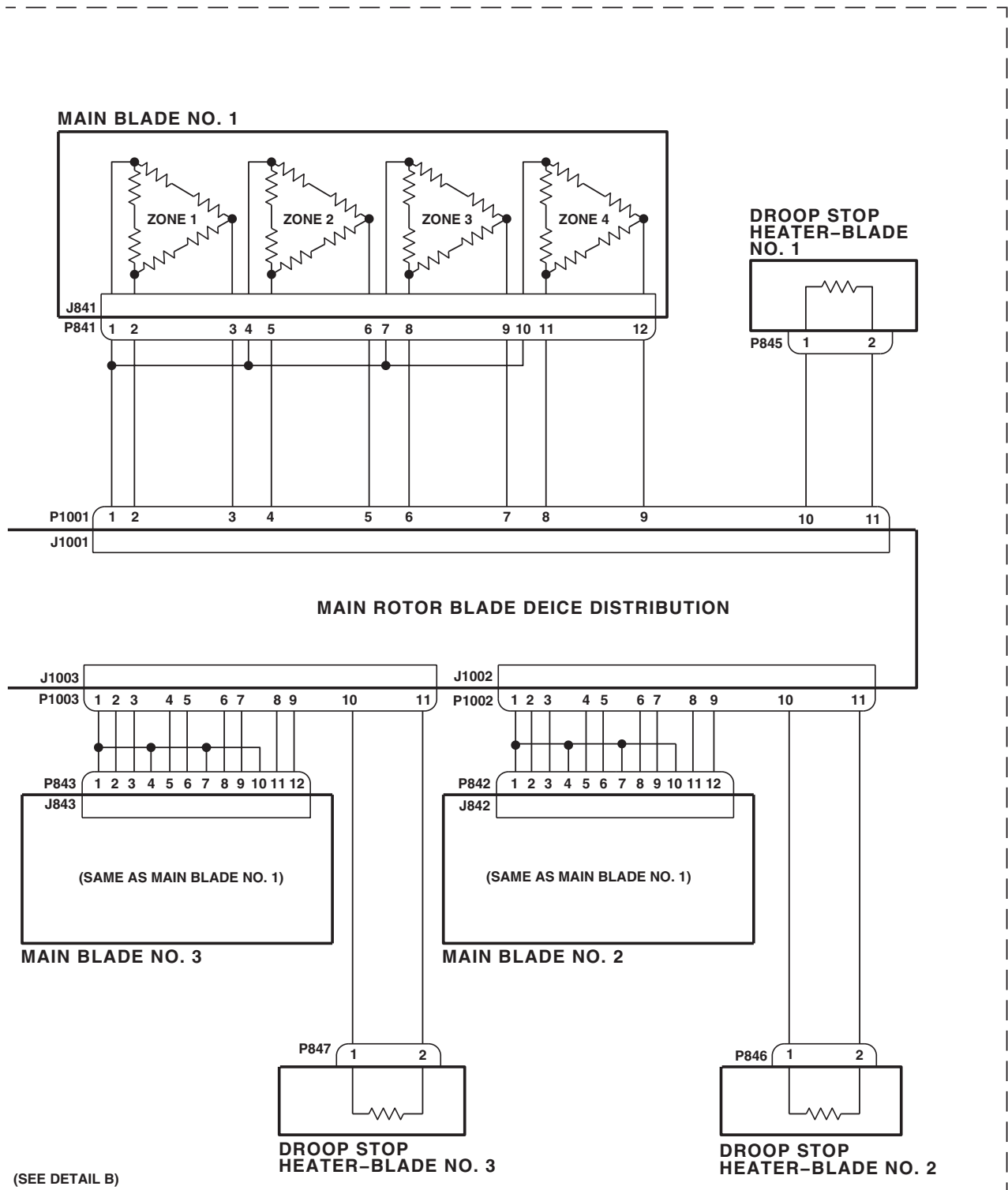
AB1525\_13  
SAFigure 2. Blade Deicing System Schematic Diagram **EH-60A** . (Sheet 13 of 19)



## SCHEMATICS - Continued

Figure 2. Blade Deicing System Schematic Diagram **EH-60A** . (Sheet 14 of 19)

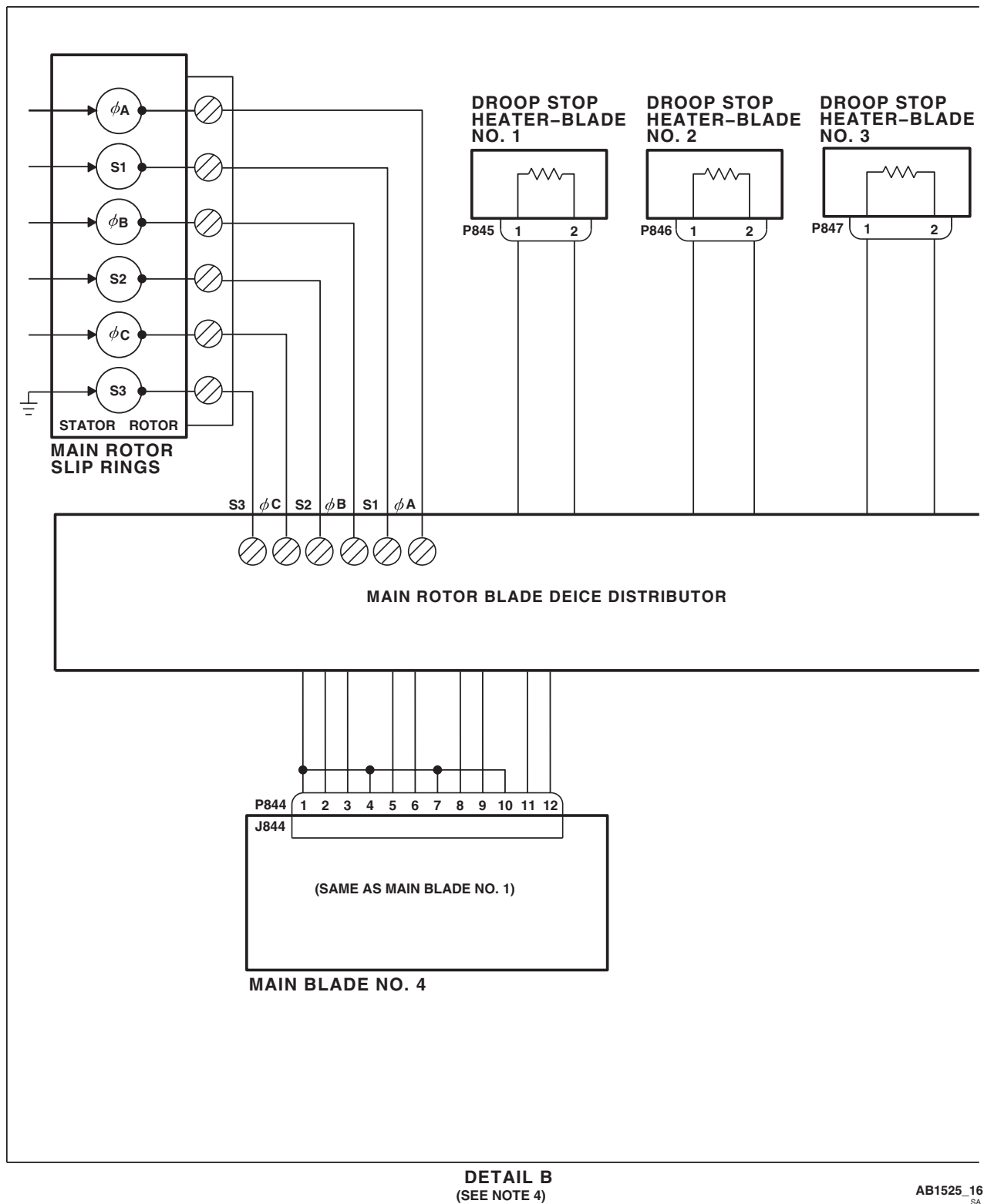
SCHEMATICS - Continued



AB1525\_15A  
SA

Figure 2. Blade Deicing System Schematic Diagram **EH-60A** . (Sheet 15 of 19)

## SCHEMATICS - Continued

Figure 2. Blade Deicing System Schematic Diagram **EH-60A** . (Sheet 16 of 19)

SCHEMATICS - Continued

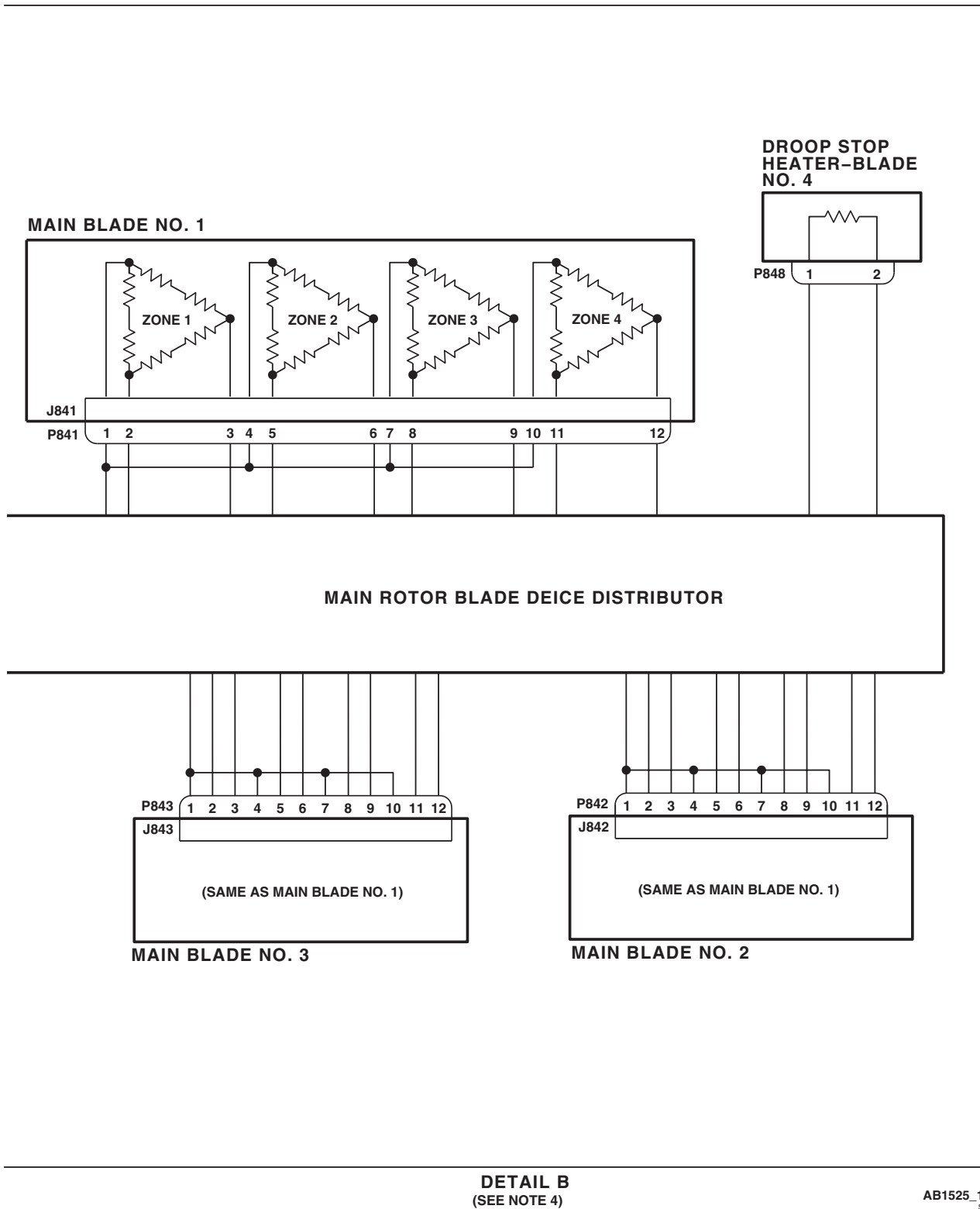
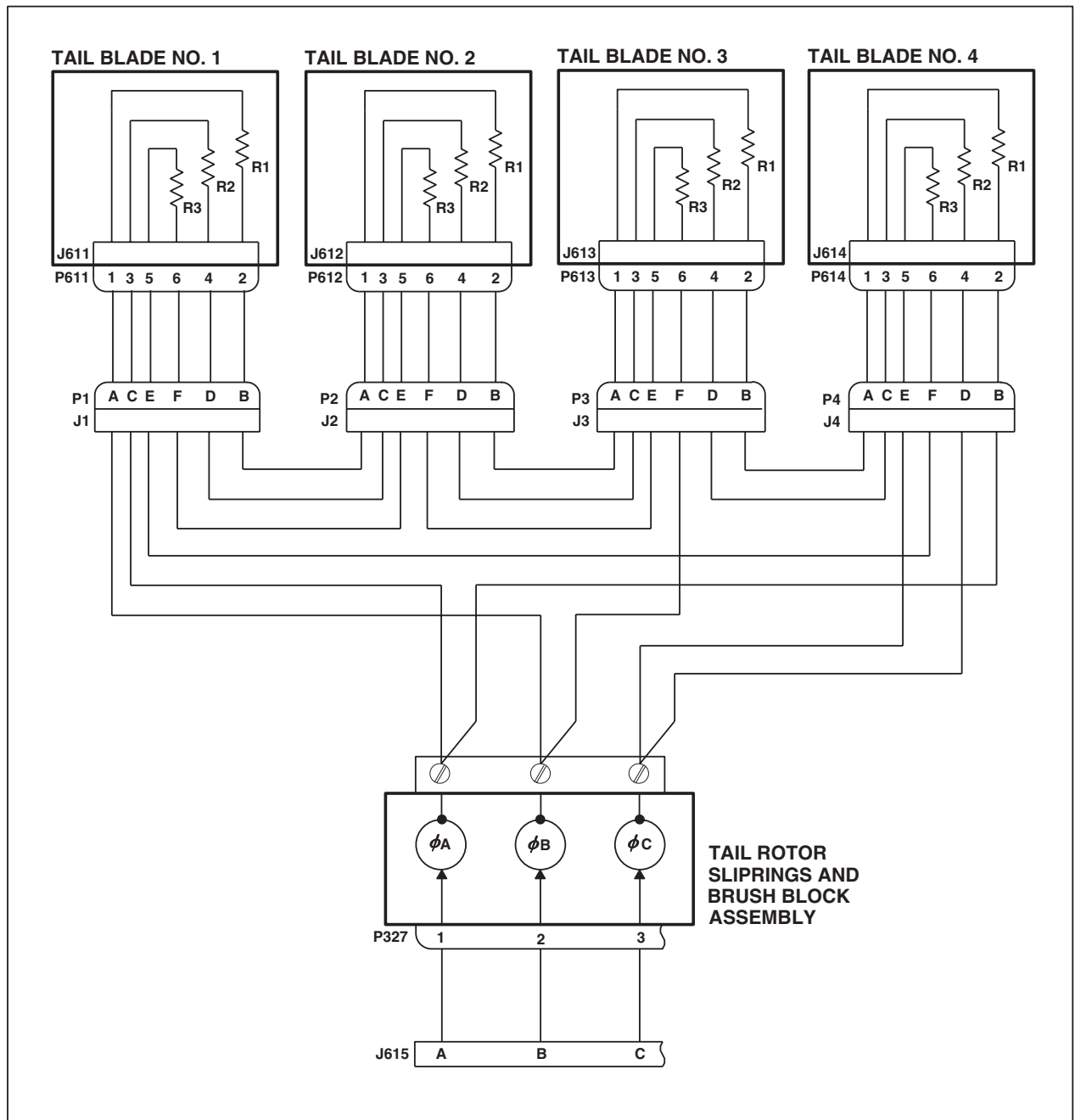


Figure 2. Blade Deicing System Schematic Diagram **EH-60A** . (Sheet 17 of 19)

## SCHEMATICS - Continued

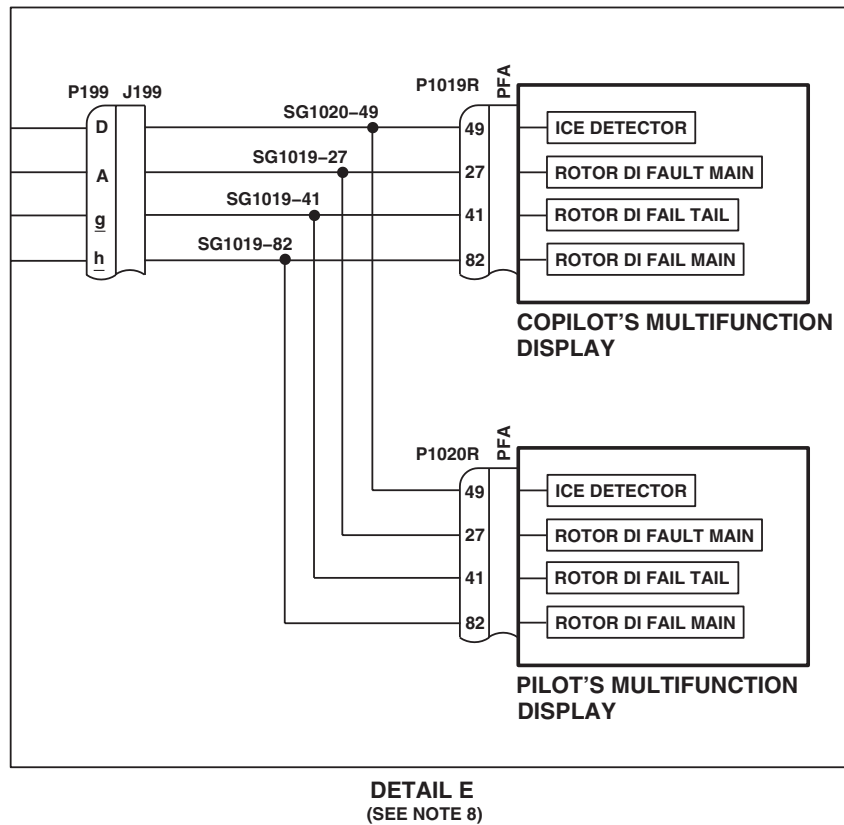


**DETAIL D**  
(SEE NOTE 7)

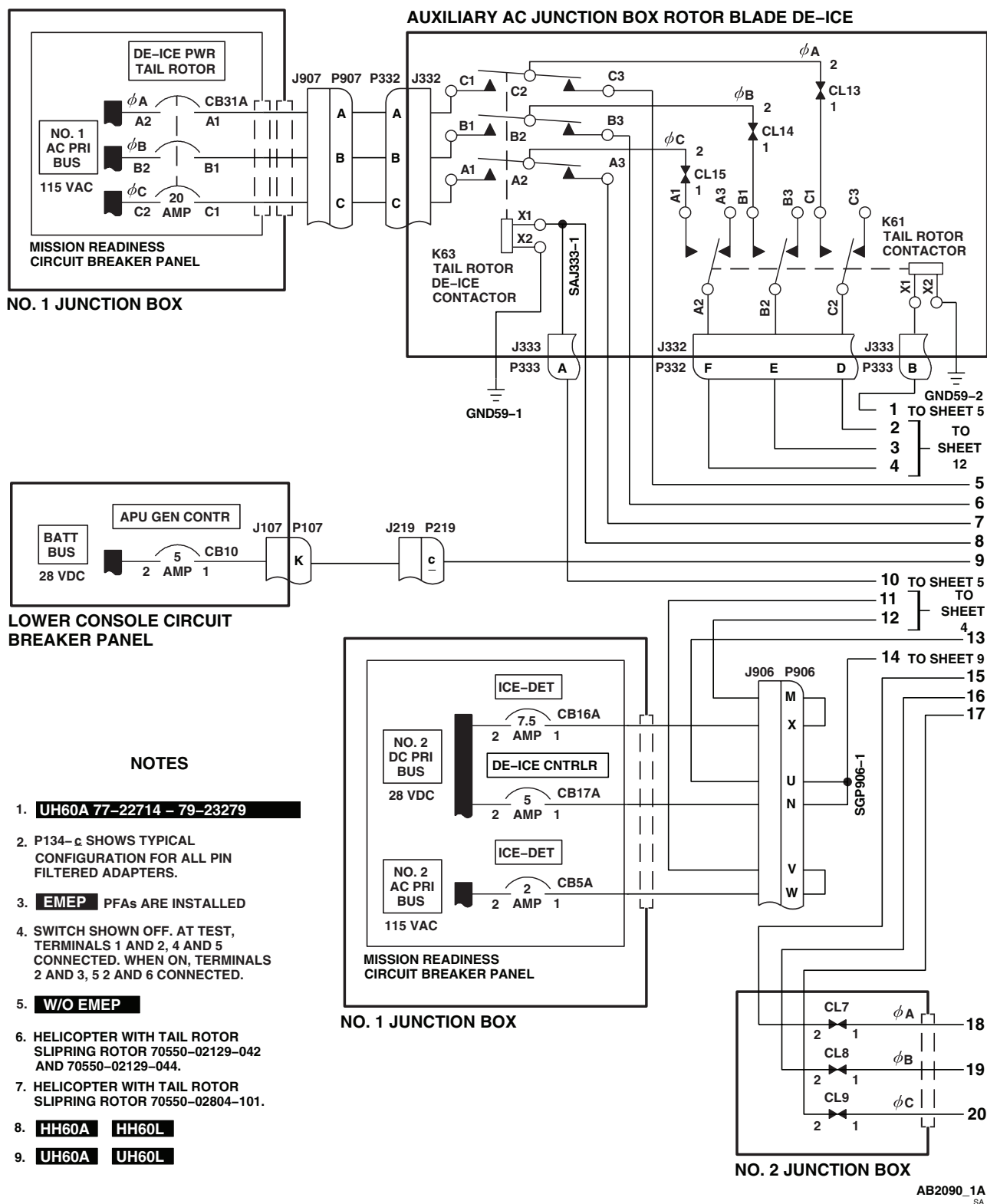
AB2090\_18  
SA

Figure 2. Blade Deicing System Schematic Diagram **EH-60A** . (Sheet 18 of 19)

## SCHEMATICS - Continued

AB2090\_19  
SAFigure 2. Blade Deicing System Schematic Diagram **EH-60A** . (Sheet 19 of 19)

## SCHEMATICS - Continued



SCHEMATICS - Continued

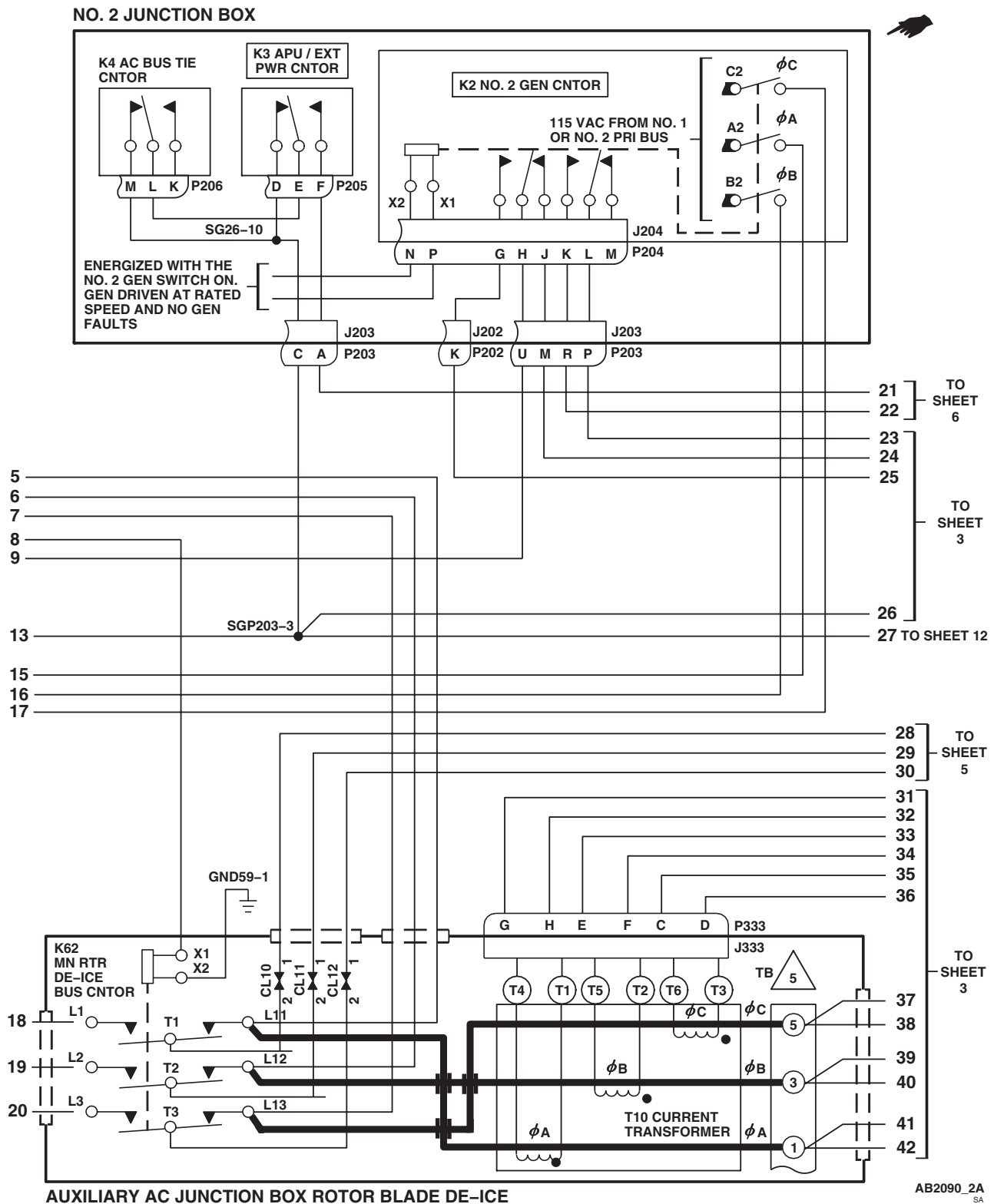
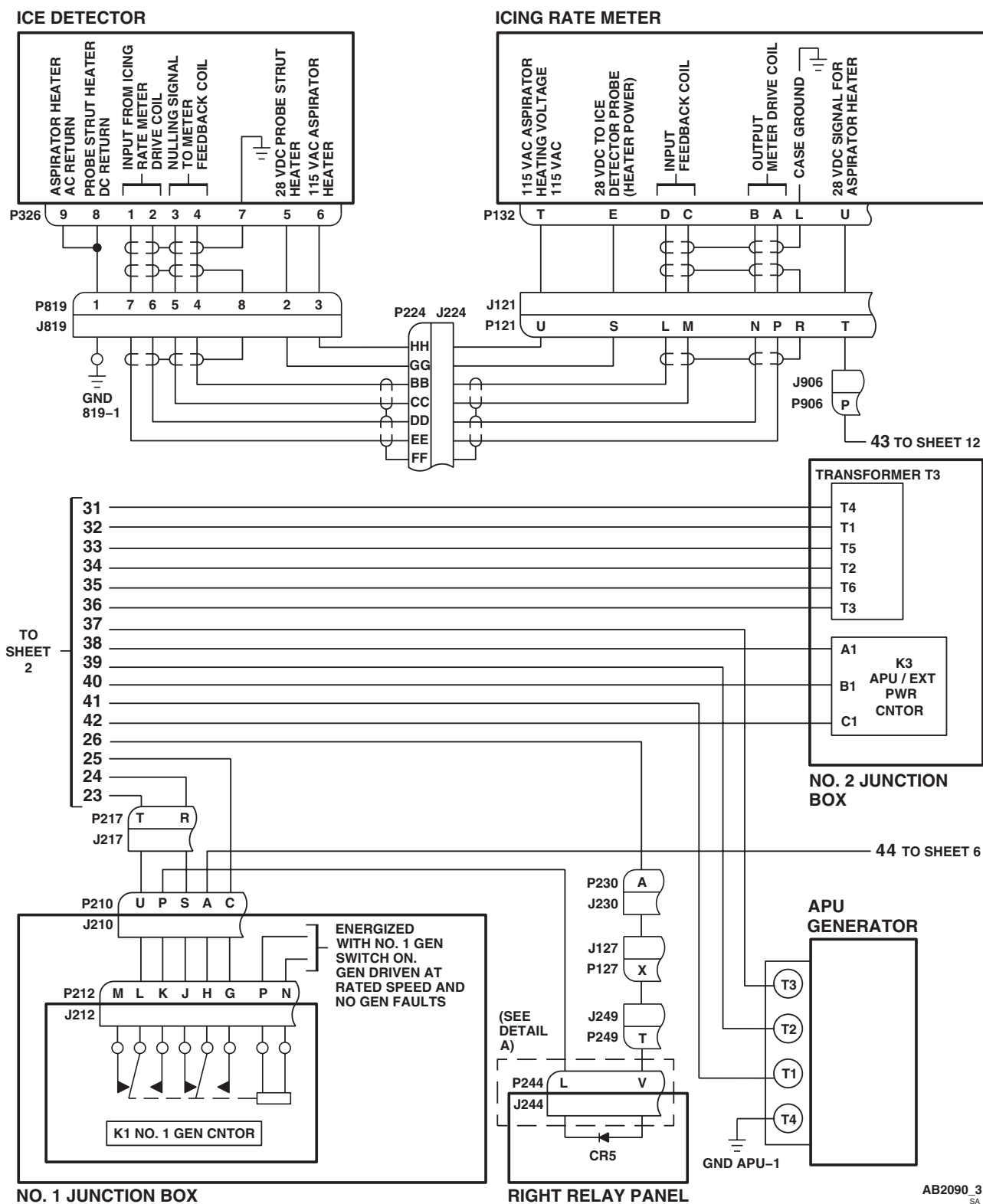


Figure 3. Blade Deicing System Schematic Diagram **HH-60A HH-60L UH-60A UH-60L** . (Sheet 2 of 17)



## SCHEMATICS - Continued

Figure 3. Blade Deicing System Schematic Diagram **HH-60A HH-60L UH-60A UH-60L** . (Sheet 3 of 17)

The diagram illustrates the electrical wiring connecting three main components:

- ICING RATE METER**: Features terminals P132 (S, F, M), H, K, J, N, G, and SG33P-1.
- UPPER CONSOLE**: Includes a transformer T7 with HV and LV windings, a ground connection G, and control switches BRT and OFF. It also has terminals J280 and P280.
- NONFLIGHT INSTRUMENT LIGHTS 5V/115V TRANSFORMER**: Shown as a box at the bottom with terminals A and C, connected to P388.

**Key Connections:**

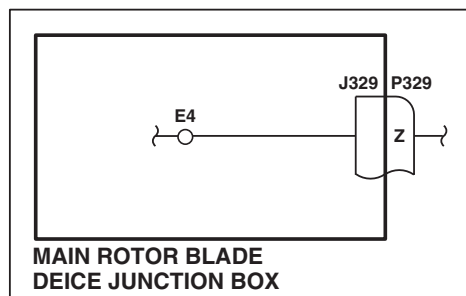
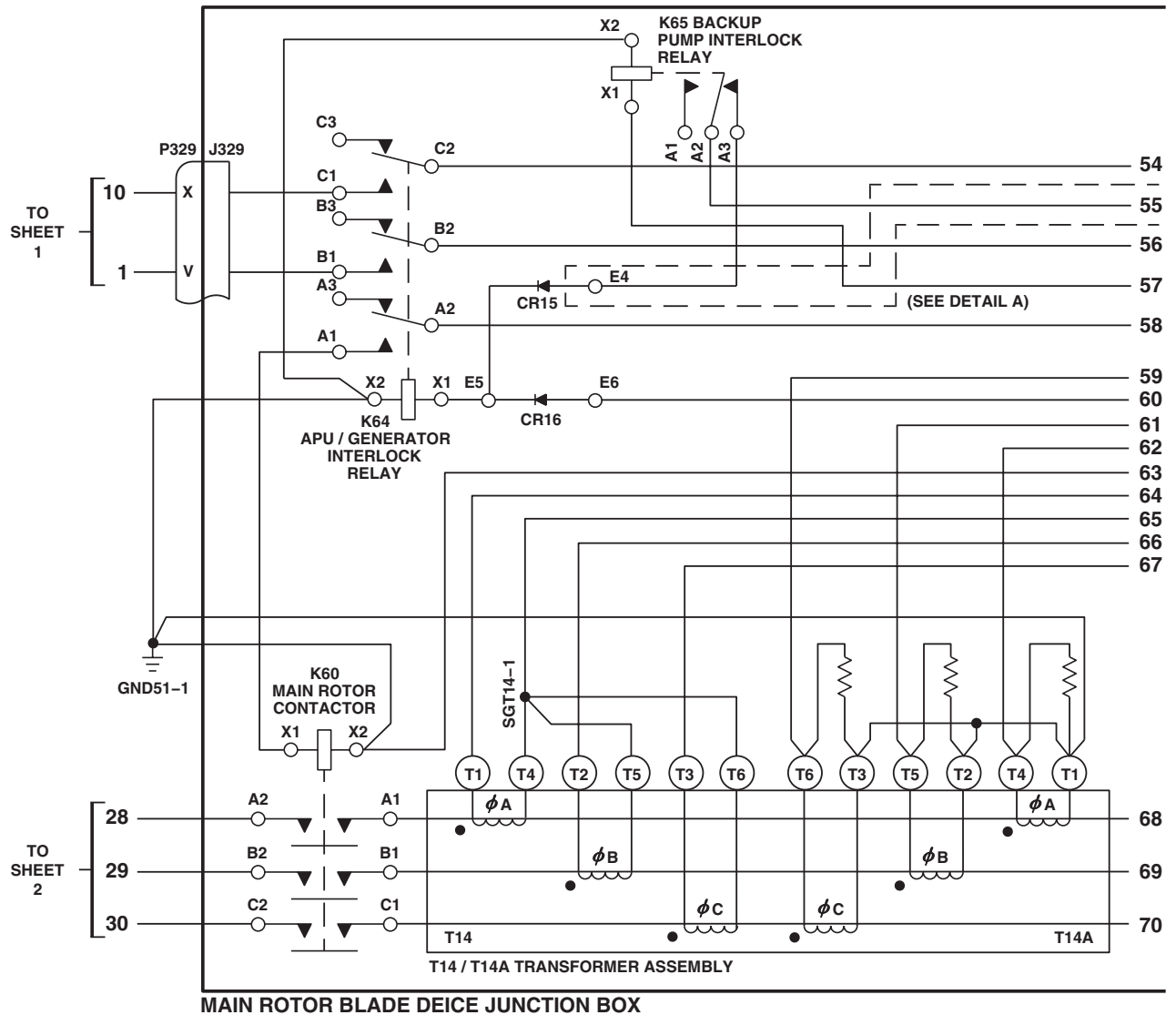
- The **ICING RATE METER**'s S terminal connects to the **UPPER CONSOLE**'s HV terminal via wire 45 (to sheet 11).
- The **ICING RATE METER**'s F terminal connects to the **UPPER CONSOLE**'s LV terminal via wire 46 (to sheet 12).
- The **ICING RATE METER**'s M terminal connects to the **UPPER CONSOLE**'s G terminal via wire 47 (to sheet 13).
- The **ICING RATE METER**'s H terminal connects to the **UPPER CONSOLE**'s BRT switch via wire 48 (to sheet 7).
- The **ICING RATE METER**'s K terminal connects to the **UPPER CONSOLE**'s OFF switch via wire 49 (to sheet 11).
- The **ICING RATE METER**'s J terminal connects to the **UPPER CONSOLE**'s J280 terminal via wire 50 (to sheet 12).
- The **ICING RATE METER**'s N terminal connects to the **UPPER CONSOLE**'s P280 terminal via wire 51 (to sheet 12).
- The **ICING RATE METER**'s G terminal connects to the **UPPER CONSOLE**'s G terminal via wire 52 (to sheet 8).
- The **ICING RATE METER**'s SG33P-1 terminal connects to the **UPPER CONSOLE**'s SG280-2 terminal via wire 53 (to sheet 7).

**Internal Connections:**

- The **UPPER CONSOLE** transformer T7 is connected to the **NONFLIGHT INSTRUMENT LIGHTS** transformer via wires 50 and 51.
- The **UPPER CONSOLE** transformer T7 is also connected to the **NONFLIGHT INSTRUMENT LIGHTS** transformer via wires 52 and 53.

Figure 3. Blade Deicing System Schematic Diagram **HH-60A HH-60L UH-60A UH-60L** . (Sheet 4 of 17)

SCHEMATICS - Continued



DETAIL A  
(SEE NOTE 1)

AB2090\_5  
SA

Figure 3. Blade Deicing System Schematic Diagram **HH-60A HH-60L UH-60A UH-60L** . (Sheet 5 of 17)

SCHEMATICS - Continued

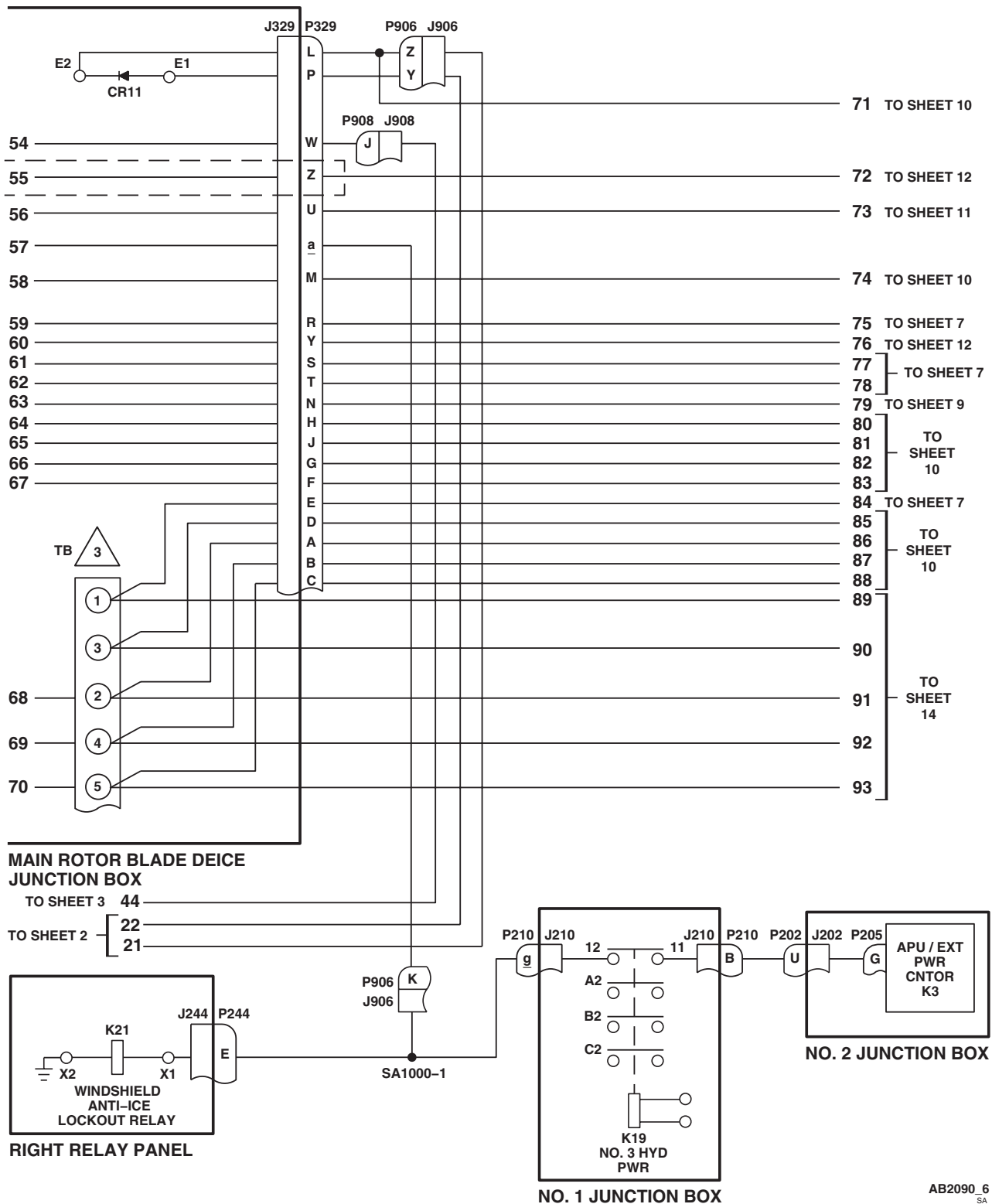
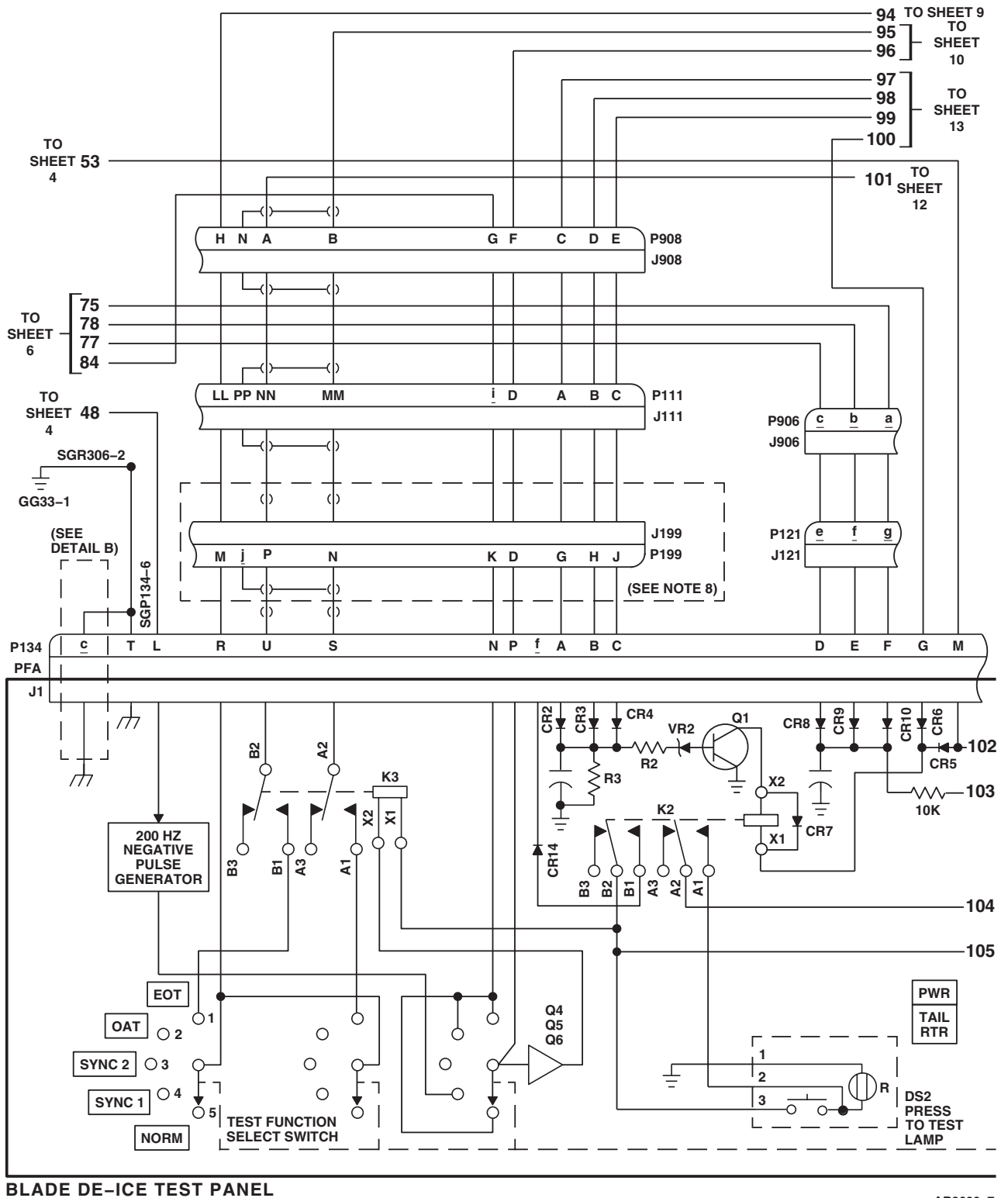


Figure 3. Blade Deicing System Schematic Diagram **HH-60A HH-60L UH-60A UH-60L** . (Sheet 6 of 17)

SCHEMATICS - Continued

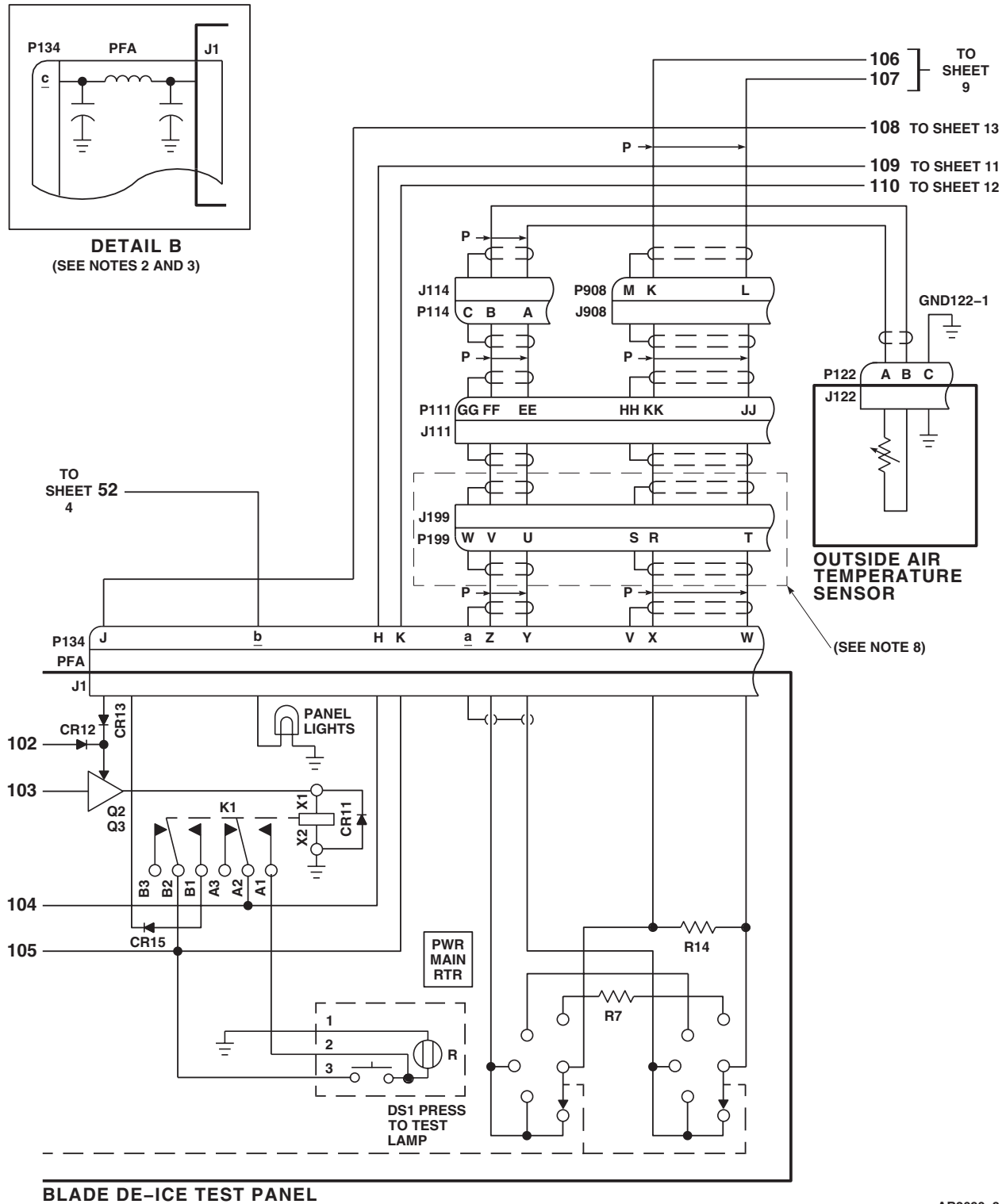


BLADE DE-ICE TEST PANEL

AB2090\_7  
SA

Figure 3. Blade Deicing System Schematic Diagram **HH-60A HH-60L UH-60A UH-60L** . (Sheet 7 of 17)

SCHEMATICS - Continued



AB2090 8  
SA

Figure 3. Blade Deicing System Schematic Diagram **HH-60A HH-60L UH-60A UH-60L** . (Sheet 8 of 17)

SCHEMATICS - Continued

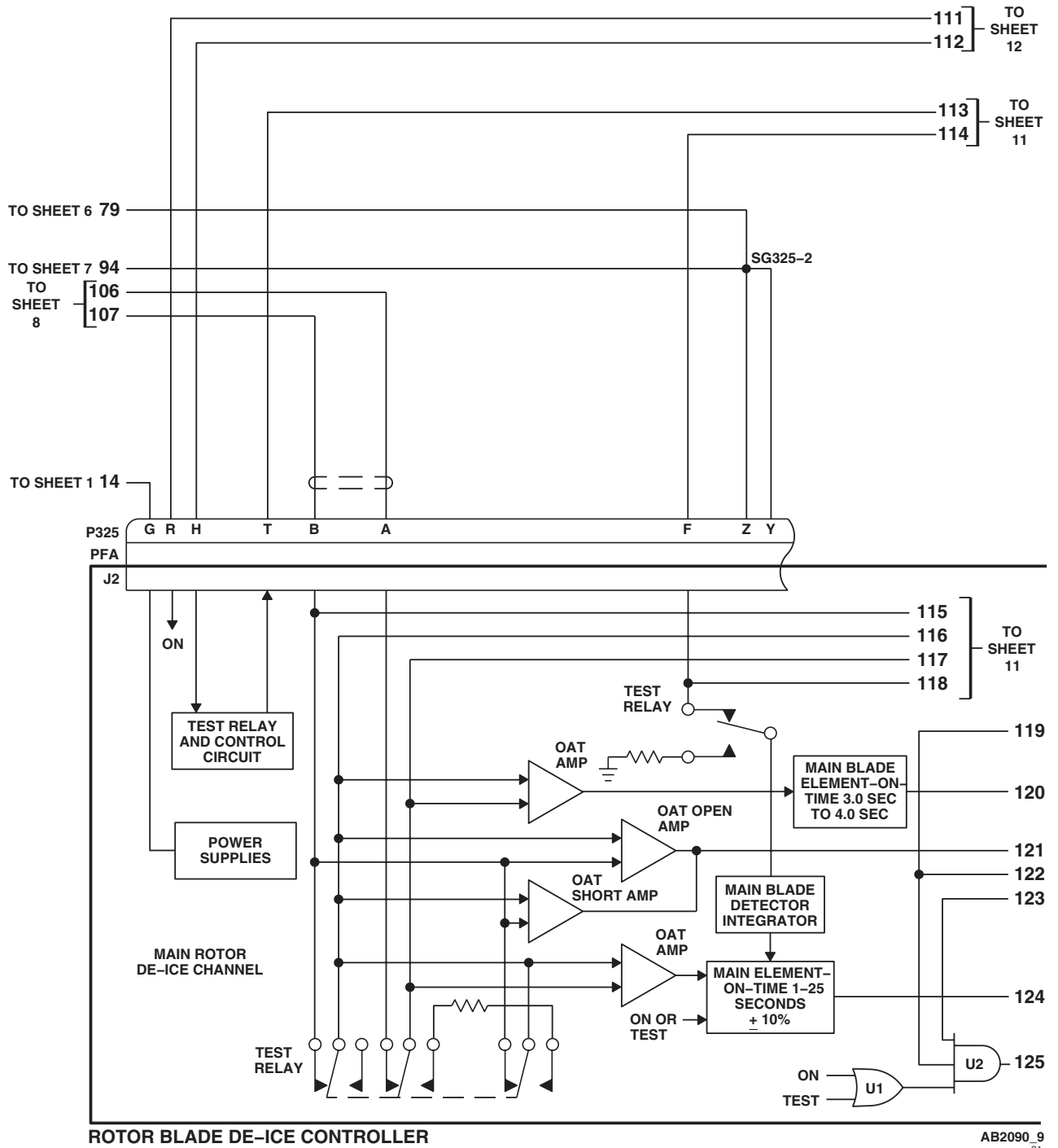
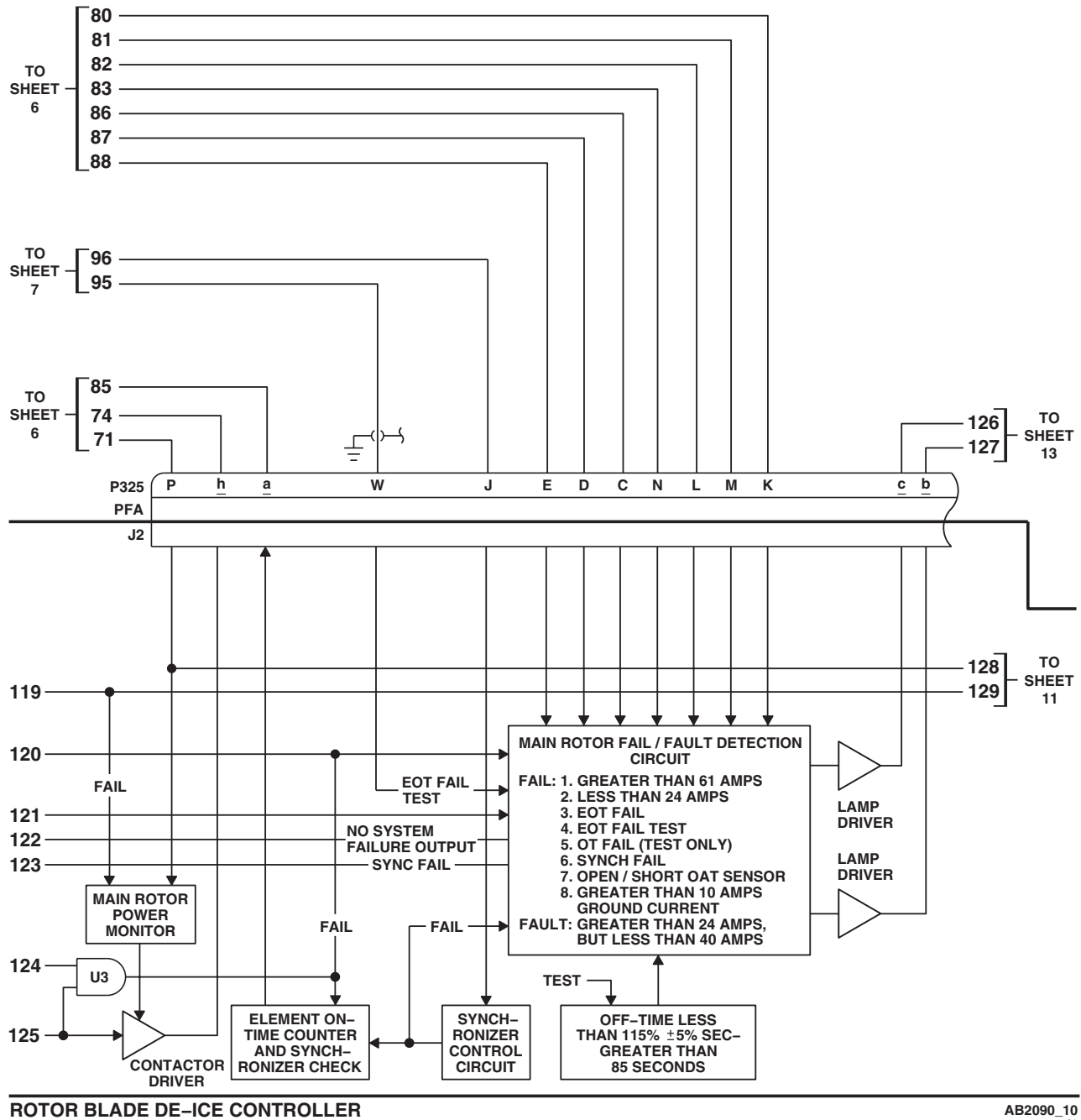


Figure 3. Blade Deicing System Schematic Diagram **HH-60A HH-60L UH-60A UH-60L** . (Sheet 9 of 17)

## SCHEMATICS - Continued

Figure 3. Blade Deicing System Schematic Diagram **HH-60A HH-60L UH-60A UH-60L** . (Sheet 10 of 17)



SCHEMATICS - Continued

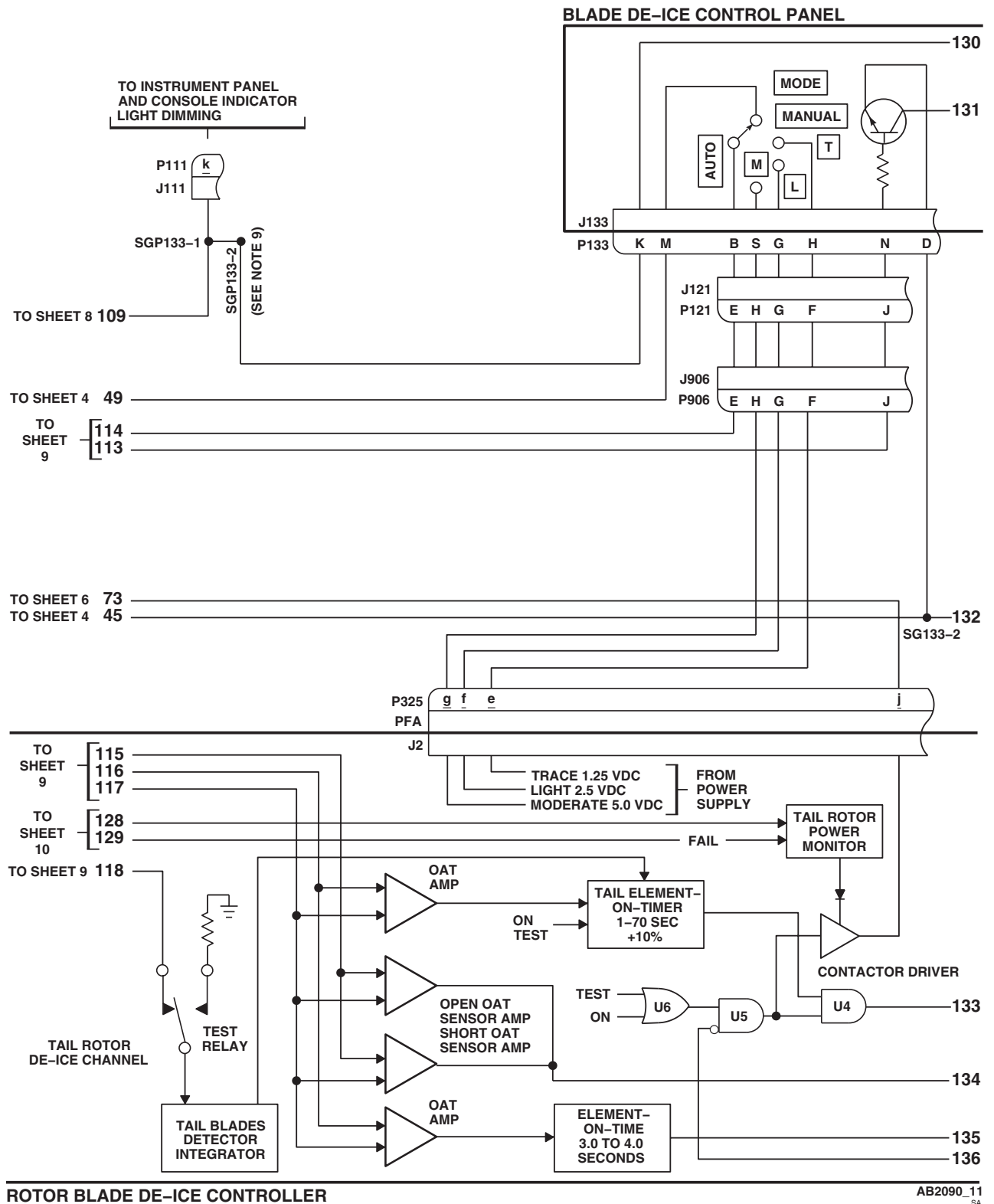


Figure 3. Blade Deicing System Schematic Diagram **HH-60A HH-60L UH-60A UH-60L** . (Sheet 11 of 17)

SCHEMATICS - Continued

BLADE DE-ICE CONTROL PANEL

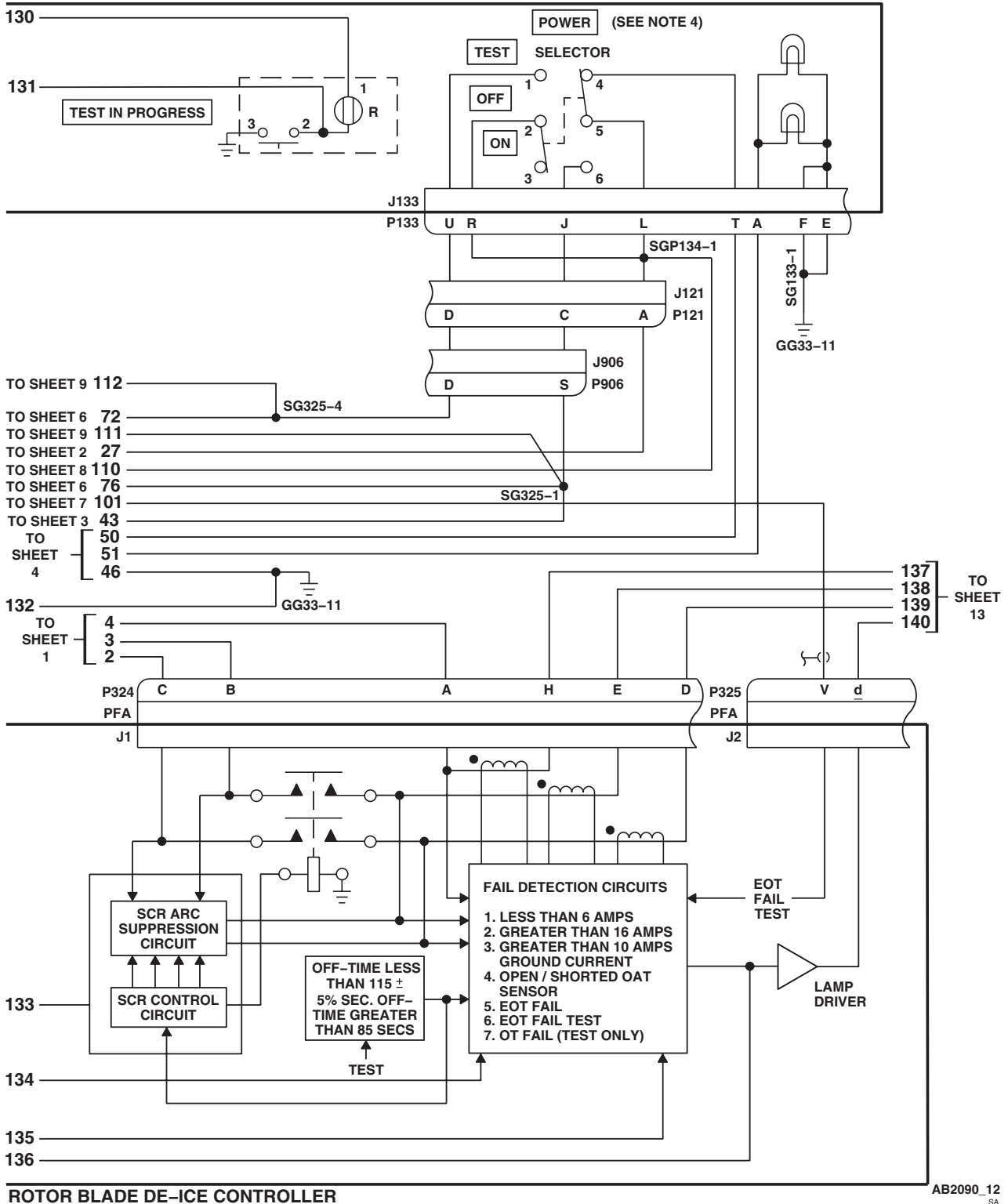
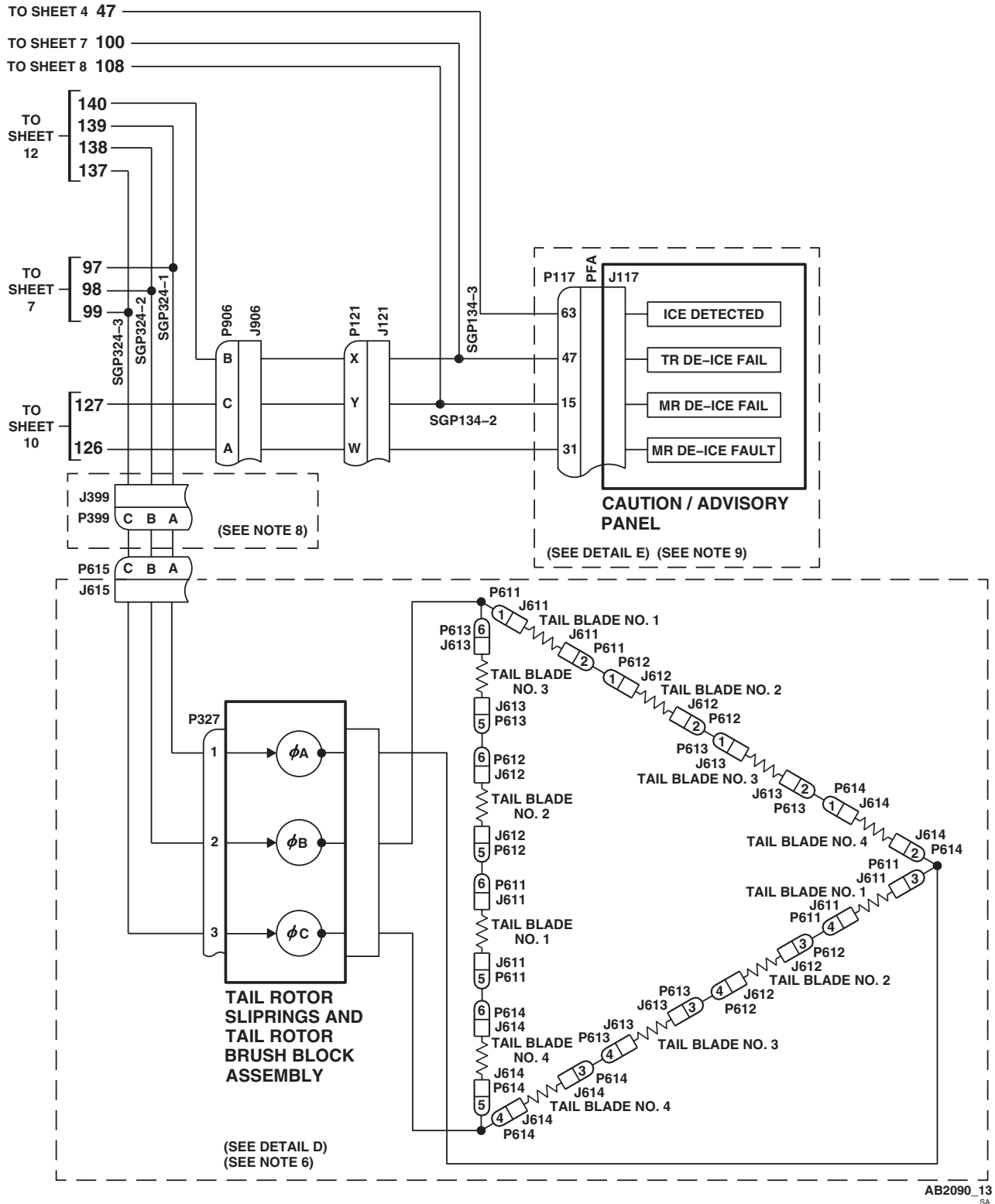
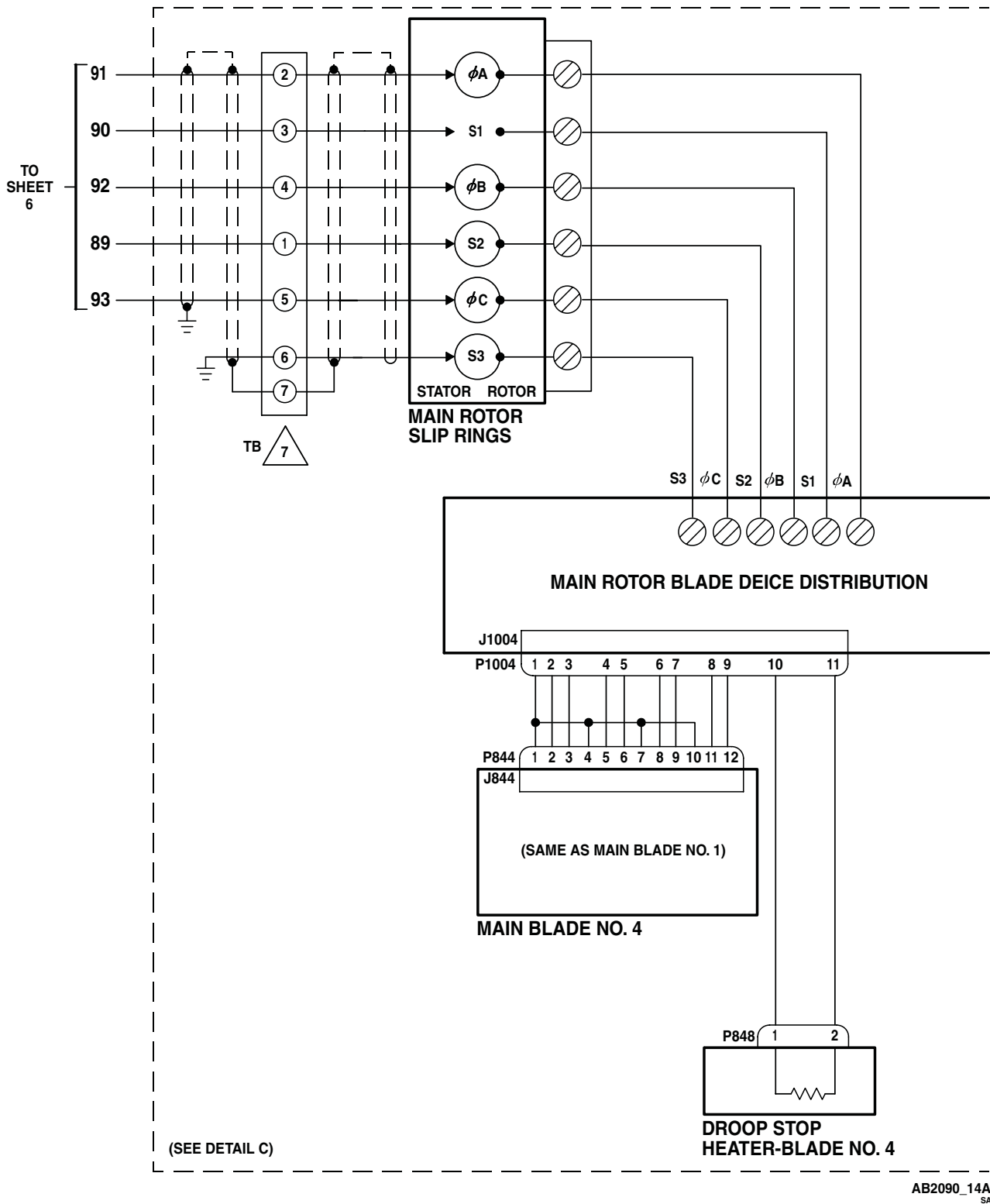


Figure 3. Blade Deicing System Schematic Diagram **HH-60A HH-60L UH-60A UH-60L** . (Sheet 12 of 17)

## SCHEMATICS - Continued

Figure 3. Blade Deicing System Schematic Diagram **HH-60A HH-60L UH-60A UH-60L** . (Sheet 13 of 17)

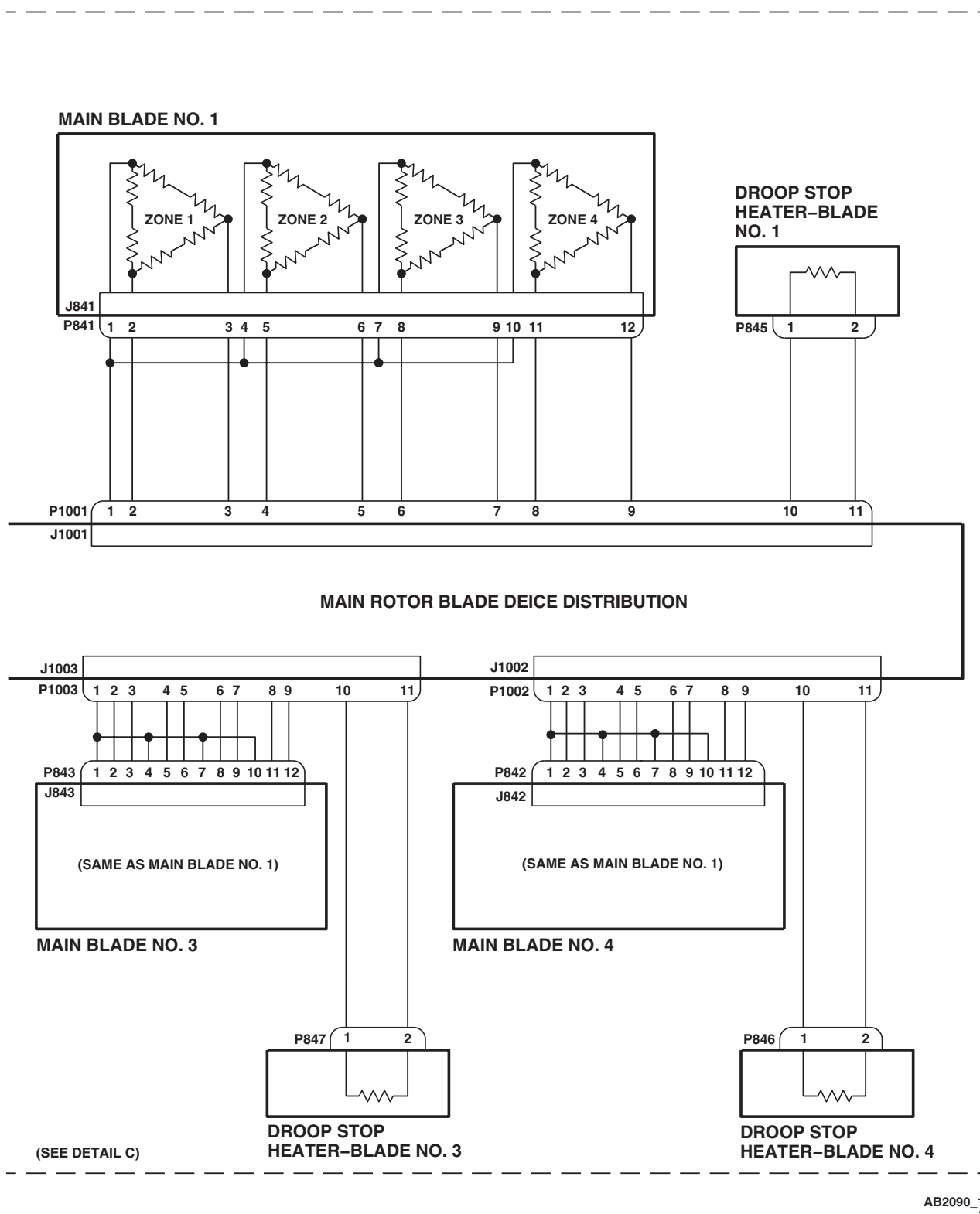
SCHEMATICS - Continued



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Figure 3. Blade Deicing System Schematic Diagram **HH-60A HH-60L UH-60A UH-60L** . (Sheet 14 of 17)

## SCHEMATICS - Continued

Figure 3. Blade Deicing System Schematic Diagram **HH-60A HH-60L UH-60A UH-60L** . (Sheet 15 of 17)

SCHEMATICS - Continued

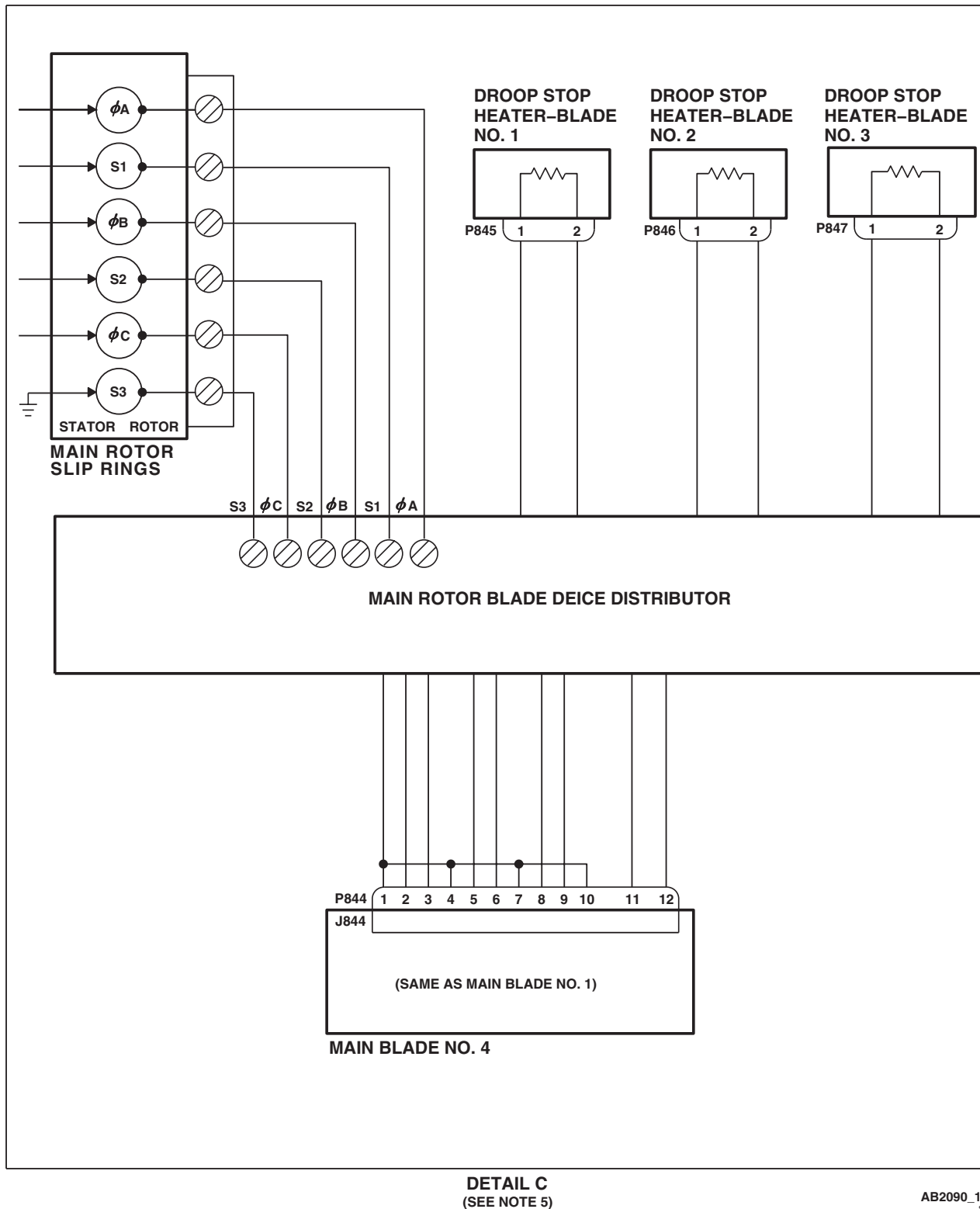
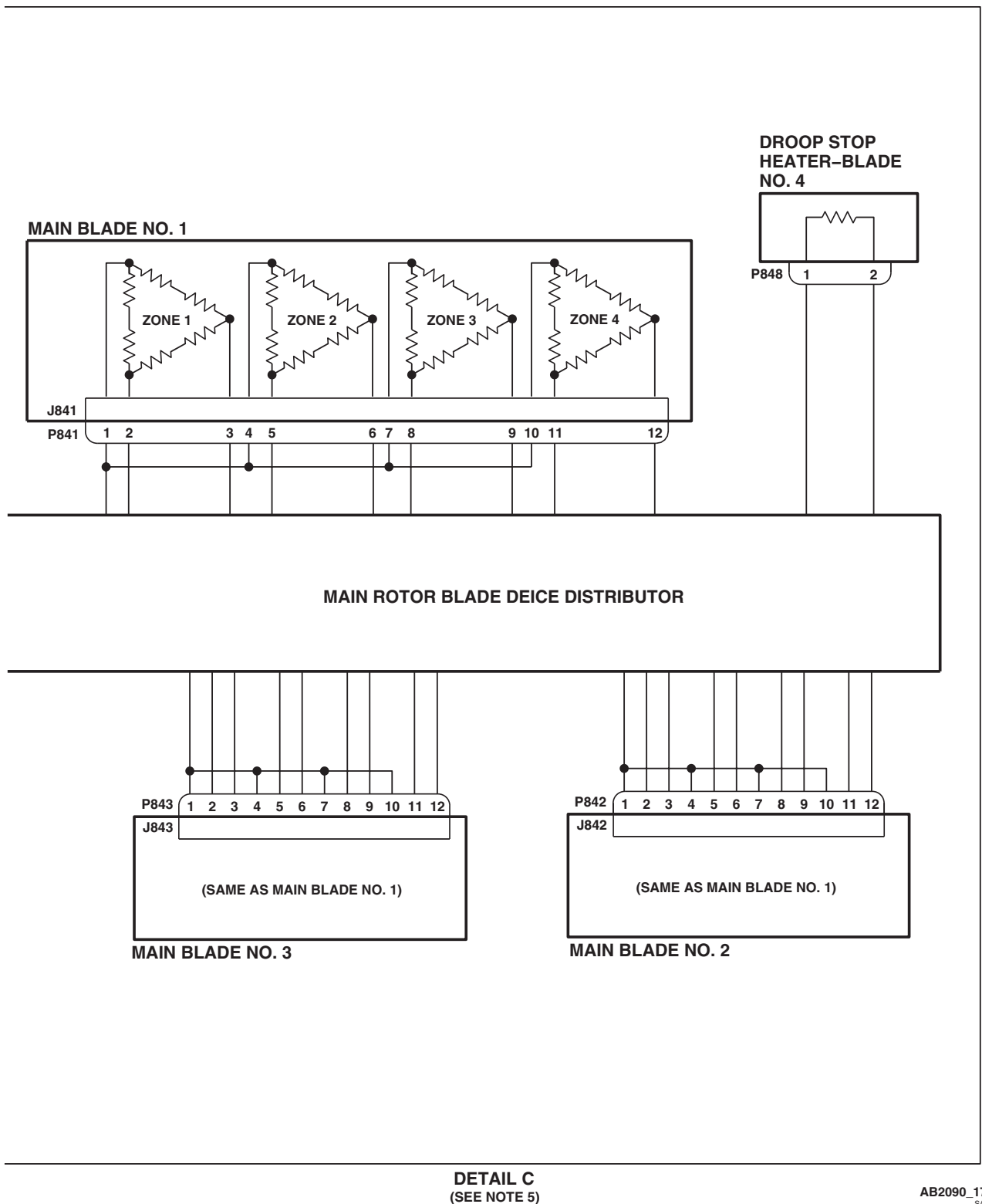
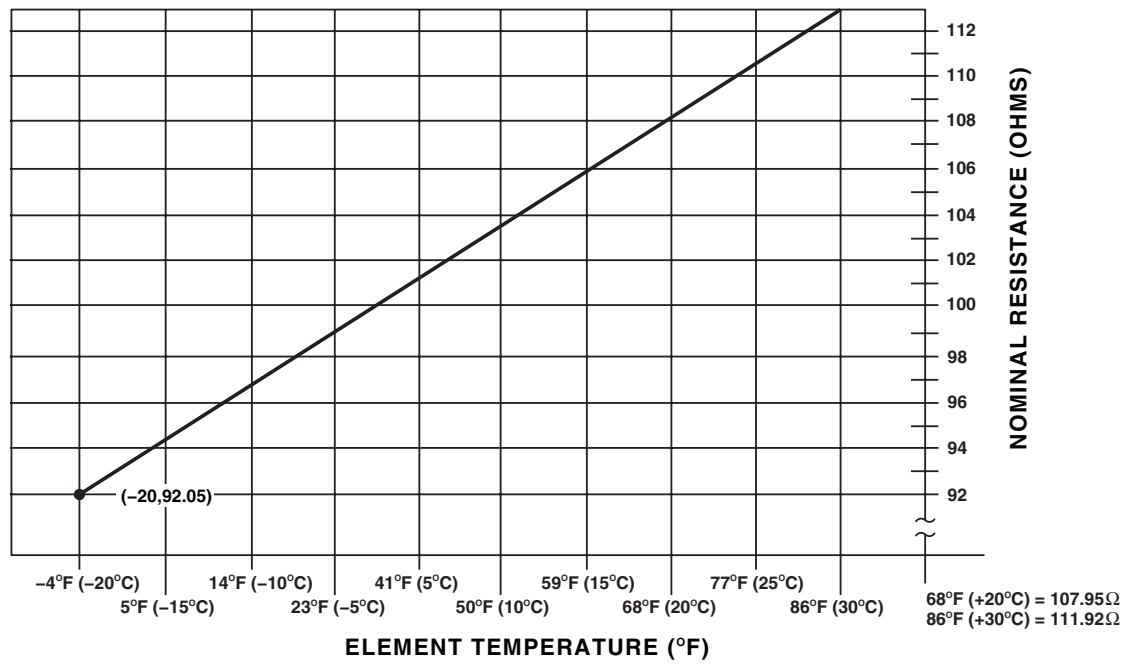


Figure 3. Blade Deicing System Schematic Diagram **HH-60A HH-60L UH-60A UH-60L** . (Sheet 16 of 17)

## SCHEMATICS - Continued

Figure 3. Blade Deicing System Schematic Diagram **HH-60A HH-60L UH-60A UH-60L** . (Sheet 17 of 17)

SCHEMATICS - Continued



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Figure 4. Outside Air Temperature/Resistance Chart.

END OF WORK PACKAGE



## UNIT LEVEL

## UTILITY SYSTEM

**CARGO HOOK SYSTEM UH-60A UH-60L HH-60A HH-60L**

This WP supersedes WP 0146 00, dated 17 April 2006.

**INITIAL SETUP:****Test Equipment**

Multimeter, AN/PSM-45A

WP 0134 00

WP 0135 00

WP 0824 00

**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01

WP 0825 00

Electrical Repairer Toolkit, SC 5180-99-B06

WP 0828 00

WP 0829 00

**Material/Parts**

20 Pound (or more) Weight

WP 0880 00

Insulated Jumper Wire

WP 0925 00

WP 1251 00

WP 1252 00

**Personnel Required**

Aircraft Electrician MOS 15F (1)

WP 1256 00

UH-60 Helicopter Repairer MOS 15T (1)

WP 1257 00

WP 1400 00

**References**

TM 1-1520-237-10

WP 1685 00

WP 0090 00

WP 1747 00

WP 0100 00

WP 0104 00

WP 0110 00

**Equipment Condition**

External Electrical Power Available, WP 1685 00

or APU Operational, TM 1-1520-237-10

**SPECIAL INSTRUCTIONS****NOTE**

See Figure 1 for component location diagram and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

The cargo hook system operational/troubleshooting procedure is subdivided as follows:

- Setup
- Normal Release
- Emergency Release
- Shutdown

**SETUP OPERATIONAL/TROUBLESHOOTING PROCEDURE****PROCEDURE****Step****WARNING**

To prevent injury to personnel, do not press EMERG RELEASE BUTTON when CARGO HOOK EMERG REL switch is in the NORMAL position, while SQUIB is installed or CARGO HOOK CAD (Cartridge Actuated Device) is connected. Remove SQUIB and disconnect CARGO HOOK CAD before any troubleshooting of the cargo hook.

**CAUTION**

Cargo hook release solenoid will be damaged if CARGO REL or NORMAL RLSE button is pressed for more than 12 seconds.

**NOTE**

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
2. Make sure upper console panel switches are placed as follows:

| SWITCH               | POSITION |
|----------------------|----------|
| CARGO HOOK ARMING    | SAFE     |
| CARGO HOOK CONTR     | CKPT     |
| CARGO HOOK EMERG REL | NORM     |

3. On cabin floor, open access panel 4B-24. Push lever to allow cargo hook to swing down.
4. Turn on electrical power. **HH-60A HH-60L** Turn pilot's or copilot's multifunction display (MFD) switch ON and press T6 switch (illuminates all MFD legends for 10 seconds, then only the active caution and advisory legends will be displayed). ■

**Indication/Condition**

Power shall be on.

**Corrective Action**

None Required

**NORMAL RELEASE OPERATIONAL/TROUBLESHOOTING PROCEDURE****PROCEDURE****Step****WARNING**

To prevent injury to personnel, do not press EMERG RELEASE BUTTON when CARGO HOOK EMERG REL switch is in the NORMAL position, while SQUIB is installed or CARGO HOOK CAD (Cartridge Actuated Device) is connected. Remove SQUIB and disconnect CARGO HOOK CAD before any troubleshooting of the cargo hook.

**CAUTION**

Cargo hook release solenoid will be damaged if CARGO REL or NORMAL RLSE button is pressed for more than 12 seconds.

**NOTE**

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Do SETUP, in this work package, if not already done.
2. Install weight on cargo hook.
3. Place upper console CARGO HOOK ARMING switch to ARMED.

**Indication/Condition**

HOOK ARMED capsule or legend shall go on.

**Corrective Action**

If Indication/Condition is not as specified, go to HOOK ARMED CAPSULE OR LEGEND DOES NOT GO ON, in this work package.

**Step****CAUTION**

Cargo hook release solenoid will be damaged if CARGO REL or NORMAL RLSE button is pressed for more than 12 seconds.

4. Press pilot's cyclic stick CARGO REL button until weight has dropped, then release.

**Indication/Condition**

- a. Cargo hook shall open and release weight.
- b. CARGO HOOK OPEN capsule or legend shall go on.
- c. CARGO HOOK OPEN capsule or legend shall go off (momentary light).

---

**NORMAL RELEASE OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Corrective Action**

1. If Indication/Condition a. is not as specified, go to CARGO HOOK DOES NOT OPEN, in this work package.
2. If Indication/Condition b. is not as specified, go to CARGO HOOK OPENS BUT CARGO HOOK OPEN CAPSULE OR LEGEND DOES NOT GO ON, in this work package.
3. If Indication/Condition c. is not as specified, go to CARGO HOOK OPEN CAPSULE OR LEGEND STAYS ON AFTER LOAD IS RELEASED, in this work package.
4. If CARGO HOOK OPEN capsule or legend goes on, but hook does not open, replace cargo hook (WP 1251 00).

**Step**

5. Install weight on cargo hook.

**CAUTION**

Cargo hook release solenoid will be damaged if CARGO REL or NORMAL RLSE button is pressed for more than 12 seconds.

6. Press copilot's cyclic stick CARGO REL button until weight has dropped, then release.

**Indication/Condition**

- a. Cargo hook shall open and release weight.
- b. CARGO HOOK OPEN capsule or legend shall go on.
- c. CARGO HOOK OPEN capsule or legend shall go off (momentary light).

**Corrective Action**

1. If Indication/Condition a. is not as specified, go to CARGO HOOK DOES NOT OPEN, in this work package.
2. If Indication/Condition b. is not as specified, go to CARGO HOOK OPENS BUT CARGO HOOK OPEN CAPSULE DOES NOT GO ON, in this work package.
3. If Indication/Condition c. is not as specified, go to CARGO HOOK OPEN CAPSULE OR LEGEND STAYS ON AFTER LOAD IS RELEASED, in this work package.
4. If CARGO HOOK OPEN capsule or legend goes on, but hook does not open, replace cargo hook (WP 1251 00).

**Step**

7. Install weight on cargo hook.
8. Place upper console CARGO HOOK CONTR switch to ALL.
9. Remove crewman's cargo hook pendant from stowage bag behind pilot's seat. Connect pendant cord to J267.

**CAUTION**

Cargo hook release solenoid will be damaged if CARGO REL or NORMAL RLSE button is pressed for more than 12 seconds.

10. Press crewman's pendant NORMAL RLSE button until weight has dropped, then release.

---

**NORMAL RELEASE OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Indication/Condition**

- a. Cargo hook shall open and release weight.
- b. CARGO HOOK OPEN capsule or legend shall go on.
- c. CARGO HOOK OPEN capsule or legend shall go off (momentary light).

**Corrective Action**

1. If Indication/Condition a. is not as specified, go to CARGO HOOK DOES NOT OPEN, in this work package.
2. If Indication/Condition b. is not as specified, go to CARGO HOOK OPENS BUT CARGO HOOK OPEN CAPSULE OR LEGEND DOES NOT GO ON, in this work package.
3. If Indication/Condition c. is not as specified, go to CARGO HOOK OPEN CAPSULE OR LEGEND STAYS ON AFTER LOAD IS RELEASED, in this work package.
4. If CARGO HOOK OPEN capsule or legend goes on, but hook does not open, replace cargo hook (WP 1251 00).

**Step**

11. Place upper console CARGO HOOK ARMING switch to SAFE.

**Indication/Condition**

HOOK ARMED capsule or legend shall go off.

**Corrective Action**

If Indication/Condition is not as specified, replace cargo hook arming switch S1 (WP 0828 00).

**Step**

12. Press pilot's cyclic stick CARGO REL button.

**Indication/Condition**

Cargo hook shall not open.

**Corrective Action**

1. If Indication/Condition is not as specified, check continuity between J104-F and terminal TB1-5, and between J104-G and terminal 1 of CARGO HOOK EMER circuit breaker.
  - a. If continuity is present, troubleshoot pilot's collective stick (WP 0135 00).
  - b. If continuity is not present, repair/replace wiring as required (WP 1747 00).

**Step**

13. Remove weight from under helicopter.
14. If no other check is to be done, go to SHUTDOWN, in this work package.

**Indication/Condition**

Shutdown shall be performed.

**Corrective Action**

None Required

**EMERGENCY RELEASE OPERATIONAL/TROUBLESHOOTING PROCEDURE****PROCEDURE****Step****WARNING**

To prevent injury to personnel, do not press EMERG RELEASE BUTTON when CARGO HOOK EMERG REL switch is in the NORMAL position, while SQUIB is installed or CARGO HOOK CAD (Cartridge Actuated Device) is connected. Remove SQUIB and disconnect CARGO HOOK CAD before any troubleshooting of the cargo hook.

**CAUTION**

Cargo hook release solenoid will be damaged if CARGO REL or NORMAL RLSE button is pressed for more than 12 seconds.

**NOTE**

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Do SETUP, in this work package, if not already done.
2. Disconnect squib, if not already done (WP 1252 00).
3. Press face of upper console CARGO HOOK EMERG REL TEST light.

**Indication/Condition**

TEST light shall go on.

**Corrective Action**

If Indication/Condition is not as specified, go to CARGO HOOK TEST LIGHT DOES NOT GO ON WHEN PRESSED, in this work package.

**Step**

4. Place upper console CARGO HOOK EMERG REL switch to OPEN.
5. With squib disconnected, install insulated jumper wire between squib connector pins 2 and 3.
6. Press crewman's pendant EMER RLSE button until TEST light goes on, then release button. Push pilot's and copilot's collective sticks HOOK EMER REL button, until TEST light goes on, then release button.

**Indication/Condition**

- a. Whenever any emergency release button is pressed, TEST light shall go on.
- b. When any emergency release button is released, TEST light shall go off.

**Corrective Action**

If Indication/Conditions are not as specified, go to CARGO HOOK EMERG REL TEST LIGHT DOES NOT GO ON DURING OPEN OR SHORT TEST, in this work package.

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**EMERGENCY RELEASE OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Step**

7. Remove insulated jumper wire from squib connector.
8. Press and release crewman's pendant EMER RLSE button. Push and release pilot's and copilot's collective sticks HOOK EMER REL button.

**Indication/Condition**

Whenever any emergency release button is pressed, TEST light shall not go on.

**Corrective Action**

If Indication/Condition is not as specified, go to CARGO HOOK EMERG REL TEST LIGHT GOES ON WHEN EMER REL BUTTON IS PRESSED/PUSHED ON, in this work package.

**Step**

9. Install insulated jumper wire between squib connector pins 2 and 3.
10. Place upper console CARGO HOOK EMERG REL switch to SHORT.
11. Press crewman's pendant EMER RLSE button until TEST light goes on, then release button. On pilot's and copilot's collective sticks, push HOOK EMER REL button until TEST light goes on, then release button.

**Indication/Condition**

- a. Whenever any emergency release button is pressed, TEST light shall go on.
- b. When any emergency release button is released, TEST light shall go off.

**Corrective Action**

If Indication/Conditions are not as specified, go to CARGO HOOK EMERG REL TEST LIGHT DOES NOT GO ON DURING OPEN OR SHORT TEST, in this work package.

**Step**

12. Remove insulated jumper wire between squib connector pins 2 and 3.
13. If no other check is to be done, go to SHUTDOWN, in this work package.

**Indication/Condition**

Shutdown shall be performed.

**Corrective Action**

None Required

**SHUTDOWN OPERATIONAL/TROUBLESHOOTING PROCEDURE  
PROCEDURE****Step**

1. Place upper console CARGO HOOK EMERG REL switch to OPEN.
2. Install crewman's cargo hook pendant behind pilot's seat.
3. **HH-60A HH-60L** Turn MFD switch OFF. 
4. Turn off electrical power.

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**SHUTDOWN OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**

5. Reconnect squib (WP 1252 00).
6. If necessary, swing cargo hook up and latch in place. Close and latch access panel.

**Indication/Condition**

Access panel shall be closed and latched.

**Corrective Action**

None Required

**HOOK ARMED CAPSULE OR LEGEND DOES NOT GO ON****SYMPTOM**

HOOK ARMED capsule or legend does not go on.

**MALFUNCTION**

HOOK ARMED capsule or legend does not go on.

**CORRECTIVE ACTION**

1. Hold caution/advisory panel BRT/DIM-TEST switch to TEST or HH-60A HH-60L press T6 switch on MFD . Go to 2.
2. Check that HOOK ARMED capsule or legend goes on; then release the caution/advisory panel BRT/DIM-TEST switch.
  - a. If HOOK ARMED capsule or legend is on, go to 3.
  - b. If HOOK ARMED capsule or legend is not on, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to 11.
3. With CARGO HOOK ARMING switch placed at ARMED, check continuity between J126-W and J200-A.
  - a. If continuity is present, go to 4.
  - b. If continuity is not present, go to 10.
4. Check for 28 vdc between P126-W and ground.
  - a. If voltage is as specified, go to 5.
  - b. If voltage is not as specified, go to 9.
5. Check helicopter configuration.
  - a. EMEP , go to 6.
  - b. W/O EMEP , go to 8.
6. EMEP Remove and check pin filtered adapter on caution/advisory panel or MFDs (WP 0925 00).
  - a. If pin filtered adapter is good, go to 7.
  - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to 11.
7. EMEP Install pin filtered adapter (WP 0925 00). Go to 8.
8. Check for 28 vdc between P118-M or HH-60A HH-60L P1019R-44 or P1020R-44 and ground.



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**HOOK ARMED CAPSULE OR LEGEND DOES NOT GO ON** - Continued

- a. If voltage is as specified, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to **11**.
  - b. If voltage is not as specified, repair/replace wiring between P200-A and P118-M or **HH-60A HH-60L** P1019R-44 or P1020R-44 (WP 1747 00). Go to **11**.
- 9. Check for 28 vdc between CARGO HOOK CONTR circuit breaker terminal 1 and ground.**
- a. If voltage is as specified, repair/replace wiring between P126-W and circuit breaker terminal 1 (WP 1747 00). Go to **11**.
  - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **11**.
- 10. Check continuity between:**
- **J126-W and S1-6**
  - **J200-A and S1-5**
  - **J200-B and S1-3**
  - **J200-C and S1-2**
- a. If continuity is present, replace switch (WP 0828 00). Go to **11**.
  - b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **11**.
- 11. Procedure completed.**

**CARGO HOOK DOES NOT OPEN****SYMPTOM**

Cargo hook does not open.

**MALFUNCTION**

Cargo hook does not open.

**CORRECTIVE ACTION**

- 1. Does cargo hook open using pilot's cyclic, copilot's cyclic, or crewman's pendant?**
  - a. If cargo hook does open, go to **2**.
  - b. If cargo hook does not open, go to **5**.
- 2. Does cargo hook open using crewman's pendant?**
  - a. If cargo hook does open, go to **3**.
  - b. If cargo hook does not open, go to **21**.
- 3. Does cargo hook open using copilot's cyclic CARGO REL button?**
  - a. If cargo hook does open, go to **4**.
  - b. If cargo hook does not open, go to **26**.
- 4. Does cargo hook open using pilot's cyclic CARGO REL button?**
  - a. If cargo hook does open, system is operational. Go to **30**.
  - b. If cargo hook does not open, go to **28**.

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**CARGO HOOK DOES NOT OPEN - Continued**

5. With any **CARGO REL** button pressed, check for 28 vdc between J218-A and J218-B.
  - a. If voltage is as specified, replace cargo hook (WP 1251 00). Go to **30**.
  - b. If voltage is not as specified, go to **6**.
6. With **CARGO REL** button pressed, check for 28 vdc between J218-A and ground.
  - a. If voltage is as specified, repair/replace wiring between J218-B and GND218-1 (WP 1747 00). Go to **30**.
  - b. If voltage is not as specified, go to **7**.
7. Check helicopter configuration.
  - a. **WINTER** , go to **8**.
  - b. **W/O WINTER** , go to **9**.
8. **WINTER** Check for 28 vdc between P244B-R and ground.
  - a. If voltage is as specified, go to **10**.
  - b. If voltage is not as specified, replace **WINTER** (WP 1400 00). Go to **30**.
  - c. If trouble remains, go to **9**.
9. With **CARGO REL** button pressed, check for 28 vdc between P244-R and ground.
  - a. If voltage is as specified, go to **10**.
  - b. If voltage is not as specified, go to **19**.
10. Check for 28 vdc between P900-r and ground.
  - a. If voltage is as specified, go to **11**.
  - b. If voltage is not as specified, go to **18**.
11. Check continuity between P241-q and P900-j.
  - a. If continuity is present, go to **12**.
  - b. If continuity is not present, repair/replace ground wire (WP 1747 00). Go to **30**.
12. Install an insulated jumper wire between P900-m and P900-r. Go to **13**.
13. Check for 28 vdc between J218-A and ground.
  - a. If voltage is as specified, go to **14**.
  - b. If voltage is not as specified, go to **16**.
14. Remove jumper wire from P900. Go to **15**.
15. Replace right relay panel (WP 0825 00). Go to **30**.
16. Remove jumper wire from P900. Go to **17**.
17. Repair/replace wiring between P900-m and J218-A (WP 1747 00). Go to **30**.
18. Check for 28 vdc between CARGO HOOK PWR circuit breaker terminal 1 and ground.
  - a. If voltage is as specified, repair/replace wiring between P266-m and P900-r (WP 1747 00). Go to **30**.

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**CARGO HOOK DOES NOT OPEN** - Continued

- b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **30**.
- 19. With CARGO HOOK ARMING switch placed to ARMED, check continuity between J200-B and J200-C.**
  - a. If continuity is present, go to **20**.
  - b. If continuity is not present, replace switch (WP 0828 00). Go to **30**.
- 20. With CARGO REL button pressed, check for 28 vdc between P200-B and ground.**
  - a. If voltage is as specified, repair/replace wiring between P200-C and P244-R (WP 1747 00). Go to **30**.
  - b. If voltage is not as specified, repair/replace wiring between P200-B and J248-R (WP 1747 00). Go to **30**.
- 21. With crewman's pendant NORMAL RLSE button pressed, check continuity between P267-B and P267-F.**
  - a. If continuity is present, go to **22**.
  - b. If continuity is not present, replace crewman's cargo hook pendant (WP 1256 00). Go to **30**.
- 22. With CARGO HOOK CONTR switch placed to ALL, check for 28 vdc between J267-F and ground.**
  - a. If voltage is as specified, repair/replace wiring between J267-B and J248-R (WP 1747 00). Go to **30**.
  - b. If voltage is not as specified, go to **23**.
- 23. With CARGO HOOK CONTR switch placed to ALL, check for 28 vdc between J200-G and ground.**
  - a. If voltage is as specified, repair/replace wiring between P200-G and J267-F (WP 1747 00). Go to **30**.
  - b. If voltage is not as specified, go to **24**.
- 24. Check continuity between J200-G and CARGO HOOK CONTR switch S2-2.**
  - a. If continuity is present, go to **25**.
  - b. If continuity is not present, repair/replace wiring between J200-G and S2-2 (WP 1747 00). Go to **30**.
- 25. With CARGO HOOK CONTR switch placed to ALL, check continuity between S2-2 and S2-3.**
  - a. If continuity is present, repair/replace wiring between S2-3 and ARMING switch S1-6 (WP 1747 00). Go to **30**.
  - b. If continuity is not present, replace switch (WP 1747 00). Go to **30**.
- 26. With copilot's cyclic CARGO REL button pressed, check continuity between P119-E and P119-F.**
  - a. If continuity is present, go to **27**.
  - b. If continuity is not present, troubleshoot copilot's cyclic stick (WP 0134 00). Go to **30**.
- 27. Check for 28 vdc between J119-F and ground.**

**CARGO HOOK DOES NOT OPEN - Continued**

- a. If voltage is as specified, repair/replace wiring between J119-E and P248-R (WP 1747 00). Go to **30**.
  - b. If voltage is not as specified, repair/replace wiring between J119-F and CARGO HOOK CONTR circuit breaker terminal 1 (WP 1747 00). Go to **30**.
- 28. With pilot's cyclic CARGO REL button pressed, check continuity between P120-E and P120-F.**
- a. If continuity is present, go to **29**.
  - b. If continuity is not present, troubleshoot pilot's cyclic stick (WP 0134 00). Go to **30**.
- 29. Check for 28 vdc between J120-F and ground.**
- a. If voltage is as specified, repair/replace wiring between J120-E and P248-R (WP 1747 00). Go to **30**.
  - b. If voltage is not as specified, repair/replace wiring between J120-F and CARGO HOOK CONTR circuit breaker terminal 1 (WP 1747 00). Go to **30**.
- 30. Procedure completed.**

**CARGO HOOK OPENS BUT CARGO HOOK OPEN CAPSULE OR LEGEND DOES NOT GO ON****SYMPTOM**

Cargo hook opens but CARGO HOOK OPEN capsule or legend does not go on.

**MALFUNCTION**

Cargo hook opens but CARGO HOOK OPEN capsule or legend does not go on.

**CORRECTIVE ACTION**

- 1. Hold caution/advisory panel or HH-60A HH-60L instrument panel INDICATOR LTS BRT/DIM-TEST switch to TEST. Go to 2.**
- 2. Check that CARGO HOOK OPEN capsule or legend goes on; then release the caution/advisory panel or HH-60A HH-60L instrument panel INDICATOR LTS BRT/DIM-TEST switch.**
  - a. If CARGO HOOK OPEN capsule or legend goes on, go to **3**.
  - b. If CARGO HOOK OPEN capsule or legend does not go on, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to **9**.
- 3. With load beam on cargo hook pulled down, check continuity between P218-C and P218-D.**
  - a. If continuity is present, go to **4**.
  - b. If continuity is not present, replace cargo hook (WP 1251 00). Go to **9**.
- 4. Check for 28 vdc between J218-D and ground.**
  - a. If voltage is as specified, go to **5**.
  - b. If voltage is not as specified, repair/replace wiring between J218-D and J119-F (WP 1747 00). Go to **9**.
- 5. Check helicopter configuration.**

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**CARGO HOOK OPENS BUT CARGO HOOK OPEN CAPSULE OR LEGEND DOES NOT GO ON** - Continued

- a. **EMEP** , go to 6.
- b. **W/O EMEP** , go to 8.
6. **EMEP** Remove and check pin filtered adapter on caution/advisory panel or MFDs (WP 0925 00).
  - a. If pin filtered adapter is good, go to 7.
  - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to 9.
7. **EMEP** Install pin filtered adapter (WP 0925 00). Go to 8.
8. With load beam pulled down, check for 28 vdc between P118-G or HH-60A HH-60L P1019R-32 or P1020R-32 and ground.
  - a. If voltage is as specified, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to 9.
  - b. If voltage is not as specified, repair/replace wiring between J218-C and P118-G, or HH-60A HH-60L P1019R-32 or P1020R-32 (WP 1747 00). Go to 9.
9. Procedure completed.

**CARGO HOOK OPEN CAPSULE OR LEGEND STAYS ON AFTER LOAD IS RELEASED****SYMPTOM**

CARGO HOOK OPEN capsule or legend stays on after load is released.

**MALFUNCTION**

CARGO HOOK OPEN capsule or legend stays on after load is released.

**CORRECTIVE ACTION**

1. Make sure no CARGO REL buttons are pressed. Go to 2.
2. Place a 20-pound (or more) weight on cargo hook. Go to 3.
3. Check that hook stays closed and holds weight.
  - a. If hook stays closed and holds weight, go to 4.
  - b. If hook does not stay closed and hold weight, replace cargo hook (WP 1251 00). Go to 5.
4. Check for 28 vdc between J218-A and ground.
  - a. If voltage is as specified, replace right relay panel (WP 0825 00). Go to 5.
  - b. If voltage is not as specified, replace cargo hook (WP 1251 00). Go to 5.
5. Procedure completed.

**CARGO HOOK TEST LIGHT DOES NOT GO ON WHEN PRESSED****SYMPTOM**

CARGO HOOK TEST light does not go on when pressed.

**MALFUNCTION**

CARGO HOOK TEST light does not go on when pressed.

**CORRECTIVE ACTION**

- 1. Check for burnt-out lamp.**
  - a. If lamp is good, go to **2**.
  - b. If lamp is not good, replace lamp in test light (WP 0829 00). Go to **5**.
- 2. Check continuity between P242-d and P902-h.**
  - a. If continuity is present, go to **3**.
  - b. If continuity is not present, go to **4**.
- 3. Check continuity between GG2-5 and test lamp pin 3.**
  - a. If continuity is present, troubleshoot light dimming control system (WP 0110 00). Go to **5**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **5**.
- 4. Check continuity between:**
  - **P902-h and test lamp pin 1**
  - **P242-d and test lamp pin 2**
  - a. If continuity is present, replace test lamp (WP 0829 00). Go to **5**.
  - b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **5**.
- 5. Procedure completed.**

**CARGO HOOK EMERG REL TEST LIGHT DOES NOT GO ON DURING OPEN OR SHORT TEST****SYMPTOM**

CARGO HOOK EMERG REL TEST light does not go on during OPEN or SHORT test.

**MALFUNCTION**

CARGO HOOK EMERG REL TEST light does not go on during OPEN or SHORT test.

**CORRECTIVE ACTION**

- 1. Does light go on using pilot's or copilot's collective stick HOOK EMER REL button, or crewman's cargo hook pendant EMER RLSE button?**
  - a. If light goes on, go to **2**.
  - b. If light does not go on, go to **5**.
- 2. Does light go on using crewman's cargo hook pendant EMER RLSE button?**
  - a. If light goes on, go to **3**.
  - b. If light does not go on, go to **17**.

---

**CARGO HOOK EMERG REL TEST LIGHT DOES NOT GO ON DURING OPEN OR SHORT TEST - Continued**

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- 3. Does light go on using copilot's collective HOOK EMER REL button?**
  - a. If light goes on, go to **4**.
  - b. If light does not go on, go to **19**.
- 4. Does light go on using pilot's collective HOOK EMER REL button?**
  - a. If light goes on, system is operational. Go to **23**.
  - b. If light does not go on, go to **21**.
- 5. Check for 28 vdc between CARGO HOOK EMER circuit breaker terminal 1 and ground.**
  - a. If voltage is as specified, go to **6**.
  - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **23**.
- 6. Check for 28 vdc between J267-E and ground.**
  - a. If voltage is as specified, go to **7**.
  - b. If voltage is not as specified, repair/replace wiring between J267-E and CARGO HOOK EMER circuit breaker terminal 1 (WP 1747 00). Go to **23**.
- 7. With crewman's EMER RLSE button pressed, check for 28 vdc between TB1-5 and ground.**
  - a. If voltage is as specified, go to **8**.
  - b. If voltage is not as specified, repair/replace wiring between TB1-5 and J267-C (WP 1747 00). Go to **23**.
- 8. Check for 1.5 to 2.5 ohms between TB1-4 and TB1-5.**
  - a. If resistance is as specified, go to **9**.
  - b. If resistance is not as specified, replace resistor R4 (WP 1747 00). Go to **23**.
- 9. Check continuity between TB1-4 and EMERG REL switch S3-2.**
  - a. If continuity is present, go to **10**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **23**.
- 10. Check continuity between EMERG REL switch S3-2 and S3-3.**
  - a. If continuity is present, go to **11**.
  - b. If continuity is not present, replace EMERG REL switch (WP 0828 00). Go to **23**.
- 11. Check continuity between EMERG REL switch S3-3 and P242-g.**
  - a. If continuity is present, go to **12**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **23**.
- 12. Check continuity between EMERG REL TEST switch terminal 2 and P242-d.**
  - a. If continuity is present, go to **13**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **23**.
- 13. Disconnect P256 on cargo hook and place an insulated jumper wire between P256-2 and P256-3. Go to 14.**

---

**CARGO HOOK EMERG REL TEST LIGHT DOES NOT GO ON DURING OPEN OR SHORT TEST - Continued**

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**14. Check continuity between P242-e and EMERG REL switch S3-4.**

- a. If continuity is present, replace cargo hook squib (WP 1252 00). Go to **23**.
- b. If trouble remains, replace left relay panel (WP 0825 00). Go to **23**.
- c. If continuity is not present, go to **15**.

**15. Remove jumper wire from P256. Go to 16.****16. Repair/replace wiring, as required, between (WP 1747 00), then go to 23:**

- P242-c and EMERG REL switch S3-1
- P256-2 and EMERG REL switch S3-4
- P256-3 and EMERG REL switch S3-1

**17. With crewman's EMER RLSE button pressed, check continuity between P267-C and P267-E.**

- a. If continuity is present, go to **18**.
- b. If continuity is not present, replace crewman's cargo hook pendant (WP 1256 00). Go to **23**.

**18. Check for 28 vdc between J267-E and ground.**

- a. If voltage is as specified, repair/replace wiring between J267-C and J248-T (WP 1747 00). Go to **23**.
- b. If voltage is not as specified, repair/replace wiring between J267-E and J248-S (WP 1747 00). Go to **23**.

**19. With copilot's HOOK EMER REL button pressed, check continuity between P103-F and P103- MOD H W/O MOD G .**

- a. If continuity is present, go to **20**.
- b. If continuity is not present, troubleshoot copilot's collective stick grip (WP 0135 00). Go to **23**.

**20. Check for 28 vdc between J103- MOD H W/O MOD G and ground.**

- a. If voltage is as specified, repair/replace wiring between J103-F and P248-T (WP 1747 00). Go to **23**.
- b. If voltage is not as specified, repair/replace wiring between J103- MOD H W/O MOD G and P248-S (WP 1747 00). Go to **23**.

**21. With pilot's HOOK EMER REL button pressed, check continuity between P104-F and P104- MOD H W/O MOD G .**

- a. If continuity is present, go to **22**.
- b. If continuity is not present, troubleshoot pilot's collective stick grip (WP 0135 00). Go to **23**.

**22. Check for 28 vdc between J104- MOD H W/O MOD G and ground.**

- a. If voltage is as specified, repair/replace wiring between J104-F and P248-T (WP 1747 00). Go to **23**.
- b. If voltage is not as specified, repair/replace wiring between J103- MOD H W/O MOD G and P248-S (WP 1747 00). Go to **23**.



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**CARGO HOOK EMERG REL TEST LIGHT DOES NOT GO ON DURING OPEN OR SHORT TEST - Continued**

**23. Procedure completed.**

**CARGO HOOK EMERG REL TEST LIGHT GOES ON WHEN EMER RLSE BUTTON IS PRESSED/PUSHED ON****SYMPTOM**

CARGO HOOK EMERG REL TEST light goes on when EMER RLSE button is pressed/pushed on.

**MALFUNCTION**

CARGO HOOK EMERG REL TEST light goes on when EMER RLSE button is pressed/pushed on.

**CORRECTIVE ACTION****1. Check that P256 is disconnected from cargo hook.**

- a. If P256 is connected to cargo hook, disconnect from cargo hook. Go to **2**.
- b. If P256 is disconnected from cargo hook, go to **11**.

**2. Disconnect P273 from J273. Check for open circuit between:**

- **P256-1 and P256-4**
- **P256-2 and P256-3**

- a. If continuity is present, repair/replace wiring (WP 1747 00). Go to **13**.
- b. If continuity is not present, reconnect J273 and P273. Go to **3**.

**3. Check continuity between:**

- **P256-1 and P256-3**
- **P256-2 and P256-4**
- **P256-2 and P273-B**
- **P256-3 and P273-A**
- **TB2-5 and P273-A**
- **TB2-6 and P273-B**

- a. If continuity is present, go to **4**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **13**.

**4. Check continuity between:**

- **S1-1 and S1-2**
- **S3-2 and S3-3**
- **S3-4 and S3-5**

- a. If continuity is present, go to **5**.
- b. If continuity is not present, replace switch (WP 1257 00). Go to **13**.

**5. Check continuity between:**

- **GG2-5 and S3-1**

# CARGO HOOK EMERG REL TEST LIGHT GOES ON WHEN EMER RLSE BUTTON IS PRESSED/PUSHED ON - Continued

- P242-c and P220-e
  - P242-e and P200-f
  - P242-g and P200-D
  - S1-1 and S3-5
  - S1-2 and J200-C
  - S3-2 and TB1-4
  - S3-3 and J200-D
  - TB1-5 and J267-C
  - TB2-5 and J200-E
- a. If continuity is present, go to **6**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **13**.
- 6. Depress crewman's cargo hook pendant EMER RLSE button. Check continuity between P267-C and P267-E.**
    - a. If continuity is present, go to **7**.
    - b. If continuity is not present, replace crewman's cargo hook pendant (WP 1256 00). Go to **13**.
  - 7. Release crewman's cargo hook pendant EMER RLSE button. Check for open circuit between P267-C and P267-E.**
    - a. If open circuit is present, go to **8**.
    - b. If open circuit is not present, replace crewman's cargo hook pendant (WP 1256 00). Go to **13**.
  - 8. Check for 1.5 to 2.5 ohms between TB1-4 and TB1-5.**
    - a. If resistance is as specified, go to **9**.
    - b. If resistance is not as specified, replace resistor R4 (WP 0880 00). Go to **13**.
  - 9. With copilot's HOOK EMER REL button pressed, check continuity between P103-F and P103- MOD H W/O MOD G .**
    - a. If continuity is present, go to **10**.
    - b. If continuity is not present, troubleshoot copilot's collective stick (WP 0135 00). Go to **13**.
  - 10. Check for 28 vdc between J103- MOD H W/O MOD G and ground.**
    - a. If voltage is as specified, repair/replace wiring between J103-F and P248-T (WP 1747 00). Go to **13**.
    - b. If voltage is not as specified, repair/replace wiring between J103- MOD H W/O MOD G and P248-S (WP 1747 00). Go to **13**.
  - 11. With pilot's HOOK EMER REL button pressed, check continuity between P104-F and P104- MOD H W/O MOD G .**
    - a. If continuity is present, go to **12**.
    - b. If continuity is not present, troubleshoot pilot's collective stick (WP 0135 00). Go to **13**.

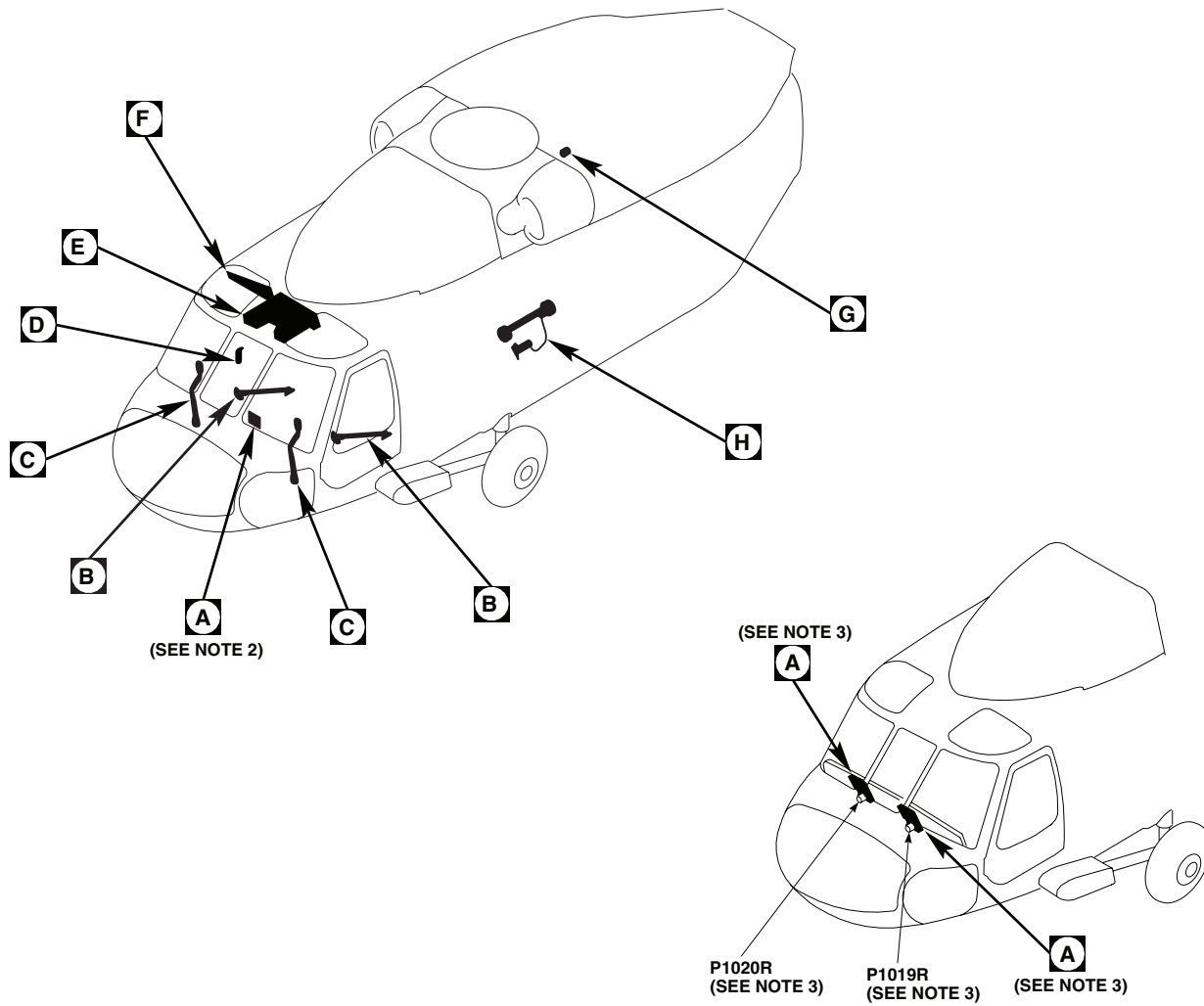
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**CARGO HOOK EMERG REL TEST LIGHT GOES ON WHEN EMER RLSE BUTTON IS PRESSED/PUSHED ON** - Continued**12. Check for 28 vdc between J104- MOD H and W/O MOD G and ground.**

- a. If voltage is as specified, repair/replace wiring between J104-F and P248-T (WP 1747 00). Go to **13**.
- b. If voltage is not as specified, repair/replace wiring between J103- MOD H and W/O MOD G and P248-S (WP 1747 00). Go to **13**.
- c. If trouble remains, replace left relay panel (WP 0825 00). Go to **13**.

**13. Procedure completed.**

# LOCATION



## NOTES

1. **WINTER**
2. **UH60A UH60L**
3. **HH60A HH60L**

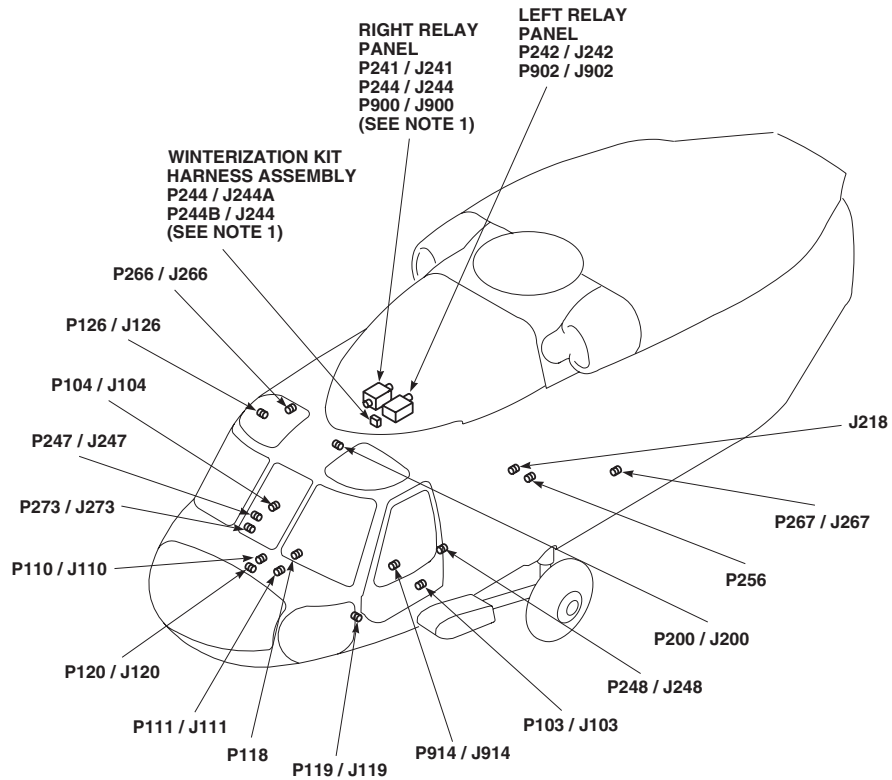


| TERMINAL BOARD /<br>DISCONNECT PLUG /<br>RECEPTACLE | LOCATION /<br>CONNECTION POINT           |
|-----------------------------------------------------|------------------------------------------|
| J218                                                | CABIN TUB<br>BL 9 RH, STA 343            |
| P103 / J103                                         | COCKPIT TUB<br>BL 32 LH, STA 247         |
| P104 / J104                                         | COCKPIT TUB<br>BL 15 RH, STA 247         |
| P110 / J110                                         | COCKPIT<br>BL 7.5 RH, STA 197            |
| P111 / J111                                         | COCKPIT<br>BL 7.5 LH, STA 197            |
| P118                                                | CAUTION / ADVISORY<br>PANEL (SEE NOTE 2) |
| P119 / J119                                         | COCKPIT TUB<br>BL 30 LH, STA 220         |
| P120 / J120                                         | COCKPIT TUB<br>BL 30 RH, STA 220         |

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Figure 1. Cargo Hook System Location Diagram. (Sheet 1 of 5)

## LOCATION - Continued



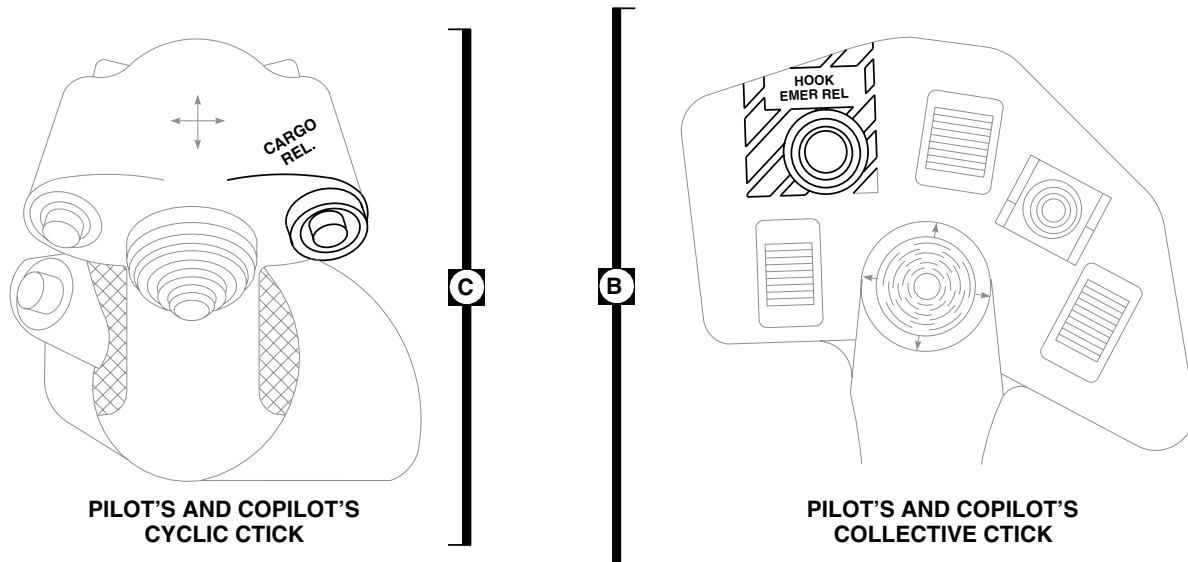
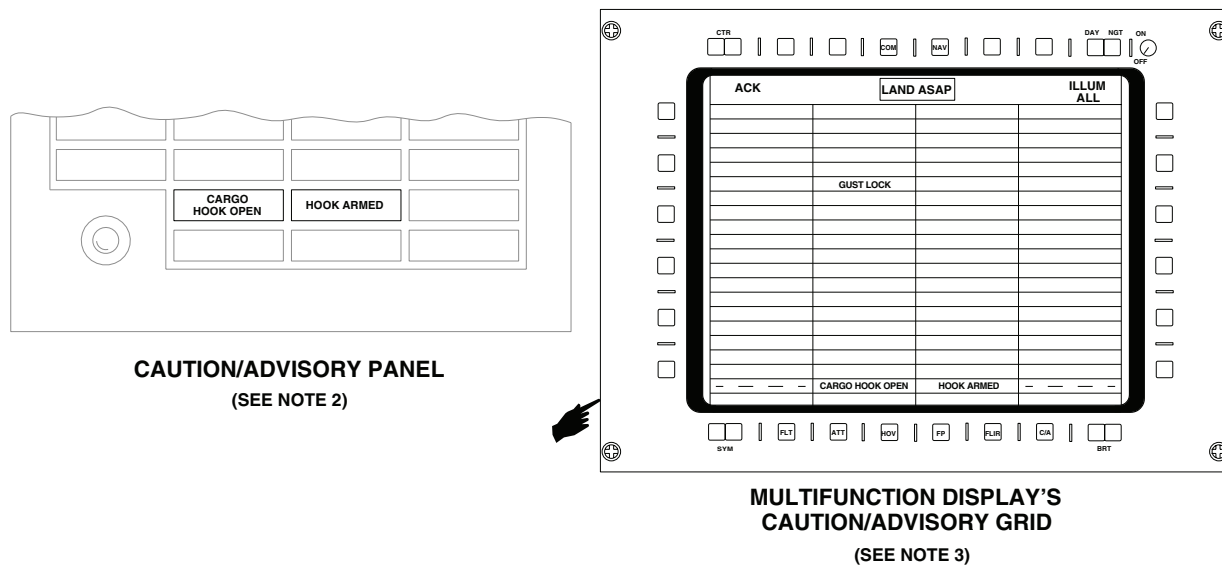
| TERMINAL BOARD /<br>DISCONNECT PLUG /<br>RECEPTACLE | LOCATION /<br>CONNECTION POINT                        |
|-----------------------------------------------------|-------------------------------------------------------|
| P126 / J126                                         | COCKPIT CEILING<br>BL 28 RH, STA 243                  |
| P200 / J200                                         | COCKPIT CEILING<br>BL 13 RH, STA 246                  |
| P241 / J241                                         | RIGHT RELAY PANEL                                     |
| P242 / J242                                         | LEFT RELAY PANEL                                      |
| P244 / J244                                         | RIGHT RELAY PANEL<br>(SEE NOTE 1)                     |
| P244 / J244A<br>P244B / J244                        | WINTERIZATION KIT<br>HARNESS ASSEMBLY<br>(SEE NOTE 1) |
| P247 / J247                                         | CABIN<br>BL 40 RH, STA 248                            |
| P248 / J248                                         | CABIN<br>BL 40 LH, STA 248                            |

| TERMINAL BOARD /<br>DISCONNECT PLUG /<br>RECEPTACLE | LOCATION /<br>CONNECTION POINT                      |
|-----------------------------------------------------|-----------------------------------------------------|
| P256                                                | CABIN TUB<br>BL 15 RH, STA 342                      |
| P266 / J266                                         | BEHIND PILOT'S CIRCUIT<br>BREAKER PANEL<br>BL 25 RH |
| P267 / J267                                         | CABIN<br>BL 14 LH, STA 381                          |
| P273 / J273                                         | CABIN<br>BL 40 RH, STA 248                          |
| P900 / J900                                         | RIGHT RELAY PANEL                                   |
| P902 / J902                                         | LEFT RELAY PANEL                                    |
| P914 / J914                                         | COCKPIT<br>BL 24 LH, STA 247                        |
| P1019R / J3                                         | MULTIFUNCTION DISPLAY<br>(SEE NOTE 3)               |
| P1020R / J3                                         | MULTIFUNCTION DISPLAY<br>(SEE NOTE 3)               |

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Figure 1. Cargo Hook System Location Diagram. (Sheet 2 of 5)

LOCATION - Continued



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Figure 1. Cargo Hook System Location Diagram. (Sheet 3 of 5)

LOCATION - Continued

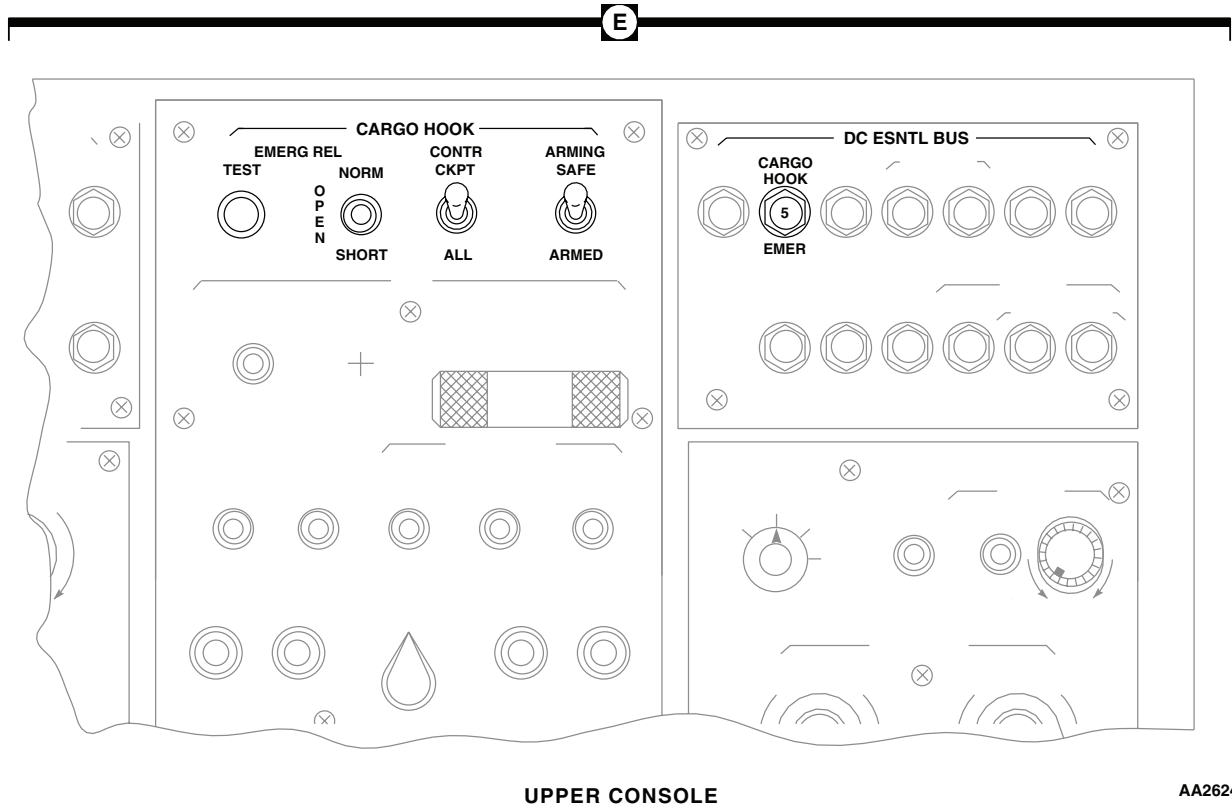
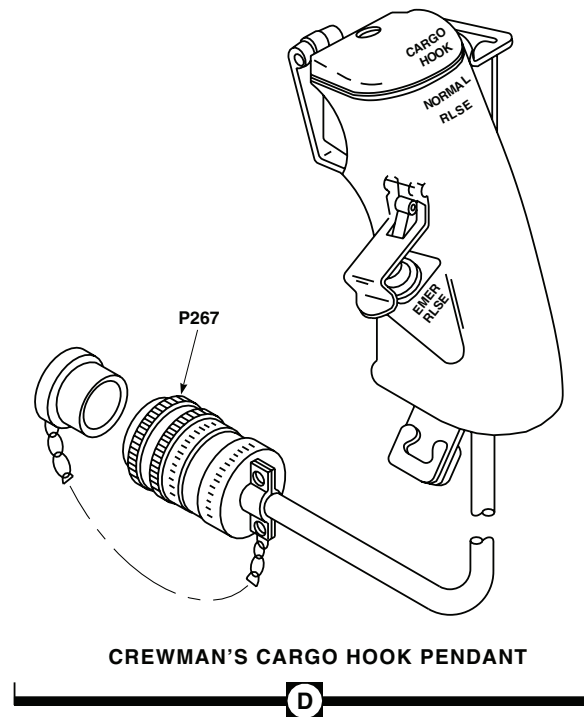
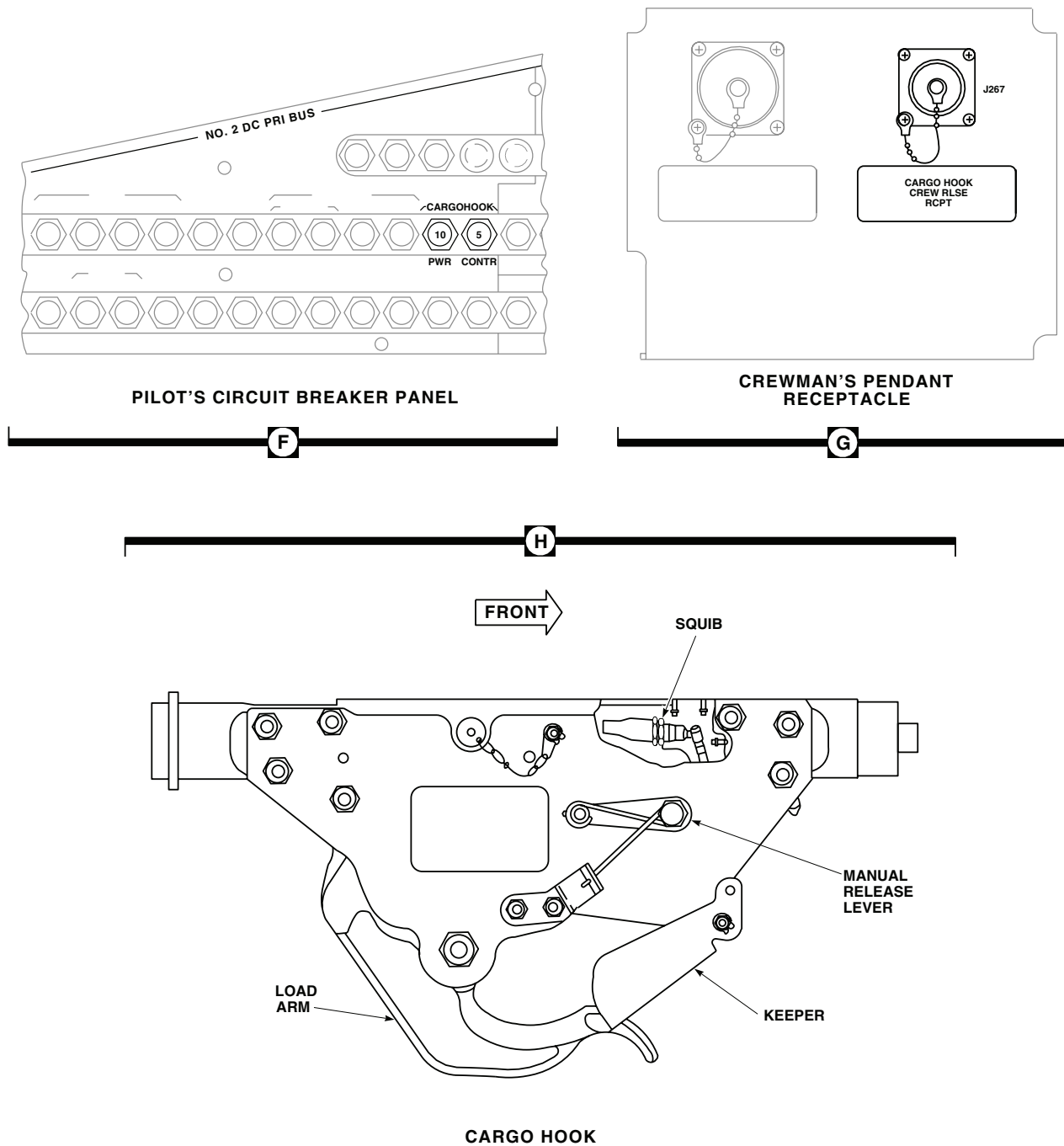


Figure 1. Cargo Hook System Location Diagram. (Sheet 4 of 5)

LOCATION - Continued



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Figure 1. Cargo Hook System Location Diagram. (Sheet 5 of 5)



## SCHEMATICS

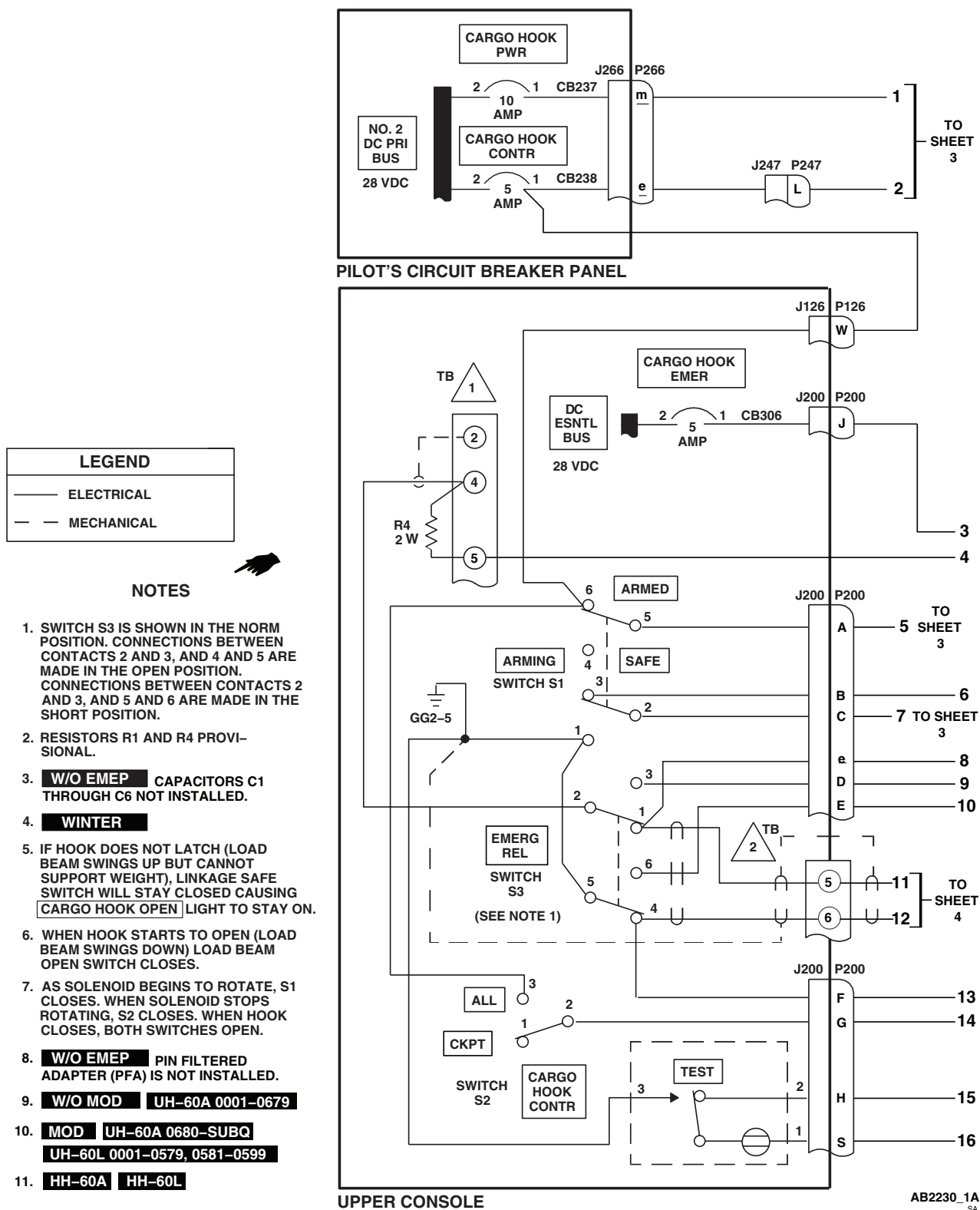


Figure 2. Cargo Hook System Schematic Diagram. (Sheet 1 of 5)

SCHEMATICS - Continued

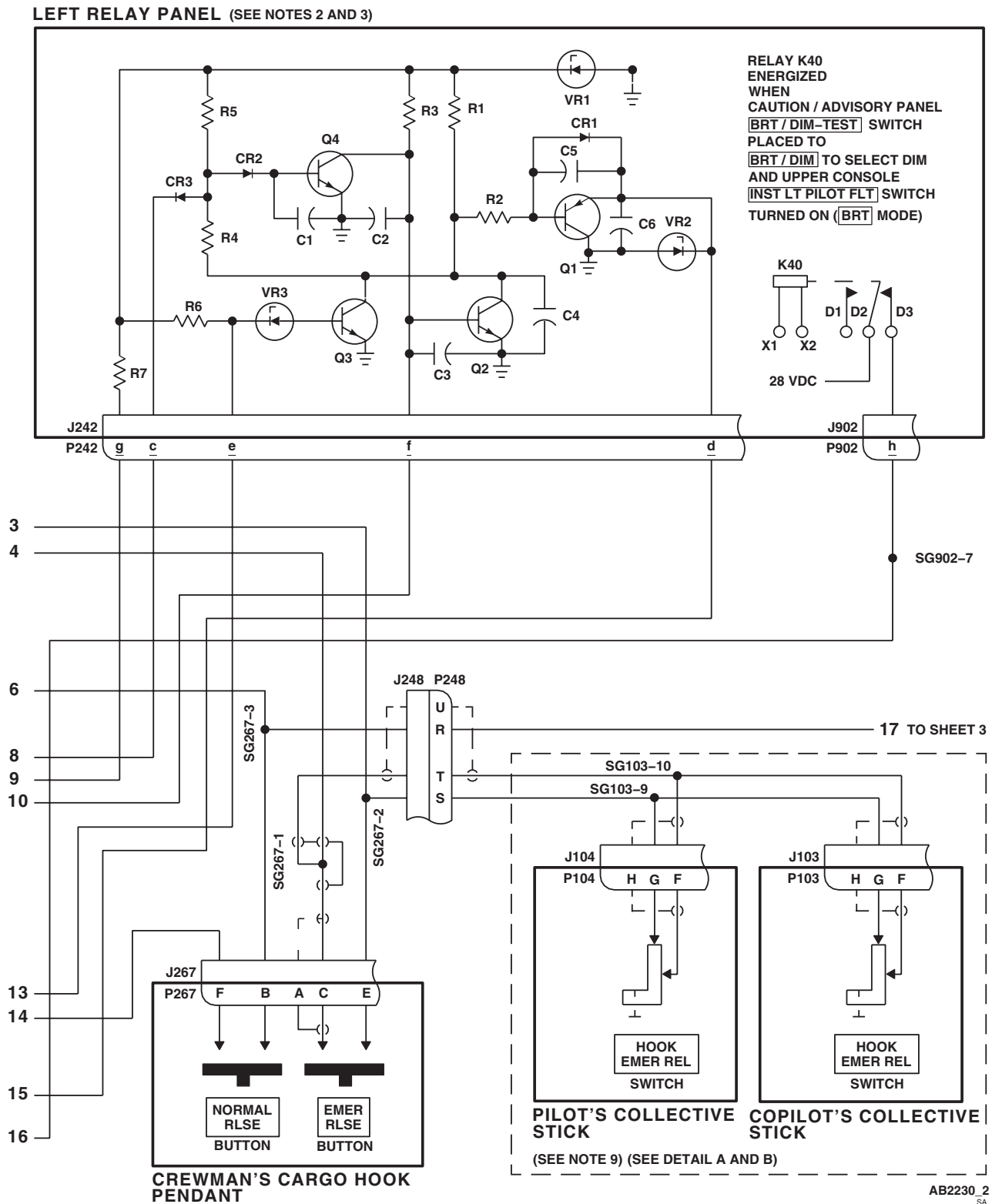
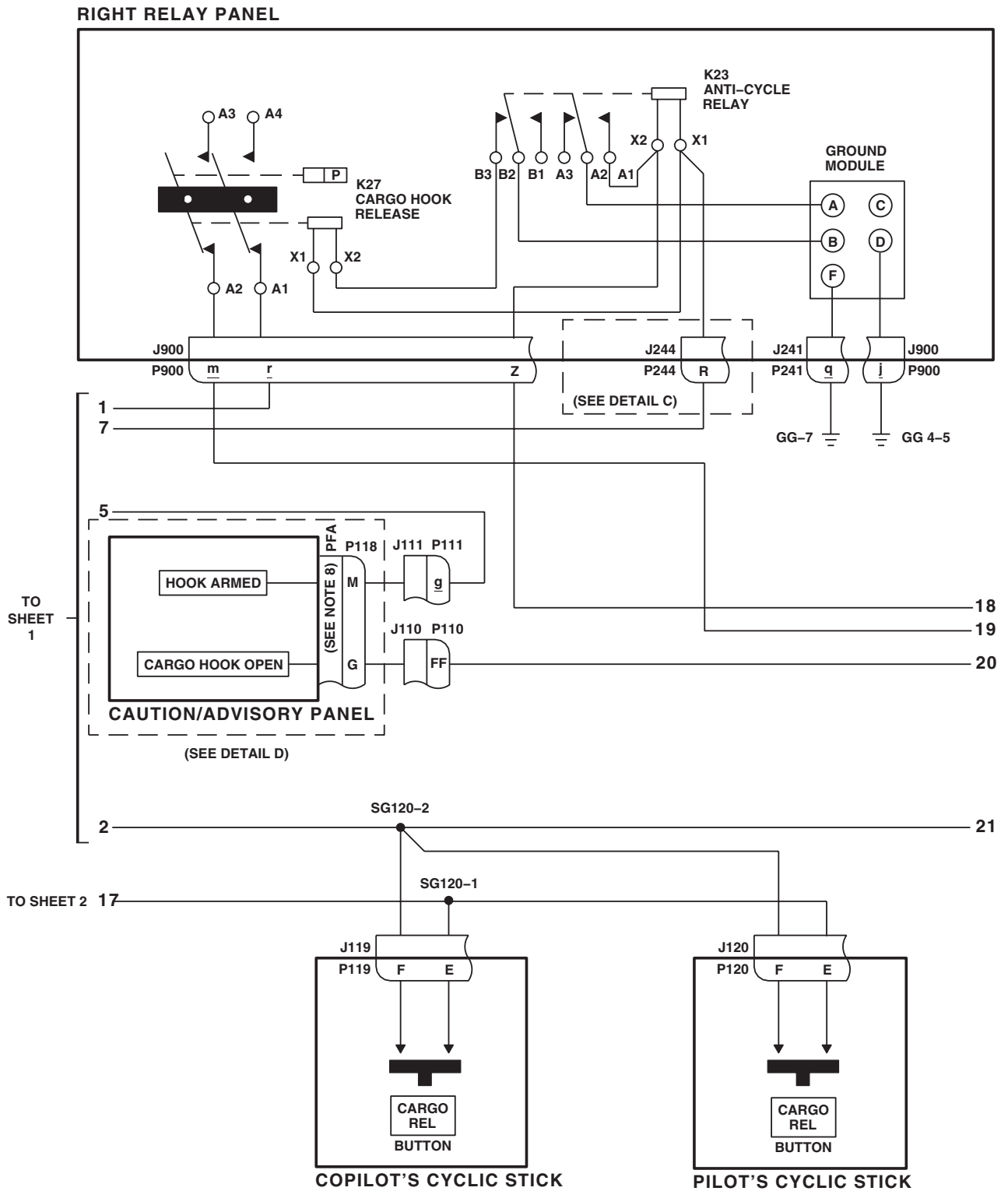


Figure 2. Cargo Hook System Schematic Diagram. (Sheet 2 of 5)

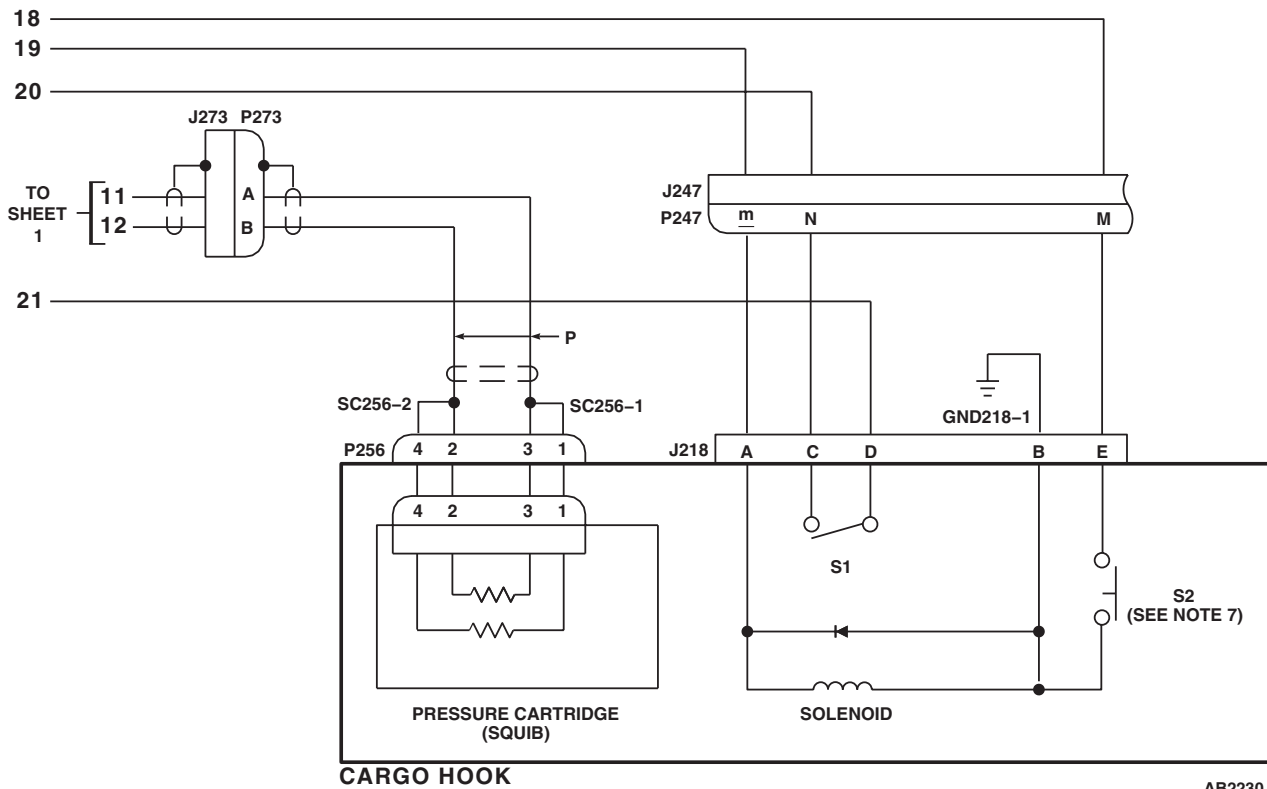
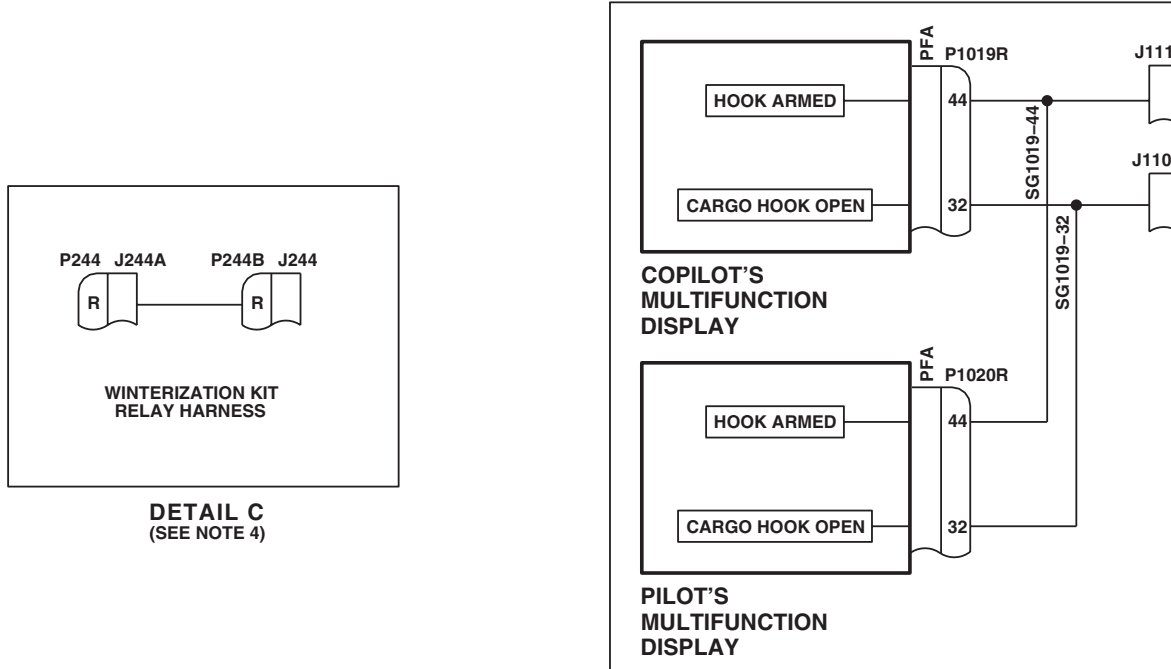
SCHEMATICS - Continued



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Figure 2. Cargo Hook System Schematic Diagram. (Sheet 3 of 5)

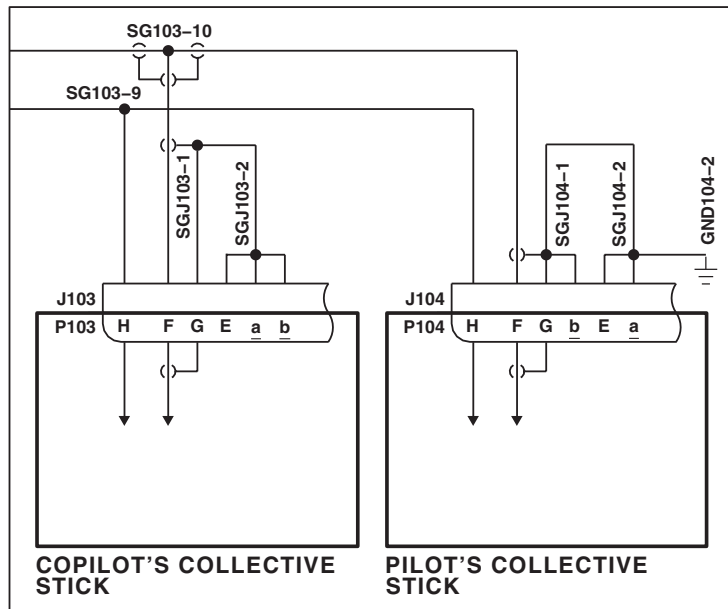
SCHEMATICS - Continued



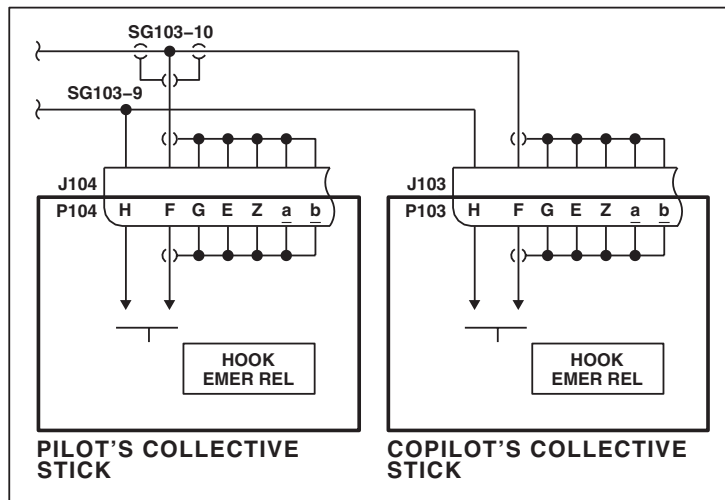
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Figure 2. Cargo Hook System Schematic Diagram. (Sheet 4 of 5)

SCHEMATICS - Continued



DETAIL B  
(SEE NOTE 11)



DETAIL A  
(SEE NOTE 10)

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Figure 2. Cargo Hook System Schematic Diagram. (Sheet 5 of 5)

END OF WORK PACKAGE



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**UNIT LEVEL****UTILITY SYSTEM**

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**CREWMAN'S CARGO HOOK PENDANT UH60A UH60L**

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**INITIAL SETUP:****Test Equipment**

Insulation Test Set, 412  
Multimeter, AN/PSM-45A

**References**

WP 1253 00  
WP 1256 00  
WP 1257 00  
WP 1747 00

**Tools and Special Tools**

Electrical Repairer Toolkit, SC 5180-99-B06

**Personnel Required**

Aircraft Electrician MOS 15F (1)

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**CREWMAN'S CARGO HOOK PENDANT OPERATIONAL/TROUBLESHOOTING PROCEDURE****UH60A UH60L****PROCEDURE****Step****NOTE**

See Figure 1 for Crewman's Cargo Hook Pendant diagram and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

1. Visually check grip for cracks.

**Indication/Condition**

Grip shall not be cracked.

**Corrective Action**

If Indication/Condition is not as specified, replace pendant assembly (WP 1256 00).

**Step**

2. Inspect switch guards for spring tension.

**Indication/Condition**

Guard tension shall be enough to keep guard in place over switch.

**Corrective Action**

If Indication/Condition is not as specified, replace guard spring (WP 1253 00).

**Step**

3. Remove cover from pendant assembly and inspect for broken or burnt wires, security of components, and general condition.

---

**CREWMAN'S CARGO HOOK PENDANT OPERATIONAL/TROUBLESHOOTING PROCEDURE**  
**UH60A UH60L** - Continued**Indication/Condition**

Wires shall not be broken or burnt, and components shall be secure and in good condition.

**Corrective Action**

If Indication/Condition is not as specified, repair/replace as required (WP 1747 00).

**Step**

4. Measure resistance between P267-B and P267-F.

**Indication/Condition**

Meter shall indicate infinite resistance.

**Corrective Action**

If Indication/Condition is not as specified, replace NORMAL RLSE switch (WP 1257 00).

**Step**

5. With NORMAL RLSE switch pressed, measure resistance between P267-B and P267-F.

**Indication/Condition**

Meter shall indicate continuity.

**Corrective Action**

1. If Indication/Condition is not as specified, check resistance between P267-B and NORMAL RLSE switch terminal 1, and between P267-F and NORMAL RLSE switch terminal 2.
  - a. If continuity is present, replace NORMAL RLSE switch (WP 1257 00).
  - b. If continuity is not present, repair/replace wiring as required (WP 1747 00).

**Step**

6. Check resistance between P267-C and P267-E.

**Indication/Condition**

Meter shall indicate infinite resistance.

**Corrective Action**

If Indication/Condition is not as specified, replace EMER RLSE switch (WP 1257 00).

**Step**

7. With EMER RLSE switch pressed, measure resistance between P267-C and P267-E.

**Indication/Condition**

Meter shall indicate continuity.

**Corrective Action**

1. If Indication/Condition is not as specified, check resistance between P267-C and EMER RLSE switch terminal 1, and between P267-E and EMER RLSE switch terminal 2.
  - a. If continuity is present, replace EMER RLSE switch (WP 1257 00).



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**CREWMAN'S CARGO HOOK PENDANT OPERATIONAL/TROUBLESHOOTING PROCEDURE**  
**UH60A UH60L** - Continued

- b. If continuity is not present, repair/replace wiring as required (WP 1747 00).

**Step**

8. Connect insulation test set to P267-B and P267-F and apply 1000 Vrms, 60 Hz.

**Indication/Condition**

The insulation resistance meter shall indicate greater than 1000 megohms.

**Corrective Action**

1. If Indication/Condition is not as specified, replace NORMAL RLSE switch (WP 1257 00).
2. If meter indicates greater than 100 megohms, but less than 1000 megohms, repair/replace wiring as required (WP 1747 00).

**Step**

9. Connect insulation test set to P267-C and P267-E and apply 1000 Vrms, 60 Hz.

**Indication/Condition**

The insulation resistance meter shall indicate greater than 1000 megohms.

**Corrective Action**

1. If Indication/Condition is not as specified, replace EMER RLSE switch (WP 1257 00).
2. If meter indicates greater than 100 megohms, but less than 1000 megohms, repair/replace wiring as required (WP 1747 00).

## LOCATION

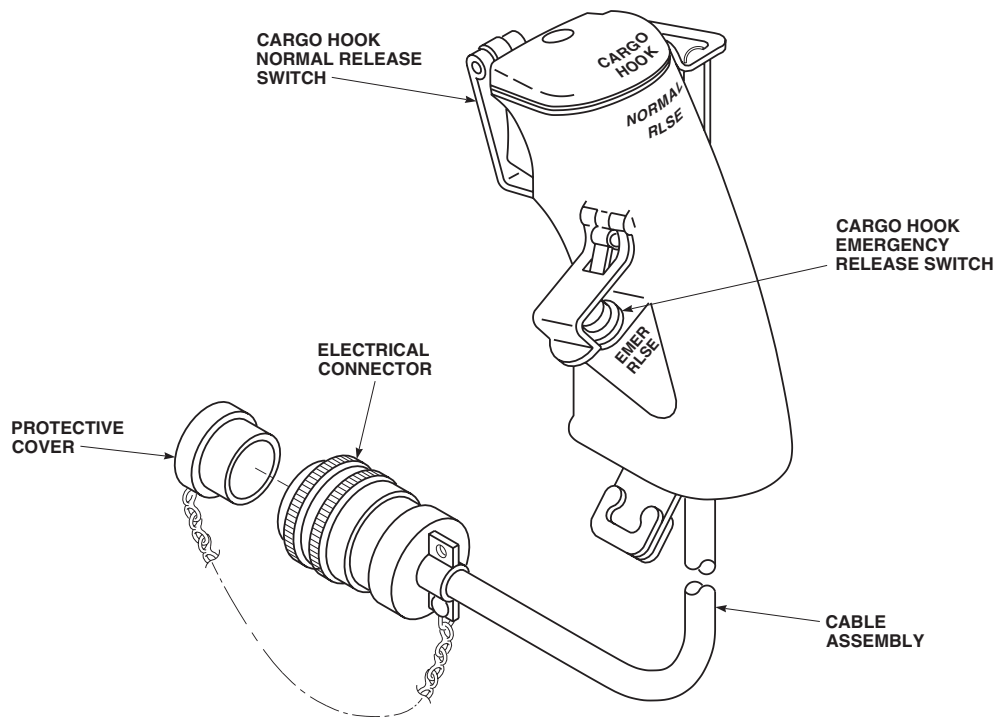
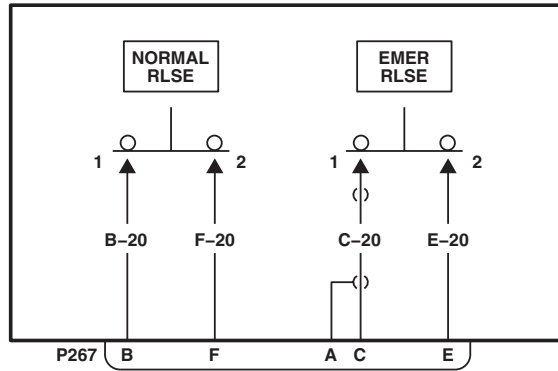
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Figure 1. Crewman's Cargo Hook Pendant.

## SCHEMATICS

**NOTE**  
SWITCHES ARE PUSHBUTTON,  
NORMALLY OPEN.



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Figure 2. Crewman's Cargo Hook Pendant Schematic Diagram.

END OF WORK PACKAGE



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**UNIT LEVEL**  
**UTILITY SYSTEM**  
**BLADE DE-ICE TEST PANEL (AVIM)**  
**This WP supersedes WP 0148 00, dated 17 April 2006.**

---

**INITIAL SETUP:****Test Equipment**

Locally-Made Blade De-ice Test Set Figure 155,  
WP 1805 00  
Multimeter, AN/PSM-45A

WP 1260 00

WP 1261 00

WP 1265 00

WP 1747 00

WP 1805 00

**Tools and Special Tools**

Electrical Repairer Toolkit, SC 5180-99-B06

**Equipment Condition**

0 to 40 vdc Bench Power Available

115 vac, 400 Hz, 3ø Bench Power Available

**Personnel Required**

Aircraft Electrician MOS 15F (1)

**References**

WP 1241 00

WP 1242 00

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**SPECIAL INSTRUCTIONS****NOTE**

See Figure 1 for component diagram and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

The blade de-ice test panel operational/troubleshooting procedure is subdivided as follows:

- Setup
- OAT Resistance
- SYNC/EOT
- Tail Rotor Fail
- Main Rotor Fail
- Shutdown

**SETUP OPERATIONAL/TROUBLESHOOTING PROCEDURE****PROCEDURE****Step**

1. Connect locally-made TEST CABLE to J3 of blade de-ice test set (test set) (WP 1805 00) and J1 of BLADE DE-ICE TEST panel (test panel).

**SETUP OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued

2. Place all test set switches to OFF or NORM, SYNC switch to 0V, and test panel switch to NORM.
3. Connect locally-made 28 VDC CABLE to J1 of test set and 28 vdc bench power.
4. Connect locally-made 115 VAC CABLE to J2 of test set and 115V, 3ø 400 Hz bench power.

**Indication/Condition**

115 vac cable shall be connected to test set and 115 v bench power.

**Corrective Action**

None Required

**OAT RESISTANCE OPERATIONAL/TROUBLESHOOTING PROCEDURE**  
**PROCEDURE**
**Step**

1. Do SETUP, in this work package, if not already done.
2. Measure resistance or check continuity per OAT RESISTANCE.

**Table 1. OAT Resistance.**

| TEST<br>PANEL<br>SWITCH<br>POSITION | TEST<br>SET,<br>TEST<br>JACKS | METER<br>INDICATION<br>(OHM) | CORRECTIVE<br>ACTION<br>(CONTINUITY)      |
|-------------------------------------|-------------------------------|------------------------------|-------------------------------------------|
| NORM                                | W - X                         | 13.167 -<br>13.433 (K)       | Replace<br>R14<br>(WP 1265 00)            |
| SYNC 1                              | W - X                         | 13.167 -<br>13.433 (K)       | Replace S1<br>(WP 1260 00)                |
| SYNC 2                              | W - X                         | 13.167 -<br>13.433 (K)       | Replace S1<br>(WP 1260 00)                |
| OAT                                 | W - X                         | 0                            | J1-W-S1D-C,<br>S1D-2-S1E-2,<br>S1E-C-J1-X |
| EOT                                 | W - X                         | 97.6 - 98.6                  | Replace R7<br>(WP 1265 00)                |
| NORM                                | W - Z                         | 0                            | J1-W-S1D-C,<br>S1D-5-J1-Z                 |

## OAT RESISTANCE OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Table 1. OAT Resistance. - Continued

| TEST<br>PANEL<br>SWITCH<br>POSITION | TEST<br>SET,<br>TEST<br>JACKS | METER<br>INDICATION<br>(OHM) | CORRECTIVE<br>ACTION<br>(CONTINUITY)                      |
|-------------------------------------|-------------------------------|------------------------------|-----------------------------------------------------------|
| SYNC 1                              | W - Z                         | 0                            | J1-W-S1D-C,<br>S1D-4-S1D-5,<br>S1D-5-J1-Z                 |
| SYNC 2                              | W - Z                         | 0                            | J1-W-S1D-C,<br>S1D-3-S1D-4,<br>S1D-4-S1D-5,<br>S1D-5-J1-Z |
| OAT                                 | W - Z                         | Infinity                     | Replace S1<br>(WP 1260 00)                                |
| EOT                                 | W - Z                         | Infinity                     | Replace S1<br>(WP 1260 00)                                |
| NORM                                | X - Y                         | 0                            | J1-X-S1E-C,<br>S1E-5-J1-V                                 |
| SYNC 1                              | X - Y                         | 0                            | J1-X-S1E-C,<br>S1E-4-S1E-5,<br>S1E-5-J1-Y                 |
| SYNC 2                              | X - Y                         | 0                            | J1-X-S1E-C,<br>S1E-3-S1E-4,<br>S1E-4-S1E-5,<br>S1E-5-J1-V |
| OAT                                 | X - Y                         | Infinity                     | Replace S1<br>(WP 1260 00)                                |
| EOT                                 | X - Z                         | 0                            | Replace S1<br>(WP 1260 00)                                |

**Indication/Condition**

Meter shall indicate corresponding resistance or continuity.

**Corrective Action**

1. If Indication/Condition is not as specified, do corresponding CORRECTIVE ACTION replacement or continuity check.

**OAT RESISTANCE OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued

- a. If continuity is present, replace switch S1 (WP 1260 00).
- b. If continuity is not present, repair/replace wiring as required (WP 1747 00).

**Step**

3. If no other check is to be done, go to SHUTDOWN, in this work package.

**Indication/Condition**

Shutdown shall be performed.

**Corrective Action**

None Required

**SYNC/EOT OPERATIONAL/TROUBLESHOOTING PROCEDURE****PROCEDURE****Step**

1. Do SETUP, in this work package, if not already done.
2. Check continuity per SYN/EOT CONTINUITY.

**Table 2. SYNC/EOT Continuity.**

| TEST<br>PANEL<br>SWITCH<br>POSITION | TEST<br>SET,<br>TEST<br>JACKS | METER<br>INDICATION<br>(OHM) | CORRECTIVE<br>ACTION<br>(CONTINUITY)                      |
|-------------------------------------|-------------------------------|------------------------------|-----------------------------------------------------------|
| NORM                                | N - P                         | 0                            | J1-N-S1A-5,<br>S1A-C-J1-P                                 |
| OAT                                 | N - P                         | 0                            | J1-N-S1A-5,<br>S1A-5-S1A-1,<br>S1A-1-S1A-1,<br>S1A-C-J1-P |
| EOT                                 | N - P                         | 0                            | J1-N-S1A-5,<br>S1A-5-S1A-1,<br>S1A-1-S1A-C,<br>S1A-C-J1-P |
| EOT                                 | R - S                         | Infinity                     | Replace<br>K3<br>(WP 1265 00)                             |
| EOT                                 | R - U                         | Infinity                     | Replace<br>K3<br>(WP 1265 00)                             |



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**SYNC/EOT OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued**Indication/Condition**

Meter shall indicate corresponding resistance.

**Corrective Action**

1. If Indication/Condition is not as specified, do corresponding CORRECTIVE ACTION replacement or continuity check.
  - a. If continuity is present, replace switch S1 (WP 1260 00).
  - b. If continuity is not present, repair/replace wiring as required (WP 1747 00).

**Step**

3. Place test panel switch to NORM.
4. Place test set POWER 28 VDC and 115 VAC switches to ON.

**Indication/Condition**

- a. Test set 28 VDC and 115 VAC indicator lamps shall go on.
- b. Test panel information plate lamps shall light evenly.

**Corrective Action**

1. If Indication/Condition a. is not as specified, replace test set.
2. If Indication/Condition b. is not as specified, replace information plate lamp (WP 1242 00).
  - a. If trouble remains, replace information plate (WP 1242 00).
3. If none of the information plate lamps light, check continuity between J1-b and J3-1, and between J1-c and J3-2.
  - a. If continuity is present, replace information plate (WP 1242 00).
  - b. If continuity is not present, repair/replace wiring (WP 1747 00).

**Step**

5. Place test set panel LTS switch 5V/115VAC to 5V or 115VAC.
6. Place test set FAULT SELECT ICE DET switch to FAULT.
7. Place test panel switch to SYNC 1.
8. Check for -33.0 to -27.0 vdc between test set TEST POINTS P and T.

**Indication/Condition**

Meter shall indicate -33.0 to -27.0 vdc.

**Corrective Action**

If Indication/Condition is not as specified, go to VOLTAGE OUTPUT MEASURED BETWEEN TEST SET TEST POINTS P AND T IS NOT BETWEEN -33.0 AND -27.0 VDC, in this work package.

**Step**

9. Place test set FAULT SELECT ICE DET switch to NORM.
10. Place test panel switch to EOT.
11. Check continuity between test set TEST POINTS R and S, and between R and U.

---

**SYNC/EOT OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued**Indication/Condition**

Meter shall not indicate continuity.

**Corrective Action**

If Indication/Condition is not as specified, go to OPEN CIRCUIT DOES NOT EXIST BETWEEN TEST SET TEST POINTS R AND S, AND R AND U WHEN FAULT SELECT ICE DET SWITCH IS PLACED TO NORM AND TEST PANEL SWITCH IW PLACED TO EOT, in this work package.

**Step**

12. Place test set SYNC switch to -5V.
13. Check resistance between test set TEST POINTS R and S, and between R and U.

**Indication/Condition**

Meter shall indicate less than 0.5 ohm resistance.

**Corrective Action**

If Indication/Condition is not as specified, go to RESISTANCE CHECK FOR LESS THAN 0.5 OHMS DOES NOT EXIST BETWEEN TEST POINTS R AND S, AND R AND U, in this work package.

**Step**

14. Place test set SYNC switch to 0V.
15. Check continuity between test set TEST POINTS R and S, and between R and U.

**Indication/Condition**

Meter shall not indicate continuity.

**Corrective Action**

If Indication/Condition is not as specified, replace relay K3 (WP 1265 00).

**Step**

16. Place test set SYNC switch to -30V.
17. Check continuity between test set TEST POINTS R and S, and between R and U.

**Indication/Condition**

Meter shall not indicate continuity.

**Corrective Action**

If Indication/Condition is not as specified, go to OPEN CIRCUIT DOES NOT EXIST BETWEEN TEST SET TEST POINTS R AND S, AND R AND U, in this work package.

**Step**

18. Place test set SYNC switch to OV.

**SYNC/EOT OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued

19. Place test panel switch to the following positions and check for continuity between the following points:

| <b>SWITCH POSITION</b> | <b>CHECK CONTINUITY BETWEEN</b>               |
|------------------------|-----------------------------------------------|
| NORM                   | TEST POINT R and S1B-5 TEST POINT R and S1C-5 |
| SYNC 1                 | TEST POINT R and S1B-4 TEST POINT R and S1C-4 |
| SYNC 2                 | TEST POINT R and S1B-3 TEST POINT R and S1C-3 |
| OAT                    | TEST POINT R and S1B-2 TEST POINT R and S1C-2 |

**Indication/Condition**

Meter shall indicate continuity.

**Corrective Action**

If Indication/Condition is not as indicated, replace switch S1 (WP 1260 00).

**Step**

20. If no other check is to be done, go to SHUTDOWN, in this work package.

**Indication/Condition**

Shutdown shall be performed.

**Corrective Action**

None Required

## **TAIL ROTOR FAIL OPERATIONAL/TROUBLESHOOTING PROCEDURE**

### **PROCEDURE**

**Step**

1. Do SETUP, in this work package, if not already done.
2. Momentarily press test panel PWR TAIL RTR lamp.

**Indication/Condition**

Lamp shall light when pressed and shall go off when released.

**Corrective Action**

1. If Indication/Condition is not as specified, replace lamp DS2 (WP 1241 00).
    - a. If trouble remains, check continuity between the following:
      - J2-23 and DS2-2
      - J2-22 and DS2-3
      - J2-14 and DS2-1
- (1) If continuity is present, replace lamp switch assembly (WP 1261 00).

---

**TAIL ROTOR FAIL OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued

- (2) If continuity is not present, repair/replace wiring (WP 1747 00).

**Step**

3. Check voltage between test set TEST POINTS f and T.

**Indication/Condition**

Meter shall indicate 0 vdc.

**Corrective Action**

If Indication/Condition is not as specified, replace relay K2 (WP 1265 00).

**Step**

4. Place test set TAIL ROTOR Aø and Bø switches to ON.  
5. Place test set FAULT SELECT CONT PNL switch to FAULT.

**Indication/Condition**

Test panel PWR TAIL RTR lamp shall light.

**Corrective Action**

If Indication/Condition is not as specified, go to TEST PANEL PWR TAIL RTR LAMP DOES NOT LIGHT WHEN TEST SET TAIL ROTOR Aø AND Bø SWITCHES ARE PLACED TO ON AND FAULT SELECT CONT PNL SWITCH IS PLACED TO FAULT, in this work package.

**Step**

6. Check voltage between test set TEST POINTS G and T, and between J and T.

**Indication/Condition**

- a. Meter shall indicate 0 vdc between G and T.  
b. Meter shall indicate 0 vdc between J and T.

**Corrective Action**

1. If Indication/Condition a. is not as specified, replace diode CR6 (WP 1265 00).  
2. If Indication/Condition b. is not as specified, replace diode CR13 (WP 1265 00).

**Step**

7. Check voltage between test set TEST POINTS K and f.

**Indication/Condition**

Meter shall indicate 0.6 to 0.8 vdc.

**Corrective Action**

1. If Indication/Condition is not as specified, check continuity between J1-K and E51, and between J1-f and E47.  
a. If continuity is present, replace diode CR14 (WP 1265 00).  
b. If continuity is not present, repair/replace wiring (WP 1747 00).

---

**TAIL ROTOR FAIL OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Step**

8. Place test set FAULT SELECT CONT PNL switch to NORM.

**Indication/Condition**

Test panel PWR TAIL RTR lamp shall go off.

**Corrective Action**

If Indication/Condition is not as specified, replace relay K2 (WP 1265 00).

**Step**

9. Check voltage between test set TEST POINTS  $\underline{f}$  and T.

**Indication/Condition**

Meter shall indicate 0 vdc.

**Corrective Action**

If Indication/Condition is not as specified, replace relay K2 (WP 1265 00).

**Step**

10. Place test set FAULT SELECT TR FAIL switch to FAULT.

**Indication/Condition**

Test set PWR TAIL RTR lamp shall light.

**Corrective Action**

1. If Indication/Condition is not as specified, check continuity between J1-G and E45.
  - a. If continuity is present, replace diode CR6 (WP 1265 00).
  - b. If continuity is not present, repair/replace wiring (WP 1747 00).

**Step**

11. Place test set CURRENT TRANSFORMER Aø switch to ON.

**Indication/Condition**

- a. Test panel PWR TAIL RTR lamp shall remain on.
- b. Test panel PWR MAIN RTR lamp shall be off.

**Corrective Action**

If Indication/Conditions are not as specified, replace diode CR5 (WP 1265 00).

**Step**

12. Place test set CURRENT TRANSFORMER Aø switch to OFF.
13. Place test set TAIL ROTOR Aø and Bø switches to OFF.

**Indication/Condition**

Test panel PWR TAIL lamp shall go off in less than 2 seconds.

---

**TAIL ROTOR FAIL OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Corrective Action**

If Indication/Condition is not as specified, replace capacitor C2 (WP 1265 00).

**Step**

14. Place test set FAULT SELECT TR FAIL switch to NORM.
15. Place test set TAIL ROTOR Bø and Cø switches to ON.
16. Place test set FAULT SELECT TR FAIL switch to FAULT.

**Indication/Condition**

Test panel PWR TAIL RTR lamp shall go ON.

**Corrective Action**

If Indication/Condition is not as specified, go to TEST PANEL PWR TAIL RTR LAMP DOES NOT LIGHT WHEN TEST SET TAIL ROTOR Bø AND Cø SWITCHES ARE PLACED TO ON AND FAULT SELECT TR FAIL SWITCH IS PLACED TO FAULT, in this work package.

**Step**

17. Place test set TAIL ROTOR Bø and Cø switches to OFF.

**Indication/Condition**

Test panel PWR TAIL RTR lamp shall go off in less than 2 seconds.

**Corrective Action**

If Indication/Condition is not as specified, replace capacitor C2 (WP 1265 00).

**Step**

18. Place test set FAULT SELECT TR FAIL switch to NORM.
19. Place test set TAIL ROTOR Aø and Cø switches to ON.
20. Place test set FAULT SELECT TR FAIL switch to FAULT.

**Indication/Condition**

Test panel PWR TAIL RTR lamp shall go on.

**Corrective Action**

If Indication/Condition is not as specified, go to TEST PANEL PWR TAIL RTR LAMP DOES NOT LIGHT WHEN TEST SET TAIL ROTOR Aø AND Bø SWITCHES ARE PLACED TO ON AND FAULT SELECT TR FAIL SWITCH IS PLACED TO FAULT, this work package.

**Step**

21. Place test set TAIL ROTOR Aø and Cø switches to OFF.

**Indication/Condition**

Test panel PWR TAIL RTR lamp shall go off in less than 2 seconds.

**Corrective Action**

If Indication/Condition is not as specified, replace capacitor C2 (WP 1265 00).

---

**TAIL ROTOR FAIL OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Step**

22. Place test set FAULT SELECT TR FAIL switch to NORM.
23. If no other check is to be done, go to SHUTDOWN, in this work package.

**Indication/Condition**

Shutdown shall be performed.

**Corrective Action**

None Required

**MAIN ROTOR FAIL OPERATIONAL/TROUBLESHOOTING PROCEDURE  
PROCEDURE****Step**

1. Do SETUP, in this work package, if not already done.
2. Momentarily press test panel PWR MAIN RTR lamp.

**Indication/Condition**

Test panel PWR MAIN RTR lamp shall go on when pressed and go off when released.

**Corrective Action**

1. If Indication/Condition is not as specified, replace lamp DS1 (WP 1241 00).
  - a. If trouble remains, check continuity between the following:
    - J2-21 and DS1-3
    - J2-24 and DS1-2
    - J2-15 and DS1-1
  - b. If continuity is present, replace lamp switch assembly (WP 1261 00).
  - c. If continuity is not present, repair/replace wiring (WP 1747 00).

**Step**

3. Check voltage between test set TEST POINTS e and T.

**Indication/Condition**

Meter shall indicate 0 vdc.

**Corrective Action**

If Indication/Condition is not as specified, replace relay K1 (WP 1265 00).

**Step**

4. Place test set CURRENT TRANSFORMER Aø switch to ON.
5. Place test set FAULT SELECT CONT PNL switch to FAULT.

**Indication/Condition**

Test panel PWR MAIN RTR lamp shall go on.

---

**MAIN ROTOR FAIL OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued**Corrective Action**

If Indication/Condition is not as specified, go to TEST PANEL PWR TAIL RTR LAMP DOES NOT LIGHT WHEN TEST SET CURRENT TRANS Aø SWITCH IS PLACED TO ON AND FAULT SELECT CONT PNL SWITCH IS PLACED TO FAULT, in this work package.

**Step**

6. Check for voltage between test set TEST POINTS K and e.

**Indication/Condition**

Meter shall indicate between 0.6 and 0.8 vdc.

**Corrective Action**

If Indication/Condition is not as specified, replace diode CR15 (WP 1265 00).

**Step**

7. Place test set FAULT SELECT CONT PNL switch to NORM.

**Indication/Condition**

Test panel PWR MAIN RTR lamp shall go off.

**Corrective Action**

If Indication/Condition is not as specified, replace relay K1 (WP 1265 00).

**Step**

8. Check for voltage between test set TEST POINTS e and T.

**Indication/Condition**

Meter shall indicate 0 vdc.

**Corrective Action**

If Indication/Condition is not as specified, replace relay K1 (WP 1265 00).

**Step**

9. Place test set FAULT SELECT MR FAIL switch to FAULT.

**Indication/Condition**

Test panel PWR MAIN RTR lamp shall light.

**Corrective Action**

1. If Indication/Condition is not as specified, check continuity between J1-J and E54.
  - a. If continuity is present, replace diode CR13 (WP 1265 00).
  - b. If continuity is not present, repair/replace wiring (WP 1747 00).

**Step**

10. Place test set TAIL ROTOR Aø and Bø switches to ON.

**Indication/Condition**

Test panel PWR MAIN RTR lamp shall remain on and test panel PWR TAIL RTR lamp shall be off.



---

**MAIN ROTOR FAIL OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued**Corrective Action**

If Indication/Condition is not as specified, replace diode CR12 (WP 1265 00).

**Step**

11. Place test set TAIL ROTOR Aø and Bø switches to OFF.
12. Place test set CURRENT TRANSFORMER Aø switch to OFF.

**Indication/Condition**

Test panel PWR MAIN RTR lamp shall go off in less than 5 seconds.

**Corrective Action**

If Indication/Condition is not as specified, replace capacitor C3 (WP 1265 00).

**Step**

13. Place test set FAULT SELECT MR FAIL switch to NORM.
14. Place test set CURRENT TRANSFORMER Bø switch to ON.
15. Place test set FAULT SELECT MR FAIL switch to FAULT.

**Indication/Condition**

Test panel PWR MAIN RTR lamp shall go on.

**Corrective Action**

If Indication/Condition is not as specified, replace diode CR9 (WP 1265 00).

**Step**

16. Place test set CURRENT TRANSFORMER Bø switch to OFF.

**Indication/Condition**

Test panel PWR MAIN RTR lamp shall go off in less than 5 seconds.

**Corrective Action**

If Indication/Condition is not as specified, replace capacitor C3 (WP 1265 00).

**Step**

17. Place test set FAULT SELECT MR FAIL switch to NORM.
18. Place test set CURRENT TRANSFORMER Cø switch to ON.
19. Place test set FAULT SELECT MR FAIL switch to FAULT.

**Indication/Condition**

Test panel PWR MAIN RTR lamp shall go on.

**Corrective Action**

If Indication/Condition is not as specified, replace diode CR10 (WP 1265 00).

**Step**

20. Place test set CURRENT TRANSFORMER Cø switch to OFF.

---

**MAIN ROTOR FAIL OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Indication/Condition**

Test panel PWR MAIN RTR lamp shall go off in less than 5 seconds.

**Corrective Action**

If Indication/Condition is not as specified, replace capacitor C3 (WP 1265 00).

**Step**

21. Place test set FAULT SELECT MR FAIL switch to NORM and then place test set DETECT switch to ON.
22. Place test set FAULT SELECT MR FAIL switch to FAULT.
23. Place test set CURRENT TRANSFORMER A $\phi$ , B $\phi$ , and C $\phi$  switches to ON.

**Indication/Condition**

Test panel PWR MAIN RTR lamp shall remain off.

**Corrective Action**

1. If Indication/Condition is not as specified, check for 95,000 to 105,000 ohms resistance across resistor R4.
  - a. If meter reading is not as specified, replace resistor R4 (WP 1265 00).
  - b. If meter reading is 100,000 ohms, replace zener diode VR8 (WP 1265 00).

**Step**

24. If no other check is to be done, go to SHUTDOWN, in this work package.

**Indication/Condition**

Shutdown shall be performed.

**Corrective Action**

None Required

**SHUTDOWN OPERATIONAL/TROUBLESHOOTING PROCEDURE  
PROCEDURE****Step**

1. Place all test set switches to OFF or NORM.
2. Disconnect test set from test panel.

**Indication/Condition**

Test set shall be disconnected.

**Corrective Action**

None Required

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**VOLTAGE OUTPUT MEASURED BETWEEN TEST SET TEST POINTS P AND T IS NOT BETWEEN -33.0 AND -27.0 VDC****SYMPTOM**

Voltage output measured between test set TEST POINTS P and T is not between -33.0 and -27.0 vdc when test set FAULT SELECT ICE DET switch is placed to FAULT and test panel switch is placed to SYNC 1.

**MALFUNCTION**

Voltage output measured between test set TEST POINTS P and T is not between -33.0 and -27.0 vdc when test set FAULT SELECT ICE DET switch is placed to FAULT and test panel switch is placed to SYNC 1.

**CORRECTIVE ACTION****1. Check for -33.0 to -27.0 vdc at zener diode VR1 anode.**

- a. If voltage is as specified, repair/replace wiring as required (WP 1747 00). Go to **7**.
- b. If voltage is not as specified, go to **2**.

**2. Is voltage output zero?**

- a. If voltage output is zero, go to **3**.
- b. If voltage output is not zero, go to **6**.

**3. Check diode CR1 for open circuit.**

- a. If open circuit is present, replace diode CR1 (WP 1265 00). Go to **7**.
- b. If open circuit is not present, go to **4**.

**4. Check resistor R1 for 41,000 to 45,000 ohms.**

- a. If resistance is as specified, go to **5**.
- b. If resistance is not as specified, replace resistor R1 (WP 1265 00). Go to **7**.

**5. Check capacitor C1 for continuity.**

- a. If continuity is present, replace capacitor C1 (WP 1265 00). Go to **7**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **7**.

**6. Check resistor R1 for 41,000 to 45,000 ohms.**

- a. If resistance is as specified, replace zener diode VR1 (WP 1265 00). Go to **7**.
- b. If resistance is not as specified, replace resistor R1 (WP 1265 00). Go to **7**.

**7. Procedure completed.**

---

**OPEN CIRCUIT DOES NOT EXIST BETWEEN TEST SET TEST POINTS R AND S, AND R AND U WHEN FAULT SELECT ICE DET SWITCH IS PLACED TO NORM AND TEST PANEL SWITCH IS PLACED TO EOT****SYMPTOM**

Open circuit does not exist between test set TEST POINTS R and S, and R and U when FAULT SELECT ICE DET switch is placed to NORM and test panel switch is placed to EOT.

**MALFUNCTION**

Open circuit does not exist between test set TEST POINTS R and S, and R and U when FAULT SELECT ICE DET switch is placed to NORM and test panel switch is placed to EOT.

**CORRECTIVE ACTION**

- 1. Check continuity between TEST POINT R and switch S1B-C.**
  - a. If continuity is present, go to **2**.
  - b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **15**.
- 2. Check continuity between S1B-C and S1B-1.**
  - a. If continuity is present, go to **3**.
  - b. If continuity is not present, replace test panel switch (WP 1260 00). Go to **15**.
- 3. Check continuity between S1B-1 and P2-20.**
  - a. If continuity is present, go to **4**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **15**.
- 4. Check continuity between S1B-C and S1C-C.**
  - a. If continuity is present, go to **5**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **15**.
- 5. Check continuity between S1C-C and S1C-1.**
  - a. If continuity is present, go to **6**.
  - b. If continuity is not present, replace test panel switch (WP 1260 00). Go to **15**.
- 6. Check continuity between S1C-1 and P2-19.**
  - a. If continuity is present, go to **7**.
  - b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **15**.
- 7. Check for 26 to 28 vdc between S1A-1 and ground.**
  - a. If voltage is as specified, go to **8**.
  - b. If voltage is not as specified, repair/replace wiring as required (WP 1747 00). Go to **15**.
- 8. Check for 14.5 to 15.5 vdc between transistor Q6 emitter and ground.**
  - a. If voltage is as specified, go to **9**.
  - b. If voltage is not as specified, replace zener diode VR7 (WP 1265 00). Go to **15**.
- 9. Check for 0.6 to 1.0 vdc between transistor Q4 base and ground.**
  - a. If voltage is as specified, go to **10**.
  - b. If voltage is not as specified, go to **14**.

**OPEN CIRCUIT DOES NOT EXIST BETWEEN TEST SET TEST POINTS R AND S, AND R AND U WHEN FAULT SELECT ICE DET SWITCH IS PLACED TO NORM AND TEST PANEL SWITCH IS PLACED TO EOT**  
- Continued

**10. Check for 0.5 to 1.5 vdc between transistor Q4 collector and ground.**

- a. If voltage is as specified, go to **11**.
- b. If voltage is not as specified, replace transistor Q4 (WP 1265 00). Go to **15**.

**11. Check for 26 to 28 vdc between transistor Q5 collector and ground.**

- a. If voltage is as specified, go to **12**.
- b. If voltage is not as specified, replace transistor Q5 (WP 1265 00). Go to **15**.

**12. Check diode CR16 for short.**

- a. If short is present, replace diode CR16 (WP 1265 00). Go to **15**.
- b. If short is not present, go to **13**.

**13. Check continuity between:**

- **Relay K3-A2 and Test Point S**
- **Relay K3-B2 and Test Point U**

- a. If continuity is present, replace relay K3 (WP 1265 00). Go to **15**.
- b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **15**.

**14. Check for 13 to 14 vdc between transistor Q6 collector and ground.**

- a. If voltage is as specified, replace transistor Q4 and/or associated circuitry (WP 1265 00). Go to **15**.
- b. If voltage is not as specified, replace transistor Q5 and/or associated circuitry (WP 1265 00). Go to **15**.

**15. Procedure completed.**

**RESISTANCE CHECK FOR LESS THAN 0.5 OHMS DOES NOT EXIST BETWEEN TEST POINTS R AND S, AND R AND U**

**SYMPTOM**

Resistance check for less than 0.5 ohms does not exist between test points R and S, and R and U when test set SYNC switch is placed to -5V, FAULT SELECT ICE DET switch is placed to NORM, and test panel switch is placed to EOT.

**MALFUNCTION**

Resistance check for less than 0.5 ohms does not exist between test points R and S, and R and U when test set SYNC switch is placed to -5V, FAULT SELECT ICE DET switch is placed to NORM, and test panel switch is placed to EOT.

**CORRECTIVE ACTION**

**1. Check for 2.0 to 4.0 vdc between transistor Q4 collector and ground.**

- a. If voltage is as specified, go to **2**.
- b. If voltage is not as specified, go to **4**.

**2. Check for 2 to 3 vdc between transistor Q5 collector and ground.**

- a. If voltage is as specified, go to **3**.
- b. If voltage is not as specified, replace transistor Q5 (WP 1265 00). Go to **6**.

---

**RESISTANCE CHECK FOR LESS THAN 0.5 OHMS DOES NOT EXIST BETWEEN TEST POINTS R AND S, AND R AND U - Continued****3. Check diode CR16 for short or open.**

- a. If short or open is present, replace diode CR16 (WP 1265 00). Go to **6**.
- b. If short or open is not present, replace relay K3 (WP 1265 00). Go to **6**.

**4. Check for -3 to -1 vdc between transistor Q4 base and ground.**

- a. If voltage is as specified, go to **5**.
- b. If voltage is not as specified, replace transistor Q4 and/or associated components (WP 1265 00). Go to **6**.

**5. Check for 15 vdc between zener diode VR7 cathode and ground.**

- a. If voltage is as specified, replace transistor Q5 and/or associated components (WP 1265 00). Go to **6**.
- b. If voltage is not as specified, replace zener diode VR7 (WP 1265 00). Go to **6**.

**6. Procedure completed.****OPEN CIRCUIT DOES NOT EXIST BETWEEN TEST SET TEST POINTS R AND S, AND R AND U WHEN FAULT SELECT ICE DET SWITCH IS PLACED TO NORM AND TEST PANEL SWITCH IS PLACED TO EOT****SYMPTOM**

Open circuit does not exist between test set TEST POINTS R and S, and R and U when test set SYNC switch is placed to -30V, FAULT SELECT ICE DET switch is placed to NORM, and test panel switch is placed to EOT.

**MALFUNCTION**

Open circuit does not exist between test set TEST POINTS R and S, and R and U when test set SYNC switch is placed to -30V, FAULT SELECT ICE DET switch is placed to NORM, and test panel switch is placed to EOT.

**CORRECTIVE ACTION****1. Check for -50 to -45 vdc across zener diode VR1.**

- a. If voltage is as specified, go to **2**.
- b. If voltage is not as specified, replace capacitor C1 and/or associated circuitry (WP 1265 00). Go to **8**.

**2. Check for 14.5 to 15.5 vdc between transistor Q6 emitter and ground.**

- a. If voltage is as specified, go to **3**.
- b. If voltage is not as specified, replace zener diode VR7 (WP 1265 00). Go to **8**.

**3. Check for 0.6 to 1.0 vdc between transistor Q4 base and ground.**

- a. If voltage is as specified, go to **4**.
- b. If voltage is not as specified, go to **7**.

**4. Check for 0.5 to 1.5 vdc between transistor Q4 collector and ground.**

- a. If voltage is as specified, go to **5**.
- b. If voltage is not as specified, replace transistor Q4 (WP 1265 00). Go to **8**.

**5. Check for 26 to 28 vdc between transistor Q5 collector and ground.**

- a. If voltage is as specified, go to **6**.

---

**OPEN CIRCUIT DOES NOT EXIST BETWEEN TEST SET TEST POINTS R AND S, AND R AND U WHEN FAULT SELECT ICE DET SWITCH IS PLACED TO NORM AND TEST PANEL SWITCH IS PLACED TO EOT**

- Continued

- b. If voltage is not as specified, replace transistor Q5 (WP 1265 00). Go to **8**.

**6. Check diode CR16 for short or open.**

- a. If short or open is present, replace diode CR16 (WP 1265 00). Go to **8**.
- b. If short or open is not present, replace relay K3 (WP 1265 00). Go to **8**.

**7. Check for 13 to 14 vdc between transistor Q6 collector and ground.**

- a. If voltage is as specified, replace transistor Q4 and/or associated circuitry (WP 1265 00). Go to **8**.
- b. If voltage is not as specified, replace transistor Q6 and/or associated circuitry (WP 1265 00). Go to **8**.

**8. Procedure completed.****TEST PANEL PWR TAIL RTR LAMP DOES NOT LIGHT WHEN TEST SET TAIL ROTOR AØ AND BØ SWITCHES ARE PLACED TO ON AND FAULT SELECT CONT PNL SWITCH IS PLACED TO FAULT****SYMPTOM**

Test panel PWR TAIL RTR lamp does not light when test set TAIL ROTOR AØ and BØ switches are placed to ON and FAULT SELECT CONT PNL switch is placed to FAULT.

**MALFUNCTION**

Test panel PWR TAIL RTR lamp does not light when test set TAIL ROTOR AØ and BØ switches are placed to ON and FAULT SELECT CONT PNL switch is placed to FAULT.

**CORRECTIVE ACTION****1. Check for 1 to 3 vdc between transistor Q1 collector and ground.**

- a. If voltage is as specified, replace transistor Q1 and/or associated circuitry (WP 1265 00). Go to **4**.
- b. If voltage is not as specified, go to **2**.

**2. Check capacitor C2 for short.**

- a. If short is present, replace capacitor C2 (WP 1265 00). Go to **4**.
- b. If short is not present, go to **3**.

**3. Check diode CR7 for short or open.**

- a. If short or open is present, replace diode CR7 (WP 1265 00). Go to **4**.
- b. If short or open is not present, replace relay K2 (WP 1265 00). Go to **4**.

**4. Procedure completed.**

**TEST PANEL PWR TAIL RTR LAMP DOES NOT LIGHT WHEN TEST SET TAIL ROTOR BØ AND CØ SWITCHES ARE PLACED TO ON AND FAULT SELECT TR FAIL SWITCH IS PLACED TO FAULT****SYMPTOM**

Test panel PWR TAIL RTR lamp does not light when test set TAIL ROTOR BØ and CØ switches are placed to ON and FAULT SELECT TR FAIL switch is placed to FAULT.

**MALFUNCTION**

Test panel PWR TAIL RTR lamp does not light when test set TAIL ROTOR BØ and CØ switches are placed to ON and FAULT SELECT TR FAIL switch is placed to FAULT.

**CORRECTIVE ACTION****1. Check continuity between:**

- J1-B and Terminal E42
- J1-C and Terminal E43

- a. If continuity is present, go to **2**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **3**.

**2. Check diode CR3 for open.**

- a. If open, replace diode CR3 (WP 1265 00). Go to **3**.
- b. If not open, replace diode CR4 (WP 1265 00). Go to **3**.

**3. Procedure completed.****TEST PANEL PWR TAIL RTR LAMP DOES NOT LIGHT WHEN TEST SET TAIL ROTOR AØ AND BØ SWITCHES ARE PLACED TO ON AND FAULT SELECT TR FAIL SWITCH IS PLACED TO FAULT****SYMPTOM**

Test panel PWR TAIL RTR lamp does not light when test set TAIL ROTOR AØ and BØ switches are placed to ON and FAULT SELECT TR FAIL switch is placed to FAULT.

**MALFUNCTION**

Test panel PWR TAIL RTR lamp does not light when test set TAIL ROTOR AØ and BØ switches are placed to ON and FAULT SELECT TR FAIL switch is placed to FAULT.

**CORRECTIVE ACTION****1. Check continuity between:**

- J1-A and E41
- J1-C and E43

- a. If continuity is present, go to **2**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **3**.

**2. Check diode CR2 for open.**

- a. If open, replace diode CR2 (WP 1265 00). Go to **3**.
- b. If not open, replace diode CR4 (WP 1265 00). Go to **3**.

**3. Procedure completed.**



---

**TEST PANEL PWR TAIL RTR LAMP DOES NOT LIGHT WHEN TEST SET CURRENT TRANS AØ SWITCH IS PLACED TO ON AND FAULT SELECT CONT PNL SWITCH IS PLACED TO FAULT****SYMPTOM**

Test panel PWR TAIL RTR lamp does not light when test set CURRENT TRANS AØ switch is placed to ON and FAULT SELECT CONT PNL switch is placed to FAULT.

**MALFUNCTION**

Test panel PWR TAIL RTR lamp does not light when test set CURRENT TRANS AØ switch is placed to ON and FAULT SELECT CONT PNL switch is placed to FAULT.

**CORRECTIVE ACTION**

- 1. Check for 2.5 to 4.5 vdc between transistor Q2 base and ground.**
  - a. If voltage is as specified, go to **2**.
  - b. If voltage is not as specified, replace capacitor C3 (WP 1265 00). Go to **5**.
- 2. Check for 3.7 to 5.7 vdc between transistor Q2 collector and ground.**
  - a. If voltage is as specified, go to **3**.
  - b. If voltage is not as specified, replace transistor Q2 and/or associated circuitry (WP 1265 00). Go to **5**.
- 3. Check for 26 to 27 vdc between transistor Q3 collector and ground.**
  - a. If voltage is as specified, go to **4**.
  - b. If voltage is not as specified, replace transistor Q3 and/or associated circuitry (WP 1265 00). Go to **5**.
- 4. Check diode CR11 for short.**
  - a. If short is present, replace diode CR11 (WP 1265 00). Go to **5**.
  - b. If short is not present, replace relay K1 (WP 1265 00). Go to **5**.
- 5. Procedure completed.**

Diagram illustrating the front panel assembly components and their connections:

- A**: Points to the front panel assembly.
- B**: Points to the front panel assembly.
- C**: Points to the front panel assembly.
- ELECTRICAL CONNECTOR J1**: Located on the top of the front panel assembly.
- ELECTRICAL CONNECTOR J2**: Located on the top of the front panel assembly.
- ELECTRICAL CONNECTOR J3**: Located on the bottom of the front panel assembly.
- ELECTRICAL CONNECTOR P2**: Located on the right side of the front panel assembly.
- DECK ASSEMBLY**: The main component being assembled onto the front panel.
- COVER ASSEMBLY**: The protective cover for the front panel.

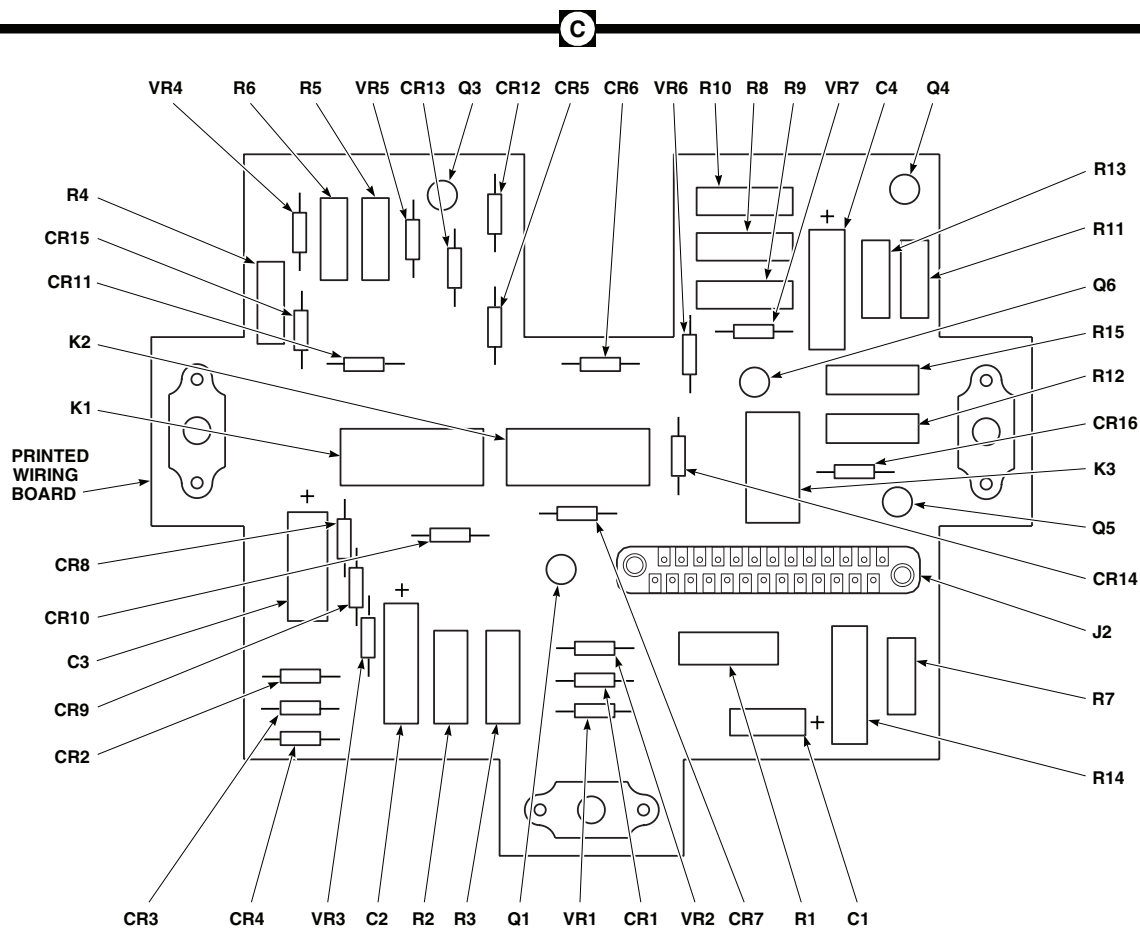
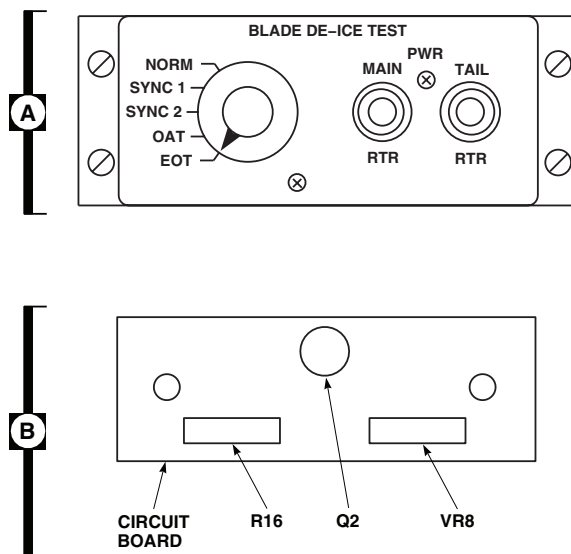
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Figure 1. Blade De-ice Test Panel.

SCHEMATICS

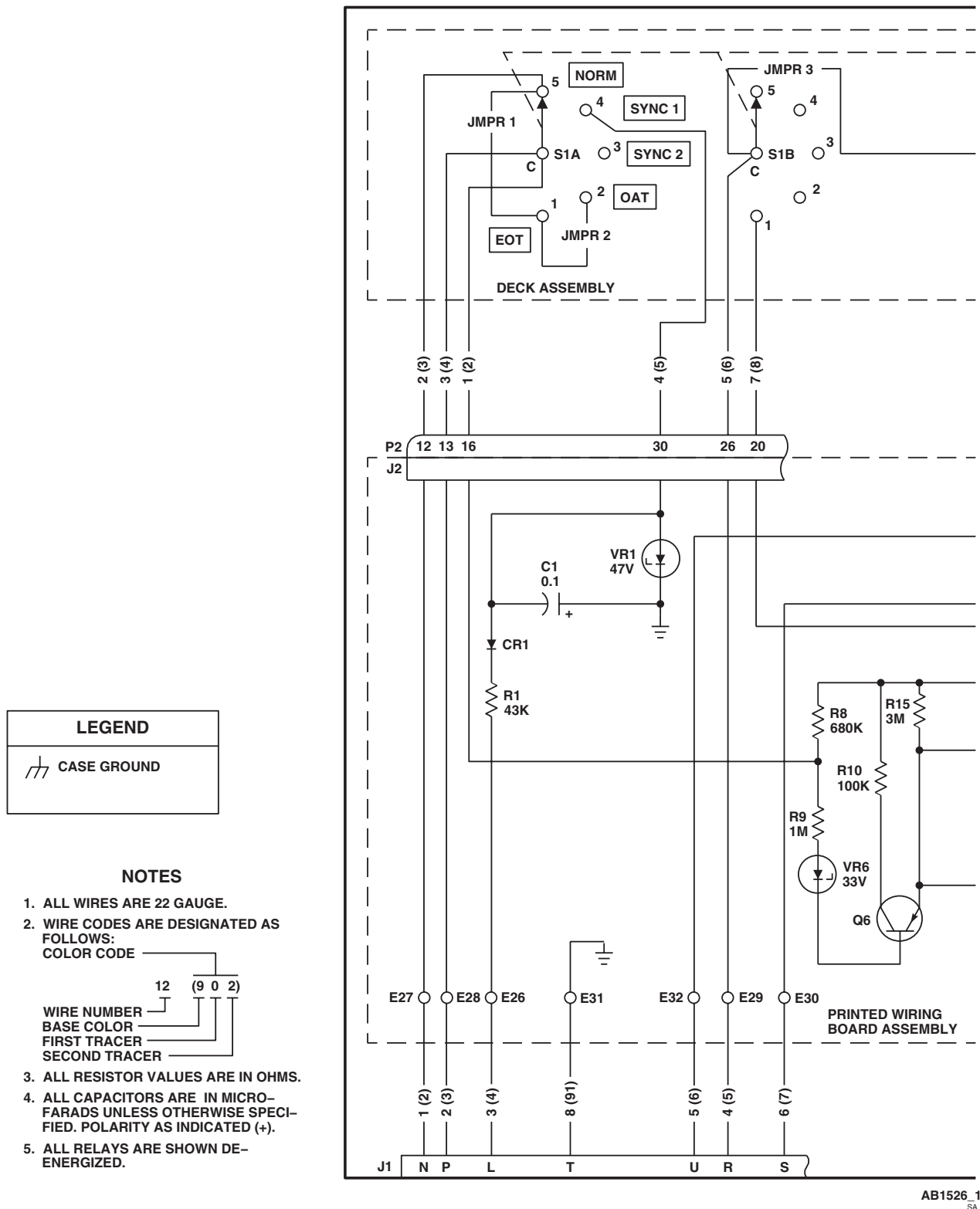


Figure 2. Blade De-ice Test Panel Schematic Diagram. (Sheet 1 of 4)

SCHEMATICS - Continued

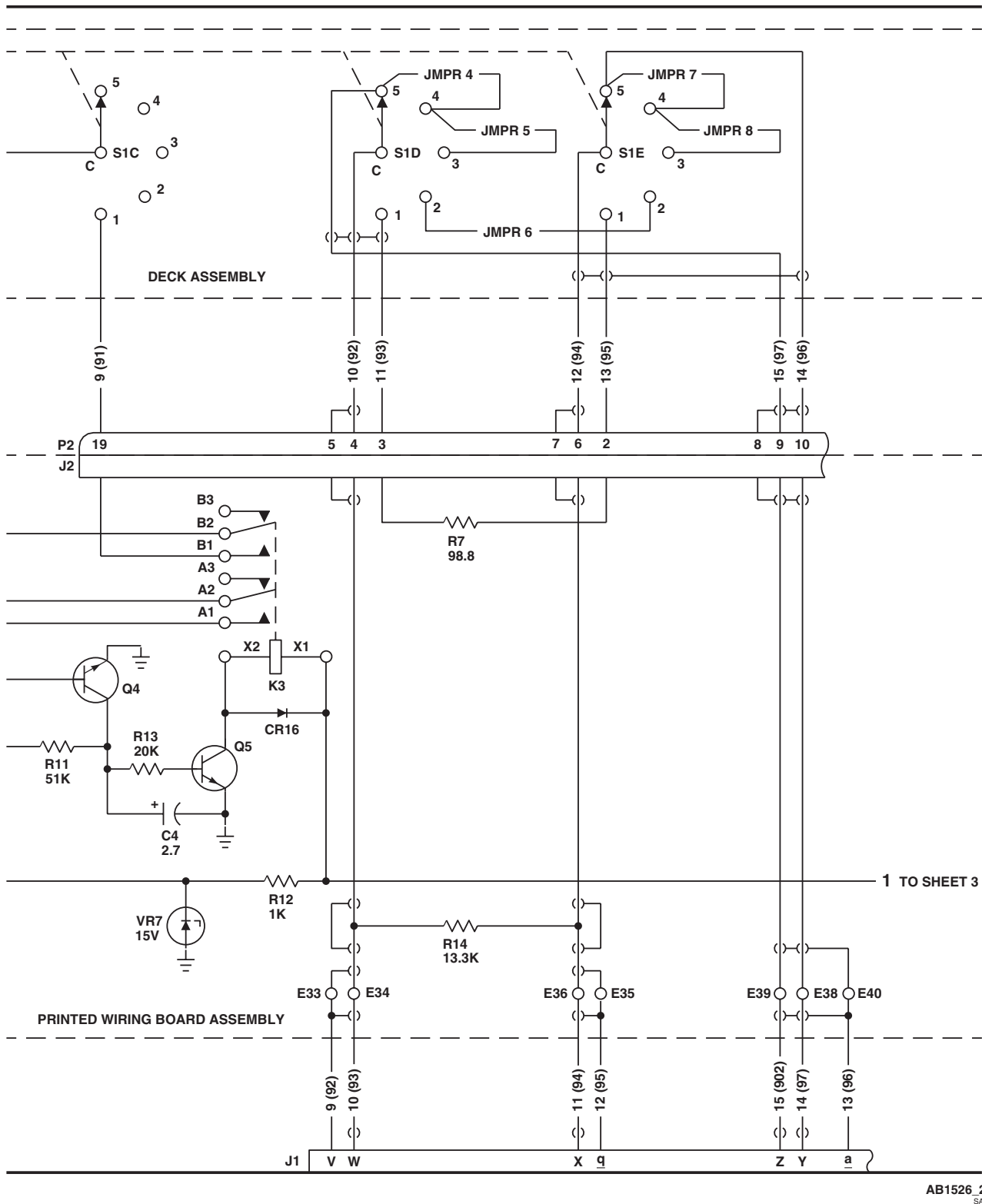


Figure 2. Blade De-ice Test Panel Schematic Diagram. (Sheet 2 of 4)



## SCHEMATICS - Continued

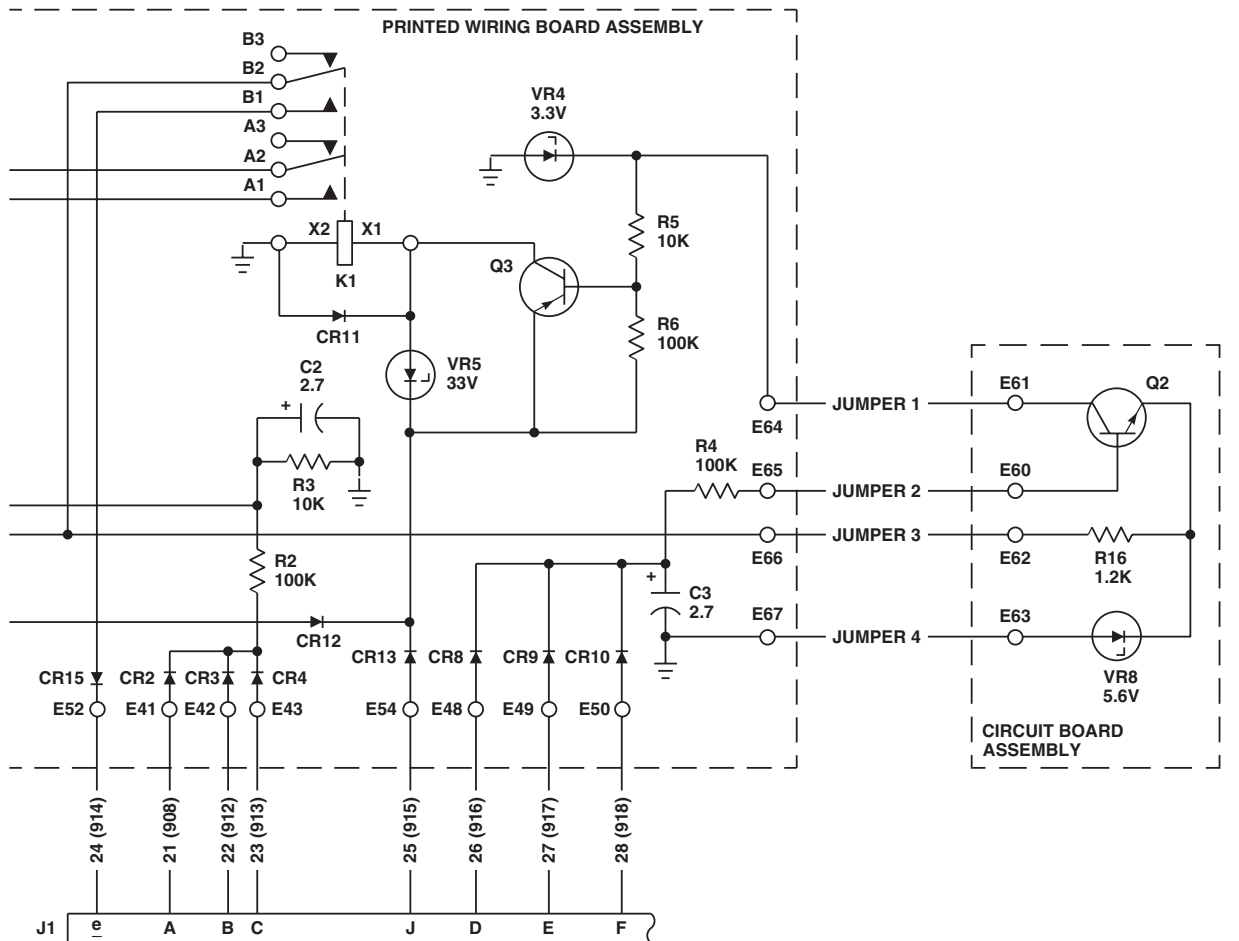
AB1526 4  
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Figure 2. Blade De-ice Test Panel Schematic Diagram. (Sheet 4 of 4)

END OF WORK PACKAGE

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**UNIT LEVEL**  
**ENVIRONMENTAL CONTROL SYSTEMS**  
**HEATING AND VENTILATION SYSTEM**  
**This WP supersedes WP 0149 00, dated 17 April 2006.**

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**INITIAL SETUP:****Test Equipment**

Multimeter, AN/PSM-45A

WP 0828 00

WP 1269 00

WP 1270 00

**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01

WP 1272 00

Electrical Repairer Toolkit, SC 5180-99-B06

WP 1273 00

WP 1275 00

**Personnel Required**

Aircraft Electrician MOS 15F (1)

WP 1276 00

UH-60 Helicopter Repairer MOS 15T (1)

WP 1279 00

WP 1280 00

**References**

TM 1-1520-237-10

WP 1399 00

WP 0104 00

WP 1400 00

WP 0697 00

WP 1747 00

WP 0698 00

WP 0824 00

WP 0825 00

**Equipment Condition**

APU Operational, TM 1-1520-237-10

No. 1 and No. 2 Engines Operational,

TM 1-1520-237-10

**SPECIAL INSTRUCTIONS****NOTE**

See Figure 1 for component location diagram, and **70309-02101-103** Figure 2  and **70309-02113-101** Figure 3  for schematic diagrams as an aid in operational/troubleshooting.

The heating and ventilation system operational/troubleshooting procedure is subdivided as follows:

- Mixing Valve 70309-02101-103
- Mixing Valve 70309-02113-101

## MIXING VALVE 70309-02101-103 OPERATIONAL/TROUBLESHOOTING PROCEDURE

### PROCEDURE

#### Step

#### NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Make sure the following circuit breakers are pushed in:

| CIRCUIT BREAKER                  | CIRCUIT BREAKER PANEL |
|----------------------------------|-----------------------|
| AIR SOURCE HEAT/START            | Copilot's             |
| HEAT & VENT                      | Pilot's               |
| HEAT VENT                        | Pilot's               |
| 2. Start APU (TM 1-1520-237-10). |                       |

#### NOTE

Open air valves in cockpit and forward cabin area before turning on blower.

3. Place upper console VENT BLOWER switch to ON.

#### Indication/Condition

- a. Ventilation blower shall operate.
- b. Air shall come out from each louver.

#### Corrective Action

1. If Indication/Condition a. is not as specified, go to VENTILATION BLOWER DOES NOT OPERATE, in this work package.
2. If Indication/Condition b. is not as specified, repair/replace ducting (WP 1279 00).

#### Step

4. Place upper console VENT BLOWER switch to OFF.

#### Indication/Condition

Air shall stop flowing out of louvers.

#### Corrective Action

If Indication/Condition is not as specified, replace right relay panel (WP 0825 00).



---

**MIXING VALVE 70309-02101-103 OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Step****NOTE**

Hot air flow may cycle on and off during operational/troubleshooting procedure.

5. Place upper console HEATER switch to ON.

**Indication/Condition**

Mixing valve shall open and warm air should come out of louvers.

**Corrective Action**

If Indication/Condition is not as specified, go to NO HEATED AIR COMING OUT OF ANY LOUVERS, in this work package.

**Step**

6. Slowly turn upper console HEATER knob from OFF to HI.

**Indication/Condition**

Temperature of air coming out of louvers shall increase.

**Corrective Action**

If Indication/Condition is not as specified, go to AIR TEMPERATURE CANNOT BE REGULATED, in this work package.

**Step**

7. Turn upper console HEATER knob to OFF.

**Indication/Condition**

Temperature of air coming out of louvers shall decrease.

**Corrective Action**

If Indication/Condition is not as specified, go to AIR TEMPERATURE CANNOT BE REGULATED, in this work package.

**Step**

8. Place upper console HEATER switch to OFF.

**Indication/Condition**

Air shall not be flowing out of louvers.

**Corrective Action**

If Indication/Condition is not as specified, replace mixing valve (WP 1269 00).

**Step**

9. With qualified personnel at controls, start No. 1 and No. 2 engines (TM 1-1520-237-10).
10. Shut down APU (TM 1-1520-237-10).
11. Place upper console AIR SOURCE HEAT/START switch to ENG.

---

**MIXING VALVE 70309-02101-103 OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued

---

**NOTE**

When HEATER switch is placed at ON (engine bleed-air shutoff valves open) and a noticeable increase in TGT (T4.5) for both engines should occur.

12. Place upper console HEATER switch to ON.

**Indication/Condition**

No. 1 engine and No. 2 engine bleed-air shutoff valves shall open. Mixer valve shall open and warm air shall be coming out of louvers.

**Corrective Action**

1. If air is coming out, but No. 1 engine TGT does not increase, go to NO. 1 ENGINE TGT DOES NOT INCREASE WHEN HEATER SWITCH IS PLACED ON, in this work package.
2. If air is coming out, but NO. 2 engine TGT does not increase, go to NO. 2 ENGINE TGT DOES NOT INCREASE WHEN HEATER SWITCH IS PLACED ON, in this work package.
3. If No. 1 and No. 2 engine TGTs do not increase, go to NO. 1 AND NO. 2 ENGINE TGTS DO NOT INCREASE WHEN HEATER SWITCH IS PLACED ON, in this work package.
4. **ESSS** If fuel fumes are noticed, clean pneumatic check valve (WP 0698 00). ■

**Step**

13. Place upper console HEATER switch to OFF.

**Indication/Condition**

Air shall stop coming out of louvers and a noticeable decrease in TGT of both engines shall occur.

**Corrective Action**

If Indication/Condition is not as specified, replace engine bleed-air shutoff valve (WP 0697 00).

**Step**

14. Shut down both engines (TM 1-1520-237-10).

**Indication/Condition**

Engines shall shut down.

**Corrective Action**

None Required

**MIXING VALVE 70309-02113-101 OPERATIONAL/TROUBLESHOOTING PROCEDURE****PROCEDURE****Step****NOTE**

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- To check the winterization mode operation of the heating and ventilation system with mixing valve 70309-02113-101 installed, the ambient air temperature must be very cold (below -25°F (-31.7°C)). If it is not cold enough, omit steps 13. thru 14.

1. Make sure the following circuit breakers are pushed in:

| <b>CIRCUIT BREAKER</b> | <b>CIRCUIT BREAKER PANEL</b> |
|------------------------|------------------------------|
| AIR SOURCE HEAT/START  | Copilot's                    |
| NO. 1 ENGINE ANTI-ICE  | Copilot's                    |
| HEAT & VENT            | Pilot's                      |
| HEAT VENT              | Pilot's                      |
| NO. 2 ENGINE ANTI-ICE  | Pilot's                      |

2. Start APU (TM 1-1520-237-10).

**NOTE**

Open air valves in cockpit and front cabin area before turning on blower.

3. Place upper console VENT BLOWER switch to ON.

**Indication/Condition**

- a. Ventilation blower shall operate.
- b. Air shall come out from each louver.

**Corrective Action**

1. If Indication/Condition a. is not as specified, go to VENTILATION BLOWER DOES NOT OPERATE, in this work package.
2. If Indication/Condition b. is not as specified, repair/replace ducting (WP 1279 00).

**Step**

4. Place upper console VENT BLOWER switch to OFF.

**Indication/Condition**

Air shall stop flowing out of louvers.

---

**MIXING VALVE 70309-02113-101 OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Corrective Action**

If Indication/Condition is not as specified, replace right relay panel (WP 0825 00).

**Step****NOTE**

Hot air may cycle on and off during operational/troubleshooting procedure.

5. Place upper console HEATER switch to ON.

**Indication/Condition**

Mixing valve shall open and warm air shall come out of louvers.

**Corrective Action**

If Indication/Condition is not as specified, go to NO HEATED AIR COMING OUT OF ANY LOUVERS, in this work package.

**Step**

6. Slowly turn upper console HEATER knob from OFF to HI.

**Indication/Condition**

Temperature of air coming out of louvers shall increase.

**Corrective Action**

If Indication/Condition is not as specified, go to AIR TEMPERATURE CANNOT BE REGULATED, in this work package.

**Step**

7. Turn upper console HEATER knob to OFF.

**Indication/Condition**

Temperature of air coming out of louvers shall decrease.

**Corrective Action**

If Indication/Condition is not as specified, go to AIR TEMPERATURE CANNOT BE REGULATED, in this work package.

**Step**

8. Place upper console HEATER switch to OFF.

**Indication/Condition**

Air shall not be flowing out of louvers.

**Corrective Action**

If Indication/Condition is not as specified, replace mixing valve (WP 1269 00).

**Step**

9. With qualified personnel at controls, start No. 1 and No. 2 engines (TM 1-1520-237-10).

---

**MIXING VALVE 70309-02113-101 OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**

10. Shut down APU (TM 1-1520-237-10).
11. Place upper console AIR SOURCE HEAT/START switch to ENG.

**NOTE**

When HEATER switch is placed at ON (engine bleed-air shutoff valves open), a noticeable increase in TGT (T4.5) for both engines should occur.

12. Place upper console HEATER switch to ON.

**Indication/Condition**

No. 1 engine and No. 2 engine bleed-air shutoff valves shall open. Mixer valve shall open and warm air shall be coming out of louvers.

**Corrective Action**

1. If air is coming out but No. 1 engine TGT does not increase, go to NO. 1 ENGINE TGT DOES NOT INCREASE WHEN HEATER SWITCH IS PLACED ON, in this work package.
2. If air is coming out but No. 2 engine TGT does not increase, go to NO. 2 ENGINE TGT DOES NOT INCREASE WHEN HEATER SWITCH IS PLACED ON, in this work package.
3. If No. 1 and No. 2 engine TGTs do not increase, go to NO. 1 AND NO. 2 ENGINE TGTS DO NOT DO NOT INCREASE WHEN HEATER SWITCH IS PLACED ON, in this work package.
4. **ESSS** If fuel fumes are noticed, clean pneumatic check valve (WP 0698 00). ■

**Step**

13. Turn upper console HEATER knob to HI and note temperature of air coming out of louvers.
14. Turn upper console ENG ANTI-ICE NO. 1 switch to ON.

**Indication/Condition**

Temperature and flow of air coming out of louvers shall decrease.

**Corrective Action**

If Indication/Condition is not as specified, go to AIR TEMPERATURE COMING OUT OF LOUVERS DOES NOT CHANGE WHEN ENG ANTI-ICE IS TURNED ON WITH MIXING VALVE PART NO. 70309-02113-101 INSTALLED, in this work package.

**Step**

15. Turn upper console HEATER knob to OFF.
16. Place upper console HEATER switch to OFF.

**Indication/Condition**

Air shall stop coming out of louvers and a noticeable decrease in TGT of both engines shall occur.

**Corrective Action**

If Indication/Condition is not as specified, replace engine bleed-air valve (WP 0697 00).

**Step**

17. Shut down both engines (TM 1-1520-237-10).

---

**MIXING VALVE 70309-02113-101 OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Indication/Condition**

Engines shall shut down.

**Corrective Action**

None Required

**VENTILATION BLOWER DOES NOT OPERATE****SYMPTOM**

Ventilation blower does not operate.

**MALFUNCTION**

Ventilation blower does not operate.

**CORRECTIVE ACTION**

- 1. Make sure VENT BLOWER switch is placed ON. Go to 2.**
- 2. Check continuity between P437-5 and P437-6.**
  - a. If continuity is present, go to **3**.
  - b. If continuity is not present, replace ventilation blower (WP 1280 00). Go to **14**.
- 3. Check for 28 vdc between J437-6 and ground.**
  - a. If voltage is as specified, go to **4**.
  - b. If voltage is not as specified, go to **11**.
- 4. Check continuity between J437-5 and P241-j.**
  - a. If continuity is present, go to **5**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 5. Check continuity between J437-4 and ground.**
  - a. If continuity is present, go to **6**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 6. Install an insulated jumper wire between J437-5 and J437-6. Go to 7.**
- 7. Check for 115 vac between:**
  - **J437-1 and J437-4**
  - **J437-2 and J437-4**
  - **J437-3 and J437-4**
  - a. If voltage is as specified, remove jumper wire and replace ventilation blower (WP 1280 00). Go to **14**.
  - b. If voltage is not as specified, remove jumper wire and go to **8**.
- 8. Check continuity between:**
  - **J437-1 and P241-g**
  - **J437-2 and P241-h**

---

**VENTILATION BLOWER DOES NOT OPERATE - Continued**

- **J437-3 and P241-i**
  - a. If continuity is present, go to **9**.
  - b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **14**.
- 9. Check for 115 vac between HEAT & VENT circuit breaker:**
  - **Terminal A1 and ground**
  - **Terminal B1 and ground**
  - **Terminal C1 and ground**
  - a. If voltage is as specified, go to **10**.
  - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **14**.
- 10. Check continuity between:**
  - **P241-E and terminal A1 (HEAT & VENT circuit breaker)**
  - **P241-n and terminal B1 (HEAT & VENT circuit breaker)**
  - **P241-N and terminal C1 (HEAT & VENT circuit breaker)**
  - a. If continuity is present, replace right relay panel (WP 0825 00). Go to **14**.
  - b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **14**.
- 11. Check for 28 vdc between VENT BLOWER switch terminal 3 and ground.**
  - a. If voltage is as specified, repair/replace wiring between switch terminal 3 and J437-6 (WP 1747 00). Go to **14**.
  - b. If voltage is not as specified, go to **12**.
- 12. Check for 28 vdc between VENT BLOWER switch terminal 2 and ground.**
  - a. If voltage is as specified, replace switch (WP 0828 00). Go to **14**.
  - b. If voltage is not as specified, go to **13**.
- 13. Check for 28 vdc between HEAT & VENT circuit breaker terminal 1 and ground.**
  - a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and VENT BLOWER switch terminal 2 (WP 1747 00). Go to **14**.
  - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **14**.
- 14. Procedure completed.**

**NO HEATED AIR COMING OUT OF ANY LOUVERS****SYMPTOM**

No heated air coming out of any louvers.

**MALFUNCTION**

No heated air coming out of any louvers.

**CORRECTIVE ACTION**

- 1. Check bleed-air line from APU check valve to mixing valve for proper connections.**

---

**NO HEATED AIR COMING OUT OF ANY LOUVERS** - Continued

- a. If proper connections are present, go to **2**.
- b. If proper connections are not present, repair/replace lines and/or rubber connections as necessary (WP 1272 00). Go to **30**.
- 2. Check helicopter configuration.**
  - a. If mixing valve part No. 70309-02101-103 installed, go to **3**.
  - b. If mixing valve part No. 70309-02113-101 installed, go to **4**.
- 3. Check for 28 vdc between P436-A and P436-B.**
  - a. If voltage is as specified, replace mixing valve (WP 1269 00). Go to **30**.
  - b. If voltage is not as specified, go to **5**.
- 4. Check for 28 vdc between P436-1 and P436-2.**
  - a. If voltage is as specified, replace mixing valve (WP 1269 00). Go to **30**.
  - b. If voltage is not as specified, go to **5**.
- 5. Check helicopter configuration.**
  - a. If mixing valve part No. 70309-02101-103 installed, go to **6**.
  - b. If mixing valve part No. 70309-02113-101 installed, go to **7**.
- 6. Check for 28 vdc between P436-A and ground.**
  - a. If voltage is as specified, repair/replace wiring (WP 1747 00). Go to **30**.
  - b. If voltage is not as specified, go to **8**.
- 7. Check for 28 vdc between P436-1 and ground.**
  - a. If voltage is as specified, repair/replace wiring (WP 1747 00). Go to **30**.
  - b. If voltage is not as specified, go to **8**.
- 8. Check helicopter configuration.**
  - a. **WINTER** , install an insulated jumper wire between J244B-K and P241-G. Go to **9**.
  - b. **W/O WINTER** , install an insulated jumper wire between P244-K and P241-G. Go to **9**.
- 9. Check helicopter configuration.**
  - a. If mixing valve part No. 70309-02101-103 installed, go to **10**.
  - b. If mixing valve part No. 70309-02113-101 installed, go to **11**.
- 10. Check for 28 vdc between P436-A and ground.**
  - a. If voltage is as specified, go to **12**.
  - b. If voltage is not as specified, go to **14**.
- 11. Check for 28 vdc between P436-1 and ground.**
  - a. If voltage is as specified, go to **12**.
  - b. If voltage is not as specified, go to **14**.
- 12. Remove jumper wire. Go to 13.**
- 13. Replace right relay panel (WP 0825 00). Go to 30.**



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**NO HEATED AIR COMING OUT OF ANY LOUVERS - Continued**

- 14. Remove jumper wire. Go to 15.**
- 15. Check helicopter configuration.**
  - a. **WINTER** , go to **16**.
  - b. **W/O WINTER** , go to **17**.
- 16. WINTER Check for 28 vdc between J244B-K and ground.**
  - a. If voltage is as specified, go to **18**.
  - b. If voltage is not as specified, replace **WINTER** (WP 1399 00). Go to **30**.
  - c. If trouble remains, go to **17**.
- 17. Check for 28 vdc between P244-K and ground.**
  - a. If voltage is as specified, go to **18**.
  - b. If voltage is not as specified, go to **19**.
- 18. Check helicopter configuration.**
  - a. Mixing valve part No. 70309-02101-103, repair/replace wiring between P241-G and P436-A (WP 1747 00). Go to **30**.
  - b. Mixing valve part No. 70309-02113-101, repair/replace wiring between P241-G and P436-1 (WP 1747 00). Go to **30**.
- 19. Install an insulated jumper wire across P902-b and P902-d. Go to 20.**
- 20. Check helicopter configuration.**
  - a. **WINTER** , go to **21**.
  - b. **W/O WINTER** , go to **22**.
- 21. WINTER Check for 28 vdc between J244B-K and ground.**
  - a. If voltage is as specified, go to **23**.
  - b. If voltage is not as specified, replace **WINTER** (WP 1400 00). Go to **30**.
  - c. If trouble remains, go to **22**.
- 22. Check for 28 vdc between P244-K and ground.**
  - a. If voltage is as specified, go to **23**.
  - b. If voltage is not as specified, go to **25**.
- 23. Remove jumper wire from P902. Go to 24.**
- 24. Replace left relay panel (WP 0825 00). Go to 30.**
- 25. Remove jumper wire from P902. Go to 26.**
- 26. Check for 28 vdc between P902-b and ground.**
  - a. If voltage is as specified, repair/replace wiring between P902-d and P244-K (WP 1747 00). Go to **30**.
  - b. If voltage is not as specified, go to **27**.
- 27. Check for 28 vdc between J231-c and ground.**

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**NO HEATED AIR COMING OUT OF ANY LOUVERS** - Continued

- a. If voltage is as specified, repair/replace wiring between P231 and P902 (WP 1747 00). Go to **30**.
- b. If voltage is not as specified, go to **28**.
- 28. Check for 28 vdc between AIR SOURCE HEAT/START circuit breaker terminal 1 and ground.**
  - a. If voltage is as specified, go to **29**.
  - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **30**.
- 29. Check wiring to HEATER switch for proper connections.**
  - a. If wiring connections are good, replace switch (WP 0828 00). Go to **30**.
  - b. If wiring connections are not good, reconnect wiring to switch (WP 1747 00). Go to **30**.
- 30. Procedure completed.**

**AIR TEMPERATURE CANNOT BE REGULATED****SYMPTOM**

Air temperature cannot be regulated.

**MALFUNCTION**

Air temperature cannot be regulated.

**CORRECTIVE ACTION**

- 1. Unlock fastener studs securing upper console panel and hinge down panel. Turn HEATER knob and check shaft for proper turning.**
  - a. If shaft turns properly, go to **2**.
  - b. If shaft does not turn properly, secure setscrews attaching shaft to adapter or shaft to HEATER knob. Adjust system as necessary (WP 1269 00). Go to **11**.
- 2. Disconnect control shaft at temperature regulator. Go to 3.**
- 3. Turn HEATER knob, visually check for turning of inner core of control shaft.**
  - a. If inner core of control shaft turns, go to **4**.
  - b. If inner core of control shaft does not turn, go to **9**.
- 4. Check temperature tube for signs of damage or loose connections.**
  - a. If damage or loose connections are present, secure temperature tube or replace (WP 1270 00). Go to **11**.
  - b. If damage or loose connections are not present, go to **5**.
- 5. Operate APU (TM 1-1520-237-10). Go to 6.**
- 6. Place HEATER switch to ON. Go to 7.**
- 7. Turn HEATER knob to HI. Go to 8.**
- 8. Check that temperature tube is warm.**
  - a. If tube is warm, replace mixture temperature sensor's pressure bellows (WP 1273 00). Go to **11**.

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**AIR TEMPERATURE CANNOT BE REGULATED - Continued**

- b. If trouble remains, replace mixture temperature sensor (WP 1273 00). Go to **11**.
- c. If tube is not warm, replace mixing valve (WP 1269 00). Go to **11**.
- 9. Disconnect control shaft at adapter. Go to 10.**
- 10. Turn HEATER knob and visually check adapter output shaft for turning.**
  - a. If shaft turns, replace control shaft (WP 1275 00). Go to **11**.
  - b. If shaft does not turn, replace adapter (WP 1276 00). Go to **11**.
- 11. Procedure completed.**

**NO. 1 ENGINE TGT DOES NOT INCREASE WHEN HEATER SWITCH IS PLACED ON  
SYMPTOM**

No. 1 engine TGT does not increase when HEATER switch is placed ON.

**MALFUNCTION**

No. 1 engine TGT does not increase when HEATER switch is placed ON.

**CORRECTIVE ACTION**

- 1. Make sure AIR/SOURCE HEAT/START switch is placed to ENG. Go to 2.**
- 2. Check for 28 vdc between P434-1 and P434-2.**
  - a. If voltage is as specified, replace No. 1 engine bleed-air shut off valve (WP 0697 00). Go to **10**.
  - b. If voltage is not as specified, go to **3**.
- 3. Check for 28 vdc between P434-1 and ground.**
  - a. If voltage is as specified, repair/replace ground wire on P434-2 (WP 1747 00). Go to **10**.
  - b. If voltage is not as specified, go to **4**.
- 4. Install an insulated jumper wire across P902-a and P902-c. Go to 5.**
- 5. Check for 28 vdc between P434-1 and ground.**
  - a. If voltage is as specified, go to **6**.
  - b. If voltage is not as specified, go to **8**.
- 6. Remove jumper wire from P902. Go to 7.**
- 7. Replace left relay panel (WP 0825 00). Go to 10.**
- 8. Remove jumper wire from P902. Go to 9.**
- 9. Check for 28 vdc between P902-a and ground.**
  - a. If voltage is as specified, repair/replace wiring between P902-c and P434-1 (WP 1747 00). Go to **10**.
  - b. If voltage is not as specified, repair/replace wiring between P902-a and SG902-1 (WP 1747 00). Go to **10**.
- 10. Procedure completed.**

**NO. 2 ENGINE TGT DOES NOT INCREASE WHEN HEATER SWITCH IS PLACED ON****SYMPTOM**

No. 2 engine TGT does not increase when HEATER switch is placed ON.

**MALFUNCTION**

No. 2 engine TGT does not increase when HEATER switch is placed ON.

**CORRECTIVE ACTION**

- 1. Make sure AIR/SOURCE HEAT/START switch is placed to ENG. Go to 2.**
- 2. Check for 28 vdc between P435-1 and P435-2.**
  - a. If voltage is as specified, replace No. 2 engine bleed-air shut off valve (WP 0697 00). Go to **12**.
  - b. If voltage is not as specified, go to **3**.
- 3. Check for 28 vdc between P435-1 and ground.**
  - a. If voltage is as specified, repair/replace ground wire on P435-2 (WP 1747 00). Go to **12**.
  - b. If voltage is not as specified, go to **4**.
- 4. Check helicopter configuration.**
  - a. **WINTER** , install an insulated jumper wire between J244B-J and P241-F. Go to **5**.
  - b. **W/O WINTER** , install an insulated jumper wire between P244-J and P241-F. Go to **5**.
- 5. Check for 28 vdc between P435-1 and ground.**
  - a. If voltage is as specified, go to **6**.
  - b. If voltage is not as specified, go to **8**.
- 6. Remove jumper wire. Go to 7.**
- 7. Replace right relay panel (WP 0825 00). Go to 12.**
- 8. Remove jumper wire. Go to 9.**
- 9. Check helicopter configuration.**
  - a. **WINTER** , go to **10**.
  - b. **W/O WINTER** , go to **11**.
- 10. **WINTER** Check for 28 vdc between J244B-J and ground.**
  - a. If voltage is as specified, repair/replace wiring between P241-F and P435-1 (WP 1747 00). Go to **12**.
  - b. If voltage is not as specified, replace **WINTER** (WP 1400 00). Go to **12**.
  - c. If trouble remains, go to **11**.
- 11. Check for 28 vdc between P244-J and ground.**
  - a. If voltage is as specified, repair/replace wiring between P241-F and P435-1 (WP 1747 00). Go to **12**.
  - b. If voltage is not as specified, repair/replace wiring between P244-J and SG902-1 (WP 1747 00). Go to **12**.

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**NO. 2 ENGINE TGT DOES NOT INCREASE WHEN HEATER SWITCH IS PLACED ON** - Continued**12. Procedure completed.****NO. 1 AND NO. 2 ENGINE TGTS DO NOT INCREASE WHEN HEATER SWITCH IS PLACED ON****SYMPTOM**

No. 1 and No. 2 engine TGTS do not increase when HEATER switch is placed ON.

**MALFUNCTION**

No. 1 and No. 2 engine TGTS do not increase when HEATER switch is placed ON.

**CORRECTIVE ACTION**

- 1. Make sure AIR/SOURCE HEAT/START switch is placed to ENG. Go to 2.**
- 2. Check for 28 vdc between J231-a and ground.**
  - a. If voltage is as specified, repair/replace wiring between P231-a and SG902-1 (WP 1747 00). Go to **4.**
  - b. If voltage is not as specified, go to **3.**
- 3. Check wiring between J231-a and switch S18 for proper connections.**
  - a. If wiring connections are good, replace switch (WP 0828 00). Go to **4.**
  - b. If wiring connections are not good, reconnect wiring to switch (WP 1747 00). Go to **4.**
- 4. Procedure completed.**

**AIR TEMPERATURE COMING OUT OF LOUVERS DOES NOT CHANGE WHEN ENG ANTI-ICE IS TURNED ON WITH MIXING VALVE PART NO. 70309-02113-101 INSTALLED****SYMPTOM**

Air temperature coming out of louvers does not change when ENG ANTI-ICE is turned ON with mixing valve part No. 70309-02113-101 installed.

**MALFUNCTION**

Air temperature coming out of louvers does not change when ENG ANTI-ICE is turned ON with mixing valve part No. 70309-02113-101 installed.

**CORRECTIVE ACTION**

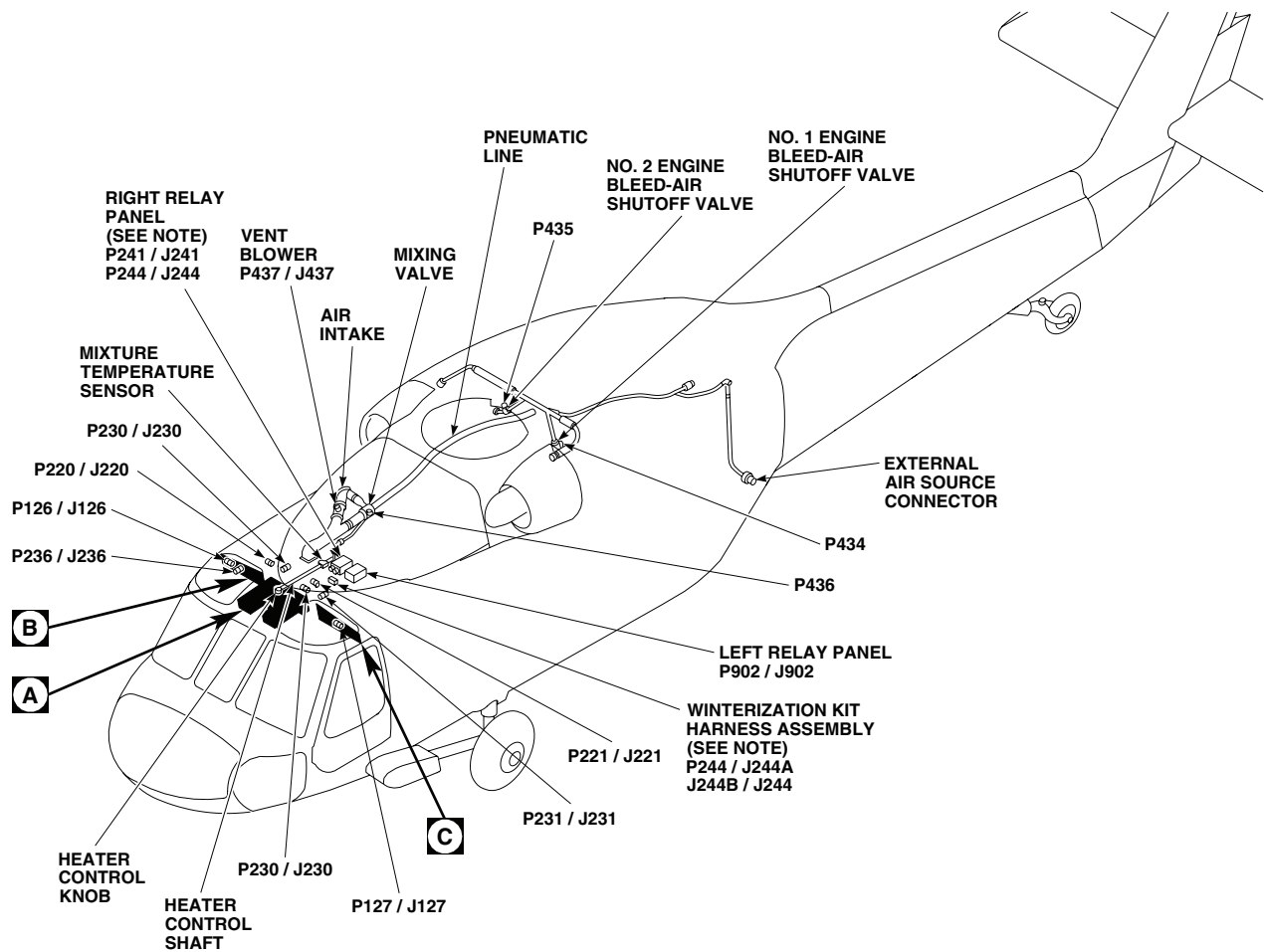
- 1. With ENG ANTI-ICE switched OFF, check for 28 vdc between J220-C and ground.**
  - a. If voltage is as specified, go to **2.**
  - b. If voltage is not as specified, go to **3.**
- 2. Check for 28 vdc between P436-4 and P436-3.**
  - a. If voltage is as specified, replace mixing valve (WP 1269 00). Go to **5.**
  - b. If voltage is not as specified, repair/replace wiring between P436 and P220 (WP 1747 00). Go to **5.**
- 3. Check for 28 vdc between J230-C and ground.**
  - a. If voltage is as specified, repair/replace wiring between J220-C and P230-C (WP 1747 00). Go to **5.**
  - b. If voltage is not as specified, go to **4.**

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**AIR TEMPERATURE COMING OUT OF LOUVERS DOES NOT CHANGE WHEN ENG ANTI-ICE IS TURNED ON WITH MIXING VALVE PART NO. 70309-02113-101 INSTALLED** - Continued

- 4. Check for broken/disconnected wires between AIR SOURCE HEAT/START switch terminal 2 and J230-C.**
  - a. If wiring connections are good, repair/replace wiring between switch S18-2 and J230-C (WP 1747 00). Go to **5.**
  - b. If wiring connections are not good, replace switch NO. 1 ENG ANTI-ICE or NO. 2 ENG ANTI-ICE (WP 0828 00). Go to **5.**
- 5. Procedure completed.**

LOCATION



NOTE  
WINTER

| TERMINAL BOARD /<br>DISCONNECT PLUG /<br>RECEPTACLE | LOCATION /<br>CONNECTION POINT                    |
|-----------------------------------------------------|---------------------------------------------------|
| J437                                                | VENTILATION BLOWER<br>VALVE                       |
| P126 / J126                                         | COCKPIT CEILING, BL 28 RH,<br>STA 243             |
| P127 / J127                                         | COCKPIT CEILING, BL 28 LH,<br>STA 243             |
| P220 / J220                                         | JUNCTION BOX, MAIN<br>ROTOR PYLON DECK            |
| P221 / J221                                         | JUNCTION BOX, MAIN<br>ROTOR PYLON DECK            |
| P222 / J222                                         | JUNCTION BOX, MAIN<br>ROTOR PYLON DECK            |
| P230 / J230                                         | CABIN CEILING, BL 4 RH,<br>STA 247                |
| P231 / J231                                         | CABIN CEILING, BL 9 LH,<br>STA 247                |
| P236 / J236                                         | BEHIND PILOT'S CIRCUIT<br>BREAKER PANEL, BL 23 RH |

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Figure 1. Heating and Ventilation System Location Diagram. (Sheet 1 of 3)

## LOCATION - Continued

| TERMINAL BOARD/<br>DISCONNECT PLUG/<br>RECEPTACLE | LOCATION/<br>CONNECTION POINT                       |
|---------------------------------------------------|-----------------------------------------------------|
| P241 / J241                                       | RIGHT RELAY PANEL<br>ASSEMBLY                       |
| P244 / J244                                       | RIGHT RELAY PANEL<br>ASSEMBLY (SEE NOTE)            |
| P244 / J244A<br>J244B / J244                      | WINTERIZATION KIT<br>HARNESS ASSEMBLY (SEE<br>NOTE) |
| P434                                              | NO. 1 ENGINE BLEED-AIR<br>SHUTOFF VALVE             |
| P435                                              | NO. 2 ENGINE BLEED-AIR<br>SHUTOFF VALVE             |
| P436                                              | MIXING VALVE                                        |
| P437 / J437                                       | VENTILATION BLOWER                                  |
| P902 / J902                                       | LEFT RELAY PANEL<br>ASSEMBLY                        |

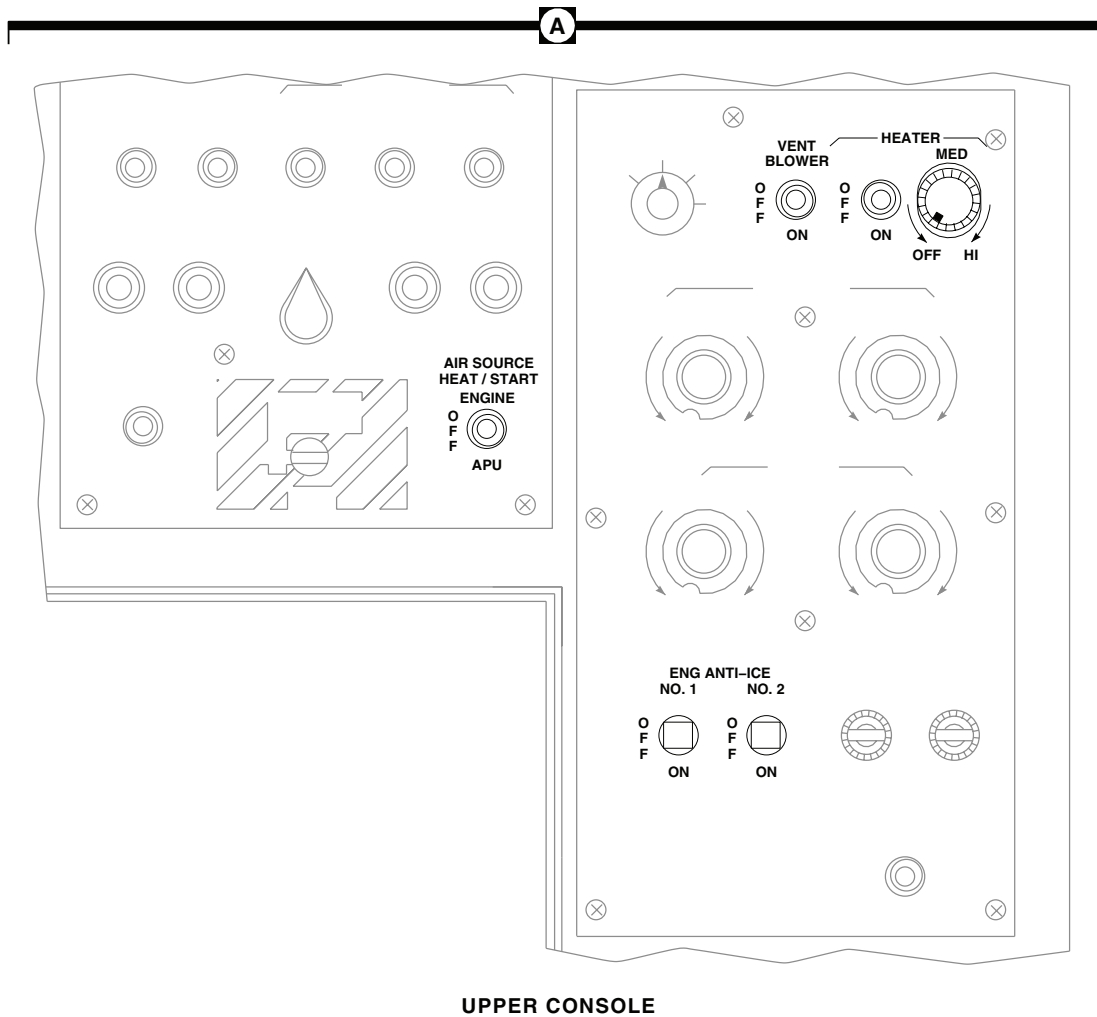
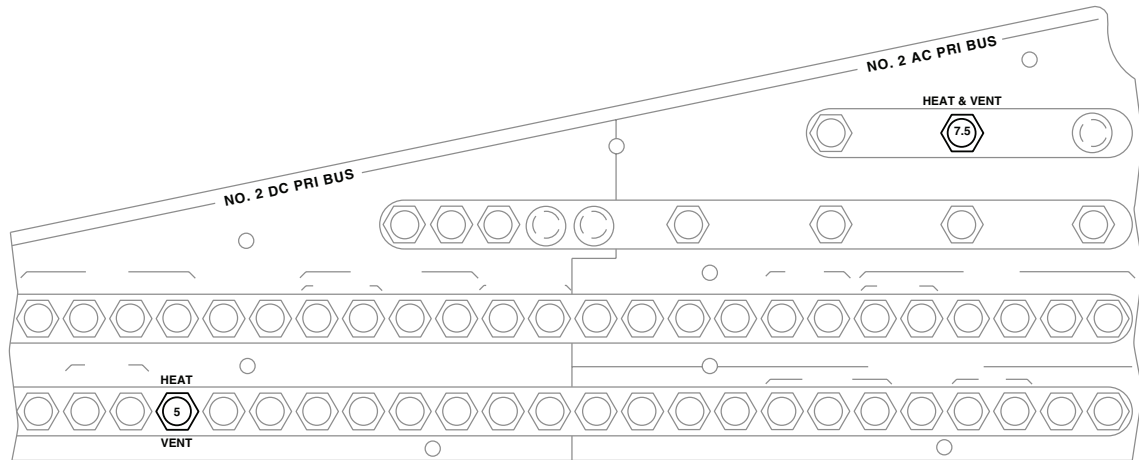


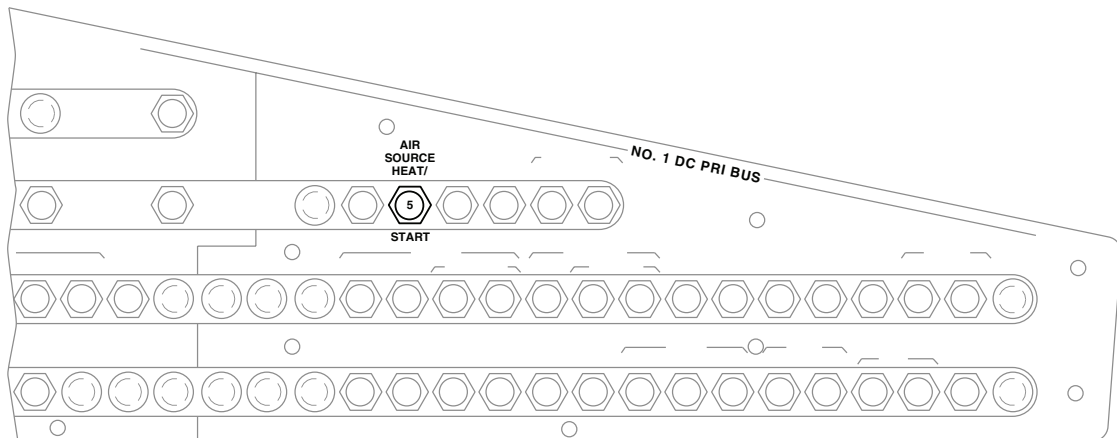
Figure 1. Heating and Ventilation System Location Diagram. (Sheet 2 of 3)



LOCATION - Continued



PILOT'S CIRCUIT BREAKER PANEL



COPLOT'S CIRCUIT BREAKER PANEL

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Figure 1. Heating and Ventilation System Location Diagram. (Sheet 3 of 3)

## SCHEMATICS

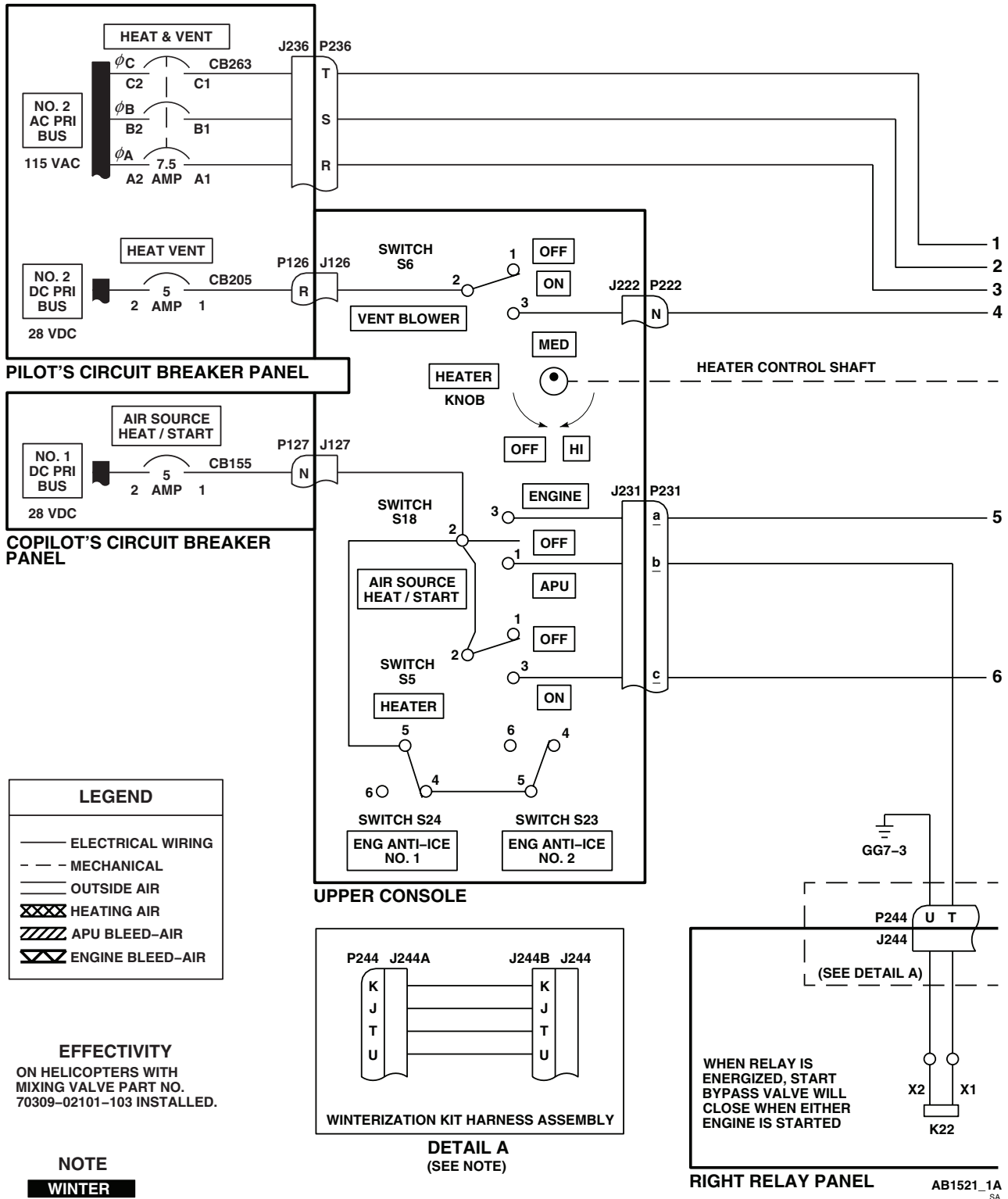


Figure 2. Heating and Ventilation System Schematic Diagram 70309-02101-103 . (Sheet 1 of 3)

SCHEMATICS - Continued

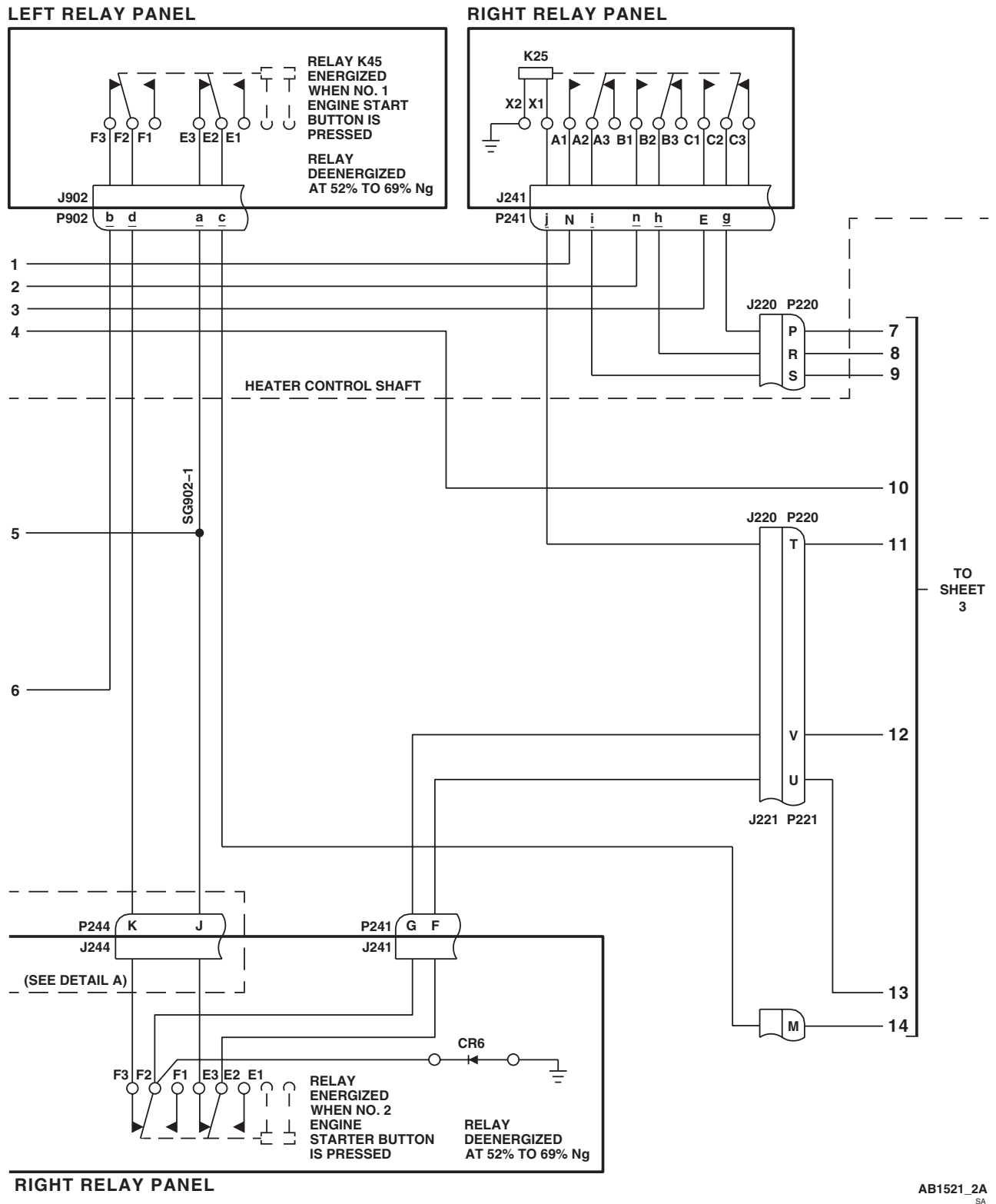
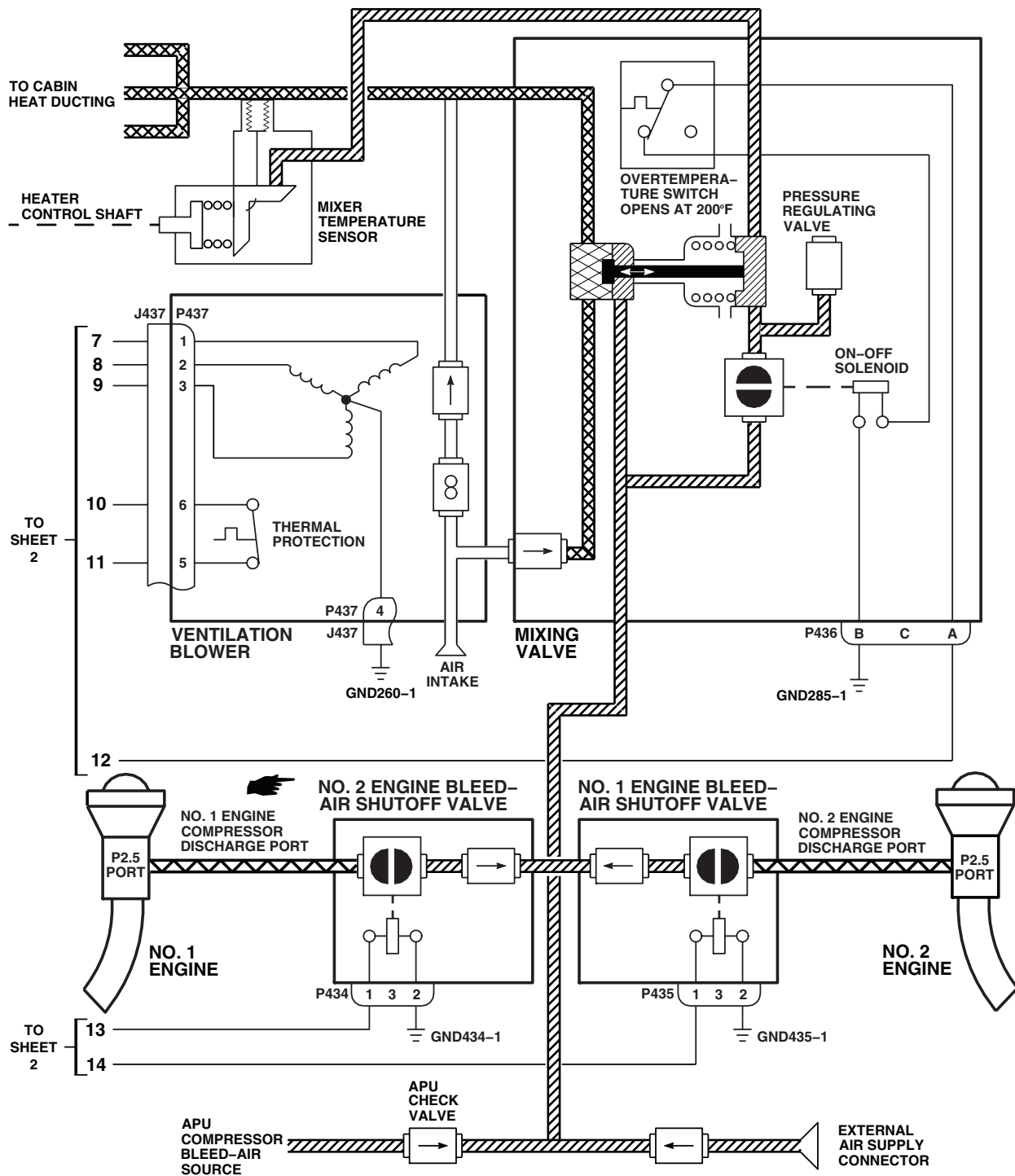


Figure 2. Heating and Ventilation System Schematic Diagram 70309-02101-103 . (Sheet 2 of 3)

SCHEMATICS - Continued



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Figure 2. Heating and Ventilation System Schematic Diagram 70309-02101-103 . (Sheet 3 of 3)

SCHEMATICS - Continued

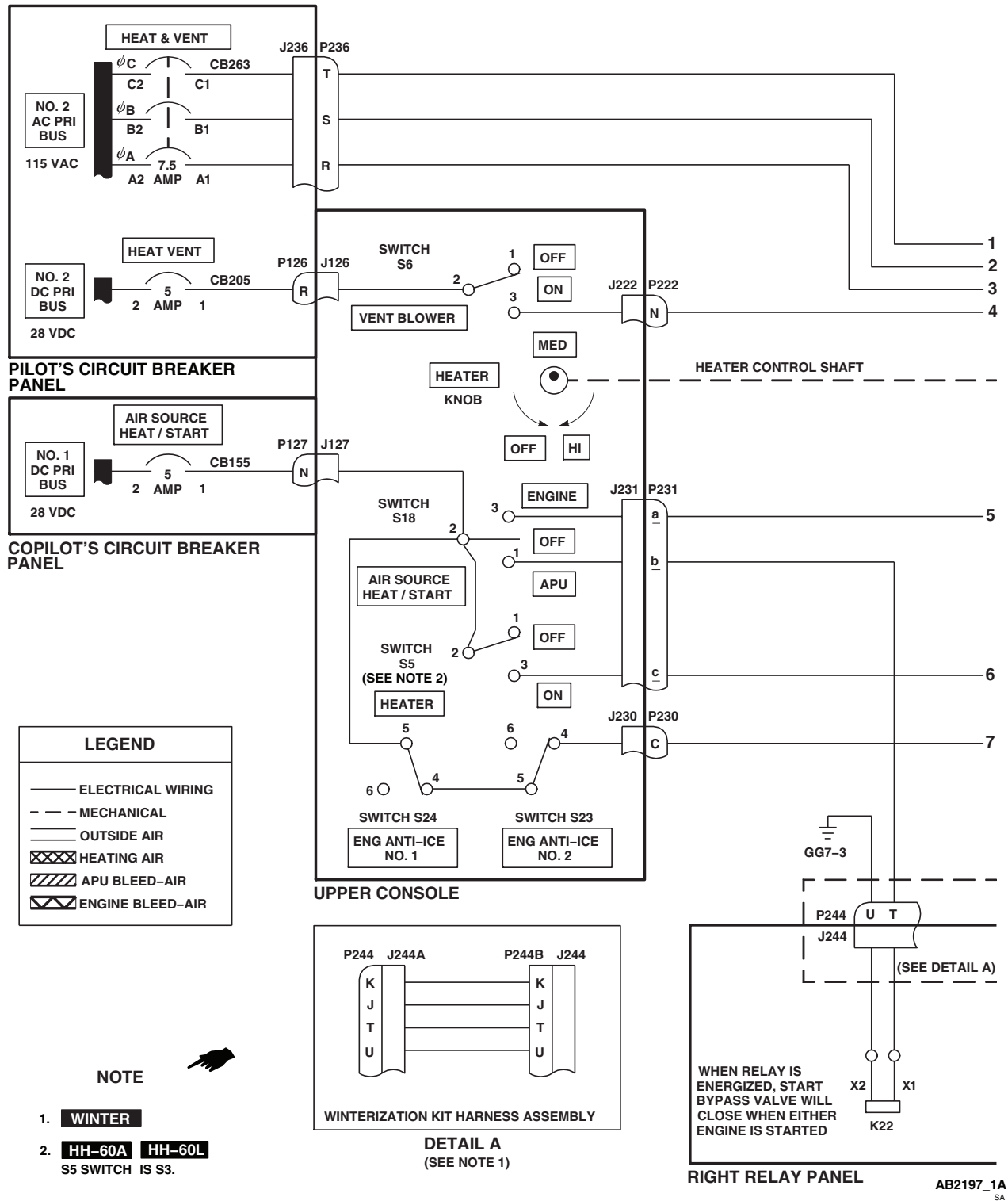


Figure 3. Heating and Ventilation System Schematic Diagram 70309-02113-101 . (Sheet 1 of 3)

SCHEMATICS - Continued

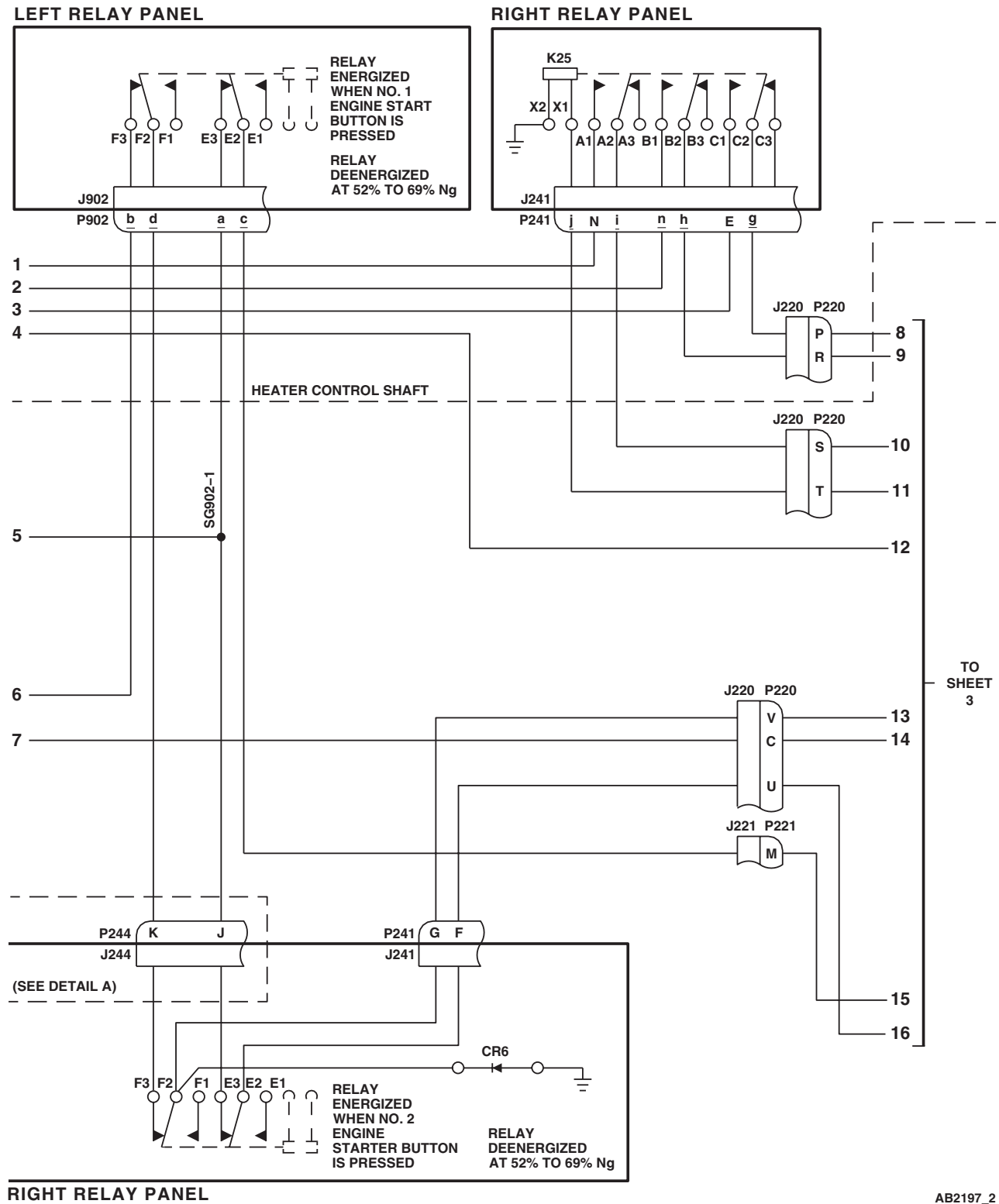


Figure 3. Heating and Ventilation System Schematic Diagram 70309-02113-101 . (Sheet 2 of 3)

## SCHEMATICS - Continued

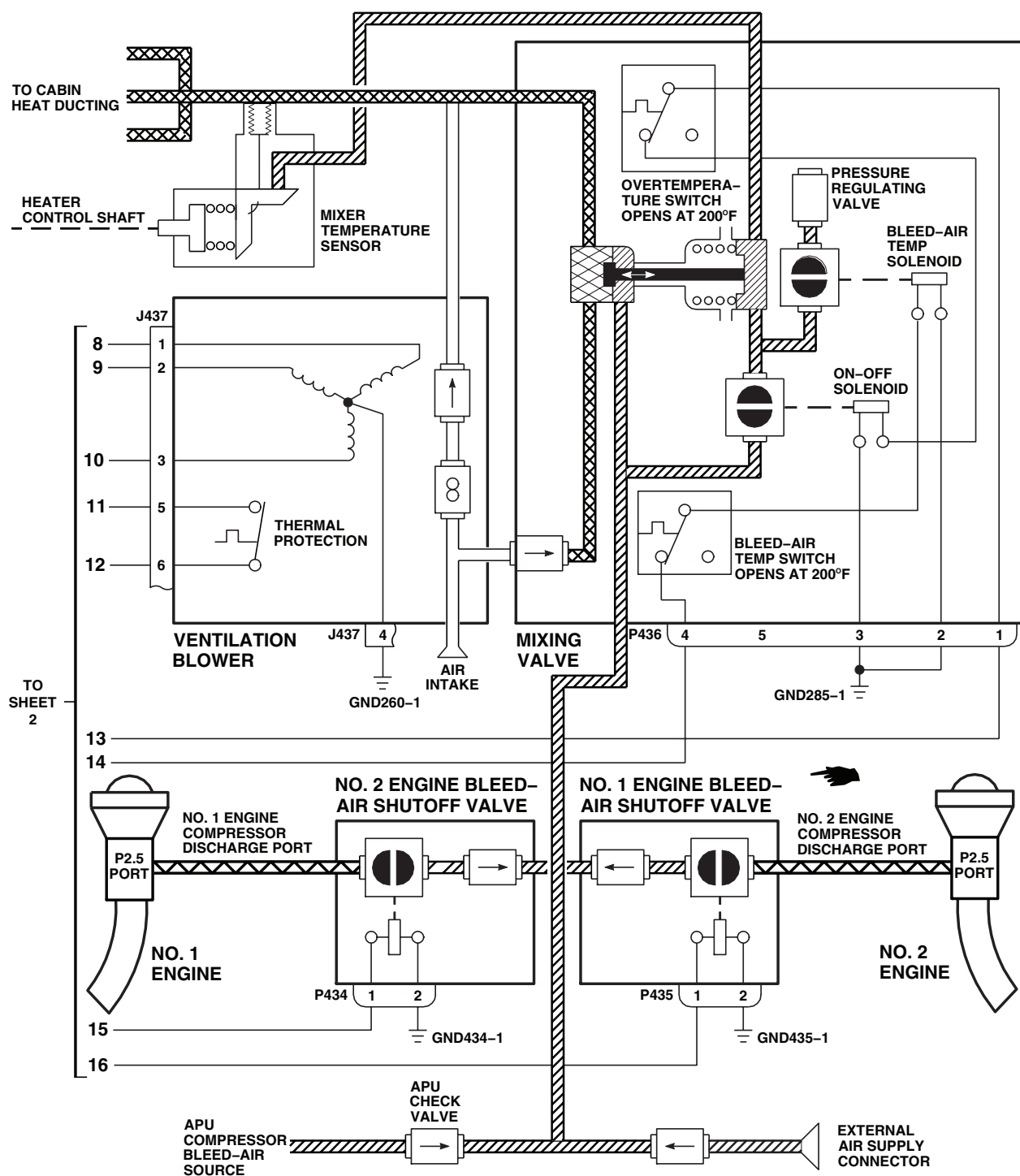
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Figure 3. Heating and Ventilation System Schematic Diagram 70309-02113-101 . (Sheet 3 of 3)

END OF WORK PACKAGE





## UNIT LEVEL

## ENVIRONMENTAL CONTROL SYSTEMS

## ENVIRONMENTAL CONTROL SYSTEM (ECS) EH60A

## INITIAL SETUP:

## Test Equipment

Multimeter, AN/PSM-45A

WP 0851 00

WP 0925 00

WP 1283 00

## Tools and Special Tools

Ammeter, Clamp On, 749

WP 1286 00

WP 1287 00

Electrical Repairer Toolkit, SC 5180-99-B06

WP 1289 00

Locally-Made Drag Beam Wedge Figure 167,

WP 1291 00

WP 1805 00

WP 1295 00

Pin Extractor, M81969114-02

WP 1315 00

Refrigeration Unit Service Toolkit

WP 1316 00

Thermometer, 389-3L

WP 1317 00

WP 1318 00

## Personnel Required

WP 1320 00

Aircraft Electrician MOS 15F (1)

WP 1323 00

Utilities Equipment Repairer MOS 52C (1)

WP 1400 00

WP 1622 00

## References

TM 1-1520-237-10

WP 1685 00

WP 0086 00

WP 1747 00

WP 0090 00

WP 1805 00

WP 0101 00

## Equipment Condition

WP 0104 00

ECS Properly Serviced, WP 1622 00

WP 0120 00

External Electrical Power Available, WP 1685 00

WP 0144 00

and APU Operational, TM 1-1520-237-10

WP 0824 00

Plenum, Not Installed, WP 1289 00

WP 0827 00

Servicing Manifold, Connected, WP 1622 00

WP 0828 00

Supply Ducting, Disconnected, WP 1291 00

## CAUTION

In the following operational/troubleshooting procedures, to prevent overheating backup pump hydraulic system, use Table 1 for limits when running backup pump.

| AIR TEMPERATURE F° (C°) | OPERATING TIME (MINUTES) | COOLDOWN TIME, (PUMP OFF) (MINUTES) |
|-------------------------|--------------------------|-------------------------------------|
| 90° (32°) max           | Unlimited                | -                                   |
| 91° (33°) to 100° (38°) | 24                       | 72                                  |

**INITIAL SETUP:** - Continued

| <b>AIR TEMPERATURE F° (C°)</b> | <b>OPERATING TIME (MINUTES)</b> | <b>COOLDOWN TIME, (PUMP OFF) (MINUTES)</b> |
|--------------------------------|---------------------------------|--------------------------------------------|
| 101° (39°) to 125° (52°)       | 16                              | 48                                         |

**NOTE**

See Figure 1 for component location diagram, Figure 2 for schematic diagram, Figure 3 for airflow diagram, Figure 4 for flow diagram, and Figure 5 current vs. temperature diagram as an aid in operational/troubleshooting.

**ENVIRONMENTAL CONTROL SYSTEM (ECS) OPERATIONAL/TROUBLESHOOTING PROCEDURE** **EH60A****PROCEDURE****Step****CAUTION**

- Make sure cabin doors are closed and personnel are clear of exhaust duct area.
- To prevent overheating backup pump hydraulic system, use Table 1 for limits when running backup pump.

**NOTE**

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Make sure the following circuit breakers are pushed in:

| <b>CIRCUIT BREAKER</b>    | <b>CIRCUIT BREAKER PANEL</b> |
|---------------------------|------------------------------|
| BACKUP PUMP PWR           | Copilot's                    |
| CPLT WSHLD ANTI-ICE       | Copilot's                    |
| ECS PWR                   | Copilot's                    |
| LIGHTS CAUT ADVSY         | Copilot's                    |
| ECS CONTR                 | Copilot's                    |
| NO. 2 SERVO WARN          | Pilot's                      |
| SEC MON BUS CONTR         | Pilot's                      |
| WINDSHIELD ANTI-ICE PILOT | Pilot's                      |
| WINDSHIELD ANTI-ICE CTR   | Pilot's                      |

**ENVIRONMENTAL CONTROL SYSTEM (ECS) OPERATIONAL/TROUBLESHOOTING PROCEDURE****EH60A** - Continued**CIRCUIT BREAKER****CIRCUIT BREAKER PANEL**

BACKUP HYD CONTR

Upper Console

CAUT/ADVSY PNL

Upper Console

2. Make sure upper console WINDSHIELD ANTI-ICE PILOT, CTR, and COPILOT switches, and BACKUP HYD PUMP switch are OFF.
3. Turn on electrical power.

**NOTE**

Backup hydraulic pump may run to charge APU accumulator. ECS will not operate until backup pump stops.

4. Hold caution/advisory panel BRT/DIM-TEST switch to TEST.

**Indication/Condition**

All capsules on caution/advisory panel shall go on.

**Corrective Action**

If Indication/Condition is not as specified, troubleshoot caution/advisory warning system (WP 0090 00).

**Step**

5. On air conditioner circuit breaker panel, pull out COMPRESSOR MOTOR, CONDENSER BLOWER, EVAPORATOR BLOWER, and HEATER ELEMENTS circuit breakers. Make sure DC POWER CONTROLS circuit breaker is pushed in.
6. On air conditioner circuit breaker panel, push in EVAPORATOR BLOWER circuit breaker.

**Indication/Condition**

Evaporator blower shall not run.

**Corrective Action**

1. If Indication/Condition is not as specified, check continuity between J246-R and J246-S.
  - a. If continuity is present, replace AIR COND switch (WP 0828 00).
  - b. If continuity is still not present, replace ECU (WP 1295 00).

**Step**

7. Place upper console ECS - AIR COND switch to FAN and then to COOL.

**Indication/Condition**

- a. Evaporator blower shall run with AIR COND switch at FAN.
- b. Evaporator blower shall run with AIR COND switch at COOL.
- c. Air shall flow from evaporator outlet duct.

**ENVIRONMENTAL CONTROL SYSTEM (ECS) OPERATIONAL/TROUBLESHOOTING PROCEDURE****EH60A** - Continued**Corrective Action**

1. If Indication/Conditions a. and b. are not as specified, go to EVAPORATOR FAN DOES NOT OPERATE WITH AIR COND SWITCH AT COOL AND FAN, in this work package.
2. If Indication/Condition a. is not as specified, replace AIR COND switch S72 (WP 0828 00).
3. If Indication/Condition b. is not as specified, check continuity between terminals S72-1 and S72-4 and between terminals S72-2 and S72-6.
  - a. If continuity is present, replace switch S72 (WP 0828 00).
  - b. If continuity is not present, repair/replace wiring between terminals S72-1 and S72-4 or between terminals S72-2 and S72-6, as required (WP 1747 00).
4. If Indication/Condition c. is not as specified, go to AIR DOES NOT FLOW FROM EVAPORATOR OUTLET DUCT, in this work package.

**Step**

8. Place upper console ECS - AIR COND switch to OFF.
9. Clamp ammeter around wire going to P7-D.
10. On air conditioning circuit breaker panel, push in HEATER ELEMENTS circuit breaker.
11. Place upper console ECS - HTR switch to ON.

**Indication/Condition**

Ammeter shall indicate zero amperes.

**Corrective Action**

If Indication/Condition is not as specified, replace ECU (WP 1295 00).

**Step**

12. Place upper console ECS - AIR COND switch to FAN.

**Indication/Condition**

- a. CABIN HEAT ON capsule on caution/advisory panel shall go on.
- b. Ammeter shall indicate between 7 and 9 amps.

**Corrective Action**

1. If Indication/Condition a. is not as specified, go to CABIN HEAT ON CAPSULE DOES NOT GO ON WITH HTR SWITCH AT ON AND AIR COND SWITCH AT FAN, in this work package.
2. If Indication/Condition b. is not as specified, go to AMMETER INDICATES LESS THAN 7 AMPS THROUGH HEATER DEMISTER, in this work package.

**Step**

13. With an air temperature no higher than 90°F (32°C), allow the evaporator to run for 30 minutes.
14. Measure evaporator inlet and outlet airstream temperatures.
15. Repeat steps 13. and 14. for a total of three measurements.

**Indication/Condition**

The average temperature differential (inlet minus outlet) shall be 10°F (-12.2°C) minimum.

---

**ENVIRONMENTAL CONTROL SYSTEM (ECS) OPERATIONAL/TROUBLESHOOTING PROCEDURE****EH60A** - Continued**Corrective Action**

If Indication/Condition is not as specified, go to AMMETER INDICATES LESS THAN 7 AMPS THROUGH HEATER DEMISTER, in this work package.

**Step**

16. Place upper console BACKUP HYD PUMP switch to ON.

**Indication/Condition**

CABIN HEAT ON capsule on caution/advisory panel shall remain on.

**Corrective Action**

If Indication/Condition is not as specified, replace No. 3 relay panel (WP 0827 00).

**Step**

17. Place upper console BACKUP HYD PUMP switch to OFF and place PILOT WINDSHIELD ANTI-ICE switch to ON.

**Indication/Condition**

CABIN HEAT ON capsule on caution/advisory panel shall remain on.

**Corrective Action**

If Indication/Condition is not as specified, replace No. 3 relay panel (WP 0827 00).

**Step**

18. Place upper console BACKUP HYD PUMP switch to ON.

**Indication/Condition**

CABIN HEAT ON capsule on caution/advisory panel shall go off.

**Corrective Action**

If Indication/Condition is not as specified, replace No. 3 relay panel (WP 0827 00).

**Step**

19. Place upper console BACKUP HYD PUMP and PILOT WINDSHIELD ANTI-ICE switches to OFF.

**Indication/Condition**

CABIN HEAT ON capsule on caution/advisory panel shall go on.

**Corrective Action**

If Indication/Condition is not as specified, go to CABIN HEAT ON CAPSULE DOES NOT GO ON WITH HTR SWITCH AT ON AND AIR COND SWITCH AT FAN, in this work package.

**Step**

20. Place upper console ECS - HTR and AIR COND switches to OFF.
21. On air conditioner circuit breaker panel, pull out HEATER ELEMENTS and EVAPORATOR BLOWER circuit breakers.
22. On air conditioner circuit breaker panel, push in CONDENSER BLOWER circuit breaker.

**ENVIRONMENTAL CONTROL SYSTEM (ECS) OPERATIONAL/TROUBLESHOOTING PROCEDURE****EH60A** - Continued**CAUTION**

To prevent damage to the helicopter from hot condenser exhaust, make sure the right side troop/cargo door is closed while the air conditioner is running.

23. Place upper console ECS - AIR COND switch to COOL.

**Indication/Condition**

- a. After five seconds, condenser blower shall go on.
- b. Air shall come out of condenser exhaust duct.

**Corrective Action**

1. If Indication/Condition a. is not as specified, go to CONDENSER BLOWER DOES NOT GO ON WITH AIR COND SWITCH AT COOL, in this work package.
2. If Indication/Condition b. is not as specified, go to AIR IS NOT COMING OUT OF CONDENSER EXHAUST DUCT, in this work package.

**Step**

24. Place upper console ECS - AIR COND switch to OFF.
25. On air conditioner circuit breaker panel, push in all circuit breakers.
26. Turn upper console ECS - TEMP CONT WARM-COOL control counterclockwise to COOL, and place AIR COND switch to COOL.

**Indication/Condition**

- a. Evaporator blower shall start immediately.
- b. After five seconds, condenser blower shall start.
- c. After an additional 5 to 10 seconds, compressor motor shall start.
- d. When compressor motor starts, AIR COND ON capsule on caution/advisory panel shall go on.
- e. Cool air shall come out of evaporator outlet duct.

**Corrective Action**

1. If Indication/Condition a. is not as specified, go to EVAPORATOR FAN DOES NOT OPERATE WITH AIR COND SWITCH AT COOL AND FAN, in this work package.
2. If Indication/Condition b. is not as specified, go to CONDENSER BLOWER DOES NOT GO ON WITH AIR COND SWITCH AT COOL, in this work package.
3. If Indication/Condition c. is not as specified, go to COMPRESSOR MOTOR DOES NOT OPERATE WITH AIR COND SWITCH AT COOL, in this work package.
4. If Indication/Condition d. is not as specified, go to AIR COND ON CAPSULE DOES NOT GO ON, in this work package.
5. If Indication/Condition e. is not as specified, go to AIR FROM EVAPORATION OUTLET DUCT DOES NOT TURN COOL, in this work package.

**Step**

27. Place upper console BACKUP HYD PUMP switch to ON.

**ENVIRONMENTAL CONTROL SYSTEM (ECS) OPERATIONAL/TROUBLESHOOTING PROCEDURE****EH60A** - Continued**Indication/Condition**

AIR COND ON capsule on caution/advisory panel shall go off.

**Corrective Action**

1. If Indication/Condition is not as specified, check continuity between P398-j and P244-A.
  - a. If continuity is present, replace No. 3 relay panel (WP 0827 00).
  - b. If continuity is not present, repair/replace wiring (WP 1747 00).
  - c. If trouble remains, troubleshoot hydraulic system (WP 0086 00).
  - d. If trouble still remains, troubleshoot right relay panel (WP 0120 00).

**Step****CAUTION**

Allow ECS to remain off for at least 2 minutes between restarts to allow internal pressures to stabilize or damage to compressor may occur.

28. Place upper console BACKUP HYD PUMP switch to OFF.

**Indication/Condition**

AIR COND ON capsule on caution/advisory panel shall go on.

**Corrective Action**

If Indication/Condition is not as specified, troubleshoot hydraulic system (WP 0086 00).

**Step**

29. Allowing at least 2 minutes between ECS restarts, alternately place upper console COPILOT, CTR, and PILOT WINDSHIELD ANTI-ICE switches to ON.

**Indication/Condition**

When each switch is ON, AIR COND ON capsule on caution/advisory panel shall go off.

**Corrective Action**

If Indication/Condition is not as specified, go to WINDSHIELD ANTI-ICE SYSTEM DOES NOT DISABLE ECS, in this work package.

**Step**

30. Make sure upper console COPILOT, CTR, and PILOT WINDSHIELD ANTI-ICE switches are OFF.

**Indication/Condition**

AIR COND ON capsule on caution/advisory panel shall go back on.

**Corrective Action**

If Indication/Condition is not as specified, troubleshoot windshield anti-ice system (WP 0144 00).

**Step**

31. Disconnect ammeter and clamp it to TEST LOOP wire of compressor harness (located near P7).

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**ENVIRONMENTAL CONTROL SYSTEM (ECS) OPERATIONAL/TROUBLESHOOTING PROCEDURE****EH60A** - Continued**Indication/Condition**

Ammeter shall indicate between upper and lower limits when cross referenced with ambient temperature (Figure 5).

**Corrective Action**

1. If ammeter indicates less than lower limit, go to COMPRESSOR DRAWS LESS THAN THE MINIMUM ALLOWABLE CURRENT, in this work package.
2. If ammeter indicates more than upper limit, go to COMPRESSOR DRAWS MORE THAN MAXIMUM ALLOWABLE CURRENT, in this work package.

**Step**

32. Check the ECS sight glass (on electrical pallet).

**Indication/Condition**

There shall be a green dot in the sight glass, and there shall be no bubbles or smears (shades of green or a wisp of bubbles are normal).

**Corrective Action**

If Indication/Condition is not as specified, purge and service the system and add freon until sight glass clears (WP 1622 00).

**Step**

33. Allow system to run for 15 minutes and stabilize.
34. Place temperature probe in evaporator outlet duct.
35. Cover about 95% of plenum inlet face.
36. On air conditioner fault indicator panel, check temperature probe reading when LOW TEMPERATURE fault indicator pops out.

**Indication/Condition**

When LOW TEMPERATURE fault indicator pops out, temperature probe shall indicate between 27°F and 43°F (-2.7° and 6.1°C) and compressor shall shut down.

**Corrective Action**

If Indication/Condition is not as specified, replace low temperature switch (WP 1315 00).

**Step**

37. Remove cover from evaporator inlet face.
38. Reset LOW TEMPERATURE fault indicator.

**Indication/Condition**

Compressor shall restart when temperature probe indicates between 48°F and 63°F (8.8°C and 17.2°C).

**Corrective Action**

If Indication/Condition is not as specified, replace low temperature switch (WP 1315 00).

**Step**

39. On air conditioner circuit breaker panel, pull out CONDENSER BLOWER circuit breaker.



**ENVIRONMENTAL CONTROL SYSTEM (ECS) OPERATIONAL/TROUBLESHOOTING PROCEDURE****EH60A** - Continued

40. On air conditioner fault indicator panel, check REF gauge reading when HIGH PRESSURE fault indicator pops out.

**Indication/Condition**

REF gauge shall indicate between 295 and 325 psig, and compressor shall shut down when HIGH PRESSURE fault indicator pops out.

**Corrective Action**

If Indication/Condition is not as specified, replace high pressure switch (WP 1320 00).

**Step**

41. On air conditioner circuit breaker panel, push in CONDENSER BLOWER circuit breaker.  
42. Reset HIGH PRESSURE fault indicator.

**Indication/Condition**

Compressor will restart when REF gauge on servicing manifold indicates 275 psig.

**Corrective Action**

If Indication/Condition is not as specified, replace high pressure switch (WP 1320 00).

**Step**

43. Remove ammeter from compressor harness and connect it to jumper wire at bottom of heater elements.  
44. Place upper console ECS - AIR COND switch to FAN.  
45. Place upper console ECS - HTR switch to ON.  
46. Place temperature probe in evaporator outlet duct.  
47. Cover about 95% of the evaporator inlet face.

**Indication/Condition**

When temperature probe indicates between 152°F and 168°F (66.6°C and 75.5°C), the heater/demister shall shut down (ammeter shall indicate zero amps) and HIGH TEMPERATURE fault indicator shall pop out.

**Corrective Action**

1. If heater/demister shuts down at a temperature outside the specified range, turn adjustment screw of temperature limiting switch first clockwise and then counterclockwise until heater shuts down.
2. If adjustment of temperature limiting switch fails to cause heater/demister shutdown, replace temperature limiting switch (WP 1315 00).

**Step**

48. Reset HIGH TEMPERATURE fault indicator.  
49. Remove cover from evaporator inlet (the rise to 180°F (82.2°C) should occur rapidly when cover is removed).

**Indication/Condition**

When temperature probe indicates 172°F to 188°F (77.7°C to 86.6°C), the HIGH TEMPERATURE fault indicator shall pop out.

**Corrective Action**

If Indication/Condition is not as specified, replace high temperature switch (WP 1315 00).

**ENVIRONMENTAL CONTROL SYSTEM (ECS) OPERATIONAL/TROUBLESHOOTING PROCEDURE****EH60A** - Continued**Step**

50. Reset HIGH TEMPERATURE fault indicator.

**CAUTION**

If only using APU generator power, make sure 60 Hz converter is not operated at full load (30 amps) or air conditioner will exceed generator rating when ECS control panel AIR COND switch is selected to COOL.

51. Place upper console ECS - AIR COND switch to COOL.  
52. Place upper console ECS - HTR switch to OFF.  
53. Make sure REF and VAC handwheels on servicing manifold are closed.  
54. Connect yellow hose to VAC side of manifold and place other end outside of work area.

**NOTE**

- Bleed freon very slowly to minimize the amount of lubricant discharged from the system.
- The following Indication/Condition may take up to an hour to occur.

55. Slowly open VAC handwheel and allow refrigerant to vent through yellow hose.

**Indication/Condition**

When REF gauge indicates between 45 and 55 psig, the LOW PRESSURE fault indicator shall pop out and the compressor shall shut down.

**Corrective Action**

If Indication/Condition is not as specified, replace low pressure switch (WP 1320 00).

**Step**

56. Service the ECS (WP 1622 00).  
57. Reset LOW PRESSURE fault indicator.  
58. With air conditioning operating (AIR COND switch on COOL and HTR switch OFF) and servicing manifold attached, record VAC gauge pressure (P1).  
59. Turn upper console ECS - TEMP CONT knob clockwise to full WARM.

**Indication/Condition**

VAC gauge indication shall increase from steady state.

**Corrective Action**

If Indication/Condition is not as specified, go to LOW PRESSURE DOES NOT INCREASE, in this work package.

**Step**

60. Record VAC gauge indication (P2).  
61. Calculate difference between first pressure (P1) and second pressure (P2).

**Indication/Condition**

Pressure differential shall be between 9 and 11 psig.

**ENVIRONMENTAL CONTROL SYSTEM (ECS) OPERATIONAL/TROUBLESHOOTING PROCEDURE****EH60A** - Continued**Corrective Action**

1. If Indication/Condition is less than specified, turn adjustment screw on hot gas bypass valve clockwise to increase pressure differential.
2. If Indication/Condition is more than specified, turn adjustment screw on hot gas bypass valve counterclockwise to decrease pressure differential.
  - a. If adjustment fails to produce change in pressure differential, replace hot gas bypass valve (WP 1315 00).
  - b. If trouble remains, replace temperature sensor in plenum air inlet (WP 1315 00).

**Step**

62. Place upper console ECS - AIR COND switch to OFF.
63. Disconnect and remove servicing manifold.
64. Turn off electrical power.
65. Make sure upper console NO. 1, NO. 2, and APU GENERATORS switches are set to OFF/RESET.
66. Make sure the following circuit breakers are pushed in:

| <b>CIRCUIT BREAKER</b>   | <b>CIRCUIT BREAKER PANEL</b> |
|--------------------------|------------------------------|
| BACKUP PUMP PWR          | Copilot's                    |
| CPLT WSHLD ANTI-ICE (dc) | Copilot's                    |
| DC ESNTL BUS SUPPLY      | Copilot's                    |
| ECS PWR                  | Copilot's                    |
| LIGHTS CAUT ADVSY        | Copilot's                    |
| NO. 1 CONVERTER          | Copilot's                    |
| NO. 1 GEN WARN           | Copilot's                    |
| APU CONTR INST           | Lower Console                |
| BATT BUS CONTR           | Lower Console                |
| ESNTL BUS AC & CONV WARN | Lower Console                |
| DC ESNTL BUS SPLY        | Pilot's                      |
| ECS CONTR                | Pilot's                      |
| NO. 2 CONVERTER          | Pilot's                      |
| NO. 2 GEN WARN           | Pilot's                      |

**ENVIRONMENTAL CONTROL SYSTEM (ECS) OPERATIONAL/TROUBLESHOOTING PROCEDURE****EH60A** - Continued

| <b>CIRCUIT BREAKER</b>    | <b>CIRCUIT BREAKER PANEL</b> |
|---------------------------|------------------------------|
| NO. 2 SERVO WARN          | Pilot's                      |
| SEC MON BUS CONTR         | Pilot's                      |
| WINDSHIELD ANTI-ICE PILOT | Pilot's                      |
| WINDSHIELD ANTI-ICE CTR   | Pilot's                      |
| BACKUP HYD CONTR          | Upper Console                |
| BATT BUS SUPPLY           | Upper Console                |
| CAUT/ADVSY PNL            | Upper Console                |

67. Start APU (TM 1-1520-237-10).

68. Place upper console APU GENERATORS switch to ON.

**Indication/Condition**

On caution/advisory panel, APU GEN ON capsule shall be on, and #1 CONV and #2 CONV capsules shall be off.

**Corrective Action**

If Indication/Condition is not as specified, troubleshoot ac electrical (APU generators) system (WP 0101 00).

**Step****NOTE**

Backup hydraulic pump may run to charge APU accumulator. ECS will not operate until backup pump stops.

69. Place upper console ECS - AIR COND switch to FAN and place HTR switch to ON.

**Indication/Condition**

CABIN HEAT ON capsule on caution/advisory panel shall go on.

**Corrective Action**

If Indication/Condition is not as specified, go to CABIN HEAT ON CAPSULE DOES NOT GO ON WITH HTR SWITCH AT ON AND AIR COND SWITCH AT FAN, in this work package.

**Step**

70. Place a locally-made drag beam wedge (WP 1805 00) between left drag beam switch plunger and switch plunger actuating bracket to simulate weight-off-wheels.

**Indication/Condition**

CABIN HEAT ON capsule on caution/advisory panel shall go off.

**Corrective Action**

1. If Indication/Condition is not as specified, check continuity between P398-c and P398-b.

**ENVIRONMENTAL CONTROL SYSTEM (ECS) OPERATIONAL/TROUBLESHOOTING PROCEDURE****EH60A** - Continued

- a. If continuity is present, troubleshoot ac electrical (interface panel ground power) system (WP 0101 00).
- b. If continuity is not present, replace No. 3 relay panel (WP 0827 00).

**Step**

71. Remove locally-made drag beam wedge from drag beam switch.
72. Place upper console ECS - AIR COND and HTR switches to OFF.
73. Place upper console APU GENERATORS switch to OFF/RESET (TM 1-1520-237-10).

**NOTE**

If #1 GEN BRG and #2 GEN BRG capsules on caution/advisory panel goes on, troubleshoot ac electrical system (WP 0101 00).

74. With qualified personnel at controls, start No. 1 and No. 2 engines and operate with main rotor turning at 100% (TM 1-1520-237-10).
75. Place upper console GENERATORS NO. 1 and NO. 2 switches to ON.

**Indication/Condition**

#1 GEN and #2 GEN capsules on caution/advisory panel shall be off.

**Corrective Action**

If Indication/Condition is not as specified, go to ECS DOES NOT OPERATE WITH BOTH MAIN GENERATORS RUNNING, in this work package.

**Step**

76. Place upper console ECS - AIR COND switch to FAN and place HTR switch to ON.

**Indication/Condition**

CABIN HEAT ON capsule on caution/advisory panel shall go on.

**Corrective Action**

If Indication/Condition is not as specified, go to CABIN HEAT ON CAPSULE DOES NOT GO ON WITH HTR SWITCH AT ON AND AIR COND SWITCH AT FAN, in this work package.

**Step**

77. Place upper console PILOT WINDSHIELD ANTI-ICE and BACKUP HYD PUMP switches to ON.

**Indication/Condition**

- a. CABIN HEAT ON capsule on caution/advisory panel shall remain on.
- b. BACK-UP PUMP ON capsule on caution/advisory panel shall go on.

**Corrective Action**

1. If Indication/Condition a. is not as specified, go to ECS DOES NOT OPERATE WITH BOTH MAIN GENERATORS RUNNING, in this work package.
2. If Indication/Condition b. is not as specified, troubleshoot hydraulic system (WP 0086 00).

**Step**

78. Place upper console PILOT WINDSHIELD ANTI-ICE and BACKUP HYD PUMP switches to OFF.

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**ENVIRONMENTAL CONTROL SYSTEM (ECS) OPERATIONAL/TROUBLESHOOTING PROCEDURE****EH60A** - Continued

79. Place upper console HTR switch to OFF and place AIR COND switch to COOL.

**Indication/Condition**

AIR COND ON capsule on caution/advisory panel shall go on.

**Corrective Action**

If Indication/Condition is not as specified, go to AIR COND ON CAPSULE DOES NOT GO ON, in this work package.

**Step**

80. Place upper console PILOT WINDSHIELD ANTI-ICE switch to ON.

**Indication/Condition**

AIR COND ON capsule on caution/advisory panel shall remain on.

**Corrective Action**

If Indication/Condition is not as specified, replace No. 3 relay panel (WP 0827 00).

**Step**

81. Place upper console PILOT WINDSHIELD ANTI-ICE switch to OFF.

82. Place upper console ECS - AIR COND switch to OFF.

83. Place upper console GENERATORS NO. 1 and NO. 2 switches to OFF/RESET.

84. Shut down No. 1 and No. 2 engines (TM 1-1520-237-10).

85. Shut down APU (TM 1-1520-237-10).

**Indication/Condition**

APU shall be shut down.

**Corrective Action**

None Required

**EVAPORATOR FAN DOES NOT OPERATE WITH AIR COND SWITCH AT COOL AND FAN****SYMPTOM**

Evaporator fan does not operate with AIR COND switch at COOL and FAN.

**MALFUNCTION**

Evaporator fan does not operate with AIR COND switch at COOL and FAN.

**CORRECTIVE ACTION****1. Check for 115 vac between:**

- P456-A and P456-D
- P456-B and P456-D
- P456-C and P456-D

a. If voltage is as specified, go to **2.**

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**EVAPORATOR FAN DOES NOT OPERATE WITH AIR COND SWITCH AT COOL AND FAN - Continued**

- b. If voltage is not as specified, go to **11**.
- 2. Check for 28 vdc between P457-B and P457-K.**
  - a. If voltage is as specified, replace electrical control unit (WP 1295 00). Go to **24**.
  - b. If trouble remains, replace evaporator fan assembly (WP 1283 00). Go to **24**.
  - c. If trouble still remains, repair/replace wiring, as required, between (WP 1747 00), then go to **24**:
    - P3-R and P9-A
    - P3-S and P9-B
    - P3-T and P9-C
    - P7-D and P9-D
    - P3-L and P9-E
    - P3-M and P9-F
  - d. If voltage is not as specified, go to **3**.
- 3. Check for 28 vdc between AIR COND switch terminal 5 and ground.**
  - a. If voltage is as specified, go to **4**.
  - b. If voltage is not as specified, go to **5**.
- 4. With AIR COND switch at FAN, check for 28 vdc between AIR COND switch terminal 6 and ground.**
  - a. If voltage is as specified, repair/replace wiring between switch terminal 6 and P457-B, or between P457-K and GNDPWR-1 (WP 1747 00). Go to **24**.
  - b. If voltage is not as specified, replace switch (WP 0828 00). Go to **24**.
- 5. Check for 28 vdc between P398-a and ground.**
  - a. If voltage is as specified, go to **6**.
  - b. If voltage is not as specified, go to **10**.
- 6. Check continuity between P398-b and P398-c.**
  - a. If continuity is present, go to **7**.
  - b. If continuity is not present, go to **8**.
- 7. Check continuity between P398-X and AIR COND switch terminal 5.**
  - a. If continuity is present, replace No. 3 relay panel (WP 0827 00). Go to **24**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **24**.
- 8. Check continuity between terminals K82-A2 and K82-A3.**
  - a. If continuity is present, repair/replace wiring between K82-A2 and P398-c, or between K82-A3 and P398-b (WP 1747 00). Go to **24**.
  - b. If continuity is not present, go to **9**.
- 9. Check for 28 vdc between K82-X1 and K82-X2.**

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**EVAPORATOR FAN DOES NOT OPERATE WITH AIR COND SWITCH AT COOL AND FAN - Continued**

- a. If voltage is as specified, troubleshoot ac electrical system (interface panel ground power procedure) (WP 0101 00). Go to **24**.
- b. If voltage is not as specified, replace relay (WP 0851 00). Go to **24**.
- 10. Check for 28 vdc between ECS CONTR circuit breaker terminal 1 and ground.**
  - a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P398-a (WP 1747 00). Go to **24**.
  - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **24**.
- 11. Check for 115 vac between ECS PWR circuit breaker:**
  - **Terminal A2 and ground**
  - **Terminal B2 and ground**
  - **Terminal C2 and ground**
  - a. If voltage is as specified, repair/replace wiring, as required, between (WP 1747 00), then go to **24**:
    - CB601-A2 and P456-A
    - CB601-B2 and P456-B
    - CB601-C2 and P456-C
    - GND456-1 and P456-D
  - b. If voltage is not as specified, go to **12**.
- 12. Check for 115 vac between ECU PWR circuit breaker:**
  - **Terminal A1 and ground**
  - **Terminal B1 and ground**
  - **Terminal C1 and ground**
  - a. If voltage is as specified, go to **13**.
  - b. If voltage is not as specified, go to **23**.
- 13. Using pin extractor part No. M83723-31-20, disconnect wire connected to ECU PWR circuit breaker terminal 5A. Go to 14.**
- 14. Check wire for continuity to ground.**
  - a. If continuity is present, go to **15**.
  - b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **24**.
- 15. Using pin extractor part No. M83723-31-20, disconnect wire connected to ECU PWR circuit breaker terminal 3. Go to 16.**
- 16. Check wire for continuity to ground.**
  - a. If continuity is present, go to **17**.
  - b. If continuity is not present, go to **19**.
- 17. Pull out ECS PWR circuit breaker. Go to 18.**



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**EVAPORATOR FAN DOES NOT OPERATE WITH AIR COND SWITCH AT COOL AND FAN - Continued****18. Check continuity between ECU PWR circuit breaker terminal 3 wire and ground.**

- a. If continuity is present, troubleshoot system circuit breaker (WP 0104 00). Go to **24**.
- b. If continuity is not present, replace circuit breaker (WP 0824 00). Go to **24**.

**19. Leave wire disconnected from ECU PWR circuit breaker terminal 3. Go to 20.****20. Check continuity between ECS PWR circuit breaker terminal 1 and ground.**

- a. If continuity is present, repair/replace wiring between circuit breaker terminal 1 and wire connected to ECU PWR CB601-3 (WP 1747 00). Go to **24**.
- b. If continuity is not present, go to **21**.

**21. Pull out ECS PWR circuit breaker. Go to 22.****22. Check continuity between ECS PWR circuit breaker terminal 2 and ground.**

- a. If continuity is present, replace circuit breaker (WP 0824 00). Go to **24**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **24**.

**23. Check for 115 vac between contactor K1:**

- **Terminal A3 and ground**
- **Terminal B3 and ground**
- **Terminal C3 and ground**

- a. If voltage is as specified, repair/replace wiring, as required, between (WP 1747 00), then go to **24**:

- K1-A3 and CB601-A1
- K1-B3 and CB601-B1
- K1-C3 and CB601-C1

- b. If voltage is not as specified, troubleshoot ac electrical system (APU generator procedure) (WP 0101 00). Go to **24**.

**24. Procedure completed.****AIR DOES NOT FLOW FROM EVAPORATOR OUTLET DUCT****SYMPTOM**

Air does not flow from evaporator outlet duct.

**MALFUNCTION**

Air does not flow from evaporator outlet duct.

**CORRECTIVE ACTION****1. Check for blocked distribution ducts.**

- a. If distribution ducts are blocked, remove obstruction. Go to **4**.
- b. If distribution ducts are not blocked, go to **2**.

**2. Check for clogged demister.**

- a. If demister is clogged, replace demister pad (WP 1315 00). Go to **4**.

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**AIR DOES NOT FLOW FROM EVAPORATOR OUTLET DUCT - Continued**

- b. If demister is not clogged, go to **3**.
- 3. Check for blocked evaporator core.**
  - a. If trouble remains, replace evaporator (WP 1315 00). Go to **4**.
- 4. Procedure completed.**

**CABIN HEAT ON CAPSULE DOES NOT GO ON WITH HTR SWITCH AT ON AND AIR COND SWITCH AT FAN****SYMPTOM**

CABIN HEAT ON capsule does not go on with HTR switch at ON and AIR COND switch at FAN.

**MALFUNCTION**

CABIN HEAT ON capsule does not go on with HTR switch at ON and AIR COND switch at FAN.

**CORRECTIVE ACTION**

- 1. Check for 28 vdc between P457-C and ground.**
  - a. If voltage is as specified, go to **2**.
  - b. If voltage is not as specified, go to **8**.
- 2. Check for 28 vdc between P457-J and ground.**
  - a. If voltage is as specified, go to **3**.
  - b. If voltage is not as specified, repair/replace wiring between ECS CONTR circuit breaker terminal 1 and P457-J (WP 1747 00). Go to **12**.
- 3. Check helicopter configuration.**
  - a. **EMEP** , go to **4**.
  - b. **W/O EMEP** , go to **6**.
- 4. EMEP Remove and check pin filtered adapter P118 (WP 0925 00).**
  - a. If pin filtered adapter is good, go to **5**.
  - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **12**.
- 5. EMEP Install pin filtered adapter (WP 0925 00). Go to 6.**
- 6. Check for 28 vdc between P118-M and ground.**
  - a. If voltage is as specified, troubleshoot caution/advisory warning system (WP 0090 00). Go to **12**.
  - b. If voltage is not as specified, go to **7**.
- 7. Check continuity between P118-M and P457-E.**
  - a. If continuity is present, replace electrical control unit (WP 1295 00). Go to **12**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **12**.
- 8. Check for 28 vdc between P398-e and ground.**
  - a. If voltage is as specified, go to **9**.
  - b. If voltage is not as specified, go to **11**.

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**CABIN HEAT ON CAPSULE DOES NOT GO ON WITH HTR SWITCH AT ON AND AIR COND SWITCH AT FAN - Continued****9. With HTR switch at ON, check continuity between P398-d and P457-C.**

- a. If continuity is present, replace No. 3 panel (WP 0827 00). Go to **12**.
- b. If continuity is not present, go to **10**.

**10. With HTR switch ON, check continuity between HTR switch terminals 2 and 3.**

- a. If continuity is present, repair/replace wiring between switch terminal 2 and P398-d, or between switch terminal 3 and P457-C (WP 1747 00). Go to **12**.
- b. If continuity is not present, replace switch (WP 0828 00). Go to **12**.

**11. With AIR COND switch at FAN, check for 28 vdc between AIR COND switch terminal 3 and ground.**

- a. If voltage is as specified, repair/replace wiring between switch terminal 3 and P398-e (WP 1747 00). Go to **12**.
- b. If voltage is not as specified, replace switch (WP 0828 00). Go to **12**.

**12. Procedure completed.****AMMETER INDICATES LESS THAN 7 AMPS THROUGH HEATER DEMISTER****SYMPTOM**

Ammeter indicates less than 7 amps through heater demister.

**MALFUNCTION**

Ammeter indicates less than 7 amps through heater demister.

**CORRECTIVE ACTION****1. Check ammeter.**

- a. If ammeter indicates more than 0 but less than 7 amps, replace heater demister assembly (WP 1315 00). Go to **3**.
- b. If ammeter indicates 0 amps, go to **2**.

**2. Check continuity between P3-D and P3-E.**

- a. If continuity is present, replace electrical control unit (WP 1295 00). Go to **3**.
- b. If trouble remains, replace heater demister assembly (WP 1315 00). Go to **3**.
- c. If continuity is not present, replace high temperature switch assembly (WP 1315 00). Go to **3**.

**3. Procedure completed.****CONDENSER BLOWER DOES NOT GO ON WITH AIR COND SWITCH AT COOL****SYMPTOM**

Condenser blower does not go on with AIR COND switch at COOL.

**MALFUNCTION**

Condenser blower does not go on with AIR COND switch at COOL.

**CORRECTIVE ACTION****1. Check for 28 vdc between P457-A and ground.**

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**CONDENSER BLOWER DOES NOT GO ON WITH AIR COND SWITCH AT COOL - Continued**

- a. If voltage is as specified, replace electrical control unit (WP 1295 00). Go to **7**.
- b. If trouble remains, go to **2**.
- c. If voltage is not as specified, go to **3**.

**2. Check continuity between:**

- **P4-A and P10-A**
- **P4-B and P10-B**
- **P4-C and P10-C**
- **P4-D and P10-D**
- **P4-E and P10-E**
- **P4-F and P10-F**

- a. If continuity is present, replace condenser blower assembly (WP 1287 00). Go to **7**.
- b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **7**.

**3. Check for 28 vdc between P398-f and ground.**

- a. If voltage is as specified, go to **4**.
- b. If voltage is not as specified, repair/replace wiring between P398-f and AIR COND switch S72 terminal 1 (WP 1747 00). Go to **7**.

**4. Check for 28 vdc between:**

- **P398-V and ground**
- **P398-W and ground**
- **P398-U and ground**

- a. If voltage is as specified, troubleshoot windshield anti-ice system (WP 0144 00). Go to **7**.
- b. If voltage is not as specified, go to **5**.

**5. Check for 28 vdc between P398-j and ground.**

- a. If voltage is as specified, troubleshoot hydraulic system (WP 0086 00). Go to **7**.
- b. If voltage is not as specified, go to **6**.

**6. Check continuity between P398-B and P457-A.**

- a. If continuity is present, replace No. 3 relay panel (WP 0827 00). Go to **7**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **7**.

**7. Procedure completed.**

**AIR IS NOT COMING OUT OF CONDENSER EXHAUST DUCT****SYMPTOM**

Air is not coming out of condenser exhaust duct.

**MALFUNCTION**

Air is not coming out of condenser exhaust duct.

**CORRECTIVE ACTION****1. Check for blocked condenser core.**

- a. If condenser core is blocked, remove obstruction. Go to **3**.
- b. If condenser core is not blocked, go to **2**.

**2. Check for blocked airduct.**

- a. If airduct is blocked, remove obstruction. Go to **3**.
- b. If airduct is not blocked, check for blocked air outlet on side of aircraft and remove obstruction. Go to **3**.

**3. Procedure completed.****COMPRESSOR MOTOR DOES NOT OPERATE WITH AIR COND SWITCH AT COOL****SYMPTOM**

Compressor motor does not operate with AIR COND switch at COOL.

**MALFUNCTION**

Compressor motor does not operate with AIR COND switch at COOL.

**CORRECTIVE ACTION****1. Check that VAC gauge reads at least 60 psig.**

- a. If gauge reads at least 60 psig, go to **2**.
- b. If gauge does not read at least 60 psig, perform leak test and service system (WP 1622 00). Go to **11**.

**2. Check for 115 vac between:**

- J7-A and ground
- J7-B and ground
- J7-C and ground

- a. If voltage is as specified, go to **3**.
- b. If voltage is not as specified, go to **4**.

**3. Check continuity between:**

- P7-A and P8-A
- P7-B and P8-B
- P7-C and P8-C

---

**COMPRESSOR MOTOR DOES NOT OPERATE WITH AIR COND SWITCH AT COOL - Continued**

- a. If continuity is present, replace compressor motor assembly (WP 1316 00). Go to **11**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **11**.
- 4. Check continuity between P3-K and P3-J.**
  - a. If continuity is present, go to **5**.
  - b. If continuity is not present, go to **10**.
- 5. Check continuity between P6-B and P6-C.**
  - a. If continuity is present, go to **6**.
  - b. If continuity is not present, go to **9**.
- 6. Check continuity between P5-A and P5-B.**
  - a. If continuity is present, go to **7**.
  - b. If continuity is not present, go to **8**.
- 7. Check continuity between P3-A and P3-B.**
  - a. If continuity is present, replace electrical control unit (WP 1295 00). Go to **11**.
  - b. If continuity is not present, replace low temperature switch (WP 1286 00). Go to **11**.
- 8. Check continuity between J11-A and J11-B.**
  - a. If continuity is present, repair/replace wiring between P5-A and P11-A, or between P5-B and P11-B (WP 1747 00). Go to **11**.
  - b. If continuity is not present, replace high pressure switch (WP 1323 00). Go to **11**.
- 9. Check continuity between J12-B and J12-C.**
  - a. If continuity is present, repair/replace wiring between P6-B and P12-B, or between P6-C and P12-C (WP 1747 00). Go to **11**.
  - b. If continuity is not present, replace low pressure switch (WP 1323 00). Go to **11**.
- 10. Check continuity between J8-D and J8-E.**
  - a. If continuity is present, repair/replace wiring between P3-K and P8-E, or between P3-J and P8-D (WP 1747 00). Go to **11**.
  - b. If continuity is not present, replace compressor assembly (WP 1316 00). Go to **11**.
- 11. Procedure completed.**

**AIR COND ON CAPSULE DOES NOT GO ON****SYMPTOM**

AIR COND ON capsule does not go on.

**MALFUNCTION**

AIR COND ON capsule does not go on.

**CORRECTIVE ACTION****1. Check helicopter configuration.**

- a. **EMEP** , go to **2**.

---

**AIR COND ON CAPSULE DOES NOT GO ON** - Continued

- b. **W/O EMEP** , go to 4.
2. **EMEP** Remove and check pin filtered adapter P118 (WP 0925 00).
  - a. If pin filtered adapter is good, go to 3.
  - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to 6.
3. **EMEP** Install pin filtered adapter (WP 0925 00). Go to 4.
4. Check for 28 vdc between P118-G and ground.
  - a. If voltage is as specified, troubleshoot caution/advisory warning system (WP 0090 00). Go to 6.
  - b. If voltage is not as specified, go to 5.
5. Check continuity between P118-G and P457-D.
  - a. If continuity is present, replace electrical control unit (WP 1295 00). Go to 6.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to 6.
6. Procedure completed.

**AIR FROM EVAPORATION OUTLET DUCT DOES NOT TURN COOL****SYMPTOM**

Air from evaporation outlet duct does not turn cool.

**MALFUNCTION**

Air from evaporation outlet duct does not turn cool.

**CORRECTIVE ACTION**

1. Over a period of five minutes, observe that vac gauge pressure decreases and increases.
  - a. If pressure increases and decreases, replace thermister (WP 1318 00). Go to 2.
  - b. If pressure does not increase or decrease, replace expansion valve (WP 1317 00). Go to 2.
2. Procedure completed.

**WINDSHIELD ANTI-ICE SYSTEM DOES NOT DISABLE ECS****SYMPTOM**

Windshield anti-ice system does not disable ECS.

**MALFUNCTION**

Windshield anti-ice system does not disable ECS.

**CORRECTIVE ACTION**

1. Is ECS disabled with WINDSHIELD ANTI-ICE COPILOT switch ON?
  - a. If ECS is disabled, go to 2.
  - b. If ECS is not disabled, go to 9.
2. Is ECS disabled with WINDSHIELD ANTI-ICE PILOT switch ON?
  - a. If ECS is disabled, go to 3.

---

**WINDSHIELD ANTI-ICE SYSTEM DOES NOT DISABLE ECS - Continued**

- b. If ECS is not disabled, go to **7**.
- 3. With WINDSHIELD ANTI-ICE CTR switch ON, check for 28 vdc between P398-U and ground.**
  - a. If voltage is as specified, replace No. 3 relay panel (WP 0827 00). Go to **11**.
  - b. If voltage is not as specified, go to **4**.
- 4. Check helicopter configuration.**
  - a. **WINTER** , go to **5**.
  - b. **W/O WINTER** , go to **6**.
- 5. **WINTER** Check continuity between P398-U and J244B-R.**
  - a. If continuity is present, troubleshoot windshield anti-ice system (WP 0144 00). Go to **11**.
  - b. If continuity is not present, replace **WINTER** (WP 1400 00). Go to **11**.
  - c. If trouble remains, go to **6**.
- 6. Check continuity between P398-U and P244-R.**
  - a. If continuity is present, troubleshoot windshield anti-ice system (WP 0144 00). Go to **11**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **11**.
- 7. With WINDSHIELD ANTI-ICE PILOT switch ON, check for 28 vdc between P398-W and ground.**
  - a. If voltage is as specified, replace No. 3 relay panel (WP 0827 00). Go to **11**.
  - b. If voltage is not as specified, go to **8**.
- 8. Check continuity between P398-W and P900-B.**
  - a. If continuity is present, troubleshoot windshield anti-ice system (WP 0144 00). Go to **11**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **11**.
- 9. With WINDSHIELD ANTI-ICE COPILOT switch ON, check for 28 vdc between P398-V and ground.**
  - a. If voltage is as specified, replace No. 3 relay panel (WP 0827 00). Go to **11**.
  - b. If voltage is not as specified, go to **10**.
- 10. Check continuity between P398-V and P900-g.**
  - a. If continuity is present, troubleshoot windshield anti-ice system (WP 0144 00). Go to **11**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **11**.
- 11. Procedure completed.**



**COMPRESSOR DRAWS LESS THAN MINIMUM ALLOWABLE CURRENT****SYMPTOM**

Compressor draws less than minimum allowable current.

**MALFUNCTION**

Compressor draws less than minimum allowable current.

**CORRECTIVE ACTION****1. Check that vac gauge indicates less than 60 psig.**

- a. If gauge indicates less than 60 psig, go to COMPRESSOR MOTOR DOES NOT OPERATE WITH AIR COND SWITCH AT COOL, in this work package.
- b. If gauge does not indicate less than 60 psig, go to **2**.

**2. Check for leak in system.**

- a. If leak is present, repair leak and service air conditioning system (WP 1622 00). Go to **3**.
- b. If leak is not present, service air conditioning system (WP 1622 00). Go to **3**.

**3. Procedure completed.****COMPRESSOR DRAWS MORE THAN MAXIMUM ALLOWABLE CURRENT****SYMPTOM**

Compressor draws more than maximum allowable current.

**MALFUNCTION**

Compressor draws more than maximum allowable current.

**CORRECTIVE ACTION****1. Check that REF gauge indicates more than 305 psig.**

- a. If gauge indicates more than 305 psig, go to **2**.
- b. If gauge indicates less than 305 psig, replace compressor (WP 1316 00). Go to **3**.

**2. Remove freon from system to reduce compressor current to below maximum allowable current.**

- a. If trouble remains, replace compressor (WP 1316 00). Go to **3**.

**3. Procedure completed.****LOW PRESSURE DOES NOT INCREASE****SYMPTOM**

Low pressure does not increase.

**MALFUNCTION**

Low pressure does not increase.

**CORRECTIVE ACTION****1. Check continuity between J246-L and J246-K.**

- a. If continuity is present, go to **2**.

---

**LOW PRESSURE DOES NOT INCREASE** - Continued

- b. If continuity is not present, replace TEMP rheostat (WP 0828 00). Go to **3**.
- 2. Check continuity between P457-F and P457-G.**
  - a. If continuity is present, replace ECU (WP 1295 00). Go to **3**.
  - b. If continuity is not present, repair/replace wiring between P457 and P246 (WP 1747 00). Go to **3**.
- 3. Procedure completed.**

**ECS DOES NOT OPERATE WITH BOTH MAIN GENERATORS RUNNING****SYMPTOM**

ECS does not operate with both main generators running.

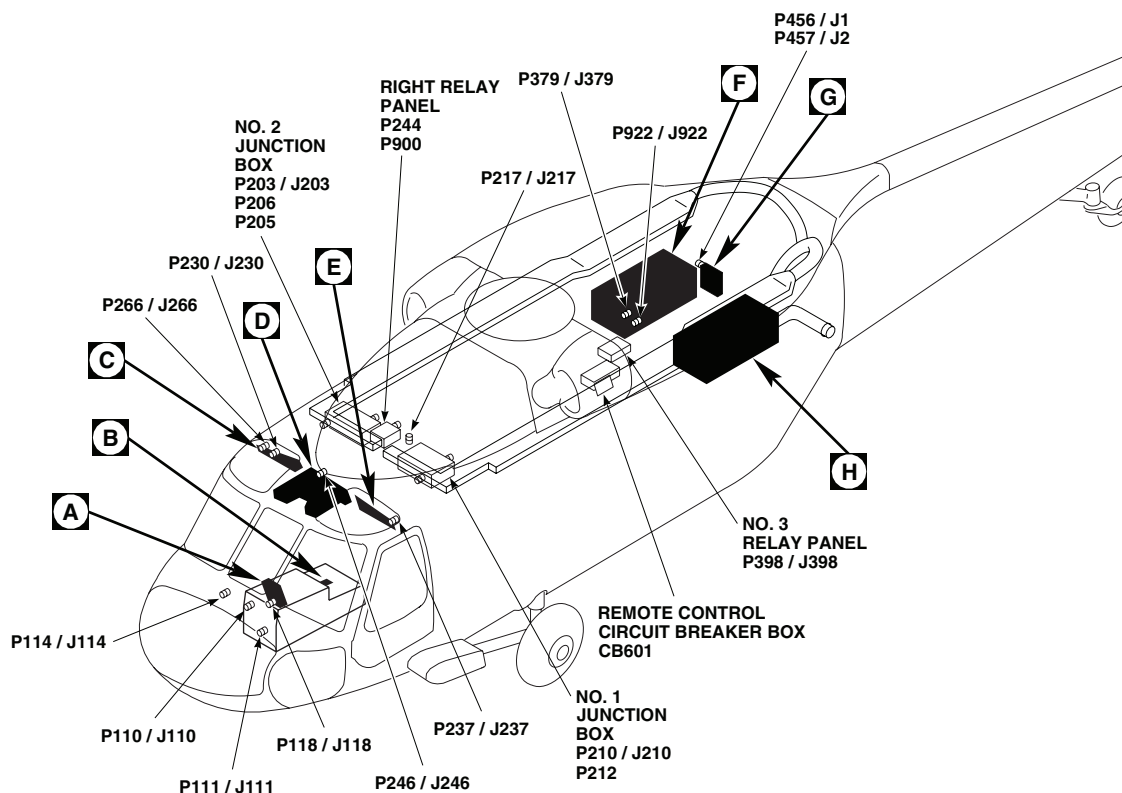
**MALFUNCTION**

ECS does not operate with both main generators running.

**CORRECTIVE ACTION**

- 1. Check for 28 vdc between P398-g and ground.**
  - a. If voltage is as specified, replace No. 3 relay panel (WP 0827 00). Go to **3**.
  - b. If voltage is not as specified, go to **2**.
- 2. Check for 28 vdc between K81-X1 and ground.**
  - a. If voltage is as specified, repair/replace wiring between P398-g and K81-X1 (WP 1747 00). Go to **3**.
  - b. If voltage is not as specified, troubleshoot ac electrical system (mission interface panel dual generators power) (WP 0101 00). Go to **3**.
- 3. Procedure completed.**

## LOCATION



| TERMINAL BOARD/<br>DISCONNECT PLUG/<br>RECEPTACLE | LOCATION/<br>CONNECTION POINT      |
|---------------------------------------------------|------------------------------------|
| P3 / J3                                           | ELECTRICAL CONTROL UNIT            |
| P4 / J4                                           | ELECTRICAL CONTROL UNIT            |
| P5 / J5                                           | ELECTRICAL CONTROL UNIT            |
| P6 / J6                                           | ELECTRICAL CONTROL UNIT            |
| P7 / J7                                           | ELECTRICAL CONTROL UNIT            |
| P8 / J8                                           | COMPRESSOR / MOTOR                 |
| P9 / J9                                           | EVAPORATOR FAN                     |
| P10 / J10                                         | CONDENSER FAN                      |
| P11 / J11                                         | HIGH PRESSURE SWITCH               |
| P12 / J12                                         | LOW PRESSURE SWITCH                |
| P110 / J110                                       | COCKPIT, BL 7.5 RH, STA 197        |
| P111 / J111                                       | COCKPIT, LB 7.5 LH, STA 197        |
| P114 / J114                                       | COCKPIT, BL 8, STA 200             |
| P118 / J118                                       | CAUTION / ADVISORY PANEL           |
| P203 / J203                                       | NO. 2 MAIN ELECTRICAL JUNCTION BOX |
| P205                                              | NO. 2 JUNCTION BOX                 |

| TERMINAL BOARD/<br>DISCONNECT PLUG/<br>RECEPTACLE | LOCATION/<br>CONNECTION POINT                  |
|---------------------------------------------------|------------------------------------------------|
| P206                                              | NO. 2 JUNCTION BOX                             |
| P210 / J210                                       | NO. 1 JUNCTION BOX                             |
| P212                                              | NO. 1 JUNCTION BOX                             |
| P217 / J217                                       | CABIN CEILING, BL 0, STA 284                   |
| P230 / J230                                       | MAIN ROTOR PYLON DECK, BL 8 RH, STA 247        |
| P237 / J237                                       | COPILLOT'S CIRCUIT BREAKER PANEL               |
| P244                                              | RIGHT RELAY PANEL                              |
| P246 / J246                                       | CABIN CEILING, BL 5 RH, STA 247                |
| P266 / J266                                       | BEHIND PILOT'S CIRCUIT BREAKER PANEL, BL 25 RH |
| P379 / J379                                       | CABIN CEILING, STA 379, WL 269, BL 5           |
| P398 / J398                                       | NO. 3 RELAY PANEL                              |
| P456 / J1                                         | ELECTRICAL CONTROL UNIT                        |
| P457 / J2                                         | ELECTRICAL CONTROL UNIT                        |
| P900                                              | RIGHT RELAY PANEL                              |
| P922 / J922                                       | CABIN CEILING, BL 10.7 LH, STA 380             |

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Figure 1. Environmental Control System Location Diagram. (Sheet 1 of 6)

LOCATION - Continued

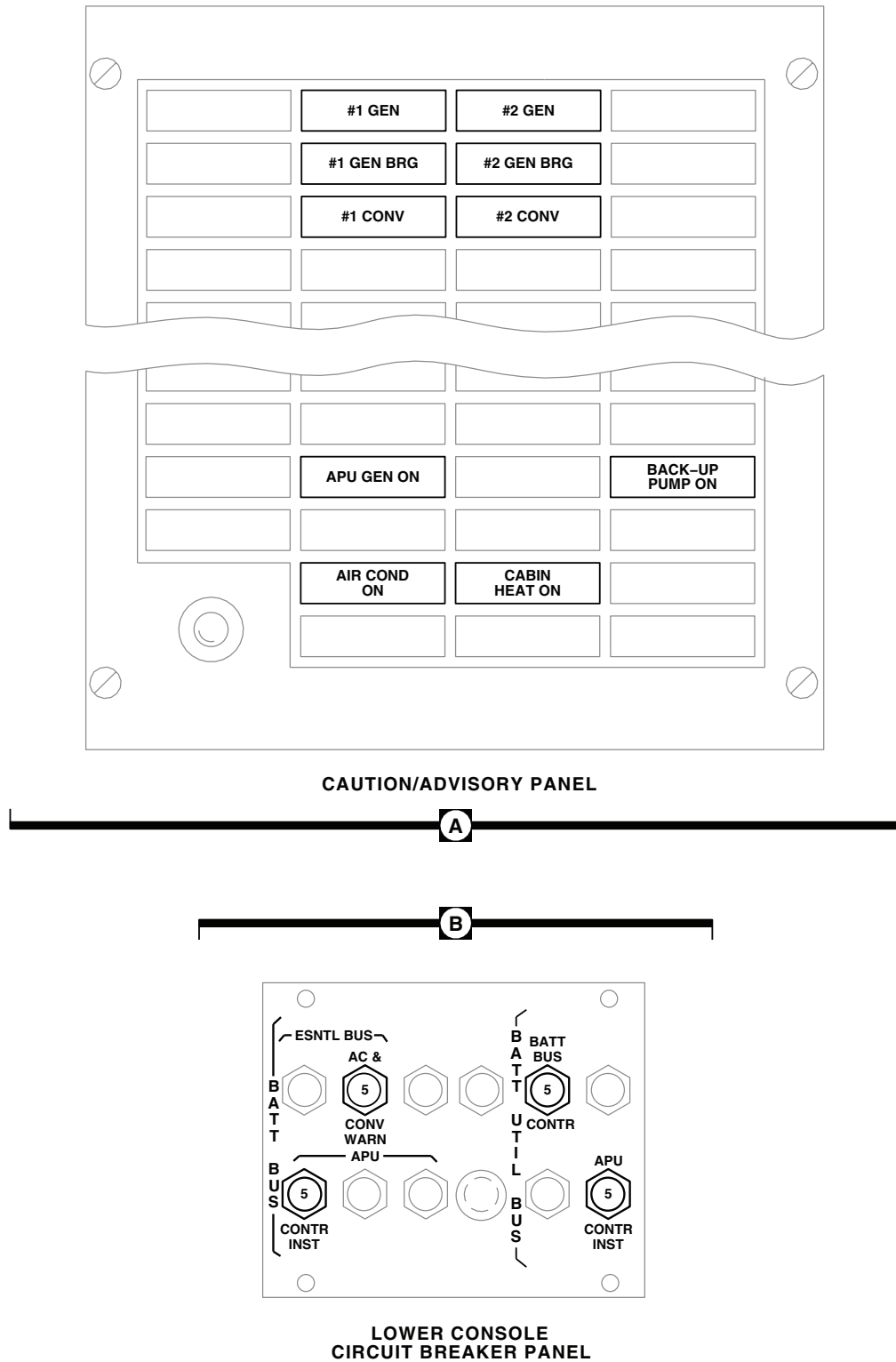
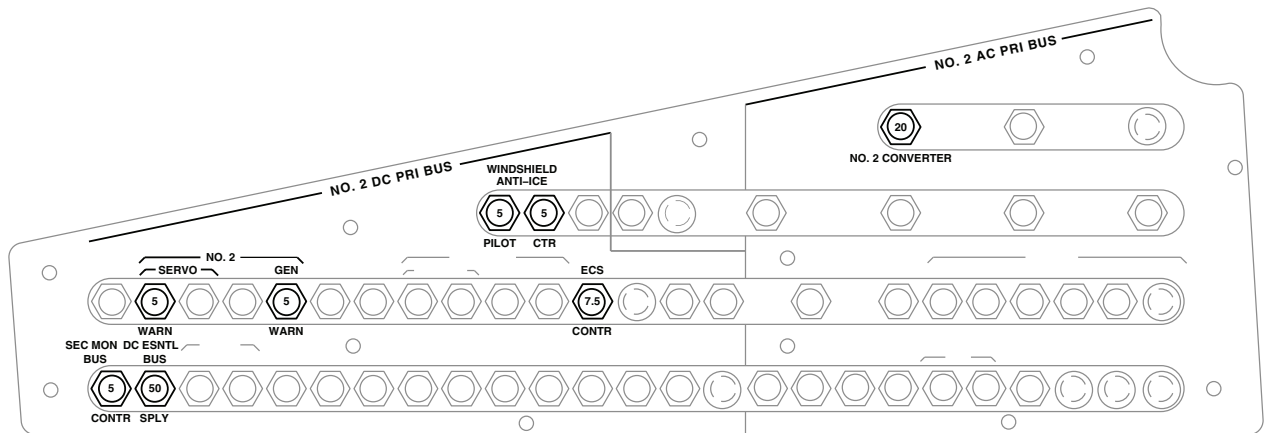


Figure 1. Environmental Control System Location Diagram. (Sheet 2 of 6)

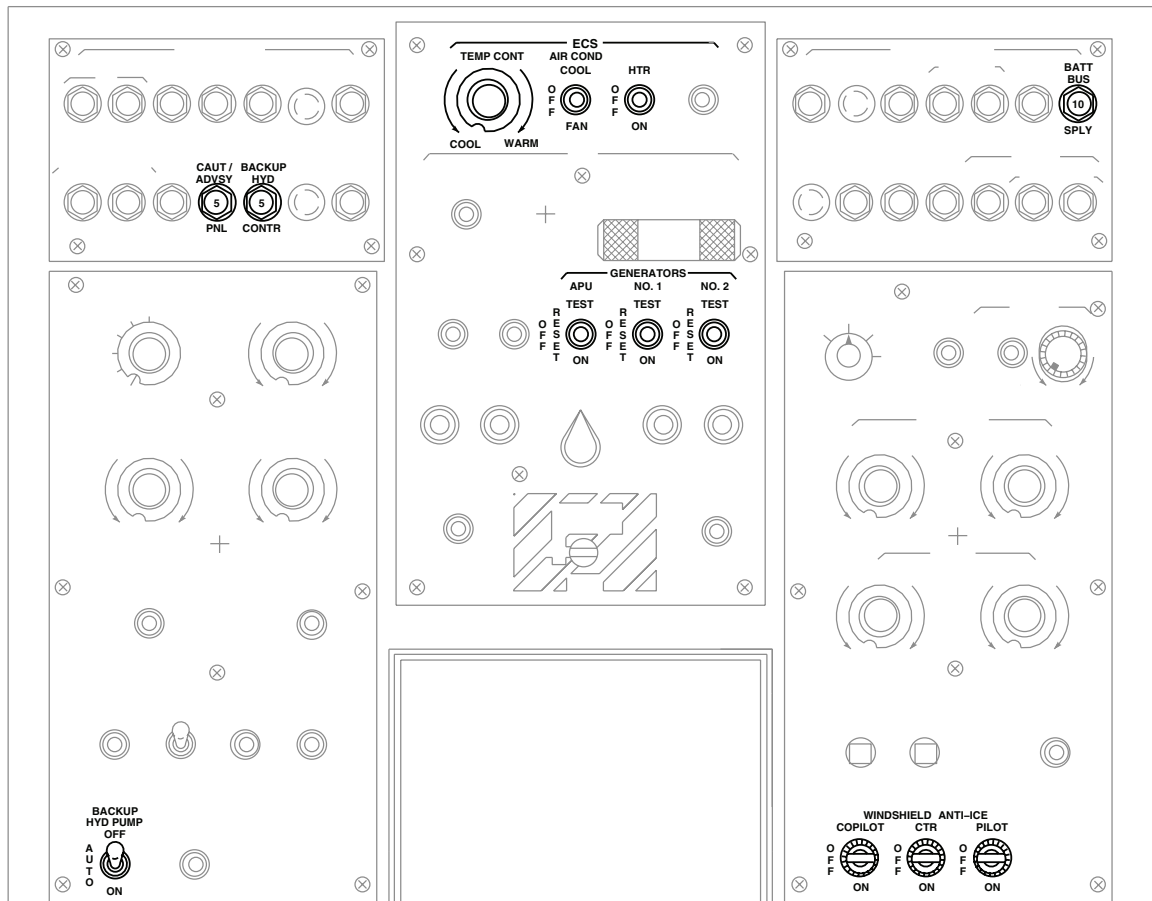
LOCATION - Continued



PILOT'S CIRCUIT BREAKER PANEL

C

D

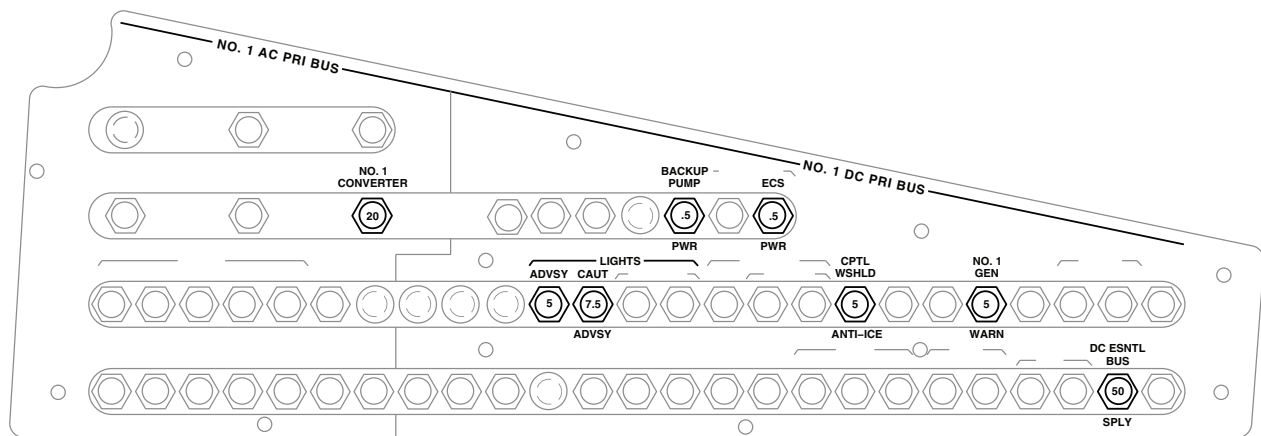


UPPER CONSOLE

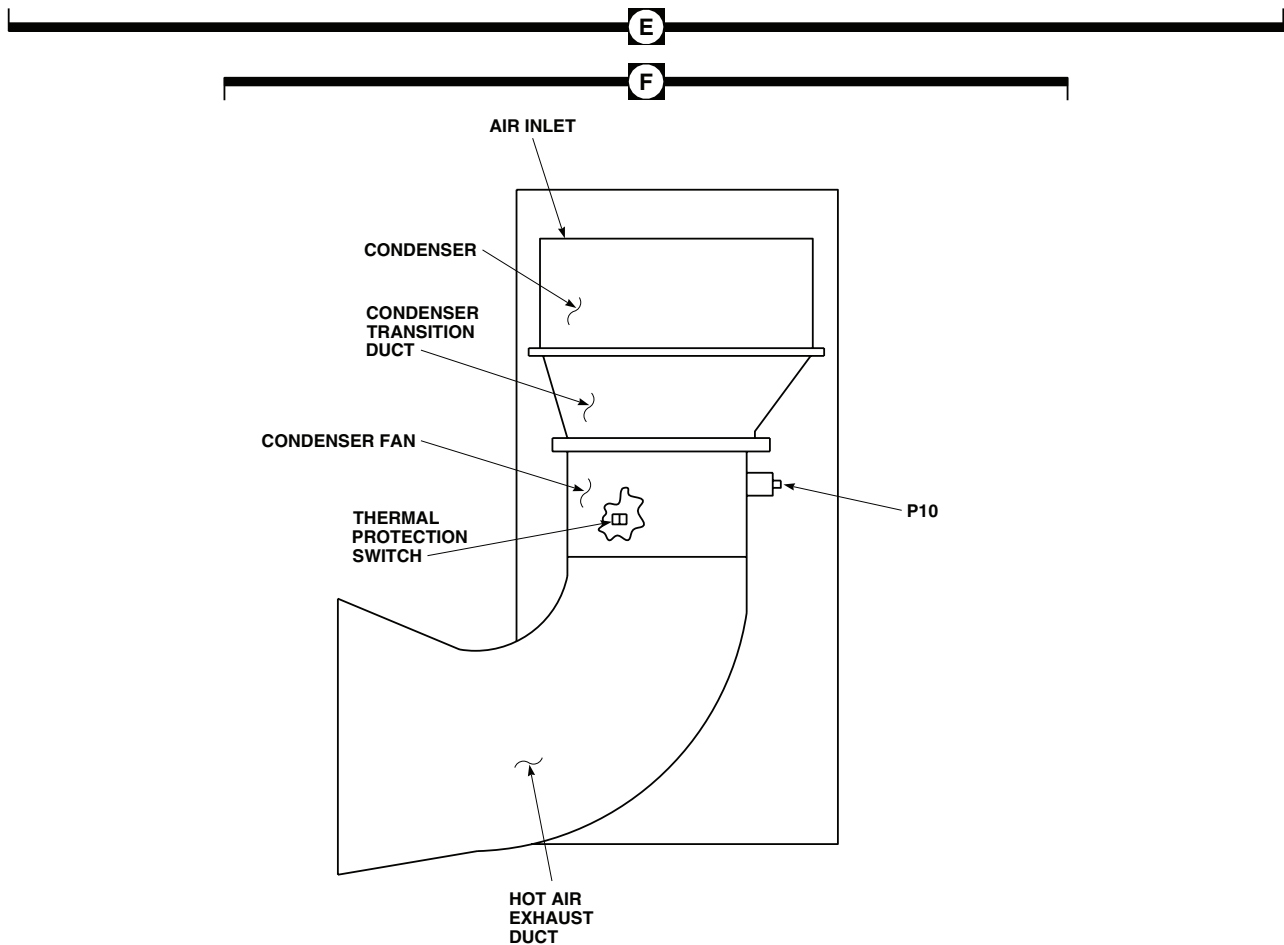
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Figure 1. Environmental Control System Location Diagram. (Sheet 3 of 6)

LOCATION - Continued



COPILOT'S CIRCUIT BREAKER PANEL



CONDENSER PALLET

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Figure 1. Environmental Control System Location Diagram. (Sheet 4 of 6)

## LOCATION - Continued

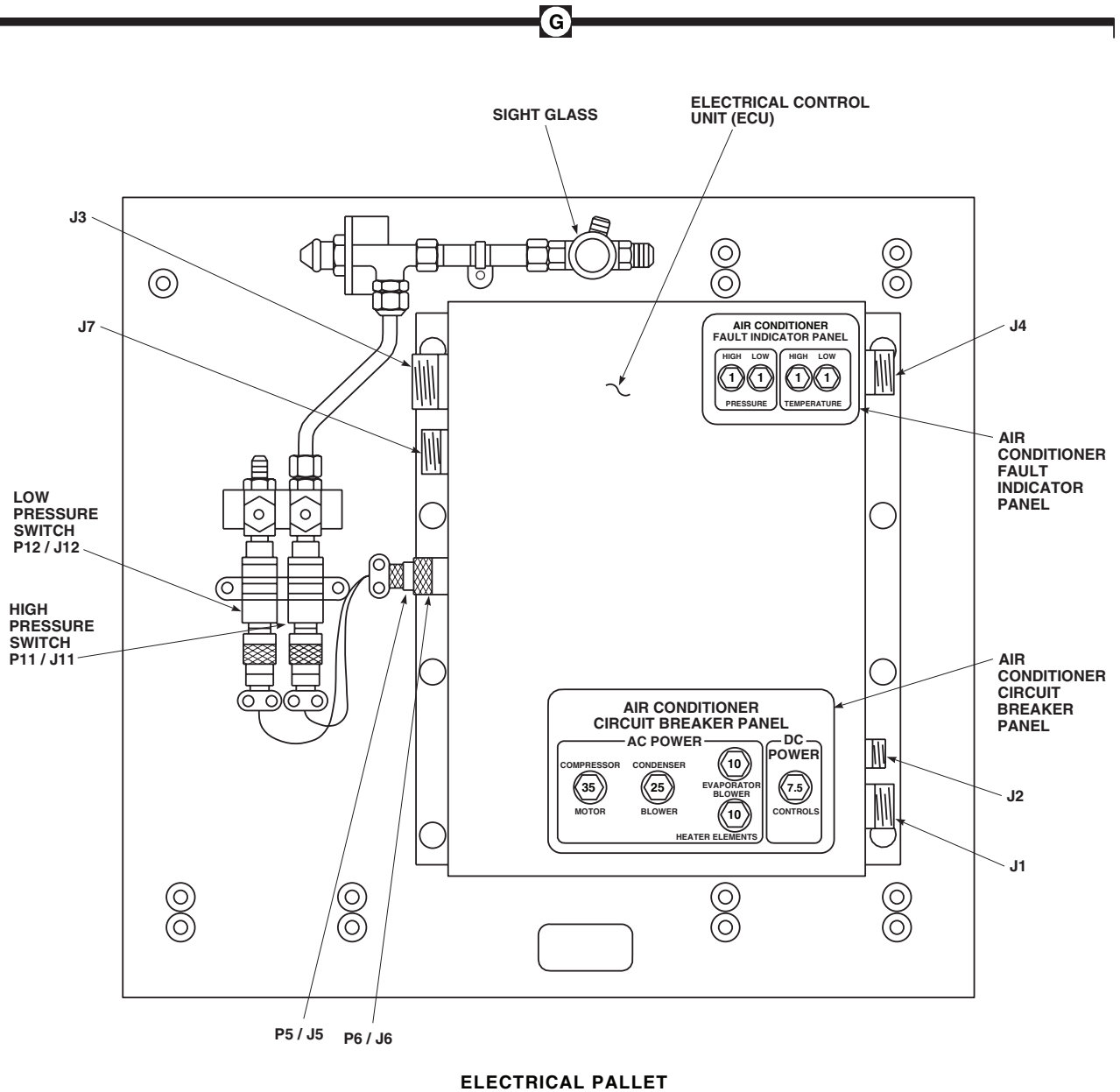
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Figure 1. Environmental Control System Location Diagram. (Sheet 5 of 6)

## LOCATION - Continued

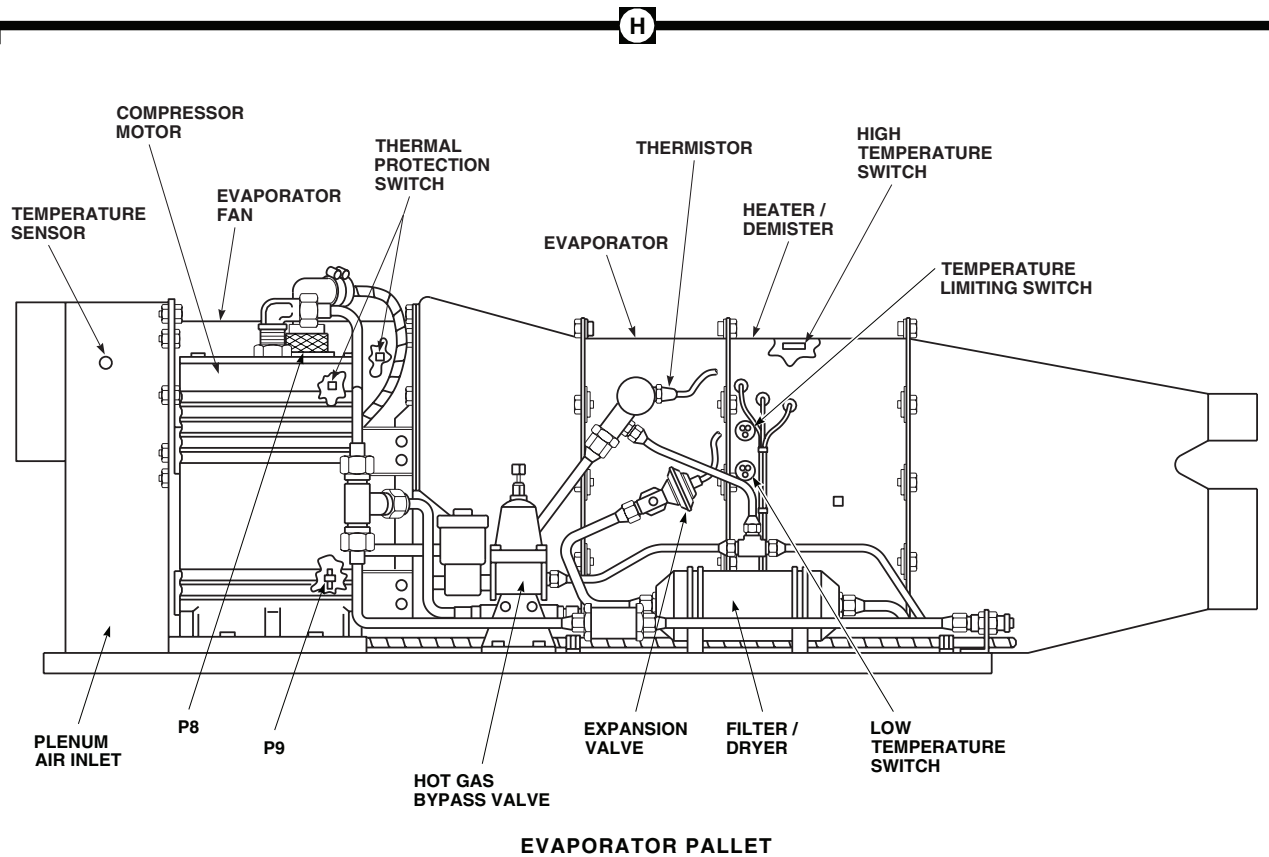
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Figure 1. Environmental Control System Location Diagram. (Sheet 6 of 6)



## SCHEMATICS

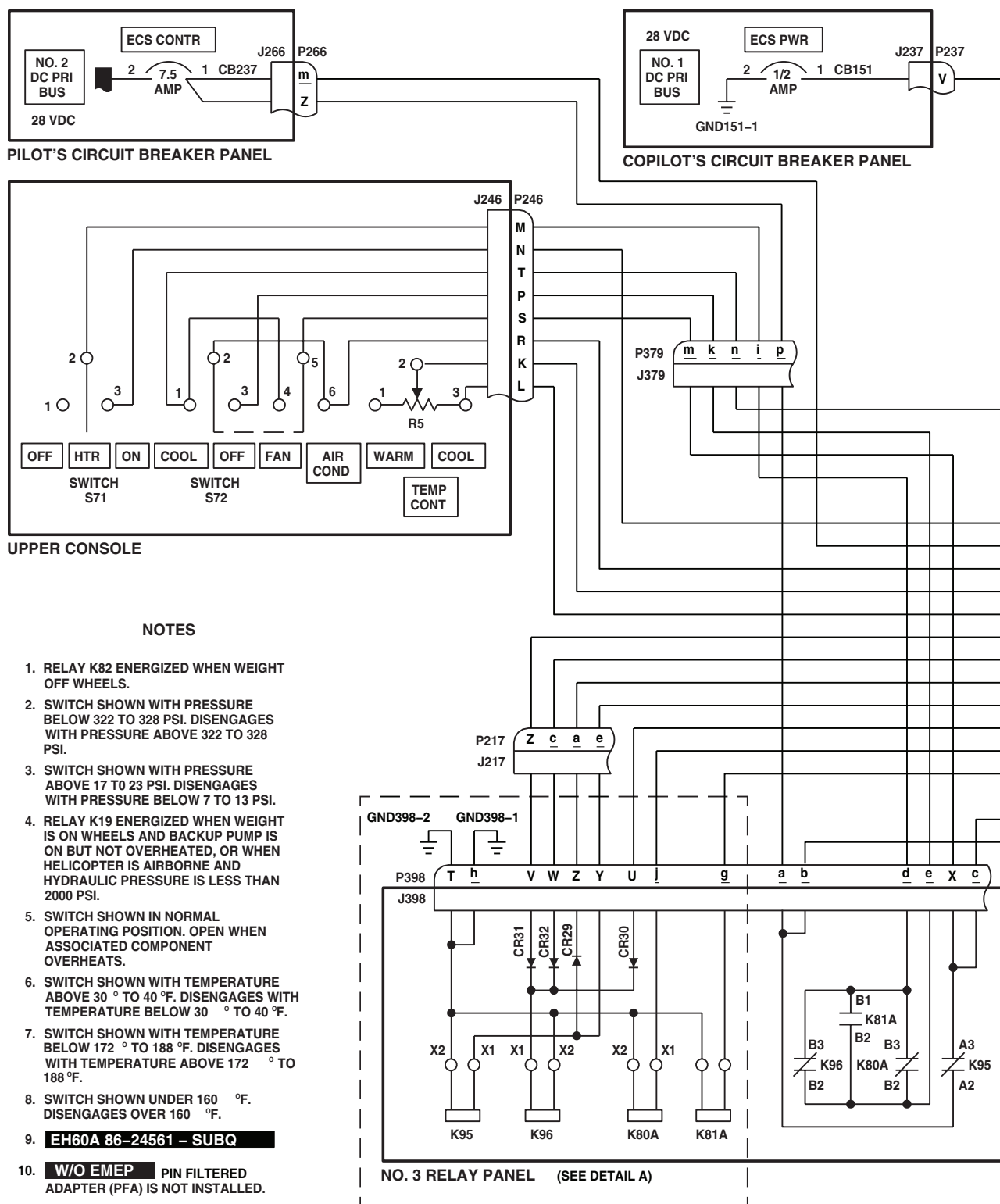
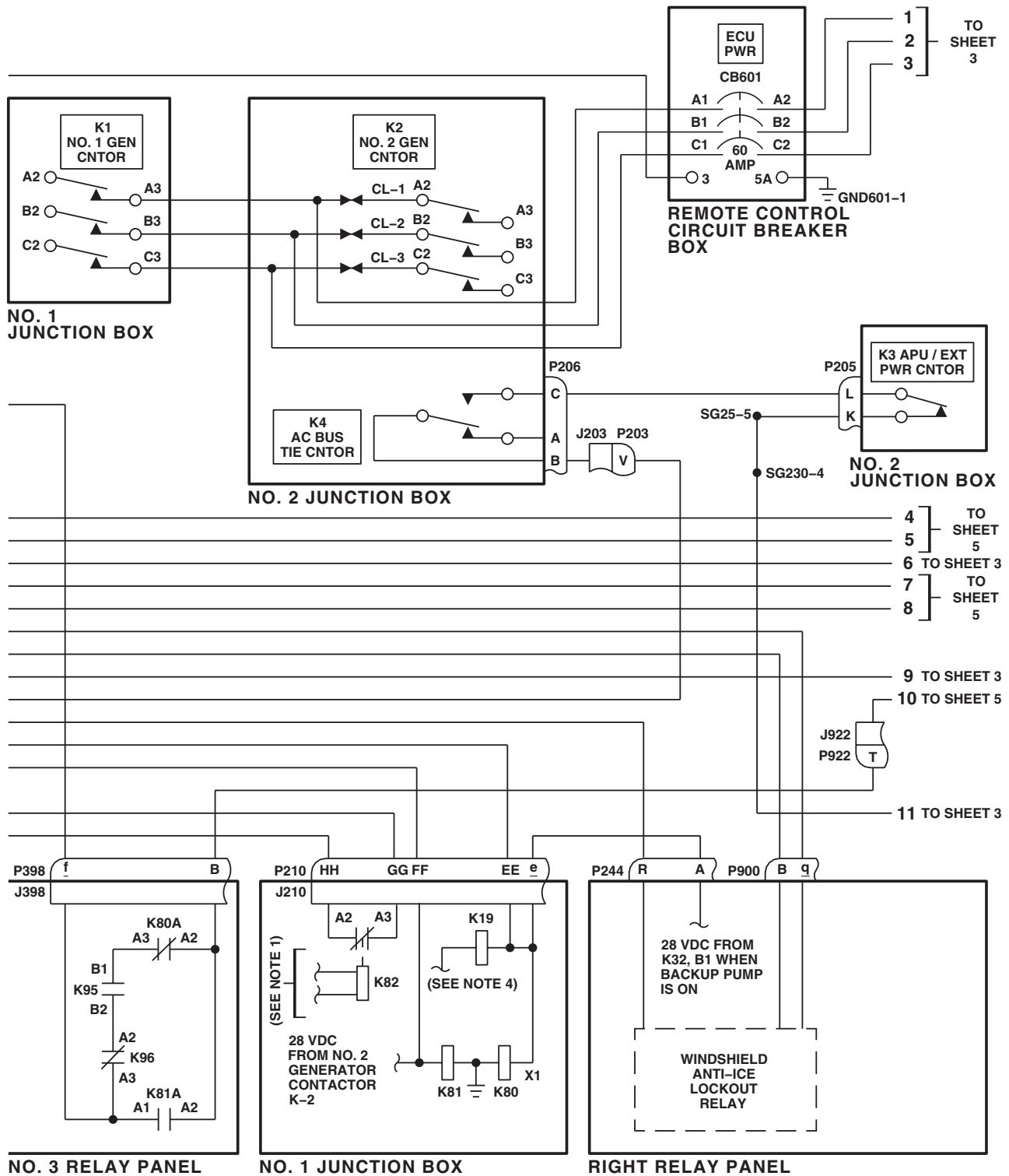
AB1522\_1  
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Figure 2. Environmental Control System Schematic Diagram. (Sheet 1 of 10)

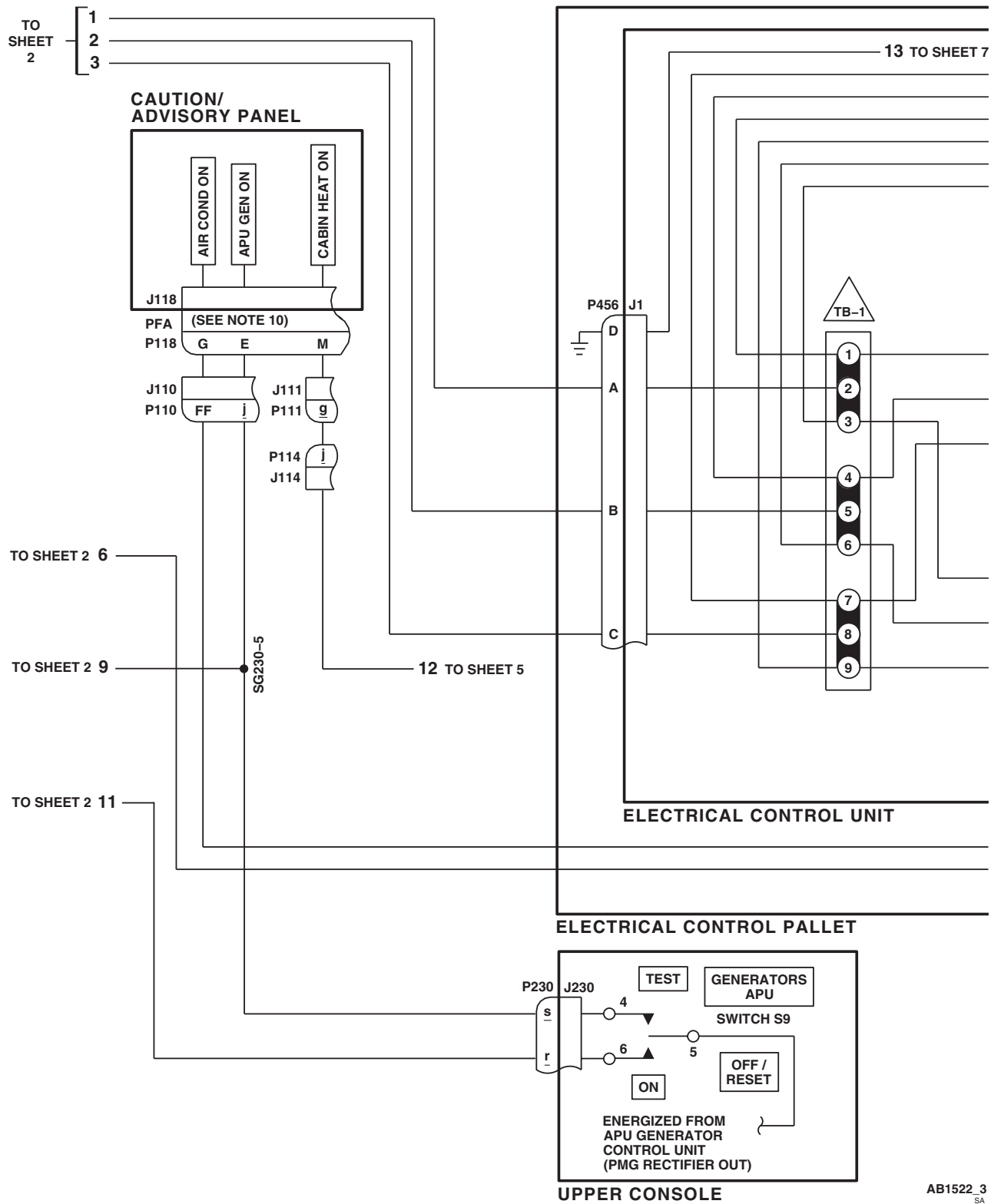
SCHEMATICS - Continued



AB1522 2  
SA

Figure 2. Environmental Control System Schematic Diagram. (Sheet 2 of 10)

SCHEMATICS - Continued



AB1522\_3  
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Figure 2. Environmental Control System Schematic Diagram. (Sheet 3 of 10)

SCHEMATICS - Continued

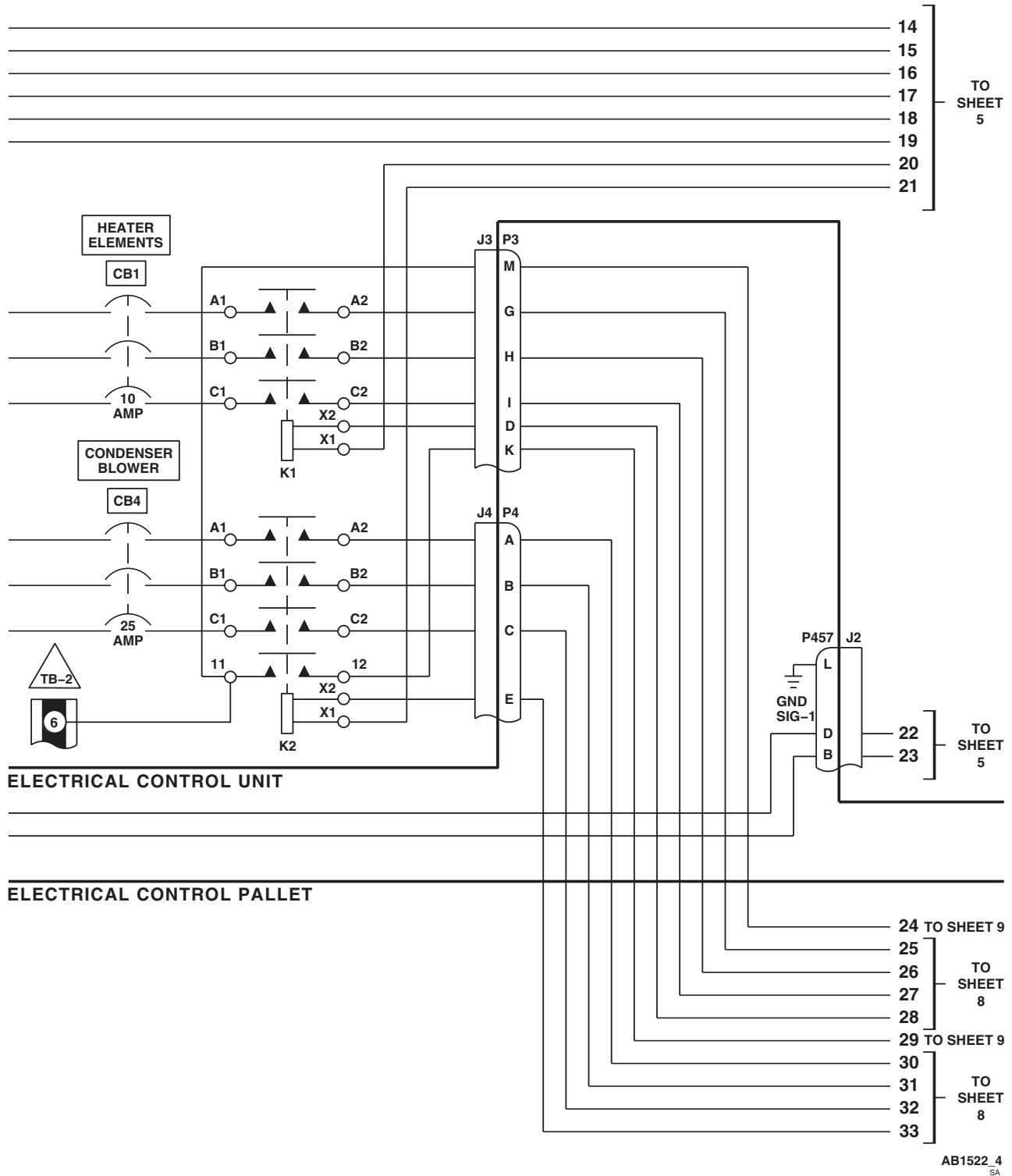


Figure 2. Environmental Control System Schematic Diagram. (Sheet 4 of 10)

SCHEMATICS - Continued

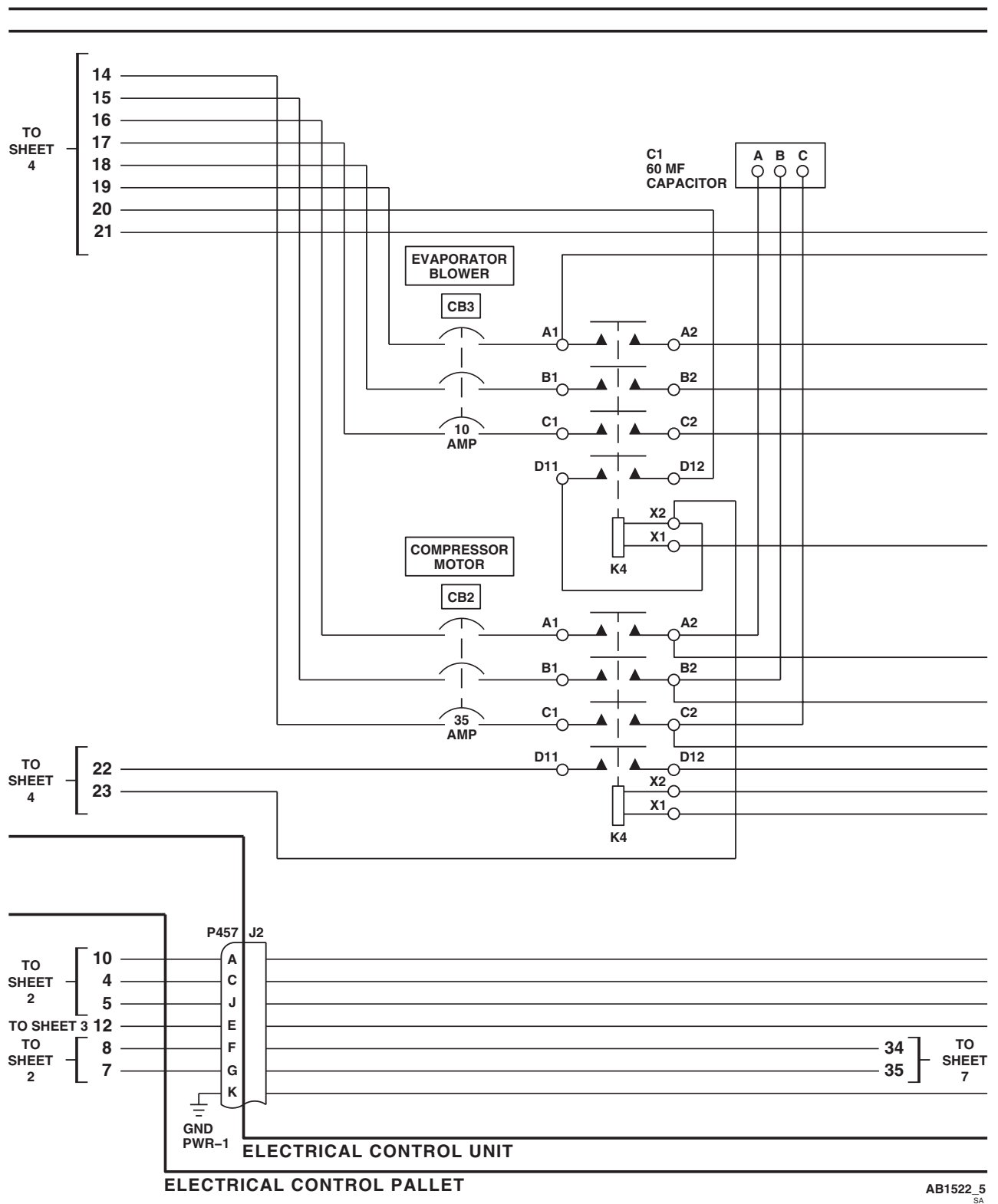
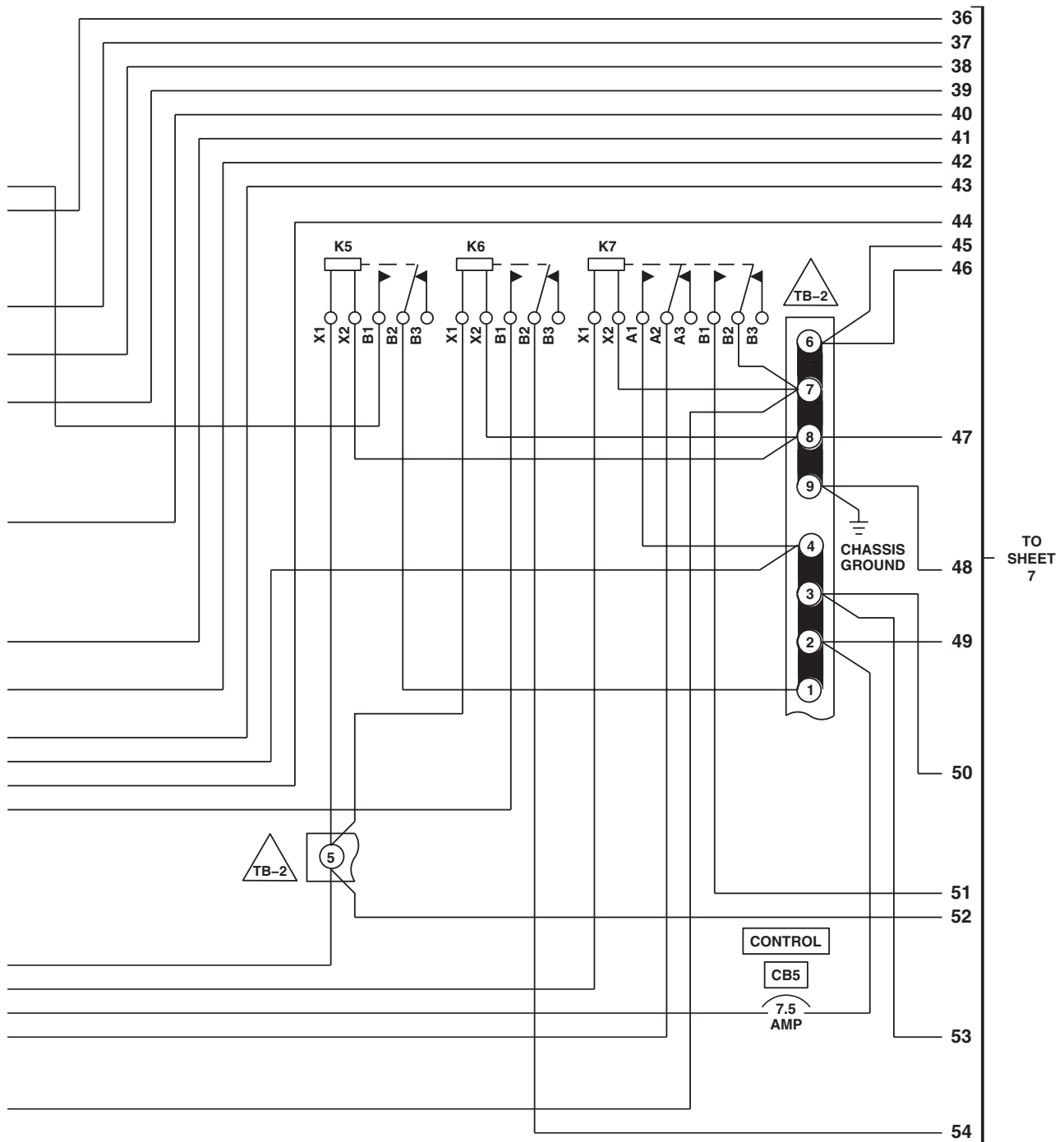


Figure 2. Environmental Control System Schematic Diagram. (Sheet 5 of 10)

## SCHEMATICS - Continued



ELECTRICAL CONTROL UNIT

ELECTRICAL CONTROL PALLET

AB1522\_6  
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Figure 2. Environmental Control System Schematic Diagram. (Sheet 6 of 10)

SCHEMATICS - Continued

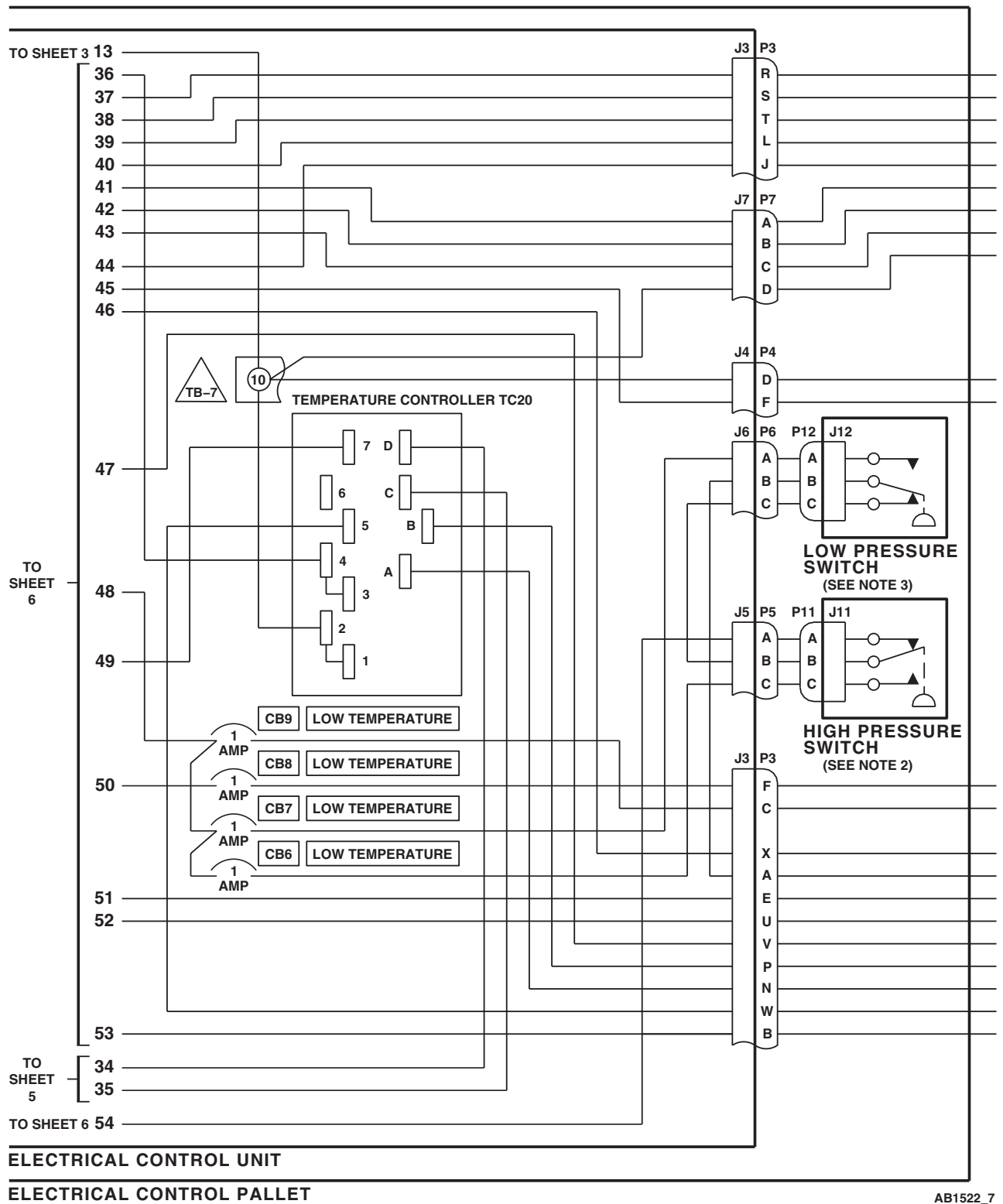


Figure 2. Environmental Control System Schematic Diagram. (Sheet 7 of 10)

SCHEMATICS - Continued

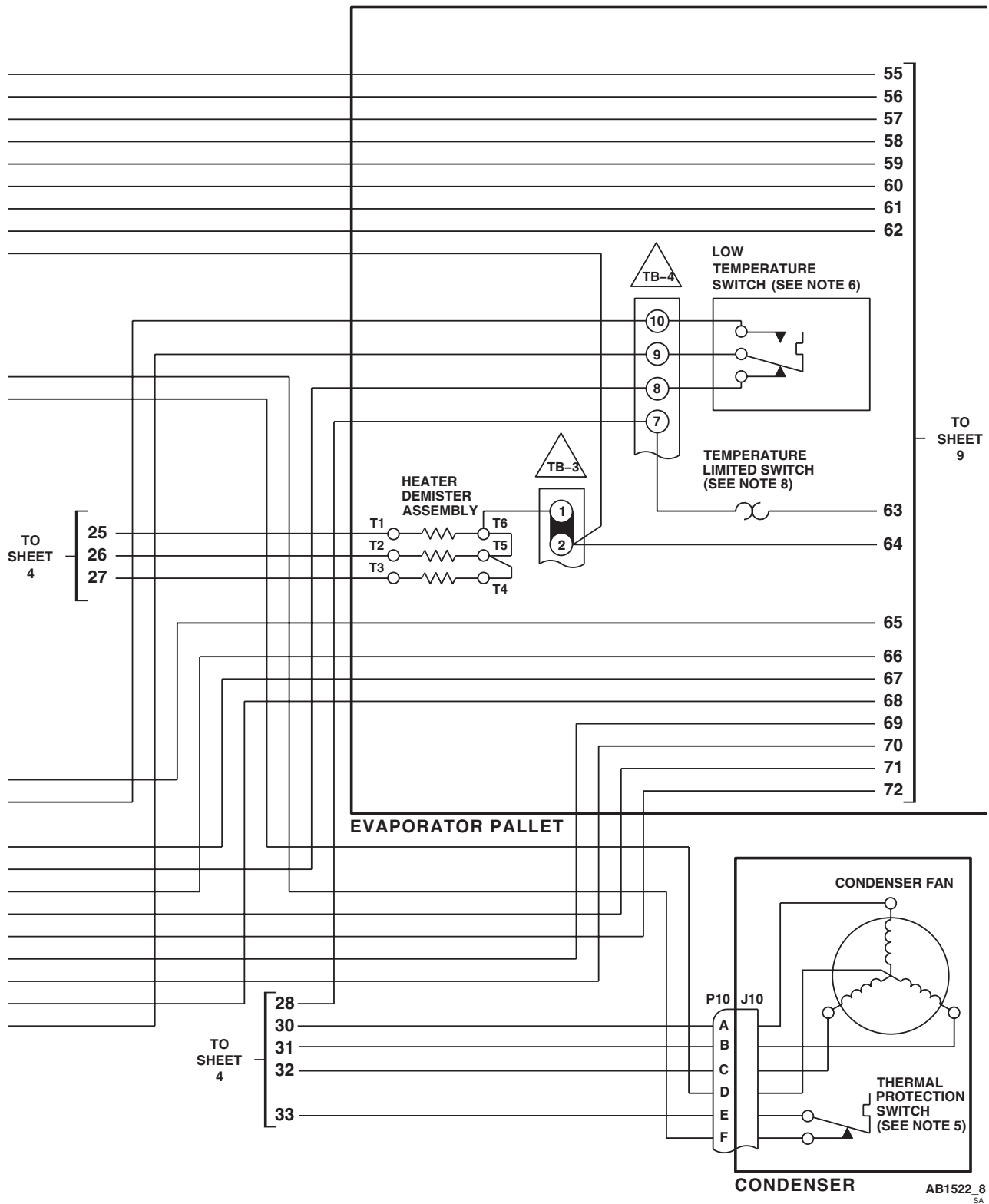


Figure 2. Environmental Control System Schematic Diagram. (Sheet 8 of 10)



SCHEMATICS - Continued

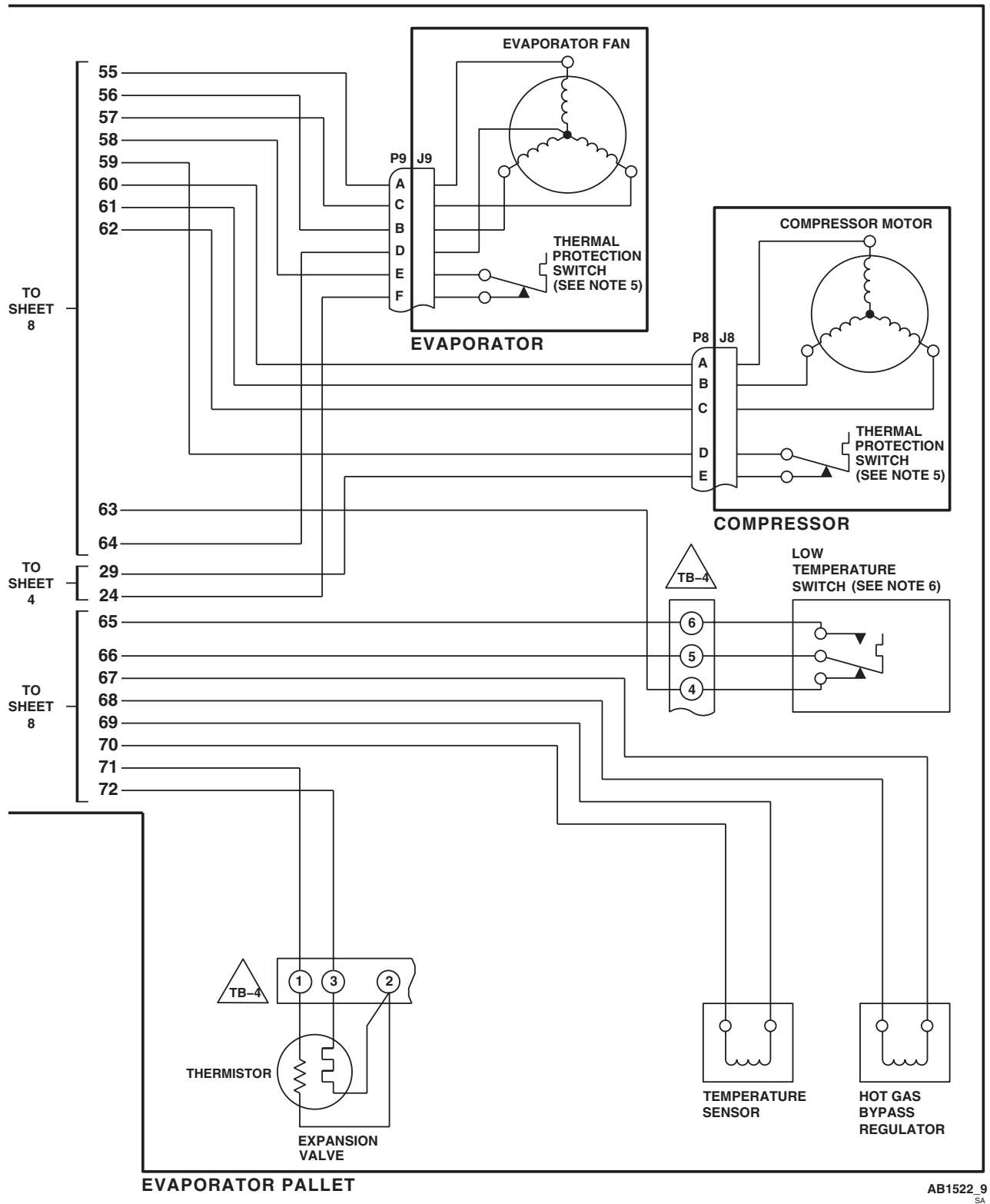
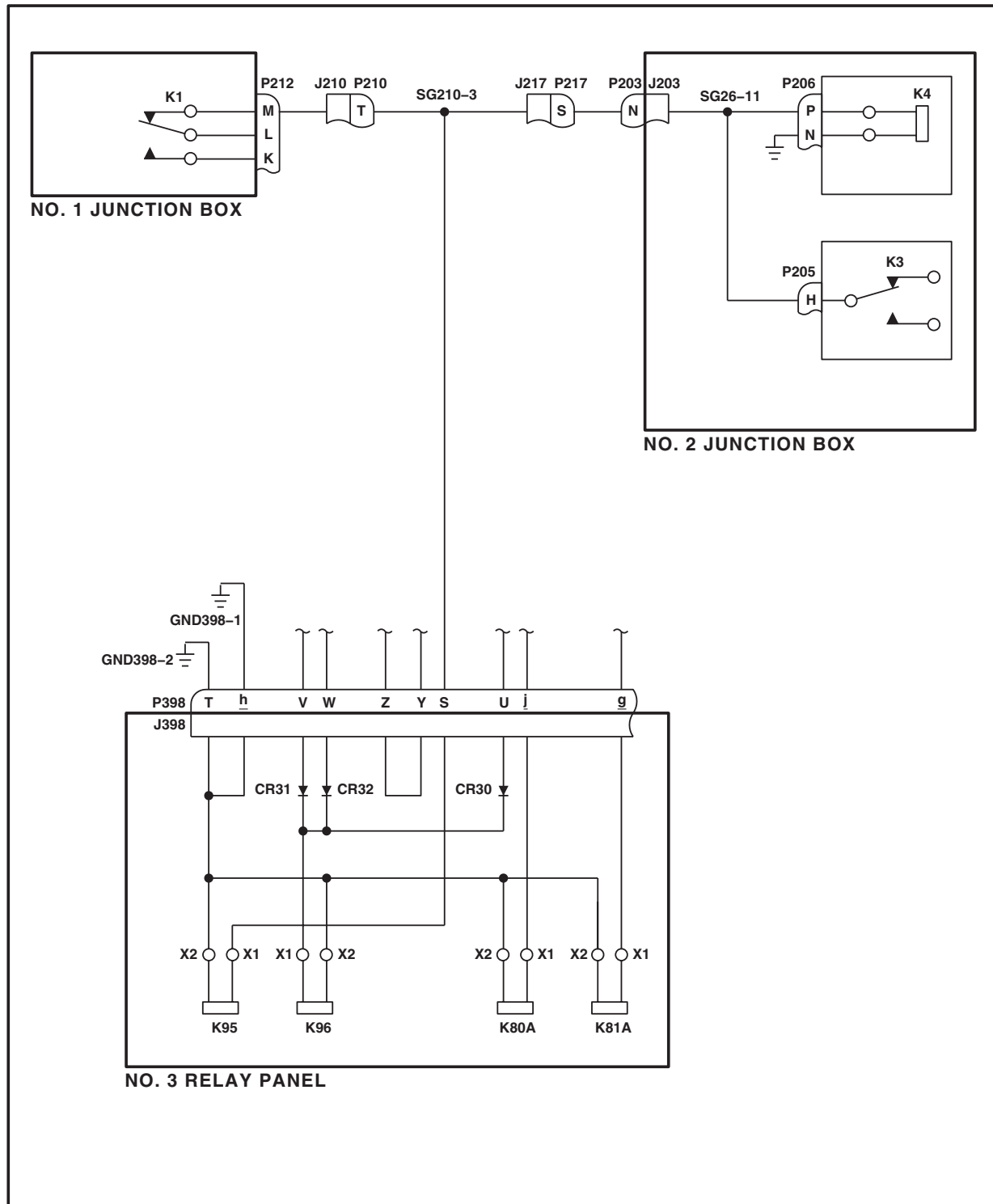


Figure 2. Environmental Control System Schematic Diagram. (Sheet 9 of 10)

SCHEMATICS - Continued

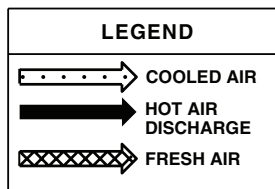
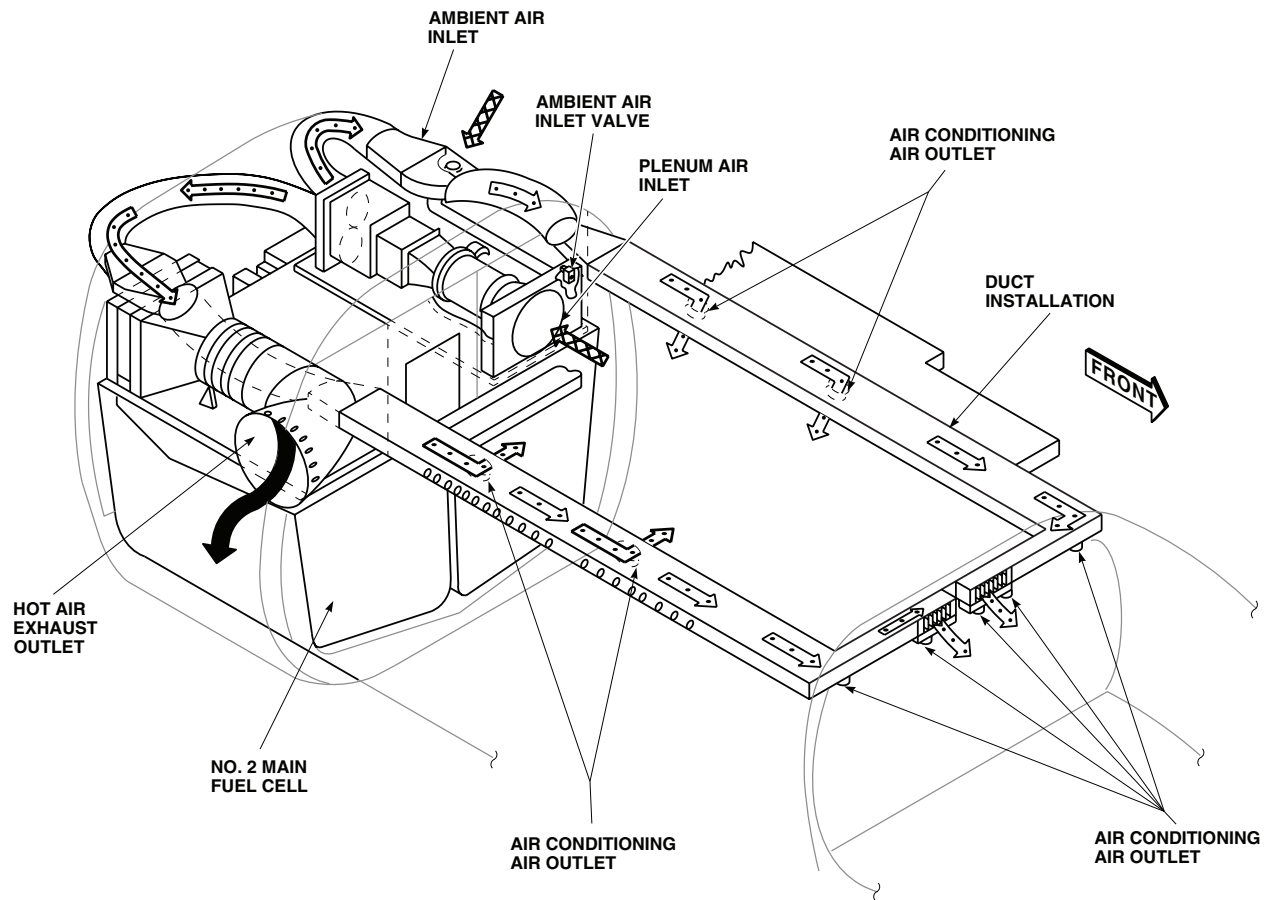


DETAIL A  
(SEE NOTE 9)

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Figure 2. Environmental Control System Schematic Diagram. (Sheet 10 of 10)

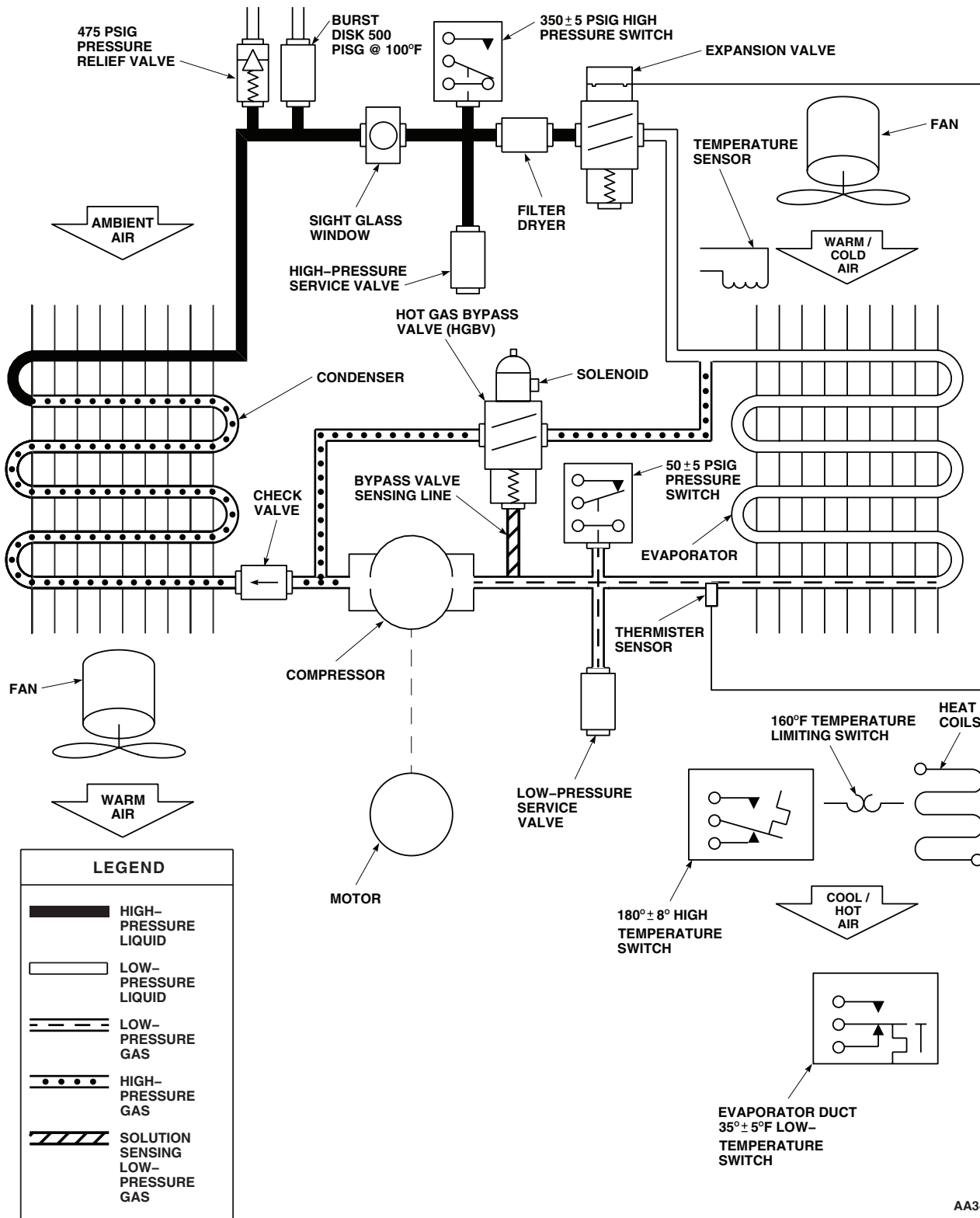
SCHEMATICS - Continued



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Figure 3. Environmental Control System Airflow Diagram.

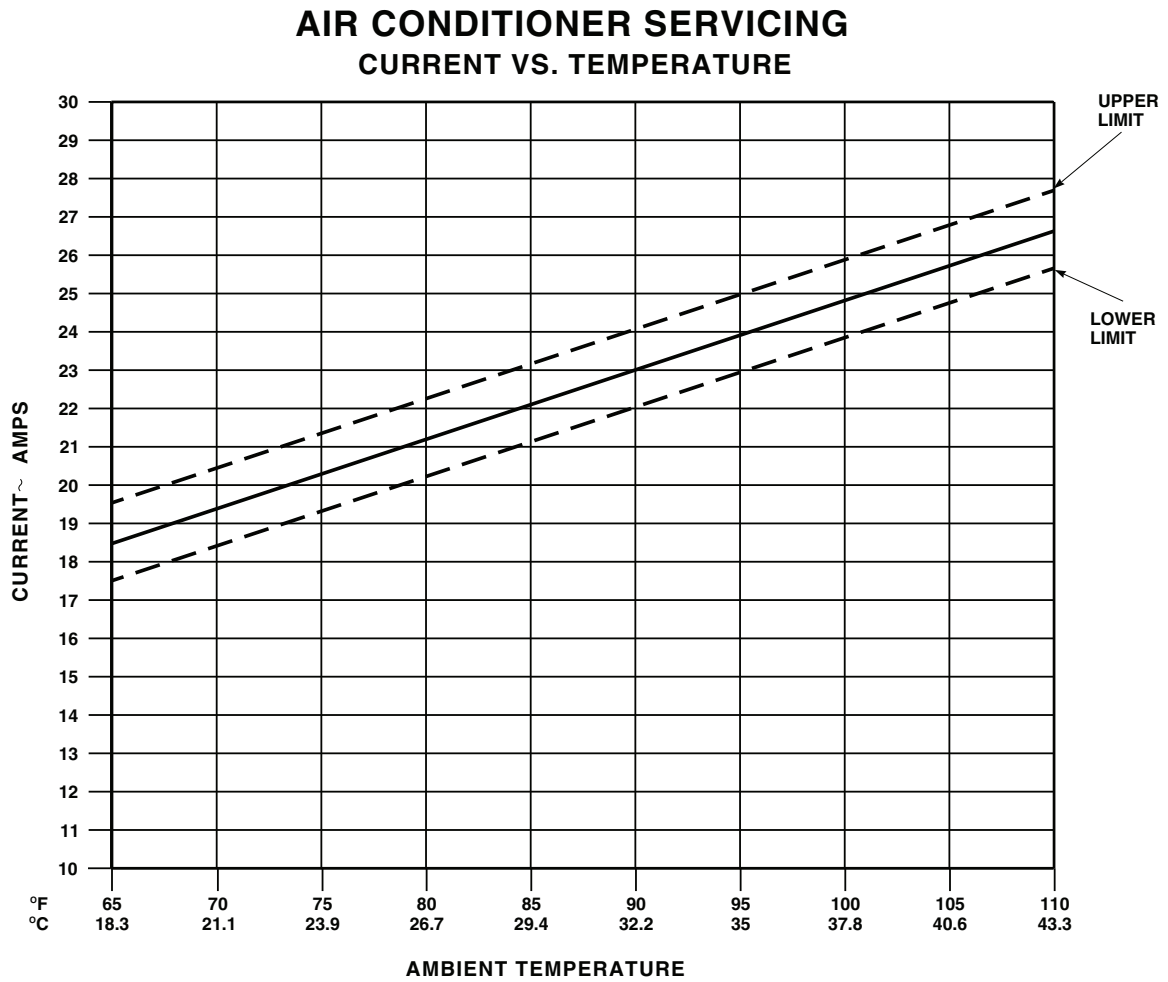
SCHEMATICS - Continued



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Figure 4. Environmental Control System Refrigerant Flow Diagram.

SCHEMATICS - Continued



FB1608  
SA

Figure 5. Air Conditioner Servicing Current vs. Temperature.

END OF WORK PACKAGE



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**UNIT LEVEL****ENVIRONMENTAL CONTROL SYSTEMS****ENVIRONMENTAL CONTROL SYSTEM (ECS) HH-60A HH-60L**

This WP supersedes WP 0151 00, dated 17 April 2006.

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**INITIAL SETUP:****Test Equipment**

Multimeter, AN/PSM-45A

WP 0156 00

WP 0824 00

WP 0825 00

**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01

WP 0827 00

Electrical Repairer Toolkit, SC 5180-99-B06

WP 1297 00

Jumper Wire

WP 1298 00

Refrigeration Unit Service Toolkit

WP 1300 00

WP 1304 00

**Personnel Required**

Aircraft Electrician MOS 15F (1)

WP 1308 00

Qualified Personnel To Run Engines (2)

WP 1309 00

UH-60 Helicopter Repairer MOS 15T (1)

WP 1318 00

WP 1626 00

**References**

TM 1-1520-237-10

WP 1685 00

WP 0104 00

WP 1747 00

WP 0122 00

WP 0126 00

WP 0153 00

WP 0155 00

**Equipment Condition**

ECS, Properly Serviced, WP 1626 00

External Electrical Power Available, WP 1685 00

and APU Operational, TM 1-1520-237-10

**ENVIRONMENTAL CONTROL SYSTEM (ECS) OPERATIONAL/TROUBLESHOOTING PROCEDURE****HH-60A HH-60L****PROCEDURE****Step****CAUTION**

Make sure cabin doors are closed and personnel are clear of exhaust duct area when ECS is operational.

**NOTE**

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- A delay of 7 to 10 seconds will occur when AC/OFF/VENT/HEAT switch is placed to VENT before indications occur. A delay of up to 1 minute will occur when AC/OFF/VENT/HEAT switch is placed to AC or HEAT before indications occur.
- The ECS heater will continue to run for approximately the same duration it was on when AC/OFF/VENT/HEAT switch is placed to OFF, pilot's hoist control panel HOIST POWER ON/OFF switch is placed to ON or BLADE DEICE POWER ON/TEST switch is placed to ON.
- See Figure 1 for component location diagram and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

1. Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.
2. On crew's rescue hoist control panel, place ARM/TEST switch to TEST.
3. On blade de-ice control panel, place POWER ON/OFF/TEST switch to OFF.
4. On pilot's hoist control panel, place HOIST POWER ON/OFF switch to OFF.

**CAUTION**

Make sure cabin doors are closed and personnel are clear of exhaust duct area when ECS is operational.

5. Turn on electrical power. Turn pilot's or copilot's multifunction display (MFD) switch ON and press T6 switch (illuminates all MFD legends for 10 seconds, then only the active caution and advisory legends will be displayed).
6. Adjust ECS control panel COOL/WARM (temperature) control to maximum COOL position.
7. Place ECS control panel AC/OFF/VENT/HEAT (mode select) switch to AC.

**Indication/Condition**

- a. AC ON capsule on ECS control panel shall go on.
- b. ECS FAULT INDICATOR PANEL HEATER indicator light on electrical control unit (ECU) distribution box shall be off.
- c. ECS FAULT INDICATOR PANEL EVAPORATOR FAN indicator light on ECU distribution box shall be off.



**ENVIRONMENTAL CONTROL SYSTEM (ECS) OPERATIONAL/TROUBLESHOOTING PROCEDURE****HH-60A HH-60L** - Continued

- d. ECS FAULT INDICATOR PANEL CONDENSER FAN indicator light on ECU distribution box shall be off.
- e. ECS FAULT INDICATOR PANEL COMPRESSOR indicator light on ECU distribution box shall be off.
- f. Evaporator blower shall turn on.
- g. Condenser blower shall turn on.
- h. Compressor motor shall turn on.
- i. Cool air shall flow from ECS duct.

**Corrective Action**

- 1. If Indication/Condition a. is not as specified, go to AC ON CAPSULE ON ECS CONTROL PANEL DOES NOT GO ON, in this work package.
- 2. If Indication/Condition b. is not as specified, go to ECS FAULT INDICATOR PANEL HEATER INDICATOR LIGHT IS ON, in this work package.
- 3. If Indication/Condition c. is not as specified, go to ECS FAULT INDICATOR PANEL EVAPORATOR FAN INDICATOR LIGHTS ON, in this work package.
- 4. If Indication/Condition d. is not as specified, go to ECS FAULT INDICATOR PANEL CONDENSER FAN INDICATOR LIGHT IS ON, in this work package.
- 5. If Indication/Condition e. is not as specified, go to ECS FAULT INDICATOR PANEL COMPRESSOR INDICATOR LIGHT IS ON, in this work package.
- 6. If Indication/Condition f. is not as specified, go to EVAPORATOR BLOWER DOES NOT TURN ON, in this work package.
- 7. If Indication/Condition g. is not as specified, go to CONDENSER BLOWER DOES NOT TURN ON, in this work package.
- 8. If Indication/Condition h. is not as specified, go to COMPRESSOR MOTOR DOES NOT TURN ON, in this work package.
- 9. If Indication/Conditions f., g., and h. are not as specified, go to ECS INOPERABLE, in this work package.
- 10. If air does not flow from ECS duct, go to AIR DOES NOT FLOW FROM ECS OUTLET DUCT, in this work package.
- 11. If air flows from ECS duct but is not cool, go to COOL AIR DOES NOT FLOW FROM ECS DUCT, in this work package.

**Step**

- 8. On ECS circuit breaker panel (ECU distribution box), pull out MOTOR COMPRESSOR circuit breaker.

**Indication/Condition**

- a. Compressor motor shall turn off.
- b. Evaporator blower shall remain on.
- c. Condenser blower shall remain on.

**Corrective Action**

If Indication/Condition a., b., or c. is not as specified, replace ECU distribution box (WP 1300 00).

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**ENVIRONMENTAL CONTROL SYSTEM (ECS) OPERATIONAL/TROUBLESHOOTING PROCEDURE**  
**HH-60A HH-60L** - Continued**Step**

9. On ECS circuit breaker panel (ECU distribution box), pull out EVAPORATOR FAN circuit breaker.

**Indication/Condition**

- a. Evaporator blower shall turn off.
- b. Condenser blower shall remain on.

**Corrective Action**

If Indication/Condition a. or b. is not as specified, replace ECU distribution box (WP 1300 00).

**Step**

10. On ECS circuit breaker panel (ECU distribution box), pull out CONDENSER FAN circuit breaker.

**Indication/Condition**

- a. Condenser blower shall turn off.
- b. ECS shall be completely inoperative.

**Corrective Action**

If Indication/Condition a. or b. is not as specified, replace ECU distribution box (WP 1300 00).

**Step****CAUTION**

Allow ECS to remain off for at least 5 minutes before restarting to allow internal pressures to stabilize or damage to compressor may occur.

11. On ECS circuit breaker panel (ECU distribution box), push in MOTOR COMPRESSOR, EVAPORATOR FAN, and CONDENSER FAN circuit breakers.

**Indication/Condition**

ECS shall turn on.

**Corrective Action**

If Indication/Condition is not as specified, replace ECU distribution box (WP 1300 00).

**Step**

12. Adjust ECS control panel mode select switch to OFF.

**Indication/Condition**

- a. AC ON capsule on ECS control panel shall go off.
- b. ECS shall turn off.

**Corrective Action**

If Indication/Condition a. or b. is not as specified, replace ECS control panel (WP 1309 00).

**Step**

13. Adjust ECS control panel mode select switch to VENT.

**ENVIRONMENTAL CONTROL SYSTEM (ECS) OPERATIONAL/TROUBLESHOOTING PROCEDURE****HH-60A HH-60L** - Continued**Indication/Condition**

- a. Evaporator blower shall turn on.
- b. Air shall flow from ECS duct.

**Corrective Action**

If Indication/Conditions a. and b. are not as specified, go to EVAPORATOR BLOWER DOES NOT GO ON WITH ECS CONTROL MODE SELECT SWITCH IN VENT POSITION, in this work package.

**Step**

14. Adjust ECS control panel mode select switch to OFF.

**Indication/Condition**

Evaporator blower shall turn off.

**Corrective Action**

If Indication/Condition is not as specified, replace ECS control panel (WP 1309 00).

**Step**

15. Place ECS control panel temperature control to maximum WARM position.
16. Place ECS control panel mode select switch to HEAT.

**Indication/Condition**

- a. HEAT ON capsule on ECS control panel shall go on.
- b. ECS shall turn on and provide warm airflow through ECS vent.

**Corrective Action**

1. If Indication/Condition a. is not as specified, go to HEAT ON CAPSULE ON ECS CONTROL PANEL DOES NOT GO ON, in this work package.
2. If Indication/Condition b. is not as specified, go to ECS HEATER INOPERABLE, in this work package.

**Step**

17. On ECS circuit breaker panel (ECU distribution box), pull out upper and lower HEATER ELEMENTS circuit breakers.

**Indication/Condition**

- a. ECS shall stay on.
- b. Cool air shall flow from ECS duct.

**Corrective Action**

If Indication/Conditions a. and b. are not as specified, replace ECU distribution box (WP 1300 00).

**Step**

18. On ECS circuit breaker panel (ECU distribution box), push in upper and lower HEATER ELEMENTS circuit breakers.

**ENVIRONMENTAL CONTROL SYSTEM (ECS) OPERATIONAL/TROUBLESHOOTING PROCEDURE****HH-60A HH-60L** - Continued**Indication/Condition**

Warm air shall flow from ECS duct.

**Corrective Action**

If Indication/Condition is not as specified, replace ECU distribution box (WP 1300 00).

**Step**

19. Place ECS control panel temperature control to COOL.

**Indication/Condition**

- a. ECS shall stay on.
- b. Cool air shall flow from ECS duct.

**Corrective Action**

If Indication/Conditions a. and b. are not as specified, replace ECS control panel (WP 1309 00).

**Step**

20. Place ECS control panel temperature control to WARM.

**Indication/Condition**

- a. ECS shall stay on.
- b. Warm air shall flow from ECS duct.

**Corrective Action**

If Indication/Conditions a. and b. are not as specified, replace ECS control panel (WP 1309 00).

**Step****NOTE**

The ECS heater will continue to run for approximately the same duration it was on when the AC/OFF/VENT/HEAT switch is placed to OFF, the pilots's hoist control panel HOIST POWER ON/OFF switch is placed to ON or the BLADE DEICE POWER ON/TEST switch is placed to ON.

21. Place lower console pilot's hoist control panel HOIST POWER ON/OFF switch to ON.

**Indication/Condition**

- a. HEAT ON capsule on ECS control panel shall go off.
- b. ECS shall turn off.

**Corrective Action**

If Indication/Conditions a. and b. are not as specified, troubleshoot rescue hoist system (WP 0153 00).

**Step**

22. Place lower console pilot's hoist control panel HOIST POWER ON/OFF switch to OFF.

**ENVIRONMENTAL CONTROL SYSTEM (ECS) OPERATIONAL/TROUBLESHOOTING PROCEDURE****HH-60A HH-60L** - Continued**Indication/Condition**

- a. HEAT ON capsule on ECS control panel shall go on.
- b. ECS shall turn on.

**Corrective Action**

If Indication/Conditions a. and b. are not as specified, troubleshoot rescue hoist system (WP 0153 00).

**Step****NOTE**

The ECS heater will continue to run for approximately the same duration it was on when the AC/OFF/VENT/HEAT switch is placed to OFF, the pilot's hoist control panel HOIST POWER ON/OFF switch is placed to ON or the BLADE DEICE POWER ON/TEST switch is placed to ON.

- 23. Place ECS control panel mode select switch to OFF.

**Indication/Condition**

- a. HEAT ON capsule on ECS control panel shall go off.
- b. ECS shall turn off.

**Corrective Action**

If Indication/Condition a. and b. is not as specified, replace ECS control panel (WP 1309 00).

**Step**

- 24. Turn off electrical power.
- 25. Make sure upper console GENERATORS NO. 1, NO. 2, and APU TEST/OFF RESET/ON switches are placed to OFF.
- 26. Start APU (TM 1-1520-237-10).
- 27. Place upper console GENERATORS APU TEST/OFF RESET/ON switch to ON.

**Indication/Condition**

APU GEN ON legend on multifunctional display (MFD) caution advisory grid shall go on, and #1 CONV and #2 CONV legends shall go off.

**Corrective Action**

If Indication/Conditions are not as specified, troubleshoot ac electrical system (APU generators) (WP 0122 00).

**Step**

- 28. Place ECS control panel mode select switch to HEAT.

**Indication/Condition**

- a. HEAT ON capsule on ECS control panel shall stay off.
- b. ECS shall stay off.

**ENVIRONMENTAL CONTROL SYSTEM (ECS) OPERATIONAL/TROUBLESHOOTING PROCEDURE****HH-60A HH-60L** - Continued**Corrective Action**

If Indication/Conditions a. and b. are not as specified, troubleshoot auxiliary power unit system (WP 0155 00) or (WP 0156 00).

**Step****NOTE**

The ECS heater will continue to run for approximately the same duration it was on when the AC/OFF/VENT/HEAT switch is placed to OFF, the pilot's hoist control panel HOIST POWER ON/OFF switch is placed to ON or the BLADE DEICE POWER ON/TEST switch is placed to ON.

29. Place ECS control panel mode select switch to OFF.
30. Place upper console GENERATORS APU TEST/OFF RESET/ON switch to OFF.
31. With qualified personnel at controls, start No. 1 and No. 2 engines and operate with main rotor turning at 100% (TM 1-1520-253-10).
32. Place upper console GENERATORS NO. 1 and NO. 2 TEST/OFF RESET/ON switches to ON.

**Indication/Condition**

#1 GEN and #2 GEN capsules on MFD caution/advisory grid shall be off.

**Corrective Action**

If Indication/Condition is not as specified, troubleshoot ac electrical system (dual generators power) (WP 0122 00).

**Step**

33. Place ECS control panel mode select switch to HEAT.

**Indication/Condition**

- a. HEAT ON capsule on ECS control panel shall go on.
- b. ECS shall turn on and provide warm airflow from ECS duct.

**Corrective Action**

If Indication/Condition is not as specified, troubleshoot ac electrical system (dual generators power) (WP 0122 00).

**Step****NOTE**

The ECS heater will continue to run for approximately the same duration it was on when the AC/OFF/VENT/HEAT switch is placed to OFF, the pilot's hoist control panel HOIST POWER ON/OFF switch is placed to ON or the BLADE DEICE POWER ON/TEST switch is placed to ON.

34. Place ECS control panel mode select switch to OFF.
35. Turn MFD switch OFF.
36. Place upper console GENERATORS NO. 1 and NO. 2 TEST/OFF/ON switches to OFF.
37. Shut down No. 1 and No. 2 engines (TM 1-1520-253-10).

**ENVIRONMENTAL CONTROL SYSTEM (ECS) OPERATIONAL/TROUBLESHOOTING PROCEDURE****HH-60A HH-60L** - Continued**Indication/Condition**

No. 1 and No. 2 engines shall be shut down.

**Corrective Action**

None Required

**AC ON CAPSULE ON ECS CONTROL PANEL DOES NOT GO ON****SYMPTOM**

AC ON capsule on ECS control panel does not go on.

**MALFUNCTION**

AC ON capsule on ECS control panel does not go on.

**CORRECTIVE ACTION**

- 1. Check for 28 vdc between P356-D and ground.**
  - a. If voltage is as specified, go to **2**.
  - b. If voltage is not as specified, go to **4**.
- 2. Check for 28 vdc between P356-G and ground.**
  - a. If voltage is as specified, go to **3**.
  - b. If voltage is not as specified, go to **9**.
- 3. Check continuity between:**
  - **P356-P and ground**
  - **P356-R and ground**
  - **P356-S and ground**
  - a. If continuity is present, replace ECS control panel (WP 1309 00). Go to **10**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **10**.
- 4. Check continuity between P356-D and K300-22 (hoist power relay).**
  - a. If continuity is present, go to **5**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **10**.
- 5. Check for 28 vdc between K300-23 and ground.**
  - a. If voltage is as specified, replace hoist power relay (WP 0827 00). Go to **10**.
  - b. If trouble remains, troubleshoot rescue hoist system (WP 0153 00). Go to **10**.
  - c. If voltage is not as specified, go to **6**.
- 6. Check continuity between K300-23 and P298-m.**
  - a. If continuity is present, go to **7**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **10**.
- 7. Check for 28 vdc between P298-A and ground.**
  - a. If voltage is as specified, replace No. 3 relay panel (WP 0827 00). Go to **10**.

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**AC ON CAPSULE ON ECS CONTROL PANEL DOES NOT GO ON - Continued**

- b. If voltage is not as specified, go to **8**.
- 8. Check continuity between P298-A and terminal 1 of ECS CONTR circuit breaker.**
  - a. If continuity is present, replace ECS CONTR circuit breaker (WP 0824 00). Go to **10**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **10**.
- 9. Check continuity between P356-G and J198-K.**
  - a. If continuity is present, troubleshoot instrument panel and consoles indicator lights dimming system (WP 0126 00). Go to **10**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **10**.
- 10. Procedure completed.**

**ECS FAULT INDICATOR PANEL HEATER INDICATOR LIGHT IS ON****SYMPTOM**

ECS FAULT INDICATOR PANEL HEATER indicator light is on.

**MALFUNCTION**

ECS FAULT INDICATOR PANEL HEATER indicator light is on.

**CORRECTIVE ACTION**

- 1. On ECS FAULT INDICATOR PANEL, press and release FAULT RESET switch.**
  - a. If HEATER indicator light goes off, go to **4**.
  - b. If HEATER indicator light stays on, go to **2**.
- 2. Check continuity between:**
  - **P352-Q and P354-R**
  - **P352-P and P354-P**
  - a. If continuity is present, go to **3**.
  - b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **4**.
- 3. Check continuity between J6-P and J6-R.**
  - a. If continuity is present, replace ECU distribution box (WP 1300 00). Go to **4**.
  - b. If continuity is not present, replace evaporator pallet (WP 1304 00). Go to **4**.
- 4. Procedure completed.**

**ECS FAULT INDICATOR PANEL EVAPORATOR FAN INDICATOR LIGHT IS ON****SYMPTOM**

ECS FAULT INDICATOR PANEL EVAPORATOR FAN indicator light is on.

**MALFUNCTION**

ECS FAULT INDICATOR PANEL EVAPORATOR FAN indicator light is on.

**CORRECTIVE ACTION**

- 1. On ECS FAULT INDICATOR PANEL, press and release FAULT RESET switch.**



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**ECS FAULT INDICATOR PANEL EVAPORATOR FAN INDICATOR LIGHT IS ON - Continued**

- a. If EVAPORATOR FAN indicator light goes off, go to **4**.
- b. If EVAPORATOR FAN indicator light stays on, go to **2**.
- 2. Check continuity between P352-N and P354-N.**
  - a. If continuity is present, go to **3**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **4**.
- 3. Check continuity between J6-N and J6-P.**
  - a. If continuity is present, replace ECU distribution box (WP 1300 00). Go to **4**.
  - b. If continuity is not present, replace evaporator pallet (WP 1304 00). Go to **4**.
- 4. Procedure completed.**

**ECS FAULT INDICATOR PANEL CONDENSER FAN INDICATOR LIGHT IS ON****SYMPTOM**

ECS FAULT INDICATOR PANEL CONDENSER FAN indicator light is on.

**MALFUNCTION**

ECS FAULT INDICATOR PANEL CONDENSER FAN indicator light is on.

**CORRECTIVE ACTION**

- 1. On ECS FAULT INDICATOR PANEL, press and release FAULT RESET switch.**
  - a. If CONDENSER FAN indicator light goes off, go to **4**.
  - b. If CONDENSER FAN indicator light stays on, go to **2**.
- 2. Check continuity between:**
  - **P352-W and P353-F**
  - **P352-V and P353-E**
  - a. If continuity is present, go to **3**.
  - b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **4**.
- 3. Check continuity between J10-E and J10-F of condenser blower.**
  - a. If continuity is present, replace ECU distribution box (WP 1300 00). Go to **4**.
  - b. If continuity is not present, replace condenser blower (WP 1298 00). Go to **4**.
- 4. Procedure completed.**

**ECS FAULT INDICATOR PANEL COMPRESSOR INDICATOR LIGHT IS ON****SYMPTOM**

ECS FAULT INDICATOR PANEL COMPRESSOR indicator light is on.

**MALFUNCTION**

ECS FAULT INDICATOR PANEL COMPRESSOR indicator light is on.

**CORRECTIVE ACTION**

- 1. Using jumper wire, jumper the following pins: go to 2.**

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**ECS FAULT INDICATOR PANEL COMPRESSOR INDICATOR LIGHT IS ON - Continued**

- **SW1-A and SW1-B**
  - **SW16-C and SW16-D**
- 2. On ECS FAULT INDICATOR PANEL, press and release FAULT RESET switch.**
- a. If COMPRESSOR indicator light goes off, replace high and low pressure switches (WP 1308 00). Go to **5**.
  - b. If COMPRESSOR indicator light stays on, go to **3**.
- 3. Check continuity between:**
- **P3-A and SW1-A (electrical pallet HIGH PRESSURE switch)**
  - **P3-B and SW1-B (electrical pallet HIGH PRESSURE switch)**
  - **P3-C and SW16-C (electrical pallet LOW PRESSURE switch)**
  - **P3-D and SW16-D (electrical pallet LOW PRESSURE switch)**
  - **P352-M and P354-M**
- a. If continuity is present, go to **4**.
  - b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **5**.
- 4. Check continuity between J6-M and J6-P.**
- a. If continuity is present, replace ECU distribution box (WP 1300 00). Go to **5**.
  - b. If continuity is not present, replace evaporator pallet (WP 1304 00). Go to **5**.
- 5. Procedure completed.**

**EVAPORATOR BLOWER DOES NOT TURN ON****SYMPTOM**

Evaporator blower does not turn on.

**MALFUNCTION**

Evaporator blower does not turn on.

**CORRECTIVE ACTION**

- 1. Verify EVAPORATOR FAN circuit breaker on ECS circuit breaker panel is pushed in.**
- a. If EVAPORATOR FAN circuit breaker is pushed in, go to **2**.
  - b. If EVAPORATOR FAN circuit breaker is pulled out, push in circuit breaker. Go to **8**.
  - c. If EVAPORATOR FAN circuit breaker will not reset, go to **7**.
- 2. Check for 115 vac between:**
- **J11-A and J11-D**
  - **J11-B and J11-D**
  - **J11-C and J11-D**
- a. If voltage is as specified, replace evaporator blower (WP 1297 00). Go to **8**.
  - b. If voltage is not as specified, go to **3**.

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**EVAPORATOR BLOWER DOES NOT TURN ON - Continued****3. Check for 115 vac between:**

- **P354-A and P354-D**
- **P354-B and P354-D**
- **P354-C and P354-D**

- a. If voltage is as specified, replace evaporator pallet (WP 1304 00). Go to **8**.
- b. If voltage is not as specified, go to **4**.

**4. Check continuity between:**

- **P352-A and P354-A**
- **P352-B and P354-B**
- **P352-C and P354-C**
- **P352-D and P354-D**

- a. If continuity is present, go to **5**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **8**.

**5. Check for 115 vac between:**

- **P349-D and ground**
- **P349-E and ground**
- **P349-F and ground**

- a. If voltage is as specified, replace ECU distribution box (WP 1300 00). Go to **8**.
- b. If voltage is not as specified, go to **6**.

**6. Check continuity between:**

- **P349-D and CB512-A2 (ECS/HEAT PWR)**
- **P349-E and CB512-B2 (ECS/HEAT POWER)**
- **P349-F and CB512-C2 (ECS/HEAT POWER)**

- a. If continuity is present, replace ECS/HEAT POWER circuit breaker (WP 0824 00). Go to **8**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **8**.

**7. Disconnect P11 from evaporator blower and push in EVAPORATOR FAN circuit breaker on ECS circuit breaker panel.**

- a. If EVAPORATOR FAN circuit breaker resets, replace evaporator blower (WP 1297 00). Go to **8**.
- b. If EVAPORATOR FAN circuit breaker does not reset, replace ECU distribution box (WP 1300 00). Go to **8**.

**8. Procedure completed.**

**CONDENSER BLOWER DOES NOT TURN ON****SYMPTOM**

Condenser blower does not turn on.

**MALFUNCTION**

Condenser blower does not turn on.

**CORRECTIVE ACTION**

- 1. Verify CONDENSER FAN circuit breaker on ECS circuit breaker panel is pushed in.**
  - a. If CONDENSER FAN circuit breaker is pushed in, go to **2**.
  - b. If CONDENSER FAN circuit breaker is pulled out, push in circuit breaker. Go to **7**.
  - c. If CONDENSER FAN circuit breaker will not reset, go to **6**.
- 2. Check for 115 vac between:**
  - **P353-A and P353-D**
  - **P353-B and P353-D**
  - **P353-C and P353-D**
  - a. If voltage is as specified, replace condenser blower (WP 1298 00). Go to **7**.
  - b. If voltage is not as specified, go to **3**.
- 3. Check for 28 vdc between P352-E and ground.**
  - a. If voltage is as specified, go to **4**.
  - b. If voltage is not as specified, go to **5**.
- 4. Check continuity between:**
  - **P353-A and P351-D**
  - **P353-B and P351-E**
  - **P353-C and P351-F**
  - **P353-D and P351-G**
  - a. If continuity is present, replace ECU distribution box (WP 1300 00). Go to **7**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **7**.
- 5. Check continuity between P356-A and P352-E.**
  - a. If continuity is present, replace ECS control panel (WP 1309 00). Go to **7**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **7**.
- 6. Disconnect P353 from condenser blower and push in CONDENSER FAN circuit breaker on ECS circuit breaker panel.**
  - a. If CONDENSER FAN circuit breaker resets, replace condenser blower (WP 1298 00). Go to **7**.
  - b. If CONDENSER FAN circuit breaker does not reset, replace ECU distribution box (WP 1300 00). Go to **7**.

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**CONDENSER BLOWER DOES NOT TURN ON** - Continued**7. Procedure completed.****COMPRESSOR MOTOR DOES NOT TURN ON****SYMPTOM**

Compressor motor does not turn on.

**MALFUNCTION**

Compressor motor does not turn on.

**CORRECTIVE ACTION**

- 1. Verify MOTOR COMPRESSOR circuit breaker on ECS circuit breaker panel is pushed in.**
  - a. If MOTOR COMPRESSOR circuit breaker is pushed in, go to **2**.
  - b. If MOTOR COMPRESSOR circuit breaker is pulled out, push in circuit breaker. Go to **13**.
  - c. If MOTOR COMPRESSOR circuit breaker will not reset, go to **12**.
- 2. Check for 115 vac between:**
  - **P357-A and ground**
  - **P357-B and ground**
  - **P357-C and ground**
  - a. If voltage is as specified, replace evaporator pallet (WP 1304 00). Go to **13**.
  - b. If voltage is not as specified, go to **3**.
- 3. Check continuity between:**
  - **P350-E and P357-A**
  - **P350-F and P357-B**
  - **P350-G and P357-C**
  - a. If continuity is present, go to **4**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **13**.
- 4. Check for 115 vac between:**
  - **P350-A and ground**
  - **P350-B and ground**
  - **P350-C and ground**
  - a. If voltage is as specified, replace ECU distribution box (WP 1300 00). Go to **13**.
  - b. If voltage is not as specified, go to **5**.
- 5. Check for 28 vdc between K310-X1 (Compressor PWR relay) and ground.**
  - a. If voltage is as specified, go to **6**.
  - b. If voltage is not as specified, go to **9**.
- 6. Check continuity between:**

**COMPRESSOR MOTOR DOES NOT TURN ON - Continued**

- **P350-A and K310-A2**
- **P350-B and K310-B2**
- **P350-C and K310-C2**
- **K310-X2 and ground**

- a. If continuity is present, go to **7**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **13**.

**7. Check for 115 vac between:**

- **K310-A1 and ground**
- **K310-B1 and ground**
- **K310-C1 and ground**

- a. If voltage is as specified, replace compressor power relay (WP 0825 00). Go to **13**.
- b. If voltage is not as specified, go to **8**.

**8. Check continuity between:**

- **K310-A1 and CL20-2**
- **K310-B1 and CL21-2**
- **K310-C1 and CL22-2**

- a. If continuity is present, troubleshoot ac electrical system (WP 0122 00). Go to **13**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **13**.

**9. Check continuity between K310-X1 and HH-60A P298-T ■ HH-60L P356-A ■ .**

- a. If continuity is present, go to **10**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **13**.

**10. Check for 28 vdc between P298-L and ground.**

- a. If voltage is as specified, replace No. 3 relay panel (WP 0827 00). Go to **13**.
- b. If voltage is not as specified, go to **11**.

**11. Check continuity between HH-60L P356-A ■ HH-60A P298-L ■ and P352-E.**

- a. If continuity is present, replace ECS control panel (WP 1309 00). Go to **13**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **13**.

**12. Disconnect P9 from compressor assembly and push in MOTOR COMPRESSOR circuit breaker on ECS circuit breaker panel.**

- a. If MOTOR COMPRESSOR circuit breaker resets, replace evaporator pallet (WP 1304 00). Go to **13**.
- b. If MOTOR COMPRESSOR circuit breaker does not reset, replace ECU distribution box (WP 1300 00). Go to **13**.

**13. Procedure completed.**

**ECS INOPERABLE****SYMPTOM**

ECS inoperable.

**MALFUNCTION**

ECS inoperable.

**CORRECTIVE ACTION****1. Check for 28 vdc between P352-L and ground.**

- a. If voltage is as specified, go to **2**.
- b. If voltage is not as specified, go to **3**.

**2. Check continuity between:**

- **P350-D and ground**
- **P352-J and ground**

- a. If continuity is present, replace ECU distribution box (WP 1300 00). Go to **4**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **4**.

**3. Check continuity between P352-L and terminal 1 of ECS CONTR circuit breaker.**

- a. If continuity is present, replace ECS CONTR circuit breaker (WP 0824 00). Go to **4**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **4**.

**4. Procedure completed.****AIR DOES NOT FLOW FROM ECS OUTLET DUCT****SYMPTOM**

Air does not flow from ECS outlet duct.

**MALFUNCTION**

Air does not flow from ECS outlet duct.

**CORRECTIVE ACTION****1. Check for blocked distribution ducts.**

- a. If distribution ducts are blocked, remove obstruction. Go to **5**.
- b. If distribution ducts are not blocked, go to **2**.

**2. Check for clogged demister.**

- a. If demister is clogged, replace demister pad (WP 1318 00). Go to **5**.
- b. If demister is not clogged, go to **3**.

**3. Check for blocked evaporator core.**

- a. If evaporator core is blocked, remove obstruction. Go to **5**.
- b. If evaporator core is not blocked, go to **4**.

**4. Check for blocked condenser core.**

---

**AIR DOES NOT FLOW FROM ECS OUTLET DUCT - Continued**

- a. If condenser core is blocked, remove obstruction. Go to **5**.
- b. If condenser core is not blocked, replace evaporator pallet (WP 1304 00). Go to **5**.

**5. Procedure completed.****COOL AIR DOES NOT FLOW FROM ECS DUCT****SYMPTOM**

Cool air does not flow from ECS duct.

**MALFUNCTION**

Cool air does not flow from ECS duct.

**CORRECTIVE ACTION**

- 1. Check for 28 vdc between P352-H and ground.**
  - a. If voltage is as specified, go to **2**.
  - b. If voltage is not as specified, go to **3**.
- 2. Check continuity between:**
  - P352-R and P354-K
  - P352-S and P354-L
  - P352-T and P354-J
  - P352-U and P354-S
  - a. If continuity is present, replace ECU distribution box (WP 1300 00). Go to **4**.
  - b. If trouble remains, replace evaporator pallet (WP 1304 00). Go to **4**.
  - c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **4**.
- 3. Check continuity between P352-H and P356-E.**
  - a. If continuity is present, replace ECS control panel (WP 1309 00). Go to **4**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **4**.
- 4. Procedure completed.**

**EVAPORATOR BLOWER DOES NOT GO ON WITH ECS CONTROL MODE SELECT SWITCH IN VENT POSITION****SYMPTOM**

Evaporator blower does not go on with ECS control mode select switch in VENT position.

**MALFUNCTION**

Evaporator blower does not go on with ECS control mode select switch in VENT position.

**CORRECTIVE ACTION**

- 1. Check for 28 vdc between P352-F and ground.**
  - a. If voltage is as specified, replace ECU distribution box (WP 1300 00). Go to **3**.
  - b. If voltage is not as specified, go to **2**.



---

**EVAPORATOR BLOWER DOES NOT GO ON WITH ECS CONTROL MODE SELECT SWITCH IN VENT POSITION** - Continued**2. Check continuity between P356-B and P352-F.**

- a. If continuity is present, replace ECS control panel (WP 1309 00). Go to **3**.
- b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **3**.

**3. Procedure completed.****HEAT ON CAPSULE ON ECS CONTROL PANEL DOES NOT GO ON****SYMPTOM**

HEAT ON capsule on ECS control panel does not go on.

**MALFUNCTION**

HEAT ON capsule on ECS control panel does not go on.

**CORRECTIVE ACTION****1. Check for 28 vdc between P356-J and ground.**

- a. If voltage is as specified, replace ECS control panel (WP 1309 00). Go to **3**.
- b. If voltage is not as specified, go to **2**.

**2. Check continuity between J198-K and P356-J.**

- a. If continuity is present, troubleshoot indicator lights dimming system (WP 0126 00). Go to **3**.
- b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **3**.

**3. Procedure completed.****ECS HEATER INOPERABLE****SYMPTOM**

ECS heater inoperable.

**MALFUNCTION**

ECS heater inoperable.

**CORRECTIVE ACTION****1. Verify upper and lower HEATER ELEMENTS circuit breakers on ECS circuit breaker panel are pushed in.**

- a. If upper and lower HEATER ELEMENTS circuit breakers are pushed in, go to **2**.
- b. If upper and lower HEATER ELEMENTS circuit breakers are pulled out, push in circuit breakers. Go to **11**.
- c. If upper and lower HEATER ELEMENTS circuit breakers will not reset, go to **5**.

**2. Check for 28 vdc between P352-G and ground.**

- a. If voltage is as specified, go to **3**.
- b. If voltage is not as specified, go to **6**.

**3. Check for 115 vac between:**

- P355-H and ground
- P355-J and ground

---

**ECS HEATER INOPERABLE - Continued**

- **P355-K and ground**
  - a. If voltage is as specified, go to **4**.
  - b. If voltage is not as specified go to **7**.
- 4. Check for 115 vac between:**
  - **P355-L and ground**
  - **P355-M and ground**
  - **P355-N and ground**
  - a. If voltage is as specified, replace evaporator pallet (WP 1304 00). Go to **11**.
  - b. If voltage is not as specified go to **8**.
- 5. Disconnect P355 from evaporator pallet assembly and reset upper and lower HEATER ELEMENTS circuit breakers.**
  - a. If upper and lower HEATER ELEMENTS circuit breakers reset, replace evaporator pallet assembly (WP 1304 00). Go to **11**.
  - b. If upper and lower HEATER ELEMENTS circuit breakers do not reset, replace ECU distribution box (WP 1300 00). Go to **11**.
- 6. Check continuity between P356-C and P352-G.**
  - a. If continuity is present, replace ECS control panel (WP 1309 00). Go to **11**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **11**.
- 7. Check continuity between:**
  - **P351-H and P355-H**
  - **P351-J and P355-J**
  - **P351-K and P355-K**
  - a. If continuity is present, replace ECU distribution box (WP 1300 00). Go to **11**.
  - b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **11**.
- 8. Check continuity between:**
  - **P351-L and P355-L**
  - **P351-M and P355-M**
  - **P351-N and P355-N**
  - a. If continuity is present, go to **9**.
  - b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **11**.
- 9. Check for 115 vac between:**
  - **P349-A and ground**
  - **P349-B and ground**
  - **P349-C and ground**

---

**ECS HEATER INOPERABLE - Continued**

- a. If voltage is as specified, replace ECU distribution box (WP 1300 00). Go to **11**.
- b. If voltage is not specified, go to **10**.

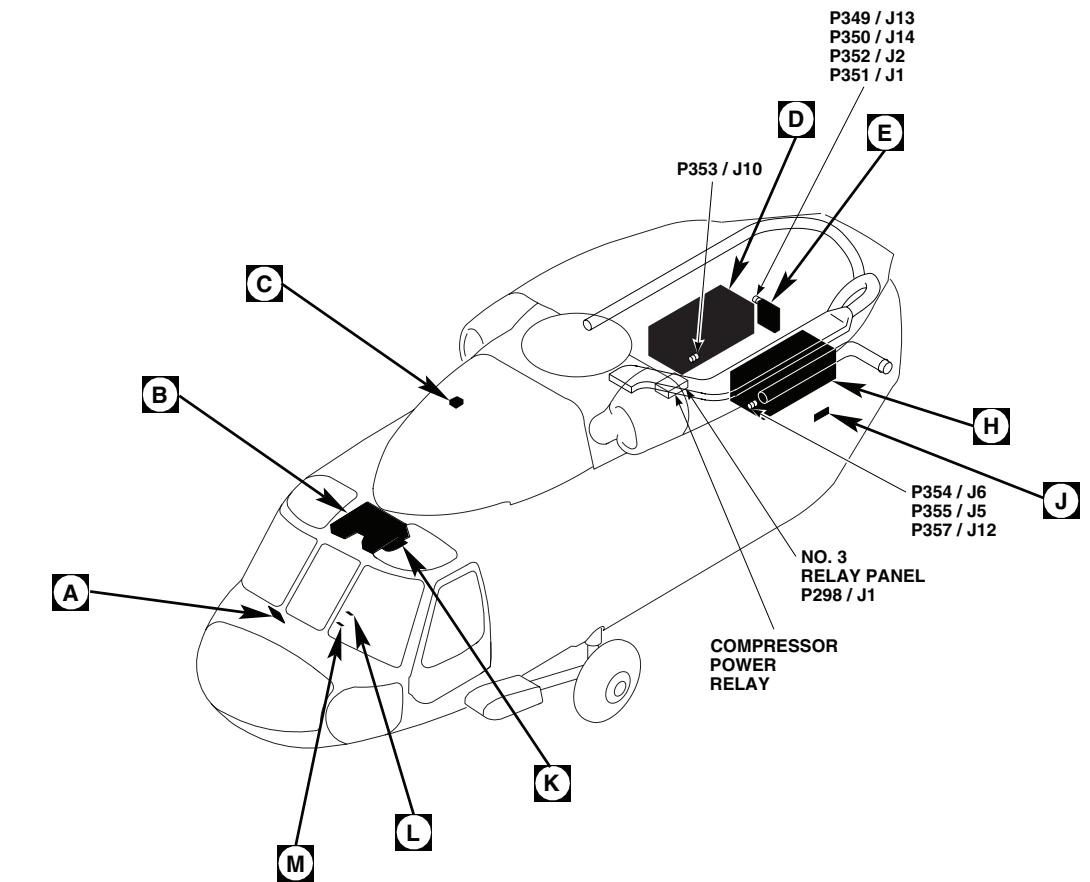
**10. Check continuity between:**

- **P349-A and CB525-A2 (HEAT POWER circuit breaker)**
- **P349-B and CB525-B2 (HEAT POWER circuit breaker)**
- **P349-C and CB525-C2 (HEAT POWER circuit breaker)**

- a. If continuity is present, troubleshoot ac electrical system (WP 0122 00). Go to **11**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **11**.

**11. Procedure completed.**

## LOCATION



| TERMINAL BOARD /<br>DISCONNECT PLUG /<br>RECEPTACLE | LOCATION /<br>CONNECTION POINT                   |
|-----------------------------------------------------|--------------------------------------------------|
| J198                                                | INDICATOR LIGHTS<br>DIMMER                       |
| J906                                                | NO. 1 JUNCTION BOX                               |
| P3 / J3                                             | ELECTRICAL CONTROL<br>UNIT                       |
| P9 / J9                                             | COMPRESSOR / MOTOR                               |
| P11 / J11                                           | EVAPORATOR BLOWER                                |
| P129 / J129                                         | COPILOT'S AUXILIARY<br>CIRCUIT BREAKER PANEL     |
| P210 / J210                                         | NO. 1 JUNCTION BOX                               |
| P212                                                | LOW TEMPERATURE<br>SWITCH SW3                    |
| P244 / J244                                         | NO. 1 JUNCTION BOX                               |
| P266 / J266                                         | BEHIND PILOT'S CIRCUIT<br>BREAKER PANEL BL 25 RH |

| TERMINAL BOARD /<br>DISCONNECT PLUG /<br>RECEPTACLE | LOCATION /<br>CONNECTION POINT              |
|-----------------------------------------------------|---------------------------------------------|
| P298 / J1                                           | NO. 3 RELAY PANEL                           |
| P329 / J329                                         | MAIN ROTOR DE-ICE<br>JUNCTION BOX           |
| P349 / J13                                          | ELECTRICAL CONTROL<br>UNIT DISTRIBUTION BOX |
| P350 / J14                                          | ELECTRICAL CONTROL<br>UNIT DISTRIBUTION BOX |
| P351 / J1                                           | ELECTRICAL CONTROL<br>UNIT DISTRIBUTION BOX |
| P352 / J2                                           | ELECTRICAL CONTROL<br>UNIT DISTRIBUTION BOX |
| P353 / J10                                          | CONDENSER BLOWER                            |
| P354 / J6                                           | EVAPORATOR PALLET                           |
| P355 / J5                                           | EVAPORATOR PALLET                           |
| P356 / J1                                           | ECS CONTROL PANEL                           |
| P357 / J12                                          | EVAPORATOR PALLET                           |

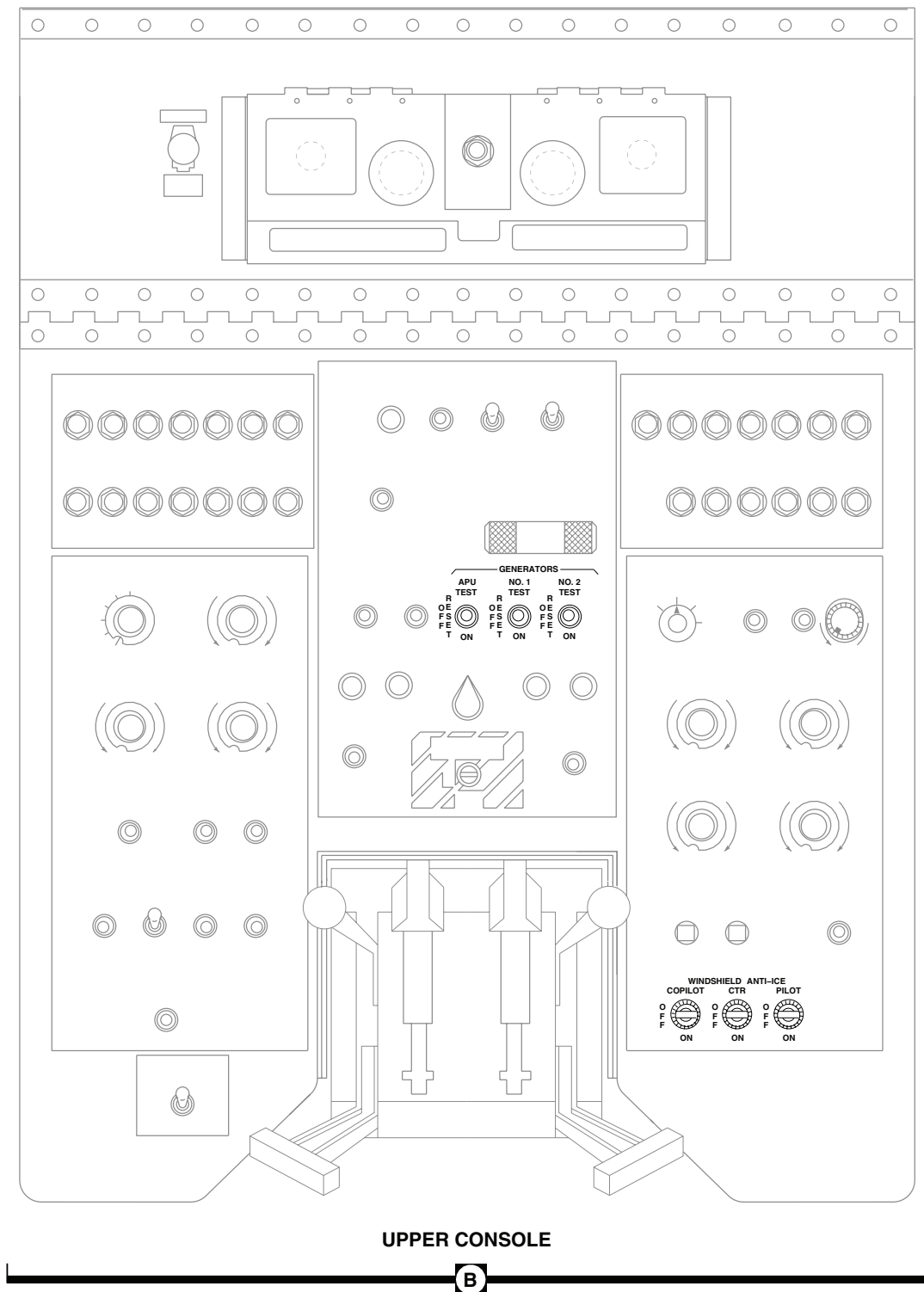
AB0216\_1A  
SA

Figure 1. Environmental Control System Location Diagram. (Sheet 1 of 7)

AB0216\_2B  
SA

### Change 1

LOCATION - Continued



AB0216 3  
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Figure 1. Environmental Control System Location Diagram. (Sheet 3 of 7)

LOCATION - Continued

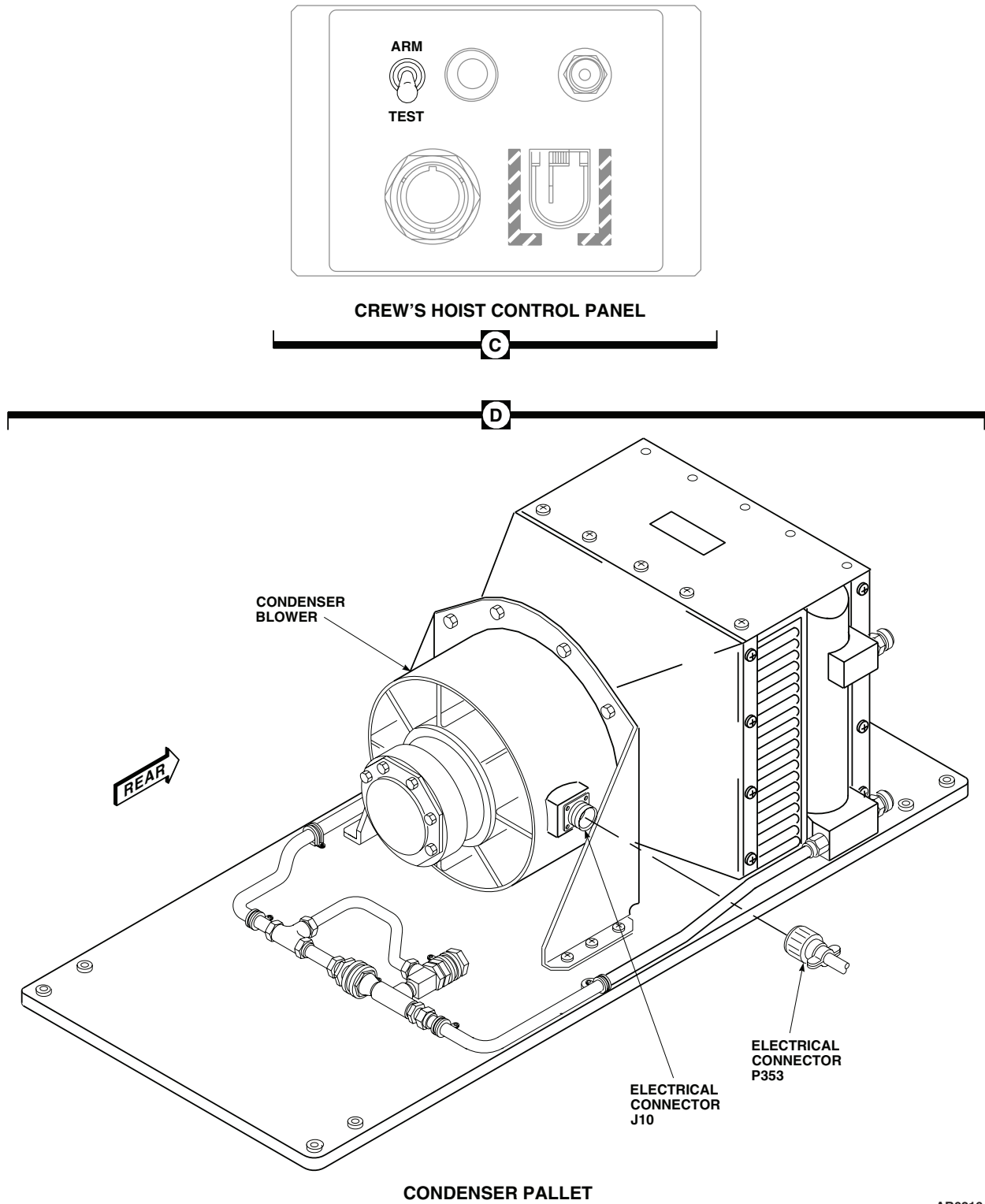


Figure 1. Environmental Control System Location Diagram. (Sheet 4 of 7)

AB0216\_4  
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LOCATION - Continued

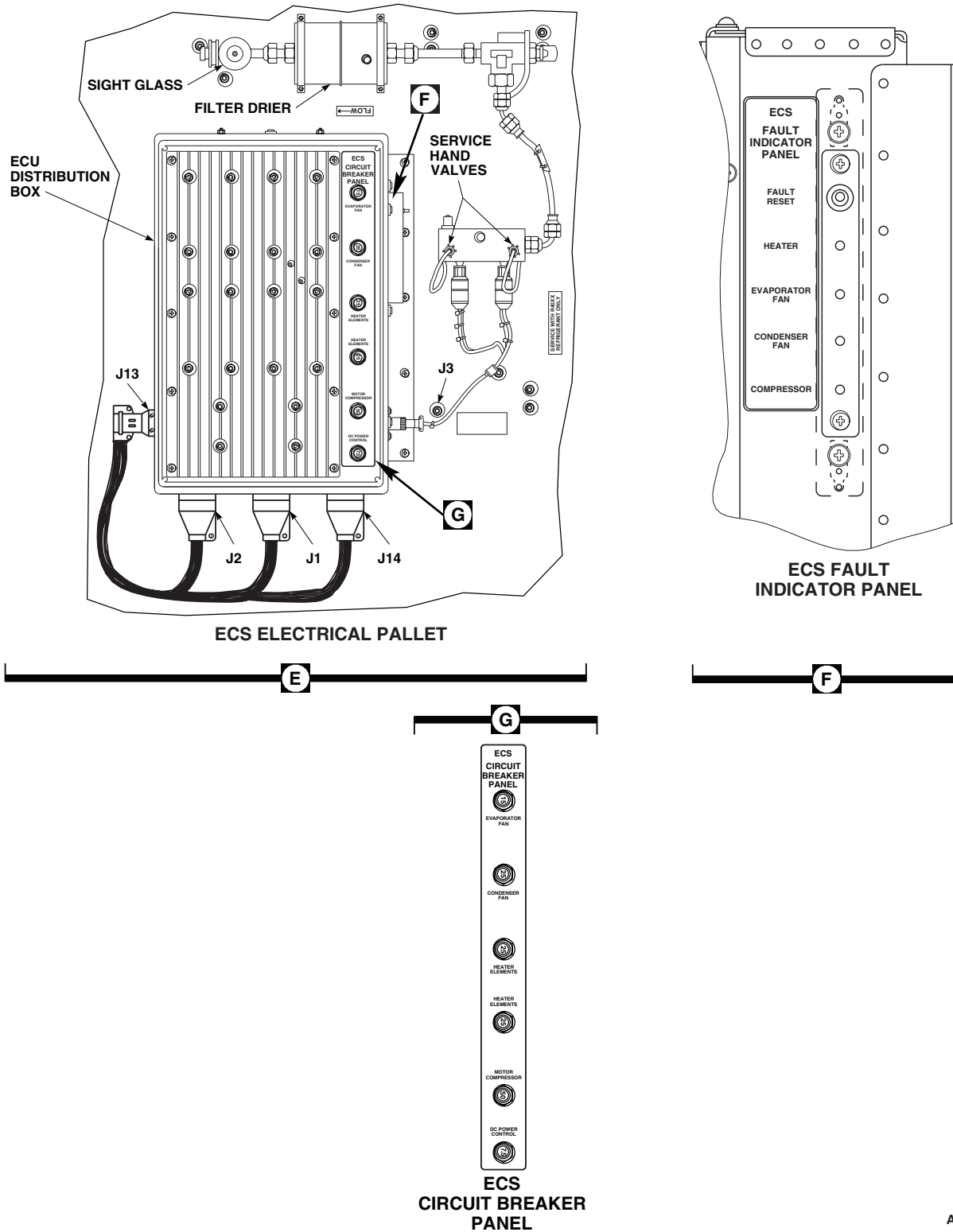
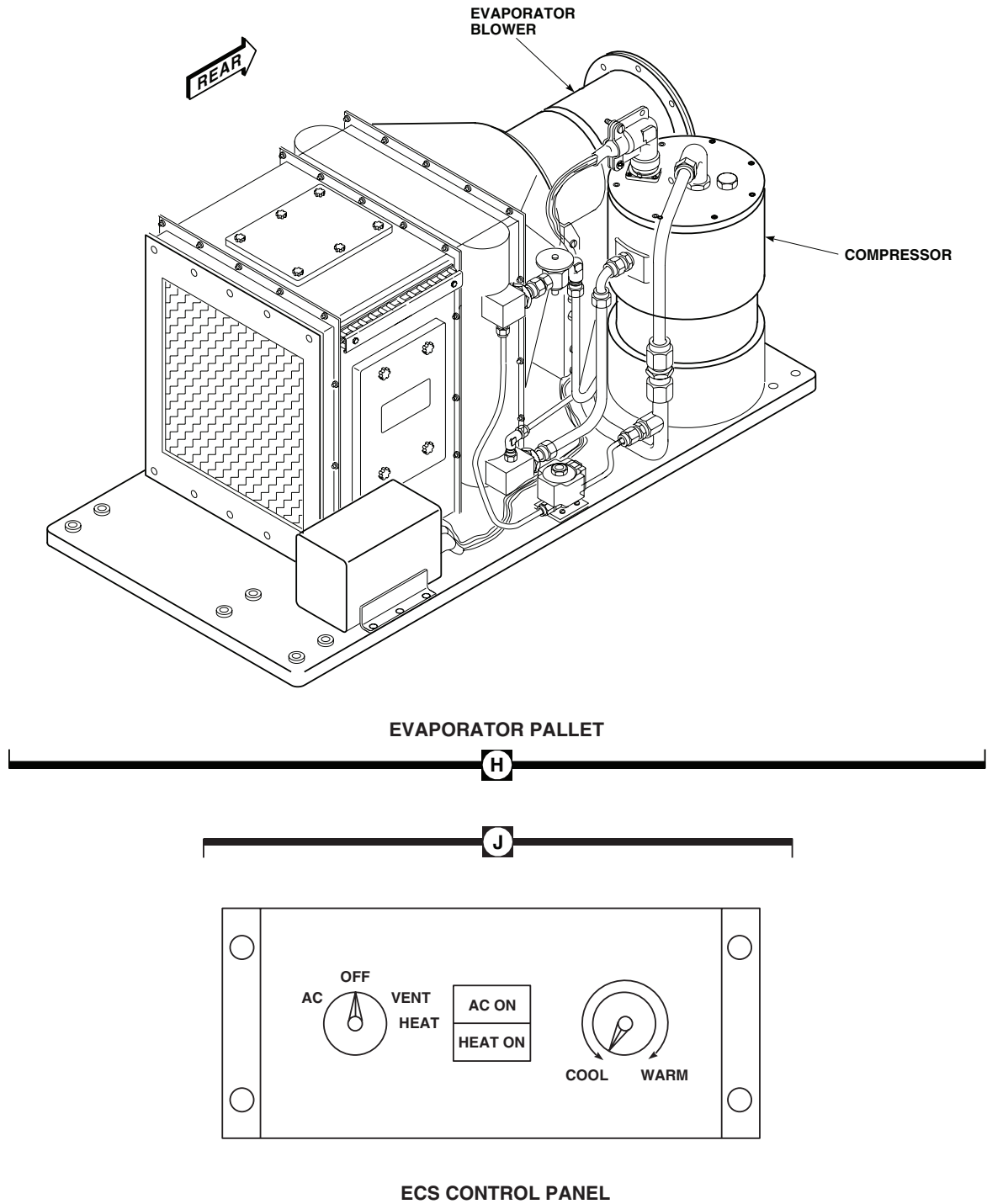


Figure 1. Environmental Control System Location Diagram. (Sheet 5 of 7)

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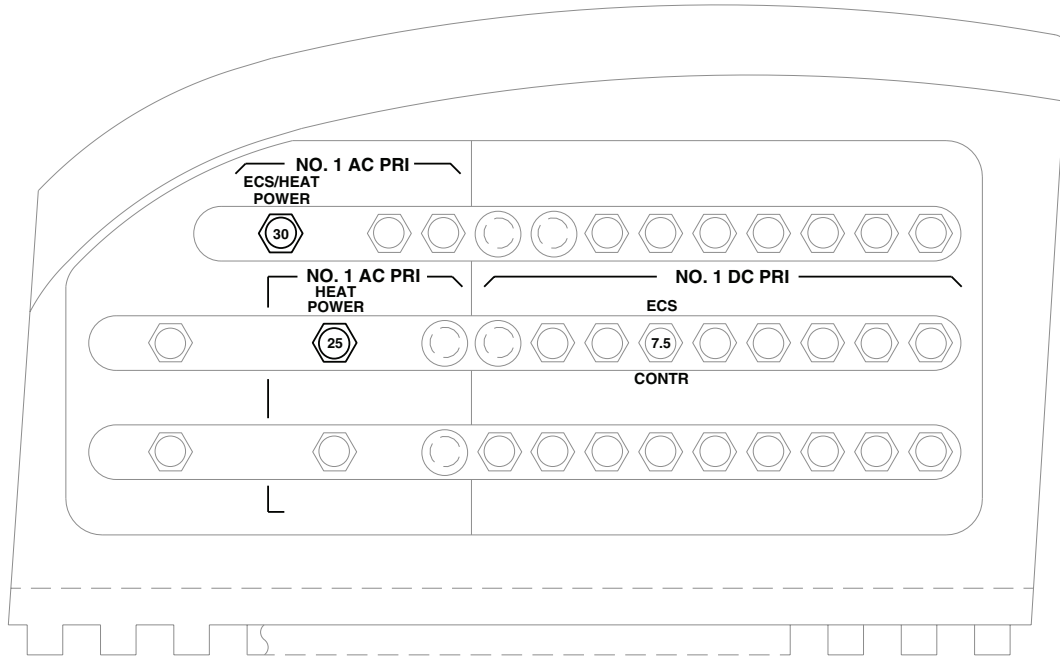
LOCATION - Continued



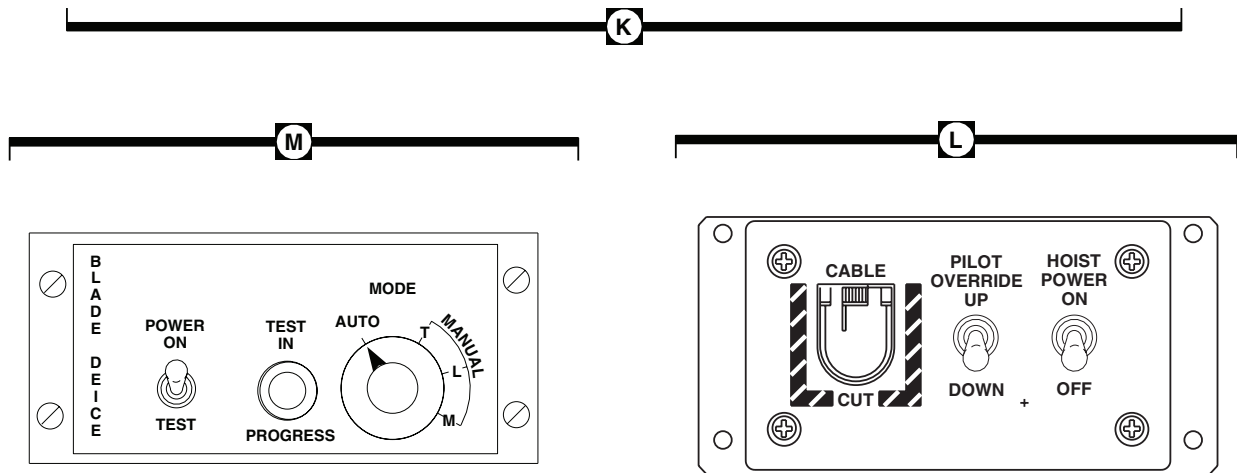
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Figure 1. Environmental Control System Location Diagram. (Sheet 6 of 7)

LOCATION - Continued



COPILOT'S AUXILIARY CIRCUIT BREAKER PANEL



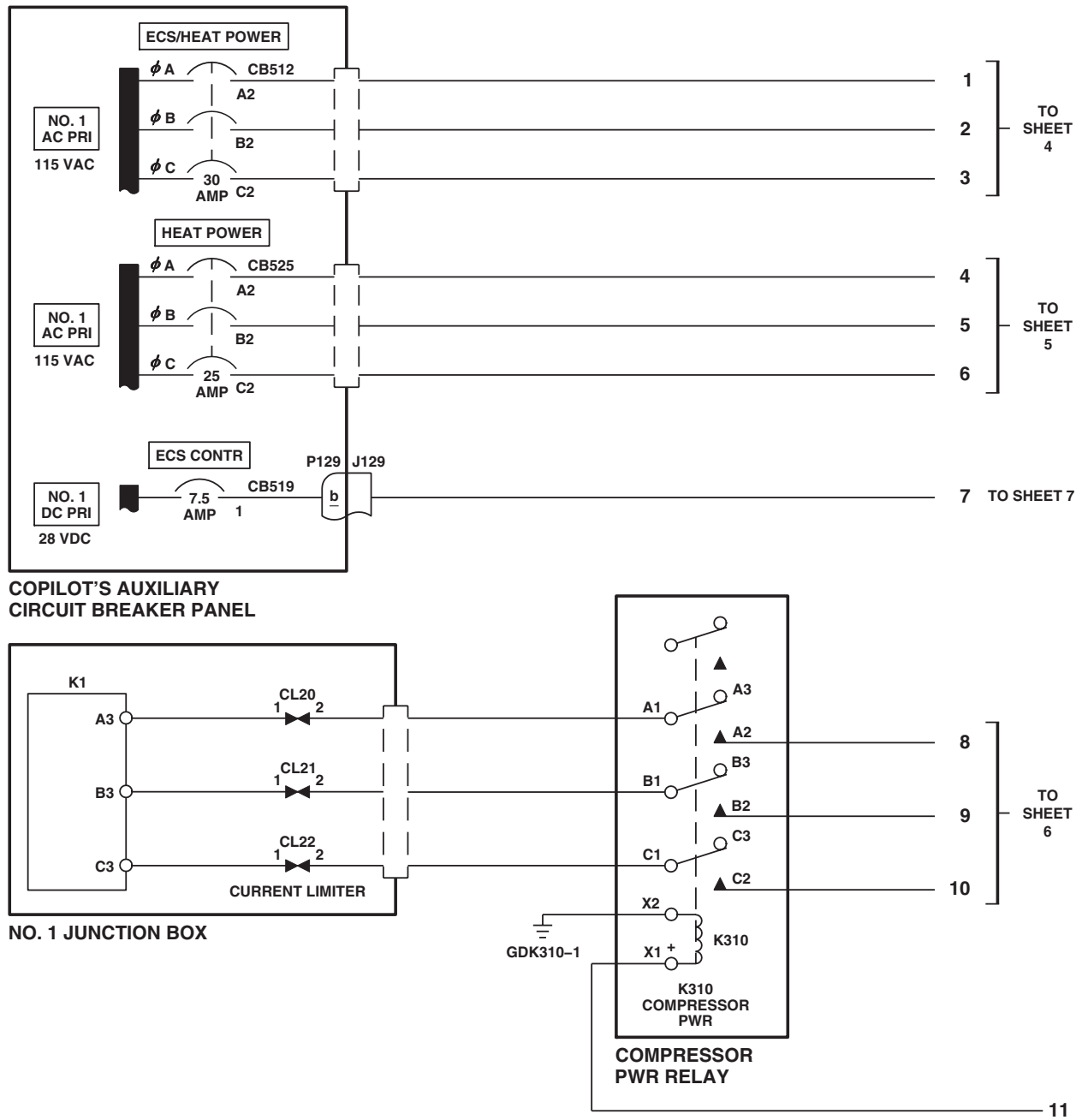
BLADE DEICE CONTROL PANEL

PILOT'S HOIST CONTROL PANEL

AB0216.7  
SA

Figure 1. Environmental Control System Location Diagram. (Sheet 7 of 7)

## SCHEMATICS



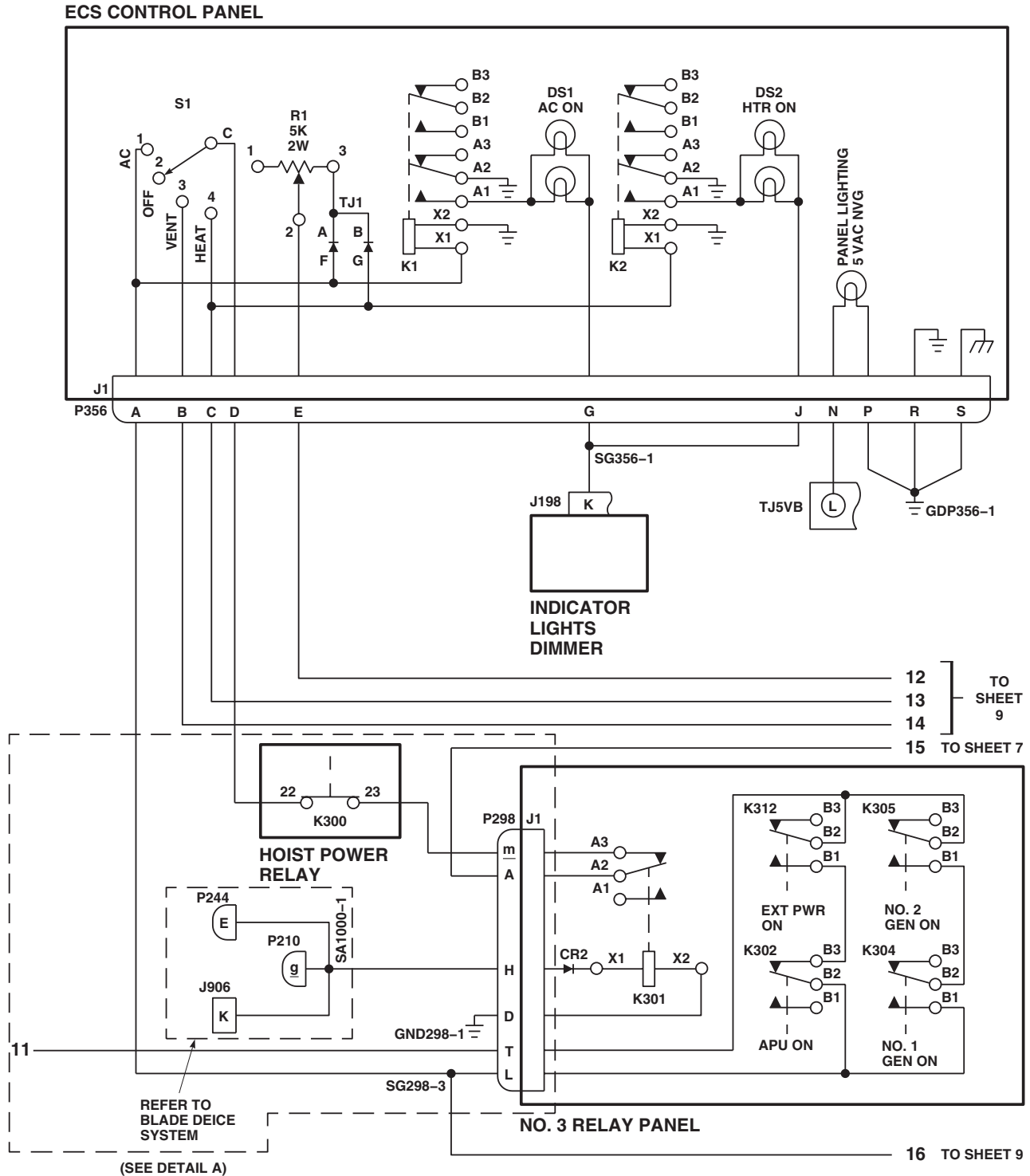
## NOTE

HH-60L

AB2104\_1  
SA

Figure 2. Environmental Control System Schematic Diagram. (Sheet 1 of 16)

SCHEMATICS - Continued



AB2104.2  
SA

Figure 2. Environmental Control System Schematic Diagram. (Sheet 2 of 16)

SCHEMATICS - Continued

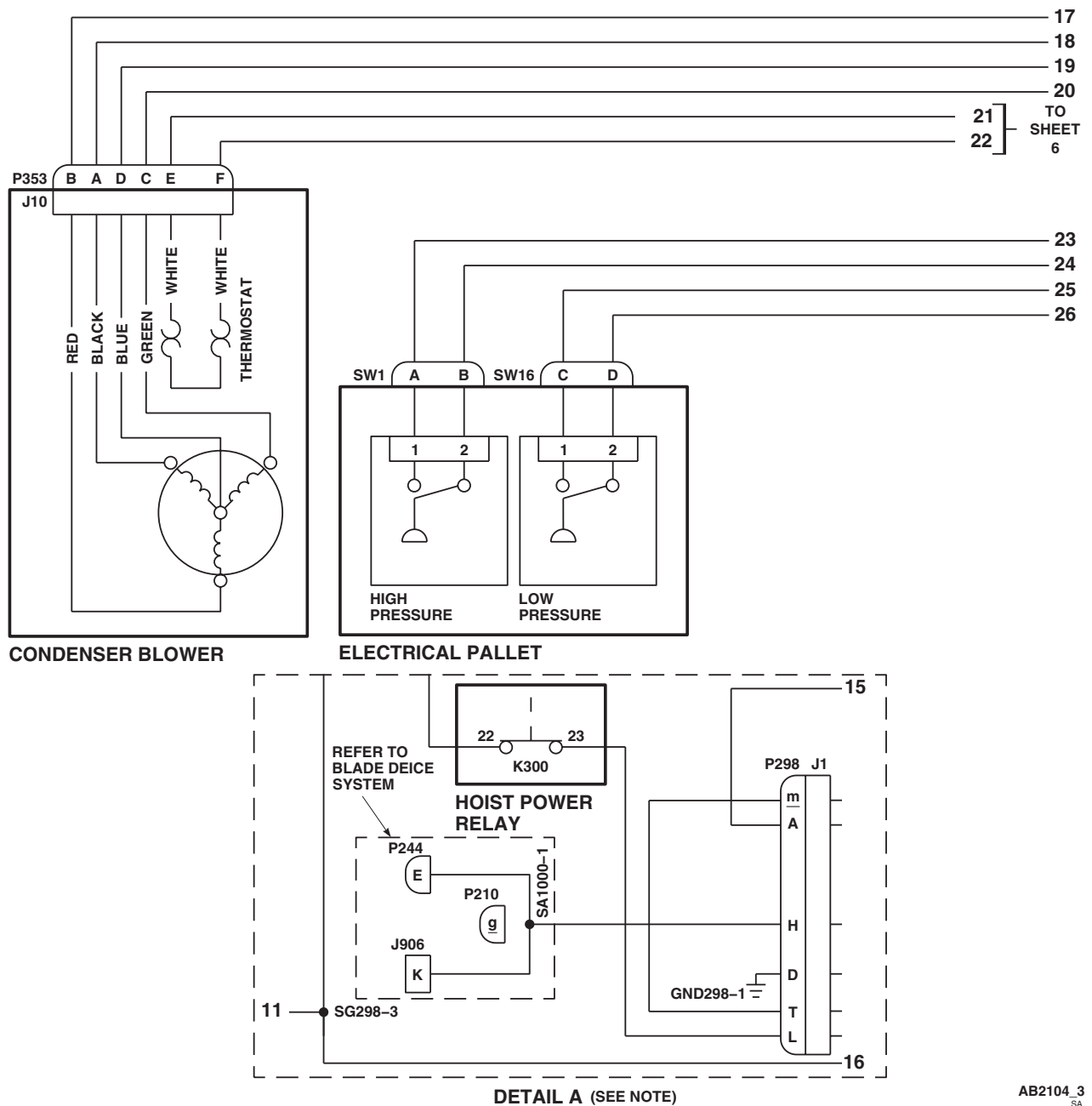
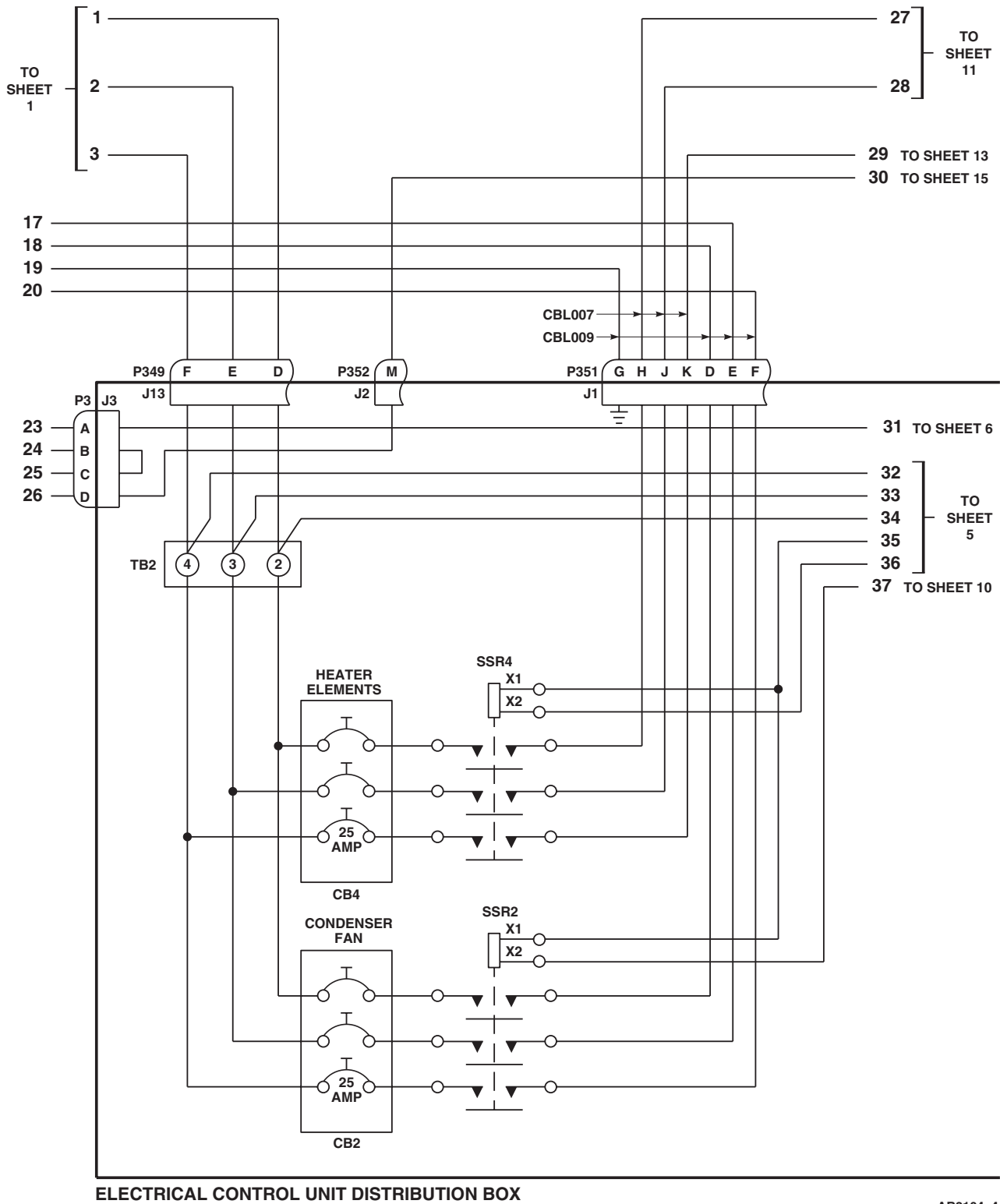


Figure 2. Environmental Control System Schematic Diagram. (Sheet 3 of 16)

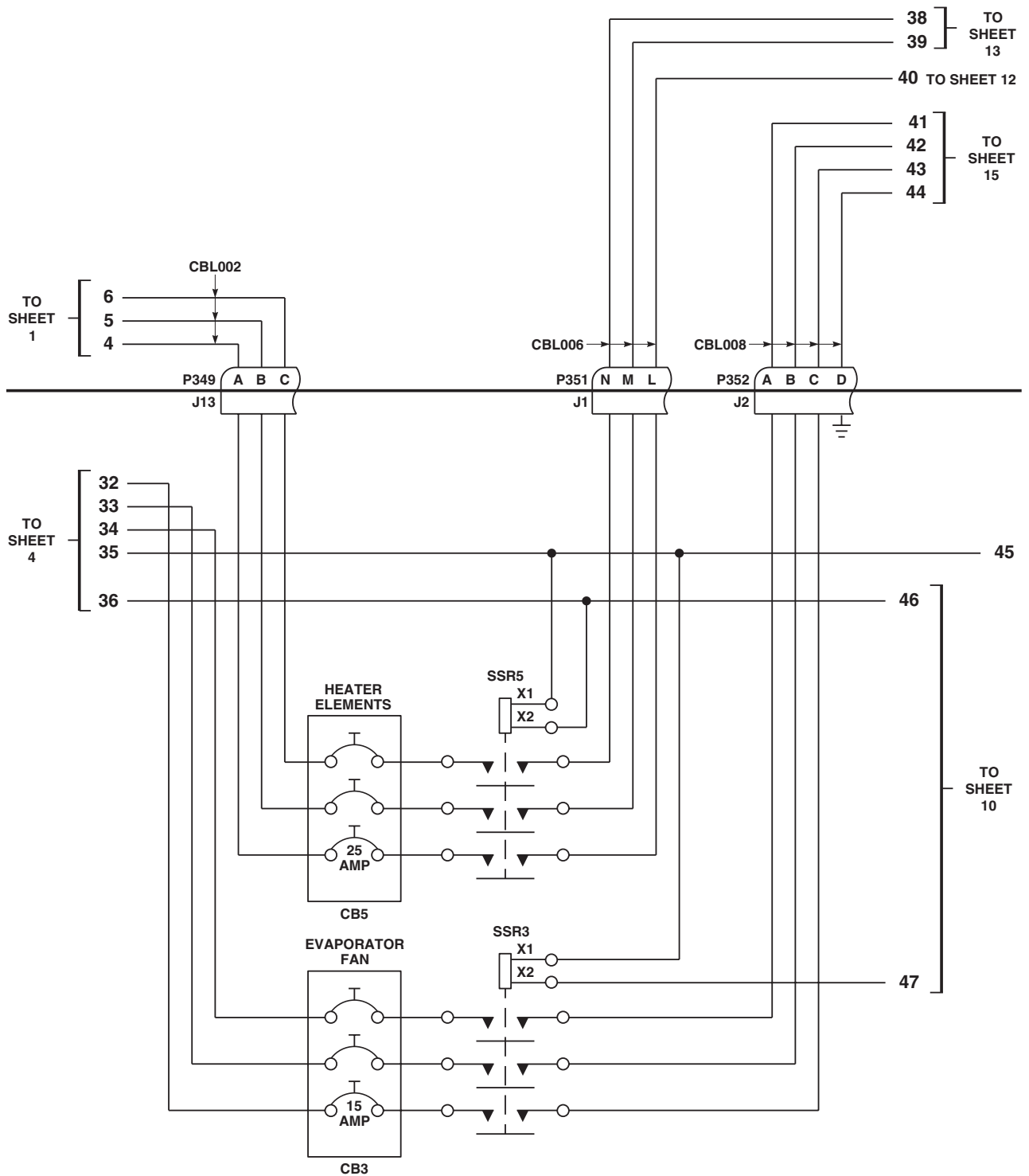
SCHEMATICS - Continued



AB2104\_4  
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Figure 2. Environmental Control System Schematic Diagram. (Sheet 4 of 16)

SCHEMATICS - Continued

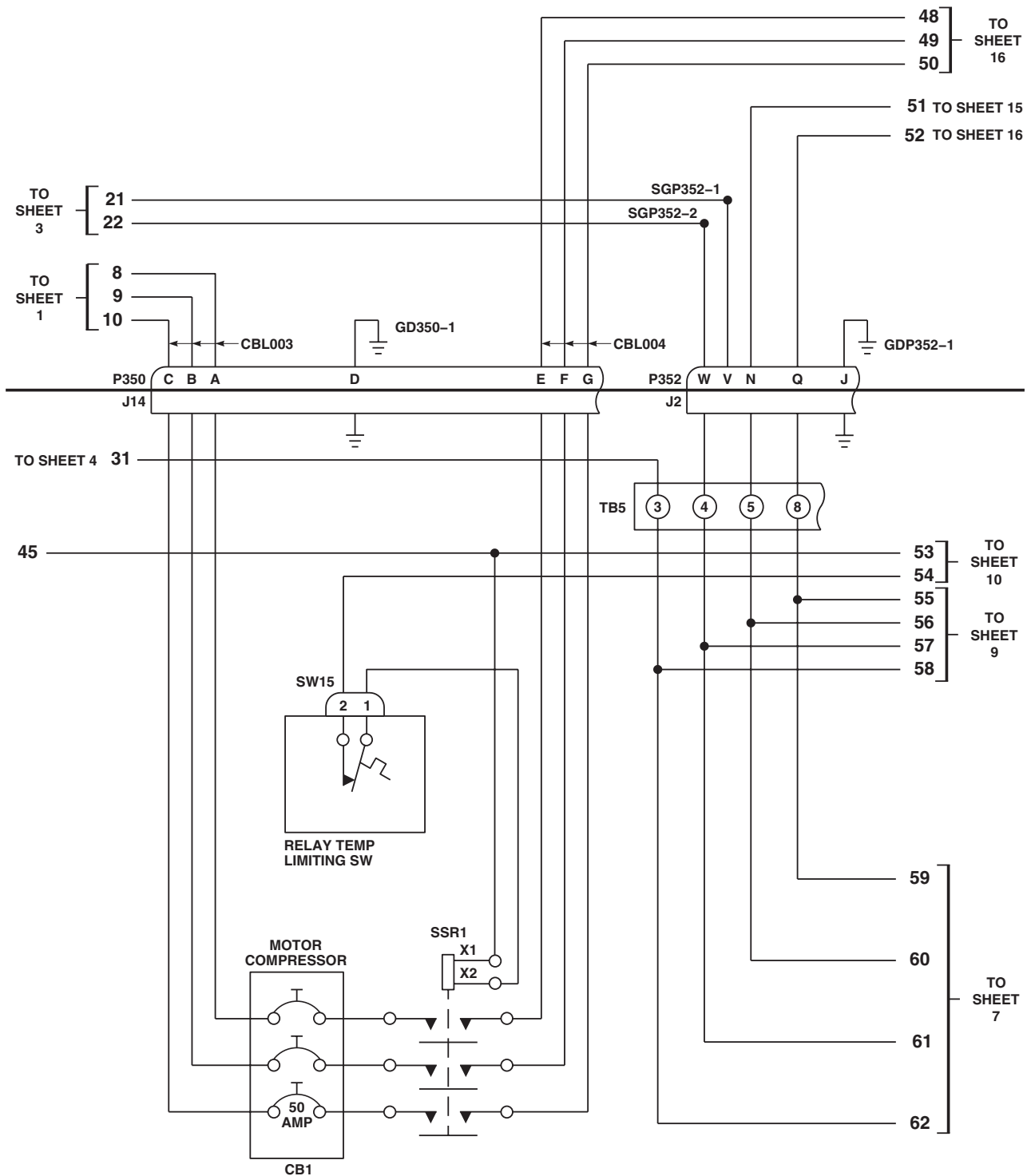


ELECTRICAL CONTROL UNIT DISTRIBUTION BOX

AB2104\_5  
SA

Figure 2. Environmental Control System Schematic Diagram. (Sheet 5 of 16)

SCHEMATICS - Continued



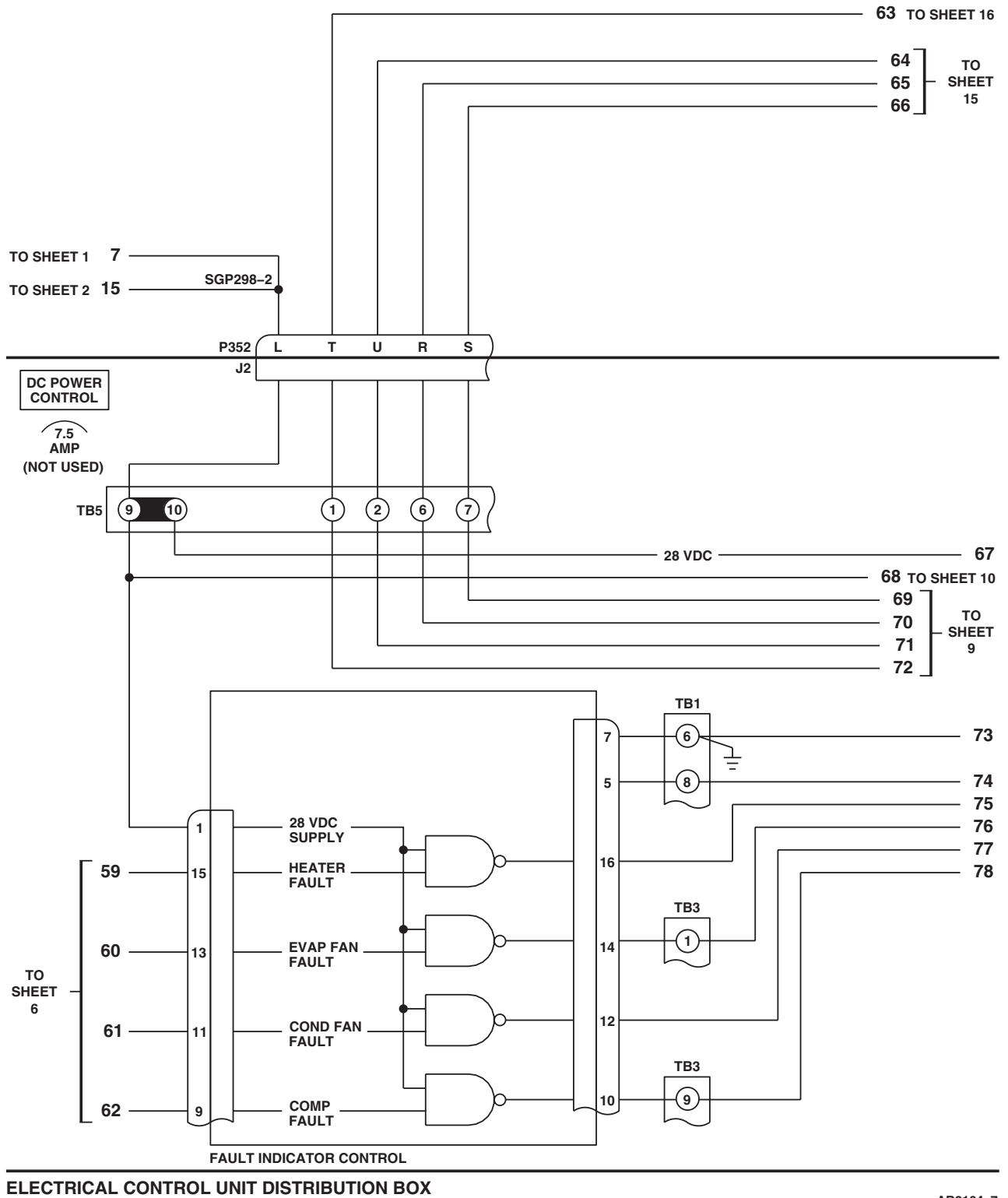
ELECTRICAL CONTROL UNIT DISTRIBUTION BOX

AB2104\_6  
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Figure 2. Environmental Control System Schematic Diagram. (Sheet 6 of 16)



SCHEMATICS - Continued

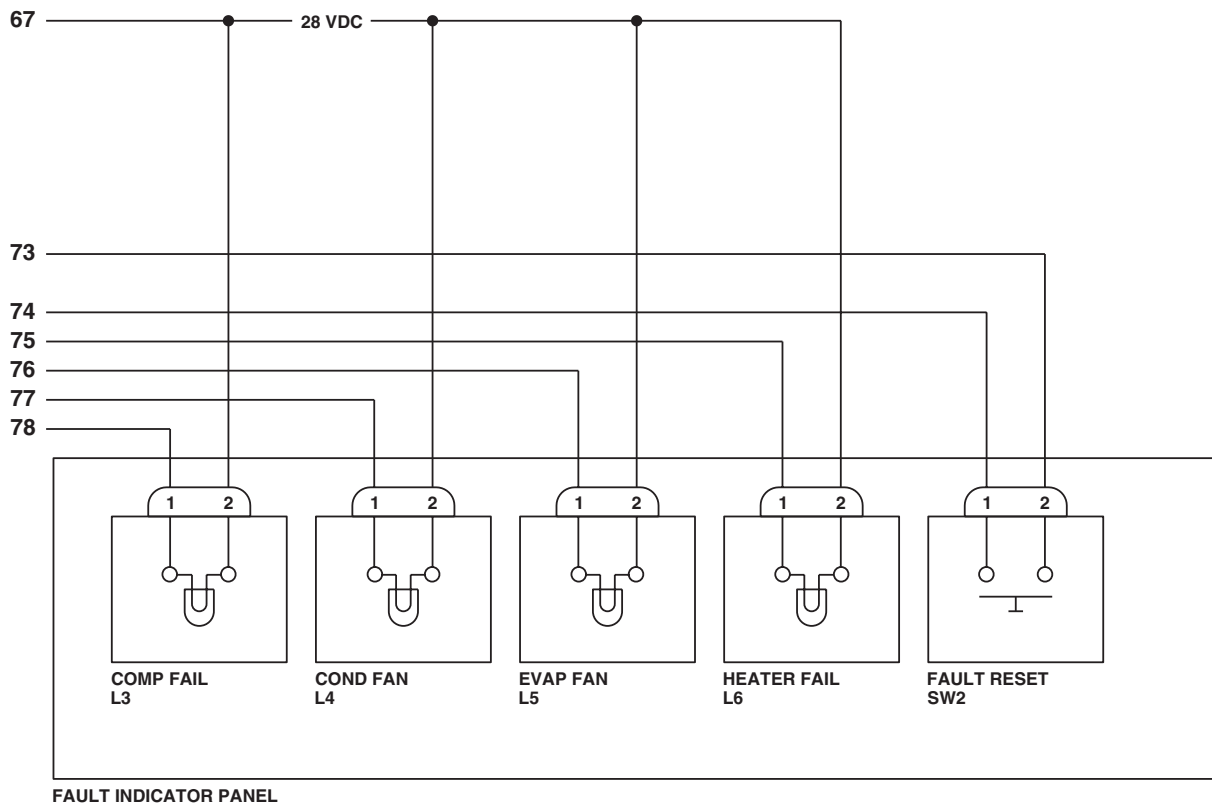


ELECTRICAL CONTROL UNIT DISTRIBUTION BOX

AB2104\_7  
SA

Figure 2. Environmental Control System Schematic Diagram. (Sheet 7 of 16)

SCHEMATICS - Continued

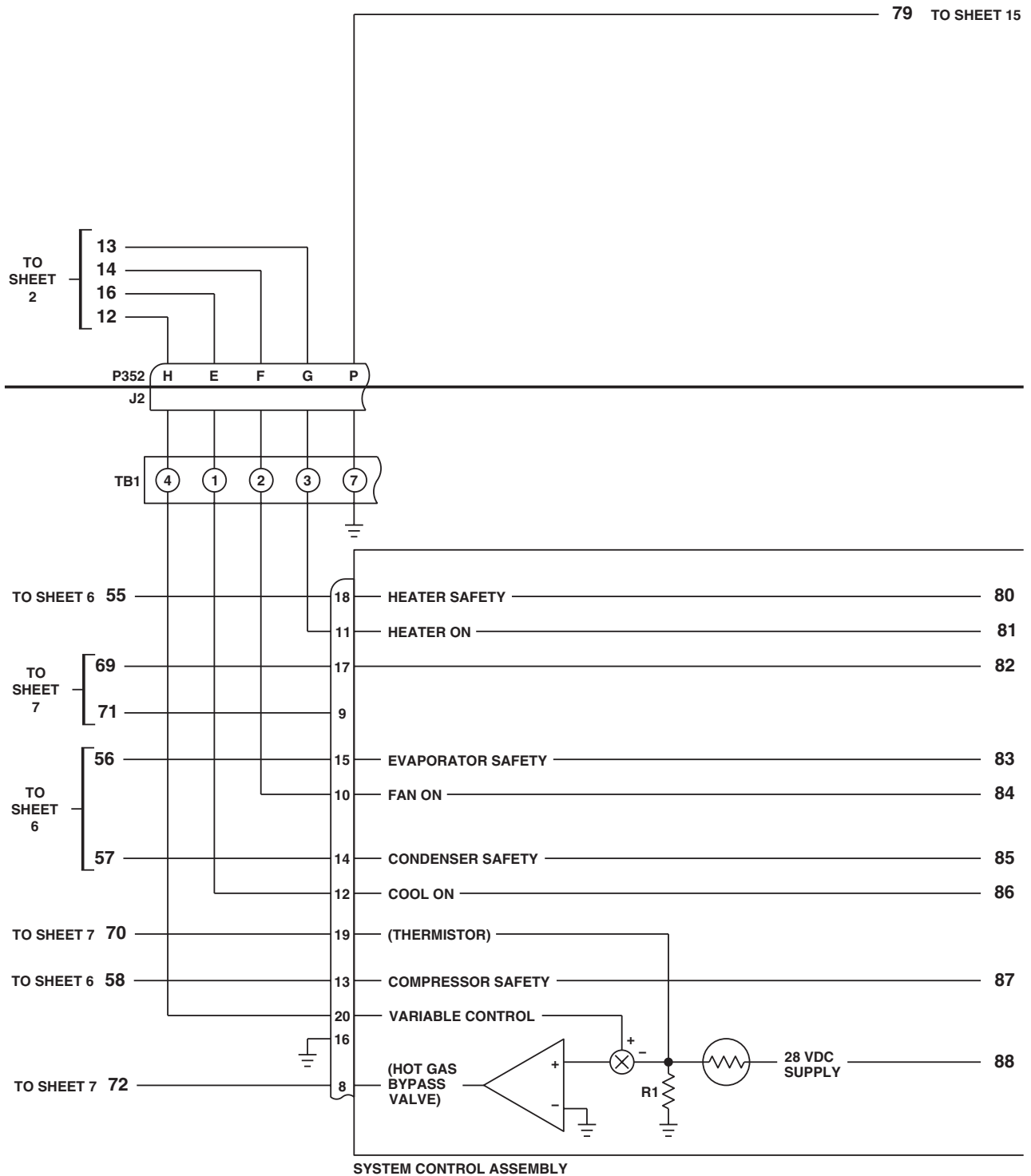


ELECTRICAL CONTROL UNIT DISTRIBUTION BOX

AB2104\_8  
SA

Figure 2. Environmental Control System Schematic Diagram. (Sheet 8 of 16)

SCHEMATICS - Continued



ELECTRICAL CONTROL UNIT DISTRIBUTION BOX

AB2104\_9  
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Figure 2. Environmental Control System Schematic Diagram. (Sheet 9 of 16)

SCHEMATICS - Continued

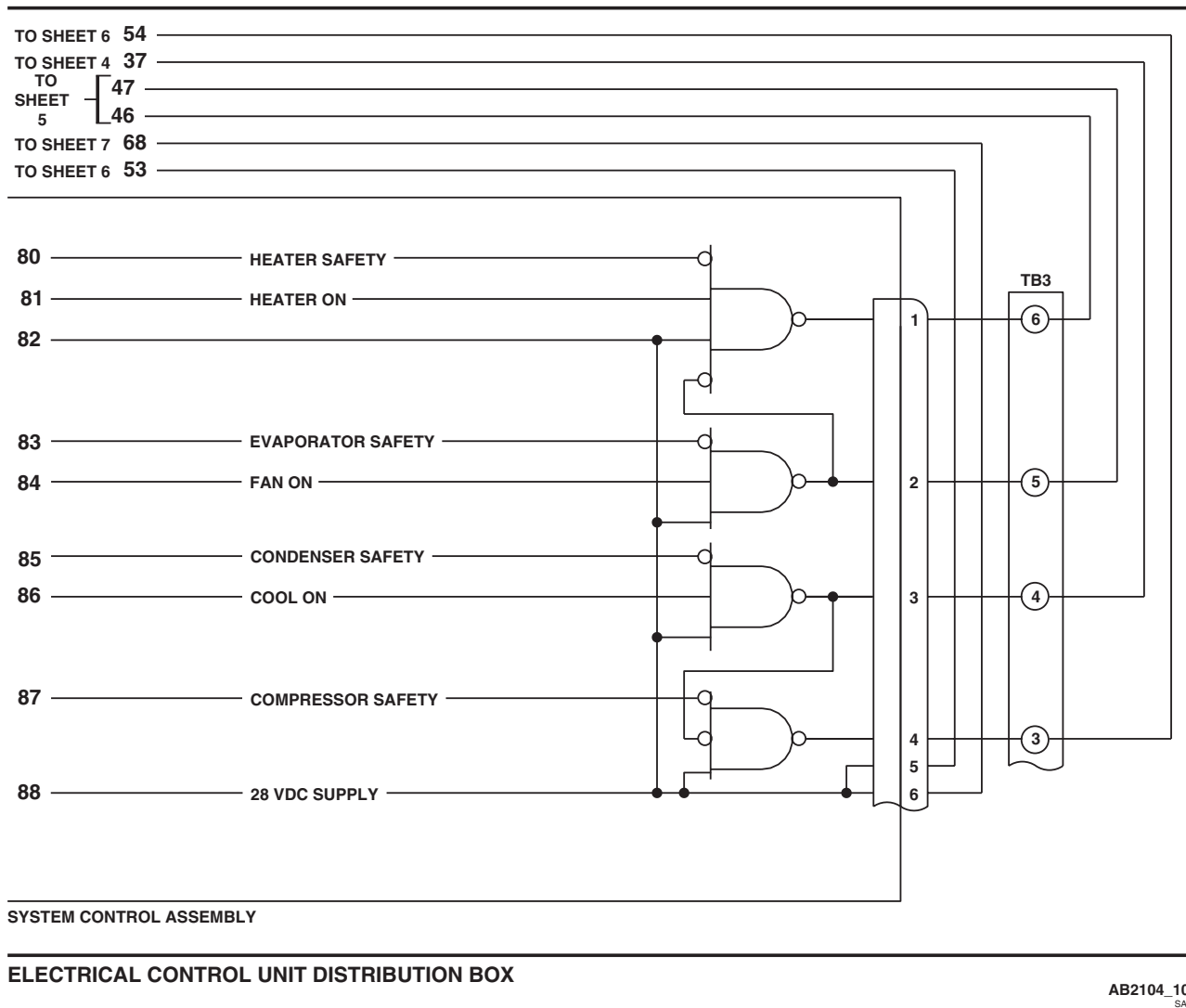


Figure 2. Environmental Control System Schematic Diagram. (Sheet 10 of 16)

SCHEMATICS - Continued

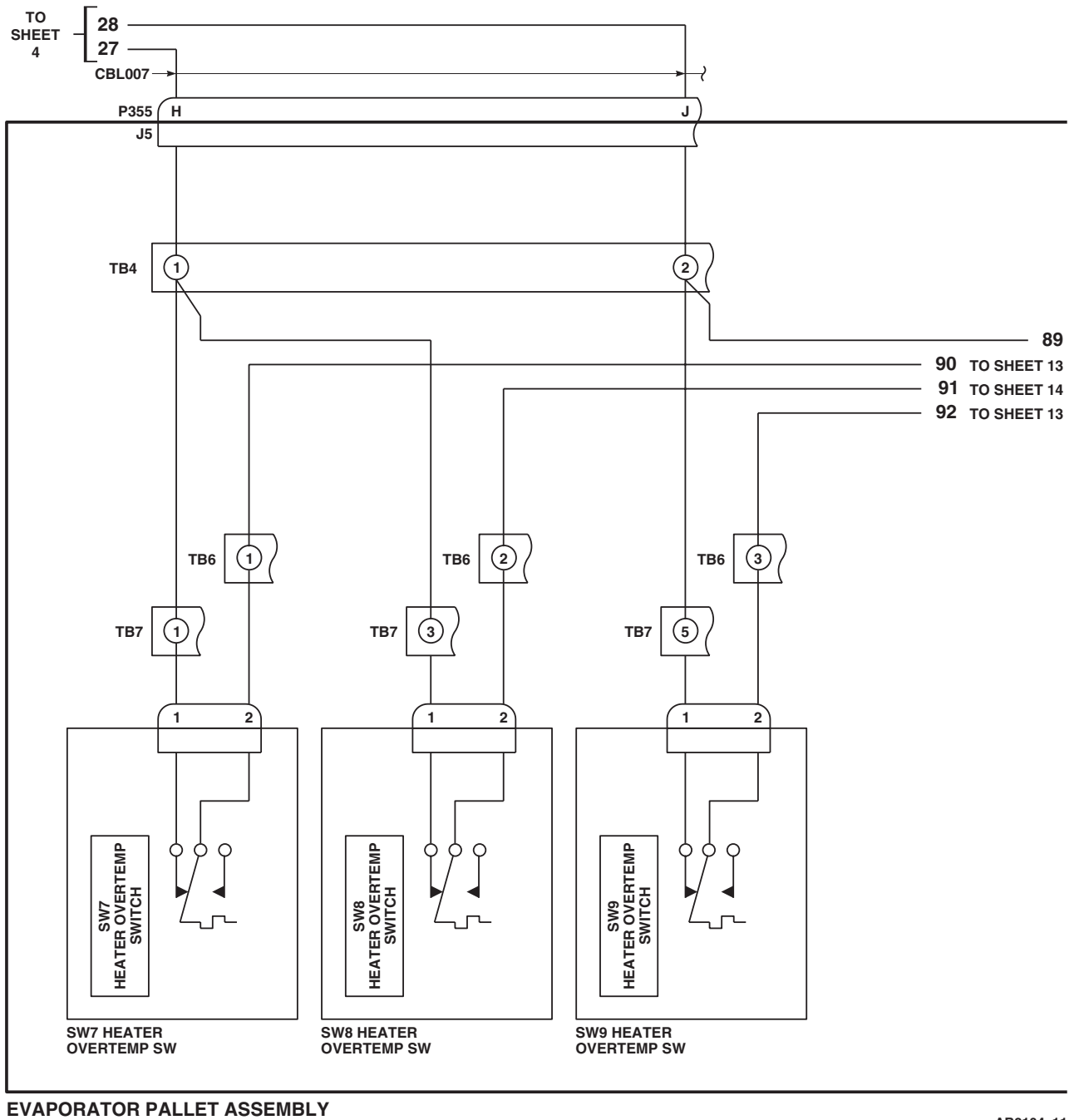
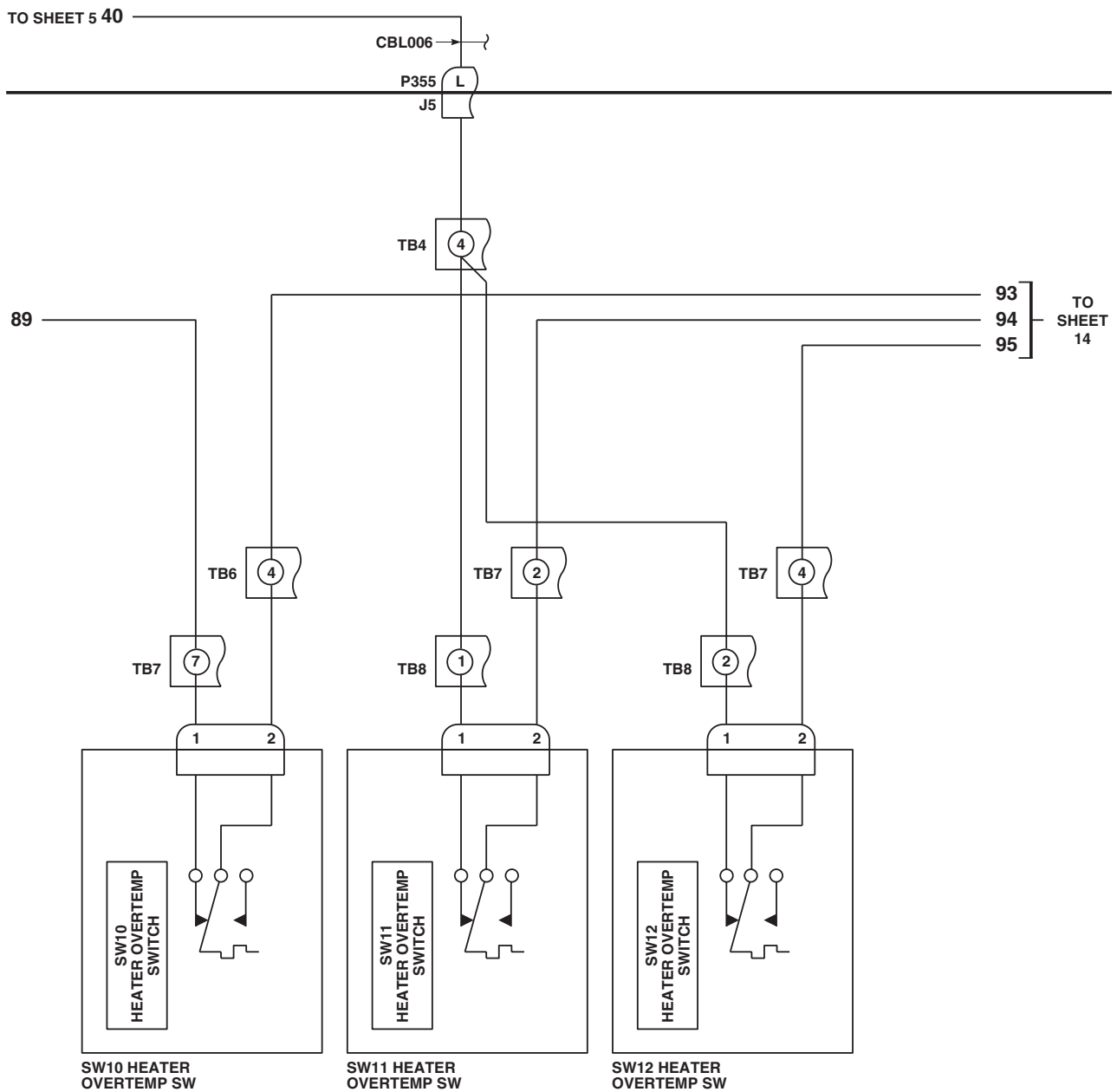


Figure 2. Environmental Control System Schematic Diagram. (Sheet 11 of 16)

SCHEMATICS - Continued

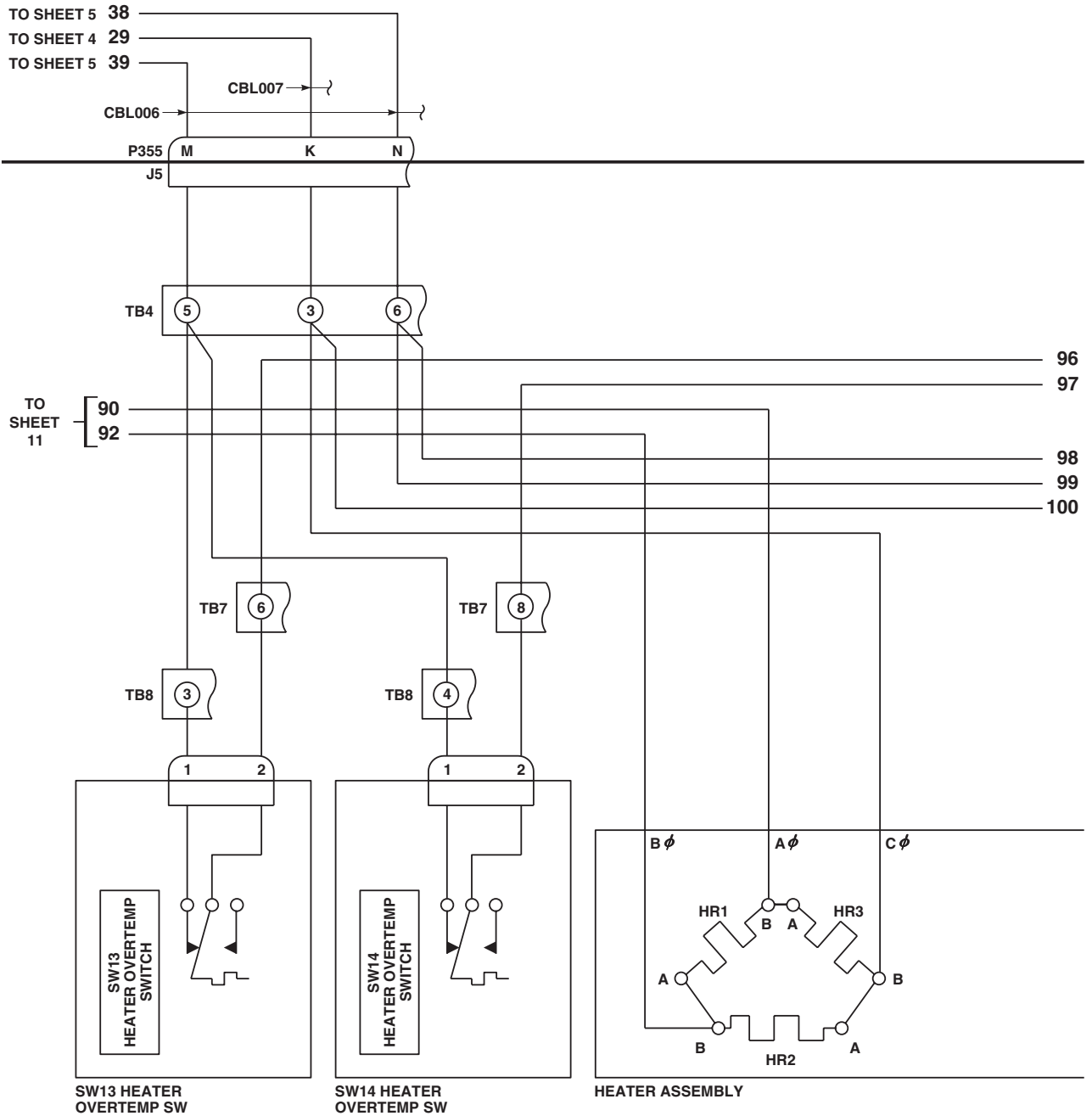


EVAPORATOR PALLET ASSEMBLY

AB2104\_12  
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Figure 2. Environmental Control System Schematic Diagram. (Sheet 12 of 16)

SCHEMATICS - Continued

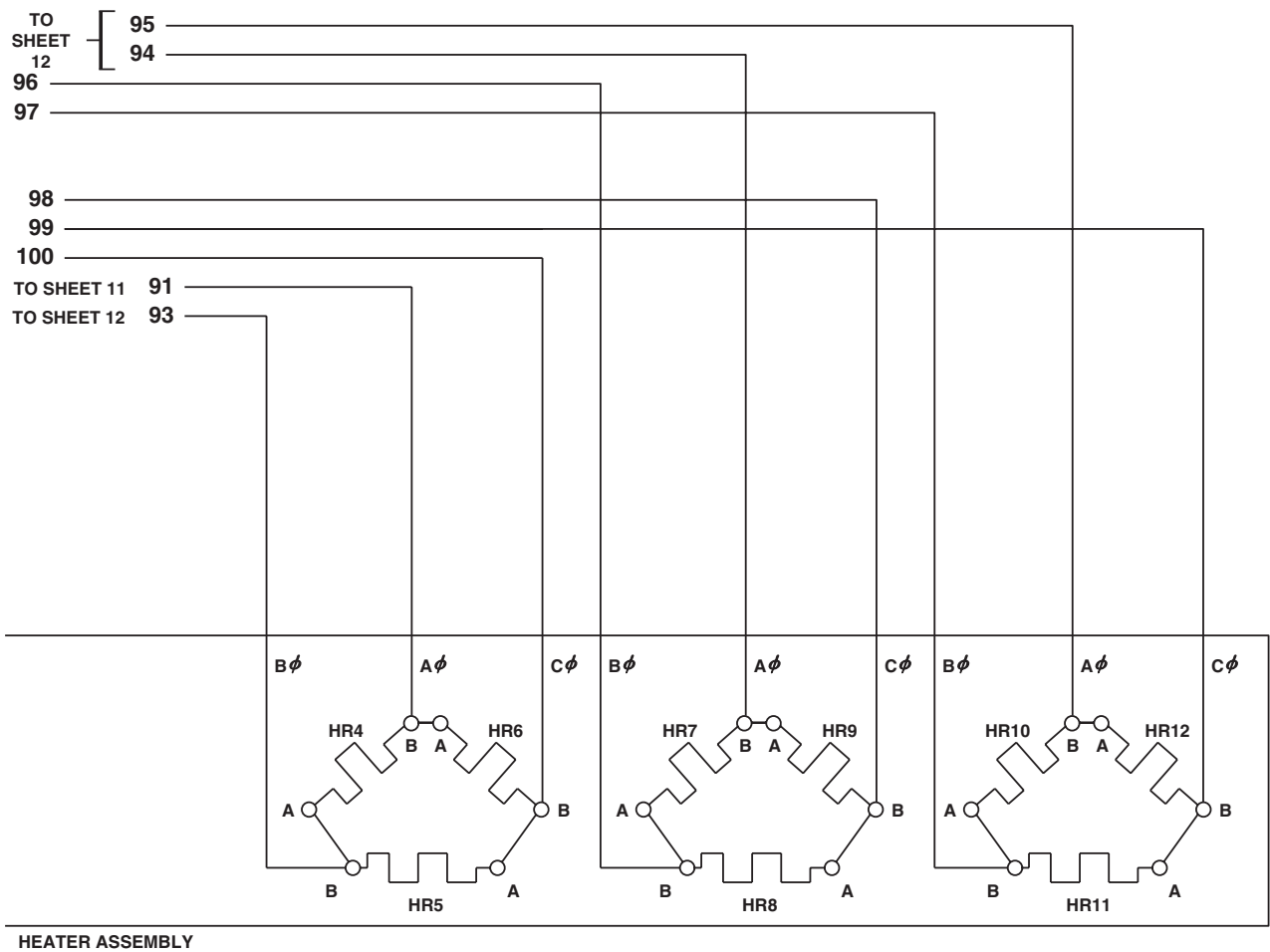


EVAPORATOR PALLET ASSEMBLY

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Figure 2. Environmental Control System Schematic Diagram. (Sheet 13 of 16)

SCHEMATICS - Continued



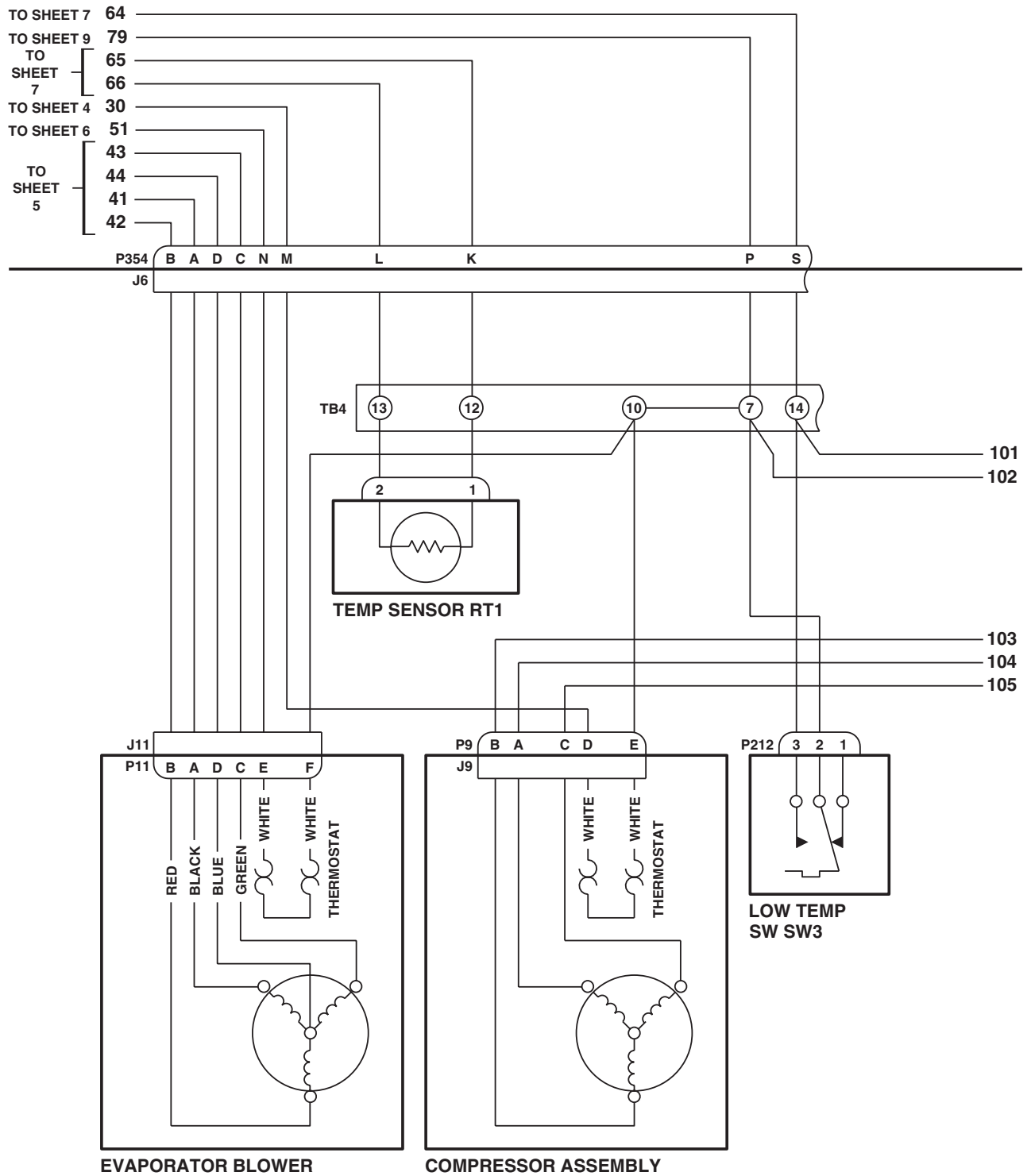
EVAPORATOR PALLET ASSEMBLY

AB2104\_14  
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Figure 2. Environmental Control System Schematic Diagram. (Sheet 14 of 16)



SCHEMATICS - Continued

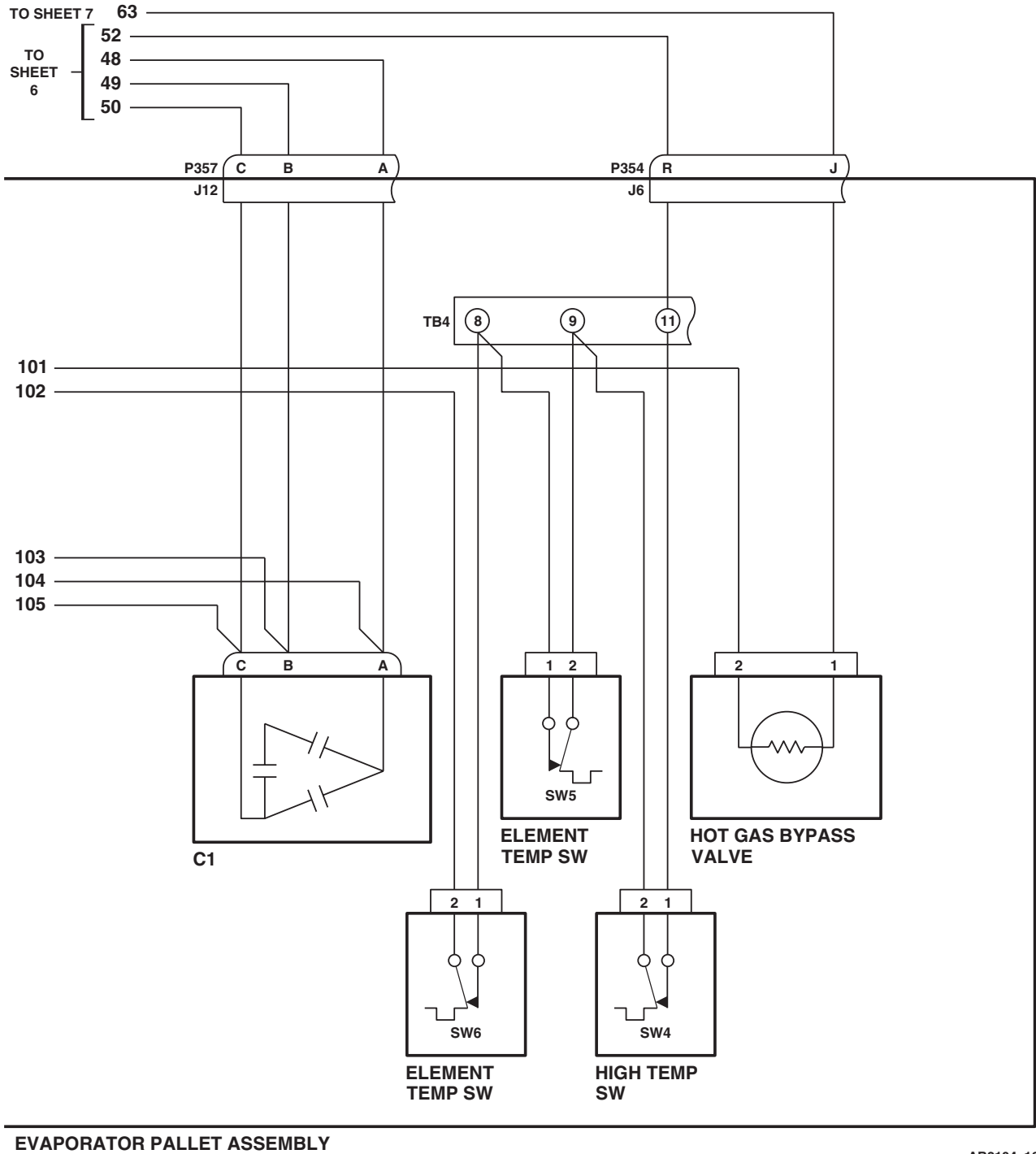


EVAPORATOR PALLET ASSEMBLY

AB2104\_15  
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Figure 2. Environmental Control System Schematic Diagram. (Sheet 15 of 16)

SCHEMATICS - Continued



AB2104\_16  
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Figure 2. Environmental Control System Schematic Diagram. (Sheet 16 of 16)

END OF WORK PACKAGE

---

**UNIT LEVEL**  
**HOISTS AND WINCHES**  
**RESCUE HOIST KIT**

**This WP supersedes WP 0152 00, dated 17 April 2006.**

---

**INITIAL SETUP:**

**Test Equipment**

Insulated Jumper Wire (5)  
Multimeter, AN/PSM-45A

WP 0104 00

WP 0110 00

WP 0824 00

WP 1333 00

**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01  
Electrical Repairer Toolkit, SC 5180-99-B06

WP 1335 00

WP 1337 00

WP 1338 00

**Personnel Required**

Aircraft Electrician MOS 15F (2)  
UH-60 Helicopter Repairer MOS 15T (1)

WP 1352 00

WP 1685 00

WP 1747 00

**References**

TM 1-1520-237-10  
TM 1-1680-320-13&P  
WP 0102 00

**Equipment Condition**

External Electrical Power Available, WP 1685 00  
or APU Operational, TM 1-1520-237-10

---

**SPECIAL INSTRUCTIONS**

**NOTE**

See Figure 1 for component location diagram and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

The rescue hoist kit operational/troubleshooting procedure is subdivided as follows:

- Cable Shear
- Rescue Hoist

**CABLE SHEAR OPERATIONAL/TROUBLESHOOTING PROCEDURE****PROCEDURE****Step****WARNING**

Rescue hoist cable cutter contains an explosive cartridge. Use extreme care when handling to prevent injury to personnel. Static producing clothing is prohibited. When disconnecting cable cutter harness, install piece of aluminum foil between cartridge pins and install shipping cap.

**NOTE**

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Disconnect connector from rescue hoist cable cutter's primer charge and install piece of aluminum foil between cartridge pins and install shipping cap.
2. Make sure HOIST CABLE SHEAR circuit breaker on upper console circuit breaker panel is pushed in.
3. Turn on electrical power.
4. On rescue hoist control panel, press face of SQUIB IND light.

**Indication/Condition**

SQUIB IND light shall go on.

**Corrective Action**

If Indication/Condition is not as specified, go to SQUIB IND LIGHT DOES NOT GO ON WHEN PRESSED, in this work package.

**Step**

5. Install insulated jumper wires between cable cutter connector pins A and B, and between pins C and D.
6. Place rescue hoist control panel SQUIB switch from NORM to TEST and hold.

**Indication/Condition**

SQUIB IND light shall go on.

**Corrective Action**

If Indication/Condition is not as specified, go to SQUIB IND LIGHT DOES NOT GO ON WHEN SQUIB SWITCH IS PLACED TO TEST, in this work package.

**Step**

7. Release rescue hoist control panel SQUIB switch. Make sure SQUIB switch returns to NORM and SQUIB IND light goes out.
8. Remove insulated jumper wires from cable cutter connector pins A and B , and from pins C and D.
9. Connect multimeter to cable cutter connector pins A and B.
10. Place rescue hoist control panel CABLE SHEAR switch to FIRE.

---

**CABLE SHEAR OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Indication/Condition**

Multimeter shall indicate 28 vdc.

**Corrective Action**

If Indication/Condition is not as specified, go to NO VOLTAGE TO CABLE CUTTER WHEN CABLE SHEAR SWITCH IN FIRE, in this work package.

**Step**

11. Connect multimeter to cable cutter connector pins C and D.
12. Place rescue hoist control panel CABLE SHEAR switch to FIRE.

**Indication/Condition**

Multimeter shall indicate 28 vdc.

**Corrective Action**

If Indication/Condition is not as specified, replace cable cutter cable (P6 to cable cutter connector) (TM 1-1680-320-13&P).

**Step**

13. Release rescue hoist control panel CABLE SHEAR switch.

**Indication/Condition**

Multimeter shall indicate 0 vdc.

**Corrective Action**

If Indication/Condition is not as specified, replace RESCUE HOIST control panel (WP 1337 00).

**Step**

14. Turn off electrical power.
15. Disconnect multimeter from cable cutter connector.
16. Remove shipping cap and aluminum foil from cable cutter's primer charge and connect hoist cable cutter connector to cable cutter.

**Indication/Condition**

Hoist cable cutter connector shall be connected to the cutter.

**Corrective Action**

None Required

**RESCUE HOIST OPERATIONAL/TROUBLESHOOTING PROCEDURE****PROCEDURE****Step****NOTE**

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Make sure RESQ HST CONTROL circuit breaker on mission readiness circuit breaker panel is pushed in.
2. Turn on electrical power.

**NOTE**

If either #1 CONV or #2 CONV lights on the caution/advisory panel are on, troubleshoot dc electrical system (WP 0102 00).

3. On rescue hoist control unit, make sure AIRCRAFT POSITION switch is in position 2 - 4, and SPEED MODE switch is in LOW. On rescue hoist control panel, place MASTER switch to ON.

**Indication/Condition****NOTE**

Listen or feel to check for fan motor operation.

- a. Fan motor in hoist shall be operating.
- b. Blue POWER ON indicator light on rescue hoist control unit shall be on.

**Corrective Action**

1. If Indication/Conditions are not as specified, go to FAN DOES NOT OPERATE AND POWER ON LIGHT DOES NOT GO ON, in this work package.
2. If Indication/Condition a. is not as specified, go to FAN DOES NOT OPERATE, BUT BLUE POWER ON LIGHT GOES ON, in this work package.
3. If Indication/Condition b. is not as specified, check for burnt-out POWER ON lamp.
  - a. If lamp is good, replace rescue hoist control unit (TM 1-1680-320-13&P).
  - b. If lamp is not good, replace lamp (WP 1352 00).

**Step**

4. Place rescue hoist control panel BOOM switch to OUT and hold.

**Indication/Condition**

Boom shall swing out.

**Corrective Action**

If Indication/Condition is not as specified, go to RESCUE HOIST BOOM DOES NOT SWING OUT, in this work package.

---

**RESCUE HOIST OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Step**

5. Place rescue hoist control panel BOOM switch to OFF.
6. Place rescue hoist control panel BOOM switch to IN and hold.

**Indication/Condition**

BOOM shall swing back into cabin.

**Corrective Action**

If Indication/Condition is not as specified, go to RESCUE HOIST BOOM DOES NOT SWING IN, in this work package.

**Step**

7. Place rescue hoist control panel BOOM switch to OFF.
8. Place crewman's pendant control BOOM switch to OUT.

**Indication/Condition**

Boom shall swing out.

**Corrective Action**

If Indication/Condition is not as specified, replace boom (TM 1-1680-320-13&P).

**Step**

9. Place crewman's pendant control BOOM switch to IN.

**Indication/Condition**

Boom shall swing back into cabin.

**Corrective Action**

If Indication/Condition is not as specified, replace boom (TM 1-1680-320-13&P).

**Step**

10. Place crewman's pendant control BOOM switch to OUT.
11. Place rescue hoist control unit SPEED MODE switch to HIGH.

**CAUTION**

To prevent damage to hook or cable, assistant should walk cable and hook away from helicopter when reeling cable.

12. Place rescue hoist control panel CABLE switch to DOWN and hold.

**Indication/Condition**

- a. First 8 to 12 feet of cable shall reel out at about 67 fpm. CAUTION light on crewman's control pendant shall be on.
- b. The next 230 feet of cable shall reel out at an increased rate of speed (about 250 fpm). CAUTION light shall be off.

---

**RESCUE HOIST OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued

- c. The last 8 to 12 feet of cable shall reel out at a decreased rate of speed (about 67 fpm). CAUTION light shall be on.
- d. Hoist down limit switch shall stop hoist when 250 feet of cable is out. There shall be no less than 5 wraps of cable remaining on drum.

**Corrective Action**

- 1. If cable does not reel out, go to RESCUE HOIST CABLE DOES NOT REEL OUT, in this work package.
- 2. If cable reels out, but does not operate correctly, replace winch and winch motor (TM 1-1680-320-13&P).

**Step**

- 13. Release rescue hoist control panel CABLE switch to OFF.
- 14. Place rescue hoist control panel CABLE switch to UP and hold.

**Indication/Condition**

- a. First 8 to 12 feet of cable shall reel in at about 67 fpm. CAUTION light on crewman's control pendant shall be on.
- b. The next 230 feet of cable shall reel in at an increased rate of speed (about 250 fpm). CAUTION light shall be on.
- c. With about 8 to 12 feet of cable left, cable speed shall decrease to about 67 fpm. CAUTION light shall be on.
- d. The last 12 to 18 inches of cable shall reel in at a much slower speed (about 12 fpm). CAUTION light shall remain on.
- e. Hoist up limit switch shall stop hoist when striker disk on top of hook hits up limit switch actuator. CAUTION light shall go out when cable stops.

**Corrective Action**

- 1. If cable does not reel in, go to RESCUE HOIST CABLE DOES NOT REEL IN, in this work package.
- 2. If cable reels in, but does not operate correctly, replace winch and winch motor (TM 1-1680-320-13&P).

**Step**

- 15. Release rescue hoist control panel CABLE switch to OFF.
- 16. Repeat steps 12 thru 14 using HOIST DOWN UP switch on crewman's control pendant instead of CABLE switch on rescue hoist control panel. Also check that cable speed can be varied when CAUTION light on pendant is out (beyond the 8 to 12 foot limits).

**Indication/Condition**

Results shall be the same as operating the hoist using rescue hoist control. Cable speed shall vary from 0 to 250 fpm.

**Corrective Action**

If Indication/Condition is not as specified, replace rescue hoist control unit (TM 1-1680-320-13&P).

**Step**

- 17. Place rescue hoist control unit SPEED MODE switch to LOW.



---

**RESCUE HOIST OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**

18. Repeat steps 12 thru 15.

**Indication/Condition**

Rescue hoist shall operate same as in steps 12 thru 15 except maximum cable speed will be 125 fpm.

**Corrective Action**

If Indication/Condition is not as specified, replace rescue hoist control unit (TM 1-1680-320-13&P).

**Step**

19. Place rescue hoist control panel BOOM switch to IN.  
20. Place rescue hoist control panel MASTER switch to OFF.  
21. Turn off electrical power.

**Indication/Condition**

Power shall be off.

**Corrective Action**

None Required

**SQUIB IND LIGHT DOES NOT GO ON WHEN PRESSED****SYMPTOM**

SQUIB IND light does not go on when pressed.

**MALFUNCTION**

SQUIB IND light does not go on when pressed.

**CORRECTIVE ACTION****1. Check for burnt-out lamp.**

- a. If lamp is good, go to **2**.  
b. If lamp is not good, replace lamp (WP 1338 00). Go to **5**.

**2. With SQUIB IND light pressed, check continuity between J228-J and J228-V.**

- a. If continuity is present, go to **3**.  
b. If continuity is not present, replace rescue hoist control panel (WP 1337 00). Go to **5**.

**3. With upper console CONSOLE LT LOWER switch set to BRT, check for 28 vdc between P228-V and ground.**

- a. If voltage is as specified, repair/replace wiring between P228-J and ground (WP 1747 00). Go to **5**.  
b. If voltage is not as specified, go to **4**.

**4. Check for 28 vdc between J902-h and ground.**

- a. If voltage is as specified, repair/replace wiring between P902-h and P228-V (WP 1747 00). Go to **5**.  
b. If voltage is not as specified, troubleshoot dimming control system (WP 0110 00). Go to **5**.

---

**SQUIB IND LIGHT DOES NOT GO ON WHEN PRESSED** - Continued**5. Procedure completed.****SQUIB IND LIGHT DOES NOT GO ON WHEN SQUIB SWITCH IS PLACED TO TEST****SYMPTOM**

SQUIB IND light does not go on when SQUIB switch is placed to test.

**MALFUNCTION**

SQUIB IND light does not go on when SQUIB switch is placed to test.

**CORRECTIVE ACTION**

- 1. Disconnect rescue hoist plug from rescue hoist. Go to 2.**
- 2. Connect insulated jumper wire between pins H and B on plug. Go to 3.**
- 3. With SQUIB switch at TEST, check to see if SQUIB IND light goes on.**
  - a. If light is on, replace rescue hoist control unit (TM 1-1680-320-13&P). Go to **7**.
  - b. If light is not on, go to **4**.
- 4. Remove jumper wire. Go to 5.**
- 5. Check for 28 vdc between P228-J and P228-L.**
  - a. If voltage is as specified, repair/replace wiring between P228-T and P1-H on rescue hoist control unit power cable (WP 1335 00). Go to **7**.
  - b. If voltage is not as specified, go to **6**.
- 6. Check for 28 vdc between HOIST CABLE SHEAR circuit breaker terminal 1 and ground.**
  - a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P228-L (WP 1747 00). Go to **7**.
  - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **7**.
- 7. Procedure completed.**

**NO VOLTAGE TO CABLE CUTTER WHEN CABLE SHEAR SWITCH IN FIRE****SYMPTOM**

No voltage to cable cutter when CABLE SHEAR switch in FIRE.

**MALFUNCTION**

No voltage to cable cutter when CABLE SHEAR switch in FIRE.

**CORRECTIVE ACTION**

- 1. Check for 28 vdc between P6-C and P6-D.**
  - a. If voltage is as specified, replace cable cutter cable (P6 to cable cutter connector) (TM 1-1680-320-13&P). Go to **3**.
  - b. If voltage is not as specified, go to **2**.
- 2. Check for 28 vdc between rescue hoist control unit power cable pins H and B.**
  - a. If voltage is as specified, replace rescue hoist control unit (TM 1-1680-320-13&P). Go to **3**.

---

**NO VOLTAGE TO CABLE CUTTER WHEN CABLE SHEAR SWITCH IN FIRE - Continued**

- b. If voltage is not as specified, replace rescue hoist control panel (WP 1337 00). Go to **3**.

**3. Procedure completed.****FAN DOES NOT OPERATE AND POWER ON LIGHT DOES NOT GO ON****SYMPTOM**

Fan does not operate and POWER ON light does not go on.

**MALFUNCTION**

Fan does not operate and POWER ON light does not go on.

**CORRECTIVE ACTION**

- 1. Check for 28 vdc between P228-D and P228-J.**
  - a. If voltage is as specified, go to **2**.
  - b. If voltage is not as specified, go to **4**.
- 2. Place MASTER switch to ON and check continuity between J228-D and J228-S.**
  - a. If continuity is present, go to **3**.
  - b. If continuity is not present, replace rescue hoist control panel (WP 1337 00). Go to **5**.
- 3. Check continuity between P228-S and P1-G.**
  - a. If continuity is present, replace rescue hoist control unit (TM 1-1680-320-13&P). Go to **5**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **5**.
- 4. Check for 28 vdc between RESQ HST CONTROL circuit breaker terminal 1 and ground.**
  - a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P228-D (WP 1747 00). Go to **5**.
  - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **5**.
- 5. Procedure completed.**

**FAN DOES NOT OPERATE, BUT BLUE POWER ON LIGHT GOES ON****SYMPTOM**

Fan does not operate, but blue POWER ON light goes on.

**MALFUNCTION**

Fan does not operate, but blue POWER ON light goes on.

**CORRECTIVE ACTION**

- 1. With MASTER switch ON, check for 28 vdc between P1-B and P1-A on rescue hoist control unit power cable.**
  - a. If voltage is as specified, replace rescue hoist control unit (TM 1-1680-320-13&P). Go to **9**.
  - b. If voltage is not as specified, go to **2**.
- 2. Check for 28 vdc between K14-A2 and ground.**

---

**FAN DOES NOT OPERATE, BUT BLUE POWER ON LIGHT GOES ON - Continued**

- a. If voltage is as specified, repair/replace wire between relay K14-A2 and P1-A on rescue hoist control unit power cable (WP 1747 00). Go to **9**.
- b. If voltage is not as specified, go to **3**.
- 3. Check for 28 vdc between relay K14-A1 and ground.**
  - a. If voltage is as specified, go to **4**.
  - b. If voltage is not as specified, repair/replace wire between relay K14-A1 and NO. 2 DC PRI BUS (WP 1747 00). Go to **9**.
- 4. Check for 28 vdc between K14-X1 and K14-X2.**
  - a. If voltage is as specified, replace relay K14 (WP 1333 00). Go to **9**.
  - b. If voltage is not as specified, go to **5**.
- 5. Check for 28 vdc between P228-D and P228-J.**
  - a. If voltage is as specified, replace rescue hoist control panel (WP 1337 00). Go to **9**.
  - b. If voltage is not as specified, go to **6**.
- 6. Check for 28 vdc between J229-B and ground.**
  - a. If voltage is as specified, go to **7**.
  - b. If voltage is not as specified, repair/replace RESQ HST CONTROL circuit breaker CB12A (WP 0824 00). Go to **9**.
- 7. Check for 28 vdc between P240-C and ground.**
  - a. If voltage is as specified, replace relay K14 (WP 1333 00). Go to **9**.
  - b. If voltage is not as specified, go to **8**.
- 8. Check continuity between P228-G and P240-C.**
  - a. If continuity is present, repair/replace wiring between P240-K and relay K14-X1 (WP 1747 00). Go to **9**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **9**.
- 9. Procedure completed.**

**RESCUE HOIST BOOM DOES NOT SWING OUT****SYMPTOM**

Rescue hoist boom does not swing out.

**MALFUNCTION**

Rescue hoist boom does not swing out.

**CORRECTIVE ACTION**

- 1. With MASTER switch ON, and BOOM switch at OUT, check for 28 vdc between P1-D and P1-B on rescue hoist control unit power cable.**
  - a. If voltage is as specified, go to **2**.
  - b. If voltage is not as specified, go to **3**.

---

**RESCUE HOIST BOOM DOES NOT SWING OUT** - Continued**2. Check that boom up limit, actuator limit, and torque limit switches are properly adjusted.**

- a. If limits and switches are properly adjusted, replace boom motor (TM 1-1680-320-13&P). Go to **4.**
- b. If limits and switches are not properly adjusted, adjust switches (TM 1-1680-320-13&P). Go to **4.**
- c. If trouble remains, replace rescue hoist control unit (TM 1-1680-320-13&P). Go to **4.**

**3. With MASTER switch ON, and BOOM switch at OUT, check continuity between J228-S and J228-R.**

- a. If continuity is present, repair/replace wire between P228-R and P1-D on rescue hoist control unit power cable (WP 1747 00). Go to **4.**
- b. If continuity is not present, replace rescue hoist control panel (WP 1337 00). Go to **4.**

**4. Procedure completed.****RESCUE HOIST BOOM DOES NOT SWING IN****SYMPTOM**

Rescue hoist boom does not swing in.

**MALFUNCTION**

Rescue hoist boom does not swing in.

**CORRECTIVE ACTION****1. With MASTER switch ON and BOOM switch at IN, check for 28 vdc between P1-F and P1-B on rescue hoist control unit power cable.**

- a. If voltage is as specified, go to **2.**
- b. If voltage is not as specified, go to **3.**

**2. Check that boom up limit, actuator limit, and torque limit switches are properly adjusted.**

- a. If limits and switches are properly adjusted, replace boom motor (TM 1-1680-320-13&P). Go to **4.**
- b. If limits and switches are not properly adjusted, adjust switches (TM 1-1680-320-13&P). Go to **4.**
- c. If trouble remains, replace rescue hoist control unit (TM 1-1680-320-13&P). Go to **4.**

**3. With MASTER switch ON and BOOM switch at IN, check continuity between J228-S and J228-P.**

- a. If continuity is present, repair/replace wire between P228-P and P1-F on rescue hoist control unit power cable (WP 1747 00). Go to **4.**
- b. If continuity is not present, replace rescue hoist control panel (WP 1337 00). Go to **4.**

**4. Procedure completed.**

**RESCUE HOIST CABLE DOES NOT REEL OUT****SYMPTOM**

Rescue hoist cable does not reel out.

**MALFUNCTION**

Rescue hoist cable does not reel out.

**CORRECTIVE ACTION**

- 1. With MASTER switch ON and CABLE switch at DOWN, check for 28 vdc between P1-C and P1-B on rescue hoist control unit power cable.**
  - a. If voltage is as specified, go to **2**.
  - b. If voltage is not as specified, go to **3**.
- 2. Check that cams in limit switch drive assembly are properly adjusted.**
  - a. If cams in limit switch drive assembly are properly adjusted, replace winch motor (TM 1-1680-320-13&P). Go to **4**.
  - b. If cams in limit switch drive assembly are not properly adjusted, adjust cams (TM 1-1680-320-13&P). Go to **4**.
  - c. If trouble remains, replace limit switch drive assembly (TM 1-1680-320-13&P). Go to **4**.
  - d. If trouble still remains, replace rescue hoist control unit (TM 1-1680-320-13&P). Go to **4**.
- 3. With MASTER switch ON and CABLE switch at DOWN, check continuity between J228-N and J228-S.**
  - a. If continuity is present, repair/replace wiring between P228-N and P1-C on rescue hoist control unit power cable (WP 1747 00). Go to **4**.
  - b. If continuity is not present, replace rescue hoist control panel (WP 1337 00). Go to **4**.
- 4. Procedure completed.**

**RESCUE HOIST CABLE DOES NOT REEL IN****SYMPTOM**

Rescue hoist cable does not reel in.

**MALFUNCTION**

Rescue hoist cable does not reel in.

**CORRECTIVE ACTION**

- 1. With MASTER switch ON and CABLE switch at UP, check for 28 vdc between P1-E and P1-B on rescue hoist control unit power cable.**
  - a. If voltage is as specified, go to **2**.
  - b. If voltage is not as specified, go to **3**.
- 2. Check that cams in limit switch drive assembly are properly adjusted.**
  - a. If cams in limit switch drive assembly are properly adjusted, replace winch motor (TM 1-1680-320-13&P). Go to **4**.

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**RESCUE HOIST CABLE DOES NOT REEL IN** - Continued

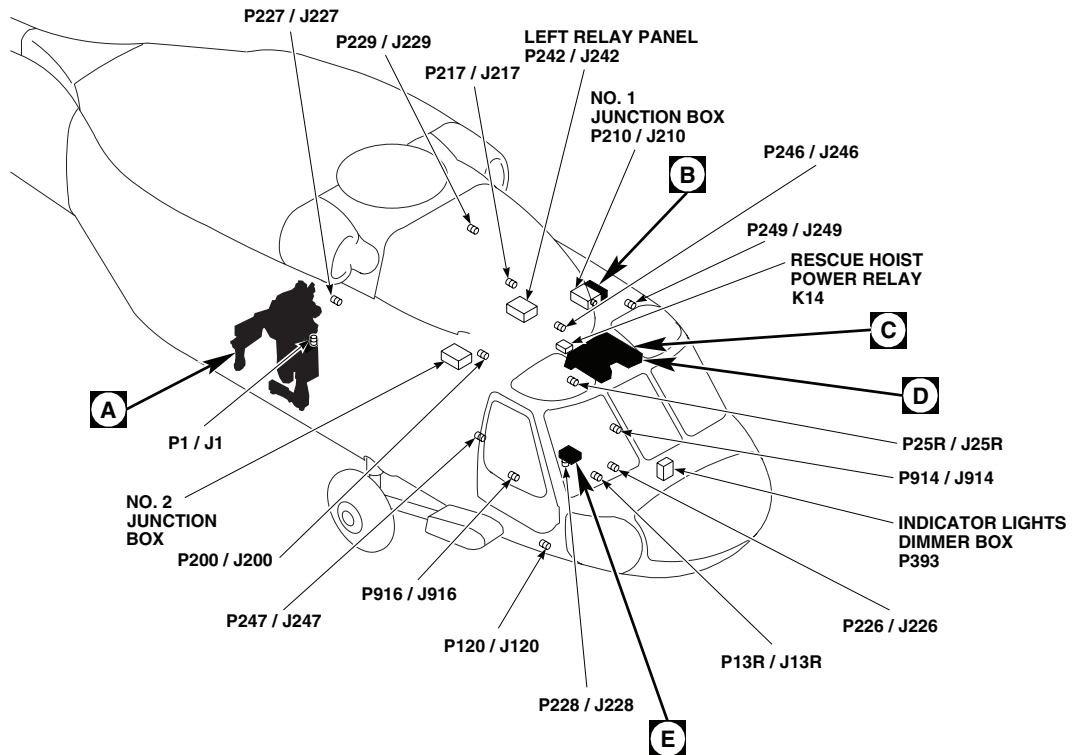
- b. If cams in limit switch drive assembly are not properly adjusted, adjust cams (TM 1-1680-320-13&P). Go to **4.**
- c. If trouble remains, replace limit switch drive assembly (TM 1-1680-320-13&P). Go to **4.**
- d. If trouble still remains, replace rescue hoist control unit (TM 1-1680-320-13&P). Go to **4.**

**3. With MASTER switch ON and CABLE switch at UP, check continuity between J228-M and J228-S.**

- a. If continuity is present, repair/replace wiring between P228-M and P1-E on rescue hoist control unit power cable (WP 1747 00). Go to **4.**
- b. If continuity is not present, replace rescue hoist control panel (WP 1337 00). Go to **4.**

**4. Procedure completed.**

## LOCATION



| TERMINAL BOARD/<br>DISCONNECT PLUG/<br>RECEPTACLE | LOCATION/<br>CONNECTION POINT      |
|---------------------------------------------------|------------------------------------|
| P1 / J1                                           | RESCUE HOIST CONTROL UNIT          |
| P13R / J13R                                       | CABIN, BL 20 RH, STA 247           |
| P25R / J25R                                       | CABIN, BL 40 RH, STA 280           |
| P120 / J120                                       | COCKPIT TUB, BL 30 RH, STA 220     |
| P126R / J126R                                     | COCKPIT CEILING, BL 28 RH, STA 243 |
| P200 / J200                                       | COCKPIT CEILING, BL 13 RH, STA 246 |
| P210 / J210                                       | NO. 1 MAIN ELECTRICAL JUNCTION BOX |
| P217 / J217                                       | CABIN CEILING, BL 0, STA 284       |
| P219 / J219                                       | CABIN FLOOR, BL 25 RH, STA 247     |
| P226 / J226                                       | LOWER CONSOLE, BL 10 LH, STA 226   |
| P227 / J227                                       | CABIN CEILING, BL 20 RH, STA 340   |

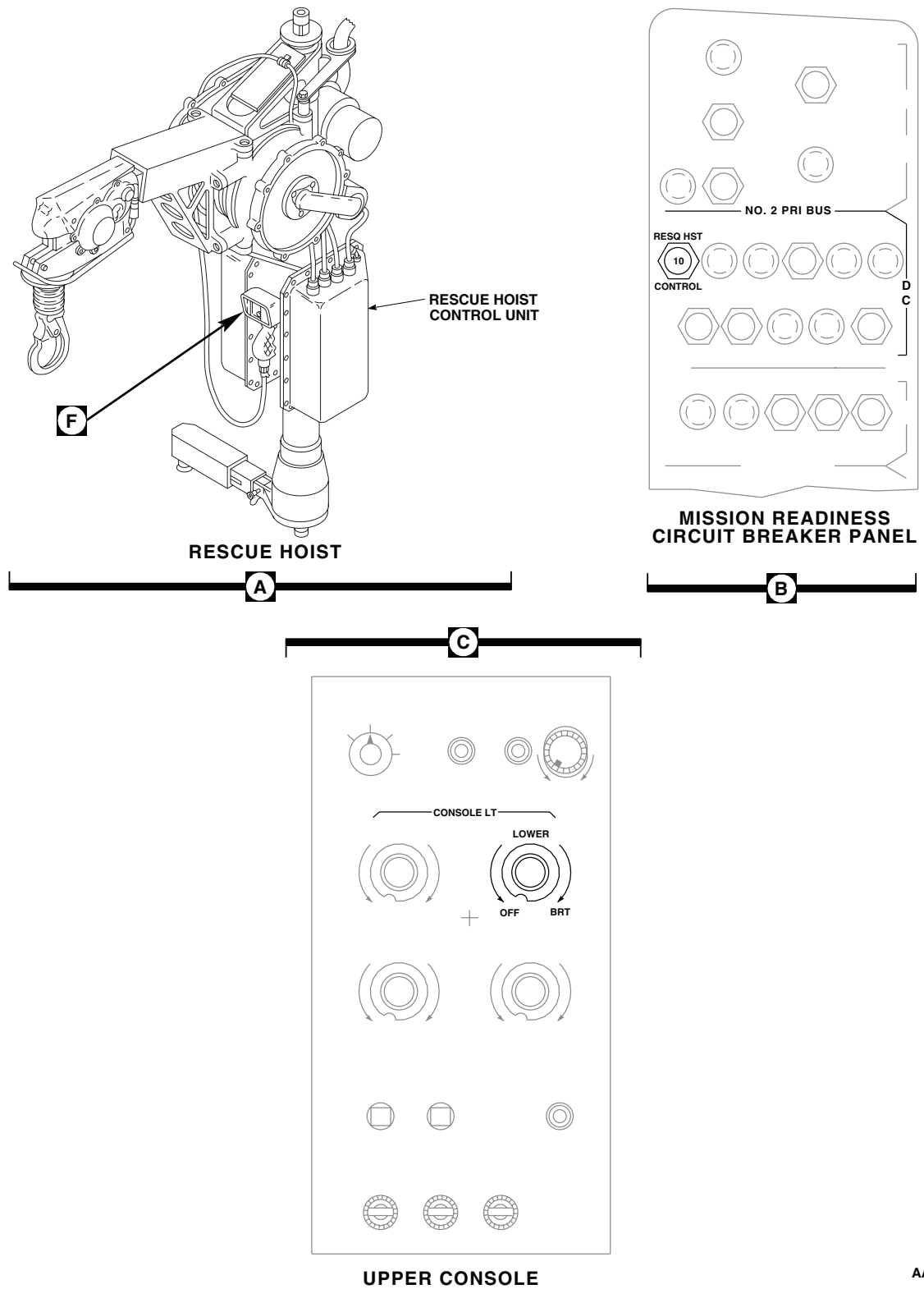
| TERMINAL BOARD/<br>DISCONNECT PLUG/<br>RECEPTACLE | LOCATION/<br>CONNECTION POINT                    |
|---------------------------------------------------|--------------------------------------------------|
| P228 / J228                                       | RESCUE HOIST CONTROL PANEL                       |
| P229 / J229                                       | CABIN CEILING, BL 5 LH, STA 300                  |
| P240                                              | NO. 1 JUNCTION BOX                               |
| P242 / J242                                       | LEFT RELAY PANEL                                 |
| P246 / J246                                       | CABIN CEILING, BL 5 RH, STA 247                  |
| P247 / J247                                       | CABIN, BL 40 RH, STA 248                         |
| P249 / J249                                       | BEHIND COPILOT'S CIRCUIT BREAKER PANEL, BL 23 LH |
| P393                                              | INDICATOR LIGHTS DIMMER BOX                      |
| P902 / J902                                       | LEFT RELAY PANEL                                 |
| P914 / J914                                       | COCKPIT, BL 24 LH, STA 247                       |
| P916 / J916                                       | COCKPIT TUB, BL 25 RH, STA 247                   |

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SA

Figure 1. Rescue Hoist Kit Location Diagram. (Sheet 1 of 3)



LOCATION - Continued



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SA

Figure 1. Rescue Hoist Kit Location Diagram. (Sheet 2 of 3)

LOCATION - Continued

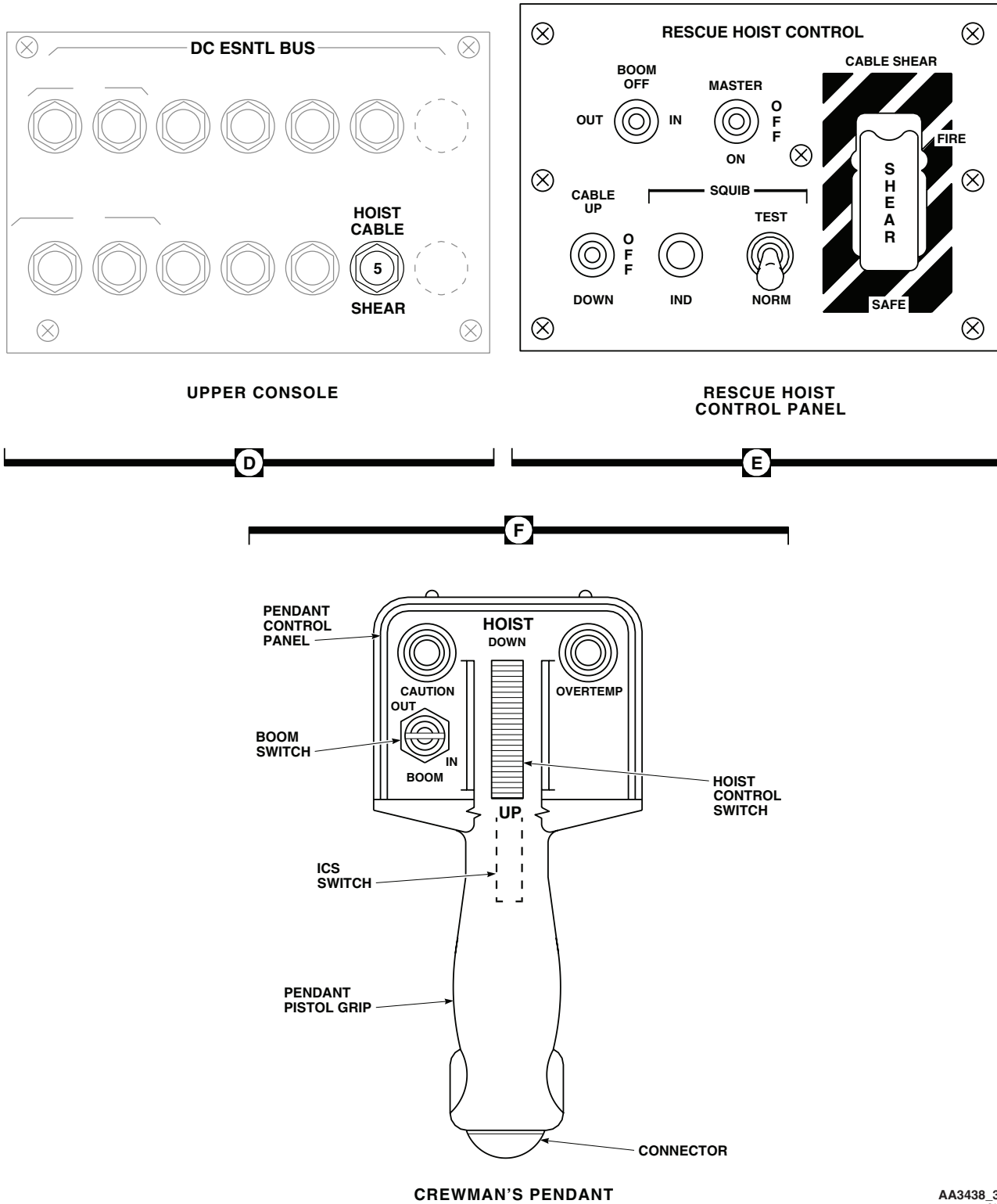


Figure 1. Rescue Hoist Kit Location Diagram. (Sheet 3 of 3)

## SCHEMATICS

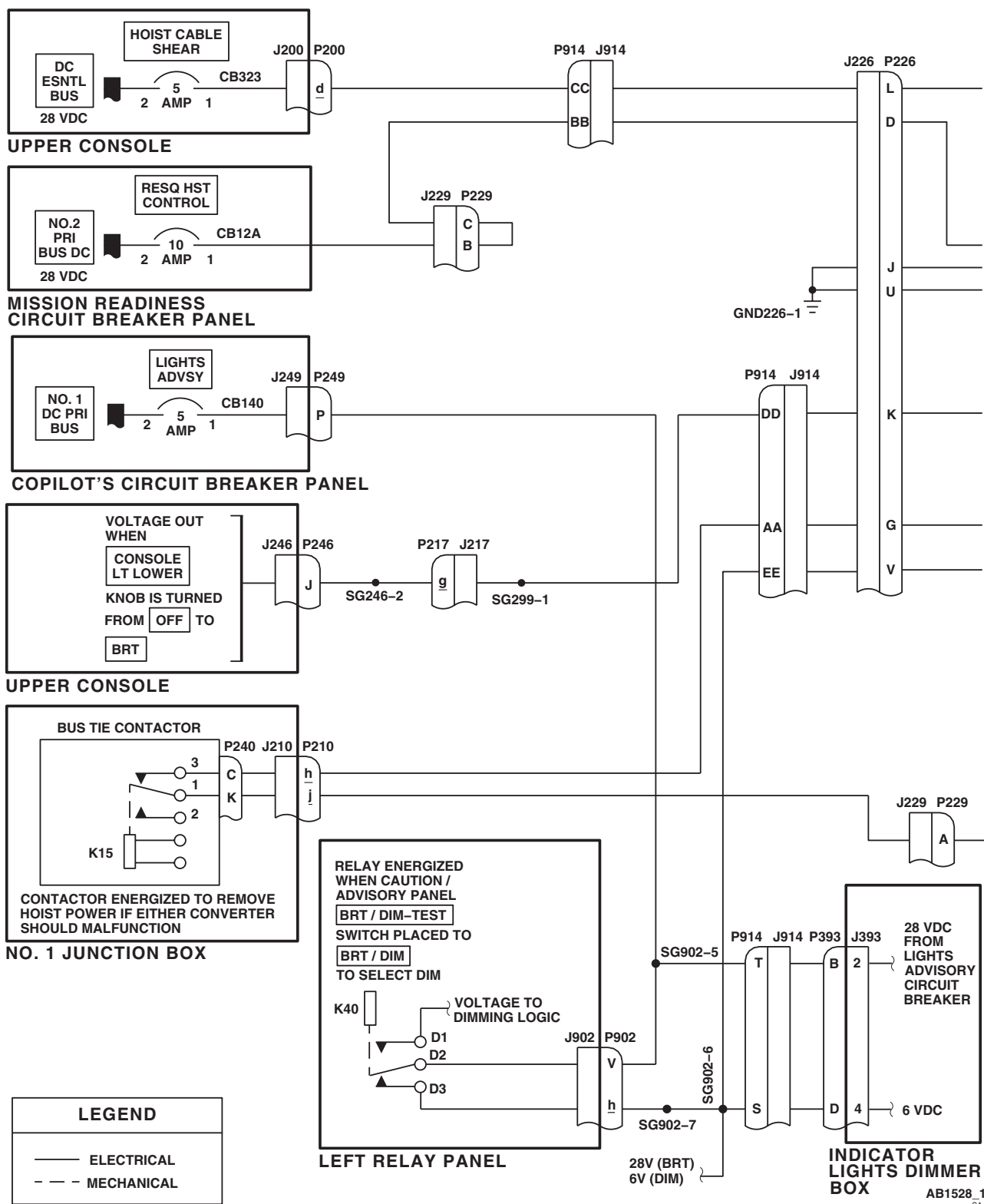


Figure 2. Rescue Hoist Kit Schematic Diagram. (Sheet 1 of 3)

SCHEMATICS - Continued

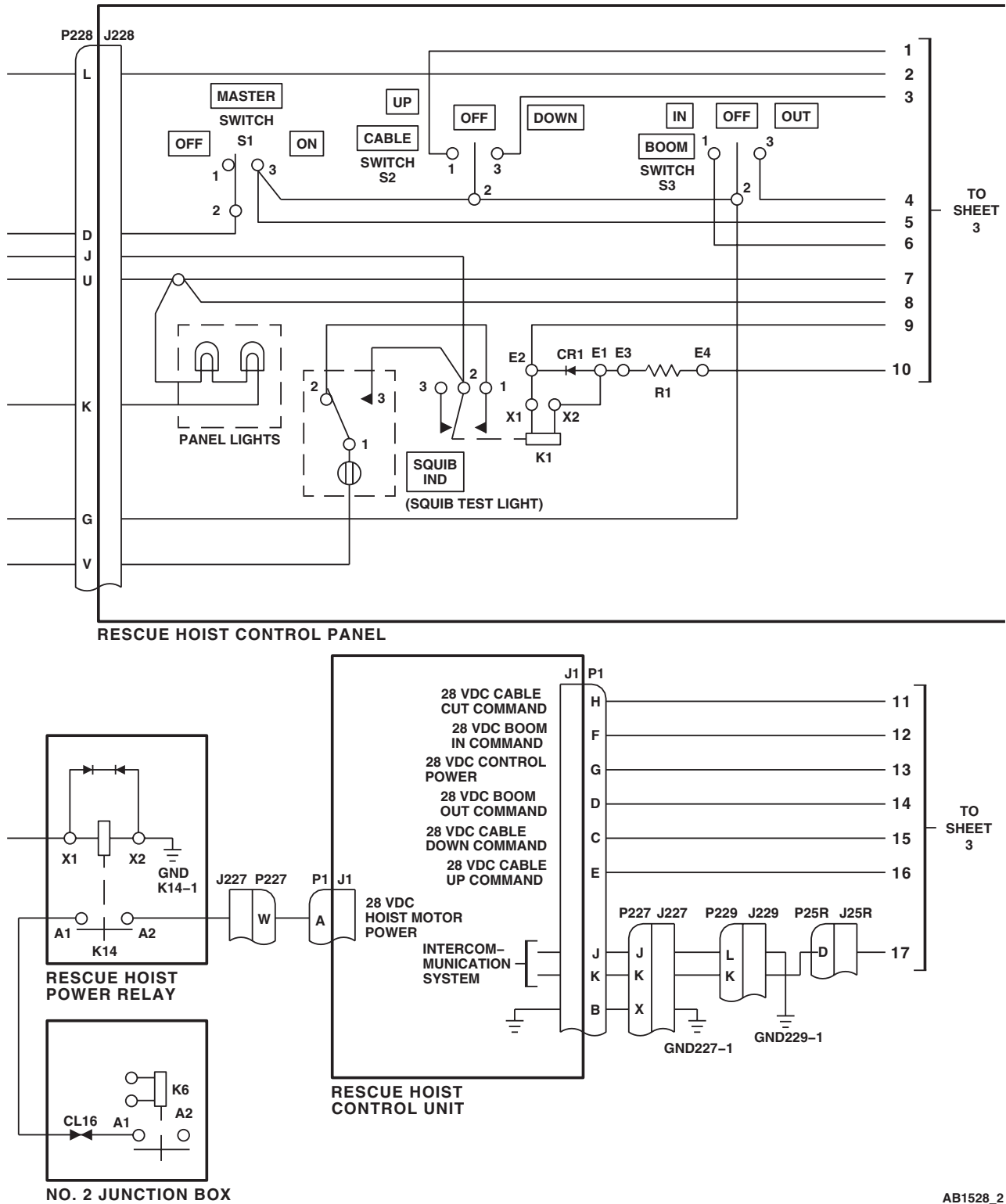


Figure 2. Rescue Hoist Kit Schematic Diagram. (Sheet 2 of 3)

## SCHEMATICS - Continued

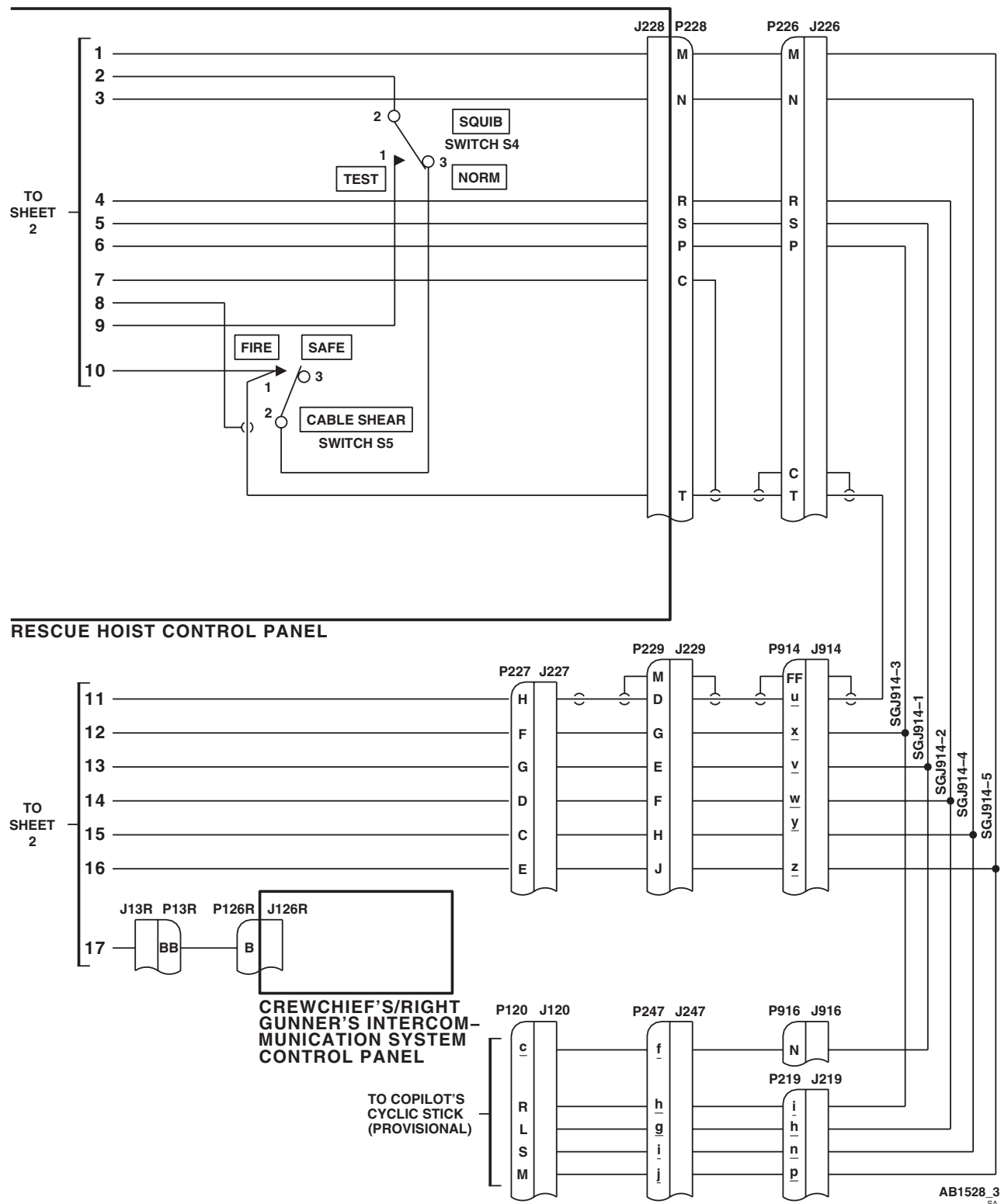


Figure 2. Rescue Hoist Kit Schematic Diagram. (Sheet 3 of 3)

END OF WORK PACKAGE



## UNIT LEVEL

## HOISTS AND WINCHES

RESCUE HOIST SYSTEM **HH-60A HH-60L**

This WP supersedes WP 0153 00, dated 17 April 2006.

**INITIAL SETUP:****Test Equipment**

|                                                |            |
|------------------------------------------------|------------|
| Container (55-gallon)                          | WP 0824 00 |
| Infrared Light Test Set Figure 149, WP 1805 00 | WP 0827 00 |
| Jumper Wires                                   | WP 0933 00 |
| Multimeter, AN/PSM-45A                         | WP 1336 00 |
| 28 vdc Test Lamp                               | WP 1342 00 |

**Tools and Special Tools**

|                                             |            |
|---------------------------------------------|------------|
| Aircraft Mechanic's Toolkit, SC 5180-99-B01 | WP 1346 00 |
| Electrical Repairer Toolkit, SC 5180-99-B06 | WP 1348 00 |
| Gloves                                      | WP 1349 00 |
|                                             | WP 1350 00 |
|                                             | WP 1685 00 |
|                                             | WP 1747 00 |
|                                             | WP 1805 00 |

**Personnel Required**

Aircraft Electrician MOS 15F (2)  
UH-60 Helicopter Repairer MOS 15T (1)

**References**

TM 1-1520-237-10  
WP 0104 00

**Equipment Condition**

External Electrical Power Available, WP 1685 00  
or APU Operational, TM 1-1520-237-10

**SPECIAL INSTRUCTIONS****NOTE**

See Figure 1 for component location diagram and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

The rescue hoist kit operational/troubleshooting procedure is subdivided as follows:

- Cable Shear
- Rescue Hoist Assembly
- Dual-Mode Controllable Searchlight

**CABLE SHEAR OPERATIONAL/TROUBLESHOOTING PROCEDURE****PROCEDURE****Step****WARNING**

The squib is an electrically activated explosive device. Improper checkout of the cable shear function could result in serious injury or death to personnel.

**NOTE**

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Disconnect connector P6 from squib on rescue hoist assembly.

**WARNING**

All squibs and squib actuated devices shall be handled as live ammunition. Squibs or squib actuated devices which have been fired may retain an explosive residue capable of presenting a hazardous condition.

2. Carefully install a piece of aluminum foil around squib pins and install shipping cap.
3. Make sure HOIST CABLE SHEAR circuit breaker on upper console circuit breaker panel is pushed in.
4. Turn on external electrical power.
5. On pilot's hoist control panel, place HOIST POWER-ON-OFF switch to ON.
6. Connect 28 vdc test lamp between pins A and B on connector P6.
7. Place crew's hoist control panel ARM-TEST switch to ARM.
8. On crew's hoist control panel, lift CABLE CUT switch guard, press and hold CABLE CUT switch.

**Indication/Condition**

Test lamp shall go on.

**Corrective Action**

If Indication/Condition is not as specified, go to TEST LAMP DOES NOT GO ON WHEN CREW'S HOIST CONTROL PANEL CABLE CUT SWITCH IS PRESSED, in this work package.

**Step**

9. Release crew's hoist control panel CABLE CUT switch and lower guard.

**Indication/Condition**

Test lamp shall go off.

**Corrective Action**

If test lamp does not go off, replace crew's hoist control panel (WP 1349 00).

**Step**

10. On pilot's hoist control panel, lift CABLE CUT switch guard and press and hold CABLE CUT switch.



---

**CABLE SHEAR OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued**Indication/Condition**

Test lamp shall go on.

**Corrective Action**

If test lamp does not go on, go to TEST LAMP DOES NOT GO ON WHEN PILOT'S HOIST CONTROL PANEL CABLE CUT SWITCH IS PRESSED, in this work package.

**Step**

11. Release pilot's hoist control panel CABLE CUT switch and lower guard.

**Indication/Condition**

Test lamp shall go off.

**Corrective Action**

If test lamp does not go off, replace pilot's hoist control panel (WP 1348 00).

**Step**

12. Place crew's hoist control panel ARM-TEST switch to TEST.
13. Turn off external electrical power.
14. Disconnect test lamp from connector P6.
15. Install jumper wires between pins A and B, and between C and D on connector P6.
16. Turn on external electrical power.
17. On pilot's hoist control panel, place HOIST POWER-ON-OFF switch to ON.
18. On crew's hoist control panel, lift CABLE CUT switch guard, press and hold CABLE CUT switch.

**Indication/Condition**

SQUIB TEST indicator shall go on.

**Corrective Action**

If Indication/Condition is not as specified, go to SQUIB TEST INDICATOR DOES NOT GO ON WHEN CREW'S HOIST CONTROL PANEL CABLE CUT SWITCH IS PRESSED, in this work package.

**Step**

19. Release crew's hoist control panel CABLE CUT switch and lower guard.

**Indication/Condition**

SQUIB TEST indicator shall go off.

**Corrective Action**

If SQUIB TEST does not go off, replace crew's hoist control panel (WP 1349 00).

**Step**

20. On pilot's hoist control panel, lift CABLE CUT switch guard and press and hold CABLE CUT switch.

**Indication/Condition**

SQUIB TEST indicator shall go on.

---

**CABLE SHEAR OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Corrective Action**

If SQUIB TEST indicator does not go on, replace crew's hoist control panel (WP 1349 00).

**Step**

21. Release pilot's hoist control panel CABLE CUT switch and lower guard.

**Indication/Condition**

SQUIB TEST indicator shall go off.

**Corrective Action**

If SQUIB TEST indicator does not go off, replace crew's hoist control panel (WP 1349 00).

**Step**

22. On pilot's hoist control panel, place HOIST POWER-ON-OFF switch to OFF.
23. Turn off external electrical power.
24. Disconnect jumper wires from connector P6.

**WARNING**

To prevent injury to personnel, do not connect electrical harness if voltage is found at connector, until problem is repaired.

25. Using multimeter check for 0 vdc at pins A and B of connector P6.
26. Remove shipping cap and aluminum foil from around squib pins and connect P6 to J6 to squib on rescue hoist assembly.

**Indication/Condition**

Connectors shall be connected.

**Corrective Action**

None Required

**RESCUE HOIST ASSEMBLY OPERATIONAL/TROUBLESHOOTING PROCEDURE****PROCEDURE****Step****NOTE**

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Make sure HOIST PWR and HOIST CONTR circuit breakers on copilot's auxiliary circuit breaker panel are pushed in.
2. Pull out HOIST CABLE SHEAR circuit breaker on upper console circuit breaker panel.
3. Connect rescue hoist pendant to crew's hoist control panel.
4. Turn on external electrical power.

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**RESCUE HOIST ASSEMBLY OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****WARNING**

Ensure that rescue hoist cable does not contact aircraft during testing or operational checkout. Failure to prevent any contact with aircraft could result in serious injury or death to personnel.

5. On pilot's hoist control panel, place HOIST POWER-ON-OFF switch to ON.

**Indication/Condition**

Hoist fan motor shall go on.

**Corrective Action**

If fan motor does not go on, go to RESCUE HOIST FAN MOTOR DOES NOT GO ON, in this work package.

**Step****WARNING**

Rescue hoist cable is stiff and abrasive. Broken cable strands are sharp. Work gloves should be worn whenever handling rescue hoist cable.

**CAUTION**

To prevent damage to hook or cable, rescue hoist cable must be kept clear of all parts of the aircraft and free from other external obstacles when operating the rescue hoist. Cable abrasion during rescue hoist operations can lead to cable failure. If cable contact or snagging occurs, suspend rescue hoist operations and inspect the cable for damage in accordance with applicable procedures.

6. Place large container under rescue hoist. Coil rescue hoist cable in container as cable is reeled out.
7. Place rescue hoist pendant UP-OFF-DN thumbwheel to full DN position and observe the following:

**Indication/Condition**

- a. Hoist cable should reel out at approximately 300 feet-per-minute until 8 to 12 feet remains from full-out limit switch.
- b. CABLE-FEET indicator on rescue hoist pendant shall show increasing total.
- c. When 8 to 12 feet of cable remains from full-out limit switch, hoist cable slows to 50 feet per minute.
- d. Hoist stops with 3-1/2 to 4-1/2 wraps of cable remaining on hoist drum.
- e. FULL OUT indicator on rescue hoist pendant is illuminated.

**Corrective Action**

1. If rescue hoist cable does not reel out, go to RESCUE HOIST CABLE DOES NOT REEL OUT WITH RESCUE HOIST PENDANT THUMBWHEEL IN DN POSITION, in this work package.
2. If rescue hoist cable reels out, but does not operate correctly, go to RESCUE HOIST CABLE REELS IN/OUT INCORRECTLY, in this work package.
3. If rescue hoist pendant CABLE-FEET indicator does not show increasing total, go to RESCUE HOIST PENDANT CABLE-FEET INDICATOR DOES NOT SHOW INCREASING/DECREASING TOTAL, in this work package.

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**RESCUE HOIST ASSEMBLY OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued

4. If rescue hoist pendant FULL OUT indicator does not illuminate, replace rescue hoist pendant (WP 1346 00).

**Step**

8. Place rescue hoist pendant UP-OFF-DN thumbwheel to full UP position and observe the following:

**Indication/Condition**

- a. Hoist cable shall reel in at approximately 300 feet-per-minute until 8 to 12 feet remains from full-in limit switch.
- b. CABLE-FEET indicator on rescue hoist pendant shall show decreasing total.
- c. When 8 to 12 feet of cable remains from full-in limit switch, hoist cable slows to 50 feet per minute.
- d. Hoist stops when cable is fully reeled in.
- e. FULL IN indicator on rescue hoist pendant is illuminated.

**Corrective Action**

1. If rescue hoist cable does not reel in, go to RESCUE HOIST CABLE DOES NOT REEL IN WITH RESCUE HOIST PENDANT THUMBWHEEL IN UP POSITION, in this work package.
2. If rescue hoist cable reels in, but does not operate correctly, go to RESCUE HOIST CABLE REELS IN/OUT INCORRECTLY, in this work package.
3. If rescue hoist pendant CABLE-FEET indicator does not show decreasing total, go to RESCUE HOIST PENDANT CABLE-FEET INDICATOR DOES NOT SHOW INCREASING/DECREASING TOTAL, in this work package.
4. If rescue hoist pendant FULL IN indicator does not illuminate, replace rescue hoist pendant (WP 1346 00).

**Step**

9. Release rescue hoist pendant UP-OFF-DN thumbwheel.

**Indication/Condition**

UP-OFF-DN thumbwheel shall return to OFF position.

**Corrective Action**

If thumbwheel does not return to OFF position, replace rescue hoist pendant (WP 1346 00).

**Step**

10. Place pilot's hoist control panel PILOT OVERRIDE UP-DOWN switch to DOWN.

**Indication/Condition**

- a. Hoist cable shall reel out at approximately 200 feet-per-minute until 8 to 12 feet remains from full-out limit switch.
- b. When 8 to 12 feet of cable remains from full-out limit switch, hoist slows to 50 feet per minute.
- c. Hoist stops with 3-1/2 to 4-1/2 wraps of cable remaining on hoist drum.

**Corrective Action**

1. If hoist cable does not reel out, go to HOIST CABLE DOES NOT REEL OUT WITH PILOT'S HOIST CONTROL PANEL PILOT OVERRIDE UP-DOWN SWITCH IN DOWN, in this work package.

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**RESCUE HOIST ASSEMBLY OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued

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2. If hoist cable reels out, but does not operate correctly, replace rescue hoist assembly (WP 1342 00).

**Step**

11. Place pilot's hoist control panel PILOT OVERRIDE UP-DOWN switch to UP.

**Indication/Condition**

- a. Hoist cable shall reel in at approximately 200 feet-per-minute until 8 to 12 feet remains from full-in limit switch.
- b. When 8 to 12 feet of cable remains from full-in limit switch, hoist slows to 50 feet per minute.
- c. Hoist stops when cable is reeled in.

**Corrective Action**

1. If hoist cable does not reel in, go to RESCUE HOIST CABLE DOES NOT REEL IN WITH PILOT'S HOIST CONTROL PANEL PILOT OVERRIDE UP-DOWN SWITCH IN UP, in this work package.
2. If hoist cable reels in, but does not operate correctly, replace rescue hoist assembly (WP 1342 00).

**Step**

12. Place rescue hoist pendant UP-OFF-DN thumbwheel to DN for 5 seconds to reel off approximately 25 feet of cable.
13. Set and hold rescue hoist pendant UP-OFF-DN thumbwheel to DN and place pilot's hoist control panel PILOT OVERRIDE UP-DOWN switch to UP.

**Indication/Condition**

Hoist cable shall reel in.

**Corrective Action**

If hoist cable does not reel in, replace rescue hoist assembly (WP 1342 00).

**Step**

14. Set and hold rescue hoist pendant UP-OFF-DN thumbwheel to UP and place pilot's hoist control panel PILOT OVERRIDE UP-DOWN switch to DOWN.

**Indication/Condition**

Hoist cable shall reel out.

**Corrective Action**

If hoist cable does not reel out, replace rescue hoist assembly (WP 1342 00).

**Step**

15. Release rescue hoist pendant UP-OFF-DN thumbwheel and pilot's hoist control panel PILOT OVERRIDE UP-DOWN switch.
16. Reel in hoist cable by setting rescue hoist pendant UP-OFF-DN thumbwheel to UP.
17. On pilot's hoist control panel, place HOIST POWER ON/OFF switch to OFF.
18. Turn off electrical power.
19. Push in HOIST CABLE SHEAR circuit breaker on upper console circuit breaker panel.

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**RESCUE HOIST ASSEMBLY OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Indication/Condition**

Circuit breaker shall be pushed in.

**Corrective Action**

None Required

**DUAL-MODE CONTROLLABLE SEARCHLIGHT OPERATIONAL/TROUBLESHOOTING PROCEDURE****Step****NOTE**

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Make certain HOIST LGHT PWR and HOIST LGHT CONTR circuit breakers on copilot's auxiliary circuit breaker panel are pushed in.
2. Pull out HOIST CABLE SHEAR circuit breaker on upper console circuit breaker panel.
3. Turn on electrical power.

**NOTE**

If light is already fully extended, no movement will be seen.

4. Set and hold rescue hoist pendant SCHLT switch to FWD position for 5 seconds.

**Indication/Condition**

Dual-mode controllable searchlight shall extend from stored position.

**Corrective Action**

If dual-mode controllable searchlight does not extend, go to DUAL-MODE CONTROLLABLE SEARCHLIGHT DOES NOT EXTEND, in this work package.

**Step**

5. Set and hold rescue hoist pendant SCHLT switch to RIGHT for 5 seconds.

**Indication/Condition**

Dual-mode controllable searchlight shall turn to the right.

**Corrective Action**

If dual-mode controllable searchlight does not turn right, go to DUAL-MODE CONTROLLABLE SEARCHLIGHT DOES NOT TURN RIGHT, in this work package.

**Step**

6. Set and hold rescue hoist pendant SCHLT switch to LEFT for 5 seconds.

**Indication/Condition**

Dual-mode controllable searchlight shall turn to the left.

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**DUAL-MODE CONTROLLABLE SEARCHLIGHT OPERATIONAL/TROUBLESHOOTING PROCEDURE -**  
Continued**Corrective Action**

If dual-mode controllable searchlight does not turn left, go to DUAL-MODE CONTROLLABLE SEARCHLIGHT DOES NOT TURN LEFT, in this work package.

**Step**

7. Set and hold rescue hoist pendant SCHLT switch to AFT position.

**Indication/Condition**

Dual-mode controllable searchlight shall retract to the stored position.

**Corrective Action**

If dual-mode controllable searchlight does not retract, go to DUAL-MODE CONTROLLABLE SEARCHLIGHT DOES NOT RETRACT, in this work package.

**Step****WARNING**

Looking directly into searchlight when it is turned on could cause damage to eyes.

8. Set crew's hoist control panel SEARCH LIGHT ON/NORMAL-OFF-ON/NVG switch to ON/NORMAL.

**Indication/Condition**

Dual-mode controllable searchlight normal lamps shall go on.

**Corrective Action**

If dual-mode controllable searchlight does not go on, go to DUAL-MODE CONTROLLABLE SEARCHLIGHT DOES NOT GO ON, in this work package.

**Step**

9. Set crew's hoist control panel SEARCH LIGHT ON/NORMAL-OFF-ON/NVG switch to OFF.

**Indication/Condition**

Dual-mode controllable searchlight shall go off.

**Corrective Action**

If dual-mode controllable searchlight does not go off, replace crew's hoist control panel (WP 1349 00).

**Step**

10. Place IR light test set (test set) (WP 1805 00) POWER switch to ON.

**Indication/Condition**

Test set POWER ON lamp shall go on.

**Corrective Action**

If test set POWER ON lamp does not go on, replace lamp, battery, or test set.

**Step**

11. Place test set IR detector lead over POWER ON lamp.

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**DUAL-MODE CONTROLLABLE SEARCHLIGHT OPERATIONAL/TROUBLESHOOTING PROCEDURE -**  
Continued**Indication/Condition**

Test set IR LIGHT ON lamp shall go on.

**Corrective Action**

If IR/DET LEDs on test set do not go on, replace test set.

**Step****WARNING**

Looking directly into searchlight when it is turned on could cause damage to eyes.

12. Place crew's hoist control panel SEARCH LIGHT ON/NORMAL-OFF-ON/NVG switch to ON/NVG.

13. Place test set IR detector lead over dual-mode controllable searchlight IR lamp.

**Indication/Condition**

Test set IR LIGHT ON lamp shall go on steady.

**Corrective Action**

If test set IR LIGHT does not go on, go to TEST SET IR LIGHT DOES NOT GO ON, in this work package.

**Step**

14. Set SEARCH LIGHT ON/NORMAL-OFF-ON/NVG switch to OFF.

15. Turn off electrical power.

16. Push in HOIST CABLE SHEAR circuit breaker on upper console circuit breaker panel.

**Indication/Condition**

Circuit breaker shall be pushed in.

**Corrective Action**

None Required

**TEST LAMP DOES NOT GO ON WHEN CREW'S HOIST CONTROL PANEL CABLE CUT SWITCH IS PRESSED****SYMPTOM**

Test lamp does not go on when crew's hoist control panel CABLE CUT switch is pressed.

**MALFUNCTION**

Test lamp does not go on when crew's hoist control panel CABLE CUT switch is pressed.

**CORRECTIVE ACTION****1. Check for 28 vdc between P226-U and ground.**

- a. If voltage is as specified, go to **2**.
- b. If voltage is not as specified, go to **3**.

**2. Check continuity between:**

- P23-A and P6-A



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**TEST LAMP DOES NOT GO ON WHEN CREW'S HOIST CONTROL PANEL CABLE CUT SWITCH IS PRESSED** - Continued

- **P23-B and P6-B**

- a. If continuity is present, replace crew's hoist control panel (WP 1349 00). Go to **4**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **4**.

- 3. Check continuity between HOIST CABLE SHEAR circuit breaker terminal 1 and P226-U.**

- a. If continuity is present, replace HOIST CABLE SHEAR circuit breaker (WP 0824 00). Go to **4**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **4**.

- 4. Procedure completed.**

**TEST LAMP DOES NOT GO ON WHEN PILOT'S HOIST CONTROL PANEL CABLE CUT SWITCH IS PRESSED****SYMPTOM**

Test lamp does not go on when pilot's hoist control panel CABLE CUT switch is pressed.

**MALFUNCTION**

Test lamp does not go on when pilot's hoist control panel CABLE CUT switch is pressed.

**CORRECTIVE ACTION**

- 1. Check continuity between:**

- **P218-N and P226-S**
- **P218-M and P226-q**

- a. If continuity is present, replace pilot's hoist control panel (WP 1348 00). Go to **2**.
- b. If trouble remains, replace crew's hoist control panel (WP 1349 00). Go to **2**.
- c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **2**.

- 2. Procedure completed.**

**SQUIB TEST INDICATOR DOES NOT GO ON WHEN CREW'S HOIST CONTROL PANEL CABLE CUT SWITCH IS PRESSED****SYMPTOM**

SQUIB TEST indicator does not go on when crew's hoist control panel CABLE CUT switch is pressed.

**MALFUNCTION**

SQUIB TEST indicator does not go on when crew's hoist control panel CABLE CUT switch is pressed.

**CORRECTIVE ACTION**

- 1. Is the backup pump operating?**

- a. If the backup is operating, turn off pump. Go to **8**.
- b. If the backup is not operating, go to **2**.

- 2. With HOIST POWER ON-OFF switch ON, check for 28 vdc between P371-A and ground.**

- a. If voltage is as specified, go to **3**.

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**SQUIB TEST INDICATOR DOES NOT GO ON WHEN CREW'S HOIST CONTROL PANEL CABLE CUT SWITCH IS PRESSED** - Continued

- b. If voltage is not as specified, go to **4**.
- 3. Check continuity between:**
  - **P371-B and ground**
  - **P371-C and ground**
  - **P372-F and P226-F**
  - **P372-L and P226-L**
  - a. If continuity is present, replace rescue hoist assembly (WP 1342 00). Go to **8**.
  - b. If problem remains, replace crew's hoist control panel (WP 1349 00). Go to **8**.
  - c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **8**.
- 4. Check for 28 vdc between K300-X1 and ground.**
  - a. If voltage is as specified, go to **5**.
  - b. If voltage is not as specified, go to **6**.
- 5. Check continuity between:**
  - **K300-X2 and P298-C**
  - **K300-12 and P371-A**
  - **K300-11 and terminal 1 (HOIST CONTR circuit breaker)**
  - a. If continuity is present, replace hoist relay panel (WP 0827 00). Go to **8**.
  - b. If trouble remains, replace No. 3 relay panel (WP 0827 00). Go to **8**.
  - c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **8**.
- 6. Check for 28 vdc between HOIST CONTR circuit breaker terminal 1 and ground.**
  - a. If voltage is as specified, go to **7**.
  - b. If voltage is not as specified, replace HOIST CONTR circuit breaker (WP 0824 00). Go to **8**.
- 7. Check continuity between:**
  - **K300-X1 and P218-E**
  - **P218-F and Terminal 1 (HOIST CONTR circuit breaker)**
  - a. If continuity is present, replace pilot's hoist control panel (WP 1348 00). Go to **8**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **8**.
- 8. Procedure completed.**

**RESCUE HOIST FAN MOTOR DOES NOT GO ON****SYMPTOM**

Rescue hoist fan motor does not go on.

**MALFUNCTION**

Rescue hoist fan motor does not go on.

**CORRECTIVE ACTION****1. With HOIST POWER ON-OFF switch ON, check for 115 vac between:**

- P370-A and P370-G
- P370-B and P370-G
- P370-C and P370-G
- P370-D and P370-G
- P370-E and P370-G
- P370-F and P370-G

a. If voltage is as specified, replace rescue hoist assembly (WP 1342 00). Go to **5**.

b. If voltage is not as specified, go to **2**.

**2. Check continuity between:**

- K300-A2 and P370-A
- K300-A2 and P370-B
- K300-B2 and P370-C
- K300-B2 and P370-D
- K300-C2 and P370-E
- K300-C2 and P370-F
- P370-G and ground

a. If continuity is present, go to **3**.

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **5**.

**3. Check for 115 vac between:**

- K300-A1 and ground
- K300-B1 and ground
- K300-C1 and ground

a. If voltage is as specified, replace hoist relay panel (WP 0827 00). Go to **5**.

b. If voltage is not as specified, go to **4**.

**4. Check continuity between:**

- K300-A1 and Terminal A2 (HOIST PWR circuit breaker)

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**RESCUE HOIST FAN MOTOR DOES NOT GO ON - Continued**

- **K300-B1 and Terminal B2 (HOIST PWR circuit breaker)**
  - **K300-C1 and Terminal C2 (HOIST PWR circuit breaker)**
  - a. If continuity is present, replace HOIST PWR circuit breaker (WP 0824 00). Go to **5**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **5**.
- 5. Procedure completed.**

**RESCUE HOIST CABLE DOES NOT REEL OUT WITH RESCUE HOIST PENDANT THUMBWHEEL IN DN POSITION****SYMPTOM**

Rescue hoist cable does not reel out with rescue hoist pendant thumbwheel in DN position.

**MALFUNCTION**

Rescue hoist cable does not reel out with rescue hoist pendant thumbwheel in DN position.

**CORRECTIVE ACTION**

- 1. With rescue hoist pendant UP-OFF-DN thumbwheel in DN position, check for 0 ohms between P20-B and P20-C.**
  - a. If resistance is as specified, go to **2**.
  - b. If resistance is not as specified, go to **4**.
- 2. Check continuity between:**
  - **P226-B and P372-B**
  - **P226-C and P372-C**
  - a. If continuity is present, go to **3**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **5**.
- 3. Check limit switches for proper adjustment (WP 1350 00).**
  - a. If limit switches are not properly adjusted, adjust limit switches (WP 1350 00). Go to **5**.
  - b. If limit switches are properly adjusted, replace crew's hoist control panel (WP 1349 00). Go to **5**.
  - c. If trouble remains, replace rescue hoist assembly (WP 1342 00). Go to **5**.
- 4. Check continuity between:**
  - **P20-B and P24-B**
  - **P20-C and P24-C**
  - a. If continuity is present, replace rescue hoist pendant (WP 1346 00). Go to **5**.
  - b. If continuity is not present, replace pendant control cable (WP 1346 00). Go to **5**.
- 5. Procedure completed.**

**RESCUE HOIST CABLE REELS IN/OUT INCORRECTLY****SYMPTOM**

Rescue hoist cable reels in/out incorrectly.

**MALFUNCTION**

Rescue hoist cable reels in/out incorrectly.

**CORRECTIVE ACTION**

- 1. Check for 235 to 265 ohms between P20-D and P20-J.**
  - a. If resistance is as specified, go to **2**.
  - b. If resistance is not as specified, go to **3**.
- 2. Check for 40 to 50 ohms between P20-E and P20-J.**
  - a. If resistance is as specified, go to **4**.
  - b. If resistance is not as specified, go to **7**.
- 3. Check continuity between:**
  - **P20-D and P24-D**
  - **P20-J and P24-J**
  - a. If continuity is present, replace rescue hoist pendant (WP 1346 00). Go to **8**.
  - b. If continuity is not present, replace pendant control cable (WP 1346 00). Go to **8**.
- 4. With hoist pendant thumbwheel in full UP then in full DN position, check for 40 to 50 ohms between P20-D and P20-E.**
  - a. If resistance is as specified, go to **5**.
  - b. If resistance is not as specified, go to **7**.
- 5. Check continuity between:**
  - **P226-D and P372-D**
  - **P226-E and P372-E**
  - **P226-J and P372-J**
  - a. If continuity is present, go to **6**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **8**.
- 6. Check limit switches for proper adjustment (WP 1336 00).**
  - a. If limit switches are not properly adjusted, adjust limit switches (WP 1350 00). Go to **8**.
  - b. If limit switches are properly adjusted, replace crew's hoist control panel (WP 1349 00). Go to **8**.
  - c. If trouble remains, replace rescue hoist assembly (WP 1342 00). Go to **8**.
- 7. Check continuity between:**
  - **P20-D and P24-D**
  - **P20-E and P24-E**

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**RESCUE HOIST CABLE REELS IN/OUT INCORRECTLY** - Continued

- a. If continuity is present, replace rescue hoist pendant (WP 1346 00). Go to **8**.
- b. If continuity is not present, replace pendant control cable (WP 1346 00). Go to **8**.

**8. Procedure completed.****RESCUE HOIST PENDANT CABLE-FEET INDICATOR DOES NOT SHOW INCREASING/DECREASING TOTAL****SYMPTOM**

Rescue hoist pendant CABLE-FEET indicator does not show increasing/decreasing total.

**MALFUNCTION**

Rescue hoist pendant CABLE-FEET indicator does not show increasing/decreasing total.

**CORRECTIVE ACTION****1. Check continuity between:**

- P20-U and P24-U
- P20-S and P24-S
- P20-R and P24-R

- a. If continuity is present, go to **2**.
- b. If continuity is not present, replace pendant control cable (WP 1346 00). Go to **3**.

**2. Check continuity between:**

- P372-R and P226-R
- P372-P and P226-P
- P372-N and P226-N

- a. If continuity is present, replace rescue hoist pendant (WP 1346 00). Go to **3**.
- b. If trouble remains, replace crew's hoist control panel (WP 1349 00). Go to **3**.
- c. If trouble still remains, replace rescue hoist assembly (WP 1342 00). Go to **3**.
- d. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **3**.

**3. Procedure completed.****RESCUE HOIST CABLE DOES NOT REEL IN WITH RESCUE HOIST PENDANT THUMBWHEEL IN UP POSITION****SYMPTOM**

Rescue hoist cable does not reel in with rescue hoist pendant thumbwheel in UP position.

**MALFUNCTION**

Rescue hoist cable does not reel in with rescue hoist pendant thumbwheel in UP position.

**CORRECTIVE ACTION**

- 1. With rescue hoist pendant UP-OFF-DN thumbwheel in UP position, check for 0 ohms between P20-A and P20-C.**

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**RESCUE HOIST CABLE DOES NOT REEL IN WITH RESCUE HOIST PENDANT THUMBWHEEL IN UP POSITION** - Continued

- a. If resistance is as specified, go to **2**.
- b. If resistance is not as specified, go to **4**.
- 2. Check continuity between P226-A and P372-A.**
  - a. If continuity is present, go to **3**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **5**.
- 3. Check limit switches for proper adjustment (WP 1350 00).**
  - a. If limit switches are not properly adjusted, adjust limit switches (WP 1350 00). Go to **5**.
  - b. If limit switches are properly adjusted, replace crew's hoist control panel (WP 1349 00). Go to **5**.
  - c. If trouble remains, replace rescue hoist assembly (WP 1342 00). Go to **5**.
- 4. Check continuity between P24-A and P20-A.**
  - a. If continuity is present, replace rescue hoist pendant (WP 1346 00). Go to **5**.
  - b. If continuity is not present, replace pendant control cable (WP 1346 00). Go to **5**.
- 5. Procedure completed.**

**HOIST CABLE DOES NOT REEL OUT WITH PILOT'S HOIST CONTROL PANEL PILOT OVERRIDE UP-DOWN SWITCH IN DOWN****SYMPTOM**

Hoist cable does not reel out with pilot's hoist control panel PILOT OVERRIDE UP-DOWN switch in DOWN.

**MALFUNCTION**

Hoist cable does not reel out with pilot's hoist control panel PILOT OVERRIDE UP-DOWN switch in DOWN.

**CORRECTIVE ACTION**

- 1. Check continuity between:**
  - **P218-H and P372-X**
  - **P218-K and P372-a**
  - **P218-D and ground**
  - a. If continuity is present, replace pilot's's hoist control panel (WP 1348 00). Go to **2**.
  - b. If trouble remains, replace rescue hoist assembly (WP 1342 00). Go to **2**.
  - c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **2**.
- 2. Procedure completed.**

**RESCUE HOIST CABLE DOES NOT REEL IN WITH PILOT'S HOIST CONTROL PANEL PILOT OVERRIDE UP-DOWN SWITCH IN UP****SYMPTOM**

Rescue hoist cable does not reel in with pilot's hoist control panel PILOT OVERRIDE UP-DOWN switch in UP.

**MALFUNCTION**

Rescue hoist cable does not reel in with pilot's hoist control panel PILOT OVERRIDE UP-DOWN switch in UP.

**CORRECTIVE ACTION****1. Check continuity between:**

- **P218-G and P372-W**
- **P218-D and ground**

- a. If continuity is present, replace pilot's hoist control panel (WP 1348 00). Go to **2**.
- b. If trouble remains, replace rescue hoist assembly (WP 1342 00). Go to **2**.
- c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **2**.

**2. Procedure completed.****DUAL-MODE CONTROLLABLE SEARCHLIGHT DOES NOT EXTEND****SYMPTOM**

Dual-mode controllable searchlight does not extend.

**MALFUNCTION**

Dual-mode controllable searchlight does not extend.

**CORRECTIVE ACTION****1. Check for 28 vdc between:**

- **P226-c and ground**
- **P226-j and ground**
- **P226-n and ground**

- a. If voltage is as specified, go to **2**.
- b. If voltage is not as specified, go to **4**.

**2. With SCHLT switch in FWD position, check for 0 ohms between P20-L and P20-M.**

- a. If resistance is as specified, go to **3**.
- b. If resistance is not as specified, go to **5**.

**3. Check continuity between:**

- **TB200-6 and P226-a**
- **TB200-1 and ground**

- a. If continuity is present, replace crew's hoist control panel (WP 1349 00). Go to **6**.



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**DUAL-MODE CONTROLLABLE SEARCHLIGHT DOES NOT EXTEND - Continued**

- b. If trouble remains, replace dual-mode controllable searchlight (WP 0933 00). Go to **6**.
- c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **6**.
- 4. Check continuity between:**
  - **P226-c and Terminal 1 (HOIST LGHT CONTR circuit breaker)**
  - **P226-j and Terminal 1 (HOIST LGHT CONTR circuit breaker)**
  - **P226-n and Terminal 1 (HOIST LGHT CONTR circuit breaker)**
- a. If continuity is present, replace HOIST LGHT CONTR circuit breaker (WP 0824 00). Go to **6**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **6**.
- 5. Check continuity between:**
  - **P20-M and P24-M**
  - **P20-L and P24-L**
- a. If continuity is present, replace rescue hoist pendant (WP 1346 00). Go to **6**.
- b. If continuity is not present, replace pendant control cable (WP 1346 00). Go to **6**.
- 6. Procedure completed.**

**DUAL-MODE CONTROLLABLE SEARCHLIGHT DOES NOT TURN RIGHT****SYMPTOM**

Dual-mode controllable searchlight does not turn right.

**MALFUNCTION**

Dual-mode controllable searchlight does not turn right.

**CORRECTIVE ACTION**

- 1. With SCHLT switch in RIGHT position, check for 0 ohms between P20-H and P20-L.**
  - a. If resistance is as specified, go to **2**.
  - b. If resistance is not as specified, go to **3**.
- 2. Check continuity between P226-Y and TB200-4.**
  - a. If continuity is present, replace crew's hoist control panel (WP 1349 00). Go to **4**.
  - b. If trouble remains, replace dual-mode controllable searchlight (WP 0933 00). Go to **4**.
  - c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **4**.
- 3. Check continuity between P20-H and P24-H.**
  - a. If continuity is present, replace rescue hoist pendant (WP 1346 00). Go to **4**.
  - b. If continuity is not present, replace pendant control cable (WP 1346 00). Go to **4**.
- 4. Procedure completed.**

**DUAL-MODE CONTROLLABLE SEARCHLIGHT DOES NOT TURN LEFT****SYMPTOM**

Dual-mode controllable searchlight does not turn left.

**MALFUNCTION**

Dual-mode controllable searchlight does not turn left.

**CORRECTIVE ACTION**

- 1. With SCHLT switch in LEFT position, check for 0 ohms between P20-K and P20-L.**
  - a. If resistance is as specified, go to **2**.
  - b. If resistance is not as specified, go to **3**.
- 2. Check continuity between P226-Z and TB200-5.**
  - a. If continuity is present, replace crew's hoist control panel (WP 1349 00). Go to **4**.
  - b. If trouble remains, replace dual-mode controllable searchlight (WP 0933 00). Go to **4**.
  - c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **4**.
- 3. Check continuity between P20-K and P24-K.**
  - a. If continuity is present, replace rescue hoist pendant (WP 1346 00). Go to **4**.
  - b. If continuity is not present, replace pendant control cable (WP 1346 00). Go to **4**.
- 4. Procedure completed.**

**DUAL-MODE CONTROLLABLE SEARCHLIGHT DOES NOT RETRACT****SYMPTOM**

Dual-mode controllable searchlight does not retract.

**MALFUNCTION**

Dual-mode controllable searchlight does not retract.

**CORRECTIVE ACTION**

- 1. With SCHLT switch in AFT position, check for 0 ohms between P20-T and P20-L.**
  - a. If resistance is as specified, go to **2**.
  - b. If resistance is not as specified, go to **3**.
- 2. Check continuity between P226-b and TB200-7.**
  - a. If continuity is present, replace crew's hoist control panel (WP 1349 00). Go to **4**.
  - b. If trouble remains, replace dual-mode controllable searchlight (WP 0933 00). Go to **4**.
  - c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **4**.
- 3. Check continuity between P20-K and P24-K.**
  - a. If continuity is present, replace rescue hoist pendant (WP 1346 00). Go to **4**.
  - b. If continuity is not present, replace pendant control cable (WP 1346 00). Go to **4**.
- 4. Procedure completed.**

**DUAL-MODE CONTROLLABLE SEARCHLIGHT DOES NOT GO ON****SYMPTOM**

Dual-mode controllable searchlight does not go on.

**MALFUNCTION**

Dual-mode controllable searchlight does not go on.

**CORRECTIVE ACTION**

- 1. Check for burnt-out normal searchlight bulb (WP 0933 00).**
  - a. If lamp is good, go to **2**.
  - b. If lamp is burnt out, replace lamp (WP 0933 00). Go to **5**.
- 2. Check for 28 vdc between TB200-3 and ground.**
  - a. If voltage is as specified, go to **3**.
  - b. If voltage is not as specified, go to **4**.
- 3. Check continuity between P226-k and TB200-2.**
  - a. If continuity is present, replace crew's hoist control panel (WP 1349 00). Go to **5**.
  - b. If trouble remains, replace dual-mode controllable searchlight (WP 0933 00). Go to **5**.
  - c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **5**.
- 4. Check continuity between TB200-3 and terminal 2 (HOIST LGHT PWR circuit breaker).**
  - a. If continuity is present, replace HOIST LGHT PWR circuit breaker (WP 0824 00). Go to **5**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **5**.
- 5. Procedure completed.**

**TEST SET IR LIGHT DOES NOT GO ON****SYMPTOM**

Test set IR LIGHT does not go on.

**MALFUNCTION**

Test set IR LIGHT does not go on.

**CORRECTIVE ACTION**

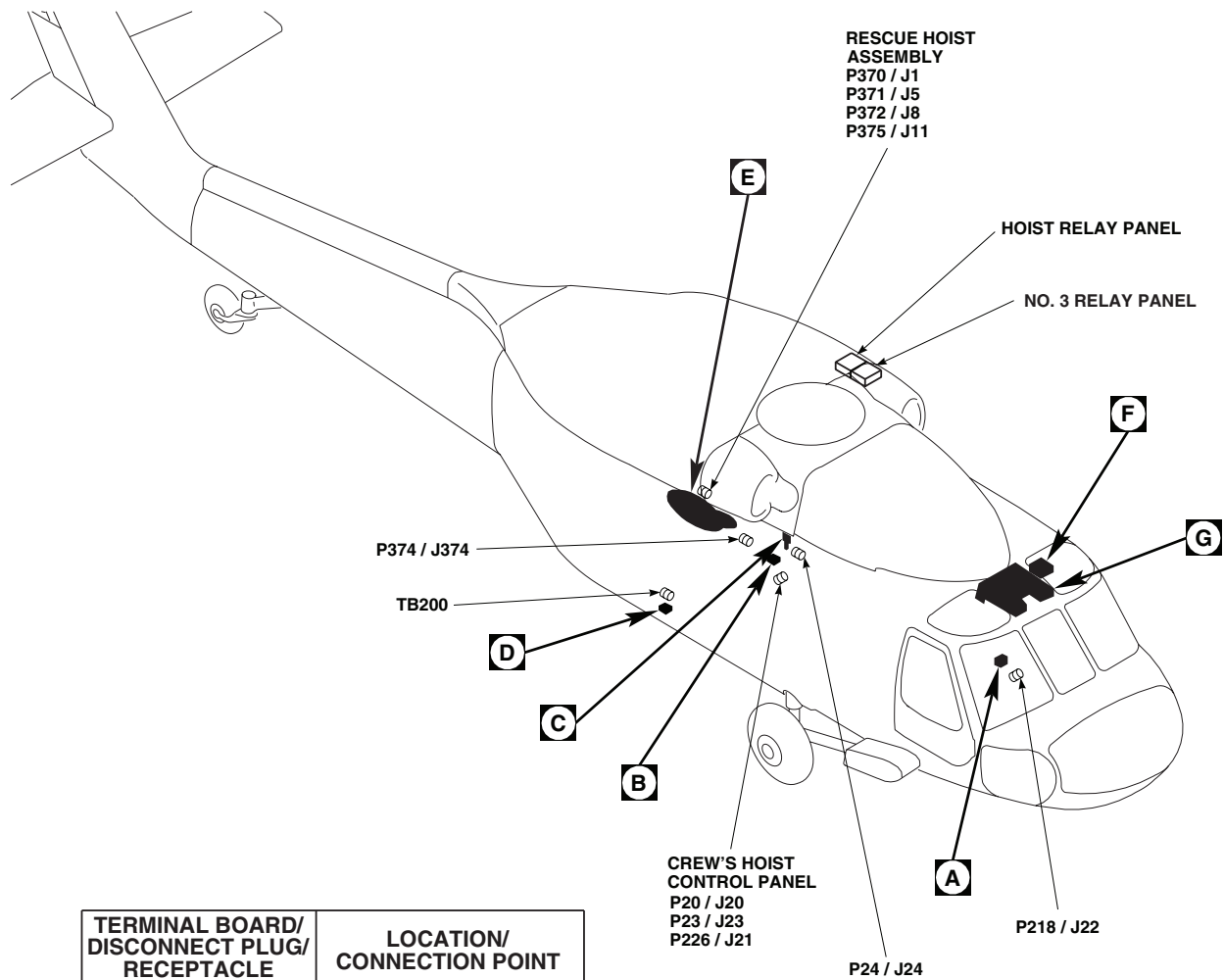
- 1. Check for burnt-out NVG searchlight bulb (WP 0933 00).**
  - a. If lamp is good, go to **2**.
  - b. If lamp is burnt out, replace lamp (WP 0933 00). Go to **5**.
- 2. Check for 28 vdc between TB200-9 and ground.**
  - a. If voltage is as specified, go to **3**.
  - b. If voltage is not as specified, go to **4**.
- 3. Check continuity between P226-p and TB200-8.**
  - a. If continuity is present, replace crew's hoist control panel (WP 1349 00). Go to **5**.

---

**TEST SET IR LIGHT DOES NOT GO ON - Continued**

- b. If trouble remains, replace dual-mode controllable searchlight (WP 0933 00). Go to **5**.
- c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **5**.
- 4. Check continuity between TB200-9 and terminal 2 (HOIST LGHT PWR circuit breaker).**
  - a. If continuity is present, replace HOIST LGHT PWR circuit breaker (WP 0824 00). Go to **5**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **5**.
- 5. Procedure completed.**

## LOCATION



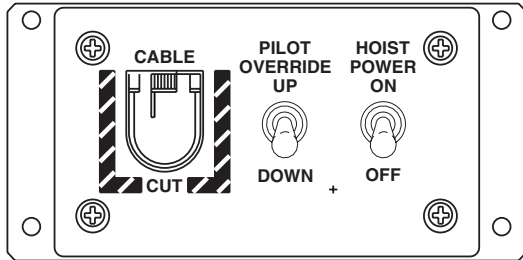
| TERMINAL BOARD/<br>DISCONNECT PLUG/<br>RECEPTACLE | LOCATION/<br>CONNECTION POINT                          |
|---------------------------------------------------|--------------------------------------------------------|
| P6 / J6                                           | RESCUE HOIST ASSEMBLY<br>PRESSURE CARTRIDGE<br>(SQUIB) |
| P20 / J20                                         | CREW'S HOIST CONTROL<br>PANEL                          |
| P23 / J23                                         | CREW'S HOIST CONTROL<br>PANEL                          |
| P24 / J24                                         | RESCUE HOIST PENDANT                                   |
| P129 / J129                                       | COPILOT'S AUXILIARY<br>CIRCUIT BREAKER PANEL           |
| P199 / J199                                       | INSTRUMENT PANEL<br>DISCONNECT                         |
| P200 / J200                                       | UPPER CONSOLE<br>DISCONNECT                            |
| P218 / J22                                        | PILOT'S HOIST CONTROL<br>PANEL                         |
| P226 / J21                                        | CREW'S HOIST CONTROL<br>PANEL                          |
| P298 / J298                                       | NO. 3 RELAY PANEL                                      |

| TERMINAL BOARD/<br>DISCONNECT PLUG/<br>RECEPTACLE | LOCATION/<br>CONNECTION POINT            |
|---------------------------------------------------|------------------------------------------|
| P356 / J356                                       | ECS CONTROL PANEL                        |
| P370 / J1                                         | RESCUE HOIST ASSEMBLY                    |
| P371 / J5                                         | RESCUE HOIST ASSEMBLY                    |
| P372 / J8                                         | RESCUE HOIST ASSEMBLY                    |
| P374 / J374                                       | TOP DECK DISCONNECT                      |
| P375 / J11                                        | RESCUE HOIST ASSEMBLY                    |
| P1000 / J1000                                     | MASTER WARNING<br>DIMMER                 |
| TB200                                             | DUAL-MODE<br>CONTROLLABLE<br>SEARCHLIGHT |

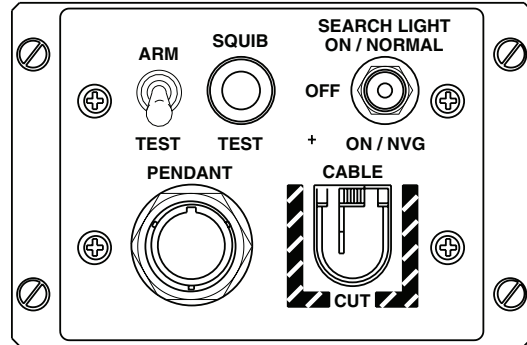
AB0143\_1  
SA

Figure 1. Rescue Hoist System Location Diagram. (Sheet 1 of 3)

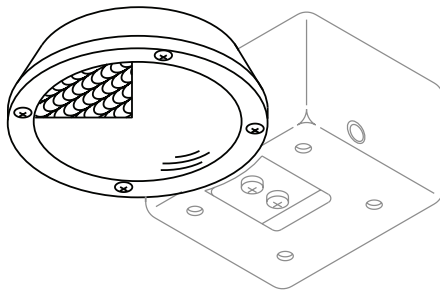
LOCATION - Continued



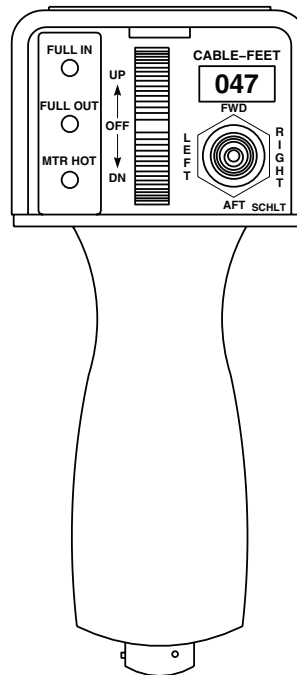
PILOT'S HOIST CONTROL PANEL



CREW'S HOIST CONTROL PANEL



DUAL-MODE  
CONTROLLABLE SEARCHLIGHT

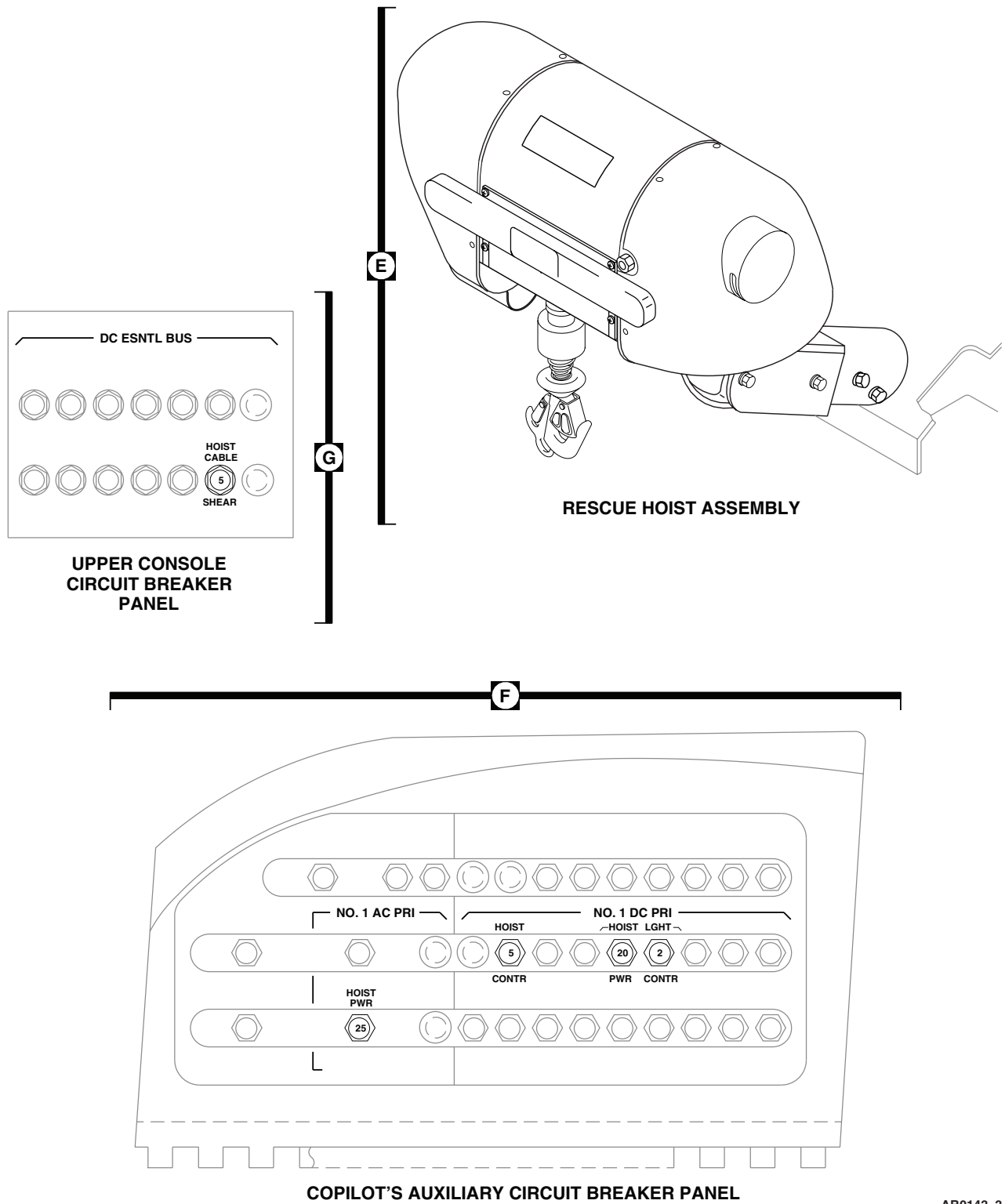


RESCUE HOIST PENDANT

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SA

Figure 1. Rescue Hoist System Location Diagram. (Sheet 2 of 3)

LOCATION - Continued



AB0143\_3  
SA

Figure 1. Rescue Hoist System Location Diagram. (Sheet 3 of 3)

SCHEMATICS

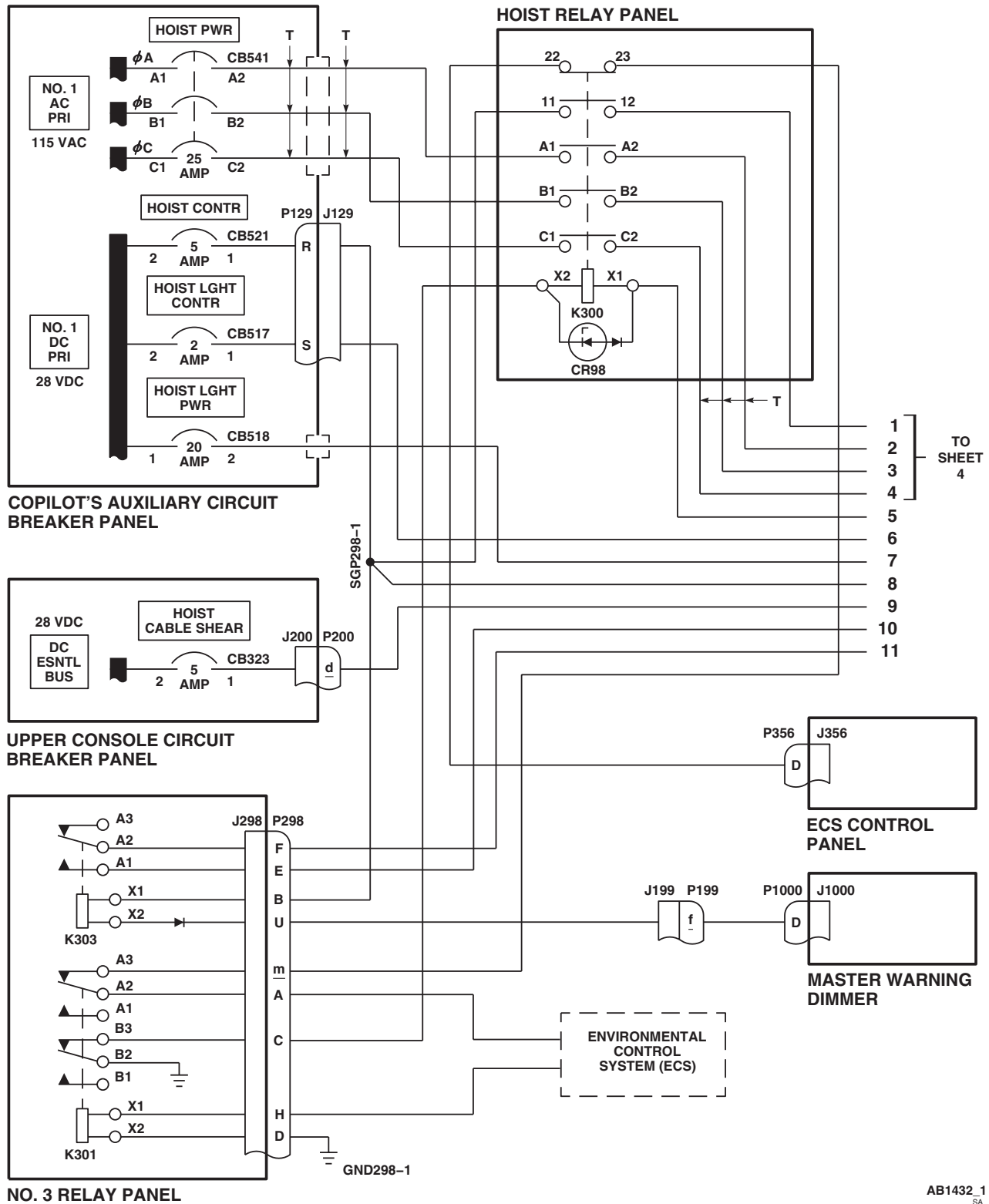
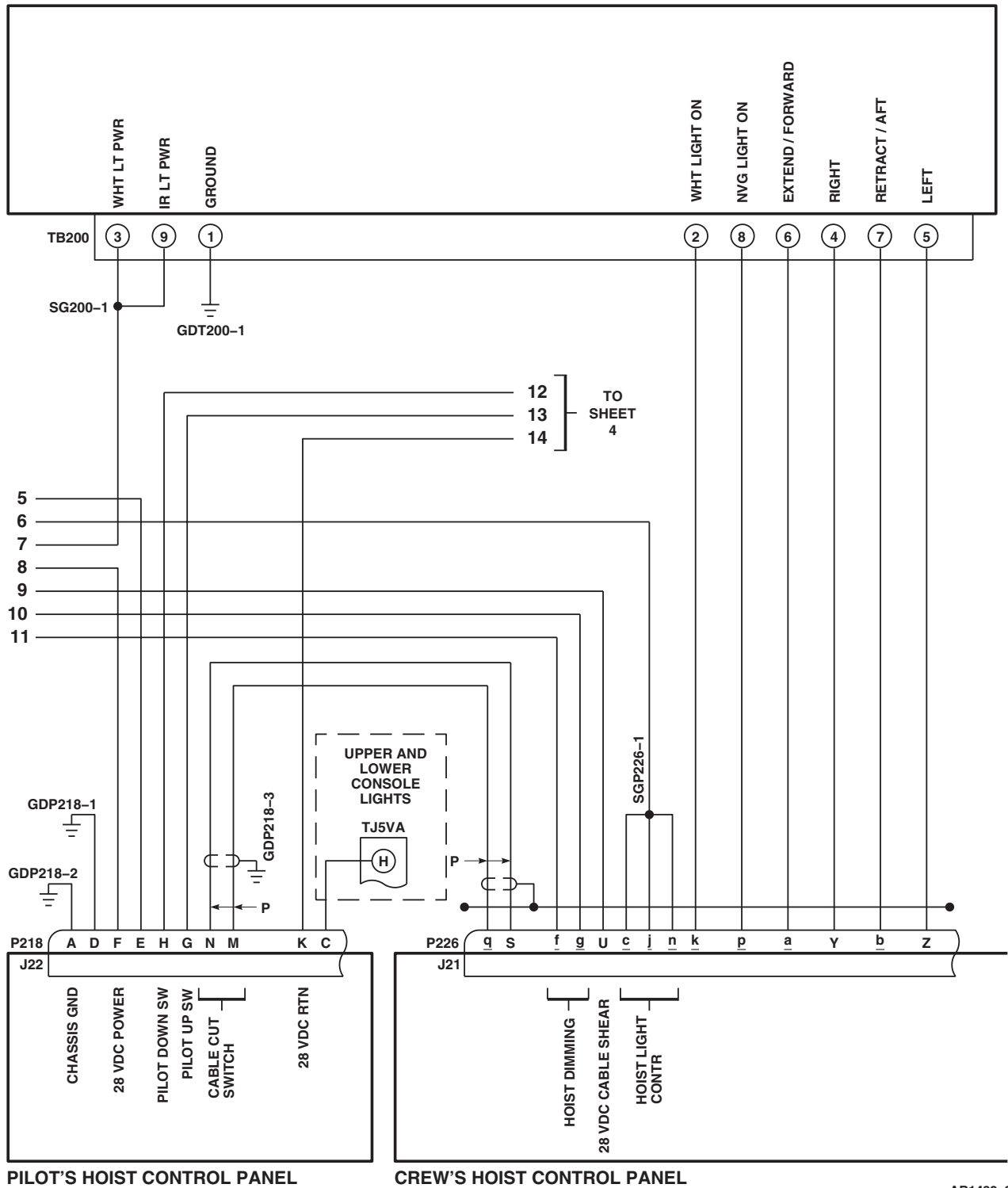


Figure 2. Rescue Hoist System Schematic Diagram. (Sheet 1 of 4)



SCHEMATICS - Continued

DUAL-MODE CONTROLLABLE SEARCHLIGHT

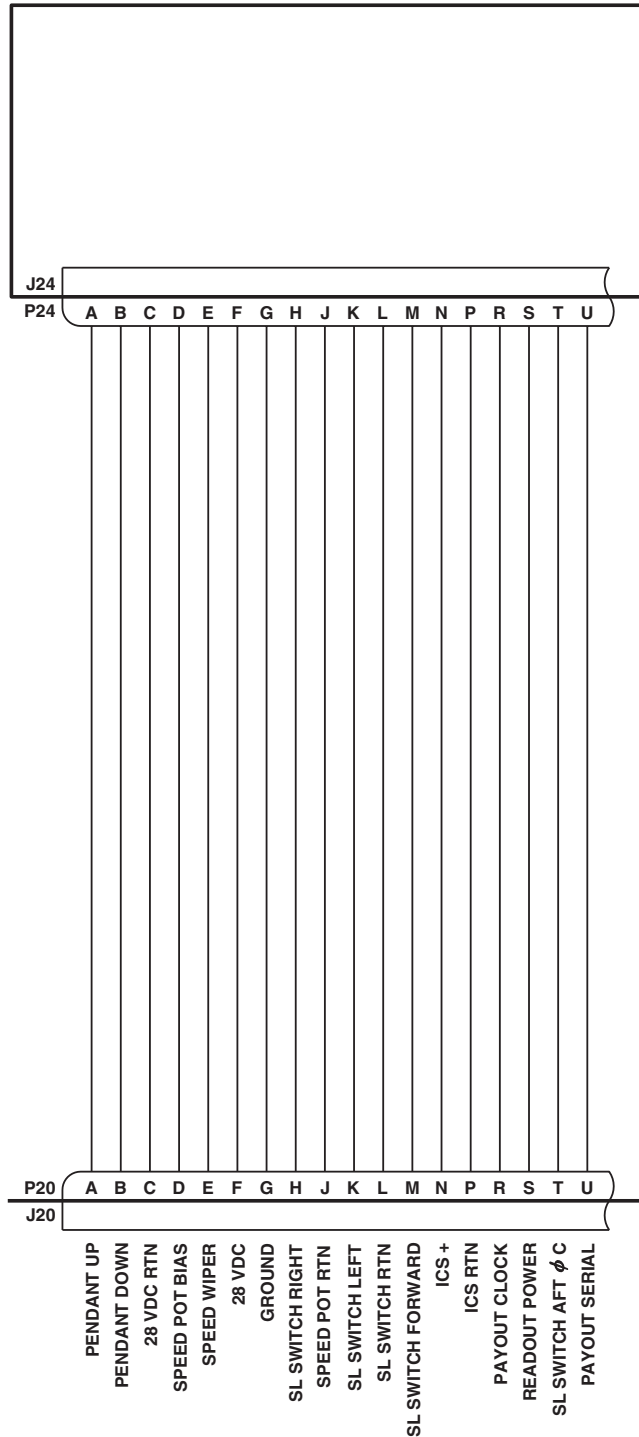


AB1432\_2  
SA

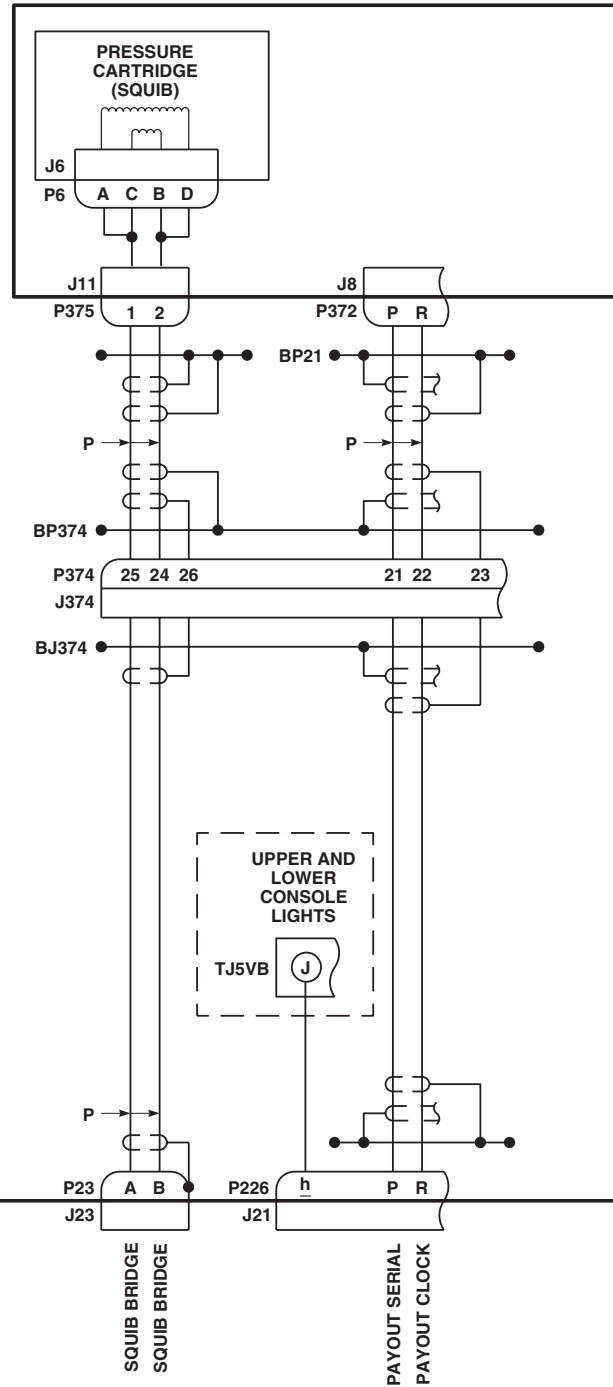
Figure 2. Rescue Hoist System Schematic Diagram. (Sheet 2 of 4)

SCHEMATICS - Continued

RESCUE HOIST PENDANT



RESCUE HOIST ASSEMBLY



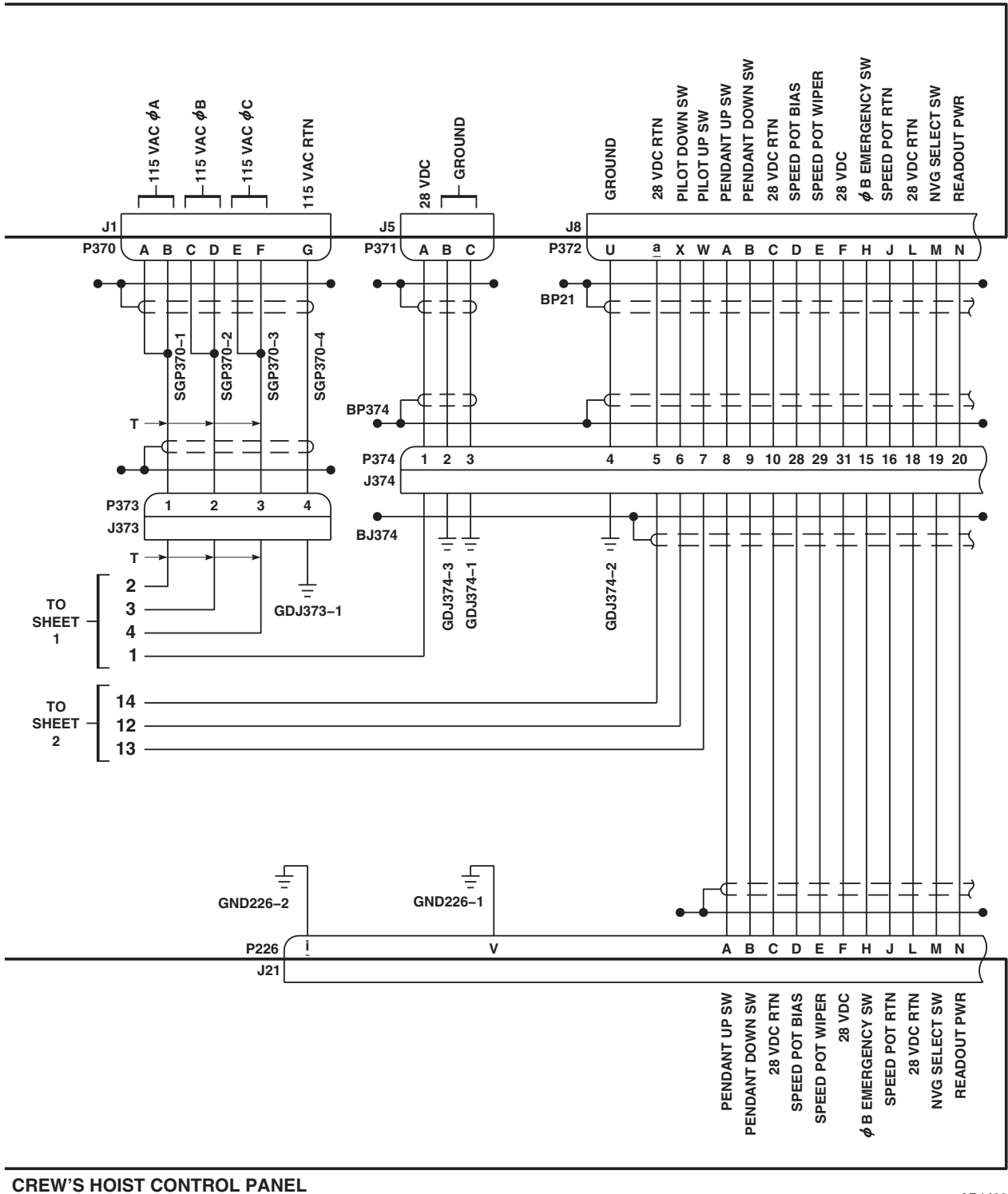
CREW'S HOIST CONTROL PANEL

AB1432\_3  
SA

Figure 2. Rescue Hoist System Schematic Diagram. (Sheet 3 of 4)

SCHEMATICS - Continued

RESCUE HOIST ASSEMBLY



CREW'S HOIST CONTROL PANEL

AB1432\_4  
SA

Figure 2. Rescue Hoist System Schematic Diagram. (Sheet 4 of 4)

END OF WORK PACKAGE



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**UNIT LEVEL****HOISTS AND WINCHES**

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**RESCUE HOIST CONTROL PANEL (AVIM)**

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**INITIAL SETUP:****Test Equipment**

Multimeter, AN/PSM-45A

WP 1353 00

WP 1354 00

WP 1356 00

**Tools and Special Tools**

Electrical Repairer Toolkit, SC 5180-99-B06

WP 1357 00

WP 1358 00

**References**

WP 1340 00

WP 1352 00

**Equipment Condition**Between 27 and 28 vdc Bench Power Available

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**SPECIAL INSTRUCTIONS****NOTE**

See Figure 1 for component location diagram and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

The rescue hoist control panel operational/troubleshooting procedure is subdivided as follows:

- Visual Inspection
- Electrical

**VISUAL INSPECTION OPERATIONAL/TROUBLESHOOTING PROCEDURE****PROCEDURE****Step**

1. Remove lower panel assembly from enclosure and inspect for broken or burnt wires, bent or broken connector pins, corroded or damaged switch terminals, security of components, and general condition.

**Indication/Condition**

Wires shall not be broken or burnt, connector pins shall not be broken or bent, switch terminals shall not be corroded or damaged, and components shall be secure and in good condition.

**Corrective Action**

If Indication/Condition is not as specified, repair/replace components as required (WP 1357 00).

## ELECTRICAL OPERATIONAL/TROUBLESHOOTING PROCEDURE

### PROCEDURE

#### Step

1. Place all rescue hoist control panel switches to OFF, SAFE, and NORM.
2. Check continuity between the following pins while operating switches listed below:

| SWITCH | SWITCH POSITION | CHECK RESISTANCE BETWEEN CONNECTOR PINS: | METER INDICATION (OHM) | CORRECTIVE ACTION (CONTINUITY)                       |
|--------|-----------------|------------------------------------------|------------------------|------------------------------------------------------|
| None   | None            | G - S                                    | 0                      | G and S3-2, S3-2 and S2-2, S2-2 and S1-3, S1-3 and S |
| CABLE  | OFF             | G - M                                    | Infinity               | Replace switch S2 (WP 1353 00)                       |
| CABLE  | OFF             | G - N                                    | Infinity               | Replace switch S2 (WP 1353 00)                       |
| CABLE  | UP              | G - M                                    | 0                      | G and S3-2, S3-2 and S2-2, M and S2-2                |
| CABLE  | DOWN            | G - N                                    | 0                      | G and S3-2, S3-2 and S2-2, N and S2-3                |
| BOOM   | OFF             | G - P                                    | Infinity               | Replace switch S3 (WP 1353 00)                       |
| BOOM   | OFF             | G - R                                    | Infinity               | Replace switch S3 (WP 1353 00)                       |
| BOOM   | IN              | G - P                                    | 0                      | G and S3-2, P and S3-1                               |
| BOOM   | OUT             | G - R                                    | 0                      | G and S3-2, R and S3-3                               |
| MASTER | OFF             | S - D                                    | Infinity               | Replace switch S1 (WP 1353 00)                       |
| MASTER | ON              | S - D                                    | 0                      | D and S1-2, S and S1-3                               |
| None   | None            | L - T                                    | Infinity               | Replace switch S5 (WP 1353 00)                       |

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**ELECTRICAL OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued**Indication/Condition**

Meter shall indicate corresponding resistance.

**Corrective Action**

1. If Indication/Condition is not as specified, do corresponding CORRECTIVE ACTION replacement or continuity check.
  - a. If continuity is present, replace switch (WP 1353 00).
  - b. If continuity is not present, repair/replace wiring.

**Step**

3. With SQUIB switch in NORM and CABLE SHEAR switch in FIRE, check for resistance between J228-L and J228-T.

**Indication/Condition**

Meter shall indicate continuity.

**Corrective Action**

If Indication/Condition is not as specified, go to METER DOES NOT INDICATE CONTINUITY BETWEEN PINS L AND T, in this work package.

**Step**

4. Connect jumper wires between J228-C and J228-U, and J228-J and J228-T.
5. Connect 28 vdc power supply, positive terminal to J228-K, and negative terminal to J228-C, J228-U, J228-J, and J228-T.

**Indication/Condition**

Information panel shall light evenly.

**Corrective Action**

1. If Indication/Condition is not as specified, replace malfunctioning lamp (WP 1352 00).
  - a. If trouble remains, replace information panel (WP 1340 00).
2. If panel does not light, replace information panel (WP 1340 00).
  - a. If trouble remains, repair/replace wiring.

**Step**

6. Turn off power supply and connect jumper wire between J228-L and J228-V.
7. Connect 28 vdc power supply + terminal to J228-L and J228-V.
8. Momentarily press SQUIB IND button.

**Indication/Condition**

SQUIB IND lamp shall go on when button is pressed.

**Corrective Action**

1. If Indication/Condition is not as specified, replace lamp (WP 1352 00).
  - a. If trouble remains, replace light assembly (WP 1354 00).

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**ELECTRICAL OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued

- b. If trouble still remains, repair/replace wiring.

**Step**

9. Momentarily place SQUIB switch to TEST.

**Indication/Condition**

SQUIB IND lamp shall go on when SQUIB switch is at TEST.

**Corrective Action**

If Indication/Condition is not as specified, go to WITH SQUIB SWITCH IN THE TEST POSITION, SQUIB IND LIGHT DOES NOT GO ON, in this work package.

**Step**

10. Turn off and remove dc power supply, and remove jumper wires from J228.

**Indication/Condition**

Jumper wires shall be removed.

**Corrective Action**

None Required

**METER DOES NOT INDICATE CONTINUITY BETWEEN PINS L AND T****SYMPTOM**

Meter does not indicate continuity between pins L and T.

**MALFUNCTION**

Meter does not indicate continuity between pins L and T.

**CORRECTIVE ACTION****1. Check continuity between:**

- J228-L and S4-2
- S4-3 and S5-2
- S5-1 and J228-T

a. If continuity is present, go to **2**.

b. If continuity is not present, repair/replace wiring as required. Go to **3**.

**2. Check continuity between switch S4-2 and S4-3.**

a. If continuity is present, replace switch S5 (WP 1353 00). Go to **3**.

b. If continuity is not present, replace switch S4 (WP 1353 00). Go to **3**.

**3. Procedure completed.**



**WITH SQUIB SWITCH IN THE TEST POSITION, SQUIB IND LIGHT DOES NOT GO ON****SYMPTOM**

With SQUIB switch in the TEST position, SQUIB IND light does not go on.

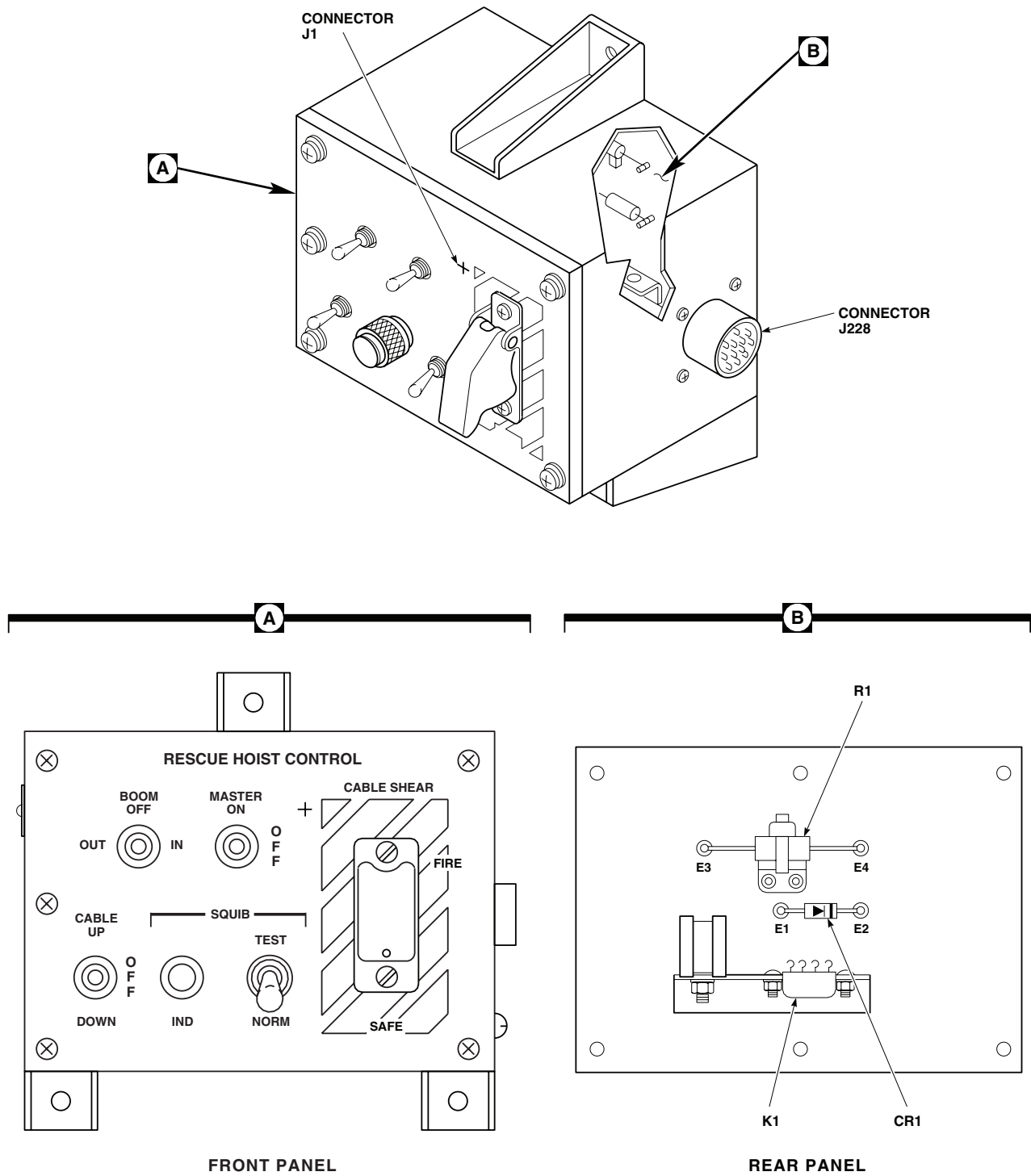
**MALFUNCTION**

With SQUIB switch in the TEST position, SQUIB IND light does not go on.

**CORRECTIVE ACTION**

- 1. With SQUIB switch in TEST position, check continuity between K1-X1 and J228-T.**
  - a. If continuity is present, go to **2**.
  - b. If continuity is not present, go to **4**.
- 2. Check for 228 to 252 ohms between K1-X2 and J228-T.**
  - a. If resistance is as specified, replace relay K1 (WP 1358 00). Go to **5**.
  - b. If resistance is not as specified, go to **3**.
- 3. Check continuity between:**
  - **K1-X2 and E3**
  - **E4 and J228-T**
  - a. If continuity is present, replace resistor R1 (WP 1356 00). Go to **5**.
  - b. If continuity is not present, repair/replace wiring. Go to **5**.
- 4. Check continuity between:**
  - **J228-L and S4-2**
  - **S4-1 and K1-X1**
  - a. If continuity is present, replace switch S4 (WP 1353 00). Go to **5**.
  - b. If continuity is not present, repair/replace wiring. Go to **5**.
- 5. Procedure completed.**

LOCATION



AA3440  
SA

Figure 1. Rescue Hoist Control Panel Components Location Diagram.

## SCHEMATICS

| WIRE LIST |      |            |        |        |
|-----------|------|------------|--------|--------|
| WIRE NO.  | GAGE | COLOR CODE | FROM   | TO     |
| 1         | 20   | 92         | J228-D | S1-2   |
| 2         | 20   | 924        | J228-G | S3-2   |
| 3         | 20   | 925        | J228-S | S1-3   |
| 4         | 20   | 923        | S1-3   | S2-2   |
| 5         | 20   | 923        | S2-2   | S3-2   |
| 6         | 20   | 926        | J228-M | S2-1   |
| 7         | 20   | 927        | J228-N | S2-3   |
| 8         | 20   | 928        | J228-P | S3-1   |
| 9         | 20   | 912        | J228-R | S3-3   |
| 10        | 20   | 8          | J228-K | J1-1   |
| 11        | 20   | 0          | J1-GND | GND 1  |
| 12        | 20   | 0          | J228-C | GND 1  |
| 13        | 20   | 2          | J228-L | S4-2   |
| 14        | 20   | 923        | S4-1   | CR1-E2 |
| 15        | 20   | 926        | R1-E3  | CR1-E1 |
| 16        | 20   | 4          | S5-1   | R1-E4  |
| 17        | 20   | 4          | J228-T | S5-1   |
| 18        | 20   | 924        | S5-2   | S4-3   |
| 19        | 24   | 0          | J228-J | K1-2   |
| 20        | 24   | 0          | K1-2   | DS1-3  |
| 21        | 24   | 901        | K1-1   | DS1-2  |
| 22        | 20   | 1          | J228-V | DS1-1  |
| 23        | 20   | 0          | J228-U | GND 1  |
| 24        | 24   | 926        | K1-X2  | CR1-E1 |
| 25        | 24   | 923        | K1-X1  | CR1-E2 |

| JUMPER LIST |     |            |              |          |
|-------------|-----|------------|--------------|----------|
| JUMPER NO.  | AWG | COLOR CODE | TERMINATIONS |          |
|             |     |            | FROM         | TO       |
| 1           | 24  | 0          | K1-2         | SPLICE 1 |

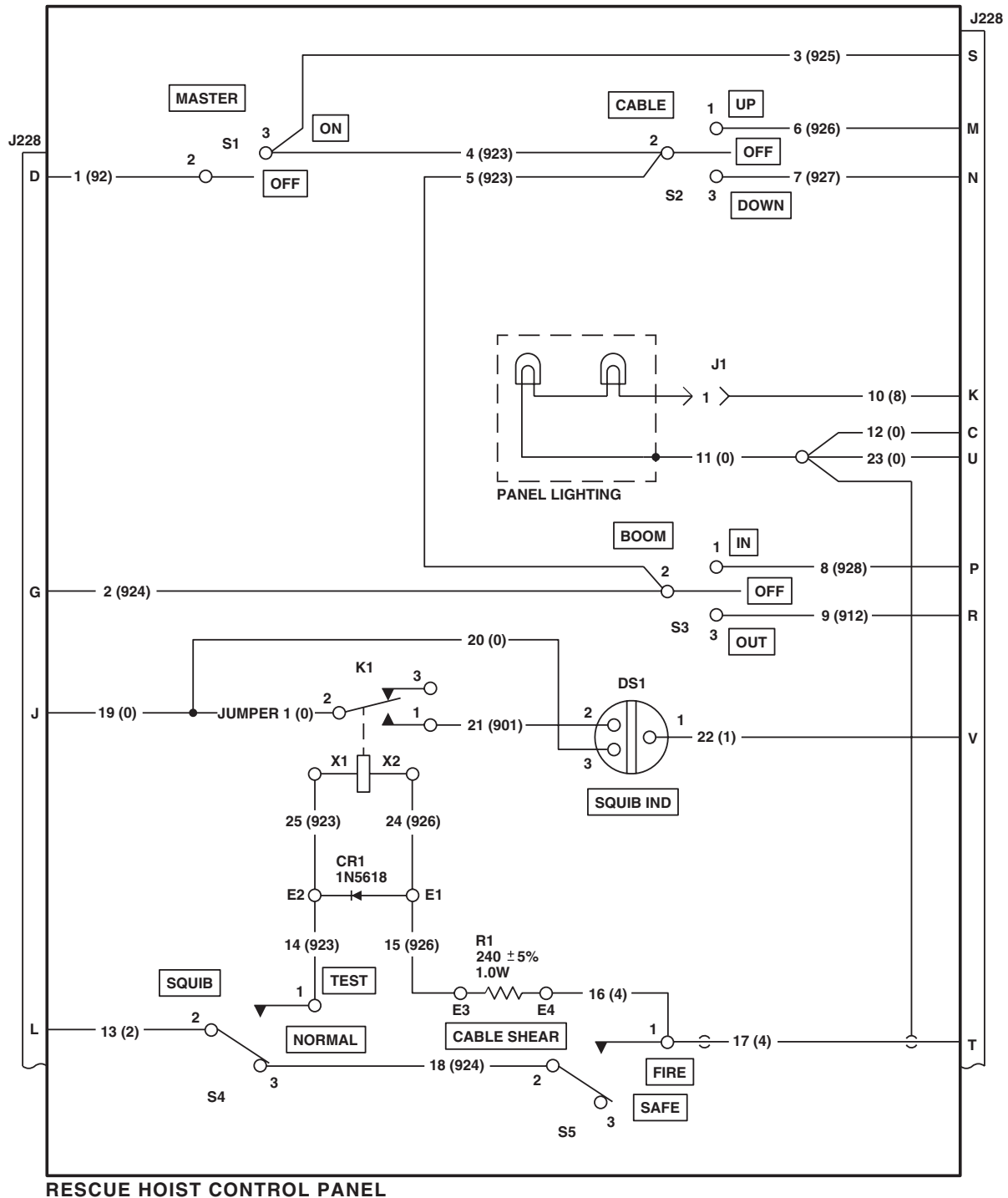
## NOTE

ALL RESISTOR VALUES ARE IN OHMS.

AB1529\_1  
SA

Figure 2. Rescue Hoist Control Panel Schematic Diagram. (Sheet 1 of 2)

SCHEMATICS - Continued



AB1529 2  
SA

Figure 2. Rescue Hoist Control Panel Schematic Diagram. (Sheet 2 of 2)

END OF WORK PACKAGE

## UNIT LEVEL

## AUXILIARY POWER PLANT SYSTEM

## AUXILIARY POWER UNIT SYSTEM (116305 SERIES)

This WP supersedes WP 0155 00, dated 17 April 2006.

## INITIAL SETUP:

## Test Equipment

APU Test Set, Simulator, Temp./Speed, H296A-1  
Multimeter, AN/PSM-45A

WP 0786 00

WP 0824 00

WP 0825 00

WP 0828 00

## Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01  
Electrical Repairer Toolkit, SC 5180-99-B06

WP 0865 00

WP 0925 00

WP 0969 00

## Personnel Required

Aircraft Electrician MOS 15F (One)  
UH-60 Helicopter Repairer MOS 15T (One)

WP 1203 00

WP 1359 00

WP 1365 00

## References

TM 1-1520-237-10

TM 55-2835-208-23

WP 0086 00

WP 0090 00

WP 0100 00

WP 0104 00

WP 0129 00

WP 0258 00

WP 0702 00

WP 0743 00

WP 1400 00

WP 1604 00

WP 1605 00

WP 1685 00

WP 1747 00

## Equipment Condition

APU Accumulator Properly Serviced, WP 1605 00

Battery Fully Charged

External Electrical Power Available, WP 1685 00

No. 1 Fuel Tank Serviced, TM 1-1520-237-10

## SPECIAL INSTRUCTIONS

## NOTE

See Figure 1 for component location diagram, Figure 2 for BITE diagram,

**UH-60A UH-60L** Figure 3 and **HH-60L HH-60A** Figure 4 for schematic diagrams, Figure 5 fuel system schematic, and Figure 6 for lubrication system schematic as an aid in operational/troubleshooting.

The auxiliary power unit system operational/troubleshooting procedure is subdivided as follows:

- APU System
- Electronic Sequence Unit (ESU)
- Simulator Setup
- Hydraulic Start Valve

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**SPECIAL INSTRUCTIONS** - Continued

- Speed Sequence
- Underspeed
- Overspeed
- Low Oil Pressure
- High Oil Temperature
- Start Bypass Valve
- Overtemperature
- Open Thermocouple
- Shorted Thermocouple
- Disconnecting Simulator

**APPLICABLE CONFIGURATION**

APU 116305-100, 116305-200, 116305-300, and 116305-302

## APU SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE

### PROCEDURE

#### Step

#### WARNING

To prevent injury to personnel and damage to helicopter, have fire guard check APU fuel drain. Wait at least two minutes after APU shutdown before attempting a restart. DO NOT RESTART APU until fuel drainage has stopped.

#### CAUTION

- To prevent possible damage to APU, do not operate APU more than 30 minutes when main engines and rotor are running and outside temperature is greater than 109°F (43°C). Continuous operation of APU is permitted at temperatures up to 124°F (51°C) only if main engines and rotors are not running.
- On helicopters with ESU 160200-201, cycling of the BATT switch during APU coast-down may cause the APU to restart with a possible overtemperature.
- If onboard fire extinguishing is required before a generator becomes operational, RESERVE fire extinguisher must be used to fight fire.

#### NOTE

- Do not connect external electrical power to helicopter before starting or during APU system operational/troubleshooting procedure.
- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- For proper Indication/Conditions, switch positions and order of operation must be strictly followed.
- When troubleshooting APU system on **HH-60L** or **HH-60A**, one multifunction display must be turned on to verify caution advisory legends.

1. Make sure all circuit breakers are pushed in except those noted being pulled for maintenance or safety.
2. Make sure upper console BATT switch and GENERATORS APU switch is placed to RESET/OFF (center position).
3. Check bleed air pneumatic tube couplings for loose clamps, cuts, and/or incorrect installation at all connecting points (WP 0702 00).

#### Indication/Condition

All couplings shall be securely installed with clamps and there shall be no cuts in pneumatic tube lines.

#### Corrective Action

If Indication/Condition is not as specified, repair/replace couplings and lines (WP 0702 00).

#### Step

4. Remove soundproofing material and panel from cabin ceiling to visually check electronic sequence unit (ESU) BITE indicators (WP 0258 00).

**APU SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued**NOTE**

- An observer must be in the cabin area to watch the ESU BITE indicators during the operational/troubleshooting procedure.
- The observer shall call out the procedural steps to the operator in the cockpit. This will allow the observer to watch for a change in ESU BITE indication as the operator does the next procedural step.

5. Visually check ESU BITE indicators.

**Indication/Condition**

All indicators shall be black.

**Corrective Action**

If Indication/Condition is not as specified, troubleshoot system for indicated malfunction.

**Step****NOTE**

- Do not attempt to start the APU if BATT LOW CHARGE capsule is on.
- NORMAL START SEQUENCE, in this work package lists the sequence of events for a normal start.
- The APU will not start if any of the following ESU internal failures appear as BITE indications:

6. Place upper console BATT switch to ON. Watch ESU BITE indicators.

| BITE INDICATION | FAILURE                   |
|-----------------|---------------------------|
| ○ ● ● ●         | Processor Board Failure   |
| ● ○ ● ●         | Sensor/Data Board Failure |
| ● ○ ○ ○         | Processor Sequence Fail   |
| ○ ○ ○ ○         | No Data                   |

**Indication/Condition**

No. 1 BITE indicator shall cycle from black to white to black. If BITE indicator is white, due to malfunction, it shall only cycle from white to black.

**Corrective Action**

If Indication/Condition is not as specified, go to ESU BITE INDICATORS DO NOT CYCLE WHEN BATT SWITCH IS PLACED ON, in this work package.



**APU SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued**Step**

7. Place upper console FUEL PUMP switch to APU BOOST.

**Indication/Condition****NOTE**

- Prime boost pump must be running for all APU motoring and start operations.
- A minimal amount of fuel may be evident during and shortly after start.
- a. PRIME BOOST PUMP ON capsule on caution/advisory panel shall go on.
- b. No fuel leakage shall occur from APU combustor drain line (below helicopter).

**Corrective Action**

1. If Indication/Condition a. is not as specified, troubleshoot fuel prime boost system (WP 0129 00).
2. If Indication/Condition b. is not as specified, go to FUEL LEAKS FROM APU COMBUSTOR DRAIN LINE WITH PRIME BOOST PUMP RUNNING, in this work package.

**Step****NOTE**

- On helicopters with ESU 160200-201, if the APU should start to motor and then shut down during an attempted start, do not place BATT switch to OFF. Any cycling of BATT switch will reset the ESU and malfunction BITE indications (indicating the cause of shutdown) will be lost.
  - If the APU shuts down, the observer shall take note of the last APU speed BITE indication before the malfunction BITE indication appears.
  - Make sure upper console APU T-handle is pushed in or APU will not start.
  - If a restart is being attempted, the original malfunction BITE indication will be shown.
8. Start APU (TM 1-1520-237-10).
9. APU shall start and BITE indicators shall cycle as follows:

| BITE INDICATION | APU SPEED SEQUENCE |
|-----------------|--------------------|
| ○ ● ● ●         | 5%                 |
| ○ ○ ● ●         | 14%                |
| ○ ○ ○ ●         | 70%                |
| ○ ○ ○ ○         | 90%                |

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**APU SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**


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**BITE INDICATION****APU SPEED SEQUENCE**

● ● ● ●

90% + 1.5 seconds (ready for service)

**Indication/Condition**

- a. APU ON capsule on caution/advisory panel shall go on.
- b. APU FAIL capsule on caution/advisory panel shall be off.

**CAUTION**

If APU OIL TEMP HI capsule is on, shut down APU immediately.

- c. APU OIL TEMP HI capsule on caution/advisory panel shall be off.
- d. No fuel leakage shall occur from APU combustor drain line (below helicopter).

**Corrective Action**

1. If shorted thermocouple (● ○ ● ○) is present and a restart attempt fails, go to )ESU BITE INDICATORS SHOW ○ ● ○ ● (SHORTED THERMOCOUPLE, in this work package.
2. On helicopters with ESU 160200-201, if fail to light (○ ● ● ○) is present after a restart attempt fails, go to ESU BITE INDICATOR SHOWS ○ ● ● ○ (FAIL TO LIGHT) FOR HELICOPTERS WITH ESU 160200-201, in this work package.
3. On helicopters with ESU 160200-600, if shorted oil pressure switch indication (○ ● ● ○) is shown, go to ESU BITE INDICATOR SHOWS ○ ● ● ○ (SHORTED OIL PRESSURE SWITCH) FOR HELICOPTERS WITH ESU 160200-600, in this work package.
4. If APU does not motor, go to APU MOTOR DOES NOT MOTOR (NO ROTATION), in this work package.
5. If APU motors, but shuts down before Indication/Condition b. occurs, and APU FAIL capsule is on, visually check ESU BITE indications and do the following:

**IF BITE INDICATION IS:****ACTION: GO TO**

● ○ ○ ● (fail to start)

ESU BITE INDICATOR SHOWS ● ○ ○ ● (FAIL TO START), in this work package.

○ ● ○ ○ (loss of speed data)

ESU BITE INDICATOR SHOWS ○ ● ○ ○ (LOSS OF SPEED DATA) FOR HELICOPTERS WITH ESU 160200-600, in this work package.

○ ○ ● ● (overtemperature)

ESU BITE INDICATOR SHOWS ○ ○ ● ● (OVERTEMPERATURE) in this work package.

○ ○ ○ ● (low oil pressure)

ESU BITE INDICATOR SHOWS ○ ○ ○ ● (LOW OIL PRESSURE), in this work package.

● ● ○ ○ (open thermocouple)

ESU BITE INDICATOR SHOWS ● ● ○ ○ (OPEN THERMOCOUPLE), in this work package.

## APU SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

| IF BITE INDICATION IS:                                                                                                                                                                                                                                                                                                                                                                                                                        | ACTION: GO TO                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ○ ○ ● ○ (external short after, ○ ● ● ● start fuel valve and exciter signal out)                                                                                                                                                                                                                                                                                                                                                               | ESU BITE INDICATOR SHOWS ○ ○ ● ○ (EXTERNAL SHORT) AFTER ○ ● ● ● (START FUEL VALVE AND EXCITER SIGNAL OUT) FOR HELICOPTERS WITH ESU 163291-100, in this work package.          |
| ○ ○ ● ○ (external short after, ○ ○ ● ● signal out; 14%)                                                                                                                                                                                                                                                                                                                                                                                       | ESU BITE INDICATOR SHOWS ○ ○ ● ○ (EXTERNAL SHORT) AFTER ○ ○ ● ● (MAIN FUEL VALVE SIGNAL OUT; 14%) FOR HELICOPTERS WITH ESU 163291-100, in this work package.                  |
| ○ ○ ● ○ (external short after, ○ ○ ○ ○ (90% rpm switch on) or ● ● ● ● (ready for service))                                                                                                                                                                                                                                                                                                                                                    | ESU BITE INDICATOR SHOWS ○ ○ ● ○ (EXTERNAL SHORT) AFTER ○ ○ ○ ○ (90% RPM SWITCH ON) OR ● ● ● ● (READY FOR SERVICE) FOR HELICOPTERS WITH ESU 163291-100, in this work package. |
| <p>6. If APU motors (attempts to start), but shuts down after Indication/Condition b. occurs (APU ON capsule goes off and APU FAIL capsule goes on), go to APU MOTORS BUT SHUTS DOWN (APU ON CAPSULE GOES OFF AND APU FAIL CAPSULE GOES ON), in this work package.</p> <p>7. If APU motors, but shuts down after APU ON capsule goes on, then off, and APU FAIL capsule is off, visually check ESU BITE indications and do the following:</p> |                                                                                                                                                                               |

| IF BITE INDICATION IS:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | ACTION: GO TO                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| ○ ● ○ ● (underspeed)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | ESU BITE INDICATOR SHOWS ○ ● ○ ● (UNDERSPEED), in this work package.        |
| ● ● ○ ● (overspeed)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | ESU BITE INDICATOR SHOWS ● ● ○ ● (OVERSPEED), in this work package.         |
| ○ ○ ● ● (overtemperature)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | ESU BITE INDICATOR SHOWS ○ ○ ● ● (OVERTEMPERATURE), in this work package.   |
| ○ ○ ○ ● (low oil pressure)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | ESU BITE INDICATOR SHOWS ○ ○ ○ ● (LOW OIL PRESSURE), in this work package.  |
| ● ● ○ ○ (open thermocouple)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | ESU BITE INDICATOR SHOWS ● ● ○ ○ (OPEN THERMOCOUPLE), in this work package. |
| <p>8. If APU operates, but Indication/Condition b. is not as specified, go to APU OPERATES, BUT APU ON CAPSULE IS OFF, in this work package.</p> <p>9. If Indication/Condition d. is not as specified, go to APU OIL TEMP HI CAPSULE GOES ON, ESU BITE INDICATOR SHOWS ● ● ● ○ (HIGH TEMPERATURE), in this work package.</p> <p>10. If Indication/Condition e. is not as specified, go to FUEL LEAKS FROM COMBUSTOR DRAIN LINE WITH APU RUNNING, in this work package.</p> <p>11. If ESU BITE indicators do not cycle as specified, go to BITE INDICATORS DO NOT CYCLE WHEN APU CONTR SWITCH PLACED ON, in this work package.</p> |                                                                             |

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**APU SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Step**

10. Recharge APU accumulator by placing GENERATORS APU switch to ON. Visually check accumulator for proper charge (pressure gage and tape indicator).

**Indication/Condition**

APU ACCUM LOW capsule on caution/advisory panel shall go off in 90 seconds and APU accumulator shall be charged.

**Corrective Action**

1. If Indication/Condition is not as specified, troubleshoot hydraulic system (WP 0086 00).
2. If APU shuts down and underspeed BITE indication (○ ● ○ ●) appears on ESU, replace left relay panel (WP 0825 00).

**Step**

11. Place upper console GENERATORS APU switch to OFF.
12. Shut down APU (TM 1-1520-237-10).

**Indication/Condition**

- a. APU shall shut down.
- b. APU ON capsule shall go off.

**Corrective Action**

1. If Indication/Condition a. is not as specified, place FUEL PUMP switch to OFF and go to APU DOES NOT SHUT DOWN WHEN APU CONTR SWITCH IS PLACED OFF, in this work package.
2. If Indication/Condition b. is not as specified, disconnect P260 on ESU.
  - a. If APU ON capsule goes off, replace ESU (WP 0865 00).
  - b. If APU ON capsule does not go off, troubleshoot caution/advisory warning system (WP 0090 00).

**Step**

13. Check ESU BITE indicators after APU is shut down.

**Indication/Condition**

All ESU BITE indicators shall be black.

**Corrective Action**

If Indication/Condition is not as specified, troubleshoot system for indicated malfunction.

**Step**

14. Place upper console FUEL PUMP switch to OFF.

**Indication/Condition**

PRIME BOOST PUMP ON capsule on caution/advisory panel shall go off.

**Corrective Action**

If Indication/Condition is not as specified, troubleshoot fuel prime boost system (WP 0129 00).

**APU SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued**Step**

15. Visually check that there is no fuel or fuel vapors coming from combustor drain line (below helicopter).
  - a. If Indication/Condition is not as specified, check APU combustor drain check valve. Replace if necessary (TM 55-2835-208-23).
16. Place upper console BATT switch to OFF.
17. If no other check is to be done, go to DISCONNECTING SIMULATOR, in this work package.

**Table 1. Normal Start Sequence.**

| <b>APU SPEED<br/>SEQUENCE (%RPM)</b> | <b>EVENT</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A. 5%                                | 1. Start fuel solenoid opens<br>2. Ignition exciter goes on<br>3. 20 - 30-second counter (inside ESU) starts<br>*4. EGT overtemperature shutdown circuit (inside ESU) limit extended to 740°C (1376°F)<br>5. Fuel pressure built up due to motoring speed<br>*6. Thermocouple checked for open condition (inside ESU)<br>7. ESU internal checks for processor sequence failure, sensor/data board failure, and data failure<br>8. Ignition starts<br>*9. Overspeed circuit activates<br>***BITE INDICATION: ○ ● ● ● |
| B. 14%                               | 1. Main fuel solenoid opens<br>2. Ignition continues<br>3. ESU internal self-checks continue<br>*4. Thermocouple checked for open condition (inside ESU)<br>***BITE INDICATION: ○ ○ ● ●                                                                                                                                                                                                                                                                                                                             |
| C. 30%                               | *1. Inside ESU - APU exhaust gas temperature is checked for being above 10°C (50°F)<br>2. ESU internal self-checks continue                                                                                                                                                                                                                                                                                                                                                                                         |

## APU SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Table 1. Normal Start Sequence. - Continued

| APU SPEED<br>SEQUENCE (%RPM) | EVENT                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D. 70%                       | <ol style="list-style-type: none"> <li>1. APU start valve (hydraulic) - closes</li> <li>2. Start fuel valve - closes</li> <li>3. Purge valve - opens (due to air pressure)</li> <li>4. Ignition - off</li> <li>*5. Oil pressure circuit - on</li> <li>6. ESU internal self-checks are made</li> <li>*7. Thermocouple checked for open condition (inside ESU)</li> <li>***BITE INDICATION: ○ ○ ○ ●</li> </ol>                                            |
| E. 90%                       | <ol style="list-style-type: none"> <li>1. This occurs in ESU: <ol style="list-style-type: none"> <li>*a. 20 - 30-second counter - disabled</li> <li>*b. APU fail circuit - deactivated</li> <li>*c. Underspeed circuit - activated</li> <li>d. 1.5-second timer - activated</li> <li>e. ESU internal self-checks continue</li> <li>*f. Thermocouple checked for open condition (inside ESU)</li> <li>***BITE INDICATION: ○ ○ ○ ○</li> </ol> </li> </ol> |
| F. 90% - 1.5 Seconds         | <ol style="list-style-type: none"> <li>1. Maximum fuel valve - opens</li> <li>2. APU ON capsule (caution/advisory panel) - on</li> <li>3. ESU internal self-checks continue</li> <li>**4. Thermocouple checked for short condition (inside ESU)</li> <li>**5. High oil temperature check</li> <li>*6. EGT shutdown limit - reduced to 656°- 679°C (1240°- 1280°F)</li> <li>***BITE INDICATION: ● ● ● ●</li> </ol>                                       |

**APU SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Table 1. Normal Start Sequence. - Continued**

| <b>APU SPEED<br/>SEQUENCE (%RPM)</b>                                                                  | <b>EVENT</b> |
|-------------------------------------------------------------------------------------------------------|--------------|
| G. APU Operating - 102% (Unloaded) APU Operating - 100% (Loaded)                                      |              |
| *ESU checks on system operation. If check is not satisfactory, APU will shut down.                    |              |
| **Warning indication that will appear as a BITE indication, but will not cause APU to shut down.      |              |
| ***BITE indication will appear on ESU when APU has reached corresponding speed during start sequence. |              |

**Indication/Condition**

Disconnecting stimulator operational/troubleshooting procedure shall be performed.

**Corrective Action**

None Required

**ELECTRONIC SEQUENCE UNIT (ESU)****PROCEDURE****Step**

1. These procedures check the ESU.
2. To check the ESU, perform the following procedures.

**Indication/Condition**

The following procedure shall be performed.

**Corrective Action**

None Required

**SIMULATOR SETUP OPERATIONAL/TROUBLESHOOTING PROCEDURE****PROCEDURE****Step****NOTE**

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
  - To prevent discharge of APU accumulator, disconnect P290 from APU start valve.
  - For proper Indication/Conditions, switch positions and order of operation must be strictly followed.
1. Connect cables (supplied with simulator) to simulator and helicopter as follows:
    - Power cable receptacle labeled P1 to J1 on simulator.

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**SIMULATOR SETUP OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**

- Test cable receptacle labeled P2 to J2 on simulator.
  - Other end of power cable to utility receptacle J257 in cabin ceiling.
  - Disconnect P289 from APU. Connect other end of test cable to P289.
2. Make sure all circuit breakers are pushed in except those noted being pulled for maintenance or safety.
  3. If not already done, remove soundproofing material and panel from cabin ceiling to visually check electronic sequence unit (ESU) BITE indicators (WP 0258 00).
  4. Turn on electrical power (WP 1685 00).
  5. Make sure upper console APU CONTR switch is placed OFF.
  6. Turn simulator TEMP OUTPUT knob until temperature indicates 400°C (752°F).

**Indication/Condition**

Temperature 400°C (752°F) shall be indicated.

**Corrective Action**

None Required

**HYDRAULIC START VALVE OPERATIONAL/TROUBLESHOOTING PROCEDURE****PROCEDURE****Step****NOTE**

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- To prevent discharge of APU accumulator, disconnect P290 from APU start valve.
- For proper Indication/Conditions, switch positions and order of operation must be strictly followed.

1. Do SIMULATOR SETUP, in this work package, if not already done.
2. Connect multimeter between P290-1 and ground.
3. Make sure simulator test set (simulator) power is off.
4. Place upper console APU CONTR switch to ON.

**Indication/Condition**

Multimeter shall read between 18 and 28 vdc.

**Corrective Action**

If Indication/Condition is not as specified, go to NO POWER AT APU START VALVE, in this work package.

**Step**

5. Turn on simulator power.



---

**HYDRAULIC START VALVE OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****NOTE**

The next step must be done within 29 seconds.

6. Turn simulator S2 switch to 70%. Turn F1 switch to +1.

**Indication/Condition**

- a. Lamp L1 shall go on, then go off.
- b. Lamp L2 shall go on, and stay on.
- c. ESU BITE indicators shall cycle to 70% speed (○ ○ ○ ●).
- d. Multimeter shall indicate less than 18 vdc.

**Corrective Action**

If Indication/Conditions are not as specified, replace ESU (WP 0865 00).

**Step**

7. If no other check is to be done, go to DISCONNECTING SIMULATOR, in this work package.

**Indication/Condition**

Disconnecting Simulator operational/troubleshooting procedure shall be performed.

**Corrective Action**

None Required

**SPEED SEQUENCE OPERATIONAL/TROUBLESHOOTING PROCEDURE****PROCEDURE****Step****NOTE**

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- To prevent discharge of APU accumulator, disconnect P290 from APU start valve.
- For proper Indication/Conditions, switch positions and order of operation must be strictly followed.

1. Do SIMULATOR SETUP, in this work package, if not already done.
2. Turn simulator power on, if not already done.
3. Turn simulator F1 switch to -1 and S2 switch to 5%. If necessary, press START/STOP RESET button as required until SECONDS display shows 00.
4. Place upper console APU CONTR switch to ON and at same time press START/STOP RESET button.

**Indication/Condition**

- a. ESU BITE indications shall remain the same.
- b. All lamps on simulator shall be off.
- c. Within 29 to 40 seconds, ESU BITE indicators shall cycle to fail to start (● ○ ○ ●).

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**SPEED SEQUENCE OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued

- d. APU FAIL capsule on caution/advisory panel shall go on.

**Corrective Action**

If Indication/Condition c. occurs, but Indication/Condition d. is not as specified, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00).

**Step**

5. Turn simulator F1 switch to 0 and S2 switch to 0. Reset SECONDS display to 00.
6. Place upper console APU CONTR switch to OFF.

**Indication/Condition**

APU FAIL capsule on caution/advisory panel shall go off.

**Corrective Action**

If Indication/Condition is not as specified, replace ESU (WP 0865 00).

**Step**

7. Turn simulator F1 switch to +1 and S2 switch to 5%.
8. Place upper console APU CONTR switch to ON and at same time press START/STOP RESET button.

**Indication/Condition**

- a. ESU BITE indicators shall cycle to 5% speed (○ ● ● ●).
- b. On simulator, lamp L1 shall be on.
- c. Within 29 to 40 seconds, ESU BITE indicators shall cycle to fail to start (● ○ ○ ●).
- d. APU FAIL capsule on caution/advisory panel shall go on.
- e. On simulator, lamp L1 shall go off.

**Corrective Action**

1. If Indication/Condition b. is not as specified, check continuity between P261-17 and P289-17.
  - a. If continuity is not present, repair/replace wiring (WP 1747 00).
  - b. If continuity is present, replace ESU (WP 0865 00).
2. If Indication/Condition d. is not as specified, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00).

**Step**

9. Turn simulator F1 switch to 0 and S2 switch to 0. Reset SECONDS display to 00.
10. Place upper console APU CONTR switch to OFF.

**Indication/Condition**

APU FAIL capsule shall go off and ESU BITE indicators shall cycle to reset (● ● ● ●).

**Corrective Action**

If Indication/Condition is not as specified, replace ESU (WP 0865 00).

---

**SPEED SEQUENCE OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Step**

11. Turn simulator F1 switch to -2 and S2 switch to 15%.
12. Place upper console APU CONTR switch to ON and at same time press START/STOP RESET button.

**Indication/Condition**

- a. ESU BITE indicators shall cycle to 5% speed (○ ● ● ●).
- b. On simulator, lamp L1 shall be on.
- c. Within 29 to 40 seconds, ESU BITE indicators shall cycle to fail to start (● ○ ○ ●).
- d. APU FAIL capsule on caution/advisory panel shall go on.
- e. On simulator, lamp L1 shall go off.

**Corrective Action**

If Indication/Conditions are not as specified, replace ESU (WP 0865 00).

**Step**

13. Turn simulator F1 switch to 0 and S2 switch to 0. Reset SECONDS display to 00.
14. Place upper console APU CONTR switch to OFF.

**Indication/Condition**

APU FAIL capsule on caution/advisory panel shall go off and ESU BITE indicators shall cycle to reset (● ● ● ●).

**Corrective Action**

If Indication/Condition is not as specified, replace ESU (WP 0865 00).

**Step**

15. Turn simulator S2 switch to 14%.
16. Place upper console APU CONTR switch to ON and at same time press START/STOP RESET button.

**Indication/Condition**

- a. ESU BITE indicators shall cycle to 14% speed (○ ○ ● ●).
- b. On simulator, lamps L1 and L2 shall be on.
- c. Within 29 to 40 seconds, ESU BITE indicators shall cycle to fail to start (● ○ ○ ●).
- d. APU FAIL capsule on caution/advisory panel shall go on.
- e. On simulator, lamps L1 and L2 shall go off.

**Corrective Action**

1. If Indication/Conditions are not as specified, check continuity between P261-9 and P289-9.
  - a. If continuity is not present, repair/replace wiring (WP 1747 00).
  - b. If continuity is present, replace ESU (WP 0865 00).

**Step**

17. Turn simulator S2 switch to 0. Reset SECONDS display to 00.

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**SPEED SEQUENCE OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued

18. Place upper console APU CONTR switch to OFF.

**Indication/Condition**

APU FAIL capsule on caution/advisory panel shall go off and ESU BITE indicators shall cycle to reset (● ● ● ●).

**Corrective Action**

If Indication/Condition is not as specified, replace ESU (WP 0865 00).

**Step**

19. Turn simulator F1 switch to -1 and S2 switch to 70%.
20. Place upper console APU CONTR switch to ON and at same time press START/STOP RESET button.

**Indication/Condition**

- a. ESU BITE indicators shall cycle to 15% speed.
- b. On simulator, lamps L1 and L2 shall be on.
- c. Within 29 to 40 seconds, ESU BITE indicators shall cycle to fail to start (● ○ ○ ●).
- d. APU FAIL capsule on caution/advisory panel shall go on.
- e. On simulator, lamps L1 and L2 shall go off.

**Corrective Action**

If Indication/Conditions are not as specified, replace ESU (WP 0865 00).

**Step**

21. Turn simulator F1 switch to 0 and S2 switch to 0. Reset SECONDS display to 00.
22. Place upper console APU CONTR switch to OFF.

**Indication/Condition**

APU FAIL capsule on caution/advisory panel shall go off and ESU BITE indicators shall cycle to reset (● ● ● ●).

**Corrective Action**

If Indication/Condition is not as specified, replace ESU (WP 0865 00).

**Step**

23. Turn simulator F1 switch to +1 and S2 switch to 70%.
24. Place upper console APU CONTR switch to ON and at same time press START/STOP RESET button.

**Indication/Condition**

- a. ESU BITE indicators shall cycle to 70% speed (○ ○ ○ ●).
- b. On simulator, lamp L1 shall go on and then go off, and lamp L2 shall go on and stay on.
- c. Within 29 to 40 seconds, ESU BITE indicators shall cycle to fail to start (● ○ ○ ●).
- d. APU FAIL capsule on caution/advisory panel shall go on.
- e. On simulator, lamp L2 shall go off.

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**SPEED SEQUENCE OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Corrective Action**

If Indication/Conditions are not as specified, replace ESU (WP 0865 00).

**Step**

25. Turn simulator S2 switch to 0. Reset SECONDS display to 00.
26. Place upper console APU CONTR switch from ON to OFF.

**Indication/Condition**

APU FAIL capsule on caution/advisory panel shall go off and ESU BITE indicators shall cycle to reset (● ● ● ●).

**Corrective Action**

If Indication/Condition is not as specified, replace ESU (WP 0865 00).

**Step**

27. If no other check is to be done, go to DISCONNECTING SIMULATOR, in this work package.

**Indication/Condition**

Disconnecting simulator operational/troubleshooting procedure shall be performed.

**Corrective Action**

None Required

**UNDERSPEED OPERATIONAL/TROUBLESHOOTING PROCEDURE  
PROCEDURE****Step****NOTE**

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- To prevent discharge of APU accumulator, disconnect P290 from APU start valve.
- For proper Indication/Conditions, switch positions and order of operation must be strictly followed.

1. Do SIMULATOR SETUP, in this work package, if not already done.
2. Turn simulator power on, if not already done.
3. Turn simulator S2 switch to 90% and make sure F1 switch is at 0.
4. Place upper console APU CONTR switch to ON.

**Indication/Condition**

In approximately 2 seconds, ESU BITE indicators shall cycle to underspeed (○ ● ○ ●) and APU FAIL capsule on caution/advisory panel shall go on.

**Corrective Action**

If Indication/Condition is not as specified, replace ESU (WP 0865 00).

---

**UNDERSPEED OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Step**

5. Turn simulator S2 switch to 0 and F1 switch to 0.
6. Place upper console APU CONTR switch to OFF.

**Indication/Condition**

APU FAIL capsule on caution/advisory panel shall go off and ESU BITE indicators shall cycle to reset (● ● ● ●).

**Corrective Action**

If Indication/Condition is not as specified, replace ESU (WP 0865 00).

**Step**

7. Turn simulator S2 switch to 90% and F1 switch to +5%.

**Indication/Condition**

- a. ESU BITE indicators shall cycle through a normal start and, in approximately 2 seconds, shall cycle to ready for service BITE indication (● ● ● ●).
- b. APU ON capsule on caution/advisory panel shall go on.

**Corrective Action**

If Indication/Conditions are not as specified, replace ESU (WP 0865 00).

**Step**

8. Slowly turn simulator F1 switch from +5 to 0.

**Indication/Condition**

- a. ESU BITE indicators shall cycle to underspeed (○ ● ○ ●) when knob F1 is at +1.
- b. APU ON capsule on caution/advisory panel shall go off.
- c. APU FAIL capsule on caution/advisory panel shall go on.

**Corrective Action**

If Indication/Conditions are not as specified, replace ESU (WP 0865 00).

**Step**

9. Turn simulator S2 switch to 0 and F1 switch to 0.
10. Place upper console APU CONTR switch to OFF.

**Indication/Condition**

APU FAIL capsule on caution/advisory panel shall go off and ESU BITE indicators shall cycle to reset (● ● ● ●).

**Corrective Action**

If Indication/Condition is not as specified, replace ESU (WP 0865 00).

**Step**

11. If no other check is to be done, go to DISCONNECTING SIMULATOR, in this work package.

**Indication/Condition**

Disconnecting simulator operational/troubleshooting procedure shall be performed.

---

**UNDERSPEED OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Corrective Action**

None Required

**OVERSPEED OPERATIONAL/TROUBLESHOOTING PROCEDURE  
PROCEDURE****Step****NOTE**

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- To prevent discharge of APU accumulator, disconnect P290 from APU start valve.
- For proper Indication/Conditions, switch positions and order of operation must be strictly followed.

1. Do SIMULATOR SETUP, in this work package, if not already done.
2. Turn simulator power on, if not already done.
3. Turn simulator F1 switch to -5, then turn S2 switch to 110% speed.
4. Place upper console APU CONTR switch to ON.

**Indication/Condition**

- a. ESU BITE indicators shall cycle from 90% speed to ready for service (● ● ● ●) in approximately 2 seconds.
- b. APU ON capsule on caution/advisory panel shall go on.
- c. On simulator, lamps L2 and L3 shall go on.

**Corrective Action**

If Indication/Conditions are not as specified, replace ESU (WP 0865 00).

**Step**

5. Turn simulator F1 switch to 0.

**Indication/Condition**

- a. ESU BITE indicators shall cycle to overspeed (● ● ○ ●).
- b. On simulator, lamps L2 and L3 shall go off.
- c. APU ON capsule on caution/advisory panel shall go off.
- d. APU FAIL capsule on caution/advisory panel shall go on.

**Corrective Action**

If Indication/Conditions are not as specified, replace ESU (WP 0865 00).

**Step**

6. Turn S2 switch to 0%.
7. Place upper console APU CONTR switch to OFF.

---

**OVERSPEED OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Indication/Condition**

APU FAIL capsule on caution/advisory panel shall go off and ESU BITE indicators shall cycle to reset (● ● ● ●).

**Corrective Action**

If Indication/Condition is not as specified, replace ESU (WP 0865 00).

**Step**

8. If no other check is to be done, go to DISCONNECTING SIMULATOR, in this work package.

**Indication/Condition**

Disconnecting simulator operational/troubleshooting procedure shall be performed.

**Corrective Action**

None Required

**LOW OIL PRESSURE OPERATIONAL/TROUBLESHOOTING PROCEDURE  
PROCEDURE****Step****NOTE**

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- To prevent discharge of APU accumulator, disconnect P290 from APU start valve.
- For proper Indication/Conditions, switch positions and order of operation must be strictly followed.

1. Do SIMULATOR SETUP, in this work package, if not already done.
2. Turn simulator power on, if not already done.

**NOTE**

Observer must watch ESU BITE indicators during the test. The next two steps must be done within 29 seconds.

3. Turn simulator F1 switch to -2. Turn S2 switch to 70%.
4. Place upper console APU CONTR switch to ON.

**Indication/Condition**

ESU BITE indicators shall cycle up to 15% speed and on simulator, lamps L1 and L2 shall go on.

**Corrective Action**

1. If Indication/Condition is not as specified, check continuity between P289-7 and P261-7.
  - a. If continuity is not present, repair/replace wiring (WP 1747 00).
  - b. If continuity is present, replace ESU (WP 0865 00).



---

**LOW OIL PRESSURE OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued**Step**

5. Turn simulator F1 switch to 0 and S2 switch to 0.
6. Place upper console APU CONTR switch to OFF.

**Indication/Condition**

APU FAIL capsule on caution/advisory panel shall go off and ESU BITE indicators shall cycle to reset (● ● ● ●).

**Corrective Action**

If Indication/Condition is not as specified, replace ESU (WP 0865 00).

**Step**

7. Turn simulator F1 switch to +1 and S2 switch to 70%.
8. Place upper console APU CONTR switch to ON.

**Indication/Condition**

- a. ESU BITE indicators shall cycle up to 70% speed (○ ○ ○ ●).
- b. On simulator, lamp L1 shall go on, then go off. Lamp L2 shall go on and stay on.

**Corrective Action**

If Indication/Conditions are not as specified, replace ESU (WP 0865 00).

**Step**

9. Place simulator S3 switch to LOP.

**Indication/Condition**

- a. ESU BITE indicators shall cycle from 70% speed to low oil pressure indication (○ ○ ○ ●) (BITEs will momentarily move, but shall remain the same).
- b. On simulator, lamp L6 shall go on when switch is pushed to LOP.
- c. APU FAIL capsule on caution/advisory panel shall go on.

**Corrective Action**

If Indication/Conditions are not as specified, replace ESU (WP 0865 00).

**Step**

10. Turn simulator F1 switch to 0 and S2 switch to 0.
11. Place upper console APU CONTR switch to OFF.

**Indication/Condition**

APU FAIL capsule on caution/advisory panel shall go off and ESU BITE indicators shall cycle to reset (● ● ● ●).

**Corrective Action**

If Indication/Condition is not as specified, replace ESU (WP 0865 00).

**Step**

12. Turn simulator F1 switch to +1 and S2 switch to 90%.
13. Place upper console APU CONTR switch to ON.

---

**LOW OIL PRESSURE OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Indication/Condition**

- a. ESU BITE indicators shall cycle to 90% speed, and in approximately 2 seconds, cycle to ready for service (● ● ● ●).
- b. APU ON capsule on caution/advisory panel shall go on.

**Corrective Action**

If Indication/Conditions are not as specified, replace ESU (WP 0865 00).

**Step**

14. Place simulator S3 switch to LOP.

**Indication/Condition**

ESU BITE indicators shall cycle to low oil pressure (○ ○ ○ ●) and APU FAIL capsule on caution/advisory panel shall go on.

**Corrective Action**

If Indication/Condition is not as specified, replace ESU (WP 0865 00).

**Step**

15. Turn simulator F1 switch to 0 and S2 switch to 0.
16. Place upper console APU CONTR switch to OFF.

**Indication/Condition**

APU FAIL capsule on caution/advisory panel shall go off and ESU BITE indicators shall cycle to reset (● ● ● ●).

**Corrective Action**

If Indication/Condition is not as specified, replace ESU (WP 0865 00).

**Step**

17. If no other check is to be done, go to DISCONNECTING SIMULATOR, in this work package.

**Indication/Condition**

Disconnecting simulator operational/troubleshooting procedure shall be performed.

**Corrective Action**

None Required

**HIGH OIL TEMPERATURE OPERATIONAL/TROUBLESHOOTING PROCEDURE****PROCEDURE****Step****NOTE**

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- To prevent discharge of APU accumulator, disconnect P290 from APU start valve.
- For proper Indication/Conditions, switch positions and order of operation must be strictly followed.

1. Do SIMULATOR SETUP, in this work package, if not already done.
2. Turn simulator power on, if not already done.
3. Turn simulator F1 switch to +1 and S2 switch to 90%.
4. Place upper console APU CONTR switch to ON.

**Indication/Condition**

- a. ESU BITE indicators shall cycle from 90% speed to ready for service (● ● ● ●) in approximately 2 seconds.
- b. On simulator, lamp L1 shall be off, lamps L2 and L3 shall be on.
- c. APU ON capsule on caution/advisory panel shall be on.

**Corrective Action**

If Indication/Conditions are not as specified, replace ESU (WP 0865 00).

**Step****NOTE**

If ESU BITE indicators indicate high oil temperature (● ● ● ○) before S3 switch is used, replace ESU.

5. Place simulator S3 switch to HI OIL TEMP.

**Indication/Condition**

- a. ESU BITE indicators shall cycle from ready for service to high oil temperature (● ● ● ○).
- b. On simulator, lamp L5 shall go on.
- c. APU OIL TEMP HI capsule on caution/advisory panel shall go on.

**Corrective Action**

1. If Indication/Condition a. is not as specified, replace ESU (WP 0865 00).
2. If Indication/Condition a. occurs, but Indication/Condition c. is not as specified, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00).

---

**HIGH OIL TEMPERATURE OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Step**

6. Place upper console APU CONTR switch to OFF.

**Indication/Condition**

APU OIL TEMP HI capsule on caution/advisory panel shall go off and ESU BITE indicators shall cycle to reset (● ● ●).

**Corrective Action**

If Indication/Condition is not as specified, replace ESU (WP 0865 00).

**Step**

7. If no other check is to be done, go to DISCONNECTING SIMULATOR, in this work package.

**Indication/Condition**

Disconnecting simulator operational/troubleshooting procedure shall be performed.

**Corrective Action**

None Required

**START BYPASS VALVE OPERATIONAL/TROUBLESHOOTING PROCEDURE  
PROCEDURE****Step****NOTE**

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- To prevent discharge of APU accumulator, disconnect P290 from APU start valve.
- For proper Indication/Conditions, switch positions and order of operation must be strictly followed.

1. Do SIMULATOR SETUP, in this work package, if not already done.
2. Turn simulator power on, if not already done.
3. Place upper console AIR SOURCE HEAT/START switch to APU.
4. Turn simulator S2 switch to 90%. Turn F1 switch to +2.
5. Place upper console APU CONTR switch to ON.

**Indication/Condition**

- a. ESU BITE indicators shall cycle from 90% speed to ready for service indication (● ● ● ●) in approximately 2 seconds.
- b. APU ON capsule on caution/advisory panel shall be on.

**Corrective Action**

1. If Indication/Conditions are not as specified, check continuity between ESU and P289.
  - a. If continuity is not present, repair/replace wiring (WP 1747 00).

---

**START BYPASS VALVE OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued

- b. If continuity is present, replace ESU (WP 0865 00).

**Step**

6. Press, then release No. 1 engine start button.

**Indication/Condition**

Whenever No. 1 engine start button is pressed, lamp L4 on simulator shall go on and #1 ENGINE STARTER capsule, on caution/advisory panel, shall go on.

**Corrective Action**

If Indication/Condition is not as specified, go to APU STALL WHEN EITHER ENGINE START BUTTON IS PRESSED (A POPPING SOUND IS HEARD), in this work package.

**Step**

7. Press, then release No. 2 engine start button.

**Indication/Condition**

Whenever No. 2 engine start button is pressed, lamp L4 on simulator shall go on and #2 ENGINE STARTER capsule, on caution/advisory panel, shall go on.

**Corrective Action**

If Indication/Condition is not as specified, go to APU STALL WHEN EITHER ENGINE START BUTTON IS PRESSED (A POPPING SOUND IS HEARD), in this work package.

**Step**

8. Place upper console APU CONTR switch to OFF.  
9. Place upper console AIR SOURCE HEAT/START switch to OFF.  
10. If no other check is to be done, go to DISCONNECTING SIMULATOR, in this work package.

**Indication/Condition**

Disconnecting simulator operational/troubleshooting procedure shall be performed.

**Corrective Action**

None Required

**OVERTEMPERATURE OPERATIONAL/TROUBLESHOOTING PROCEDURE****PROCEDURE****Step****NOTE**

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- To prevent discharge of APU accumulator, disconnect P290 from APU start valve.
- For proper Indication/Conditions, switch positions and order of operation must be strictly followed.

1. Do SIMULATOR SETUP, in this work package, if not already done.
2. Turn simulator power on, if not already done.

**NOTE**

The next three steps must be done in less than 29 seconds.

3. Turn simulator F1 switch to -5. Turn S2 switch to 90%.
4. Place upper console APU CONTR switch to ON.
5. Slowly turn simulator TEMP OUTPUT - R1 knob until ESU BITE indicators cycle to overtemperature (○ ○ ● ●).

**Indication/Condition**

- a. ESU BITE indicators shall cycle up to 70% BITE indication.
- b. As knob is turned, ESU BITE indicators shall cycle to overtemperature between 1364° and 1429°F (740° and 776°C).
- c. APU FAIL capsule on caution/advisory panel shall go on.

**Corrective Action**

If Indication/Conditions are not as specified, replace ESU (WP 0865 00).

**Step**

6. Turn simulator F1 switch to 0 and S2 to 0. Turn TEMP OUTPUT knob R1 until temperature indicates 400°C (752°F).
7. Place upper console APU CONTR switch to OFF.

**Indication/Condition**

APU OIL TEMP HI capsule on caution/advisory panel shall go off and ESU BITE indicators shall cycle to reset (● ● ● ●).

**Corrective Action**

If Indication/Condition is not as specified, replace ESU (WP 0865 00).

**Step**

8. Turn simulator F1 switch to +5. Turn S2 switch to 90%.

---

**OVERTEMPERATURE OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**

9. Place upper console APU CONTR switch to ON.

**Indication/Condition**

ESU BITE indicators shall cycle up to ready for service indication (● ● ● ●) and APU ON capsule on caution/advisory panel shall be on.

**Corrective Action**

If Indication/Condition is not as specified, replace ESU (WP 0865 00).

**Step**

10. Slowly turn simulator TEMP OUTPUT knob until ESU BITE indicators cycle to overtemperature.

**Indication/Condition**

As knob is turned, ESU BITE indicators shall cycle to overtemperature between 1242° and 1282°F (672° and 694°C), and APU FAIL capsule on caution/advisory panel shall be on.

**Corrective Action**

If Indication/Condition is not as specified, replace ESU (WP 0865 00).

**Step**

11. Turn simulator F1 switch to 0 and S2 switch to 0.  
12. Place upper console APU CONTR switch to OFF.

**Indication/Condition**

APU OIL TEMP HI capsule on caution/advisory panel shall go off and ESU BITE indicators shall cycle to reset (● ● ● ●).

**Corrective Action**

If Indication/Condition is not as specified, replace ESU (WP 0865 00).

**Step**

13. Turn TEMP OUTPUT knob until temperature indicates approximately 400°C (752°F).  
14. If no other check is to be done, go to DISCONNECTING SIMULATOR, in this work package.

**Indication/Condition**

Disconnecting simulator operational/troubleshooting procedure shall be performed.

**Corrective Action**

None Required

## OPEN THERMOCOUPLE OPERATIONAL/TROUBLESHOOTING PROCEDURE

### PROCEDURE

#### Step

#### NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- To prevent discharge of APU accumulator, disconnect P290 from APU start valve.
- For proper Indication/Conditions, switch positions and order of operation must be strictly followed.

1. Do SIMULATOR SETUP, in this work package, if not already done.
2. Turn simulator power on, if not already done.
3. Place upper console APU CONTR switch to ON.

#### Indication/Condition

- a. ESU BITE indicators shall cycle up to ready for service indication (● ● ● ●).
- b. APU ON capsule on caution/advisory panel shall go on.
- c. On simulator, lamps L2 and L3 shall go on.

#### Corrective Action

If Indication/Conditions are not as specified, replace ESU (WP 0865 00).

#### Step

#### NOTE

Disregard initial ESU BITE indications when connector is disconnected.

4. Disconnect P2 from test receptacle J2.

#### Indication/Condition

- a. Simulator lamps L2 and L3 shall go off.
- b. APU ON capsule on caution/advisory panel shall go off.
- c. Approximately 12 to 24 seconds after connector was pulled, BITE indicators shall cycle to open thermocouple indication (● ● ○ ○).
- d. APU FAIL capsule on caution/advisory panel shall be on.

#### Corrective Action

1. If Indication/Conditions are not as specified, check continuity between the following points:
  - P289-1 and P289-2
  - P261-1 and P261-2
- a. If continuity is not present, repair/replace wiring (WP 1747 00).
- b. If continuity is present, replace ESU (WP 0865 00).



**OPEN THERMOCOUPLE OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Step**

5. Reconnect P2 to J2 receptacle on simulator.
6. Turn simulator F1 switch to 0 and S2 switch to 0.
7. Place upper console APU CONTR switch to OFF.

**Indication/Condition**

APU OIL TEMP HI capsule on caution/advisory panel shall go off and ESU BITE indicators shall cycle to reset (● ● ● ●).

**Corrective Action**

If Indication/Condition is not as specified, replace ESU (WP 0865 00).

**Step**

8. If no other check is to be done, go to DISCONNECTING SIMULATOR, in this work package.

**Indication/Condition**

Disconnecting simulator operational/troubleshooting procedure shall be performed.

**Corrective Action**

None Required

**SHORTED THERMOCOUPLE OPERATIONAL/TROUBLESHOOTING PROCEDURE**  
**PROCEDURE**
**Step****NOTE**

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- To prevent discharge of APU accumulator, disconnect P290 from APU start valve.
- For proper Indication/Conditions, switch positions and order of operation must be strictly followed.

1. Do SIMULATOR SETUP, in this work package, if not already done.
2. Turn simulator power on, if not already done.
3. Turn simulator F1 switch to +5 and S2 switch to 90%.
4. Place upper console APU CONTR switch to ON.
5. On helicopters with APU ESU 160200-201, on simulator, rotate R1 switch to 10° - 13°C (50°-55°F).

**Indication/Condition**

ESU BITE indicator shall not be (● ○ ● ○) shorted thermocouple.

**Corrective Action**

If Indication/Condition is not as specified, go to ESU BITE INDICATORS SHOW ● ○ ● ○ (SHORTED THERMOCOUPLE), in this work package.

---

**SHORTED THERMOCOUPLE OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Step**

6. On helicopters with ESU 160200-600, on simulator, rotate R1 switch to 118° - 130°C (244°-266°F).

**Indication/Condition**

ESU BITE indicator shall not be (● ○ ● ○) shorted thermocouple.

**Corrective Action**

If Indication/Condition is not as specified, go to ESU BITE INDICATORS SHOW ● ○ ● ○ (SHORTED THERMOCOUPLE), in this work package.

**Step**

7. Place upper console APU CONTR switch to OFF.

**Indication/Condition**

APU OIL TEMP HI capsule on caution/advisory panel shall go off and ESU BITE indicators shall cycle to reset (● ● ● ●).

**Corrective Action**

If Indication/Condition is not as specified, replace ESU (WP 0865 00).

**Step**

8. If no other check is to be done, go to DISCONNECTING SIMULATOR, in this work package.

**Indication/Condition**

Disconnecting simulator operational/troubleshooting procedure shall be performed.

**Corrective Action**

None Required

**DISCONNECTING SIMULATOR OPERATIONAL/TROUBLESHOOTING PROCEDURE****Step**

1. Place upper console APU CONTR switch to OFF.

**Indication/Condition**

ESU BITE indicators shall cycle to reset (● ● ● ●).

**Corrective Action**

If Indication/Condition is not as specified, replace ESU (WP 0865 00).

**Step**

2. Turn off electrical power.
3. Make sure simulator switches are placed as follows:
  - F1-0
  - S2-0%

---

**DISCONNECTING SIMULATOR OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued

- R1-400°C
  - S3-NORM
  - S1-OFF
4. Make sure upper console AIR SOURCE/HEAT START switch is placed OFF.
  5. Disconnect power cable from receptacle in cabin ceiling and simulator. Disconnect test cable from P289 and simulator. Connect P289 to APU.
  6. Install test cables in simulator.
  7. Make sure P290 is connected to APU hydraulic start valve.
  8. If necessary, go back to troubleshooting chart.
  9. On cabin ceiling, install panel and soundproofing material over ESU (WP 0258 00).

**Indication/Condition**

Panel and soundproofing material shall be installed.

**Corrective Action**

None Required

**ESU BITE INDICATORS DO NOT CYCLE WHEN BATT SWITCH IS PLACED ON****SYMPTOM**

ESU BITE indicators do not cycle when BATT switch is placed ON.

**MALFUNCTION**

ESU BITE indicators do not cycle when BATT switch is placed ON.

**CORRECTIVE ACTION**

- 1. Check for 28 vdc between P260-5 and ground.**
  - a. If voltage is as specified, go to **2**.
  - b. If voltage is not as specified, go to **3**.
- 2. Check continuity between P260-6 and ground.**
  - a. If continuity is present, replace ESU (WP 0865 00). Go to **10**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **10**.
- 3. Install an insulated jumper wire between P242-n and P242-q. Go to 4.**
- 4. Check for 28 vdc between P260-5 and ground.**
  - a. If voltage is as specified, go to **5**.
  - b. If voltage is not as specified, go to **7**.
- 5. Remove jumper wire from P242. Go to 6.**
- 6. Replace left relay panel (WP 0825 00). Go to 10.**
- 7. Remove jumper wire from P242. Go to 8.**
- 8. Check for 28 vdc between P242-q and ground.**

---

**ESU BITE INDICATORS DO NOT CYCLE WHEN BATT SWITCH IS PLACED ON - Continued**

- a. If voltage is as specified, repair/replace wiring between P242-n and P260-5 (WP 1747 00). Go to **10**.
- b. If voltage is not as specified, go to **9**.
- 9. Check for 28 vdc between APU CONTR INST circuit breaker terminal 1 and ground.**
  - a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P242-q (WP 1747 00). Go to **10**.
  - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **10**.
- 10. Procedure completed.**

**FUEL LEAKS FROM APU COMBUSTOR DRAIN LINE WITH PRIME BOOST PUMP RUNNING****SYMPTOM**

Fuel leaks from APU combustor drain line with prime boost pump running.

**MALFUNCTION**

Fuel leaks from APU combustor drain line with prime boost pump running.

**CORRECTIVE ACTION**

- 1. Disconnect P305 from start fuel valve, P306 from main fuel valve, and P303 from ignition exciter. Go to 2.**
- 2. Check for fuel leakage.**
  - a. If any fuel leakage is present, go to **3**.
  - b. If no fuel leakage is present, replace ESU (WP 0865 00). Go to **5**.
- 3. Disconnect line to main fuel manifold and purge fuel into container. Go to 4.**
- 4. Using APU CONTR switch, motor APU and check for fuel flow.**
  - a. If fuel flows, replace main fuel valve (TM 55-2835-208-23). Go to **5**.
  - b. If fuel does not flow, replace fuel start valve (TM 55-2835-208-23). Go to **5**.
- 5. Procedure completed.**

**ESU BITE INDICATORS SHOW ● ○ ● ○ (SHORTED THERMOCOUPLE)****SYMPTOM**

ESU BITE indicators show ● ○ ● ○ (shorted thermocouple).

**MALFUNCTION**

ESU BITE indicators show ● ○ ● ○ (shorted thermocouple).

**CORRECTIVE ACTION**

- 1. Disconnect P289 from APU. Go to 2.**
- 2. Check continuity between P261-1 and P261-2.**
  - a. If continuity is present, repair/replace wiring, as required, between (WP 1747 00), then go to **6: P261-1 and P289-1, P261-2 and P289-2**
  - b. If continuity is not present, go to **3**.

---

**ESU BITE INDICATORS SHOW ● ○ ● ○ (SHORTED THERMOCOUPLE) - Continued**

- 3. Replace APU thermocouple harness (TM 55-2835-208-23). Go to 4.**
- 4. Start and operate APU (TM 1-1520-237-10). Go to 5.**
- 5. Does APU operate normally?**
  - a. If APU operates normally, system is operational. Go to **6**.
  - b. If APU does not operate normally and ESU BITE indicators show ○ ● ● ○ (fail to light), replace ESU for helicopters with ESU 160200-201 (WP 0865 00). Go to **6**.
- 6. Procedure completed.**

**ESU BITE INDICATOR SHOWS ○ ● ● ○ (FAIL TO LIGHT) FOR HELICOPTERS WITH ESU 160200-201****SYMPTOM**

ESU BITE indicator shows ○ ● ● ○ (fail to light) for helicopters with ESU 160200-201.

**MALFUNCTION**

ESU BITE indicator shows ○ ● ● ○ (fail to light) for helicopters with ESU 160200-201.

**CORRECTIVE ACTION**

- 1. Visually check for fuel vapor at exhaust duct or overboard drain.**
  - a. If vapor is present, go to **2**.
  - b. If vapor is not present, go to **13**.
- 2. Do ESU SPEED SEQUENCE procedure, in this work package, to verify signal M to ignition exciter.**
  - a. If speed sequence is good, replace igniter plug (TM 55-2835-208-23). Go to **3**.
- 3. Start and operate APU (TM 1-1520-237-10). Go to 4.**
- 4. Does APU operate normally?**
  - a. If APU operates normally, system is operational. Go to **24**.
  - b. If APU does not operate normally and ESU BITE indicator shows fail to light, go to **5**.
- 5. Install an insulated jumper wire between P303-A and P303-B. Go to 6.**
- 6. Check continuity between J289-17 and ground.**
  - a. If continuity is present, go to **7**.
  - b. If continuity is not present, go to **11**.
- 7. Remove jumper wire. Go to 8.**
- 8. Replace exciter cable (TM 55-2835-208-23). Go to 9.**
- 9. Start and operate APU (TM 1-1520-237-10). Go to 10.**
- 10. Does APU operate normally?**
  - a. If APU operates normally, system is operational. Go to **24**.
  - b. If APU does not operate normally and ESU BITE indicator shows fail to light, replace ignition exciter (TM 55-2835-208-23). Go to **24**.
- 11. Remove jumper wire. Go to 12.**

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**ESU BITE INDICATOR SHOWS ○ ● ● ○ (FAIL TO LIGHT) FOR HELICOPTERS WITH ESU 160200-201 -**  
Continued

12. Repair/replace wiring between J289-17 and ignition exciter (WP 1747 00). Go to 13.
13. Do ESU SPEED SEQUENCE procedure, in this work package, to verify signal M to start fuel valve.
  - a. If speed sequence is good, replace start fuel nozzle (TM 55-2835-208-23). Go to 14.
14. Start and operate APU (TM 1-1520-237-10). Go to 15.
15. Does APU operate normally?
  - a. If APU operates normally, system is operational. Go to 24.
  - b. If APU does not operate normally and ESU BITE indicator shows fail to light, go to 16.
16. Install an insulated jumper wire between P305-A and P305-B. Go to 17.
17. Check continuity between J289-17 and ground.
  - a. If continuity is present, go to 18.
  - b. If continuity is not present, go to 22.
18. Remove jumper wire. Go to 19.
19. Replace fuel start valve (TM 55-2835-208-23). Go to 20.
20. Start and operate APU (TM 1-1520-237-10). Go to 21.
21. Does APU operate normally?
  - a. If APU operates normally, system is operational. Go to 24.
  - b. If APU does not operate normally and ESU BITE indicator shows fail to light, replace start nozzle purge valve (TM 55-2835-208-23). Go to 24.
22. Remove jumper wire. Go to 23.
23. Repair/replace wiring between J289-17 and start fuel valve (WP 1747 00). Go to 24.
24. Procedure completed.

**ESU BITE INDICATOR SHOWS ○ ● ● ○ (SHORTED OIL PRESSURE SWITCH) FOR HELICOPTERS WITH ESU 160200-600****SYMPTOM**

ESU BITE indicator shows ○ ● ● ○ (shorted oil pressure switch) for helicopters with ESU 160200-600.

**MALFUNCTION**

ESU BITE indicator shows ○ ● ● ○ (shorted oil pressure switch) for helicopters with ESU 160200-600.

**CORRECTIVE ACTION**

1. Check continuity between: J289-7 and J289-16, J289-7 and J289-23.
  - a. If continuity is present, go to 2.
  - b. If continuity is not present, repair/replace wiring between J289 and P304 (WP 1747 00). Go to 7.
  - c. If trouble remains, replace low oil pressure switch (TM 55-2835-208-23). Go to 7.
2. Check for 28 vdc between P304-C and ground.
  - a. If voltage is as specified, replace low oil pressure switch (TM 55-2835-208-23). Go to 7.

## ESU BITE INDICATOR SHOWS ○ ● ● ○ (SHORTED OIL PRESSURE SWITCH) FOR HELICOPTERS WITH ESU 160200-600 - Continued

- b. If voltage is not as specified, go to **3**.
- 3. Check for 28 vdc between P289-16 and ground.**
  - a. If voltage is as specified, repair/replace wiring between J289 and P304 (WP 1747 00). Go to **7**.
  - b. If trouble remains, repair/replace wiring between P261 and P304 (WP 1747 00). Go to **7**.
  - c. If voltage is not as specified, go to **4**.
- 4. Check for 28 vdc between ESU J1-16 and ground.**
  - a. If voltage is as specified, repair/replace wiring between P261 and TB301-1 (WP 1747 00). Go to **7**.
  - b. If voltage is not as specified, go to **5**.
- 5. Check for 28 vdc between P260-5 and ground.**
  - a. If voltage is as specified, replace ESU (WP 0865 00). Go to **7**.
  - b. If voltage is not as specified, go to **6**.
- 6. Check continuity between J242-q and J242-n.**
  - a. If continuity is present, repair/replace wiring between P242 and P260 (WP 1747 00). Go to **7**.
  - b. If continuity is not present, replace left relay panel (TM 1-1520-237-23). Go to **7**.
- 7. Procedure completed.**

## APU MOTOR DOES NOT MOTOR (NO ROTATION)

### SYMPTOM

APU motor does not motor (no rotation).

### MALFUNCTION

APU motor does not motor (no rotation).

### CORRECTIVE ACTION

- 1. Visually check for any of the following BITE indications:**
  - ● ● ●
  - ○ ● ●
  - ○ ○ ○
  - ○ ○ ○
  - a. If any indication listed is present, replace ESU (WP 0865 00). Go to **9**.
  - b. If any indication listed is not present, go to **2**.
- 2. Make sure APU CONTR switch is placed OFF. Go to 3.**
- 3. Press APU start valve manual override lever. Go to 4.**
- 4. Check that APU start motor operates.**
  - a. If motor operates, go to **5**.
  - b. If motor does not operate, go to **6**.

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**APU MOTOR DOES NOT MOTOR (NO ROTATION) - Continued**

- 5. Do ESU HYDRAULIC START VALVE procedure, in this work package.**
  - a. If ESU hydraulic start valve is good, replace APU hydraulic start valve (WP 0743 00). Go to **9**.
- 6. Disconnect start motor from APU accessory pad. Go to 7.**
- 7. Using a 1/2 inch allen wrench, turn splined shaft on accessory pad. Check that shaft turns.**
  - a. If shaft turns, go to **8**.
  - b. If shaft does not turn, replace seized APU (TM 1-1520-237-23). Go to **9**.
- 8. Attempt to turn splined shaft of APU start motor.**
  - a. If shaft turns, replace APU hydraulic start valve (WP 0743 00). Go to **9**.
  - b. If shaft does not turn, replace APU start motor (WP 1365 00). Go to **9**.
- 9. Procedure completed.**

**ESU BITE INDICATOR SHOWS ● ○ ○ ● (FAIL TO START)****SYMPTOM**

ESU BITE indicator shows ● ○ ○ ● (fail to start).

**MALFUNCTION**

ESU BITE indicator shows ● ○ ○ ● (fail to start).

**CORRECTIVE ACTION**

- 1. What was APU speed BITE indication before it changed to fail to start indication?**
  - a. If APU speed was over 14% and less than 70% ○ ○ ● ●, go to **2**.
  - b. If APU speed was over 70% and less than 90% ○ ○ ○ ●, go to **15**.
- 2. Check that main fuel valve actuates (determined by feeling the valve for a thump) during start sequence.**
  - a. If valve actuates, go to **3**.
  - b. If valve does not actuate, go to **13**.
- 3. Loosen B-nut on inlet fuel filter inlet. Place rag around nut. Go to 4.**
- 4. On upper console panel, place FUEL PUMP switch to APU BOOST. Go to 5.**
- 5. Check that fuel is going into rag.**
  - a. If fuel is present, go to **6**.
  - b. If no fuel is present, go to **11**.
- 6. Tighten B-nut. Go to 7.**
- 7. Loosen B-nut on inlet fuel filter outlet. Place rag around nut. Go to 8.**
- 8. Place upper console FUEL PUMP switch to APU BOOST. Go to 9.**
- 9. Check that fuel is going into rag.**
  - a. If fuel is present, go to **10**.



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**ESU BITE INDICATOR SHOWS ● ○ ○ ● (FAIL TO START) - Continued**

b. If fuel is not present, replace filter (TM 55-2835-208-23). Go to **24**.

**10. Check for restricted outlet filter.**

a. If filter is restricted, replace fuel pump (TM 55-2835-208-23). Go to **24**.

b. If filter is not restricted, replace main fuel nozzles/manifold assembly (TM 55-2835-208-23). Go to **24**.

**11. Tighten B-nut. Go to 12.**

**12. Troubleshoot fuel prime/boost system (WP 0129 00). Go to 24.**

**13. Check for 28 vdc between P306-A and ground.**

a. If voltage is as specified, go to **14**.

b. If voltage is not as specified, do ESU SPEED SEQUENCE procedure steps 1. through 13. and step 39., in this work package, to verify power to main fuel valve. Go to **24**.

**14. Check continuity between P306-B and ground.**

a. If continuity is present, replace main fuel valve (TM 55-2835-208-23). Go to **24**.

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **24**.

**15. Make sure all bleed-air ducts are closed and driven equipment is unloaded. Check air inlet for foreign object debris (FOD). Go to 16.**

**16. Restart APU (TM 1-1520-237-10). Go to 17.**

**17. Does APU operate normally?**

a. If APU operates normally, system is operational. Go to **24**.

b. If APU does not operate normally, go to **18**.

**18. Replace fuel manifold (TM 55-2835-208-23). Go to 19.**

**19. Restart APU (TM 1-1520-237-10). Go to 20.**

**20. Does APU operate normally?**

a. If APU operates normally, system is operational. Go to **24**.

b. If APU does not operate normally, go to **21**.

**21. Replace fuel pump/acceleration control (TM 55-2835-208-23). Go to 22.**

**22. Restart APU (TM 1-1520-237-10). Go to 23.**

**23. Does APU operate normally?**

a. If APU operates normally, system is operational. Go to **24**.

b. If APU does not operate normally, replace APU (WP 1359 00). Go to **24**.

**24. Procedure completed.**

**ESU BITE INDICATOR SHOWS ○ ● ○ ○ (LOSS OF SPEED DATA) FOR HELICOPTERS WITH ESU 160200-600****SYMPTOM**

ESU BITE indicator shows ○ ● ○ ○ (loss of speed data) for helicopters with ESU 160200-600.

**MALFUNCTION**

ESU BITE indicator shows ○ ● ○ ○ (loss of speed data) for helicopters with ESU 160200-600.

**CORRECTIVE ACTION**

- 1. Check for 1,190 to 1,450 ohms between P261-12 and P261-13.**
  - a. If resistance is as specified, replace ESU (WP 0865 00). Go to **8**.
  - b. If resistance is not as specified, go to **2**.
- 2. Install an insulated jumper wire between P302-A and P302-B. Go to 3.**
- 3. Check continuity between P261-12 and P261-13.**
  - a. If continuity is present, go to **4**.
  - b. If continuity is not present, go to **6**.
- 4. Remove jumper wire from P302. Go to 5.**
- 5. Replace magnetic pickup (TM 55-2835-208-23). Go to 8.**
- 6. Remove jumper wire from P302. Go to 7.**
- 7. Repair/replace wiring between P261 and P302 (WP 1747 00). Go to 8.**
- 8. Procedure completed.**

**ESU BITE INDICATOR SHOWS ○ ○ ● ● (OVERTEMPERATURE)****SYMPTOM**

ESU BITE indicator shows ○ ○ ● ● (overtemperature).

**MALFUNCTION**

ESU BITE indicator shows ○ ○ ● ● (overtemperature).

**CORRECTIVE ACTION**

- 1. Do ESU OVERTEMPERATURE procedure.**
  - a. If APU overtemperature is good, go to **2**.
- 2. Replace combustor drain valve (TM 55-2835-208-23). Go to 3.**
- 3. Start and operate APU (TM 1-1520-237-10). Go to 4.**
- 4. Does APU operate normally?**
  - a. If APU operates normally, system is operational. Go to **8**.
  - b. If APU does not operate normally and ESU BITE indicator shows overtemperature, go to **5**.
- 5. Replace maximum fuel valve (TM 55-2835-208-23). Go to 6.**
- 6. Start and operate APU (TM 1-1520-237-10). Go to 7.**
- 7. Does APU operate normally?**

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**ESU BITE INDICATOR SHOWS ○ ○ ● ● (OVERTEMPERATURE) - Continued**

- a. If APU operates normally, system is operational. Go to **8**.
- b. If APU does not operate normally, replace fuel pump/acceleration control (TM 55-2835-208-23). Go to **8**.

**8. Procedure completed.****ESU BITE INDICATOR SHOWS ○ ○ ○ ● (LOW OIL PRESSURE)****SYMPTOM**

ESU BITE indicator shows ○ ○ ○ ● (low oil pressure).

**MALFUNCTION**

ESU BITE indicator shows ○ ○ ○ ● (low oil pressure).

**CORRECTIVE ACTION****1. Check APU oil sump for proper level.**

- a. If oil level is good, go to **2**.
- b. If oil level is not good, service APU oil sump as required (WP 1604 00). Go to **5**.

**2. Do ESU LOW OIL PRESSURE procedure, in this work package.**

- a. If APU oil pressure is good, replace APU low oil pressure switch (TM 55-2835-208-23). Go to **3**.

**3. Start and operate APU (TM 1-1520-237-10). Go to 4.****4. Does APU operate normally?**

- a. If APU operates normally, system is operational. Go to **5**.
- b. If APU does not operate normally and ESU BITE indicator shows low oil pressure, repair/replace wiring between J289-7 and P304-B (WP 1747 00). Go to **5**.

**5. Procedure completed.****ESU BITE INDICATOR SHOWS ● ● ○ ○ (OPEN THERMOCOUPLE)****SYMPTOM**

ESU BITE indicator shows ● ● ○ ○ (open thermocouple).

**MALFUNCTION**

ESU BITE indicator shows ● ● ○ ○ (open thermocouple).

**CORRECTIVE ACTION****1. Check continuity between P261-1 and P261-2.**

- a. If continuity is present, replace ESU (WP 0865 00). Go to **3**.
- b. If continuity is not present, go to **2**.

**2. Check continuity between J289-1 and J289-2.**

- a. If continuity is present, repair/replace wiring as required (WP 1747 00) between, then go to **3**:
  - P261-1 and P289-1

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**ESU BITE INDICATOR SHOWS ● ● ○ ○ (OPEN THERMOCOUPLE) - Continued**

- P261-2 and P289-2

- b. If continuity is not present, replace APU thermocouple and harness (TM 55-2835-208-23). Go to **3**.

**3. Procedure completed.****ESU BITE INDICATOR SHOWS ○ ○ ● ○ (EXTERNAL SHORT) AFTER ○ ● ● ● (START FUEL VALVE AND EXCITER SIGNAL OUT) FOR HELICOPTERS WITH ESU 163291-100****SYMPTOM**

ESU BITE indicator shows ○ ○ ● ○ (external short) after ○ ● ● ● (start fuel valve and exciter signal out) for helicopters with ESU 163291-100.

**MALFUNCTION**

ESU BITE indicator shows ○ ○ ● ○ (external short) after ○ ● ● ● (start fuel valve and exciter signal out) for helicopters with ESU 163291-100.

**CORRECTIVE ACTION****1. Check continuity between J303-B and ground.**

- a. If continuity is present, replace ignition exciter (TM 55-2835-208-23). Go to **5**.
- b. If continuity is not present, go to **2**.

**2. Check continuity between J305-A and ground.**

- a. If continuity is present, replace fuel start valve (TM 55-2835-208-23). Go to **5**.
- b. If continuity is not present, go to **3**.

**3. With P303 and P305 disconnected, check continuity between J289-17 and ground.**

- a. If continuity is present, replace APU electrical harness (TM 55-2835-208-23). Go to **5**.
- b. If continuity is not present, go to **4**.

**4. Check continuity between P261-17 and ground.**

- a. If continuity is present, repair/replace wiring as required (WP 1747 00) between, then go to **5**:
  - P261 and P303
  - P261 and P305
- b. If continuity is not present, replace ESU (WP 0865 00). Go to **5**.

**5. Procedure completed.**

**ESU BITE INDICATOR SHOWS ○ ○ ● ○ (EXTERNAL SHORT) AFTER ○ ○ ● ● (MAIN FUEL VALVE SIGNAL OUT; 14%) FOR HELICOPTERS WITH ESU 163291-100****SYMPTOM**

ESU BITE indicator shows ○ ○ ● ○ (external short) after ○ ○ ● ● (main fuel valve signal out; 14%) for helicopters with ESU 163291-100.

**MALFUNCTION**

ESU BITE indicator shows ○ ○ ● ○ (external short) after ○ ○ ● ● (main fuel valve signal out; 14%) for helicopters with ESU 163291-100.

**CORRECTIVE ACTION****1. Check continuity between J306-A and ground.**

- a. If continuity is present, replace main fuel valve (TM 55-2835-208-23). Go to **4**.
- b. If continuity is not present, go to **2**.

**2. With P306 disconnected, check continuity between J289-9 and ground.**

- a. If continuity is present, replace APU electrical harness (TM 55-2835-208-23). Go to **4**.
- b. If continuity is not present, go to **3**.

**3. With J306 disconnected and J289 connected, check continuity between P261-9 and ground.**

- a. If continuity is present, repair/replace wiring between P261-9 and P306-A (WP 1747 00). Go to **4**.
- b. If continuity is not present, replace ESU (WP 0865 00). Go to **4**.

**4. Procedure completed.****ESU BITE INDICATOR SHOWS ○ ○ ● ○ (EXTERNAL SHORT) AFTER ○ ○ ○ ○ (90% RPM SWITCH ON) OR ● ● ● (READY FOR SERVICE) FOR HELICOPTERS WITH ESU 163291-100****SYMPTOM**

ESU BITE indicator shows ○ ○ ● ○ (external short) after ○ ○ ○ ○ (90% rpm switch on) or ● ● ● ● (ready for service) for helicopters with ESU 163291-100.

**MALFUNCTION**

ESU BITE indicator shows ○ ○ ● ○ (external short) after ○ ○ ○ ○ (90% rpm switch on) or ● ● ● ● (ready for service) for helicopters with ESU 163291-100.

**CORRECTIVE ACTION****1. Check continuity between J307-A and ground.**

- a. If continuity is present, replace maximum fuel valve (TM 55-2835-208-23). Go to **4**.
- b. If continuity is not present, go to **2**.

**2. With P307 disconnected, check continuity between J289-15 and ground.**

- a. If continuity is present, replace APU electrical harness (TM 55-2835-208-23). Go to **4**.
- b. If continuity is not present, go to **3**.

**3. Check continuity between P261-15 and ground.**

**ESU BITE INDICATOR SHOWS ○ ○ ● ○ (EXTERNAL SHORT) AFTER ○ ○ ○ ○ (90% RPM SWITCH ON) OR ● ● ● (READY FOR SERVICE) FOR HELICOPTERS WITH ESU 163291-100 - Continued**

- a. If continuity is present, repair/replace wiring between P261-15 and P307-A (WP 1747 00). Go to **4**.
- b. If continuity is not present, replace ESU (WP 0865 00). Go to **4**.

**4. Procedure completed.**

**APU MOTORS BUT SHUTS DOWN (APU ON CAPSULE GOES OFF AND APU FAIL CAPSULE GOES ON)  
SYMPTOM**

APU motors but shuts down (APU ON capsule goes off and APU FAIL capsule goes on).

**MALFUNCTION**

APU motors but shuts down (APU ON capsule goes off and APU FAIL capsule goes on).

**CORRECTIVE ACTION**

**1. Visually check for any of the following BITE indications:**

- ● ● ●
- ○ ● ●
- ○ ○ ○
- ○ ○ ○

- a. If any indication listed is present, replace ESU (WP 0865 00). Go to **10**.
- b. If any indication listed is not present, go to **2**.

**2. Reset ESU. Go to 3.**

**3. Check for 1,190 to 1,450 ohms between P261-12 and P261-13.**

- a. If resistance is as specified, replace ESU (WP 0865 00). Go to **10**.
- b. If resistance is not as specified, go to **4**.

**4. Install an insulated jumper wire between P302-A and P302-B. Go to 5.**

**5. Check continuity between P261-12 and P261-13.**

- a. If continuity is present, go to **6**.
- b. If continuity is not present, go to **8**.

**6. Remove jumper wire from P302. Go to 7.**

**7. Remove and reinstall magnetic pickup to be sure of correct adjustment (TM 55-2835-208-23).**

- a. If trouble remains, replace magnetic pickup (TM 55-2835-208-23). Go to **10**.

**8. Remove jumper wire from P302. Go to 9.**

**9. Repair/replace wiring between P261 and P302 (WP 1747 00). Go to 10.**

**10. Procedure completed.**

**ESU BITE INDICATOR SHOWS ○ ● ○ ● (UNDERSPEED)****SYMPTOM**

ESU BITE indicator shows ○ ● ○ ● (underspeed).

**MALFUNCTION**

ESU BITE indicator shows ○ ● ○ ● (underspeed).

**CORRECTIVE ACTION**

- 1. Check helicopter configuration.**
  - a. ESU 160200-600, go to **2**.
  - b. All other ESU's, go to **3**.
- 2. Troubleshoot system for loss of speed data ○ ● ○ ○ ( ESU BITE INDICATOR SHOWS ○ ● ○ ● (LOSS OF SPEED DATA) FOR HELICOPTERS WITH ESU 160200-600, in this work package).**
  - a. If trouble remains, go to **3**.
- 3. Make sure all bleed-air ducts are closed and driven equipment is unloaded. Check air inlet for FOD. Go to 4.**
- 4. Do ESU SPEED SEQUENCE procedure, in this work package, to verify signal to maximum fuel valve.**
  - a. If speed sequence is good, go to **5**.
- 5. Loosen B-nut on inlet fuel filter. Place rag around nut. Go to 6.**
- 6. On upper console, place FUEL PUMP switch to APU BOOST. Go to 7.**
- 7. Check that fuel is going into rag.**
  - a. If fuel is present, go to **8**.
  - b. If fuel is not present, go to **19**.
- 8. Install an insulated jumper wire between P307-A and P307-D. Go to 9.**
- 9. Check continuity between P289-15 and ground.**
  - a. If continuity is present, go to **10**.
  - b. If continuity is not present, go to **17**.
- 10. Remove jumper wire. Go to 11.**
- 11. Replace maximum fuel valve (TM 55-2835-208-23). Go to 12.**
- 12. Start and operate APU (TM 1-1520-237-10). Go to 13.**
- 13. Does APU operate normally?**
  - a. If APU operates normally, system is operational. Go to **21**.
  - b. If APU does not operate normally and ESU BITE indications shows underspeed, go to **14**.
- 14. Replace APU fuel manifold (TM 55-2835-208-23). Go to 15.**
- 15. Start and operate APU (TM 1-1520-237-10). Go to 16.**
- 16. Does APU operate normally?**

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**ESU BITE INDICATOR SHOWS ○ ● ○ ● (UNDERSPEED) - Continued**

- a. If APU operates normally, system is operational. Go to **21**.
- b. If APU does not operate normally, replace APU fuel pump/acceleration control (TM 55-2835-208-23). Go to **21**.
- 17. Remove jumper wire. Go to 18.**
- 18. Repair/replace wiring between J289-15 and maximum fuel valve (TM 55-2835-208-23). Go to 21.**
- 19. Tighten B-nut. Go to 20.**
- 20. Troubleshoot fuel prime/boost system (WP 0129 00). Go to 21.**
- 21. Procedure completed.**

**ESU BITE INDICATOR SHOWS ● ● ○ ● (OVERSPEED)****SYMPTOM**

ESU BITE indicator shows ● ● ○ ● (overspeed).

**MALFUNCTION**

ESU BITE indicator shows ● ● ○ ● (overspeed).

**CORRECTIVE ACTION**

- 1. Do ESU SPEED SEQUENCE procedure, in this work package. to verify signal to start fuel valve.**
  - a. If speed sequence is good, go to **2**.
- 2. Install an insulated jumper wire between P305-A and P305-B. Go to 3.**
- 3. Check continuity between J289-17 and ground.**
  - a. If continuity is present, go to **4**.
  - b. If continuity is not present, go to **10**.
- 4. Remove jumper wire. Go to 5.**
- 5. Start and operate APU (TM 1-1520-237-10). Go to 6.**
- 6. Does APU operate normally?**
  - a. If APU operates normally, system is operational. Go to **12**.
  - b. If APU does not operate normally and ESU BITE indicator shows overspeed, go to **7**.
- 7. Replace APU start fuel valve (TM 55-2835-208-23). Go to 8.**
- 8. Start and operate APU (TM 1-1520-237-10). Go to 9.**
- 9. Does APU operate normally?**
  - a. If APU operates normally, system is operational. Go to **12**.
  - b. If APU does not operate normally, replace APU fuel pump/acceleration control (TM 55-2835-208-23). Go to **12**.
- 10. Remove jumper wire. Go to 11.**
- 11. Repair/replace wiring (WP 1747 00), as required between, then go to 12:**
  - P261 and P303



---

**ESU BITE INDICATOR SHOWS ● ● ○ ● (OVERSPEED) - Continued**

- P261 and P305

12. Procedure completed.

**ESU BITE INDICATOR SHOWS ○ ○ ● ● (OVERTEMPERATURE)****SYMPTOM**

ESU BITE indicator shows ○ ○ ● ● (overtemperature).

**MALFUNCTION**

ESU BITE indicator shows ○ ○ ● ● (overtemperature).

**CORRECTIVE ACTION****WARNING**

Failure to disconnect electrical connectors will permit fuel to enter combustion chamber and create a fire hazard.

1. **Make sure all bleed-air ducts are closed and driven equipment is unloaded. Check air inlet for FOD. Go to 2.**
2. **Do ESU OVERTEMPERATURE procedure, in this work package.**
  - a. If APU overtemperature and overspeed is good, go to 3.
3. **Replace combustor drain valve (TM 55-2835-208-23). Go to 4.**
4. **Start and operate APU (TM 1-1520-237-10). Go to 5.**
5. **Does APU operate normally?**
  - a. If APU operates normally, system is operational. Shut down APU (TM 1-1520-237-10). Go to 15.
  - b. If APU does not operate normally and ESU BITE indicator shows overtemperature, shut down APU (TM 1-1520-237-10). Go to 6.
6. **Disconnect P305, P306, and P307 from fuel valves. Disconnect P303 from ignition excitor. Disconnect fuel line from main fuel valve to the fuel manifold. Go to 7.**
7. **Place suitable containers under open fuel lines. Go to 8.**
8. **Motor APU and check for fuel flow from main fuel valve.**
  - a. If fuel is present, replace main fuel valve (TM 55-2835-208-23). Go to 15.
  - b. If fuel is not present, go to 9.
9. **Leave electrical connectors disconnected. Reconnect fuel line to fuel manifold. Disconnect fuel line at outlet of max fuel valve. Go to 10.**
10. **Place suitable containers under open fuel lines. Go to 11.**
11. **Motor APU and check for fuel at maximum fuel valve outlet.**
  - a. If fuel is present, replace maximum fuel valve (TM 55-2835-208-23). Go to 15.
  - b. If fuel is not present, go to 12.
12. **Reconnect all electrical connectors and fuel lines. Go to 13.**
13. **Start and operate APU (TM 1-1520-237-10). Go to 14.**

---

**ESU BITE INDICATOR SHOWS ○ ○ ● ● (OVERTEMPERATURE) - Continued****14. Does APU operate normally?**

- a. If APU operates normally, system is operational. Shut down APU (TM 1-1520-237-10). Go to **15**.
- b. If APU does not operate normally, shut down APU (TM 1-1520-237-10). Replace fuel pump/acceleration control (TM 55-2835-208-23). Go to **15**.

**15. Procedure completed.****ESU BITE INDICATOR SHOWS ○ ○ ○ ● (LOW OIL PRESSURE)****SYMPTOM**

ESU BITE indicator shows ○ ○ ○ ● (low oil pressure).

**MALFUNCTION**

ESU BITE indicator shows ○ ○ ○ ● (low oil pressure).

**CORRECTIVE ACTION****1. Check APU oil sump for proper level.**

- a. If oil level is good, go to **2**.
- b. If oil level is not good, service APU oil sump as required (WP 1604 00). Go to **6**.

**2. Do ESU LOW OIL PRESSURE procedure, in this work package.**

- a. If APU low oil pressure is good, go to **3**.

**3. Replace APU low oil pressure switch (TM 55-2835-208-23). Go to 4.****4. Start and operate APU (TM 1-1520-237-10). Go to 5.****5. Does APU operate normally?**

- a. If APU operates normally, system is operational. Go to **6**.
- b. If APU does not operate normally and ESU BITE indicator shows low oil pressure, repair/replace wiring between J289-7 and P304-B (WP 1747 00). Go to **6**.

**6. Procedure completed.****ESU BITE INDICATOR SHOWS ● ● ○ ○ (OPEN THERMOCOUPLE)****SYMPTOM**

ESU BITE indicator shows ● ● ○ ○ (open thermocouple).

**MALFUNCTION**

ESU BITE indicator shows ● ● ○ ○ (open thermocouple).

**CORRECTIVE ACTION****1. Check continuity between P261-1 and P261-2.**

- a. If continuity is present, replace ESU (WP 0865 00). Go to **3**.
- b. If continuity is not present, go to **2**.

**2. Check continuity between J289-1 and J289-2.**

---

**ESU BITE INDICATOR SHOWS ● ● ○ ○ (OPEN THERMOCOUPLE) - Continued**

- a. If continuity is present, repair/replace wiring, as required, between (WP 1747 00), then go to **3**: P289-1 and P261-1 P289-2 and P261-2
- b. If continuity is not present, replace APU thermocouple and harness (TM 55-2835-208-23). Go to **3**.

**3. Procedure completed.****APU OPERATES, BUT APU ON CAPSULE IS OFF****SYMPTOM**

APU operates, but APU ON capsule is off.

**MALFUNCTION**

APU operates, but APU ON capsule is off.

**CORRECTIVE ACTION**

- 1. Place and hold caution/advisory panel BRT/DIM-TEST switch to TEST. Go to 2.**
- 2. Check that APU ON capsule goes on.**
  - a. If capsule goes on, go to **3**.
  - b. If capsule does not go on, go to **10**.
- 3. Install an insulated jumper wire between P260-5 and P260-11. Go to 4.**
- 4. Place BATT switch to ON. Go to 5.**
- 5. Check that APU ON capsule goes on.**
  - a. If capsule goes on, remove jumper and replace ESU (WP 0865 00). Go to **11**.
  - b. If capsule does not go on, remove jumper and go to **6**.
- 6. Check helicopter configuration.**
  - a. **EMEP** , go to **7**.
  - b. **W/O EMEP** , go to **9**.
- 7. EMEP Remove and check pin filtered adapter P118 or HH-60A HH-60L P1019R or P1020R (WP 0925 00).**
  - a. If pin filtered adapter is good, go to **8**.
  - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **11**.
- 8. EMEP Install pin filtered adapter (WP 0925 00). Go to 9.**
- 9. Check for 28 vdc between P118-B or HH-60A HH-60L P1019R-84 or P1020R-84 and ground.**
  - a. If voltage is as specified, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to **11**.
  - b. If voltage is not as specified, repair/replace wiring between P118-B or HH-60A HH-60L P1019R-84 or P1020R-84 and P260-11 (WP 1747 00). Go to **11**.
- 10. Check APU ON capsule lamps.**
  - a. If lamps are good, troubleshoot caution/advisory warning system (WP 0090 00). Go to **11**.

---

**APU OPERATES, BUT APU ON CAPSULE IS OFF - Continued**

- b. If lamps are not good, replace lamps (WP 0786 00). Go to **11**.

**11. Procedure completed.****APU OIL TEMP HI CAPSULE GOES ON, ESU BITE INDICATOR SHOWS ● ● ● ○ (HIGH OIL TEMPERATURE)****SYMPTOM**

APU OIL TEMP HI capsule goes on, ESU BITE indicator shows ● ● ● ○ (high oil temperature).

**MALFUNCTION**

APU OIL TEMP HI capsule goes on, ESU BITE indicator shows ● ● ● ○ (high oil temperature).

**CORRECTIVE ACTION****1. Check APU oil sump for proper oil level.**

- a. If oil level is good, go to **2**.
- b. If oil level is not good, service APU oil sump (WP 1604 00). Go to **3**.

**2. Do ESU HIGH OIL TEMPERATURE procedure, in this work package.**

- a. If APU high oil temperature is good, replace APU high oil temperature switch (TM 55-2835-208-23). Go to **3**.

**3. Procedure completed.****FUEL LEAKS FROM COMBUSTOR DRAIN LINE WITH APU RUNNING****SYMPTOM**

Fuel leaks from combustor drain line with APU running.

**MALFUNCTION**

Fuel leaks from combustor drain line with APU running.

**CORRECTIVE ACTION****1. Disconnect P305 from start fuel valve. Go to 2.****2. Check for fuel leakage.**

- a. If fuel leakage is found, go to **3**.
- b. If no fuel leaks are found, replace ESU (WP 0865 00). Go to **5**.

**3. Cycle valve five times and check that leakage stops.**

- a. If leakage stops, go to **4**.
- b. If leakage does not stop, replace fuel start valve (TM 55-2835-208-23). Go to **5**.

**4. Check main fuel supply for contamination.**

- a. If contamination is found, purge fuel system. Go to **5**.
- b. If no contamination is found, replace fuel start valve (TM 55-2835-208-23). Go to **5**.

**5. Procedure completed.**

**BITE INDICATORS DO NOT CYCLE WHEN APU CONTR SWITCH PLACED ON****SYMPTOM**

BITE indicators do not cycle when APU CONTR switch placed ON.

**MALFUNCTION**

BITE indicators do not cycle when APU CONTR switch placed ON.

**CORRECTIVE ACTION****1. Check for 28 vdc between P260-1 and ground.**

- a. If voltage is as specified, replace ESU (WP 0865 00). Go to **12**.
- b. If voltage is not as specified, go to **2**.

**2. Check for 28 vdc between J200-R and ground.**

- a. If voltage is as specified, repair/replace wiring between P200-R and P260-1 (WP 1747 00). Go to **12**.
- b. If voltage is not as specified, go to **3**.

**3. Check for 28 vdc between P231-V and ground.**

- a. If voltage is as specified, go to **4**.
- b. If voltage is not as specified, go to **8**.

**4. Check for 28 vdc between APU T-handle terminal C and ground.**

- a. If voltage is as specified, go to **5**.
- b. If voltage is not as specified, repair/replace wiring between APU T-handle terminal C and J231-V (WP 1747 00). Go to **12**.

**5. Check for 28 vdc between APU T-handle switch terminal NC and ground.**

- a. If voltage is as specified, go to **6**.
- b. If voltage is not as specified, replace switch (WP 1203 00). Go to **12**.

**6. Check for 28 vdc between APU CONTR switch terminal 2 and ground.**

- a. If voltage is as specified, go to **7**.
- b. If voltage is not as specified, repair/replace wiring between APU T-handle switch terminal NC and APU CONTR switch (WP 1747 00). Go to **12**.

**7. Check for 28 vdc between APU CONTR switch terminal 3 and ground.**

- a. If voltage is as specified, repair/replace wiring between switch terminal 3 and J200-R (WP 1747 00). Go to **12**.
- b. If voltage is not as specified, replace switch (WP 0828 00). Go to **12**.

**8. Check continuity between P231-V and P242-n.**

- a. If continuity is present, go to **9**.
- b. If continuity is not present, repair/replace wiring between P242-n and P231-V (WP 1747 00). Go to **12**.

**9. Check continuity between P242-k and ground.**

---

**BITE INDICATORS DO NOT CYCLE WHEN APU CONTR SWITCH PLACED ON** - Continued

- a. If continuity is present, go to **10**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **12**.
- 10. Check for 28 vdc between APU CONTR INST circuit breaker terminal 1 and ground.**
  - a. If voltage is as specified, go to **11**.
  - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **12**.
- 11. Check continuity between APU CONTR INST circuit breaker terminal 1 and P242-p.**
  - a. If continuity is present, replace left relay panel (WP 0825 00). Go to **12**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **12**.
- 12. Procedure completed.**

**APU DOES NOT SHUT DOWN WHEN APU CONTR SWITCH IS PLACED OFF****SYMPTOM**

APU does not shut down when APU CONTR switch is placed OFF.

**MALFUNCTION**

APU does not shut down when APU CONTR switch is placed OFF.

**CORRECTIVE ACTION**

- 1. Check for 28 vdc between P260-2 and P260-6.**
  - a. If voltage is as specified, replace ESU (WP 0865 00). Go to **5**.
  - b. If trouble remains, replace main fuel valve or maximum fuel valve (WP 0969 00). Go to **5**.
  - c. If voltage is not as specified, go to **2**.
- 2. Check for 28 vdc between P260-2 and ground.**
  - a. If voltage is as specified, repair/replace wiring (WP 1747 00). Go to **5**.
  - b. If voltage is not as specified, go to **3**.
- 3. Check for 28 vdc between J200-P and ground.**
  - a. If voltage is as specified, repair/replace wiring between P200-P and P260-2 (WP 1747 00). Go to **5**.
  - b. If voltage is not as specified, go to **4**.
- 4. Check for 28 vdc between APU CONTR switch terminal 1 and ground.**
  - a. If voltage is as specified, repair/replace wiring between switch terminal 1 and J200-P (WP 1747 00). Go to **5**.
  - b. If voltage is not as specified, replace switch (WP 0828 00). Go to **5**.
- 5. Procedure completed.**

**NO POWER AT APU START VALVE****SYMPTOM**

No power at APU start valve.

**MALFUNCTION**

No power at APU start valve.

**CORRECTIVE ACTION****1. Check continuity between P260-9 and P290-1.**

- a. If continuity is present, go to **2**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **3**.

**2. Check for 28 vdc between P260-1 and ground.**

- a. If voltage is as specified, replace ESU (WP 0865 00). Go to **3**.
- b. If voltage is not as specified, go to BITE INDICATORS DO NOT CYCLE WHEN APU CONTR SWITCH PLACED ON, in this work package.

**3. Procedure completed.****APU STALL WHEN EITHER ENGINE START BUTTON IS PRESSED (A POPPING SOUND IS HEARD)****SYMPTOM**

APU stall when either engine START button is pressed (a popping sound is heard).

**MALFUNCTION**

APU stall when either engine START button is pressed (a popping sound is heard).

**CORRECTIVE ACTION****1. With APU SOURCE/HEAT START switch placed to APU and with either NO. 1 or NO. 2 ENGINE START button pressed, check for 28 vdc between P260-10 and ground.**

- a. If voltage is as specified, go to **2**.
- b. If voltage is not as specified, go to **14**.

**2. Install an insulated jumper wire between P261-18 and P261-19. Go to 3.****3. Check continuity between P289-18 and P289-19.**

- a. If continuity is present, go to **4**.
- b. If continuity is not present, go to **12**.

**4. Remove jumper wire from P261. Go to 5.****5. Install an insulated jumper wire between J289-18 and J289-19. Go to 6.****6. Check continuity between P308-A and P308-B.**

- a. If continuity is present, go to **7**.
- b. If continuity is not present, go to **10**.

**7. Remove jumper wire from P289. Go to 8.****8. Do ESU START BYPASS VALVE procedure, in this work package. Go to 9.**

---

**APU STALL WHEN EITHER ENGINE START BUTTON IS PRESSED (A POPPING SOUND IS HEARD) -**  
Continued

9. Check that lamp L4 on test set comes on.
  - a. If lamp comes on, replace APU start bypass valve (TM 55-2835-208-23). Go to 27.
  - b. If lamp does not come on, replace ESU (WP 0865 00). Go to 27.
10. Remove jumper wire from P289. Go to 11.
11. Repair/replace wiring, as required, between (WP 1747 00), then go to 27:
  - J289-18 and P308-A
  - J289-19 and P308-B
12. Remove jumper wire from P261. Go to 13.
13. Repair/replace wiring, as required, between (WP 1747 00), then go to 27:
  - P261-18 and P289-18
  - P261-19 and P289-19
14. Check helicopter configuration.
  - a. **WINTER** , go to 15.
  - b. **W/O WINTER** , go to 16.
15. **WINTER** Check continuity between J244-A and J244-B.
  - a. If continuity is present, go to 16.
  - b. If continuity is not present, replace **WINTER** (WP 1400 00). Go to 27.
16. Check for 28 vdc between P244-T and P244-U.
  - a. If voltage is as specified, go to 17.
  - b. If voltage is not as specified, go to 23.
17. Install an insulated jumper wire between P244-T and P244-S. Go to 18.
18. Check for 28 vdc between P260-10 and ground.
  - a. If voltage is as specified, go to 19.
  - b. If voltage is not as specified, go to 21.
19. Remove jumper wire from P244. Go to 20.
20. Replace right relay panel (WP 0825 00). Go to 27.
21. Remove jumper wire from P244. Go to 22.
22. Repair/replace wiring between P244-S and P260-10 (WP 1747 00). Go to 27.
23. Check continuity between P244-U and ground.
  - a. If continuity is present, go to 24.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to 27.
24. Check for 28 vdc between APU SOURCE HEAT/START circuit breaker terminal 1 and ground.
  - a. If voltage is as specified, go to 25.
  - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to 27.

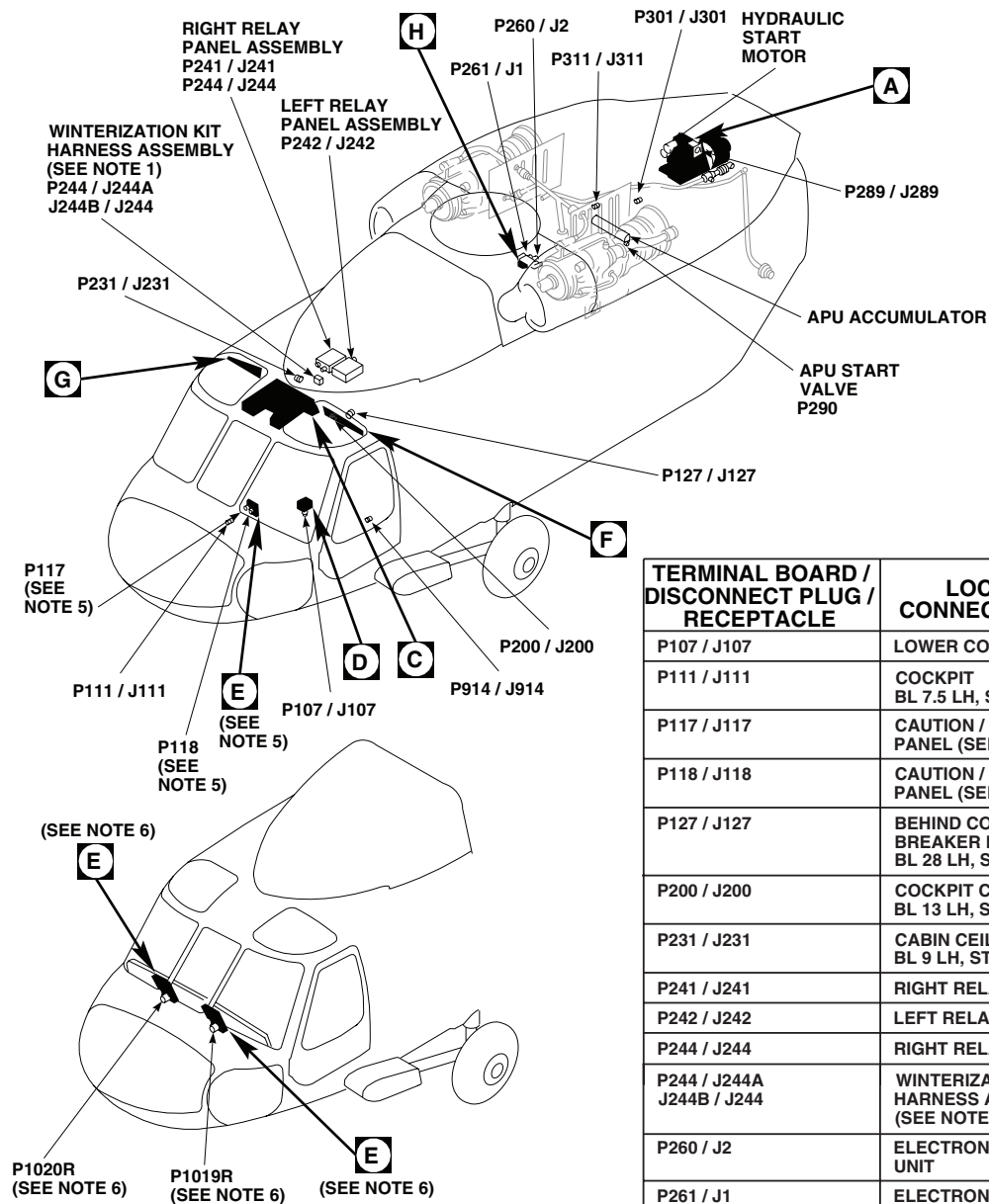


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**APU STALL WHEN EITHER ENGINE START BUTTON IS PRESSED (A POPPING SOUND IS HEARD) -**  
Continued

- 25. Check continuity between P244-T and AIR SOURCE HEAT/START switch terminal 1.**
  - a. If continuity is present, go to **26**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **27**.
- 26. Check continuity between APU SOURCE HEAT/START circuit breaker terminal 1 and AIR SOURCE HEAT/START switch terminal 2.**
  - a. If continuity is present, replace switch (WP 0828 00). Go to **27**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **27**.
- 27. Procedure completed.**

## LOCATION



| TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE | LOCATION / CONNECTION POINT                                    |
|-----------------------------------------------|----------------------------------------------------------------|
| P107 / J107                                   | LOWER CONSOLE                                                  |
| P111 / J111                                   | COCKPIT<br>BL 7.5 LH, STA 197                                  |
| P117 / J117                                   | CAUTION / ADVISORY<br>PANEL (SEE NOTE 5)                       |
| P118 / J118                                   | CAUTION / ADVISORY<br>PANEL (SEE NOTE 5)                       |
| P127 / J127                                   | BEHIND COPILOT'S CIRCUIT<br>BREAKER PANEL<br>BL 28 LH, STR 243 |
| P200 / J200                                   | COCKPIT CEILING<br>BL 13 LH, STA 246                           |
| P231 / J231                                   | CABIN CEILING<br>BL 9 LH, STA 247                              |
| P241 / J241                                   | RIGHT RELAY PANEL                                              |
| P242 / J242                                   | LEFT RELAY PANEL                                               |
| P244 / J244                                   | RIGHT RELAY PANEL                                              |
| P244 / J244A<br>J244B / J244                  | WINTERIZATION KIT<br>HARNESS ASSEMBLY<br>(SEE NOTE 1)          |
| P260 / J2                                     | ELECTRONIC SEQUENCE<br>UNIT                                    |
| P261 / J1                                     | ELECTRONIC SEQUENCE<br>UNIT                                    |
| P289 / J289                                   | APU                                                            |
| P290                                          | APU START VALVE                                                |
| P301 / J301                                   | TRANSITION DECK<br>BL 12.88 LH, STA 415                        |
| P311 / J311                                   | CABIN CEILING<br>BL 16.50 LH, STA 383                          |
| P914 / J914                                   | COCKPIT<br>BL 24 LH, STA 247                                   |
| P1019R / J3                                   | COPILOT'S MULTIFUNCTION<br>DISPLAY (SEE NOTE 6)                |
| P1020R / J3                                   | PILOT'S MULTIFUNCTION<br>DISPLAY (SEE NOTE 6)                  |

## NOTES

1. **WINTER**
2. HELICOPTERS WITH APU ESU  
160200-201.
3. HELICOPTERS WITH APU ESU  
160200-600.
4. HELICOPTERS WITH APU ESU  
163290-100.
5. **UH60A UH60L EH60A**
6. **HH60A HH60L**

AA2759\_1C  
SA

Figure 1. Auxiliary Power Unit System Location Diagram. (Sheet 1 of 9)

## LOCATION - Continued

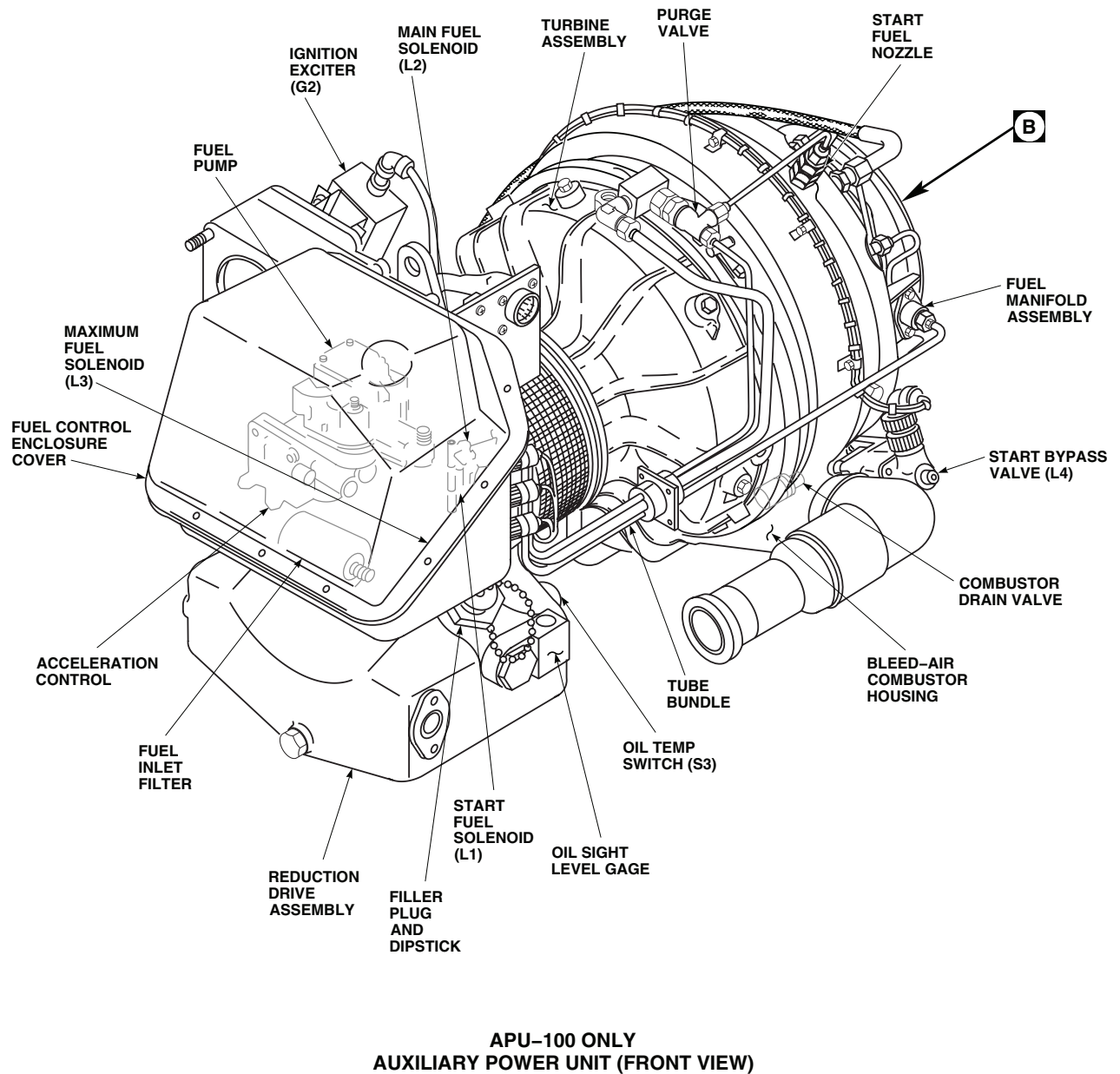
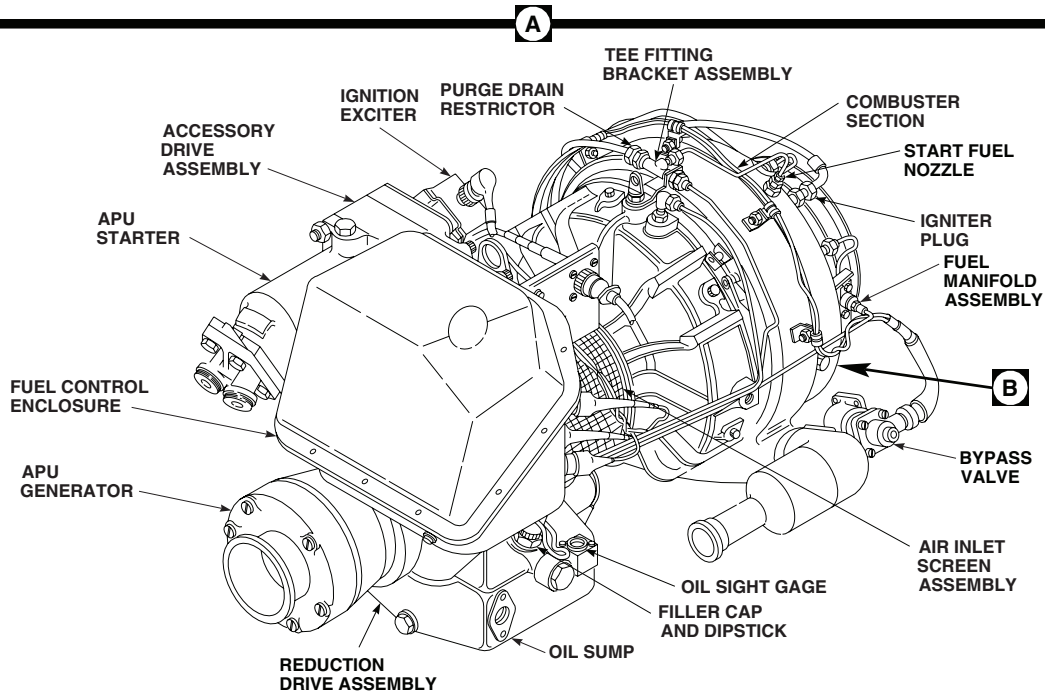
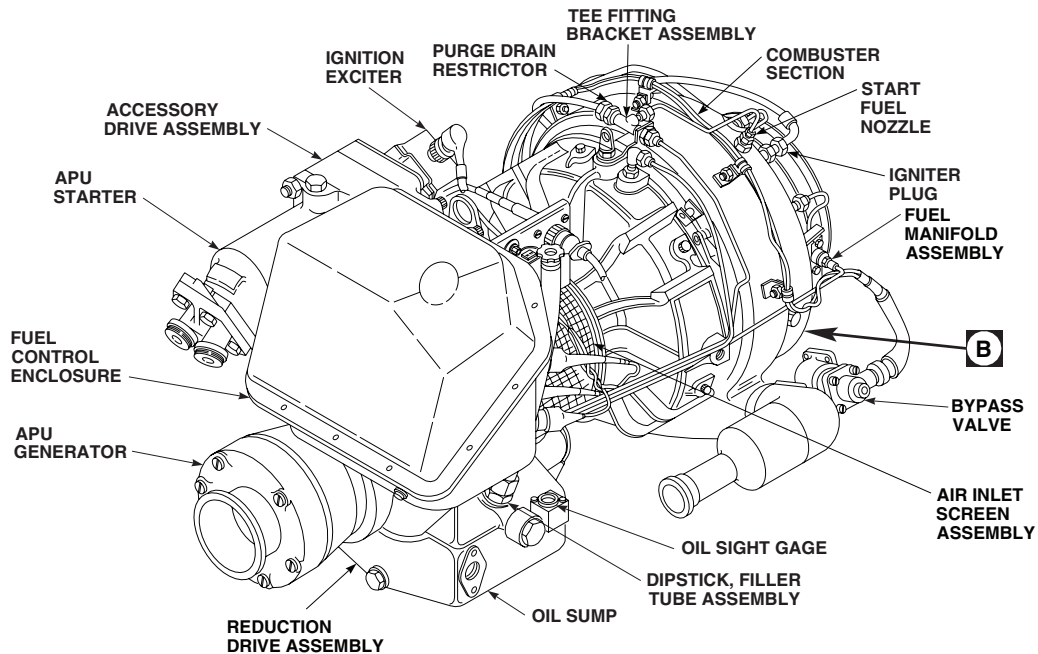
AA2759\_2A  
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Figure 1. Auxiliary Power Unit System Location Diagram. (Sheet 2 of 9)

LOCATION - Continued



**APU-200 ONLY**  
**AUXILIARY POWER UNIT (FRONT VIEW)**

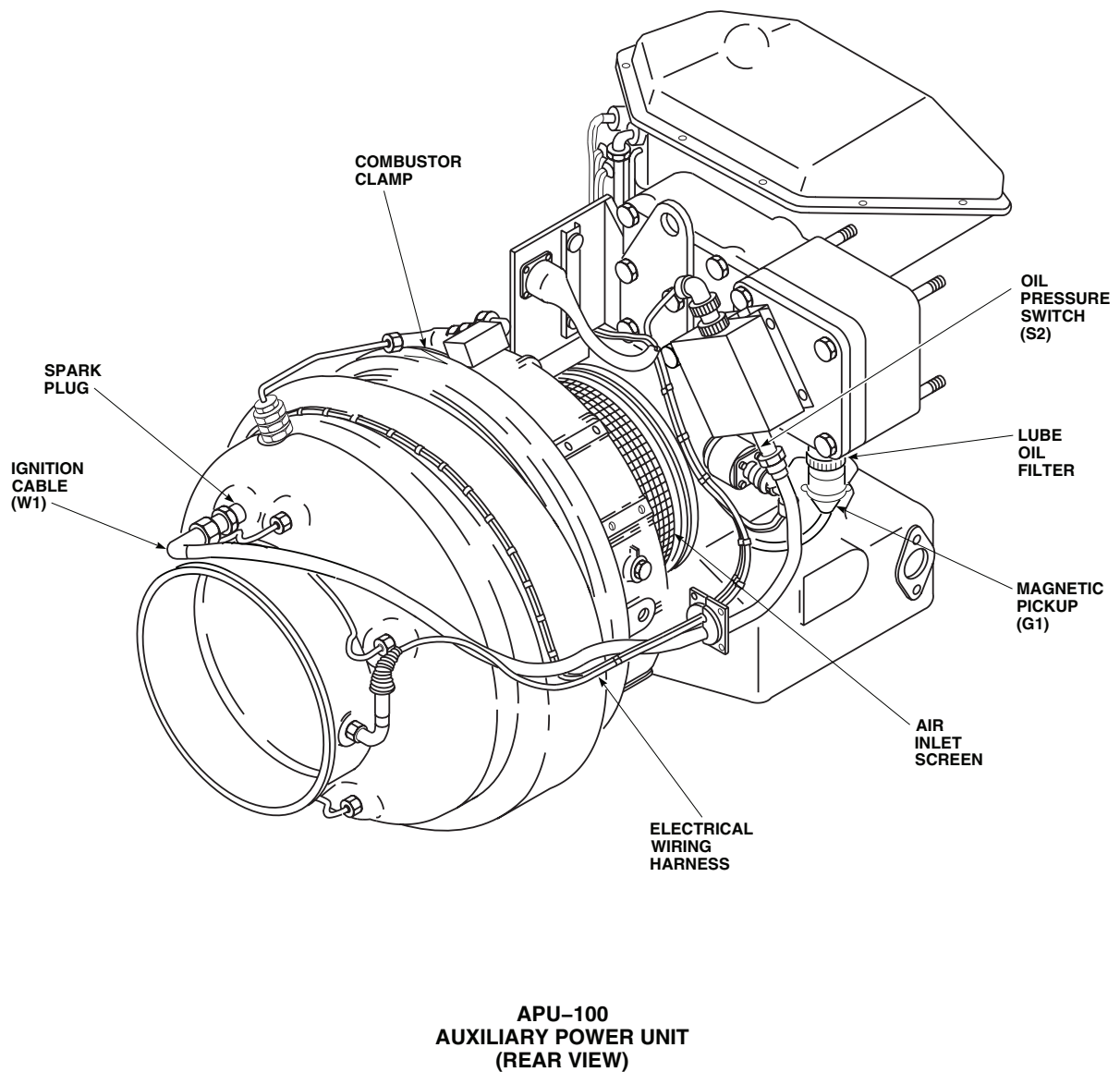


**APU-300/302 ONLY**  
**AUXILIARY POWER UNIT (FRONT VIEW)**

AA2759\_3B  
SA

Figure 1. Auxiliary Power Unit System Location Diagram. (Sheet 3 of 9)

## LOCATION - Continued



B

AA2759\_4A  
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Figure 1. Auxiliary Power Unit System Location Diagram. (Sheet 4 of 9)

## LOCATION - Continued

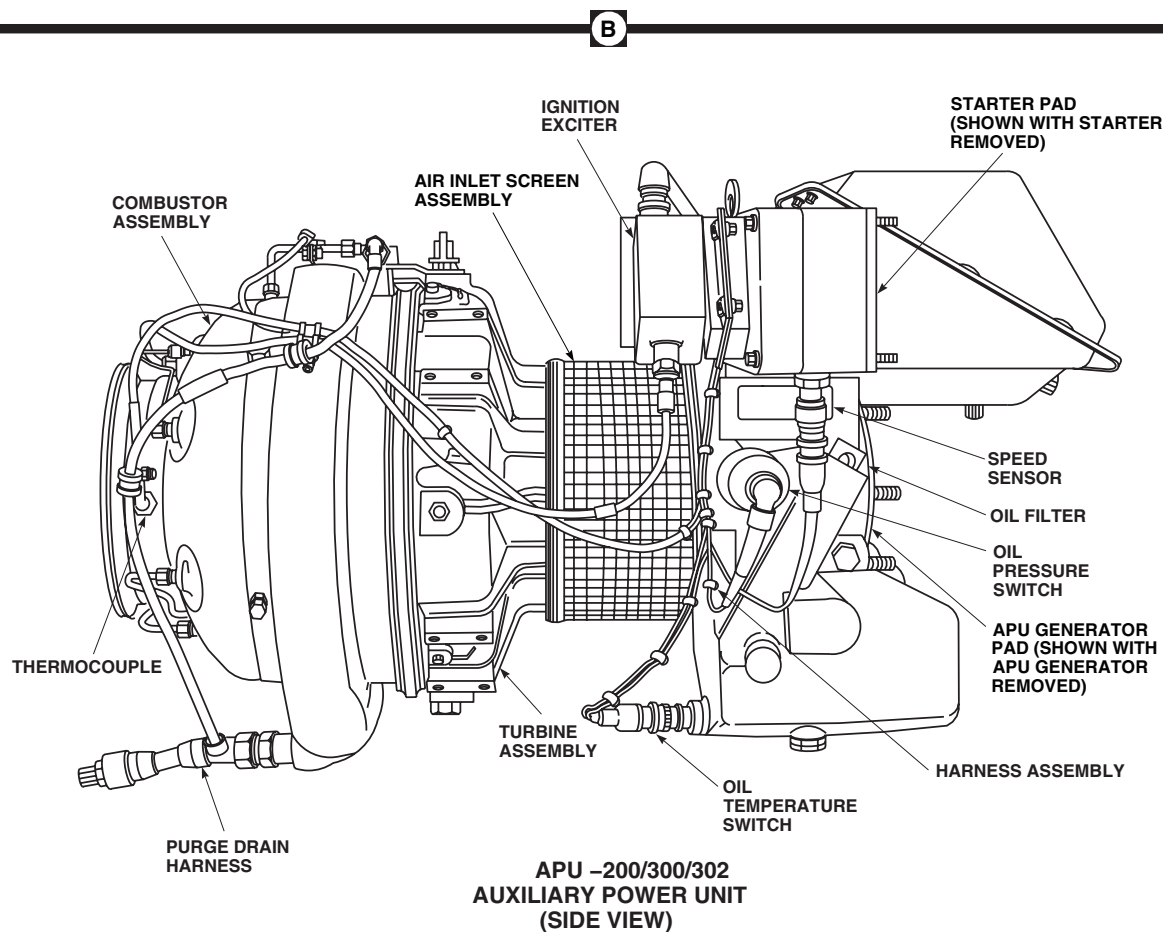
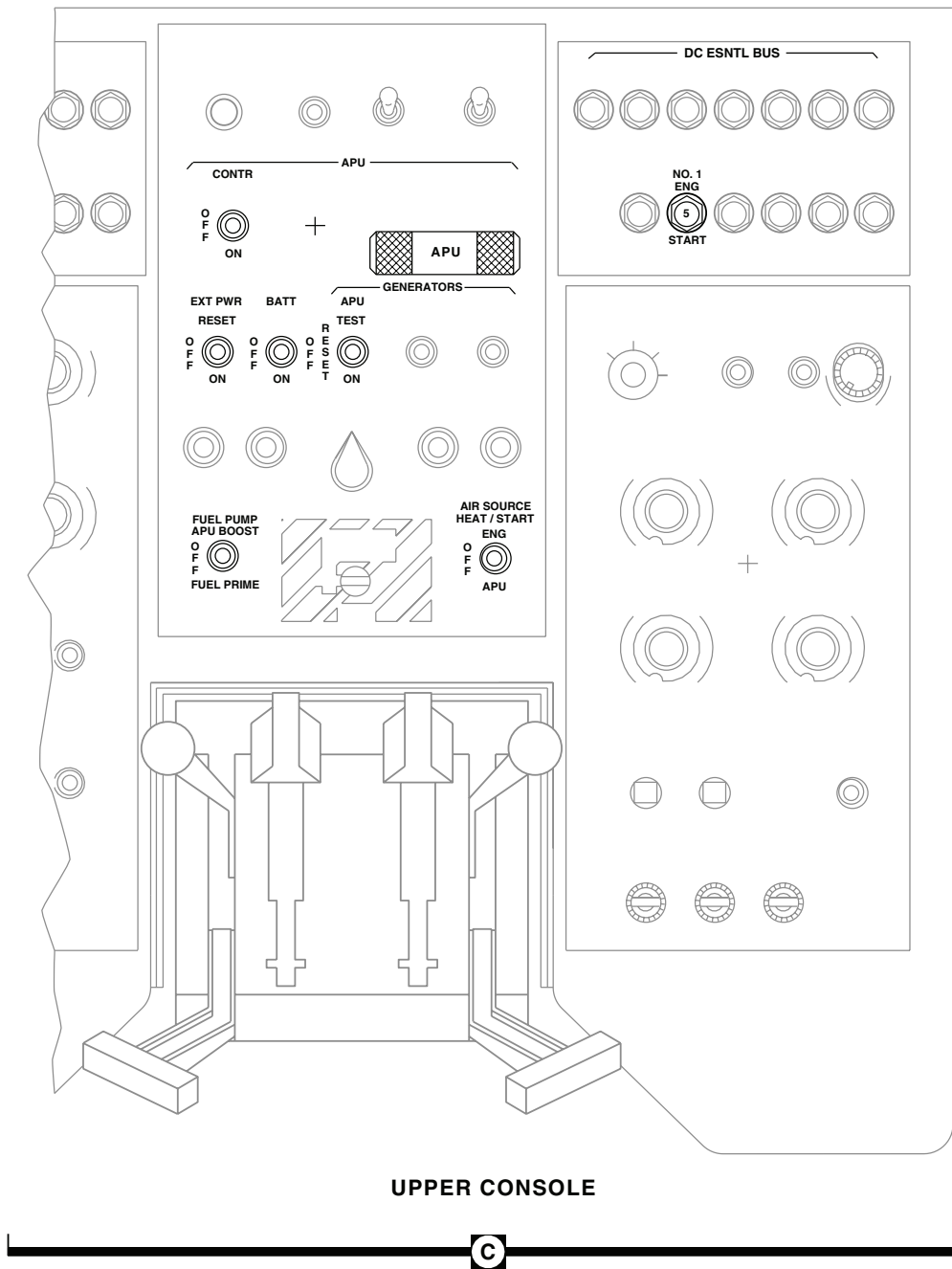
AA2759\_5B  
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Figure 1. Auxiliary Power Unit System Location Diagram. (Sheet 5 of 9)

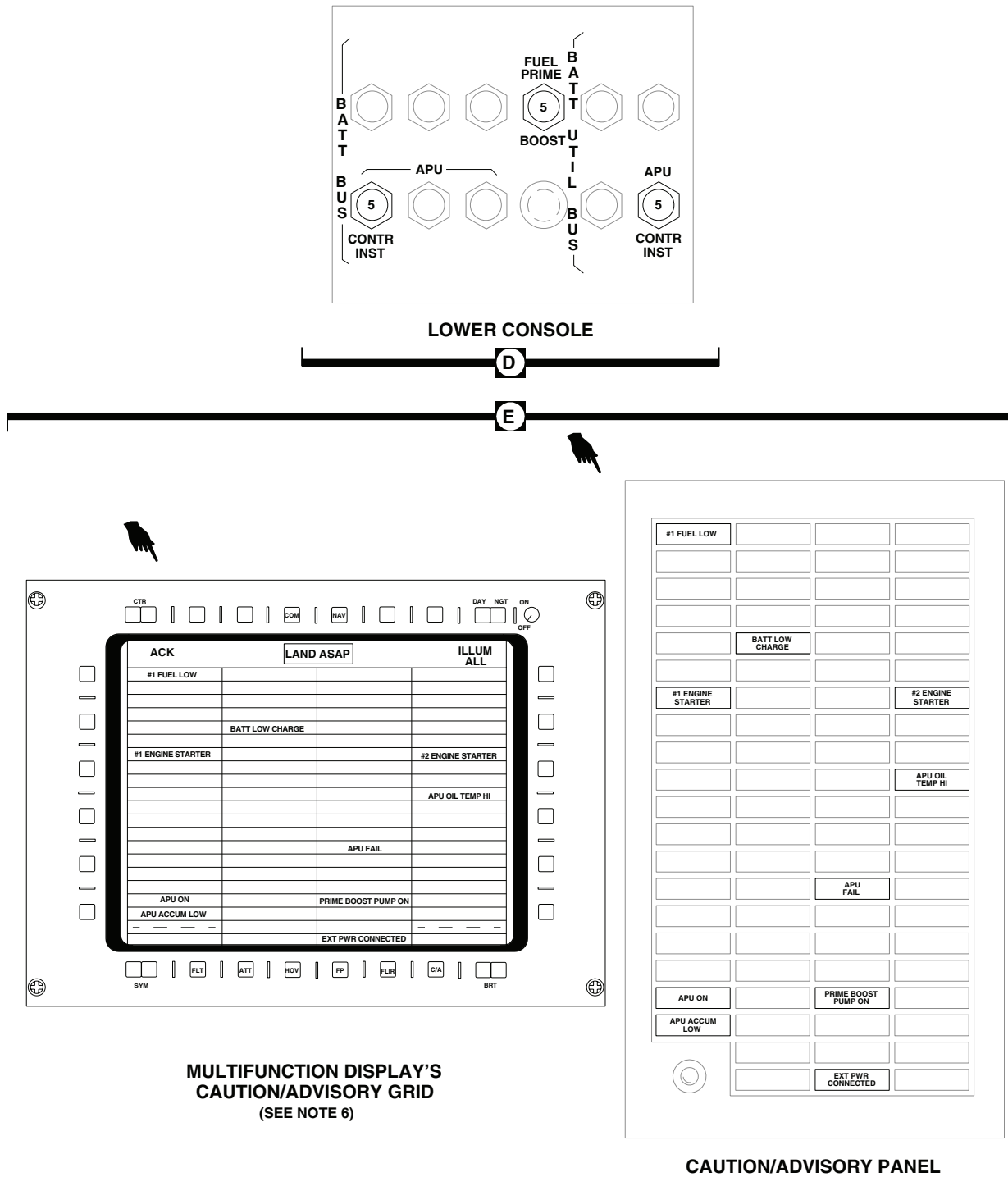
LOCATION - Continued



AA2759\_6A  
SA

Figure 1. Auxiliary Power Unit System Location Diagram. (Sheet 6 of 9)

LOCATION - Continued



**MULTIFUNCTION DISPLAY'S  
CAUTION/ADVISORY GRID**  
(SEE NOTE 6)

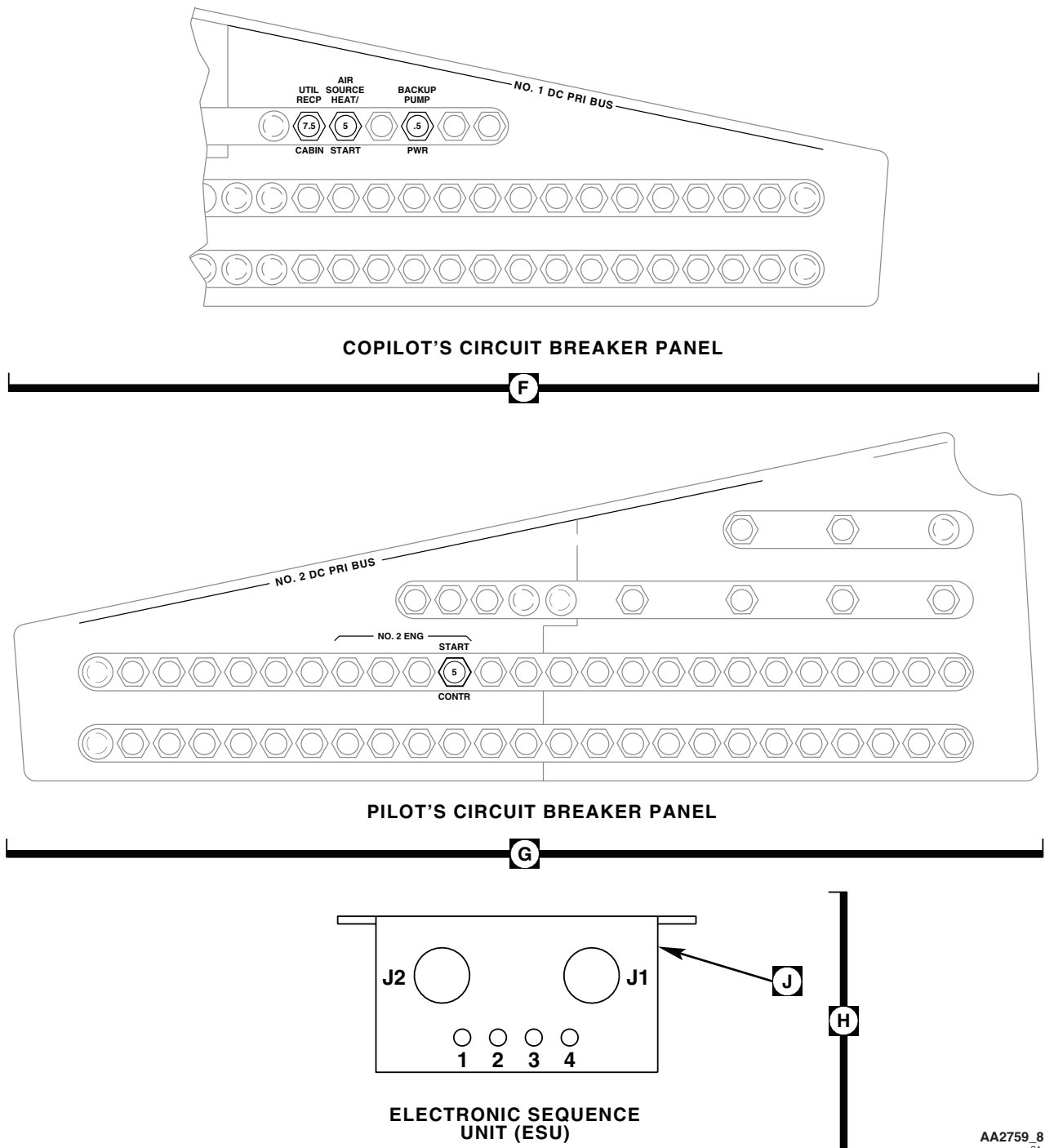
**CAUTION/ADVISORY PANEL**

AA2759\_7C  
SA

Figure 1. Auxiliary Power Unit System Location Diagram. (Sheet 7 of 9)



LOCATION - Continued



AA2759\_8  
SA

Figure 1. Auxiliary Power Unit System Location Diagram. (Sheet 8 of 9)

## LOCATION - Continued

| T-62T-40-1        |         |         |         |                                            |
|-------------------|---------|---------|---------|--------------------------------------------|
| BITE #<br>1 2 3 4 |         |         |         | DECODED BITE INFORMATION                   |
| ● ● ● ●           | ● ● ● ● | ● ● ● ● | ● ● ● ● | RESET                                      |
| ○ ● ● ●           | ○ ● ● ● | ○ ● ● ● | ○ ● ● ● | START FUEL VALVE & EXCITER SIGNAL OUT (5%) |
| ○ ○ ● ●           | ○ ○ ● ● | ○ ○ ● ● | ○ ○ ● ● | MAIN FUEL VALVE SIGNAL OUT (14%)           |
| ○ ○ ○ ●           | ○ ○ ○ ● | ○ ○ ○ ● | ○ ○ ○ ● | START FUEL VALVE & EXCITER SIGNAL OFF(70%) |
| ○ ○ ○ ○           | ○ ○ ○ ○ | ○ ○ ○ ○ | ○ ○ ○ ○ | 90% RPM SWITCH ON                          |
| ● ● ● ●           | ● ● ● ● | ● ● ● ● | ● ● ● ● | READY FOR SERVICE (90% + 1.5 SEC)          |
| ○ ● ● ●           | ○ ● ● ● | ○ ● ● ● | ○ ● ● ● | PROCESSOR BOARD FAILURE                    |
| ● ○ ● ●           | ● ○ ● ● | ● ○ ● ● | ● ○ ● ● | SENSOR / DATA BOARD FAILURE                |
| ○ ○ ● ●           | ○ ○ ● ● | ○ ○ ● ● | ○ ○ ● ● | OVERTEMPERATURE                            |
| ● ● ○ ●           | ● ● ○ ● | ● ● ○ ● | ● ● ○ ● | OVERSPEED                                  |
| ○ ● ○ ●           | ○ ● ○ ● | ○ ● ○ ● | ○ ● ○ ● | UNDERSPEED                                 |
| ● ○ ○ ●           | ● ○ ○ ● | ● ○ ○ ● | ● ○ ○ ● | FAIL TO START                              |
| ○ ○ ○ ●           | ○ ○ ○ ● | ○ ○ ○ ● | ○ ○ ○ ● | LOW OIL PRESSURE                           |
| ● ● ● ○           | ● ● ● ○ | ● ● ● ○ | ● ● ● ○ | HIGH OIL TEMPERATURE (WARNING)             |
| ○ ● ● ○           | ○ ● ● ○ | ○ ● ● ○ | ○ ● ● ○ | FAIL TO LIGHT                              |
| ● ○ ● ○           | ● ○ ● ○ | ● ○ ● ○ | ● ○ ● ○ | SHORTED THERMOCOUPLE PROBE (WARNING)       |
| ● ● ○ ○           | ● ● ○ ○ | ● ● ○ ○ | ● ● ○ ○ | OPEN THERMOCOUPLE                          |
| ● ○ ○ ○           | ● ○ ○ ○ | ● ○ ○ ○ | ● ○ ○ ○ | PROCESSOR SEQUENCE FAIL                    |
| ○ ○ ○ ○           | ○ ○ ○ ○ | ○ ○ ○ ○ | ○ ○ ○ ○ | NO DATA                                    |

MALFUNCTION BITE INDICATIONS/  
START SEQUENCE DECAL

(SEE NOTE 2)

| T-62T-40-1        |         |         |         |                                            |
|-------------------|---------|---------|---------|--------------------------------------------|
| BITE #<br>1 2 3 4 |         |         |         | DECODED BITE INFORMATION                   |
| ● ● ● ●           | ● ● ● ● | ● ● ● ● | ● ● ● ● | RESET                                      |
| ○ ● ● ●           | ○ ● ● ● | ○ ● ● ● | ○ ● ● ● | START FUEL VALVE & EXCITER SIGNAL OUT (5%) |
| ○ ○ ● ●           | ○ ○ ● ● | ○ ○ ● ● | ○ ○ ● ● | MAIN FUEL VALVE SIGNAL OUT (14%)           |
| ○ ○ ○ ●           | ○ ○ ○ ● | ○ ○ ○ ● | ○ ○ ○ ● | START FUEL VALVE & EXCITER SIGNAL OFF(70%) |
| ○ ○ ○ ○           | ○ ○ ○ ○ | ○ ○ ○ ○ | ○ ○ ○ ○ | 90% RPM SWITCH ON                          |
| ● ● ● ●           | ● ● ● ● | ● ● ● ● | ● ● ● ● | READY FOR SERVICE (90% + 1.5 SEC)          |
| ○ ● ● ●           | ○ ● ● ● | ○ ● ● ● | ○ ● ● ● | PROCESSOR BOARD FAILURE                    |
| ● ○ ● ●           | ● ○ ● ● | ● ○ ● ● | ● ○ ● ● | SENSOR / DATA BOARD FAILURE                |
| ○ ○ ● ●           | ○ ○ ● ● | ○ ○ ● ● | ○ ○ ● ● | OVERTEMPERATURE                            |
| ● ● ○ ●           | ● ● ○ ● | ● ● ○ ● | ● ● ○ ● | OVERSPEED                                  |
| ○ ● ○ ●           | ○ ● ○ ● | ○ ● ○ ● | ○ ● ○ ● | UNDERSPEED                                 |
| ● ○ ○ ●           | ● ○ ○ ● | ● ○ ○ ● | ● ○ ○ ● | FAIL TO START                              |
| ○ ○ ○ ●           | ○ ○ ○ ● | ○ ○ ○ ● | ○ ○ ○ ● | LOW OIL PRESSURE                           |
| ● ● ● ○           | ● ● ● ○ | ● ● ● ○ | ● ● ● ○ | HIGH OIL TEMPERATURE (WARNING)             |
| ○ ● ● ○           | ○ ● ● ○ | ○ ● ● ○ | ○ ● ● ○ | SHORTED OIL PRESSURE SWITCH                |
| ● ○ ● ○           | ● ○ ● ○ | ● ○ ● ○ | ● ○ ● ○ | SHORTED THERMOCOUPLE PROBE (WARNING)       |
| ● ● ○ ○           | ● ● ○ ○ | ● ● ○ ○ | ● ● ○ ○ | OPEN THERMOCOUPLE                          |
| ● ○ ○ ○           | ● ○ ○ ○ | ● ○ ○ ○ | ● ○ ○ ○ | PROCESSOR SEQUENCE FAIL                    |
| ○ ○ ○ ○           | ○ ○ ○ ○ | ○ ○ ○ ○ | ○ ○ ○ ○ | NO DATA                                    |
| ○ ● ○ ○           | ○ ● ○ ○ | ○ ● ○ ○ | ○ ● ○ ○ | LOSS OF SPEED DATA                         |

MALFUNCTION BITE INDICATIONS/  
START SEQUENCE DECAL

(SEE NOTE 3)

| T-62T-40-1        |         |         |         |                                             |
|-------------------|---------|---------|---------|---------------------------------------------|
| BITE #<br>1 2 3 4 |         |         |         | DECODED BITE INFORMATION                    |
| ● ● ● ●           | ● ● ● ● | ● ● ● ● | ● ● ● ● | RESET                                       |
| ○ ● ● ●           | ○ ● ● ● | ○ ● ● ● | ○ ● ● ● | START FUEL VALVE & EXCITER SIGNAL OUT (5%)  |
| ○ ○ ● ●           | ○ ○ ● ● | ○ ○ ● ● | ○ ○ ● ● | MAIN FUEL VALVE SIGNAL OUT (14%)            |
| ○ ○ ○ ●           | ○ ○ ○ ● | ○ ○ ○ ● | ○ ○ ○ ● | START FUEL VALVE & EXCITER SIGNAL OFF (70%) |
| ○ ○ ○ ○           | ○ ○ ○ ○ | ○ ○ ○ ○ | ○ ○ ○ ○ | 90% RPM SWITCH ON                           |
| ● ● ● ●           | ● ● ● ● | ● ● ● ● | ● ● ● ● | READY FOR SERVICE (90% + 1.5 SEC)           |
| ○ ● ● ●           | ○ ● ● ● | ○ ● ● ● | ○ ● ● ● | PROCESSOR BOARD FAILURE                     |
| ● ○ ● ●           | ● ○ ● ● | ● ○ ● ● | ● ○ ● ● | SENSOR / DATA BOARD FAILURE                 |
| ○ ○ ● ●           | ○ ○ ● ● | ○ ○ ● ● | ○ ○ ● ● | OVERTEMPERATURE                             |
| ● ● ○ ●           | ● ● ○ ● | ● ● ○ ● | ● ● ○ ● | OVERSPEED                                   |
| ○ ● ○ ●           | ○ ● ○ ● | ○ ● ○ ● | ○ ● ○ ● | UNDERSPEED                                  |
| ● ○ ○ ●           | ● ○ ○ ● | ● ○ ○ ● | ● ○ ○ ● | FAIL TO START                               |
| ○ ○ ○ ●           | ○ ○ ○ ● | ○ ○ ○ ● | ○ ○ ○ ● | LOW OIL PRESSURE                            |
| ● ● ● ○           | ● ● ● ○ | ● ● ● ○ | ● ● ● ○ | HIGH OIL TEMPERATURE (WARNING)              |
| ○ ● ● ○           | ○ ● ● ○ | ○ ● ● ○ | ○ ● ● ○ | SHORTED OIL PRESSURE SWITCH                 |
| ● ○ ● ○           | ● ○ ● ○ | ● ○ ● ○ | ● ○ ● ○ | SHORTED THERMOCOUPLE PROBE (WARNING)        |
| ● ● ○ ○           | ● ● ○ ○ | ● ● ○ ○ | ● ● ○ ○ | OPEN THERMOCOUPLE                           |
| ● ○ ○ ○           | ● ○ ○ ○ | ● ○ ○ ○ | ● ○ ○ ○ | PROCESSOR SEQUENCE FAIL                     |
| ○ ○ ○ ○           | ○ ○ ○ ○ | ○ ○ ○ ○ | ○ ○ ○ ○ | NO DATA                                     |
| ○ ○ ○ ●           | ○ ○ ○ ● | ○ ○ ○ ● | ○ ○ ○ ● | EXTERNAL SHORT                              |
| ○ ● ○ ○           | ○ ● ○ ○ | ○ ● ○ ○ | ○ ● ○ ○ | LOSS OF SPEED DATA                          |

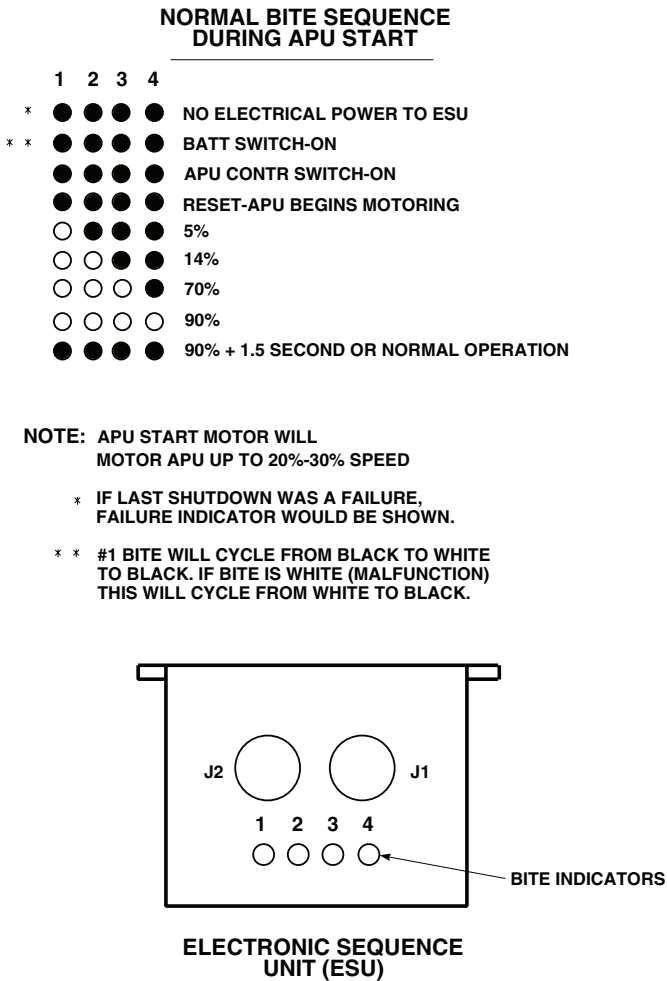
MALFUNCTION BITE INDICATIONS/  
START SEQUENCE DECAL

(SEE NOTE 4)

J

Figure 1. Auxiliary Power Unit System Location Diagram. (Sheet 9 of 9)

SCHEMATICS



## SCHEMATICS - Continued

### FAILURE BITE INDICATIONS ASSOCIATED WITH SHUTDOWN DURING START OF OPERATION

(SEE NOTE 1)

|     | 1 | 2 | 3 | 4 |                             |
|-----|---|---|---|---|-----------------------------|
|     | ○ | ○ | ● | ● | OVERTEMPERATURE             |
|     | ● | ● | ○ | ● | OVERSPEED                   |
|     | ○ | ● | ○ | ● | UNDERSPEED                  |
|     | ● | ○ | ○ | ● | FAIL TO START               |
|     | ○ | ○ | ○ | ● | LOW OIL PRESSURE            |
|     | ○ | ● | ● | ○ | FAIL TO LIGHT               |
|     | ● | ● | ○ | ○ | OPEN THERMOCOUPLE           |
| + * | ○ | ● | ● | ● | PROCESSOR BOARD FAILURE     |
| *   | ● | ○ | ● | ● | SENSOR / DATA BOARD FAILURE |
| *   | ● | ○ | ○ | ○ | PROCESSOR SEQUENCE FAIL     |
| *   | ○ | ○ | ○ | ○ | NO DATA                     |

\* ESU RELATED FAILURE ONLY  
+ APU FAIL CAPSULE WILL NOT BE ON

(SEE NOTE 2)

|     | 1 | 2 | 3 | 4 |                             |
|-----|---|---|---|---|-----------------------------|
|     | ○ | ○ | ● | ● | OVERTEMPERATURE             |
|     | ● | ● | ○ | ● | OVERSPEED                   |
|     | ○ | ● | ○ | ● | UNDERSPEED                  |
|     | ● | ○ | ○ | ● | FAIL TO START               |
|     | ○ | ○ | ○ | ● | LOW OIL PRESSURE            |
|     | ○ | ● | ● | ○ | SHORTED OIL PRESSURE SWITCH |
|     | ● | ● | ○ | ○ | OPEN THERMOCOUPLE           |
| + * | ○ | ● | ● | ● | PROCESSOR BOARD FAILURE     |
| *   | ● | ○ | ● | ● | SENSOR / DATA BOARD FAILURE |
| *   | ● | ○ | ○ | ○ | PROCESSOR SEQUENCE FAIL     |
| *   | ○ | ○ | ○ | ○ | NO DATA                     |
|     | ○ | ● | ○ | ○ | LOSS OF SPEED DATA          |

\* ESU RELATED FAILURE ONLY  
+ APU FAIL CAPSULE WILL NOT BE ON

### WARNING BITE INDICATORS

|   | 1 | 2 | 3 | 4 |                      |
|---|---|---|---|---|----------------------|
|   | ● | ● | ● | ○ | HIGH OIL TEMPERATURE |
| + | ● | ○ | ● | ○ | SHORTED THERMOCOUPLE |

THESE INDICATIONS WILL NOT CAUSE APU TO SHUT DOWN; ALSO, INDICATIONS WILL NOT BE SHOWN AFTER A NORMAL SHUTDOWN

+ THIS MALFUNCTION WILL CAUSE A SHUTDOWN DURING NEXT ATTEMPTED START WITH A FAIL LIGHT BITE INDICATION (SEE NOTE 1)

(SEE NOTE 3)

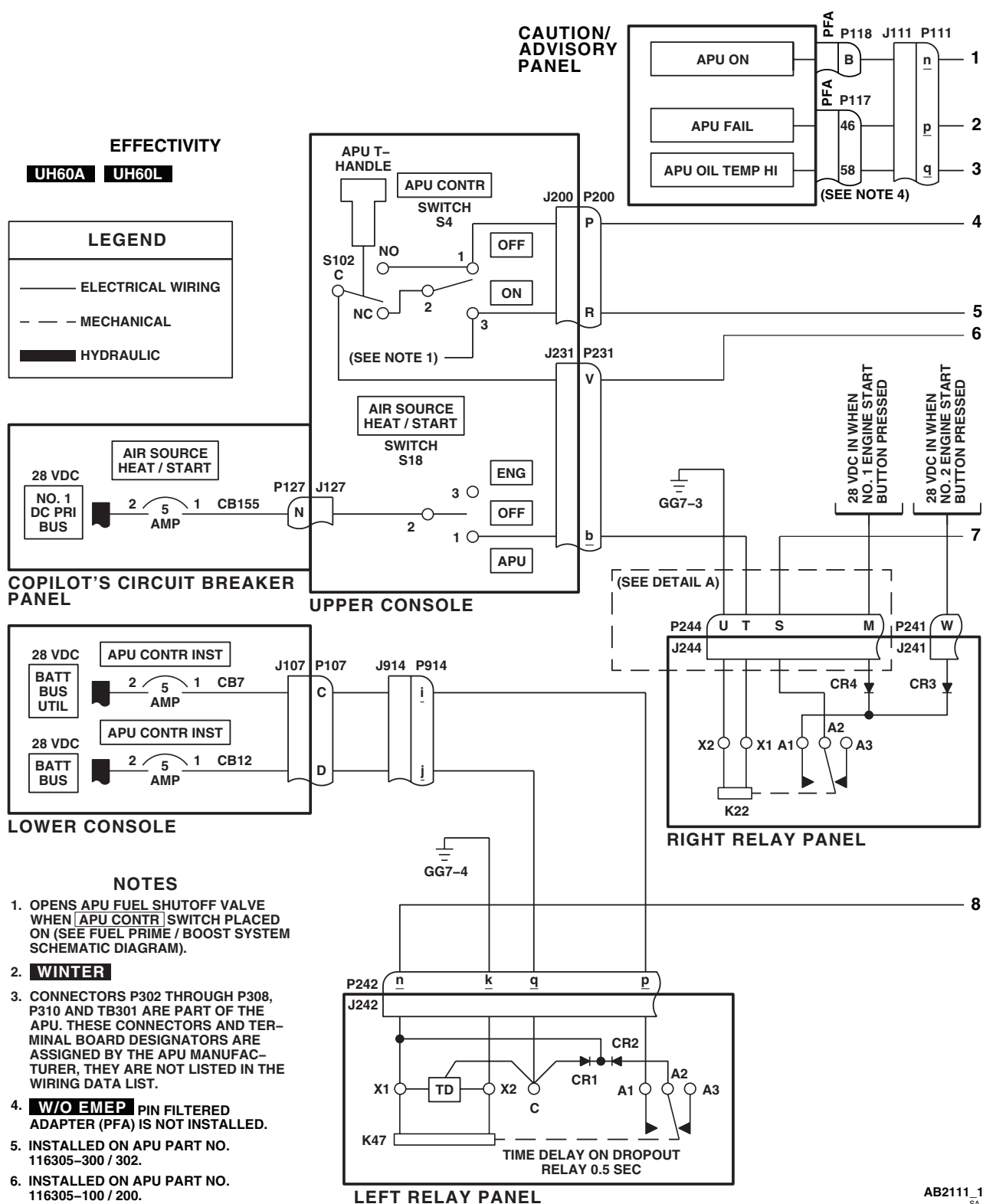
|     | 1 | 2 | 3 | 4 |                             |
|-----|---|---|---|---|-----------------------------|
|     | ○ | ○ | ● | ● | OVERTEMPERATURE             |
|     | ● | ● | ○ | ● | OVERSPEED                   |
|     | ○ | ● | ○ | ● | UNDERSPEED                  |
|     | ● | ○ | ○ | ● | FAIL TO START               |
|     | ○ | ○ | ○ | ● | LOW OIL PRESSURE            |
|     | ○ | ● | ● | ○ | SHORTED OIL PRESSURE SWITCH |
|     | ● | ● | ○ | ○ | OPEN THERMOCOUPLE           |
| + * | ○ | ● | ● | ● | PROCESSOR BOARD FAILURE     |
| *   | ● | ○ | ● | ● | SENSOR / DATA BOARD FAILURE |
| *   | ● | ○ | ○ | ○ | PROCESSOR SEQUENCE FAIL     |
| *   | ○ | ○ | ○ | ○ | NO DATA                     |
|     | ○ | ○ | ● | ○ | EXTERNAL SHORT              |
|     | ○ | ● | ○ | ○ | LOSS OF SPEED DATA          |

\* ESU RELATED FAILURE ONLY  
+ APU FAIL CAPSULE WILL NOT BE ON

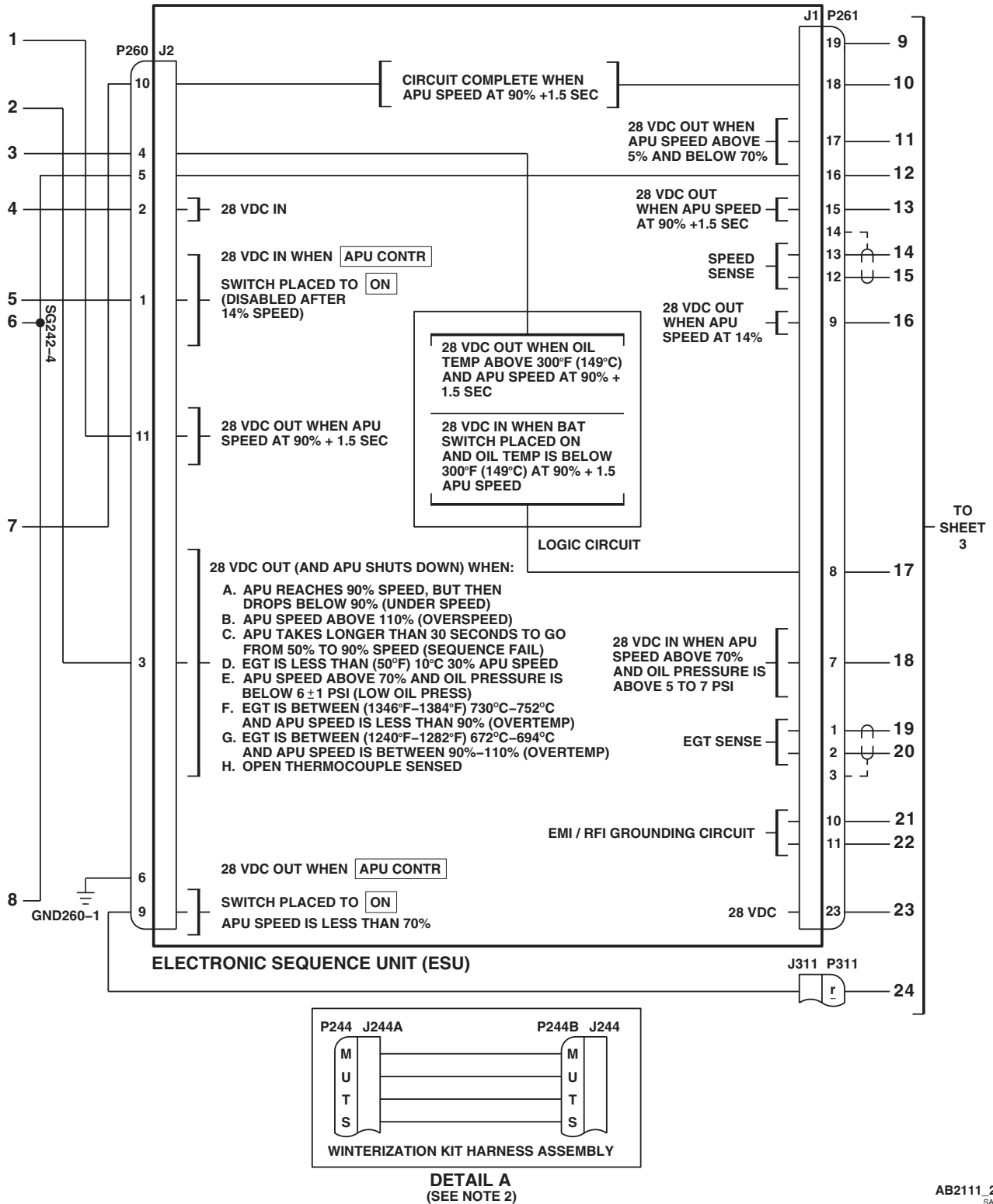
AA2760\_2A  
SA

Figure 2. Electronic Sequence Unit (ESU) BITE Indicator Conditions. (Sheet 2 of 2)

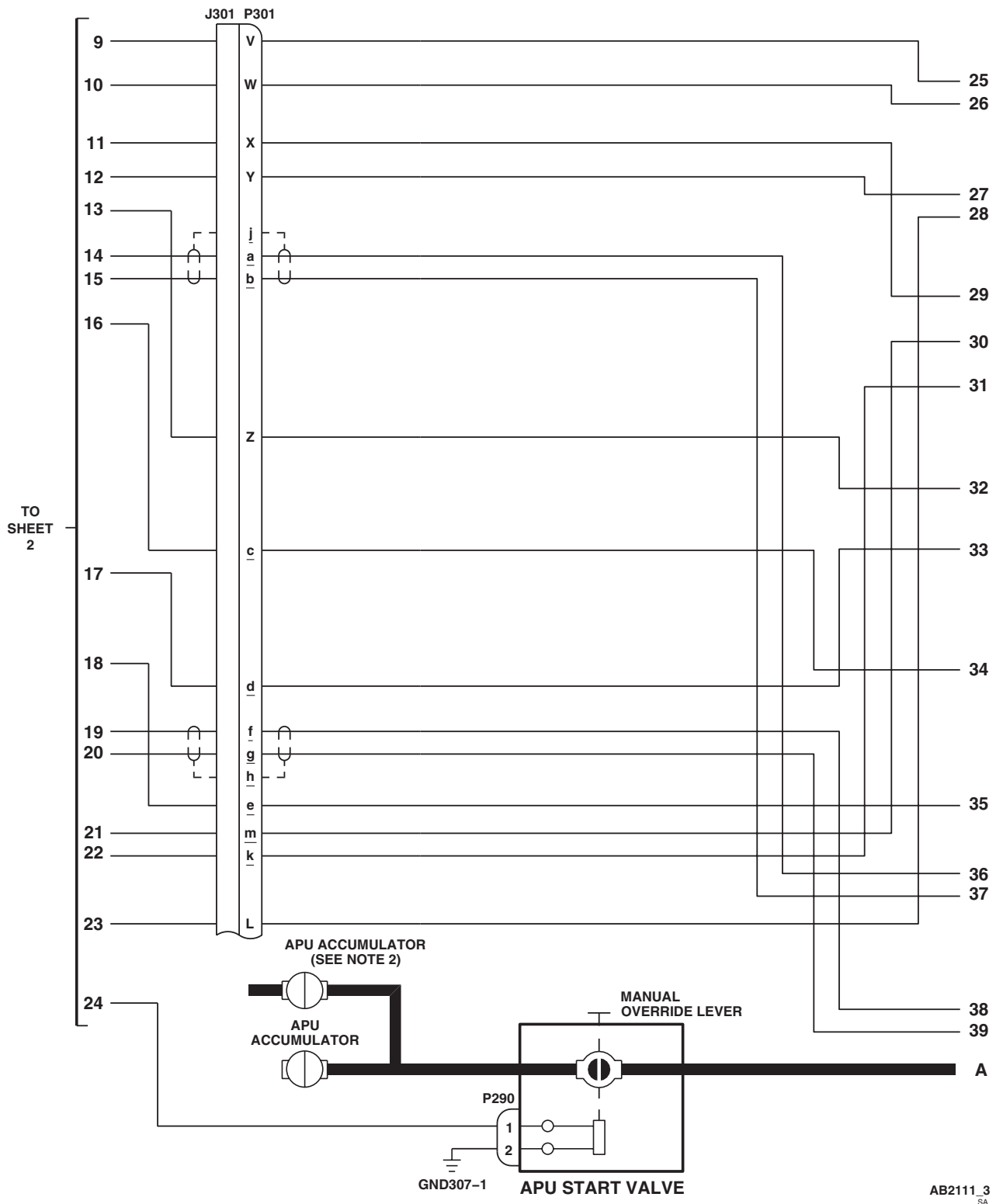
## SCHEMATICS - Continued

Figure 3. Auxiliary Power Unit System Schematic Diagram **UH-60A UH-60L** . (Sheet 1 of 5)

SCHEMATICS - Continued



SCHEMATICS - Continued



AB2111\_3  
SA

Figure 3. Auxiliary Power Unit System Schematic Diagram **UH-60A UH-60L** . (Sheet 3 of 5)

SCHEMATICS - Continued

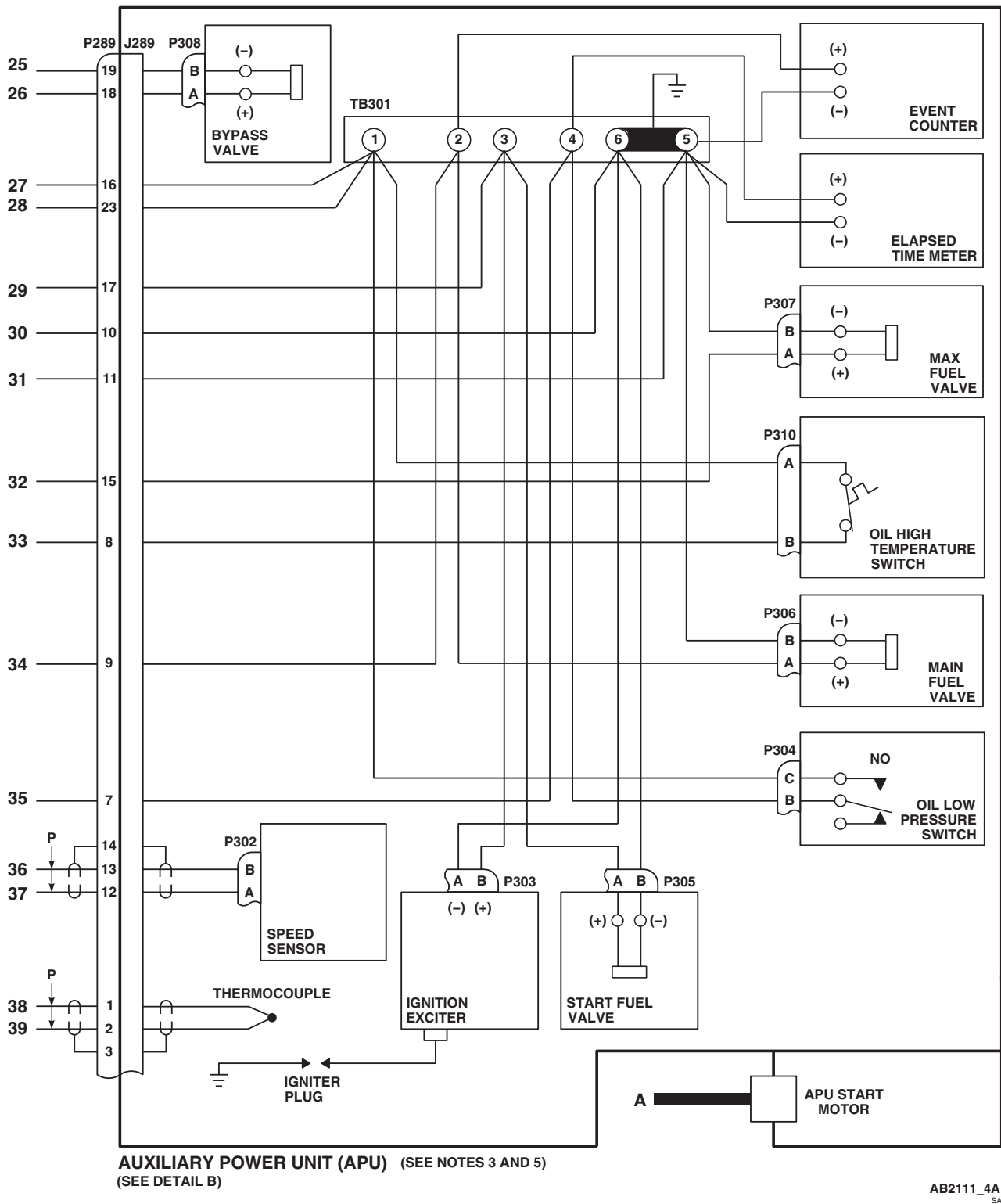


Figure 3. Auxiliary Power Unit System Schematic Diagram UH-60A UH-60L . (Sheet 4 of 5)



## SCHEMATICS - Continued

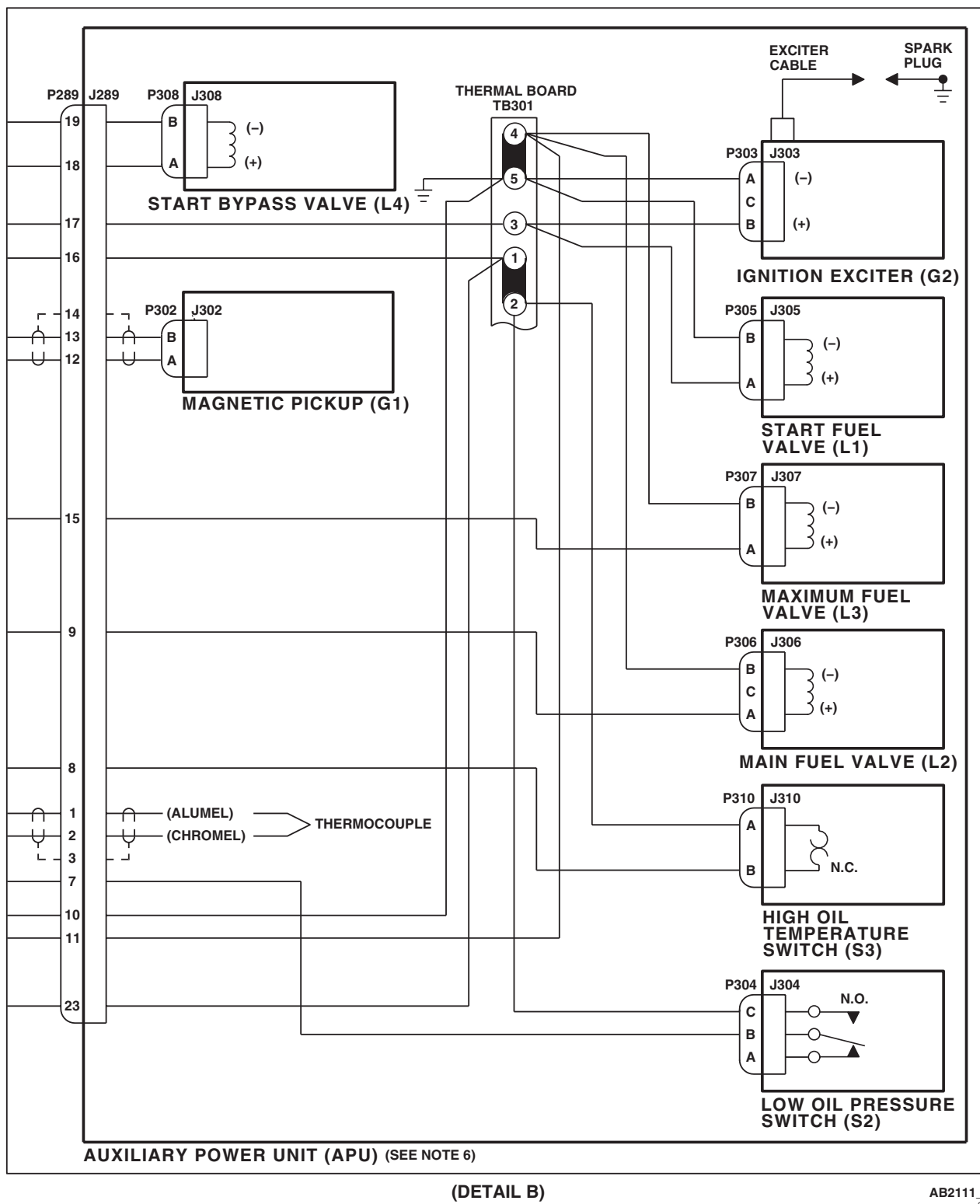
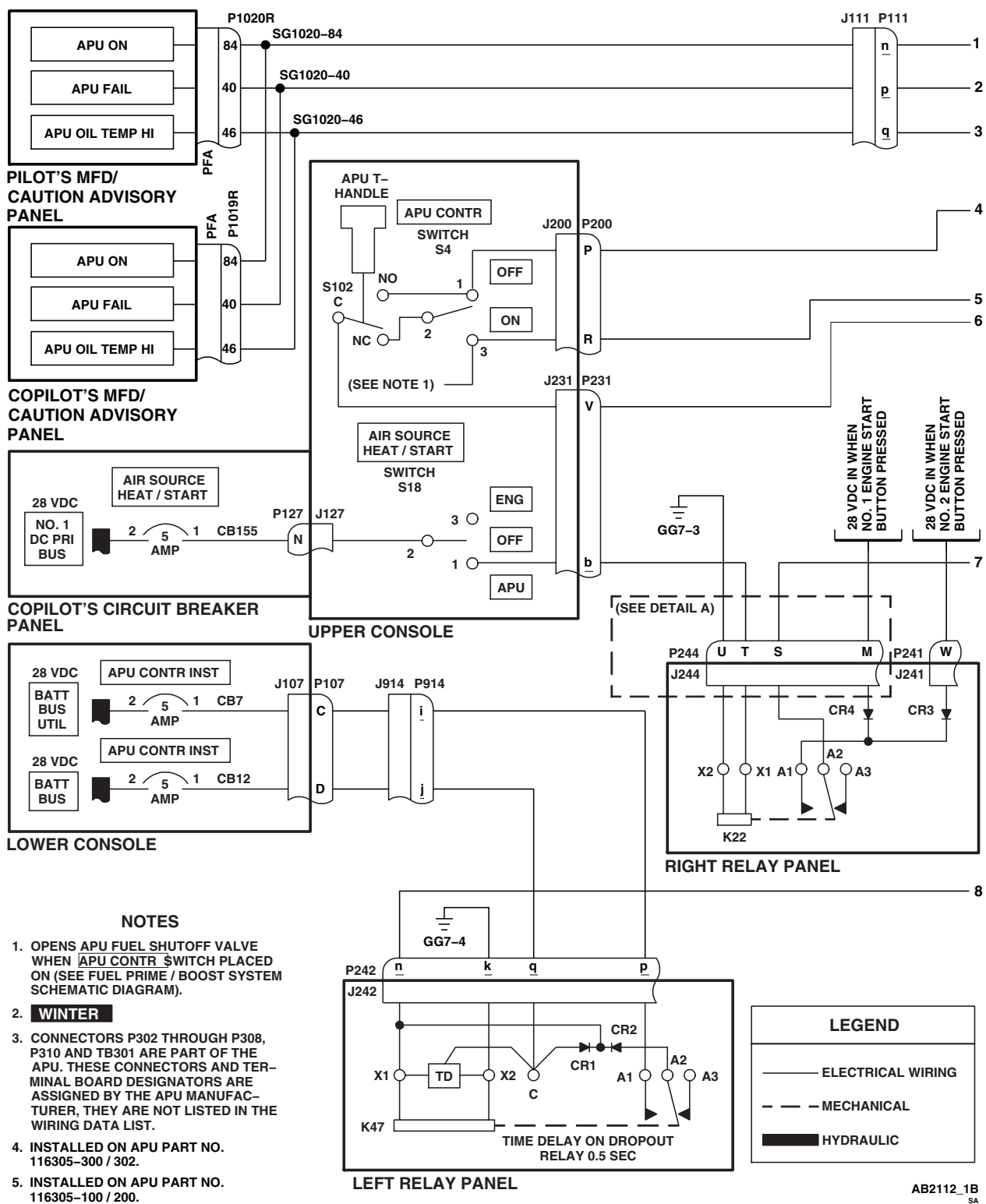


Figure 3. Auxiliary Power Unit System Schematic Diagram UH-60A UH-60L . (Sheet 5 of 5)

## SCHEMATICS - Continued

Figure 4. Auxiliary Power Unit System Schematic Diagram **HH-60L HH-60A** . (Sheet 1 of 5)

**Wiring Diagram for Electronic Sequence Unit (ESU)**

**Connections:**

- P260 J2:** 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11
- J1 P261:** 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24

**Logic and Timing Conditions:**

- CIRCUIT COMPLETE WHEN APU SPEED AT 90% +1.5 SEC**
- 28 VDC OUT WHEN APU SPEED ABOVE 5% AND BELOW 70%**
- 28 VDC IN WHEN APU CONTR**
- SWITCH PLACED TO (DISABLED AFTER 14% SPEED)**
- 28 VDC OUT WHEN APU SPEED AT 90% + 1.5 SEC**
- 28 VDC OUT WHEN OIL TEMP ABOVE 300°F (149°C) AND APU SPEED AT 90% +1.5 SEC**
- 28 VDC IN WHEN BAT SWITCH PLACED ON AND OIL TEMP IS BELOW 300°F (149°C) AT 90% +1.5 SEC APU SPEED**
- LOGIC CIRCUIT**
- 28 VDC OUT (AND APU SHUTS DOWN) WHEN:**
  - A. APU REACHES 90% SPEED, BUT THEN DROPS BELOW 90% (UNDER SPEED)
  - B. APU SPEED ABOVE 110% (OVERSPEED)
  - C. APU TAKES LONGER THAN 30 SECONDS TO GO FROM 50% TO 90% SPEED (SEQUENCE FAIL)
  - D. EGT IS LESS THAN 50° AT 30% APU SPEED
  - E. APU SPEED ABOVE 70% AND OIL PRESSURE IS BELOW 5 TO 7 PSI (LOW OIL PRESS)
  - F. EGT IS BETWEEN 1346°F – 1385°F (730°C – 752°C) AND APU SPEED IS LESS THAN 90% (OVERTEMP)
  - G. EGT IS BETWEEN 1241°F – 1281°F (672°C – 694°C) AND APU SPEED IS BETWEEN 90% – 110% (OVERTEMP)
  - H. OPEN THERMOCOUPLE SENSED
- 28 VDC IN WHEN APU SPEED ABOVE 70% AND OIL PRESSURE IS ABOVE 5 TO 7 PSI**
- EXHAUST GAS TEMPERATURE SENSE**
- EMI / RFI GROUNDING CIRCUIT**
- 28 VDC OUT WHEN APU CONTR**
- SWITCH PLACED TO ON AND APU SPEED IS LESS THAN 70%**

**Winterization Kit Harness Assembly (DETAIL A)**

**Connections:**

- P244 J244A:** M, U, T, S
- J244B J244:** M, U, T, S

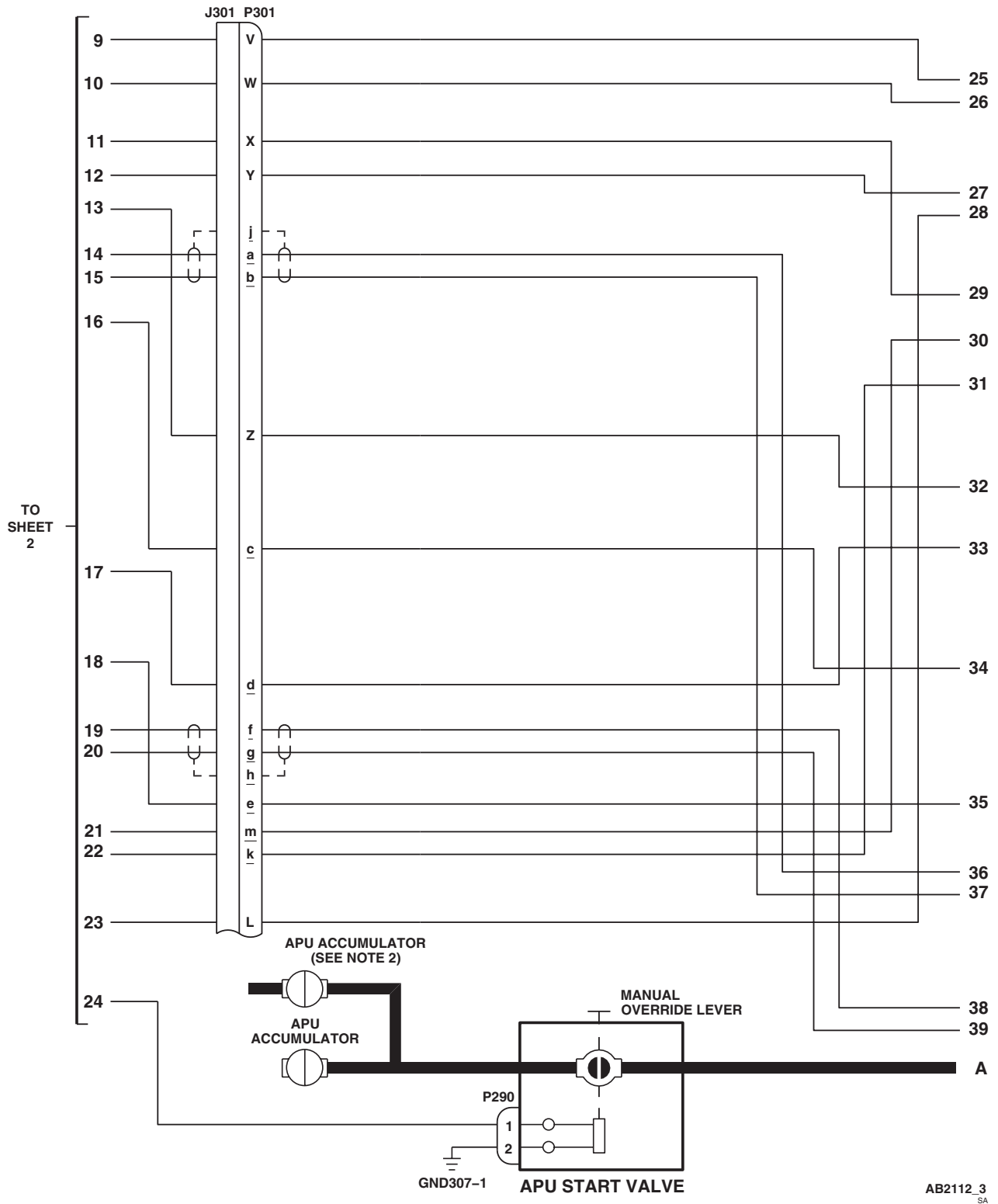
**WINTERIZATION KIT HARNESS ASSEMBLY**

**DETAIL A (SEE NOTE 2)**

**AB2112**

Figure 4. Auxiliary Power Unit System Schematic Diagram **HH-60L HH-60A**. (Sheet 2 of 5)

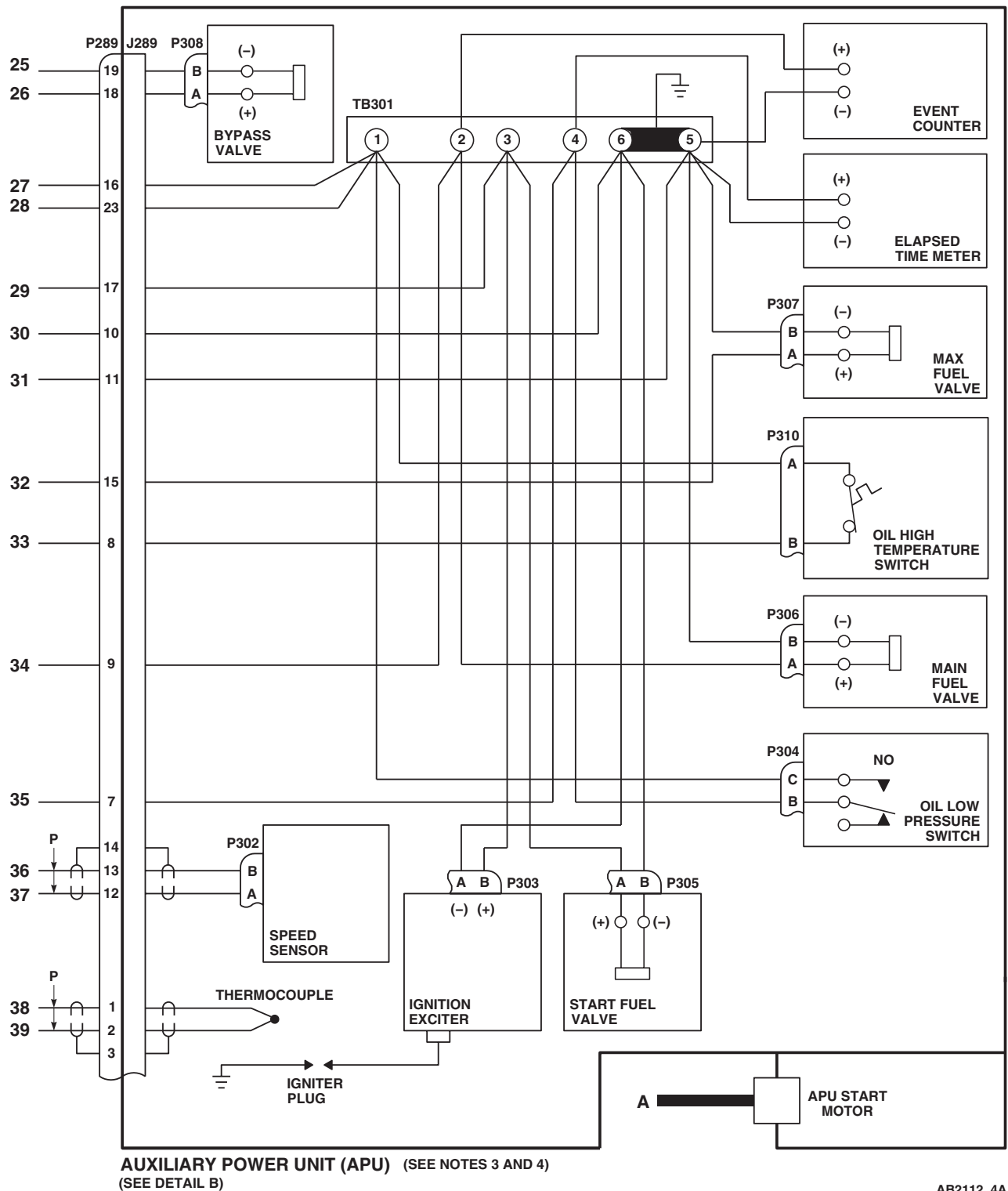
SCHEMATICS - Continued



AB2112\_3  
SA

Figure 4. Auxiliary Power Unit System Schematic Diagram **HH-60L HH-60A** . (Sheet 3 of 5)

## SCHEMATICS - Continued

Figure 4. Auxiliary Power Unit System Schematic Diagram **HH-60L HH-60A** . (Sheet 4 of 5)

SCHEMATICS - Continued

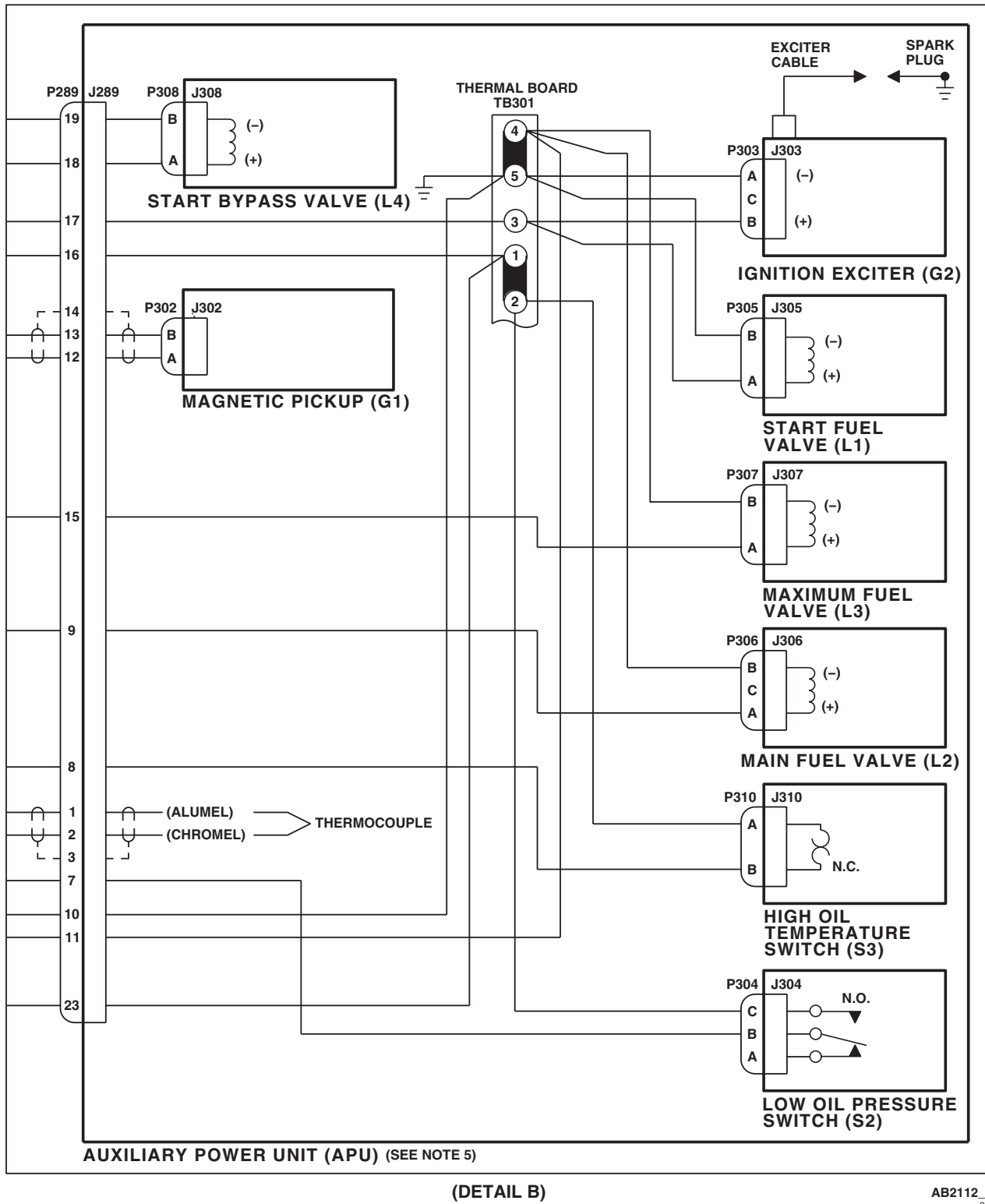


Figure 4. Auxiliary Power Unit System Schematic Diagram **HH-60L HH-60A** . (Sheet 5 of 5)

SCHEMATICS - Continued

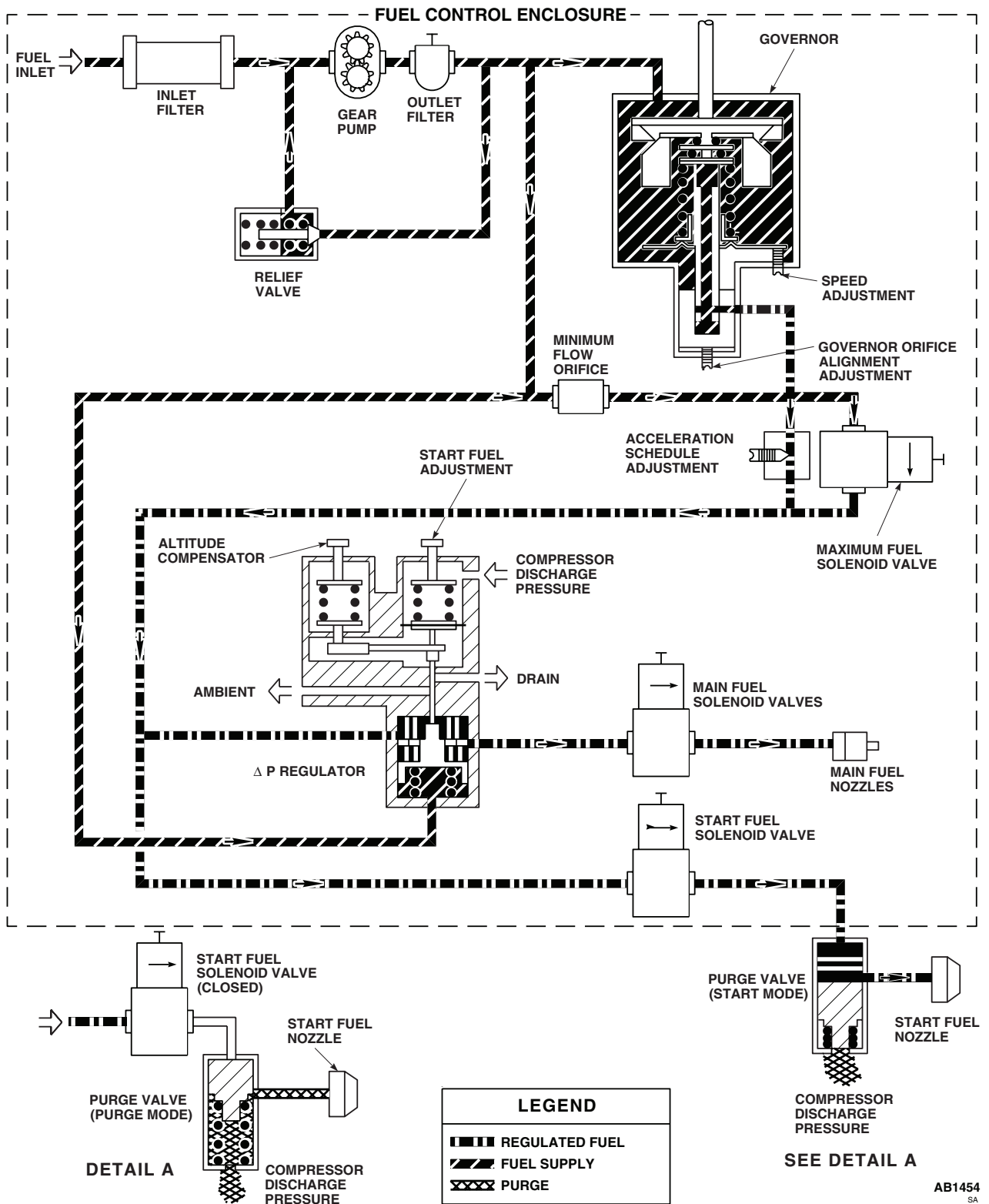


Figure 5. Auxiliary Power Unit Fuel System Schematic.

## SCHEMATICS - Continued

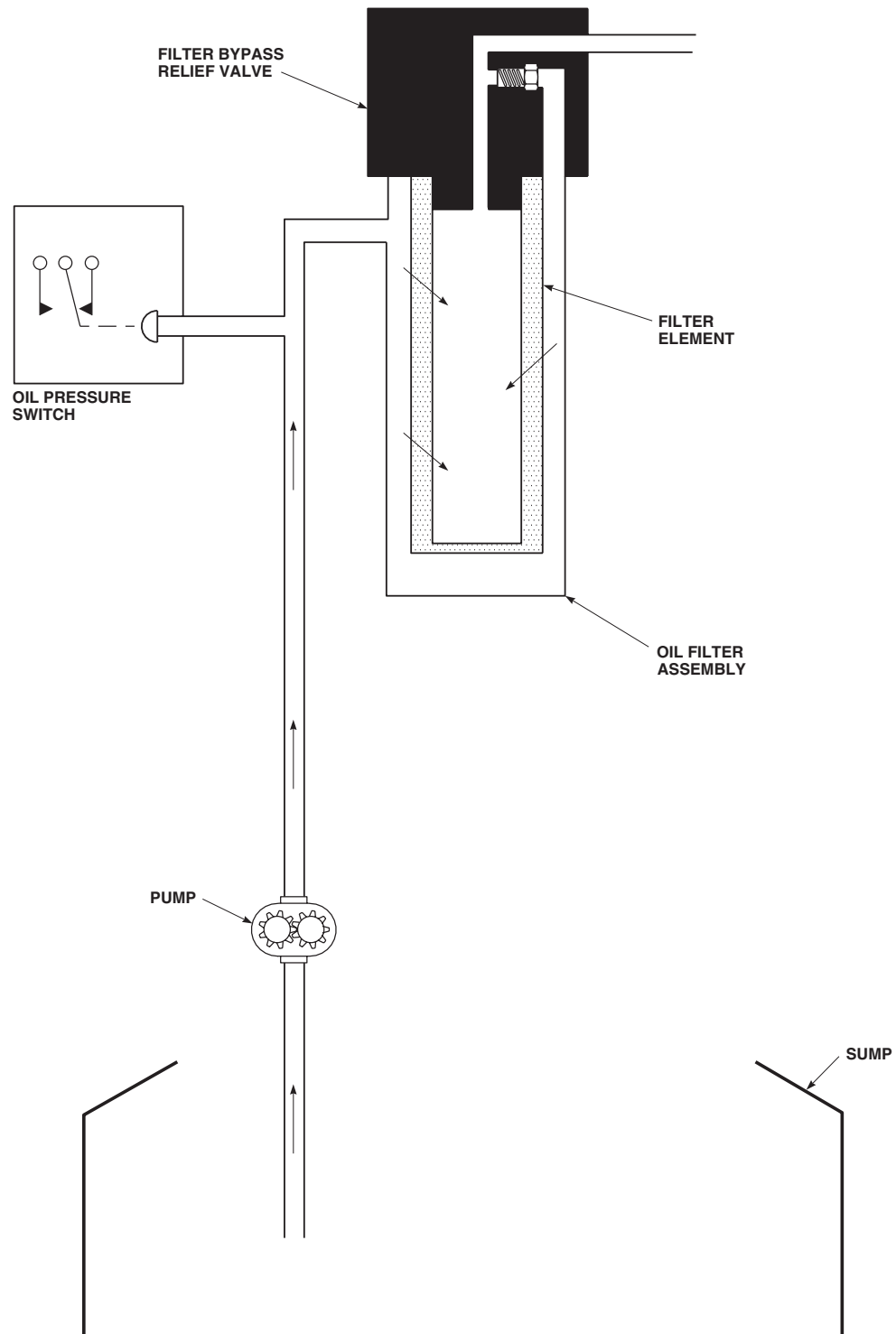
AB1455  
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Figure 6. Auxiliary Power Unit Lubrication System Schematic.

END OF WORK PACKAGE



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**UNIT LEVEL****AUXILIARY POWER PLANT SYSTEM****AUXILIARY POWER UNIT SYSTEM (3800480 SERIES)****This WP supersedes WP 0156 00, dated 17 April 2006.**

---

**INITIAL SETUP:****Test Equipment**

Multimeter, AN/PSM-45A

WP 0824 00

WP 0825 00

WP 0828 00

**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01

Electrical Repairer Toolkit, SC 5180-99-B06

WP 0842 00

WP 0865 00

WP 0925 00

**Personnel Required**

Aircraft Electrician MOS 15F (1)

UH-60 Helicopter Repairer MOS 15T (1)

WP 1359 00

WP 1360 00

WP 1365 00

**References**

TM 1-1520-237-10

TM 55-2835-209-23

WP 0086 00

WP 0101 00

WP 0102 00

WP 0104 00

WP 0129 00

WP 0258 00

WP 0702 00

WP 0743 00

WP 0785 00

WP 1366 00

WP 1604 00

WP 1605 00

WP 1685 00

WP 1747 00

**Equipment Condition**

Accumulator Reservoir Properly Serviced,

WP 1605 00

Battery Serviceable and Fully Charged,

External Power Available, WP 1685 00

No. 1 Fuel Tank Serviced, TM 1-1520-237-10

---

**APPLICABLE CONFIGURATION**

APU 3800480-1 and 3800480-2

## AUXILIARY POWER UNIT SYSTEM (3800480 SERIES) OPERATIONAL/TROUBLESHOOTING PROCEDURE

### PROCEDURE

#### Step

#### CAUTION

- To prevent possible damage to APU, do not operate APU more than 30 minutes when main engines and rotor are running and outside temperature is greater than 109°F. Continuous operation of APU is permitted at temperatures up to 124°F only if main engines and rotors are not running.
- If onboard fire extinguishing is required before a generator becomes operational, RESERVE fire extinguisher must be used to fight fire.

#### NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- See Figure 1 for component location diagram, Figure 2 for system schematic diagram, Figure 3 for lubrication system schematic, and Figure 4 for fuel system schematic as an aid in operational/troubleshooting.

1. Check bleed air pneumatic tube couplings for loose clamps, cuts, and/or incorrect installation at all connecting points (WP 0702 00).

#### Indication/Condition

All couplings shall be securely installed with clamps and there shall be no cuts in pneumatic tube lines.

#### Corrective Action

If Indication/Condition is not as specified, repair/replace couplings and lines (WP 0702 00).

#### Step

2. Check for oil leaks at the APU components:
  - Oil pressure plug and packing
  - Oil fill cap and gasket
  - Starter motor flange gasket
  - Breather vent fitting and packing
  - Oil filter cover and packing
  - Low oil pressure switch and packing
  - Monopole and packing
  - Generator flange gasket
  - Hot sensor (oil temp bulb) and packing
  - Fill-to-spill plug and packing

---

**AUXILIARY POWER UNIT SYSTEM (3800480 SERIES) OPERATIONAL/TROUBLESHOOTING PROCEDURE**

- Continued

- Magnetic drain plug and packing
- Fuel control upper cover and packing
- Compression housing
- Spar idler gearshift retainer
- Gear box housing

**Indication/Condition**

No leaks shall occur.

**Corrective Action**

If any component listed above leaks oil, replace the component and its packing or gasket (TM 55-2835-209-23).

**Step**

3. Check for fuel leaks at the APU components:

- Fuel inlet fitting
- Fuel control assembly
- Fuel control-to-solenoid valve tube
- Fuel solenoid valve-to-manifold assembly tube
- Fuel control lower cover fuel drain adapter and packing
- Fuel manifold assembly
- Primary fuel nozzle and seal plate
- Fuel filter cover and packing
- Combuster housing fuel drain tube
- Orifice fitting and packing

**Indication/Condition**

No fuel leaks shall occur.

**Corrective Action**

If any component listed leaks fuel, replace the component and its packing or gasket (TM 55-2835-209-23).

**Step**

4. Make sure upper console BATT, GENERATORS APU, and APU CONTR switches are placed OFF.
5. Make sure the following circuit breakers are pushed in:

---

**CIRCUIT BREAKER**

NO. 2 ENG START CONTR

---

**CIRCUIT BREAKER PANEL**

Pilot's

---

**AUXILIARY POWER UNIT SYSTEM (3800480 SERIES) OPERATIONAL/TROUBLESHOOTING PROCEDURE**  
- Continued

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| <b>CIRCUIT BREAKER</b>  | <b>CIRCUIT BREAKER PANEL</b> |
|-------------------------|------------------------------|
| UTIL RECP CABIN         | Copilot's                    |
| AIR SOURCE HEAT / START | Copilot's                    |
| PUMP PWR                | Copilot's                    |
| NO. 1 ENG START         | Upper Console                |
| APU CONTR INST          | Lower Console                |
| FUEL PRIME BOOST        | Lower Console                |
| APU CONTR INST          | Lower Console                |

6. Make sure upper console APU T-handle is pushed in.

7. Remove soundproofing material and open panel below ESU in cabin ceiling (WP 0258 00).

**NOTE**

- An observer must be in the cabin area to watch the ESU BITE indicators during the operational/troubleshooting procedure.
- The observer shall call out the procedural steps to the operator in the cockpit. This will allow the observer to watch for a change in ESU BITE indication as the operator does the next procedural step.

8. Check ESU BITE indicators.

**Indication/Condition**

All indicators shall be white unless a previous failure has been detected.

**Corrective Action**

If Indication/Condition is not as specified, troubleshoot system per BITE INDICATION SYMPTOMS, in this work package and NON-BITE INDICATION SYMPTOMS, in this work package, as required.

**Step****NOTE**

Do not attempt to start the APU if BATT LOW CHARGE capsule on caution/advisory panel is on.

9. Place upper console BATT switch to ON.

## AUXILIARY POWER UNIT SYSTEM (3800480 SERIES) OPERATIONAL/TROUBLESHOOTING PROCEDURE

- Continued

### NOTE

- The ESU BITE indicators will not change status.
- If the APU shuts down, the observer shall take note of the last ESU speed BITE indication before the malfunction BITE indication appears.
- To clear BITE code on ESU, cycle APU switch. The APU should start if malfunction has been fixed. If malfunction still exists, the ESU will not allow the hydraulic accumulator to dump and the APU starter will not engage.

10. Start APU (TM 1-1520-237-10).

### Indication/Condition

a. APU shall start and BITE indicators shall cycle as follows:

- ● ● ● -RESET
- ● ● ● -5%
- ○ ● ● -70%
- ○ ○ ● -95% + 1.5 sec

b. APU ON capsule on caution/advisory panel shall go on.

c. APU FAIL capsule on caution/advisory panel shall be off.

### CAUTION

If APU OIL TEMP HI capsule on caution/advisory panel is on, shut down APU immediately.

d. APU OIL TEMP HI capsule on caution/advisory panel shall be off.

### Corrective Action

If Indication/Conditions are not as specified, the APU CONT switch should be left in the ON position unless an emergency condition exists, until the APU FAIL capsule goes on.

### Step

11. On upper console, place GENERATORS APU switch to ON to recharge APU accumulator.

### Indication/Condition

APU ACCUM LOW capsule on caution/advisory panel shall go off in 90 seconds and APU accumulator shall be charged. Visually check accumulator for proper charge (pressure gage and tape indicator).

### Corrective Action

1. If Indication/Condition is not as specified, troubleshoot hydraulic system (WP 0086 00).
2. If APU shuts down and loss of DC power indication (● ● ● ●) appears on ESU, go to LOSS OF DC POWER (● ● ● ●), in this work package.

### Step

12. Shut down APU (TM 1-1520-237-10).

13. Check ESU BITE indicators after APU is shut down.

**AUXILIARY POWER UNIT SYSTEM (3800480 SERIES) OPERATIONAL/TROUBLESHOOTING PROCEDURE**

- Continued

**Indication/Condition**

All ESU BITE indicators shall be white.

**Corrective Action**

If Indication/Condition is not as specified, troubleshoot the system for the malfunction indicated (BITE INDICATION SYMPTOMS, in this work package).

**Step**

14. Check combustor drain line (below helicopter).

**Indication/Condition**

Small amount of fuel shall be draining.

**Corrective Action**

If Indication/Condition is not as specified, check APU combustor drain valve. Replace if necessary (TM 55-2835-209-23).

**Step**

15. On upper console, place BATT switch to OFF.  
16. On cabin ceiling, close panel below ESU and install soundproofing material (WP 0258 00).

**Table 1. BITE Indication Symptoms**

| ESU BITE INDICATION | BITE DEFINITION            | REFER TO                                                   |
|---------------------|----------------------------|------------------------------------------------------------|
| ● ● ● ●             | Loss of DC Power           | LOSS OF DC POWER (● ● ● ●), in this work package           |
| ○ ● ● ●             | ESU Failure                | ESU FAILURE (○ ● ● ●), in this work package                |
| ● ○ ● ●             | A/C Start System Failure   | A/C START SYSTEM FAILURE (● ○ ● ●), in this work package   |
| ○ ○ ● ●             | Overtemperature            | OVERTEMPERATURE (○ ○ ● ●), in this work package            |
| ● ● ○ ●             | Overspeed                  | OVERSPEED (● ● ○ ●), in this work package                  |
| ○ ● ○ ●             | Underspeed                 | UNDERSPEED (○ ● ○ ●), in this work package                 |
| ● ○ ○ ●             | Fail to Start              | FAIL TO START (● ○ ○ ●), in this work package              |
| ○ ○ ○ ●             | Low Oil Pressure           | LOW OIL PRESSURE (○ ○ ○ ●), in this work package           |
| ● ● ● ○             | Oil Pressure Switch Failed | OIL PRESSURE SWITCH FAILED (● ● ● ○), in this work package |

**AUXILIARY POWER UNIT SYSTEM (3800480 SERIES) OPERATIONAL/TROUBLESHOOTING PROCEDURE**  
 - Continued

**Table 1. BITE Indication Symptoms - Continued**

| ESU BITE INDICATION | BITE DEFINITION             | REFER TO                                                 |
|---------------------|-----------------------------|----------------------------------------------------------|
| ○ ● ● ○             | Thermocouple Failed         | THERMOCOUPLE FAILED (○ ● ● ○), in this work package      |
| ● ○ ● ○             | Monopole Failed             | MONOPOLE FAILED (● ○ ● ○), in this work package          |
| ○ ○ ● ○             | Ignition Unit Failed        | IGNITION UNIT FAILED (○ ○ ● ○), in this work package     |
| ● ● ○ ○             | Fuel Solenoid Failed        | FUEL SOLENOID FAILED (● ● ○ ○), in this work package     |
| ○ ● ○ ○             | Hot Sensor Failed           | HOT SENSOR FAILED (○ ● ○ ○), in this work package        |
| ● ○ ○ ○             | Fuel Torque Motor Failed    | FUEL TORQUE MOTOR FAILED (● ○ ○ ○), in this work package |
| ○ ○ ○ ○             | No Data (while APU running) | NO DATA ○ ○ ○ ○, in this work package                    |

**Table 2. Non-BITE Indication Symptoms.**

| APU TROUBLE SYMPTOM                                                                  | REFER TO                                                                                  |
|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| APU running; APU ON advisory capsule will not go on.                                 | APU ON ADVISORY CAPSULE FAILURE, in this work package                                     |
| APU running; APU FAIL advisory capsule remains on.                                   | APU FAIL ADVISORY CAPSULE FAILURE, in this work package                                   |
| APU shut down; APU ON advisory capsule remains on.                                   | APU ON ADVISORY CAPSULE FAILURE, in this work package                                     |
| APU emergency shutdown; APU FAIL advisory capsule will not go on.                    | APU FAIL ADVISORY CAPSULE FAILURE, in this work package                                   |
| APU oil temperature is less than 325°F, APU OIL TEMP HI advisory capsule remains on. | APU OIL TEMP HI ADVISORY CAPSULE FAILURE, in this work package                            |
| APU does not shut down when APU CONTR switch is in off position.                     | APU DOES NOT SHUT DOWN WHEN APU CONTR SWITCH IS IN THE OFF POSITION, in this work package |

**AUXILIARY POWER UNIT SYSTEM (3800480 SERIES) OPERATIONAL/TROUBLESHOOTING PROCEDURE**

- Continued

**Table 3. Normal Start Sequence.**

| <b>APU SPEED SEQUENCE (% RPM)</b> | <b>EVENT</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A. 0%                             | <p>*1. ESU powers up and performs prestart BITE.</p> <p>*2. ESU internally checks for processor sequence failure, data failure and conditional driver failure.</p> <p>*3. EGT overtemperature shutdown limit set to 1810°F (988°C).</p> <p>*4. Fail to start timer in ESU starts (30 seconds).</p> <p>*5. Oil pressure switch fault is enabled.</p> <p>*6. APU start valve is energized if prestart BITE and internal test Indication/Conditions were okay.</p> <p>**BITE INDICATION: ● ● ● ●</p> |
| B. 5%                             | <p>*1. Fuel solenoid - on.</p> <p>*2. Ignition unit - on.</p> <p>*3. Fail to start timer is increased to 70 seconds.</p> <p>*4. Fuel pressure built up due to motoring speed.</p> <p>**BITE INDICATION: ○ ● ● ●</p>                                                                                                                                                                                                                                                                               |
| C. 10%                            | *Oil pressure switch fault is disabled.                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| D. 40%                            | *ESU checks for indication of ignition.                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| E. 50%                            | *EGT overtemperature shutdown limit begins decreasing to 1350°F (732°C) at 90% speed.                                                                                                                                                                                                                                                                                                                                                                                                             |
| F. 70%                            | <p>*1. APU start valve (hydraulic/electrical) - off.</p> <p>**BITE INDICATION: ○ ○ ● ●</p>                                                                                                                                                                                                                                                                                                                                                                                                        |
| G. 95%                            | <p>*1. Ignition - off.</p> <p>*2. Fail to start circuit is deactivated.</p> <p>*3. Underspeed fault detection is activated and will shut down the APU if speed decreases below 90% for 10 seconds.</p> <p>*4. Low oil pressure fault detection is activated and will shut down the APU if oil pressure switch remains closed for 15 seconds.</p> <p>*5. ESU checks for EGT to be above 300°F. Shuts down APU if not.</p>                                                                          |



# AUXILIARY POWER UNIT SYSTEM (3800480 SERIES) OPERATIONAL/TROUBLESHOOTING PROCEDURE

- Continued

**Table 3. Normal Start Sequence. - Continued**

| APU SPEED SEQUENCE (% RPM) | EVENT                                                                                                                                                                                   |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            | **BITE INDICATION: ○ ○ ○ ●                                                                                                                                                              |
| H. 95% + 1.5 seconds       | *1. APU ON capsule - on.                                                                                                                                                                |
|                            | *2. Hourmeter/start counter - on.                                                                                                                                                       |
|                            | **BITE INDICATION: ● ● ● ●                                                                                                                                                              |
| I. 99%                     | *EGT overtemperature shutdown limit per following table:                                                                                                                                |
|                            | <b>EGT Range Time Delay</b>                                                                                                                                                             |
|                            | 1265 - 1328°F 8.9 sec.                                                                                                                                                                  |
|                            | 1329 - 1379°F 6.0 sec.                                                                                                                                                                  |
|                            | 1380 - 1419°F 3.7 sec.                                                                                                                                                                  |
|                            | 1420 - 1439°F 2.0 sec.                                                                                                                                                                  |
|                            | 1440 - 1459°F 1.0 sec.                                                                                                                                                                  |
|                            | 1460+°F 0.1 sec.                                                                                                                                                                        |
|                            | *ESU checks on system operation. If check is not satisfactory, APU will shut down. **BITE indication will appear on ESU when APU has reached corresponding speed during start sequence. |

## Indication/Condition

Panel shall be closed and soundproofing material shall be installed.

## Corrective Action

None Required

## LOSS OF DC POWER (● ● ● ●)

### SYMPTOM

Loss of dc power (● ● ● ●).

### MALFUNCTION

Loss of dc power (● ● ● ●).

### CORRECTIVE ACTION

#### 1. Check for 28 vdc between P260-5 and ground.

- If voltage is as specified, go to **2**.
- If voltage is not as specified, go to **3**.

**LOSS OF DC POWER (● ● ● ●) - Continued****2. Check continuity between P260-6 and ground.**

- a. If continuity is present, replace ESU (WP 0865 00). Go to **6**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **6**.

**3. Check continuity between P242-n and P260-5.**

- a. If continuity is present, go to **4**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **6**.

**4. Check for 28 vdc between P242-g and ground.**

- a. If voltage is as specified, replace left relay panel (WP 0825 00). Go to **6**.
- b. If voltage is not as specified, go to **5**.

**5. Check for 28 vdc between BATT BUS APU CONTR INST circuit breaker terminal 1 and ground.**

- a. If voltage is as specified, repair/replace wiring between circuit breaker and P242-g (WP 1747 00). Go to **6**.
- b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **6**.

**6. Procedure completed.****ESU FAILURE (○ ● ● ●)****SYMPTOM**

ESU failure (○ ● ● ●).

**MALFUNCTION**

ESU failure (○ ● ● ●).

**CORRECTIVE ACTION**

- 1. Replace ESU (WP 0865 00). Go to 2.**
- 2. Procedure completed.**

**A/C START SYSTEM FAILURE (● ○ ● ●)****SYMPTOM**

A/C start system failure (● ○ ● ●).

**MALFUNCTION**

A/C start system failure (● ○ ● ●).

**CORRECTIVE ACTION**

- 1. Make sure APU start valve connectors P290 and P311 are installed. Go to 2.**
- 2. Does APU start?**
  - a. If APU starts, system is operational. Go to **6**.
  - b. If APU does not start, go to **3**.
- 3. Check for 20 to 500 ohms between P260-9 and ground.**
  - a. If resistance is as specified, replace ESU (WP 0865 00). Go to **6**.

---

**A/C START SYSTEM FAILURE (● ○ ● ●) - Continued**

- b. If resistance is not as specified, go to **4**.
- 4. Check continuity between P260-9 and P290-1.**
  - a. If continuity is present, go to **5**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **6**.
- 5. Check continuity between P290-2 and ground.**
  - a. If continuity is present, replace APU start valve (WP 0743 00). Go to **6**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **6**.
- 6. Procedure completed.**

**OVERTEMPERATURE (○ ○ ● ●)****SYMPTOM**

Overtemperature (○ ○ ● ●).

**MALFUNCTION**

Overtemperature (○ ○ ● ●).

**CORRECTIVE ACTION**

- 1. Using a 1 inch socket, hand rotate generator fan clockwise (viewed aft). Does fan rotate with normal resistance?**
  - a. If fan rotates with normal resistance, go to **2**.
  - b. If fan does not rotate with normal resistance, go to **9**.
- 2. Check for leakage in aircraft pneumatic ducts.**
  - a. If leakage is present, repair/replace ducts, as required (WP 1360 00). Go to **13**.
  - b. If leakage is not present, go to **3**.
- 3. Is there abnormal noise during fan rotation?**
  - a. If abnormal noise is heard, go to **4**.
  - b. If abnormal noise is not heard, go to **8**.
- 4. During fan rotation, does APU compressor rotate?**
  - a. If APU compressor rotates, go to **5**.
  - b. If APU compressor does not rotate, go to **7**.
- 5. Is noise from APU?**
  - a. If noise is heard, replace APU (WP 1359 00). Go to **13**.
  - b. If noise is not heard, go to **6**.
- 6. Is noise coming from APU start motor?**
  - a. If noise is heard, replace APU start motor (WP 1365 00). Go to **13**.
  - b. If noise is not heard, replace APU generator (WP 0842 00). Go to **13**.
- 7. Does APU generator fan turn counterclockwise freely?**
  - a. If fan turns counterclockwise freely, replace vespal spline adapter (WP 1366 00). Go to **13**.

**OVERTEMPERATURE (○ ○ ● ●) - Continued**

b. If fan does not turn counterclockwise freely, replace APU (WP 1359 00). Go to **13**.

**8. Check combustor housing flange. Is there evidence of leakage?**

a. If evidence of leakage is present, replace APU (WP 1359 00). Go to **13**.

b. If evidence of leakage is not present, replace fuel control assembly (TM 55-2835-209-23). Go to **13**.

**9. Remove APU start motor (WP 1365 00). Go to 10.****10. Rotate generator fan. Does fan rotate with normal resistance?**

a. If fan rotates with normal resistance, replace APU start motor (WP 1365 00). Go to **13**.

b. If fan does not rotate with normal resistance, go to **11**.

**11. Remove APU generator (WP 0842 00). Go to 12.****12. Does generator shaft spin freely?**

a. If generator shaft spins freely, replace APU (WP 1359 00). Go to **13**.

b. If generator shaft does not spin freely, replace APU generator (WP 0842 00). Go to **13**.

**13. Procedure completed.****OVERSPEED (● ● ○ ●)****SYMPTOM**

Overspeed (● ● ○ ●).

**MALFUNCTION**

Overspeed (● ● ○ ●).

**CORRECTIVE ACTION****1. Replace fuel control assembly (TM 55-2835-209-23). Go to 2.****2. Check APU for normal operation.**

a. If APU operates normally, system is operational. Go to **3**.

b. If APU does not operate normally, replace APU (WP 1359 00). Go to **3**.

**3. Procedure completed.****UNDERSPEED (○ ● ○ ●)****SYMPTOM**

Underspeed (○ ● ○ ●).

**MALFUNCTION**

Underspeed (○ ● ○ ●).

**CORRECTIVE ACTION****1. Is APU fuel filter assembly serviceable?**

a. If APU fuel filter assembly is serviceable, go to **2**.

b. If APU fuel filter assembly is not serviceable, replace APU fuel filter assembly (TM 55-2835-209-23). Go to **3**.

---

**UNDERSPEED (○● ○ ●) - Continued****2. Troubleshoot fuel prime boost system (WP 0129 00).**

- a. If fuel prime boost system is good, go to **Low Oil Pressure (○ ○ ○ ●)**.
- b. If fuel prime boost system is not good, repair/replace as required (WP 0129 00). Go to **3**.

**3. Procedure completed.****FAIL TO START (● ○ ○ ●)****SYMPTOM**

Fail to start (● ○ ○ ●).

**MALFUNCTION**

Fail to start (● ○ ○ ●).

**CORRECTIVE ACTION****1. Does APU motor?**

- a. If APU motors, go to **2**.
- b. If APU does not motor, go to **21**.

**2. Does APU light off?**

- a. If APU lights off, go to **3**.
- b. If APU does not light off, go to **8**.

**3. Check combustor housing flange. Is there evidence of excessive leakage?**

- a. If excessive leakage is present, replace APU (WP 1359 00). Go to **27**.
- b. If excessive leakage is not present, go to **4**.

**4. Make sure APU bleed valve is closed.**

- a. If valve is closed, go to **5**.
- b. If valve is not closed, troubleshoot APU bleed valve and associated electrical wiring (WP 0101 00 and WP 0102 00). Go to **27**.

**5. Check APU fuel filter assembly. Is it serviceable?**

- a. If APU fuel filter assembly is serviceable, go to **6**.
- b. If APU fuel filter assembly is not serviceable, replace APU fuel filter assembly (TM 55-2835-209-23). Go to **27**.

**6. Replace fuel control assembly (TM 55-2835-209-23). Go to 7.****7. Does APU fail to start?**

- a. If APU fails, replace APU (WP 1359 00). Go to **27**.
- b. If APU does not fail, system is operational. Go to **27**.

**8. Is igniter plug assembly firing audibly during APU motoring?**

- a. If igniter plug assembly fires audibly, go to **9**.
- b. If igniter plug assembly does not fire audibly, go to **13**.

**9. Is fuel present at APU fuel drain?**

---

**FAIL TO START (● ○ ○ ●) - Continued**

- a. If fuel is present, go to **10**.
- b. If fuel is not present, go to **12**.
- 10. With APU CONTR switch placed in OFF position and APU accumulator serviced, press APU start valve manual override lever. Go to 11.**
- 11. When accumulator is fully recharged, attempt start. Go to 27.**
- 12. Is fuel present at APU fuel inlet?**
  - a. If fuel is present, replace fuel solenoid valve (TM 55-2835-209-23). Go to **27**.
  - b. If fuel is not present, troubleshoot fuel prime boost system (WP 0129 00). Go to **27**.
- 13. During APU motoring, do BITE indicators sequence, as described in A/C START SYSTEM FAILURE (● ○ ● ●), in this work package, through 5% speed description?**
  - a. If BITE indicators sequence during motoring, check/replace igniter plug assembly, igniter plug lead, and ignition unit (TM 55-2835-209-23). Go to **27**.
  - b. If BITE indicators do not sequence during motoring, go to **14**.
- 14. Disconnect W1P6 from J6 on the monopole. Go to 15.**
- 15. Check for open circuit between P261-12 and P261-13.**
  - a. If open circuit is present, replace monopole (TM 55-2835-209-23). Go to **27**.
  - b. If open circuit is not present, go to **16**.
- 16. Check for open circuit between J289-12 and J289-13.**
  - a. If open circuit is present, go to **17**.
  - b. If open circuit is not present, go to **19**.
- 17. Repair/replace wiring between P261 and P289 (TM 55-2835-209-23). Go to 18.**
- 18. Reconnect W1P6 to J6 on monopole. Go to 27.**
- 19. Repair/replace wiring between J289 and W1P6 (WP 1747 00). Go to 20.**
- 20. Reconnect W1P6 to J6 on monopole. Go to 27.**
- 21. Does APU start valve open after APU CONTR switch is placed in ON position?**
  - a. If valve opens after switch is on, go to **22**.
  - b. If valve does not open after switch is on, go to **26**.
- 22. Remove APU start motor (WP 1365 00). Go to 23.**
- 23. Does APU motor rotate freely by hand?**
  - a. If motor rotates freely by hand, go to **24**.
  - b. If motor does not rotate freely by hand, replace APU start motor (WP 1365 00). Go to **27**.
- 24. Remove APU generator (WP 0842 00). Go to 25.**
- 25. Does APU generator rotate freely by hand?**
  - a. If generator rotates freely by hand, replace APU (WP 1359 00). Go to **27**.
  - b. If generator does not rotate freely by hand, replace APU generator (WP 0842 00). Go to **27**.

**FAIL TO START (● ○ ○ ●) - Continued**

**26. With APU CONTR switch placed in OFF position and APU accumulation serviced, press APU start valve manual override lever. Does accumulator discharge?**

- a. If accumulator discharges, replace APU start valve (WP 0743 00). Go to **27**.
- b. If accumulator does not discharge, troubleshoot APU hydraulic start system (WP 0086 00). Go to **27**.

**27. Procedure completed.**

**LOW OIL PRESSURE (○ ○ ○ ●)****SYMPTOM**

Low oil pressure (○ ○ ○ ●).

**MALFUNCTION**

Low oil pressure (○ ○ ○ ●).

**CORRECTIVE ACTION****NOTE**

This failure indicates that ESU has sensed a short circuit between low oil pressure (LOP) switch pins for 15 seconds. This can only occur above 95% speed.

**1. Check APU oil sump for proper servicing. Is APU oil sump serviced?**

- a. If APU oil sump is good, go to **2**.
- b. If APU oil sump is not good, service APU oil sump (WP 1604 00). Go to **12**.

**2. Check APU oil filter element. Is it serviceable?**

- a. If APU oil filter is serviceable, go to **3**.
- b. If APU oil filter is not serviceable, replace APU oil filter element (TM 55-2835-209-23). Go to **12**.

**3. Disconnect W1P8 from low oil pressure switch. Go to 4.**

**4. Check for open circuit between P261-23 and P261-7.**

- a. If open circuit is present, go to **5**.
- b. If open circuit is not present, go to **11**.

**5. Check for open circuit between P261-7 and ground.**

- a. If open circuit is present, go to **6**.
- b. If open circuit is not present, go to **9**.

**6. Replace low oil pressure switch (TM 55-2835-209-23) and restart APU (TM 1-1520-237-10). Does APU operate properly?**

- a. If APU operates properly, system is operational. Go to **12**.
- b. If APU does not operate properly, remove APU (WP 1359 00), perform tasks in Chapter 6 Sections 3 through 6 (TM 55-2835-209-23), then go to **7**.

**7. Is lube pump drive (shear) pin found intact?**

- a. If pin is found intact, replace APU (WP 1359 00). Go to **12**.

**LOW OIL PRESSURE (○ ○ ○ ●) - Continued**

- b. If pin is not found intact, replace pin, service lubrication system, and replace oil filter per Chapter 6 Sections 12 through 19 (TM 55-2835-209-23), then go to **8**.

**8. Install APU (WP 1359 00). Does APU operate properly?**

- a. If APU operates properly, go to **12**.
- b. If APU does not operate properly, replace APU (WP 1359 00). Go to **12**.

**9. Check for open circuit between J289-7 and ground.**

- a. If open circuit is present, repair/replace wiring between P261 and P289 (WP 1747 00). Go to **12**.
- b. If open circuit is not present, go to **10**.

**10. Check for open circuit between J8-B and ground.**

- a. If open circuit is present, repair/replace wiring between J289 and W1P8 (WP 1747 00). Go to **12**.
- b. If open circuit is not present, replace low oil pressure switch (TM 55-2835-209-23). Go to **12**.

**11. Check for open circuit between J289-23 and J289-7.**

- a. If open circuit is present, repair/replace wiring between P261 and P289 (WP 1747 00). Go to **12**.
- b. If open circuit is not present, repair/replace wiring between J289 and W1P8 (WP 1747 00). Go to **12**.

**12. Procedure completed.****OIL PRESSURE SWITCH FAILED (● ● ● ○)****SYMPTOM**

Oil pressure switch failed (● ● ● ○).

**MALFUNCTION**

Oil pressure switch failed (● ● ● ○).

**CORRECTIVE ACTION****1. Check for maximum of 10 ohms between P261-7 and P261-23.**

- a. If resistance is as specified, replace ESU (WP 0865 00). Go to **4**.
- b. If resistance is not as specified, go to **2**.

**2. Check for maximum of 10 ohms between J8-A and J8-B.**

- a. If resistance is as specified, go to **3**.
- b. If resistance is not as specified, replace low oil pressure switch (TM 55-2835-209-23). Go to **4**.

**3. Check for maximum of 10 ohms between J289-7 and J289-23.**

- a. If resistance is as specified, repair/replace wiring between P261 and P289 (WP 1747 00). Go to **4**.
- b. If resistance is not as specified, repair/replace wiring between J289 and W1P8 (WP 1747 00). Go to **4**.



---

**OIL PRESSURE SWITCH FAILED (● ● ● ○) - Continued****4. Procedure completed.****THERMOCOUPLE FAILED (○ ● ● ○)****SYMPTOM**

Thermocouple failed (○ ● ● ○).

**MALFUNCTION**

Thermocouple failed (○ ● ● ○).

**CORRECTIVE ACTION**

- 1. Check for maximum of 50 ohms between P261-1 and P261-2.**
  - a. If resistance is as specified, go to **2**.
  - b. If resistance is not as specified, go to **5**.
- 2. Disconnect APU wiring harness assembly from thermocouple assembly terminals (TM 55-2835-209-23). Go to 3.**
- 3. Check for open circuit between P261-1 and P261-2.**
  - a. If open circuit is present, replace ESU (WP 0865 00). Go to **7**.
  - b. If open circuit is not present, go to **4**.
- 4. Check for open circuit between J289-1 and J289-2.**
  - a. If open circuit is present, repair/replace wiring between P261 and P289 (WP 1747 00). Go to **7**.
  - b. If open circuit is not present, repair/replace wiring between J289 and thermocouple assembly terminals (WP 1747 00). Go to **7**.
- 5. Check for maximum of 1 ohm between thermocouple assembly terminals AL and CH.**
  - a. If resistance is as specified, go to **6**.
  - b. If resistance is not as specified, replace thermocouple assembly (TM 55-2835-209-23). Go to **7**.
- 6. Check for maximum of 50 ohms between J289-1 and J289-2.**
  - a. If resistance is as specified, repair/replace wiring between P261 and P289 (WP 1747 00). Go to **7**.
  - b. If resistance is not as specified, repair/replace wiring between J289 and thermocouple assembly terminals (WP 1747 00). Go to **7**.
- 7. Procedure completed.**

**MONOPOLE FAILED (● ○ ● ○)****SYMPTOM**

Monopole failed (● ○ ● ○).

**MALFUNCTION**

Monopole failed (● ○ ● ○).

**CORRECTIVE ACTION**

- 1. Check for 3 to 20 ohms between P261-12 and P261-13.**
  - a. If resistance is as specified, go to **2**.
  - b. If resistance is not as specified, go to **5**.
- 2. Check for open circuit between:**
  - **P261-12 and P261-14**
  - **P261-13 and P261-14**
  - a. If open circuit is present, replace ESU (WP 0865 00). Go to **7**.
  - b. If open circuit is not present, go to **3**.
- 3. Check for open circuit between:**
  - **J289-12 and J289-14**
  - **J289-13 and J289-14**
  - a. If open circuit is present, repair/replace wiring between P261 and P289 (WP 1747 00). Go to **7**.
  - b. If open circuit is not present, go to **4**.
- 4. Check for open circuit between:**
  - **J6-A and J6-B**
  - **J6-C and J6-B**
  - a. If open circuit is present, repair/replace wiring between J289 and W1P6 (WP 1747 00). Go to **7**.
  - b. If open circuit is not present, replace monopole (motional pickup transducer) (TM 55-2835-209-23). Go to **7**.
- 5. Check for 3 to 20 ohms between J6-A and J6-C.**
  - a. If resistance is as specified, go to **6**.
  - b. If resistance is not as specified, replace monopole (motional pickup transducer) (TM 55-2835-209-23). Go to **7**.
- 6. Check for 3 to 20 ohms between J289-12 and J289-13.**
  - a. If resistance is as specified, repair/replace wiring between P261 and P289 (WP 1747 00). Go to **7**.
  - b. If resistance is not as specified, repair/replace wiring between J289 and W1P6 (WP 1747 00). Go to **7**.

---

**MONOPOLE FAILED (● ○ ● ○) - Continued****7. Procedure completed.****IGNITION UNIT FAILED (○ ○ ● ○)****SYMPTOM**

Ignition unit failed (○ ○ ● ○).

**MALFUNCTION**

Ignition unit failed (○ ○ ● ○).

**CORRECTIVE ACTION**

- 1. Disconnect W1P7 from ignition unit. Go to 2.**
- 2. Check for open circuit between P261-17 and P261-15.**
  - a. If open circuit is present, go to **3**.
  - b. If open circuit is not present, go to **8**.
- 3. Check for open circuit between P261-17 and ground.**
  - a. If open circuit is present, go to **4**.
  - b. If open circuit is not present, go to **7**.
- 4. Install an insulated jumper wire between W1P7-A and W1P7-B. Go to 5.**
- 5. Check for short circuit between P261-17 and P261-15.**
  - a. If short circuit is present, replace ignition unit (TM 55-2835-209-23). Go to **13**.
  - b. If short circuit is not present, go to **6**.
- 6. Check for short circuit between J289-17 and J289-15.**
  - a. If short circuit is present, repair/replace wiring between P261 and P289 (WP 1747 00). Go to **13**.
  - b. If short circuit is not present, repair/replace wiring between J289 and W1P7 (WP 1747 00). Go to **13**.
- 7. Check for open circuit between J289-17 and ground.**
  - a. If open circuit is present, go to **9**.
  - b. If open circuit is not present, go to **11**.
- 8. Check for open circuit between J289-17 and J289-15.**
  - a. If open circuit is present, go to **9**.
  - b. If open circuit is not present, go to **11**.
- 9. Repair/replace wiring between P261 and P289 (WP 1747 00). Go to 10.**
- 10. Reconnect W1P7 to J7 on the ignition unit. Go to 13.**
- 11. Repair/replace wiring between J289 and W1P7 (WP 1747 00). Go to 12.**
- 12. Reconnect W1P7 to J7 on the ignition unit. Go to 13.**
- 13. Procedure completed.**

**FUEL SOLENOID FAILED (● ● ○ ○)****SYMPTOM**

Fuel solenoid failed (● ● ○ ○).

**MALFUNCTION**

Fuel solenoid failed (● ● ○ ○).

**CORRECTIVE ACTION**

- 1. Check for 10 to 50 ohms between P261-10 and P261-11.**
  - a. If resistance is as specified, go to **2**.
  - b. If resistance is not as specified, go to **12**.
- 2. Check for open circuit between:**
  - **P261-11 and ground**
  - **P261-10 and ground**
  - a. If open circuit is present, replace ESU (WP 0865 00). Go to **14**.
  - b. If open circuit is not present, go to **3**.
- 3. Leave P261 disconnected. Go to 4.**
- 4. Check for open circuit between:**
  - **J4-A and ground**
  - **J4-B and ground**
  - a. If open circuit is present, go to **5**.
  - b. If open circuit is not present, go to **10**.
- 5. Check for open circuit between:**
  - **P289-11 and ground**
  - **P289-10 and ground**
  - a. If open circuit is present, go to **6**.
  - b. If open circuit is not present, go to **8**.
- 6. Repair/replace wiring between J289 and W2P4 (WP 1747 00). Go to 7.**
- 7. Reconnect P261. Go to 14.**
- 8. Repair/replace wiring between P261 and P289 (WP 1747 00). Go to 9.**
- 9. Reconnect P261. Go to 14.**
- 10. Replace fuel solenoid valve (TM 55-2835-209-23). Go to 11.**
- 11. Reconnect P261. Go to 14.**
- 12. Check for 10 to 50 ohms between J4-A and J4-B.**
  - a. If resistance is as specified, go to **13**.
  - b. If resistance is not as specified, replace fuel solenoid valve (TM 55-2835-209-23). Go to **14**.

---

**FUEL SOLENOID FAILED (● ● ○ ○) - Continued****13. Check for 10 to 50 ohms between J289-10 and J289-11.**

- a. If resistance is as specified, repair/replace wiring between P261 and P289 (WP 1747 00). Go to **14**.
- b. If resistance is not as specified, repair/replace wiring between J289 and W2P4 (WP 1747 00). Go to **14**.

**14. Procedure completed.****HOT SENSOR FAILED (○ ● ○ ○)****SYMPTOM**

Hot sensor failed (○ ● ○ ○).

**MALFUNCTION**

Hot sensor failed (○ ● ○ ○).

**CORRECTIVE ACTION****1. Check for 68 to 242 ohms between P261-8 and P261-9.**

- a. If resistance is as specified, go to **2**.
- b. If resistance is not as specified, go to **5**.

**2. Check for open circuit between:**

- **P261-8 and ground**
- **P261-9 and ground**

- a. If open circuit is present, replace ESU (WP 0865 00). Go to **7**.
- b. If open circuit is not present, go to **3**.

**3. Check for open circuit between:**

- **J5-A and ground**
- **J5-B and ground**

- a. If open circuit is present, replace oil temperature bulb (TM 55-2835-209-23). Go to **7**.
- b. If open circuit is not present, go to **4**.

**4. Check for open circuit between:**

- **J289-8 and ground**
- **J289-9 and ground**

- a. If open circuit is present, repair/replace wiring between P261 and P289 (WP 1747 00). Go to **7**.
- b. If open circuit is not present, repair/replace wiring between J289 and W1P5 (WP 1747 00). Go to **7**.

**5. Check for 68 to 242 ohms between J5-A and J5-B.**

- a. If resistance is as specified, go to **6**.
- b. If resistance is not as specified, replace oil temperature bulb (TM 55-2835-209-23). Go to **7**.

---

**HOT SENSOR FAILED (○ ● ○ ○) - Continued****6. Check for 68 to 242 ohms between J289-8 and J289-9.**

- a. If resistance is as specified, repair/replace wiring between connectors P261 and P289 (WP 1747 00). Go to **7**.
- b. If resistance is not as specified, repair/replace wiring between J289 and W1P5 (WP 1747 00). Go to **7**.

**7. Procedure completed.****FUEL TORQUE MOTOR FAILED (● ○ ○ ○)****SYMPTOM**

Fuel torque motor failed (● ○ ○ ○).

**MALFUNCTION**

Fuel torque motor failed (● ○ ○ ○).

**CORRECTIVE ACTION****1. Check for 20 to 55 ohms between P261-18 and P261-19.**

- a. If resistance is as specified, go to **2**.
- b. If resistance is not as specified, go to **10**.

**2. Check for open circuit between:**

- P261-18 and ground
- P261-19 and ground

- a. If open circuit is present, replace ESU (WP 0865 00). Go to **12**.
- b. If open circuit is not present, go to **3**.

**3. Leave P261 disconnected. Go to 4.****4. Check for open circuit between:**

- J9-1 and ground
- J9-3 and ground

- a. If open circuit is present, go to **5**.
- b. If open circuit is not present, replace fuel control assembly, reconnect P261 (TM 55-2835-209-23). Go to **12**.

**5. Check for open circuit between:**

- P289-18 and ground
- P289-19 and ground

- a. If open circuit is present, go to **6**.
- b. If open circuit is not present, go to **8**.

**6. Repair/replace wiring between J289 and W2P9 (WP 1747 00). Go to 7.****7. Reconnect P261. Go to 12.****8. Repair/replace wiring between P261 and P289 (WP 1747 00). Go to 9.**

---

**FUEL TORQUE MOTOR FAILED (● ○ ○ ○) - Continued**

**9. Reconnect P261. Go to 12.**

**10. Check for 20 to 55 ohms between J9-1 and J9-3.**

- a. If resistance is as specified, go to **11**.
- b. If resistance is not as specified, replace fuel control assembly (TM 55-2835-209-23). Go to **12**.

**11. Check for 20 to 55 ohms between J289-18 and J289-19**

- a. If resistance is as specified, repair/replace wiring between P261 and P289 (WP 1747 00). Go to **12**.
- b. If resistance is not as specified, repair/replace wiring between J289 and W2P9 (WP 1747 00). Go to **12**.

**12. Procedure completed.**

**NO DATA (○ ○ ○ ○)**

**SYMPTOM**

No data (○ ○ ○ ○).

**MALFUNCTION**

No data (○ ○ ○ ○).

**CORRECTIVE ACTION**

**NOTE**

Make sure APU is not accelerating and speed has stabilized before checking BITE code.

**1. Restart APU (TM 1-1520-237-10).**

- a. If APU starts, system is operational. Go to **3**.
- b. If APU does not start, go to **2**.

**2. Does no data failure condition still exist?**

- a. If no data failure is present, replace ESU (WP 0865 00). Go to **3**.
- b. If data failure is present, troubleshoot based on the failure BITEs shown. Go to **3**.

**3. Procedure completed.**

**APU ON ADVISORY CAPSULE FAILURE****SYMPTOM**

APU ON advisory capsule failure.

**MALFUNCTION**

APU ON advisory capsule failure.

**CORRECTIVE ACTION****NOTE**

Make sure APU is not accelerating and speed has stabilized before checking BITE code.

1. **Does ready for service come on (● ● ● ●)?**
  - a. If ready for service comes on, go to **2**.
  - b. If ready for service does not come on, replace ESU (WP 0865 00). Go to **9**.
2. **Shut down APU (TM 1-1520-237-10). Go to 3.**
3. **Check helicopter configuration.**
  - a. **EMEP** , go to **4**.
  - b. **W/O EMEP** , go to **6**.
4. **EMEP Remove and check pin filtered adapter P118 (WP 0925 00).**
  - a. If pin filtered adapter is good, go to **5**.
  - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **10**.
5. **EMEP Install pin filtered adapter (WP 0925 00). Go to 6.**
6. **Check continuity between P118-B and P260-11.**
  - a. If continuity is present, go to **7**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **10**.
7. **Replace ESU (WP 0865 00). Go to 8.**
8. **Restart APU (TM 1-1520-237-10). Go to 9.**
9. **Does APU ON advisory capsule go on?**
  - a. If APU ON advisory capsule is on, system is operational. Go to **10**.
  - b. If APU ON advisory capsule is not on, replace caution/advisory panel (WP 0785 00). Go to **10**.
10. **Procedure completed.**



**APU FAIL ADVISORY CAPSULE FAILURE****SYMPTOM**

APU FAIL advisory capsule failure.

**MALFUNCTION**

APU FAIL advisory capsule failure.

**CORRECTIVE ACTION**

1. **Disconnect P117. Go to 2.**
2. **Check helicopter configuration.**
  - a. **EMEP** , go to 3.
  - b. **W/O EMEP** , go to 5.
3. **EMEP Remove and check pin filtered adapter P117 (WP 0925 00).**
  - a. If pin filtered adapter is good, go to 4.
  - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to 7.
4. **EMEP Install pin filtered adapter (WP 0925 00). Go to 5.**
5. **Does APU FAIL advisory capsule go off?**
  - a. If APU FAIL advisory capsule is off, go to 6.
  - b. If APU FAIL advisory capsule is not off, replace caution/advisory panel (WP 0785 00). Go to 7.
6. **Check continuity between P117-46 and P260-3.**
  - a. If continuity is present, replace ESU (WP 0865 00). Go to 7.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to 7.
7. **Procedure completed.**

**APU ON ADVISORY CAPSULE FAILURE****SYMPTOM**

APU ON advisory capsule failure.

**MALFUNCTION**

APU ON advisory capsule failure.

**CORRECTIVE ACTION**

1. **Disconnect P260. Go to 2.**
2. **Does APU ON advisory capsule go off?**
  - a. If APU ON advisory capsule is out, replace ESU (WP 0865 00). Go to 7.
  - b. If APU ON advisory capsule is not out, go to 3.
3. **Check helicopter configuration.**
  - a. **EMEP** , go to 4.
  - b. **W/O EMEP** , go to 6.

---

**APU ON ADVISORY CAPSULE FAILURE - Continued**

4. **EMEP** Remove and check pin filtered adapter P118 (WP 0925 00).
  - a. If pin filtered adapter is good, go to 5.
  - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to 7.
5. **EMEP** Install pin filtered adapter (WP 0925 00). Go to 6.
6. Check continuity between P260-11 and P118-B.
  - a. If continuity is present, replace caution/advisory panel (WP 0785 00). Go to 7.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to 7.
7. Procedure completed.

**APU FAIL ADVISORY CAPSULE FAILURE****SYMPTOM**

APU FAIL advisory capsule failure.

**MALFUNCTION**

APU FAIL advisory capsule failure.

**CORRECTIVE ACTION**

1. Disconnect P117. Go to 2.
2. Does APU FAIL advisory capsule go off?
  - a. If APU FAIL advisory capsule goes off, go to 3.
  - b. If APU FAIL advisory capsule does not go off, replace caution/advisory panel (WP 0785 00). Go to 7.
3. Check helicopter configuration.
  - a. **EMEP** , go to 4.
  - b. **W/O EMEP** , go to 6.
4. **EMEP** Remove and check pin filtered adapter P117 (WP 0925 00).
  - a. If pin filtered adapter is good, go to 5.
  - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to 7.
5. **EMEP** Install pin filtered adapter (WP 0925 00). Go to 6.
6. Check continuity between P117-46 and P260-3.
  - a. If continuity is present, replace ESU (WP 0865 00). Go to 7.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to 7.
7. Procedure completed.

**APU OIL TEMP HI ADVISORY CAPSULE FAILURE****SYMPTOM**

APU OIL TEMP HI advisory capsule failure.

**MALFUNCTION**

APU OIL TEMP HI advisory capsule failure.

**CORRECTIVE ACTION**

- 1. Disconnect P260. Go to 2.**
- 2. Does APU OIL TEMP HI advisory capsule go off?**
  - a. If APU OIL TEMP HI advisory capsule goes off, replace ESU (WP 0865 00). Go to **7**.
  - b. If APU OIL TEMP HI advisory capsule does not go off, go to **3**.
- 3. Check helicopter configuration.**
  - a. **EMEP** , go to **4**.
  - b. **W/O EMEP** , go to **6**.
- 4. EMEP Remove and check pin filtered adapter P117 (WP 0925 00).**
  - a. If pin filtered adapter is good, go to **5**.
  - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **7**.
- 5. EMEP Install pin filtered adapter (WP 0925 00). Go to 6.**
- 6. Check continuity between P260-4 and P117-58.**
  - a. If continuity is present, replace caution/advisory panel (WP 0785 00). Go to **7**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **7**.
- 7. Procedure completed.**

**APU DOES NOT SHUT DOWN WHEN APU CONTR SWITCH IS IN THE OFF POSITION****SYMPTOM**

APU does not shut down when APU CONTR switch is in the OFF position.

**MALFUNCTION**

APU does not shut down when APU CONTR switch is in the OFF position.

**CORRECTIVE ACTION**

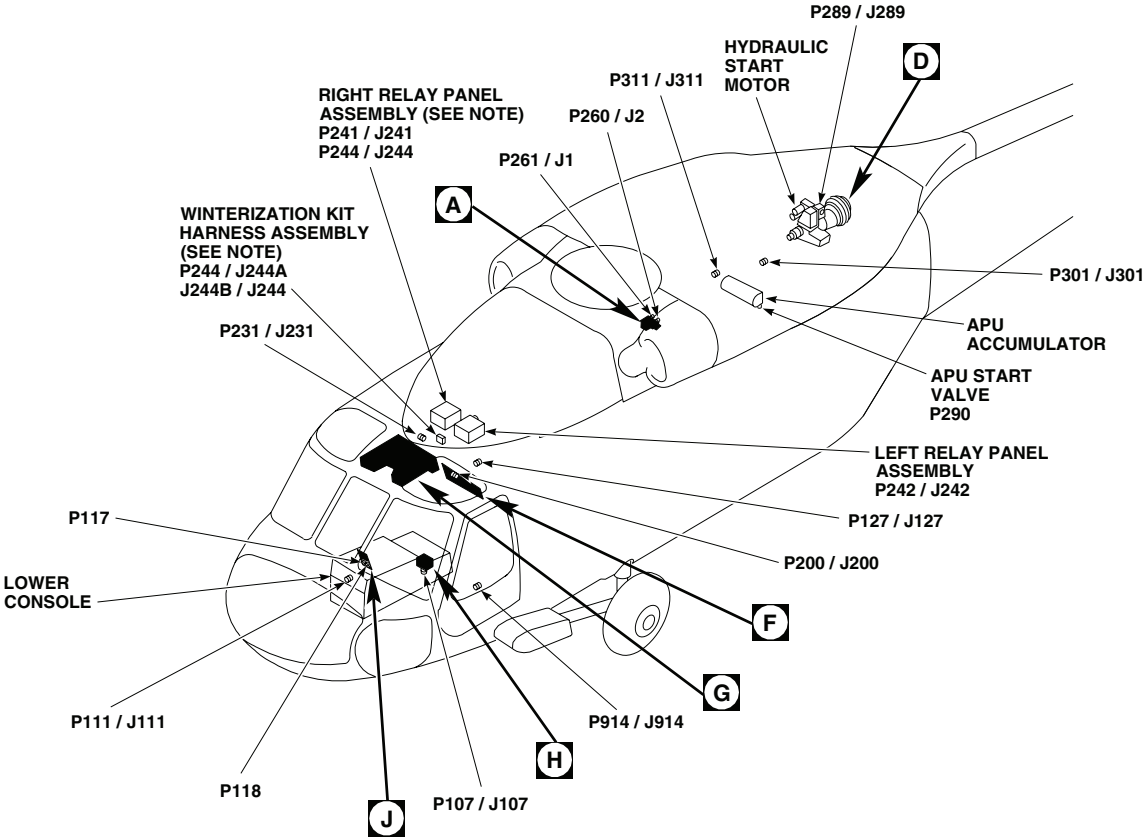
- 1. Set APU CONTR switch to the OFF position. Go to 2.**
- 2. Check for 28 vdc between P260-2 and ground.**
  - a. If voltage as specified, replace ESU (WP 0865 00). Go to **4**.
  - b. If voltage is not as specified, go to **3**.
- 3. Check continuity between APU CONTR switch terminal 1 and P260- 2.**
  - a. If continuity is present, replace switch (WP 0828 00). Go to **4**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **4**.

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**APU DOES NOT SHUT DOWN WHEN APU CONTR SWITCH IS IN THE OFF POSITION - Continued**

**4. Procedure completed.**

LOCATION



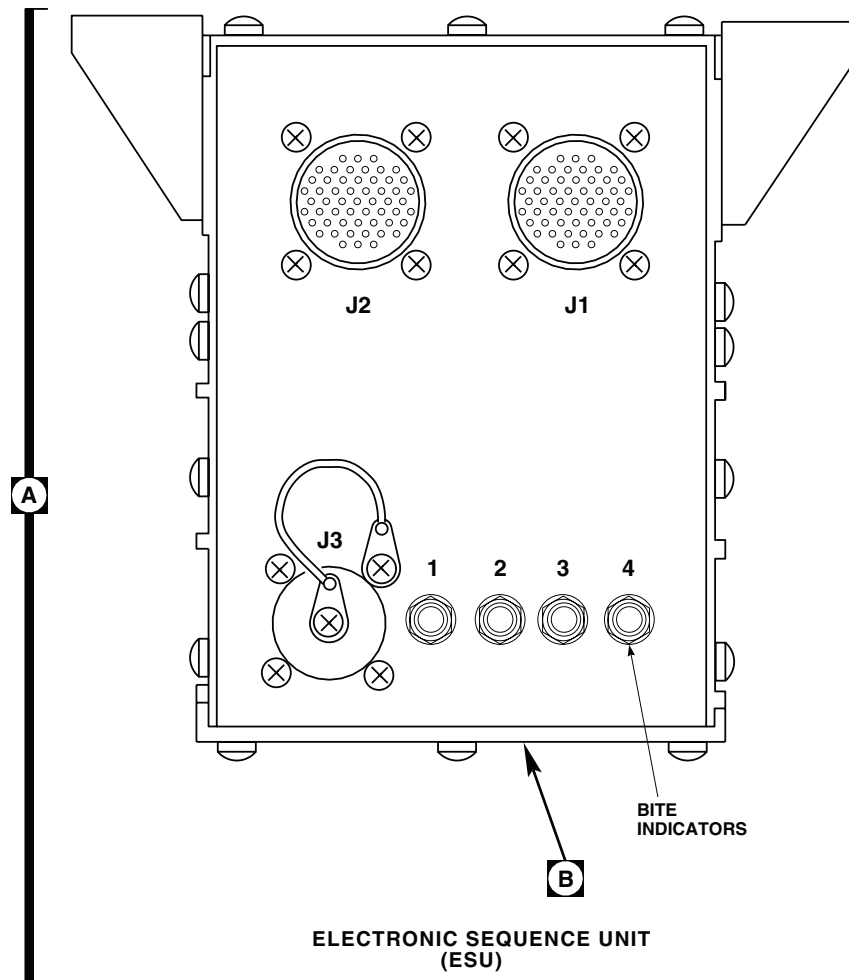
| TERMINAL BOARD /<br>DISCONNECT PLUG /<br>RECEPTACLE | LOCATION /<br>CONNECTION POINT                                  |
|-----------------------------------------------------|-----------------------------------------------------------------|
| P107 / J107                                         | LOWER CONSOLE                                                   |
| P111 / J111                                         | COCKPIT, BL 7.5 LH, STA 197                                     |
| P117                                                | CAUTION / ADVISORY<br>PANEL                                     |
| P118                                                | CAUTION / ADVISORY<br>PANEL                                     |
| P127 / J127                                         | BEHIND COPILOT'S CIRCUIT<br>BREAKER PANEL, BL 28 LH,<br>STA 243 |
| P200 / J200                                         | COCKPIT CEILING, BL 13 LH,<br>STA 246                           |
| P231 / J231                                         | CABIN CEILING, BL 9 LH,<br>STA 247                              |
| P241 / J241                                         | RIGHT RELAY PANEL                                               |
| P242 / J242                                         | LEFT RELAY PANEL                                                |

NOTE  
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Figure 1. Auxiliary Power Unit System Location Diagram. (Sheet 1 of 8)

LOCATION - Continued



| TERMINAL BOARD/<br>DISCONNECT PLUG/<br>RECEPTACLE | LOCATION/<br>CONNECTION POINT                       |
|---------------------------------------------------|-----------------------------------------------------|
| P244 / J244                                       | RIGHT RELAY PANEL<br>(SEE NOTE)                     |
| P244 / J244A<br>J244B / J244                      | WINTERIZATION KIT<br>HARNESS ASSEMBLY<br>(SEE NOTE) |
| P260 / J2                                         | ELECTRONIC SEQUENCE<br>UNIT                         |
| P261 / J1                                         | ELECTRONIC SEQUENCE<br>UNIT                         |
| P289 / J289                                       | APU                                                 |
| P290                                              | APU START VALVE                                     |
| P301 / J301                                       | TRANSITION DECK,<br>BL 12.88 LH, STA 415            |
| P311 / J311                                       | CABIN CEILING,<br>BL 16.50 LH, STA 383              |
| P914 / J914                                       | COCKPIT, BL 24 LH, STA 247                          |

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Figure 1. Auxiliary Power Unit System Location Diagram. (Sheet 2 of 8)

LOCATION - Continued

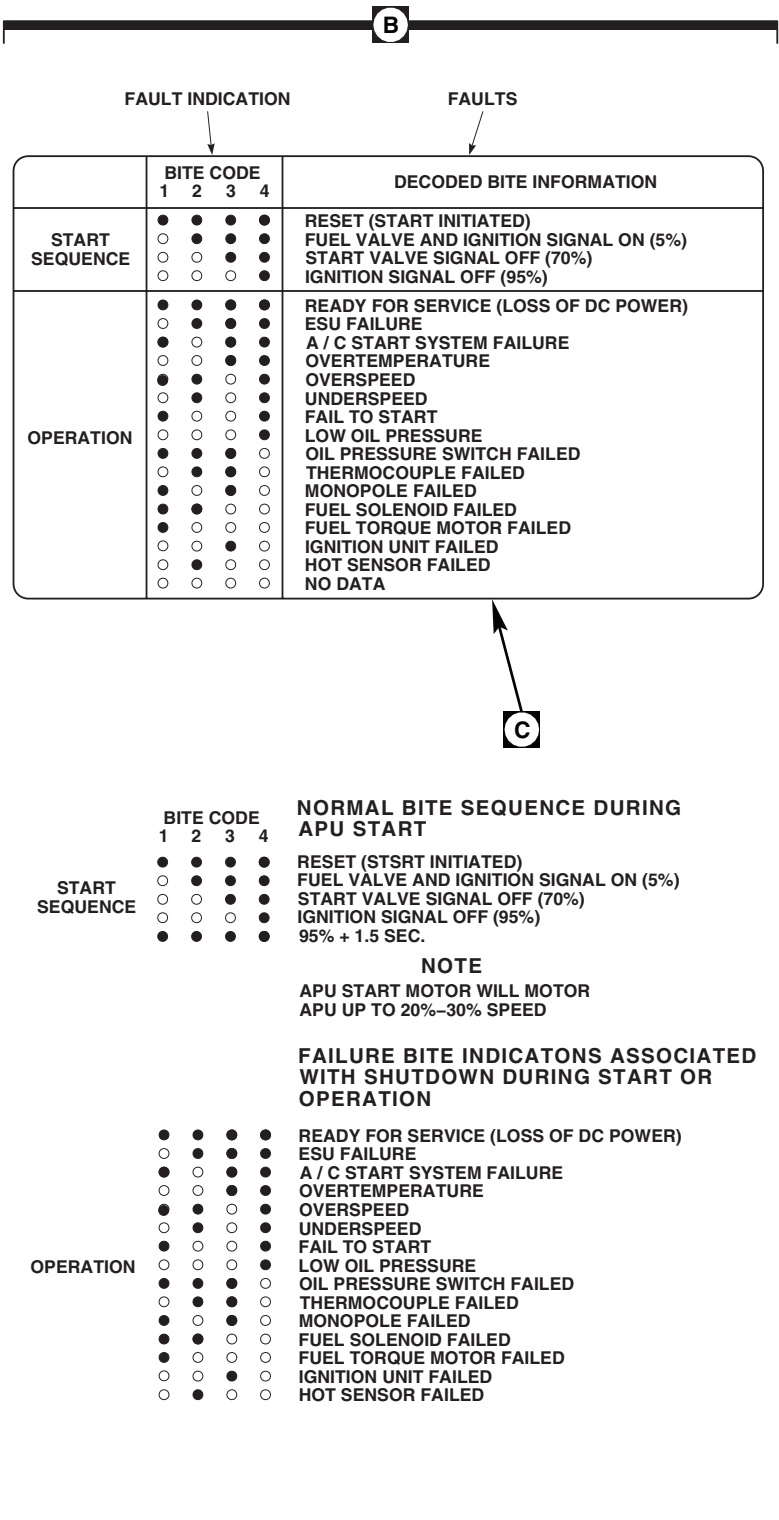


Figure 1. Auxiliary Power Unit System Location Diagram. (Sheet 3 of 8)

## LOCATION - Continued

| C                                  |                                                                                                                                                                                            |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BITE INDICATIONS                   |                                                                                                                                                                                            |
| ○ ● ● ● ESU FAILURE                | INTERNAL ESU FAILURE.                                                                                                                                                                      |
| ● ○ ● ● A / C START SYSTEM FAILURE | APU START VALVE OPEN / SHORTED.                                                                                                                                                            |
| ○ ○ ● ● OVERTEMPERATURE            | EGT GREATER THAN 1810° FROM 0-50% SPEED DECREASING LINEARY TO 1350° AT 90%. AT 100% EGT LIMIT PER TABLE 36.1-3, ITEM I.                                                                    |
| ● ● ○ ● OVERSPEED                  | APU SPEED INCREASES TO 107%.                                                                                                                                                               |
| ○ ● ○ ● UNDERSPEED                 | APU SPEED DECREASED BELOW 90% FOR 10 SECONDS AFTER ACHIEVING 95%.                                                                                                                          |
| ● ○ ○ ● FAIL TO START              | APU SPEED DOES NOT ACHIEVE 5% IN 30 SECONDS OR 95% IN 70 SECONDS. APU FUEL SOLENOID VALVE IS ON, EGT IS LESS THAN 300°F AND EGT HAS NOT INCREASED 200°F OVER START INITIATION TEMPERATURE. |
| ○ ○ ○ ● LOW OIL PRESSURE           | LOW OIL PRESSURE SWITCH IS CLOSED FOR 15 SECONDS WITH APU ABOVE 95% SPEED.                                                                                                                 |
| ● ● ● ○ OIL PRESSURE SWITCH FAILED | LOW OIL PRESSURE SWITCH IS OPEN AT START INITIATION OF BELOW 10% SPEED.                                                                                                                    |
| ○ ● ● ○ THERMOCOUPLE FAILED        | EGT DOES NOT INCREASE 200°F BEFORE 40% SPEED. EGT IS LESS THAN 300°F AT 95% SPEED. EGT IS OUT OF RANGE HIGH OR LOW.                                                                        |
| ● ○ ● ○ MONOPOLE FAILED            | MONOPOLE CIRCUIT IS OPEN.                                                                                                                                                                  |
| ● ● ○ ○ FUEL SOLENOID FAILED       | FUEL SOLENOID IS OPEN, SHORTED, OR SHORTED TO AIRCRAFT GROUND.                                                                                                                             |
| ○ ○ ● ○ IGNITION UNIT FAILED       | IGNITION UNIT IS OPEN, SHORTED, OR SHORTED TO AIRCRAFT GROUND.                                                                                                                             |
| ● ○ ○ ○ FUEL TORQUE MOTOR FAILED   | FUEL CONTROL ASSEMBLY IS OPEN, SHORTED, OR SHORTED TO AIRCRAFT GROUND.                                                                                                                     |

AA2685\_4  
SA

Figure 1. Auxiliary Power Unit System Location Diagram. (Sheet 4 of 8)



## LOCATION - Continued

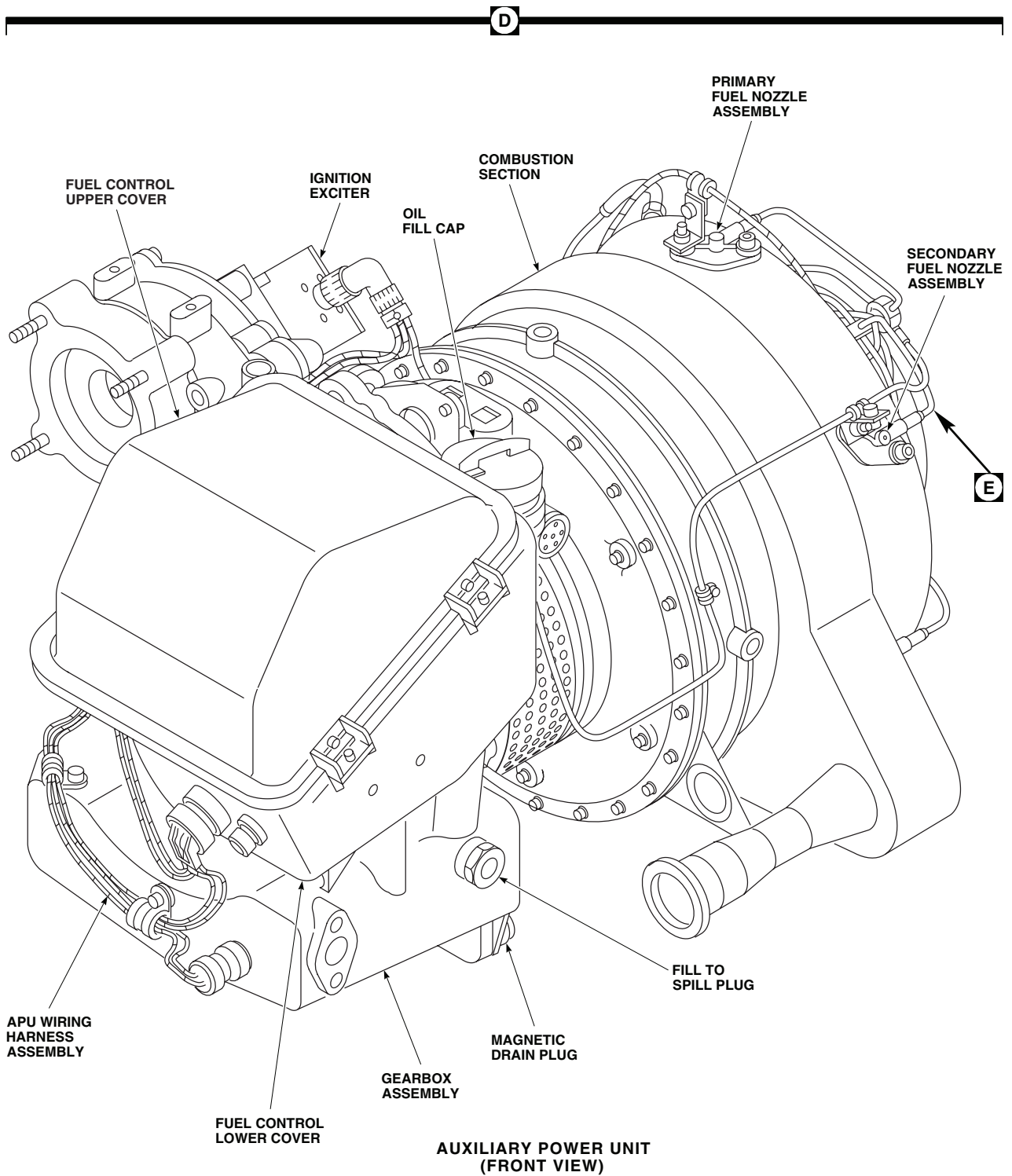
AA2685\_5  
SA

Figure 1. Auxiliary Power Unit System Location Diagram. (Sheet 5 of 8)

## LOCATION - Continued

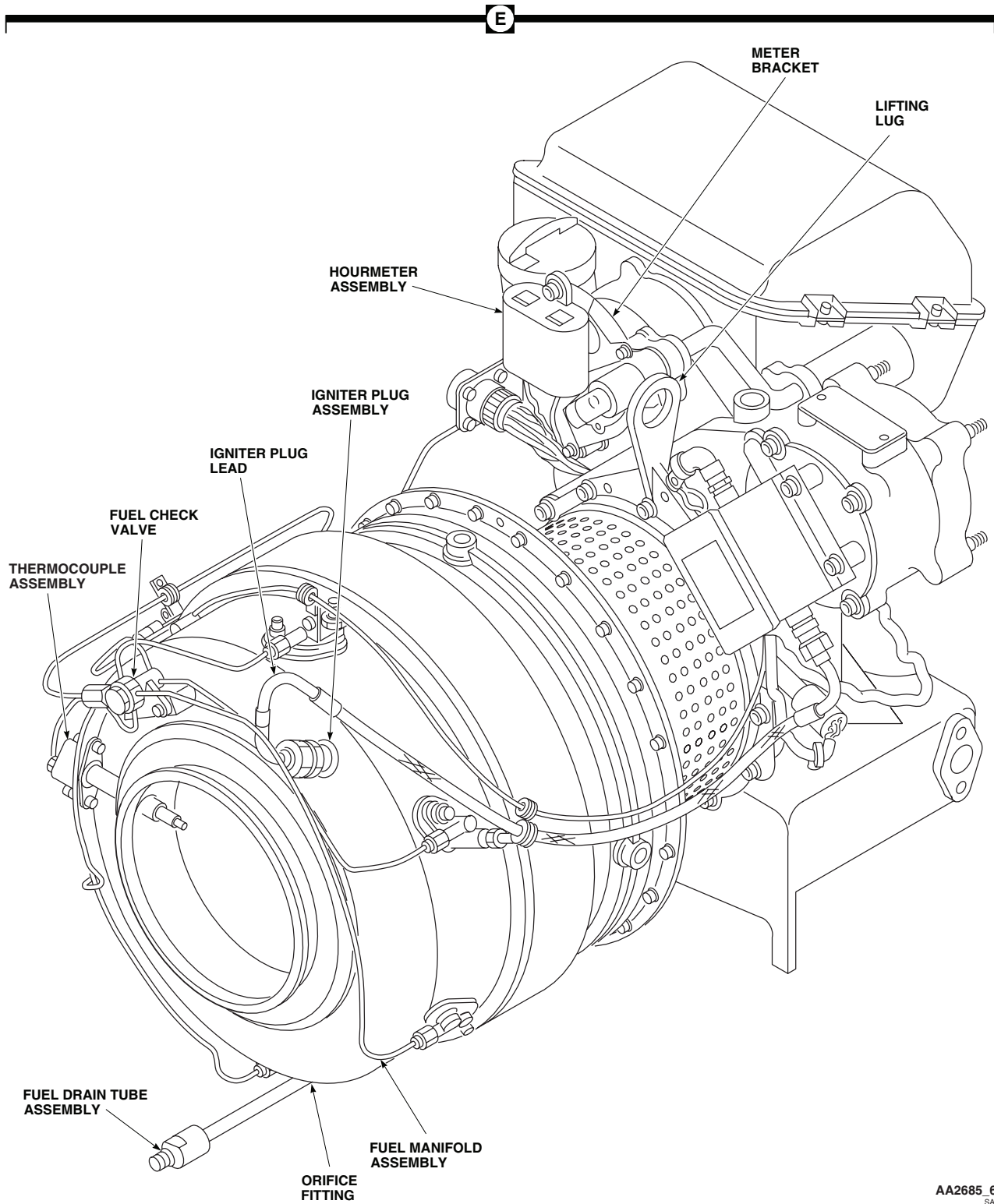
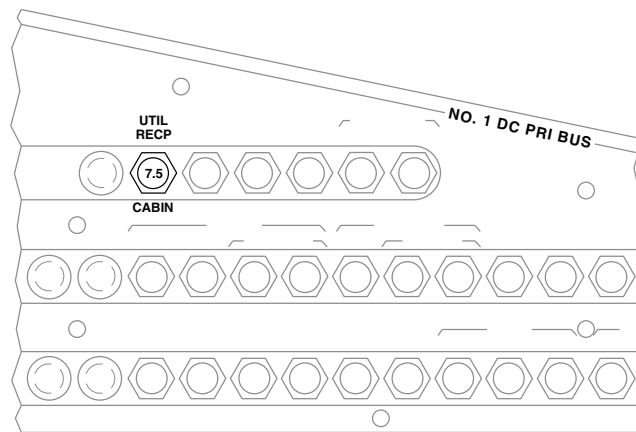


Figure 1. Auxiliary Power Unit System Location Diagram. (Sheet 6 of 8)

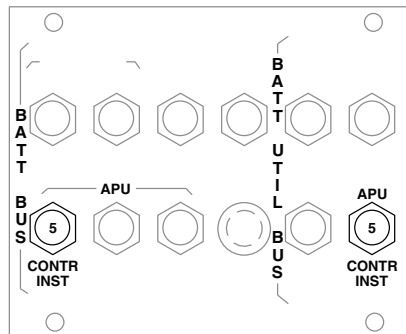
LOCATION - Continued



COPLOT'S CIRCUIT BREAKER PANEL

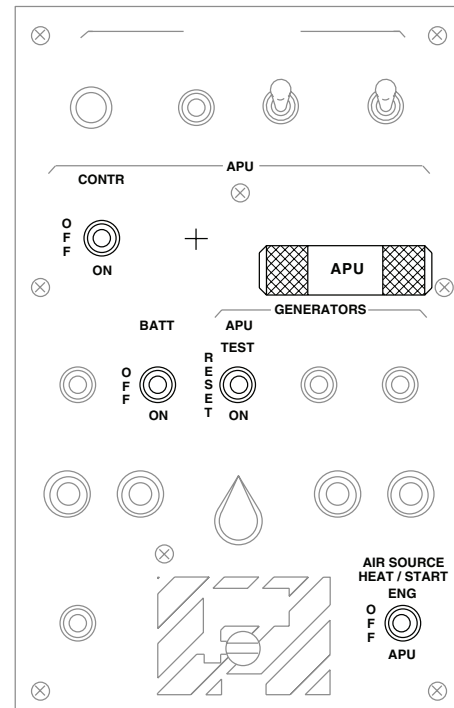
F

H



LOWER CONSOLE

G

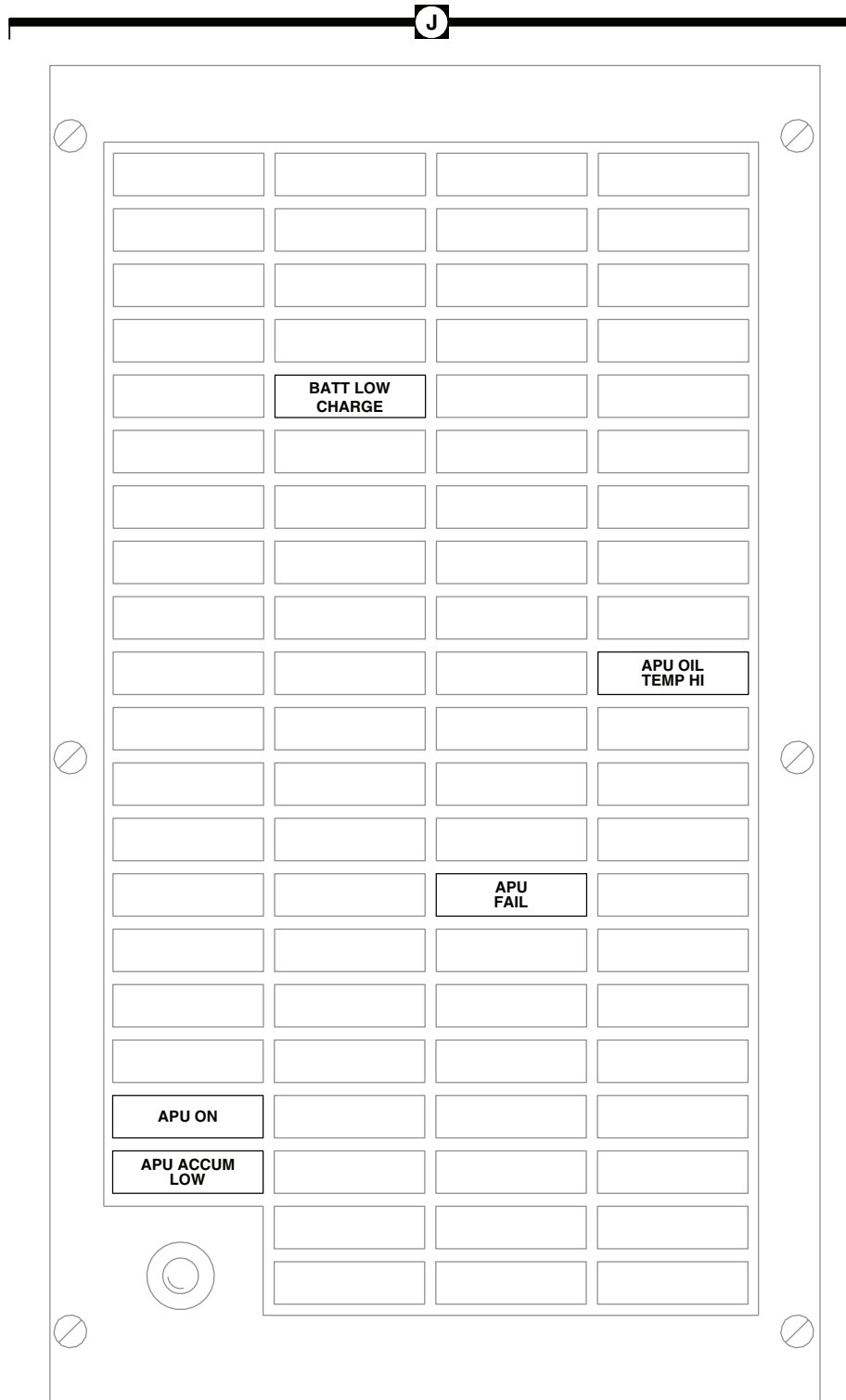


UPPER CONSOLE

AA2685\_7  
SA

Figure 1. Auxiliary Power Unit System Location Diagram. (Sheet 7 of 8)

LOCATION - Continued



CAUTION/ADVISORY PANEL

AA2685\_8  
SA

Figure 1. Auxiliary Power Unit System Location Diagram. (Sheet 8 of 8)

## SCHEMATICS

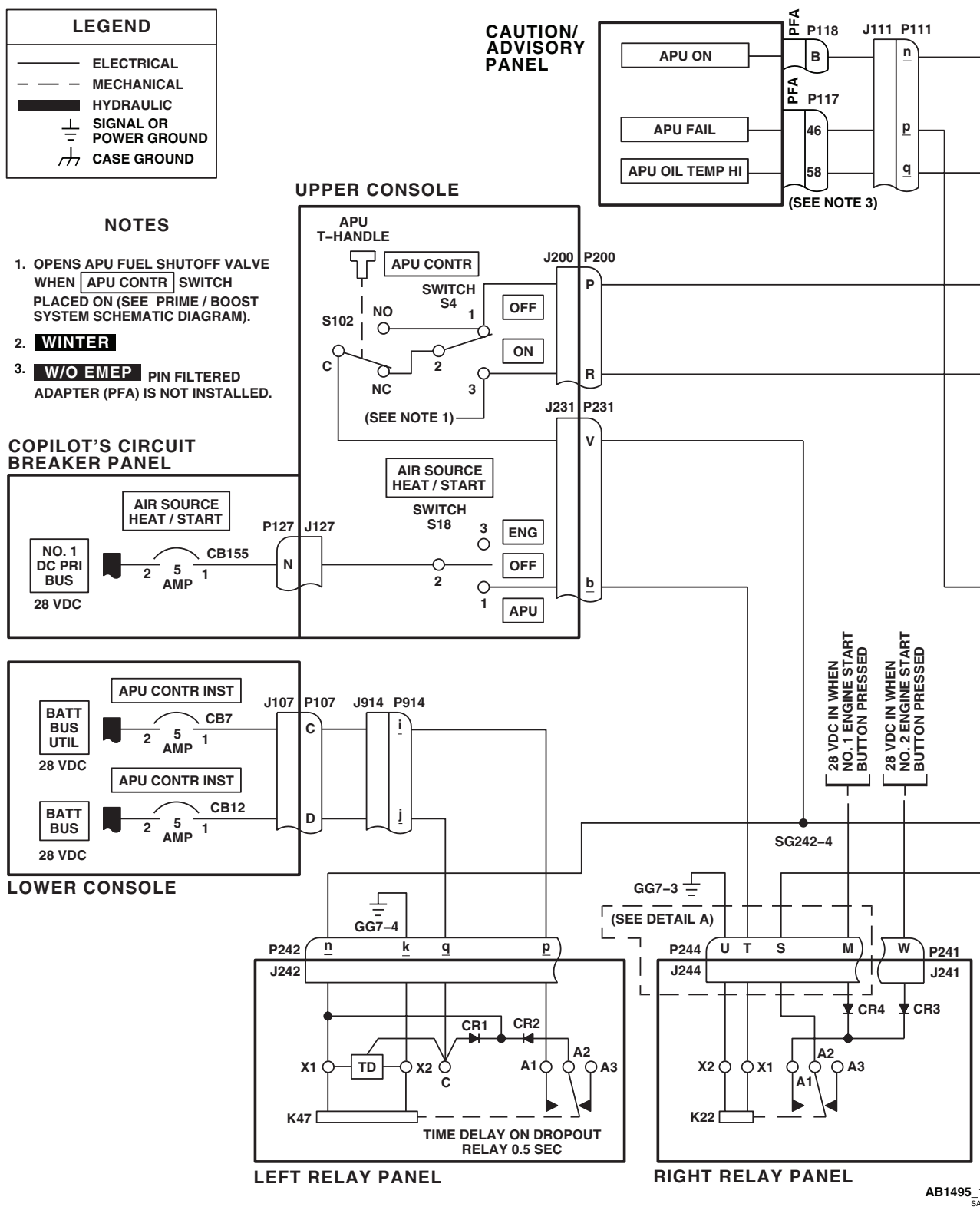


Figure 2. Auxiliary Power Unit System Schematic Diagram. (Sheet 1 of 4)

SCHEMATICS - Continued

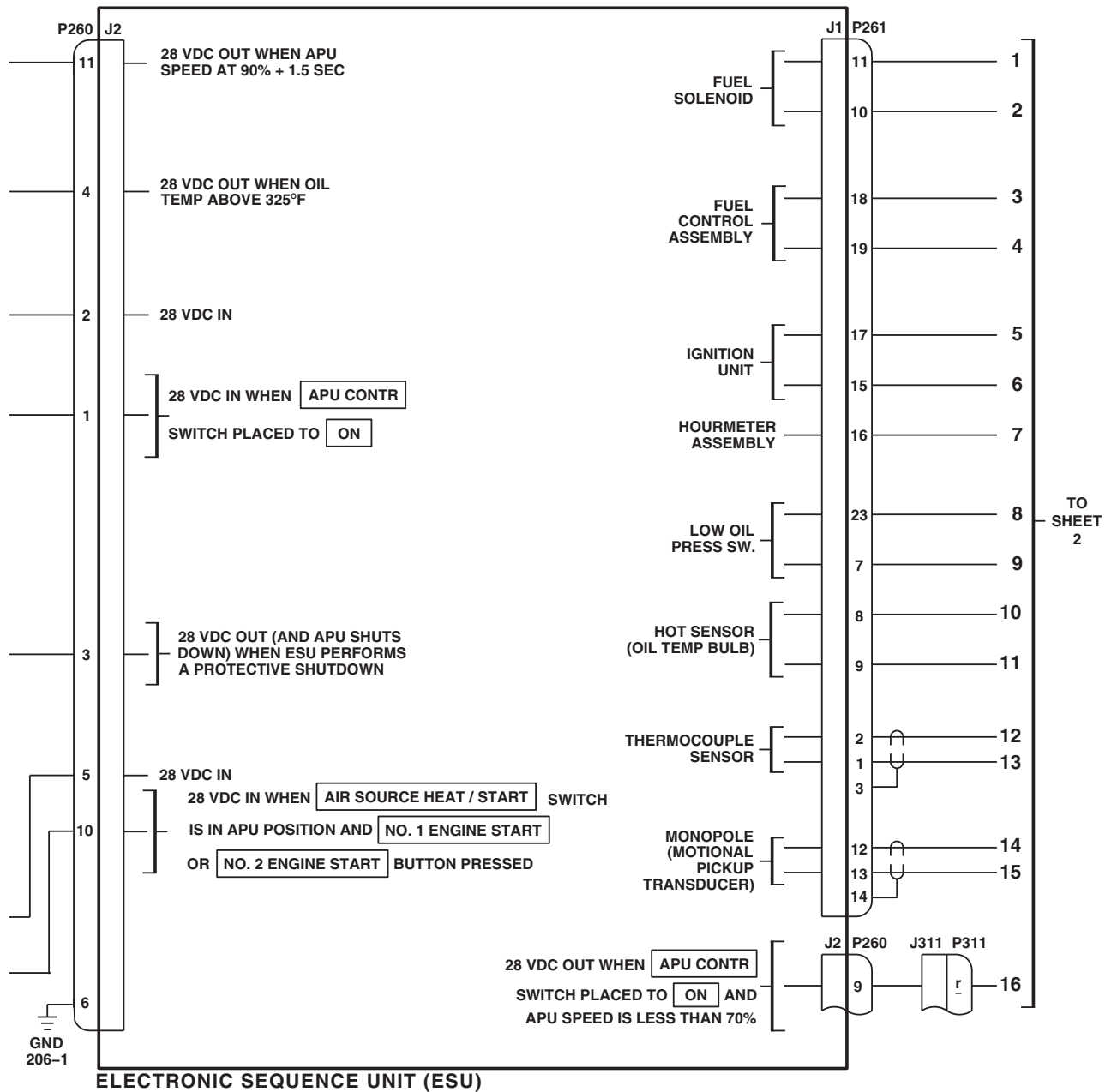
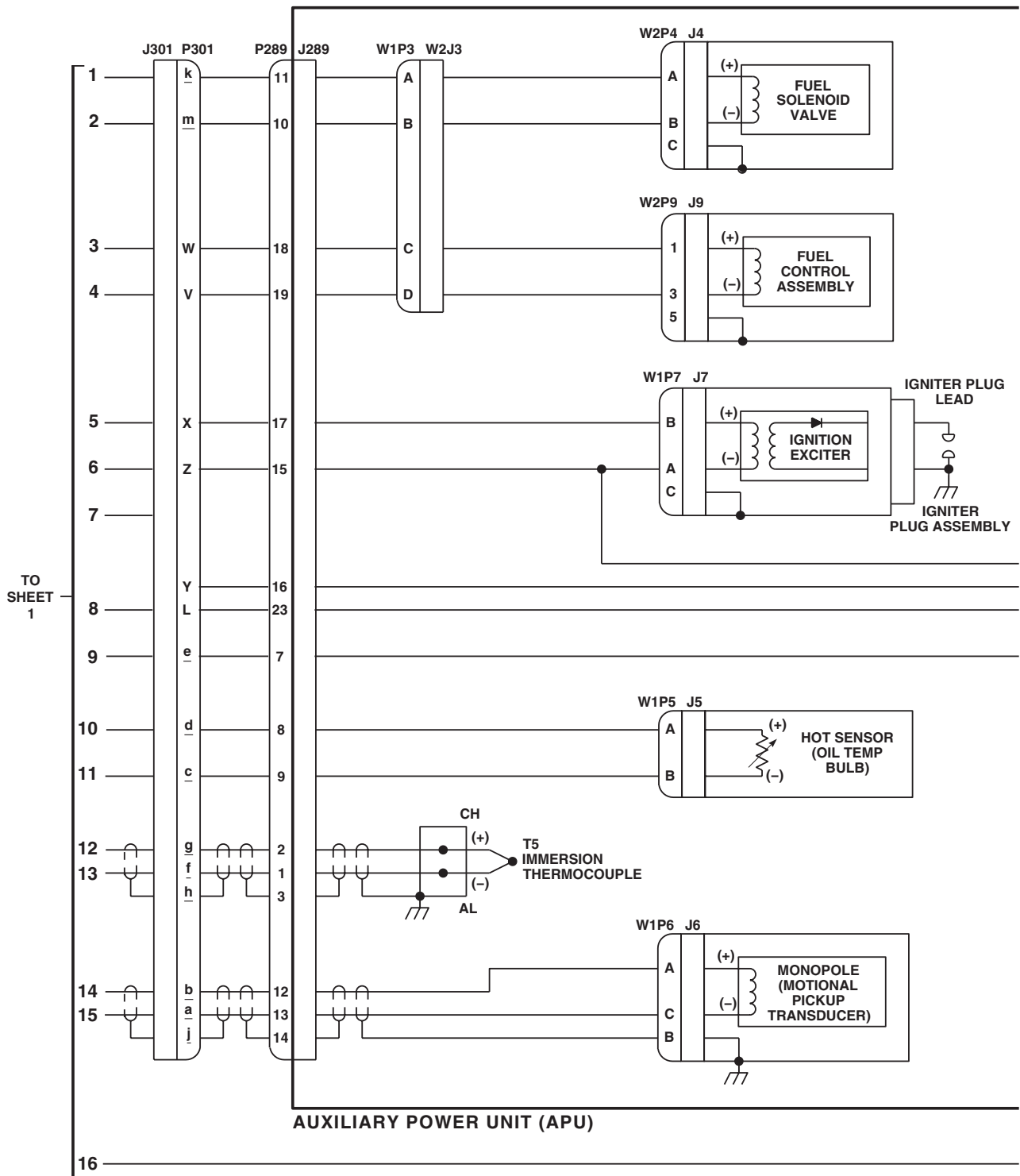


Figure 2. Auxiliary Power Unit System Schematic Diagram. (Sheet 2 of 4)

SCHEMATICS - Continued



AB1495\_3  
SA

Figure 2. Auxiliary Power Unit System Schematic Diagram. (Sheet 3 of 4)

## SCHEMATICS - Continued

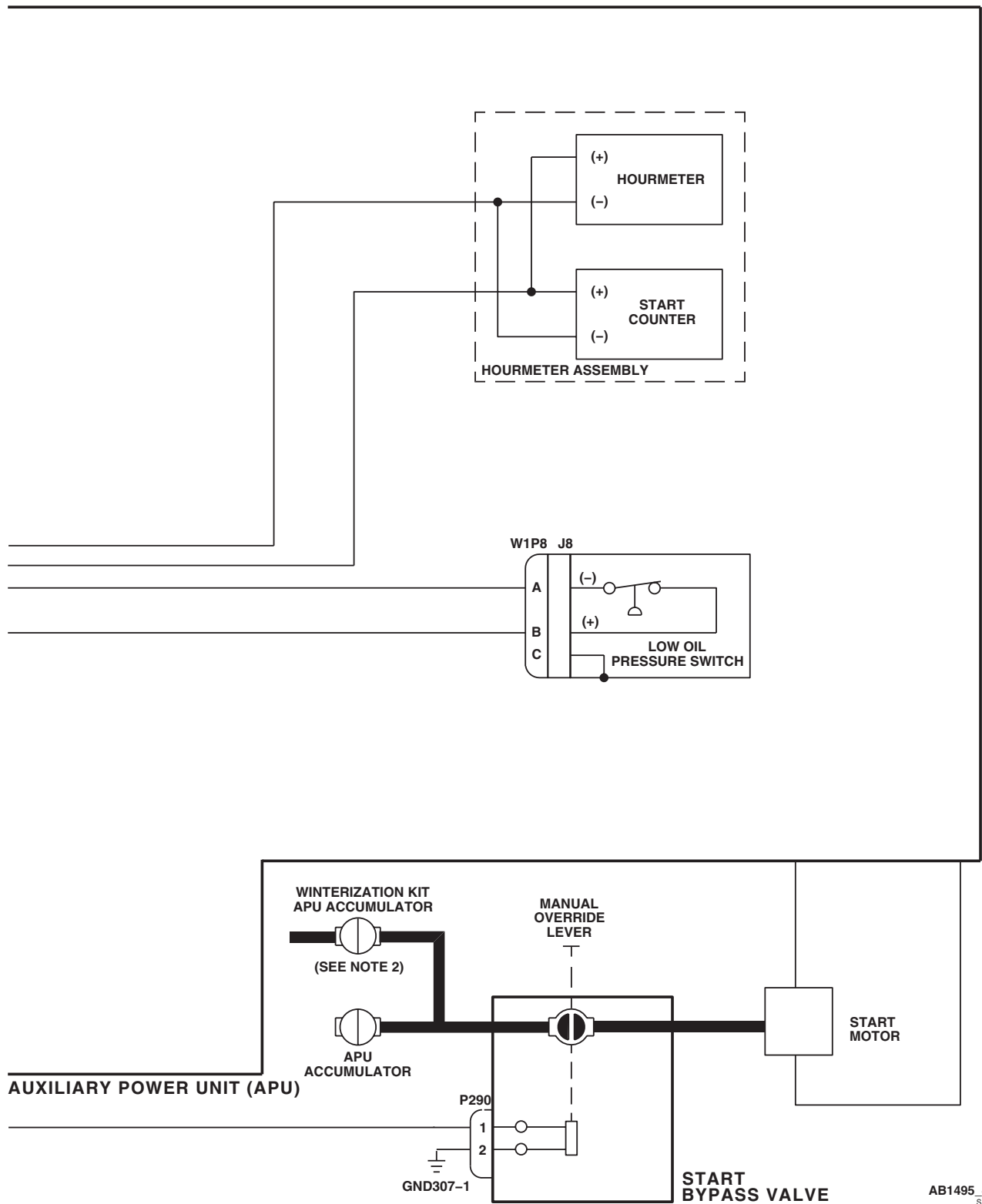
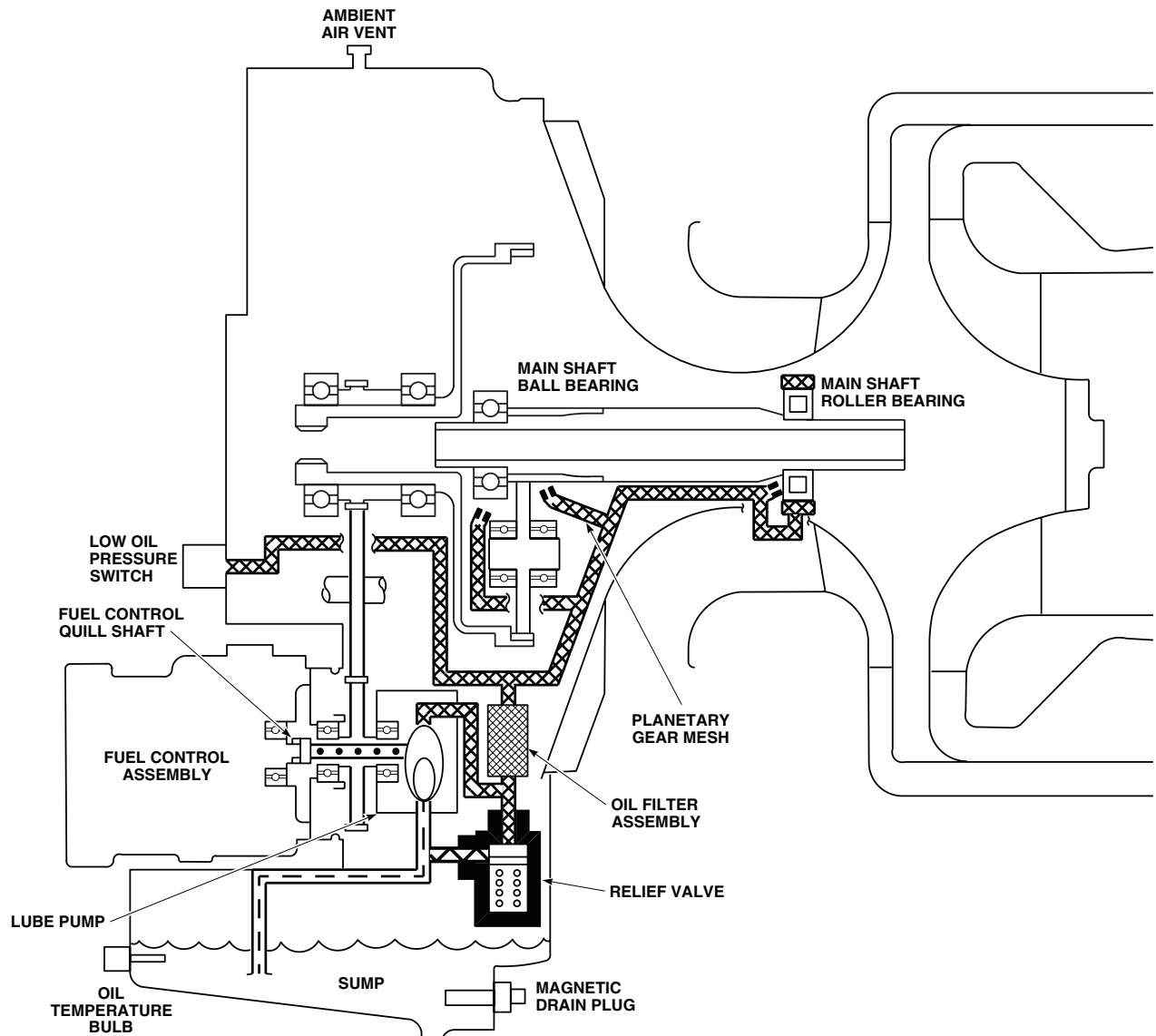


Figure 2. Auxiliary Power Unit System Schematic Diagram. (Sheet 4 of 4)



SCHEMATICS - Continued

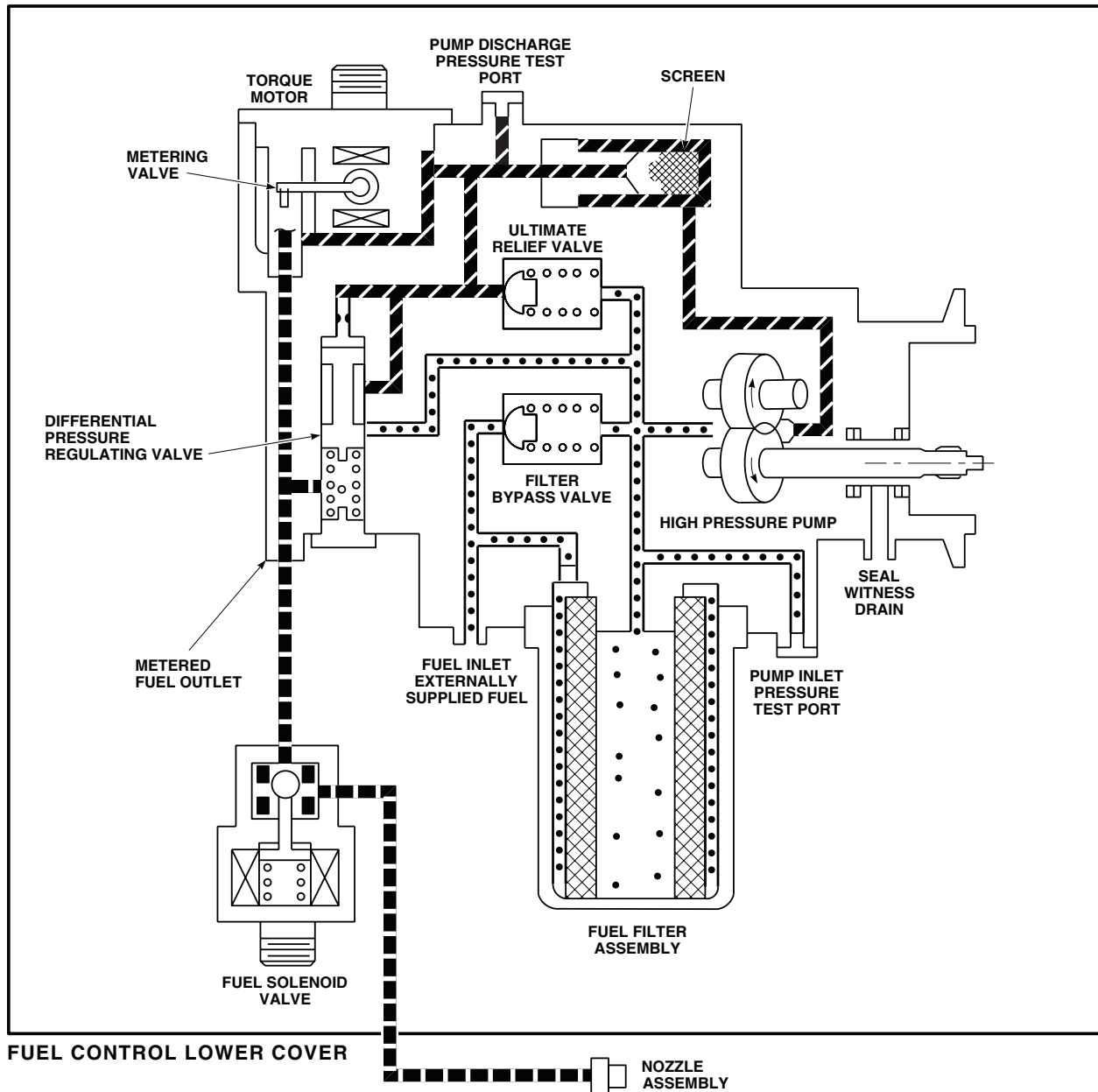


| LEGEND |                     |
|--------|---------------------|
|        | PRESSURE OIL        |
|        | PRESSURE RELIEF OIL |
|        | OIL PUMP INLET      |
|        | SPLINE LUBE         |
|        | OIL JET             |

AB1496  
SA

Figure 3. Auxiliary Power Unit Lubrication System Schematic.

SCHEMATICS - Continued



AB1497  
SA

Figure 4. Auxiliary Power Unit Fuel System Schematic.

END OF WORK PACKAGE

---

**UNIT LEVEL**  
**MISSION EQUIPMENT**  
**AEROMEDICAL EVACUATION KIT**  
**This WP supersedes WP 0157 00, dated 17 April 2006.**

---

**INITIAL SETUP:****Test Equipment**

Multimeter, AN/PSM-45A

WP 1382 00

WP 1384 00

WP 1385 00

**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01

Electrical Repairer Toolkit, SC 5180-99-B06

WP 1390 00

WP 1391 00

WP 1396 00

**Personnel Required**

Aircraft Electrician MOS 15F (1)

UH-60 Helicopter Repairer MOS 15T (1)

WP 1685 00

WP 1747 00

**References**

TM 1-1520-237-10

WP 0104 00

WP 0824 00

**Equipment Condition**

External Electrical Power Available, WP 1685 00

or APU Operational, TM 1-1520-237-10

Kit Installed in Helicopter, WP 1382 00

---

**AEROMEDICAL EVACUATION KIT OPERATIONAL/TROUBLESHOOTING PROCEDURE,**  
**PROCEDURE**

**Step****NOTE**

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- See Figure 1 for component location diagram and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

1. On helicopters equipped with floor plate 70802-12215-041, pull out and hold one pedestal lock handle. Turn pedestal 90° clockwise to ground load (lateral) position. Release handle and lock pedestal in place.

**Indication/Condition**

Lockpin shall disengage, pedestal shall turn smoothly, and lockpin shall lock pedestal in new position.

**Corrective Action**

1. If Indication/Condition is not as specified, inspect components for damage.
  - a. Repair/replace components as required.
  - b. If lockpin does not engage, check stop on floor plate for damage and/or adjustment (WP 1391 00).

---

**AEROMEDICAL EVACUATION KIT OPERATIONAL/TROUBLESHOOTING PROCEDURE, - Continued****Step**

2. On helicopters equipped with floor plate 70802-12215-041, pull out and hold same pedestal lock handle. Turn pedestal 90° counterclockwise to flight (longitudinal) position. Release handle and lock pedestal in place.

**Indication/Condition**

Lockpin shall disengage, pedestal shall turn smoothly, and lockpin shall lock pedestal in flight position.

**Corrective Action**

1. If Indication/Condition is not as specified, inspect components for damage.
  - a. Repair/replace components as required.
  - b. If lockpin does not engage, check adjustment of ceiling support, or condition and adjustment of stop on floor plate (WP 1391 00).

**Step**

3. On helicopters equipped with floor plate 70802-12215-045, pull out and hold one pedestal lock handle. Turn pedestal 60° clockwise to ground load position. Release handle and lock pedestal in place.

**Indication/Condition**

Lockpin shall disengage, pedestal shall turn smoothly, and lockpin shall lock pedestal in new position.

**Corrective Action**

1. If Indication/Condition is not as specified, inspect components for damage.
  - a. Repair/replace components as required.
  - b. If lockpin does not engage, check stop on floor plate for damage and/or adjustment (WP 1391 00).

**Step**

4. On helicopters equipped with floor plate 70802-12215-045, pull out and hold same pedestal lock handle. Turn pedestal 120° counterclockwise to second ground load position. Release handle and lock pedestal in place.

**Indication/Condition**

Lockpin shall disengage, pedestal shall turn smoothly, and lockpin shall lock pedestal in second ground load position.

**Corrective Action**

1. If Indication/Condition is not as specified, inspect components for damage.
  - a. Repair/replace components as required.
  - b. If lockpin does not engage, check adjustment of ceiling support, or condition and adjustment of stop on floor plate (WP 1391 00).

**Step**

5. On helicopters equipped with floor plate 70802-12215-045, pull out and hold same pedestal lock handle. Turn pedestal 60° clockwise to flight (longitudinal) position. Release handle and lock pedestal in place.

**Indication/Condition**

Lockpin shall disengage, pedestal shall turn smoothly, and lockpin shall lock pedestal in flight position.

---

**AEROMEDICAL EVACUATION KIT OPERATIONAL/TROUBLESHOOTING PROCEDURE, - Continued****Corrective Action**

1. If Indication/Condition is not as specified, inspect components for damage.
  - a. Repair/replace components as required.
  - b. If lockpin does not engage, check adjustment of ceiling support or condition and adjustment of stop on floor plate (WP 1391 00).

**Step**

6. Repeat steps 1.through 5., using pedestal lock handle on opposite end of pedestal.
7. Make sure CTR PIVOT handle of upper litter support is at UNLOCKED, and pin is out of fitting.
8. Place either end lever to TILT.

**Indication/Condition**

Litter shall move to tilted position, coming to rest on stops.

**Corrective Action**

If Indication/Condition is not as specified, check end pivot components for damage.

**Step**

9. Place end lever to TILT. Raise litter support to level position.

**Indication/Condition**

Litter support shall rise to level position, and lock in place.

**Corrective Action**

1. If Indication/Condition is not as specified, check end pivot components and guides.
  - a. Repair/replace components as required (WP 1390 00).

**Step**

10. Place end lever to LOCK.

**Indication/Condition**

End of litter shall lock in place.

**Corrective Action**

If Indication/Condition is not as specified, repair/replace components as required (WP 1391 00).

**Step**

11. Repeat steps 7. through 10. with other end of litter support.
12. Unlock CTR PIVOT handle, slide center pivot pin into pedestal pivot fitting by lining up pin with slot in fitting, and pushing on CTR PIVOT handle. Place handle to LOCK.
13. Place both end pivot handles to UNLOCK.
14. Place either end pivot handle to TILT.

**Indication/Condition**

Litter support shall pivot around center support tube, and litter shall rest on stop.

**AEROMEDICAL EVACUATION KIT OPERATIONAL/TROUBLESHOOTING PROCEDURE, - Continued****Corrective Action**

1. If Indication/Condition is not as specified, check center pivot components.
  - a. Repair/replace components as required (WP 1396 00).

**Step**

15. Place end pivot handle in TILT position, and pull up on litter support. Place both end pivot handles to LOCKED.

**Indication/Condition**

Litter support shall lock in place.

**Corrective Action**

1. If Indication/Condition is not as specified, check guides.
  - a. Repair/replace components as required (WP 1390 00).

**Step**

16. Repeat steps 13. through 15. with other end of litter support.
17. Repeat steps 7. through 15. with other side upper litter support.
18. Make sure the following circuit breakers are pushed in:

**CIRCUIT BREAKER****CIRCUIT BREAKER PANEL**

NO. 1 LTR LTS

Mission Readiness

NO. 2 LTR LTS

Mission Readiness

19. Turn on electrical power.
20. Press and then release each PUSH-ON/PUSH-OFF LITTER LIGHT switch. Press each switch again.

**Indication/Condition**

Each light shall go on when switch is pushed first time, and then go off when switch is pressed second time.

**Corrective Action**

1. If one litter light does not go on, go to ONE OF THE LITTER LIGHTS DOES NOT GO ON, in this work package.
2. If no aft LITTER LIGHTS go on, go to NONE OF THE FOUR AFT LITTER LIGHTS GO ON, in this work package.
3. If no forward LITTER LIGHTS go on, go to NONE OF THE FOUR FORWARD LITTER LIGHTS GO ON, in this work package.
4. If Indication/Condition is not as specified, replace GG7 (WP 1747 00).

**PROCEDURE****Step**

21. Turn off electrical power.

---

**AEROMEDICAL EVACUATION KIT OPERATIONAL/TROUBLESHOOTING PROCEDURE, - Continued****Indication/Condition**

Power shall be off.

**Corrective Action**

None Required

**ONE OF THE LITTER LIGHTS DOES NOT GO ON****SYMPTOM**

One of the litter lights does not go on.

**MALFUNCTION**

One of the litter lights does not go on.

**CORRECTIVE ACTION**

- 1. Replace lamp (WP 1384 00). Go to 2.**
- 2. Check light for proper operation.**
  - a. If light is good, system is operational. Go to **4**.
  - b. If light is not good, go to **3**.
- 3. Check for broken wire between switch and lamp housing.**
  - a. If broken wire is present, repair/replace wiring between lamp housing and switch (WP 1747 00). Go to **4**.
  - b. If broken wire is not present, replace light assembly (WP 1385 00). Go to **4**.
- 4. Procedure completed.**

**NONE OF THE FOUR AFT LITTER LIGHTS GO ON****SYMPTOM**

None of the four aft litter lights go on.

**MALFUNCTION**

None of the four aft litter lights go on.

**CORRECTIVE ACTION**

- 1. Check for 28 vdc between J233-A and ground.**
  - a. If voltage is as specified, go to **2**.
  - b. If voltage is not as specified, go to **3**.
- 2. Check continuity between J233-C and ground.**
  - a. If continuity is present, replace wiring harness (WP 1747 00). Go to **4**.
  - b. If continuity is not present, repair/replace ground wire (WP 1747 00). Go to **4**.
- 3. Check for 28 vdc between NO. 1 LTR LTS circuit breaker terminal 1 and ground.**
  - a. If voltage is as specified, repair/replace wiring between J233-A and circuit breaker terminal 1 (WP 1747 00). Go to **4**.
  - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **4**.

---

**NONE OF THE FOUR AFT LITTER LIGHTS GO ON** - Continued**4. Procedure completed.****NONE OF THE FOUR FORWARD LITTER LIGHTS GO ON****SYMPTOM**

None of the four forward litter lights go on.

**MALFUNCTION**

None of the four forward litter lights go on.

**CORRECTIVE ACTION****1. Check for 28 vdc between J233-D and ground.**

- a. If voltage is as specified, go to **2**.
- b. If voltage is not as specified, go to **3**.

**2. Check continuity between J233-D and ground.**

- a. If continuity is present, replace wiring harness (WP 1747 00). Go to **4**.
- b. If continuity is not present, repair/replace ground wire (WP 1747 00). Go to **4**.

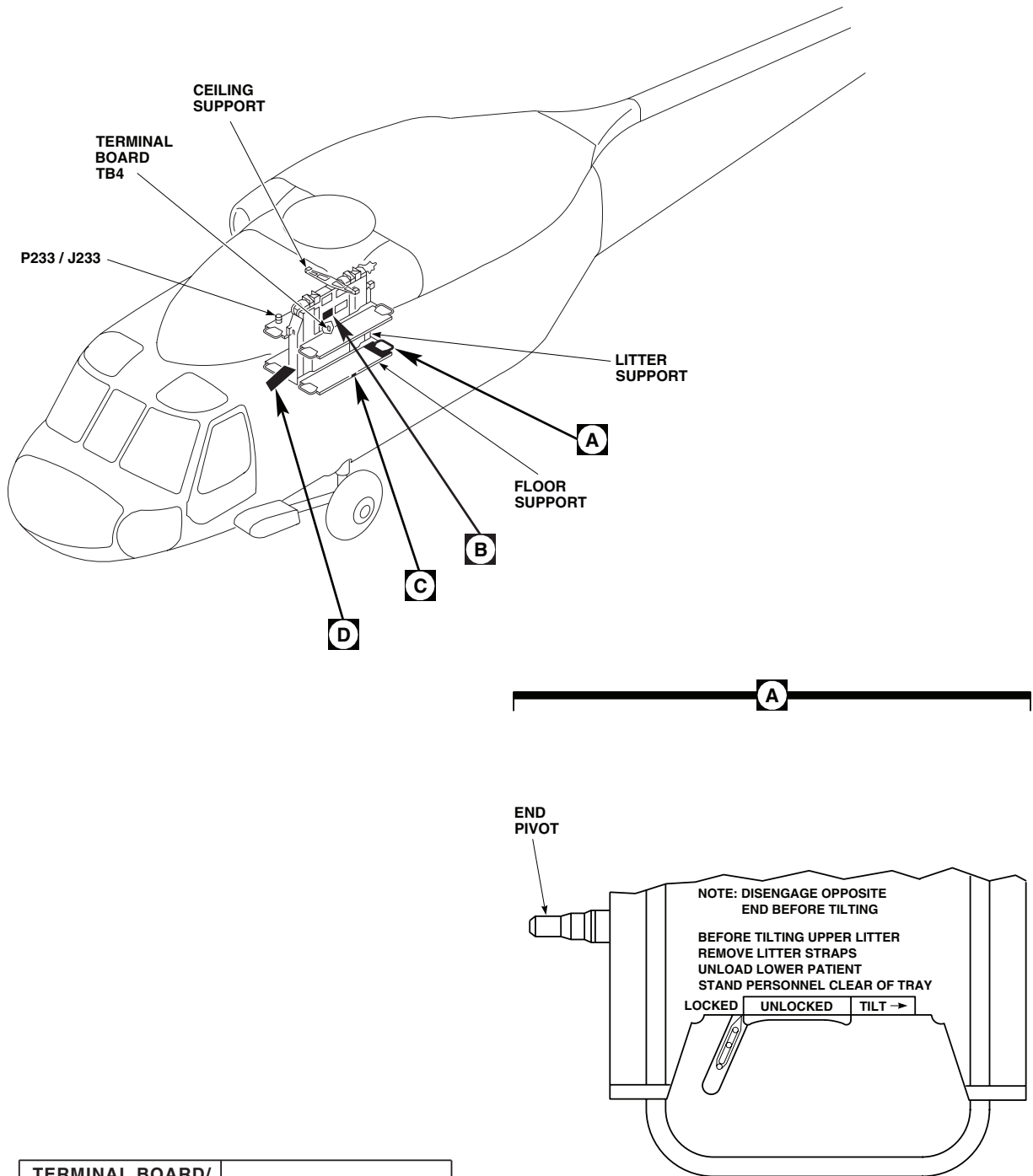
**3. Check for 28 vdc between NO. 2 LTR LTS circuit breaker terminal 1 and ground.**

- a. If voltage is as specified, repair/replace wiring between J233-D and circuit breaker terminal 1 (WP 1747 00). Go to **4**.
- b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **4**.

**4. Procedure completed.**



# LOCATION

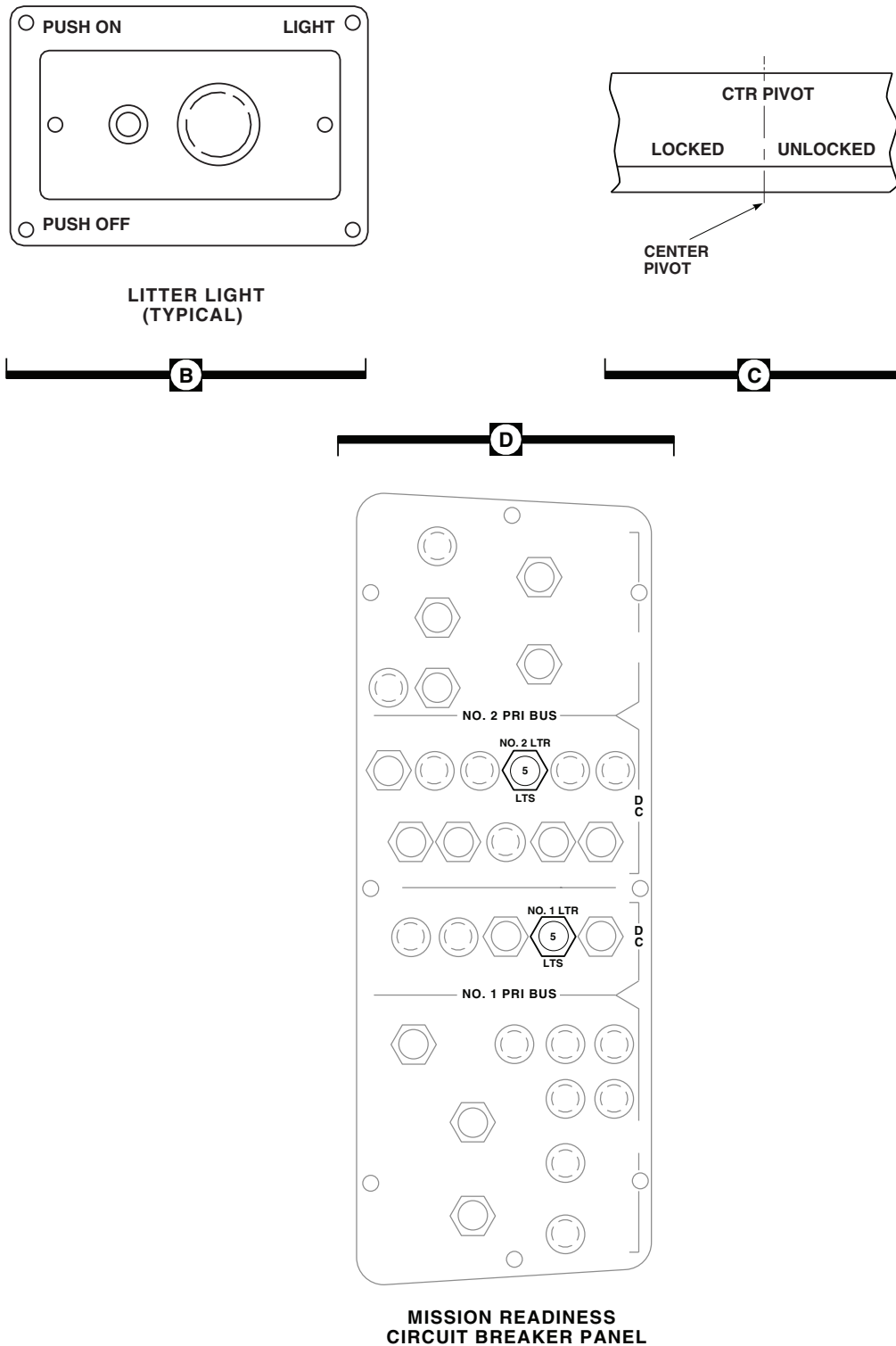


| TERMINAL BOARD/<br>DISCONNECT PLUG/<br>RECEPTACLE | LOCATION/<br>CONNECTION POINT        |
|---------------------------------------------------|--------------------------------------|
| P233 / J233                                       | CABIN CEILING, BL 5 LH,<br>STA 304.5 |
| TB4                                               | MEDEVAC PEDESTAL, BL 0,<br>STA 341   |

AA3436\_1  
SA

Figure 1. Aeromedical Evacuation Kit Location Diagram. (Sheet 1 of 2)

LOCATION - Continued



AA3436 2  
SA

Figure 1. Aeromedical Evacuation Kit Location Diagram. (Sheet 2 of 2)

## SCHEMATICS

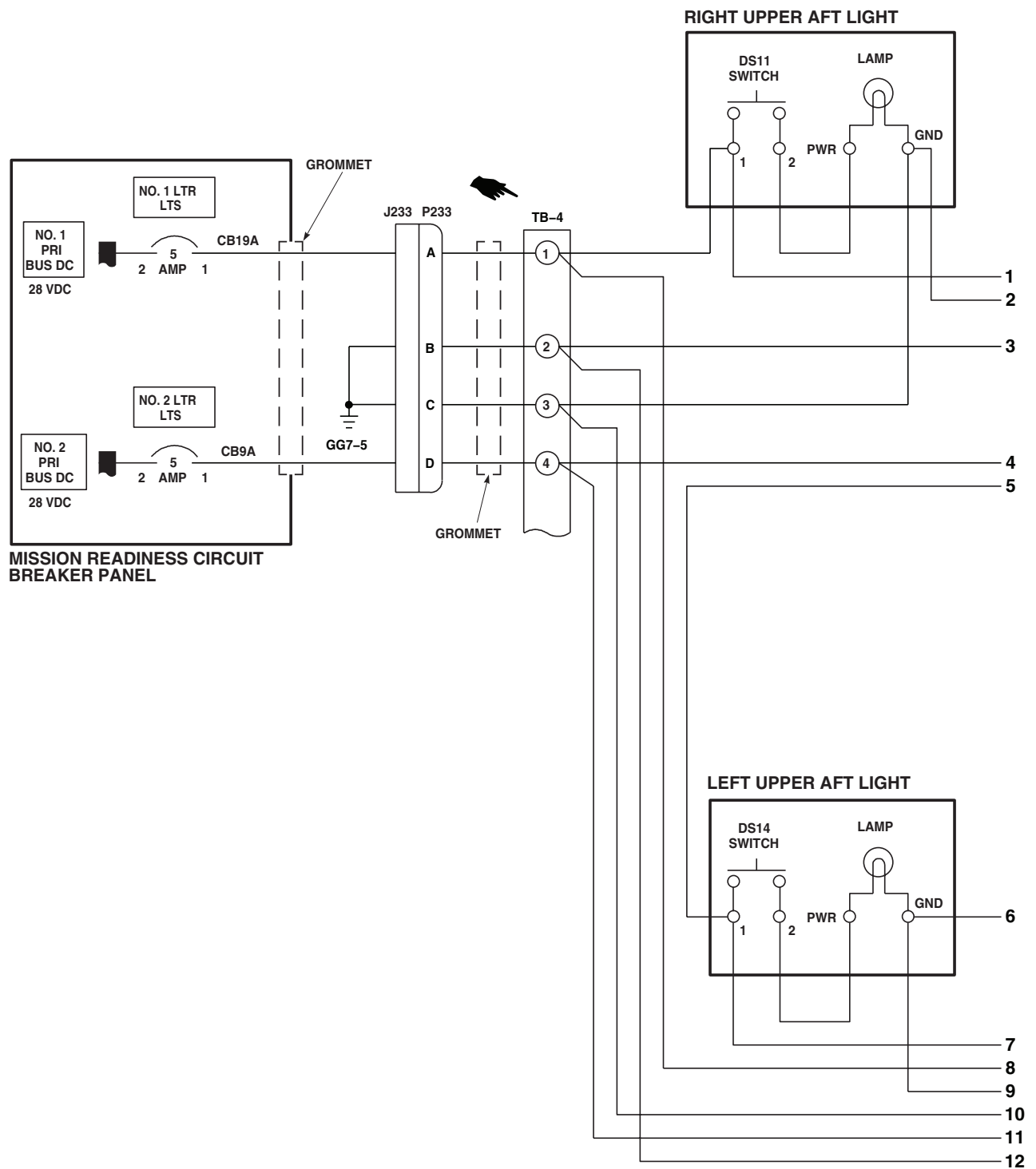
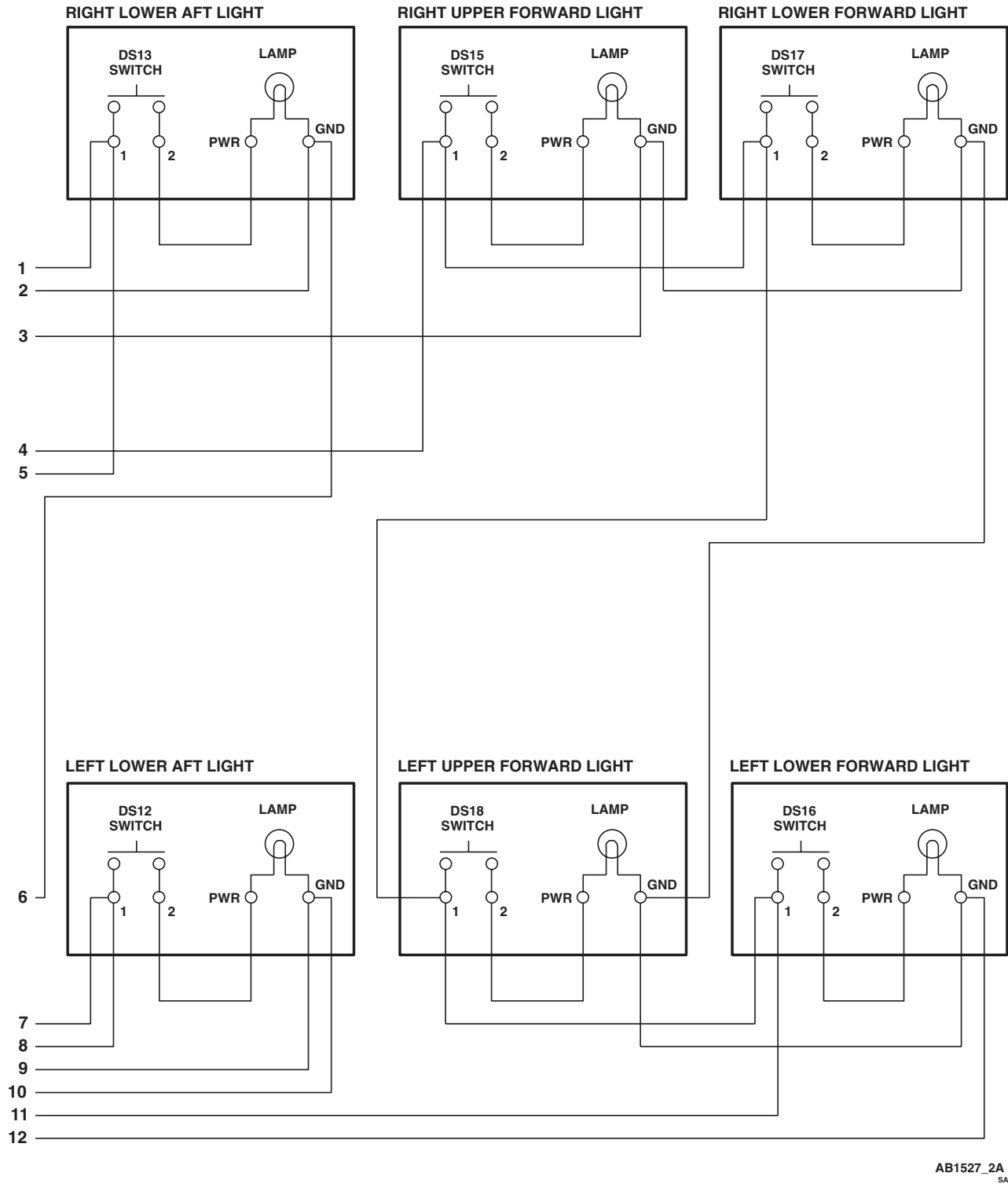
AB1527\_1B  
SA

Figure 2. Aeromedical Evacuation Kit Schematic Diagram. (Sheet 1 of 2)

SCHEMATICS - Continued



AB1527\_2A  
SA

Figure 2. Aeromedical Evacuation Kit Schematic Diagram. (Sheet 2 of 2)

END OF WORK PACKAGE

## UNIT LEVEL

## MISSION EQUIPMENT

EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM **W/O MWO 50-78**

## INITIAL SETUP:

**Test Equipment**

Compressor Unit, MIL-C-13874  
 Locally-Made ESSS Gate Valve Test Harness, Figure  
 156, WP 1805 00  
 Locally-Made Fuel Quantity Simulator Test Set, Figure  
 152, WP 1805 00  
 Multimeter, AN/PSM-45A (2 required)

WP 1428 00  
 WP 1431 00  
 WP 1437 00  
 WP 1438 00  
 WP 1439 00  
 WP 1440 00  
 WP 1442 00  
 WP 1445 00

**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01  
 Container, Suitable  
 Electrical Repairer Toolkit, SC 5180-99-B06  
 Locally-Made Drag Beam Wedge, Figure 167,  
 WP 1805 00  
 Locally-Made External Stores Support System (ESSS)  
 Pneumatic Adapter, Figure 151, WP 1805 00  
 Stopwatch  
 Torque Wrench, GG-W-686

WP 1447 00  
 WP 1448 00  
 WP 1451 00  
 WP 1453 00  
 WP 1455 00  
 WP 1457 00  
 WP 1465 00  
 WP 1476 00  
 WP 1612 00  
 WP 1613 00

**Personnel Required**

Aircraft Electrician MOS 15F (1)  
 UH-60 Helicopter Repairer MOS 15T (1)

WP 1620 00  
 WP 1685 00  
 WP 1747 00  
 WP 1805 00

**References**

TM 1-1520-237-10  
 WP 0090 00  
 WP 0104 00  
 WP 0106 00  
 WP 0110 00  
 WP 0131 00  
 WP 0159 00  
 WP 0443 00  
 WP 0785 00  
 WP 0824 00  
 WP 0825 00  
 WP 0925 00  
 WP 0996 00  
 WP 1377 00  
 WP 1424 00  
 WP 1425 00

**Equipment Condition**

All Explosive Cartridges Removed from Ejector Racks,  
 WP 1442 00  
 or All Explosive Cartridges Removed from Ejector  
 Racks, WP 1445 00  
 External Electrical Power Available, WP 1685 00  
 or APU Operational, TM 1-1520-237-10  
 External Range Extension Tanks (Inboard or Outboard)  
**EMPTY**, WP 1620 00  
 Main Fuel Tanks, Not More Than **650** Pounds of Fuel  
 in Each Tank, WP 1612 00  
 or Main Fuel Tanks, Not More Than **650** Pounds of  
 Fuel in Each Tank, WP 1613 00  
 Rear Fairings on All Vertical Stores (Pylons) Removed,  
 WP 1447 00

## APPLICABLE CONFIGURATION

This operational/troubleshooting procedure is designed to be utilized when the horizontal (Wings) and/or vertical (Pylons) stores are initially installed or replaced. When only tanks are removed, refer to WP 0159 00.

## EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE **W/O MWO 50-78**

### PROCEDURE

#### Step

#### NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- Special Icons used in this operational/troubleshooting procedure:
  - CU** Steps that are to be done if compressor unit is used.
  - INBD** Steps that are to be done if inboard tanks are installed.
  - OUTBD** Steps that are to be done if outboard tanks are installed.
- As an alternate to using the compressor unit to pressurize the ESSS Range Extension System, the Aviation Ground Power Unit (AGPU) or the APU may be used. If the alternate is selected, steps with **CU** may be skipped.
- The operational/troubleshooting procedure will check the ESSS Range Extension System with any tank configuration, installed on the helicopter, authorized by the Aircraft Operator's Manual (TM 1-1520-237-10).
- See Figure 1 for component location diagram and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

1. **CU** Unfasten fasteners on left inboard stores support rear fairing and hinge fairing open to gain access to bleed-air quick-disconnect (QD). ■
2. **CU** Disconnect hose from outboard side of bleed-air QD. ■
3. **CU** Using locally-made pneumatic adapter (WP 1805 00), tee air compressor into helicopter bleed-air QD and ESSS bleed-air hose. ■
4. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
5. Pull out the following circuit breakers:

---

**EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE W/O MWO 50-78** - Continued
 

---

**CIRCUIT BREAKER****CIRCUIT BREAKER PANEL****UH60A UH60L**

|                    |                                |
|--------------------|--------------------------------|
| ESSS JTSN OUTBD    | Copilot's                      |
| ESSS JTSN INBD     | Copilot's                      |
| BACKUP PUMP PWR    | Copilot's                      |
| UTIL RECP CABIN    | Copilot's                      |
| ESSS JTSN OUTBD    | Upper Console (copilot's side) |
| ESSS JTSN INBD     | Upper Console (copilot's side) |
| EXT FUEL RH        | Mission Readiness              |
| EXT FUEL LH        | Mission Readiness              |
| AUX FUEL QTY       | Mission Readiness              |
| NO. 2 XFER CONTROL | Mission Readiness              |

**EH60A**

|                 |                                |
|-----------------|--------------------------------|
| ESSS JTSN OUTBD | Copilot's                      |
| ESSS JTSN INBD  | Copilot's                      |
| UTIL RECP CABIN | Copilot's                      |
| EXT FUEL LH     | Copilot's                      |
| EXT FUEL RH     | Pilot's                        |
| AUX FUEL CONTR  | Pilot's                        |
| AUX FUEL QTY    | Pilot's                        |
| ESSS JTSN OUTBD | Upper Console (copilot's side) |
| ESSS JTSN INBD  | Upper Console (copilot's side) |

---

**EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE W/O MWO 50-78** - Continued
**NOTE**

To ensure proper readings are obtained during operational check of the ESSS fuel system, verify dip switches on the back of the auxiliary fuel management panel are set to the type of fuel used.

6. On back of auxiliary fuel management panel, verify dip switches are set to the type of fuel used.

**CAUTION**

Prior to activating fuel gate valves that have been out of service for more than 30 days, it is recommended to open and close fuel gate valves a few times by hand prior to applying electrical power to reduce the possibility of damage to auxiliary fuel management panel and circuitry.

7. Disconnect P360 from auxiliary fuel management control panel (control panel).
8. Connect P1 of locally-made ESSS gate valve test harness (test harness) to UTIL RCPT +28V receptacle J258 in cabin ceiling.
9. Turn on electrical power.
10. Push in UTIL RECP CABIN circuit breaker on copilot's circuit breaker panel.
11. Connect locally-made test harness pin E to P360-B.
12. Connect pins A and D of locally-made test harness to the following pairs of pins on P360, and alternately place locally-made test harness switch to OPEN and CLOSE:

| TEST HARNESS PIN | P360- | VALVE OPERATION      |
|------------------|-------|----------------------|
| A                | F     | LEFT OUTBD - OPEN    |
| D                | E     | LEFT OUTBD - CLOSED  |
| A                | Y     | LEFT INBD - OPEN     |
| D                | Z     | LEFT INBD - CLOSED   |
| A                | g     | RIGHT OUTBD - OPEN   |
| D                | P     | RIGHT OUTBD - CLOSED |
| A                | j     | RIGHT INBD - OPEN    |
| D                | i     | RIGHT INBD - CLOSED  |

**Indication/Condition**

Gate valves shall open and close as indicated.



---

**EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE W/O MWO 50-78** - Continued
**Corrective Action**

1. If Indication/Condition is not as specified, connect locally-made test harness pins A, D, and E to pins A, D, and E of malfunctioning gate valve. Alternately place locally-made test harness switch to OPEN and CLOSED.
  - a. If gate valve operates normally, repair/replace wiring, as required, between P360 and gate valve's electrical connector (WP 1747 00).
  - b. If gate valve does not operate normally, replace gate valve (WP 1440 00).

**Step**

13. Turn off electrical power.
14. Connect P360 to control panel.
15. Turn on electrical power.
16. Adjust upper console INST LT - PILOT FLT control to at least one quarter turn clockwise.
17. Adjust upper console CONSOLE LT - LOWER control to BRT.

**Indication/Condition**

Information on plate lighting control panel shall go on at full brightness.

**Corrective Action**

If Indication/Condition is not as specified, troubleshoot upper and lower console lights (WP 0106 00).

**Step**

18. Push in the following circuit breakers:

**CIRCUIT BREAKER****CIRCUIT BREAKER PANEL****UH60A UH60L**

AUX FUEL QTY

Mission Readiness

NO. 2 XFER CONTROL

Mission Readiness

**EH60A**

AUX FUEL QTY

Pilot's

19. Momentarily place caution/advisory panel BRT/DIM-TEST switch to BRT/DIM.

**Indication/Condition**

AUX FUEL QTY POUNDS digital display (digital display) on control panel shall dim.

**Corrective Action**

If Indication/Condition is not as specified, troubleshoot instrument panel and console indicator lights dimming (WP 0110 00) or caution/advisory warning system (WP 0090 00), as required.

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**EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE W/O MWO 50-78** - Continued
**Step**

20. Adjust control panel QTY BRIGHTNESS control from DECR to INCR, then back to DECR.

**Indication/Condition**

AUX FUEL QTY POUNDS digital display (digital display) shall go to full brightness, then back to dim.

**Corrective Action**

If Indication/Condition is not as specified, replace control panel (WP 1448 00).

**Step**

21. Adjust upper console CONSOLE LT - LOWER and INST LT - PILOT FLT controls to OFF.  
 22. Place control panel switches as indicated:

| SWITCH               | POSITION |
|----------------------|----------|
| FUEL XFR PRESS OUTBD | OFF      |
| FUEL XFR PRESS INBD  | OFF      |
| TANKS                | OUTBD    |
| MODE                 | OFF      |
| MANUAL XFR RIGHT     | OFF      |
| MANUAL XFR LEFT      | OFF      |
| AUX FUEL QTY Rotary  | TOTAL    |

23. Push in the following circuit breakers:

| CIRCUIT BREAKER    | CIRCUIT BREAKER PANEL |
|--------------------|-----------------------|
| <b>UH60A UH60L</b> |                       |
| EXT FUEL RH        | Mission Readiness     |
| EXT FUEL LH        | Mission Readiness     |
| <b>EH60A</b>       |                       |
| EXT FUEL LH        | Copilot's             |
| EXT FUEL RH        | Pilot's               |
| AUX FUEL CONTR     | Pilot's               |

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**EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE W/O MWO 50-78** - Continued

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24. Press and hold control panel TEST button.

**Indication/Condition**

- a. Digital display shall indicate 8888.
- b. DEGRADED and VENT SENSOR lights shall illuminate.

**Corrective Action**

If Indication/Conditions a. and b. are not as specified, go to WHEN CONTROL PANEL TEST BUTTON IS PRESSED, ALL CONTROL PANEL INDICATORS ARE NOT ON AND/OR DIGITAL DISPLAY DOES NOT INDICATE 8888, in this work package.

**Step**

25. Release TEST button.

**Indication/Condition**

- a. Digital display shall cycle the indication 8 from left to right 3 times.
- b. Digital display shall indicate the word Good for about 5 seconds.
- c. Digital display shall indicate either 4, 5, or 8 (preset fuel type) in left-most position for about 3 seconds.
- d. Digital display may indicate some value of auxiliary fuel remaining (unknown at this time).

**Corrective Action**

- 1. If Indication/Condition a. and/or c. are not as specified, replace control panel (WP 1448 00).
- 2. If Indication/Condition b. is not as specified, but display E01, E04, E05, or E06, replace control panel (WP 1448 00).
- 3. If Indication/Condition b. is not as specified, but displays E02, go to DURING SELF-TEST, CONTROL PANEL DIGITAL DISPLAY INDICATES E02 INSTEAD OF GOOD, in this work package.
- 4. If Indication/Condition b. is not as specified, but displays E03, go to DURING SELF-TEST, CONTROL PANEL DIGITAL DISPLAY INDICATES E03 INSTEAD OF GOOD, in this work package.

**Step****NOTE**

Disregard any EXTERNAL indicator displays at this time.

26. Press and hold control panel STATUS button.

**Indication/Condition**

EXTERNAL RIGHT INBD EMPTY, RIGHT OUTBD EMPTY, LEFT INBD EMPTY, and LEFT OUTBD EMPTY indicators shall go on.

**Corrective Action**

If Indication/Condition is not as specified, replace control panel (WP 1448 00).

**Step**

27. Release STATUS button.

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**EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE W/O MWO 50-78** - Continued

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**Indication/Condition**

All control panel indicators and caution/advisory panel AUX FUEL capsule shall be off.

**Corrective Action**

1. If EXTERNAL RIGHT NO FLOW and/or EXTERNAL LEFT NO FLOW indicators are on, replace control panel (WP 1448 00).
2. If EXTERNAL RIGHT INBD EMPTY, RIGHT OUTBD EMPTY, LEFT INBD EMPTY, or LEFT OUTBD EMPTY indicator is on (tank installed in that position), go to CONTROL PANEL EMPTY INDICATOR IS ON WITH ADEQUATE FUEL IN FUEL TANK, in this work package.
3. If Indication/Condition is not as specified, go to WHEN CONTROL PANEL STATUS BUTTON IS PRESSED, AND THEN RELEASED, CONTROL PANEL INDICATORS AND/OR CAUTION/ADVISORY PANEL AUX FUEL CAPSULE IS NOT OFF, in this work package.

**Step**

28. Turn off electrical power.
29. **OUTBD** Fill each outboard tank, if installed, with **160 - 190** pounds of fuel. ■
30. **INBD** Fill each inboard tank, if installed, with **220 - 250** pounds of fuel. ■
31. Turn on electrical power.
32. **CU** Apply air source, regulated to 25 - 45 psi to helicopter's external tanks. ■
33. Record CDU TOTAL FUEL indication.
34. **OUTBD** Place control panel PRESSURE - OUTBD switch to ON. ■
35. **INBD** Place control panel PRESSURE - INBD switch to ON. ■
36. Record time.
37. Inspect fuel and pneumatic hoses and fittings for signs of leaking or damage (WP 1377 00).

**Indication/Condition**

There shall be no signs of leakage or damage.

**Corrective Action**

If Indication/Condition is not as specified, replace hoses and fittings as necessary (WP 1431 00).

**Step****NOTE**

- While doing the following procedure, inspect fuel and pneumatic hoses and fittings as specified in step 37.
- The K FACTOR called out in the following steps is a calibration adjustment used for calibrating the fuel system and is stamped on the fuel flow transmitter. Record this value.
- Disregard digit to the right of decimal for K FACTOR entry.

38. Record K FACTOR.

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**EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE W/O MWO 50-78** - Continued

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39. Pull up and turn control panel AUX FUEL QTY switch to CAL.
40. Place control panel INCR-DECR switch to INCR, then to DECR.

**Indication/Condition**

Digital display indication shall increase, and then decrease.

**Corrective Action**

If Indication/Condition is not as specified, go to WITH WEIGHT-ON-WHEELS CONTROL PANEL CAL INDICATION CANNOT BE CHANGED, in this work package.

**Step**

41. Place control panel INCR-DECR switch to either position to enter K FACTOR on digital display.
42. Using locally-made wedge, push plunger down on left drag beam switch.
43. Momentarily place control panel INCR-DECR switch to INCR, then DECR.

**Indication/Condition**

Digital display indication shall not change.

**Corrective Action**

If Indication/Condition is not as specified, go to DIGITAL DISPLAY INDICATION CHANGES WITH DRAG BEAM SWITCH IN IN-FLIGHT POSITION, in this work package.

**Step**

44. Remove locally-made wedge.
45. Turn control panel AUX FUEL QTY switch to OUTBD.
46. **OUTBD** Disconnect P2/J849 and P2/J850. ■

**Indication/Condition**

Digital display shall indicate cccc.

**Corrective Action**

If Indication/Condition is not as specified, replace control panel (WP 1448 00).

**Step**

47. Connect P2/J849 and P2/J850 if outboard tanks are installed, or install insulated jumper wire between J1-1 and J1-2.
48. Place control panel INCR-DECR switch to either position to set digital display indication to 1000.

**Indication/Condition**

Digital display shall indicate 1000 pounds.

**Corrective Action**

1. If Indication/Condition is not as specified, replace control panel (WP 1448 00).
  - a. If trouble remains, replace left relay panel (WP 0825 00).

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**EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE W/O MWO 50-78** - Continued

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**Step**

49. Turn control panel AUX FUEL QTY switch to INBD.
50. **INBD** Disconnect P2/J835 and P2/J836. ■

**Indication/Condition**

Digital display shall indicate cccc.

**Corrective Action**

If Indication/Condition is not as specified, replace control panel (WP 1448 00).

**Step**

51. Connect P2/J835 and P2/J836 if inboard tanks are installed, or install insulated jumper wire between inboard J1-1 and J1-2.
52. Place control panel INCR-DECR switch to either position to set digital display indication to 1000 pounds.

**Indication/Condition**

Digital display shall indicate 1000 pounds.

**Corrective Action**

1. If Indication/Condition is not as specified, replace control panel (WP 1448 00).
  - a. If trouble remains, replace left relay panel (WP 0825 00).

**Step**

53. Turn control panel AUX FUEL QTY switch to TOTAL.

**Indication/Condition**

Digital display shall indicate 2000 pounds.

**Corrective Action**

If Indication/Condition is not as specified, replace control panel (WP 1448 00).

**Step**

54. **OUTBD** Turn control panel AUX FUEL QTY switch to OUTBD. ■
55. **OUTBD** Place control panel FUEL XFR - MODE switch to MANUAL. ■
56. **OUTBD** Check CDU TOTAL FUEL indication. ■

**Indication/Condition**

CDU TOTAL FUEL indication shall be the same as recorded in step 32.

**Corrective Action**

If Indication/Condition is not as specified, go to CDU TOTAL FUEL INDICATION INCREASES WHEN FUEL TRANSFER IS NOT ACTUATED, in this work package.

**Step**

57. **OUTBD** Before continuing, make sure that at least 5 minutes have elapsed from time recorded in step 36. ■

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**EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE W/O MWO 50-78** - Continued

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58. **OUTBD** Place control panel MANUAL XFR - RIGHT switch to ON until CDU TOTAL FUEL indication increases by 20 pounds. ■

**Indication/Condition**

- a. Elapsed time required to increase indication by 20 pounds shall be less than 1 1/2 minutes.

**NOTE**

Disregard caution/advisory panel AUX FUEL capsule if it goes on momentarily.

- b. Control panel EXTERNAL RIGHT NO FLOW indicator shall momentarily go on, then go off.  
c. Remaining control panel EXTERNAL indicators shall remain off.  
d. Control panel digital display indication shall decrease.

**Corrective Action**

If Indication/Conditions are not as specified, go to WITH MANUAL MODE SELECTED, CDU TOTAL FUEL INDICATION DOES NOT INCREASE AND/OR RESULT IS NOT AS SPECIFIED WHEN ATTEMPTING TO TRANSFER FUEL FROM RIGHT OUTBOARD FUEL TANK, in this work package.

**Step**

59. **OUTBD** Place control panel MANUAL XFR - RIGHT switch to OFF. ■  
60. **OUTBD** Place control panel MANUAL XFR - LEFT switch to ON until CDU TOTAL FUEL indication increases by 20 pounds. ■

**Indication/Condition**

- a. Elapsed time required to increase indication by 20 pounds shall be less than 1 1/2 minutes.

**NOTE**

Disregard caution/advisory panel AUX FUEL capsule if it goes on momentarily.

- b. Control panel EXTERNAL LEFT NO FLOW indicator shall momentarily go on, then go off.  
c. Remaining control panel EXTERNAL indicators shall remain off.  
d. Control panel digital display indication shall decrease.

**Corrective Action**

If Indication/Conditions are not as specified, go to WITH MANUAL MODE SELECTED, CDU TOTAL FUEL INDICATION DOES NOT INCREASE AND/OR RESULT IS NOT AS SPECIFIED WHEN ATTEMPTING TO TRANSFER FUEL FROM LEFT OUTBOARD FUEL TANK, in this work package.

**Step**

61. **OUTBD** Place control panel MANUAL XFER - LEFT switch to OFF. ■  
62. **INBD** Turn control panel AUX FUEL QTY switch to INBD. ■  
63. **INBD** Place control panel FUEL XFR - TANKS switch to INBD. ■

**Indication/Condition**

Digital display indication shall not change.

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**EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE W/O MWO 50-78** - Continued

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**Corrective Action**

If Indication/Condition is not as specified, replace control panel (WP 1448 00).

**Step**

64. **INBD** Place control panel MANUAL XFR - RIGHT switch to ON until CDU TOTAL FUEL indication increases by 20 pounds. ■

**Indication/Condition**

- a. Elapsed time required to increase indication by 20 pounds shall be less than 1 1/2 minutes.

**NOTE**

Disregard caution/advisory panel AUX FUEL capsule if it goes on momentarily.

- b. Control panel EXTERNAL RIGHT NO FLOW indicator shall momentarily go on, then go off.  
c. Remaining control panel EXTERNAL indicators shall remain off.  
d. Control panel digital display indication shall decrease.

**Corrective Action**

If CDU TOTAL FUEL indication does not increase, and/or Indication/Conditions are not as specified, go to WITH MANUAL MODE SELECTED, CDU TOTAL FUEL INDICATION DOES NOT INCREASE AND/OR RESULT IS NOT AS SPECIFIED WHEN ATTEMPTING TO TRANSFER FUEL FROM RIGHT INBOARD FUEL TANK, in this work package.

**Step**

65. **INBD** Place control panel MANUAL XFR - RIGHT switch to OFF. ■  
66. **INBD** Place control panel MANUAL XFR - LEFT switch to ON until CDU TOTAL FUEL indication increases by 20 pounds. ■

**Indication/Condition**

- a. Elapsed time required to increase indication by 20 pounds shall be less than 1 1/2 minutes.

**NOTE**

Disregard caution/advisory panel AUX FUEL capsule if it goes on momentarily.

- b. Control panel EXTERNAL LEFT NO FLOW indicator shall momentarily go on, then go off.  
c. Remaining control panel EXTERNAL indicators shall remain off.  
d. Control panel digital display indication shall decrease.

**Corrective Action**

If Indication/Conditions are not as specified, go to WITH MANUAL MODE SELECTED, CDU TOTAL FUEL INDICATION DOES NOT INCREASE AND/OR RESULT IS NOT AS SPECIFIED WHEN ATTEMPTING TO TRANSFER FUEL FROM LEFT INBOARD FUEL TANK, in this work package.

**Step**

67. **INBD** Place control panel MANUAL XFR - LEFT switch to OFF. ■  
68. **OUTBD** Place control panel FUEL XFR - TANKS and AUX FUEL QTY switches to OUTBD. ■



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**EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE W/O MWO 50-78** - Continued

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69. **OUTBD** Record CDU TOTAL FUEL indication. ■
70. **OUTBD** Record control panel digital display indication. ■
71. **OUTBD** Place control panel MANUAL XFR - LEFT and RIGHT switches to ON. ■
72. **OUTBD** When CDU TOTAL FUEL indication is 100 pounds more than value recorded in step 69., place control panel FUEL XFR - MODE switch to OFF. ■
73. **OUTBD** Place control panel MANUAL XFR - LEFT and RIGHT switches to OFF. ■
74. **OUTBD** Record control panel digital display indication. ■
75. **OUTBD** Subtract recorded indication of step 74. from recorded indication of step 70. ■

**Indication/Condition**

The difference shall be between 90 and 110 pounds.

**Corrective Action**

If Indication/Condition is not as specified, go to FUEL TRANSFER TIME AND/OR GAGING ACCURACY IS NOT AS SPECIFIED, in this work package.

**Step**

76. Turn off electrical power.
77. Connect locally-made fuel quantity test simulator (simulator) power cord to UTIL RCPT +28V (J258) in cabin ceiling.
78. Disconnect helicopter P905 from J905.
79. Connect simulator J905 to helicopter P905.
80. Turn simulator TANK 1 and TANK 2 controls fully counterclockwise.
81. Connect No. 1 digital multimeter to simulator TANK 1 OUTPUT.
82. Connect No. 2 digital multimeter to simulator TANK 2 OUTPUT.
83. Turn on electrical power.
84. Turn simulator TANK 1 control to obtain 6.0 vdc No. 1 multimeter indication.
85. Turn simulator TANK 2 control to obtain 6.0 vdc No. 2 multimeter indication.
86. Turn control panel AUX FUEL QTY switch to INBD.
87. Place control panel INCR-DECR switch to INCR to obtain display of 6000 pounds.
88. Turn control panel AUX FUEL QTY switch to OUTBD.
89. Place control panel INCR-DECR switch to INCR to obtain display of 1500 pounds.
90. Place control panel FUEL XFR - MODE switch to AUTO.

**Indication/Condition**

Digital display indication shall not change.

**Corrective Action**

If Indication/Condition is not as specified, replace control panel (WP 1448 00).

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**EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE W/O MWO 50-78** - Continued

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**Step****NOTE**

- Read steps 91. through 94. completely before continuing.
- Do steps 91. through 94. as quickly as possible.

91. Slowly turn simulator TANK 1 control counterclockwise until control panel digital display indication begins to decrease.
92. Record No. 1 multimeter indication.
93. Slowly turn simulator TANK 1 control clockwise until control panel digital display indication stops changing.
94. Record No. 1 multimeter indication.

**Indication/Condition**

- a. Recorded indication of step 92. shall be between 4.65 and 4.85 vdc.
- b. Recorded indication of step 94. shall be between 4.90 and 5.10 vdc.
- c. EXTERNAL RIGHT NO FLOW indicator on control panel shall remain off during fuel transfer.
- d. EXTERNAL LEFT NO FLOW indicator on control panel shall remain off during fuel transfer.

**Corrective Action**

1. If Indication/Condition a. and/or b. are not as specified, go to AUTO MODE FUEL TRANSFER LIMITS ARE NOT AS SPECIFIED, in this work package.
2. If Indication/Condition c. is not as specified, go to CONTROL PANEL RIGHT NO FLOW INDICATOR REMAINS ON DURING FUEL TRANSFER, in this work package.
3. If Indication/Condition d. is not as specified, go to CONTROL PANEL LEFT NO FLOW INDICATOR REMAINS ON DURING FUEL TRANSFER, in this work package.

**Step**

95. Turn simulator TANK 1 control clockwise to obtain 6.0 vdc No. 1 multimeter indication.

**NOTE**

- Read steps 96. through 99. completely before continuing.
- Do steps 96. through 99. as quickly as possible.

96. Slowly turn simulator TANK 2 control counterclockwise until control panel digital display indication begins to decrease.
97. Record No. 2 multimeter indication.
98. Slowly turn simulator TANK 2 control clockwise until control panel digital display indication stops changing.
99. Record No. 2 multimeter indication.

**Indication/Condition**

- a. Recorded indication of step 97. shall be between 4.65 and 4.85 vdc.
- b. Recorded indication of step 99. shall be between 4.9 and 5.1 vdc.
- c. EXTERNAL RIGHT NO FLOW indicator on control panel shall remain off during fuel transfer.

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**EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE W/O MWO 50-78** - Continued

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- d. EXTERNAL LEFT NO FLOW indicator on control panel shall remain off during fuel transfer.

**Corrective Action**

1. If Indication/Conditions a. and/or b. are not as specified, go to AUTO MODE FUEL TRANSFER LIMITS ARE NOT AS SPECIFIED, in this work package.
2. If Indication/Condition c. is not as specified, go to CONTROL PANEL RIGHT NO FLOW INDICATOR REMAINS ON DURING FUEL TRANSFER, in this work package.
3. If Indication/Condition d. is not as specified, go to CONTROL PANEL LEFT NO FLOW INDICATOR REMAINS ON DURING FUEL TRANSFER, in this work package.

**Step**

100. Turn simulator TANK 2 control to obtain 6.0 vdc No. 2 multimeter indication.
101. Place control panel FUEL XFR - MODE switch to OFF.
102. Place control panel INCR-DECR switch to either position to obtain digital display indication of 1000 pounds.
103. Place control panel FUEL XFR - MODE switch to AUTO.

**Indication/Condition**

Digital display indication shall not change.

**Corrective Action**

If Indication/Condition is not as specified, replace control panel (WP 1448 00).

**Step****NOTE**

- Read steps 104. through 107. completely before continuing.
- Do steps 104. through 107. as quickly as possible.

104. Slowly turn simulator TANK 1 control counterclockwise until control panel digital display indication begins to decrease.
105. Record No. 1 multimeter indication.
106. Slowly turn simulator TANK 1 control clockwise until control panel digital display indication stops changing.
107. Record No. 1 multimeter indication.

**Indication/Condition**

- a. Recorded indication of step 105. shall be between 3.65 and 3.85 vdc.
- b. Recorded indication of step 107. shall be between 4.9 and 5.1 vdc.

**Corrective Action**

If Indication/Conditions are not as specified, replace control panel (WP 1448 00).

**Step**

108. Turn simulator TANK 1 control clockwise to obtain 6.0 vdc No. 1 multimeter indication.

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**EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE W/O MWO 50-78** - Continued

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**NOTE**

- Read steps 109. through 112. completely before continuing.
- Do steps 109. through 112. as quickly as possible.

109. Slowly turn simulator TANK 2 control counterclockwise until control panel digital display indication begins to decrease.
110. Record No. 2 multimeter indication.
111. Slowly turn simulator TANK 2 counterclockwise until control panel digital display indication stops changing.
112. Record No. 2 multimeter indication.

**Indication/Condition**

- a. Recorded indication of step 110. shall be between 3.65 and 3.85 vdc.
- b. Recorded indication of step 112. shall be between 4.9 and 5.1 vdc.

**Corrective Action**

If Indication/Conditions are not as specified, replace control panel (WP 1448 00).

**Step**

113. Turn simulator TANK 2 control clockwise to obtain 6.0 vdc No. 2 multimeter indication.
114. Place control panel FUEL XFR - MODE switch to OFF.
115. Turn control panel AUX FUEL QTY switch to INBD.
116. Place control panel INCR-DECR switch to either position to obtain digital display indication of 2000 pounds.
117. Turn control panel AUX FUEL QTY switch to OUTBD.
118. Place control panel FUEL XFR - MODE switch to AUTO.

**Indication/Condition**

Digital display indication shall not change.

**Corrective Action**

If Indication/Condition is not as specified, replace control panel (WP 1448 00).

**Step****NOTE**

- Read steps 119. through 122. completely before continuing.
- Do steps 119. through 122. as quickly as possible.

119. Slowly turn simulator TANK 1 control counterclockwise until control panel digital display indication begins to decrease.
120. Record No. 1 multimeter indication.
121. Slowly turn simulator TANK 1 control clockwise until control panel digital display indication stops changing.
122. Record No. 1 multimeter indication.

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**EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE W/O MWO 50-78** - Continued

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**Indication/Condition**

- a. Recorded indication of step 120. shall be between 2.9 and 3.1 vdc.
- b. Recorded indication of step 122. shall be between 4.9 and 5.1 vdc.

**Corrective Action**

If Indication/Conditions are not as specified, replace control panel (WP 1448 00).

**Step**

123. Slowly turn simulator TANK 1 control to obtain 6.0 vdc No. 1 multimeter indication.

**NOTE**

- Read steps 124. through 127. completely before continuing.
- Do steps 124. through 127. as quickly as possible.

124. Slowly turn simulator TANK 2 control counterclockwise until control panel digital display indication begins to decrease.
125. Record No. 2 multimeter indication.
126. Slowly turn simulator TANK 2 control clockwise until control panel digital display indication stops changing.
127. Record No. 2 multimeter indication.

**Indication/Condition**

- a. Recorded indication of step 125. shall be between 2.9 and 3.1 vdc.
- b. Recorded indication of step 127. shall be between 4.9 and 5.1 vdc.

**Corrective Action**

If Indication/Conditions are not as specified, replace control panel (WP 1448 00).

**Step**

128. Turn simulator TANK 2 control to obtain 2.7 vdc No. 2 multimeter indication.
129. Place control panel FUEL XFR - MODE switch to OFF.
130. **INBD** Place control panel FUEL XFR - TANKS switch to INBD. ■
131. **INBD** Place control panel FUEL XFR - MODE switch to AUTO until EXTERNAL - RIGHT INBD EMPTY and LEFT INBD EMPTY indicators go on. ■

**Indication/Condition**

- a. EXTERNAL - RIGHT INBD EMPTY and LEFT INBD EMPTY indicators shall go on.
- b. Caution/advisory panel AUX FUEL capsule shall go on immediately after either INBD EMPTY indicator goes on.
- c. Control panel EXTERNAL - RIGHT NO FLOW and LEFT NO FLOW indicators shall go on immediately after the corresponding INBD EMPTY indicator goes on.

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**EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE W/O MWO 50-78** - Continued

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**Corrective Action**

1. If Indication/Condition a. is not as specified, go to CONTROL PANEL EXTERNAL RIGHT INBD EMPTY AND/OR LEFT INBD EMPTY INDICATORS DO NOT GO ON WITH INADEQUATE FUEL IN TANK, in this work package.
2. If Indication/Condition b. and/or c. is not as specified, go to CAUTION/ADVISORY PANEL AUX FUEL CAPSULE AND/OR CONTROL PANEL EXTERNAL RIGHT NO FLOW AND LEFT NO FLOW INDICATORS DO NOT GO ON, in this work package.

**Step**

132. **INBD** Turn simulator TANK 2 control to obtain 6.0 vdc No. 2 multimeter indication. ■
133. **INBD** Turn simulator TANK 1 control to obtain 6.0 vdc No. 1 multimeter indication. ■
134. **INBD** Place control panel FUEL XFR - MODE switch to MANUAL. ■
135. **INBD** Place control panel MANUAL XFR - LEFT and RIGHT switches to ON. ■

**Indication/Condition**

- a. Control panel EXTERNAL - RIGHT NO FLOW and LEFT NO FLOW indicators shall go off, then go on within 5 minutes.
- b. Control panel digital display indication shall not change when EXTERNAL - RIGHT NO FLOW and LEFT NO FLOW indicators are on.

**Corrective Action**

If Indication/Conditions are not as specified, replace corresponding fuel tank (WP 1465 00).

**Step**

136. **INBD** Momentarily press control panel STATUS button. ■

**Indication/Condition**

- a. Control panel EXTERNAL - INBD EMPTY and OUTBD EMPTY indicators shall go off.
- b. Caution/advisory panel AUX FUEL capsule shall go off.

**Corrective Action**

If Indication/Conditions are not as specified, go to WHEN CONTROL PANEL STATUS BUTTON IS PRESSED, AND THEN RELEASED, CONTROL PANEL INDICATORS AND/OR CAUTION/ADVISORY PANEL AUX FUEL CAPSULE IS NOT OFF, in this work package.

**Step**

137. **INBD** Turn simulator TANK 2 control to obtain 2.7 vdc No. 2 multimeter indication. ■
138. **OUTBD** Make sure control panel FUEL XFR - MODE switch to OFF. ■
139. **OUTBD** Place control panel FUEL XFR - TANKS switch to OUTBD. ■
140. **OUTBD** Place control panel FUEL XFR - MODE switch to AUTO until EXTERNAL - RIGHT OUTBD EMPTY and LEFT OUTBD EMPTY indicators go on. ■

**Indication/Condition**

- a. EXTERNAL - RIGHT OUTBD EMPTY and LEFT OUTBD EMPTY indicators shall go on.

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**EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE W/O MWO 50-78** - Continued

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- b. Caution/advisory panel AUX FUEL capsule shall go on immediately after either OUTBD EMPTY indicator goes on.
- c. Control panel EXTERNAL - RIGHT NO FLOW and LEFT NO FLOW indicators shall go on immediately after corresponding tank OUTBD EMPTY indicator goes on.

**Corrective Action**

- 1. If Indication/Condition a. is not as specified, go to CONTROL PANEL EXTERNAL RIGHT OUTBD EMPTY AND/OR LEFT OUTBD EMPTY INDICATORS DO NOT GO ON WITH INADEQUATE FUEL IN TANK, in this work package.
- 2. If Indication/Condition b. and/or c. is not as specified, go to CAUTION/ADVISORY PANEL AUX FUEL CAPSULE AND/OR CONTROL PANEL EXTERNAL RIGHT NO FLOW And LEFT NO FLOW INDICATORS DO NOT GO ON IMMEDIATELY AFTER CORRESPONDING TANK EMPTY INDICATOR GOES ON, in this work package.

**Step**

141. **OUTBD** Momentarily press control panel STATUS button. ■

**Indication/Condition**

- a. Control panel EXTERNAL - INBD EMPTY and OUTBD EMPTY indicators shall go off.
- b. Caution/advisory panel AUX FUEL capsule shall go off.

**Corrective Action**

- 1. If Indication/Conditions are not as specified, go to WHEN CONTROL PANEL STATUS BUTTON IS PRESSED, AND THEN RELEASED, CONTROL PANEL INDICATORS AND/OR CAUTION/ADVISORY PANEL AUX FUEL CAPSULE IS NOT OFF, in this work package.

**Step**

142. **OUTBD** Place control panel FUEL XFR - MODE switch to MANUAL. ■
143. **OUTBD** Place control panel MANUAL XFR - LEFT and RIGHT switches to ON. ■

**Indication/Condition**

- a. Control panel EXTERNAL - RIGHT NO FLOW and LEFT NO FLOW indicators shall go off, then come back on within 5 minutes.
- b. Control panel digital display indication shall not change when the EXTERNAL RIGHT NO FLOW and LEFT NO FLOW indicators are on.

**Corrective Action**

If Indication/Conditions are not as specified, replace corresponding fuel tank (WP 1465 00).

**Step**

144. Place control panel FUEL XFR - MODE, MANUAL XFR - RIGHT and LEFT, and PRESS - INBD and OUTBD switches to OFF.

**Indication/Condition**

All four ESSS bleed-air regulator valves shall be venting air.

## EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE **W/O MWO 50-78** - Continued

### Corrective Action

1. If Indication/Condition is not as specified, disconnect connector from bleed-air regulator.
  - a. If bleed-air valve vents air, replace control panel (WP 1448 00).
  - b. If bleed-air valve does not vent air, clean valve, then do steps **CU** 32. ■ , **OUTBD** 34. ■ , **INBD** 35. ■ , and 144.
- (1) If trouble remains, replace bleed-air regulator valve (WP 1439 00).

### Step

145. **UH60A UH60L** Pull out NO. 2 XFER CONTROL circuit breaker on mission readiness circuit breaker panel. ■  
**EH60A** Pull out AUX FUEL CONTR circuit breaker on pilot's circuit breaker panel. ■
146. Remove overflow sensor from vent tube elbow in rear of main fuel tanks (WP 1425 00).
147. Disconnect P376 from J376.
148. Push in circuit breaker pulled out in step 145.

### Indication/Condition

Control panel VENT SENSOR FAIL capsule shall go on.

### Corrective Action

If Indication/Condition is not as specified, go to VENT SENSOR FAIL INDICATOR DOES NOT GO ON, in this work package.

### Step

149. Pull out circuit breaker pushed in step 148.
150. Connect P376 to J376.
151. Immerse thermistor portion of overflow sensor in container of fuel.
152. Push in circuit breaker pulled out in step 149.

### Indication/Condition

Control panel VENT SENSOR OVFL capsule shall go on.

### Corrective Action

If Indication/Condition is not as specified, replace control panel (WP 1448 00).

### Step

153. Install overflow sensor (WP 1424 00).
154. Turn off electrical power.
155. **CU** Turn off air source. ■
156. Allow bleed-air regulator valves to vent air to ambient pressure.
157. **CU** Disconnect air compressor, hoses, and locally-made pneumatic adapter. ■
158. **CU** Connect pneumatic hose to outboard side of helicopter bleed-air QD. ■
159. **CU** **TORQUE NUTS TO 135 - 150 INCH-POUNDS.** ■



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**EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE W/O MWO 50-78** - Continued

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- 160. Disconnect test equipment.
- 161. Connect helicopter electrical connectors.
- 162. Install explosive cartridges (WP 1442 00 or WP 1445 00).
- 163. Fasten helicopter fairings (WP 1428 00).

**Indication/Condition**

Helicopter fairings shall be fastened.

**Corrective Action**

None Required

**WHEN CONTROL PANEL TEST BUTTON IS PRESSED, ALL CONTROL PANEL INDICATORS ARE NOT ON AND/OR DIGITAL DISPLAY DOES NOT INDICATE 8888****SYMPTOM**

When control panel TEST button is pressed, all control panel indicators are not on and/or digital display does not indicate 8888.

**MALFUNCTION**

When control panel TEST button is pressed, all control panel indicators are not on and/or digital display does not indicate 8888.

**CORRECTIVE ACTION****1. Does digital display indicate 8888?**

- a. If 8888 is displayed, replace applicable auxiliary fuel management control panel indicator lamp (WP 1451 00). Go to **8**.
- b. If trouble remains, replace auxiliary fuel management control panel (WP 1448 00). Go to **8**.
- c. If 8888 is not displayed, go to **2**.

**2. Do some indicators go on?**

- a. If indicators go on, replace auxiliary fuel management control panel digital display segment (WP 1451 00). Go to **8**.
- b. If trouble remains, replace auxiliary fuel management control panel (WP 1448 00). Go to **8**.
- c. If indicators do not go on, go to **3**.

**3. Check for 28 vdc between P360-W and ground.**

- a. If voltage is as specified, replace auxiliary fuel management control panel (WP 1448 00). Go to **8**.
- b. If voltage is not as specified, go to **4**.

**4. Check helicopter configuration.**

- a. **UH60A UH60L** , go to **5**.
- b. **EH60A** , go to **6**.

**5. UH60A UH60L Check for 28 vdc between NO. 2 XFER CONTROL circuit breaker terminal 1 and ground.**

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**WHEN CONTROL PANEL TEST BUTTON IS PRESSED, ALL CONTROL PANEL INDICATORS ARE NOT ON AND/OR DIGITAL DISPLAY DOES NOT INDICATE 8888 - Continued**

- a. If voltage is as specified, go to **7**.
- b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **8**.
- 6. EH60A Check for 28 vdc between AUX FUEL CONTR circuit breaker terminal 1 and ground.**
  - a. If voltage is as specified, go to **7**.
  - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **8**.
- 7. Check for 28 vdc between J346-35 and ground.**
  - a. If voltage is as specified, replace fuel system main wiring harness (WP 1453 00). Go to **8**.
  - b. If voltage is not as specified, repair/replace wiring between J346-35 and circuit breaker terminal 1 (WP 1747 00). Go to **8**.
- 8. Procedure completed.**

**DURING SELF-TEST, CONTROL PANEL DIGITAL DISPLAY INDICATES E02 INSTEAD OF GOOD****SYMPTOM**

During self-test, control panel digital display indicates E02 instead of GOOD.

**MALFUNCTION**

During self-test, control panel digital display indicates E02 instead of GOOD.

**CORRECTIVE ACTION**

- 1. Check for 115 vac between P360-f and ground.**
  - a. If voltage is as specified, go to **2**.
  - b. If voltage is not as specified, go to **4**.
- 2. Check continuity between:**
  - **P361-U and J524-2**
  - **P361-S and J524-1**
  - a. If continuity is present, replace fuel temperature transducer (WP 1437 00). Go to **6**.
  - b. If continuity is not present, go to **3**.
- 3. Check continuity between:**
  - **P346-20 and J361-U**
  - **P346-21 and J361-S**
  - a. If continuity is present, repair/replace wiring between (WP 1747 00), then go to **6**:
    - **J346-20 and J524-2**
    - **J346-21 and J524-1**
  - b. If continuity is not present, replace fuel system main wiring harness (WP 1453 00). Go to **6**.
- 4. Check for 115 vac between AUX FUEL QTY circuit breaker terminal 1 and ground.**
  - a. If voltage is as specified, go to **5**.

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**DURING SELF-TEST, CONTROL PANEL DIGITAL DISPLAY INDICATES E02 INSTEAD OF GOOD** - Continued

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b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **6**.

**5. Check for 115 vac between J346-38 and ground.**

a. If voltage is as specified, replace fuel harness system main wiring harness (WP 1453 00). Go to **6**.

b. If voltage is not as specified, repair/replace wiring between J346-38 and circuit breaker terminal 1 (WP 1747 00). Go to **6**.

**6. Procedure completed.**

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**DURING SELF-TEST, CONTROL PANEL DIGITAL DISPLAY INDICATES E03 INSTEAD OF GOOD**

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**SYMPTOM**

During self-test, control panel digital display indicates E03 instead of GOOD.

**MALFUNCTION**

During self-test, control panel digital display indicates E03 instead of GOOD.

**CORRECTIVE ACTION**

**1. Check continuity between:**

- **P361-X and P523-A**
- **P361-Y and P523-B**

a. If continuity is present, go to **2**.

b. If continuity is not present, go to **3**.

**2. Check for 2,000 ohms to 3,000 ohms between fuel flow transmitter connector pins A and B.**

a. If resistance is as specified, replace auxiliary fuel management control panel (WP 1448 00). Go to **4**.

b. If resistance is not as specified, replace fuel flow transmitter (WP 1438 00). Go to **4**.

**3. Check continuity between:**

- **P361-X and P346-23**
- **P361-Y and P346-24**

a. If continuity is present, repair/replace wiring between (WP 1747 00), then go to **4**:

- **J346-23 and P523-A**
- **J346-24 and P523-B**

b. If continuity is not present, replace fuel system main wiring harness (WP 1453 00). Go to **4**.

**4. Procedure completed.**

**CONTROL PANEL EMPTY INDICATOR IS ON WITH ADEQUATE FUEL IN FUEL TANK****SYMPTOM**

Control panel EMPTY indicator is on with adequate fuel in fuel tank.

**MALFUNCTION**

Control panel EMPTY indicator is on with adequate fuel in fuel tank.

**CORRECTIVE ACTION**

- 1. Check for infinite resistance between these pins on applicable fuel tank connector:**
  - 3 and 5
  - 3 and 4
  - a. If resistance is as specified, go to **2**.
  - b. If resistance is not as specified, remove/replace low level sensor assembly (WP 1476 00). Go to **3**.
  - c. If trouble remains, replace/send fuel tank to depot (WP 1424 00). Go to **3**.
- 2. Check continuity between the applicable points:**
  - J1-3 (Left Outboard) and P361-G
  - J1-4 (Left Outboard) and P361-F
  - J1-5 (Left Outboard) and P361-E
  - J1-3 (Left Inboard) and P361-D
  - J1-4 (Left Inboard) and P361-C
  - J1-5 (Left Inboard) and P361-B
  - J1-3 (Right Outboard) and P361-K
  - J1-4 (Right Outboard) and P361-L
  - J1-5 (Right Outboard) and P361-M
  - J1-3 (Right Inboard) and P361-N
  - J1-4 (Right Inboard) and P361-P
  - J1-5 (Right Inboard) and P361-R
  - a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to **3**.
  - b. If continuity is not present, troubleshoot for open wiring, replace harness as required (WP 1747 00). Go to **3**.
- 3. Procedure completed.**

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**WHEN CONTROL PANEL STATUS BUTTON IS PRESSED, AND THEN RELEASED, CONTROL PANEL INDICATORS AND/OR CAUTION/ADVISORY PANEL AUX FUEL CAPSULE IS NOT OFF**

**SYMPTOM**

When control panel STATUS button is pressed, and then released, control panel indicators and/or caution/advisory panel AUX FUEL capsule is not off.

**MALFUNCTION**

When control panel STATUS button is pressed, and then released, control panel indicators and/or caution/advisory panel AUX FUEL capsule is not off.

**CORRECTIVE ACTION**

**1. Are all control panel indicators off?**

- a. If all indicators are off, go to **2**.
- b. If all indicators are not off, replace auxiliary fuel management control panel (WP 1448 00). Go to **8**.

**2. Check helicopter configuration.**

- a. **EMEP** , go to **3**.
- b. **W/O EMEP** , go to **5**.

**3. EMEP Remove and check pin filtered adapter P117 (WP 0925 00).**

- a. If pin filtered adapter is good, go to **4**.
- b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **8**.

**4. EMEP Install pin filtered adapter (WP 0925 00). Go to 5.**

**5. Check for 28 vdc between P117-41 and ground.**

- a. If voltage is as specified, replace caution/advisory panel (WP 0785 00). Go to **8**.
- b. If voltage is not as specified, go to **6**.

**6. Check continuity between P117-41 and P360-X.**

- a. If continuity is present, replace auxiliary fuel management panel (WP 1448 00). Go to **8**.
- b. If continuity is not present, go to **7**.

**7. Check continuity between P346-10 and P360-X.**

- a. If continuity is present, repair/replace wiring between J346-10 and P117-41 (WP 1747 00). Go to **8**.
- b. If continuity is not present, replace fuel system main wiring harness (WP 1453 00). Go to **8**.

**8. Procedure completed.**

**WITH WEIGHT-ON-WHEELS CONTROL PANEL CAL INDICATION CANNOT BE CHANGED****SYMPTOM**

With weight-on-wheels control panel CAL indication cannot be changed.

**MALFUNCTION**

With weight-on-wheels control panel CAL indication cannot be changed.

**CORRECTIVE ACTION****1. Check continuity between P360-r and P360-n .**

- a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to **4.**
- b. If continuity is not present, go to **2.**

**2. Check continuity between:**

- **P360-r and P902-C**
- **P360-n and P902-D**

- a. If continuity is present, replace left relay panel (WP 0825 00). Go to **4.**
- b. If continuity is not present, go to **3.**

**3. Check continuity between:**

- **P346-2 and P360-n**
- **P346-3 and P360-r**

- a. If continuity is present, repair/replace wiring, as required, between (WP 1747 00), then go to **4:**
  - **J346-2 and P902-D**
  - **J346-3 and P902-C**
- b. If continuity is not present, replace fuel system main wiring harness (WP 1455 00). Go to **4.**

**4. Procedure completed.****DIGITAL DISPLAY INDICATION CHANGES WITH DRAG BEAM SWITCH IN IN-FLIGHT POSITION****SYMPTOM**

Digital display indication changes with drag beam switch in in-flight position.

**MALFUNCTION**

Digital display indication changes with drag beam switch in in-flight position.

**CORRECTIVE ACTION****1. Remove locally-made drag beam wedge from left drag beam switch. Go to 2.****2. Check continuity between P360-r and P360-n.**

- a. If continuity is present, go to **3.**

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**DIGITAL DISPLAY INDICATION CHANGES WITH DRAG BEAM SWITCH IN IN-FLIGHT POSITION -**  
Continued

- b. If continuity is not present, replace auxiliary fuel management control panel (WP 1448 00). Go to **8**.
- 3. Check continuity between J163-A and ground.**
  - a. If continuity is present, go to **4**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **8**.
- 4. Check continuity between J163-A and J163-C.**
  - a. If continuity is present, replace left drag beam switch (WP 0443 00). Go to **8**.
  - b. If continuity is not present, go to **5**.
- 5. Check continuity between J163-C and P242-L.**
  - a. If continuity is present, go to **6**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **8**.
- 6. Check for 28 vdc between P902-CC and ground.**
  - a. If voltage is as specified, replace left relay panel (WP 0825 00). Go to **8**.
  - b. If voltage is not as specified, go to **7**.
- 7. Check for 28 vdc between BACKUP HYD CONTR circuit breaker terminal 1 and ground.**
  - a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P902-CC (WP 1747 00). Go to **8**.
  - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **8**.
- 8. Procedure completed.**

**CDU TOTAL FUEL INDICATION INCREASES WHEN FUEL TRANSFER IS NOT ACTUATED****SYMPTOM**

CDU TOTAL FUEL indication increases when fuel transfer is not actuated.

**MALFUNCTION**

CDU TOTAL FUEL indication increases when fuel transfer is not actuated.

**CORRECTIVE ACTION**

- 1. Is the manual lever of all the four shutoff gate valves in the full CLSD position?**
  - a. If manual levers are in the full CLSD position, replace auxiliary fuel management control panel (WP 1448 00). Go to **9**.
  - b. If manual levers are not in the full CLSD position, go to **2**.
- 2. For any valve that manual lever is not in full CLSD position, check for 28 vdc between applicable connector:**
  - **P840-D and P840-E Left Outboard**
  - **P809-D and P809-E Left Inboard**
  - **P838-D and P838-E Right Outboard**

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CDU TOTAL FUEL INDICATION INCREASES WHEN FUEL TRANSFER IS NOT ACTUATED - Continued

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- **P810-D and P810-E Right Inboard**
  - a. If voltage is as specified, go to **3**.
  - b. If voltage is not as specified, go to **4**.
- 3. Check for 0 vdc between applicable connector:**
  - **P840-A and P840-E Left Outboard**
  - **P809-A and P809-E Left Inboard**
  - **P838-A and P838-E Right Outboard**
  - **P810-A and P810-E Right Inboard**
  - a. If voltage is as specified, replace shutoff gate valve (WP 1440 00). Go to **9**.
  - b. If voltage is not as specified, replace auxiliary fuel management control panel (WP 1448 00). Go to **9**.
- 4. Check for which valve voltage was not as specified.**
  - a. If left outboard valve, go to **5**.
  - b. If left inboard valve, go to **6**.
  - c. If right inboard valve, go to **7**.
  - d. If right outboard valve, go to **8**.
- 5. For left outboard valve, check continuity between:**
  - **P840-A and P360-F**
  - **P840-D and P360-E**
  - **P840-E and ground**
  - a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to **9**.
  - b. If continuity is not present, troubleshoot for open wiring, replace harness as required (WP 1747 00). Go to **9**.
- 6. For left inboard valve, check continuity between:**
  - **P809-A and P360-Y**
  - **P809-D and P360-Z**
  - **P809-E and ground**
  - a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to **9**.
  - b. If continuity is not present, troubleshoot for open wiring, replace harness as required (WP 1747 00). Go to **9**.
- 7. For right inboard valve, check continuity between:**
  - **P810-A and P360-j**
  - **P810-D and P360-i**



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CDU TOTAL FUEL INDICATION INCREASES WHEN FUEL TRANSFER IS NOT ACTUATED - Continued

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- **P810-E and ground**

- a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to **9**.
- b. If continuity is not present, troubleshoot for open wiring, replace harness as required (WP 1747 00). Go to **9**.

**8. For right outboard valve, check continuity between:**

- **P838-A and P360-g**
- **P838-D and P360-P**
- **P838-E and ground**

- a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to **9**.
- b. If continuity is not present, troubleshoot for open wiring, replace harness as required (WP 1747 00). Go to **9**.

**9. Procedure completed.**

**WITH MANUAL MODE SELECTED, CDU TOTAL FUEL INDICATION DOES NOT INCREASE AND/OR RESULT IS NOT AS SPECIFIED WHEN ATTEMPTING TO TRANSFER FUEL FROM RIGHT OUTBOARD FUEL TANK**

**SYMPTOM**

With MANUAL mode selected, CDU TOTAL FUEL indication does not increase and/or Indication/Condition is not as specified when attempting to transfer fuel from right outboard fuel tank.

**MALFUNCTION**

With MANUAL mode selected, CDU TOTAL FUEL indication does not increase and/or Indication/Condition is not as specified when attempting to transfer fuel from right outboard fuel tank.

**CORRECTIVE ACTION**

**1. Does CDU TOTAL FUEL indication increase?**

- a. If CDU TOTAL FUEL indication does increase, go to **2**.
- b. If CDU TOTAL FUEL indication does not increase, go to **7**.

**2. Does control panel digital display indication decrease?**

- a. If control panel digital display indication does decrease, go to **3**.
- b. If control panel digital display indication does not decrease, go to **6**.

**3. Does control panel EXTERNAL RIGHT NO FLOW and caution/advisory panel AUX FUEL capsule momentarily go on, then off?**

- a. If capsules momentarily go on, then off, go to **4**.
- b. If capsules do not momentarily go on, then off, go to **5**.

**4. Do the remaining control panel EXTERNAL indicators remain off?**

- a. If remaining indicators remain off, go to **FUEL TRANSFER TIME AND/OR GAGING ACCURACY IS NOT AS SPECIFIED, in this work package**.

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**WITH MANUAL MODE SELECTED, CDU TOTAL FUEL INDICATION DOES NOT INCREASE AND/OR RESULT IS NOT AS SPECIFIED WHEN ATTEMPTING TO TRANSFER FUEL FROM RIGHT OUTBOARD FUEL TANK - Continued**

- b. If remaining indicators do not remain off, replace auxiliary fuel management control panel (WP 1448 00). Go to **21**.
- 5. Does control panel RIGHT NO FLOW indicator remain on?**
  - a. If indicator remains on, go to **CONTROL PANEL RIGHT NO FLOW INDICATOR REMAINS ON DURING FUEL TRANSFER, in this work package**.
  - b. If indicator does not remain on, replace auxiliary fuel management control panel (WP 1448 00). Go to **21**.
- 6. With control panel MANUAL XFR RIGHT switch at ON, check for more than 20 mvac between fuel flow transmitter connector pins A and B.**
  - a. If voltage is not as specified, replace auxiliary fuel management control panel (WP 1448 00). Go to **21**.
  - b. If voltage is not as specified, replace fuel flow transmitter (WP 1438 00). Go to **21**.
- 7. Does control panel digital display indication decrease?**
  - a. If indication decreases, go to **8**.
  - b. If indication does not decrease, go to **9**.
- 8. Does RIGHT NO FLOW indicator go on?**
  - a. If indicator is on, replace auxiliary fuel management control panel (WP 1448 00). Go to **21**.
  - b. If indicator is not on, troubleshoot fuel quantity system (WP 0131 00). Go to **21**.
- 9. Is manual lever of right outboard fuel shutoff gate valve in full OPEN position?**
  - a. If manual lever is in full OPEN position, go to **10**.
  - b. If manual lever is not in full OPEN position, go to **12**.
- 10. Check for 28 vdc between P837-A and P837-B.**
  - a. If voltage is as specified, replace right outboard bleed-air regulator valve (WP 1439 00). Go to **21**.
  - b. If voltage is not as specified, go to **11**.
- 11. Check continuity between:**
  - **P837-A and P360-R**
  - **P837-B and ground**
  - a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to **21**.
  - b. If continuity is not present, troubleshoot for open wiring, replace harness as required (WP 1747 00). Go to **21**.
- 12. Is right outboard bleed air regulator valve continuously venting air?**
  - a. If valve is continuously venting air, go to **13**.
  - b. If valve is not continuously venting air, go to **18**.
- 13. Check for 28 vdc between P360-V and ground.**
  - a. If voltage is as specified, go to **14**.

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**WITH MANUAL MODE SELECTED, CDU TOTAL FUEL INDICATION DOES NOT INCREASE AND/OR RESULT IS NOT AS SPECIFIED WHEN ATTEMPTING TO TRANSFER FUEL FROM RIGHT OUTBOARD FUEL TANK - Continued**

- b. If voltage is not as specified, go to **16**.

**14. Check for 28 vdc between P837-A and P837-B.**

- a. If voltage is as specified, replace right outboard bleed-air regulator valve (WP 1439 00). Go to **21**.
- b. If voltage is not as specified, go to **15**.

**15. Check continuity between:**

- **P837-A and P360-R**
- **P837-B and ground**

- a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to **21**.
- b. If continuity is not present, troubleshoot for open wiring, replace harness as required (WP 1747 00). Go to **21**.

**16. Check for 28 vdc between EXT FUEL RH circuit breaker terminal 1 and ground.**

- a. If voltage is as specified, go to **17**.
- b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **21**.

**17. Check for 28 vdc between J346-33 and ground.**

- a. If voltage is as specified, replace fuel system main wiring harness (WP 1455 00). Go to **21**.
- b. If voltage is not as specified, repair/replace wiring between J346-33 and EXT FUEL RH circuit breaker terminal 1 (WP 1747 00). Go to **21**.

**18. Check for 28 vdc between P838-A and P838-E.**

- a. If voltage is as specified, go to **19**.
- b. If voltage is not as specified, go to **20**.

**19. Check for 0 vdc between P838-D and P838-E.**

- a. If voltage is as specified, replace right outboard fuel shutoff gate valve (WP 1440 00). Go to **21**.
- b. If voltage is not as specified, replace auxiliary fuel management control panel (WP 1448 00). Go to **21**.

**20. Check continuity between these points:**

- **P838-A and P360-g**
- **P838-D and P360-P**
- **P838-E and ground**

- a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to **21**.
- b. If continuity is not present, troubleshoot for open wiring, replace harness as required (WP 1747 00). Go to **21**.

**21. Procedure completed.**

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**WITH MANUAL MODE SELECTED, CDU TOTAL FUEL INDICATION DOES NOT INCREASE AND/OR RESULT IS NOT AS SPECIFIED WHEN ATTEMPTING TO TRANSFER FUEL FROM LEFT OUTBOARD FUEL TANK**

**SYMPTOM**

With MANUAL mode selected, CDU TOTAL FUEL indication does not increase and/or Indication/Condition is not as specified when attempting to transfer fuel from left outboard fuel tank.

**MALFUNCTION**

With MANUAL mode selected, CDU TOTAL FUEL indication does not increase and/or Indication/Condition is not as specified when attempting to transfer fuel from left outboard fuel tank.

**CORRECTIVE ACTION**

- 1. Does CDU TOTAL FUEL indication increase?**
  - a. If indication does increase, go to **2**.
  - b. If indication does not increase, go to **7**.
- 2. Does control panel digital display indication decrease?**
  - a. If indication does decrease, go to **3**.
  - b. If indication does not decrease, go to **6**.
- 3. Does control panel EXTERNAL LEFT NO FLOW and caution/advisory panel AUX FUEL capsule momentarily go on, then off?**
  - a. If capsules momentarily go on, then off, go to **4**.
  - b. If capsules do not momentarily go on, then off, go to **5**.
- 4. Do the remaining control panel EXTERNAL indicators remain off?**
  - a. If remaining EXTERNAL indicators remain off, go to **FUEL TRANSFER TIME AND/OR GAGING ACCURACY IS NOT AS SPECIFIED, in this work package**.
  - b. If remaining EXTERNAL indicators do not remain off, replace auxiliary fuel management control panel (WP 1448 00). Go to **21**.
- 5. Does control panel LEFT NO FLOW indicator remain on?**
  - a. If indicator remains on, go to **CONTROL PANEL RIGHT NO FLOW INDICATOR REMAINS ON DURING FUEL TRANSFER, in this work package**.
  - b. If indicator does not remain on, replace auxiliary fuel management control panel (WP 1448 00). Go to **21**.
- 6. With control panel MANUAL XFR LEFT switch at ON, check for more than 20 mvac between fuel flow transmitter connector pins A and B.**
  - a. If voltage is as specified, replace auxiliary fuel management control panel (WP 1448 00). Go to **21**.
  - b. If voltage is not as specified, replace fuel flow transmitter (WP 1438 00). Go to **21**.
- 7. Does control panel digital display indication decrease?**
  - a. If indication decreases, go to **8**.
  - b. If indication does not decrease, go to **9**.

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**WITH MANUAL MODE SELECTED, CDU TOTAL FUEL INDICATION DOES NOT INCREASE AND/OR RESULT IS NOT AS SPECIFIED WHEN ATTEMPTING TO TRANSFER FUEL FROM LEFT OUTBOARD FUEL TANK - Continued**

**8. Does LEFT NO FLOW indicator go on?**

- a. If indicator is on, replace auxiliary fuel management control panel (WP 1448 00). Go to **21**.
- b. If indicator is not on, troubleshoot fuel quantity system (WP 0131 00). Go to **21**.

**9. Is manual lever of left outboard fuel shutoff gate valve in full OPEN position?**

- a. If manual lever is in full OPEN position, go to **10**.
- b. If manual lever is not in full OPEN position, go to **12**.

**10. Check for 28 vdc between P839-A and P839-B.**

- a. If voltage is as specified, replace left outboard bleed-air regulator valve (WP 1439 00). Go to **21**.
- b. If voltage is not as specified, go to **11**.

**11. Check continuity between:**

- **P839-A and P360-G**
- **P839-B and ground**

- a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to **21**.
- b. If continuity is not present, troubleshoot for open wiring, replace harness as required (WP 1747 00). Go to **21**.

**12. Is left outboard bleed air regulator valve continuously venting air?**

- a. If valve is continuously venting air, go to **13**.
- b. If valve is not continuously venting air, go to **18**.

**13. Check for 28 vdc between P360-C and ground.**

- a. If voltage is as specified, go to **14**.
- b. If voltage is not as specified, go to **16**.

**14. Check for 28 vdc between P839-A and P839-B.**

- a. If voltage is as specified, replace left outboard bleed-air regulator valve (WP 1439 00). Go to **21**.
- b. If voltage is not as specified, go to **15**.

**15. Check continuity between:**

- **P839-A and P360-G**
- **P839-B and ground**

- a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to **21**.
- b. If continuity is not present, troubleshoot for open wiring, replace harness as required (WP 1747 00). Go to **21**.

**16. Check for 28 vdc between EXT FUEL LH circuit breaker terminal 1 and ground.**

- a. If voltage is as specified, go to **17**.

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**WITH MANUAL MODE SELECTED, CDU TOTAL FUEL INDICATION DOES NOT INCREASE AND/OR RESULT IS NOT AS SPECIFIED WHEN ATTEMPTING TO TRANSFER FUEL FROM LEFT OUTBOARD FUEL TANK** - Continued

b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **21**.

**17. Check for 28 vdc between J346-45 and ground.**

a. If voltage is as specified, replace fuel system main wiring harness (WP 1455 00). Go to **21**.

b. If voltage is not as specified, repair/replace wiring between J346-45 and circuit breaker terminal 1 (WP 1747 00). Go to **21**.

**18. Check for 28 vdc between P840-A and P840-E.**

a. If voltage is as specified, go to **19**.

b. If voltage is not as specified, go to **20**.

**19. Check for 0 vdc between P840-D and P840-E.**

a. If voltage is as specified, replace left outboard fuel shutoff gate valve (WP 1440 00). Go to **21**.

b. If voltage is not as specified, replace auxiliary fuel management control panel (WP 1448 00). Go to **21**.

**20. Check continuity between:**

- **P840-A and P360-F**

- **P840-D and P360-E**

- **P840-E and ground**

a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to **21**.

b. If continuity is not present, troubleshoot for open wiring, replace harness as required (WP 1747 00). Go to **21**.

**21. Procedure completed.**

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**WITH MANUAL MODE SELECTED, CDU TOTAL FUEL INDICATION DOES NOT INCREASE AND/OR RESULT IS NOT AS SPECIFIED WHEN ATTEMPTING TO TRANSFER FUEL FROM RIGHT INBOARD FUEL TANK**

**SYMPTOM**

With MANUAL mode selected, CDU TOTAL FUEL indication does not increase and/or Indication/Condition is not as specified when attempting to transfer fuel from right inboard fuel tank.

**MALFUNCTION**

With MANUAL mode selected, CDU TOTAL FUEL indication does not increase and/or Indication/Condition is not as specified when attempting to transfer fuel from right inboard fuel tank.

**CORRECTIVE ACTION**

**1. Does CDU TOTAL FUEL indication increase?**

a. If indication does increase, go to **2**.

b. If indication does not increase, go to **7**.

**2. Does control panel digital display indication decrease?**

a. If indication does decrease, go to **3**.

---

**WITH MANUAL MODE SELECTED, CDU TOTAL FUEL INDICATION DOES NOT INCREASE AND/OR RESULT IS NOT AS SPECIFIED WHEN ATTEMPTING TO TRANSFER FUEL FROM RIGHT INBOARD FUEL TANK** - Continued

- b. If indication does not decrease, go to **6**.
- 3. Does control panel EXTERNAL RIGHT NO FLOW and caution/advisory panel AUX FUEL capsule momentarily go on, then off?**
  - a. If capsules momentarily go on, then off, go to **4**.
  - b. If capsules does not momentarily go on, then off, go to **5**.
- 4. Do the remaining control panel EXTERNAL indicators remain off?**
  - a. If remaining EXTERNAL indicators remain off, go to **FUEL TRANSFER TIME AND/OR GAGING ACCURACY IS NOT AS SPECIFIED, in this work package**.
  - b. If remaining EXTERNAL indicators do not remain off, replace auxiliary fuel management control panel (WP 1448 00). Go to **18**.
- 5. Does control panel RIGHT NO FLOW indicator remain on?**
  - a. If indicator remains on, go to **CONTROL PANEL RIGHT NO FLOW INDICATOR REMAINS ON DURING FUEL TRANSFER, in this work package**.
  - b. If indicator does not remain on, replace auxiliary fuel management control panel (WP 1448 00). Go to **18**.
- 6. With control panel MANUAL XFR RIGHT switch at ON, check for more than 20 mvac between P523-A and P523-B of fuel flow transmitter.**
  - a. If voltage is as specified, replace auxiliary fuel management control panel (WP 1448 00). Go to **18**.
  - b. If voltage is not as specified, replace fuel flow transmitter (WP 1438 00). Go to **18**.
- 7. Does control panel digital display indication decrease?**
  - a. If indication decreases, go to **8**.
  - b. If indication does not decrease, go to **9**.
- 8. Does RIGHT NO FLOW indicator go on?**
  - a. If indicator is on, replace auxiliary fuel management control panel (WP 1448 00). Go to **18**.
  - b. If indicator is not on, troubleshoot fuel quantity system (WP 0131 00). Go to **18**.
- 9. Is manual lever of right inboard fuel shutoff gate valve in full OPEN position?**
  - a. If manual lever is in full OPEN position, go to **10**.
  - b. If manual lever is not in full OPEN position, go to **12**.
- 10. Check for 28 vdc between P811-A and P811-B.**
  - a. If voltage is as specified, replace right inboard bleed-air regulator valve (WP 1439 00). Go to **18**.
  - b. If voltage is not as specified, go to **11**.
- 11. Check continuity between:**
  - **P811-A and P360-s**
  - **P811-B and ground**

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**WITH MANUAL MODE SELECTED, CDU TOTAL FUEL INDICATION DOES NOT INCREASE AND/OR RESULT IS NOT AS SPECIFIED WHEN ATTEMPTING TO TRANSFER FUEL FROM RIGHT INBOARD FUEL TANK** - Continued

- a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to **18**.
  - b. If continuity is not present, troubleshoot for open wiring, replace harness as required (WP 1747 00). Go to **18**.
- 12. Is right inboard bleed air regulator valve continuously venting air?**
- a. If valve is continuously venting air, go to **13**.
  - b. If valve is not continuously venting air, go to **15**.
- 13. Check for 28 vdc between P811-A and P811-B.**
- a. If voltage is as specified, replace right inboard bleed-air regulator valve (WP 1439 00). Go to **18**.
  - b. If voltage is not as specified, go to **14**.
- 14. Check continuity between:**
- **P811-A and P360-s**
  - **P811-B and ground**
- a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to **18**.
  - b. If continuity is not present, troubleshoot for open wiring, replace harness as required (WP 1747 00). Go to **18**.
- 15. Check for 28 vdc between P810-A and P810-E.**
- a. If voltage is as specified, go to **16**.
  - b. If voltage is not as specified, go to **17**.
- 16. Check for 0 vdc between P810-D and P810-E.**
- a. If voltage is as specified, replace left inboard fuel shutoff gate valve (WP 1440 00). Go to **18**.
  - b. If voltage is not as specified, replace auxiliary fuel management control panel (WP 1448 00). Go to **18**.
- 17. Check continuity between:**
- **P810-A and P360-j**
  - **P810-D and P360-i**
  - **P810-E and ground**
- a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to **18**.
  - b. If continuity is not present, troubleshoot for open wiring, replace harness as required (WP 1747 00). Go to **18**.
- 18. Procedure completed.**



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**WITH MANUAL MODE SELECTED, CDU TOTAL FUEL INDICATION DOES NOT INCREASE AND/OR RESULT IS NOT AS SPECIFIED WHEN ATTEMPTING TO TRANSFER FUEL FROM LEFT INBOARD FUEL TANK**

**SYMPTOM**

With MANUAL mode selected, CDU TOTAL FUEL indication does not increase and/or Indication/Condition is not as specified when attempting to transfer fuel from left inboard fuel tank.

**MALFUNCTION**

With MANUAL mode selected, CDU TOTAL FUEL indication does not increase and/or Indication/Condition is not as specified when attempting to transfer fuel from left inboard fuel tank.

**CORRECTIVE ACTION**

**1. Does CDU TOTAL FUEL indication increase?**

- a. If indication increases, go to **2**.
- b. If indication does not increase, go to **7**.

**2. Does control panel digital display indication decrease?**

- a. If indication decreases, go to **3**.
- b. If indication does not decrease, go to **6**.

**3. Does control panel EXTERNAL LEFT NO FLOW and caution/advisory panel AUX FUEL capsule momentarily go on, then off?**

- a. If capsules momentarily go on, then off, go to **4**.
- b. If capsules does not momentarily go on, then off, go to **5**.

**4. Do the remaining control panel EXTERNAL indicators remain off?**

- a. If remaining EXTERNAL indicators remain off, go to **FUEL TRANSFER TIME AND/OR GAGING ACCURACY IS NOT AS SPECIFIED, in this work package**.
- b. If remaining EXTERNAL indicators do not remain off, replace auxiliary fuel management control panel (WP 1448 00). Go to **18**.

**5. Does control panel LEFT NO FLOW indicator remain on?**

- a. If indicator remains on, go to **CONTROL PANEL LEFT NO FLOW INDICATOR REMAINS ON DURING FUEL TRANSFER, in this work package**.
- b. If indicator does not remain on, replace auxiliary fuel management control panel (WP 1448 00). Go to **18**.

**6. With control panel MANUAL XFR LEFT switch at ON, check for more than 20 mvac between fuel flow transmitter connector pins A and B.**

- a. If voltage is as specified, replace auxiliary fuel management control panel (WP 1448 00). Go to **18**.
- b. If voltage is not as specified, replace fuel flow transmitter (WP 1438 00). Go to **18**.

**7. Does control panel digital display indication decrease?**

- a. If indication decreases, go to **8**.
- b. If indication does not decrease, go to **9**.

---

**WITH MANUAL MODE SELECTED, CDU TOTAL FUEL INDICATION DOES NOT INCREASE AND/OR RESULT IS NOT AS SPECIFIED WHEN ATTEMPTING TO TRANSFER FUEL FROM LEFT INBOARD FUEL TANK** - Continued

**8. Does LEFT NO FLOW indicator go on?**

- a. If indicator is on, replace auxiliary fuel management control panel (WP 1448 00). Go to **18**.
- b. If indicator is not on, troubleshoot fuel quantity system (WP 0131 00). Go to **18**.

**9. Is manual lever of left inboard fuel shutoff gate valve in full OPEN position?**

- a. If manual lever is in full OPEN position, go to **10**.
- b. If manual lever is not in full OPEN position, go to **12**.

**10. Check for 28 vdc between P808-A and P808-B.**

- a. If voltage is as specified, replace left inboard bleed-air regulator valve (WP 1439 00). Go to **18**.
- b. If voltage is not as specified, go to **11**.

**11. Check continuity between:**

- **P808-A and P360-m**
- **P808-B and ground**

- a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to **18**.
- b. If continuity is not present, troubleshoot for open wiring, replace harness as required (WP 1747 00). Go to **18**.

**12. Is left inboard bleed air regulator valve continuously venting air?**

- a. If valve is continuously venting air, go to **13**.
- b. If valve is not continuously venting air, go to **15**.

**13. Check for 28 vdc between P808-A and P808-B.**

- a. If voltage is as specified, replace left inboard bleed-air regulator valve (WP 1439 00). Go to **18**.
- b. If voltage is not as specified, go to **14**.

**14. Check continuity between:**

- **P808-A and P360-m**
- **P808-B and ground**

- a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to **18**.
- b. If continuity is not present, troubleshoot for open wiring, replace harness as required (WP 1747 00). Go to **18**.

**15. Check for 28 vdc between P809-A and P809-E.**

- a. If voltage is as specified, go to **16**.
- b. If voltage is not as specified, go to **17**.

**16. Check for 0 vdc between P809-D and P809-E.**

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**WITH MANUAL MODE SELECTED, CDU TOTAL FUEL INDICATION DOES NOT INCREASE AND/OR RESULT IS NOT AS SPECIFIED WHEN ATTEMPTING TO TRANSFER FUEL FROM LEFT INBOARD FUEL TANK** - Continued

- a. If voltage is as specified, replace left inboard fuel shutoff gate valve (WP 1440 00). Go to **18.**
- b. If voltage is not as specified, replace auxiliary fuel management control panel (WP 1448 00). Go to **18.**

**17. Check continuity between:**

- **P809-A and P360-Y**
- **P809-D and P360-Z**
- **P809-E and ground**

- a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to **18.**
- b. If continuity is not present, troubleshoot for open wiring, replace harness as required (WP 1747 00). Go to **18.**

**18. Procedure completed.****FUEL TRANSFER TIME AND/OR GAGING ACCURACY IS NOT AS SPECIFIED****SYMPTOM**

Fuel transfer time and/or gaging accuracy is not as specified.

**MALFUNCTION**

Fuel transfer time and/or gaging accuracy is not as specified.

**CORRECTIVE ACTION**

- 1. With control panel MANUAL XFR LEFT and/or RIGHT switch, as applicable, at ON, inspect fuel and pneumatic hoses on fittings for signs of leakage or damage.**
  - a. If leakage or damage is present, replace damaged or leaking hoses and/or fittings as necessary. Go to **6.**
  - b. If leakage or damage is not present, go to **2.**
- 2. Pull and turn control panel AUX FUEL QTY switch to CAL. Go to 3.**
- 3. Does control panel digital display indication equal K FACTOR value on fuel flow transmitter?**
  - a. If indication is equal to K FACTOR, go to **4.**
  - b. If indication is not equal to K FACTOR, replace auxiliary fuel management control panel (WP 1448 00). Go to **6.**
- 4. With control panel MANUAL XFR LEFT and/or RIGHT switch, as applicable, at ON, check for more than 20 mvac between fuel flow transmitter connector pins A and B.**
  - a. If voltage is as specified, replace applicable bleed-air regulator valve (WP 1439 00). Go to **6.**
  - b. If trouble remains, go to **5.**
  - c. If voltage is not as specified, replace fuel flow transmitter (WP 1438 00). Go to **6.**
- 5. Do operational/troubleshooting procedure of fuel quantity system (WP 0131 00).**

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**FUEL TRANSFER TIME AND/OR GAGING ACCURACY IS NOT AS SPECIFIED - Continued**

- a. If fuel quantity system is good, replace auxiliary fuel management control panel (WP 1448 00). Go to **6**.
- b. If fuel quantity system is not good, adjust/troubleshoot system as required (WP 0131 00). Go to **6**.

**6. Procedure completed.****AUTO MODE FUEL TRANSFER LIMITS ARE NOT AS SPECIFIED****SYMPTOM**

Auto mode fuel transfer limits are not as specified.

**MALFUNCTION**

Auto mode fuel transfer limits are not as specified.

**CORRECTIVE ACTION****1. Did digital display indication increase or decrease?**

- a. If indication changes, replace auxiliary fuel management control panel (WP 1448 00). Go to **3**.
- b. If indication does not change, go to **2**.

**2. Check continuity between:**

- P905-K (Tank 1) and J346-30
- P905-L (Tank 1) and J346-31
- P905-V (Tank 2) and J346-27
- P905-U (Tank 2) and J346-28

- a. If continuity is present, replace fuel system main wiring harness (WP 1455 00). Go to **3**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **3**.

**3. Procedure completed.****CONTROL PANEL RIGHT NO FLOW INDICATOR REMAINS ON DURING FUEL TRANSFER****SYMPTOM**

Control panel RIGHT NO FLOW indicator remains on during fuel transfer.

**MALFUNCTION**

Control panel RIGHT NO FLOW indicator remains on during fuel transfer.

**CORRECTIVE ACTION**

- 1. Disconnect J520 from right external fuel flow sensor. Go to 2.**
- 2. Connect insulated jumper wire between J520-1 and J520-2. Go to 3.**
- 3. Place control panel MANUAL XFR RIGHT switch to ON. Go to 4.**
- 4. Does RIGHT NO FLOW indicator go on momentarily?**

- a. If indicator goes on momentarily, replace right external fuel flow sensor (WP 1437 00). Go to **8**.

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**CONTROL PANEL RIGHT NO FLOW INDICATOR REMAINS ON DURING FUEL TRANSFER - Continued**

- b. If indicator does not go on momentarily, go to **5**.
- 5. Place control panel MANUAL XFR RIGHT switch to OFF. Go to 6.**
- 6. Check continuity between:**
  - **J520-1 and P360-e**
  - **P520-2 and P360-d**
- a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to **8**.
- b. If continuity is not present, go to **7**.
- 7. Check continuity between:**
  - **J346-16 and J520-1**
  - **J346-17 and J520-2**
- a. If continuity is present, replace fuel system main wiring harness (WP 1455 00). Go to **8**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **8**.
- 8. Procedure completed.**

**CONTROL PANEL LEFT NO FLOW INDICATOR REMAINS ON DURING FUEL TRANSFER****SYMPTOM**

Control panel LEFT NO FLOW indicator remains on during fuel transfer.

**MALFUNCTION**

Control panel LEFT NO FLOW indicator remains on during fuel transfer.

**CORRECTIVE ACTION**

- 1. Disconnect J521 from left external fuel flow sensor. Go to 2.**
- 2. Connect insulated jumper wire between J521-1 and J521-2. Go to 3.**
- 3. Place control panel MANUAL XFR LEFT switch to ON. Go to 4.**
- 4. Does LEFT NO FLOW indicator go on momentarily?**
  - a. If indicator goes on momentarily, replace left external fuel flow sensor (WP 1437 00). Go to **8**.
  - b. If indicator does not go on momentarily, go to **5**.
- 5. Place control panel MANUAL XFR LEFT switch to OFF. Go to 6.**
- 6. Check continuity between:**
  - **J521-1 and P360-q**
  - **J521-2 and P360-d**
- a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to **8**.
- b. If continuity is not present, go to **7**.

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**CONTROL PANEL LEFT NO FLOW INDICATOR REMAINS ON DURING FUEL TRANSFER - Continued****7. Check continuity between:**

- J346-14 and J521-1
- J346-15 and J521-2

- a. If continuity is present, replace fuel system main wiring harness (WP 1455 00). Go to **8**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **8**.

**8. Procedure completed.****CONTROL PANEL EXTERNAL RIGHT INBD EMPTY AND/OR LEFT INBD EMPTY INDICATORS DO NOT GO ON WITH INADEQUATE FUEL IN TANK****SYMPTOM**

Control panel EXTERNAL RIGHT INBD EMPTY and/or LEFT INBD EMPTY indicators do not go on with inadequate fuel in tank.

**MALFUNCTION**

Control panel EXTERNAL RIGHT INBD EMPTY and/or LEFT INBD EMPTY indicators do not go on with inadequate fuel in tank.

**CORRECTIVE ACTION****1. Disconnect J1 from applicable inboard fuel tank. Go to 2.****2. Does applicable INBD EMPTY indicator go on?**

- a. If indicator is on, remove/replace low level sensor assembly (WP 0996 00). Go to **4**.
- b. If indicator is not on, go to **3**.

**3. Check continuity between:**

- J1-3 (Left) and P361-D
- J1-4 (Left) and P361-C
- J1-5 (Left) and P361-B
- J1-3 (Right) and P361-N
- J1-4 (Right) and P361-P
- J1-5 (Right) and P361-R

- a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to **4**.
- b. If continuity is not present, troubleshoot for open wiring, replace harness as required (WP 1453 00). Go to **4**.

**4. Procedure completed.**

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**CAUTION/ADVISORY PANEL AUX FUEL CAPSULE AND/OR CONTROL PANEL EXTERNAL RIGHT NO FLOW AND LEFT NO FLOW INDICATORS DO NOT GO ON****SYMPTOM**

Caution/advisory panel AUX FUEL capsule and/or control panel EXTERNAL RIGHT NO FLOW and LEFT NO FLOW indicators do not go on immediately after corresponding tank EMPTY indicator goes on.

**MALFUNCTION**

Caution/advisory panel AUX FUEL capsule and/or control panel EXTERNAL RIGHT NO FLOW and LEFT NO FLOW indicators do not go on immediately after corresponding tank EMPTY indicator goes on.

**CORRECTIVE ACTION**

- 1. Do control panel EXTERNAL RIGHT NO FLOW and LEFT NO FLOW indicators go on?**
  - a. If indicators are on, go to **2**.
  - b. If indicators are not on, go to **6**.
- 2. Momentarily place caution/advisory panel BRT/DIM-TEST switch to TEST. Go to 3.**
- 3. Does caution/advisory panel AUX FUEL capsule go on, then off?**
  - a. If caution/advisory panel AUX FUEL capsule goes on, then off, go to **4**.
  - b. If capsule does not go on, then off, troubleshoot caution/advisory warning system (WP 0090 00). Go to **10**.
- 4. Check for 28 vdc between P117-41 and ground.**
  - a. If voltage is as specified, replace caution/advisory panel (WP 0785 00). Go to **10**.
  - b. If voltage is not as specified, go to **5**.
- 5. Check continuity between P346-10 and P360-X.**
  - a. If continuity is present, repair/replace wiring between J346-10 and P117-41 (WP 1747 00). Go to **10**.
  - b. If continuity is not present, replace fuel system main wiring harness (WP 1453 00). Go to **10**.
- 6. Disconnect applicable connector, J521 (Left) and J520 (Right), from external fuel flow sensor. Go to 7.**
- 7. With control panel mode switch in AUTO, is correct corresponding NO FLOW indicator on?**
  - a. If indicator is on, replace external fuel flow sensor (WP 1437 00). Go to **10**.
  - b. If indicator is not on, go to **8**.
- 8. Check for infinite resistance between:**
  - **J360-q (Left) and J360-d**
  - **J360-e (Right) and J360-d**
  - a. If resistance is as specified, replace auxiliary fuel management control panel (WP 1448 00). Go to **10**.
  - b. If resistance is not as specified, go to **9**.

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**CAUTION/ADVISORY PANEL AUX FUEL CAPSULE AND/OR CONTROL PANEL EXTERNAL RIGHT NO FLOW AND LEFT NO FLOW INDICATORS DO NOT GO ON - Continued****9. Check for infinite resistance between:**

- **J346-14 (Left) and J346-15**
- **J346-16 (Right) and J346-17**
- a. If resistance is as specified, replace fuel system main wiring harness (WP 1453 00). Go to **10**.
- b. If resistance is not as specified, troubleshoot wiring for short circuit. Repair/replace wiring as required (WP 1747 00). Go to **10**.

**10. Procedure completed.****CONTROL PANEL EXTERNAL RIGHT OUTBD EMPTY AND/OR LEFT OUTBD EMPTY INDICATORS DO NOT GO ON WITH INADEQUATE FUEL IN TANK****SYMPTOM**

Control panel EXTERNAL RIGHT OUTBD EMPTY and/or LEFT OUTBD EMPTY indicators do not go on with inadequate fuel in tank.

**MALFUNCTION**

Control panel EXTERNAL RIGHT OUTBD EMPTY and/or LEFT OUTBD EMPTY indicators do not go on with inadequate fuel in tank.

**CORRECTIVE ACTION****1. Disconnect J1 from applicable outboard fuel tank. Go to 2.****2. Does applicable OUTBOARD EMPTY indicator go on?**

- a. If indicator is on, remove/replace low level sensor assembly (WP 0996 00). Go to **4**.
- b. If indicator is not on, go to **3**.

**3. Check continuity between:**

- **J1-3 (Left) and P361-G**
- **J1-4 (Left) and P361-F**
- **J1-5 (Left) and P361-E**
- **J1-3 (Right) and P361-K**
- **J1-4 (Right) and P361-L**
- **J1-5 (Right) and P361-M**

- a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to **4**.
- b. If continuity is not present, troubleshoot for open wiring. Replace harness as required (WP 1453 00). Go to **4**.

**4. Procedure completed.**



**VENT SENSOR FAIL INDICATOR DOES NOT GO ON****SYMPTOM**

VENT SENSOR FAIL indicator does not go on.

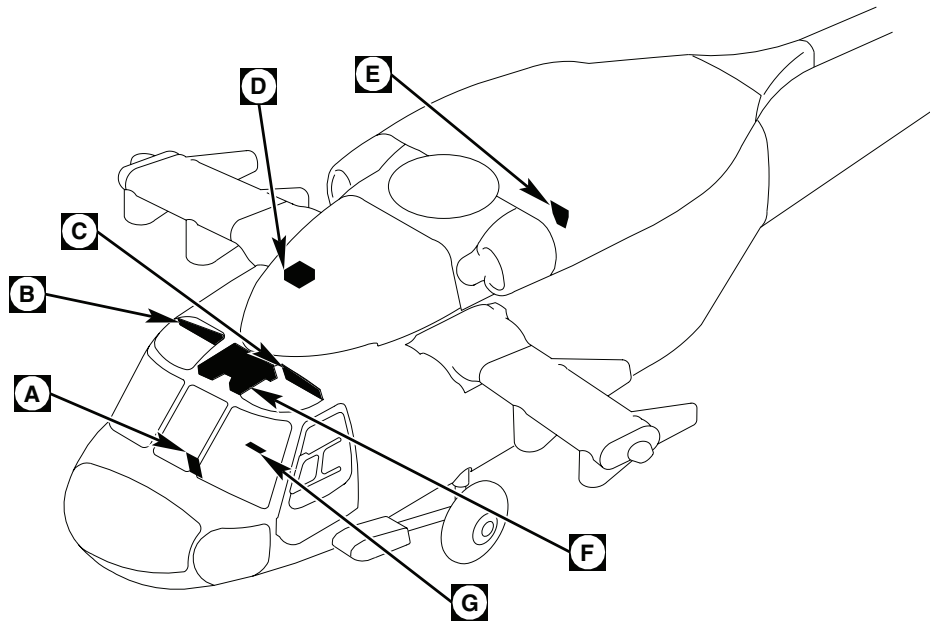
**MALFUNCTION**

VENT SENSOR FAIL indicator does not go on.

**CORRECTIVE ACTION**

- 1. Check that helicopter wiring harness J376 is connected to fuel overflow sensor P376.**
  - a. If J376 is connected to P376, go to **2**.
  - b. If J376 is not connected to P376, replace auxiliary fuel management control panel (WP 1448 00). Go to **4**.
- 2. Check overflow sensor resistance as follows:**
  - **P361-W and P361-Z 650-1500 ohms**
  - **P361-Z and P361-V 360-440 ohms**
  - a. If resistance is as specified, replace auxiliary fuel management control panel (WP 1448 00). Go to **4**.
  - b. If resistance is not as specified, go to **3**.
- 3. Check overflow sensor resistance as follows:**
  - **P376-C and P376-A 650-1500 ohms**
  - **P376-C and P376-B 360-440 ohms**
  - a. If resistance is as specified, repair/replace wiring (WP 1747 00). Go to **4**.
  - b. If resistance is not as specified, replace fuel overflow sensor (WP 1457 00). Go to **4**.
- 4. Procedure completed.**

## LOCATION



| TERMINAL BOARD /<br>DISCONNECT PLUG /<br>RECEPTACLE | LOCATION /<br>CONNECTION POINT        |
|-----------------------------------------------------|---------------------------------------|
| J299<br>(SEE NOTE 1)                                | CABIN CEILING<br>BL 10 LH, STA 302    |
| J520                                                | RIGHT EXT FUEL FLOW<br>SENSOR         |
| J521                                                | LEFT EXT FUEL FLOW<br>SENSOR          |
| J522                                                | INT AUX FUEL FLOW<br>SENSOR           |
| J524                                                | FUEL TEMPERATURE<br>TRANSDUCER        |
| P1 / J1                                             | RIGHT INBOARD FUEL<br>TANK CONNECTOR  |
| P1 / J1                                             | LEFT INBOARD FUEL TANK<br>CONNECTOR   |
| P1 / J1                                             | LEFT OUTBOARD FUEL<br>TANK CONNECTOR  |
| P1 / J1                                             | RIGHT OUTBOARD FUEL<br>TANK CONNECTOR |
| P2 / J835                                           | LEFT INBOARD FUEL TANK<br>CONNECTOR   |

| TERMINAL BOARD /<br>DISCONNECT PLUG /<br>RECEPTACLE | LOCATION /<br>CONNECTION POINT                                 |
|-----------------------------------------------------|----------------------------------------------------------------|
| P2 / J836                                           | RIGHT INBOARD FUEL TANK<br>CONNECTOR                           |
| P2 / J849                                           | LEFT OUTBOARD FUEL<br>TANK CONNECTOR                           |
| P2 / J850                                           | RIGHT OUTBOARD FUEL<br>TANK CONNECTOR                          |
| P113 / J113                                         | COCKPIT<br>BL 6 LH, STA 197                                    |
| P114 / J114                                         | COCKPIT<br>BL 8 RH, STA 200                                    |
| P117                                                | CAUTION / ADVISORY<br>PANEL                                    |
| P137                                                | NO. 2 SIGNAL DATA<br>CONVERTER                                 |
| P138                                                | NO. 1 SIGNAL DATA<br>CONVERTER                                 |
| P163 / J163                                         | LEFT DRAG BEAM SWITCH                                          |
| P217 / J217                                         | MAIN ROTOR PYLON DECK<br>BL 0, STA 284                         |
| P236 / J236<br>(SEE NOTE 2)                         | BEHIND PILOT'S CIRCUIT<br>BREAKER PANEL<br>BL 23 RH            |
| P237 / J237<br>(SEE NOTE 2)                         | BEHIND COPILOT'S<br>CIRCUIT BREAKER PANEL<br>BL 25 LH, STA 247 |
| P246 / J246                                         | CABIN CEILING<br>BL 5 RH, STA 247                              |
| P334 / J334                                         | CABIN CEILING<br>BL 16.5 LH, STA 390                           |

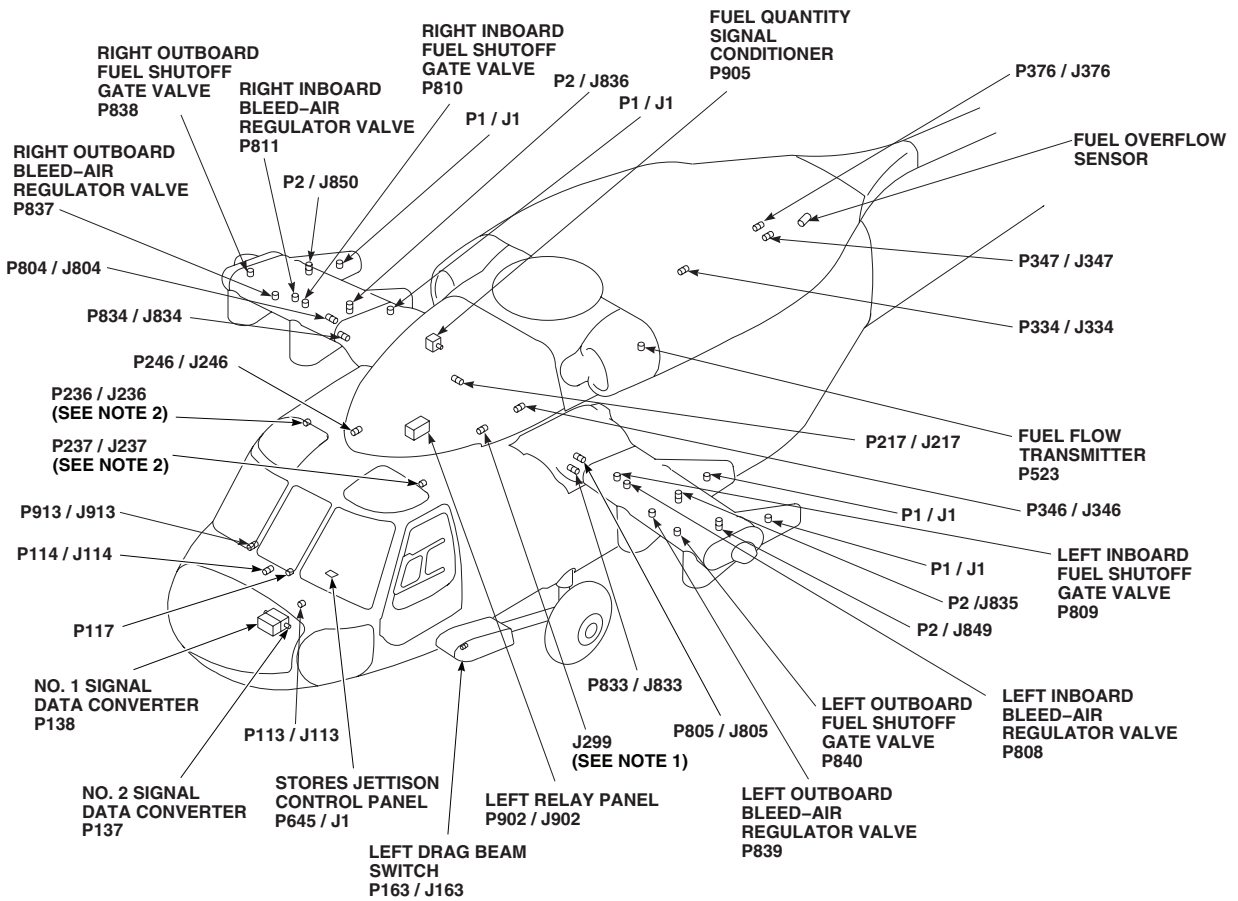
## NOTES

1. UH60A UH60L
2. EH60A

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Figure 1. ESSS Range Extension System Location Diagram. (Sheet 1 of 5)

## LOCATION - Continued



| TERMINAL BOARD/<br>DISCONNECT PLUG/<br>RECEPTACLE | LOCATION/<br>CONNECTION POINT                     |
|---------------------------------------------------|---------------------------------------------------|
| P346 / J346                                       | CABIN CEILING<br>BL 20 LH, STA 304                |
| P347 / J347                                       | TRANSITION SECTION<br>BL 10 LH, STA 425           |
| P360 / J360                                       | AUXILIARY FUEL MANAGE-<br>MENT CONTROL PANEL      |
| P361 / J361                                       | AUXILIARY FUEL MANAGE-<br>MENT CONTROL PANEL      |
| P376 / J376                                       | FUEL OVERFLOW SENSOR                              |
| P523                                              | FUEL FLOW TRANSMITTER                             |
| P645 / J1                                         | STORES JETTISON<br>CONTROL PANEL                  |
| P804 / J804                                       | HORIZONTAL STORES<br>SUPPORT<br>BL 59 RH, STA 295 |
| P805 / J805                                       | HORIZONTAL STORES<br>SUPPORT<br>BL 59 LH, STA 295 |
| P808                                              | LEFT INBOARD BLEED-AIR<br>REGULATOR VALVE         |

| TERMINAL BOARD/<br>DISCONNECT PLUG/<br>RECEPTACLE | LOCATION/<br>CONNECTION POINT               |
|---------------------------------------------------|---------------------------------------------|
| P809                                              | LEFT INBOARD FUEL SHUT-<br>OFF GATE VALVE   |
| P810                                              | RIGHT INBOARD FUEL<br>SHUTOFF GATE VALVE    |
| P811                                              | RIGHT INBOARD BLEED-AIR<br>REGULATOR VALVE  |
| P833 / J833                                       | HSS ATTACHMENT FITTING<br>BL 59 LH, STA 295 |
| P834 / J834                                       | HSS ATTACHMENT FITTING<br>BL 59 LH, STA 295 |
| P837                                              | RIGHT OUTBOARD BLEED-AIR<br>REGULATOR VALVE |
| P838                                              | RIGHT OUTBOARD FUEL<br>SHUTOFF GATE VALVE   |
| P839                                              | LEFT OUTBOARD BLEED-AIR<br>REGULATOR VALVE  |
| P840                                              | LEFT OUTBOARD FUEL<br>SHUTOFF GATE VALVE    |
| P902 / J902                                       | LEFT RELAY PANEL                            |
| P905                                              | FUEL QUANTITY SIGNAL<br>CONDITIONER         |
| P913 / J913                                       | COCKPIT<br>BL 24 RH, STA 247                |

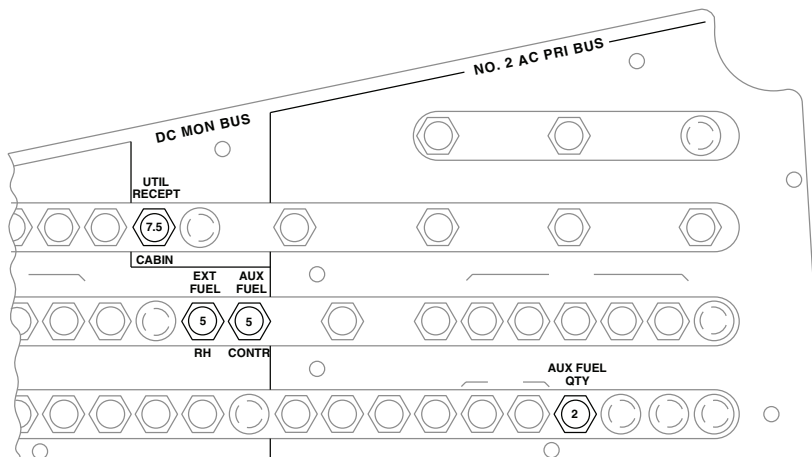
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Figure 1. ESSS Range Extension System Location Diagram. (Sheet 2 of 5)

LOCATION - Continued



CAUTION/ADVISORY PANEL



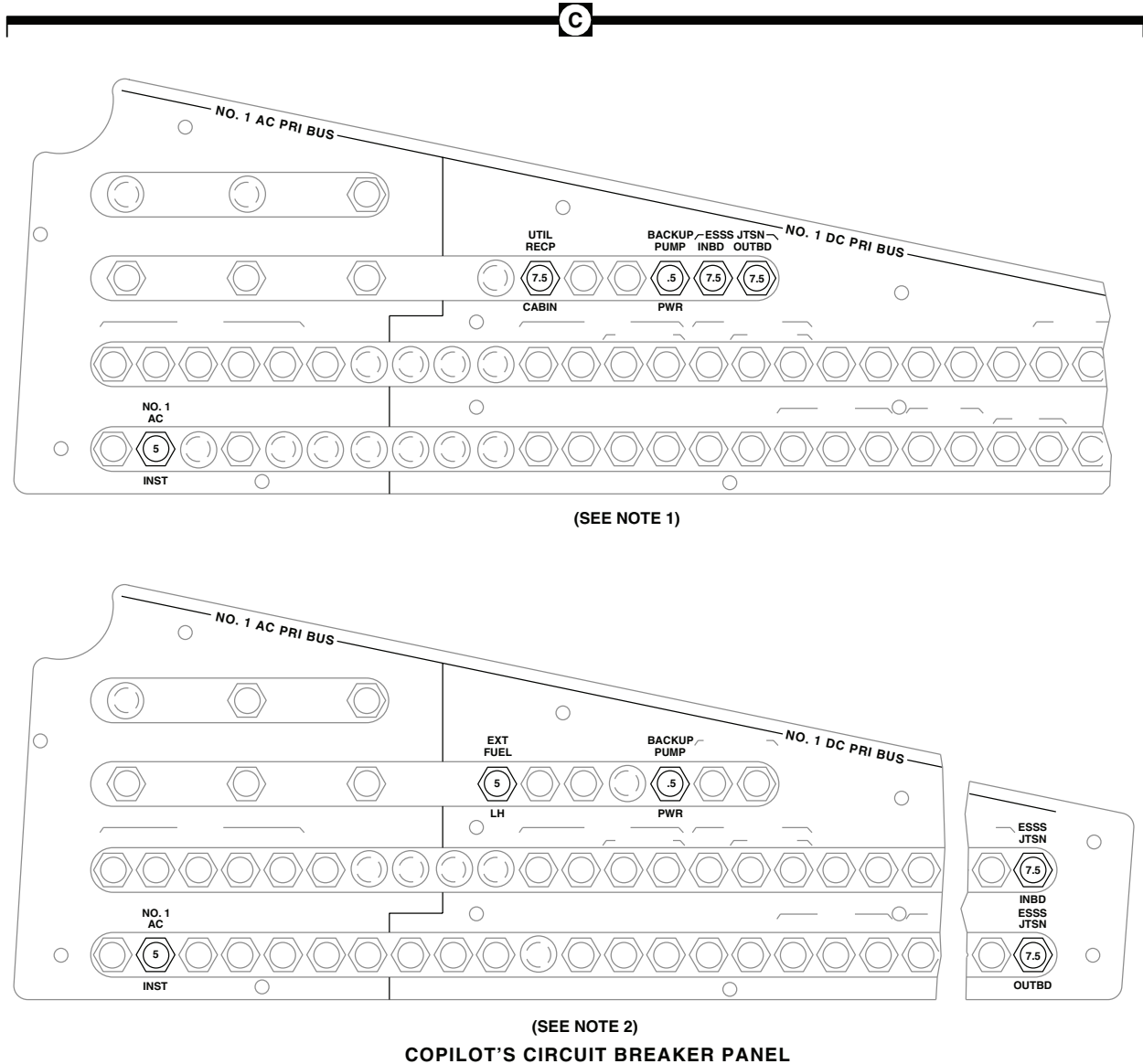
(SEE NOTE 2)

PILOT'S CIRCUIT BREAKER PANEL

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SA

Figure 1. ESSS Range Extension System Location Diagram. (Sheet 3 of 5)

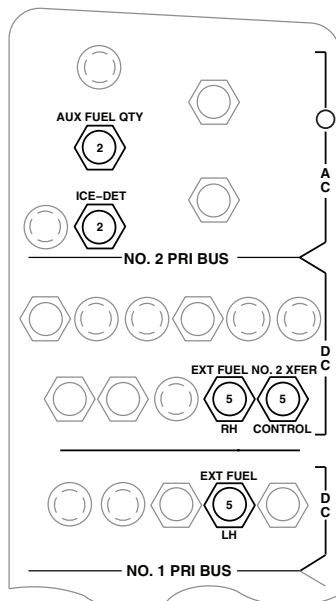
LOCATION - Continued



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Figure 1. ESSS Range Extension System Location Diagram. (Sheet 4 of 5)

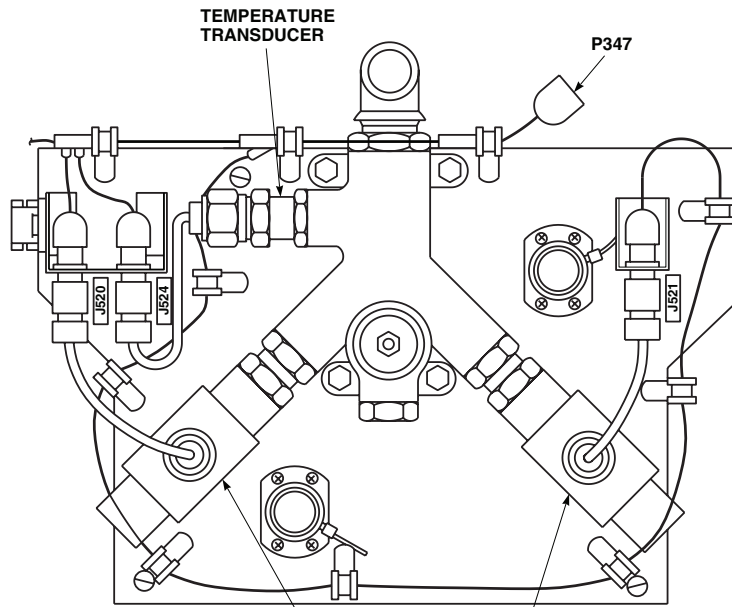
LOCATION - Continued



(SEE NOTE 1)

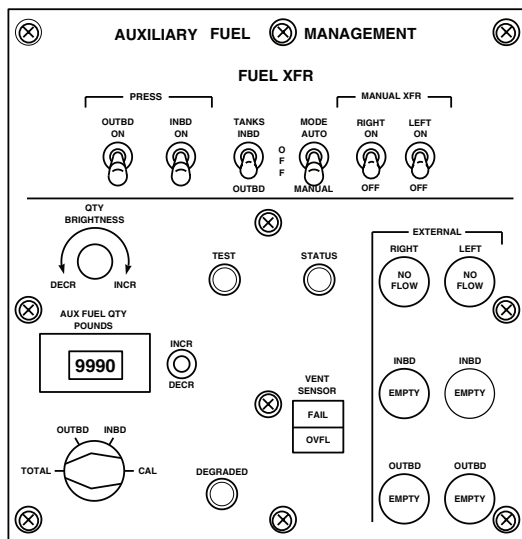
**MISSION READINESS  
CIRCUIT BREAKER PANEL**

**D**



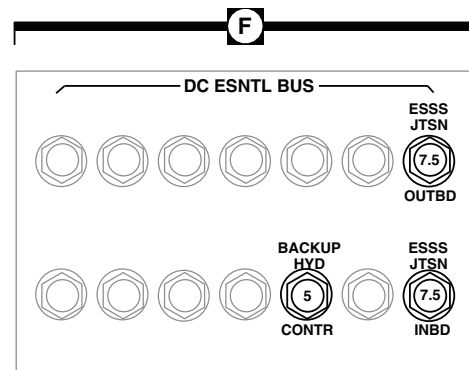
**FLOW SENSOR PANEL**

**E**



**AUXILIARY FUEL MANAGEMENT  
CONTROL PANEL**

**G**



**UPPER CONSOLE  
CIRCUIT BREAKER PANEL**

**F**

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Figure 1. ESSS Range Extension System Location Diagram. (Sheet 5 of 5)

## SCHEMATICS

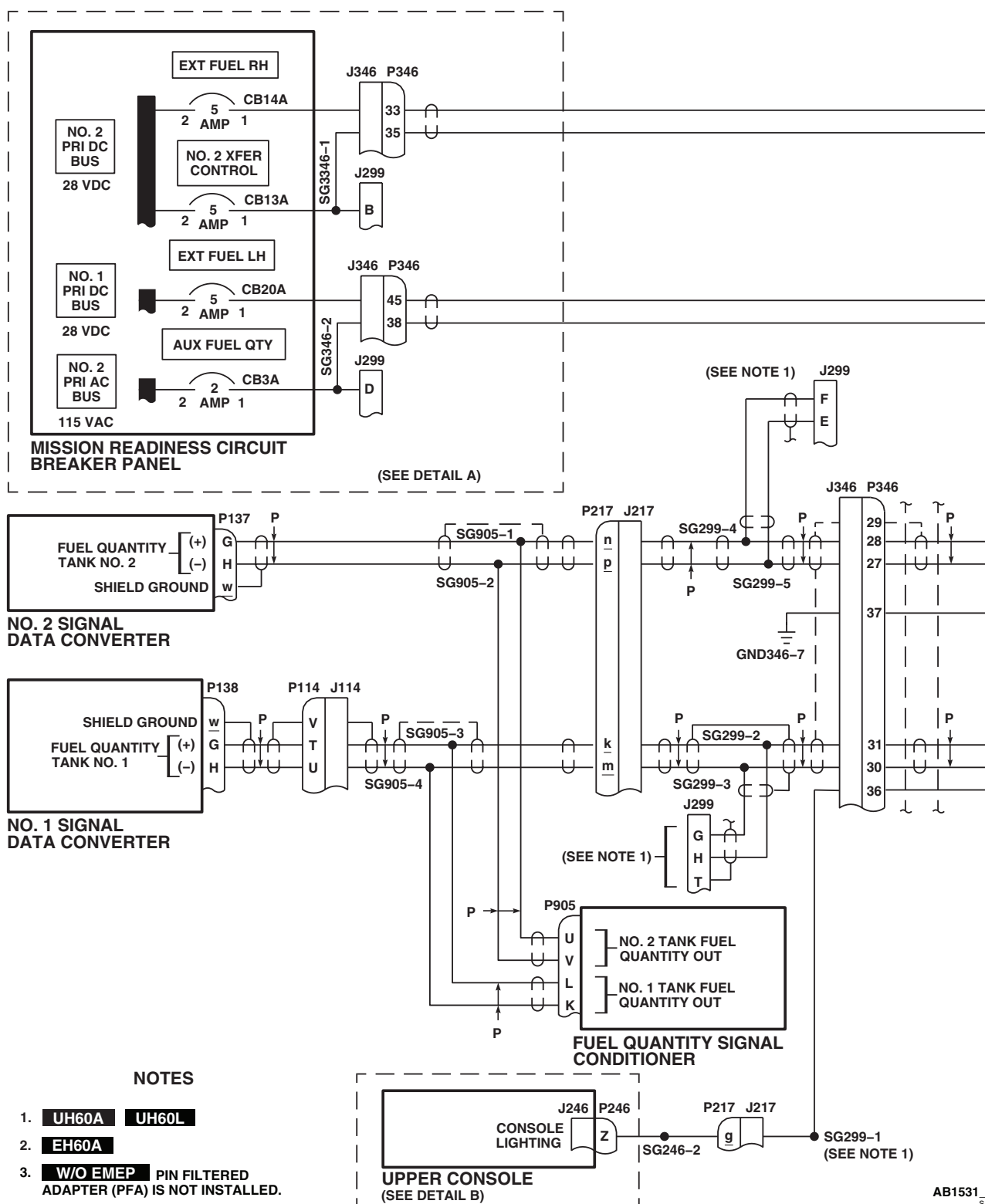


Figure 2. ESSS Range Extension System Schematic Diagram. (Sheet 1 of 7)

SCHEMATICS - Continued

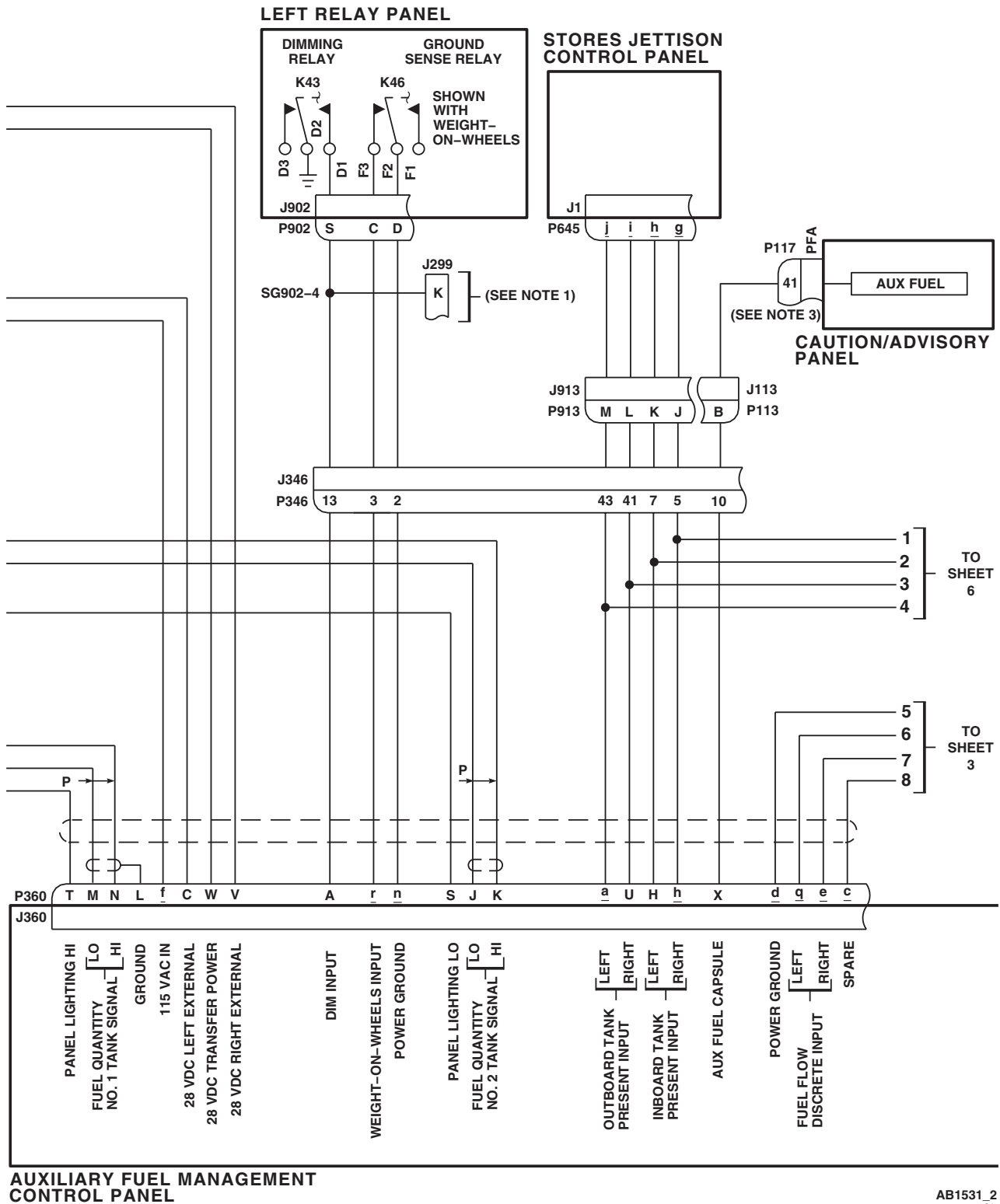
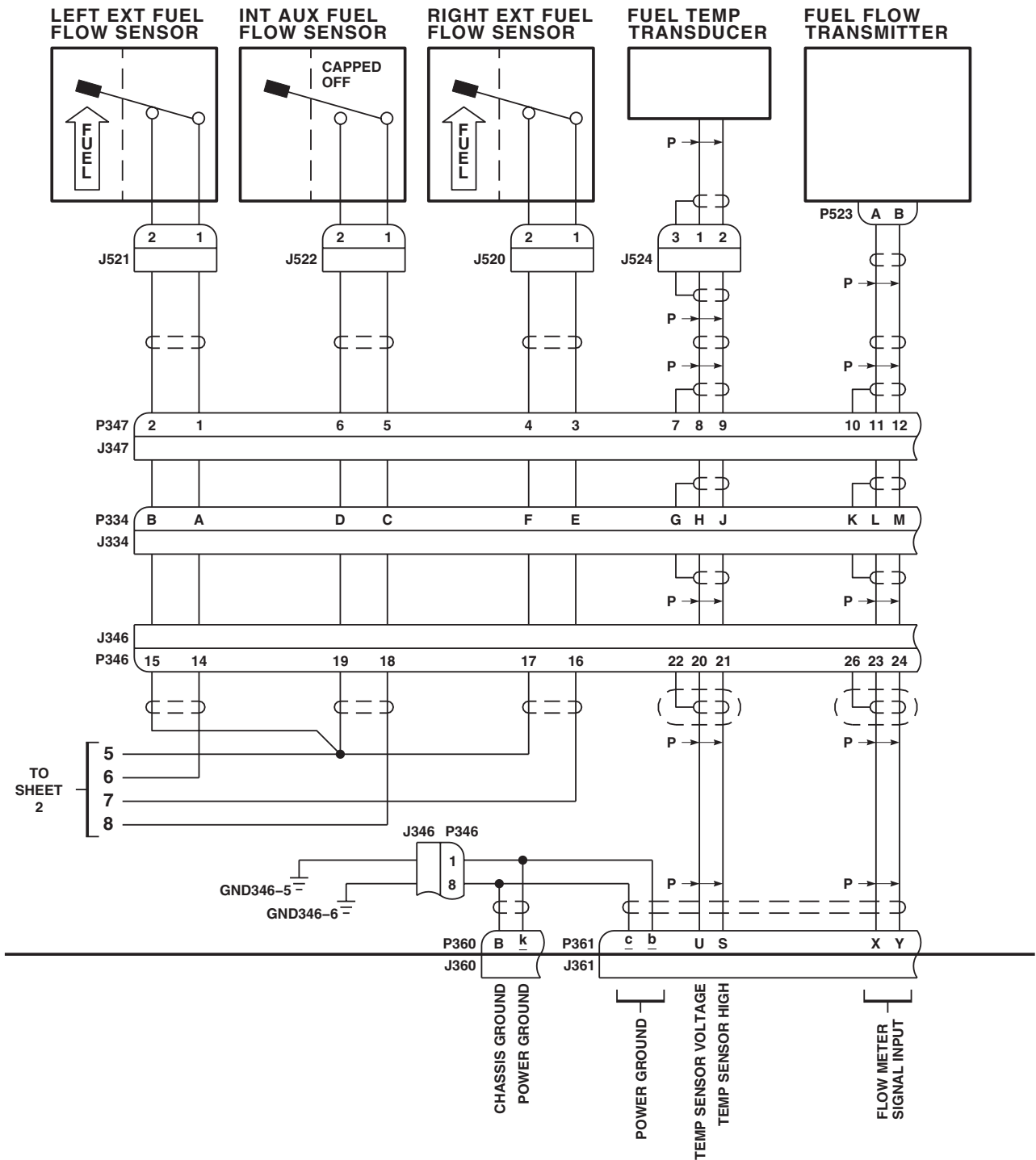


Figure 2. ESSS Range Extension System Schematic Diagram. (Sheet 2 of 7)



SCHEMATICS - Continued

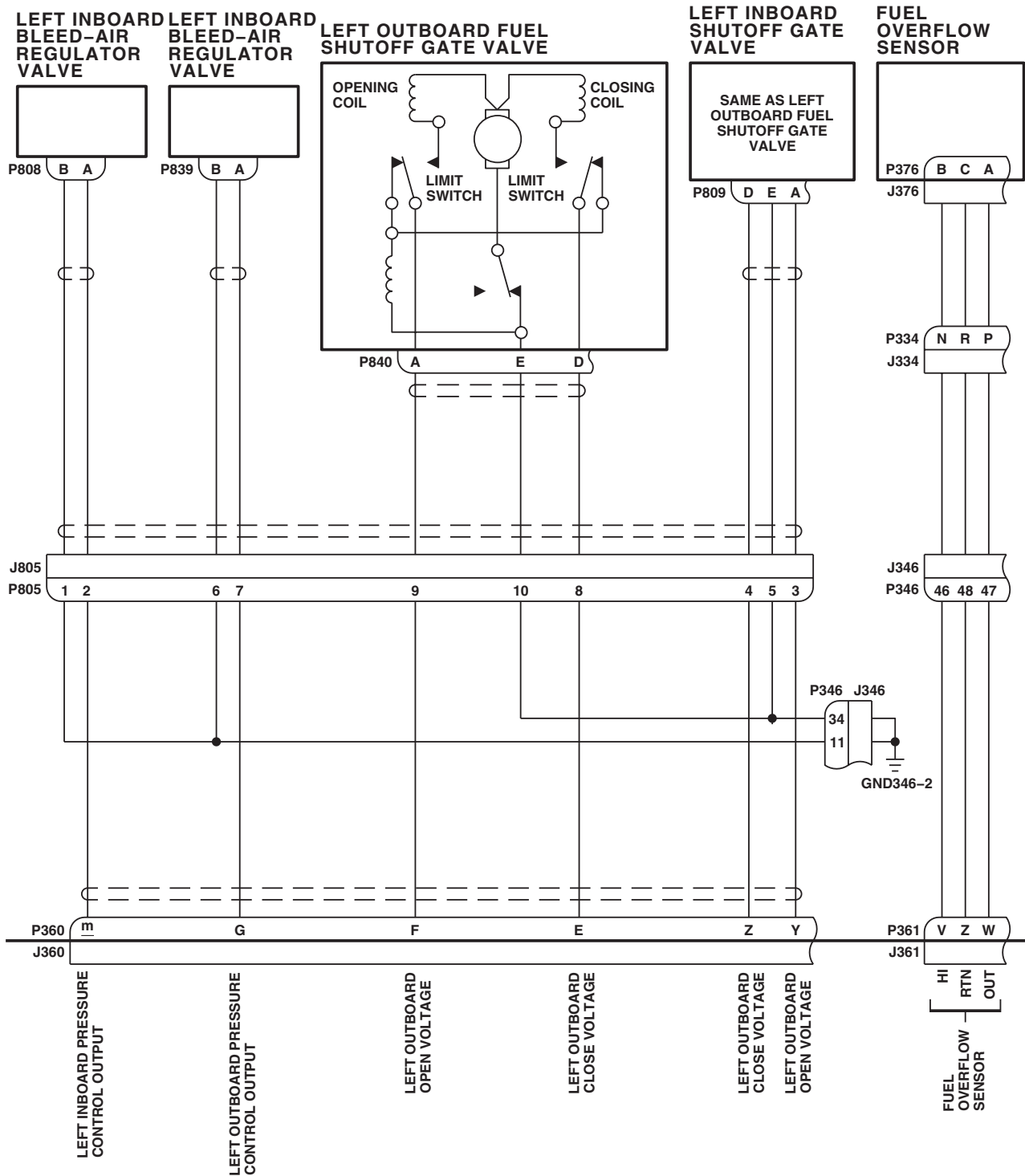


AUXILIARY FUEL MANAGEMENT CONTROL PANEL

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SA

Figure 2. ESSS Range Extension System Schematic Diagram. (Sheet 3 of 7)

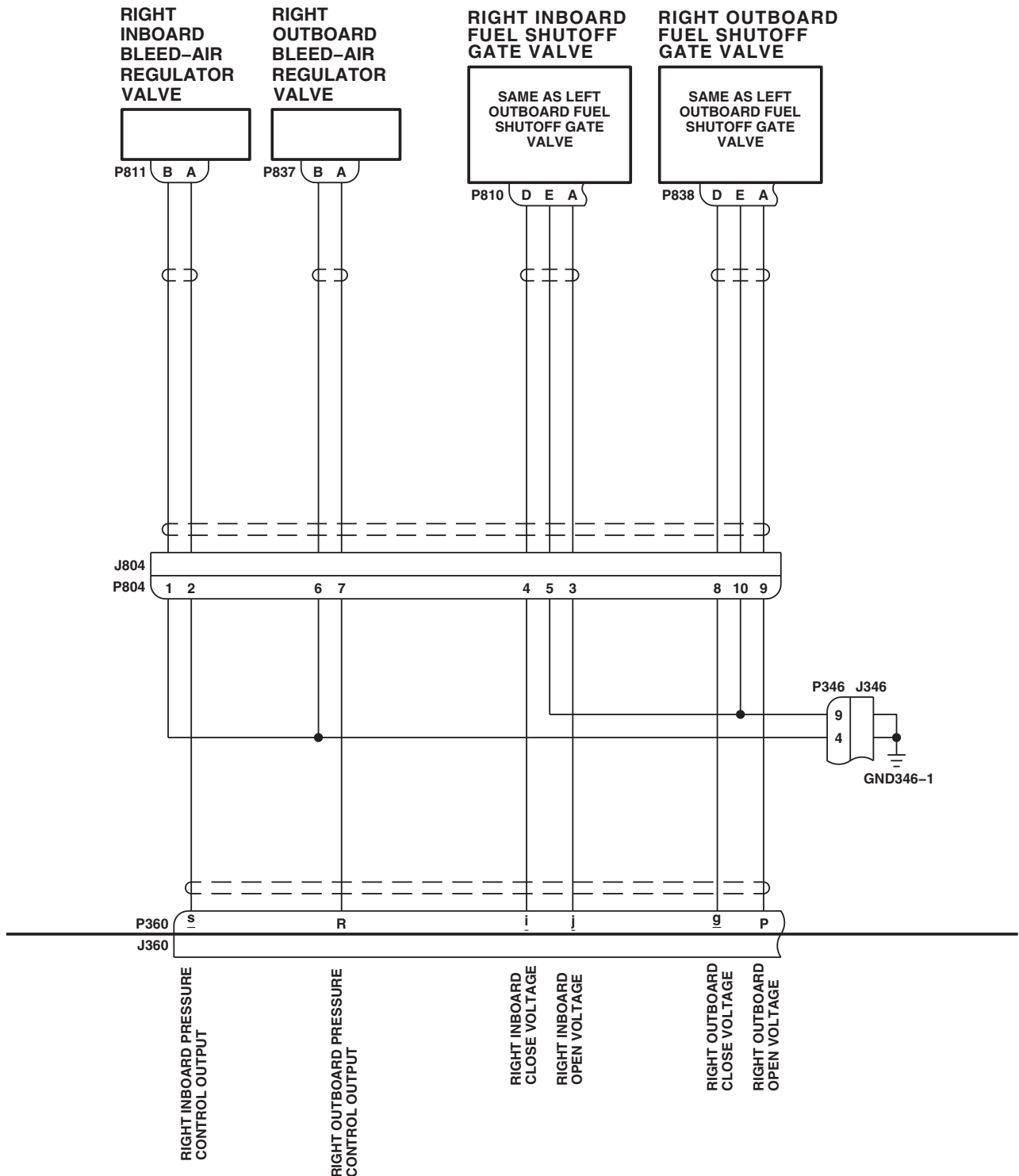
SCHEMATICS - Continued



AB1531\_4  
SA

Figure 2. ESSS Range Extension System Schematic Diagram. (Sheet 4 of 7)

SCHEMATICS - Continued



AUXILIARY FUEL MANAGEMENT CONTROL PANEL

AB1531\_5  
SA

Figure 2. ESSS Range Extension System Schematic Diagram. (Sheet 5 of 7)

SCHEMATICS - Continued

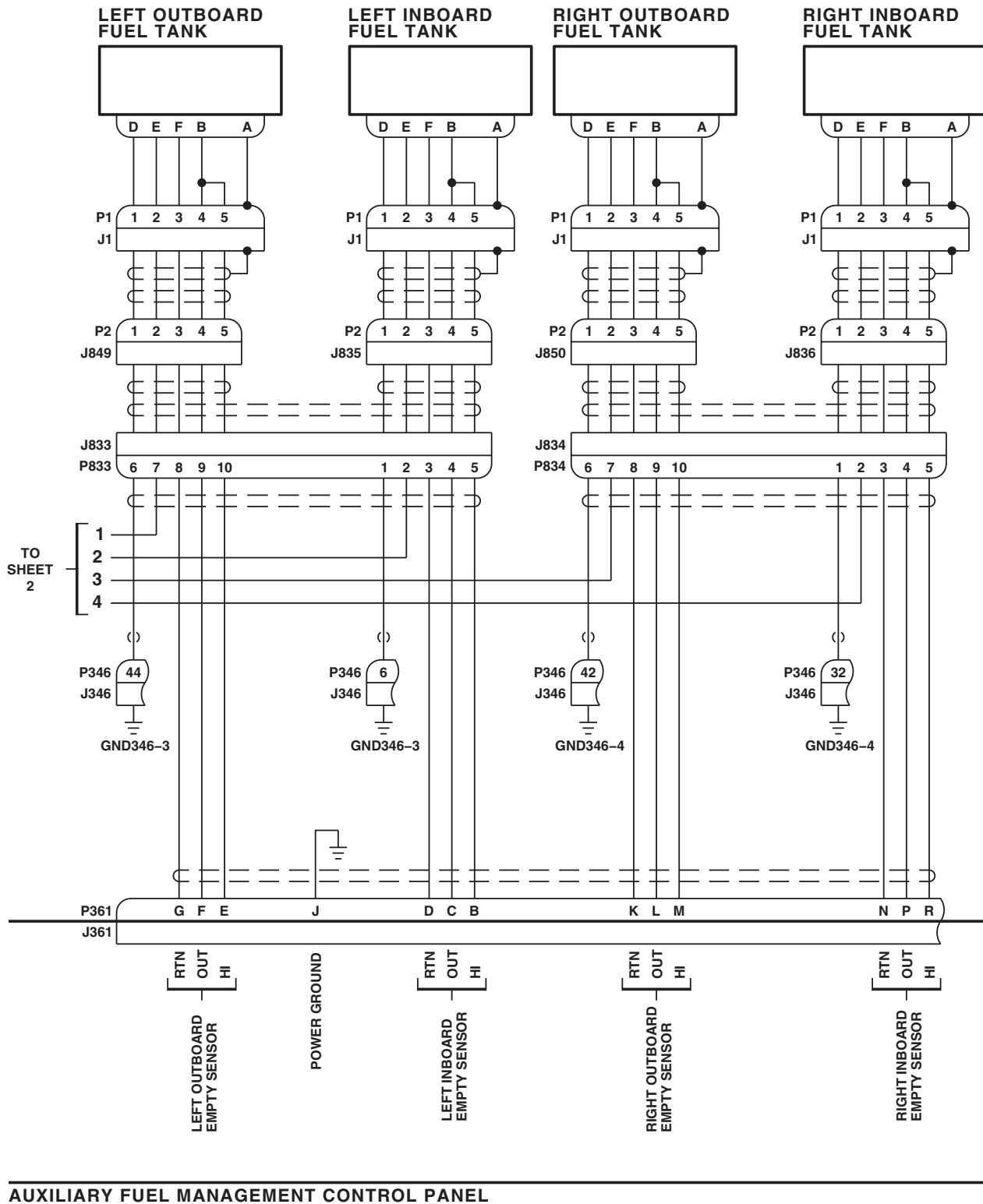
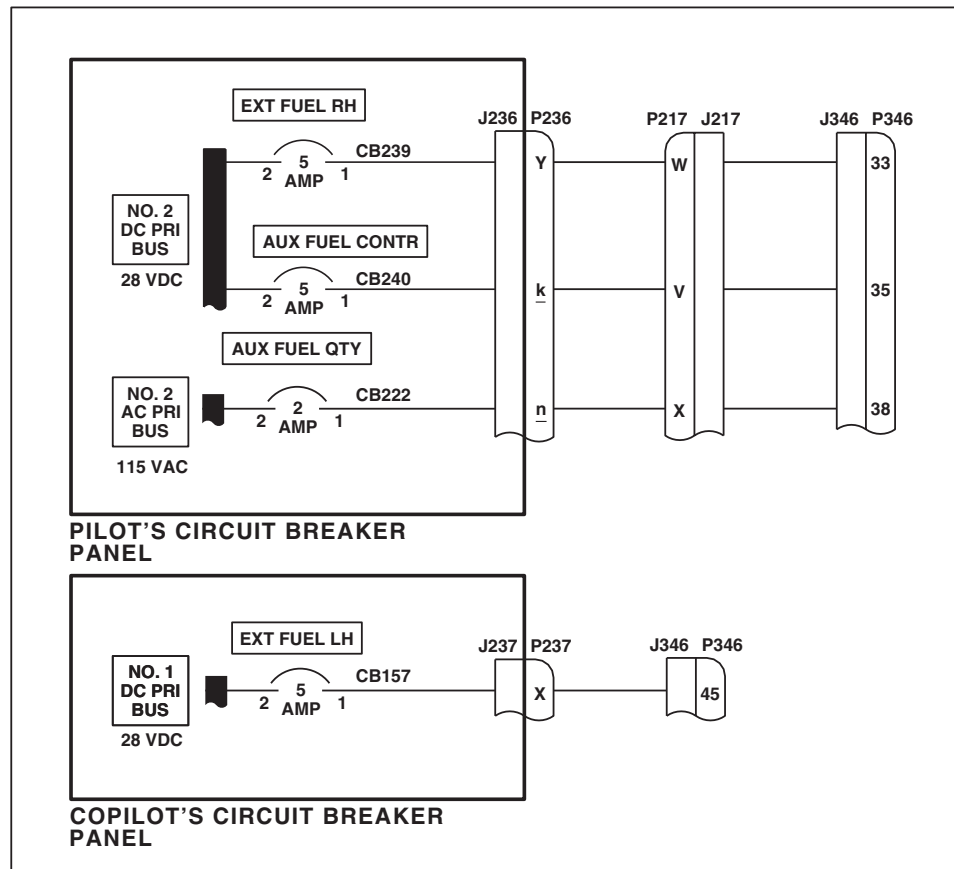
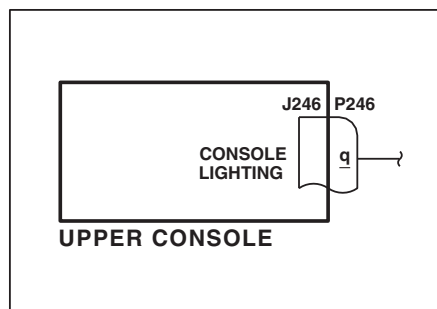


Figure 2. ESSS Range Extension System Schematic Diagram. (Sheet 6 of 7)

SCHEMATICS - Continued



DETAIL A  
(SEE NOTE 2)



DETAIL B  
(SEE NOTE 2)

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SA

Figure 2. ESSS Range Extension System Schematic Diagram. (Sheet 7 of 7)

END OF WORK PACKAGE



## UNIT LEVEL

## MISSION EQUIPMENT

EXTERNAL STORES SUPPORT SYSTEM (ESSS) EXTERNAL TANK CHECK **W/O MWO 50-78**

## INITIAL SETUP:

## Test Equipment

Aviation Ground Power Unit (AGPU), 83-36A  
 Compressor Unit, MIL-C-13874  
 Locally-Made Fuel Quantity Simulator Test Set, Figure 152, WP 1805 00  
 Multimeter, AN/PSM-45A (2 required)

WP 1447 00  
 WP 1612 00  
 WP 1613 00  
 WP 1620 00  
 WP 1685 00  
 WP 1805 00

## Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01  
 Container, Suitable  
 Electrical Repairer Toolkit, SC 5180-99-B06  
 Locally-Made External Stores Support System (ESSS)  
 Pneumatic Adapter, Figure 151, WP 1805 00  
 Torque Wrench, GG-W-686

## Personnel Required

Aircraft Electrician MOS 15F (1)  
 UH-60 Helicopter Repairer MOS 15T (1)

## References

TM 1-1500-204-23  
 TM 1-1520-237-10  
 WP 0104 00  
 WP 0158 00  
 WP 1442 00  
 WP 1445 00

## Equipment Condition

All Explosive Cartridges Removed from Ejector Racks, WP 1442 00  
 or All Explosive Cartridges Removed from Ejector Racks, WP 1445 00  
 External Electrical Power Available, WP 1685 00  
 or APU Operational, TM 1-1520-237-10  
 External Range Extension Tanks **220-250** Pounds of Fuel in Each Inboard, if Installed, WP 1620 00  
 External Range Extension Tanks **160-190** Pounds of Fuel in Each Outboard, if Installed, WP 1620 00  
 Main Fuel Tanks, **Not More Than 650** Pounds of Fuel in Each Tank, WP 1612 00  
 or Main Fuel Tanks, not more than **650** Pounds of Fuel in Each Tank, WP 1613 00  
 Rear Fairings on Vertical Stores (Pylons), Which Have Tanks Installed, Removed, WP 1447 00

## SPECIAL INSTRUCTIONS

## NOTE

See Figure 1 for component location diagram.

This operational/troubleshooting procedure is subdivided:

- Setup
- Outboard Tanks Installed
- Inboard Tanks Installed
- Shutdown

**SPECIAL INSTRUCTIONS - Continued****APPLICABLE CONFIGURATION**

This operational/troubleshooting procedure is designed to be utilized when the horizontal (Wings) and vertical (Pylons) stores are installed on a permanent basis, and only the inboard or outboard external tanks are removed and installed.

**SETUP OPERATIONAL/TROUBLESHOOTING PROCEDURE****PROCEDURE****Step****NOTE**

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- As an alternate to using the compressor unit to pressurize the ESSS Range Extension System, the Aviation Ground Power Unit (AGPU) or the APU may be used. If the alternate is selected, steps with **CU** may be skipped.
- The operational/troubleshooting procedure will check the ESSS Range Extension System with any tank configuration, installed on the helicopter, authorized by the Aircraft Operator's Manual (TM 1-1520-237-10).
- If APU is selected for electrical power, it will provide air pressure.
- If AGPU is use for external electrical power, it can provide an external air source when pneumatic line is connected to helicopter.
- Special Icons used in this operational/troubleshooting procedure:

**CU** Steps that are to be done if compressor unit is used.

**INBD** Steps that are to be done if inboard tanks are installed.

**OUTBD** Steps that are to be done if outboard tanks are installed.

**AGPU** Steps that are to be done if AGPU is to be used.

1. **CU** Using locally-made pneumatic adapter (WP 1805 00), tee air compressor into helicopter bleed-air QD and ESSS bleed-air hose. ■
2. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
3. Connect locally-made fuel quantity test simulator (simulator) (WP 1805 00) power cord to UTIL RCPT +28V (J258) in cabin ceiling.
4. Disconnect helicopter connector P905 from fuel quantity signal conditioner connector J905.
5. Connect simulator connector J905 to helicopter connector P905.
6. Turn simulator TANK 1 and TANK 2 controls fully counterclockwise.
7. Connect No. 1 multimeter to simulator TANK 1 OUTPUT.
8. Connect No. 2 multimeter to simulator TANK 2 OUTPUT.
9. **AGPU** Connect AGPU power cable to helicopter external power receptacle. ■
10. **AGPU** Connect AGPU pneumatic hose to helicopter. ■



---

**SETUP OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**

11. Turn on electrical power.
12. **CU** Apply air source, regulated at 35-45 psi, to helicopter's external tanks. ■
13. **AGPU** Apply air source, regulated at 35-45 psi, to helicopter's external tanks. ■
14. **Record** time.
15. **INBD** Place control panel PRESSURE INBD switch to ON. ■
16. **OUTBD** Place control panel PRESSURE OUTBD switch to ON. ■
17. Inspect fuel and pneumatic hoses and fittings for signs of leaking or damage (TM 1-1500-204-23).
18. Turn simulator TANK 1 control to get 6.0 vdc No. 1 multimeter indication.
19. Turn simulator TANK 2 control to get 6.0 vdc No. 2 multimeter indication.
20. **INBD** Turn control panel AUX FUEL QTY switch to INBD. ■
21. **INBD** Place control panel INCR-DECR switch to INCR to get display of 6000 pounds. ■
22. **OUTBD** Turn control panel AUX FUEL QTY switch to OUTBD. ■
23. **OUTBD** Place control panel INCR-DECR switch to INCR to get display of 1500 pounds. ■

**Indication/Condition**

1500 pounds shall be displayed.

**Corrective Action**

None Required

**OUTBOARD TANKS INSTALLED OPERATIONAL/TROUBLESHOOTING PROCEDURE****PROCEDURE****Step****NOTE**

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- As an alternate to using the compressor unit to pressurize the ESSS Range Extension System, the Aviation Ground Power Unit (AGPU) or the APU may be used. If the alternate is selected, steps with **CU** may be skipped.
- The fault isolation procedure will check the ESSS Range Extension System with any tank configuration, installed on the helicopter, authorized by the Aircraft Operator's Manual (TM 1-1520-237-10).
- If APU is selected for electrical power, it will provide air pressure.
- If AGPU is use for external electrical power, it can provide an external air source when pneumatic line is connected to helicopter.
- Special Icons used in this FIP:

**CU** Steps that are to be done if compressor unit is used.

**INBD** Steps that are to be done if inboard tanks are installed.

**OUTBD** Steps that are to be done if outboard tanks are installed.

**AGPU** Steps that are to be done if AGPU is to be used.

- If any trouble is encountered during this fault isolation procedure, go to WP 0158 00.

1. Do SETUP, in this work package, if not already done.
2. Turn simulator TANK 2 control to get 2.7 vdc No. 2 multimeter indication.
3. Place control panel FUEL XFR - MODE switch to OFF.
4. Place control panel FUEL XFR - TANKS switch to OUTBD.
5. Before continuing, make sure that at least 5 minutes have elapsed from time recorded in SETUP, step 14, in this work package.
6. Place control panel FUEL XFR - MODE switch to AUTO until EXTERNAL - RIGHT OUTBD EMPTY and LEFT OUTBD EMPTY indicators go on.
7. Momentarily press control panel STATUS button.
8. Place control panel FUEL XFR - MODE switch to MANUAL.
9. Place control panel MANUAL XFR - LEFT and RIGHT switches to ON.

**Indication/Condition**

- a. Control panel EXTERNAL - RIGHT NO FLOW and LEFT NO FLOW indicators shall go off, then come back on within 5 minutes.

---

**OUTBOARD TANKS INSTALLED OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued

- b. Control panel digital display indication shall not change when the EXTERNAL RIGHT NO FLOW and LEFT NO FLOW indicators are on.

**Corrective Action**

None Required

**Step**

10. Place control panel FUEL XFR - MODE, MANUAL XFR - RIGHT and LEFT, and PRESS - OUTBD switch to OFF.
11. Check for less than 50 mvac and between -50 and 50 mvdc between J705-A and J705-B (left outboard).
12. Check for less than 50 mvac and between -50 and 50 mvdc between J710-A and J710-B (right outboard).
13. If no other check is to be done, go to SHUTDOWN, in this work package.

**Indication/Condition**

Shut down shall be performed.

**Corrective Action**

None Required

**INBOARD TANKS INSTALLED OPERATIONAL/TROUBLESHOOTING PROCEDURE****PROCEDURE****Step****NOTE**

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- As an alternate to using the compressor unit to pressurize the ESSS Range Extension System, the Aviation Ground Power Unit (AGPU) or the APU may be used. If the alternate is selected, steps with **CU** may be skipped.
- The fault isolation procedure will check the ESSS Range Extension System with any tank configuration, installed on the helicopter, authorized by the Aircraft Operator's Manual (TM 1-1520-237-10).
- If APU is selected for electrical power, it will provide air pressure.
- If AGPU is use for external electrical power, it can provide an external air source when pneumatic line is connected to helicopter.
- Special Icons used in this FIP:
  - CU** Steps that are to be done if compressor unit is used.
  - INBD** Steps that are to be done if inboard tanks are installed.
  - OUTBD** Steps that are to be done if outboard tanks are installed.
  - AGPU** Steps that are to be done if AGPU is to be used.
- If any trouble is encountered during this Fault Isolation Procedure, go to WP 0158 00.

1. Do SETUP, in this work package, if not already done.
2. Turn simulator TANK 2 control to get 2.7 vdc No. 2 multimeter indication, if not already done.
3. Make sure control panel FUEL XFR - MODE switch is OFF.
4. Place control panel FUEL XFR - TANKS switch to INBD.
5. Before continuing, make sure that at least 5 minutes have elapsed from time recorded in SETUP, step 14.
6. Place control panel FUEL XFR - MODE switch to AUTO until EXTERNAL - RIGHT INBD EMPTY and LEFT INBD EMPTY indicators go on.
7. Momentarily press control panel STATUS button.
8. Place control panel FUEL XFR - MODE switch to MANUAL.
9. Place control panel MANUAL XFR - LEFT and RIGHT switches to ON.

**Indication/Condition**

- a. Control panel EXTERNAL - RIGHT NO FLOW and LEFT NO FLOW indicators shall go off, then come back on within 5 minutes.

---

**INBOARD TANKS INSTALLED OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued

- b. Control panel digital display indication shall not change when the EXTERNAL RIGHT NO FLOW and LEFT NO FLOW indicators are on.

**Corrective Action**

None Required

**Step**

10. Place control panel FUEL XFR - MODE, MANUAL XFR - RIGHT and LEFT, and PRESS - INBD switch to OFF.
11. Check for less than 50 mvac and between -50 and 50 mvdc between: J709-B and J709-A (left inboard).
12. Check for less than 50 mvac and between -50 and 50 mvdc between: J709-C and J709-N (left inboard).
13. Check for less than 50 mvac and between -50 and 50 mvdc between J706-B and J706-A (right inboard).
14. Check for less than 50 mvac and between -50 and 50 mvdc between J706-C and J706-N (right inboard).
15. If no other check is to be done, go to SHUTDOWN, in this work package.

**Indication/Condition**

Shut down shall be performed.

**Corrective Action**

None Required

**SHUTDOWN OPERATIONAL/TROUBLESHOOTING PROCEDURE****PROCEDURE****Step**

1. Turn off electrical power.
2. **CU** Turn off air source. ■
3. **AGPU** Turn off air source. ■
4. Allow bleed-air regulator valves to vent air to ambient pressure.
5. **CU** Disconnect air compressor, hoses, and locally-made pneumatic adapter. ■
6. **CU** Connect pneumatic hose to outboard side of helicopter bleed-air QD. ■
7. **CU** **TORQUE NUTS TO 135 - 150 INCH-POUNDS.** ■
8. Disconnect test equipment.
9. Connect helicopter electrical connectors.
10. Install all explosive cartridges in ejector racks.
11. Install all vertical stores (pylon) rear fairings.

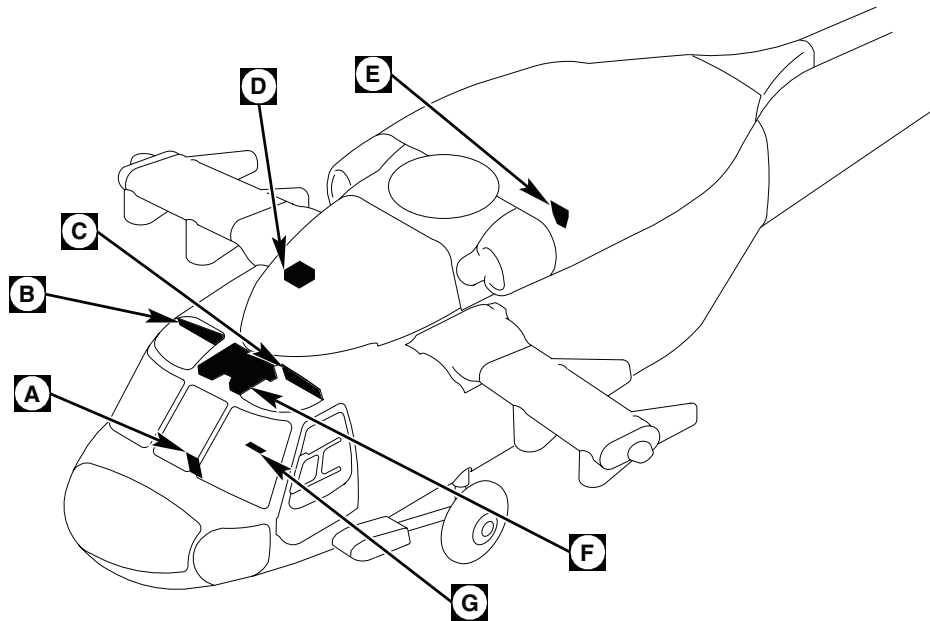
**Indication/Condition**

All vertical stores (pylon) rear fairings shall be installed.

**Corrective Action**

None Required

## LOCATION



| TERMINAL BOARD /<br>DISCONNECT PLUG /<br>RECEPTACLE | LOCATION /<br>CONNECTION POINT        |
|-----------------------------------------------------|---------------------------------------|
| J299<br>(SEE NOTE 1)                                | CABIN CEILING<br>BL 10 LH, STA 302    |
| J520                                                | RIGHT EXT FUEL FLOW<br>SENSOR         |
| J521                                                | LEFT EXT FUEL FLOW<br>SENSOR          |
| J522                                                | INT AUX FUEL FLOW<br>SENSOR           |
| J524                                                | FUEL TEMPERATURE<br>TRANSDUCER        |
| P1 / J1                                             | RIGHT INBOARD FUEL<br>TANK CONNECTOR  |
| P1 / J1                                             | LEFT INBOARD FUEL TANK<br>CONNECTOR   |
| P1 / J1                                             | LEFT OUTBOARD FUEL<br>TANK CONNECTOR  |
| P1 / J1                                             | RIGHT OUTBOARD FUEL<br>TANK CONNECTOR |
| P2 / J835                                           | LEFT INBOARD FUEL TANK<br>CONNECTOR   |

| TERMINAL BOARD /<br>DISCONNECT PLUG /<br>RECEPTACLE | LOCATION /<br>CONNECTION POINT                                 |
|-----------------------------------------------------|----------------------------------------------------------------|
| P2 / J836                                           | RIGHT INBOARD FUEL TANK<br>CONNECTOR                           |
| P2 / J849                                           | LEFT OUTBOARD FUEL<br>TANK CONNECTOR                           |
| P2 / J850                                           | RIGHT OUTBOARD FUEL<br>TANK CONNECTOR                          |
| P113 / J113                                         | COCKPIT<br>BL 6 LH, STA 197                                    |
| P114 / J114                                         | COCKPIT<br>BL 8 RH, STA 200                                    |
| P117                                                | CAUTION / ADVISORY<br>PANEL                                    |
| P137                                                | NO. 2 SIGNAL DATA<br>CONVERTER                                 |
| P138                                                | NO. 1 SIGNAL DATA<br>CONVERTER                                 |
| P163 / J163                                         | LEFT DRAG BEAM SWITCH                                          |
| P217 / J217                                         | MAIN ROTOR PYLON DECK<br>BL 0, STA 284                         |
| P236 / J236<br>(SEE NOTE 2)                         | BEHIND PILOT'S CIRCUIT<br>BREAKER PANEL<br>BL 23 RH            |
| P237 / J237<br>(SEE NOTE 2)                         | BEHIND COPILOT'S<br>CIRCUIT BREAKER PANEL<br>BL 25 LH, STA 247 |
| P246 / J246                                         | CABIN CEILING<br>BL 5 RH, STA 247                              |
| P334 / J334                                         | CABIN CEILING<br>BL 16.5 LH, STA 390                           |

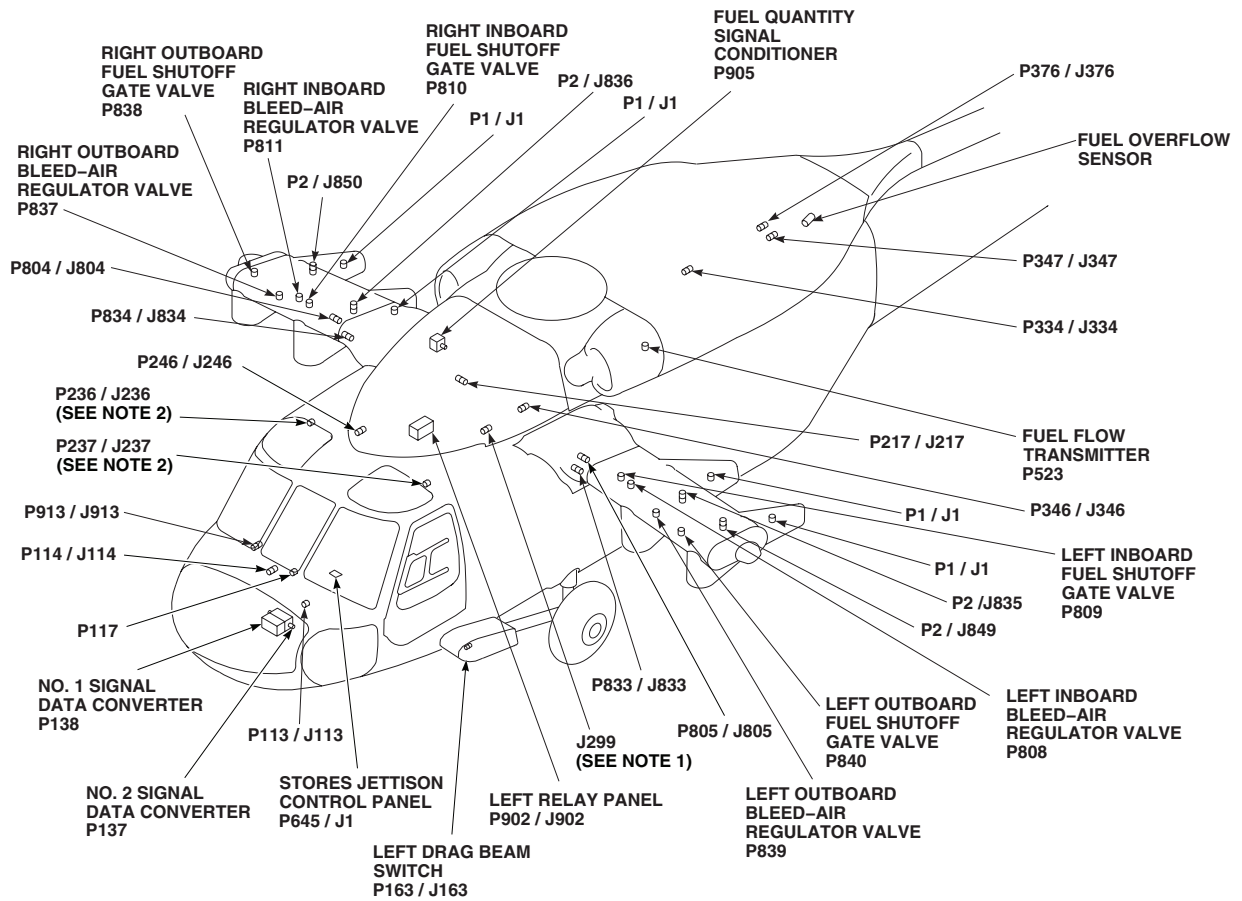
## NOTES

1. UH60A UH60L
2. EH60A

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Figure 1. ESSS External Tank Check Location Diagram. (Sheet 1 of 5)

## LOCATION - Continued



| TERMINAL BOARD/<br>DISCONNECT PLUG/<br>RECEPTACLE | LOCATION/<br>CONNECTION POINT                     |
|---------------------------------------------------|---------------------------------------------------|
| P346 / J346                                       | CABIN CEILING<br>BL 20 LH, STA 304                |
| P347 / J347                                       | TRANSITION SECTION<br>BL 10 LH, STA 425           |
| P360 / J360                                       | AUXILIARY FUEL MANAGE-<br>MENT CONTROL PANEL      |
| P361 / J361                                       | AUXILIARY FUEL MANAGE-<br>MENT CONTROL PANEL      |
| P376 / J376                                       | FUEL OVERFLOW SENSOR                              |
| P523                                              | FUEL FLOW TRANSMITTER                             |
| P645 / J1                                         | STORES JETTISON<br>CONTROL PANEL                  |
| P804 / J804                                       | HORIZONTAL STORES<br>SUPPORT<br>BL 59 RH, STA 295 |
| P805 / J805                                       | HORIZONTAL STORES<br>SUPPORT<br>BL 59 LH, STA 295 |
| P808                                              | LEFT INBOARD BLEED-AIR<br>REGULATOR VALVE         |

| TERMINAL BOARD/<br>DISCONNECT PLUG/<br>RECEPTACLE | LOCATION/<br>CONNECTION POINT               |
|---------------------------------------------------|---------------------------------------------|
| P809                                              | LEFT INBOARD FUEL SHUT-<br>OFF GATE VALVE   |
| P810                                              | RIGHT INBOARD FUEL<br>SHUTOFF GATE VALVE    |
| P811                                              | RIGHT INBOARD BLEED-AIR<br>REGULATOR VALVE  |
| P833 / J833                                       | HSS ATTACHMENT FITTING<br>BL 59 LH, STA 295 |
| P834 / J834                                       | HSS ATTACHMENT FITTING<br>BL 59 LH, STA 295 |
| P837                                              | RIGHT OUTBOARD BLEED-AIR<br>REGULATOR VALVE |
| P838                                              | RIGHT OUTBOARD FUEL<br>SHUTOFF GATE VALVE   |
| P839                                              | LEFT OUTBOARD BLEED-AIR<br>REGULATOR VALVE  |
| P840                                              | LEFT OUTBOARD FUEL<br>SHUTOFF GATE VALVE    |
| P902 / J902                                       | LEFT RELAY PANEL                            |
| P905                                              | FUEL QUANTITY SIGNAL<br>CONDITIONER         |
| P913 / J913                                       | COCKPIT<br>BL 24 RH, STA 247                |

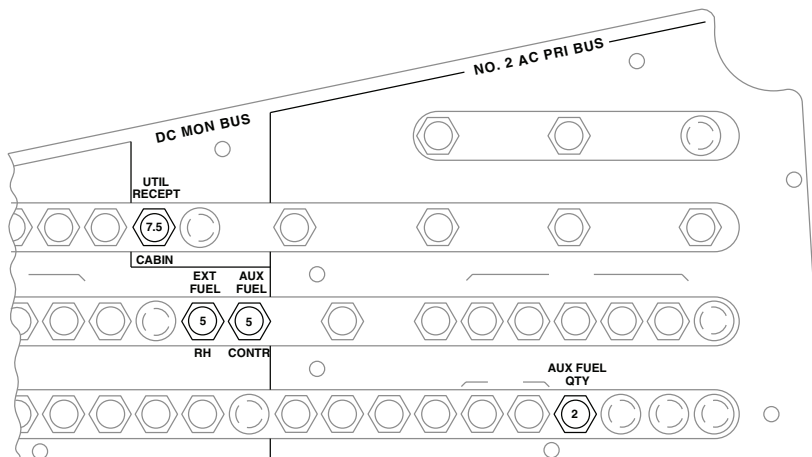
AB1101\_2  
SA

Figure 1. ESSS External Tank Check Location Diagram. (Sheet 2 of 5)

LOCATION - Continued



CAUTION/ADVISORY PANEL



(SEE NOTE 2)

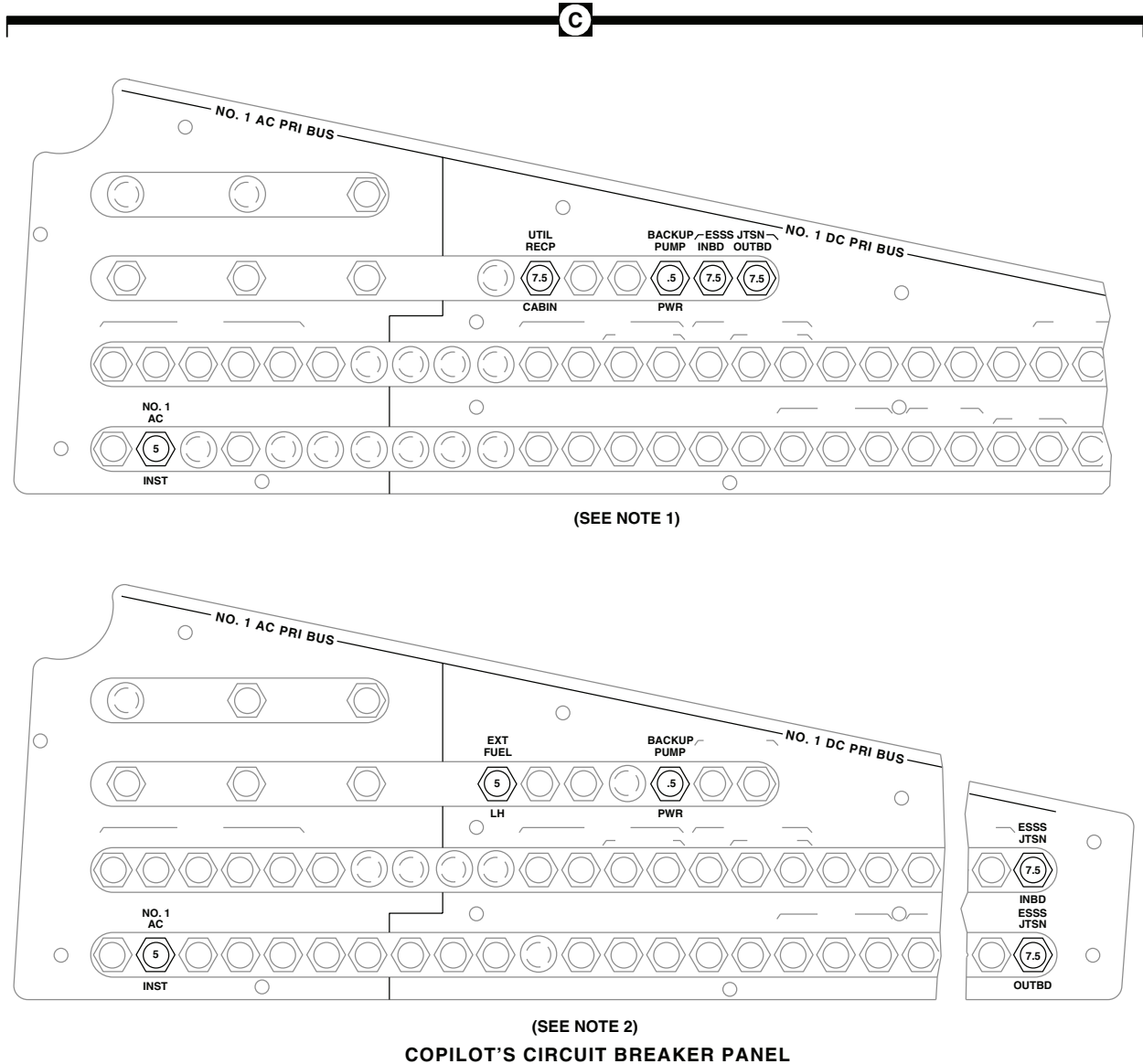
PILOT'S CIRCUIT BREAKER PANEL

AA3446 3  
SA

Figure 1. ESSS External Tank Check Location Diagram. (Sheet 3 of 5)



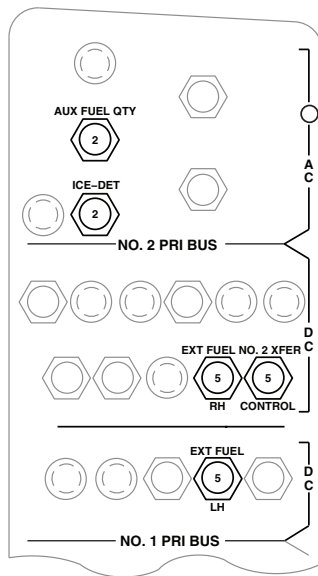
LOCATION - Continued



AA3446\_4  
SA

Figure 1. ESSS External Tank Check Location Diagram. (Sheet 4 of 5)

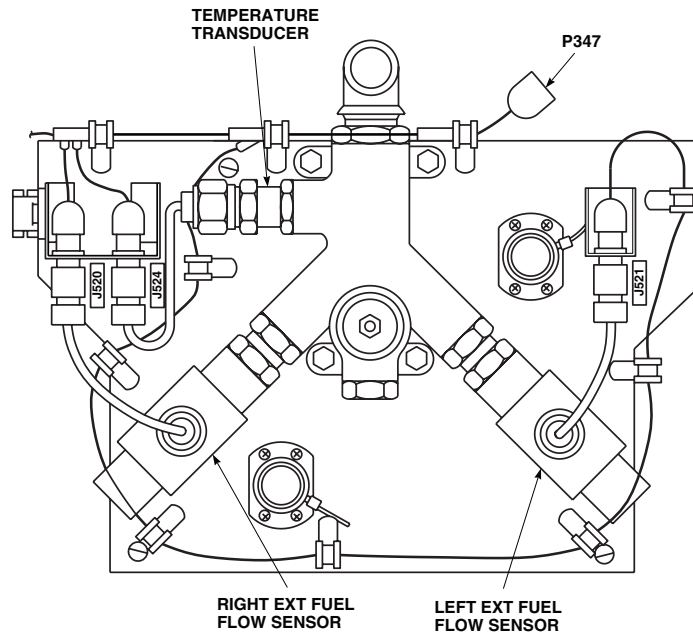
LOCATION - Continued



(SEE NOTE 1)

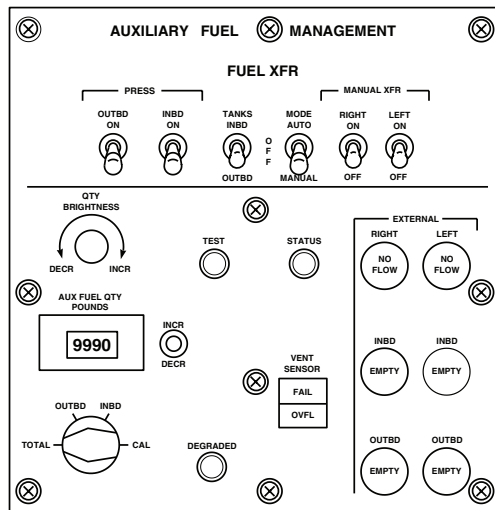
MISSION READINESS  
CIRCUIT BREAKER PANEL

D



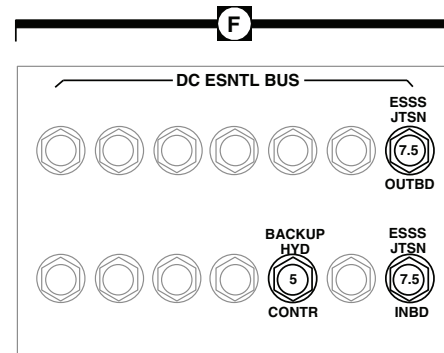
FLOW SENSOR PANEL

E



AUXILIARY FUEL MANAGEMENT  
CONTROL PANEL

G



UPPER CONSOLE  
CIRCUIT BREAKER PANEL

F

AA3446\_5A  
SA

Figure 1. ESSS External Tank Check Location Diagram. (Sheet 5 of 5)

END OF WORK PACKAGE

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**UNIT LEVEL****MISSION EQUIPMENT****EXTERNAL STORES SUPPORT SYSTEM (ESSS) JETTISON SYSTEM**

**This WP supersedes WP 0160 00, dated 17 April 2006.**

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**INITIAL SETUP:****Test Equipment**

Locally-Made Test Lamps Figure 150, WP 1805 00 (8  
required)  
Multimeter, AN/PSM-45A

WP 0443 00

WP 0824 00

WP 1425 00

WP 1428 00

WP 1442 00

WP 1445 00

WP 1448 00

WP 1460 00

WP 1462 00

WP 1463 00

WP 1465 00

**Tools and Special Tools**

Aircraft Mechanics Toolkit, SC 5180-99-B01  
Electrical Repairer Toolkit, SC 5180-99-B06  
Locally-Made Drag Beam Wedge, Figure 167,  
WP 1805 00 (2 required)  
Stopwatch

WP 1685 00

WP 1747 00

WP 1805 00

**Personnel Required**

Aircraft Electrician MOS 15F (1)  
UH-60 Helicopter Repairer MOS 15T (1)

**References**

TM 1-1520-237-10  
WP 0104 00  
WP 0110 00

**Equipment Condition**

External Electrical Power Available, WP 1685 00  
or APU Operational, TM 1-1520-237-10

## EXTERNAL STORES SUPPORT SYSTEM (ESSS) JETTISON SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE

### PROCEDURE

#### Step

#### WARNING

- To prevent accidental firing of an explosive cartridge, observe all safety precautions regarding the handling and storage of explosives.
- Failure to do stray voltage check would result in injury if stray voltage is present when an explosive cartridge is installed.

#### NOTE

- Stray voltage check is contained in steps 63. through 69.
- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- If fuel tanks are connected to the left, right, or both stores stations, the BOTH mode of jettison is automatically selected even if the selector switch is at L or R. This fault isolation procedure isolates each tank, if installed. This enables the maintenance technician to check the selector switch in each position (INBD L, INBD R, INBD BOTH OUTBD BOTH, OUTBD R, OUTBD L, and ALL).
- The operational/troubleshooting procedure checks all four stores stations, regardless of helicopter's external tank configuration.
- See Figure 1 for component location diagram and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

1. Remove all explosive cartridges installed in ejector racks (WP 1442 00 or WP 1445 00).
2. Remove rear fairing on all vertical stores pylons (WP 1425 00).
3. Disconnect electrical connectors, if connected to rack harnesses:
  - J705 (left outboard)
  - J707 (left inboard)
  - P709 (left inboard)
  - P706 (right inboard)
  - J708 (right inboard)
  - J710 (right outboard)
4. Check continuity between the following:
  - J705-B and ground (left outboard)
  - J707-B and ground (left inboard)
  - P709-A and ground (left inboard)

## EXTERNAL STORES SUPPORT SYSTEM (ESSS) JETTISON SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE - Continued

- P709-N and ground (left inboard)
- P706-A and ground (right inboard)
- P706-N and ground (right inboard)
- J708-B and ground (right inboard)
- J710-B and ground (right outboard)

### Indication/Condition

Continuity shall be present.

### Corrective Action

If Indication/Condition is not as specified, repair/replace wiring as required (WP 1747 00).

### Step

5. **W/O MWO 50-78** Disconnect P1 from J1 on installed external fuel tanks.
6. **MWO 50-78** Disconnect W4P2 from J1 on installed external fuel tanks.
7. Connect a locally-made test lamp between the following:
  - J705-A and J705-B (left outboard)
  - J707-A and J707-B (left inboard)
  - P709-B and P709-A (left inboard)
  - P709-C and P709-N (left inboard)
  - P706-B and P706-A (right inboard)
  - P706-C and P706-N (right inboard)
  - J708-A and J708-B (right inboard)
  - J710-A and J710-B (right outboard)
8. Make sure the following circuit breakers are pushed in:

#### CIRCUIT BREAKER

#### CIRCUIT BREAKER PANEL

|                 |               |
|-----------------|---------------|
| ESSS JTSN OUTBD | Copilot's     |
| ESSS JTSN INBD  | Copilot's     |
| ESSS JTSN OUTBD | Upper Console |
| ESSS JTSN INBD  | Upper Console |

9. Pull out BACKUP HYD CONTR circuit breaker (upper console circuit breaker panel).
10. Turn on electrical power.
11. Turn upper console CONSOLE LT/LOWER control to BRT.

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**EXTERNAL STORES SUPPORT SYSTEM (ESSS) JETTISON SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE - Continued****Indication/Condition**

Panel lights on stores jettison control panel (control panel) shall go on at full brightness.

**Corrective Action**

If Indication/Condition is not as specified, troubleshoot instrument panel and console indicator lights dimming (WP 0110 00).

**PROCEDURE****Step**

12. Place control panel EMER JETT ALL switch up.

**Indication/Condition**

All locally-made test lamps shall be off.

**Corrective Action**

If Indication/Condition is not as specified, replace control panel (WP 1460 00).

**Step**

13. Place control panel EMER JETT ALL switch down.
14. Turn control panel selector switch to ALL.
15. Momentarily hold control panel JETT switch up.

**Indication/Condition**

All locally-made test lamps shall be off.

**Corrective Action**

If Indication/Condition is not as specified, replace control panel (WP 1460 00).

**Step**

16. Turn control panel selector switch to OFF.
17. Insert locally-made drag beam wedge (WP 1805 00) in left drag beam switch to simulate in-flight condition.
18. Place control panel EMER JETT ALL switch up and start stopwatch.

**Indication/Condition**

- a. All outboard locally-made test lamps shall go on.
- b. All inboard locally-made test lamps shall go on 0.5 to 1.5 seconds later.

**Corrective Action**

1. If Indication/Conditions a. and b. are not as specified, go to NONE OF THE LIGHTS GO ON WHEN EMER JETT ALL SWITCH IS PLACED UP, in this work package.
2. If only right outboard locally-made test lamps do not go on, go to EITHER OUTBOARD TEST LIGHTS DOES NOT GO ON WHEN EMER JETT ALL SWITCH IS PLACED UP, in this work package.
3. If only left outboard locally-made test lamps do not go on, go to EITHER OUTBOARD TEST LIGHT DOES NOT GO ON WHEN EMER JETT ALL SWITCH IS PLACED UP, in this work package.

---

**EXTERNAL STORES SUPPORT SYSTEM (ESSS) JETTISON SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE - Continued**

4. If Indication/Condition a. is not as specified, check for 28 vdc between:
  - P646-f and P646-e
  - P646-g and P646-e
  - a. If voltage is as specified, replace control panel (WP 1460 00).
  - b. If voltage is not as specified, check 28 vdc between ESSS JTSN OUTBD circuit breaker (upper console circuit breaker panel) and ground.
    - (1) If voltage is not as specified, replace circuit breaker (WP 0824 00).
    - (2) If voltage is as specified, repair/replace wiring between circuit breaker and P646-f, or circuit breaker and P646-g, as required (WP 1747 00).
5. If only right inboard locally-made test lamps do not go on, replace control panel (WP 1460 00).
6. If only left inboard locally-made test lamps do not go on, replace control panel (WP 1460 00).
7. If inboard locally-made test lamps do not go on, check 28 vdc between:
  - P646-h and P646-e
  - P646-j and P646-e
  - a. If voltage is as specified, replace control panel (WP 1460 00).
  - b. If voltage is not as specified, check 28 vdc between ESSS JTSN INBD circuit breaker (upper console circuit breaker panel) and ground.
    - (1) If voltage is not as specified, replace circuit breaker (WP 0824 00).
    - (2) If voltage is as specified, repair/replace wiring between circuit breaker and P646-h, or circuit breaker and P646-j, as required (WP 1747 00).
8. If inboard locally-made test lamps go on, but not in time specified, replace control panel (WP 1460 00).

**Step**

19. Stop and reset stopwatch.
20. Place control panel EMER JETT ALL switch down.
21. Remove locally-made wedge from left drag beam switch.
22. Insert locally-made drag beam wedge in right drag beam switch to simulate in-flight condition.
23. Turn control panel selector switch to ALL.
24. Momentarily hold (for at least 2 seconds) control panel JETT switch up and start stop watch.

**Indication/Condition**

- a. All outboard locally-made test lamps shall go on.
- b. All inboard locally-made test lamps shall go on 0.5 to 1.5 seconds later.

**Corrective Action**

1. If Indication/Conditions a. and b. are not as specified, go to NONE OF THE TEST LIGHTS GO ON WHEN SELECTOR SWITCH IS SET TO ALL AND JETT SWITCH IS HELD UP, in this work package.

---

**EXTERNAL STORES SUPPORT SYSTEM (ESSS) JETTISON SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE** - Continued

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2. If only right outboard locally-made test lamps do not go on, go to EITHER OUTBOARD TEST LIGHT DOES NOT GO ON WHEN SELECTOR SWITCH IS SET TO ALL AND JETT SWITCH IS HELD UP, in this work package.
3. If only left outboard locally-made test lamps do not go on, go to EITHER OUTBOARD TEST LIGHT DOES NOT GO ON WHEN SELECTOR SWITCH IS SET TO ALL AND JETT SWITCH IS HELD UP, in this work package.
4. If Indication/Condition a. is not as specified, check 28 vdc between:
  - P645-f and P645-k
  - P645-q and P645-k
  - a. If voltage is as specified, replace control panel (WP 1460 00).
  - b. If voltage is not as specified, check 28 vdc between ESSS JTSN OUTBD circuit breaker (copilot's circuit breaker panel) and ground.
    - (1) If voltage is not as specified, replace circuit breaker (WP 0824 00).
    - (2) If voltage is as specified, repair/replace wiring between circuit breaker and P645-f, or circuit breaker and P645-q, as required (WP 1747 00).
5. If only right inboard locally-made test lamps do not go on, replace control panel (WP 1460 00).
6. If only left inboard locally-made test lamps do not go on, replace control panel (WP 1460 00).
7. If inboard locally-made test lamps do not go on, check 28 vdc between:
  - P645-n and P645-k
  - P645-p and P645-k
  - a. If voltage is as specified, replace control panel (WP 1460 00).
  - b. If voltage is not as specified, check 28 vdc between ESSS JTSN INBD circuit breaker (upper console circuit breaker panel) and ground.
    - (1) If voltage is not as specified, replace circuit breaker (WP 0824 00).
    - (2) If voltage is as specified, repair/replace wiring between circuit breaker and P645-n, or circuit breaker and P645-p, as required (WP 1747 00).
8. If inboard locally-made test lamps go on, but not in time specified, replace control panel (WP 1460 00).

**Step**

25. Stop and reset stopwatch.
26. Turn control panel selector switch to OUTBD L.
27. Momentarily hold control panel JETT switch up.

**Indication/Condition**

- a. Left outboard locally-made test lamps shall go on.
- b. All other locally-made test lamps shall be off.

**Corrective Action**

If Indication/Conditions are not as specified, replace control panel (WP 1460 00).



---

**EXTERNAL STORES SUPPORT SYSTEM (ESSS) JETTISON SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE - Continued****Step**

28. Turn control panel selector switch to OUTBD R.
29. Momentarily hold control panel JETT switch up.

**Indication/Condition**

- a. Right outboard locally-made test lamps shall go on.
- b. All other locally-made test lamps shall be off.

**Corrective Action**

If Indication/Conditions are not as specified, replace control panel (WP 1460 00).

**Step**

30. Turn control panel selector switch to OUTBD BOTH.
31. Momentarily hold control panel JETT switch up.

**Indication/Condition**

- a. Right and left outboard locally-made test lamps shall go on.
- b. All other locally-made test lamps shall be off.

**Corrective Action**

If Indication/Conditions are not as specified, replace control panel (WP 1460 00).

**Step**

32. Turn control panel selector switch to INBD BOTH.
33. Momentarily hold control panel JETT switch up.

**Indication/Condition**

- a. Right and left inboard locally-made test lamps shall go on.
- b. All other locally-made test lamps shall be off.

**Corrective Action**

If Indication/Conditions are not as specified, replace control panel (WP 1460 00).

**Step**

34. Turn control panel selector switch to INBD R.
35. Momentarily hold control panel JETT switch up.

**Indication/Condition**

- a. Right inboard locally-made test lamps shall go on.
- b. All other locally-made test lamps shall be off.

**Corrective Action**

If Indication/Conditions are not as specified, replace control panel (WP 1460 00).

---

**EXTERNAL STORES SUPPORT SYSTEM (ESSS) JETTISON SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE - Continued****Step**

36. Turn control panel selector switch to INBD L.
37. Momentarily hold control panel JETT switch up.

**Indication/Condition**

- a. Left inboard locally-made test lamps shall go on.
- b. All other locally-made test lamps shall be off.

**Corrective Action**

If Indication/Conditions are not as specified, replace control panel (WP 1460 00).

**Step**

38. Turn control panel selector switch to OFF.
39. **W/O MWO 50-78** Connect right outboard fuel tank connector P1 to J1, or if external fuel tank is not installed, install a jumper wire between J1-1 and J1-2.
40. **MWO 50-78** Connect right outboard fuel tank connector W4P2 to J1, or if external fuel tank is not installed, install a jumper wire between W4P2-C or E and W4P2-F.
41. Turn control panel selector switch to OUTBD R.
42. Momentarily hold control panel JETT switch up.

**Indication/Condition**

Right and left outboard locally-made test lamps shall go on.

**Corrective Action**

1. **W/O MWO 50-78** If Indication/Condition is not as specified, go to W/O MWO 50-78 RIGHT AND LEFT OUTBOARD TEST LIGHTS DO NOT GO ON WHEN THE SELECTOR SWITCH IS SET TO OUTBD R AND THE JETT SWITCH IS HELD UP, in this work package.
2. **MWO 50-78** If Indication/Condition is not as specified, go to MWO 50-78 RIGHT AND LEFT OUTBOARD TEST LIGHTS DO NOT GO ON WHEN THE SELECTOR SWITCH IS SET TO OUTBD R AND THE JETT SWITCH IS HELD UP, in this work package.

**Step**

43. **W/O MWO 50-78** Disconnect right outboard external fuel tank connector P1 from J1, or if external fuel tank is not installed, remove jumper wire.
44. **MWO 50-78** Disconnect right outboard external fuel tank connector W4P2 from J1, or if external fuel tank is not installed, remove jumper wire installed in step 40.
45. **W/O MWO 50-78** Connect right inboard external fuel tank connector P1 to J1, or if external fuel tank is not installed, install an insulated jumper wire between J1-1 and J1-2.
46. **MWO 50-78** Connect right inboard external fuel tank connector W4P2 to J1, or if external fuel tank is not installed, install an insulated jumper wire between W4P2-C or -E and W4P2-F.
47. Turn control panel selector switch to INBD R.
48. Momentarily hold control panel JETT switch up, then release.

---

**EXTERNAL STORES SUPPORT SYSTEM (ESSS) JETTISON SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE - Continued****Indication/Condition**

Right and left inboard locally-made test lamps shall go on.

**Corrective Action**

1. **W/O MWO 50-78** If Indication/Condition is not as specified, go to W/O MWO 50-78 RIGHT AND LEFTLIGHTS DO NOT GO ON WHEN THE SELECTOR SWITCH IS SET TO INBD R AND THE JETT SWITCH IS HELD UP, in this work package.
2. **MWO 50-78** If Indication/Condition is not as specified, go to MWO 50-78 RIGHT AND LEFT INBOARD TEST LIGHTS DO NOT GO ON WHEN THE SELECTOR SWITCH IS SET TO INBD R AND THE JETT SWITCH IS HELD UP, in this work package.

**Step**

49. **W/O MWO 50-78** Disconnect right inboard external fuel tank connector P1 from J1, or if external fuel tank is not installed, remove jumper wire.
50. **MWO 50-78** Disconnect right inboard external fuel tank connector W4P2 from J1, or if external fuel tank is not installed, remove jumper wire installed in step 46.
51. **W/O MWO 50-78** Connect left outboard external fuel tank connector P1 to J1, or if external fuel tank is not installed, install an insulated jumper wire between J1-1 and J1-2.
52. **MWO 50-78** Connect left outboard external fuel tank connector W4P2 to J1, or if external fuel tank is not installed, install an insulated jumper wire between W4P2-C or -E and W4P2-F.
53. Turn control panel selector switch to OUTBD L.
54. Momentarily hold control panel JETT switch up.

**Indication/Condition**

Right and left outboard locally-made test lamps shall go on.

**Corrective Action**

1. **W/O MWO 50-78** If Indication/Condition is not as specified, go to W/O MWO 50-78 RIGHT AND LEFT OUTBOARD TEST LIGHTS DO NOT GO ON WHEN THE SELECTOR SWITCH IS SET TO OUTBD L AND THE JETT SWITCH IS HELD UP, in this work package.
2. **MWO 50-78** If Indication/Condition is not as specified, go to MWO 50-78 RIGHT AND LEFT OUTBOARD TEST LIGHTS DO NOT GO ON WHEN THE SELECTION SWITCH IS SET TO OUTBD L AND THE JETT SWITCH IS HELD UP, in this work package.

**Step**

55. **W/O MWO 50-78** Disconnect left outboard external fuel tank connector P1 from J1, or if external fuel tank is not installed, remove jumper wire.
56. **MWO 50-78** Disconnect left outboard external fuel tank connector W4P2 from J1, or if external fuel tank is not installed, remove jumper wire.
57. **W/O MWO 50-78** Connect left inboard external fuel tank connector P1 to J1, or if external fuel tank is not installed, install an insulated jumper wire between J1-1 and J1-2.
58. **MWO 50-78** Connect left inboard external fuel tank connector W4P2 to J1, or if external fuel tank is not installed, install an insulated jumper wire between W4P2-C or -E and W4P2-F.
59. Turn control panel selector switch to INBD L.

---

**EXTERNAL STORES SUPPORT SYSTEM (ESSS) JETTISON SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE** - Continued

60. Momentarily hold control panel JETT switch up.

**Indication/Condition**

Right and left inboard locally-made test lamps go on.

**Corrective Action**

1. **W/O MWO 50-78** If Indication/Condition is not as specified, go to W/O MWO 50-78 RIGHT AND LEFT INBOARD TEST LIGHTS DO NOT GO ON WHEN THE SELECTOR SWITCH IS SET TO INBD L AND THE JETT SWITCH IS HELD UP, in this work package.
2. **MWO 50-78** If Indication/Condition is not as specified, go to MWO 50-78 RIGHT AND LEFT INBOARD TEST LIGHTS DO NOT GO ON WHEN THE SELECTOR SWITCH IS SET TO INBD L AND THE JETT SWITCH IS HELD UP, in this work package.

**Step**

61. **W/O MWO 50-78** Disconnect left inboard external fuel tank connector P1 from J1, or if external fuel tank is not installed, remove jumper wire.
62. **MWO 50-78** Disconnect left inboard external fuel tank connector W4P2 from J1, or if external fuel tank is not installed, remove jumper wire.
63. Turn control panel selector switch to OFF.
64. Install locally-made drag beam wedge in left drag beam switch. **DO NOT REMOVE LOCALLY-MADE DRAG BEAM WEDGE FROM RIGHT DRAG BEAM SWITCH.**
65. Turn control panel selector switch to ALL.
66. Make sure that control panel EMER JETT ALL and JETT switches are down and their switch guards are down.
67. Check for ac and dc voltage between the following:
  - J705-A and J705-B (left outboard)
  - J707-A and J707-B (left inboard)
  - P709-B and P709-A (left inboard)
  - P709-C and P709-N (left inboard)
  - P706-B and P706-A (right inboard)
  - P706-C and P706-N (right inboard)
  - J708-A and J708-B (right inboard)
  - J710-A and J710-B (right outboard)

**Indication/Condition**

- a. AC voltage shall be -50 to +50 mvac.
- b. DC voltage shall be between -10 and +10 mvdc.

**Corrective Action**

If Indication/Conditions are not as specified, go to MULTIMETER DOES NOT INDICATE LESS THAN 10 MVDC AND 50 MVAC, in this work package.

## EXTERNAL STORES SUPPORT SYSTEM (ESSS) JETTISON SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE - Continued

### Step

68. Remove locally-made drag beam wedges from left and right drag beam switches.

69. Check for ac and dc voltage between the following:

- J705-A and J705-B (left outboard)
- J707-A and J707-B (left inboard)
- P709-B and P709-A (left inboard)
- P709-C and P709-N (left inboard)
- P706-B and P706-A (right inboard)
- P706-C and P706-N (right inboard)
- J708-A and J708-B (right inboard)
- J710-A and J710-B (right outboard)

### Indication/Condition

- a. AC voltage shall be between -50 to +50 mvac.
- b. DC voltage shall be between -10 and +10 mvdc.

### Corrective Action

If Indication/Condition is not as specified, replace control panel (WP 1460 00).

### Step

70. Install locally-made drag beam wedges in both left and right drag beam switches to simulate in-flight condition.

71. Check the following:

| CONNECTOR | BREECH     | RESULT        |
|-----------|------------|---------------|
| P707-A    | Center Pin | Continuity    |
| P707-A    | Shell      | No Continuity |
| P707-B    | Center Pin | No Continuity |
| P707-B    | Shell      | Continuity    |
| P705-A    | Center Pin | Continuity    |
| P705-A    | Shell      | No Continuity |
| P705-B    | Center Pin | No Continuity |
| P705-B    | Shell      | Continuity    |
| P708-A    | Center Pin | Continuity    |

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**EXTERNAL STORES SUPPORT SYSTEM (ESSS) JETTISON SYSTEM OPERATIONAL/**  
**TROUBLESHOOTING PROCEDURE - Continued**

| CONNECTOR | BREECH     | RESULT        |
|-----------|------------|---------------|
| P708-A    | Shell      | No Continuity |
| P708-B    | Center Pin | No Continuity |
| P708-B    | Shell      | Continuity    |
| P710-A    | Center Pin | Continuity    |
| P710-A    | Shell      | No Continuity |
| P710-B    | Center Pin | No Continuity |
| P710-B    | Shell      | Continuity    |

**Indication/Condition**

Results shall be as indicated.

**Corrective Action**

If Indication/Conditions are not as specified, replace breech harness as required (WP 1460 00).

**Step**

72. Turn off electrical power.
73. Disconnect all test equipment.
74. Remove locally-made drag beam wedges from drag beam switches.
75. Remove locally-made test lamps.
76. Remove all jumper wires.
77. Connect all electrical connectors to their receptacles.
78. Install rear fairings on all vertical stores pylons (WP 1428 00).
79. Install explosive cartridges in ejector racks (WP 1445 00).
80. Push in BACKUP HYD CONTR circuit breaker (upper console circuit breaker panel).

**Indication/Condition**

Circuit breaker shall be pushed in.

**Corrective Action**

None Required

**NONE OF THE TEST LIGHTS GO ON WHEN EMER JETT ALL SWITCH IS PLACED UP****SYMPTOM**

None of the test lights go on when EMER JETT ALL switch is placed up.

**MALFUNCTION**

None of the test lights go on when EMER JETT ALL switch is placed up.

**CORRECTIVE ACTION**

- 1. Check continuity between P646-e and ground.**
  - a. If continuity is present, replace stores jettison control panel (WP 1460 00). Go to **9**.
  - b. If continuity is not present, go to **2**.
- 2. Check continuity between J163-D and ground.**
  - a. If continuity is present, go to **3**.
  - b. If continuity is not present, repair/replace ground wire as required (WP 1747 00). Go to **9**.
- 3. Install an insulated jumper wire between J163-F and J163-D. Go to 4.**
- 4. Check continuity between P646-e and ground.**
  - a. If continuity is present, go to **5**.
  - b. If continuity is not present, go to **7**.
- 5. Remove jumper wire from J163. Go to 6.**
- 6. Replace left drag beam switch (WP 0443 00). Go to 9.**
- 7. Remove jumper wire from J163. Go to 8.**
- 8. Repair/replace wiring between P646-e and J163-F (WP 1747 00). Go to 9.**
- 9. Procedure completed.**

**EITHER OUTBOARD TEST LIGHT DOES NOT GO ON WHEN EMER JETT ALL SWITCH IS PLACED UP****SYMPTOM**

Either outboard test light does not go on when EMER JETT ALL switch is placed up.

**MALFUNCTION**

Either outboard test light does not go on when EMER JETT ALL switch is placed up.

**CORRECTIVE ACTION**

- 1. Does right outboard test light go on?**
  - a. If test light is on, go to **2**.
  - b. If test light is not on, go to **6**.
- 2. Check for 28 vdc between J705-A and ground.**
  - a. If voltage is as specified, repair ground wire or replace harness as required (WP 1747 00). Go to **13**.
  - b. If voltage is not as specified, go to **3**.
- 3. Check for 28 vdc between P680-8 and ground.**

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**EITHER OUTBOARD TEST LIGHT DOES NOT GO ON WHEN EMER JETT ALL SWITCH IS PLACED UP -**  
Continued

- a. If voltage is as specified, replace harness (WP 1462 00). Go to **13**.
  - b. If voltage is not as specified, go to **4**.
- 4. Check for 28 vdc between J661-M and ground.**
  - a. If voltage is as specified, replace harness (WP 1462 00). Go to **13**.
  - b. If voltage is not as specified, go to **5**.
- 5. Check continuity between J661-M and P646-G.**
  - a. If continuity is present, replace stores jettison control panel (WP 1460 00). Go to **13**.
  - b. If continuity is not present, replace harness (WP 1462 00). Go to **13**.
- 6. Check for 28 vdc between J710-A and ground.**
  - a. If voltage is as specified, repair ground wire or replace harness as required (WP 1747 00). Go to **13**.
  - b. If voltage is not as specified, go to **7**.
- 7. Check for 28 vdc between P677-1 and ground.**
  - a. If voltage is as specified, replace harness (WP 1462 00). Go to **13**.
  - b. If voltage is not as specified, go to **8**.
- 8. Check for 28 vdc between J660-A and ground.**
  - a. If voltage is as specified, replace harness (WP 1462 00). Go to **13**.
  - b. If voltage is not as specified, go to **9**.
- 9. Check continuity between J660-A and P645-R.**
  - a. If continuity is present, replace stores jettison control panel (WP 1460 00). Go to **13**.
  - b. If continuity is not present, go to **10**.
- 10. Check continuity between P660-A and P677-1.**
  - a. If continuity is present, go to **11**.
  - b. If continuity is not present, replace harness (WP 1462 00). Go to **13**.
- 11. Check continuity between P677-1 and P679-12.**
  - a. If continuity is present, go to **12**.
  - b. If continuity is not present, replace harness (WP 1462 00). Go to **13**.
- 12. Check continuity between P679-12 and P646-R.**
  - a. If continuity is present, replace stores jettison control panel (WP 1460 00). Go to **13**.
  - b. If continuity is not present, replace harness (WP 1462 00). Go to **13**.
- 13. Procedure completed.**



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**NONE OF THE TEST LIGHTS GO ON WHEN SELECTOR SWITCH IS SET TO ALL AND JETT SWITCH IS HELD UP****SYMPTOM**

None of the test lights go on when selector switch is set to ALL and JETT switch is held up.

**MALFUNCTION**

None of the test lights go on when selector switch is set to ALL and JETT switch is held up.

**CORRECTIVE ACTION****1. Check continuity between P645-k and ground.**

- a. If continuity is present, replace stores jettison control panel (WP 1460 00). Go to **4**.
- b. If continuity is not present, go to **2**.

**2. Check continuity between J164-D and ground.**

- a. If continuity is present, go to **3**.
- b. If continuity is not present, repair/replace ground wire as required (WP 1747 00). Go to **4**.

**3. Check continuity between J164-F and P645-k.**

- a. If continuity is present, adjust or replace right drag beam switch (WP 0443 00). Go to **4**.
- b. If continuity is not present, repair/replace ground wire as required (WP 1747 00). Go to **4**.

**4. Procedure completed.****EITHER OUTBOARD TEST LIGHT DOES NOT GO ON WHEN SELECTOR SWITCH IS SET TO ALL AND JETT SWITCH IS HELD UP****SYMPTOM**

Either outboard test light does not go on when selector switch is set to ALL and JETT switch is held up.

**MALFUNCTION**

Either outboard test light does not go on when selector switch is set to ALL and JETT switch is held up.

**CORRECTIVE ACTION****1. Does right outboard test light go on?**

- a. If light is on, go to **2**.
- b. If light is not on, go to **5**.

**2. Check for 28 vdc between P678-3 and ground.**

- a. If voltage is as specified, replace harness (WP 1462 00). Go to **8**.
- b. If voltage is not as specified, go to **3**.

**3. Check for 28 vdc between J660-L and ground.**

- a. If voltage is as specified, replace harness (WP 1462 00). Go to **8**.
- b. If voltage is not as specified, go to **4**.

**4. Check continuity between J660-L and P645-G.**

- a. If continuity is present, replace stores jettison control panel (WP 1460 00). Go to **8**.

---

**EITHER OUTBOARD TEST LIGHT DOES NOT GO ON WHEN SELECTOR SWITCH IS SET TO ALL AND JETT SWITCH IS HELD UP** - Continued

- b. If continuity is not present, replace harness (WP 1462 00). Go to **8**.
- 5. Check for 28 vdc between P677-1 and ground.**
  - a. If voltage is as specified, replace harness (WP 1462 00). Go to **8**.
  - b. If voltage is not as specified, go to **6**.
- 6. Check for 28 vdc between J660-A and ground.**
  - a. If voltage is as specified, replace harness (WP 1462 00). Go to **8**.
  - b. If voltage is not as specified, go to **7**.
- 7. Check continuity between J660-A and P645-R.**
  - a. If continuity is present, replace stores jettison control panel (WP 1460 00). Go to **8**.
  - b. If continuity is not present, replace harness (WP 1462 00). Go to **8**.
- 8. Procedure completed.**

**W/O MWO 50-78 RIGHT AND LEFT OUTBOARD TEST LIGHTS DO NOT GO ON WHEN THE SELECTOR SWITCH IS SET TO OUTBD R AND THE JETT SWITCH IS HELD UP****SYMPTOM**

**W/O MWO 50-78** Right and left outboard test lights do not go on when the selector switch is set to OUTBD R and the JETT switch is held up.

**MALFUNCTION**

**W/O MWO 50-78** Right and left outboard test lights do not go on when the selector switch is set to OUTBD R and the JETT switch is held up.

**CORRECTIVE ACTION**

- 1. Check continuity between P645-i and ground.**
  - a. If continuity is present, replace stores jettison control panel (WP 1460 00). Go to **7**.
  - b. If continuity is not present, go to **2**.
- 2. Is jumper wire being used in place of right outboard fuel tank?**
  - a. If jumper wire is being used, go to **3**.
  - b. If jumper wire is not being used, go to **5**.
- 3. Remove jumper wire. Go to 4.**
- 4. Check continuity between J1-1 and ground.**
  - a. If continuity is present, troubleshoot open wiring between J1-2 and P645-i. Replace harness as required (WP 1462 00). Go to **7**.
  - b. If continuity is not present, troubleshoot open wiring between J1-1 and ground. Replace harness as required (WP 1462 00). Go to **7**.
- 5. Check continuity between right outboard fuel tank P1-1 and P1-2.**
  - a. If continuity is present, go to **6**.
  - b. If continuity is not present, replace fuel tank (WP 1465 00). Go to **7**.

---

**W/O MWO 50-78 RIGHT AND LEFT OUTBOARD TEST LIGHTS DO NOT GO ON WHEN THE SELECTOR SWITCH IS SET TO OUTBD R AND THE JETT SWITCH IS HELD UP - Continued****6. Check continuity between J1-1 and ground.**

- a. If continuity is present, troubleshoot open wiring between J1-2 and P645-j. Replace harness as required (WP 1462 00). Go to **7**.
- b. If continuity is not present, troubleshoot open wiring between J1-1 and ground. Replace harness as required (WP 1462 00). Go to **7**.

**7. Procedure completed.****MWO 50-78 RIGHT AND LEFT OUTBOARD TEST LIGHTS DO NOT GO ON WHEN THE SELECTOR SWITCH IS SET TO OUTBD R AND THE JETT SWITCH IS HELD UP****SYMPTOM**

**MWO 50-78** Right and left outboard test lights do not go on when the selector switch is set to OUTBD R and the JETT switch is held up.

**MALFUNCTION**

**MWO 50-78** Right and left outboard test lights do not go on when the selector switch is set to OUTBD R and the JETT switch is held up.

**CORRECTIVE ACTION****1. Check continuity between P645-j and ground.**

- a. If continuity is present, replace stores jettison control panel (WP 1460 00). Go to **7**.
- b. If continuity is not present, go to **2**.

**2. Is jumper wire being used in place of right outboard fuel tank?**

- a. If jumper wire is being used, go to **3**.
- b. If jumper wire is not being used, go to **5**.

**3. Remove jumper wire. Go to 4.****4. Check continuity between right outboard fuel tank W4P2-F and ground.**

- a. If continuity is present, go to **5**.
- b. If continuity is not present, troubleshoot open wiring between W4P2-F and P856-32. Replace harness as required (WP 1462 00). Go to **7**.

**5. Check continuity between right outboard fuel tank:**

- **W4P2-C and P856-31**
- **W4P2-E and P856-33**
- **P857-1 and P645-j**

- a. If continuity is present, go to **6**.
- b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **7**.

**6. Check continuity between right outboard fuel tank:**

- **W4P1-C and W4P1-F**
- **W4P1-D and W4P1-F**

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**MWO 50-78 RIGHT AND LEFT OUTBOARD TEST LIGHTS DO NOT GO ON WHEN THE SELECTOR SWITCH IS SET TO OUTBD R AND THE JETT SWITCH IS HELD UP - Continued**

- a. If continuity is present, replace auxiliary fuel management panel (WP 1448 00). Go to **7**.
- b. If continuity is not present, replace fuel tank or harness as required (WP 1465 00). Go to **7**.

**7. Procedure completed.****W/O MWO 50-78 RIGHT AND LEFT INBOARD TEST LIGHTS DO NOT GO ON WHEN THE SELECTOR SWITCH IS SET TO INBD R AND THE JETT SWITCH IS HELD UP****SYMPTOM**

**W/O MWO 50-78** Right and left inboard test lights do not go on when the selector switch is set to INBD R and the JETT switch is held up.

**MALFUNCTION**

**W/O MWO 50-78** Right and left inboard test lights do not go on when the selector switch is set to INBD R and the JETT switch is held up.

**CORRECTIVE ACTION****1. Check continuity between P645-g and ground.**

- a. If continuity is present, replace stores jettison control panel (WP 1460 00). Go to **7**.
- b. If continuity is not present, go to **2**.

**2. Is jumper wire being used in place of right inboard fuel tank?**

- a. If jumper wire is being used, go to **3**.
- b. If jumper wire is not being used, go to **5**.

**3. Remove jumper wire. Go to 4.****4. Check continuity between J1-1 and ground.**

- a. If continuity is present, troubleshoot open wiring between J1-2 and P645-g. Replace harness as required (WP 1462 00). Go to **7**.
- b. If continuity is not present, troubleshoot open wiring between J1-1 and ground. Replace harness as required (WP 1462 00). Go to **7**.

**5. Check continuity between right inboard fuel tank P1-1 and P1-2.**

- a. If continuity is present, go to **6**.
- b. If continuity is not present, replace fuel tank (WP 1465 00). Go to **7**.

**6. Check continuity between J1-1 and ground.**

- a. If continuity is present, troubleshoot open wiring between J1-2 and P645-g. Replace harness as required (WP 1462 00). Go to **7**.
- b. If continuity is not present, troubleshoot open wiring between J1-1 and ground. Replace harness as required (WP 1462 00). Go to **7**.

**7. Procedure completed.**

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**MWO 50-78** RIGHT AND LEFT INBOARD TEST LIGHTS DO NOT GO ON WHEN THE SELECTOR SWITCH IS SET TO INBD R AND THE JETT SWITCH IS HELD UP

**SYMPTOM**

**MWO 50-78** Right and left inboard test lights do not go on when the selector switch is set to INBD R and the JETT switch is held up.

**MALFUNCTION**

**MWO 50-78** Right and left inboard test lights do not go on when the selector switch is set to INBD R and the JETT switch is held up.

**CORRECTIVE ACTION**

1. Check continuity between P645-g and ground.
  - a. If continuity is present, replace stores jettison control panel (WP 1460 00). Go to 7.
  - b. If continuity is not present, go to 2.
2. Is jumper wire being used in place of right inboard fuel tank?
  - a. If jumper wire is being used, go to 3.
  - b. If jumper wire is not being used, go to 5.
3. Remove jumper wire. Go to 4.
4. Check continuity between right inboard fuel tank W4P2-F and ground.
  - a. If continuity is present, go to 5.
  - b. If continuity is not present, troubleshoot open wiring between W4P2-F and P856-35. Replace harness as required (WP 1462 00). Go to 7.
5. Check continuity between right inboard fuel tank:
  - W4P2-C and P856-34
  - W4P2-E and P856-36
  - P857-2 and P645-g
  - a. If continuity is present, go to 6.
  - b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to 7.
6. Check continuity between right inboard fuel tank:
  - W4P1-C and W4P1-F
  - W4P1-D and W4P1-F
  - a. If continuity is present, replace auxiliary fuel management panel (WP 1448 00). Go to 7.
  - b. If continuity is not present, replace fuel tank or harness as required (WP 1465 00). Go to 7.
7. Procedure completed.

**W/O MWO 50-78** RIGHT AND LEFT OUTBOARD TEST LIGHTS DO NOT GO ON WHEN THE SELECTOR SWITCH IS SET TO OUTBD L AND THE JETT SWITCH IS HELD UP

#### SYMPTOM

**W/O MWO 50-78** Right and left outboard test lights do not go on when the selector switch is set to OUTBD L and the JETT switch is held up.

#### MALFUNCTION

**W/O MWO 50-78** Right and left outboard test lights do not go on when the selector switch is set to OUTBD L and the JETT switch is held up.

#### CORRECTIVE ACTION

- 1. Check continuity between P645-j and ground.**
  - a. If continuity is present, replace stores jettison control panel (WP 1460 00). Go to **7**.
  - b. If continuity is not present, go to **2**.
- 2. Is jumper wire being used in place of left outboard fuel tank?**
  - a. If jumper wire is being used, go to **3**.
  - b. If jumper wire is not being used, go to **5**.
- 3. Remove jumper wire. Go to 4.**
- 4. Check continuity between J1-1 and ground.**
  - a. If continuity is present, troubleshoot open wiring between J1-2 and P645-j. Replace harness as required (WP 1462 00). Go to **7**.
  - b. If continuity is not present, troubleshoot open wiring between J1-1 and ground. Replace harness as required (WP 1462 00). Go to **7**.
- 5. Check continuity between left outboard fuel tank P1-1 and P1-2.**
  - a. If continuity is present, go to **6**.
  - b. If continuity is not present, replace fuel tank (WP 1465 00). Go to **7**.
- 6. Check continuity between J1-1 and ground.**
  - a. If continuity is present, troubleshoot open wiring between J1-2 and P645-j. Replace harness as required (WP 1462 00). Go to **7**.
  - b. If continuity is not present, troubleshoot open wiring between J1-1 and ground. Replace harness as required (WP 1462 00). Go to **7**.
- 7. Procedure completed.**

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**MWO 50-78 RIGHT AND LEFT OUTBOARD TEST LIGHTS DO NOT GO ON WHEN THE SELECTOR SWITCH IS SET TO OUTBD L AND THE JETT SWITCH IS HELD UP**

**SYMPTOM**

**MWO 50-78** Right and left outboard test lights do not go on when the selector switch is set to OUTBD L and the JETT switch is held up.

**MALFUNCTION**

**MWO 50-78** Right and left outboard test lights do not go on when the selector switch is set to OUTBD L and the JETT switch is held up.

**CORRECTIVE ACTION**

- 1. Check continuity between P645-g and ground.**
  - a. If continuity is present, replace stores jettison control panel (WP 1460 00). Go to **7**.
  - b. If continuity is not present, go to **2**.
- 2. Is jumper wire being used in place of right inboard fuel tank?**
  - a. If jumper wire is being used, go to **3**.
  - b. If jumper wire is not being used, go to **5**.
- 3. Remove jumper wire. Go to 4.**
- 4. Check continuity between left outboard fuel tank W4P2-F and ground.**
  - a. If continuity is present, go to **5**.
  - b. If continuity is not present, troubleshoot open wiring between W4P2-F and P856-41. Replace harness as required (WP 1462 00). Go to **7**.
- 5. Check continuity between left outboard fuel tank:**
  - **W4P2-C and P856-40**
  - **W4P2-E and P856-42**
  - **P857-18 and P645-j**
  - a. If continuity is present, go to **6**.
  - b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **7**.
- 6. Check continuity between left outboard fuel tank:**
  - **W4P1-C and W4P1-F**
  - **W4P1-D and W4P1-F**
  - a. If continuity is present, replace auxiliary fuel management panel (WP 1448 00). Go to **7**.
  - b. If continuity is not present, replace fuel tank or harness as required (WP 1465 00). Go to **7**.
- 7. Procedure completed.**

**W/O MWO 50-78** RIGHT AND LEFT INBOARD TEST LIGHTS DO NOT GO ON WHEN THE SELECTOR SWITCH IS SET TO INBD L AND THE JETT SWITCH IS HELD UP

#### SYMPTOM

**W/O MWO 50-78** Right and left inboard test lights do not go on when the selector switch is set to INBD L and the JETT switch is held up.

#### MALFUNCTION

**W/O MWO 50-78** Right and left inboard test lights do not go on when the selector switch is set to INBD L and the JETT switch is held up.

#### CORRECTIVE ACTION

- 1. Check continuity between P645-h and ground.**
  - a. If continuity is present, replace stores jettison control panel (WP 1460 00). Go to **7**.
  - b. If continuity is not present, go to **2**.
- 2. Is jumper wire being used in place of left inboard fuel tank?**
  - a. If jumper wire is being used, go to **3**.
  - b. If jumper wire is not being used, go to **5**.
- 3. Remove jumper wire. Go to 4.**
- 4. Check continuity between J1-1 and ground.**
  - a. If continuity is present, troubleshoot open wiring between J1-2 and P645-h. Replace harness as required (WP 1462 00). Go to **7**.
  - b. If continuity is not present, troubleshoot open wiring between J1-1 and ground. Replace harness as required (WP 1462 00). Go to **7**.
- 5. Check continuity between left inboard fuel tank P1-1 and P1-2.**
  - a. If continuity is present, go to **6**.
  - b. If continuity is not present, replace fuel tank (WP 1465 00). Go to **7**.
- 6. Check continuity between J1-1 and ground.**
  - a. If continuity is present, troubleshoot open wiring between J1-2 and P645-h. Replace harness as required (WP 1462 00). Go to **7**.
  - b. If continuity is not present, troubleshoot open wiring between J1-1 and ground. Replace harness as required (WP 1462 00). Go to **7**.
- 7. Procedure completed.**



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**MWO 50-78 RIGHT AND LEFT INBOARD TEST LIGHTS DO NOT GO ON WHEN THE SELECTOR SWITCH IS SET TO INBD L AND THE JETT SWITCH IS HELD UP**

**SYMPTOM**

**MWO 50-78** Right and left inboard test lights do not go on when the selector switch is set to INBD L and the JETT switch is held up.

**MALFUNCTION**

**MWO 50-78** Right and left inboard test lights do not go on when the selector switch is set to INBD L and the JETT switch is held up.

**CORRECTIVE ACTION**

- 1. Check continuity between P645-h and ground.**
  - a. If continuity is present, replace stores jettison control panel (WP 1460 00). Go to **7**.
  - b. If continuity is not present, go to **2**.
- 2. Is jumper wire being used in place of left inboard fuel tank?**
  - a. If jumper wire is being used, go to **3**.
  - b. If jumper wire is not being used, go to **5**.
- 3. Remove jumper wire. Go to 4.**
- 4. Check continuity between left inboard fuel tank W4P2-F and ground.**
  - a. If continuity is present, go to **5**.
  - b. If continuity is not present, troubleshoot open wiring between W4P2-F and P856-38. Replace harness as required (WP 1462 00). Go to **7**.
- 5. Check continuity between left inboard fuel tank:**
  - **W4P2-C and P856-37**
  - **W4P2-E and P856-39**
  - **P857-19 and P645-h**
  - a. If continuity is present, go to **6**.
  - b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **7**.
- 6. Check continuity between left inboard fuel tank:**
  - **W4P1-C and W4P1-F**
  - **W4P1-D and W4P1-F**
  - a. If continuity is present, replace auxiliary fuel management panel (WP 1448 00). Go to **7**.
  - b. If continuity is not present, replace fuel tank or harness as required (WP 1465 00). Go to **7**.
- 7. Procedure completed.**

**MULTIMETER DOES NOT INDICATE LESS THAN 10 MVDC AND 50 MVAC****SYMPTOM**

Multimeter does not indicate less than 10 mvdc and 50 mvac.

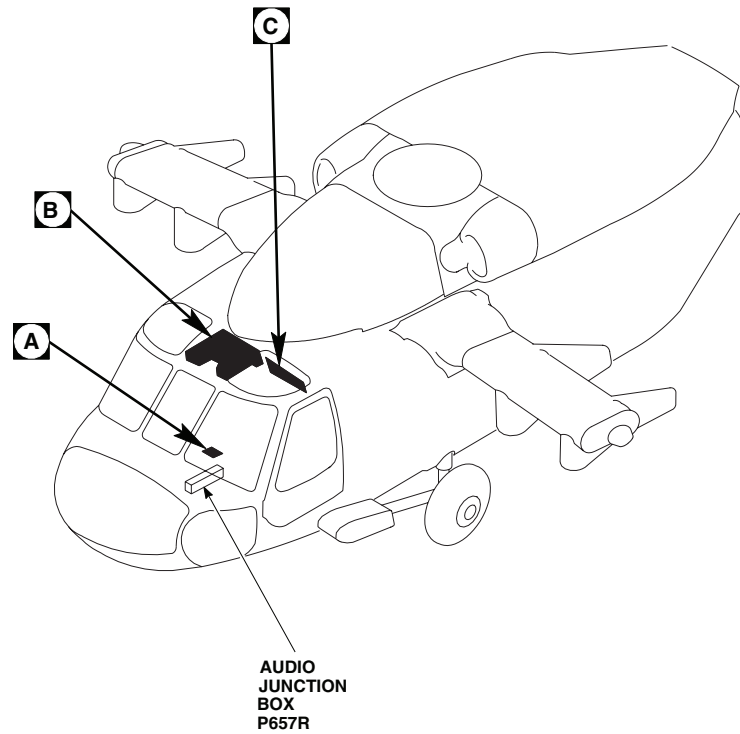
**MALFUNCTION**

Multimeter does not indicate less than 10 mvdc and 50 mvac.

**CORRECTIVE ACTION**

- 1. Check associated electrical circuit bonding jumpers for security and integrity. Go to 2.**
- 2. Ensure that bonding jumpers and mounting surfaces are prepared for good electrical bonding.**
  - a. If jumpers and mounting surfaces are prepared, go to **3**.
  - b. If jumpers and mounting surfaces are not prepared, repair bonding jumper or replace harness as required (WP 1462 00). Go to **8**.
- 3. Check associated electrical circuit harnesses for damage to alphabraid shielding.**
  - a. If associated electrical circuit harness is damaged, replace malfunctioning harness (WP 1463 00). Go to **8**.
  - b. If associated electrical circuit harness is not damaged, go to **4**.
- 4. Disconnect P645 and P646 from stores jettison control panel. Is voltage still greater than 10 mvdc or 50 mvac?**
  - a. If voltage is still greater than 10 mvdc or 50 mvac, go to **5**.
  - b. If voltage is not still greater than 10 mvdc or 50 mvac, replace stores jettison control panel (WP 1460 00). Go to **8**.
- 5. Starting from control panel, sequentially disconnect harness leading to the ejector rack disconnect where 10 mvdc or 50 mvac is exceeded. Go to 6.**
- 6. After each harness is disconnected, recheck voltage with multimeter. Go to 7.**
- 7. When voltage is less than 10 mvdc or 50 mvac , last harness disconnected is known to be malfunctioning. Repair/replace harness as required (WP 1747 00). Go to 8.**
- 8. Procedure completed.**

## LOCATION



## NOTES

1. UH-60A UH-60L HH-60A   
HH-60L
2. EH-60A
3. W / O MWO 50-78
4. MWO 50-78

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Figure 1. ESSS Stores Jettison System Location Diagram. (Sheet 1 of 4)

## LOCATION - Continued

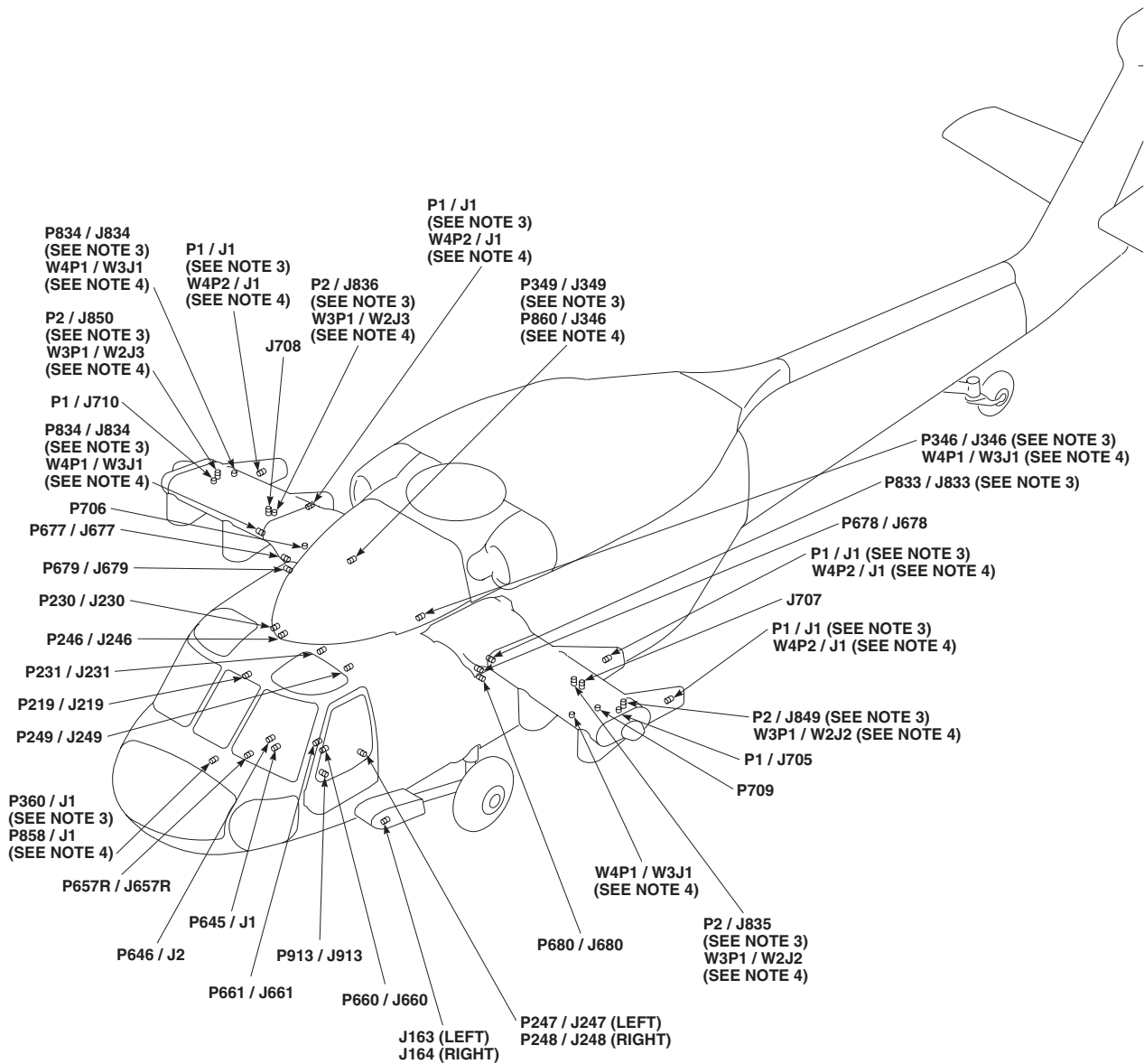
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Figure 1. ESSS Stores Jettison System Location Diagram. (Sheet 2 of 4)

## LOCATION - Continued

| TERMINAL BOARD /<br>DISCONNECT PLUG /<br>RECEPTACLE | LOCATION /<br>CONNECTION POINT                         |
|-----------------------------------------------------|--------------------------------------------------------|
| P1 / J1<br>(SEE NOTE 3)                             | LEFT INBOARD FUEL TANK<br>CONNECTOR                    |
| P1 / J1<br>(SEE NOTE 3)                             | LEFT OUTBOARD FUEL TANK<br>CONNECTOR                   |
| P1 / J1<br>(SEE NOTE 3)                             | RIGHT INBOARD FUEL<br>TANK CONNECTOR                   |
| P1 / J1<br>(SEE NOTE 3)                             | RIGHT OUTBOARD FUEL<br>TANK CONNECTOR                  |
| J707                                                | LEFT INBOARD EJECTOR<br>RACK CONNECTOR                 |
| J708                                                | RIGHT INBOARD EJECTOR<br>RACK CONNECTOR                |
| P1 / J705                                           | LEFT OUTBOARD EJECTOR<br>RACK CONNECTOR                |
| P1 / J710                                           | RIGHT OUTBOARD EJECTOR<br>RACK CONNECTOR               |
| P2 / J835<br>(SEE NOTE 3)                           | LEFT INBOARD FUEL TANK<br>DISCONNECT                   |
| P2 / J836<br>(SEE NOTE 3)                           | RIGHT INBOARD FUEL TANK<br>DISCONNECT                  |
| P2 / J849<br>(SEE NOTE 3)                           | LEFT OUTBOARD FUEL TANK<br>DISCONNECT                  |
| P2 / J850<br>(SEE NOTE 3)                           | RIGHT OUTBOARD FUEL<br>TANK DISCONNECT                 |
| P103R                                               | TRANSPONDER<br>COMPUTER                                |
| P163 / J163                                         | LEFT DRAG BEAM SWITCH                                  |
| P164 / J164                                         | RIGHT DRAG BEAM SWITCH                                 |
| P219 / J219                                         | CABIN FLOOR, BL 25 RH,<br>STA 247                      |
| P230 / J230                                         | CABIN CEILING, BL 8 RH,<br>STA 247                     |
| P231 / J231                                         | CABIN CEILING, BL 9 LH,<br>STA 247                     |
| P246 / J246                                         | CABIN CEILING, BL 5 RH                                 |
| P247 / J247                                         | CABIN, BL 40 LH, STA 248                               |
| P248 / J248                                         | CABIN, BL 40 RH, STA 248                               |
| P249 / J249                                         | BEHIND COPILOT'S<br>CIRCUIT BREAKER PANEL,<br>BL 23 LH |
| P346 / J346<br>(SEE NOTE 3)                         | CABIN CEILING, BL 20 LH,<br>STA 304                    |
| P349 / J349<br>(SEE NOTE 3)                         | CABIN CEILING, BL 20 RH,<br>STA 304                    |
| P360 / J1<br>(SEE NOTE 3)                           | AUXILIARY FUEL<br>MANAGEMENT CONTROL<br>PANEL          |
| P645 / J1                                           | STORES JETTISON<br>CONTROL PANEL                       |
| P646 / J2                                           | STORES JETTISON<br>CONTROL PANEL                       |
| P657R / J657R                                       | AVIONICS JUNCTION BOX<br>ASSEMBLY                      |

| TERMINAL BOARD /<br>DISCONNECT PLUG /<br>RECEPTACLE | LOCATION /<br>CONNECTION POINT                  |
|-----------------------------------------------------|-------------------------------------------------|
| P660 / J660                                         | LEFT SIDE LOWER CON-<br>SOLE, BL 11 LH, STA 244 |
| P661 / J661                                         | LEFT SIDE LOWER CON-<br>SOLE, BL 11 LH, STA 244 |
| P677 / J677                                         | ATTACHMENT FITTING,<br>BL 59 RH, STA 291        |
| P678 / J678                                         | ATTACHMENT FITTING,<br>BL 59 LH, STA 289        |
| P679 / J679                                         | ATTACHMENT FITTING,<br>BL 59 RH, STA 289        |
| P680 / J680                                         | ATTACHMENT FITTING,<br>BL 59 LH, STA 291        |
| P706                                                | RIGHT INBOARD EJECTOR<br>RACK CONNECTOR         |
| P709 / J1                                           | LEFT INBOARD EJECTOR<br>RACK CONNECTOR          |
| P833 / J833<br>(SEE NOTE 3)                         | ATTACHMENT FITTING,<br>BL 59 LH, STA 295        |
| P834 / J834<br>(SEE NOTE 3)                         | ATTACHMENT FITTING,<br>BL 59 RH, STA 295        |
| P856 / J1                                           | AUXILIARY FUEL<br>MANAGEMENT CONTROL<br>PANEL   |
| P860 / J346                                         | CABIN CEILING, BL 20 LH,<br>STA 304             |
| P861 / W2J1<br>(SEE NOTE 4)                         | ATTACHMENT FITTING,<br>BL 59 LH, STA 295        |
| P862 / W2J1<br>(SEE NOTE 4)                         | ATTACHMENT FITTING,<br>BL 59 RH, STA 295        |
| P913 / J913                                         | COCKPIT, BL 16 LH, STA 247                      |
| W3P1 / W2J2<br>(SEE NOTE 4)                         | LEFT INBOARD FUEL TANK<br>DISCONNECT            |
| W3P1 / W2J3<br>(SEE NOTE 4)                         | RIGHT INBOARD FUEL TANK<br>DISCONNECT           |
| W3P1 / W2J2<br>(SEE NOTE 4)                         | LEFT OUTBOARD FUEL<br>TANK DISCONNECT           |
| W3P1 / W2J3<br>(SEE NOTE 4)                         | RIGHT OUTBOARD FUEL<br>TANK DISCONNECT          |
| W4P1 / W3J1<br>(SEE NOTE 4)                         | LEFT INBOARD FUEL TANK<br>DISCONNECT            |
| W4P1 / W3J1<br>(SEE NOTE 4)                         | RIGHT INBOARD FUEL TANK<br>DISCONNECT           |
| W4P1 / W3J1<br>(SEE NOTE 4)                         | LEFT OUTBOARD FUEL<br>TANK DISCONNECT           |
| W4P1 / W3J1<br>(SEE NOTE 4)                         | RIGHT OUTBOARD FUEL<br>TANK DISCONNECT          |
| W4P2 / J1<br>(SEE NOTE 4)                           | LEFT INBOARD FUEL TANK<br>CONNECTOR             |
| W4P2 / J1<br>(SEE NOTE 4)                           | LEFT OUTBOARD FUEL<br>TANK CONNECTOR            |
| W4P2 / J1<br>(SEE NOTE 4)                           | RIGHT INBOARD FUEL<br>TANK CONNECTOR            |
| W4P2 / J1<br>(SEE NOTE 4)                           | RIGHT OUTBOARD FUEL<br>TANK CONNECTOR           |

AA3448 3B  
SA

Figure 1. ESSS Stores Jettison System Location Diagram. (Sheet 3 of 4)

LOCATION - Continued

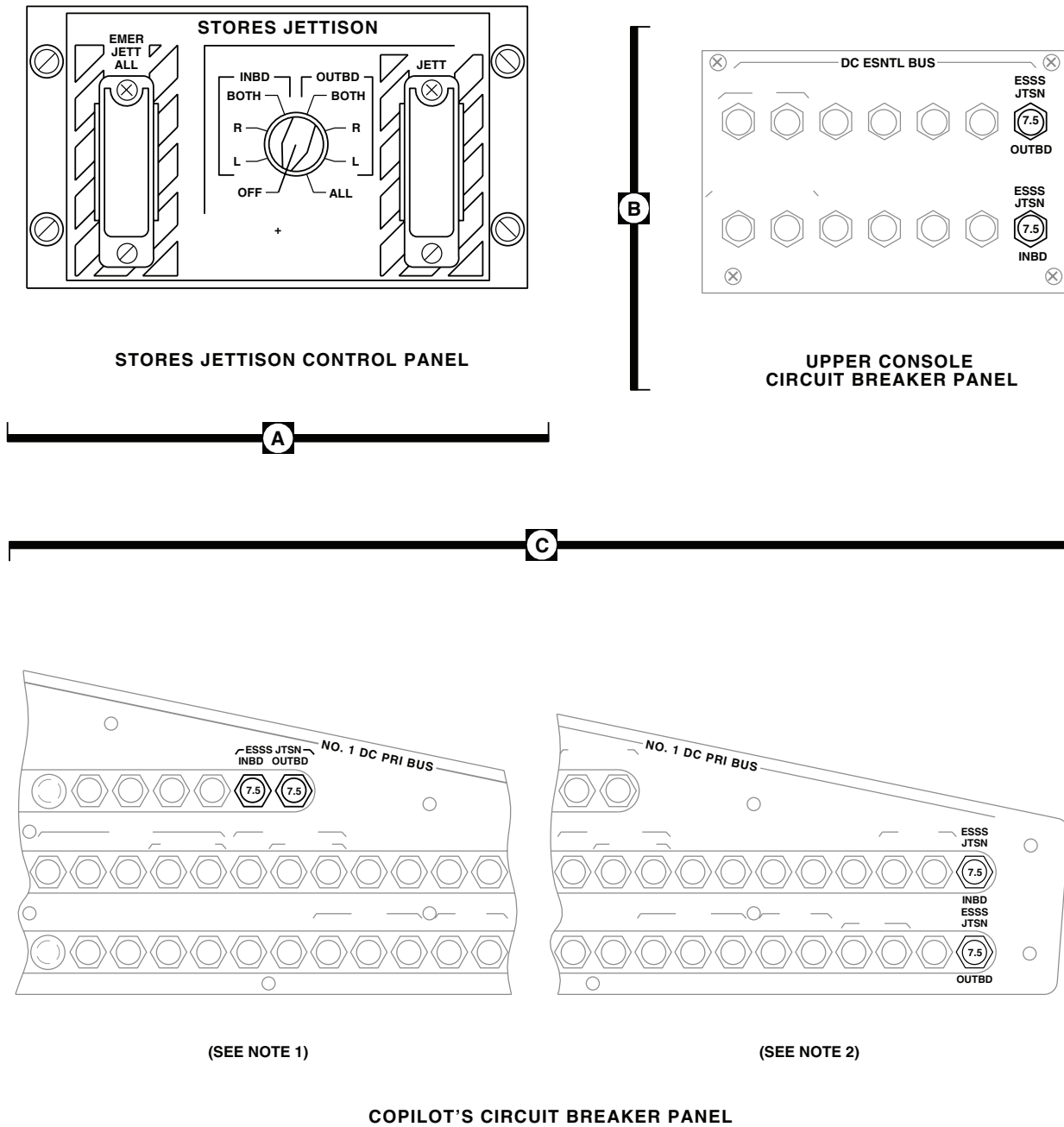
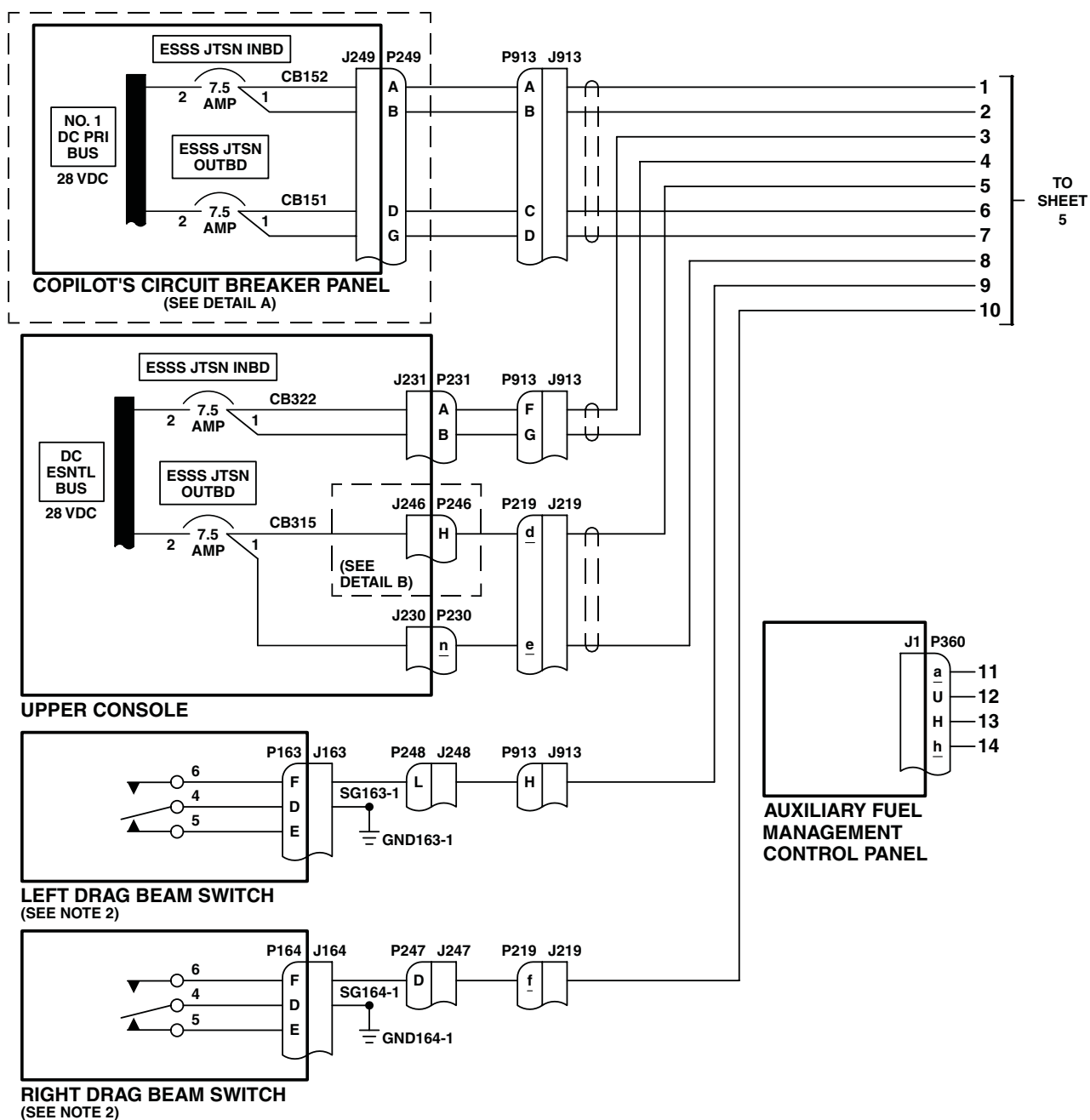


Figure 1. ESSS Stores Jettison System Location Diagram. (Sheet 4 of 4)

## SCHEMATICS



## NOTES

1. **EH60A**
2. BOTH DRAG BEAM SWITCHES SHOWN WITH WEIGHT-ON-WHEELS.
3. WIRING TO ALTERNATE INBOARD EJECTOR RACK BRU22A / A CONNECTOR IS PROVISIONAL.
4. **MWO 50-56**

## EFFECTIVITY

W / O MWO 50-78

AB2199\_1A  
SA

Figure 2. ESSS Stores Jettison System Schematic Diagram. (Sheet 1 of 8)

**LEFT OUTBOARD FUEL TANK**

**LEFT INBOARD FUEL TANK**

**RIGHT OUTBOARD FUEL TANK**

**RIGHT INBOARD FUEL TANK**

P1 1 2  
J1

P2 1 2  
J849

J833 6 7 1 2  
P833

SGP346-5

SGP346-6

SGP346-7

SGP346-8

P346 44 6 42 32  
J346

GND346-3

GND346-4

P346 43 41 7 5  
J346

P913 M L K J  
J913 15 16 17 18

**TO SHEET 6**

**MAJOR CHANGE**

**COPILOT'S CIRCUIT BREAKER PANEL**

ESSS JTSN INBD

ESSS JTSN OUTBD

NO. 1 DC PRI BUS  
28 VDC

7.5 AMP

7.5 AMP

CB126

CB101

J249 A B D G  
P249

**DETAIL A**  
(SEE NOTE 1)

**UPPER CONSOLE**

J246 h  
P246

**DETAIL B**  
(SEE NOTE 1)

**ACTIVITY**

**MWO 50-78**

**AB2199\_2A**

### Change 1



## SCHEMATICS - Continued

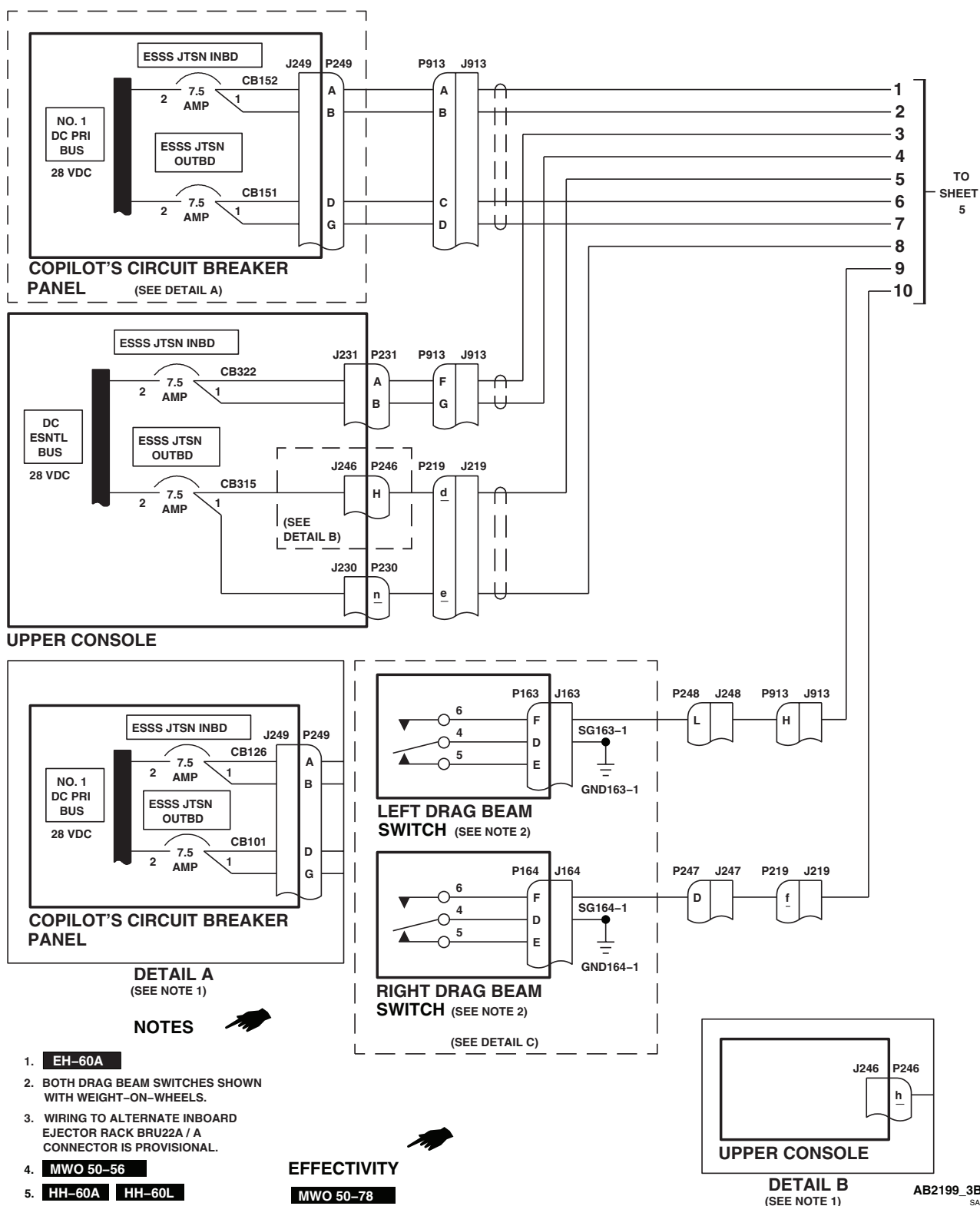


Figure 2. ESSS Stores Jettison System Schematic Diagram. (Sheet 3 of 8)

## SCHEMATICS - Continued

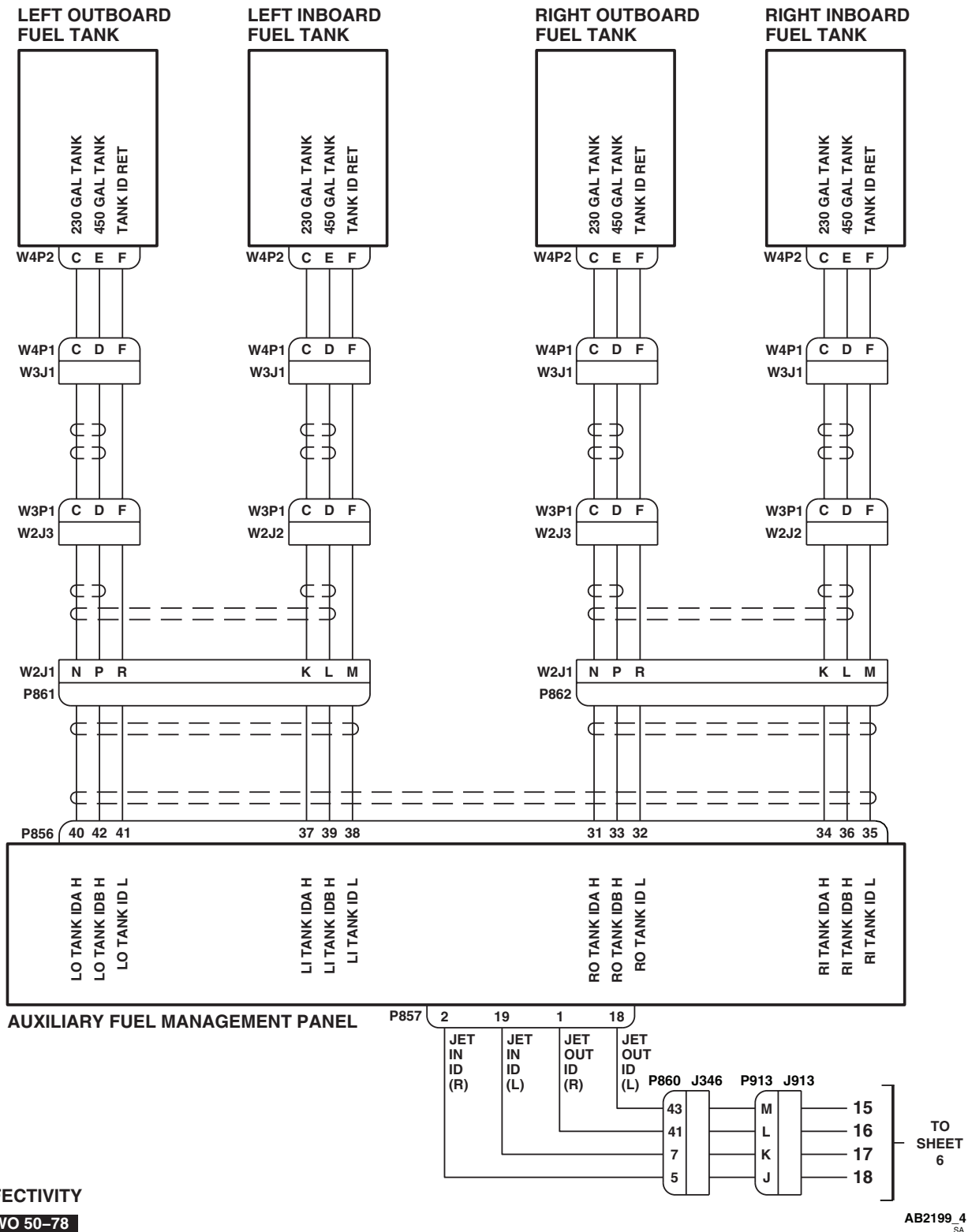


Figure 2. ESSS Stores Jettison System Schematic Diagram. (Sheet 4 of 8)

SCHEMATICS - Continued

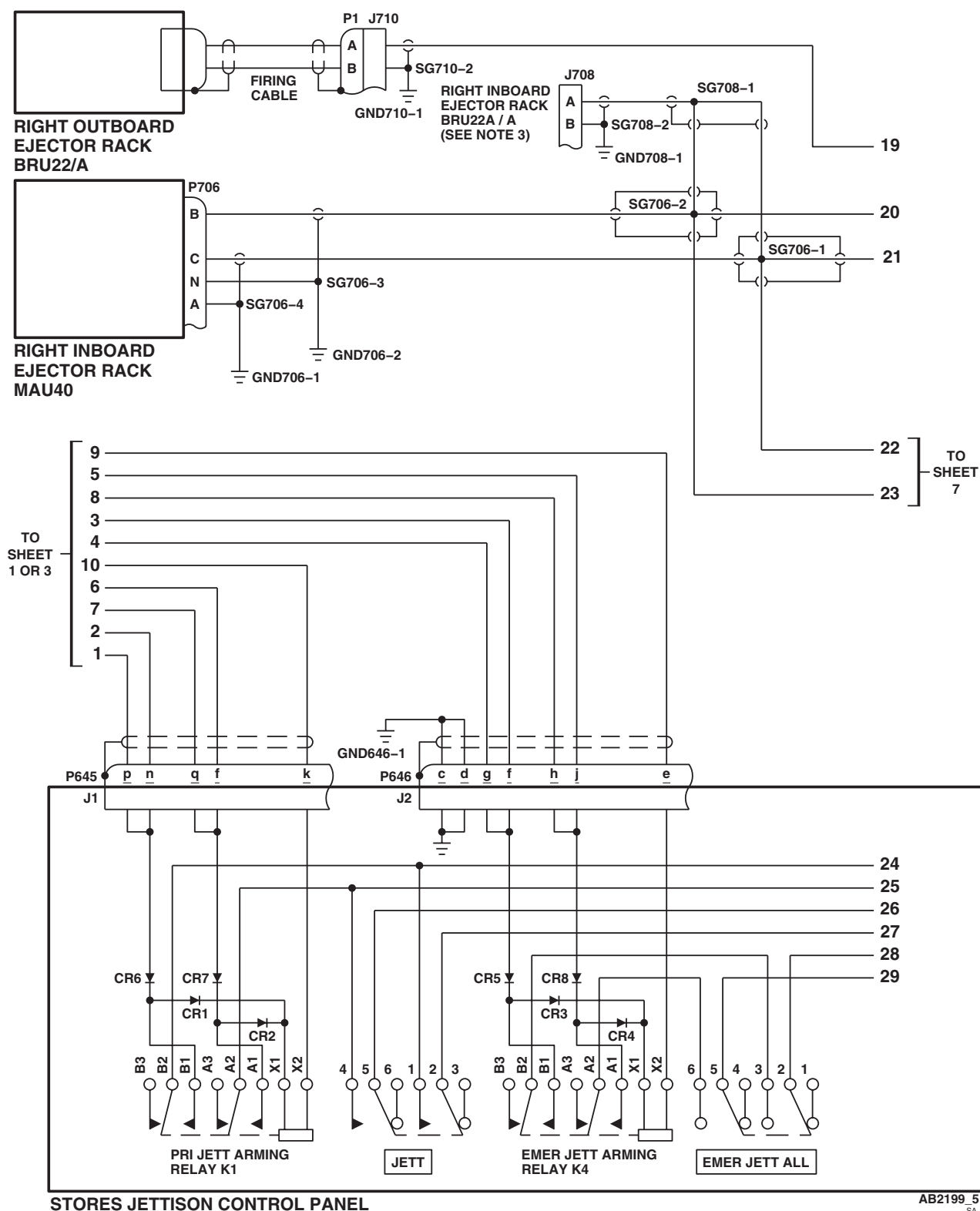
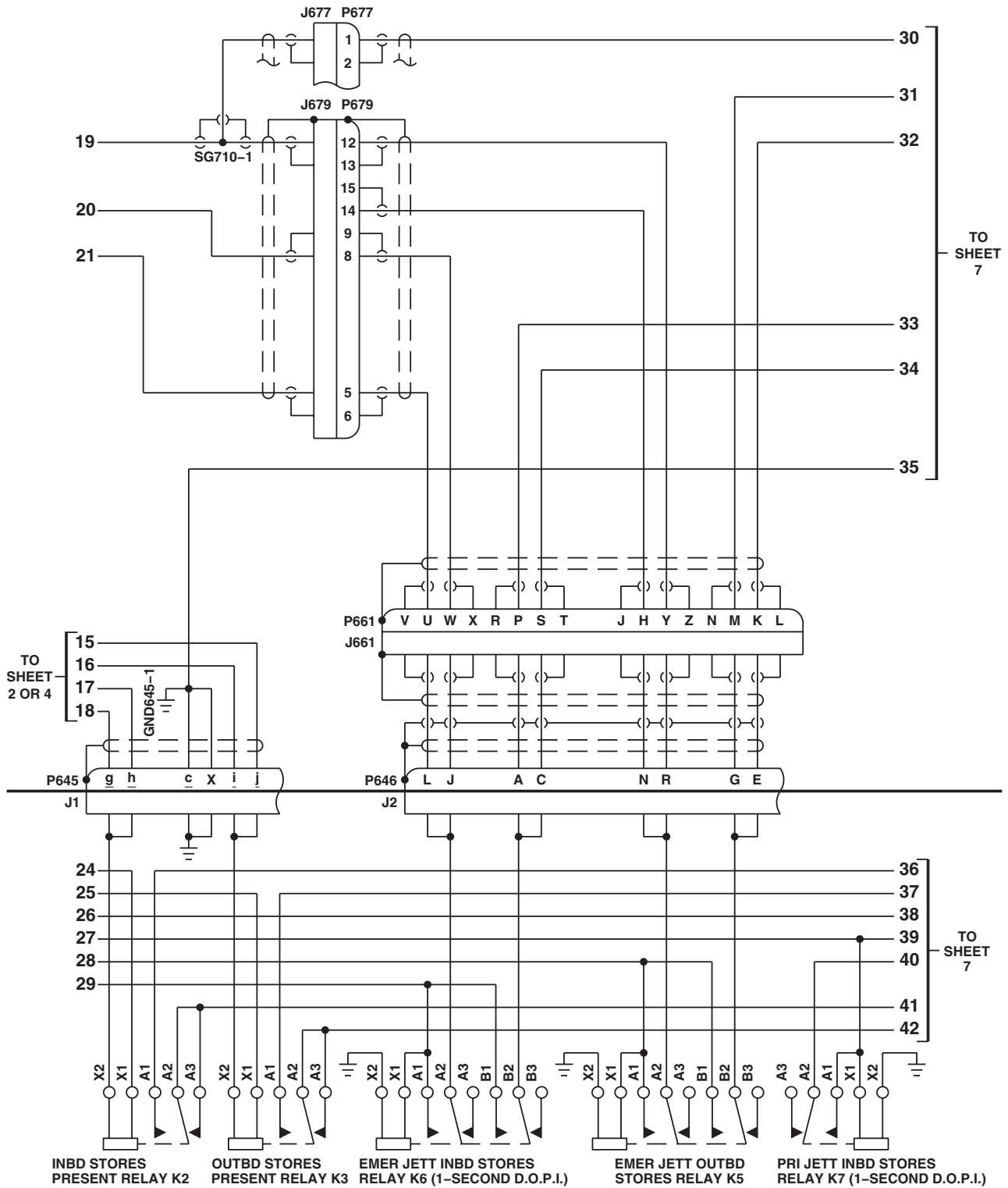


Figure 2. ESSS Stores Jettison System Schematic Diagram. (Sheet 5 of 8)

SCHEMATICS - Continued

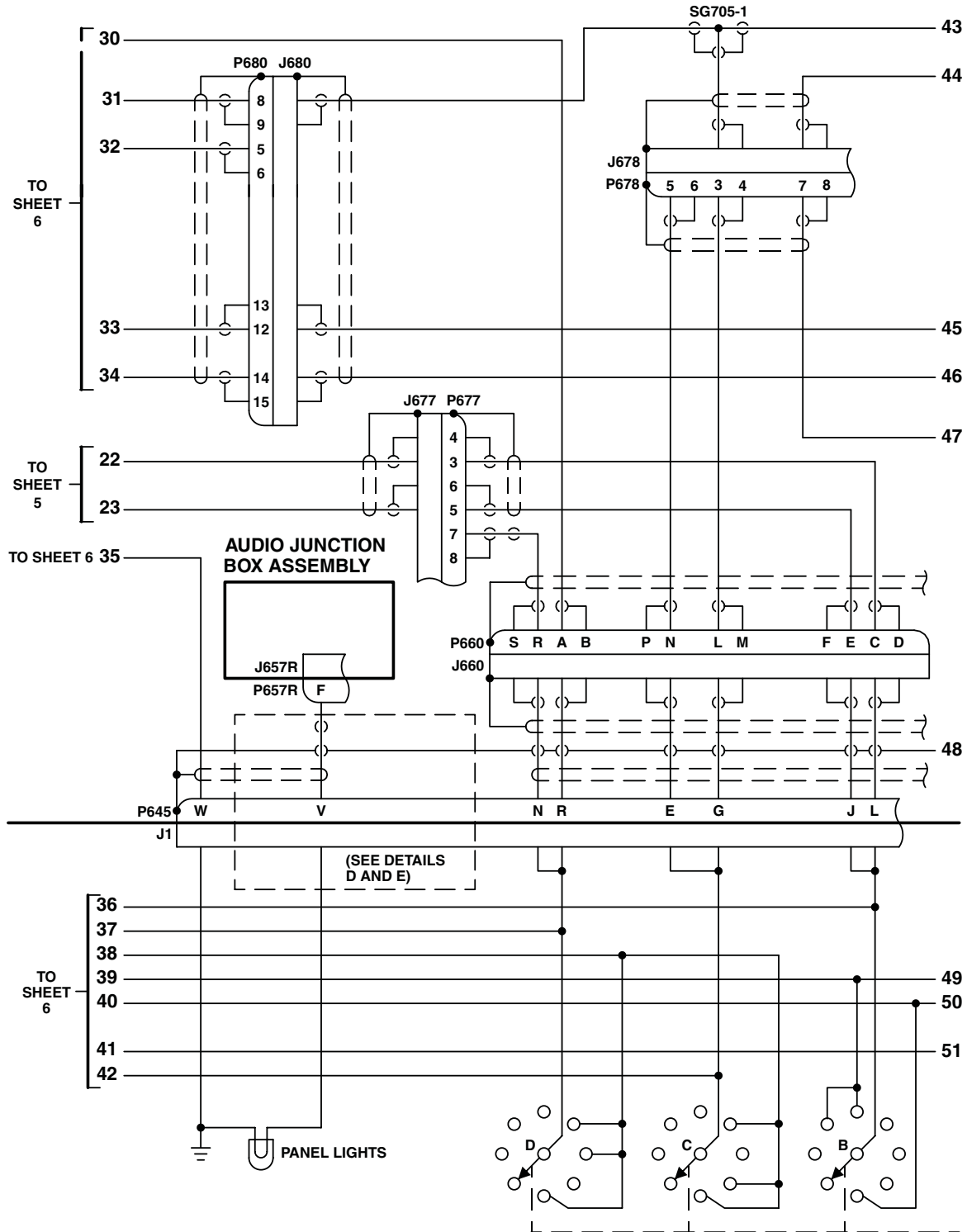


STORES JETTISON CONTROL PANEL

AB2199 6  
SA

Figure 2. ESSS Stores Jettison System Schematic Diagram. (Sheet 6 of 8)

SCHEMATICS - Continued



STORES JETTISON CONTROL PANEL

AB2199\_7A  
SA

Figure 2. ESSS Stores Jettison System Schematic Diagram. (Sheet 7 of 8)

SCHEMATICS - Continued

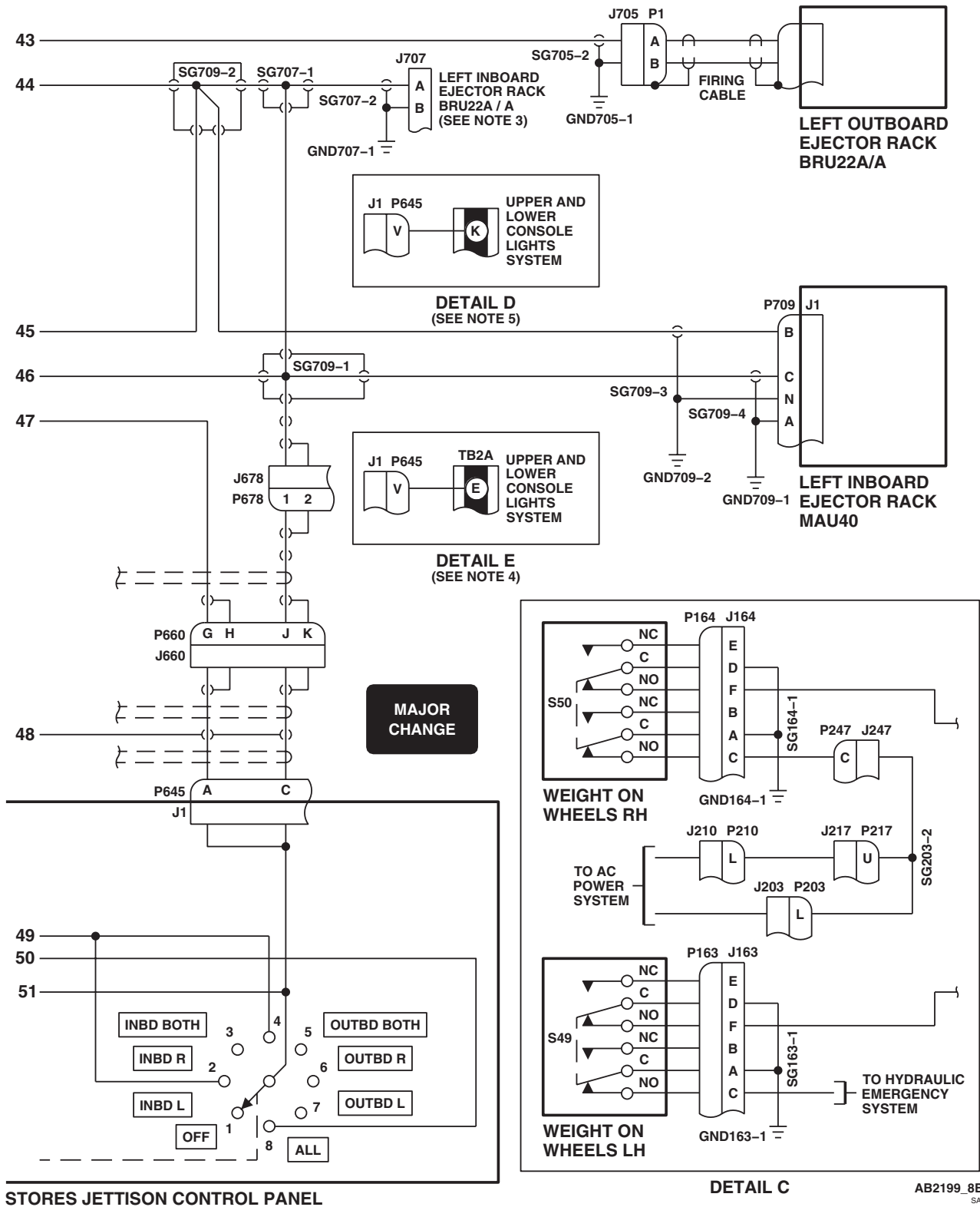


Figure 2. ESSS Stores Jettison System Schematic Diagram. (Sheet 8 of 8)

END OF WORK PACKAGE

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**UNIT LEVEL****MISSION EQUIPMENT**

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**AUXILIARY CABIN HEATER**

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**INITIAL SETUP:****Test Equipment**

Multimeter, AN/PSM-45A

WP 0827 00

WP 0836 00

WP 1448 00

**Tools and Special Tools**

Electrical Repairer Toolkit, SC 5180-99-B06

WP 1493 00

Thermometer, 389-3L

WP 1502 00

WP 1503 00

**Personnel Required**

Aircraft Electrician MOS 15F (1)

WP 1504 00

WP 1513 00

WP 1586 00

**References**

TM 1-1520-237-10

WP 1685 00

WP 0086 00

WP 1747 00

WP 0090 00

WP 0101 00

WP 0104 00

WP 0106 00

WP 0110 00

WP 0824 00

**Equipment Condition**

External Electrical Power Available, WP 1685 00

or APU Operational, TM 1-1520-237-10

No. 1 and No. 2 Engines Operational,

TM 1-1520-237-10

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**SPECIAL INSTRUCTIONS****NOTE**

See Figure 1 for component location diagram and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

The auxiliary cabin heater operational/troubleshooting procedure is subdivided as follows:

- Using External or APU Power
- Using Main Generator Power

## USING EXTERNAL OR APU POWER OPERATIONAL/TROUBLESHOOTING PROCEDURE

### PROCEDURE

#### Step

#### NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Make sure the following circuit breakers are pushed in:

| CIRCUIT BREAKER | CIRCUIT BREAKER PANEL |
|-----------------|-----------------------|
| AUX HTR BLOWER  | Mission Readiness     |
| AUX HTR CONTROL | Mission Readiness     |
| LIGHTS ADVSY    | Copilot's             |
| LIGHTS LWR CSL  | Copilot's             |

2. Pull out BACKUP HYD CONTR circuit breaker (upper console circuit breaker panel).
3. Turn on electrical power.
4. Turn upper console INST LT PILOT FLT and CONSOLE LT LOWER control fully clockwise to BRT.

#### Indication/Condition

AUX CABIN HEATER control panel (control panel) on lower console shall light to full brightness.

#### Corrective Action

If Indication/Condition is not as specified, troubleshoot upper and lower console lights (WP 0106 00).

#### Step

5. Push in control panel HTR ON and HTR INOP lights.

#### Indication/Condition

Lights shall go on.

#### Corrective Action

If Indication/Condition is not as specified, troubleshoot instrument panel and consoles indicator lights dimming (WP 0110 00).

#### Step

6. With HTR ON and HTR INOP lights pushed in, place caution/advisory panel BRT/DIM-TEST switch to BRT/DIM and hold.

#### Indication/Condition

HTR ON and HTR INOP lights shall dim.



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**USING EXTERNAL OR APU POWER OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Corrective Action**

If Indication/Condition is not as specified, troubleshoot instrument panel and consoles indicator lights dimming (WP 0110 00) or caution/advisory warning system (WP 0090 00).

**Step**

7. Release caution/advisory panel BRT/DIM-TEST switch and release HTR ON and HTR INOP lights.
8. Record ambient temperature of cabin area.
9. Set air temperature switch under left gunner's window to a value greater than the temperature noted in step 8.
10. Place control panel OFF-ON-RESET switch to ON.

**Indication/Condition**

Blower shall operate, heater shall not operate.

**Corrective Action**

If Indication/Condition is not as specified, go to BLOWER DOES NOT RUN WITH OFF-ON-RESET SWITCH AT ON, in this work package.

**Step**

11. Place control panel OFF-ON-RESET switch to RESET and hold.

**Indication/Condition**

HTR ON and HTR INOP indicator lights shall go on.

**Corrective Action**

1. If both lights do not go on, replace control panel (WP 1493 00).
2. If HTR ON light does not go on, go to HTR ON LIGHT DOES NOT GO ON WITH OFF-ON-RESET SWITCH AT RESET, in this work package.
3. If HTR INOP light does not go on, check continuity between P274-K and P273-C.
  - a. If continuity is present, replace control panel (WP 1493 00).
    - (1) If trouble remains, replace No. 3 relay panel (WP 0827 00).
  - b. If continuity is not present, repair/replace wiring (WP 1747 00).

**Step**

12. Release control panel OFF-ON-RESET switch.

**Indication/Condition**

- a. HTR INOP light shall go off.
- b. HTR ON light shall remain on.
- c. Heater and blower shall operate.

**Corrective Action**

1. If HTR INOP light does not go off, replace No. 3 relay panel (WP 0827 00).
2. If HTR ON light does not remain on, go to HTR ON LIGHT DOES NOT GO ON WITH OFF-ON-RESET SWITCH AT RESET, in this work package.

**USING EXTERNAL OR APU POWER OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued

3. If heater does not go on, go to HEATER DOES NOT GO ON WITH OFF-ON-RESET SWITCH AT ON (AFTER RESET), in this work package.

**Step**

13. Record ambient temperature of cabin area.
14. Set air temperature switch to a value less than the ambient temperature of cabin.

**Indication/Condition**

Heater shall blow cool air (HTR ON light remains on).

**Corrective Action**

1. If Indication/Condition is not as specified, make sure P728 and P731 are connected to their proper connectors.
  - a. If connector are not properly connected, connect P728 to J728 and P731 to J731.
  - b. If connectors are properly connected, replace adjustable air temperature switch (WP 1586 00).
  - c. If trouble remains, replace muffler (WP 1503 00).

**Step**

15. Place control panel OFF-ON-RESET switch to OFF.
16. Turn off electrical power.

**Indication/Condition**

Power shall be off.

**Corrective Action**

None Required

**USING MAIN GENERATOR POWER OPERATIONAL/TROUBLESHOOTING PROCEDURE****Step****NOTE**

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Make sure the following circuit breakers are pushed in:

**CIRCUIT BREAKER****CIRCUIT BREAKER PANEL**

AC ESNTL BUS SPLY

Pilot's

AC ESNTL BUS WARN

AC Essential Bus

APU CONTR INST

Lower Console

**USING MAIN GENERATOR POWER OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**

| <b>CIRCUIT BREAKER</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>CIRCUIT BREAKER PANEL</b> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| AUX HTR BLOWER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Mission Readiness            |
| AUX HTR CONTROL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Mission Readiness            |
| BACKUP HYD CONTR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Upper Console                |
| BATT BUS CONTR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Lower Console                |
| BATT BUS SPLY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Upper Console                |
| CAUT/ADVSY PNL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Upper Console                |
| DC ESNTL BUS SPLY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Pilot's                      |
| ESNTL BUS AC & CONV WARN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Lower Console                |
| ESNTL BUS DC SPLY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Lower Console                |
| LIGHTS ADVSY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Copilot's                    |
| LIGHTS CAUT ADVSY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Copilot's                    |
| LIGHTS LWR CSL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Copilot's                    |
| NO. 1 CONVERTER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Copilot's                    |
| NO. 2 CONVERTER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Pilot's                      |
| NO. 1 GEN WARN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Copilot's                    |
| NO. 2 GEN WARN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Pilot's                      |
| <ol style="list-style-type: none"> <li>2. Make sure upper console GENERATORS NO. 1, NO. 2, and APU switches are at OFF/RESET.</li> <li>3. Start APU (TM 1-1520-237-10).</li> <li>4. With qualified personnel at the controls, start both engines and operate with main rotor turning at 100% (TM 1-1520-237-10).</li> <li>5. Place upper console GENERATORS NO. 1 and NO. 2 switches to ON.</li> <li>6. Pull out BACKUP HYD CONTR circuit breaker (upper console circuit breaker panel).</li> <li>7. Record ambient temperature of cabin area.</li> <li>8. Set air temperature switch under left gunner's window to a value greater than ambient temperature of cabin.</li> <li>9. Place AUX CABIN HEATER control panel OFF-ON-RESET switch to RESET, then ON, and then release.</li> </ol> |                              |
| <b>Indication/Condition</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                              |
| <ol style="list-style-type: none"> <li>a. Blower and heater shall operate.</li> <li>b. HTR ON light on control panel shall go on.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                              |

---

**USING MAIN GENERATOR POWER OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Corrective Action**

1. If Indication/Condition a. is not as specified, go to BLOWER DOES RUN WITH OFF-ON-RESET SWITCH AT ON, then HEATER DOES NOT GO ON WITH OFF-ON-RESET SWITCH AT ON (AFTER RESET), in this work package.
2. If Indication/Condition b. is not as specified, go to HTR ON LIGHT DOES NOT GO ON WITH OFF-ON-RESET SWITCH AT RESET, in this work package.

**Step**

10. Push in BACKUP HYD CONTR circuit breaker (upper console circuit breaker panel) and place upper console BACKUP HYD PUMP switch to ON.

**Indication/Condition**

Blower and heater shall go off (HTR INOP light shall go on).

**Corrective Action**

If Indication/Condition is not as specified, immediately pull out BACKUP HYD CONTR circuit breaker and go to AUXILIARY CABIN HEATER CONTINUES TO OPERATE WHEN BACKUP HYDRAULIC PUMP IS RUNNING, in this work package.

**Step**

11. Pull out BACKUP HYD CONTR circuit breaker (upper console circuit breaker panel) and place upper console BACKUP HYD PUMP switch to OFF.
12. Momentarily place control panel OFF-ON-RESET switch to RESET.
13. Place upper console GENERATORS NO. 1 switch to OFF/RESET.

**Indication/Condition**

Blower and heater shall go off (HTR INOP light shall go on).

**Corrective Action**

If Indication/Condition is not as specified, go to AUXILIARY CABIN HEATER CONTINUES TO OPERATE WHEN GENERATORS NO.1 SWITCH PLACED TO OFF/RESET, in this work package.

**Step**

14. Place upper console GENERATORS NO. 1 switch to ON.
15. Momentarily place control panel OFF-ON-RESET switch to RESET.
16. Place upper console GENERATORS NO. 2 switch to OFF/RESET.

**Indication/Condition**

Blower and heater shall go off (HTR INOP light shall go on).

**Corrective Action**

If Indication/Condition is not as specified, go to AUXILIARY CABIN HEATER CONTINUES TO OPERATE WHEN GENERATORS NO. 2 SWITCH PLACED TO OFF/RESET, in this work package.

**Step**

17. Place upper console GENERATORS NO. 2 switch to ON.

---

**USING MAIN GENERATOR POWER OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**

18. Momentarily place control panel OFF-ON-RESET switch to RESET.
19. Record ambient temperature of cabin area.
20. Set air temperature switch to a value less than the ambient temperature of cabin.

**Indication/Condition**

Heater shall blow cool air (HTR ON light shall remain on).

**Corrective Action**

1. If Indication/Condition is not as specified, make sure P728 and P731 are connected to their proper connectors.
  - a. If connectors are not properly connected, connect P728 to J728 and P731 to J731.
  - b. If connectors are properly connected, replace adjustable air temperature switch (WP 1586 00).
  - c. If trouble remains, replace muffler (WP 1503 00).

**Step**

21. Place control panel OFF-ON-RESET switch to OFF.
22. Place upper console GENERATORS NO. 1, NO. 2, and APU switches to OFF/RESET.
23. Shut down engines and APU (TM 1-1520-237-10).
24. Turn off electrical power.

**Indication/Condition**

Power shall be off.

**Corrective Action**

None Required

**BLOWER DOES NOT RUN WITH OFF-ON-RESET SWITCH AT ON****SYMPTOM**

Blower does not run with OFF-ON-RESET switch at ON.

**MALFUNCTION**

Blower does not run with OFF-ON-RESET switch at ON.

**CORRECTIVE ACTION**

1. Install an insulated jumper wire between P727-E and P727-F. Go to 2.
2. Check for 115 vac between:
  - P727-A and P727-D
  - P727-B and P727-D
  - P727-C and P727-D
  - a. If voltage is as specified, replace blower (WP 1504 00). Go to 15.
  - b. If voltage is not as specified, go to 3.
3. Check continuity between P727-D and ground.

---

**BLOWER DOES NOT RUN WITH OFF-ON-RESET SWITCH AT ON - Continued**

- a. If continuity is present, go to **4**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **15**.
- 4. Check for 115 vac between:**
  - **P722-D and ground**
  - **P722-E and ground**
  - **P722-F and ground**
  - a. If voltage is as specified, go to **5**.
  - b. If voltage is not as specified, go to **14**.
- 5. Check continuity between:**
  - **P722-A and P727-A**
  - **P722-B and P727-B**
  - **P722-C and P727-C**
  - a. If continuity is present, go to **6**.
  - b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **15**.
- 6. Check for 28 vdc between P723-R and P723-S.**
  - a. If voltage is as specified, replace No. 3 relay panel (WP 0827 00). Go to **15**.
  - b. If voltage is not as specified, go to **7**.
- 7. Check for 28 vdc between P723-R and ground.**
  - a. If voltage is as specified, go to **8**.
  - b. If voltage is not as specified, go to **10**.
- 8. Check continuity between P723-S and P727-F.**
  - a. If continuity is present, go to **9**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **15**.
- 9. Check continuity between P727-E and ground.**
  - a. If continuity is present, replace blower (WP 1504 00). Go to **15**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **15**.
- 10. Check continuity between P724-A and P723-R.**
  - a. If continuity is present, go to **11**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **15**.
- 11. Check for 28 vdc between P724-C and ground.**
  - a. If voltage is as specified, replace AUX CABIN HEATER control panel (WP 1493 00). Go to **15**.
  - b. If voltage is not as specified, go to **12**.
- 12. Check for 28 vdc between P723-G and ground.**

---

**BLOWER DOES NOT RUN WITH OFF-ON-RESET SWITCH AT ON** - Continued

- a. If voltage is as specified, repair/replace wiring between P723-M and P724-C (WP 1747 00). Go to **15**.
  - b. If trouble remains, replace No. 3 relay panel (WP 0827 00). Go to **15**.
  - c. If voltage is not as specified, go to **13**.
- 13. Check for 28 vdc between AUX HTR CONTROL circuit breaker terminal 1 and ground.**
- a. If voltage is as specified, repair/replace wiring as required (WP 1747 00). Go to **15**.
  - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **15**.
- 14. Check for 115 vac between AUX HTR BLOWER circuit breaker:**
- **Terminal A1 and ground**
  - **Terminal B1 and ground**
  - **Terminal C1 and ground**
- a. If voltage is as specified, repair/replace wiring as required (WP 1747 00). Go to **15**.
  - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **15**.
- 15. Procedure completed.**

**HTR ON LIGHT DOES NOT GO ON WITH OFF-ON-RESET SWITCH AT RESET****SYMPTOM**

HTR ON light does not go on with OFF-ON-RESET switch at RESET.

**MALFUNCTION**

HTR ON light does not go on with OFF-ON-RESET switch at RESET.

**CORRECTIVE ACTION**

- 1. Check continuity between P724-H and P723-D.**
  - a. If continuity is present, go to **2**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **4**.
- 2. Check continuity between P723-L and P299-M.**
  - a. If continuity is present, go to **3**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **4**.
- 3. Check continuity between J299-M and ground.**
  - a. If continuity is present, replace range extension provision control panel (WP 1448 00). Go to **4**.
  - b. If trouble remains, replace No. 3 relay panel (WP 0827 00). Go to **4**.
  - c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **4**.
- 4. Procedure completed.**

**HEATER DOES NOT GO ON WITH OFF-ON-RESET SWITCH AT ON (AFTER RESET)****SYMPTOM**

Heater does not go on with OFF-ON-RESET switch at ON (after reset).

**MALFUNCTION**

Heater does not go on with OFF-ON-RESET switch at ON (after reset).

**CORRECTIVE ACTION****1. Check for 115 vac between:**

- P726-A and ground
- P726-C and ground
- P726-E and ground

a. If voltage is as specified, replace heater (WP 1502 00). Go to **30**.

b. If voltage is not as specified, go to **2**.

**2. Check for 115 vac between:**

- P721-B and ground
- P721-D and ground
- P721-F and ground

a. If voltage is as specified, go to **3**.

b. If voltage is not as specified, go to **28**.

**3. Check continuity between:**

- P721-A and P726-A
- P721-C and P726-C
- P721-E and P726-E

a. If continuity is present, go to **4**.

b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **30**.

**4. Check for 28 vdc between P723-N and P723-P.**

a. If voltage is as specified, replace No. 3 relay panel (WP 0827 00). Go to **30**.

b. If voltage is not as specified, go to **5**.

**5. Check for 28 vdc between P723-N and ground.**

a. If voltage is as specified, go to **6**.

b. If voltage is not as specified, go to **9**.

**6. Check for 28 vdc between:**

- P725-B and ground
- P725-C and ground



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**HEATER DOES NOT GO ON WITH OFF-ON-RESET SWITCH AT ON (AFTER RESET) - Continued**

- a. If voltage is as specified, replace adjustable air temperature switch (WP 1586 00). Go to **30**.
  - b. If trouble remains, go to **7**.
  - c. If voltage is not as specified, go to **8**.
- 7. Check continuity between:**
- **P728-A and P725-D**
  - **P728-B and P725-E**
- a. If continuity is present, replace cabin temperature sensor (WP 1503 00). Go to **30**.
  - b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **30**.
- 8. Check for 28 vdc between P725-C and ground.**
- a. If voltage is as specified, repair/replace wiring between P725-B and P727-F (WP 1747 00). Go to **30**.
  - b. If voltage is not as specified, repair/replace wiring between P725-C and P723-P (WP 1747 00). Go to **30**.
- 9. Check for 28 vdc between J730-B and ground.**
- a. If voltage is as specified, go to **10**.
  - b. If voltage is not as specified, go to **11**.
- 10. Check continuity between J730-A and P723-N.**
- a. If continuity is present, replace cycling duct sensor (WP 1513 00). Go to **30**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **30**.
- 11. Check continuity between J730-B and P723-A.**
- a. If continuity is present, go to **12**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **30**.
- 12. Check for 28 vdc between P723-B and P723-K.**
- a. If voltage is as specified, replace No. 3 relay panel (WP 0827 00). Go to **30**.
  - b. If voltage is not as specified, go to **13**.
- 13. Check for 28 vdc between P723-B and ground.**
- a. If voltage is as specified, go to **14**.
  - b. If voltage is not as specified, go to **15**.
- 14. Check continuity between P723-K and P724-J.**
- a. If continuity is present, replace AUX CABIN HEATER control panel (WP 1493 00). Go to **30**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **30**.
- 15. Check for 28 vdc between J729-A and ground.**
- a. If voltage is as specified, go to **16**.
  - b. If voltage is not as specified, go to **17**.
- 16. Check continuity between J729-B and ground.**

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**HEATER DOES NOT GO ON WITH OFF-ON-RESET SWITCH AT ON (AFTER RESET) - Continued**

- a. If continuity is present, replace duct overheat sensor (WP 1513 00). Go to **30**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **30**.
- 17. Check for 28 vdc between P726-B and ground.**
  - a. If voltage is as specified, go to **18**.
  - b. If voltage is not as specified, go to **23**.
- 18. Check continuity between P726-D and J729-A.**
  - a. If continuity is present, go to **19**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **30**.
- 19. Check continuity between J726-B and J726-D.**
  - a. If continuity is present, go to **20**.
  - b. If continuity is not present, replace heater (WP 1502 00). Go to **30**.
- 20. Check for 28 vdc between P725-A and P725-B.**
  - a. If voltage is as specified, replace adjustable air temperature switch (WP 1586 00). Go to **30**.
  - b. If trouble remains, go to **21**.
  - c. If voltage is not as specified, go to **22**.
- 21. Check continuity between:**
  - **J728-A and P725-D**
  - **J728-B and P725-E**
  - a. If continuity is present, replace cabin temperature sensor (WP 1503 00). Go to **30**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **30**.
- 22. Check for 28 vdc between P725-A and ground.**
  - a. If voltage is as specified, repair/replace wiring between P725-B and P727-F (WP 1747 00). Go to **30**.
  - b. If voltage is not as specified, repair/replace wiring between P725-A and P726-B (WP 1747 00). Go to **30**.
- 23. Check for 28 vdc between P724-C and ground.**
  - a. If voltage is as specified, go to **24**.
  - b. If voltage is not as specified, go to **25**.
- 24. Check continuity between P724-A and P726-B.**
  - a. If continuity is present, replace AUX CABIN HEATER control panel (WP 1493 00). Go to **30**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **30**.
- 25. Check for 28 vdc between P723-G and ground.**
  - a. If voltage is as specified, go to **26**.
  - b. If voltage is not as specified, go to **27**.
- 26. Check continuity between P723-H and P724-C.**

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**HEATER DOES NOT GO ON WITH OFF-ON-RESET SWITCH AT ON (AFTER RESET) - Continued**

- a. If continuity is present, replace No. 3 relay panel (WP 0827 00). Go to **30**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **30**.
- 27. Check for 28 vdc between AUX HTR CONTROL circuit breaker terminal 1 and ground.**
- a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P723-G (WP 1747 00). Go to **30**.
  - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **30**.
- 28. Check for 115 vac between:**
- **CL20 Terminal 1 and ground**
  - **CL21 Terminal 1 and ground**
  - **CL22 Terminal 1 and ground**
- a. If voltage is as specified, repair/replace wiring between CL20, CL21, CL22, and P721 as required (WP 1747 00). Go to **30**.
  - b. If voltage is not as specified, go to **29**.
- 29. Check for 115 vac between:**
- **CL20 Terminal 2 and ground**
  - **CL21 Terminal 2 and ground**
  - **CL22 Terminal 2 and ground**
- a. If voltage is as specified, replace current limiters as required (WP 0836 00). Go to **30**.
  - b. If voltage is not as specified, troubleshoot AC electrical system (WP 0101 00). Go to **30**.
- 30. Procedure completed.**

**AUXILIARY CABIN HEATER CONTINUES TO OPERATE WHEN BACKUP HYDRAULIC PUMP IS RUNNING**  
**SYMPTOM**

Auxiliary cabin heater continues to operate when backup hydraulic pump is running.

**MALFUNCTION**

Auxiliary cabin heater continues to operate when backup hydraulic pump is running.

**CORRECTIVE ACTION**

- 1. Push in BACKUP HYD CONTR circuit breaker. Go to 2.**
- 2. Check for 28 vdc between P723-E and ground.**
  - a. If voltage is as specified, replace No. 3 relay panel (WP 0827 00). Go to **5**.
  - b. If voltage is as specified, go to **3**.
- 3. Pull out BACKUP HYD CONTR circuit breaker. Go to 4.**
- 4. Check continuity between P723-E and K19 NO. 3 HYD PWR PUMP RELAY terminal 1.**
  - a. If continuity is present, troubleshoot hydraulics system (WP 0086 00). Go to **5**.

---

**AUXILIARY CABIN HEATER CONTINUES TO OPERATE WHEN BACKUP HYDRAULIC PUMP IS RUNNING**  
- Continued

- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **5**.

**5. Procedure completed.****AUXILIARY CABIN HEATER CONTINUES TO OPERATE WHEN GENERATORS NO. 1 SWITCH PLACED TO OFF/RESET****SYMPTOM**

Auxiliary cabin heater continues to operate when GENERATORS NO. 1 switch placed to OFF/RESET.

**MALFUNCTION**

Auxiliary cabin heater continues to operate when GENERATORS NO. 1 switch placed to OFF/RESET.

**CORRECTIVE ACTION**

- 1. Check for 28 vdc between P723-H and ground.**
  - a. If voltage is as specified, replace No. 3 relay panel (WP 0827 00). Go to **3**.
  - b. If voltage is not as specified, go to **2**.
- 2. Check continuity between P723-H and P212-C.**
  - a. If continuity is present, troubleshoot AC electrical system (WP 0101 00). Go to **3**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **3**.
- 3. Procedure completed.**

**AUXILIARY CABIN HEATER CONTINUES TO OPERATE WHEN GENERATORS NO. 2 SWITCH PLACED TO OFF/RESET****SYMPTOM**

Auxiliary cabin heater continues to operate when GENERATORS NO. 2 switch placed to OFF/RESET.

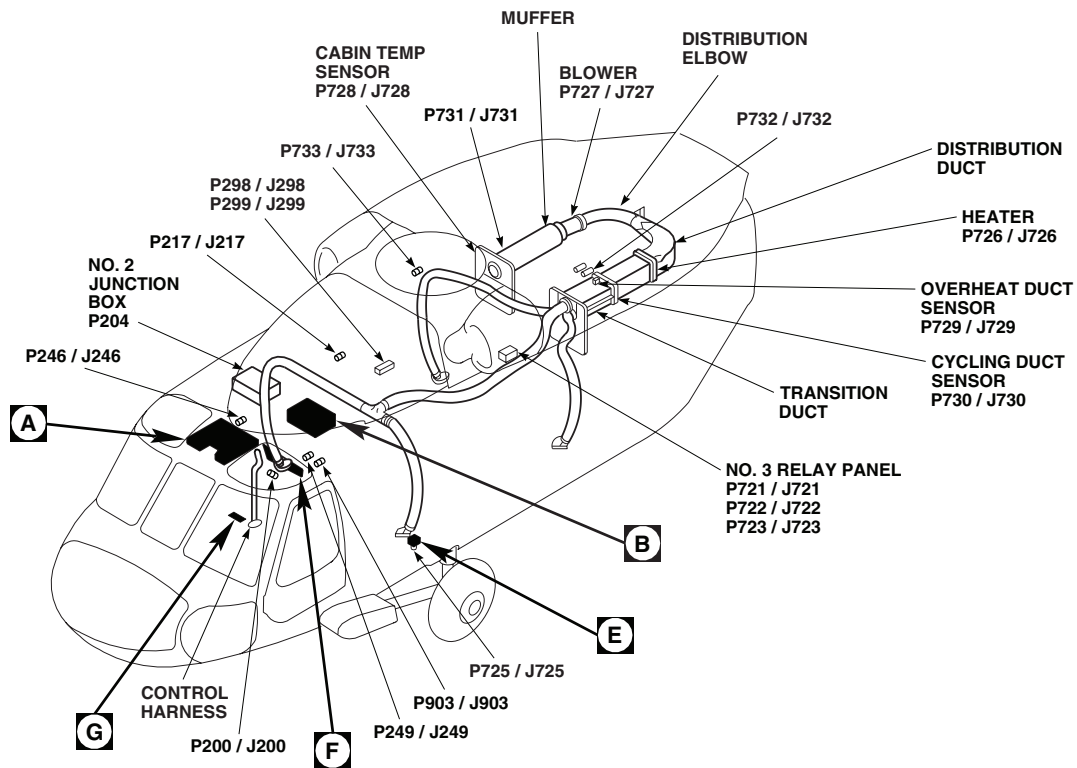
**MALFUNCTION**

Auxiliary cabin heater continues to operate when GENERATORS NO. 2 switch placed to OFF/RESET.

**CORRECTIVE ACTION**

- 1. Check for 28 vdc between P723-F and ground.**
  - a. If voltage is as specified, replace No. 3 relay panel (WP 0827 00). Go to **3**.
  - b. If voltage is not as specified, go to **2**.
- 2. Check continuity between P723-F and P204-C.**
  - a. If continuity is present, troubleshoot AC electrical system (WP 0101 00). Go to **3**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **3**.
- 3. Procedure completed.**

## LOCATION



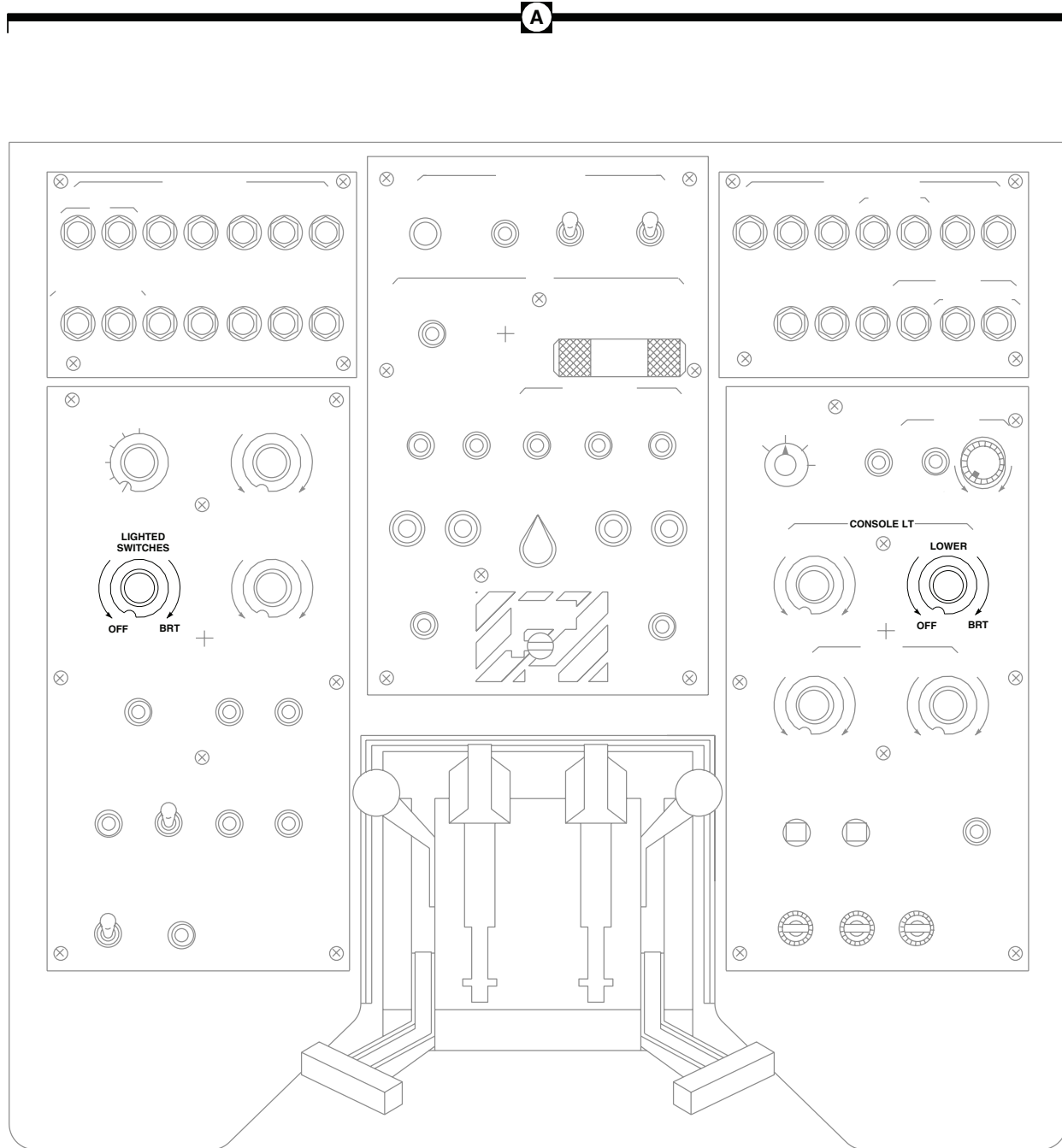
| TERMINAL BOARD/<br>DISCONNECT PLUG/<br>RECEPTACLE | LOCATION/<br>CONNECTION POINT                                  |
|---------------------------------------------------|----------------------------------------------------------------|
| J729                                              | OVERHEAT DUCT SENSOR<br>STA 421, WL 250                        |
| J730                                              | CYCLING DUCT SENSOR<br>STA 421, WL 250                         |
| P200 / J200                                       | COCKPIT CEILING<br>BL 13 LH, STA 246                           |
| P204                                              | NO. 2 JUNCTION BOX                                             |
| P212                                              | NO. 1 JUNCTION BOX                                             |
| P217 / J217                                       | CABIN CEILING, BL 0, STA<br>287                                |
| P246 / J246                                       | CABIN CEILING, BL 5 RH                                         |
| P249 / J249                                       | BEHIND COPILOT'S CIRCUIT<br>BREAKER PANEL<br>BL 23 LH, STA 247 |
| P298 / J298                                       | STA 296, WL 265                                                |
| P299 / J299                                       | STA 295, WL 265                                                |
| P315 / J315                                       | LIGHTED SWITCHES                                               |

| TERMINAL BOARD/<br>DISCONNECT PLUG/<br>RECEPTACLE | LOCATION/<br>CONNECTION POINT                |
|---------------------------------------------------|----------------------------------------------|
| P721 / J721                                       | NO. 3 RELAY PANEL                            |
| P722 / J722                                       | NO. 3 RELAY PANEL                            |
| P723 / J723                                       | NO. 3 RELAY PANEL                            |
| P724 / J724                                       | AUX CABIN HEATER<br>CONTROL PANEL            |
| P725 / J725                                       | ADJUSTABLE AIR<br>TEMPERATURE SWITCH         |
| P726 / J726                                       | HEATER, STA 443.5, WL 260                    |
| P727 / J727                                       | BLOWER, STA 443.5,<br>WL 260                 |
| P728 / J728                                       | CABIN TEMPERATURE<br>SENSOR, STA 398, WL 254 |
| P731 / J731                                       | BLOWER O / H SENSOR<br>STA 421               |
| P732 / J732                                       | STA 421, WL 250                              |
| P733 / J733                                       | BL 24.71, STA 360, WL 265.30                 |
| P902 / J902                                       | LEFT RELAY PANEL                             |

AA6092\_1A  
SA

Figure 1. Auxiliary Cabin Heater Location Diagram. (Sheet 1 of 5)

## LOCATION - Continued

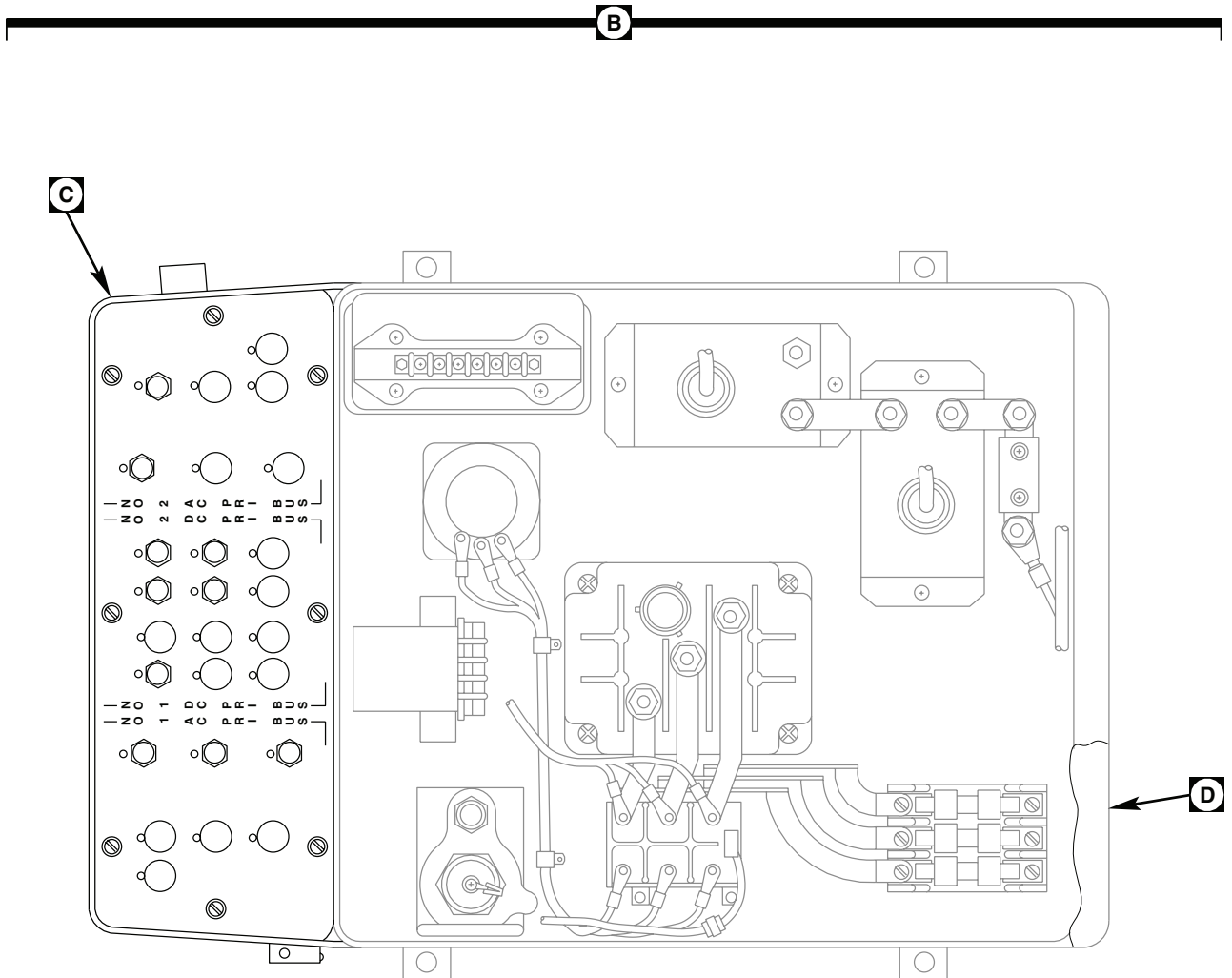


## UPPER CONSOLE

AA6092\_2  
SA

Figure 1. Auxiliary Cabin Heater Location Diagram. (Sheet 2 of 5)

LOCATION - Continued

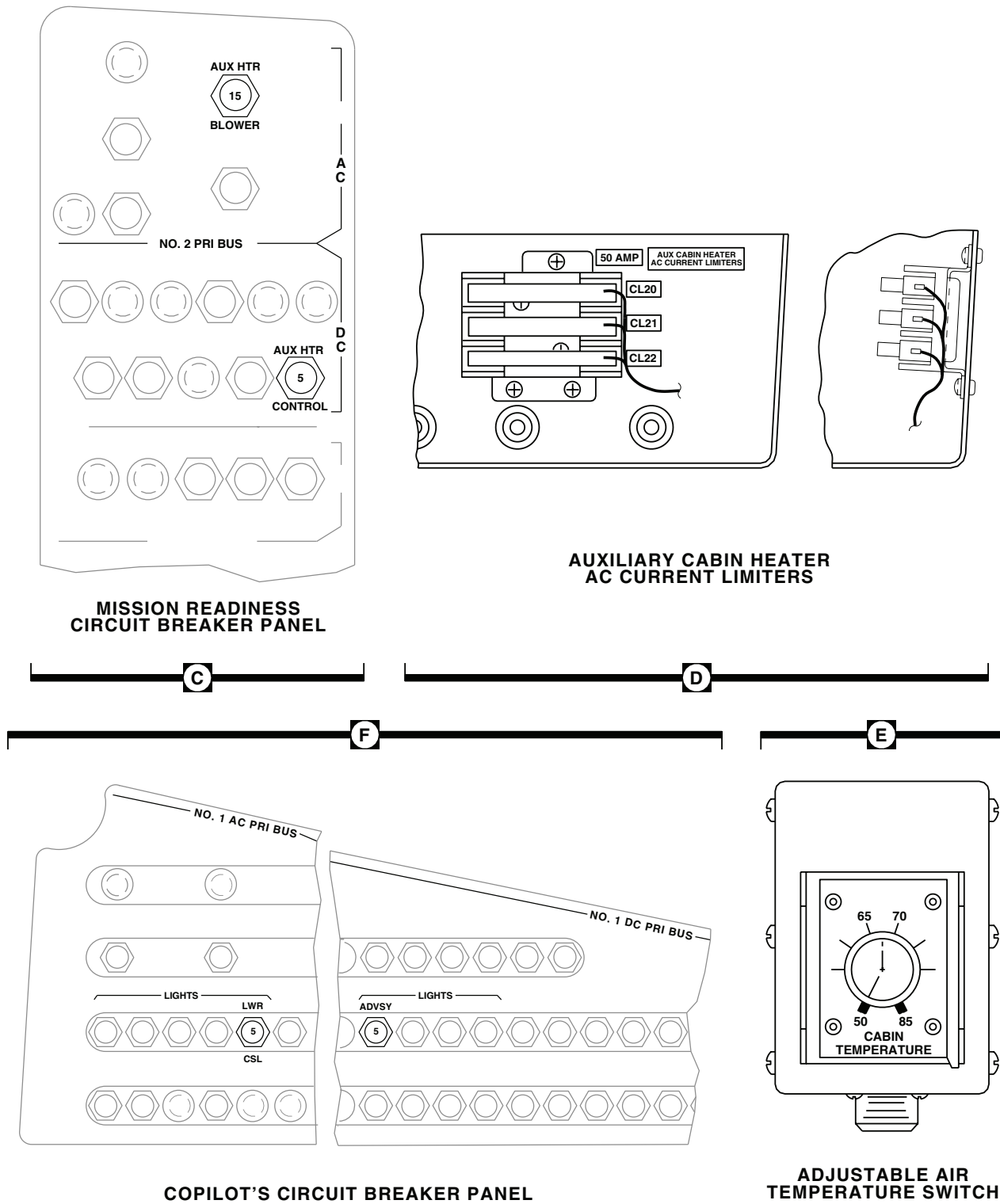


NO. 1 JUNCTION BOX

AA6092\_3  
SA

Figure 1. Auxiliary Cabin Heater Location Diagram. (Sheet 3 of 5)

LOCATION - Continued

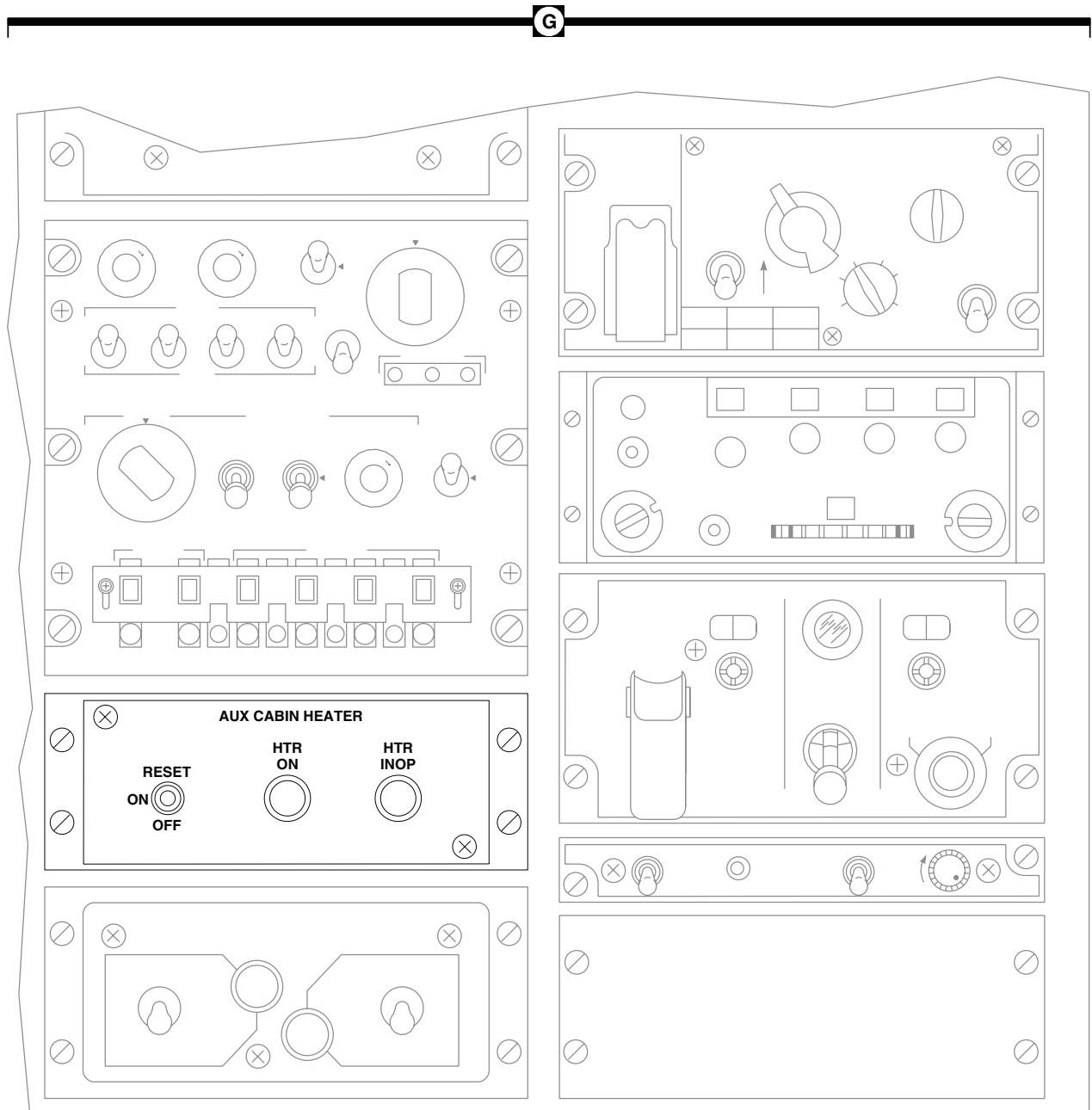


AA6092\_4A  
SA

Figure 1. Auxiliary Cabin Heater Location Diagram. (Sheet 4 of 5)



LOCATION - Continued

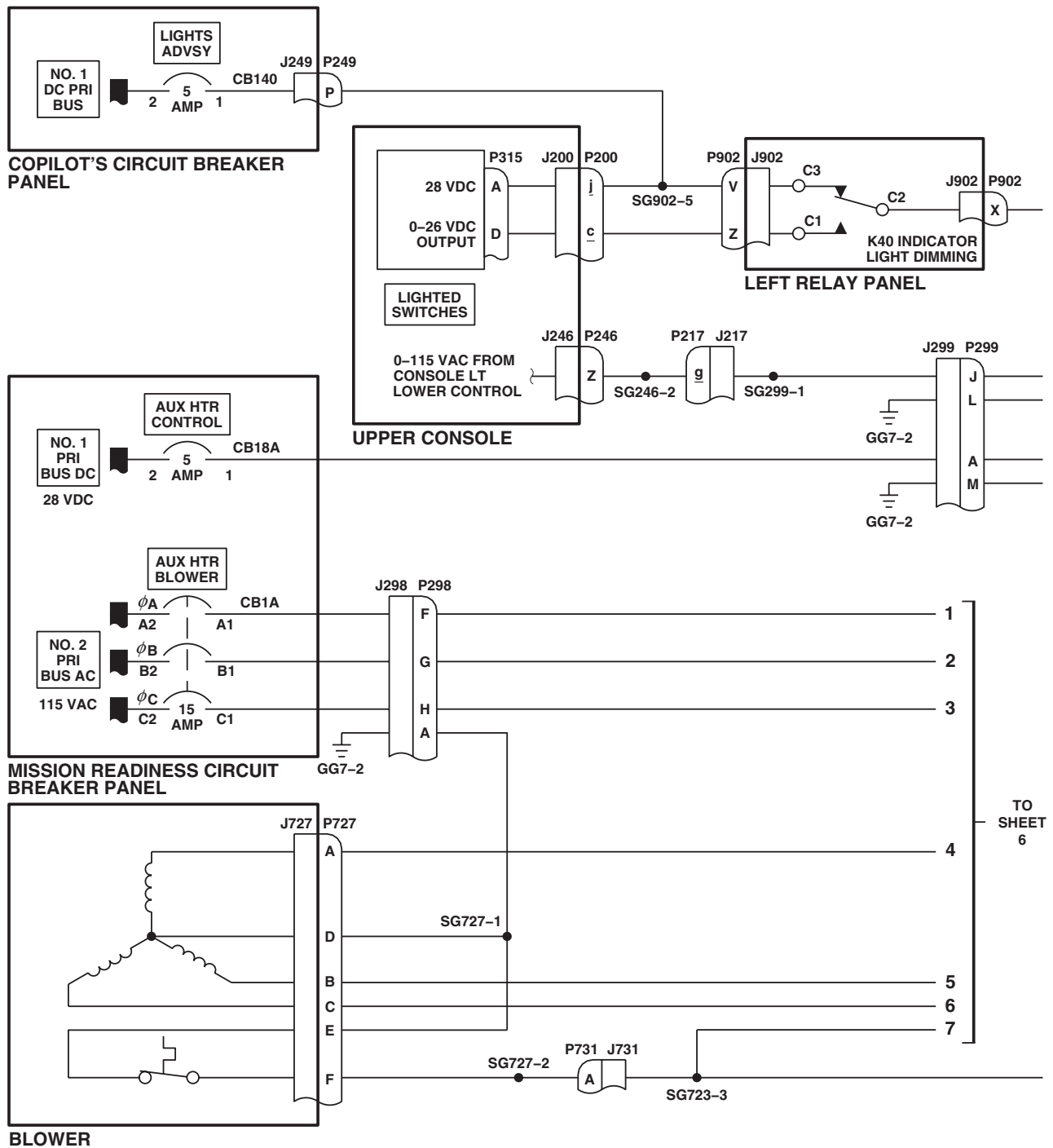


LOWER CENTER CONSOLE

AA6092\_5  
SA

Figure 1. Auxiliary Cabin Heater Location Diagram. (Sheet 5 of 5)

## SCHEMATICS



**NOTE**  
RELAYS K102 AND K103 SHOWN WITH  
GENERATORS OFF.

AB1535\_1A  
SA

Figure 2. Auxiliary Cabin Heater Schematic Diagram. (Sheet 1 of 6)

SCHEMATICS - Continued

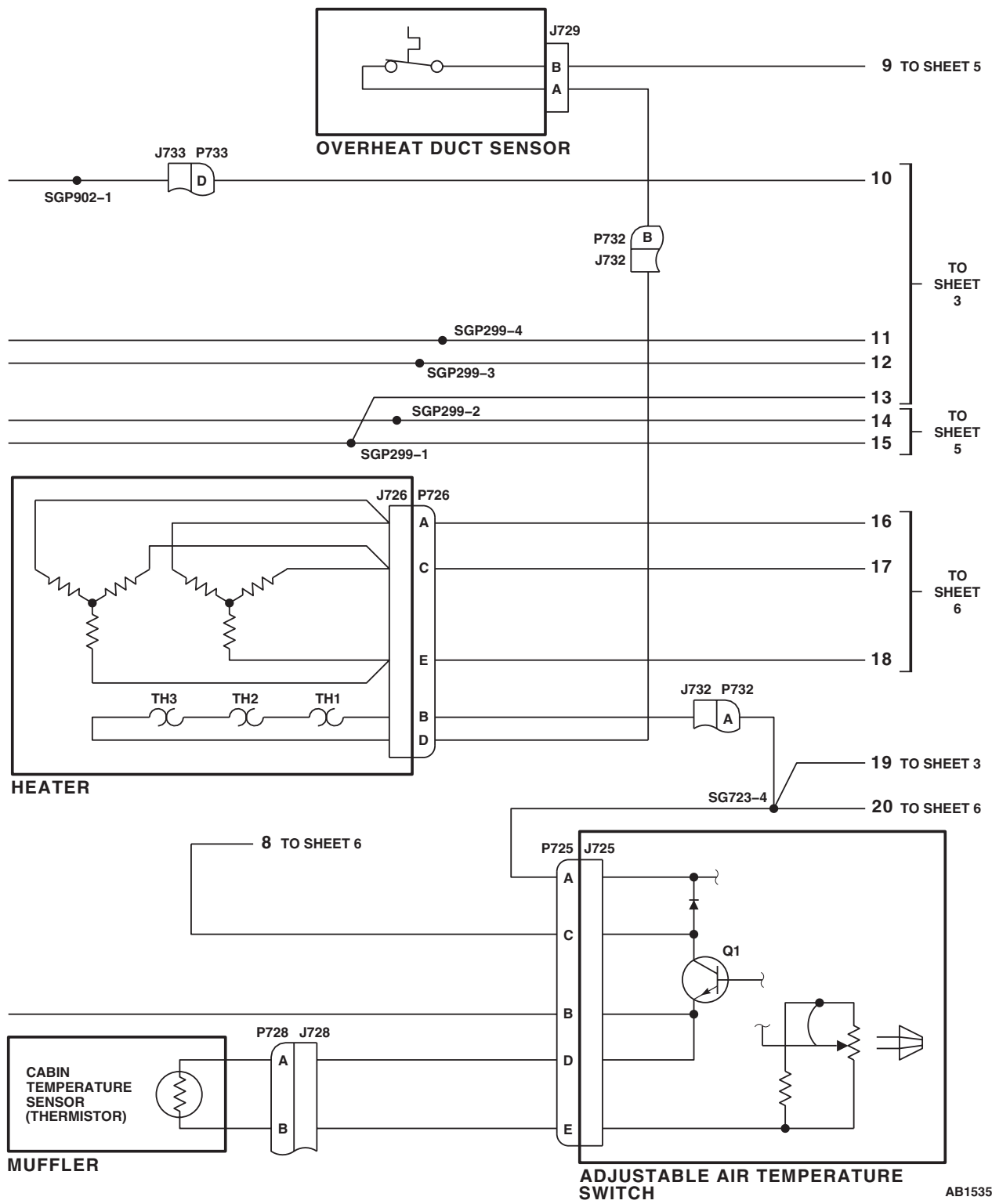


Figure 2. Auxiliary Cabin Heater Schematic Diagram. (Sheet 2 of 6)

SCHEMATICS - Continued

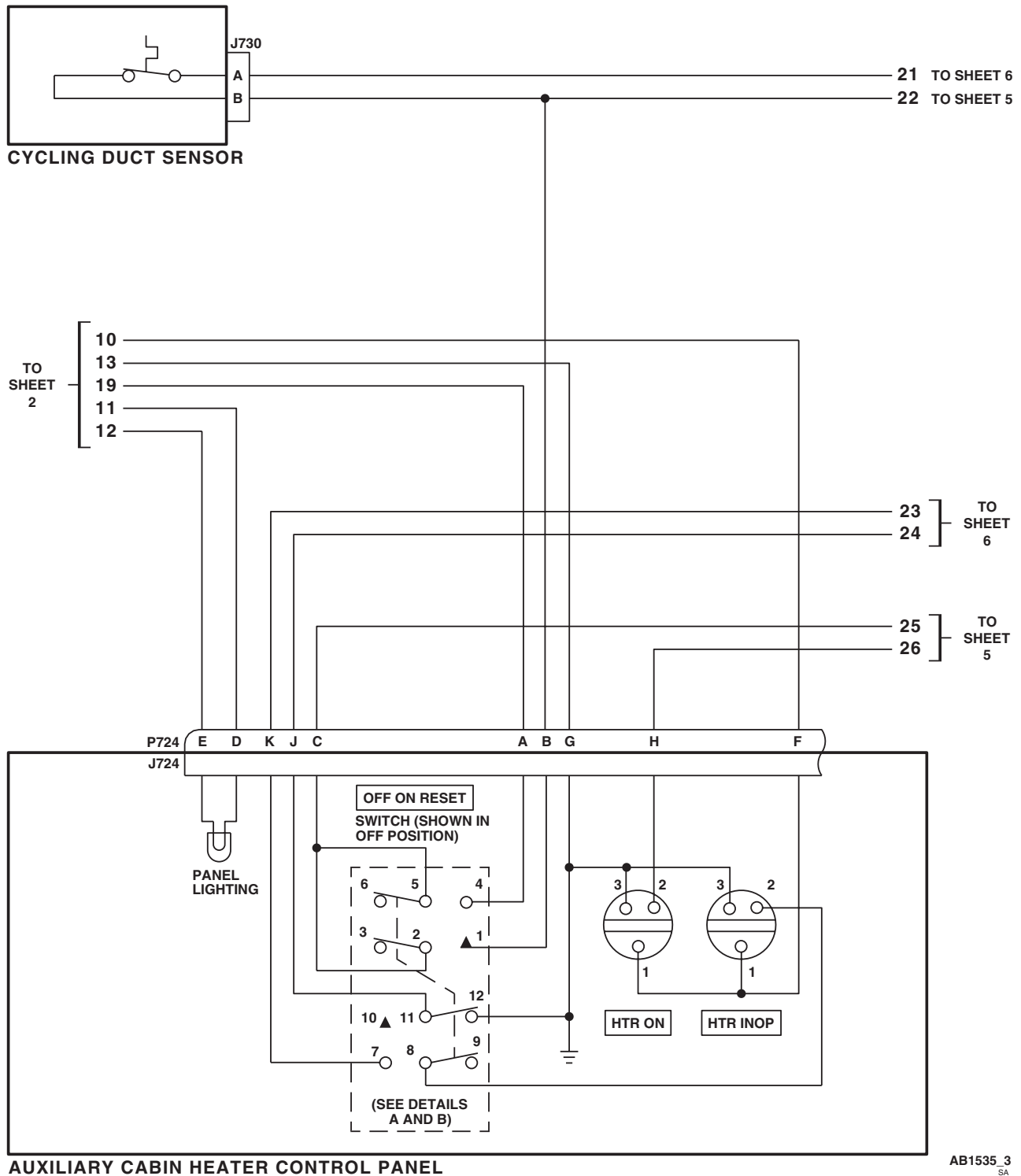


Figure 2. Auxiliary Cabin Heater Schematic Diagram. (Sheet 3 of 6)

SCHEMATICS - Continued

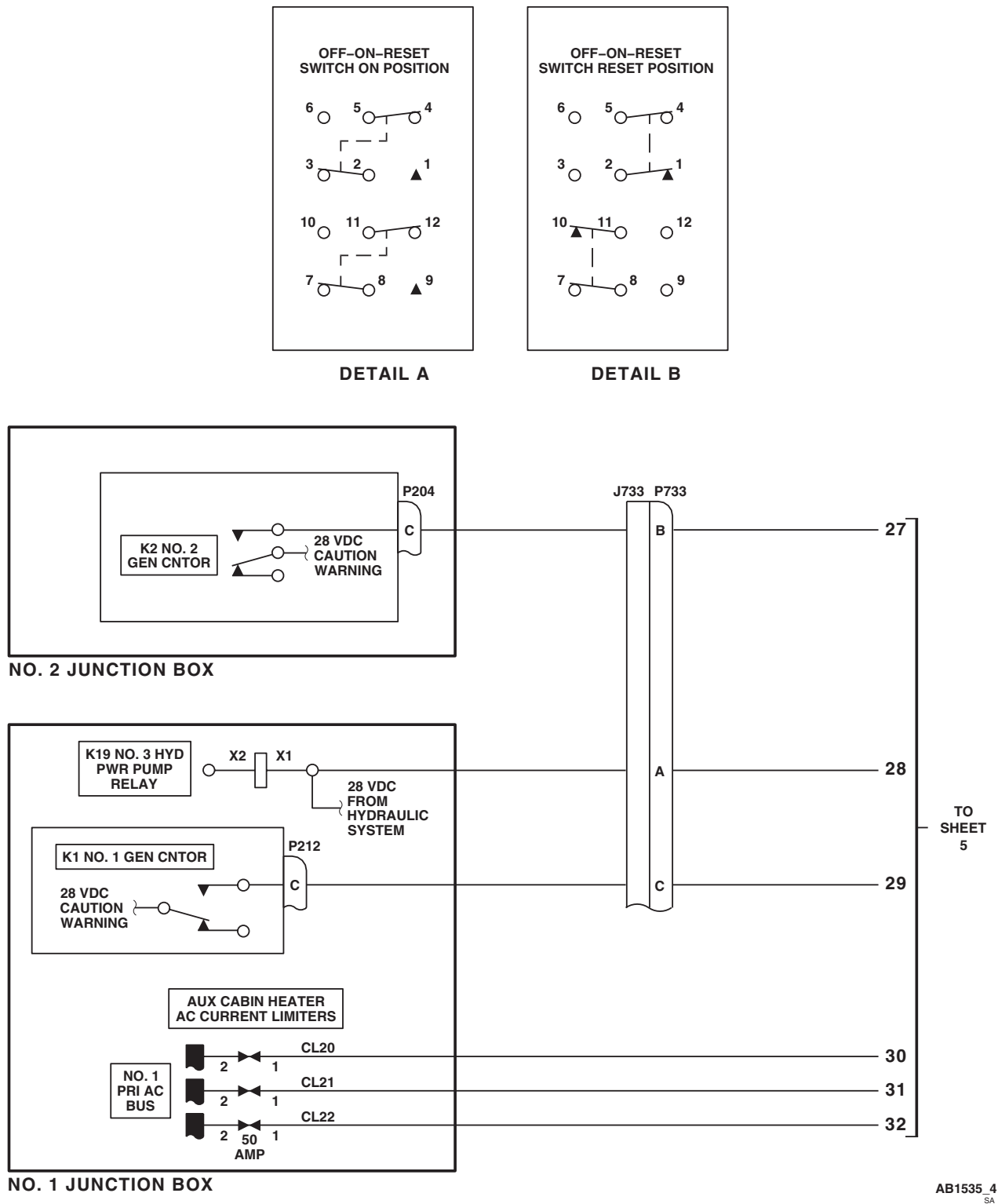


Figure 2. Auxiliary Cabin Heater Schematic Diagram. (Sheet 4 of 6)

SCHEMATICS - Continued

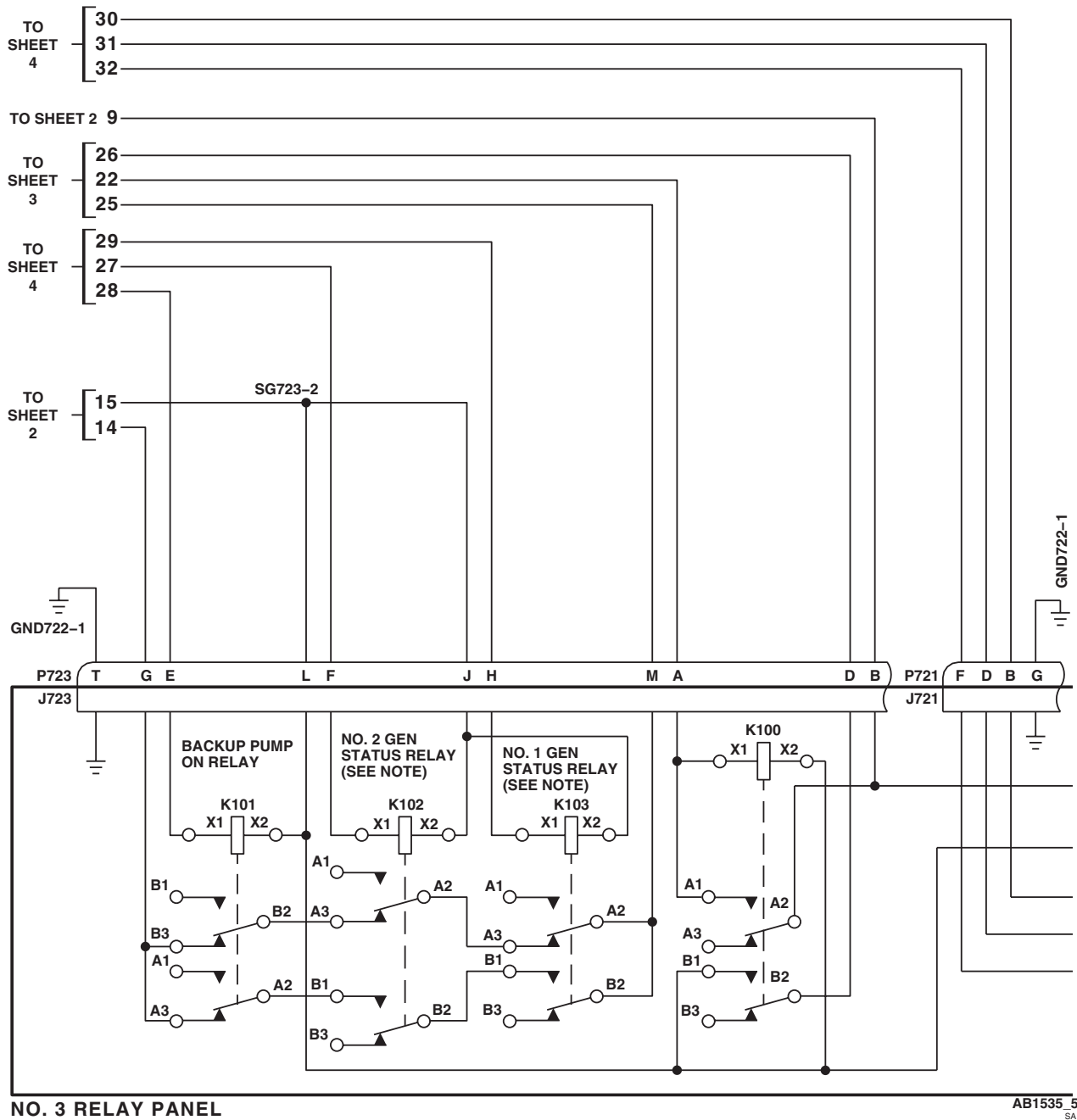


Figure 2. Auxiliary Cabin Heater Schematic Diagram. (Sheet 5 of 6)

SCHEMATICS - Continued

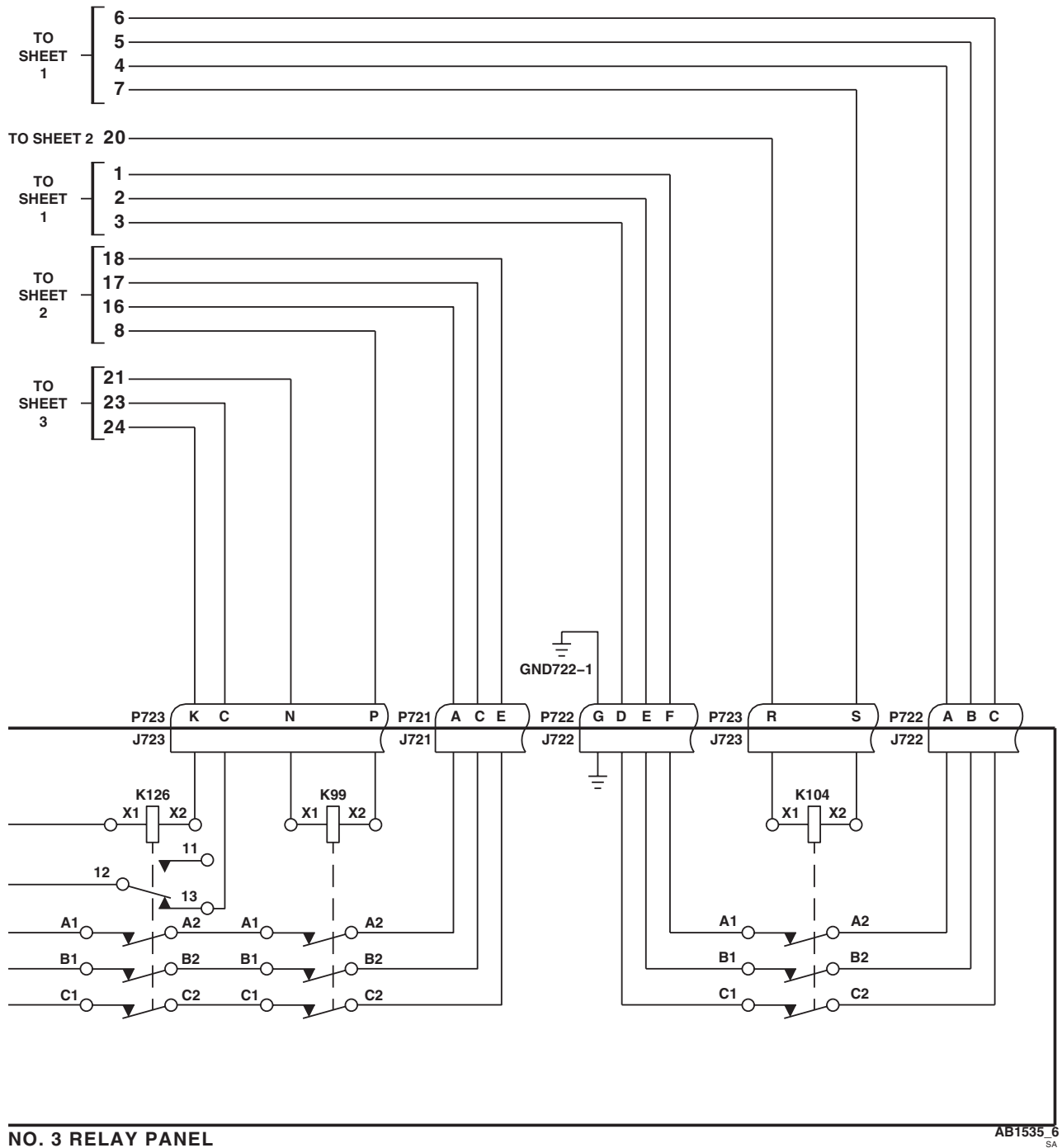


Figure 2. Auxiliary Cabin Heater Schematic Diagram. (Sheet 6 of 6)

END OF WORK PACKAGE





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**UNIT LEVEL****MISSION EQUIPMENT****CHAFF DISPENSER (M130) SYSTEM UH-60A UH-60L HH-60A HH-60L**

This WP supersedes WP 0162 00, dated 17 April 2006.

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**INITIAL SETUP:****Test Equipment**

Test Set, Electronic Systems (test set), M91

WP 0443 00

WP 0824 00

WP 1520 00

**Tools and Special Tools**

Electronic Equipment Toolkit, TK-101/G

WP 1522 00

Locally-Made Drag Beam Wedge, Figure 167,

WP 1523 00

WP 1805 00

WP 1524 00

WP 1525 00

**Personnel Required**

Avionics Mechanic MOS 15N (1)

WP 1685 00

WP 1747 00

WP 1805 00

**References**

TM 11-1520-237-10

TM 11-1520-237-23

TM 9-1095-206-12&P

WP 0104 00

**Equipment Condition**

External Electrical Power Available, WP 1685 00

or APU Operational, TM 1-1520-237-10

**CHAFF DISPENSER (M130) SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE****UH-60A UH-60L HH-60A HH-60L****PROCEDURE****Step****WARNING**

- The flare and chaff dispenser payload modules must be removed from the dispensers before doing any checks or maintenance on the dispenser system.
- Make sure RIPPLE FIRE switch guard is in the down position on dispenser control panel.

**NOTE**

- On test set, TS PWR ON lamp lights (clear) and remains on throughout the test sequence until helicopter power to test set is disconnected or shut off.
- When the test set is installed on the dispenser assembly and 28 vdc helicopter power has been applied, the sequencer switch inside the dispenser assembly resets, making an audible sound as it rotates. There will be no such sound if the sequencer switch has been previously reset or if switch is at 12 or 24.
- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- See Figure 1 for component location diagram, and **UH-60L MOD HH-60A HH-60L** Figure 2 , **UH-60A W/OMOD** Figure 3  for schematic diagram, and Figure 4 for system test setup as an aid in operational/troubleshooting.

1. Pull out CHAFF DISP circuit breaker on copilot's circuit breaker panel.
2. Loosen screws and remove payload module.
3. **UH-60A 77-22714 - 77-22723** Insert locally-made drag beam wedge (WP 1805 00) between left drag beam switch plunger and plunger actuating bracket, to simulate weight-off-wheels. 
4. Turn on electrical power.
5. Turn chaff dispenser assembly C-F (chaff-flare) switch to C.
6. Connect test set to chaff dispenser breech assembly with captive mounting studs. Secure mounting studs uniformly using hexagonal screwdriver provided in test set carrying case.
7. Make sure test set TEST SEQUENCE switch is at START/HOME.
8. Connect test set power cable NO. 1 between J1 on electronics module and 28 VDC ACFT PWR receptacle on test set.
9. Remove safety pin from SAFETY PIN receptacle of electronics module.
10. Turn electronics module switches as follows:

|                   |   |
|-------------------|---|
| CHAFF SALVO COUNT | 1 |
|-------------------|---|

|             |   |
|-------------|---|
| BURST COUNT | 6 |
|-------------|---|

---

**CHAFF DISPENSER (M130) SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE**  
**UH-60A UH-60L HH-60A HH-60L** - Continued

|                |     |
|----------------|-----|
| SALVO INTERVAL | 1   |
| BURST INTERVAL | 0.1 |

11. Push in CHAFF DISP circuit breaker (copilot's circuit breaker panel).
12. Turn dispenser control panel MAN-PGRM switch to MAN.
13. Turn dispenser control panel CHAFF counter knob until counter indicates 30.
14. Place dispenser control panel SAFE-ARM switch to ARM.

**Indication/Condition**

ARM light shall go on.

**Corrective Action**

If Indication/Condition is not as specified, go to ARM LIGHT DOES NOT GO ON, in this work package.

**Step**

15. Press to test all test set lamps.

**Indication/Condition**

Test set lamps shall go on when pressed and go off when released.

**Corrective Action**

1. If Indication/Condition is not as specified, replace any lamp that does not go on when pressed.
  - a. If none of the indicating lamps light, troubleshoot test set (TM 9-1095-206-12&P).

**Step**

16. Turn test set TEST SEQUENCE switch to TS RESET.

**Indication/Condition**

No visual indication shall occur.

**Corrective Action**

If Indication/Condition is not as specified, troubleshoot test set (TM 9-1095-206-12&P).

**Step**

17. Turn test set TEST SEQUENCE selector switch to SV SELF TEST.

**Indication/Condition**

Test set STRAY VOLTAGE lamp (red) shall go on.

**Corrective Action**

If Indication/Condition is not as specified, troubleshoot test set (TM 9-1095-206-12&P).

**Step**

18. Turn test set TEST SEQUENCE selector switch clockwise to TS RESET.

---

**CHAFF DISPENSER (M130) SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE**  
**UH-60A UH-60L HH-60A HH-60L** - Continued**Indication/Condition**

Test set STRAY VOLTAGE lamp (red) shall go off.

**Corrective Action**

If Indication/Condition is not as specified, troubleshoot test set (TM 9-1095-206-12&P).

**Step**

19. Turn test set TEST SEQUENCE selector switch to STRAY VOLTAGE.

**Indication/Condition**

Test set STRAY VOLTAGE lamp (red) shall not go on.

**Corrective Action**

If Indication/Condition is not as specified, go to STRAY VOLTAGE LIGHT GOES ON, in this work package.

**Step****NOTE**

When test set MANUAL SYSTEM RESET switch is pressed and released, the sequencer switch inside the dispenser assembly resets, making an audible sound as it rotates. There will be no such sound if the sequencer switch has previously reset or if switch is at 12 or 24.

20. Turn test set TEST SEQUENCE selector switch clockwise to SYS NOT RESET.

**Indication/Condition**

Test set SYS NOT RESET lamp (amber) shall not go on.

**Corrective Action**

1. If Indication/Condition is not as specified, press and release test set MANUAL SYSTEM RESET switch, SYS NOT RESET lamp shall go off.
2. If trouble remains, troubleshoot test set (TM 9-1095-206-12&P).

**Step**

21. Turn test set TEST SEQUENCE switch to DISP COMPL.
22. Press and release CHAFF DISPENSE pushbutton once.

**Indication/Condition**

The CHAFF counter dispenser control panel shall count down by one.

**Corrective Action**

If Indication/Condition is not as specified, go to CHAFF COUNTER DOES NOT COUNT DOWN, in this work package.

**Step**

23. Place dispenser control panel MAN-PGRM switch to PGRM.
24. Momentarily press lower console CHAFF DISPENSE pushbutton once.

---

**CHAFF DISPENSER (M130) SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE**  
**UH-60A UH-60L HH-60A HH-60L** - Continued**Indication/Condition**

The CHAFF counter shall count down from 29 to 23.

**Corrective Action**

1. If Indication/Condition is not as specified, check continuity between P1-B and J673R-B.
  - a. If continuity is present, replace dispenser control panel (WP 1522 00).
  - b. If continuity is not present, repair/replace wiring (WP 1747 00).

**Step**

25. Repeat step 24. until CHAFF counter reads 00.

**Indication/Condition**

- a. CHAFF counter shall read 00.
- b. Test set DISPENSE COMPLETE lamp (green) shall go on.

**Corrective Action**

1. If Indication/Condition a. is not as specified, replace dispenser control panel (WP 1522 00).
2. If Indication/Condition b. is not as specified, troubleshoot test set (TM 9-1095-206-12&P).

**Step**

26. Turn test set TEST SEQUENCE selector switch to SYS NOT RESET.

**Indication/Condition**

The test set SYS NOT RESET lamp (amber) shall go on.

**Corrective Action**

If Indication/Condition is not as specified, troubleshoot test set (TM 9-1095-206-12&P).

**Step****NOTE**

When the test set MANUAL SYSTEM RESET switch is pressed and released, the sequencer switch inside the dispenser assembly resets, making an audible sound as it rotates. There will be no such sound if the sequencer switch has been previously reset or is in position 12 or 24.

27. Momentarily press test set MANUAL SYSTEM RESET switch.

**Indication/Condition**

Test set SYS NOT RESET lamp (amber) shall go off.

**Corrective Action**

If Indication/Condition is not as specified, replace chaff dispenser assembly (WP 1520 00).

**Step**

28. Turn test set TEST SEQUENCE switch to STRAY VOLT.

**CHAFF DISPENSER (M130) SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE****UH-60A UH-60L HH-60A HH-60L** - Continued**Indication/Condition**

STRAY VOLTAGE lamp (red) on test set shall not go on.

**Corrective Action**

If Indication/Condition is not as specified, go to STRAY VOLTAGE LIGHT GOES ON, in this work package.

**Step****NOTE**

When the TEST SEQUENCE switch is turned to START/HOME, the DISPENSE COMPLETE lamp goes off and the STRAY VOLTAGE lamp goes on momentarily when passing through TS RESET.

29. Turn test set TEST SEQUENCE selector switch to START/HOME.
30. Install safety pin flag assembly in electronics module SAFETY PIN receptacle.
31. Turn off electrical power by pulling out CHAFF DISP circuit breaker (copilot's circuit breaker panel).
32. Place dispenser control panel SAFE-ARM switch to SAFE.
33. Reset dispenser control panel CHAFF counter to 30.
34. Disconnect test set power cable NO. 1 from electronics module and test set.
35. Place receptacle covers on electronics module and test set receptacles and secure.
36. Loosen captive mounting studs securing test set to chaff dispenser assembly.
37. Remove test set from chaff dispenser assembly and store in carrying case along with power cable NO. 1 and hexagonal screwdriver.
38. **UH-60A 77-22714 - 77-22723** Remove locally-made drag beam wedge from left drag beam switch. ■
39. Turn off electrical power.
40. Install payload module and tighten screws.

**Indication/Condition**

Module shall be installed.

**Corrective Action**

None Required

**ARM LIGHT DOES NOT GO ON****SYMPTOM**

ARM light does not go on.

**MALFUNCTION**

ARM light does not go on.

**CORRECTIVE ACTION****1. Check helicopter configuration.**

- a. **UH-60L 92-26408 - SUBQ MOD HH-60A HH-60L** , go to **2.**

## ARM LIGHT DOES NOT GO ON - Continued

- b. **UH-60A 78-22987 - SUBQ W/O MOD UH-60L 89-26149 - 89-26154 AND 89-26179 - 92-26407**, go to **10**.
- c. **UH-60A 77-22724 - 78-22986 W/O MOD**, go to **18**.
- d. **UH-60A 77-22714 - 77-22723 W/O MOD**, go to **28**.
2. **UH-60L 92-26408 - SUBQ AND MOD HH-60A HH-60L** Press ARM light.
  - a. If ARM light goes on, go to **3**.
  - b. If ARM light does not go on, go to **5**.
3. **UH-60L 92-26408 - SUBQ AND MOD HH-60A HH-60L** Check for 28 vdc between J673R-P and ground.
  - a. If voltage is as specified, replace electronic module (WP 1520 00). Go to **41**.
  - b. If voltage is not as specified, go to **4**.
4. **UH-60L 92-26408 - SUBQ AND MOD HH-60A HH-60L** Check continuity between P669R-F and J673R-F.
  - a. If continuity is present, replace dispenser control panel (WP 1522 00). Go to **41**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **41**.
5. **UH-60L 92-26408 - SUBQ AND MOD HH-60A HH-60L** Check ARM light lamp.
  - a. If ARM light lamp is good, go to **6**.
  - b. If ARM light lamp is burnt out, replace lamp (WP 1523 00). Go to **41**.
6. **UH-60L 92-26408 - SUBQ AND MOD HH-60A HH-60L** Check fuse in dispenser control panel.
  - a. If fuse is good, go to **7**.
  - b. If fuse is blown, replace fuse (WP 1524 00). Go to **41**.
7. **UH-60L 92-26408 - SUBQ AND MOD HH-60A HH-60L** Check for 28 vdc between P669R-H and ground.
  - a. If voltage is as specified, go to **8**.
  - b. If voltage is not as specified, go to **9**.
8. **UH-60L 92-26408 - SUBQ AND MOD HH-60A HH-60L** Check continuity between P669R-R and ground.
  - a. If continuity is present, replace chaff dispenser control panel (WP 1522 00). Go to **41**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **41**.
9. **UH-60L 92-26408 - SUBQ AND MOD HH-60A HH-60L** Check for 28 vdc between CHAFF DISP circuit breaker terminal 1 and ground.
  - a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P669R-H (WP 1747 00). Go to **41**.
  - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **41**.
10. **UH-60A 78-22987 - SUBQ W/O MOD UH-60L 89-26149 - 89-26154 AND 89-26179 - 92-26407** Press ARM light.
  - a. If ARM light goes on, go to **11**.

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**ARM LIGHT DOES NOT GO ON - Continued**

- b. If ARM light does not go on, go to **13**.
- 11. **UH-60A 78-22987 - SUBQ W/O MOD UH-60L 89-26149 - 89-26154 AND 89-26179 - 92-26407**  
**Check for 28 vdc between J673R-P and ground.**
  - a. If voltage is as specified, replace electronic module (WP 1520 00). Go to **41**.
  - b. If voltage is not as specified, go to **12**.
- 12. **UH-60A 78-22987 - SUBQ W/O MOD UH-60L 89-26149 - 89-26154 AND 89-26179 - 92-26407**  
**Check continuity between P1-F and J673R-F.**
  - a. If continuity is present, replace dispenser control panel (WP 1522 00). Go to **41**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **41**.
- 13. **UH-60A 78-22987 - SUBQ W/O MOD UH-60L 89-26149 - 89-26154 AND 89-26179 - 92-26407**  
**Check ARM light lamp.**
  - a. If ARM light lamp is good, go to **14**.
  - b. If ARM light lamp is burnt out, replace lamp (WP 1523 00). Go to **41**.
- 14. **UH-60A 78-22987 - SUBQ W/O MOD UH-60L 89-26149 - 89-26154 AND 89-26179 - 92-26407**  
**Check fuse in dispenser control panel.**
  - a. If fuse is good, go to **15**.
  - b. If fuse is blown, replace fuse (WP 1524 00). Go to **41**.
- 15. **UH-60A 78-22987 - SUBQ W/O MOD UH-60L 89-26149 - 89-26154 AND 89-26179 - 92-26407**  
**Check for 28 vdc between P1-H and ground.**
  - a. If voltage is as specified, go to **16**.
  - b. If voltage is not as specified, go to **17**.
- 16. **UH-60A 78-22987 - SUBQ W/O MOD UH-60L 89-26149 - 89-26154 AND 89-26179 - 92-26407**  
**Check continuity between P1-R and ground.**
  - a. If continuity is present, replace dispenser control panel (WP 1522 00). Go to **41**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **41**.
- 17. **UH-60A 78-22987 - SUBQ W/O MOD UH-60L 89-26149 - 89-26154 AND 89-26179 - 92-26407**  
**Check for 28 vdc between CHAFF DISP circuit breaker terminal 1 and ground.**
  - a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P1-H (WP 1747 00). Go to **41**.
  - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **41**.
- 18. **UH-60A 77-22724 - 78-22986 W/O MOD** **Press ARM light.**
  - a. If ARM light goes on, go to **19**.
  - b. If ARM light does not go on, go to **21**.
- 19. **UH-60A 77-22724 - 78-22986 W/O MOD** **Check for 28 vdc between J673R-P and ground.**
  - a. If voltage is as specified, replace electronic module (WP 1520 00). Go to **41**.
  - b. If voltage is not as specified, go to **20**.
- 20. **UH-60A 77-22724 - 78-22986 W/O MOD** **Check continuity between P1-F and J673R-F.**



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**ARM LIGHT DOES NOT GO ON - Continued**

- a. If continuity is present, replace dispenser control panel (WP 1522 00). Go to **41**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **41**.
- 21. UH-60A 77-22724 - 78-22986 W/O MOD Check ARM light lamp.**
  - a. If ARM light lamp is good, go to **22**.
  - b. If ARM light lamp is burnt out, replace lamp (WP 1523 00). Go to **41**.
- 22. UH-60A 77-22724 - 78-22986 W/O MOD Check fuse in dispenser control panel.**
  - a. If fuse is good, go to **23**.
  - b. If fuse is blown, replace fuse (WP 1524 00). Go to **41**.
- 23. UH-60A 77-22724 - 78-22986 W/O MOD Check for 28 vdc between P1-H and ground.**
  - a. If voltage is as specified, go to **24**.
  - b. If voltage is not as specified, go to **25**.
- 24. UH-60A 77-22724 - 78-22986 W/O MOD Check continuity between P1-R and ground.**
  - a. If continuity is present, replace dispenser control panel (WP 1522 00). Go to **41**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **41**.
- 25. UH-60A 77-22724 - 78-22986 W/O MOD Check for 28 vdc between P656R-EE and ground.**
  - a. If voltage is as specified, go to **26**.
  - b. If voltage is not as specified, go to **27**.
- 26. UH-60A 77-22724 - 78-22986 W/O MOD Check continuity between P656R-FF and P1-H.**
  - a. If continuity is present, replace audio junction box assembly (TM 11-1520-237-23). Go to **41**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **41**.
- 27. UH-60A 77-22724 - 78-22986 W/O MOD Check for 28 vdc between CHAFF DISP circuit breaker terminal 1 and ground.**
  - a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P656R-EE (WP 1747 00). Go to **41**.
  - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **41**.
- 28. UH-60A-77-22714 - 77-22723 W/O MOD Press ARM light.**
  - a. If ARM light goes on, go to **29**.
  - b. If ARM light does not go on, go to **31**.
- 29. UH-60A-77-22714 - 77-22723 W/O MOD Check for 28 vdc between J673R-P and ground.**
  - a. If voltage is as specified, replace electronic module (WP 1520 00). Go to **41**.
  - b. If voltage is not as specified, go to **30**.
- 30. UH-60A-77-22714 - 77-22723 W/O MOD Check continuity between P1-F and J673R-F.**
  - a. If continuity is present, replace dispenser control panel (WP 1522 00). Go to **41**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **41**.

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**ARM LIGHT DOES NOT GO ON - Continued**

31. **UH-60A-77-22714 - 77-22723 W/O MOD** Check ARM light lamp.
  - a. If ARM light lamp is good, go to **32**.
  - b. If ARM light lamp is burnt out, replace lamp (WP 1523 00). Go to **41**.
32. **UH-60A-77-22714 - 77-22723 W/O MOD** Check fuse in dispenser control panel.
  - a. If fuse is good, go to **33**.
  - b. If fuse is blown, replace fuse (WP 1524 00). Go to **41**.
33. **UH-60A-77-22714 - 77-22723 W/O MOD** Check for 28 vdc between P1-H and ground.
  - a. If voltage is as specified, go to **34**.
  - b. If voltage is not as specified, go to **35**.
34. **UH-60A-77-22714 - 77-22723 W/O MOD** Check continuity between P1-V and ground.
  - a. If continuity is present, replace dispenser control panel (WP 1522 00). Go to **41**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **41**.
35. **UH-60A-77-22714 - 77-22723 W/O MOD** Check for 28 vdc between P656R-EE and ground.
  - a. If voltage is as specified, go to **36**.
  - b. If voltage is not as specified, go to **40**.
36. **UH-60A-77-22714 - 77-22723 W/O MOD** Check continuity between P656R-FF and P1-H.
  - a. If continuity is present, go to **37**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **41**.
37. **UH-60A-77-22714 - 77-22723 W/O MOD** Check for 28 vdc between J163-A and J163-B.
  - a. If voltage is as specified, check for proper adjustment on left drag beam switch (WP 0443 00).
  - b. If adjustment can not be done, replace left drag beam switch (WP 0443 00). Go to **41**.
  - c. If voltage is not as specified, go to **38**.
38. **UH-60A-77-22714 - 77-22723 W/O MOD** Check continuity between J163-A and ground.
  - a. If continuity is present, go to **39**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **41**.
39. **UH-60A-77-22714 - 77-22723 W/O MOD** Check continuity between J163-B and P656R-GG.
  - a. If continuity is present, replace audio junction box assembly (TM 11-1520-237-23). Go to **41**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **41**.
40. Check for 28 vdc between CHAFF DISP circuit breaker terminal 1 and ground.
  - a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P656R-EE (WP 1747 00). Go to **41**.
  - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **41**.
41. Procedure completed.

**STRAY VOLTAGE LIGHT GOES ON****SYMPTOM**

STRAY VOLTAGE light goes on.

**MALFUNCTION**

STRAY VOLTAGE light goes on.

**CORRECTIVE ACTION**

- 1. Check for 28 vdc between J673R-A and ground.**
  - a. If voltage is as specified, go to **2**.
  - b. If voltage is not as specified, troubleshoot test set (TM 9-1095-206-12&P). Go to **3**.
- 2. Check continuity between J673R-A and remaining pins of J673R.**
  - a. If continuity is present, repair/replace wiring as required (WP 1747 00). Go to **3**.
  - b. If continuity is not present, replace dispenser control panel (WP 1522 00). Go to **3**.
- 3. Procedure completed.**

**CHAFF COUNTER DOES NOT COUNT DOWN****SYMPTOM**

Chaff counter does not count down.

**MALFUNCTION**

Chaff counter does not count down.

**CORRECTIVE ACTION**

- 1. Check helicopter configuration.**
  - a. **UH-60L 92-26408 - SUBQ MOD HH-60A HH-60L** , go to **2**.
  - b. **UH-60L 89-26149 UH-60L 89-26154 W/O MOD UH-60L 89-26179 - 92-26407** , go to **8**.
- 2. UH-60L 92-26408 - SUBQ AND MOD HH-60A HH-60L** Press and hold CHAFF DISPENSE switch. Go to **3**.
- 3. UH-60L 92-26408 - SUBQ AND MOD HH-60A HH-60L** Check continuity between P669R-C and P669R-D.
  - a. If continuity is present, go to **4**.
  - b. If continuity is not present, go to **6**.
- 4. UH-60L 92-26408 - SUBQ AND MOD HH-60A HH-60L** Check continuity between P669R-A and J673R-A.
  - a. If continuity is present, go to **5**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **17**.
- 5. UH-60L 92-26408 - SUBQ AND MOD HH-60A HH-60L** Check continuity between P669R-T and J673R-T.
  - a. If continuity is present, replace dispenser control panel (WP 1522 00). Go to **17**.
  - b. If trouble remains, replace electronics module (WP 1520 00). Go to **17**.

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**CHAFF COUNTER DOES NOT COUNT DOWN - Continued**

- c. If trouble still remains, replace dispenser assembly (WP 1520 00). Go to **17**.
- d. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **17**.
- 6. **UH-60L 92-26408 - SUBQ AND MOD HH-60A HH-60L** Check continuity between P669R-D and CHAFF DISPENSE switch terminal 3.
  - a. If continuity is present, go to **7**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **17**.
- 7. **UH-60L 92-26408 - SUBQ AND MOD HH-60A HH-60L** Check continuity between P669R-C and CHAFF DISPENSE switch terminal 4.
  - a. If continuity is present, replace switch (WP 1525 00). Go to **17**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **17**.
- 8. **UH-60L 89-26149 - 89-26154 AND 89-26179 - 92-26407 W/O MOD** Press and hold CHAFF DISPENSE switch. Go to **9**.
- 9. **UH-60L 89-26149 - 89-26154 AND 89-26179 - 92-26407 W/O MOD** Check continuity between P1-C and P1-D.
  - a. If continuity is present, go to **10**.
  - b. If continuity is not present, go to **12**.
- 10. **UH-60L 89-26149 - 89-26154 AND 89-26179 - 92-26407 W/O MOD** Check continuity between P1-A and J673R-A.
  - a. If continuity is present, go to **11**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **17**.
- 11. **UH-60L 89-26149 - 89-26154 AND 89-26179 - 92-26407 W/O MOD** Check continuity between P1-T and J673R-T.
  - a. If continuity is present, replace dispenser control panel (WP 1522 00). Go to **17**.
  - b. If trouble remains, replace electronics module (WP 1520 00). Go to **17**.
  - c. If trouble still remains, replace dispenser assembly (WP 1520 00). Go to **17**.
  - d. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **17**.
- 12. Check helicopter configuration.
  - a. **UH-60L 89-26149, 89-26154 UH-60L 89-26179-92-26407** , go to **13**.
  - b. **W/O MOD** , go to **15**.
- 13. **UH-60L 89-26149 - 89-26154 AND 89-26179 - 92-26407** Check continuity between P1-D and P20R-S.
  - a. If continuity is present, go to **14**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **17**.
- 14. **UH-60L 89-26149 - 89-26154 AND 89-26179 - 92-26407** Check continuity between P1-C and P20R-M.
  - a. If continuity is present, replace switch (WP 1525 00). Go to **17**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **17**.

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**CHAFF COUNTER DOES NOT COUNT DOWN** - Continued

15. **W/O MOD** Check continuity between P1-D and CHAFF DISP switch terminal 3.
  - a. If continuity is present, go to **16**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **17**.
16. **W/O MOD** Check continuity between P1-C and CHAFF DISP switch terminal 4.
  - a. If continuity is present, replace switch (WP 1525 00). Go to **17**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **17**.
17. Procedure completed.

**STRAY VOLTAGE LIGHT GOES ON****SYMPTOM**

STRAY VOLTAGE light goes on.

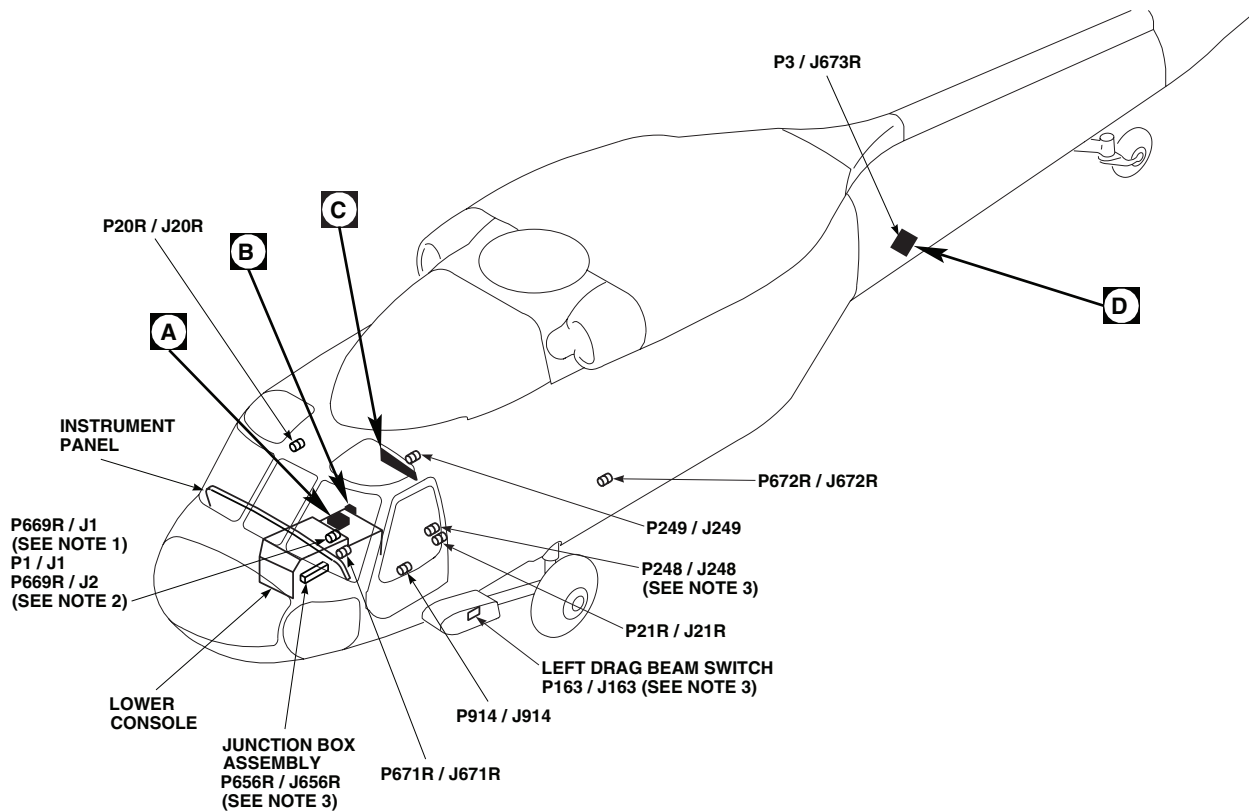
**MALFUNCTION**

STRAY VOLTAGE light goes on.

**CORRECTIVE ACTION**

1. Check for 28 vdc between J673R-B and ground.
  - a. If voltage is as specified, go to **2**.
  - b. If voltage is not as specified, troubleshoot test set (TM 9-1095-206-12&P). Go to **3**.
2. Check continuity between J673R-B and remaining pins of J673R.
  - a. If continuity is present, repair/replace wiring as required (WP 1747 00). Go to **3**.
  - b. If continuity is not present, replace dispenser control panel (WP 1522 00). Go to **3**.
3. Procedure completed.

## LOCATION



## NOTES

1. UH-60L 92-26408 - SUBQ  
MOD HH-60A HH-60L
2. UH-60L 89-26149  
89-26154  
AND 89-26179 -  
92-26407 W/O MOD
3. W/O MOD
4. UH-60A UH-60L PRIOR TO  
92-26408 .

| TERMINAL BOARD /<br>DISCONNECT PLUG /<br>RECEPTACLE | LOCATION /<br>CONNECTION POINT                         |
|-----------------------------------------------------|--------------------------------------------------------|
| P1 / J1                                             | DISPENSER CONTROL<br>PANEL<br>(SEE NOTE 2)             |
| P3 / J673R                                          | ELECTRONIC MODULE                                      |
| P20R / J20R                                         | CABIN, BL 40 RH, STA 248<br>(SEE NOTE 4)               |
| P21R / J21R                                         | CABIN, BL 40 LH, STA 248                               |
| P163 / J163                                         | LEFT DRAG BEAM SWITCH<br>(SEE NOTE 3)                  |
| P248 / J248                                         | CABIN, BL 40 LH, STA 248<br>(SEE NOTE 3)               |
| P249 / J249                                         | BEHIND COPILOT'S<br>CIRCUIT BREAKER PANEL,<br>BL 23 LH |
| P656R / J656R                                       | JUNCTION BOX ASSEMBLY                                  |
| P669R / J1                                          | DISPENSER CONTROL<br>PANEL<br>(SEE NOTE 1)             |
| P669R / J2                                          | COCKPIT, BL 4 LH, STA 245<br>(SEE NOTE 2)              |
| P671R / J671R                                       | COCKPIT, BL 0, STA 235                                 |
| P672R / J672R                                       | CABIN, BL 37 LH, STA 389                               |
| P914 / J914                                         | COCKPIT, BL 24 LH,<br>STA 247                          |

AA2619\_1B  
SA

Figure 1. Chaff Dispenser System Location Diagram. (Sheet 1 of 4)

LOCATION - Continued

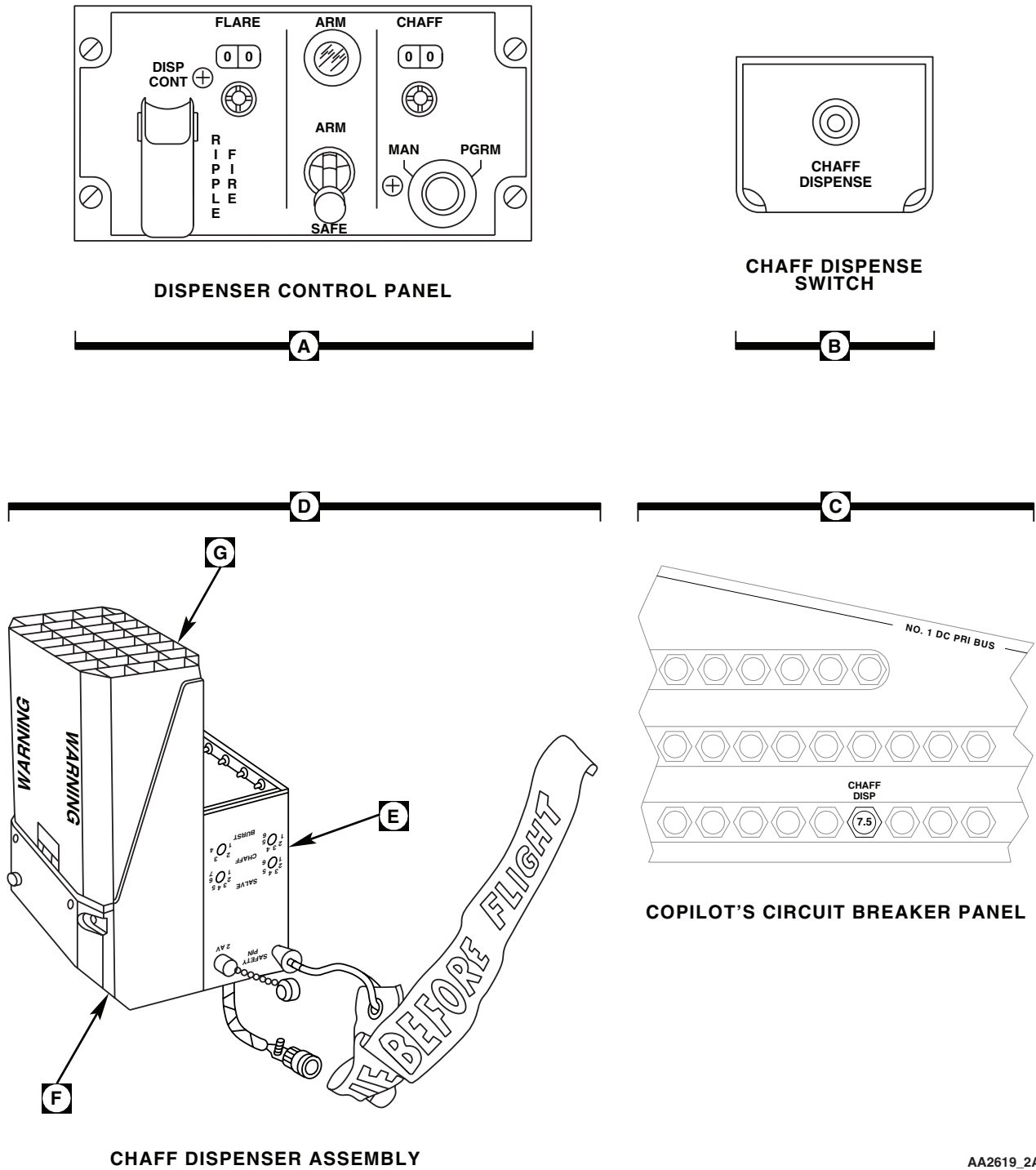


Figure 1. Chaff Dispenser System Location Diagram. (Sheet 2 of 4)

AA2619\_2A  
SA

SAFETY PIN AND WARNING FLAG

SAFETY PIN

24 V

SAFETY PIN

3 5 8

1 2 9 SALVO

1 2 9 LOUD CHAFF

3 5 8 BURST

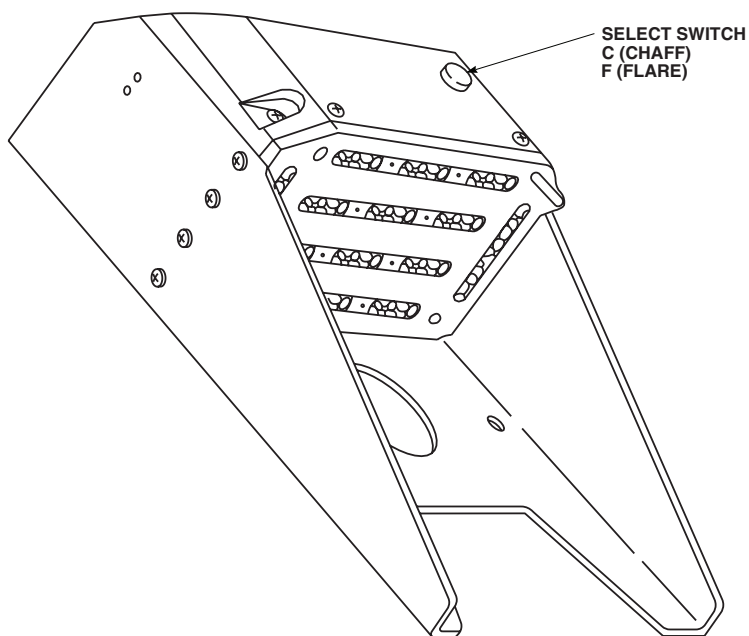
BEFORE FLIGHT

ELECTRONICS MODULE

## ELECTRONICS MODULE

**E**

**F**



SELECT SWITCH  
C (CHAFF)  
F (FLARE)

## DISPENSER ASSEMBLY

AA2619\_3A  
SA

**0162 00-16**

### Change 1



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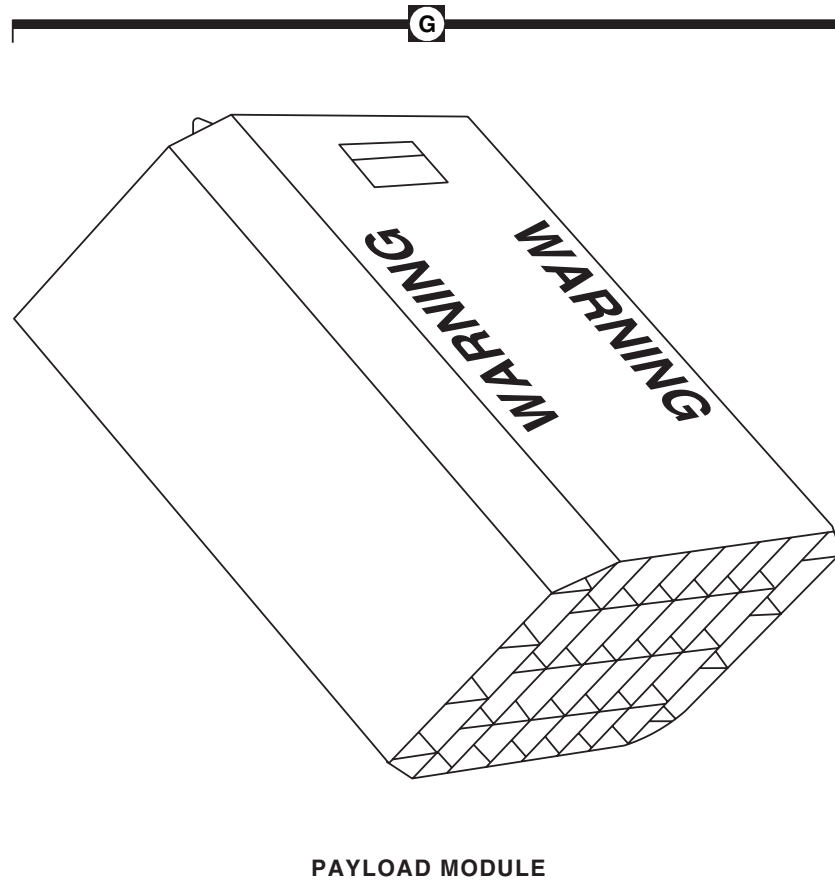
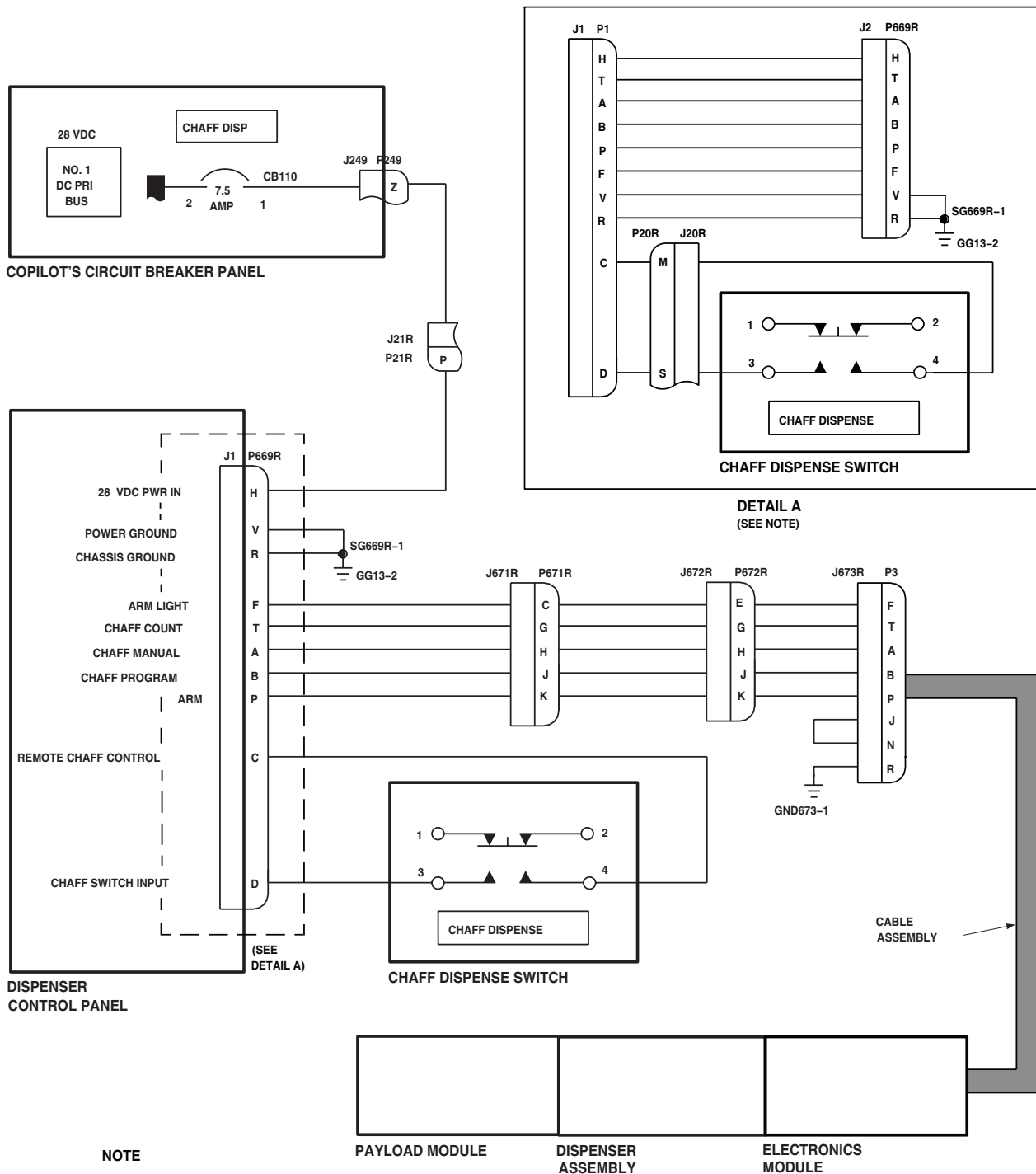
**LOCATION** - ContinuedAA2619\_4A  
SA

Figure 1. Chaff Dispenser System Location Diagram. (Sheet 4 of 4)

# SCHEMATICS



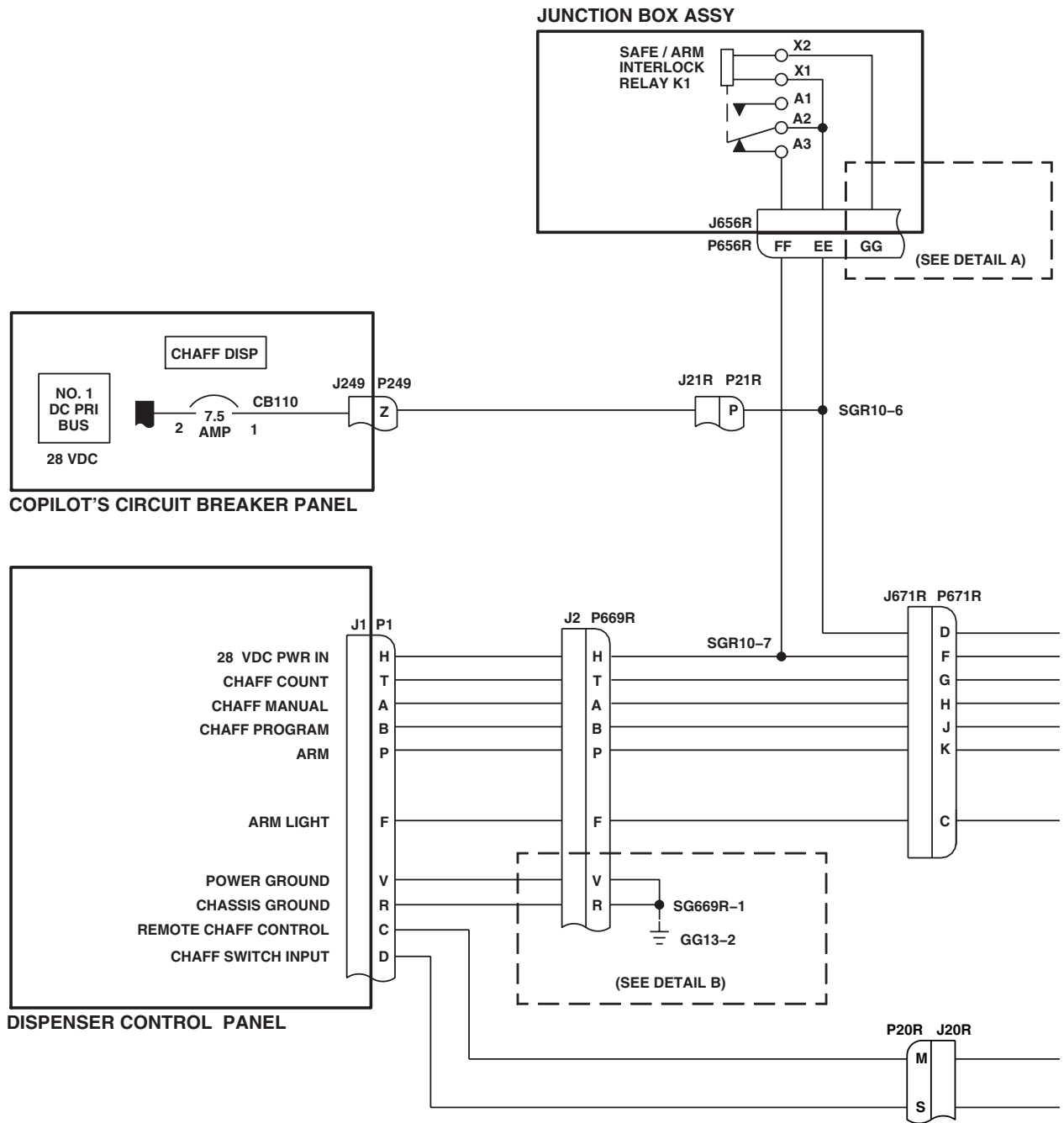
## NOTE

UH-60L 89-26149  
89-26154  
AND 89-26179 -  
92-26407

AB1472B  
SA

Figure 2. Chaff Dispenser System Schematic Diagram UH-60L MOD HH-60A HH-60L .

## SCHEMATICS - Continued



## NOTES

1. UH-60A 77-22714 - 77-22723
2. DRAG BEAM SWITCH SHOWN WITH WEIGHT - ON - WHEELS.

## EFFECTIVITY

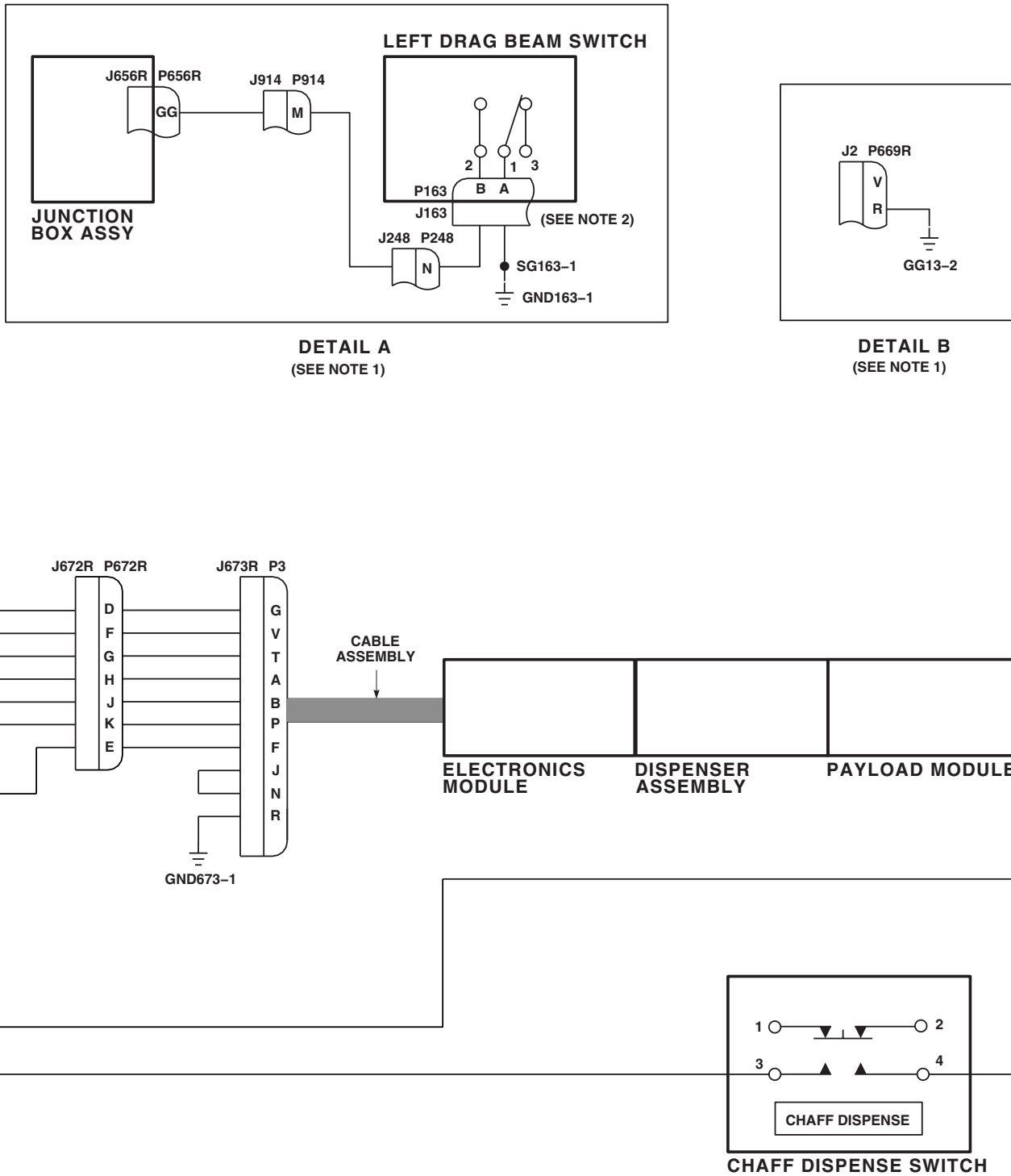
UH-60A

W/O MOD

AB1473\_1A  
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Figure 3. Chaff Dispenser System Schematic Diagram UH-60A W/OMOD . (Sheet 1 of 2)

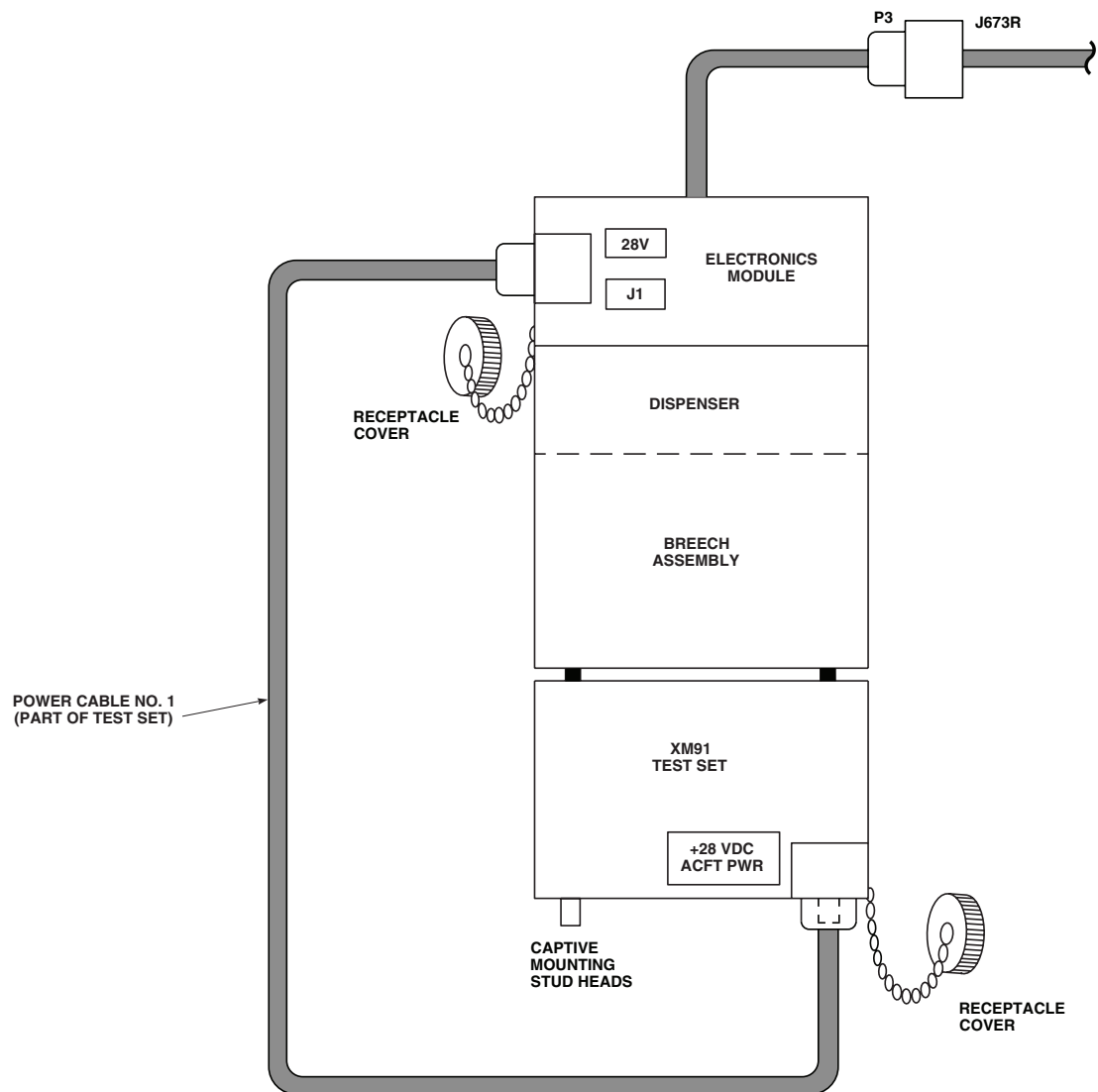
SCHEMATICS - Continued



AB1473\_2  
SA

Figure 3. Chaff Dispenser System Schematic Diagram **UH-60A W/OMOD** . (Sheet 2 of 2)

TEST SETUP



AA2623  
SA

Figure 4. Chaff Dispenser System Test Setup.

END OF WORK PACKAGE



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**UNIT LEVEL****MISSION EQUIPMENT****CHAFF/FLARE DISPENSER (XM130) SYSTEM EH-60A****This WP supersedes WP 0163 00, dated 17 April 2006.**

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**INITIAL SETUP:****Test Equipment**

Multimeter, AN/PSM-45A

Test Set, Electronics System, M92

TM 11-5865-200-12

WP 0104 00

WP 0824 00

WP 1520 00

**Tools and Special Tools**

Electronic Equipment Toolkit, TK-101/G

Locally-Made Dispenser Test Cable Figure 175,

WP 1805 00

Locally-Made Drag Beam Wedge Figure 167,

WP 1805 00

WP 1522 00

WP 1523 00

WP 1524 00

WP 1525 00

WP 1526 00

WP 1527 00

**Material/Parts**

Isopropyl Alcohol, Technical, Item 170, WP 1803 00

Machinery Towel, Item 211, WP 1803 00

WP 1685 00

WP 1747 00

WP 1803 00

WP 1805 00

**Personnel Required**

Avionics Mechanic MOS 15N (1)

**Equipment Condition**

External Electrical Power Available, WP 1685 00

or APU Operational, TM 1-1520-237-10

**References**

TM 1-1520-237-10

TM 9-1095-206-12&P

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**SPECIAL INSTRUCTIONS****NOTE**

See Figure 1 for component location diagram, Figure 2 for schematic diagram, and Figure 3 for system test setup as an aid in operational/troubleshooting.

The chaff/flare dispenser system operational/troubleshooting procedure is subdivided as follows:

- Helicopter/Dispenser System Interface
- Flare Dispenser System
- Chaff Dispenser System
- Shutdown

## HELICOPTER/DISPENSER SYSTEM INTERFACE OPERATIONAL/TROUBLESHOOTING PROCEDURE

### PROCEDURE

#### Step

#### WARNING

Injury to personnel can occur if maintenance or checks on dispenser system is done with flare and chaff dispenser payload modules installed. Remove flare and chaff dispenser payload modules.

#### NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Make sure that the CHAFF/FLARE DISP circuit breaker on the copilot's circuit breaker panel is pushed in.
2. Turn on electrical power.
3. Remove safety pin from dispenser system electronics module.
4. Turn dispenser control panel ARM indicator DIM control fully counterclockwise.
5. Place dispenser control panel ARM-SAFE switch to ARM.

#### Indication/Condition

ARM indicator shall go on.

#### Corrective Action

If Indication/Condition is not as specified, go to ARM INDICATOR DOES NOT GO ON, in this work package.

#### Step

6. With dispenser control panel arm indicator pressed, turn dispenser control panel ARM indicator DIM control fully clockwise, and then fully counterclockwise.

#### Indication/Condition

ARM indicator shall go from bright to dim and then bright again.

#### Corrective Action

If Indication/Condition is not as specified, replace dispenser control panel (WP 1522 00).



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**FLARE DISPENSER SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE****PROCEDURE****Step****WARNING**

Injury to personnel can occur if maintenance or checks on dispenser system is done with flare and chaff dispenser payload modules installed. Remove flare and chaff dispenser payload modules.

**NOTE**

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Do HELICOPTER/DISPENSER SYSTEM INTERFACE, in this work package, if not already done.
2. Pull out CHAFF/FLARE DISP circuit breaker (copilot's circuit breaker panel).
3. Pull out ALQ-156 and LIGHTS CAUT/ADVSY circuit breakers (copilot's circuit breaker panel).

**CAUTION**

Make sure antennas are connected to the ALQ-156 receiver-transmitter before turning on control indicator. Operation without antennas connected will damage transmitter.

4. Connect test set (M92) to chaff/flare dispenser system as follows (Figure 3):
  - Connect P1 of test cable (70600-45008-041) to helicopter J673R.
  - Connect J1 of test cable (70600-45008-041) to P3 of electronics module.
  - Connect P1 of test cable No. 2 (9325992) to J1 of dispenser test adapter assembly (9326211).
  - Connect P2 of power cable No. 1 (9325989) to J1 of test set.
  - Connect P1 of power cable No. 1 (9325989) to J128-V on electronics module.
  - Skip this step when doing chaff dispenser check. Insert flare simulator (9325117) mounting rod into faceplate of dispenser test adapter assembly. Secure mounting rod with thumbscrew (fingertight).
  - Connect dispenser test adapter assembly to breech assembly of dispenser. Using a 5/32-inch hexagonal wrench, handtighten both mounting studs evenly.
  - Place dispenser assembly C-F selector switch to F for flare checks, or C for chaff checks.
  - Skip this step when doing chaff dispenser checks. Connect P1 of test cable No. 1 (9325991) to J3 of test set.
  - Skip this step when doing chaff dispenser checks. Support flare simulator assembly and connect P7 of test cable No. 1 to J1 of flare simulator assembly.
  - Connect red lead and black lead of multimeter to test jacks J8 and J9 of test set.
  - Position electronic module CHAFF switches as follows:
    - SALVO COUNT 4 (bursts)
    - SALVO INTERVAL 2 (seconds)

**FLARE DISPENSER SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**

- BURST COUNT 3 (firings)
- BURST INTERVAL 0.4 (seconds)
- If installed, remove safety pin from electronics module SAFETY PIN receptacle.

5. Place dispenser control panel switches as follows:

|                          |      |
|--------------------------|------|
| ARM-SAFE                 | SAFE |
| RIPPLE FIRE              | Down |
| FLARE and CHAFF Counters | 30   |

6. Push in CHAFF/FLARE DISP circuit breaker (copilot's circuit breaker panel).
7. Make sure PUSH FOR STANDBY pushbutton on the ALQ-156 control indicator panel is at "out" position (not depressed).

**NOTE**

Warmup time is approximately eight to ten minutes. It may be as long as 30 minutes or as little as 15 seconds depending upon equipment and environmental temperature.

8. Set ALQ-156 control indicator POWER ON/OFF switch to ON position.

**Indication/Condition**

WRMUP STATUS lamp shall light.

**Corrective Action**

If Indication/Condition is not as specified, troubleshoot ALQ-156 system (TM 11-5865-200-12).

**Step****WARNING**

When the ALQ-156 is on, high frequency electromagnetic radiation is present. High frequency electromagnetic radiation can cause internal burns without causing any sensation of heat. When the WRMUP (warm-up) lamp goes out, press PUSH FOR STANDBY pushbutton to disable transmitter.

9. When the WRMUP lamp goes out, the ALQ-156 system is on. Depress PUSH FOR STANDBY pushbutton on control indicator.

**Indication/Condition**

STBY STATUS lamp shall light.

**Corrective Action**

If Indication/Condition is not as specified, troubleshoot ALQ-156 system (TM 11-5865-200-12).

**Step**

10. Place dispenser control panel ARM-SAFE switch to ARM.

---

**FLARE DISPENSER SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****NOTE**

With dispenser test adapter assembly installed and 28 vdc power applied, sequencer switch in dispenser assembly may automatically reset. This results in an audible sound in the immediate vicinity.

11. Momentarily press test set POWER ON indicator.

**Indication/Condition**

POWER ON indicator shall go on when pressed and go off when released.

**Corrective Action**

1. If Indication/Condition is not as specified, troubleshoot test set (TM 9-1095-206-12&P).
  - a. If trouble remains, replace electronics module (WP 1520 00).

**Step**

12. Place test set UNIT POWER switch to OFF.
13. Place test set POWER ON switch to ON.

**Indication/Condition**

POWER ON indicator shall go on.

**Corrective Action**

If Indication/Condition is not as specified, troubleshoot test set (TM 9-1095-206-12&P).

**Step**

14. Position test set RANGE switch to 1 to 15 position and SEQUENCE switch to OFF.
15. Momentarily press test set STRAY VOLTAGE SELF TEST switch.

**Indication/Condition**

Multimeter shall indicate 28 vdc while switch is pressed.

**Corrective Action**

If Indication/Condition is not as specified, check test setup cable connections (Figure 3).

**Step**

16. Turn test set SEQUENCE switch clockwise, stopping for 5 seconds at each position to read multimeter, until SEQUENCE switch is turned to OFF.

**Indication/Condition**

At each switch position, multimeter shall indicate 0.25 vdc or less for 3 seconds or less.

**Corrective Action**

1. If Indication/Condition is not as specified, replace dispenser assembly (WP 1520 00).
  - a. If trouble remains, replace electronics module (WP 1520 00).

**Step**

17. Place test set RANGE switch within 16 to 30 position.

---

**FLARE DISPENSER SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued

18. Turn test set SEQUENCE switch clockwise, stopping for 5 seconds at each position to read multimeter, until SEQUENCE switch is turned to OFF.

**Indication/Condition**

At each switch position, multimeter shall indicate 0.25 vdc or less for 3 seconds or less.

**Corrective Action**

1. If Indication/Condition is not as specified, replace dispenser assembly (WP 1520 00).
  - a. If trouble remains, replace electronics module (WP 1520 00).

**Step**

19. Place multimeter to lowest dc voltage range and repeat steps 11., 12., and 13.
20. Place test set POWER switch to OFF.

**CAUTION**

To prevent damage to multimeter, do not press test set STRAY VOLTAGE SELF TEST switch while doing resistance checks.

21. Set multimeter to read resistance on RX1000 ohms scale.
22. Place test set RANGE switch within 1 to 15 position.
23. Turn test set SEQUENCE switch clockwise, stopping at each position to read multimeter, until SEQUENCE switch is turned to OFF.

**Indication/Condition**

Multimeter shall read between 800 and 900 ohms in each switch position except for first position, which shall read between 250 and 400 ohms.

**Corrective Action**

If Indication/Condition is not as specified, press and release MANUAL SYSTEM RESET switch on dispenser test adapter.

**Step**

24. Place test set RANGE switch within 16 to 30 position and repeat step 22.
25. Place test set POWER switch to ON.
26. Press test set FIRING ORDER RESET pushbutton.

**Indication/Condition**

Firing order indicators 1 through 30 shall change from white to red.

**Corrective Action**

1. If Indication/Condition is not as specified, repeat step 24.
  - a. If trouble remains, troubleshoot test set (TM 9-1095-206-12&P).

---

**FLARE DISPENSER SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued**Step****CAUTION**

Step 25. may require operator to press FLARE SIMULATOR RESET pushbutton. Do not press and hold pushbutton, or damage to test set may result.

27. Momentarily press test set FLARE SIMULATOR RESET pushbutton, if flare simulator assembly indicator (amber) is not on.

**Indication/Condition**

Flare simulator assembly indicator (amber) shall go on.

**Corrective Action**

If Indication/Condition is not as specified, troubleshoot test set (TM 9-1095-206-12&P).

**Step**

28. Press instrument panel FLARES pushbutton once.

**Indication/Condition**

- a. Dispenser control panel FLARE counter shall count down to 29.
- b. Test set firing order No. 1 indicator changes from red to white.

**Corrective Action**

1. If Indication/Conditions a. and b. are not as specified, remove dispenser test adapter assembly from flare dispenser and clean contacts of both units with machinery towel, Item 211, WP 1803 00 dampened with isopropyl alcohol, Item 170, WP 1803 00. Reinstall test adapter assembly and repeat operational/troubleshooting procedure from step 20.
  - a. If trouble remains, go to FLARE COUNTER AND TEST SET DO NOT REGISTER SIMULATED FLARE LAUNCH, in this work package.
2. If Indication/Condition a. is not as specified, replace dispenser control panel (WP 1522 00).
  - a. If trouble remains, repair/replace wiring between J998R-5 and J669R-U (WP 1747 00).
3. If Indication/Condition b. is not as specified, troubleshoot test set (TM 9-1095-206-12&P).
4. If dispenser control panel FLARE counter counts down to 27 and test set firing order indicators 1 through 3 change from red to white, replace flare dispenser (WP 1527 00).
  - a. If trouble remains, replace flare simulator and/or test set.

**Step**

29. Press instrument panel FLARES pushbutton once.

**Indication/Condition**

- a. Dispenser control panel FLARE counter shall count down to 26.
- b. Test set firing order No. 2, 3, and 4 indicators shall change from red to white.

**Corrective Action**

1. If Indication/Conditions a. and b. are not as specified, replace flare dispenser assembly (WP 1520 00).
  - a. If trouble remains, replace electronics module (WP 1520 00).

---

**FLARE DISPENSER SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued

- b. If trouble still remains, repair/replace wiring between flare dispenser and electronics module (WP 1747 00).

**Step**

30. Raise dispenser control panel RIPPLE FIRE switch guard and place switch in up position until FLARE counter reads 00.

**Indication/Condition**

- a. FLARE counter shall count down to 00.
- b. Test set firing order indicators No. 5 through 30 shall change from red to white.

**Corrective Action**

1. If Indication/Condition a. is not as specified, replace dispenser control panel (WP 1522 00).
2. If Indication/Conditions a. and b. are not as specified, go to FLARE COUNTER DOES NOT COUNT DOWN WHEN RIPPLE FIRE SWITCH IS IN UP POSITION, in this work package.

**Step****NOTE**

When MANUAL SYSTEM RESET switch on dispenser test adapter assembly is pressed, dispenser assembly sequencer switch makes an audible sound as the switch turns.

31. Press and release dispenser test adapter assembly MANUAL SYSTEM RESET switch.
32. Press test set FIRING ORDER RESET pushbutton.
33. Set dispenser control panel FLARE counter to read 30.
34. Place dispenser control panel ARM-SAFE switch to SAFE.

**Indication/Condition**

Dispenser control panel ARM indicator shall go off.

**Corrective Action**

If Indication/Condition is not as specified, replace dispenser control panel (WP 1522 00).

**Step**

35. Raise dispenser control panel RIPPLE FIRE switch guard and place switch to up position until FLARE counter reads 00.

**Indication/Condition**

- a. Dispenser control panel ARM indicator shall go on.
- b. Dispenser control panel FLARE counter shall count down to 00.

**Corrective Action**

If Indication/Condition a. and/or b. are not as specified, replace dispenser control panel (WP 1522 00).

**Step**

36. Place dispenser control panel RIPPLE FIRE switch guard to down position.

---

**FLARE DISPENSER SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued**Indication/Condition**

Dispenser control panel ARM indicator shall go off.

**Corrective Action**

If Indication/Condition is not as specified, replace dispenser control panel (WP 1522 00).

**Step**

37. Set dispenser control panel FLARE counter to 30 and ARM-SAFE switch to ARM.
38. Press and release dispenser test adapter assembly MANUAL SYSTEM RESET switch.
39. Press test set FIRING ORDER RESET pushbutton.
40. Depress PUSH FOR STANDBY pushbutton on ALQ-156 control indicator.

**Indication/Condition**

STBY lamp shall go out.

**Corrective Action**

If Indication/Condition is not as specified, troubleshoot ALQ-156 system (TM 11-5865-200-12).

**Step**

41. Momentarily place ALQ-156 control indicator TEST FLARE switch to TEST (up) position.

**Indication/Condition**

FLARE counter on dispenser control panel shall read 29 and firing order indicator No. 1 on test set shall change to WHITE.

**Corrective Action**

1. If Indication/Condition is not as specified, replace electronics module (WP 1520 00).
  - a. If trouble remains, check continuity between control indicator, dispenser control panel, and electronics module.
    - (1) If continuity is not present, repair/replace wiring as required (WP 1747 00).
    - (2) If continuity is present, troubleshoot ALQ-156 system (TM 11-5865-200-12).

**Step**

42. Place ALQ-156 control indicator POWER ON/OFF switch to OFF.
43. Place dispenser control panel ARM-SAFE switch to SAFE.
44. Disconnect P7 of test cable No. 1 from J1 at flare simulator assembly.
45. Remove dispenser test adapter assembly from flare dispenser assembly.
46. Remove flare simulator assembly from dispenser test adapter assembly.
47. If no other check is to be done, go to SHUTDOWN, in this work package.

**Indication/Condition**

Shutdown shall be performed.

---

**FLARE DISPENSER SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Corrective Action**

None Required

**CHAFF DISPENSER SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE  
PROCEDURE****Step****WARNING**

Injury to personnel can occur if maintenance or checks on dispenser system is done with flare and chaff dispenser payload modules installed. Remove flare and chaff dispenser payload modules.

**NOTE**

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Do HELICOPTER/DISPENSER SYSTEM INTERFACE, in this work package, if not already done.
2. Do steps 2. through 24. of Flare Dispenser System, in this work package. Skip all steps that apply to flare dispenser system as noted.
3. Place dispenser control panel MAN-PGRM switch to MAN.
4. Place dispenser control panel ARM-SAFE switch to ARM.

**Indication/Condition**

ARM indicator shall go on.

**Corrective Action**

If Indication/Condition is not as specified, go to ARM INDICATOR DOES NOT GO ON, in this work package.

**Step**

5. Press lower console CHAFF DISP pushbutton once.

**Indication/Condition**

- a. Dispenser control panel CHAFF counter shall count down to 29.
- b. Test set firing order indicator No. 1 shall change from red to white.

**Corrective Action**

1. If Indication/Conditions a. and b. are not as specified, go to CHAFF COUNTER AND TEST SET DO NOT REGISTER SIMULATED CHAFF LAUNCH, in this work package.
2. If Indication/Condition a. is not as specified, replace dispenser control panel (WP 1522 00).
3. If Indication/Condition b. is not as specified, troubleshoot test set (TM 9-1095-206-12&P).

**Step**

6. Place dispenser control panel MAN-PGRM switch to PGRM.
7. Press lower console CHAFF DISP pushbutton once.



---

**CHAFF DISPENSER SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued**Indication/Condition**

- a. Dispenser control panel CHAFF counter shall count down to 17.
- b. Test set firing order indicators shall change from red to white to read same count as dispenser control panel CHAFF counter.

**Corrective Action**

1. If Indication/Conditions a. and/or b. are not as specified, go to CHAFF DISPENSER PROGRAM MODE OF OPERATION DOES NOT OPERATE, in this work package.

**Step**

8. Repeat step 58. until all test set firing order indicators change from red to white.
9. Press and release dispenser test adapter assembly MANUAL SYSTEM RESET pushbutton.
10. Press test set FIRING ORDER RESET pushbutton.

**Indication/Condition**

All test set firing order indicators shall change from white to red.

**Corrective Action**

If Indication/Condition is not as specified, troubleshoot test set (TM 9-1095-206-12&P).

**Step**

11. Place dispenser control panel ARM-SAFE switch to SAFE.
12. Remove dispenser test adapter assembly from chaff dispenser assembly.
13. If no other check is to be done, go to SHUTDOWN, in this work package.

**Indication/Condition**

Shutdown shall be performed.

**Corrective Action**

None Required

**SHUTDOWN OPERATIONAL/TROUBLESHOOTING PROCEDURE****PROCEDURE****Step**

1. Turn off electrical power.
2. Disconnect power cable No. 1 from electronics module.
3. Disconnect test cable (70600-45008-041) from electronics module P3 and helicopter J673R.
4. Connect electronics module P3 to helicopter J673R.
5. Insert safety-pin flag assembly into electronics module SAFETY PIN receptacle.

**Indication/Condition**

Safety-pin flag assembly shall be inserted into the electronics module SAFETY PIN receptacle.

---

**SHUTDOWN OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued**Corrective Action**

None Required

**ARM INDICATOR DOES NOT GO ON****SYMPTOM**

ARM indicator does not go on.

**MALFUNCTION**

ARM indicator does not go on.

**CORRECTIVE ACTION**

- 1. Press ARM indicator Press-To-Test lamp. Does indicator go on?**
  - a. If indicator goes on, go to **2**.
  - b. If indicator does not go on, replace lamp (WP 1523 00). Go to **7**.
  - c. If trouble remains, go to **3**.
- 2. Pullout CHAFF/FLARE DISP circuit breaker and connect a jumper wire between J673R-P and P673R-F. Check continuity between P669R-P and P669R-F.**
  - a. If continuity is present, replace electronics module (WP 1520 00). Go to **7**.
  - b. If trouble remains, replace dispenser control panel (WP 1522 00). Go to **7**.
  - c. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **7**.
- 3. Check dispenser control panel fuse.**
  - a. If fuse is good, go to **4**.
  - b. If fuse is not good, replace fuse (WP 1524 00). Go to **7**.
- 4. Check for 28 vdc between P669R-H and ground.**
  - a. If voltage is as specified, replace dispenser control panel (WP 1522 00). Go to **7**.
  - b. If voltage is not as specified, go to **5**.
- 5. Check for 28 vdc between P656R-EE and ground.**
  - a. If voltage is as specified, replace junction box assembly (TM 11-1520-249-23-3). Go to **7**.
  - b. If voltage is not as specified, go to **6**.
- 6. Check for 28 vdc between CHAFF/FLARE DISP circuit breaker terminal 1 and ground.**
  - a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P656R-EE (WP 1747 00). Go to **7**.
  - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **7**.
- 7. Procedure completed.**

---

**FLARE COUNTER AND TEST SET DO NOT REGISTER SIMULATED FLARE LAUNCH****SYMPTOM**

FLARE counter and test set do not register simulated flare launch.

**MALFUNCTION**

FLARE counter and test set do not register simulated flare launch.

**CORRECTIVE ACTION**

- 1. With FLARES pushbutton pressed, check for 28 vdc between P999R-5 and ground.**
  - a. If voltage is as specified, replace flare dispenser (WP 1527 00). Go to **8**.
  - b. If voltage is not as specified, go to **2**.
- 2. With FLARES pushbutton pressed, check for 28 vdc between J998R-5 and ground.**
  - a. If voltage is as specified, repair/replace wiring between P998R-5 and P999R-5 (WP 1747 00). Go to **8**.
  - b. If voltage is not as specified, go to **3**.
- 3. With FLARES pushbutton pressed, check for 28 vdc between J673R-M and ground.**
  - a. If voltage is as specified, replace electronics module (WP 1520 00). Go to **8**.
  - b. If voltage is not as specified, go to **4**.
- 4. Check for 28 vdc between P16R-B and ground.**
  - a. If voltage is as specified, go to **5**.
  - b. If voltage is not as specified, go to **7**.
- 5. With FLARES pushbutton pressed, check continuity between J16R-B and J16R-C.**
  - a. If continuity is present, repair/replace wiring between P16R-C and J673R-M (WP 1747 00). Go to **8**.
  - b. If continuity is not present, go to **6**.
- 6. Check continuity between:**
  - **J16R-B and FLARE switch terminal 4**
  - **J16R-C and FLARE switch terminal 3**
  - a. If continuity is present, replace switch (WP 1526 00). Go to **8**.
  - b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **8**.
- 7. Check continuity between P669R-M and P16R-B.**
  - a. If continuity is present, replace dispenser control panel (WP 1522 00). Go to **8**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **8**.
- 8. Procedure completed.**

**FLARE COUNTER DOES NOT COUNT DOWN WHEN RIPPLE FIRE SWITCH IS IN UP POSITION****SYMPTOM**

FLARE counter does not count down when RIPPLE FIRE switch is in up position.

**MALFUNCTION**

FLARE counter does not count down when RIPPLE FIRE switch is in up position.

**CORRECTIVE ACTION**

- 1. With RIPPLE FIRE switch in up position, check for 28 vdc between J673R-S and ground.**
  - a. If voltage is as specified, replace electronics module (WP 1520 00). Go to **3**.
  - b. If trouble remains, replace flares dispenser assembly (WP 1527 00). Go to **3**.
  - c. If voltage is not as specified, go to **2**.
- 2. Check continuity between P669R-S and J673R-S.**
  - a. If continuity is present, replace dispenser control panel (WP 1522 00). Go to **3**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **3**.
- 3. Procedure completed.**

**CHAFF COUNTER AND TEST SET DO NOT REGISTER SIMULATED CHAFF LAUNCH****SYMPTOM**

Chaff counter and test set do not register simulated chaff launch.

**MALFUNCTION**

Chaff counter and test set do not register simulated chaff launch.

**CORRECTIVE ACTION**

- 1. With CHAFF DISPENSE pushbutton pressed, check continuity between P669R-C and P669R-D.**
  - a. If continuity is present, go to **2**.
  - b. If continuity is not present, replace CHAFF DISPENSE pushbutton (WP 1525 00). Go to **4**.
  - c. If trouble remains, repair/replace wiring to CHAFF DISPENSE pushbutton (WP 1747 00). Go to **4**.
- 2. Check continuity between P669R-A and J673R-A.**
  - a. If continuity is present, go to **3**.
  - b. If continuity is not present, repair/replace wiring between P669R-A and J673R-A (WP 1747 00). Go to **4**.
- 3. With CHAFF DISPENSE pushbutton depressed, check for 28 vdc between J1-A of test cable 70600-45008 and ground.**
  - a. If voltage is as specified, replace electronics module (WP 1520 00). Go to **4**.
  - b. If trouble remains, replace chaff dispenser (WP 1520 00). Go to **4**.
  - c. If voltage is not as specified, replace dispenser control panel (WP 1522 00). Go to **4**.

---

**CHAFF COUNTER AND TEST SET DO NOT REGISTER SIMULATED CHAFF LAUNCH - Continued****4. Procedure completed.****CHAFF DISPENSER PROGRAM MODE OF OPERATION DOES NOT OPERATE****SYMPTOM**

Chaff dispenser program mode of operation does not operate.

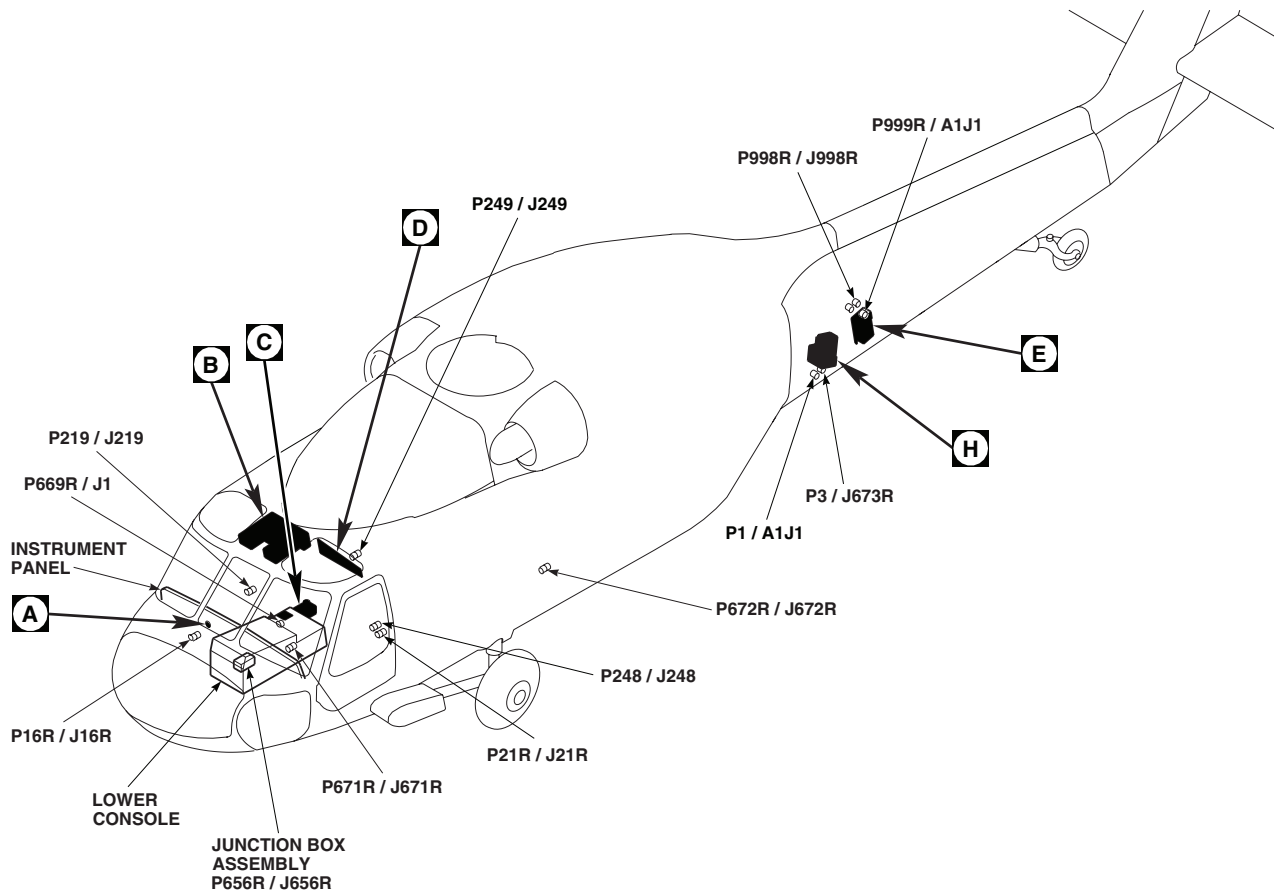
**MALFUNCTION**

Chaff dispenser program mode of operation does not operate.

**CORRECTIVE ACTION**

- 1. With CHAFF DISPENSE pushbutton pressed, check for 28 vdc between J1-B of test cable 70600-45008 and ground.**
  - a. If voltage is as specified, replace electronics module (WP 1520 00). Go to **3**.
  - b. If trouble remains, replace chaff dispenser (WP 1520 00). Go to **3**.
  - c. If voltage is not as specified, go to **2**.
- 2. Check continuity between P669R-B and J673R-B.**
  - a. If continuity is present, replace dispenser control panel (WP 1522 00). Go to **3**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **3**.
- 3. Procedure completed.**

## LOCATION



| TERMINAL BOARD /<br>DISCONNECT PLUG /<br>RECEPTACLE | LOCATION /<br>CONNECTION POINT |
|-----------------------------------------------------|--------------------------------|
| P1 / A1J1                                           | DISPENSER ASSEMBLY             |
| P2 / A2J2                                           | ALQ-156 CONTROL<br>INDICATOR   |
| P3 / J673R                                          | ELECTRONICS MODULE             |
| P16R / J16R                                         | COCKPIT, BL 7.5 RH, STA<br>197 |
| P21R / J21R                                         | CABIN, BL 40 RH, STA 248       |

| TERMINAL BOARD /<br>DISCONNECT PLUG /<br>RECEPTACLE | LOCATION /<br>CONNECTION POINT      |
|-----------------------------------------------------|-------------------------------------|
| P219 / J219                                         | CABIN FLOOR, BL 25 RH,<br>STA 247   |
| P246 / J246 (SEE NOTE)                              | UPPER CONSOLE                       |
| P248 / J248                                         | CABIN, BL 40 LH, STA 248            |
| P249 / J249                                         | CABIN CEILING, BL 23 LH,<br>STA 247 |
| P656R / J656R                                       | AUDIO JUNCTION BOX<br>ASSEMBLY      |
| P669R / J1                                          | DISPENSER CONTROL<br>PANEL          |
| P671R / J671R                                       | COCKPIT, BL 0, STA 235              |
| P672R / J672R                                       | CABIN, BL 37 LH, STA 389            |
| P998R / J998R                                       | LEFT TAIL CONE, STA 525             |
| P999R / A1J1                                        | FLARE DISPENSER                     |

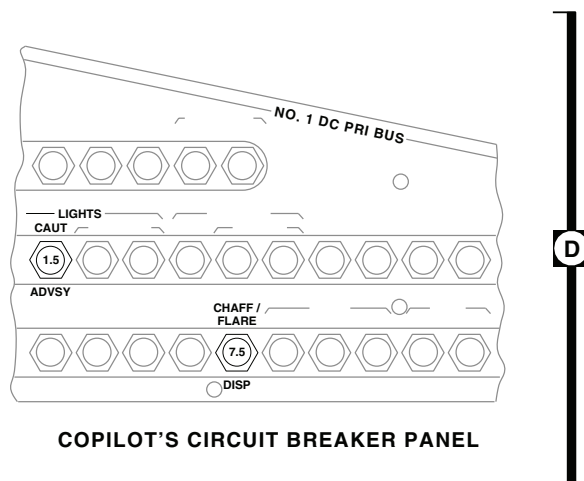
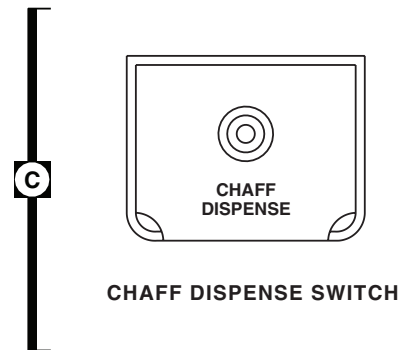
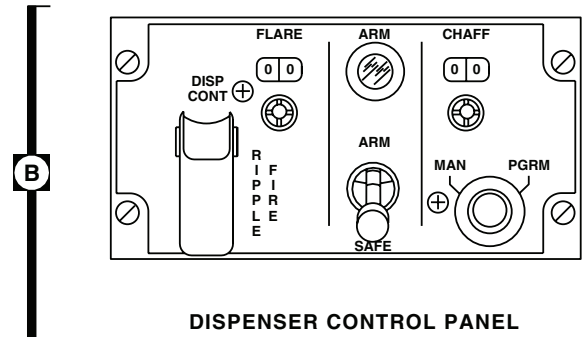
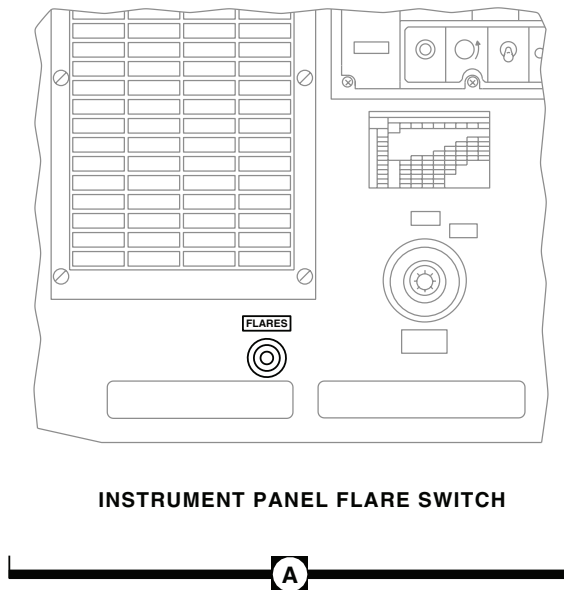
## NOTE

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Figure 1. Chaff/Flare Dispenser System Location Diagram. (Sheet 1 of 4)

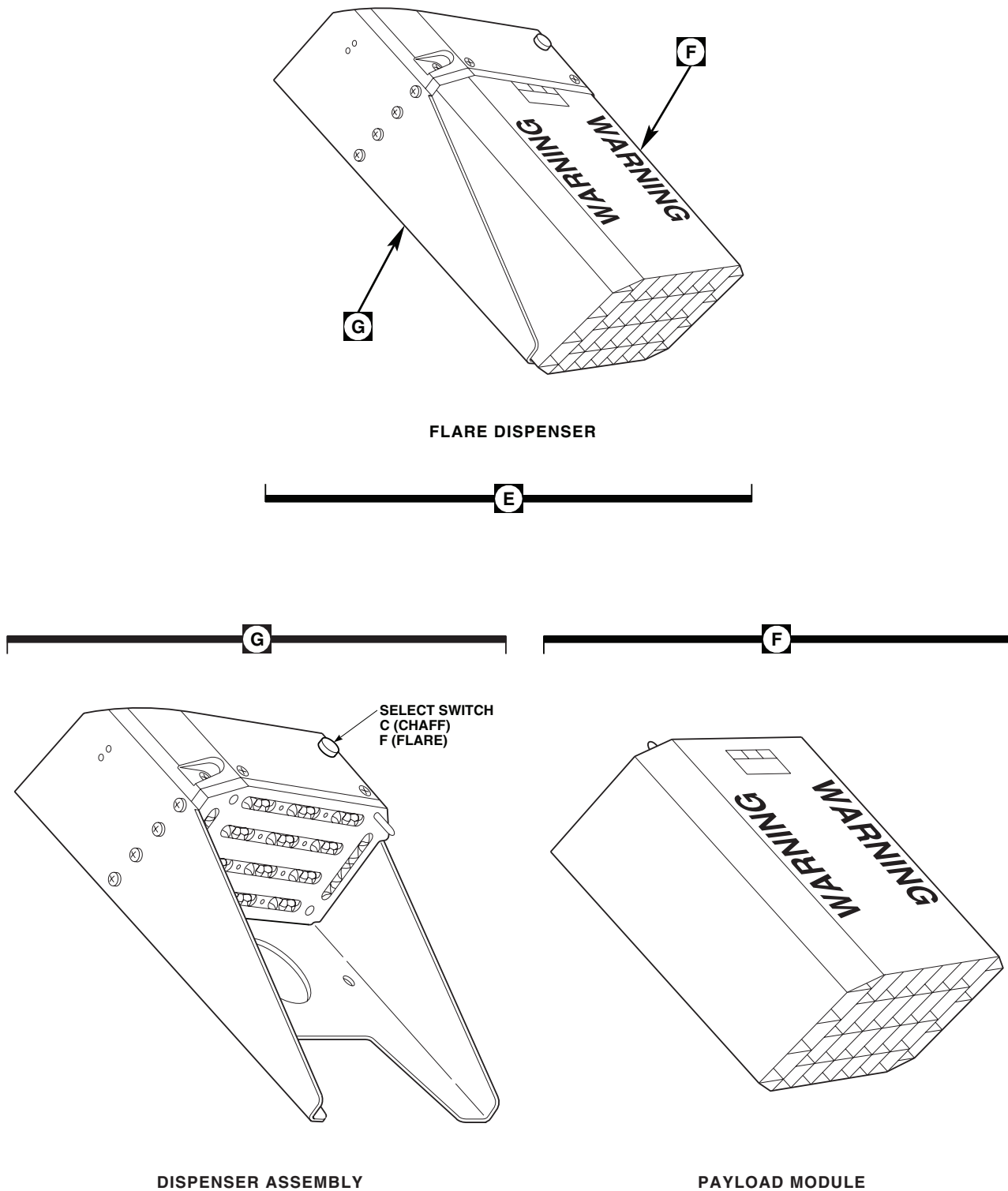
LOCATION - Continued



AA3422\_2  
SA

Figure 1. Chaff/Flare Dispenser System Location Diagram. (Sheet 2 of 4)

LOCATION - Continued

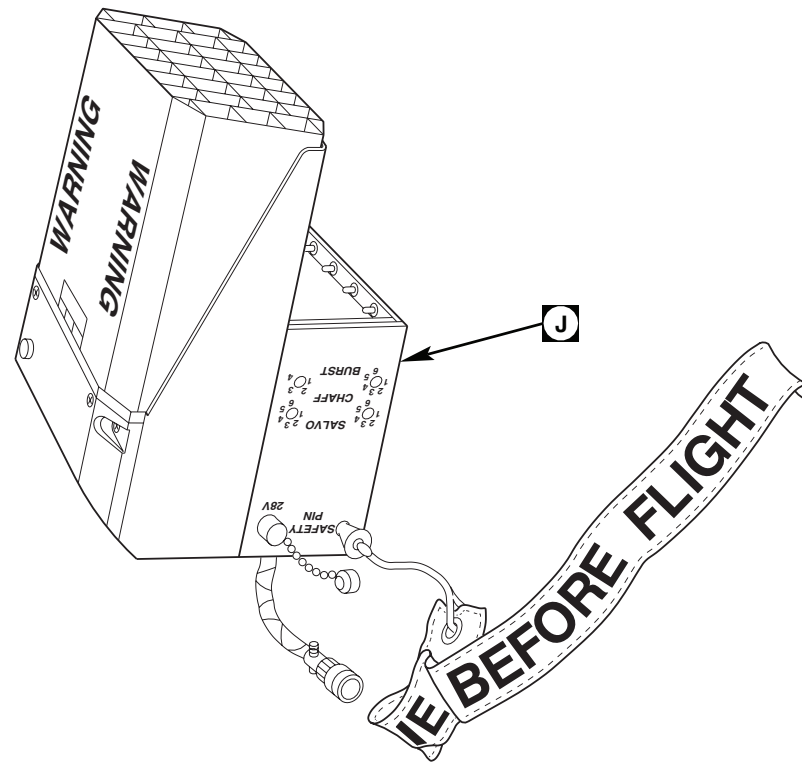


AA3422 3  
SA

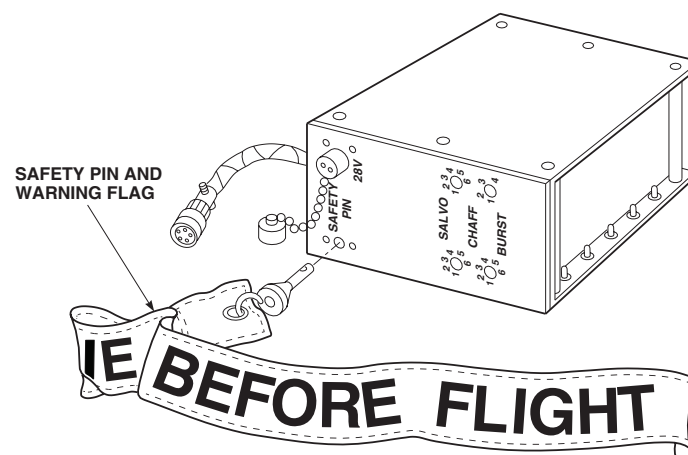
Figure 1. Chaff/Flare Dispenser System Location Diagram. (Sheet 3 of 4)



LOCATION - Continued



CHAFF DISPENSER



ELECTRONICS MODULE

AA3422\_4  
SA

Figure 1. Chaff/Flare Dispenser System Location Diagram. (Sheet 4 of 4)

SCHEMATICS

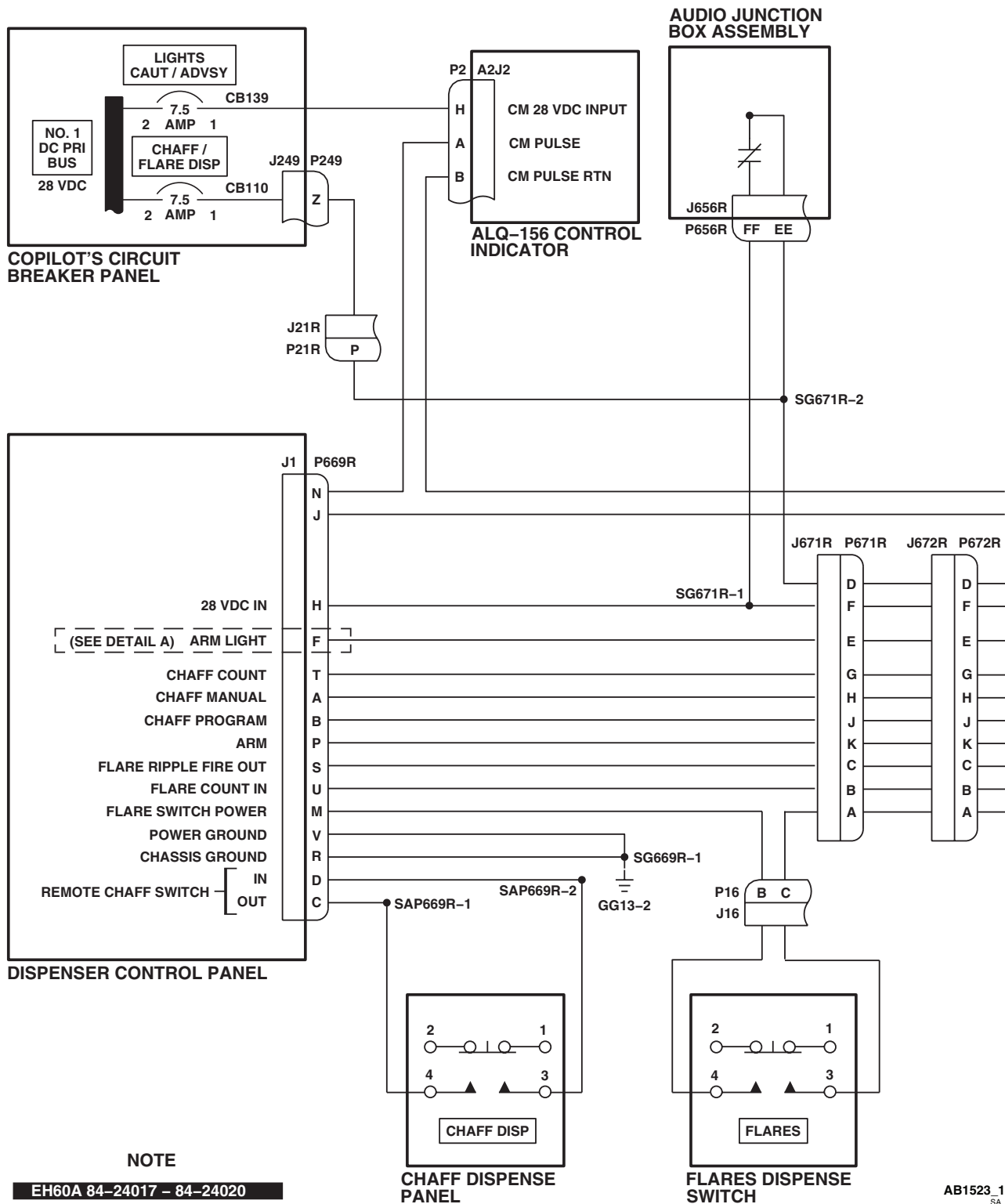


Figure 2. Chaff/Flare Dispenser System Schematic Diagram. (Sheet 1 of 2)

SCHEMATICS - Continued

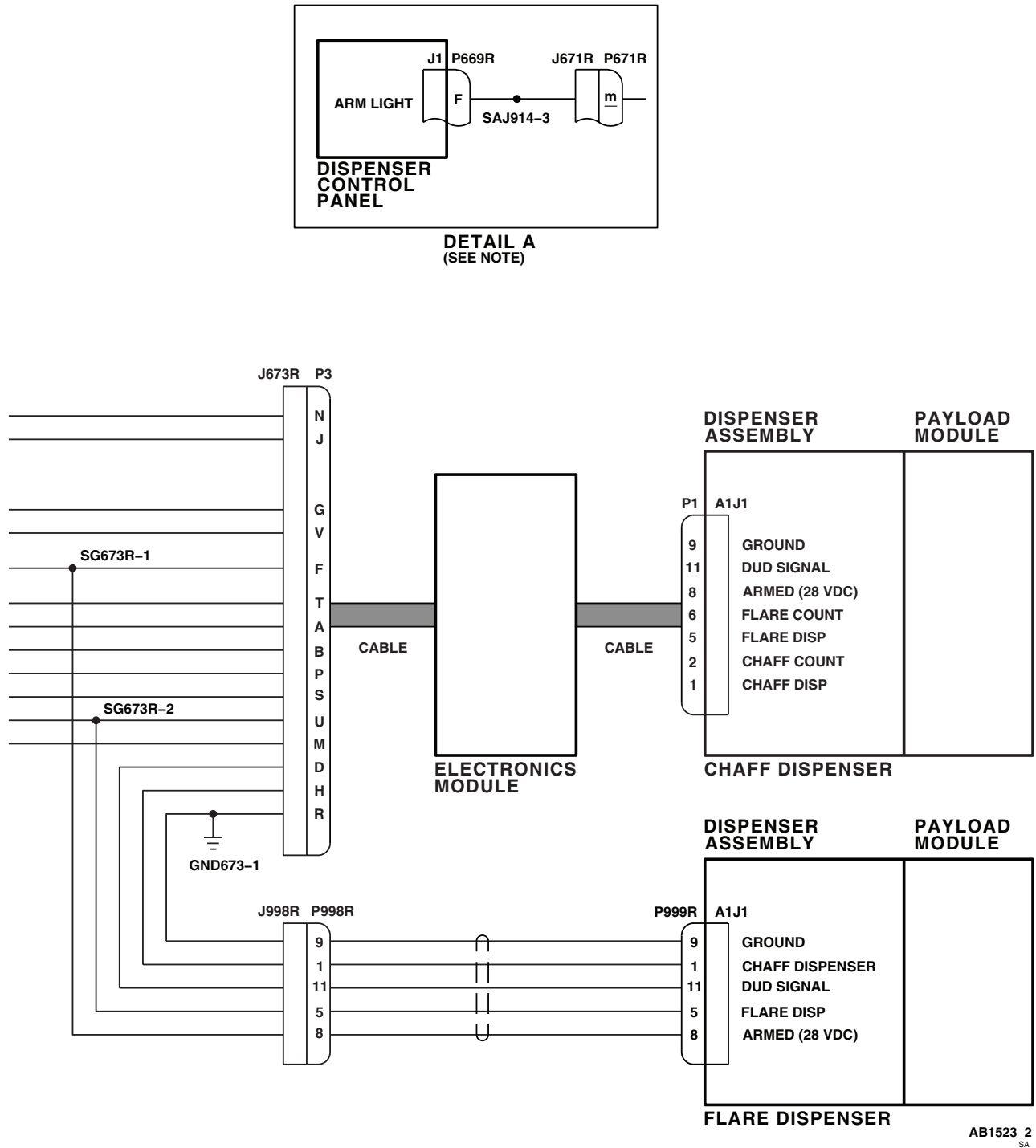


Figure 2. Chaff/Flare Dispenser System Schematic Diagram. (Sheet 2 of 2)

The diagram illustrates the test setup for the M92 test set. Key components and their connections are as follows:

- Helicopter Connector J673R** is connected to **P1** of the **Dispenser Test Cable**.
- The **Dispenser Test Cable** connects to **J1 P3** of the **Electronics Module**.
- The **Electronics Module** has a connection at **J1 P1** to a power source (indicated by a line).
- The **Electronics Module** is connected to **P1 A1J1** of the **Chaff Dispenser Assembly**.
- The **Chaff Dispenser Assembly** is connected to **P1** of the **Chaff Dispenser Electrical Checks**.
- The **Flare Dispenser Assembly** is connected to **P1** of the **Flare Dispenser Electrical Checks**.
- The **Chaff Dispenser Electrical Checks** and **Flare Dispenser Electrical Checks** are connected via a dashed line to the **Dispenser Test Adapter Assembly**.
- The **Dispenser Test Adapter Assembly** is connected to **J1 P1** of the **Dispenser Test Cable**.
- The **Dispenser Test Adapter Assembly** is connected to **J1 P7** of the **Flare Simulator Assembly (See Note)**.
- The **Flare Simulator Assembly** is connected to **P1 P7** of the **Test Cable No. 1**.
- The **Test Cable No. 1** connects to **P1 J3** of the **M92 Test Set**.
- The **Test Cable No. 2** connects to **P2 J2** of the **M92 Test Set**.
- The **M92 Test Set** is connected to **J8 (+)** and **J9 (-)** of the **Multimeter**.
- The **Multimeter** is connected to **DC (+)** and **COM (-)** terminals.
- The **Multimeter Test Leads** are connected to the **Multimeter** and the **M92 Test Set**.
- The **Mounting Rod** is used to secure the **Dispenser Test Adapter Assembly** and the **Flare Simulator Assembly**.
- The **Power Cable No. 1 (9325989)** is connected to the **Electronics Module**.

**FLARE SIMULATOR ASSEMBLY AND TEST CABLE NO. 1 ARE NOT USED TO TEST CHAFF DISPENSER.**

Figure 3. Chaff/Flare Dispenser System Test Setup.

### Change 1

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**UNIT LEVEL**  
**MISSION EQUIPMENT**  
**AIR TRANSPORT HYDRAULIC CART**

---

**INITIAL SETUP:****Test Equipment**

Multimeter, AN/PSM-45A

**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01

Electrical Repairer Toolkit, SC 5180-99-B06

**Material/Parts**

Hydraulic Fluid, Fire Resistant, Item 144, WP 1803 00

**Personnel Required**

Aircraft Electrician MOS 15F (1)

UH-60 Helicopter Repairer MOS 15T (1)

**References**

WP 0104 00

WP 0824 00

WP 1411 00

WP 1412 00

WP 1413 00

WP 1416 00

WP 1417 00

WP 1803 00

**Equipment Condition**

28 vdc Electrical Power Available,

---

**AIR TRANSPORT HYDRAULIC CART OPERATIONAL/TROUBLESHOOTING PROCEDURE****PROCEDURE****Step****CAUTION**

Do not disturb factory set flow control valve settings. Adjustment could cause helicopter damage.

**NOTE**

- If circuit breaker pops out during operational/troubleshooting procedure, a short circuit in hydraulic cart wiring is indicated (WP 0104 00).
- See **70700-20437-041** Figure 1  and **70700-81650-041** Figure 2  for component location diagram, and **70700-20437-041** Figure 3  and **70700-81650-041** Figure 4  for schematic diagram as an aid in operational/troubleshooting.

1. Connect hydraulic hoses from reservoir can to cart.
2. Connect all three landing gear hose assemblies to quick-disconnects on cart.
3. Connect ground strap to a suitable ground.
4. Layout electrical cable and connect to suitable 28 vdc electrical source.
5. **DO NOT TURN ELECTRICAL POWER ON AT THIS TIME.**
6. Make sure one reservoir can is filled with about 4 gallons of hydraulic fluid.
7. Open breather cap on top of can. If reservoir is not filled, fill with hydraulic fluid.
8. Crack open bleeder cap on top of the pump.

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**AIR TRANSPORT HYDRAULIC CART OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**

9. Bleed until air-free oil comes out.
10. Tighten bleeder cap and check for leaks.
11. Make sure MOTOR/PUMP POWER and CONTROL POWER circuit breakers on control box are pushed in.

**NOTE**

If red button on top of filter should pop up during pump operation, stop pump and replace filter element before doing next step.

12. On control pendant, press face of START button, then release.

**Indication/Condition**

- a. Pump shall start to operate when START button is pressed and continue to operate after button is released.
- b. POWER ON light on control panel and control pendant shall be on.
- c. Allow pump to operate for about 2 minutes. Visually check for any leaks.

**Corrective Action**

1. If motor does not operate, go to CART MOTOR DOES NOT OPERATE WHEN START BUTTON IS PRESSED, in this work package.
2. If motor stops after START button is released, go to CART MOTOR DOES NOT CONTINUE TO OPERATE AFTER START BUTTON IS RELEASED, in this work package.
3. If motor operates, but POWER ON light is not on, replace light (WP 1411 00).

**Step**

13. On control box, push EMERGENCY STOP button.

**Indication/Condition**

Motor shall stop operating and both POWER ON lights shall go out.

**Corrective Action**

If Indication/Condition is not as specified, replace EMERGENCY STOP switch (WP 1412 00).

**Step**

14. Disconnect return hose of reservoir can at SYSTEM RETURN quick-disconnect on cart frame.
15. On control pendant, press, then release START button.

**Indication/Condition**

- a. Both POWER ON lights shall be on and the pump shall momentarily start to operate.
- b. Then, the MOTOR STOP light shall go on, the POWER ON lights shall go off, and the pump shall stop operating.
- c. MOTOR STOP light shall go off once pressure has dropped.

**Corrective Action**

If Indication/Conditions are not as specified, replace pressure switch (WP 1416 00).

**Step**

16. Connect hose from reservoir can to cart.

---

**AIR TRANSPORT HYDRAULIC CART OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**

17. On control pendant, press, then release START button, to turn on cart.
18. On control pendant, push and hold RIGHT switch to UP.
19. Check cart and right main landing gear hose for leaks.

**Indication/Condition**

Sound of motor RPM decreasing shall be heard.

**Corrective Action**

If Indication/Condition is not as specified, go to HOSE DOES NOT PRESSURIZE WHEN RIGHT SWITCH PLACED UP (MOTOR RPM DID NOT DECREASE), in this work package.

**Step**

20. Release switch to center position.
21. On control pendant, push and hold TAIL switch to UP.
22. Check cart and tail landing gear hose for leaks.

**Indication/Condition**

Hose shall pressurize as indicated by sound of motor RPM decreasing.

**Corrective Action**

If Indication/Condition is not as specified, go to HOSE DOES NOT PRESSURIZE WHEN TAIL SWITCH IS PLACED UP (MOTOR RPM DID NOT DECREASE), in this work package.

**Step**

23. Release switch to center position.
24. On control pendant, push and hold LEFT switch to UP.
25. Check cart and left main landing gear hose for leaks.

**Indication/Condition**

Hose shall pressurize as indicated by sound of motor RPM decreasing.

**Corrective Action**

If Indication/Condition is not as specified, go to HOSE DOES NOT PRESSURIZE WHEN LEFT SWITCH IS PLACED UP (MOTOR RPM DID NOT DECREASE), in this work package.

**Step**

26. Release switch to center position.
27. Remove either quick-disconnect fitting from end of hose connected to RIGHT MAIN coupling on cart.
28. Route hose into reservoir can.
29. On control pendant, push and hold RIGHT switch to UP.

**Indication/Condition**

Fluid shall be flowing into reservoir can.

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**AIR TRANSPORT HYDRAULIC CART OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Corrective Action**

None Required

**Step**

30. On control pendant, push and hold RIGHT switch to DOWN.

**Indication/Condition**

- a. Fluid shall stop flowing into reservoir can.
- b. On cart 70700-81650-041, sound of motor RPM increasing shall be heard.

**Corrective Action**

1. If Indication/Condition a. is not as specified, go to FLUID DOES NOT STOP FLOWING WHEN RIGHT SWITCH IS PLACED DOWN, in this work package.
2. If Indication/Condition b. is not as specified, go to HOSE DOES NOT PRESSURIZE WHEN RIGHT SWITCH IS PLACED DOWN (MOTOR RPM DID NOT DECREASE), in this work package.

**Step**

31. Release RIGHT switch to center position.
32. Remove hose from top of reservoir can.
33. Install quick-disconnect fitting into end of hose.
34. Remove either quick-disconnect fitting from end of hose coming from TAIL coupling on cart.
35. Route hose into reservoir can.
36. On control pendant, push and hold TAIL switch to UP.

**Indication/Condition**

Fluid shall flow into reservoir can.

**Corrective Action**

None Required

**Step**

37. On control pendant, push and hold TAIL switch to DOWN.

**Indication/Condition**

- a. Fluid shall stop flowing into reservoir can.
- b. On cart 70700-81650-041, sound of motor RPM increasing shall be heard.

**Corrective Action**

1. If Indication/Condition a. is not as specified, go to FLUID DOES NOT STOP FLOWING WHEN TAIL SWITCH IS PLACED DOWN, in this work package.
2. If Indication/Condition b. is not as specified, go to HOSE DOES NOT DEPRESSURIZE WHEN TAIL SWITCH IS PLACED DOWN (MOTOR RPM DID NOT DECREASE), in this work package.

**Step**

38. Release TAIL switch to center position.



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**AIR TRANSPORT HYDRAULIC CART OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**

39. Remove hose from top of reservoir can.
40. Install quick-disconnect fitting into end of hose.
41. Remove either quick-disconnect fitting from end of hose coming from LEFT coupling on cart.
42. Route hose into reservoir can.
43. On control pendant, push and hold LEFT switch to UP.

**Indication/Condition**

Fluid shall flow into reservoir can.

**Corrective Action**

None Required

**Step**

44. On control pendant, push and hold LEFT switch to DOWN.

**Indication/Condition**

- a. Fluid shall stop flowing into reservoir can.
- b. On cart 70700-81650-041, sound of motor RPM increasing shall be heard.

**Corrective Action**

1. If Indication/Condition a. is not as specified, go to FLUID DOES NOT STOP FLOWING WHEN LEFT SWITCH IS PLACED DOWN, in this work package.
2. If Indication/Condition b. is not as specified, go to HOSE DOES NOT DEPRESSURIZE WHEN LEFT SWITCH IS PLACED DOWN (MOTOR RPM DID NOT INCREASE), in this work package.

**Step**

45. Release LEFT switch to center position.
46. Remove hose from top of reservoir can.
47. Install quick-disconnect fitting into end of hose.
48. On control pendant, press face of STOP switch.

**Indication/Condition**

Pump motor shall stop operating.

**Corrective Action**

If Indication/Condition is not as specified, replace STOP switch (WP 1411 00).

**Step**

49. If necessary, disconnect electrical power and ground strap.

**Indication/Condition**

Electrical power and ground strap shall be disconnected.

**Corrective Action**

None Required

**CART MOTOR DOES NOT OPERATE WHEN START BUTTON IS PRESSED****SYMPTOM**

Cart motor does not operate when START button is pressed.

**MALFUNCTION**

Cart motor does not operate when START button is pressed.

**CORRECTIVE ACTION**

- 1. With START button pressed, check for 28 vdc at terminals of motor.**
  - a. If voltage is as specified, replace pump and motor assembly (WP 1413 00). Go to **22**.
  - b. If voltage is not as specified, go to **2**.
- 2. With START button pressed, check for 28 vdc between motor (+) terminal and cart ground.**
  - a. If voltage is as specified, go to **3**.
  - b. If voltage is not as specified, go to **5**.
- 3. Check continuity between motor (-) terminal and P1-B.**
  - a. If continuity is present, go to **4**.
  - b. If continuity is not present, repair/replace wiring. Go to **22**.
- 4. Check continuity between J1-B and TB1-2.**
  - a. If continuity is present, repair/replace wiring between TB1-1 and J2-C. Go to **22**.
  - b. If continuity is not present, repair/replace wiring. Go to **22**.
- 5. With START button pressed, check for 28 vdc between J1-A and cart ground.**
  - a. If voltage is as specified, repair/replace wiring between P1-A and motor (+) terminal. Go to **22**.
  - b. If voltage is not as specified, go to **6**.
- 6. Check continuity between J2-B and J4-B.**
  - a. If continuity is present, go to **7**.
  - b. If continuity is not present, go to **20**.
- 7. With START button pressed, check continuity between P4-B and P4-D.**
  - a. If continuity is present, go to **8**.
  - b. If continuity is not present, go to **12**.
- 8. Check continuity between J4-D and relay K1-X1.**
  - a. If continuity is present, go to **9**.
  - b. If continuity is not present, go to **11**.
- 9. Check continuity between relay K1-X2 and TB1-1.**
  - a. If continuity is present, go to **10**.
  - b. If continuity is not present, repair/replace wiring. Go to **22**.

---

**CART MOTOR DOES NOT OPERATE WHEN START BUTTON IS PRESSED - Continued****10. Check continuity between TB1-1 and J2-C.**

- a. If continuity is present, replace relay K1 (WP 1412 00). Go to **22**.
- b. If continuity is not present, repair/replace wiring. Go to **22**.

**11. Check continuity between relay K1-A1 and K1-X1.**

- a. If continuity is present, repair/replace wiring between J4-D and relay K1-A1. Go to **22**.
- b. If continuity is not present, repair/replace wiring. Go to **22**.

**12. Check continuity between START button terminal 1 and P4-D.**

- a. If continuity is present, go to **13**.
- b. If continuity is not present, go to **15**.

**13. Check continuity between START button terminal 2 and P4-B.**

- a. If continuity is present, replace start switch (WP 1411 00). Go to **22**.
- b. If continuity is not present, go to **14**.

**14. Check continuity between START button terminal 2 and STOP button terminal 2.**

- a. If continuity is present, replace wiring between STOP button terminal 2 and P4-B. Go to **22**.
- b. If continuity is not present, repair/replace wiring. Go to **22**.

**15. Check hydraulic cart configuration.**

- a. If hydraulic cart is 70700-81650-041, go to **16**.
- b. If hydraulic cart is 70700-20437-041, go to **17**.

**16. Check continuity between POWER ON terminal 1 and START button terminal 1.**

- a. If continuity is present, repair/replace wiring between POWER ON terminal 1 and P4-D. Go to **22**.
- b. If continuity is not present, repair/replace wiring. Go to **22**.

**17. Check continuity between START button terminal 1 and LEFT switch terminal 2.**

- a. If continuity is present, go to **18**.
- b. If continuity is not present, repair/replace wiring. Go to **22**.

**18. Check continuity between LEFT switch terminal 2 and TAIL switch terminal 2.**

- a. If continuity is present, go to **19**.
- b. If continuity is not present, repair/replace wiring. Go to **22**.

**19. Check continuity between RIGHT switch terminal 2 and TAIL switch terminal 2.**

- a. If continuity is present, repair/replace wiring between RIGHT switch terminal 2 and P4-D. Go to **22**.
- b. If continuity is not present, repair/replace wiring. Go to **22**.

**20. Check continuity between CONTROL POWER circuit breaker terminal 2 and J4-B.**

- a. If continuity is present, go to **21**.
- b. If continuity is not present, repair/replace wiring. Go to **22**.

---

**CART MOTOR DOES NOT OPERATE WHEN START BUTTON IS PRESSED - Continued****21. Check continuity between CONTROL POWER circuit breaker terminal 1 and J2-B.**

- a. If continuity is present, replace circuit breaker (WP 0824 00). Go to **22**.
- b. If continuity is not present, repair/replace wiring. Go to **22**.

**22. Procedure completed.****CART MOTOR DOES NOT CONTINUE TO OPERATE AFTER START BUTTON IS RELEASED****SYMPTOM**

Cart motor does not continue to operate after START button is released.

**MALFUNCTION**

Cart motor does not continue to operate after START button is released.

**CORRECTIVE ACTION**

- 1. Check for 28 vdc between pressure switch terminal NC and terminal C.**
  - a. If voltage is as specified, replace pressure switch (WP 1416 00). Go to **10**.
  - b. If voltage is not as specified, go to **2**.
- 2. Check for 28 vdc between pressure switch terminal NC and cart ground.**
  - a. If voltage is as specified, go to **3**.
  - b. If voltage is not as specified, go to **5**.
- 3. Check continuity between pressure switch terminal C and P3-C.**
  - a. If continuity is present, go to **4**.
  - b. If continuity is not present, repair/replace wiring. Go to **10**.
- 4. Check continuity between J3-C and relay K1-A2.**
  - a. If continuity is present, replace relay (WP 1412 00). Go to **10**.
  - b. If continuity is not present, repair/replace wiring. Go to **10**.
- 5. Check continuity between pressure switch terminal NC and P3-D.**
  - a. If continuity is present, go to **6**.
  - b. If continuity is not present, repair/replace wiring. Go to **10**.
- 6. Check continuity between J3-D and J4-C.**
  - a. If continuity is present, go to **7**.
  - b. If continuity is not present, go to **8**.
- 7. Check continuity between P4-C and STOP switch terminal 1.**
  - a. If continuity is present, replace STOP switch (WP 1411 00). Go to **10**.
  - b. If continuity is not present, repair/replace wiring. Go to **10**.
- 8. Check continuity between EMERGENCY STOP switch and J3-D.**
  - a. If continuity is present, go to **9**.
  - b. If continuity is not present, repair/replace wiring. Go to **10**.

---

**CART MOTOR DOES NOT CONTINUE TO OPERATE AFTER START BUTTON IS RELEASED** - Continued**9. Check continuity between EMERGENCY STOP switch NC and J4-C.**

- a. If continuity is present, replace EMERGENCY STOP switch (WP 1412 00). Go to **10**.
- b. If continuity is not present, repair/replace wiring. Go to **10**.

**10. Procedure completed.****HOSE DOES NOT PRESSURIZE WHEN RIGHT SWITCH PLACED UP (MOTOR RPM DID NOT DECREASE)****SYMPTOM**

Hose does not pressurize when RIGHT switch placed UP (motor rpm did not decrease).

**MALFUNCTION**

Hose does not pressurize when RIGHT switch placed UP (motor rpm did not decrease).

**CORRECTIVE ACTION**

- 1. With cart motor operating, push right directional valve manual button on up side. Go to 2.**
- 2. Hose shall pressurize and sound of motor RPM decreasing shall be heard.**
  - a. If hose does pressurize and motor RPM decreases, go to **3**.
  - b. If hose does not pressurize and motor RPM does not decrease, replace directional control valve assembly (WP 1417 00). Go to **8**.
- 3. Check if TAIL and LEFT switches pressurize their hoses.**
  - a. If they pressurize their hoses, go to **4**.
  - b. If they do not pressurize their hoses, repair/replace wiring between RIGHT switch terminal 2 and P4-D. Go to **8**.
- 4. With RIGHT switch placed UP, check for 28 vdc between UP solenoid terminal 1 and terminal 2.**
  - a. If voltage is as specified, replace directional control valve assembly (WP 1417 00). Go to **8**.
  - b. If voltage is not as specified, go to **5**.
- 5. With RIGHT switch placed UP, check for 28 vdc between UP solenoid terminal 1 and cart ground.**
  - a. If voltage is as specified, go to **6**.
  - b. If voltage is not as specified, go to **7**.
- 6. Check continuity between UP solenoid terminal 2 and cart ground.**
  - a. If continuity is present, replace directional control valve assembly (WP 1417 00). Go to **8**.
  - b. If continuity is not present, repair/replace wiring. Go to **8**.
- 7. Check continuity between UP solenoid terminal 1 and RIGHT switch terminal 3.**
  - a. If continuity is present, replace switch (WP 1411 00). Go to **8**.
  - b. If continuity is not present, repair/replace wiring. Go to **8**.
- 8. Procedure completed.**

---

**HOSE DOES NOT PRESSURIZE WHEN TAIL SWITCH IS PLACED UP (MOTOR RPM DID NOT DECREASE)****SYMPTOM**

Hose does not pressurize when TAIL switch is placed UP (motor rpm did not decrease).

**MALFUNCTION**

Hose does not pressurize when TAIL switch is placed UP (motor rpm did not decrease).

**CORRECTIVE ACTION**

- 1. With cart motor operating, push tail wheel directional valve manual button on up side. Go to 2.**
- 2. Hose shall pressurize and sound of motor RPM decreasing shall be heard.**
  - a. If hose does pressurize and motor RPM decreases, go to **3**.
  - b. If hose does not pressurize and motor RPM does not decrease, replace directional control valve assembly (WP 1417 00). Go to **10**.
- 3. With TAIL switch placed UP, check for 28 vdc between UP solenoid terminal 1 and terminal 2.**
  - a. If voltage is as specified, replace directional control valve assembly (WP 1417 00). Go to **10**.
  - b. If voltage is not as specified, go to **4**.
- 4. With TAIL switch placed UP, check for 28 vdc between UP solenoid terminal 1 and cart ground.**
  - a. If voltage is as specified, go to **5**.
  - b. If voltage is not as specified, go to **6**.
- 5. Check continuity between UP solenoid terminal 2 and cart ground.**
  - a. If continuity is present, replace directional control valve assembly (WP 1417 00). Go to **10**.
  - b. If continuity is not present, repair/replace wiring. Go to **10**.
- 6. Check continuity between UP solenoid terminal 1 and TAIL switch terminal 3.**
  - a. If continuity is present, go to **7**.
  - b. If continuity is not present, repair/replace wiring. Go to **10**.
- 7. Check hydraulic cart configuration.**
  - a. If hydraulic cart is 70700-81650-041, go to **8**.
  - b. If hydraulic cart is 70700-20437-041, repair/replace wiring. Go to **10**.
- 8. Check continuity between TAIL switch terminal 2 and RIGHT switch terminal 5.**
  - a. If continuity is present, go to **9**.
  - b. If continuity is not present, repair/replace wiring. Go to **10**.
- 9. Check continuity between TAIL switch terminal 2 and terminal 5.**
  - a. If continuity is present, replace TAIL switch (WP 1411 00). Go to **10**.
  - b. If continuity is not present, repair/replace wiring. Go to **10**.
- 10. Procedure completed.**

---

**HOSE DOES NOT PRESSURIZE WHEN LEFT SWITCH IS PLACED UP (MOTOR RPM DID NOT DECREASE)****SYMPTOM**

Hose does not pressurize when LEFT switch is placed UP (motor rpm did not decrease).

**MALFUNCTION**

Hose does not pressurize when LEFT switch is placed UP (motor rpm did not decrease).

**CORRECTIVE ACTION**

- 1. With cart motor operating, push left directional valve manual button on up side. Go to 2.**
- 2. Hose shall pressurize and sound of motor RPM decreasing shall be heard.**
  - a. If hose does pressurize and motor RPM decreases, go to **3**.
  - b. If hose does not pressurize and motor RPM does not decrease, replace directional control valve assembly (WP 1417 00). Go to **10**.
- 3. With LEFT switch placed UP, check for 28 vdc between UP solenoid terminal 1 and terminal 2.**
  - a. If voltage is as specified, replace directional control valve assembly (WP 1417 00). Go to **10**.
  - b. If voltage is not as specified, go to **4**.
- 4. With LEFT switch placed UP, check for 28 vdc between UP solenoid terminal 1 and cart ground.**
  - a. If voltage is as specified, go to **5**.
  - b. If voltage is not as specified, go to **6**.
- 5. Check continuity between UP solenoid terminal 2 and cart ground.**
  - a. If continuity is present, replace directional control valve assembly (WP 1417 00). Go to **10**.
  - b. If continuity is not present, repair/replace wiring. Go to **10**.
- 6. Check continuity between UP solenoid terminal 1 and LEFT switch terminal 3.**
  - a. If continuity is present, go to **7**.
  - b. If continuity is not present, repair/replace wiring. Go to **10**.
- 7. Check hydraulic cart configuration.**
  - a. If hydraulic cart is 70700-81650-041, go to **8**.
  - b. If hydraulic cart is 70700-20437-041, repair/replace wiring. Go to **10**.
- 8. Check continuity between LEFT switch terminal 2 and TAIL switch terminal 5.**
  - a. If continuity is present, go to **9**.
  - b. If continuity is not present, repair/replace wiring. Go to **10**.
- 9. Check continuity between LEFT switch terminal 2 and terminal 5.**
  - a. If continuity is present, replace LEFT switch (WP 1411 00). Go to **10**.
  - b. If continuity is not present, repair/replace wiring. Go to **10**.

---

**HOSE DOES NOT PRESSURIZE WHEN LEFT SWITCH IS PLACED UP (MOTOR RPM DID NOT DECREASE) - Continued****10. Procedure completed.****FLUID DOES NOT STOP FLOWING WHEN RIGHT SWITCH IS PLACED DOWN****SYMPTOM**

Fluid does not stop flowing when RIGHT switch is placed DOWN.

**MALFUNCTION**

Fluid does not stop flowing when RIGHT switch is placed DOWN.

**CORRECTIVE ACTION**

- 1. With cart motor operating, push right directional valve manual button on down side. Go to 2.**
- 2. Fluid shall stop flowing into can.**
  - a. If fluid stops flowing, go to **3**.
  - b. If fluid does not stop flowing, replace directional control valve assembly (WP 1417 00). Go to **12**.
- 3. With RIGHT switch placed DOWN, check for 28 vdc between DOWN solenoid terminal 1 and terminal 2.**
  - a. If voltage is as specified, replace directional control valve assembly (WP 1417 00). Go to **12**.
  - b. If voltage is not as specified, go to **4**.
- 4. With RIGHT switch placed DOWN, check for 28 vdc between DOWN solenoid terminal 1 and cart ground.**
  - a. If voltage is as specified, go to **5**.
  - b. If voltage is not as specified, go to **6**.
- 5. Check continuity between DOWN solenoid terminal 2 and cart ground.**
  - a. If continuity is present, replace directional control valve assembly (WP 1417 00). Go to **12**.
  - b. If continuity is not present, repair/replace wiring. Go to **12**.
- 6. Check continuity between DOWN solenoid terminal 1 and P3-J.**
  - a. If continuity is present, go to **7**.
  - b. If continuity is not present, repair/replace wiring. Go to **12**.
- 7. Check hydraulic cart configuration.**
  - a. If hydraulic cart is 70700-81650-041, go to **8**.
  - b. If hydraulic cart is 70700-20437-041, go to **10**.
- 8. Check continuity between J3-J and J4-N.**
  - a. If continuity is present, go to **9**.
  - b. If continuity is not present, repair/replace wiring. Go to **12**.
- 9. Check continuity between RIGHT switch terminal 1 and P4-N.**
  - a. If continuity is present, replace switch (WP 1411 00). Go to **12**.
  - b. If continuity is not present, repair/replace wiring. Go to **12**.



---

**FLUID DOES NOT STOP FLOWING WHEN RIGHT SWITCH IS PLACED DOWN** - Continued**10. Check continuity between J3-J and J4-K.**

- a. If continuity is present, go to **11**.
- b. If continuity is not present, repair/replace wiring. Go to **12**.

**11. Check continuity between RIGHT switch terminal 1 and P4-K.**

- a. If continuity is present, replace switch (WP 1411 00). Go to **12**.
- b. If continuity is not present, repair/replace wiring. Go to **12**.

**12. Procedure completed.****HOSE DOES NOT PRESSURIZE WHEN RIGHT SWITCH IS PLACED DOWN (MOTOR RPM DID NOT DECREASE)****SYMPTOM**

Hose does not pressurize when RIGHT switch is placed DOWN (motor rpm did not decrease).

**MALFUNCTION**

Hose does not pressurize when RIGHT switch is placed DOWN (motor rpm did not decrease).

**CORRECTIVE ACTION****1. With cart motor operating, push bleed valve manual button. Go to 2.****2. Hose shall depressurize.**

- a. If hose does depressurize, go to **3**.
- b. If hose does not depressurize, replace directional control valve assembly (WP 1417 00). Go to **8**.

**3. With RIGHT switch placed DOWN, check for 28 vdc between BLEED solenoid terminal 1 and terminal 2.**

- a. If voltage is as specified, replace directional control valve assembly (WP 1417 00). Go to **8**.
- b. If voltage is not as specified, go to **4**.

**4. With RIGHT switch placed DOWN, check for 28 vdc between BLEED solenoid terminal 1 and cart ground.**

- a. If voltage is as specified, repair/replace wiring between BLEED solenoid terminal 2 and cart ground. Go to **8**.
- b. If voltage is not as specified, go to **5**.

**5. Check continuity between BLEED solenoid terminal 1 and P3-R.**

- a. If continuity is present, go to **6**.
- b. If continuity is not present, repair/replace wiring. Go to **8**.

**6. Check continuity between TB2-2 and J4-L.**

- a. If continuity is present, go to **7**.
- b. If continuity is not present, repair/replace wiring. Go to **8**.

**7. Check continuity between RIGHT switch terminal 6 and P3-L.**

- a. If continuity is present, replace RIGHT switch (WP 1411 00). Go to **8**.

---

**HOSE DOES NOT PRESSURIZE WHEN RIGHT SWITCH IS PLACED DOWN (MOTOR RPM DID NOT DECREASE) - Continued**

- b. If continuity is not present, repair/replace wiring. Go to **8**.

**8. Procedure completed.****FLUID DOES NOT STOP FLOWING WHEN TAIL SWITCH IS PLACED DOWN****SYMPTOM**

Fluid does not stop flowing when TAIL switch is placed DOWN.

**MALFUNCTION**

Fluid does not stop flowing when TAIL switch is placed DOWN.

**CORRECTIVE ACTION**

- 1. With cart motor operating, push right directional valve manual button on down side. Go to 2.**
- 2. Fluid shall stop flowing into can.**
  - a. If fluid stops flowing, go to **3**.
  - b. If fluid does not stop flowing, replace directional control valve assembly (WP 1417 00). Go to **13**.
- 3. With TAIL switch placed DOWN, check for 28 vdc between DOWN solenoid terminal 1 and terminal 2.**
  - a. If voltage is as specified, replace directional control valve assembly (WP 1417 00). Go to **13**.
  - b. If voltage is not as specified, go to **4**.
- 4. With TAIL switch placed DOWN, check for 28 vdc between DOWN solenoid terminal 1 and cart ground.**
  - a. If voltage is as specified, go to **5**.
  - b. If voltage is not as specified, go to **6**.
- 5. Check continuity between DOWN solenoid terminal 2 and cart ground.**
  - a. If continuity is present, replace directional control valve assembly (WP 1417 00). Go to **13**.
  - b. If continuity is not present, repair/replace wiring. Go to **13**.
- 6. Check hydraulic cart configuration.**
  - a. If hydraulic cart is 70700-81650-041, go to **7**.
  - b. If hydraulic cart is 70700-20437-041, go to **10**.
- 7. Check continuity between DOWN solenoid terminal 1 and P3-E.**
  - a. If continuity is present, go to **8**.
  - b. If continuity is not present, repair/replace wiring. Go to **13**.
- 8. Check continuity between J3-E and J4-E.**
  - a. If continuity is present, go to **9**.
  - b. If continuity is not present, repair/replace wiring. Go to **13**.
- 9. Check continuity between TAIL switch terminal 1 and P4-E.**
  - a. If continuity is present, replace switch (WP 1411 00). Go to **13**.

---

**FLUID DOES NOT STOP FLOWING WHEN TAIL SWITCH IS PLACED DOWN - Continued**

- b. If continuity is not present, repair/replace wiring. Go to **13**.
- 10. Check continuity between DOWN solenoid terminal 1 and P3-M.**
  - a. If continuity is present, go to **11**.
  - b. If continuity is not present, repair/replace wiring. Go to **13**.
- 11. Check continuity between J3-M and J4-G.**
  - a. If continuity is present, go to **12**.
  - b. If continuity is not present, repair/replace wiring. Go to **13**.
- 12. Check continuity between TAIL switch terminal 1 and P3-G.**
  - a. If continuity is present, replace switch (WP 1411 00). Go to **13**.
  - b. If continuity is not present, repair/replace wiring. Go to **13**.
- 13. Procedure completed.**

**HOSE DOES NOT DEPRESSURIZE WHEN TAIL SWITCH IS PLACED DOWN (MOTOR RPM DID NOT INCREASE)****SYMPTOM**

Hose does not depressurize when TAIL switch is placed DOWN (motor rpm did not increase).

**MALFUNCTION**

Hose does not depressurize when TAIL switch is placed DOWN (motor rpm did not increase).

**CORRECTIVE ACTION**

- 1. Check continuity between TB2-3 and J4-G.**
  - a. If continuity is present, go to **2**.
  - b. If continuity is not present, repair/replace wiring. Go to **3**.
- 2. Check continuity between TAIL switch terminal 6 and P4-G.**
  - a. If continuity is present, replace switch (WP 1411 00). Go to **3**.
  - b. If continuity is not present, repair/replace wiring. Go to **3**.
- 3. Procedure completed.**

**FLUID DOES NOT STOP FLOWING WHEN LEFT SWITCH IS PLACED DOWN****SYMPTOM**

Fluid does not stop flowing when left switch is placed DOWN.

**MALFUNCTION**

Fluid does not stop flowing when left switch is placed DOWN.

**CORRECTIVE ACTION**

- 1. With cart motor operating, push tail wheel directional valve manual button on down side. Go to 2.**
- 2. Fluid shall stop flowing into can.**
  - a. If fluid stops flowing, go to **3**.

---

**FLUID DOES NOT STOP FLOWING WHEN LEFT SWITCH IS PLACED DOWN** - Continued

---

- b. If fluid does not stop flowing, replace directional control valve assembly (WP 1417 00). Go to **13**.
- 3. With LEFT switch placed DOWN, check for 28 vdc between DOWN solenoid terminal 1 and terminal 2.**
  - a. If voltage is as specified, replace directional control valve assembly (WP 1417 00). Go to **13**.
  - b. If voltage is not as specified, go to **4**.
- 4. With LEFT switch placed DOWN, check for 28 vdc between DOWN solenoid terminal 1 and cart ground.**
  - a. If voltage is as specified, go to **5**.
  - b. If voltage is not as specified, go to **6**.
- 5. Check continuity between DOWN solenoid terminal 2 and cart ground.**
  - a. If continuity is present, replace directional control valve assembly (WP 1417 00). Go to **13**.
  - b. If continuity is not present, repair/replace wiring. Go to **13**.
- 6. Check hydraulic cart configuration.**
  - a. If hydraulic cart is 70700-81650-041, go to **7**.
  - b. If hydraulic cart is 70700-20437-041, go to **10**.
- 7. Check continuity between DOWN solenoid terminal 1 and P3-N.**
  - a. If continuity is present, go to **8**.
  - b. If continuity is not present, repair/replace wiring. Go to **13**.
- 8. Check continuity between J3-N and J4-J.**
  - a. If continuity is present, go to **9**.
  - b. If continuity is not present, repair/replace wiring. Go to **13**.
- 9. Check continuity between LEFT switch terminal 1 and P4-J.**
  - a. If continuity is present, replace switch (WP 1411 00). Go to **13**.
  - b. If continuity is not present, repair/replace wiring. Go to **13**.
- 10. Check continuity between DOWN solenoid terminal 1 and P3-K.**
  - a. If continuity is present, go to **11**.
  - b. If continuity is not present, repair/replace wiring. Go to **13**.
- 11. Check continuity between J3-K and J4-E.**
  - a. If continuity is present, go to **12**.
  - b. If continuity is not present, repair/replace wiring. Go to **13**.
- 12. Check continuity between LEFT switch terminal 1 and P3-E.**
  - a. If continuity is present, replace switch (WP 1411 00). Go to **13**.
  - b. If continuity is not present, repair/replace wiring. Go to **13**.
- 13. Procedure completed.**

**HOSE DOES NOT DEPRESSURIZE WHEN LEFT SWITCH IS PLACED DOWN (MOTOR RPM DID NOT INCREASE)****SYMPTOM**

Hose does not depressurize when LEFT switch is placed DOWN (motor rpm did not increase).

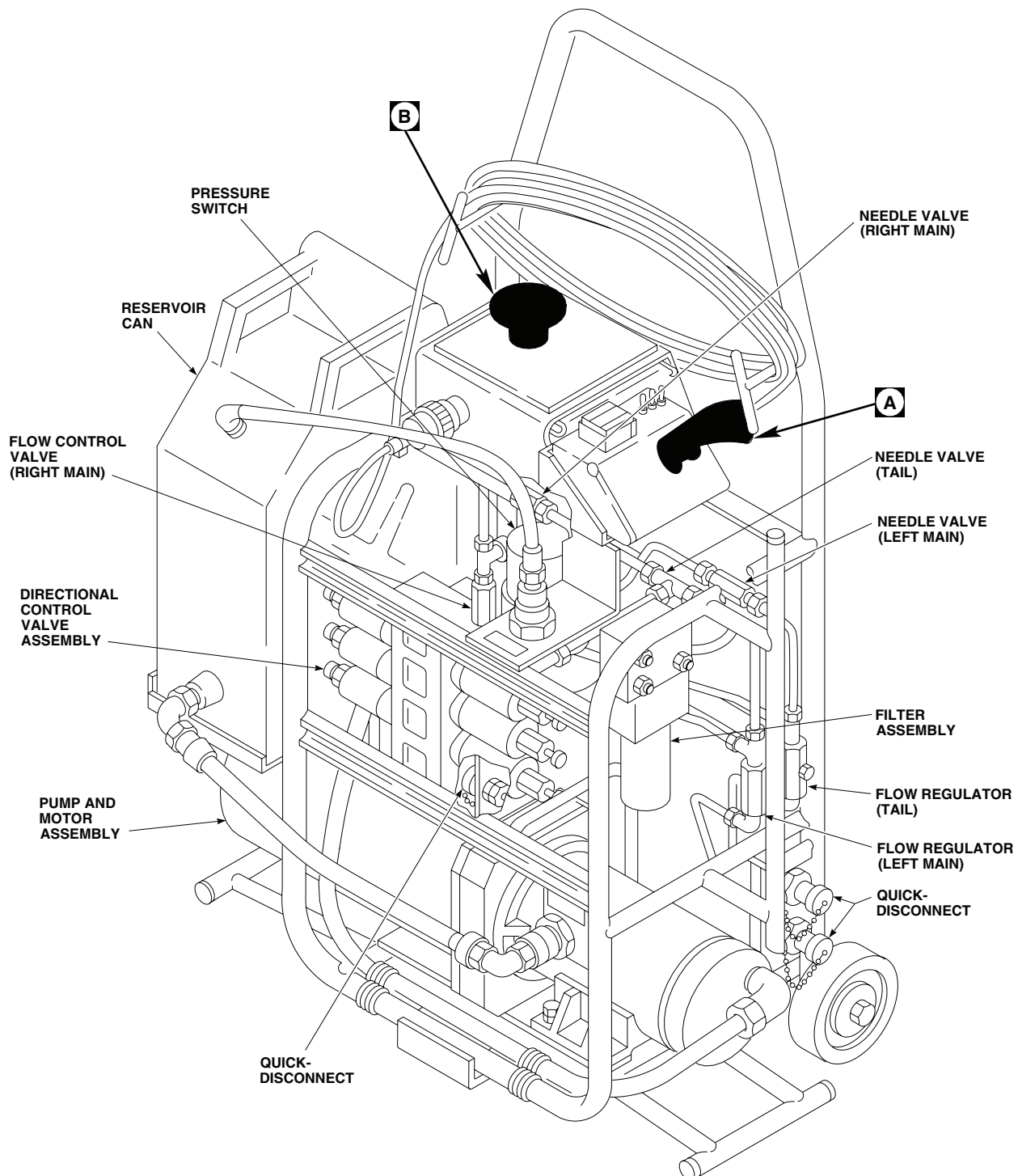
**MALFUNCTION**

Hose does not depressurize when LEFT switch is placed DOWN (motor rpm did not increase).

**CORRECTIVE ACTION**

- 1. Check continuity between TB2-4 and J4-F.**
  - a. If continuity is present, go to **2**.
  - b. If continuity is not present, repair/replace wiring. Go to **3**.
- 2. Check continuity between LEFT switch terminal 6 and P4-F.**
  - a. If continuity is present, replace switch (WP 1411 00). Go to **3**.
  - b. If continuity is not present, repair/replace wiring. Go to **3**.
- 3. Procedure completed.**

## LOCATION



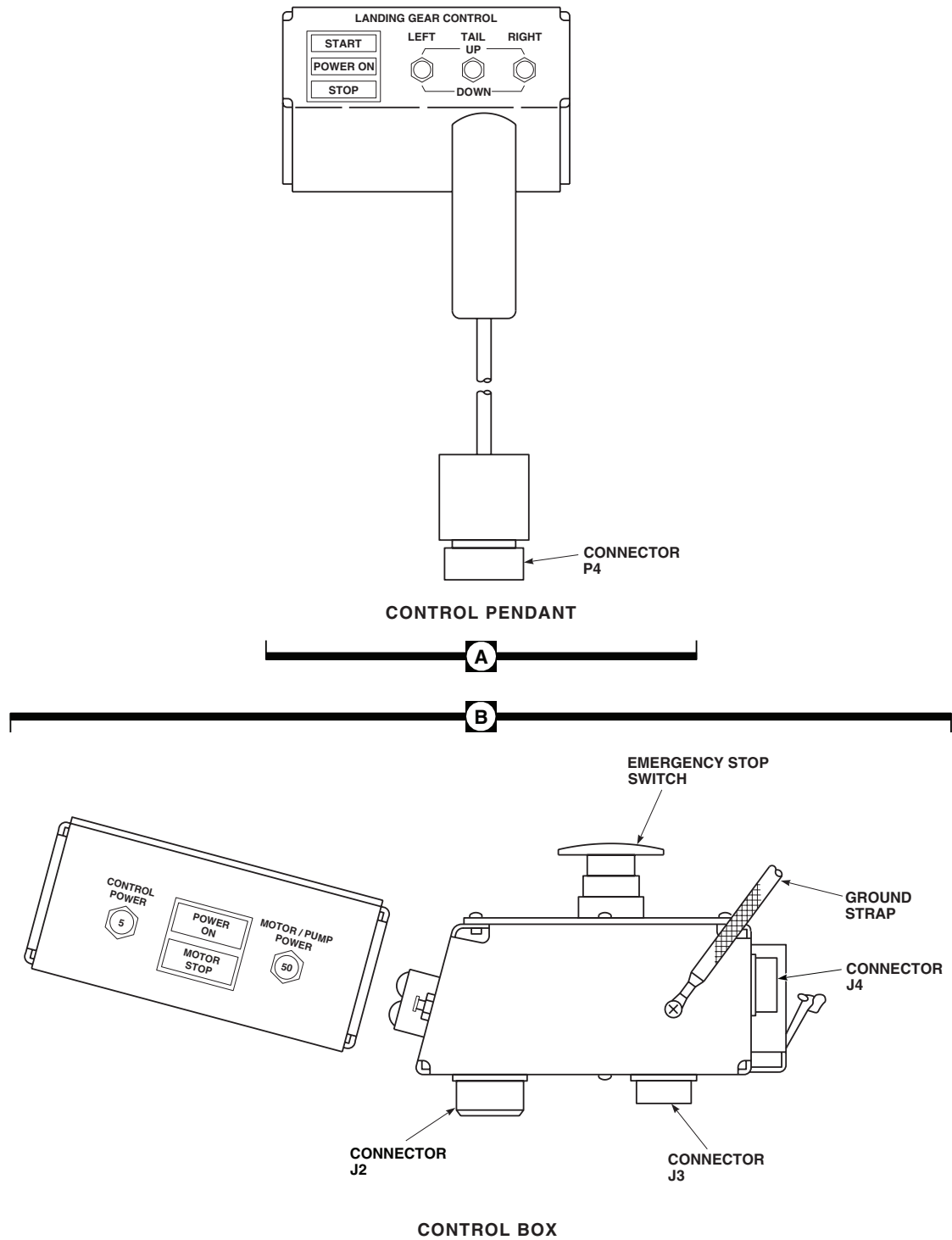
## EFFECTIVITY

70700-20437-041.

AA8704\_1  
SA

Figure 1. Air Transport Hydraulic Cart Location Diagram 70700-20437-041 . (Sheet 1 of 2)

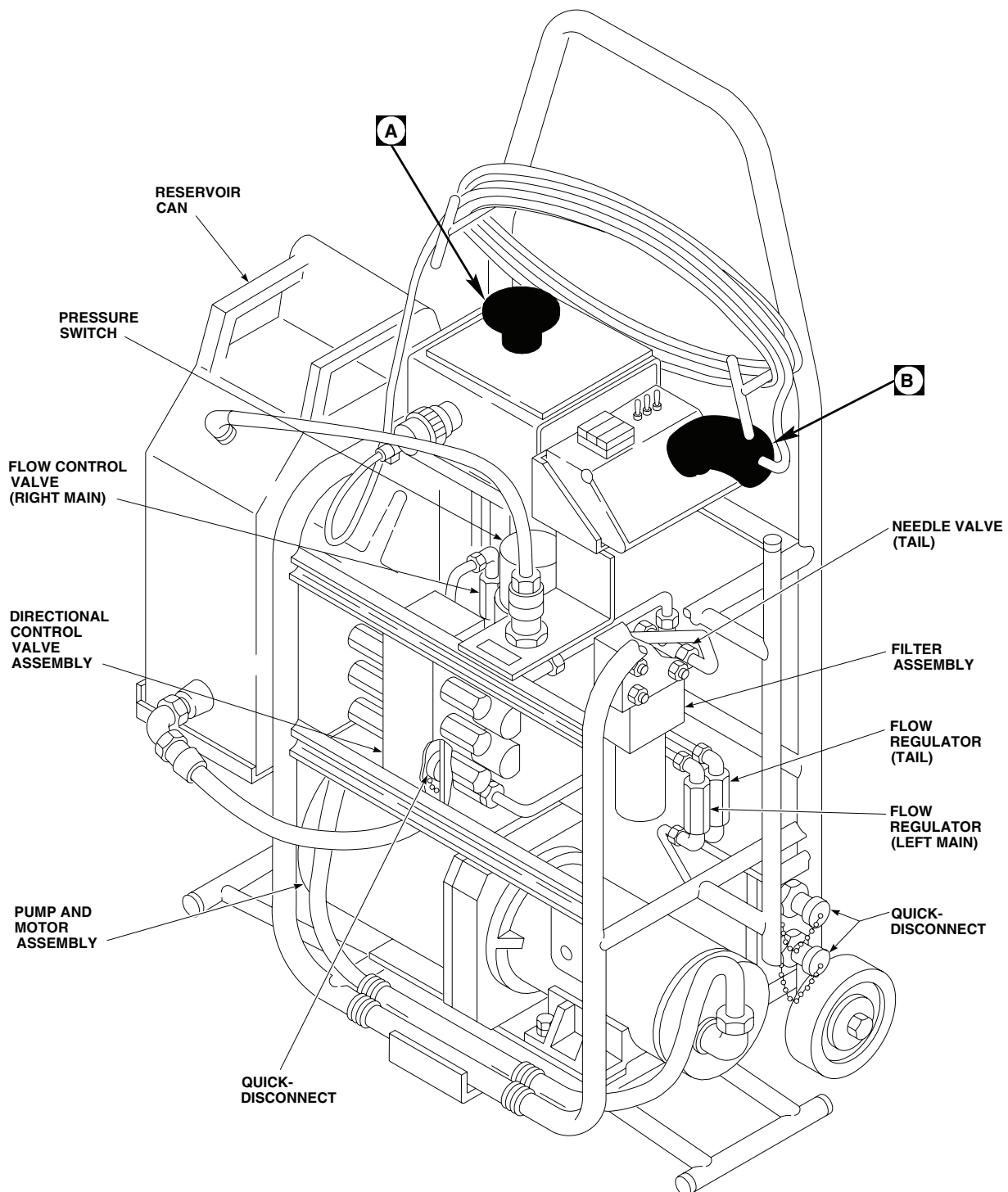
LOCATION - Continued



AA8704\_2  
SA

Figure 1. Air Transport Hydraulic Cart Location Diagram 70700-20437-041 . (Sheet 2 of 2)

## LOCATION - Continued



## EFFECTIVITY

70700-81650-041.

AA8707\_1  
SA

Figure 2. Air Transport Hydraulic Cart Location Diagram 70700-81650-041 . (Sheet 1 of 2)



## LOCATION - Continued

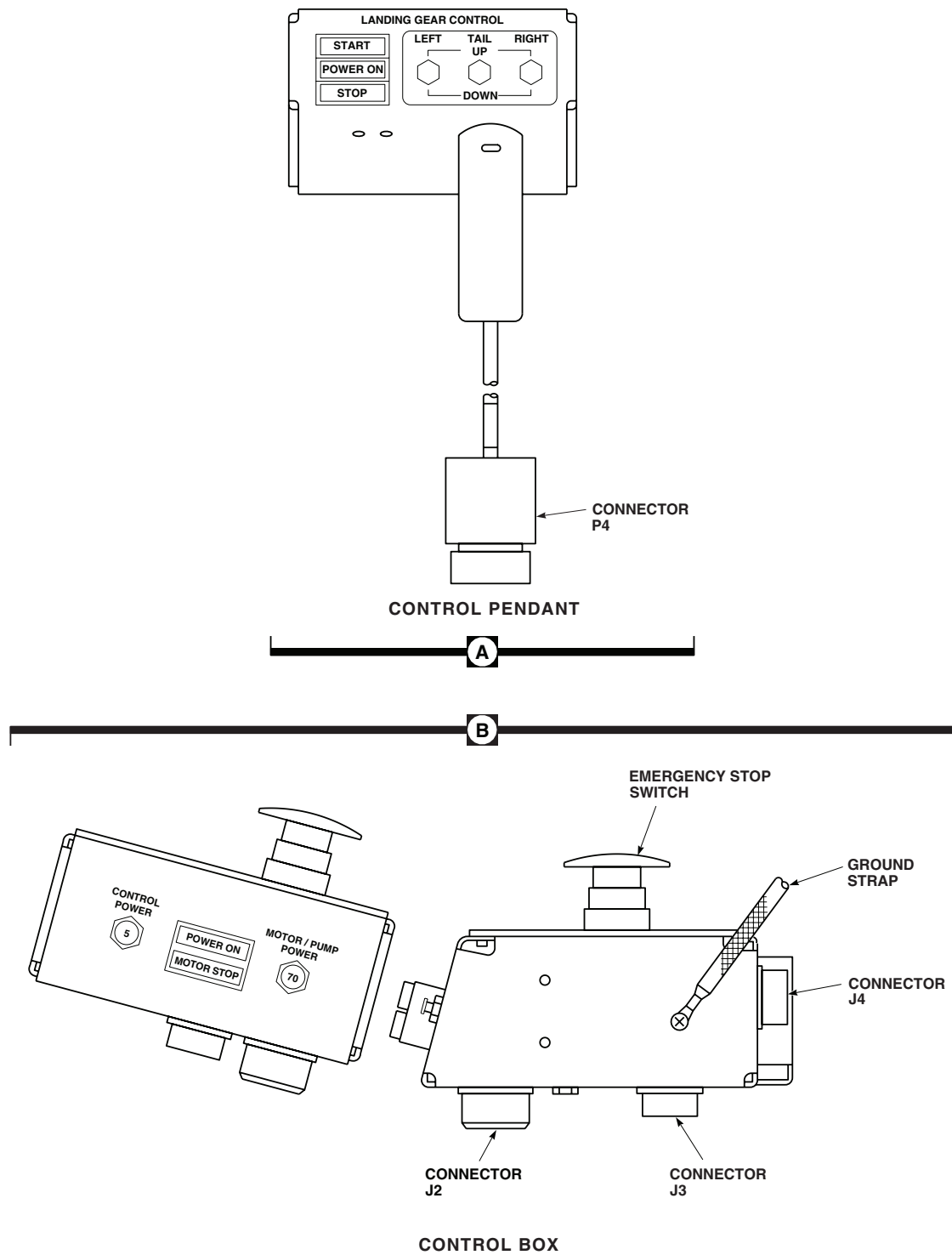
AA8707\_2  
SA

Figure 2. Air Transport Hydraulic Cart Location Diagram 70700-81650-041 . (Sheet 2 of 2)

SCHEMATICS

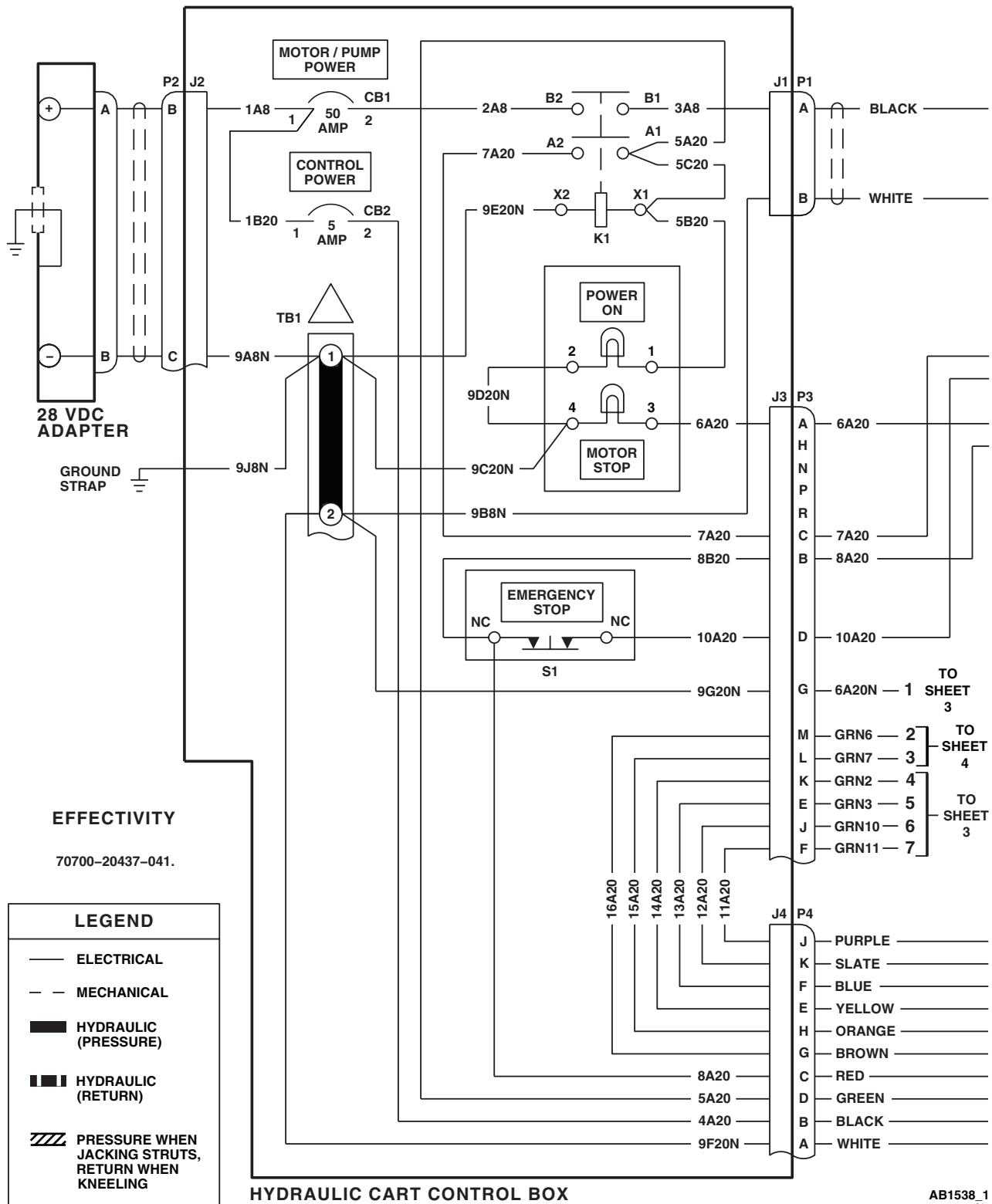
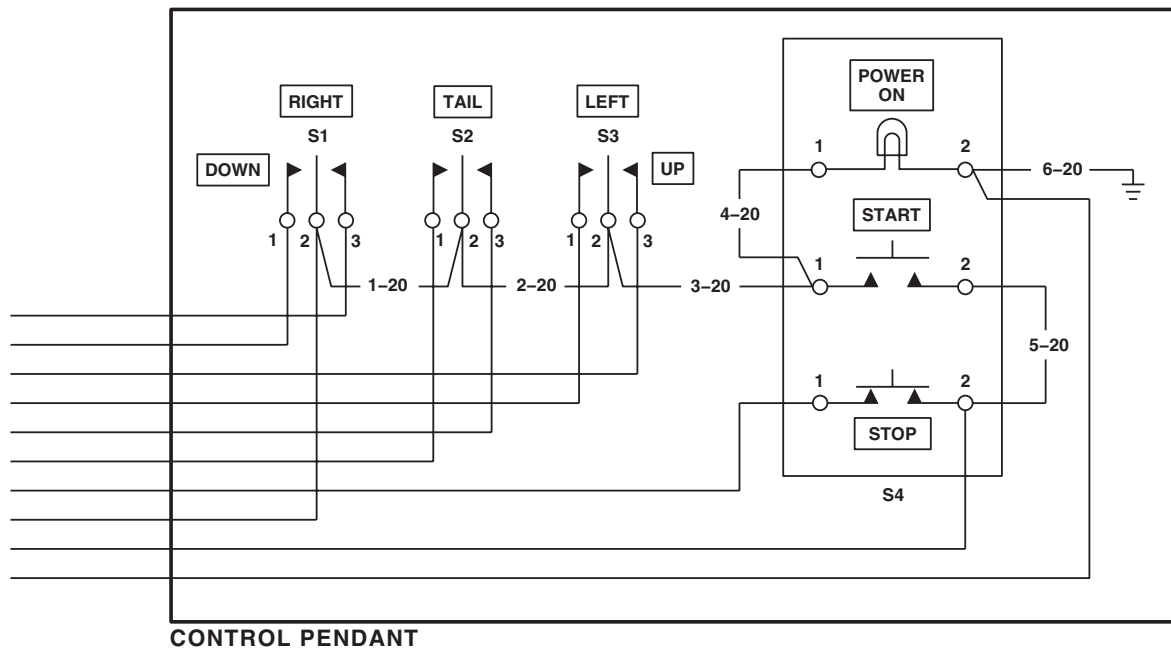
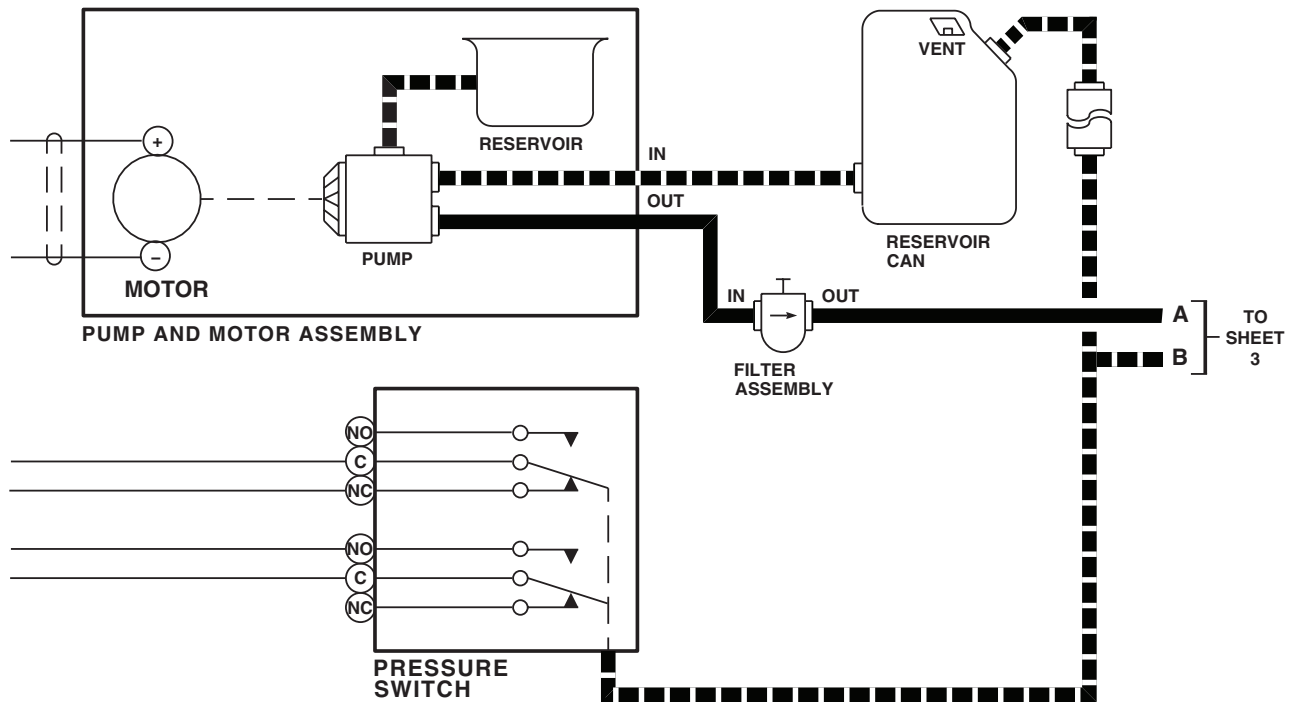


Figure 3. Air Transport Hydraulic Cart Schematic Diagram **70700-20437-041** . (Sheet 1 of 4)

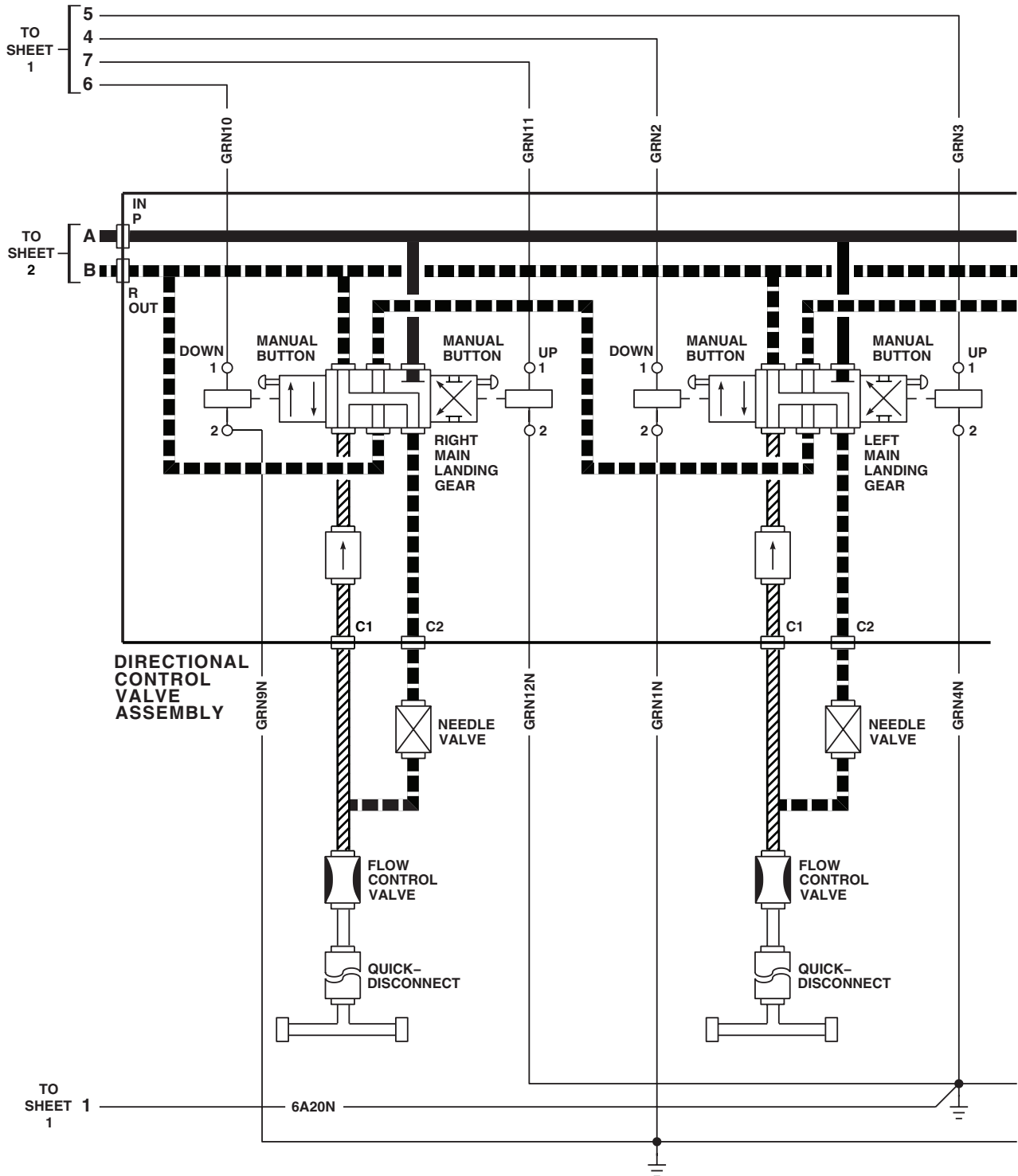
SCHEMATICS - Continued



AB1538\_2  
SA

Figure 3. Air Transport Hydraulic Cart Schematic Diagram 70700-20437-041 . (Sheet 2 of 4)

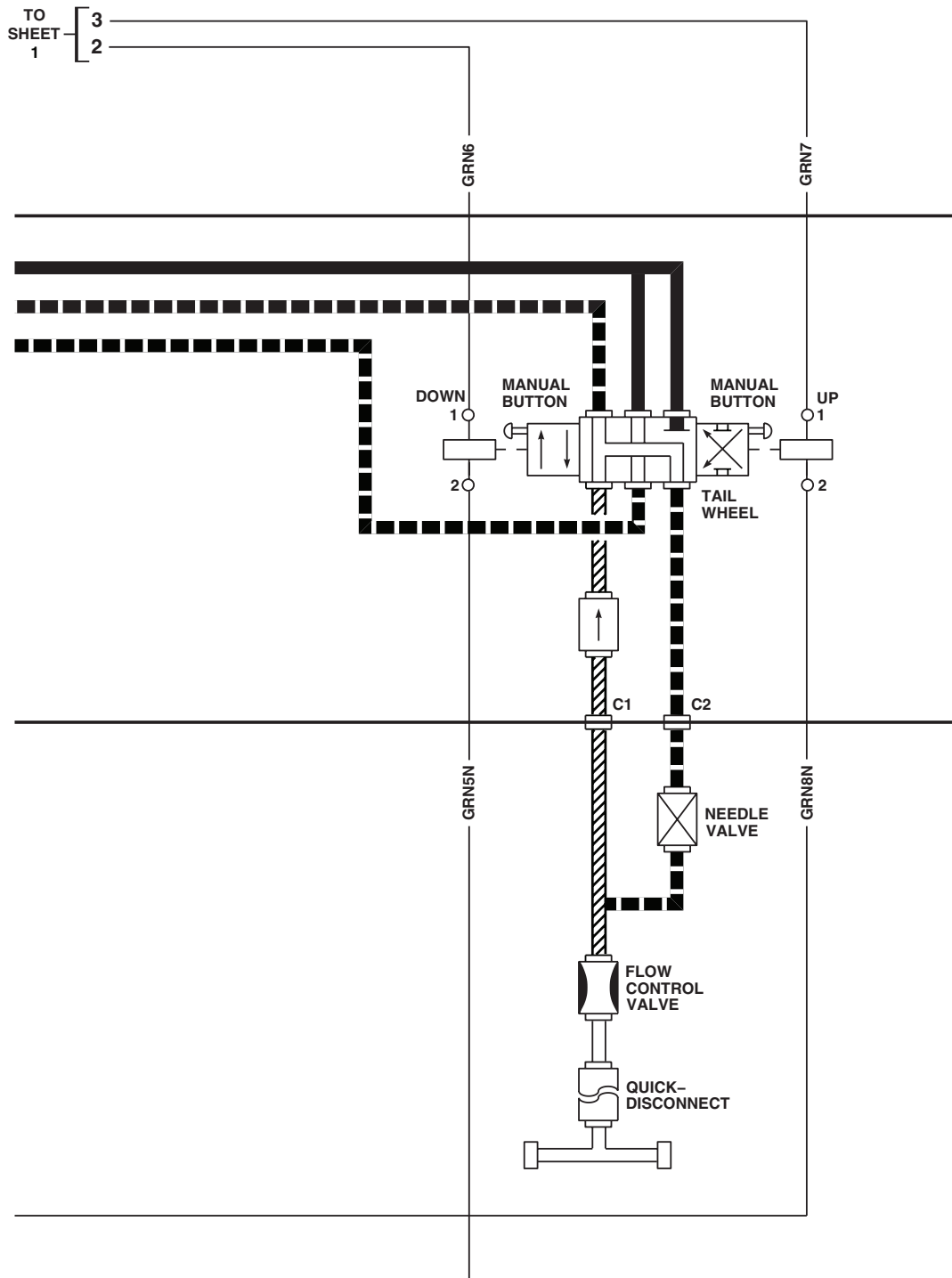
SCHEMATICS - Continued



AB1538 3  
SA

Figure 3. Air Transport Hydraulic Cart Schematic Diagram 70700-20437-041 . (Sheet 3 of 4)

SCHEMATICS - Continued



AB1538\_4  
SA

Figure 3. Air Transport Hydraulic Cart Schematic Diagram 70700-20437-041 . (Sheet 4 of 4)

**28 VDC ADAPTER**

**GROUND STRAP**

**EFFECTIVITY**  
70700-81650-041.

**LEGEND**

- ELECTRICAL
- MECHANICAL
- HYDRAULIC (PRESSURE)
- HYDRAULIC (RETURN)
- PRESSURE WHEN JACKING STRUTS, RETURN WHEN KNEELING

**HYDRAULIC CART CONTROL BOX**

**AB1539.1**

AB1539\_1

## SCHEMATICS - Continued

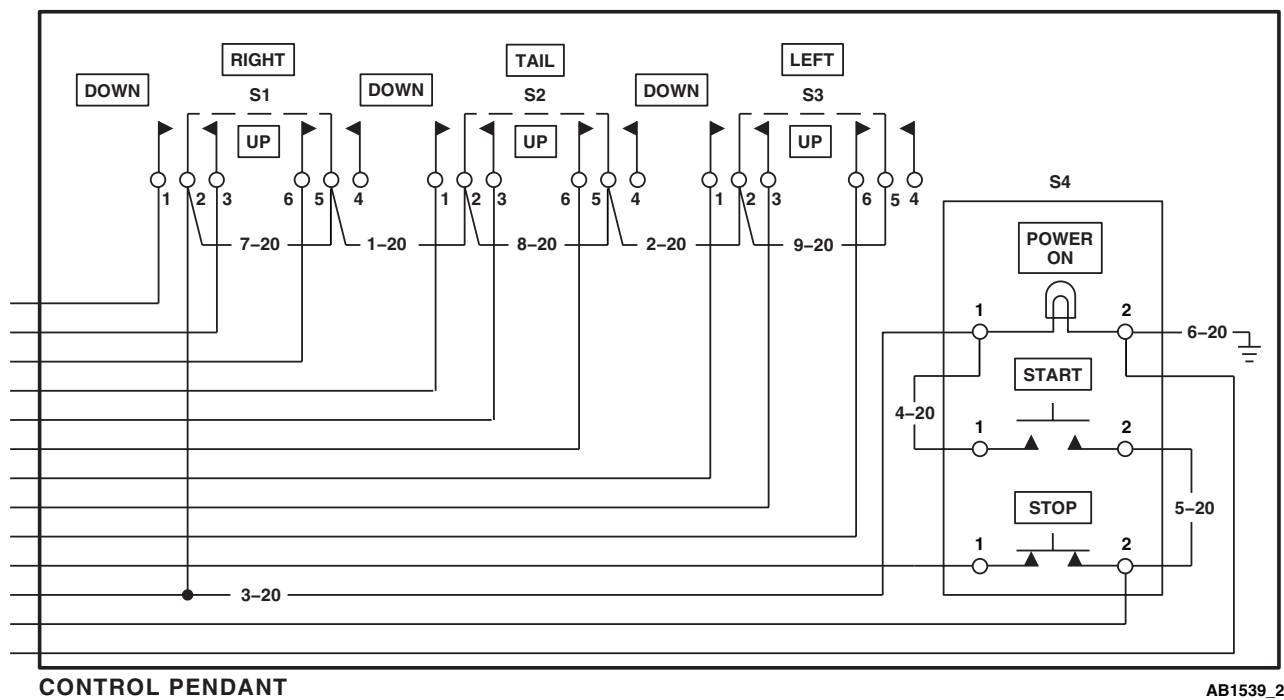
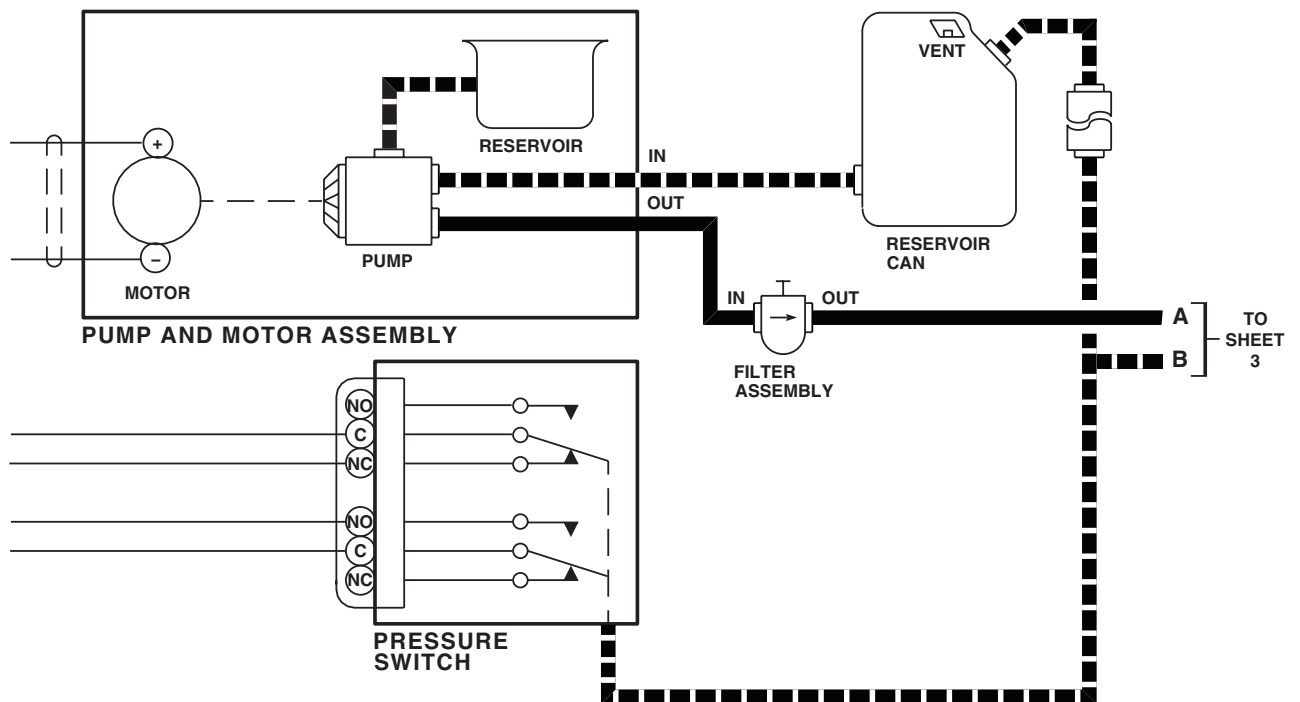
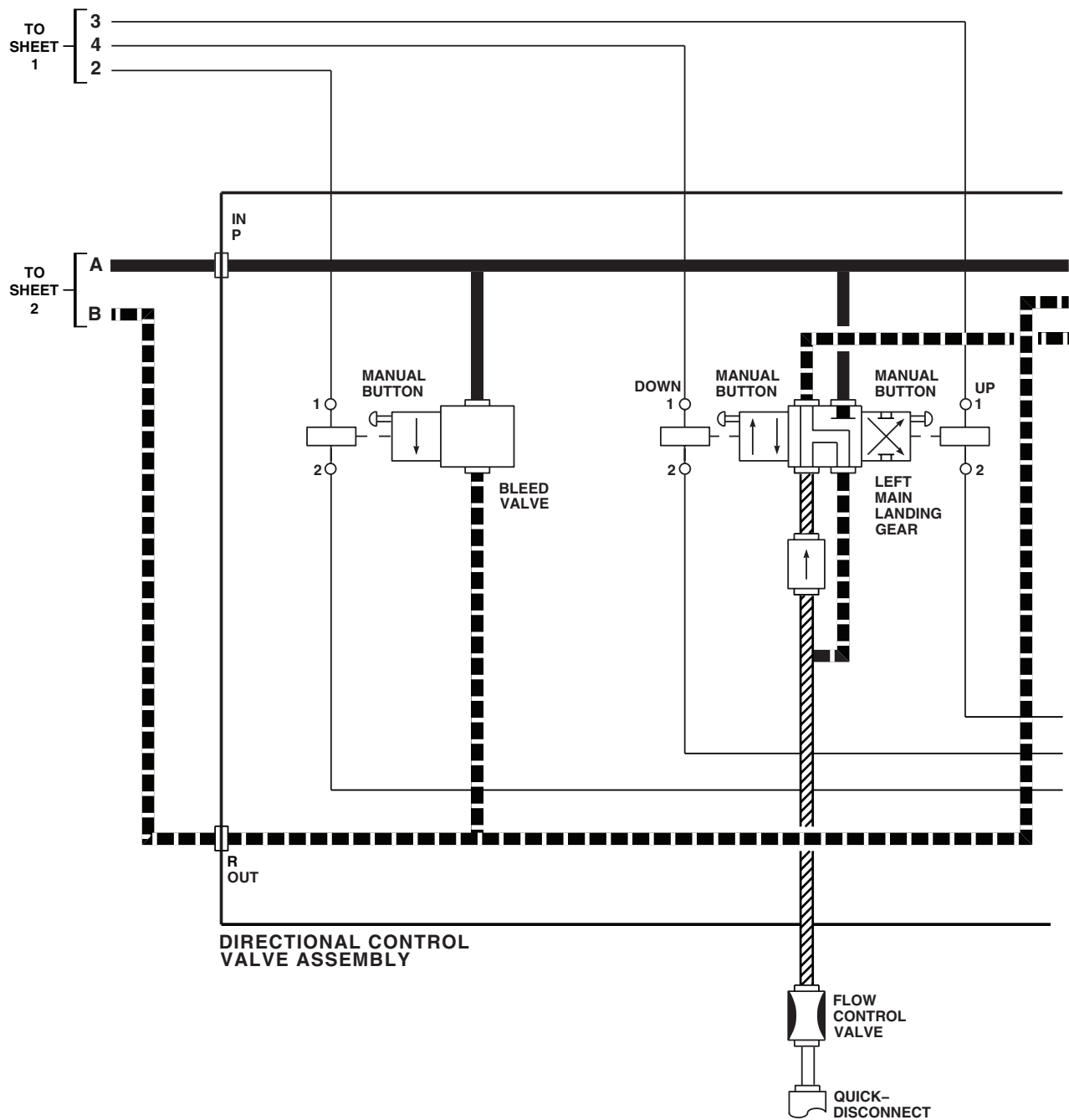


Figure 4. Air Transport Hydraulic Cart Schematic Diagram 70700-81650-041 . (Sheet 2 of 4)

SCHEMATICS - Continued

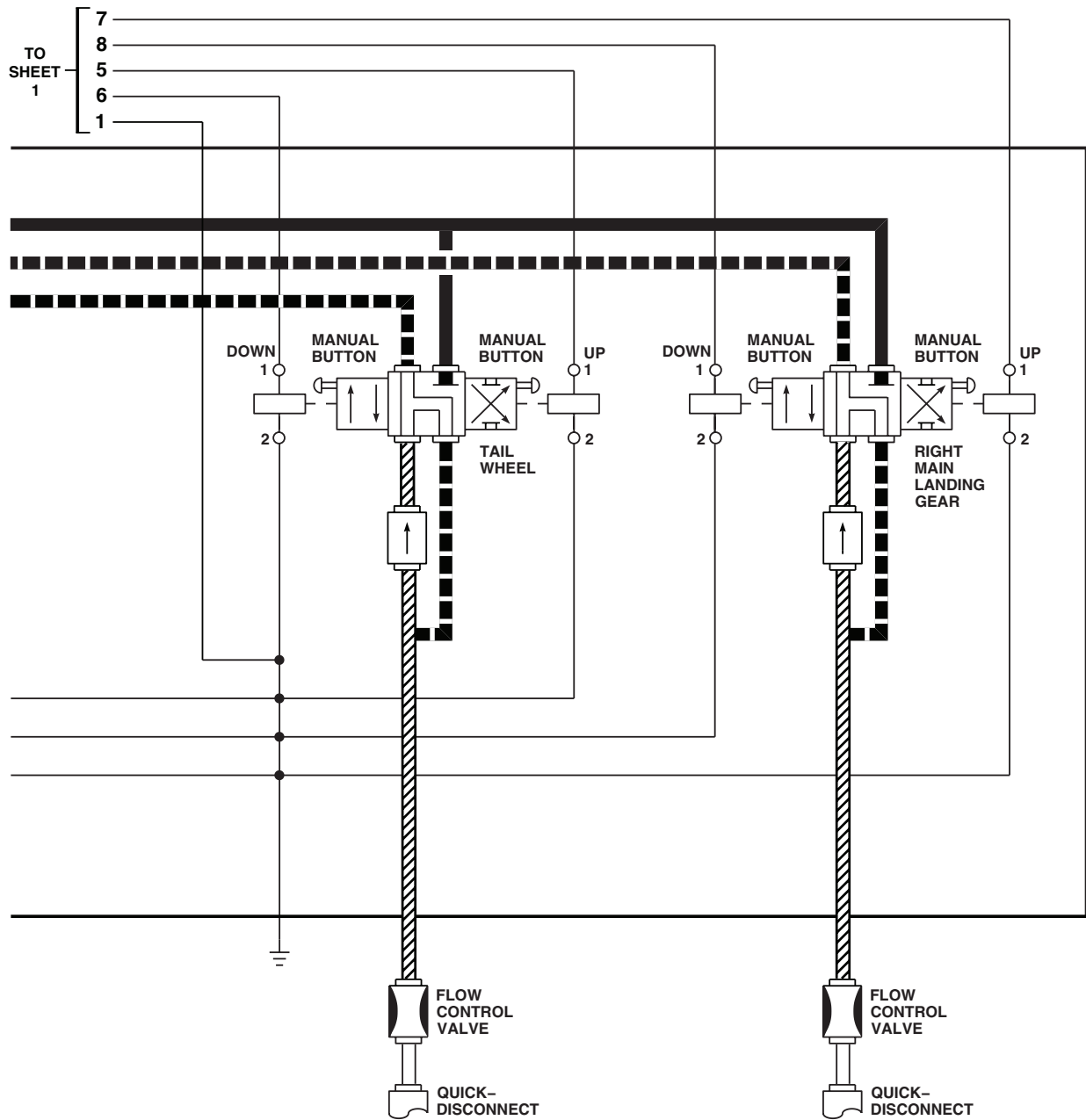


AB1539 3  
SA

Figure 4. Air Transport Hydraulic Cart Schematic Diagram 70700-81650-041 . (Sheet 3 of 4)



SCHEMATICS - Continued



AB1539\_4  
SA

Figure 4. Air Transport Hydraulic Cart Schematic Diagram 70700-81650-041 . (Sheet 4 of 4)

END OF WORK PACKAGE

0164 00-29/30 Blank



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**UNIT LEVEL****MISSION EQUIPMENT**

---

**AUXILIARY CABIN HEATER (AVIM)**

---

**INITIAL SETUP:****Test Equipment**

Multimeter, AN/PSM-45A (2 required)

**Tools and Special Tools**

Electrical Repairer Toolkit, SC 5180-99-B06

Locally-Made Jumper Plug, Figure 158, WP 1805 00

**References**

TM 55-1500-323-24

TM 55-1500-344-23

WP 1585 00

WP 1586 00

WP 1805 00

**Personnel Required**

Aircraft Electrician MOS 15F (1)

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**SPECIAL INSTRUCTIONS****NOTE**

See Figure 1 for component location diagram and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

The auxiliary cabin heater operational/troubleshooting procedure is subdivided as follows:

- Visual Inspection
- Ohms

**VISUAL INSPECTION OPERATIONAL/TROUBLESHOOTING PROCEDURE****PROCEDURE****Step**

1. Inspect auxiliary cabin heater for broken or burnt wires, bent or broken connector pins, corroded or damaged electrical connections, security of components, and general condition.

**Indication/Condition**

- a. Wires shall not be broken or burnt.
- b. Connector pins shall not be broken or bent.
- c. Electrical terminals shall not be corroded or damaged.
- d. Components shall be secured and in good condition.

**Corrective Action**

1. If Indication/Condition a. is not as specified, repair/replace wiring.
2. If Indication/Condition b. is not as specified, repair/replace connector (TM 55-1500-323-24).

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**VISUAL INSPECTION OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**

3. If Indication/Condition c. is not as specified, repair/replace terminals (TM 55-1500-323-24 and TM 55-1500-344-23).
4. If Indication/Condition d. is not as specified, repair/replace components (TM 55-1500-344-23).

**OHMS OPERATIONAL/TROUBLESHOOTING PROCEDURE****PROCEDURE****Step**

1. Connect multimeter between the following and record readings:

- J726-A and J726-C
- J726-A and J726-E
- J726-E and J726-C

**Indication/Condition**

All three readings shall be approximately the same.

**Corrective Action**

1. If Indication/Condition is not as specified, check continuity between the heater elements and J726.
  - a. If continuity is present, replace heater element as required (WP 1585 00).
  - b. If continuity is not present, repair/replace wiring as required.

**Step**

2. Check continuity between J726-B and J726-D.

**Indication/Condition**

Multimeter shall indicate continuity.

**Corrective Action**

1. If Indication/Condition is not as specified, check continuity between thermostat and J726-B and J726-D.
  - a. If continuity is present, replace thermostat (WP 1586 00).
  - b. If continuity is not present, repair/replace wiring as required.

**Step****NOTE**

The following steps must be accomplished with air inlet temperature between 60°F (15.6°C) and 95°F (35°C).

3. Connect locally-made jumper plug (WP 1805 00) to J726.

---

**OHMS OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued**NOTE**

Multimeter probes must be held in place for 10 to 20 seconds to obtain an accurate reading.

4. Connect multimeter to jumper plug as follows:

| <b>FROM</b>     | <b>TO</b>           |
|-----------------|---------------------|
| Black test jack | Red test jack       |
| Black test jack | Housing ground stud |
| Red test jack   | Housing ground stud |

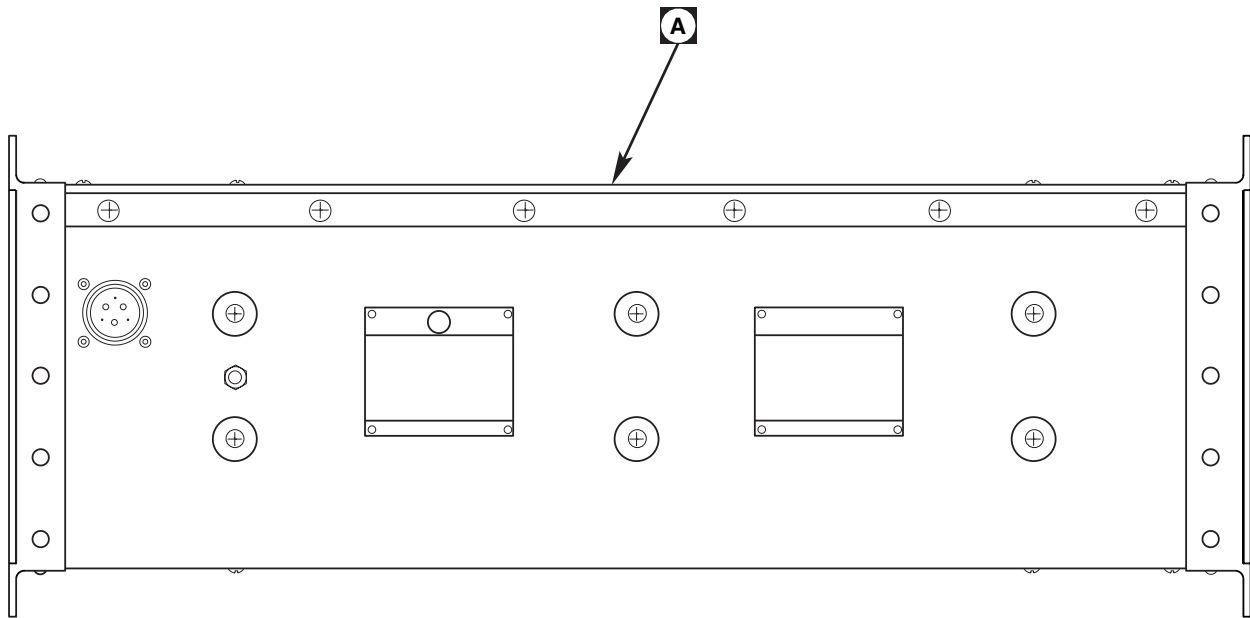
**Indication/Condition**

Multimeter shall indicate at least 10 megohms.

**Corrective Action**

1. If Indication/Condition is not as specified, check continuity and inspect wiring between heater elements and J726.
  - a. If continuity is present, and wiring inspection is good, replace J726.
  - b. If continuity is not present, or wiring does not pass inspection, repair/replace wiring as required.

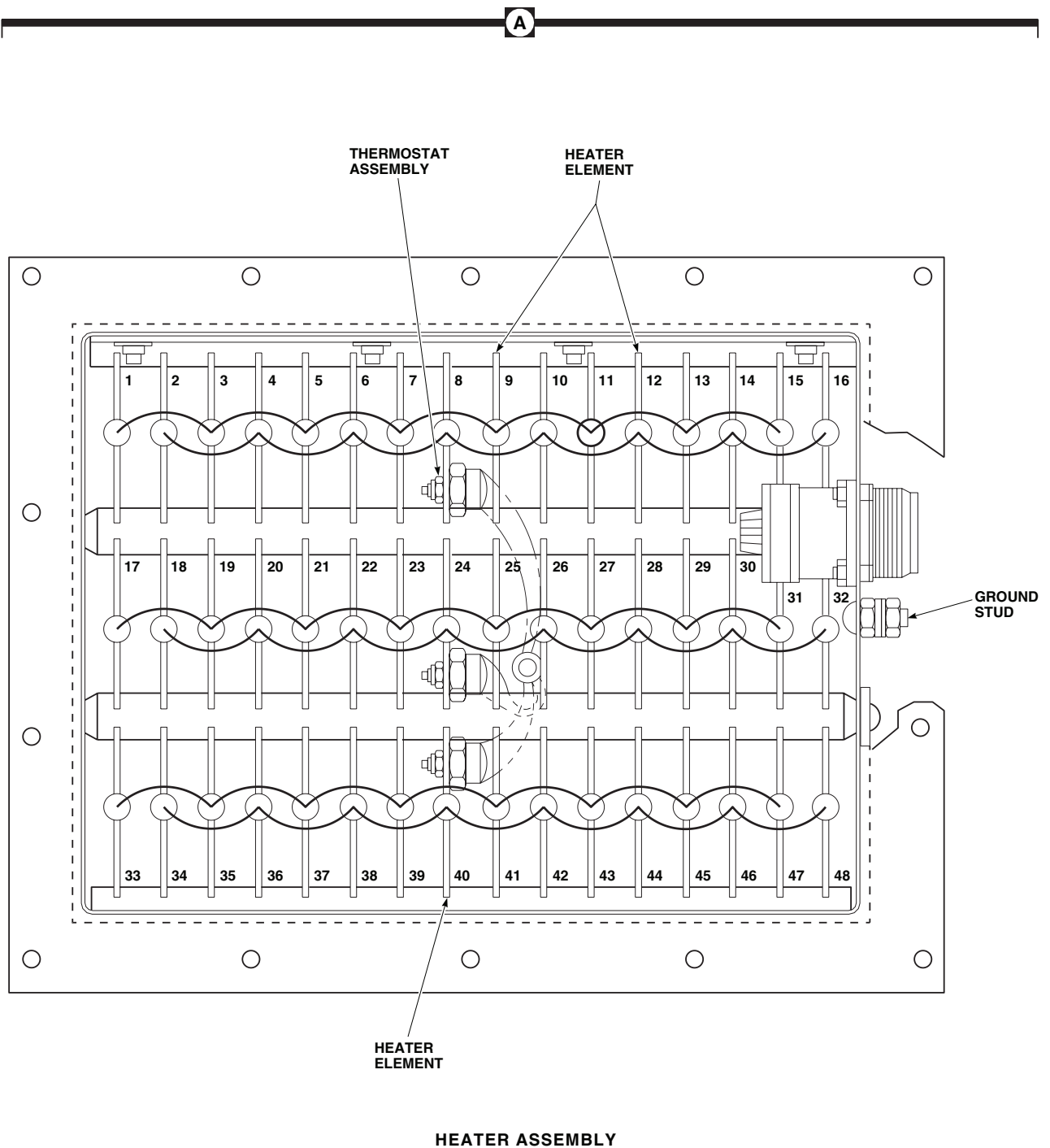
LOCATION



AA6094\_1  
SA

Figure 1. Auxiliary Cabin Heater Location Diagram. (Sheet 1 of 2)

LOCATION - Continued



AA6094\_2  
SA

Figure 1. Auxiliary Cabin Heater Location Diagram. (Sheet 2 of 2)

## SCHEMATICS

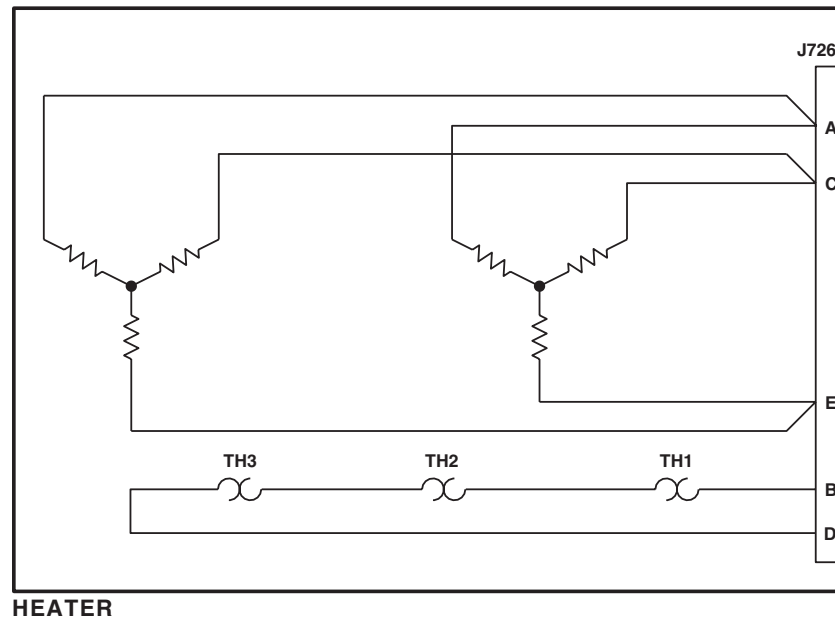
AB1536  
SA

Figure 2. Auxiliary Cabin Heater Schematic Diagram.

END OF WORK PACKAGE



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**UNIT LEVEL**  
**MISSION EQUIPMENT**  
**STORES JETTISON CONTROL PANEL (AVIM)**

---

**INITIAL SETUP:****Test Equipment**

Multimeter, AN/PSM-45A

WP 1577 00

WP 1578 00

WP 1579 00

**Tools and Special Tools**

Electrical Repairer Toolkit, SC 5180-99-B06

**Equipment Condition**

Between 27 and 28 vdc, Bench Power Available

**MWO 50-56** 115 vac, 400 Hz Bench Power Available**References**

WP 1461 00

WP 1576 00

---

**STORES JETTISON CONTROL PANEL (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE**

**PROCEDURE****Step****NOTE**

See Figure 1 for component location diagram and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

1. Remove cover from stores jettison control panel (control panel) and inspect for broken or burnt wires, security of components, and general condition.

**Indication/Condition**

Wires shall not be broken or burnt, and components shall be secure and in good condition.

**Corrective Action**

If Indication/Condition is not as specified, repair/replace as required.

**Step**

2. Check continuity between:

- J1-c and J1-X
- J1-c and J2-c
- J1-c and J2-d
- J1-j and J1-i
- J1-g and J1-h
- J1-n and J1-p
- J1-f and J1-q

---

**STORES JETTISON CONTROL PANEL (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE -**  
Continued

- J1-J and J1-L
- J1-G and J1-E
- J1-A and J1-C
- J2-f and J2-g
- J2-j and J2-h
- J1-R and J1-N
- J2-J and J2-L
- J2-A and J2-C
- J2-N and J2-R
- J2-E and J2-G

**Indication/Condition**

Continuity shall be present.

**Corrective Action**

If Indication/Condition is not as specified, repair/replace wiring as required.

**Step****NOTE**

There are two different stores jettison control panels. One uses 28 vdc and the other uses 115 vac, 400 Hz.

3. Connect 28 vdc between J1-V and J1-W.

**Indication/Condition**

Information plate legend shall light evenly.

**Corrective Action**

1. If information plate lights, but not evenly, replace malfunctioning information plate lamp (WP 1461 00).
  - a. If trouble remains, replace information plate (WP 1461 00).
2. If no information plate lamps go on, check continuity between J1-V and J3-(center contact), and J1-W and J3-(ground contact).
  - a. If continuity is not present, repair/replace wiring.
  - b. If continuity is present, replace information plate (WP 1461 00).

**Step**

4. Disconnect 28 vdc from control panel.
5. Make sure control panel JETT and EMER JETT ALL switches are down.
6. Turn control panel rotary switch to INBD BOTH.
7. Connect 28 vdc between J1-n and J1-k, and J1-n and J1-X.

---

**STORES JETTISON CONTROL PANEL (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE -**  
Continued

8. With JETT switch held up, check for 28 vdc between J1-A and J1-c. Release switch.

**Indication/Condition**

Multimeter shall indicate 28 vdc.

**Corrective Action**

If Indication/Condition is not as specified, go to MULTIMETER DOES NOT INDICATE 28 VDC BETWEEN J1-A AND J1-c, in this work package.

**Step**

9. Disconnect jumper wire from J1-k.  
10. With JETT switch held up, check for 0 vdc between J1-A and J1-c. Release switch.

**Indication/Condition**

Multimeter shall indicate 0 vdc.

**Corrective Action**

If Indication/Condition is not as specified, replace relay K1 (WP 1578 00).

**Step**

11. Connect jumper wire between 28 vdc supply ground and J1-k.  
12. Check for 0 vdc between J1-A and J1-c.

**Indication/Condition**

Multimeter shall indicate 0 vdc.

**Corrective Action**

If Indication/Condition is not as specified, replace JETT switch S1 (WP 1577 00).

**Step**

13. Turn rotary switch to OUTBD ALL.  
14. With JETT switch held up, check for 28 vdc between J1-L and J1-c. Release switch.

**Indication/Condition**

Multimeter shall indicate 28 vdc in about 1 second after JETT switch is placed up.

**Corrective Action**

1. If multimeter indicates 28 vdc, but time delay is not as specified, replace relay K7 (WP 1578 00).
2. If multimeter does not indicate 28 vdc, go to MULTIMETER DOES NOT INDICATE 28 VDC BETWEEN J1-L AND J1-c, in this work package.

**Step**

15. Disconnect jumper wire from J1-X.  
16. With JETT switch held up, check for 0 vdc between J1-L and J1-K. Release switch.

## STORES JETTISON CONTROL PANEL (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

### Indication/Condition

Multimeter shall indicate 0 vdc.

### Corrective Action

If Indication/Condition is not as specified, replace relay K7 (WP 1578 00).

### Step

17. Connect jumper wire between 28 vdc supply ground and J1-X.
18. With JETT switch held up, check for indicated voltage between the following points and J1-c while turning rotary switch as indicated. Release switch after taking measurement.

| ROTARY SWITCH POSITION | J1-A   | MULTIMETER INDICATION AT J1-L | J1-G  |
|------------------------|--------|-------------------------------|-------|
| OFF                    | 0 VDC  | 0 VDC                         |       |
| INBD L                 | 28 VDC | 0 VDC                         |       |
| INBD R                 | 0 VDC  | 28 VDC                        |       |
| INBD BOTH              | 28 VDC | 28 VDC                        |       |
| OUTBD BOTH             | 0 VDC  | 0 VDC                         |       |
| OUTBD R                | 0 VDC  | 0 VDC                         |       |
| OUTBD L                | 0 VDC  | 0 VDC                         |       |
| ALL                    | 28 VDC | 28 VDC                        | 0 VDC |

### Indication/Condition

Multimeter shall indicate corresponding dc voltage.

### Corrective Action

1. If Indication/Condition is not as specified at J1-A and J1-L, replace rotary switch S3 (WP 1576 00).
  - a. If trouble remains, repair/replace wiring.
2. If Indication/Condition is not as specified at J1-G, replace diode CR2 (WP 1579 00).

### Step

19. Turn rotary switch to INBD L.
20. Connect jumper wire between 28 vdc supply ground and J1-h.
21. With JETT switch held up, check for indicated voltage between the following points. Release switch after taking measurement.
  - J1-C and J1-c
  - J1-J and J1-c

---

**STORES JETTISON CONTROL PANEL (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE -**  
Continued**Indication/Condition**

Multimeter shall indicate 28 vdc.

**Corrective Action**

1. If multimeter does not indicate 28 vdc between J1-C and J1-c, repair/replace wiring between terminals A2 and A3 of relay K2.
2. If multimeter does not indicate 28 vdc between J1-J and J1-c, measure dc voltage between terminals X1 and X2 of relay K2.
  - a. If voltage is as specified, replace relay K2 (WP 1578 00).
  - b. If voltage is not as specified, repair/replace wiring as required.

**Step**

22. Remove jumper wire from J1-h.
23. With JETT switch held up, check for 0 vdc between J1-J and J1-c. Release switch.

**Indication/Condition**

Multimeter shall indicate 0 vdc.

**Corrective Action**

If Indication/Condition is not as specified, replace relay K2 (WP 1578 00).

**Step**

24. Remove jumper wire from J1-n and connect it to J1-f.
25. Turn rotary switch to OUTBD ALL.
26. With JETT switch held up, check for 28 vdc between J1-N and J1-c. Release switch.

**Indication/Condition**

Multimeter shall indicate 28 vdc.

**Corrective Action**

If Indication/Condition is not as specified, go to MULTIMETER DOES NOT INDICATE 28 VDC BETWEEN J1-N AND J1-c, in this work package.

**Step**

27. Disconnect jumper wire from J1-k.
28. With JETT switch held up, check for 0 vdc between J1-N and J1-c. Release switch.

**Indication/Condition**

Multimeter shall indicate 0 vdc.

**Corrective Action**

If Indication/Condition is not as specified, replace relay K1 (WP 1578 00).

**Step**

29. Connect jumper wire between 28 vdc supply ground and J1-k.

## STORES JETTISON CONTROL PANEL (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

30. Check for 0 vdc between J1-N and J1-c.

### Indication/Condition

Multimeter shall indicate 0 vdc.

### Corrective Action

If Indication/Condition is not as specified, replace JETT switch S1 (WP 1577 00).

### Step

31. With JETT switch held up, check indicated voltage between the following points and J1-c while operating rotary switch as indicated. Release switch after taking measurement.

| ROTARY SWITCH POSITION | J1-N   | MULTIMETER INDICATION AT J1-G | J1-A  |
|------------------------|--------|-------------------------------|-------|
| OFF                    | 0 VDC  | 0 VDC                         |       |
| INBD L                 | 0 VDC  | 0 VDC                         |       |
| INBD R                 | 0 VDC  | 0 VDC                         |       |
| INBD BOTH              | 0 VDC  | 0 VDC                         | 0 VDC |
| OUTBD BOTH             | 28 VDC | 28 VDC                        |       |
| OUTBD R                | 28 VDC | 0 VDC                         |       |
| OUTBD L                | 0 VDC  | 28 VDC                        |       |
| OUTBD ALL              | 28 VDC | 28 VDC                        |       |

### Indication/Condition

Multimeter shall indicate corresponding dc voltage.

### Corrective Action

1. If Indication/Condition is not as specified for J1-N and J1-G, replace rotary switch S3 (WP 1576 00).
  - a. If trouble remains, repair/replace wiring.
2. If Indication/Condition is not as specified for J1-A, replace diode CR1 (WP 1579 00).

### Step

32. Turn rotary switch to OUTBD L.
33. Connect jumper wire between 28 vdc supply ground and J1-j.
34. With JETT switch held up, check for 28 vdc between the following points. Release switch after taking measurement.
  - J1-E and J1-c
  - J1-R and J1-c

---

**STORES JETTISON CONTROL PANEL (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE -**  
Continued**Indication/Condition**

Multimeter shall indicate 28 vdc.

**Corrective Action**

1. If multimeter does not indicate 28 vdc between J1-E and J1-c, repair/replace wiring between terminals A2 and A3 of relay K3.
2. If multimeter does not indicate 28 vdc between J1-R and J1-c, check for 28 vdc between terminals X1 and X2 of relay K3.
  - a. If voltage is as specified, replace relay K3 (WP 1578 00).
  - b. If voltage is not as specified, repair/replace wiring.

**Step**

35. Remove jumper wire from J1-j.
36. With JETT switch held up, check for 0 vdc between J1-R and J1-c. Release switch.

**Indication/Condition**

Multimeter shall indicate 0 vdc.

**Corrective Action**

If Indication/Condition is not as specified, replace relay K3 (WP 1578 00).

**Step**

37. Disconnect all jumper wires from control panel.
38. Install an insulated jumper wire between J2-d and J2-e.
39. Connect 28 vdc positive lead to J2-f and negative lead to J2-c.
40. Place EMER JETT ALL switch up and measure dc voltage between the following points:
  - J2-J and J2-d
  - J2-A and J2-d
  - J2-G and J2-d

**Indication/Condition**

- a. Multimeter shall indicate 28 vdc between J2-J and J2-d and J2-A and J2-d in about 1 second after EMER JETT ALL switch is placed up.
- b. Multimeter shall indicate 0 vdc between J2-G and J2-d.

**Corrective Action**

1. If multimeter indicates 28 vdc between either J2-J and J2-d and J2-A and J2-d, but time delay is not as specified, replace relay K6 (WP 1578 00).
2. If multimeter does not indicate 28 vdc between J2-J and J2-d and J2-A and J2-d, go to MULTIMETER DOES NOT INDICATE 28 VDC BETWEEN J2-J AND J2-d AND/OR BETWEEN J2-A AND J2-d, in this work package.
3. If multimeter does not indicate 0 vdc between J2-G and J2-d, replace diode CR4 (WP 1579 00).

## STORES JETTISON CONTROL PANEL (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

### Step

41. Place EMER JETT ALL switch down and check for 0 vdc between J2-J and J2-d.

### Indication/Condition

Multimeter shall indicate 0 vdc.

### Corrective Action

If Indication/Condition is not as specified, replace EMER JETT ALL switch (WP 1577 00).

### Step

42. Disconnect 28 vdc from J2-c and J2-f.
43. Disconnect jumper wire from J2-d and J2-e.
44. Place EMER JETT ALL switch in the up position and check for 0 vdc between the following points:
- J2-J and J2-d
  - J2-A and J2-d

### Indication/Condition

Multimeter shall indicate 0 vdc.

### Corrective Action

If Indication/Condition is not as specified, replace relay K6 (WP 1578 00).

### Step

45. Remove jumper wire from J2-f and connect 28 vdc between J2-h and J2-c, and between J2-h and J2-e.
46. Measure dc voltage between the following points:

| FROM | TO           | MULTIMETER INDICATION |
|------|--------------|-----------------------|
| J2-N | J2- <u>d</u> | 28 VDC                |
| J2-E | J2- <u>d</u> | 28 VDC                |
| J2-A | J2- <u>d</u> | 0 VDC                 |

### Indication/Condition

Multimeter shall indicate corresponding dc voltage.

### Corrective Action

1. If multimeter does not indicate 28 vdc between J2-N and J2-d and/or J2-E and J2-d, go to MULTIMETER DOES NOT INDICATE 28 VDC BETWEEN J2-N AND J2-d AND/OR BETWEEN J2-E AND J2-d, in this work package.
2. If multimeter does not indicate 0 vdc between J2-A and J2-d, replace diode CR3 (WP 1579 00).



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**STORES JETTISON CONTROL PANEL (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE -**  
Continued**Step**

47. Place EMER JETT ALL switch down and check for 0 vdc between J2-N and J2-d.

**Indication/Condition**

Multimeter shall indicate 0 vdc.

**Corrective Action**

If Indication/Condition is not as specified, replace EMER JETT ALL switch (WP 1577 00).

**Step**

48. Disconnect jumper wire from J2-c.

49. Place EMER JETT ALL switch up and measure dc voltage between the following points:

- J2-N and J2-d
- J2-E and J2-d

**Indication/Condition**

Multimeter shall indicate 0 vdc.

**Corrective Action**

If Indication/Condition is not as specified, replace relay K5 (WP 1578 00).

**Step**

50. Disconnect jumper wire from J2-e and connect it to J2-c.

51. Check for 0 vdc between J2-N and J2-d.

**Indication/Condition**

Multimeter shall indicate 0 vdc.

**Corrective Action**

If Indication/Condition is not as specified, replace relay K4 (WP 1578 00).

**Step**

52. Disconnect all jumper wires from control panel.

**Indication/Condition**

All jumper wires shall be disconnected from control panel.

**Corrective Action**

None Required

**MULTIMETER DOES NOT INDICATE 28 VDC BETWEEN J1-A AND J1-C****SYMPTOM**

Multimeter does not indicate 28 vdc between J1-A and J1-c.

**MALFUNCTION**

Multimeter does not indicate 28 vdc between J1-A and J1-c.

**CORRECTIVE ACTION**

- 1. Check for 28 vdc between JETT switch terminal 1 and J1-c.**
  - a. If voltage is as specified, go to **2**.
  - b. If voltage is not as specified, go to **4**.
- 2. With JETT switch held in up position, check for 28 vdc between switch terminal 2 and J1-c.**
  - a. If voltage is as specified, go to **3**.
  - b. If voltage is not as specified, replace switch (WP 1577 00). Go to **9**.
- 3. Check continuity between JETT switch terminal 2 and INBD/OUTBD switch terminal A4.**
  - a. If continuity is present, replace INBD/OUTBD switch S3 (WP 1577 00). Go to **9**.
  - b. If continuity is not present, repair/replace wiring. Go to **9**.
- 4. Check for 28 vdc between relay K1-B1 and K1-X2.**
  - a. If voltage is as specified, go to **5**.
  - b. If voltage is not as specified, go to **8**.
- 5. Check for 28 vdc between K1-X1 and K1-X2.**
  - a. If voltage is as specified, go to **6**.
  - b. If voltage is not as specified, go to **7**.
- 6. Check continuity between K1-B2 and JETT switch terminal 1.**
  - a. If continuity is present, replace relay K1 (WP 1578 00). Go to **9**.
  - b. If continuity is not present, repair/replace wiring. Go to **9**.
- 7. Check diode CR1.**
  - a. If diode is good, repair/replace wiring. Go to **9**.
  - b. If diode is not good, replace diode CR1 (WP 1579 00). Go to **9**.
- 8. Check diode CR6.**
  - a. If diode is good, repair/replace wiring. Go to **9**.
  - b. If diode is not good, replace diode CR6 (WP 1579 00). Go to **9**.
- 9. Procedure completed.**

**MULTIMETER DOES NOT INDICATE 28 VDC BETWEEN J1-L AND J1-C****SYMPTOM**

Multimeter does not indicate 28 vdc between J1-L and J1-c.

**MALFUNCTION**

Multimeter does not indicate 28 vdc between J1-L and J1-c.

**CORRECTIVE ACTION****1. Check continuity between:**

- S3-A4 and K7-X1
- S3-B8 and K7-A2
- J1-X and K7-X2

a. If continuity is present, go to **2**.

b. If continuity is not present, repair/replace wiring. Go to **3**.

**2. Check continuity between INBD/OUTBD switch terminals B8 and BC.**

a. If continuity is present, replace relay K7 (WP 1578 00). Go to **3**.

b. If continuity is not present, replace rotary switch S3 (WP 1576 00). Go to **3**.

**3. Procedure completed.****MULTIMETER DOES NOT INDICATE 28 VDC BETWEEN J1-N AND J1-C****SYMPTOM**

Multimeter does not indicate 28 vdc between J1-N and J1-c.

**MALFUNCTION**

Multimeter does not indicate 28 vdc between J1-N and J1-c.

**CORRECTIVE ACTION****1. Check for 28 vdc between JETT switch terminal 4 and J1-c.**

a. If voltage is as specified, go to **2**.

b. If voltage is not as specified, go to **4**.

**2. With JETT switch held in up position, check for 28 vdc between switch terminal 5 and J1-c.**

a. If voltage is as specified, go to **3**.

b. If voltage is not as specified, replace switch (WP 1577 00). Go to **9**.

**3. Check continuity between JETT switch terminal 5 and INBD/OUTBD switch terminal D8.**

a. If continuity is present, replace INBD/OUTBD switch S3 (WP 1577 00). Go to **9**.

b. If continuity is not present, repair/replace wiring. Go to **9**.

**4. Check for 28 vdc between relay K1-A1 and K1-X2.**

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**MULTIMETER DOES NOT INDICATE 28 VDC BETWEEN J1-N AND J1-C** - Continued

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- a. If voltage is as specified, go to **5**.
- b. If voltage is not as specified, go to **8**.
- 5. Check for 28 vdc between K1-X1 and K1-X2.**
  - a. If voltage is as specified, go to **6**.
  - b. If voltage is not as specified, go to **7**.
- 6. Check continuity between K1-A2 and JETT switch terminal 4.**
  - a. If continuity is present, replace relay K1 (WP 1578 00). Go to **9**.
  - b. If continuity is not present, repair/replace wiring. Go to **9**.
- 7. Check diode CR2.**
  - a. If diode is good, repair/replace wiring. Go to **9**.
  - b. If diode is not good, replace diode CR2 (WP 1579 00). Go to **9**.
- 8. Check diode CR7.**
  - a. If diode is good, repair/replace wiring. Go to **9**.
  - b. If diode is not good, replace diode CR7 (WP 1579 00). Go to **9**.
- 9. Procedure completed.**

**MULTIMETER DOES NOT INDICATE 28 VDC BETWEEN J2-J AND J2-D And/Or Between J2-A And J2-d****SYMPTOM**

Multimeter does not indicate 28 vdc between J2-J and J2-d and/or between J2-A and J2-d.

**MALFUNCTION**

Multimeter does not indicate 28 vdc between J2-J and J2-d and/or between J2-A and J2-d.

**CORRECTIVE ACTION**

- 1. Check for 28 vdc between:**
  - **J2-J and J2-d**
  - **J2-A and J2-d**
  - a. If voltage is as specified, go to **2**.
  - b. If voltage is not as specified, go to **3**.
- 2. Check continuity between:**
  - **K6-A1 and K6-B1**
  - **K6-A1 and K6-X1**
  - a. If continuity is present, replace relay K6 (WP 1578 00). Go to **11**.
  - b. If continuity is not present, repair/replace wiring. Go to **11**.
- 3. Check for 28 vdc between EMER JETT ALL switch terminal 3 and J2-d.**
  - a. If voltage is as specified, go to **4**.
  - b. If voltage is not as specified, go to **6**.

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**MULTIMETER DOES NOT INDICATE 28 VDC BETWEEN J2-J AND J2-D And/Or Between J2-A And J2-d -**  
Continued

- 4. With EMER JETT ALL switch in up position, check for 28 vdc between switch terminal 2 and J2-d.**
  - a. If voltage is as specified, go to **5**.
  - b. If voltage is not as specified, replace EMER JETT ALL switch S2 (WP 1577 00). Go to **11**.
- 5. Check continuity between:**
  - **S2-2 and K6-X1**
  - **K6-A1 and K6-X1**
  - **J2-c and K6-X2**
  - a. If continuity is present, replace relay K6 (WP 1578 00). Go to **11**.
  - b. If continuity is not present, repair/replace wiring. Go to **11**.
- 6. Check for 28 vdc between K4-B1 and K4-X2.**
  - a. If voltage is as specified, go to **7**.
  - b. If voltage is not as specified, go to **10**.
- 7. Check for 28 vdc between K4-X1 and K4-X2.**
  - a. If voltage is as specified, go to **8**.
  - b. If voltage is not as specified, go to **9**.
- 8. Check continuity between K4-B2 and INBD/OUTBD switch terminal 3.**
  - a. If continuity is present, replace relay K4 (WP 1578 00). Go to **11**.
  - b. If continuity is not present, repair/replace wiring. Go to **11**.
- 9. Check diode CR3.**
  - a. If diode is good, repair/replace wiring. Go to **11**.
  - b. If diode is not good, replace diode CR3 (WP 1579 00). Go to **11**.
- 10. Check diode CR5.**
  - a. If diode is good, repair/replace wiring. Go to **11**.
  - b. If diode is not good, replace diode CR5 (WP 1579 00). Go to **11**.
- 11. Procedure completed.**

**MULTIMETER DOES NOT INDICATE 28 VDC BETWEEN J2-N AND J2-D And/Or Between J2-E And J2-d****SYMPTOM**

Multimeter does not indicate 28 vdc between J2-N and J2-d and/or between J2-E and J2-d.

**MALFUNCTION**

Multimeter does not indicate 28 vdc between J2-N and J2-d and/or between J2-E and J2-d.

**CORRECTIVE ACTION**

- 1. Check for 28 vdc between:**
  - **J2-N and J2-d**

**MULTIMETER DOES NOT INDICATE 28 VDC BETWEEN J2-N AND J2-D And/Or Between J2-E And J2-d -**  
Continued

- **J2-E and J2-d**
  - a. If voltage is as specified, go to **2**.
  - b. If voltage is not as specified, go to **3**.
- 2. Check continuity between:**
  - **K5-A1 and K5-B1**
  - **K5-A1 and K5-X1**
    - a. If continuity is present, replace relay K5 (WP 1578 00). Go to **11**.
    - b. If continuity is not present, repair/replace wiring. Go to **11**.
- 3. Check for 28 vdc between EMER JETT ALL switch terminal 6 and J2-d.**
  - a. If voltage is as specified, go to **4**.
  - b. If voltage is not as specified, go to **6**.
- 4. With EMER JETT ALL switch in up position, check for 28 vdc between switch terminal 5 and J2-d.**
  - a. If voltage is as specified, go to **5**.
  - b. If voltage is not as specified, replace EMER JETT ALL switch S2 (WP 1577 00). Go to **11**.
- 5. Check continuity between:**
  - **S2-5 and K5-X1**
  - **K5-A1 and K5-X1**
  - **J2-c and K5-X2**
    - a. If continuity is present, replace relay K5 (WP 1578 00). Go to **11**.
    - b. If continuity is not present, repair/replace wiring. Go to **11**.
- 6. Check for 28 vdc between K4-A1 and K4-X2.**
  - a. If voltage is as specified, go to **7**.
  - b. If voltage is not as specified, go to **10**.
- 7. Check for 28 vdc between K4-X1 and K4-X2.**
  - a. If voltage is as specified, go to **8**.
  - b. If voltage is not as specified, go to **9**.
- 8. Check continuity between K4-A2 and INBD/OUTBD switch terminal 6.**
  - a. If continuity is present, replace relay K4 (WP 1578 00). Go to **11**.
  - b. If continuity is not present, repair/replace wiring. Go to **11**.
- 9. Check diode CR4.**
  - a. If diode is good, repair/replace wiring. Go to **11**.
  - b. If diode is not good, replace diode CR4 (WP 1579 00). Go to **11**.
- 10. Check diode CR8.**

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**MULTIMETER DOES NOT INDICATE 28 VDC BETWEEN J2-N AND J2-D And/Or Between J2-E And J2-d -**  
Continued

- a. If diode is good, repair/replace wiring. Go to **11**.
- b. If diode is not good, replace diode CR8 (WP 1579 00). Go to **11**.

**11. Procedure completed.**

LOCATION

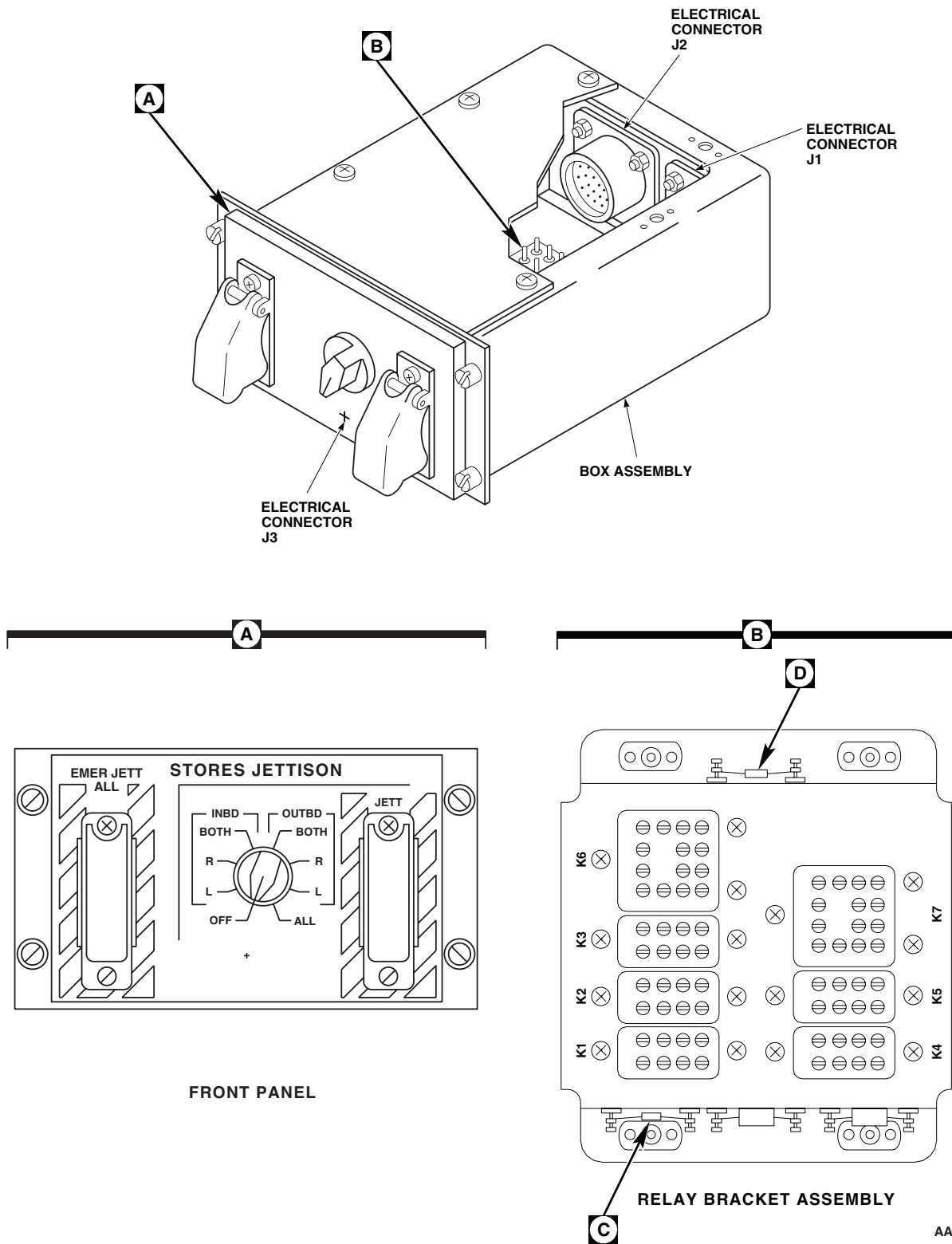


Figure 1. Stores Jettison Control Panel Components Location Diagram. (Sheet 1 of 2)



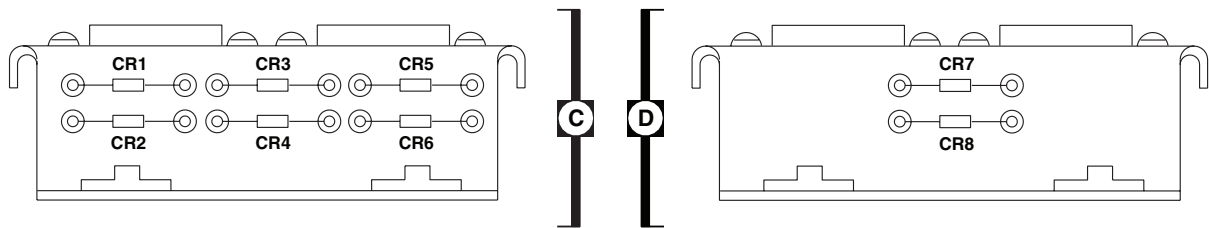
**LOCATION** - ContinuedAA3450\_2  
SA

Figure 1. Stores Jettison Control Panel Components Location Diagram. (Sheet 2 of 2)

## SCHEMATICS

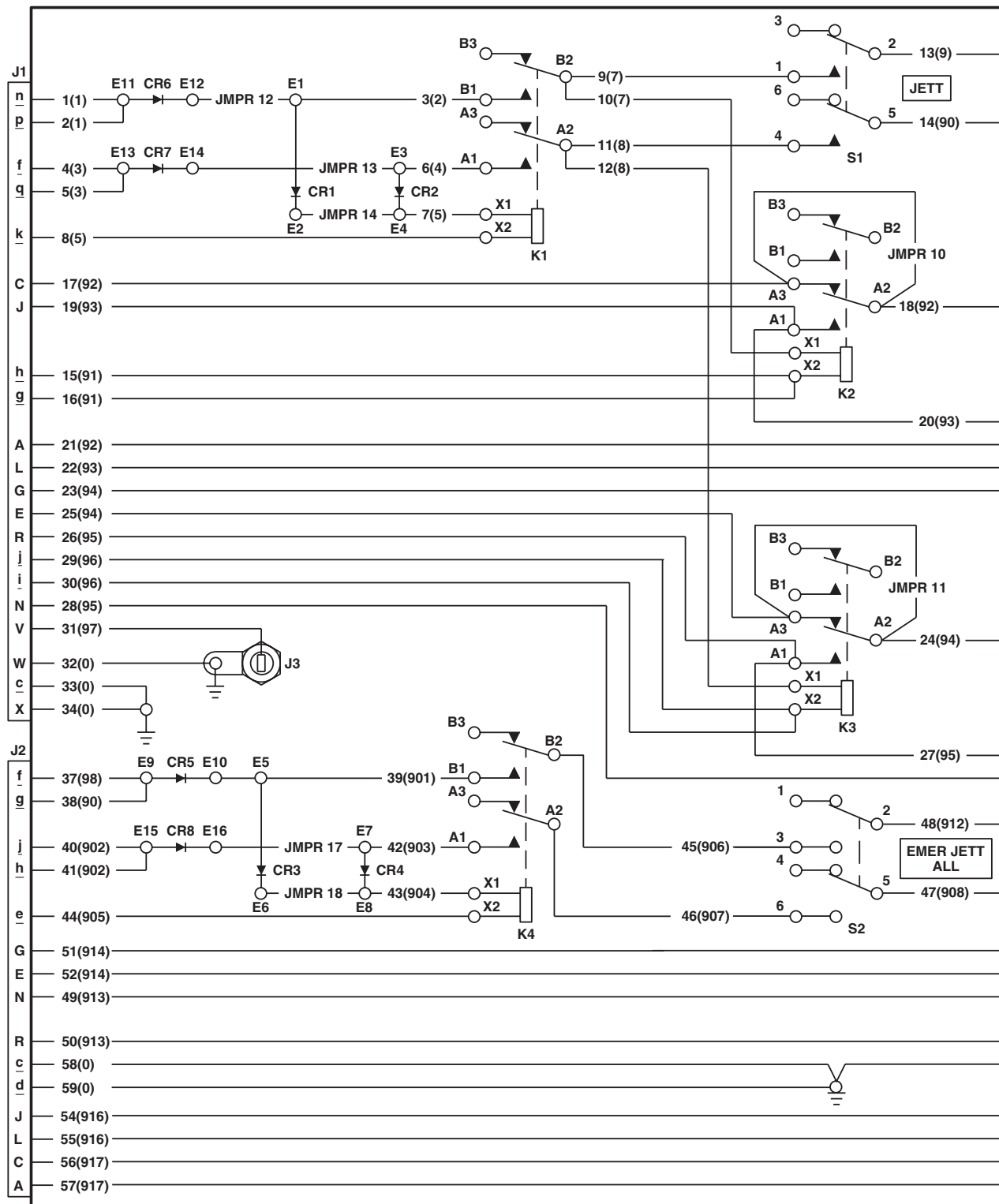
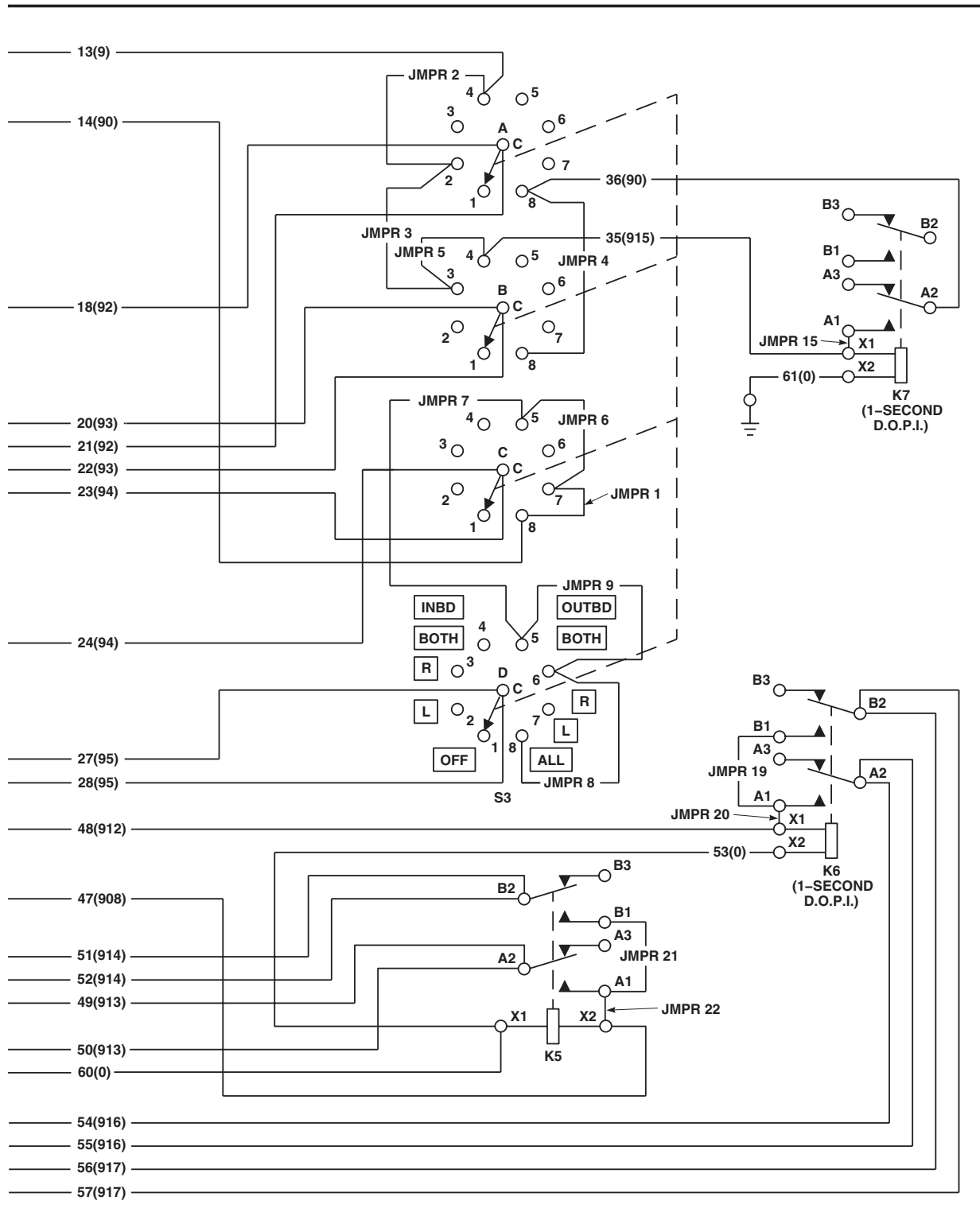
AB1533.1  
SA

Figure 2. Stores Jettison Control Panel Schematic Diagram. (Sheet 1 of 2)

SCHEMATICS - Continued



AB1533\_2  
SA

Figure 2. Stores Jettison Control Panel Schematic Diagram. (Sheet 2 of 2)

END OF WORK PACKAGE



## UNIT LEVEL

## MISSION EQUIPMENT

EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM **HH-60A HH-60L**

This WP supersedes WP 0167 00, dated 17 April 2006.

## INITIAL SETUP:

**Test Equipment**

|                                                                       |            |
|-----------------------------------------------------------------------|------------|
| Compressor Unit                                                       | WP 1439 00 |
| Locally-Made Fuel Quantity Simulator Test Set, Figure 152, WP 1805 00 | WP 1440 00 |
| Locally-Made Gate Valve Test Harness, Figure 156, WP 1805 00          | WP 1442 00 |
| Multimeter, AN/PSM-45A                                                | WP 1445 00 |
|                                                                       | WP 1447 00 |
|                                                                       | WP 1448 00 |
|                                                                       | WP 1450 00 |

**Tools and Special Tools**

|                                                                                              |            |
|----------------------------------------------------------------------------------------------|------------|
| Aircraft Mechanic's Toolkit, SC 5180-99-B01                                                  | WP 1451 00 |
| Container, Suitable                                                                          | WP 1453 00 |
| Electrical Repairer Toolkit, SC 5180-99-B06                                                  | WP 1454 00 |
| Locally-Made Drag Beam Wedge, Figure 167, WP 1805 00                                         | WP 1455 00 |
| Locally-Made External Stores Support System (ESSS) Pneumatic Adapter, Figure 151, WP 1805 00 | WP 1457 00 |
| Stopwatch                                                                                    | WP 1465 00 |
| Torque Wrench, GG-W-686                                                                      | WP 1469 00 |
|                                                                                              | WP 1475 00 |
|                                                                                              | WP 1476 00 |
|                                                                                              | WP 1612 00 |
|                                                                                              | WP 1613 00 |

**Personnel Required**

|                                       |            |
|---------------------------------------|------------|
| Aircraft Electrician MOS 15F (1)      | WP 1620 00 |
| UH-60 Helicopter Repairer MOS 15T (1) | WP 1685 00 |
|                                       | WP 1747 00 |
|                                       | WP 1805 00 |

**References**

TM 1-1520-237-10  
 WP 0086 00  
 WP 0100 00  
 WP 0104 00  
 WP 0124 00  
 WP 0125 00  
 WP 0131 00  
 WP 0799 00  
 WP 0824 00  
 WP 0825 00  
 WP 0961 00  
 WP 0972 00  
 WP 1433 00  
 WP 1435 00  
 WP 1437 00  
 WP 1438 00

**Equipment Condition**

All Explosive Cartridges Removed from Ejector Racks, WP 1442 00  
 All Explosive Cartridges Removed from Ejector Racks, WP 1445 00  
 External Electrical Power Available, WP 1685 00 or APU Operational, TM 1-1520-237-10  
 External Range Extension Tanks (Inboard or Outboard) EMPTY, WP 1620 00  
 Main Fuel Tanks, Not More Than 650 Pounds of Fuel in Each Tank, WP 1612 00  
 or Main Fuel Tanks, Not More Than 650 Pounds of Fuel in Each Tank, WP 1613 00  
 Rear Fairings on All Vertical Stores (Pylons) Removed, WP 1447 00

## APPLICABLE CONFIGURATION

This troubleshooting procedure is designed to be used when the horizontal (Wings) and/or vertical (Pylons) stores are initially installed or replaced. When only tanks are removed, refer to WP 1465 00.

## EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE

### PROCEDURE

#### Step

#### NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

- Special Icons used in this operational/troubleshooting procedure:

**CU** Steps that are to be done if compressor unit is used.

**INBD** Steps that are to be done if inboard tanks are installed.

**OUTBD** Steps that are to be done if outboard tanks are installed.

- As an alternate to using the compressor unit pressurize the ESSS Range Extension System, the Aviation Ground Power Unit (AGPU) or the APU may be used. If the alternate is selected, steps with **CU** may be skipped.
  - The operational/troubleshooting procedure will check the ESSS Range Extension System with any tank configuration, installed on the helicopter, authorized by the Aircraft Operator's Manual (TM 1-1520-237-10).
  - See Figure 1 for component location diagram and Figure 2 for schematic diagram as an aid in operational/troubleshooting.
1. **CU** Unfasten fasteners on left inboard stores support rear fairing and hinge fairing open to gain access to bleed-air quick-disconnect (QD). ■
  2. **CU** Disconnect hose from outboard side of bleed-air QD. ■
  3. **CU** Using locally-made pneumatic adapter (WP 1805 00), tee air compressor into helicopter bleed-air QD and ESSS bleed-air hose. ■
  4. Make sure all circuit breakers are pushed in except those noted being pulled for maintenance or safety.
  5. Pull out the following circuit breakers:

#### CIRCUIT BREAKER

#### CIRCUIT BREAKER PANEL

ESSS JTSN OUTBD

Copilot's

ESSS JTSN INBD

Copilot's

# EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE - Continued

| CIRCUIT BREAKER    | CIRCUIT BREAKER PANEL          |
|--------------------|--------------------------------|
| BACKUP PUMP PWR    | Copilot's                      |
| UTIL RECP CABIN    | Copilot's                      |
| ESSS JTSN OUTBD    | Upper Console (copilot's side) |
| ESSS JTSN INBD     | Upper Console (copilot's side) |
| EXT FUEL RH        | Mission Readiness              |
| EXT FUEL LH        | Mission Readiness              |
| AUX FUEL QTY       | Mission Readiness              |
| NO. 2 XFER CONTROL | Mission Readiness              |

## CAUTION

Prior to activating fuel gate valves that have been out of service for more than 30 days, it is recommended to open and close fuel gate valves a few times by hand prior to applying electrical power to reduce the possibility of damage to auxiliary fuel management panel and circuitry.

6. Disconnect P360 from auxiliary fuel management control panel.
7. Connect P1 of locally-made ESSS gate shutoff valve test harness to UTIL RCPT +28V receptacle J258 in cabin ceiling.
8. Turn on electrical power.
9. Push in UTIL RECP CABIN circuit breaker on copilot's circuit breaker panel.
10. Connect locally-made test harness (WP 1805 00) pin E to P360-B.
11. Connect pins A and D of locally-made test harness to the following pairs of pins on P360, and alternately place locally-made test harness switch to OPEN and CLOSE:

| TEST HARNESS PIN | P360- | VALVE OPERATION     |
|------------------|-------|---------------------|
| A                | F     | LEFT OUTBD - OPEN   |
| D                | E     | LEFT OUTBD - CLOSED |
| A                | Y     | LEFT INBD - OPEN    |
| D                | Z     | LEFT INBD - CLOSED  |
| A                | g     | RIGHT OUTBD - OPEN  |

## EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE - Continued

| TEST HARNESS PIN | P360- | VALVE OPERATION      |
|------------------|-------|----------------------|
| D                | P     | RIGHT OUTBD - CLOSED |
| A                | i     | RIGHT INBD - OPEN    |
| D                | i     | RIGHT INBD - CLOSED  |

### Indication/Condition

Gate valves shall open and close as indicated.

### Corrective Action

1. If Indication/Condition is not as specified, connect locally-made test harness pins A, D, and E to pins A, D, and E of malfunctioning gate valve. Alternately place locally-made test harness switch to OPEN and CLOSED.
  - a. If gate valve operates normally, repair/replace wiring between P360 and gate valve's electrical connector (WP 1747 00).
  - b. If gate valve does not operate normally, replace gate valve (WP 1440 00).

### Step

12. Turn off electrical power.
13. Connect P360 to auxiliary fuel management control panel.
14. Turn on electrical power.
15. Turn pilot's or copilot's multifunction display (MFD) switch ON and press T6 switch (illuminates all MFD legends for 10 seconds, then only the active caution and advisory legends will be displayed).
16. Turn upper console INST LT - PILOT FLT control at least one quarter turn clockwise.
17. Turn upper console CONSOLE LT - LOWER control to BRT.

### Indication/Condition

Information plate lighting on control panels and upper console shall go on at full brightness.

### Corrective Action

If Indication/Condition is not as specified, troubleshoot upper and lower console lights (WP 0124 00).

### Step

18. Push in the following circuit breakers:

| CIRCUIT BREAKER    | CIRCUIT BREAKER PANEL |
|--------------------|-----------------------|
| AUX FUEL QTY       | Mission Readiness     |
| NO. 2 XFER CONTROL | Mission Readiness     |



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**EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE - Continued**

19. Momentarily place INDICATOR LTS BRT/DIM - TEST switch to TEST.

**Indication/Condition**

AUX FUEL QTY POUNDS digital display on auxiliary fuel management control panel shall dim.

**Corrective Action**

If Indication/Condition is not as specified, troubleshoot instrument panel and console indicator lights dimming (WP 0125 00).

**Step**

20. Adjust auxiliary fuel management control panel QTY BRIGHTNESS control from DECR to INCR, then back to DECR.

**Indication/Condition**

AUX FUEL QTY POUNDS digital display shall go on to full brightness, then back to dim.

**Corrective Action**

If Indication/Condition is not as specified, replace auxiliary fuel management control panel (WP 1448 00).

**Step**

21. Adjust upper console CONSOLE LT - LOWER and INST LT - PILOT FLT controls to OFF.  
22. Place auxiliary fuel management control panel switches as indicated:

| SWITCH           | POSITION |
|------------------|----------|
| PRESS OUTB       | OFF      |
| PRESS INBD       | OFF      |
| TANKS            | OUTBD    |
| MODE             | OFF      |
| MANUAL XFR RIGHT | OFF      |
| MANUAL XFR LEFT  | OFF      |
| AUX FUEL QTY     | TOTAL    |

23. Press and hold auxiliary fuel management control panel TEST pushbutton.

**Indication/Condition**

All auxiliary fuel management control panel indicators shall go on and digital display shall indicate 8888.

**Corrective Action**

If Indication/Condition is not as specified, go to WHEN AUXILIARY FUEL MANAGEMENT CONTROL PANEL TEST PUSHBUTTON IS PRESSED, ALL INDICATORS ARE NOT ON AND/OR DIGITAL DISPLAY DOES NOT INDICATE 8888, in this work package.

---

**EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE - Continued****Step**

24. Release TEST pushbutton.

**Indication/Condition**

- a. Digital display shall cycle the indication 8 from left to right 3 times.
- b. Digital display shall indicate the word GOOD for about 5 seconds.
- c. Digital display shall indicate either 4, 5, or 8 (preset fuel type) in left-most position for about 3 seconds.
- d. Digital display may indicate some value of auxiliary fuel remaining (unknown at this time).

**Corrective Action**

1. If Indication/Condition a. and/or c. are not as specified, replace auxiliary fuel management control panel (WP 1448 00).
2. If Indication/Condition b. is not as specified, but display E01, E04, E05, or E06, replace control panel (WP 1448 00).
3. If Indication/Condition b. is not as specified, but displays E02, go to DURING SELF-TEST, AUXILIARY FUEL MANAGEMENT DIGITAL DISPLAY INDICATES E02 INSTEAD OF GOOD, in this work package.
4. If Indication/Condition b. is not as specified, but displays E03, go to DURING SELF-TEST, AUXILIARY FUEL MANAGEMENT DIGITAL DISPLAY INDICATES E03 INSTEAD OF GOOD, in this work package.

**Step****NOTE**

Disregard any EXTERNAL indicator displayed at this time.

25. Press and hold auxiliary fuel management control panel STATUS pushbutton.

**Indication/Condition**

EXTERNAL RIGHT INBD EMPTY, EXTERNAL RIGHT OUTBD EMPTY, EXTERNAL LEFT INBD EMPTY, and EXTERNAL LEFT OUTBD EMPTY indicators shall go on.

**Corrective Action**

If Indication/Condition is not as specified, replace auxiliary fuel management control panel (WP 1448 00).

**Step**

26. Release STATUS pushbutton.

**Indication/Condition**

All auxiliary fuel management control panel indicators and MFD AUX FUEL legend shall be off.

**Corrective Action**

1. If EXTERNAL RIGHT NO FLOW and/or EXTERNAL LEFT NO FLOW indicators are on, replace auxiliary fuel management control panel (WP 1448 00).
2. If EXTERNAL RIGHT INBD EMPTY, EXTERNAL RIGHT OUTBD EMPTY, EXTERNAL LEFT INBD EMPTY or EXTERNAL LEFT OUTBD EMPTY indicator is on (tank installed in that position), go to AUXILIARY FUEL MANAGEMENT CONTROL PANEL EMPTY INDICATOR IS ON WITH ADEQUATE FUEL IN FUEL TANK, in this work package.

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**EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE - Continued**

3. If Indication/Condition is not as specified, go to WHEN AUXILIARY FUEL MANAGEMENT CONTROL PANEL STATUS PUSHBUTTON IS PRESSED AND RELEASED, INDICATORS AND/OR PILOT'S/ COPILOT'S MFD AUX FUEL LEGEND ARE NOT OFF, in this work package.

**Step**

27. Turn off electrical power.
28. **OUTBD** Fill each outboard tank, if installed with 160 - 190 pounds of fuel. ■
29. **INBD** Fill each inboard tank, if installed with 160 - 190 pounds of fuel. ■
30. Turn on electrical power.
31. **CU** Apply air source, regulated to 25 - 45 psi to helicopter's external tanks. ■
32. Record central display unit (CDU) TOTAL FUEL indication.
33. **OUTBD** Place auxiliary fuel management control panel PRESS OUTBD switch to ON. ■
34. **INBD** Place auxiliary fuel management control panel PRESS INBD switch to ON. ■
35. **Record** time.
36. Inspect fuel and pneumatic hoses and fittings for signs of leaking or damage (WP 0961 00).

**Indication/Condition**

There shall be no signs of leakage or damage.

**Corrective Action**

If Indication/Condition is not as specified, replace hoses and fittings as necessary (WP 0972 00).

**Step****NOTE**

- While doing the following procedure, inspect fuel and pneumatic hoses and fittings as specified in step 37.
- The K FACTOR called out in the following steps is a calibration adjustment used for calibrating the fuel system and is stamped on the fuel flow transmitter. Record this value.
- Disregard digit to the right of decimal for K FACTOR entry.

37. **Record** K FACTOR.
38. Turn auxiliary fuel management control panel AUX FUEL QTY switch to CAL.
39. Place auxiliary fuel management control panel INCR-DECR switch to INCR, then to DECR.

**Indication/Condition**

Digital display indication shall increase and then decrease.

**Corrective Action**

If Indication/Condition is not as specified, go to WITH AUXILIARY FUEL MANAGEMENT CONTROL PANEL AUX FUEL QTY SWITCH TO CAL DIGITAL DISPLAY INDICATION CANNOT BE CHANGED, in this work package.

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**EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE - Continued****Step**

40. Place auxiliary fuel management control panel INCR-DECR switch to either position to enter K FACTOR on digital display.
41. Using locally-made drag beam wedge push plunger down on left drag beam switch.
42. Momentarily place auxiliary fuel management control panel INCR-DECR switch to INCR, then to DECR.

**Indication/Condition**

Digital display indication shall not change.

**Corrective Action**

If Indication/Condition is not as specified, go to DIGITAL DISPLAY INDICATION CHANGES WITH DRAG BEAM SWITCH IN IN-FLIGHT POSITION, in this work package.

**Step**

43. Remove locally-made drag beam wedge.
44. Turn auxiliary fuel management control panel AUX FUEL QTY switch to OUTBD.
45. **OUTBD** Disconnect P2/J849 and P2/J850. ■

**Indication/Condition**

Digital display shall indicate cccc.

**Corrective Action**

If Indication/Condition is not as specified, replace auxiliary fuel management control panel (WP 1448 00).

**Step**

46. Connect P2/J849 and P2/J850 if outboard tanks are installed, or install insulated jumper wire between J1-1 and J1-2.
47. Place auxiliary fuel management control panel INCR-DECR switch to either position to set digital display indication to 1000.

**Indication/Condition**

Digital display shall indicate 1000 pounds.

**Corrective Action**

1. If Indication/Condition is not as specified, replace auxiliary fuel management control panel (WP 1448 00).
2. If trouble remains, replace left relay panel (WP 0825 00).

**Step**

48. Turn auxiliary fuel management control panel AUX FUEL QTY switch to INBD.
49. **INBD** Disconnect P2/J835 and P2/J836. ■

**Indication/Condition**

Digital display shall indicate cccc.

**Corrective Action**

If Indication/Condition is not as specified, replace auxiliary fuel management control panel (WP 1448 00).

---

**EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE - Continued****Step**

50. Connect P2/J835 and P2/J836 if inboard tanks are installed, or install insulated jumper wire between inboard J1-1 and J1-2.
51. Place auxiliary fuel management control panel INCR-DECR switch to either position to set digital display indication to 1000 pounds.

**Indication/Condition**

Digital display shall indicate 1000 pounds.

**Corrective Action**

If Indication/Condition is not as specified, replace auxiliary fuel management control panel (WP 1448 00).

**Step**

52. Turn auxiliary fuel management control panel AUX FUEL QTY switch to TOTAL.

**Indication/Condition**

Digital display shall indicate 2000 pounds.

**Corrective Action**

If Indication/Condition is not as specified, replace auxiliary fuel management control panel (WP 1448 00).

**Step**

53. **OUTBD** Turn auxiliary fuel management control panel AUX FUEL QTY switch to OUTBD. ■
54. **OUTBD** Turn auxiliary fuel management control panel MODE switch to MANUAL. ■
55. **OUTBD** Check CDU TOTAL FUEL indication. ■

**Indication/Condition**

CDU TOTAL FUEL indication shall be the same as recorded in step 32.

**Corrective Action**

If Indication/Condition is not as specified, go to CDU TOTAL FUEL INDICATION INCREASES WHEN FUEL TANK TRANSFER IS NOT ACTIVE, in this work package.

**Step**

56. Turn pilot's or copilot's MFD switch ON and press T6 switch (illuminates all MFD legends for 10 seconds; then only the active caution and advisory legends will be displayed).
57. **OUTBD** Before continuing, make sure that at least 5 minutes have elapsed from time recorded in step 35. ■

**NOTE**

Disregard MFD AUX FUEL legend if it goes on momentarily.

58. **OUTBD** Place auxiliary fuel management control panel MANUAL XFR - RIGHT switch to ON until CDU TOTAL FUEL indication increases by 20 pounds. ■

**Indication/Condition**

- a. Elapsed time required to increase indication by 20 pounds shall be less than 1-1/2 minutes.

---

**EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE - Continued**

- b. Auxiliary fuel management control panel EXTERNAL RIGHT NO FLOW indicator shall momentarily go on, then go off.
- c. Remaining auxiliary fuel management control panel EXTERNAL indicators shall remain off.
- d. Auxiliary fuel management control panel digital display indication shall decrease.

**Corrective Action**

If Indication/Conditions are not as specified, go to WITH MANUAL MODE SELECTED, CDU TOTAL FUEL INDICATION DOES NOT INCREASE AND/OR RESULT IS NOT AS SPECIFIED WHEN ATTEMPTING TO TRANSFER FUEL FROM RIGHT OUTBOARD FUEL TANK, in this work package.

**Step**

59. **OUTBD** Place auxiliary fuel management control panel MANUAL XFR - RIGHT switch to OFF. ■

**NOTE**

Disregard MFD AUX FUEL legend if it goes on momentarily.

60. **OUTBD** Place auxiliary fuel management control panel MANUAL XFR - LEFT switch to ON until CDU TOTAL FUEL indication increases by 20 pounds. ■

**Indication/Condition**

- a. Elapsed time required to increase indication by 20 pounds shall be less than 1-1/2 minutes.
- b. Auxiliary fuel management control panel EXTERNAL LEFT NO FLOW indicator shall momentarily go on, then go off.
- c. Remaining auxiliary fuel management control panel EXTERNAL indicators shall remain off.
- d. Auxiliary fuel management control panel digital display indication shall decrease.

**Corrective Action**

If Indication/Conditions are not as specified, go to WITH MANUAL MODE SELECTED, CDU TOTAL FUEL INDICATION DOES NOT INCREASE AND/OR RESULT IS NOT AS SPECIFIED WHEN ATTEMPTING TO TRANSFER FUEL FROM LEFT OUTBOARD FUEL TANK, in this work package.

**Step**

61. **OUTBD** Place auxiliary fuel management control panel MANUAL XFR - LEFT switch to OFF. ■
62. **INBD** Turn auxiliary fuel management control panel AUX FUEL QTY switch to INBD. ■
63. **INBD** Place auxiliary fuel management control panel TANKS switch to INBD. ■

**Indication/Condition**

Digital display indication shall not change.

**Corrective Action**

If Indication/Condition is not as specified, replace auxiliary fuel management control panel (WP 1448 00).

---

**EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE - Continued****Step****NOTE**

Disregard MFD AUX FUEL legend if it goes on momentarily.

64. **INBD** Place auxiliary fuel management control panel MANUAL XFR - RIGHT switch to ON until CDU TOTAL FUEL indication increases by 20 pounds. ■

**Indication/Condition**

- Elapsed time required to increase indication by 20 pounds shall be less than 1-1/2 minutes.
- Auxiliary fuel management control panel EXTERNAL RIGHT NO FLOW indicator shall momentarily go on, then go off.
- Remaining auxiliary fuel management control panel EXTERNAL indicators shall remain off.
- Auxiliary fuel management control panel digital display indication shall decrease.

**Corrective Action**

If CDU TOTAL FUEL indication does not increase, and/or Indication/Conditions are not as specified, go to WITH MANUAL MODE SELECTED, CDU TOTAL FUEL INDICATION DOES NOT INCREASE AND/OR RESULT IS NOT AS SPECIFIED WHEN ATTEMPTING TO TRANSFER FUEL FROM RIGHT INBOARD FUEL TANK, in this work package.

**Step**

65. **INBD** Place auxiliary fuel management control panel MANUAL XFR - RIGHT switch to OFF. ■

**NOTE**

Disregard MFD AUX FUEL legend if it goes on momentarily.

66. **INBD** Place auxiliary fuel management control panel MANUAL XFR - LEFT switch to ON until CDU TOTAL FUEL indication increases by 20 pounds. ■

**Indication/Condition**

- Elapsed time required to increase indication by 20 pounds shall be less than 1-1/2 minutes.
- Control panel EXTERNAL LEFT NO FLOW indicator shall momentarily go on, then go off.
- Remaining auxiliary fuel management control panel EXTERNAL indicators shall remain off.
- Auxiliary fuel management control panel digital display indication shall decrease.

**Corrective Action**

If Indication/Conditions are not as specified, go to WITH MANUAL MODE SELECTED, CDU TOTAL FUEL INDICATION DOES NOT INCREASE AND/OR RESULT IS NOT AS SPECIFIED WHEN ATTEMPTING TO TRANSFER FUEL FROM LEFT INBOARD FUEL TANK, in this work package.

**Step**

67. **INBD** Place auxiliary fuel management control panel MANUAL XFR - LEFT switch to OFF. ■
68. **OUTBD** Place auxiliary fuel management control panel TANKS and AUX FUEL QTY switches to OUTBD. ■
69. **OUTBD Record** CDU TOTAL FUEL indication. ■
70. **OUTBD Record** auxiliary fuel management control panel digital display indication. ■

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**EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE - Continued**

71. **OUTBD** Place auxiliary fuel management control panel MANUAL XFR - LEFT and RIGHT switches to ON. ■
72. **OUTBD** When CDU TOTAL FUEL indication is 100 pounds more than value recorded in step 70., place auxiliary fuel management control panel MODE switch to OFF. ■
73. **OUTBD** Place auxiliary fuel management control panel MANUAL XFR - LEFT and RIGHT switches to OFF. ■
74. **OUTBD Record** auxiliary fuel management control panel digital display indication. ■
75. **OUTBD** Subtract recorded indication of step 74. from recorded indication of step 70. ■

**Indication/Condition**

The difference shall be between 90 and 110 pounds.

**Corrective Action**

If Indication/Condition is not as specified, go to FUEL TRANSFER TIME AND/OR GAGING ACCURACY IS NOT AS SPECIFIED, in this work package.

**Step**

76. Turn off electrical power.
77. Connect locally-made fuel quantity simulator test set (simulator) power cord to UTIL RCPT +28V (J258) in cabin ceiling.
78. Disconnect P905 from J905.
79. Connect simulator J905 to helicopter P905.
80. Turn simulator TANK 1 and TANK 2 controls fully counterclockwise.
81. Connect No. 1 digital multimeter to simulator TANK 1 OUTPUT.
82. Connect No. 2 digital multimeter to simulator TANK 2 OUTPUT.
83. Turn on electrical power.
84. Turn simulator Tank 1 control to obtain 6.0 vdc indication on No. 1 multimeter.
85. Turn simulator Tank 2 control to obtain 6.0 vdc indication on No. 2 multimeter.
86. Turn auxiliary fuel management control panel AUX FUEL QTY switch to INBD.
87. Place auxiliary fuel management control panel INCR-DECR switch to INCR to obtain display of 6000 pounds.
88. Turn auxiliary fuel management control panel AUX FUEL QTY switch to OUTBD.
89. Place auxiliary fuel management control panel INCR-DECR switch to INCR to obtain display of 1500 pounds.
90. Place auxiliary fuel management control panel MODE switch to AUTO.

**Indication/Condition**

Digital display indication shall not change.

**Corrective Action**

If Indication/Condition is not as specified, replace auxiliary fuel management control panel (WP 1448 00).



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**EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE - Continued****Step****NOTE**

- Read steps 91. thru 94. completely before continuing.
- Do steps 91. thru 94. as quickly as possible.

91. Slowly turn simulator TANK 1 control counter-clockwise until auxiliary fuel management control panel digital display indication begins to decrease.
92. **Record** No. 1 multimeter indication.
93. Slowly turn simulator TANK 1 control clockwise until auxiliary fuel management control panel digital display indication stops changing.
94. **Record** No. 1 multimeter indication.

**Indication/Condition**

- a. Recorded indication of step 92. shall be between 4.65 and 4.85 vdc.
- b. Recorded indication of step 94. shall be between 4.90 and 5.10 vdc.
- c. EXTERNAL RIGHT NO FLOW indicator on auxiliary fuel management control panel shall remain off during fuel transfer.
- d. EXTERNAL LEFT NO FLOW indicator on auxiliary fuel management control panel shall remain off during fuel transfer.

**Corrective Action**

1. If Indication/Conditions a. and/or b. are not as specified, go to AUTO MODE FUEL TRANSFER LIMITS ARE NOT AS SPECIFIED, in this work package.
2. If Indication/Condition c. is not as specified, go to AUXILIARY FUEL MANAGEMENT CONTROL PANEL EXTERNAL RIGHT NO FLOW INDICATOR REMAINS ON DURING FUEL TRANSFER, in this work package.
3. If Indication/Condition d. is not as specified, go to AUXILIARY FUEL MANAGEMENT CONTROL PANEL EXTERNAL LEFT NO FLOW INDICATOR REMAINS ON DURING FUEL TRANSFER, in this work package.

**Step**

95. Turn simulator TANK 1 control clockwise to obtain 6.0 vdc indication on No. 1 multimeter.

**NOTE**

- Read steps 96. thru 99. completely before continuing.
- Do. steps 96. thru 99. as quickly as possible.

96. Slowly turn simulator TANK 2 control counter-clockwise until auxiliary fuel management control panel digital display indication begins to decrease.
97. **Record** No. 2 multimeter indication.
98. Slowly turn simulator TANK 2 control clockwise until auxiliary fuel management control panel digital display indication stops changing.
99. **Record** No. 2 multimeter indication.

---

**EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE - Continued****Indication/Condition**

- a. Recorded indication of step 97. shall be between 4.65 and 4.85 vdc.
- b. Recorded indication of step 99. shall be between 4.90 and 5.10 vdc.
- c. EXTERNAL RIGHT NO FLOW indicator on auxiliary fuel management control panel shall remain off during fuel transfer.
- d. EXTERNAL LEFT NO FLOW indicator on auxiliary fuel management control panel shall remain off during fuel transfer.

**Corrective Action**

1. If Indication/Conditions a. and/or b. are not as specified, go to AUTO MODE FUEL TRANSFER LIMITS ARE NOT AS SPECIFIED, in this work package.
2. If Indication/Condition c. is not as specified, go to AUXILIARY FUEL MANAGEMENT CONTROL PANEL EXTERNAL RIGHT NO FLOW INDICATOR REMAINS ON DURING FUEL TRANSFER, in this work package.
3. If Indication/Condition d. is not as specified, go to AUXILIARY FUEL MANAGEMENT CONTROL PANEL EXTERNAL LEFT NO FLOW INDICATOR REMAINS ON DURING FUEL TRANSFER, in this work package.

**Step**

100. Turn simulator TANK 2 control to obtain 6.0 vdc indication on No. 2 multimeter.
101. Place auxiliary fuel management control panel MODE switch to AUTO.
102. Place auxiliary fuel management control panel INCR-DECR switch to either position to obtain digital display indication of 1000 pounds.
103. Place auxiliary fuel management control panel MODE switch to AUTO.

**Indication/Condition**

Digital display indication shall not change.

**Corrective Action**

If Indication/Condition is not as specified, replace auxiliary fuel management control panel (WP 1448 00).

**Step****NOTE**

- Read steps 104. thru 107. completely before continuing.
- Do steps 104. thru 107. as quickly as possible.

104. Slowly turn simulator TANK 1 control counter-clockwise until auxiliary fuel management control panel digital display indication begins to decrease.
105. **Record** No. 1 multimeter indication.
106. Slowly turn simulator TANK 1 control clockwise until auxiliary fuel management control panel digital display indication stops changing.
107. **Record** No. 1 multimeter indication.

---

**EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE - Continued****Indication/Condition**

- a. Recorded indication of step 105. shall be between 3.65 and 3.85 vdc.
- b. Recorded indication of step 107. shall be between 4.90 and 5.10 vdc.

**Corrective Action**

If Indication/Conditions are not as specified, replace auxiliary fuel management control panel (WP 1448 00).

**Step**

108. Turn simulator TANK 1 control clockwise to obtain 6.0 vdc No. 1 multimeter indication.

**NOTE**

- Read steps 109. thru 112. completely before continuing.
- Do steps 109. thru 112. as quickly as possible.

109. Slowly turn simulator TANK 2 control counter-clockwise until auxiliary fuel management control panel digital display indication begins to decrease.

110. **Record** No. 2 multimeter indication.

111. Slowly turn simulator TANK 2 control clockwise until auxiliary fuel management control panel digital display indication stops changing.

112. **Record** No. 2 multimeter indication.

**Indication/Condition**

- a. Recorded indication of step 110. shall be between 3.65 and 3.85 vdc.
- b. Recorded indication of step 112. shall be between 4.90 and 5.10 vdc.

**Corrective Action**

If Indication/Conditions are not as specified, replace auxiliary fuel management control panel (WP 1448 00).

**Step**

113. Turn simulator TANK 2 control clockwise to obtain 6.0 vdc indication on No. 2 multimeter.

114. Place auxiliary fuel management control panel MODE switch to OFF.

115. Turn auxiliary fuel management control panel AUX FUEL QTY switch to INBD.

116. Place auxiliary fuel management control panel INCR-DECR switch to either position to obtain digital display indication of 2000 pounds.

117. Turn auxiliary fuel management control panel AUX FUEL QTY switch to OUTBD.

118. Place auxiliary fuel management control panel MODE switch to AUTO.

**Indication/Condition**

Digital display indication shall not change.

**Corrective Action**

If Indication/Condition is not as specified, replace auxiliary fuel management control panel (WP 1448 00).

---

**EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE - Continued****Step****NOTE**

- Read steps 119. thru 122. completely before continuing.
- Do steps 119. thru 122. as quickly as possible.

119. Slowly turn simulator TANK 1 control counter-clockwise until auxiliary fuel management control panel digital display indication begins to decrease.
120. **Record** No. 1 multimeter indication.
121. Slowly turn simulator TANK 1 control clockwise until auxiliary fuel management control panel digital display indication stops changing.
122. **Record** No. 1 multimeter indication.

**Indication/Condition**

- a. Recorded indication of step 120. shall be between 2.90 and 3.10 vdc.
- b. Recorded indication of step 122. shall be between 4.90 and 5.10 vdc.

**Corrective Action**

If Indication/Conditions are not as specified, replace auxiliary fuel management control panel (WP 1448 00).

**Step**

123. Slowly turn simulator TANK 1 control to obtain 6.0 vdc indication on No. 1 multimeter.

**NOTE**

- Read steps 124. thru 127. completely before continuing.
- Do steps 124. thru 127. as quickly as possible.

124. Slowly turn simulator TANK 2 control counter-clockwise until auxiliary fuel management control panel digital display indication begins to decrease.
125. **Record** No. 2 multimeter indication.
126. Slowly turn simulator TANK 2 control clockwise until auxiliary fuel management control panel digital display indication stops changing.
127. **Record** No. 2 multimeter indication.

**Indication/Condition**

- a. Recorded indication of step 125. shall be between 2.90 and 3.10 vdc.
- b. Recorded indication of step 127. shall be between 4.90 and 5.10 vdc.

**Corrective Action**

If Indication/Conditions are not as specified, replace auxiliary fuel management control panel (WP 1448 00).

**Step**

128. Turn simulator TANK 2 control to obtain 2.70 vdc indication on No. 2 multimeter.
129. Place auxiliary fuel management control panel MODE switch to OFF.

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**EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE - Continued**

130. **INBD** Place auxiliary fuel management control panel TANKS switch to INBD. ■
131. **INBD** Place auxiliary fuel management control panel MODE switch to AUTO until EXTERNAL - RIGHT INBD EMPTY and EXTERNAL - LEFT INBD EMPTY indicators go on. ■

**Indication/Condition**

- EXTERNAL - RIGHT INBD EMPTY and EXTERNAL - LEFT INBD EMPTY indicators shall go on.
- MFD AUX FUEL legend shall go immediately after either INBD EMPTY indicator goes on.
- Auxiliary fuel management control panel EXTERNAL - RIGHT INBD EMPTY and EXTERNAL - LEFT INBD EMPTY indicators shall go on immediately after the corresponding INBD EMPTY indicator goes on.

**Corrective Action**

- If Indication/Condition a. is not as specified, go to AUXILIARY FUEL MANAGEMENT CONTROL PANEL EXTERNAL RIGHT INBD EMPTY AND/OR EXTERNAL LEFT INBD EMPTY INDICATORS DO NOT GO ON WITH INADEQUATE FUEL IN TANK, in this work package.
- If Indication/Conditions b. and/or c. is not as specified, go to MFD AUX FUEL LEGEND AND/OR AUXILIARY FUEL MANAGEMENT CONTROL PANEL EXTERNAL RIGHT NO FLOW AND EXTERNAL LEFT NO FLOW INDICATORS DO NOT GO ON, in this work package.

**Step**

132. **INBD** Turn simulator TANK 2 control to obtain 6.00 vdc indication on No. 2 multimeter. ■
133. **INBD** Turn simulator TANK 1 control to obtain 6.00 vdc indication on No. 1 multimeter. ■
134. **INBD** Place auxiliary fuel management control panel MODE switch to MANUAL. ■
135. **INBD** Place auxiliary fuel management control panel MANUAL XFR - LEFT and RIGHT switches to ON. ■

**Indication/Condition**

- Auxiliary fuel management control panel EXTERNAL - RIGHT NO FLOW and EXTERNAL - LEFT NO FLOW indicators shall go off, then go on within 5 minutes.
- Auxiliary fuel management control panel digital display indication shall not change when EXTERNAL - RIGHT NO FLOW and EXTERNAL - LEFT NO FLOW indicators are on.

**Corrective Action**

If Indication/Conditions are not as specified, replace corresponding fuel tank (WP 1465 00).

**Step**

136. **INBD** Momentarily press auxiliary fuel management control panel STATUS pushbutton. ■

**Indication/Condition**

- Auxiliary fuel management control panel EXTERNAL - INBD EMPTY and EXTERNAL - OUTBD EMPTY indicators shall go off.
- MFD AUX FUEL legend shall go off.

**Corrective Action**

- If Indication/Condition a. is not as specified, go to AUXILIARY FUEL MANAGEMENT CONTROL PANEL EXTERNAL INBD EMPTY AND/OR EXTERNAL OUTBD EMPTY INDICATORS ARE NOT OFF, in this work package.

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**EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE - Continued**

2. If Indication/Condition b. is not as specified, go to WHEN AUXILIARY FUEL MANAGEMENT CONTROL PANEL STATUS PUSHBUTTON IS PRESSED AND RELEASED, INDICATORS AND/OR PILOT'S/COPILOT'S MFD AUX FUEL LEGEND ARE NOT OFF, in this work package.

**Step**

137. **INBD** Turn simulator TANK 2 control to obtain 2.70 vdc indication on No. 2 multimeter. ■
138. **OUTBD** Place auxiliary fuel management control panel MODE switch to OFF. ■

**Indication/Condition**

- a. EXTERNAL - RIGHT OUTBD EMPTY and EXTERNAL - LEFT OUTBD EMPTY indicators shall go on.
- b. MFD AUX FUEL legend shall go on immediately after either EXTERNAL - RIGHT OUTBD EMPTY or EXTERNAL - LEFT OUTBD EMPTY indicator goes on.
- c. Auxiliary fuel management control panel EXTERNAL - RIGHT OUTBD EMPTY and EXTERNAL - LEFT OUTBD EMPTY indicators shall go on immediately after corresponding tank OUTBD EMPTY indicator goes on.

**Corrective Action**

1. If Indication/Condition a. is not as specified, go to WHEN AUXILIARY FUEL MANAGEMENT CONTROL PANEL STATUS PUSHBUTTON IS PRESSED AND RELEASED, INDICATORS AND/OR PILOT'S/COPILOT'S MFD AUX FUEL LEGEND ARE NOT OFF, in this work package.
2. If Indication/Conditions b. and/or c. are not as specified, go to MFD AUX FUEL LEGEND AND/OR AUXILIARY FUEL MANAGEMENT CONTROL PANEL EXTERNAL RIGHT NO FLOW AND EXTERNAL LEFT NO FLOW INDICATORS DO NOT GO ON, in this work package.

**Step**

139. **OUTBD** Press auxiliary fuel management control panel STATUS pushbutton. ■

**Indication/Condition**

- a. Auxiliary fuel management control panel EXTERNAL - INBD EMPTY and EXTERNAL - OUTBD EMPTY indicators shall go off.
- b. MFD AUX FUEL legend shall go off.

**Corrective Action**

1. If Indication/Condition a. is not as specified, go to AUXILIARY FUEL MANAGEMENT CONTROL PANEL EXTERNAL INBD EMPTY AND/OR EXTERNAL OUTBD EMPTY INDICATORS ARE NOT OFF, in this work package.
2. If Indication/Condition b. is not as specified, go to WHEN AUXILIARY FUEL MANAGEMENT CONTROL PANEL STATUS PUSHBUTTON IS PRESSED AND RELEASED, INDICATORS AND/OR PILOT'S/COPILOT'S MFD AUX FUEL LEGEND ARE NOT OFF, in this work package.

**Step**

140. **OUTBD** Place auxiliary fuel management control panel MODE switch to MANUAL. ■
141. **OUTBD** Place auxiliary fuel management control panel MANUAL XFR - LEFT and MANUAL XFR - RIGHT switches to ON. ■

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**EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE - Continued****Indication/Condition**

- a. Auxiliary fuel management control panel EXTERNAL - RIGHT NO FLOW and EXTERNAL - LEFT NO FLOW indicators shall go off, then come back on within 5 minutes.
- b. Auxiliary fuel management control panel digital display indication shall not change when the EXTERNAL - RIGHT NO FLOW and EXTERNAL - LEFT NO FLOW indicators are on.

**Corrective Action**

If Indication/Conditions are not as specified, replace corresponding external fuel tank (WP 1465 00).

**Step**

142. Place auxiliary fuel management control panel MODE, MANUAL XFR - LEFT and MANUAL XFR - RIGHT, PRESS - INBD, and PRESS OUTBD switches to OFF.

**Indication/Condition**

All four ESSS bleed-air regulator valves shall be venting air.

**Corrective Action**

1. If Indication/Condition is not as specified, disconnect connector from bleed-air regulator.
  - a. If bleed-air valve vents air, replace auxiliary fuel management control panel (WP 1448 00).
  - b. If bleed-air valve does not vent air, clean valve, then do steps **CU** 31. **■** , **OUTBD** 33. **■** , **INBD** 34. **■** , and 142.
  - c. If trouble remains, replace bleed-air regulator valve (WP 1439 00).

**Step**

143. Pull out NO. 2 XFER CONTROL circuit breaker on mission readiness circuit breaker panel.
144. Remove overflow sensor from vent tube elbow in rear of main fuel tanks (WP 1457 00).
145. Disconnect P376 from J376.
146. Push in NO. 2 XFER CONTROL circuit breaker.

**Indication/Condition**

Auxiliary fuel management control panel VENT SENSOR FAIL indicator shall go on.

**Corrective Action**

If Indication/Condition is not as specified, replace auxiliary fuel management control panel (WP 1448 00).

**Step**

147. Pull out NO. 2 XFER CONTROL circuit breaker.
148. Connect P376 to J376.
149. Immerse thermistor portion of overflow sensor in container of fuel.
150. Push in NO. 2 XFER CONTROL circuit breaker.

**Indication/Condition**

Auxiliary fuel management control panel VENT SENSOR OVFL indicator shall go on.

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**EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE - Continued****Corrective Action**

If Indication/Condition is not as specified, go to VENT SENSOR OVFL INDICATOR DOES NOT GO ON, in this work package.

**Step**

151. Install overflow sensor (WP 1457 00).
152. Turn off electrical power.
153. **CU** Turn off air source. ■
154. Allow bleed-air regulator valves to vent air to ambient pressure.
155. **CU** Disconnect air compressor, hoses, and locally-made pneumatic adapter. ■
156. **CU** Connect pneumatic hose to outboard side of helicopter bleed-air QD. ■
157. **CU** **TORQUE NUTS TO 135 - 150 INCH-POUNDS.** ■
158. Disconnect test equipment.
159. Connect helicopter electrical connectors.
160. Install explosive cartridges.
161. Fasten helicopter fairings (QD).

**Indication/Condition**

Helicopter fairings shall be fastened.

**Corrective Action**

None Required

**WHEN AUXILIARY FUEL MANAGEMENT CONTROL PANEL TEST PUSHBUTTON IS PRESSED, ALL INDICATORS ARE NOT ON AND/OR DIGITAL DISPLAY DOES NOT INDICATE 8888****SYMPTOM**

When auxiliary fuel management control panel TEST pushbutton is pressed, all indicators are not on and/or digital display does not indicate 8888.

**MALFUNCTION**

When auxiliary fuel management control panel TEST pushbutton is pressed, all indicators are not on and/or digital display does not indicate 8888.

**CORRECTIVE ACTION****1. Does digital display indicate 8888?**

- a. If 8888 is displayed, replace applicable auxiliary fuel management control panel indicator lamp (WP 1451 00). Go to **6**.
- b. If trouble remains, replace auxiliary fuel management control panel (WP 1448 00). Go to **6**.
- c. If 8888 is not displayed, go to **2**.

**2. Do some indicators go on?**

- a. If indicators go on, replace auxiliary fuel management control panel digital display segment (WP 1450 00). Go to **6**.



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**WHEN AUXILIARY FUEL MANAGEMENT CONTROL PANEL TEST PUSHBUTTON IS PRESSED, ALL INDICATORS ARE NOT ON AND/OR DIGITAL DISPLAY DOES NOT INDICATE 8888** - Continued

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- b. If trouble remains, replace auxiliary fuel management control panel (WP 1448 00). Go to **6**.
- c. If indicators do not go on, go to **3**.
- 3. Check for 28 vdc between P360-W and ground.**
  - a. If voltage is as specified, replace auxiliary fuel management control panel (WP 1448 00). Go to **6**.
  - b. If voltage is not as specified, go to **4**.
- 4. Check for 28 vdc between NO. 2 XFER CONTROL circuit breaker terminal 1 and ground.**
  - a. If voltage is as specified, go to **5**.
  - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **6**.
- 5. Check for 28 vdc between J346-35 and ground.**
  - a. If voltage is as specified, replace fuel system main wiring harness (WP 1453 00). Go to **6**.
  - b. If voltage is not as specified, repair/replace wiring between J346-35 and circuit breaker terminal 1 (WP 1747 00). Go to **6**.
- 6. Procedure completed.**

**DURING SELF-TEST, AUXILIARY FUEL MANAGEMENT CONTROL PANEL DIGITAL DISPLAY INDICATES E02 INSTEAD OF GOOD****SYMPTOM**

During self-test, auxiliary fuel management control panel digital display indicates E02 instead of GOOD.

**MALFUNCTION**

During self-test, auxiliary fuel management control panel digital display indicates E02 instead of GOOD.

**CORRECTIVE ACTION**

- 1. Check for 115 vac between P360-f and ground.**
  - a. If voltage is as specified, go to **2**.
  - b. If voltage is not as specified, go to **4**.
- 2. Check continuity between:**
  - **P361-S and J524-2**
  - **P361-U and J524-1**
  - a. If continuity is present, replace fuel temperature transducer (WP 1437 00). Go to **6**.
  - b. If continuity is not present, go to **3**.
- 3. Check continuity between:**
  - **P346-20 and J361-U**
  - **P346-21 and J361-S**
  - a. If continuity is present, repair/replace wiring between (WP 1747 00), then go to **6**:
    - **J346-20 and J524-2**

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**DURING SELF-TEST, AUXILIARY FUEL MANAGEMENT CONTROL PANEL DIGITAL DISPLAY INDICATES E02 INSTEAD OF GOOD - Continued**

- J346-21 and J524-1
- b. If continuity is not present, replace fuel system main wiring harness (WP 1453 00). Go to **6**.
- 4. Check for 115 vac between AUX FUEL QTY circuit breaker terminal 1 and ground.**
  - a. If voltage is as specified, go to **5**.
  - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **6**.
- 5. Check for 115 vac between J346-38 and ground.**
  - a. If voltage is as specified, replace fuel system main wiring harness (WP 1453 00). Go to **6**.
  - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **6**.
- 6. Procedure completed.**

**DURING SELF-TEST, AUXILIARY FUEL MANAGEMENT CONTROL PANEL DIGITAL DISPLAY INDICATES E03 INSTEAD OF GOOD****SYMPTOM**

During self-test, auxiliary fuel management control panel digital display indicates E03 instead of GOOD.

**MALFUNCTION**

During self-test, auxiliary fuel management control panel digital display indicates E03 instead of GOOD.

**CORRECTIVE ACTION**

- 1. Check continuity between:**
  - **P361-X and P523-A**
  - **P361-Y and P523-B**
  - a. If continuity is present, go to **2**.
  - b. If continuity is not present, go to **3**.
- 2. Check for 2000 ohms to 3000 ohms between fuel flow transmitter connector pins A and B.**
  - a. If resistance is as specified, replace auxiliary fuel management control panel (WP 1448 00). Go to **4**.
  - b. If resistance is not as specified, replace fuel flow transmitter (WP 1437 00). Go to **4**.
- 3. Check continuity between:**
  - **P361-X and P346-23**
  - **P361-Y and P346-24**
  - a. If continuity is present, repair/replace wiring or components (WP 1747 00) then go to **4**:
    - P523-A and J346-23
    - P533-B and J346-24
  - b. If continuity is not present, replace fuel system main wiring harness (WP 1453 00). Go to **4**.
- 4. Procedure completed.**

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**AUXILIARY FUEL MANAGEMENT CONTROL PANEL EMPTY INDICATOR IS ON WITH ADEQUATE FUEL IN FUEL TANK****SYMPTOM**

Auxiliary fuel management control panel EMPTY indicator is on with adequate fuel in fuel tank.

**MALFUNCTION**

Auxiliary fuel management control panel EMPTY indicator is on with adequate fuel in fuel tank.

**CORRECTIVE ACTION****1. Check for infinite resistance between these pins on applicable fuel tank connector:**

- 3 and 5
- 3 and 4

- a. If resistance is as specified, go to **3**.
- b. If resistance is not as specified, go to **2**.

**2. Check continuity between these points:**

- P1-3 (Left outboard) and pin F
- P1-4 (Left outboard) and pin B
- P1-5 (Left outboard) and pin B
- P1-3 (Left inboard) and pin F
- P1-4 (Left inboard) and pin B
- P1-5 (Left inboard) and pin B
- P1-3 (Right outboard) and pin F
- P1-4 (Right outboard) and pin B
- P1-5 (Right outboard) and pin B
- P1-3 (Right inboard) and pin F
- P1-4 (Right inboard) and pin B
- P1-5 (Right inboard) and pin B

- a. If continuity is present, remove/replace low level sensor assembly (WP 1476 00). Go to **4**.
- b. If trouble remains, replace/send fuel tank to depot (WP 1465 00). Go to **4**.
- c. If continuity is not present, replace electrical interface cable (WP 1469 00). Go to **4**.

**3. Check continuity between:**

- J1-3 (Left outboard) and J833-8
- J1-4 (Left outboard) and J833-9
- J1-5 (Left outboard) and J833-10
- J1-3 (Left inboard) and J833-3

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**AUXILIARY FUEL MANAGEMENT CONTROL PANEL EMPTY INDICATOR IS ON WITH ADEQUATE FUEL IN FUEL TANK - Continued**

- J1-4 (Left inboard) and J833-4
- J1-5 (Left inboard) and J833-5
- J1-3 (Right outboard) and J834-8
- J1-4 (Right outboard) and J834-9
- J1-5 (Right outboard) and J834-10
- J1-3 (Right inboard) and J834-3
- J1-4 (Right inboard) and J834-4
- J1-5 (Right inboard) and J834-5

- a. If continuity is present, replace fuel system main wiring harness (WP 1453 00). Go to **4**.
- b. If continuity is not present, replace front electrical harness (WP 1455 00). Go to **4**.

**4. Procedure completed.****WHEN AUXILIARY FUEL MANAGEMENT CONTROL PANEL STATUS PUSHBUTTON IS PRESSED AND RELEASED, INDICATORS AND/OR PILOT'S/COPILOT'S MFD AUX FUEL LEGEND ARE NOT OFF****SYMPTOM**

When auxiliary fuel management control panel STATUS pushbutton is pressed and released, indicators and/or pilot's/copilot's MFD AUX FUEL legend are not off.

**MALFUNCTION**

When auxiliary fuel management control panel STATUS pushbutton is pressed and released, indicators and/or pilot's/copilot's MFD AUX FUEL legend are not off.

**CORRECTIVE ACTION****1. Are all auxiliary fuel management control panel indicators off?**

- a. If all indicators are off, go to **2**.
- b. If all indicators are not off, repair/replace auxiliary fuel management control panel (WP 1448 00). Go to **5**.

**2. Check for 28 vdc between:**

- P1019R-37 and ground
- P1020R-37 and ground

- a. If voltage is as specified, replace copilot's or pilot's MFD (WP 0799 00). Go to **5**.
- b. If voltage is not as specified, go to **3**.

**3. With P346 and P360 disconnected, check continuity between P360-X and ground.**

- a. If continuity is present, replace fuel system main wiring harness (WP 1453 00). Go to **5**.
- b. If continuity is not present, go to **4**.

**4. With P346, P1019R, and P1020R disconnected, check continuity between J346-10 and ground.**

- a. If continuity is present, repair/replace wiring (WP 1747 00). Go to **5**.

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**WHEN AUXILIARY FUEL MANAGEMENT CONTROL PANEL STATUS PUSHBUTTON IS PRESSED AND RELEASED, INDICATORS AND/OR PILOT'S/COPILOT'S MFD AUX FUEL LEGEND ARE NOT OFF -**

Continued

- b. If continuity is not present, replace auxiliary fuel management control panel (WP 1448 00). Go to **5**.

**5. Procedure completed.****WITH AUXILIARY FUEL MANAGEMENT CONTROL PANEL AUX FUEL QTY SWITCH TO CAL DIGITAL DISPLAY INDICATION CANNOT BE CHANGED****SYMPTOM**

With auxiliary fuel management control panel AUX FUEL QTY switch to CAL digital display indication cannot be changed.

**MALFUNCTION**

With auxiliary fuel management control panel AUX FUEL QTY switch to CAL digital display indication cannot be changed.

**CORRECTIVE ACTION****1. Check continuity between P360-r and P360-n.**

- a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to **4**.
- b. If continuity is not present, go to **2**.

**2. Check continuity between:**

- **P360-r and P902-C**
- **P360-n and P902-D**

- a. If continuity is present, replace left relay panel (WP 0825 00). Go to **4**.
- b. If continuity is not present, go to **3**.

**3. Check continuity between:**

- **P360-r and P346-3**
- **P360-n and P346-2**

- a. If continuity is present, repair/replace wiring, as required, between (WP 1747 00), then go to **4**:
  - J346-2 and P902-D
  - J346-3 and P902-C

- b. If continuity is not present, replace fuel system main wiring harness (WP 1453 00). Go to **4**.

**4. Procedure completed.**

**DIGITAL DISPLAY INDICATION CHANGES WITH DRAG BEAM SWITCH IN IN-FLIGHT POSITION****SYMPTOM**

Digital display indication changes with drag beam switch in in-flight position.

**MALFUNCTION**

Digital display indication changes with drag beam switch in in-flight position.

**CORRECTIVE ACTION**

- 1. Remove locally-made drag beam wedge from left drag beam switch. Go to 2.**
- 2. Check continuity between P360-r and P360-n.**
  - a. If continuity is present, go to **3**.
  - b. If continuity is not present, replace auxiliary fuel management control panel (WP 1448 00). Go to **5**.
- 3. Check continuity between:**
  - **J346-2 and P902-D**
  - **J346-3 and P902-C**
  - a. If continuity is present, repair/replace wiring (WP 1747 00). Go to **5**.
  - b. If continuity is not present, go to **4**.
- 4. Check continuity between:**
  - **P346-2 and P360-n**
  - **P346-3 and P360-r**
  - a. If continuity is present, replace fuel system main wiring harness (WP 1453 00). Go to **5**.
  - b. If continuity is not present, troubleshoot hydraulic systems (WP 0086 00). Go to **5**.
- 5. Procedure completed.**

**CDU TOTAL FUEL INDICATION INCREASES WHEN FUEL TRANSFER IS NOT ACTUATED****SYMPTOM**

CDU TOTAL FUEL indication increases when fuel transfer is not actuated.

**MALFUNCTION**

CDU TOTAL FUEL indication increases when fuel transfer is not actuated.

**CORRECTIVE ACTION**

- 1. Is the manual lever of all four shutoff gate valves in the fully closed position?**
  - a. If manual levers are in the fully closed position, replace auxiliary fuel management control panel (WP 1448 00). Go to **12**.
  - b. If manual levers are not in the fully closed position, go to **2**.
- 2. For any valve not in fully closed position, check for 28 vdc between**
  - **P840-D and P840-E (left outboard)**

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CDU TOTAL FUEL INDICATION INCREASES WHEN FUEL TRANSFER IS NOT ACTUATED - Continued

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- P809-D and P809-E (left inboard)
  - P838-D and P838-E (right outboard)
  - P810-D and P810-E (right inboard)
- a. If voltage is as specified, go to **3**.
  - b. If voltage is not as specified, go to **4**.
- 3. Check for 0 vdc between:**
- P840-A and P840-E (left outboard)
  - P809-A and P809-E (left inboard)
  - P838-A and P838-E (right outboard)
  - P810-A and P810-E (right inboard)
- a. If voltage is as specified, replace shutoff gate valve (WP 1440 00). Go to **12**.
  - b. If voltage is not as specified, replace auxiliary fuel management control panel (WP 1448 00). Go to **12**.
- 4. For left outboard valve, check continuity between:**
- P840-A and P360-F
  - P840-D and P360-E
  - P840-E and ground
- a. If continuity is present, go to **6**.
  - b. If continuity is not present, go to **5**.
- 5. Check continuity between:**
- P805-8 and P360-E
  - P805-9 and P360-F
  - P805-10 and P364-34
- a. If continuity is present, replace fuel system rear electrical harness (WP 1454 00). Go to **12**.
  - b. If continuity is not present, replace fuel system main wiring harness (WP 1453 00). Go to **12**.
- 6. For left inboard valve, check continuity between:**
- P809-A and P360-Y
  - P809-D and P360-Z
  - P809-E and ground
- a. If continuity is present, go to **8**.
  - b. If continuity is not present, go to **7**.
- 7. Check continuity between:**
- P805-3 and P360-Y

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CDU TOTAL FUEL INDICATION INCREASES WHEN FUEL TRANSFER IS NOT ACTUATED - Continued

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- **P805-4 and P360-Z**
  - **P805-5 and P346-34**
  - a. If continuity is present, replace fuel system rear electrical harness (WP 1454 00). Go to **12**.
  - b. If continuity is not present, replace fuel system main wiring harness (WP 1453 00). Go to **12**.
- 8. For right inboard valve, check continuity between:**
- **P810-A and P360-j**
  - **P810-D and P360-i**
  - **P810-E and ground**
  - a. If continuity is present, go to **10**.
  - b. If continuity is not present, go to **9**.
- 9. Check continuity between:**
- **P804-3 and P360-j**
  - **P804-4 and P360-i**
  - **P804-5 and P346-9**
  - a. If continuity is present, replace fuel system rear electrical harness (WP 1454 00). Go to **12**.
  - b. If continuity is not present, replace fuel system main wiring harness (WP 1453 00). Go to **12**.
- 10. For right outboard valve, check continuity between:**
- **P838-A and P360-j**
  - **P838-D and P360-i**
  - **P838-E and ground**
  - a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to **12**.
  - b. If continuity is not present, go to **11**.
- 11. Check continuity between:**
- **P804-8 and P360-g**
  - **P804-9 and P360-P**
  - **P804-10 and P346-9**
  - a. If continuity is present, replace fuel system rear electrical harness (WP 1454 00). Go to **12**.
  - b. If continuity is not present, replace fuel system main wiring harness (WP 1453 00). Go to **12**.
- 12. Procedure completed.**



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**WITH MANUAL MODE SELECTED, CDU TOTAL FUEL INDICATION DOES NOT INCREASE AND/OR RESULT IS NOT AS SPECIFIED WHEN ATTEMPTING TO TRANSFER FUEL FROM RIGHT OUTBOARD FUEL TANK**

**SYMPTOM**

With MANUAL mode selected, CDU TOTAL FUEL indication does not increase and/or Indication/Condition is not as specified when attempting to transfer fuel from right outboard fuel tank.

**MALFUNCTION**

With MANUAL mode selected, CDU TOTAL FUEL indication does not increase and/or Indication/Condition is not as specified when attempting to transfer fuel from right outboard fuel tank.

**CORRECTIVE ACTION**

**1. Does CDU TOTAL FUEL indication increase?**

- a. If CDU TOTAL FUEL indication does increase, go to **2**.
- b. If CDU TOTAL FUEL indication does not increase, go to **6**.

**2. Does auxiliary fuel management control panel digital display indication decrease?**

- a. If auxiliary fuel management control panel digital display indication does decrease, go to **3**.
- b. If auxiliary fuel management control panel digital display indication does not decrease, go to **5**.

**3. Does auxiliary fuel management control panel EXTERNAL RIGHT NO FLOW indicator momentarily go on, then off?**

- a. If indicator momentarily goes on, then off, go to **4**.
- b. If indicator remains on, go to AUXILIARY FUEL MANAGEMENT CONTROL PANEL EXTERNAL RIGHT NO FLOW INDICATOR REMAINS ON DURING FUEL TRANSFER, in this work package.

**4. Do the remaining auxiliary fuel management control panel EXTERNAL indicators remain off?**

- a. If the remaining indicators remain off, go to AUXILIARY FUEL MANAGEMENT CONTROL PANEL EXTERNAL RIGHT NO FLOW INDICATOR REMAINS ON DURING FUEL TRANSFER, in this work package.
- b. If the remaining indicators do not remain off, replace auxiliary fuel management control panel (WP 1448 00). Go to **16**.

**5. With auxiliary fuel management control panel MANUAL XFR RIGHT switch at ON, check for more than 20 mvac between fuel flow transmitter connector pins A and B.**

- a. If voltage is as specified, replace fuel flow transmitter (WP 1438 00). Go to **16**.
- b. If voltage is not as specified, replace auxiliary fuel management control panel (WP 1448 00). Go to **16**.

**6. Does auxiliary fuel management control panel digital display indication decrease?**

- a. If indication decreases, go to **7**.
- b. If indication does not decrease, go to **8**.

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**WITH MANUAL MODE SELECTED, CDU TOTAL FUEL INDICATION DOES NOT INCREASE AND/OR RESULT IS NOT AS SPECIFIED WHEN ATTEMPTING TO TRANSFER FUEL FROM RIGHT OUTBOARD FUEL TANK - Continued**

**7. Does EXTERNAL RIGHT NO FLOW indicator go on?**

- a. If indicator is on, replace auxiliary fuel management control panel (WP 1448 00). Go to **16**.
- b. If indicator is not on, troubleshoot fuel quantity system (WP 0131 00). Go to **16**.

**8. Is manual lever of right outboard fuel shutoff gate valve in full OPEN position?**

- a. If manual lever is in full OPEN position, go to **9**.
- b. If manual lever is not in full OPEN position, go to **12**.

**9. Check for 28 vdc between P837-A and P837-B.**

- a. If voltage is as specified, replace right outboard bleed-air regulator valve (WP 1439 00). Go to **16**.
- b. If voltage is not as specified, go to **10**.

**10. Check continuity between:**

- **P837-A and P360-R**
- **P837-B and ground**

- a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to **16**.
- b. If continuity is not present, go to **11**.

**11. Check continuity between:**

- **P804-6 and P364-4**
- **P804-7 and P360-R**

- a. If continuity is present, replace rear electrical harness (WP 1454 00). Go to **16**.
- b. If continuity is not present, replace fuel system main wiring harness (WP 1453 00). Go to **16**.

**12. Is right outboard bleed air regulator valve continuously venting air?**

- a. If valve is continuously venting air, go to **13**.
- b. If valve is not continuously venting air, replace auxiliary fuel management control panel (WP 1448 00). Go to **16**.

**13. Check for 28 vdc between P360-V and ground.**

- a. If voltage is as specified, replace auxiliary fuel management control panel (WP 1448 00). Go to **16**.
- b. If voltage is not as specified, go to **14**.

**14. Check for 28 vdc between EXT FUEL RH circuit breaker terminal 1 and ground.**

- a. If voltage is as specified, go to **15**.
- b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **16**.

**15. Check for 28 vdc between J346-33 and ground.**

- a. If voltage is as specified, replace fuel system main wiring harness (WP 1453 00). Go to **16**.
- b. If voltage is not as specified, repair/replace wiring between J346-33 and EXT FUEL RH circuit breaker terminal 1 (WP 1747 00). Go to **16**.

**WITH MANUAL MODE SELECTED, CDU TOTAL FUEL INDICATION DOES NOT INCREASE AND/OR RESULT IS NOT AS SPECIFIED WHEN ATTEMPTING TO TRANSFER FUEL FROM RIGHT OUTBOARD FUEL TANK** - Continued

**16. Procedure completed.**

**WITH MANUAL MODE SELECTED, CDU TOTAL FUEL INDICATION DOES NOT INCREASE AND/OR RESULT IS NOT AS SPECIFIED WHEN ATTEMPTING TO TRANSFER FUEL FROM LEFT OUTBOARD FUEL TANK**

**SYMPTOM**

With MANUAL mode selected, CDU TOTAL FUEL indication does not increase and/or Indication/Condition is not as specified when attempting to transfer fuel from left outboard fuel tank.

**MALFUNCTION**

With MANUAL mode selected, CDU TOTAL FUEL indication does not increase and/or Indication/Condition is not as specified when attempting to transfer fuel from left outboard fuel tank.

**CORRECTIVE ACTION**

**1. Does CDU TOTAL FUEL indication increase?**

- a. If CDU TOTAL FUEL indication does increase, go to **2**.
- b. If CDU TOTAL FUEL indication does not increase, go to **6**.

**2. Does auxiliary fuel management control panel digital display indication decrease?**

- a. If auxiliary fuel management control panel digital display indication does decrease, go to **3**.
- b. If auxiliary fuel management control panel digital display indication does not decrease, go to **5**.

**3. Does auxiliary fuel management control panel EXTERNAL LEFT NO FLOW indicator momentarily go on, then off?**

- a. If indicator momentarily goes on, then off, go to **4**.
- b. If indicator remains on, go to AUXILIARY FUEL MANAGEMENT CONTROL PANEL EXTERNAL RIGHT NO FLOW INDICATOR REMAINS ON DURING FUEL TRANSFER, in this work package.

**4. Do the remaining auxiliary fuel management control panel EXTERNAL indicators remain off?**

- a. If the remaining indicators remain off, go to AUXILIARY FUEL MANAGEMENT CONTROL PANEL EXTERNAL RIGHT NO FLOW INDICATOR REMAINS ON DURING FUEL TRANSFER, in this work package.
- b. If the remaining indicators do not remain off, replace auxiliary fuel management control panel (WP 1448 00). Go to **16**.

**5. With auxiliary fuel management control panel MANUAL XFR RIGHT switch at ON, check for more than 20 mvac between fuel flow transmitter connector pins A and B.**

- a. If voltage is as specified, replace fuel flow transmitter (WP 1438 00). Go to **16**.
- b. If voltage is not as specified, replace auxiliary fuel management control panel (WP 1448 00). Go to **16**.

**6. Does auxiliary fuel management control panel digital display indication decrease?**

- a. If indication decreases, go to **7**.

---

**WITH MANUAL MODE SELECTED, CDU TOTAL FUEL INDICATION DOES NOT INCREASE AND/OR RESULT IS NOT AS SPECIFIED WHEN ATTEMPTING TO TRANSFER FUEL FROM LEFT OUTBOARD FUEL TANK** - Continued

- b. If indication does not decrease, go to **8**.

**7. Does EXTERNAL LEFT NO FLOW indicator go on?**

- a. If indicator is on, replace auxiliary fuel management control panel (WP 1448 00). Go to **16**.
- b. If indicator is not on, troubleshoot fuel quantity system (WP 0131 00). Go to **16**.

**8. Is manual lever of left outboard fuel shutoff gate valve in full OPEN position?**

- a. If manual lever is in full OPEN position, go to **9**.
- b. If manual lever is not in full OPEN position, go to **12**.

**9. Check for 28 vdc between P839-A and P839-B.**

- a. If voltage is as specified, replace right outboard bleed-air regulator valve (WP 1439 00). Go to **16**.
- b. If voltage is not as specified, go to **10**.

**10. Check continuity between:**

- **P839-A and P360-G**
- **P839-B and ground**

- a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to **16**.
- b. If continuity is not present, go to **11**.

**11. Check continuity between:**

- **P805-6 and P346-11**
- **P805-7 and P360-G**

- a. If continuity is present, replace rear electrical harness (WP 1454 00). Go to **16**.
- b. If continuity is not present, replace fuel system main wiring harness (WP 1453 00). Go to **16**.

**12. Is left outboard bleed air regulator valve continuously venting air?**

- a. If valve is continuously venting air, go to **13**.
- b. If valve is not continuously venting air, replace auxiliary fuel management control panel (WP 1448 00). Go to **16**.

**13. Check for 28 vdc between P360-C and ground.**

- a. If voltage is as specified, replace auxiliary fuel management control panel (WP 1448 00). Go to **16**.
- b. If voltage is not as specified, go to **14**.

**14. Check for 28 vdc between EXT FUEL LH circuit breaker terminal 1 and ground.**

- a. If voltage is as specified, go to **15**.
- b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **16**.

**15. Check for 28 vdc between J346-45 and ground.**

- a. If voltage is as specified, replace fuel system main wiring harness (WP 1453 00). Go to **16**.

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**WITH MANUAL MODE SELECTED, CDU TOTAL FUEL INDICATION DOES NOT INCREASE AND/OR RESULT IS NOT AS SPECIFIED WHEN ATTEMPTING TO TRANSFER FUEL FROM LEFT OUTBOARD FUEL TANK - Continued**

- b. If voltage is not as specified, repair/replace wiring between J346-33 and EXT FUEL RH circuit breaker terminal 1 (WP 1747 00). Go to **16**.

**16. Procedure completed.****WITH MANUAL MODE SELECTED, CDU TOTAL FUEL INDICATION DOES NOT INCREASE AND/OR RESULT IS NOT AS SPECIFIED WHEN ATTEMPTING TO TRANSFER FUEL FROM RIGHT INBOARD FUEL TANK****SYMPTOM**

With MANUAL mode selected, CDU TOTAL FUEL indication does not increase and/or Indication/Condition is not as specified when attempting to transfer fuel from right inboard fuel tank.

**MALFUNCTION**

With MANUAL mode selected, CDU TOTAL FUEL indication does not increase and/or Indication/Condition is not as specified when attempting to transfer fuel from right inboard fuel tank.

**CORRECTIVE ACTION****1. Does CDU TOTAL FUEL indication increase?**

- a. If CDU TOTAL FUEL indication does increase, go to **2**.
- b. If CDU TOTAL FUEL indication does not increase, go to **6**.

**2. Does auxiliary fuel management control panel digital display indication decrease?**

- a. If auxiliary fuel management control panel digital display indication does decrease, go to **3**.
- b. If auxiliary fuel management control panel digital display indication does not decrease, go to **5**.

**3. Does auxiliary fuel management control panel EXTERNAL RIGHT NO FLOW indicator momentarily go on, then off?**

- a. If indicator momentarily goes on, then off, go to **4**.
- b. If indicator remains on, go to AUXILIARY FUEL MANAGEMENT CONTROL PANEL EXTERNAL RIGHT NO FLOW INDICATOR REMAINS ON DURING FUEL TRANSFER, in this work package.

**4. Do the remaining auxiliary fuel management control panel EXTERNAL indicators remain off?**

- a. If the remaining indicators remain off, go to AUXILIARY FUEL MANAGEMENT CONTROL PANEL EXTERNAL RIGHT NO FLOW INDICATOR REMAINS ON DURING FUEL TRANSFER, in this work package.
- b. If the remaining indicators do not remain off, replace auxiliary fuel management control panel (WP 1448 00). Go to **12**.

**5. With auxiliary fuel management control panel MANUAL XFR RIGHT switch at ON, check for more than 20 mvac between fuel flow transmitter connector pins A and B.**

- a. If voltage is as specified, replace fuel flow transmitter (WP 1438 00). Go to **12**.
- b. If voltage is not as specified, replace auxiliary fuel management control panel (WP 1448 00). Go to **12**.

**WITH MANUAL MODE SELECTED, CDU TOTAL FUEL INDICATION DOES NOT INCREASE AND/OR RESULT IS NOT AS SPECIFIED WHEN ATTEMPTING TO TRANSFER FUEL FROM RIGHT INBOARD FUEL TANK** - Continued

- 6. Does auxiliary fuel management control panel digital display indication decrease?**
  - a. If indication decreases, go to **7**.
  - b. If indication does not decrease, go to **8**.
- 7. Does EXTERNAL RIGHT NO FLOW indicator go on?**
  - a. If indicator is on, replace auxiliary fuel management control panel (WP 1448 00). Go to **12**.
  - b. If indicator is not on, troubleshoot fuel quantity system (WP 0131 00). Go to **12**.
- 8. Is manual lever of right outboard fuel shutoff gate valve in full OPEN position?**
  - a. If manual lever is in full OPEN position, go to **9**.
  - b. If manual lever is not in full OPEN position, go to **12**.
- 9. Check for 28 vdc between P811-A and P811-B.**
  - a. If voltage is as specified, replace right outboard bleed-air regulator valve (WP 1439 00). Go to **12**.
  - b. If voltage is not as specified, go to **10**.
- 10. Check continuity between:**
  - **P811-A and P360-s**
  - **P811-B and ground**
  - a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to **12**.
  - b. If continuity is not present, go to **11**.
- 11. Check continuity between:**
  - **P804-1 and P346-4**
  - **P804-2 and P360-s**
  - a. If continuity is present, replace rear electrical harness (WP 1454 00). Go to **12**.
  - b. If continuity is not present, replace fuel system main wiring harness (WP 1453 00). Go to **12**.
- 12. Procedure completed.**

**WITH MANUAL MODE SELECTED, CDU TOTAL FUEL INDICATION DOES NOT INCREASE AND/OR RESULT IS NOT AS SPECIFIED WHEN ATTEMPTING TO TRANSFER FUEL FROM LEFT INBOARD FUEL TANK**

#### **SYMPTOM**

With MANUAL mode selected, CDU TOTAL FUEL indication does not increase and/or Indication/Condition is not as specified when attempting to transfer fuel from left inboard fuel tank.

#### **MALFUNCTION**

With MANUAL mode selected, CDU TOTAL FUEL indication does not increase and/or Indication/Condition is not as specified when attempting to transfer fuel from left inboard fuel tank.

#### **CORRECTIVE ACTION**

- 1. Does CDU TOTAL FUEL indication increase?**

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**WITH MANUAL MODE SELECTED, CDU TOTAL FUEL INDICATION DOES NOT INCREASE AND/OR RESULT IS NOT AS SPECIFIED WHEN ATTEMPTING TO TRANSFER FUEL FROM LEFT INBOARD FUEL TANK** - Continued

- a. If CDU TOTAL FUEL indication does increase, go to **2**.
- b. If CDU TOTAL FUEL indication does not increase, go to **6**.
- 2. Does auxiliary fuel management control panel digital display indication decrease?**
  - a. If auxiliary fuel management control panel digital display indication does decrease, go to **3**.
  - b. If auxiliary fuel management control panel digital display indication does not decrease, go to **5**.
- 3. Does auxiliary fuel management control panel EXTERNAL LEFT NO FLOW indicator momentarily go on, then off?**
  - a. If indicator momentarily goes on, then off, go to **4**.
  - b. If indicator remains on, go to AUXILIARY FUEL MANAGEMENT CONTROL PANEL EXTERNAL RIGHT NO FLOW INDICATOR REMAINS ON DURING FUEL TRANSFER, in this work package.
- 4. Do the remaining auxiliary fuel management control panel EXTERNAL indicators remain off?**
  - a. If the remaining indicators remain off, go to AUXILIARY FUEL MANAGEMENT CONTROL PANEL EXTERNAL RIGHT NO FLOW INDICATOR REMAINS ON DURING FUEL TRANSFER, in this work package.
  - b. If the remaining indicators do not remain off, replace auxiliary fuel management control panel (WP 1448 00). Go to **12**.
- 5. With auxiliary fuel management control panel MANUAL XFR RIGHT switch at ON, check for more than 20 mvac between fuel flow transmitter connector pins A and B.**
  - a. If voltage is as specified, replace fuel flow transmitter (WP 1438 00). Go to **12**.
  - b. If voltage is not as specified, replace auxiliary fuel management control panel (WP 1448 00). Go to **12**.
- 6. Does auxiliary fuel management control panel digital display indication decrease?**
  - a. If indication decreases, go to **7**.
  - b. If indication does not decrease, go to **8**.
- 7. Does EXTERNAL LEFT NO FLOW indicator go on?**
  - a. If indicator is on, replace auxiliary fuel management control panel (WP 1448 00). Go to **12**.
  - b. If indicator is not on, troubleshoot fuel quantity system (WP 0131 00). Go to **12**.
- 8. Is manual lever of left outboard fuel shutoff gate valve in full OPEN position?**
  - a. If manual lever is in full OPEN position, go to **9**.
  - b. If manual lever is not in full OPEN position, go to **12**.
- 9. Check for 28 vdc between P808-A and P808-B.**
  - a. If voltage is as specified, replace right outboard bleed-air regulator valve (WP 1439 00). Go to **12**.
  - b. If voltage is not as specified, go to **10**.

**WITH MANUAL MODE SELECTED, CDU TOTAL FUEL INDICATION DOES NOT INCREASE AND/OR RESULT IS NOT AS SPECIFIED WHEN ATTEMPTING TO TRANSFER FUEL FROM LEFT INBOARD FUEL TANK** - Continued

**10. Check continuity between:**

- **P808-A and P360-m**
- **P808-B and ground**

- a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to **12**.
- b. If continuity is not present, go to **11**.

**11. Check continuity between:**

- **P805-1 and ground**
- **P805-2 and P360-m**

- a. If continuity is present, replace rear electrical harness (WP 1454 00). Go to **12**.
- b. If continuity is not present, replace fuel system main wiring harness (WP 1453 00). Go to **12**.

**12. Procedure completed.**

**FUEL TRANSFER TIME AND/OR GAGING ACCURACY IS NOT AS SPECIFIED**

**SYMPTOM**

Fuel transfer time and/or gaging accuracy is not as specified.

**MALFUNCTION**

Fuel transfer time and/or gaging accuracy is not as specified.

**CORRECTIVE ACTION**

- 1. With auxiliary fuel management control panel MANUAL XFR LEFT and/or MANUAL XFR RIGHT switch at ON, inspect fuel and pneumatic hoses on fittings for signs of leakage or damage.**
  - a. If leakage or damage is present, replace damaged or leaking hoses and/or fittings as necessary (WP 1433 00 or WP 1435 00). Go to **6**.
  - b. If leakage or damage is not present, go to **2**.
- 2. Pull and turn auxiliary fuel management control panel AUX FUEL QTY switch to CAL. Go to 3.**
- 3. Does auxiliary fuel management control panel digital display indication equal K FACTOR value on fuel flow transmitter?**
  - a. If indication is equal to K FACTOR, go to **4**.
  - b. If indication is not equal to K FACTOR, replace auxiliary fuel management control panel (WP 1448 00). Go to **6**.
- 4. With auxiliary fuel management control panel MANUAL XFR LEFT and/or MANUAL XFR RIGHT switch at ON, check for more than 20 mvac between fuel flow transmitter connector pins A and B.**
  - a. If voltage is as specified, replace bleed-air regulator valve (WP 1439 00). Go to **6**.
  - b. If trouble remains, go to **5**.
  - c. If voltage is not as specified, replace fuel flow transmitter (WP 1438 00). Go to **5**.



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**FUEL TRANSFER TIME AND/OR GAGING ACCURACY IS NOT AS SPECIFIED - Continued****5. Do operational/troubleshooting procedure of fuel quantity system (WP 0131 00).**

- a. If fuel quantity system is good, replace auxiliary fuel management control panel (WP 1448 00). Go to **6**.
- b. If fuel quantity system is not good, adjust/troubleshoot system as required (WP 0131 00). Go to **6**.

**6. Procedure completed.****AUTO MODE FUEL TRANSFER LIMITS ARE NOT AS SPECIFIED****SYMPTOM**

Auto mode fuel transfer limits are not as specified.

**MALFUNCTION**

Auto mode fuel transfer limits are not as specified.

**CORRECTIVE ACTION****1. Did digital display indication increase or decrease?**

- a. If indication changes, replace auxiliary fuel management control panel (WP 1448 00). Go to **3**.
- b. If indication does not change, go to **2**.

**2. Check continuity between:**

- P905-K (tank 1) and J346-30
- P905-L (tank 1) and J346-31
- P905-V (tank 2) and J346-27
- P905-U (tank 2) and J346-28

- a. If continuity is present, replace fuel system main wiring harness (WP 1453 00). Go to **3**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **3**.

**3. Procedure completed.****AUXILIARY FUEL MANAGEMENT CONTROL PANEL EXTERNAL RIGHT NO FLOW INDICATOR REMAINS ON DURING FUEL TRANSFER****SYMPTOM**

Auxiliary fuel management control panel EXTERNAL RIGHT NO FLOW indicator remains on during fuel transfer.

**MALFUNCTION**

Auxiliary fuel management control panel EXTERNAL RIGHT NO FLOW indicator remains on during fuel transfer.

**CORRECTIVE ACTION**

- 1. Disconnect J520 from right external fuel flow sensor. Go to 2.**
- 2. Connect insulated jumper wire between J520-1 and J520-2. Go to 3.**

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**AUXILIARY FUEL MANAGEMENT CONTROL PANEL EXTERNAL RIGHT NO FLOW INDICATOR REMAINS ON DURING FUEL TRANSFER - Continued**

3. Place auxiliary fuel management control panel MANUAL XFR RIGHT switch to ON. Go to 4.
4. Does EXTERNAL RIGHT NO FLOW indicator go on momentarily?
  - a. If indicator goes on momentarily, replace right external fuel flow sensor (WP 1437 00). Go to 8.
  - b. If indicator does not go on momentarily, go to 5.
5. Place auxiliary fuel management control panel MANUAL XFR RIGHT switch to OFF. Go to 6.
6. Check continuity between:
  - J520-1 and P360-e
  - J520-2 and P360-d
  - a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to 8.
  - b. If continuity is not present, go to 7.
7. Check continuity between:
  - J520-1 and P346-16
  - J520-2 and P346-17
  - a. If continuity is present, replace fuel system main wiring harness (WP 1453 00). Go to 8.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to 8.
8. Procedure completed.

**AUXILIARY FUEL MANAGEMENT CONTROL PANEL EXTERNAL LEFT NO FLOW INDICATOR REMAINS ON DURING FUEL TRANSFER****SYMPTOM**

Auxiliary fuel management control panel EXTERNAL LEFT NO FLOW indicator remains on during fuel transfer.

**MALFUNCTION**

Auxiliary fuel management control panel EXTERNAL LEFT NO FLOW indicator remains on during fuel transfer.

**CORRECTIVE ACTION**

1. Disconnect J521 from right external fuel flow sensor. Go to 2.
2. Connect insulated jumper wire between J521-1 and J521-2. Go to 3.
3. Place auxiliary fuel management control panel MANUAL XFR LEFT switch to ON. Go to 4.
4. Does EXTERNAL LEFT NO FLOW indicator go on momentarily?
  - a. If indicator goes on momentarily, replace right external fuel flow sensor (WP 1437 00). Go to 8.
  - b. If indicator does not go on momentarily, go to 5.

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**AUXILIARY FUEL MANAGEMENT CONTROL PANEL EXTERNAL LEFT NO FLOW INDICATOR REMAINS ON DURING FUEL TRANSFER - Continued**

5. Place auxiliary fuel management control panel **MANUAL XFR LEFT** switch to **OFF**. Go to 6.
6. Check continuity between:
  - J521-1 and P360-q
  - J521-2 and P360-d
  - a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to 8.
  - b. If continuity is not present, go to 7.
7. Check continuity between:
  - J521-1 and P346-14
  - J521-2 and P346-15
  - a. If continuity is present, replace fuel system main wiring harness (WP 1453 00). Go to 8.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to 8.
8. Procedure completed.

**AUXILIARY FUEL MANAGEMENT CONTROL PANEL EXTERNAL RIGHT INBD EMPTY AND/OR EXTERNAL LEFT INBD EMPTY INDICATORS DO NOT GO ON WITH INADEQUATE FUEL IN TANK****SYMPTOM**

Auxiliary fuel management control panel EXTERNAL RIGHT INBD EMPTY and/or EXTERNAL LEFT INBD EMPTY indicators do not go on with inadequate fuel in tank.

**MALFUNCTION**

Auxiliary fuel management control panel EXTERNAL RIGHT INBD EMPTY and/or EXTERNAL LEFT INBD EMPTY indicators do not go on with inadequate fuel in tank.

**CORRECTIVE ACTION**

1. Disconnect J1 from applicable inboard fuel tank. Go to 2.
2. Does applicable EXTERNAL INBD EMPTY indicator go on?
  - a. If indicator is on, remove/replace low level sensor assembly (WP 1476 00). Go to 3.
  - b. If indicator is not on, replace auxiliary fuel management control panel (WP 1448 00). Go to 3.
3. Procedure completed.

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**MFD AUX FUEL LEGEND AND/OR AUXILIARY FUEL MANAGEMENT CONTROL PANEL EXTERNAL RIGHT NO FLOW AND EXTERNAL LEFT NO FLOW INDICATORS DO NOT GO ON****SYMPTOM**

MFD AUX FUEL legend and/or auxiliary fuel management control panel EXTERNAL RIGHT NO FLOW and EXTERNAL LEFT NO FLOW indicators do not go on immediately after corresponding tank empty indicator goes on.

**MALFUNCTION**

MFD AUX FUEL legend and/or auxiliary fuel management control panel EXTERNAL RIGHT NO FLOW and EXTERNAL LEFT NO FLOW indicators do not go on immediately after corresponding tank empty indicator goes on.

**CORRECTIVE ACTION**

- 1. Do auxiliary fuel management control panel EXTERNAL RIGHT NO FLOW and EXTERNAL LEFT NO FLOW indicators go on?**
  - a. If indicators are on, go to **2**.
  - b. If indicators are not on, go to **5**.
- 2. Press MFD T6 switch, does AUX FUEL legend go on, for approximately 10 seconds then off?**
  - a. If legend goes on, than off, go to **3**.
  - b. If legend does not go on, then off, troubleshoot MFD/caution/advisory warning system (WP 0100 00). Go to **9**.
- 3. Check for 28 vdc between:**
  - **P1019R-37 and ground**
  - **P1020R-37 and ground**
  - a. If voltage is as specified, replace copilot's or pilot's MFD (WP 0799 00). Go to **9**.
  - b. If voltage is not as specified, go to **4**.
- 4. Check continuity between P346-10 and P360-X.**
  - a. If continuity is present, repair/replace wiring between J346-10 and P1019R-37 or P1020R-37 (WP 1747 00). Go to **9**.
  - b. If continuity is not present, replace fuel system main wiring harness (WP 1453 00). Go to **9**.
- 5. Disconnect connector, J521 (left) or J520 (right), from external fuel flow sensor. Go to 6.**
- 6. With auxiliary fuel management control panel MODE switch at AUTO, is correct corresponding no flow indicator on?**
  - a. If indicator is on, replace external fuel flow sensor (WP 1437 00). Go to **9**.
  - b. If indicator is not on, go to **7**.
- 7. Check for infinite resistance between:**
  - **P360-g (left) and J360-d**
  - **P360-e (right) and J360-d**

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**MFD AUX FUEL LEGEND AND/OR AUXILIARY FUEL MANAGEMENT CONTROL PANEL EXTERNAL  
RIGHT NO FLOW AND EXTERNAL LEFT NO FLOW INDICATORS DO NOT GO ON - Continued**

- a. If resistance is as specified, replace auxiliary fuel management control panel (WP 1448 00). Go to **9**.
  - b. If resistance is not as specified, go to **8**.
- 8. Check for infinite resistance between:**
- **P346-14 (left) and J346-15**
  - **P346-16 (right) and J346-17**
- a. If resistance is as specified, replace fuel system main wiring harness (WP 1453 00). Go to **9**.
  - b. If resistance is not as specified, troubleshoot wiring for short circuit. Repair/replace wiring as required (WP 1747 00). Go to **9**.
- 9. Procedure completed.**

**AUXILIARY FUEL MANAGEMENT CONTROL PANEL EXTERNAL INBD EMPTY AND/OR EXTERNAL  
OUTBD EMPTY INDICATORS ARE NOT OFF****SYMPTOM**

Auxiliary fuel management control panel EXTERNAL INBD EMPTY and/or EXTERNAL OUTBD EMPTY indicators are not off.

**MALFUNCTION**

Auxiliary fuel management control panel EXTERNAL INBD EMPTY and/or EXTERNAL OUTBD EMPTY indicators are not off.

**CORRECTIVE ACTION**

- 1. Disconnect J1 from applicable outboard fuel tank. Go to 2.**
- 2. Does applicable indicator go on?**
  - a. If indicator is on, remove/replace low level sensor assembly (WP 1475 00). Go to **3**.
  - b. If indicator does not go on, replace auxiliary fuel management control panel (WP 1448 00). Go to **3**.
- 3. Procedure completed.**

**VENT SENSOR OVFL INDICATOR DOES NOT GO ON****SYMPTOM**

VENT SENSOR OVFL indicator does not go on.

**MALFUNCTION**

VENT SENSOR OVFL indicator does not go on.

**CORRECTIVE ACTION**

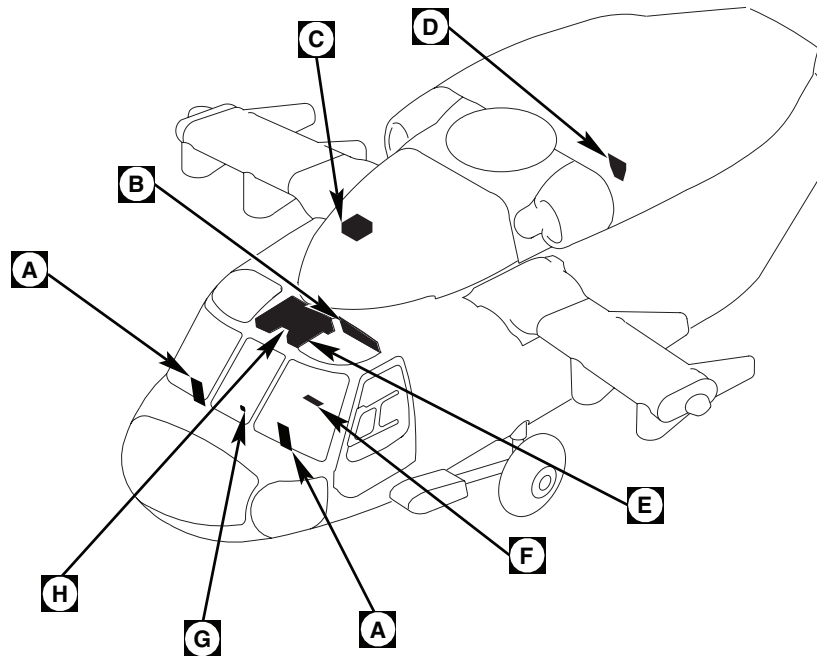
- 1. Check connector J376 is connected to P376 on vent sensor.**
  - a. If connector J376 and P376 are connected, go to **2**.
  - b. If connector J376 is not connected to connector P376, connect J376 to P376. Go to **5**.
- 2. Check overflow sensor resistance as follows:**

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**VENT SENSOR OVFL INDICATOR DOES NOT GO ON** - Continued

- **P361-W and P361-Z (650-1500 ohms)**
  - **P361-V and P361-Z (360-440 ohms)**
  - a. If resistance is as specified, replace auxiliary fuel management control panel (WP 1448 00). Go to **5**.
  - b. If resistance is not as specified, go to **3**.
- 3. Check overflow sensor resistance as follows:**
- **P376-C and P376-A (650-1500 ohms)**
  - **P376-C and P376-B (360-440 ohms)**
  - a. If resistance is as specified, go to **4**.
  - b. If resistance is not as specified, replace vent sensor (WP 1457 00). Go to **5**.
- 4. Check continuity between:**
- **P346-46 and P361-V**
  - **P346-47 and P361-W**
  - **P346-48 and P361-Z**
  - a. If continuity is present, repair/replace wiring (WP 1747 00) between, then go to **5**:
    - J346-46 and J376-B
    - J346-47 and J376-A
    - J346-48 and J376-C
  - b. If continuity is not present, replace fuel system main wiring harness (WP 1453 00). Go to **5**.
- 5. Procedure completed.**

## LOCATION



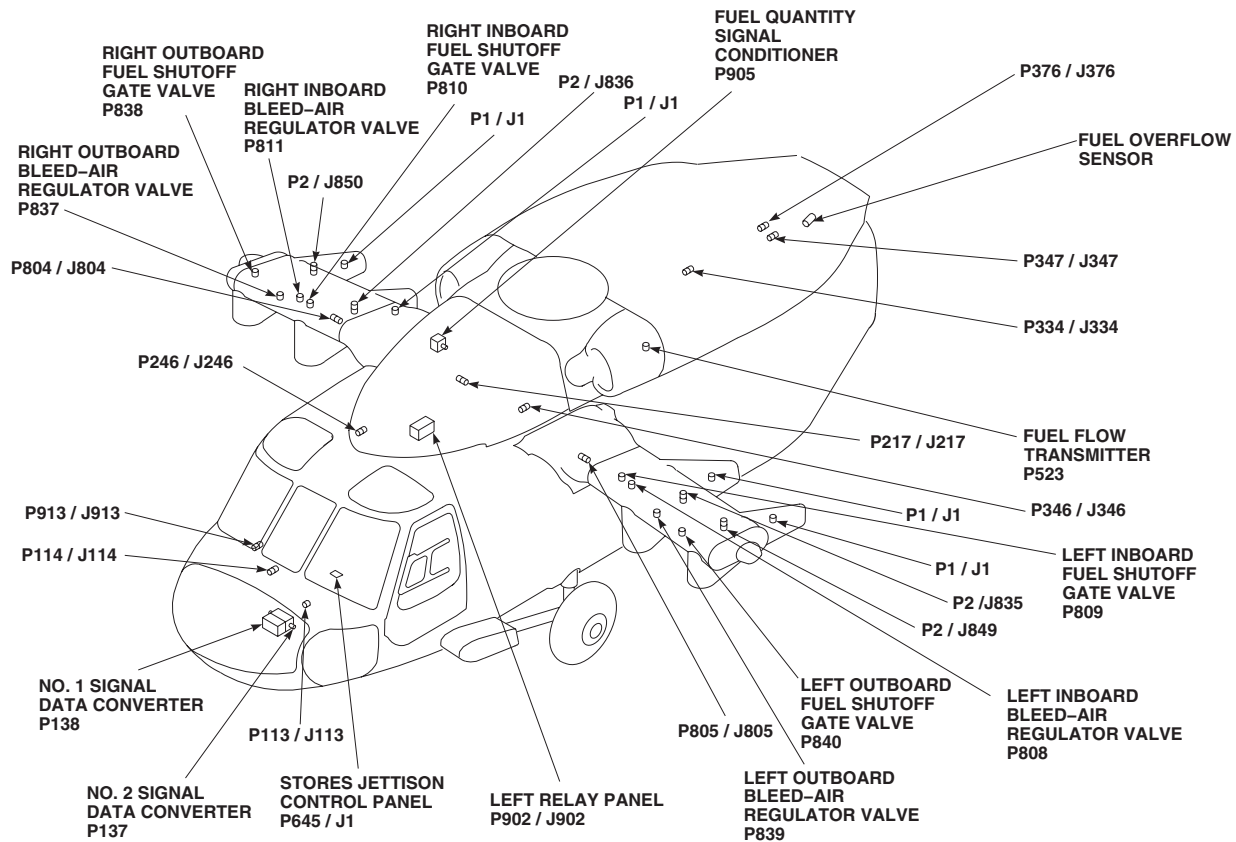
| TERMINAL BOARD /<br>DISCONNECT PLUG /<br>RECEPTACLE | LOCATION /<br>CONNECTION POINT        |
|-----------------------------------------------------|---------------------------------------|
| J520                                                | RIGHT EXT FUEL FLOW<br>SENSOR         |
| J521                                                | LEFT EXT FUEL FLOW<br>SENSOR          |
| J524                                                | FUEL TEMPERATURE<br>TRANSDUCER        |
| P1 / J1                                             | RIGHT INBOARD FUEL<br>TANK CONNECTOR  |
| P1 / J1                                             | LEFT INBOARD FUEL TANK<br>CONNECTOR   |
| P1 / J1                                             | LEFT OUTBOARD FUEL<br>TANK CONNECTOR  |
| P1 / J1                                             | RIGHT OUTBOARD FUEL<br>TANK CONNECTOR |
| P2 / J835                                           | LEFT INBOARD FUEL TANK<br>CONNECTOR   |

| TERMINAL BOARD /<br>DISCONNECT PLUG /<br>RECEPTACLE | LOCATION /<br>CONNECTION POINT         |
|-----------------------------------------------------|----------------------------------------|
| P2 / J836                                           | RIGHT INBOARD FUEL TANK<br>CONNECTOR   |
| P2 / J849                                           | LEFT OUTBOARD FUEL<br>TANK CONNECTOR   |
| P2 / J850                                           | RIGHT OUTBOARD FUEL<br>TANK CONNECTOR  |
| P113 / J113                                         | COCKPIT<br>BL 6 LH, STA 197            |
| P114 / J114                                         | COCKPIT<br>BL 8 RH, STA 200            |
| P137                                                | NO. 2 SIGNAL DATA<br>CONVERTER         |
| P138                                                | NO. 1 SIGNAL DATA<br>CONVERTER         |
| P217 / J217                                         | MAIN ROTOR PYLON DECK<br>BL 0, STA 284 |
| P246 / J246                                         | CABIN CEILING<br>BL 5 RH, STA 247      |
| P334 / J334                                         | CABIN CEILING<br>BL 16.5 LH, STA 390   |

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Figure 1. ESSS Range Extension System Location Diagram. (Sheet 1 of 5)

## LOCATION - Continued



| TERMINAL BOARD /<br>DISCONNECT PLUG /<br>RECEPTACLE | LOCATION /<br>CONNECTION POINT                    |
|-----------------------------------------------------|---------------------------------------------------|
| P346 / J346                                         | CABIN CEILING<br>BL 20 LH, STA 304                |
| P347 / J347                                         | TRANSITION SECTION<br>BL 10 LH, STA 425           |
| P360 / J360                                         | AUXILIARY FUEL MANAGE-<br>MENT CONTROL PANEL      |
| P361 / J361                                         | AUXILIARY FUEL MANAGE-<br>MENT CONTROL PANEL      |
| P376 / J376                                         | AFT TRANSITION<br>BL 18.25 LH, STA 443.5          |
| P523                                                | FUEL FLOW TRANSMITTER                             |
| P645 / J1                                           | STORES JETTISON<br>CONTROL PANEL                  |
| P804 / J804                                         | HORIZONTAL STORES<br>SUPPORT<br>BL 59 RH, STA 295 |
| P805 / J805                                         | HORIZONTAL STORES<br>SUPPORT<br>BL 59 LH, STA 295 |
| P808                                                | LEFT INBOARD BLEED-AIR<br>REGULATOR VALVE         |
| P809                                                | LEFT INBOARD FUEL SHUT-<br>OFF GATE VALVE         |

| TERMINAL BOARD /<br>DISCONNECT PLUG /<br>RECEPTACLE | LOCATION /<br>CONNECTION POINT               |
|-----------------------------------------------------|----------------------------------------------|
| P810                                                | RIGHT INBOARD FUEL<br>SHUTOFF GATE VALVE     |
| P811                                                | RIGHT INBOARD BLEED-AIR<br>REGULATOR VALVE   |
| P837                                                | RIGHT OUTBOARD BLEED-<br>AIR REGULATOR VALVE |
| P838                                                | RIGHT OUTBOARD FUEL<br>SHUTOFF GATE VALVE    |
| P839                                                | LEFT OUTBOARD BLEED-<br>AIR REGULATOR VALVE  |
| P840                                                | LEFT OUTBOARD FUEL<br>SHUTOFF GATE VALVE     |
| P902 / J902                                         | LEFT RELAY PANEL                             |
| P905                                                | FUEL QUANTITY SIGNAL<br>CONDITIONER          |
| P913 / J913                                         | COCKPIT<br>BL 24 RH, STA 247                 |
| P1019R / J1019R                                     | MULTIFUNCTION DISPLAY<br>(MFD)               |
| P1020R / J1020R                                     | MULTIFUNCTION DISPLAY<br>(MFD)               |

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SA

Figure 1. ESSS Range Extension System Location Diagram. (Sheet 2 of 5)



LOCATION - Continued

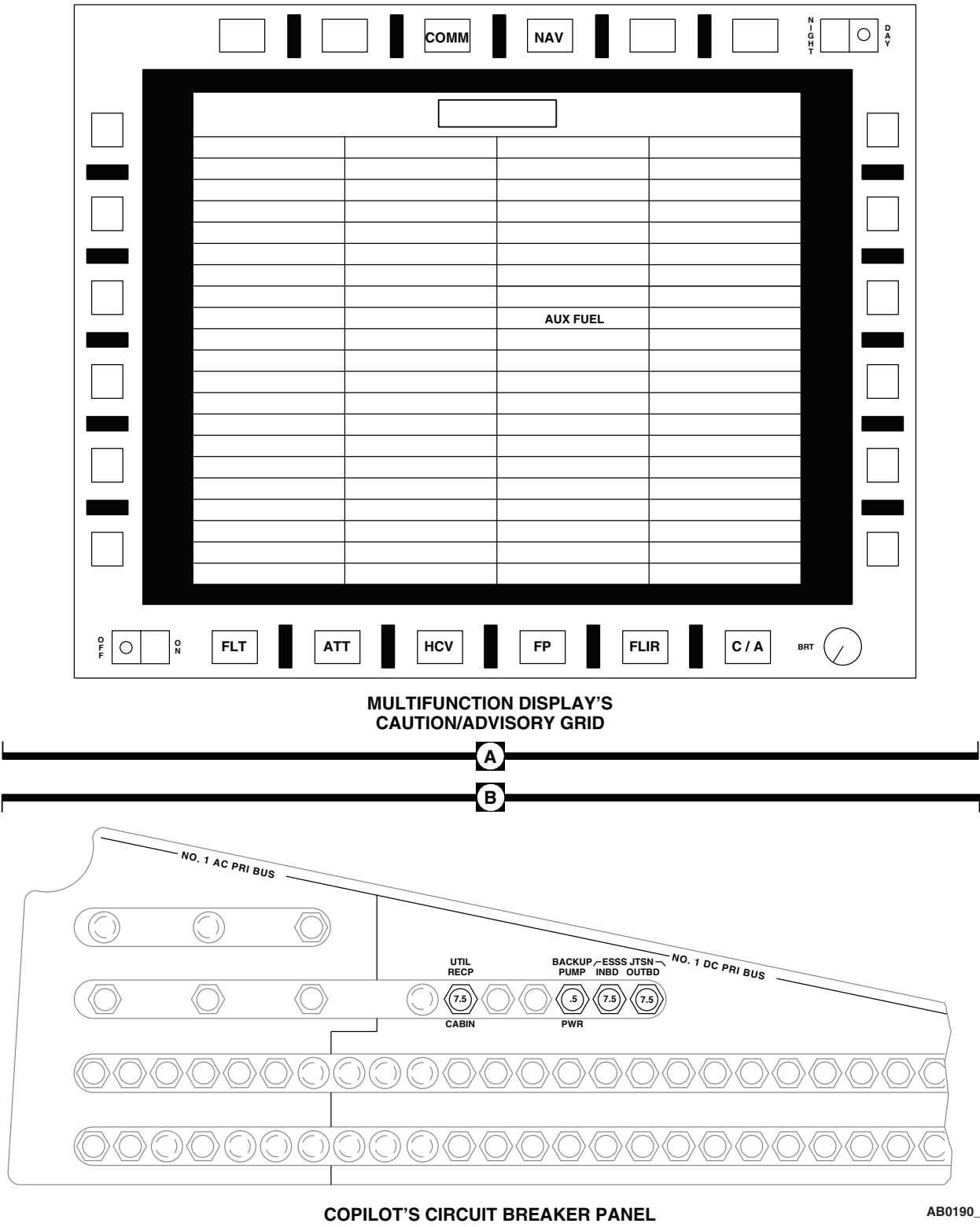
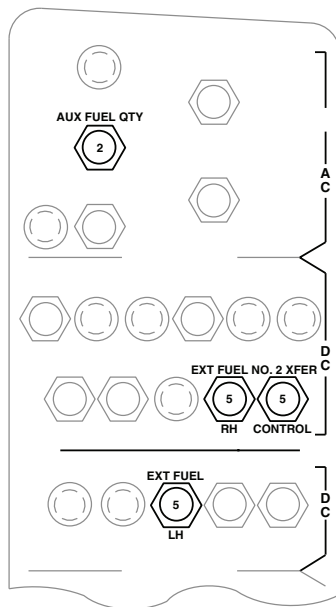


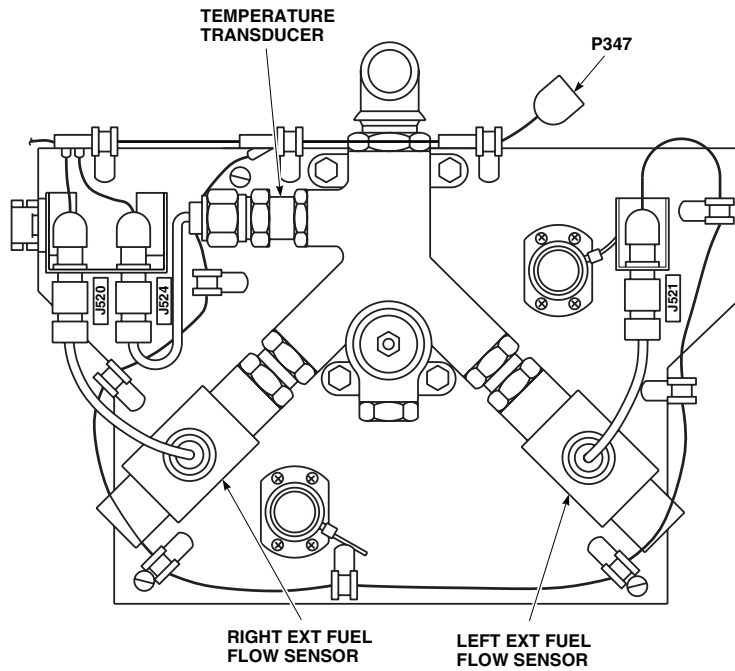
Figure 1. ESSS Range Extension System Location Diagram. (Sheet 3 of 5)

LOCATION - Continued



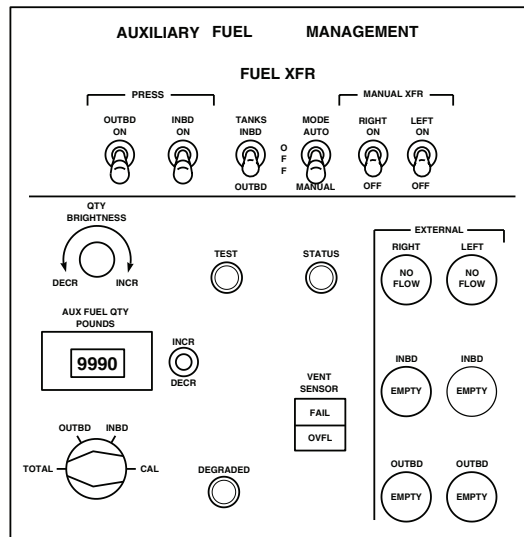
**MISSION READINESS  
CIRCUIT BREAKER PANEL**

**C**



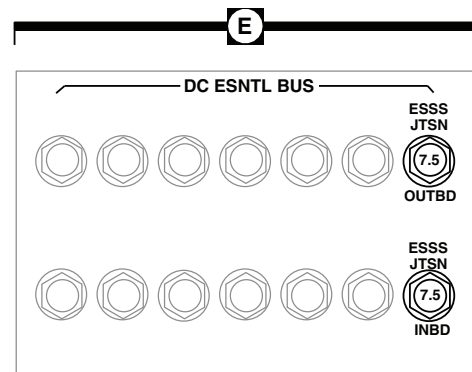
**FLOW SENSOR PANEL**

**D**



**AUXILIARY FUEL MANAGEMENT  
CONTROL PANEL**

**F**



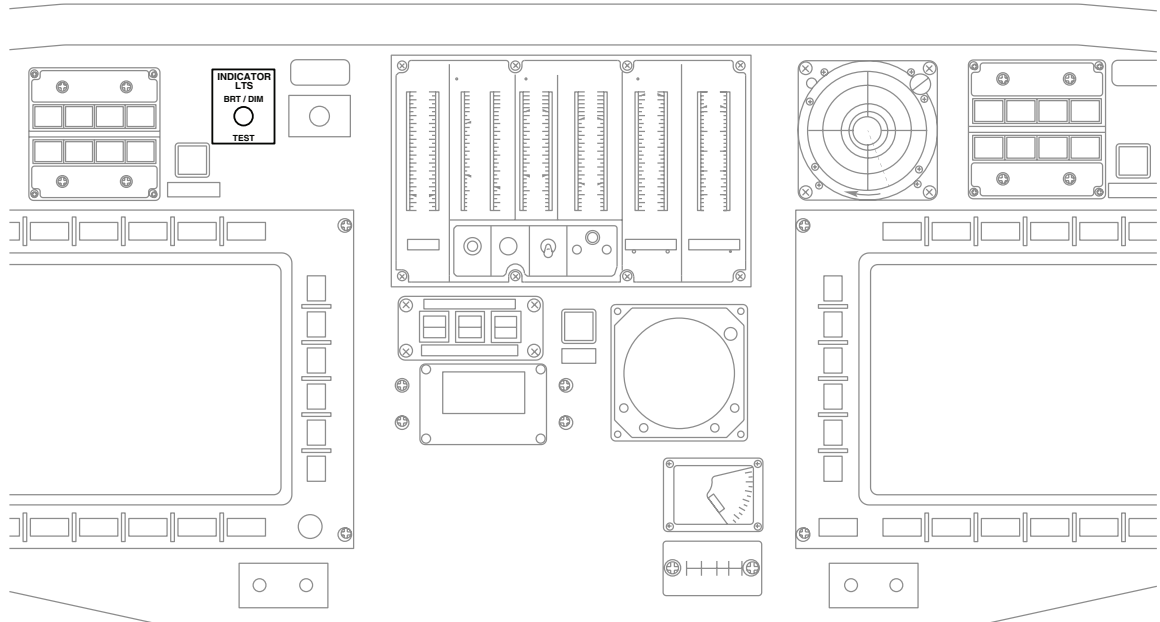
**UPPER CONSOLE  
CIRCUIT BREAKER PANEL**

**E**

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SA

Figure 1. ESSS Range Extension System Location Diagram. (Sheet 4 of 5)

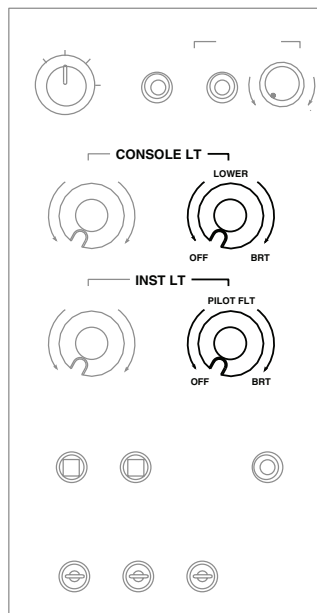
LOCATION - Continued



INSTRUMENT PANEL

G

H



UPPER CONSOLE

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SA

Figure 1. ESSS Range Extension System Location Diagram. (Sheet 5 of 5)

## SCHEMATICS

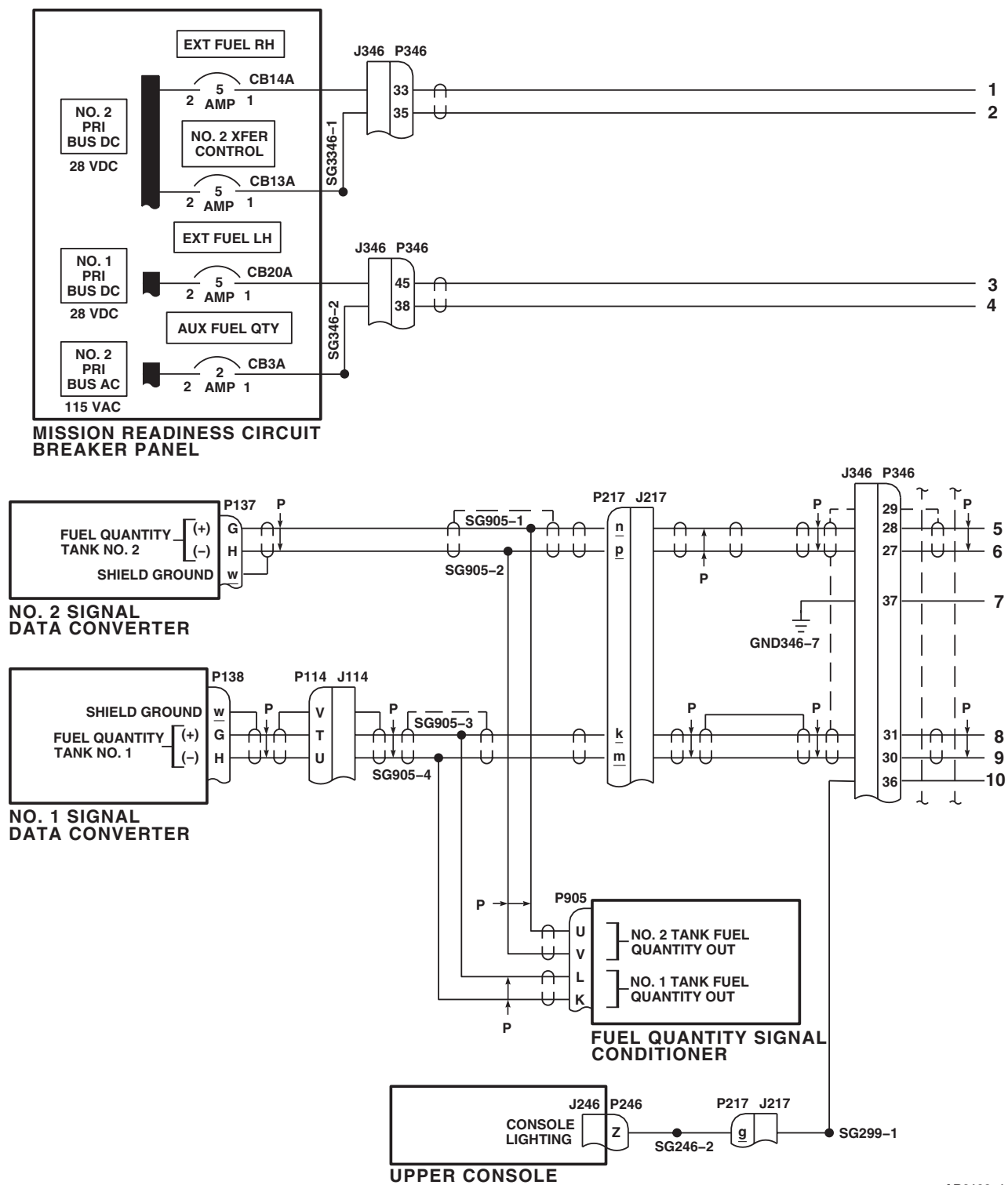
AB2103.1  
SA

Figure 2. ESSS Range Extension System Schematic Diagram. (Sheet 1 of 6)

**LEFT RELAY PANEL**

**STORES JETTISON CONTROL PANEL**

**PILOTS MFD/ CAUTION ADVISORY PANEL**

**COPILOTS MFD/ CAUTION ADVISORY PANEL**

**AUXILIARY FUEL MANAGEMENT CONTROL PANEL**

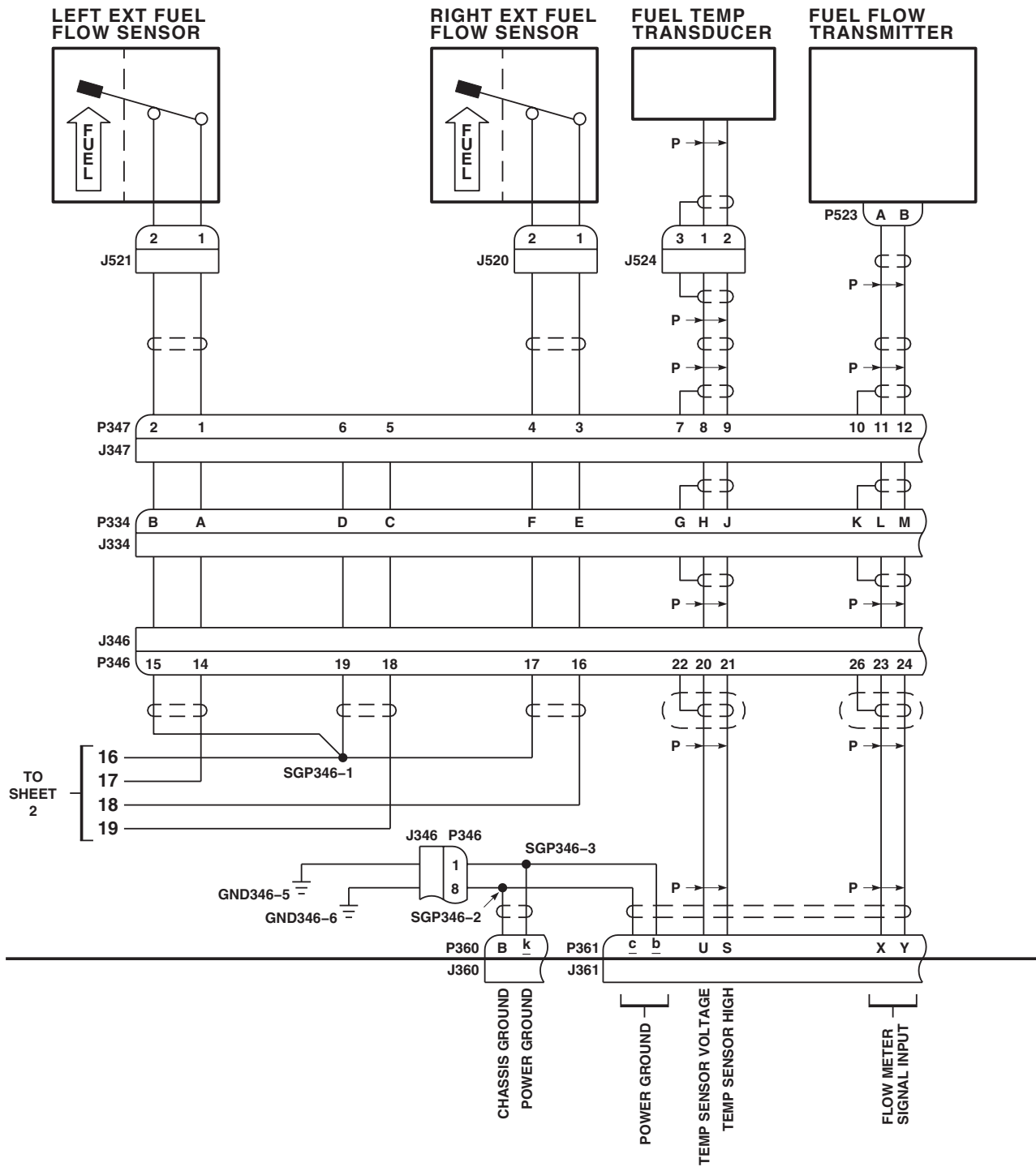
**TO SHEET 6**

**TO SHEET 3**

**AB2103\_2**

### Change 1

SCHEMATICS - Continued

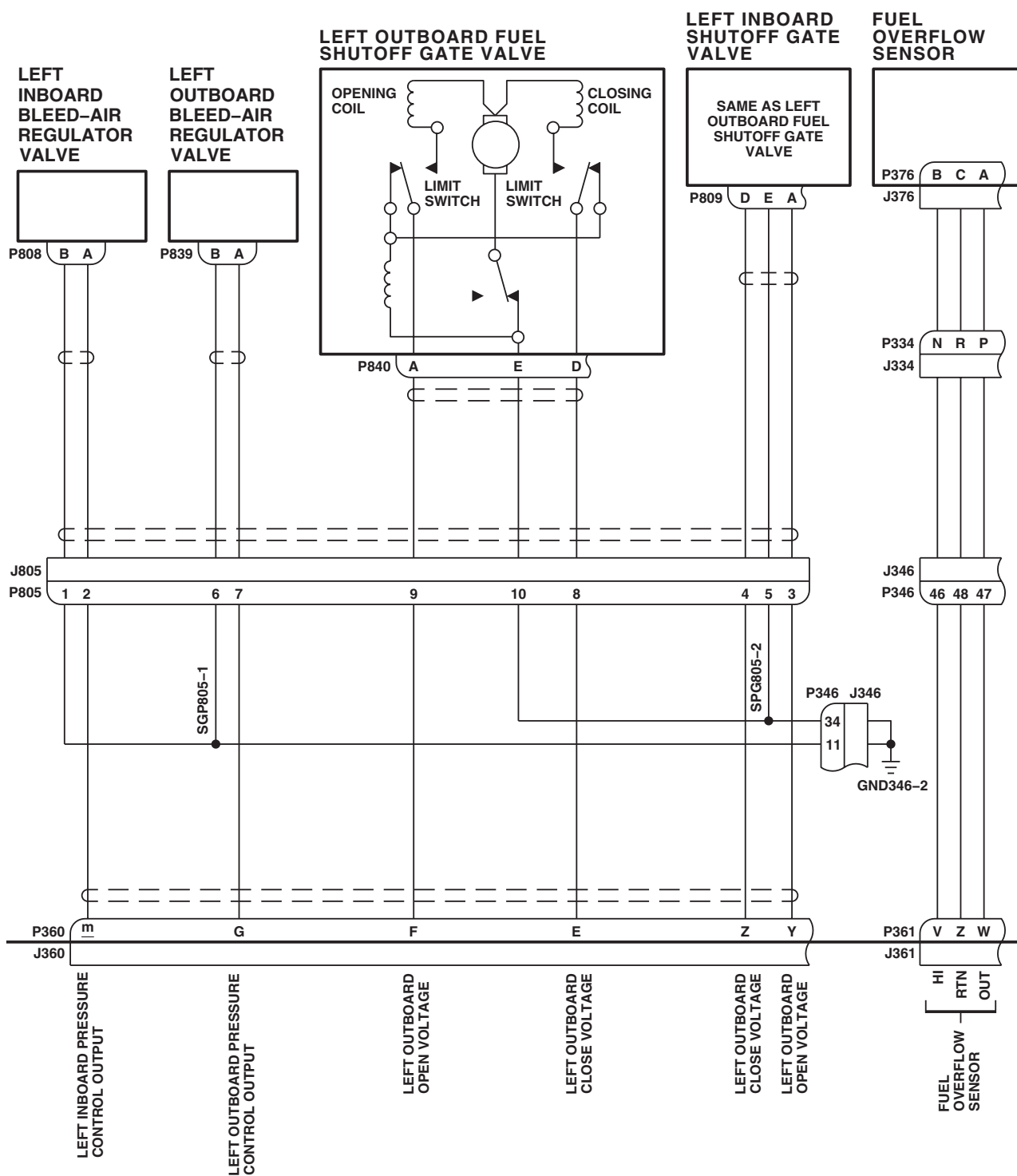


AUXILIARY FUEL MANAGEMENT CONTROL PANEL

AB2103\_3  
SA

Figure 2. ESSS Range Extension System Schematic Diagram. (Sheet 3 of 6)

## SCHEMATICS - Continued

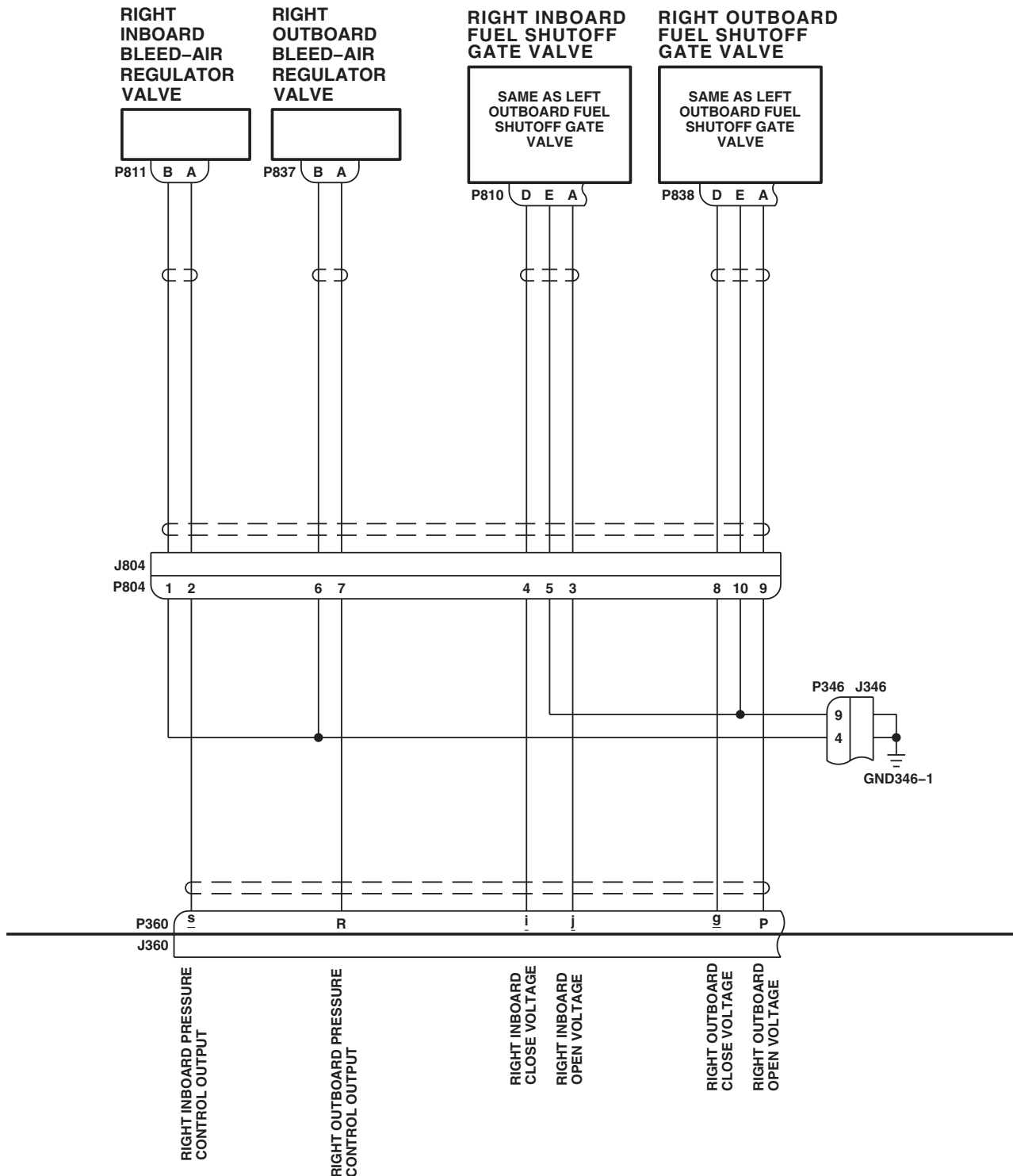


## AUXILIARY FUEL MANAGEMENT CONTROL PANEL

AB2103\_4  
SA

Figure 2. ESSS Range Extension System Schematic Diagram. (Sheet 4 of 6)

## SCHEMATICS - Continued



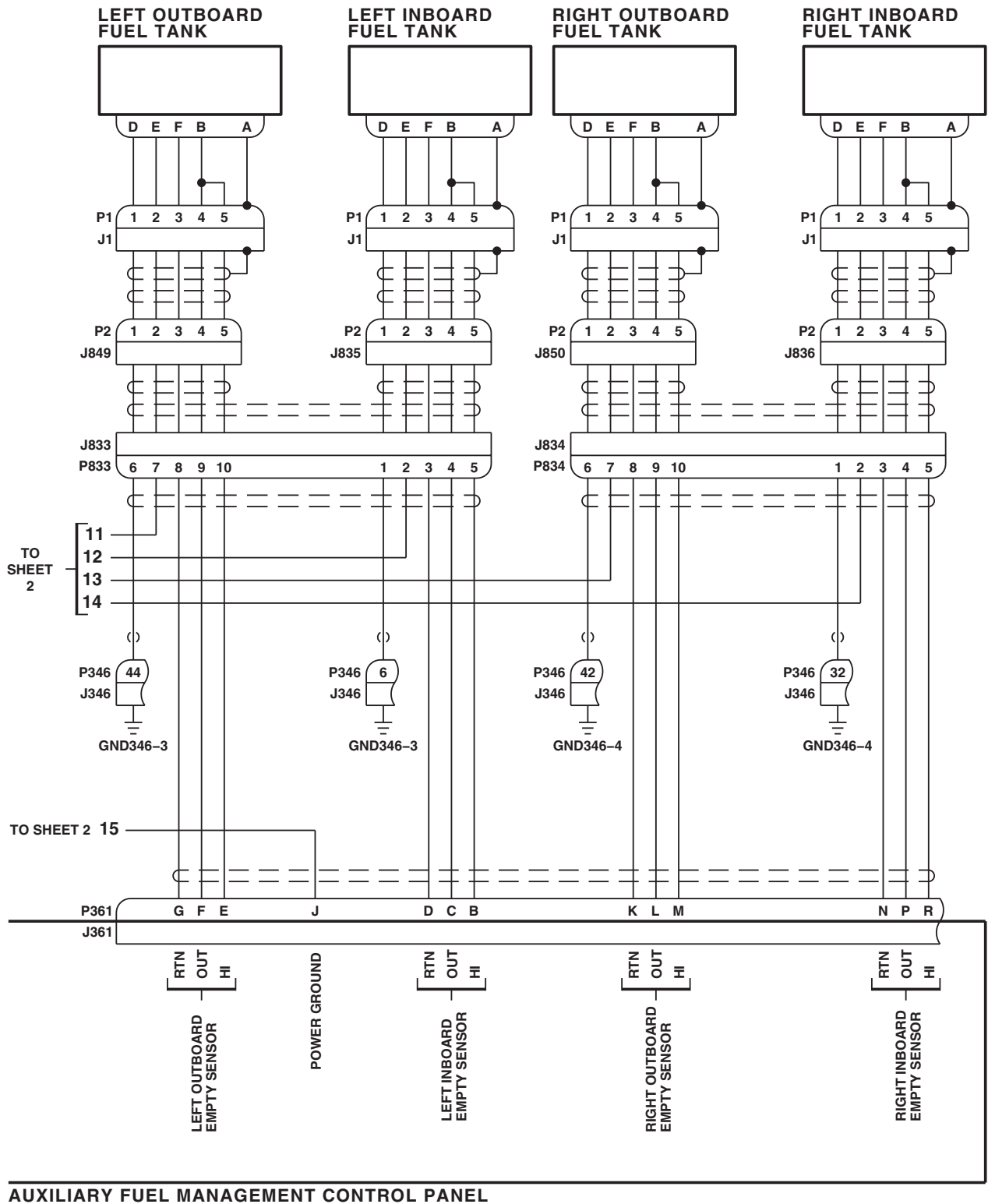
## AUXILIARY FUEL MANAGEMENT CONTROL PANEL

AB2103 5  
SA

Figure 2. ESSS Range Extension System Schematic Diagram. (Sheet 5 of 6)



SCHEMATICS - Continued



AB2103\_6  
SA

Figure 2. ESSS Range Extension System Schematic Diagram. (Sheet 6 of 6)

END OF WORK PACKAGE



## UNIT LEVEL

## MISSION EQUIPMENT

## EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM MWO 50-78

This WP supersedes WP 0168 00, dated 17 April 2006.

## INITIAL SETUP:

## Test Equipment

|                                                                       |            |
|-----------------------------------------------------------------------|------------|
| Compressor Unit, MIL-C-13874                                          | WP 1428 00 |
| Locally-Made Fuel Quantity Simulator Test Set, Figure 152, WP 1805 00 | WP 1431 00 |
| Locally-Made Gate Valve Test Harness, Figure 156, WP 1805 00          | WP 1432 00 |
| Multimeter, AN/PSM-45A (2 required)                                   | WP 1437 00 |
|                                                                       | WP 1438 00 |
|                                                                       | WP 1439 00 |
|                                                                       | WP 1440 00 |

## Tools and Special Tools

|                                                      |            |
|------------------------------------------------------|------------|
| Aircraft Mechanic's Toolkit, SC 5180-99-B01          | WP 1442 00 |
| Container, Suitable                                  | WP 1445 00 |
| Electrical Repairer Toolkit, SC 5180-99-B06          | WP 1447 00 |
| Locally-Made Drag Beam Wedge, Figure 167, WP 1805 00 | WP 1448 00 |
| Locally-Made External Stores Support System (ESSS)   | WP 1457 00 |
| Pneumatic Adapter, Figure 151, WP 1805 00            | WP 1465 00 |
| Stopwatch                                            | WP 1552 00 |
| Torque Wrench, GG-W-686                              | WP 1612 00 |
|                                                      | WP 1613 00 |
|                                                      | WP 1620 00 |
|                                                      | WP 1685 00 |

## Personnel Required

|                                       |            |
|---------------------------------------|------------|
| Aircraft Electrician MOS 15F (1)      | WP 1747 00 |
| UH-60 Helicopter Repairer MOS 15T (1) | WP 1805 00 |

## References

TM 1-1520-237-10  
 TM 11-1520-237-23  
 WP 0090 00  
 WP 0104 00  
 WP 0105 00  
 WP 0110 00  
 WP 0131 00  
 WP 0159 00  
 WP 0160 00  
 WP 0443 00  
 WP 0824 00  
 WP 0825 00  
 WP 0925 00

## Equipment Condition

All Explosive Cartridges Removed from Ejector Racks, WP 1442 00  
 or All Explosive Cartridges Removed from Ejector Racks, WP 1445 00  
 External Electrical Power Available, WP 1685 00  
 or APU Operational, TM 1-1520-237-10  
 External Range Extension Tanks (Inboard or Outboard) EMPTY, WP 1620 00  
 Main Fuel Tanks, Not More Than 650 Pounds of Fuel in Each Tank, WP 1612 00  
 or Main Fuel Tanks, Not More Than 650 Pounds of Fuel in Each Tank, WP 1613 00  
 Rear Fairings on All Vertical Stores (Pylons) Removed, WP 1447 00

## APPLICABLE CONFIGURATION

This operational/troubleshooting procedure is designed to be utilized when the horizontal (Wings) and/or Vertical (Pylons) stores are initially installed or replaced. When only tanks are removed, refer to WP 0159 00.

## EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE **MWO 50-78**

### PROCEDURE

#### Step

#### NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

- Special icons used in this operational/troubleshooting procedure:

**CU** Steps that are to be done if compressor unit is used.

**INBD** Steps that are to be done if inboard tanks are installed.

**OUTBD** Steps that are to be done if outboard tanks are installed.

- As an alternate to using the compressor unit to pressurize the ESSS Range Extension System, the Aviation Ground Power Unit (AGPU) or the APU may be used. If the alternate is selected, steps with **CU** may be skipped.
  - The operational/troubleshooting procedure will check the ESSS Range Extension System with any tank configuration, installed on the helicopter, authorized by the Aircraft Operator's Manual (TM 1-1520-237-10).
  - See Figure 1 for component location diagram and Figure 2 for schematic diagram as an aid in operational/troubleshooting.
1. **CU** Unfasten fasteners on left inboard stores support rear fairing and hinge fairing open to gain access to bleed-air quick-disconnect (QD). ■
  2. **CU** Disconnect hose from outboard side of bleed-air QD. ■
  3. **CU** Using locally-made pneumatic adapter (WP 1805 00), tee air compressor into helicopter bleed-air QD and ESSS bleed-air hose. ■
  4. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
  5. Pull out the following circuit breakers:

#### CIRCUIT BREAKER

#### CIRCUIT BREAKER PANEL

**UH-60A UH-60L**

ESSS JTSN OUTBD

Copilot's

---

**EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE** **MWO 50-78** - Continued

| <b>CIRCUIT BREAKER</b> | <b>CIRCUIT BREAKER PANEL</b>   |
|------------------------|--------------------------------|
| ESSS JTSN INBD         | Copilot's                      |
| BACKUP PUMP PWR        | Copilot's                      |
| UTIL RECP CABIN        | Copilot's                      |
| ESSS JTSN OUTBD        | Upper Console (copilot's side) |
| ESSS JTSN INBD         | Upper Console (copilot's side) |
| EXT FUEL RH            | Mission Readiness              |
| EXT FUEL LH            | Mission Readiness              |
| NO. 2 XFER CONTROL     | Mission Readiness              |
| <b>EH-60A</b>          |                                |
| ESSS JTSN OUTBD        | Copilot's                      |
| ESSS JTSN INBD         | Copilot's                      |
| UTIL RECP CABIN        | Copilot's                      |
| EXT FUEL LH            | Copilot's                      |
| EXT FUEL RH            | Pilot's                        |
| AUX FUEL CONTR         | Pilot's                        |
| AUX FUEL QTY           | Pilot's                        |
| ESSS JTSN OUTBD        | Upper Console (copilot's side) |
| ESSS JTSN INBD         | Upper Console (copilot's side) |

6. Turn on electrical power.

7. Adjust upper console INST LT NON FLT control to BRT.

**Indication/Condition**

Integrally illuminated panel lighting on front of auxiliary fuel management control panel shall go on at full brightness.

**Corrective Action**

If Indication/Condition is not as specified, troubleshoot instrument panel lights (WP 0105 00).

# EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE **MWO 50-78** - Continued

## Step

8. Push in the following circuit breakers:

| CIRCUIT BREAKER    | CIRCUIT BREAKER PANEL |
|--------------------|-----------------------|
| NO. 2 EXT FUEL RH  | Mission Readiness     |
| NO. 2 XFER CONTROL | Mission Readiness     |
| AUX FUEL QTY       | Pilot's               |
| LIGHTS ADVSY       | Copilot's             |

9. Adjust upper console INST LT NON FLT control to BRT.

## Indication/Condition

All lit fuel quantity displays and all lit annunciators on front of auxiliary fuel management control panel shall go on at full brightness.

## Corrective Action

1. If Indication/Condition is not as specified, troubleshoot instrument panel lights (WP 0105 00).
2. "blank display" - Run 1-BIT to confirm presence of 28 vdc, then check for open between P856 and W4P2.

## Step

10. Momentarily place caution/advisory panel BRT/DIM-TEST switch to BRT/DIM.

## Indication/Condition

All lit fuel quantity displays and all lit annunciators on front of auxiliary fuel management control panel shall dim.

## Corrective Action

If Indication/Condition is not as specified, troubleshoot instrument panel and console indicator lights dimming (WP 0110 00) or caution/advisory warning system (WP 0090 00).

## Step

11. Push in the following circuit breakers:

| CIRCUIT BREAKER      | CIRCUIT BREAKER PANEL |
|----------------------|-----------------------|
| <b>UH-60A UH-60L</b> |                       |
| EXT FUEL RH          | Mission Readiness     |
| NO. 2 XFER CONTROL   | Mission Readiness     |
| EXT FUEL LH          | Mission Readiness     |

# EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE **MWO 50-78** - Continued

## CIRCUIT BREAKER

## CIRCUIT BREAKER PANEL

### **EH-60A**

|                |           |
|----------------|-----------|
| EXT FUEL LH    | Copilot's |
| EXT FUEL RH    | Pilot's   |
| AUX FUEL CONTR | Pilot's   |

12. Auxiliary fuel management control panel TEST/RESET button - press and release.

### Indication/Condition

- All auxiliary fuel management control panel annunciators and fuel quantity displays illuminate for three seconds - all 35 pixels of each of the four fuel quantity 5 x 7 array displays illuminate momentarily.
- Following the lamp test BIT, none of the error codes shown in ERROR CODES, in this work package are displayed.

### Corrective Action

- If some of the pixels or annunciators in Indication/Condition a. fail to light, replace auxiliary fuel management control panel (WP 1448 00).
- If none of the pixels or annunciators in Indication/Condition a. light, go to WITH WEIGHT-ON-WHEELS AUXILIARY FUEL MANAGEMENT CONTROL PANEL SELF-TEST DOES NOT RUN, in this work package if no fault is found, replace auxiliary fuel management control panel (WP 1448 00).
- If Indication/Condition b. is not as specified, go to PRESSING AUXILIARY FUEL MANAGEMENT CONTROL PANEL TEST/RESET PUSHBUTTON RESULTS IN ERROR CODES, in this work package.

**Table 1. Error Codes.**

| CODE | MEANING                           |
|------|-----------------------------------|
| E01  | Internal, Panel CPU test fail     |
| E02  | Internal, Panel memory test fail  |
| E03  | Internal, Panel display test fail |
| E04  | Internal, Tank gaging fail        |
| E05  | External, Probe test fail, open   |
| E06  | External, Probe test fail, short  |
| E07  | External, Attitude sensor fail    |

### Step

13. Turn off electrical power.

# EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE **MWO 50-78** - Continued

14. Fill each outboard tank, if installed to 190 pounds of fuel.
15. Fill each inboard tank, if installed to 250 pounds of fuel.
16. Check that EMPTY annunciators on the auxiliary fuel management control panel do not light for the fueled tanks.

## Indication/Condition

Annunciators shall not light for fueled tanks.

## Corrective Action

If Indication/Condition is not specified, go to AUXILIARY FUEL MANAGEMENT CONTROL PANEL EMPTY ANNUNCIATOR IS ON FOR A TANK WITH ADEQUATE FUEL, in this work package.

## Step

### CAUTION

Prior to activating fuel gate valves that have been out of service for more than 30 days, it is recommended to open and close fuel gate valves a few times by hand prior to applying electrical power to reduce the possibility of damage to auxiliary fuel management control panel and circuitry.

17. Disconnect P857 from auxiliary fuel management control panel.
18. Connect P1 of locally-made ESSS gate valve test harness to UTIL RCPT +28V receptacle J258 in cabin ceiling.
19. Turn on electrical power.
20. Push in UTIL RECP CABIN circuit breaker on copilot's circuit breaker panel.
21. Connect locally-made test harness pin E to P857-33.
22. Connect pins A and D of locally-made test harness to the following pairs of pins on P857, and alternately place locally-made test harness switch to OPEN and CLOSE.

| TEST HARNESS PIN | P857- | VALVE OPERATION         |
|------------------|-------|-------------------------|
| A                | 3     | LEFT OUTBD - OPEN       |
| D                | 5     | LEFT OUTBD -<br>CLOSED  |
| A                | 8     | LEFT INBD - OPEN        |
| D                | 7     | LEFT INBD -<br>CLOSED   |
| A                | 14    | RIGHT OUTBD -<br>OPEN   |
| D                | 13    | RIGHT OUTBD -<br>CLOSED |
| A                | 6     | RIGHT INBD - OPEN       |



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**EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE MWO 50-78** - Continued
 

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| TEST HARNESS PIN | P857- | VALVE OPERATION        |
|------------------|-------|------------------------|
| D                | 4     | RIGHT INBD -<br>CLOSED |

**Indication/Condition**

Gate valves shall open and close as indicated.

**Corrective Action**

1. If Indication/Condition is not as specified, connect locally-made test harness pins A, D, and E to pins A, D, and E of malfunctioning gate valve.

**Step**

23. Alternately place locally-made test harness switch to OPEN and CLOSED.

**Indication/Condition**

Gate valves shall open and close as indicated.

**Corrective Action**

1. If gate valve operates normally, repair/replace wiring, as required, between P857 and gate valve's electrical connector (WP 1747 00).
2. If gate valve does not operate normally, replace gate valve (WP 1440 00).

**Step**

24. Turn off electrical power.
25. Disconnect locally-made test harness pin E from P857-33.
26. Connect P857 to auxiliary fuel management control panel.
27. Disconnect P1 of locally-made ESSS gate valve test harness from UTIL RCPT +28V receptacle J258 in cabin ceiling.
28. Pull out UTIL RECP CABIN circuit breaker on copilot's circuit breaker panel.
29. Set XFER MODE switch to OFF.
30. Turn on electrical power.
31. Apply air source, regulated to 25 - 45 psi to helicopter's bleed air manifold.
32. Set auxiliary fuel management control panel PRESS switch to ALL (or to tank set installed), allow tanks to pressurize.
33. Set XFER FROM to INBD or OUTBD, as installed.
34. Check auxiliary fuel management control panel and CDU fuel quantity displays.

**Indication/Condition**

Fuel quantities do not change.

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**EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE MWO 50-78** - Continued**Corrective Action**

If Indication/Condition is not as specified, turn off air, go to CDU TOTAL FUEL DISPLAY INCREASES WHEN XFER MODE SWITCH IS SET OFF, in this work package.

**Step**

35. Record CDU TOTAL FUEL quantity and auxiliary fuel management control panel fuel quantities (should be **650** pounds in each main tank; **190** pounds in each OUTBD tank; and **250** pounds in each INBD tank).
36. Inspect fuel and pneumatic hoses and fittings for signs of leaking or damage (WP 1431 00 or WP 1432 00).

**Indication/Condition**

There shall be no signs of leakage or damage.

**Corrective Action**

If Indication/Condition is not as specified, replace hoses and fittings as necessary (WP 1431 00 or WP 1432 00).

**Step****NOTE**

- Continue inspecting ESSS components throughout this procedure. If Indication/Condition is not as specified in step 36., halt the procedure, correct the problem, then continue with the procedure.
- Disregard caution/advisory panel AUX FUEL capsule if it goes on momentarily.

37. Set auxiliary fuel management control panel XFER MODE switch to OFF.
38. Turn on electrical power, if off.
39. Apply compressed air, if off.
40. Set auxiliary fuel management control panel as follows:
  - PRESS - OUTBD
  - MAN XFER - RIGHT
  - XFER FROM - OUTBD
  - XFER MODE - MAN
41. Allow CDU TOTAL FUEL indication to increase by 20 pounds.
42. Set auxiliary fuel management control panel XFER MODE switch to OFF.
43. Check auxiliary fuel management control panel R OUTBD fuel quantity display and CDU TOTAL FUEL indicator.

**Indication/Condition**

- a. Elapsed time required to increase CDU TOTAL FUEL indication by 20 pounds shall be less than 1 1/2 minutes.
- b. Auxiliary fuel management control panel R OUTBD fuel quantity display decrease equals CDU TOTAL FUEL quantity increase. Remaining auxiliary fuel management control panel annunciators unchanged.

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**EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE MWO 50-78** - Continued**Corrective Action**

If Indication/Conditions are not as specified, go to R OUTBD TO MAIN, XFER MODE SWITCH TO MAN, CDU TOTAL FUEL DISPLAY DOES NOT INCREASE OR THE RESULT IS NOT AS SPECIFIED, in this work package.

**Step**

44. Set auxiliary fuel management control panel MAN XFER switch to LEFT.
45. Set auxiliary fuel management control panel XFER MODE switch to MAN.
46. Allow CDU TOTAL FUEL indication to increase by 20 pounds.
47. Set auxiliary fuel management control panel XFER MODE switch to OFF.
48. Check auxiliary fuel management control panel L OUTBD fuel quantity display and CDU TOTAL FUEL indicators.

**Indication/Condition**

- a. Elapsed time required to increase CDU TOTAL FUEL indication by 20 pounds shall be less than 1 1/2 minutes.
- b. Auxiliary fuel management control panel L OUTBD fuel quantity display decrease equals CDU TOTAL FUEL increase. Remaining auxiliary fuel management control panel annunciators unchanged.

**Corrective Action**

If Indication/Conditions are not as specified, go to L OUTBD TO MAIN, XFER MODE SWITCH TO MAN, CDU TOTAL FUEL DISPLAY DOES NOT INCREASE OR THE RESULT IS NOT AS SPECIFIED WHEN ATTEMPTING TO TRANSFER FUEL, in this work package.

**Step**

49. Set auxiliary fuel management control panel PRESS switch to INBD, allow pressurization.
50. Set auxiliary fuel management control panel XFER FROM switch to INBD.
51. Set auxiliary fuel management control panel XFER MODE switch to MAN.
52. Allow CDU TOTAL FUEL indication to increase by 20 pounds.
53. Set auxiliary fuel management control panel XFER MODE switch to OFF.
54. Check auxiliary fuel management control panel L INBD fuel quantity display and CDU TOTAL FUEL indicator.

**Indication/Condition**

- a. Elapsed time required to increase CDU TOTAL FUEL indication by 20 pounds shall be less than 1 1/2 minutes.
- b. Auxiliary fuel management control panel L INBD fuel quantity display decrease equals CDU TOTAL FUEL increase. Remaining auxiliary fuel management control panel annunciators unchanged.

**Corrective Action**

If Indication/Conditions are not as specified, go to L INBD TO MAIN, XFER MODE SWITCH TO MAN, CDU TOTAL FUEL DISPLAY DOES NOT INCREASE OR THE RESULT IS NOT AS SPECIFIED, in this work package.

**Step**

55. Set auxiliary fuel management control panel MAN XFER switch to RIGHT.

---

**EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE MWO 50-78** - Continued

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56. Set auxiliary fuel management control panel XFER MODE switch to MAN.
57. Allow CDU TOTAL FUEL indication to increase by 20 pounds.
58. Set auxiliary fuel management control panel XFER MODE switch to OFF.
59. Check auxiliary fuel management control panel R INBD fuel quantity display and CDU TOTAL FUEL indicator.

**Indication/Condition**

- a. Elapsed time required to increase CDU TOTAL FUEL indication by 20 pounds shall be less than 1 1/2 minutes.
- b. Auxiliary fuel management control panel R INBD fuel quantity display decrease equals CDU TOTAL FUEL increase. Remaining auxiliary fuel management control panel annunciators unchanged.

**Corrective Action**

If Indication/Conditions are not as specified, go to R INBD TO MAIN, XFER MODE SWITCH TO MAN, CDU TOTAL FUEL DISPLAY DOES NOT INCREASE OR RESULT IS NOT AS SPECIFIED, in this work package.

**Step**

60. Set auxiliary fuel management control panel MAN XFER switch to BOTH.
61. Set auxiliary fuel management control panel XFER MODE switch to MAN.
62. Allow CDU TOTAL FUEL indication to increase by 20 pounds.
63. Set auxiliary fuel management control panel XFER MODE switch to OFF.
64. Check auxiliary fuel management control panel L and R INBD fuel quantity displays and CDU TOTAL FUEL indicator.

**Indication/Condition**

- a. Elapsed time required to increase CDU TOTAL FUEL indication by 20 pounds shall be less than 1 1/2 minutes.
- b. The sum of auxiliary fuel management control panel L and R INBD fuel quantity displays decrease equal CDU TOTAL FUEL increase. Auxiliary fuel management control panel OUTBD annunciators unchanged.

**Corrective Action**

If Indication/Conditions are not as specified, go to L INBD TO MAIN, XFER MODE SWITCH TO MAN, CDU TOTAL FUEL DISPLAY DOES NOT INCREASE OR THE RESULT IS NOT AS SPECIFIED and R INBD TO MAIN, XFER MODE SWITCH TO MAN, CDU TOTAL FUEL DISPLAY DOES NOT INCREASE OR RESULT IS NOT AS SPECIFIED, in this work package.

**Step**

65. Set auxiliary fuel management control panel PRESS switch to OUTBD.
66. Set auxiliary fuel management control panel MAN XFER switch to BOTH.
67. Set auxiliary fuel management control panel XFER MODE switch to MAN.
68. Allow CDU TOTAL FUEL indication to increase by 20 pounds.
69. Set auxiliary fuel management control panel XFER MODE switch to OFF.
70. Check auxiliary fuel management control panel L and R OUTBD fuel quantity displays and CDU TOTAL FUEL indicator.

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**EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE MWO 50-78** - Continued

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**Indication/Condition**

- a. Elapsed time required to increase CDU TOTAL FUEL indication by 20 pounds shall be less than 1 1/2 minutes.
- b. The sum of auxiliary fuel management control panel L and R OUTBD fuel quantity displays decrease equal CDU TOTAL FUEL increase. Auxiliary fuel management control panel OUTBD annunciators unchanged.

**Corrective Action**

If Indication/Conditions are not as specified, go to R OUTBD TO MAIN, XFER MODE SWITCH TO MAN, CDU TOTAL FUEL DISPLAY DOES NOT INCREASE OR THE RESULT IS NOT AS SPECIFIED and L OUTBD TO MAIN, XFER MODE SWITCH TO MAN, CDU TOTAL FUEL DISPLAY DOES NOT INCREASE OR THE RESULT IS NOT AS SPECIFIED WHEN ATTEMPTING TO TRANSFER FUEL, in this work package.

**Step**

71. Set auxiliary fuel management control panel XFER MODE switch to AUTO until CDU TOTAL FUEL indication increases by 40 pounds, then, set XFER MODE switch to OFF.
72. Check auxiliary fuel management control panel L and R INBD or OUTBD displays and CDU TOTAL FUEL indicator.

**Indication/Condition**

- a. Elapsed time required to increase CDU TOTAL FUEL indication by 40 pounds shall be less than 3 minutes.
- b. Quantity in auxiliary fuel management control panel L and R INBD or OUTBD displays shall decrease, CDU TOTAL FUEL quantity shall increase.

**Corrective Action**

If Indication/Conditions are not as specified, replace auxiliary fuel management control panel (WP 1448 00).

**Step**

73. Set auxiliary fuel management control panel PRESS switch to OFF.
74. Turn off electrical power.
75. Connect locally-made fuel quantity test simulator power cord to UTIL RCPT +28V (J258) in cabin ceiling.
76. Disconnect helicopter electrical connector P905 from J905.
77. Connect simulator J905 to helicopter electrical connector P905.
78. Turn simulator TANK 2 and TANK 2 controls fully counterclockwise.
79. Set multimeters to read volts 0-10 vdc range, then connect multimeters to simulator TANK 1 and 2 meter HI/LO jacks.
80. Turn on electrical power and push in UTIL RECP CABIN circuit breaker on copilot's circuit breaker panel.
81. Simulate full main tanks by adjusting simulator TANK 1 and TANK 2 controls to obtain 6.0 vdc indication on both multimeters.
82. Turn on compressed air supply, allow tanks to pressurize.
83. Set auxiliary fuel management control panel XFER FROM switch to INBD.
84. Set auxiliary fuel management control panel PRESS switch to ALL.
85. Set auxiliary fuel management control panel XFER MODE switch to AUTO.

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**EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE MWO 50-78** - Continued

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**Indication/Condition**

No fuel transfer, auxiliary fuel management control panel fuel quantity displays shall not decrease and CDU TOTAL FUEL indicator shall not increase.

**Corrective Action**

If Indication/Condition is not as specified, go to XFER MODE SWITCH AT AUTO, FUEL TRANSFER LIMITS ARE NOT AS SPECIFIED, in this work package.

**Step****NOTE**

The following procedures cause fuel to flow from auxiliary tanks to main tanks when auxiliary tanks are pressurized. Read steps 86. through 88. first, then perform steps as quickly as possible. Conditions that stop AUTO mode fuel transfer:

Main tanks 1 and 2 total quantity greater than 2300 pounds.

Either tank empty of an inboard or outboard pair.

Fuel overflowing from a vent detected.

XFER MODE switch at OFF.

Any BIT error except attitude sensor.

Tanks not installed or detected.

86. Slowly turn simulator TANK 1 control counterclockwise until the fuel quantity of any auxiliary fuel management control panel display begins to decrease. Record voltage reading.
87. Slowly turn simulator TANK 1 control clockwise until all auxiliary fuel management control panel displays stop changing. Record voltage reading.

**Indication/Condition**

- a. Reading recorded in step 85. shall be 5.5 vdc or less.
- b. Reading recorded in step 86. shall be greater than 5.5 vdc.
- c. Auxiliary fuel management control panel L and R NO FLOW annunciator shall remain off.

**Corrective Action**

1. If Indication/Conditions a. and b. are not as specified, go to XFER MODE SWITCH AT AUTO, FUEL TRANSFER LIMITS ARE NOT AS SPECIFIED, in this work package.
2. If Indication/Condition c. is not as specified, go to AUXILIARY FUEL MANAGEMENT CONTROL PANEL R NO FLOW ANNUNCIATOR REMAINS ON DURING FUEL TRANSFER and AUXILIARY FUEL MANAGEMENT CONTROL PANEL L NO FLOW ANNUNCIATOR REMAINS ON DURING FUEL TRANSFER, in this work package.

**Step**

88. Turn simulator TANK 1 control clockwise to obtain 6.0 vdc indication on No. 1 multimeter.

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**EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE MWO 50-78** - Continued**NOTE**

Repeat steps 86. through 88. for tank 2 using TANK 2 control and multimeter.

89. Place auxiliary fuel management control panel XFER MODE switch to OFF.
90. Turn simulator TANK 2 control to obtain 0.7 vdc on No. 2 multimeter.
91. Place auxiliary fuel management control panel XFER FROM switch to INBD.
92. Place auxiliary fuel management control panel XFER MODE switch to AUTO until L or R INBD EMPTY annunciator goes on.

**Indication/Condition**

- a. L or R INBD EMPTY annunciator shall go on.
- b. Auxiliary fuel management control panel L and/or R NO FLOW, and caution/advisory panel AUX FUEL capsule shall go on immediately after either L or R INBD EMPTY annunciator goes on.
- c. Auxiliary fuel management control panel L and R NO FLOW annunciators shall go on immediately after either L or R INBD EMPTY annunciator goes on.

**Corrective Action**

1. If Indication/Condition a. is not as specified, go to AUXILIARY FUEL MANAGEMENT CONTROL PANEL R INBD EMPTY AND/OR L INBD EMPTY ANNUNCIATORS DO NOT GO ON WITH INADEQUATE FUEL IN THE TANK, in this work package.
2. If Indication/Condition b. and/or c. is not as specified, go to CAUTION/ADVISORY PANEL AUX FUEL CAPSULE AND/OR AUXILIARY FUEL MANAGEMENT CONTROL PANEL R NO FLOW AND L NO FLOW ANNUNCIATOR DO NOT GO ON, in this work package.

**Step**

93. Turn simulator TANK 2 control to obtain 6.0 vdc No. 2 multimeter indication.
94. Turn simulator TANK 1 control to obtain 6.0 vdc No. 1 multimeter indication.
95. Place auxiliary fuel management control panel XFER MODE switch to MAN.
96. Place auxiliary fuel management control panel MAN XFER switch to BOTH.

**Indication/Condition**

- a. Auxiliary fuel management control panel L and R NO FLOW annunciators shall go off, then on within 5 minutes.
- b. Auxiliary fuel management control panel L and R fuel quantity displays shall not change when R NO FLOW annunciator is on.

**Corrective Action**

If Indication/Condition is not as specified, replace corresponding fuel tank (WP 1465 00).

**Step**

97. Set auxiliary fuel management control panel XFER MODE switch to OFF.
98. Turn simulator TANK 2 control to obtain 2.7 vdc No. 2 multimeter indication.
99. Set auxiliary fuel management control panel XFER FROM switch to OUTBD.

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**EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE MWO 50-78** - Continued

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100. Set auxiliary fuel management control panel XFER MODE switch to AUTO until L and R OUTBD EMPTY annunciators go on.

**Indication/Condition**

- a. L and R OUTBD EMPTY annunciators shall go on.
- b. Caution/advisory panel AUX FUEL capsule shall go on immediately after any EMPTY annunciator goes on.
- c. Auxiliary fuel management control panel R and L NO FLOW annunciators shall go on immediately after corresponding OUTBD EMPTY annunciator goes on.

**Corrective Action**

1. If Indication/Condition a. is not as specified, go to AUXILIARY FUEL MANAGEMENT CONTROL PANEL R OUTBD EMPTY AND/OR L OUTBD EMPTY ANNUNCIATORS DO NOT GO ON WITH INADEQUATE FUEL IN TANK, in this work package.
2. If Indication/Condition b. and/or c. is not as specified, go to CAUTION/ADVISORY PANEL AUX FUEL CAPSULE AND/OR AUXILIARY FUEL MANAGEMENT CONTROL PANEL R NO FLOW AND L NO FLOW ANNUNCIATORS DO NOT GO ON, in this work package.

**Step**

101. Set auxiliary fuel management control panel XFER MODE switch to OFF, and PRESS switch to OFF.

**Indication/Condition**

All four ESSS bleed-air regulator valves shall be venting air.

**Corrective Action**

If Indication/Condition is not as specified, replace bleed-air regulator valve as required (WP 1439 00).

**Step****NOTE**

Fill container with clean water in preparation for testing the fuel vent overflow sensor in steps 102. through 109.

102. Pull out NO. 2 EXT FUEL RH and NO. 2 XFER CONTROL circuit breakers on mission readiness circuit breaker panel. Pull out AUX FUEL CONTR circuit breaker on pilot's circuit breaker panel.
103. Remove overflow sensor from vent tube elbow in rear of main fuel tank (WP 1457 00).
104. Disconnect electrical connector P376 from J376.
105. Push in either circuit breaker pulled out in step 102.

**Indication/Condition**

Auxiliary fuel management control panel VENT FAIL annunciator shall go on.

**Corrective Action**

If Indication/Condition is not as specified, go to AUXILIARY FUEL MANAGEMENT CONTROL PANEL VENT FAIL OR VENT OVFL ANNUNCIATOR DOES NOT GO ON, in this work package.



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**EXTERNAL STORES SUPPORT SYSTEM (ESSS) RANGE EXTENSION SYSTEM OPERATIONAL/ TROUBLESHOOTING PROCEDURE MWO 50-78** - Continued**Step**

106. Pull out circuit breakers pushed in step 105.
107. Connect electrical connector P376 to J376.
108. Immerse thermistor portion of overflow sensor in container of fuel.
109. Push in circuit breaker pulled out in step 106.

**Indication/Condition**

Auxiliary fuel management control panel VENT OVFL annunciator shall go on.

**Corrective Action**

If Indication/Condition is not as specified, go to AUXILIARY FUEL MANAGEMENT CONTROL PANEL VENT FAIL OR VENT OVFL ANNUNCIATOR DOES NOT GO ON, in this work package.

**Step**

110. Install overflow sensor (WP 1457 00).
111. Turn off electrical power.
112. **CU** Turn off compressed air source. ■
113. Allow bleed-air regulator valves to vent air to ambient pressure.
114. **CU** Disconnect air compressor, hoses, and locally-made pneumatic adapter. ■
115. **CU** Connect pneumatic hose to outboard side of helicopter bleed-air QD. ■
116. **CU** TORQUE NUTS TO 135 -150 INCH-POUNDS. ■
117. Disconnect test equipment.
118. Connect helicopter electrical connectors.
119. For check of auxiliary fuel management control panel jettison ID A and B outputs, refer to WP 0160 00.
120. Install explosive cartridges (WP 1445 00).
121. Fasten helicopter fairings (WP 1428 00).

**Indication/Condition**

Helicopter fairings shall be fastened.

**Corrective Action**

None Required

**WITH WEIGHT-ON-WHEELS AUXILIARY FUEL MANAGEMENT CONTROL PANEL SELF-TEST DOES NOT RUN****SYMPTOM**

With weight-on-wheels auxiliary fuel management control panel self-test does not run.

**MALFUNCTION**

With weight-on-wheels auxiliary fuel management control panel self-test does not run.

**CORRECTIVE ACTION**

- 1. Disconnect P856 from rear of auxiliary fuel management control panel. Check continuity between P856-11 and P856-12.**
  - a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to **5**.
  - b. If continuity is not present go to **2**.
- 2. Check continuity between:**
  - **P856-11 and P902-C**
  - **P856-12 and P902-D**
  - a. If continuity is present, go to **3**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **5**.
- 3. Check continuity between:**
  - **P242-L and J163-C**
  - **J163 and ground**
  - a. If continuity is present, go to **4**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **5**.
- 4. Check continuity between P163-A and P163-C.**
  - a. If continuity is present, replace left relay panel (WP 0825 00). Go to **5**.
  - b. If continuity is not present, replace left drag beam switch (WP 0443 00). Go to **5**.
- 5. Procedure completed.**

**PRESSING AUXILIARY FUEL MANAGEMENT CONTROL PANEL TEST/RESET PUSHBUTTON RESULTS IN ERROR CODES****SYMPTOM**

Pressing auxiliary fuel management control panel TEST/RESET pushbutton results in error codes.

**MALFUNCTION**

Pressing auxiliary fuel management control panel TEST/RESET pushbutton results in error codes.

**CORRECTIVE ACTION**

- 1. Does I-BIT result in error codes?**
  - a. If error codes are present, go to **2**.

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**PRESSING AUXILIARY FUEL MANAGEMENT CONTROL PANEL TEST/RESET PUSHBUTTON RESULTS IN ERROR CODES - Continued**

- b. If error codes are not present, go to **4**.

**2. Check error codes.**

- a. If error codes E01, E02, E03, E04, or ---- are present, replace auxiliary fuel management control panel (WP 1448 00). Go to **6**.
- b. If error codes E05, E06, or FP, go to **3**.
- c. If error codes E07, go to **4**.

**3. Check continuity between P856 and W4P2 for each external fuel tank.**

- a. If continuity is present, replace tank probe (WP 1552 00). Go to **6**.
- b. If continuity is not present, repair/replace wiring or wiring harness (WP 1747 00). Go to **6**.

**4. Check operation of copilot's gyro (TM 11-1520-237-23).**

- a. If gyro is operating properly, go to **5**.
- b. If gyro is not operating properly, troubleshoot gyro (TM 11-1520-237-23). Go to **6**.

**5. Check continuity between:**

- P856-25 and P867-R
- P856-26 and P867-H
- P856-27 and P867-B
- P856-28 and P867-T
- P856-29 and P867-P
- P856-30 and P867-J

- a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to **6**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **6**.

**6. Procedure completed.****AUXILIARY FUEL MANAGEMENT CONTROL PANEL EMPTY ANNUNCIATOR IS ON FOR A TANK WITH ADEQUATE FUEL****SYMPTOM**

Auxiliary fuel management control panel EMPTY annunciator is on for a tank with adequate fuel.

**MALFUNCTION**

Auxiliary fuel management control panel EMPTY annunciator is on for a tank with adequate fuel.

**CORRECTIVE ACTION****1. For the tank with the EMPTY indication measure resistance between:**

- W4P2-C and W4P2-F
- W4P2-E and W4P2-F

- a. If resistance is less than 300 ohms or greater than 500 ohms, replace tank probe (WP 1552 00). Go to **4**.

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**AUXILIARY FUEL MANAGEMENT CONTROL PANEL EMPTY ANNUNCIATOR IS ON FOR A TANK WITH ADEQUATE FUEL** - Continued

b. If resistance is greater than 300 ohms but less than 500 ohms, go to **2**.

**2. Check continuity between:**

- **P856-31 and W4P2-C (Right Outboard)**
- **P856-32 and W4P2-F (Right Outboard)**
- **P856-33 and W4P2-E (Right Outboard)**
- **P856-34 and W4P2-C (Right Inboard)**
- **P856-35 and W4P2-F (Right Inboard)**
- **P856-36 and W4P2-E (Right Inboard)**
- **P856-40 and W4P2-C (Left Outboard)**
- **P856-41 and W4P2-F (Left Outboard)**
- **P856-42 and W4P2-E (Left Outboard)**
- **P856-37 and W4P2-C (Left Inboard)**
- **P856-38 and W4P2-F (Left Inboard)**
- **P856-39 and W4P2-E (Left Inboard)**

a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to **4**.

b. If continuity is not present, repair/replace fuel flow transmitter (WP 1438 00). Go to **4**.

**3. Check continuity between:**

- **P361-X and P346-23**
- **P361-Y and P346-24**

a. If continuity is present, repair/replace wiring between (WP 1747 00), then go to **4**:

- **J346-23 and P523-A**
- **J346-24 and P523-B**

b. If continuity is not present, repair/replace wiring or wiring harness (WP 1747 00). Go to **4**.

**4. Procedure completed.****CDU TOTAL FUEL DISPLAY INCREASES WHEN XFER MODE SWITCH IS SET OFF****SYMPTOM**

CDU TOTAL FUEL display increases when XFER MODE switch is set OFF.

**MALFUNCTION**

CDU TOTAL FUEL display increases when XFER MODE switch is set OFF.

**CORRECTIVE ACTION**

- 1. Do any auxiliary fuel management control panel fuel quantity displays decrease when CDU TOTAL FUEL display increases?**

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**CDU TOTAL FUEL DISPLAY INCREASES WHEN XFER MODE SWITCH IS SET OFF - Continued**

- a. If auxiliary fuel management control panel fuel quantity displays decrease, go to **2**.
  - b. If auxiliary fuel management control panel fuel quantity displays do not decrease, troubleshoot fuel quantity system (WP 0131 00). Go to **6**.
- 2. Check manual lever of applicable shutoff gate valve in closed position.**
- a. If manual lever is in closed position, replace shutoff gate valve (WP 1440 00). Go to **6**.
  - b. If manual lever is not in closed position, go to **3**.
- 3. Check for 28 vdc at valve with manual lever not at closed position:**
- **P840-D and P840-E Left Outboard**
  - **P809-D and P809-E Left inboard**
  - **P838-D and P838-E Right Outboard**
  - **P810-D and P810-E Right inboard**
- a. If voltage is as specified, go to **4**.
  - b. If voltage is not as specified, go to **5**.
- 4. Check for 0 vdc between:**
- **P840-A and P840-E Left Outboard**
  - **P809-A and P809-E Left inboard**
  - **P838-A and P838-E Right Outboard**
  - **P810-A and P810-E Right inboard**
- a. If voltage is as specified, replace shutoff gate valve (WP 1440 00). Go to **6**.
  - b. If voltage is not as specified, replace auxiliary fuel management control panel (WP 1448 00). Go to **5**.
- 5. Check continuity between:**
- **P809-A and P857-8**
  - **P809-D and P857-7**
  - **P809-E and ground**
  - **P810-A and P857-6**
  - **P810-D and P857-4**
  - **P810-E and ground**
  - **P838-A and P857-14**
  - **P838-D and P857-13**
  - **P838-E and ground**
  - **P840-A and P857-3**
  - **P840-D and P857-5**

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**CDU TOTAL FUEL DISPLAY INCREASES WHEN XFER MODE SWITCH IS SET OFF** - Continued

- **P840-E and ground**

- a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to **6**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **6**.

**6. Procedure completed.**

**R OUTBD TO MAIN, XFER MODE SWITCH TO MAN, CDU TOTAL FUEL DISPLAY DOES NOT INCREASE OR THE RESULT IS NOT AS SPECIFIED**

**SYMPTOM**

R OUTBD to main, XFER MODE switch to MAN, CDU TOTAL FUEL display does not increase or the Indication/Condition is not as specified.

**MALFUNCTION**

R OUTBD to main, XFER MODE switch to MAN, CDU TOTAL FUEL display does not increase or the Indication/Condition is not as specified.

**CORRECTIVE ACTION**

**1. Does CDU TOTAL FUEL display increase?**

- a. If CDU TOTAL FUEL display does increase, go to **2**.
- b. If CDU TOTAL FUEL display does not increase, go to **8**.

**2. Does auxiliary fuel management control panel R OUTBD display decrease?**

- a. If auxiliary fuel management control panel R OUTBD display decreases, go to **3**.
- b. If auxiliary fuel management control panel R OUTBD display does not decrease, go to **10**.

**3. EMEP Does auxiliary fuel management control panel R NO FLOW annunciator and the caution/advisory panel AUX FUEL capsule momentarily go on, then off.**

- a. If AUX FUEL capsule went on then off, go to **4**.
- b. If AUX FUEL capsule did not go on then off, go to **7**.

**4. EMEP Do the remaining auxiliary fuel management control panel lights remain off?**

- a. If remaining lights remain off, go to **5**.
- b. If remaining lights do not remain off, replace auxiliary fuel management control panel (WP 1448 00). Go to **21**.

**5. Does the transfer of 20 pounds of fuel exceed 1.5 minutes?**

- a. If the transfer exceeds 1.5 minutes, go to **6**.
- b. If the transfer is 1.5 minutes or less, go to **21**.

**6. Do any related fuel or pneumatic hoses leak?**

- a. If a leak is found, replace hose and connector as required (WP 1431 00 or WP 1432 00). Go to **21**.
- b. If no leaks are found, go to **10**.

**7. Does auxiliary fuel management control panel R NO FLOW annunciator remain on?**

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**R OUTBD TO MAIN, XFER MODE SWITCH TO MAN, CDU TOTAL FUEL DISPLAY DOES NOT INCREASE OR THE RESULT IS NOT AS SPECIFIED - Continued**

- a. If annunciator remains on, go to **AUXILIARY FUEL MANAGEMENT CONTROL PANEL R NO FLOW ANNUNCIATOR REMAINS ON DURING FUEL TRANSFER, in this work package.**
  - b. If annunciator is not on, replace auxiliary fuel management control panel (WP 1448 00). Go to **21.**
- 8. Does auxiliary fuel management control panel R OUTBD decrease?**
- a. If R OUTBD decreases, go to **9.**
  - b. If R OUTBD decreases, go to **10.**
- 9. Does auxiliary fuel management control panel R NO FLOW annunciator go on?**
- a. If R NO FLOW annunciator goes on, replace auxiliary fuel management control panel (WP 1448 00). Go to **21.**
  - b. If R NO FLOW annunciator is not on, troubleshoot fuel quantity system (WP 0131 00). Go to **21.**
- 10. Is manual lever of right outboard fuel shutoff valve in full OPEN position?**
- a. If manual lever is in full OPEN position, go to **11.**
  - b. If manual lever is not in full OPEN position, go to **16.**
- 11. Is right outboard bleed-air regulator valve continuously venting air?**
- a. If valve is continuously venting air, go to **12.**
  - b. If valve is not continuously venting air, replace right bleed-air regulator valve (WP 1439 00). Go to **21.**
- 12. Check for 28 vdc between P837-A and P837-B.**
- a. If voltage is as specified, replace right bleed-air regulator valve (WP 1439 00). Go to **21.**
  - b. If voltage is not as specified, go to **13.**
- 13. Check for 28 vdc between:**
- **J346-33 and ground**
  - **J346-35 and ground**
- a. If voltage is as specified, go to **15.**
  - b. If voltage is not as specified, go to **14.**
- 14. Check for 28 vdc between:**
- **CB14A-1 and ground**
  - **CB13A-1 and ground**
- a. If voltage is as specified, repair/replace wiring between, then go to **21.**
    - **J346-33 and CB14A-1**
    - **J346-35 and CB13A-1**
  - b. If voltage is not as specified, replace circuit breaker as necessary. Go to **21.**
- 15. Check continuity between:**

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**R OUTBD TO MAIN, XFER MODE SWITCH TO MAN, CDU TOTAL FUEL DISPLAY DOES NOT INCREASE OR THE RESULT IS NOT AS SPECIFIED - Continued**

- **P837-A and P857-9**
  - **P837-B and ground**
  - **P860-33 and P857-17**
  - **P860-35 and P857-16**
  - **P857-32 and ground**
  - **P857-33 and ground**
- a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to **21**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **21**.
- 16. Check 28 vdc between P838-A and P838-E.**
- a. If voltage is as specified, go to **17**.
- b. If voltage is not as specified, go to **18**.
- 17. Check 0 vdc between P838-D and P838-E.**
- a. If voltage is as specified, replace right outboard fuel shutoff gate valve (WP 1440 00). Go to **21**.
- b. If voltage is not as specified, go to **18**.
- 18. Check 28 vdc between:**
- **J346-33 and ground**
  - **J346-35 and ground**
- a. If voltage is as specified, go to **20**.
- b. If voltage is not as specified, go to **19**.
- 19. Check 28 vdc between:**
- **CB14A-1 and ground**
  - **CB13A-1 and ground**
- a. If voltage is as specified, repair/replace wiring between then go to **21**.
- **CB14A-1 and J346-33**
  - **CB13A-1 and J346-35**
- b. If voltage is not as specified, replace circuit breaker as necessary (WP 0824 00). Go to **21**.
- 20. Check continuity between:**
- **P838-A and P857-14**
  - **P838-D and P857-13**
  - **P838-E and ground**
  - **P857-32 and ground**
  - **P857-33 and ground**



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**R OUTBD TO MAIN, XFER MODE SWITCH TO MAN, CDU TOTAL FUEL DISPLAY DOES NOT INCREASE OR THE RESULT IS NOT AS SPECIFIED - Continued**

- a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to **21**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **21**.

**21. Procedure completed.****L OUTBD TO MAIN, XFER MODE SWITCH TO MAN, CDU TOTAL FUEL DISPLAY DOES NOT INCREASE OR THE RESULT IS NOT AS SPECIFIED WHEN ATTEMPTING TO TRANSFER FUEL****SYMPTOM**

L OUTBD to main, XFER MODE switch to MAN, CDU TOTAL FUEL display does not increase or the Indication/Condition is not as specified when attempting to transfer fuel.

**MALFUNCTION**

L OUTBD to main, XFER MODE switch to MAN, CDU TOTAL FUEL display does not increase or the Indication/Condition is not as specified when attempting to transfer fuel.

**CORRECTIVE ACTION****1. Does CDU TOTAL FUEL display increase?**

- a. If CDU TOTAL FUEL display does increase, go to **2**.
- b. If CDU TOTAL FUEL display does not increase, go to **8**.

**2. Does auxiliary fuel management control panel R INBD display decrease?**

- a. If auxiliary fuel management control panel L OUTBD display decreases, go to **3**.
- b. If auxiliary fuel management control panel L OUTBD display does not decrease, go to **10**.

**3. EMEP Does auxiliary fuel management control panel L NO FLOW annunciator and the caution/advisory panel AUX FUEL capsule momentarily go on, then off.**

- a. If AUX FUEL capsule went on then off, go to **4**.
- b. If AUX FUEL capsule did not go on then off, go to **7**.

**4. EMEP Do the remaining auxiliary fuel management control panel lights remain off?**

- a. If remaining lights remain off, go to **5**.
- b. If remaining lights do not remain off, replace auxiliary fuel management control panel (WP 1448 00). Go to **21**.

**5. Does the transfer of 20 pounds of fuel exceed 1.5 minutes?**

- a. If the transfer exceeds 1.5 minutes, go to **6**.
- b. If the transfer is 1.5 minutes or less, go to **21**.

**6. Do any related fuel or pneumatic hoses leak?**

- a. If a leak is found, replace hose and connector as required (WP 1431 00 or WP 1432 00). Go to **21**.
- b. If no leaks are found, go to **10**.

**7. Does auxiliary fuel management control panel L NO FLOW annunciator remain on?**

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**L OUTBD TO MAIN, XFER MODE SWITCH TO MAN, CDU TOTAL FUEL DISPLAY DOES NOT INCREASE OR THE RESULT IS NOT AS SPECIFIED WHEN ATTEMPTING TO TRANSFER FUEL - Continued**

- a. If annunciator remains on, go to **AUXILIARY FUEL MANAGEMENT CONTROL PANEL R NO FLOW ANNUNCIATOR REMAINS ON DURING FUEL TRANSFER**, in this work package.
  - b. If annunciator is not on, replace auxiliary fuel management control panel (WP 1448 00). Go to **21**.
- 8. Does auxiliary fuel management control panel L OUTBD decrease?**
- a. If L OUTBD decreases, go to **9**.
  - b. If L OUTBD decreases, go to **10**.
- 9. Does auxiliary fuel management control panel L NO FLOW annunciator go on?**
- a. If L NO FLOW annunciator goes on, replace auxiliary fuel management control panel (WP 1448 00). Go to **21**.
  - b. If L NO FLOW annunciator is not on, troubleshoot fuel quantity system (WP 0131 00). Go to **21**.
- 10. Is manual lever of left inboard fuel shutoff gate valve in full OPEN position?**
- a. If manual lever is in full OPEN position, go to **11**.
  - b. If manual lever is not in full OPEN position, go to **16**.
- 11. Is left outboard bleed-air regulator valve continuously venting air?**
- a. If valve is continuously venting air, go to **12**.
  - b. If valve is not continuously venting air, replace right bleed-air regulator valve (WP 1439 00). Go to **21**.
- 12. Check for 28 vdc between P839-A and P839-B.**
- a. If voltage is as specified, replace right bleed-air regulator valve (WP 1439 00). Go to **21**.
  - b. If voltage is not as specified, go to **13**.
- 13. Check for 28 vdc between:**
- **J346-33 and ground**
  - **J346-35 and ground**
- a. If voltage is as specified, go to **15**.
  - b. If voltage is not as specified, go to **14**.
- 14. Check for 28 vdc between:**
- **CB14A-1 and ground**
  - **CB13A-1 and ground**
- a. If voltage is as specified, repair/replace wiring between, then go to **21**.
    - **J346-33 and CB14A-1**
    - **J346-35 and CB13A-1**
  - b. If voltage is not as specified, replace circuit breaker as necessary (WP 0824 00). Go to **21**.
- 15. Check continuity between:**

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**L OUTBD TO MAIN, XFER MODE SWITCH TO MAN, CDU TOTAL FUEL DISPLAY DOES NOT INCREASE OR THE RESULT IS NOT AS SPECIFIED WHEN ATTEMPTING TO TRANSFER FUEL - Continued**

- **P839-A and P857-10**
- **P839-B and ground**
- **P860-33 and P857-17**
- **P860-35 and P857-16**
- **P857-32 and ground**
- **P857-33 and ground**

- a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to **21**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **21**.

**16. Check 28 vdc between P840-A and P840-E.**

- a. If voltage is as specified, go to **17**.
- b. If voltage is not as specified, go to **18**.

**17. Check 0 vdc between P840-D and P840-E.**

- a. If voltage is as specified, replace right outboard fuel shutoff gate valve (WP 1440 00). Go to **21**.
- b. If voltage is not as specified, go to **18**.

**18. Check 28 vdc between:**

- **J346-33 and ground**
- **J346-35 and ground**

- a. If voltage is as specified, go to **20**.
- b. If voltage is not as specified, go to **19**.

**19. Check 28 vdc between:**

- **CB14A-1 and ground**
- **CB13A-1 and ground**

- a. If voltage is as specified, repair/replace wiring between then go to **21**.
  - **CB14A-1 and J346-33**
  - **CB13A-1 and J346-35**
- b. If voltage is not as specified, replace circuit breaker as necessary (WP 0824 00). Go to **21**.

**20. Check continuity between:**

- **P840-A and P857-3**
- **P840-D and P857-5**
- **P840-E and ground**
- **P857-32 and ground**
- **P857-33 and ground**

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**L OUTBD TO MAIN, XFER MODE SWITCH TO MAN, CDU TOTAL FUEL DISPLAY DOES NOT INCREASE OR THE RESULT IS NOT AS SPECIFIED WHEN ATTEMPTING TO TRANSFER FUEL - Continued**

- a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to **21**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **21**.

**21. Procedure completed.****L INBD TO MAIN, XFER MODE SWITCH TO MAN, CDU TOTAL FUEL DISPLAY DOES NOT INCREASE OR THE RESULT IS NOT AS SPECIFIED****SYMPTOM**

L INBD to main, XFER MODE switch to MAN, CDU TOTAL FUEL display does not increase or the Indication/Condition is not as specified.

**MALFUNCTION**

L INBD to main, XFER MODE switch to MAN, CDU TOTAL FUEL display does not increase or the Indication/Condition is not as specified.

**CORRECTIVE ACTION****1. Does CDU TOTAL FUEL display increase?**

- a. If CDU TOTAL FUEL display does increase, go to **2**.
- b. If CDU TOTAL FUEL display does not increase, go to **8**.

**2. Does auxiliary fuel management control panel L INBD display decrease?**

- a. If auxiliary fuel management control panel L INBD display decreases, go to **3**.
- b. If auxiliary fuel management control panel L INBD display does not decrease, go to **10**.

**3. EMEP Does auxiliary fuel management control panel L NO FLOW annunciator and the caution/advisory panel AUX FUEL capsule momentarily go on, then off.**

- a. If AUX FUEL capsule went on then off, go to **4**.
- b. If AUX FUEL capsule did not go on then off, go to **7**.

**4. EMEP Do the remaining auxiliary fuel management control panel lights remain off?**

- a. If remaining lights remain off, go to **5**.
- b. If remaining lights do not remain off, replace auxiliary fuel management control panel (WP 1448 00). Go to **21**.

**5. Does the transfer of 20 pounds of fuel exceed 1.5 minutes?**

- a. If the transfer exceeds 1.5 minutes, go to **6**.
- b. If the transfer is 1.5 minutes or less, go to **21**.

**6. Do any related fuel or pneumatic hoses leak?**

- a. If a leak is found, replace hose and connector as required (WP 1431 00 or WP 1432 00). Go to **21**.
- b. If no leaks are found, go to **10**.

**7. Does auxiliary fuel management control panel L NO FLOW annunciator remain on?**

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**L INBD TO MAIN, XFER MODE SWITCH TO MAN, CDU TOTAL FUEL DISPLAY DOES NOT INCREASE OR THE RESULT IS NOT AS SPECIFIED - Continued**

- a. If annunciator remains on, go to **AUXILIARY FUEL MANAGEMENT CONTROL PANEL R NO FLOW ANNUNCIATOR REMAINS ON DURING FUEL TRANSFER**, in this work package.
- b. If annunciator is not on, replace auxiliary fuel management control panel (WP 1448 00). Go to **21**.

**8. Does auxiliary fuel management control panel L INBD decrease?**

- a. If L INBD decreases, go to **9**.
- b. If L INBD decreases, go to **10**.

**9. Does auxiliary fuel management control panel L NO FLOW annunciator go on?**

- a. If L NO FLOW annunciator goes on, replace auxiliary fuel management control panel (WP 1448 00). Go to **21**.
- b. If L NO FLOW annunciator is not on, troubleshoot fuel quantity system (WP 0131 00). Go to **21**.

**10. Is manual lever of left inboard fuel shutoff gate valve in full OPEN position?**

- a. If manual lever is in full OPEN position, go to **11**.
- b. If manual lever is not in full OPEN position, go to **16**.

**11. Is left inboard bleed-air regulator valve continuously venting air?**

- a. If valve is continuously venting air, go to **12**.
- b. If valve is not continuously venting air, replace right bleed-air regulator valve (WP 1439 00). Go to **21**.

**12. Check for 28 vdc between P808-A and P808-B.**

- a. If voltage is as specified, replace right bleed-air regulator valve (WP 1439 00). Go to **21**.
- b. If voltage is not as specified, go to **13**.

**13. Check for 28 vdc between:**

- **J346-33 and ground**
  - **J346-35 and ground**
- a. If voltage is as specified, go to **15**.
  - b. If voltage is not as specified, go to **14**.

**14. Check for 28 vdc between:**

- **CB14A-1 and ground**
  - **CB13A-1 and ground**
- a. If voltage is as specified, repair/replace wiring between, then go to **21**.
    - **J346-33 and CB14A-1**
    - **J346-35 and CB13A-1**
  - b. If voltage is not as specified, replace circuit breaker as necessary (WP 0824 00). Go to **21**.

**15. Check continuity between:**

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**L INBD TO MAIN, XFER MODE SWITCH TO MAN, CDU TOTAL FUEL DISPLAY DOES NOT INCREASE OR THE RESULT IS NOT AS SPECIFIED - Continued**

- **P808-A and P857-11**
  - **P808-B and ground**
  - **P860-33 and P857-17**
  - **P860-35 and P857-16**
  - **P857-32 and ground**
  - **P857-33 and ground**
- a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to **21**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **21**.
- 16. Check 28 vdc between P809-A and P809-E.**
- a. If voltage is as specified, go to **17**.
- b. If voltage is not as specified, go to **18**.
- 17. Check 0 vdc between P809-D and P809-E.**
- a. If voltage is as specified, replace right outboard fuel shutoff gate valve (WP 1440 00). Go to **21**.
- b. If voltage is not as specified, go to **18**.
- 18. Check 28 vdc between:**
- **J346-33 and ground**
  - **J346-35 and ground**
- a. If voltage is as specified, go to **20**.
- b. If voltage is not as specified, go to **19**.
- 19. Check 28 vdc between:**
- **CB14A-1 and ground**
  - **CB13A-1 and ground**
- a. If voltage is as specified, repair/replace wiring between then go to **21**.
- **CB14A-1 and J346-33**
  - **CB13A-1 and J346-35**
- b. If voltage is not as specified, replace circuit breaker as necessary (WP 0824 00). Go to **21**.
- 20. Check continuity between:**
- **P809-A and P857-8**
  - **P809-D and P857-7**
  - **P809-E and ground**
  - **P857-32 and ground**
  - **P857-33 and ground**

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**L INBD TO MAIN, XFER MODE SWITCH TO MAN, CDU TOTAL FUEL DISPLAY DOES NOT INCREASE OR THE RESULT IS NOT AS SPECIFIED - Continued**

- a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to **21**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **21**.

**21. Procedure completed.****R INBD TO MAIN, XFER MODE SWITCH TO MAN, CDU TOTAL FUEL DISPLAY DOES NOT INCREASE OR RESULT IS NOT AS SPECIFIED****SYMPTOM**

R INBD to main, XFER MODE switch to MAN, CDU TOTAL FUEL display does not increase or Indication/Condition is not as specified.

**MALFUNCTION**

R INBD to main, XFER MODE switch to MAN, CDU TOTAL FUEL display does not increase or Indication/Condition is not as specified.

**CORRECTIVE ACTION****1. Does CDU TOTAL FUEL display increase?**

- a. If CDU TOTAL FUEL display does increase, go to **2**.
- b. If CDU TOTAL FUEL display does not increase, go to **8**.

**2. Does auxiliary fuel management control panel R INBD display decrease?**

- a. If auxiliary fuel management control panel R INBD display decreases, go to **3**.
- b. If auxiliary fuel management control panel R INBD display does not decrease, go to **10**.

**3. EMEP Does auxiliary fuel management control panel R NO FLOW annunciator and the caution/advisory panel AUX FUEL capsule momentarily go on, then off.**

- a. If AUX FUEL capsule went on then off, go to **4**.
- b. If AUX FUEL capsule did not go on then off, go to **7**.

**4. EMEP Do the remaining auxiliary fuel management control panel lights remain off?**

- a. If remaining lights remain off, go to **5**.
- b. If remaining lights do not remain off, replace auxiliary fuel management control panel (WP 1448 00). Go to **21**.

**5. Does the transfer of 20 pounds of fuel exceed 1.5 minutes?**

- a. If the transfer exceeds 1.5 minutes, go to **6**.
- b. If the transfer is 1.5 minutes or less, go to **21**.

**6. Do any related fuel or pneumatic hoses leak?**

- a. If a leak is found, replace hose and connector as required (WP 1431 00 or WP 1432 00). Go to **21**.
- b. If no leaks are found, go to **10**.

**7. Does auxiliary fuel management control panel R NO FLOW annunciator remain on?**

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**R INBD TO MAIN, XFER MODE SWITCH TO MAN, CDU TOTAL FUEL DISPLAY DOES NOT INCREASE OR RESULT IS NOT AS SPECIFIED - Continued**

- a. If annunciator remains on, go to **AUXILIARY FUEL MANAGEMENT CONTROL PANEL R NO FLOW ANNUNCIATOR REMAINS ON DURING FUEL TRANSFER**, in this work package.
  - b. If annunciator is not on, replace auxiliary fuel management control panel (WP 1448 00). Go to **21**.
- 8. Does auxiliary fuel management control panel R INBD decrease?**
- a. If R INBD decreases, go to **9**.
  - b. If R INBD decreases, go to **10**.
- 9. Does auxiliary fuel management control panel R NO FLOW annunciator go on?**
- a. If R NO FLOW annunciator goes on, replace auxiliary fuel management control panel (WP 1448 00). Go to **21**.
  - b. If R NO FLOW annunciator is not on, troubleshoot fuel quantity system (WP 0131 00). Go to **21**.
- 10. Is manual lever of left inboard fuel shutoff gate valve in full OPEN position?**
- a. If manual lever is in full OPEN position, go to **11**.
  - b. If manual lever is not in full OPEN position, go to **16**.
- 11. Is right inboard bleed-air regulator valve continuously venting air?**
- a. If valve is continuously venting air, go to **12**.
  - b. If valve is not continuously venting air, replace right bleed-air regulator valve (WP 1439 00). Go to **21**.
- 12. Check for 28 vdc between P811-A and P811-B.**
- a. If voltage is as specified, replace right bleed-air regulator valve (WP 1439 00). Go to **21**.
  - b. If voltage is not as specified, go to **13**.
- 13. Check for 28 vdc between:**
- **J346-33 and ground**
  - **J346-35 and ground**
- a. If voltage is as specified, go to **15**.
  - b. If voltage is not as specified, go to **14**.
- 14. Check for 28 vdc between:**
- **CB14A-1 and ground**
  - **CB13A-1 and ground**
- a. If voltage is as specified, repair/replace wiring between, then go to **21**.
    - **J346-33 and CB14A-1**
    - **J346-35 and CB13A-1**
  - b. If voltage is not as specified, replace circuit breaker as necessary (WP 0824 00). Go to **21**.
- 15. Check continuity between:**



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**R INBD TO MAIN, XFER MODE SWITCH TO MAN, CDU TOTAL FUEL DISPLAY DOES NOT INCREASE OR RESULT IS NOT AS SPECIFIED - Continued**

- **P811-A and P857-12**
  - **P811-B and ground**
  - **P860-33 and P857-17**
  - **P860-35 and P857-16**
  - **P857-32 and ground**
  - **P857-33 and ground**
- a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to **21**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **21**.
- 16. Check 28 vdc between P809-A and P809-E.**
- a. If voltage is as specified, go to **17**.
- b. If voltage is not as specified, go to **18**.
- 17. Check 0 vdc between P809-D and P809-E.**
- a. If voltage is as specified, replace right outboard fuel shutoff gate valve (WP 1440 00). Go to **21**.
- b. If voltage is not as specified, go to **18**.
- 18. Check 28 vdc between:**
- **J346-33 and ground**
  - **J346-35 and ground**
- a. If voltage is as specified, go to **20**.
- b. If voltage is not as specified, go to **19**.
- 19. Check 28 vdc between:**
- **CB14A-1 and ground**
  - **CB13A-1 and ground**
- a. If voltage is as specified, repair/replace wiring between then go to **21**.
- **CB14A-1 and J346-33**
  - **CB13A-1 and J346-35**
- b. If voltage is not as specified, replace circuit breaker as necessary (WP 0824 00). Go to **21**.
- 20. Check continuity between:**
- **P810-A and P857-6**
  - **P810-D and P857-4**
  - **P810-E and ground**
  - **P857-32 and ground**
  - **P857-33 and ground**

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**R INBD TO MAIN, XFER MODE SWITCH TO MAN, CDU TOTAL FUEL DISPLAY DOES NOT INCREASE OR RESULT IS NOT AS SPECIFIED - Continued**

- a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to **21**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **21**.

**21. Procedure completed.****XFER MODE SWITCH AT AUTO, FUEL TRANSFER LIMITS ARE NOT AS SPECIFIED****SYMPTOM**

XFER MODE switch at AUTO, fuel transfer limits are not as specified.

**MALFUNCTION**

XFER MODE switch at AUTO, fuel transfer limits are not as specified.

**CORRECTIVE ACTION**

- 1. With either TANK 1 or TANK 2 indicating 1100 pounds or less (or either TANK 1 or TANK 2 simulated input is less than or equal 5.5 volts), and the CDU TOTAL FUEL display showing less than 2300 pounds (or the sum of TANK 1 and TANK 2 simulated inputs is greater than 11.5 volts), does fuel transfer begin.**
  - a. If fuel transfer begins, go to **3**.
  - b. If fuel transfer does not begin, go to **2**.
- 2. Check continuity between:**
  - **P856-6 and P905-L (tank 1)**
  - **P856-7 and P905-K (tank 1)**
  - **P856-13 and P905-U (tank 2)**
  - **P856-14 and P905-V (tank 2)**
  - a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to **4**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **4**.
- 3. Does fuel transfer stop when CDU TOTAL FUEL display has increased to 2300 pounds (or TANK 1 and TANK 2 simulated inputs are greater than 11.5 volts).**
  - a. If fuel transfer stops, go to **4**.
  - b. If fuel transfer does not stop, replace auxiliary fuel management control panel (WP 1448 00). Go to **4**.
- 4. Procedure completed.**

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**AUXILIARY FUEL MANAGEMENT CONTROL PANEL R NO FLOW ANNUNCIATOR REMAINS ON DURING FUEL TRANSFER****SYMPTOM**

Auxiliary fuel management control panel R NO FLOW annunciator remains on during fuel transfer.

**MALFUNCTION**

Auxiliary fuel management control panel R NO FLOW annunciator remains on during fuel transfer.

**CORRECTIVE ACTION**

1. Disconnect J520 from right auxiliary fuel flow sensor. Go to 2.
2. Connect insulated jumper wire between J520-1 and J520-2. Go to 3.
3. Place auxiliary fuel management control panel MAN XFER switch to RIGHT. Go to 4.
4. Does auxiliary fuel management control panel R NO FLOW annunciator go on momentarily?
  - a. If annunciator goes on momentarily, replace right auxiliary fuel flow sensor (WP 1437 00). Go to 8.
  - b. If annunciator does not go on momentarily, go to 5.
5. Disconnect P856 from auxiliary fuel management control panel. Go to 6.
6. Leave jumper connected to J520. Go to 7.
7. Check continuity between P856-10 and P856-15.
  - a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to 8.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to 8.
8. Procedure completed.

**AUXILIARY FUEL MANAGEMENT CONTROL PANEL L NO FLOW ANNUNCIATOR REMAINS ON DURING FUEL TRANSFER****SYMPTOM**

Auxiliary fuel management control panel L NO FLOW annunciator remains on during fuel transfer.

**MALFUNCTION**

Auxiliary fuel management control panel L NO FLOW annunciator remains on during fuel transfer.

**CORRECTIVE ACTION**

1. Disconnect J521 from right auxiliary fuel flow sensor. Go to 2.
2. Connect insulated jumper wire between J521-1 and J521-2. Go to 3.
3. Place auxiliary fuel management control panel MAN XFER switch to RIGHT. Go to 4.
4. Does auxiliary fuel management control panel R NO FLOW annunciator go on momentarily?

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**AUXILIARY FUEL MANAGEMENT CONTROL PANEL L NO FLOW ANNUNCIATOR REMAINS ON DURING FUEL TRANSFER - Continued**

- a. If annunciator goes on momentarily, replace left auxiliary fuel flow sensor (WP 1437 00). Go to **8**.
- b. If annunciator does not go on momentarily, go to **5**.
- 5. Disconnect P856 from auxiliary fuel management control panel. Go to 6.**
- 6. Leave jumper connected to J521. Go to 7.**
- 7. Check continuity between P856-8 and P856-9.**
  - a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to **8**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **8**.
- 8. Procedure completed.**

**AUXILIARY FUEL MANAGEMENT CONTROL PANEL R INBD EMPTY AND/OR L INBD EMPTY ANNUNCIATORS DO NOT GO ON WITH INADEQUATE FUEL IN THE TANK****SYMPTOM**

Auxiliary fuel management control panel R INBD EMPTY and/or L INBD EMPTY annunciators do not go on with inadequate fuel in the tank.

**MALFUNCTION**

Auxiliary fuel management control panel R INBD EMPTY and/or L INBD EMPTY annunciators do not go on with inadequate fuel in the tank.

**CORRECTIVE ACTION**

- 1. Disconnect W4P2 from applicable inboard fuel tank. Go to 2.**
- 2. Does L INBD EMPTY or R INBD EMPTY annunciator go on?**
  - a. If annunciator goes on, replace tank probe (WP 1552 00). Go to **4**.
  - b. If annunciator is on, go to **3**.
- 3. Check continuity between:**
  - **P856-34 and W4P2-C (right)**
  - **P856-35 and W4P2-F (right)**
  - **P856-36 and W4P2-E (right)**
  - **P856-37 and W4P2-C (right)**
  - **P856-38 and W4P2-F (right)**
  - **P856-39 and W4P2-E (right)**
  - a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to **4**.
  - b. If continuity is not present, repair/replace wiring. Go to **4**.
- 4. Procedure completed.**

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**CAUTION/ADVISORY PANEL AUX FUEL CAPSULE AND/OR AUXILIARY FUEL MANAGEMENT CONTROL PANEL R NO FLOW AND L NO FLOW ANNUNCIATORS DO NOT GO ON****SYMPTOM**

Caution/advisory panel AUX FUEL capsule and/or auxiliary fuel management control panel R NO FLOW and L NO FLOW annunciators do not go on immediately after corresponding tank EMPTY annunciator goes on.

**MALFUNCTION**

Caution/advisory panel AUX FUEL capsule and/or auxiliary fuel management control panel R NO FLOW and L NO FLOW annunciators do not go on immediately after corresponding tank EMPTY annunciator goes on.

**CORRECTIVE ACTION**

- 1. Do auxiliary fuel management control panel R NO FLOW and L NO FLOW annunciator go on?**
  - a. If annunciators are on, go to **2**.
  - b. If annunciators are not on, go to **9**.
- 2. Momentarily place caution/advisory panel BRT/DIM-TEST switch to TEST. Go to 3.**
- 3. Does caution/advisory panel AUX FUEL capsule go on then off.**
  - a. If capsule goes on then off, go to **4**.
  - b. If capsule does not go on then off, troubleshoot caution/advisory warning system (WP 0090 00). Go to **13**.
- 4. Check helicopter configuration.**
  - a. **EMEP** , go to **5**.
  - b. **W/O EMEP** , go to **7**.
- 5. Remove and check pin filtered adapter P117 (WP 0925 00).**
  - a. If pin filtered adapter is good, go to **6**.
  - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **13**.
- 6. Install pin filtered adapter (WP 0925 00). Go to 7.**
- 7. Check for 28 vdc between P117-41 and ground.**
  - a. If voltage is as specified, troubleshoot caution/advisory system (WP 0090 00). Go to **13**.
  - b. If voltage is not as specified, go to **8**.
- 8. Check continuity between P117-41 and P856-24.**
  - a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to **13**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **13**.
- 9. Disconnect connector J521 (left) or J520 (right) from auxiliary fuel flow sensor. Go to 10.**
- 10. With auxiliary fuel management control panel switch at AUTO, is correct NO FLOW annunciator on?**
  - a. If annunciator is on, replace auxiliary fuel flow sensor (WP 1437 00). Go to **13**.

---

**CAUTION/ADVISORY PANEL AUX FUEL CAPSULE AND/OR AUXILIARY FUEL MANAGEMENT CONTROL PANEL R NO FLOW AND L NO FLOW ANNUNCIATORS DO NOT GO ON - Continued**

b. If annunciator is not on, go to **11**.

**11. Check for infinite resistance between:**

- **P856-8 and P856-9 (left)**
- **P856-10 and P856-15 (right)**

a. If resistance is as specified, replace auxiliary fuel management control panel (WP 1448 00). Go to **13**.

b. If resistance is not as specified, go to **12**.

**12. Check for infinite resistance between:**

- **J346-14 and J346-9 (left)**
- **J346-16 and J346-17 (right)**

a. If resistance is as specified, repair/replace wiring (WP 1747 00). Go to **13**.

b. If resistance is not as specified, repair/replace wiring (WP 1747 00). Go to **13**.

**13. Procedure completed.**

**AUXILIARY FUEL MANAGEMENT CONTROL PANEL R OUTBD EMPTY AND/OR L OUTBD EMPTY ANNUNCIATORS DO NOT GO ON WITH INADEQUATE FUEL IN TANK****SYMPTOM**

Auxiliary fuel management control panel R OUTBD EMPTY and/or L OUTBD EMPTY annunciators do not go on with inadequate fuel in tank.

**MALFUNCTION**

Auxiliary fuel management control panel R OUTBD EMPTY and/or L OUTBD EMPTY annunciators do not go on with inadequate fuel in tank.

**CORRECTIVE ACTION**

**1. Disconnect W4P2 from outboard fuel tank. Go to 2.**

**2. Does L OUTBD EMPTY or R OUTBD EMPTY annunciator go on?**

- a. If annunciator goes on, replace tank probe (WP 1552 00). Go to **4**.
- b. If annunciator is on, go to **3**.

**3. Check continuity between:**

- **P856-31 and W4P2-C (right)**
- **P856-32 and W4P2-F (right)**
- **P856-33 and W4P2-E (right)**
- **P856-40 and W4P2-C (right)**
- **P856-41 and W4P2-F (right)**
- **P856-42 and W4P2-E (right)**

a. If continuity is present, replace auxiliary fuel management control panel (WP 1448 00). Go to **4**.

---

**AUXILIARY FUEL MANAGEMENT CONTROL PANEL R OUTBD EMPTY AND/OR L OUTBD EMPTY ANNUNCIATORS DO NOT GO ON WITH INADEQUATE FUEL IN TANK - Continued**

- b. If continuity is not present, repair/replace wiring. Go to **4**.

**4. Procedure completed.****AUXILIARY FUEL MANAGEMENT CONTROL PANEL VENT FAIL OR VENT OVFL ANNUNCIATOR DOES NOT GO ON****SYMPTOM**

Auxiliary fuel management control panel VENT FAIL or VENT OVFL annunciator does not go on.

**MALFUNCTION**

Auxiliary fuel management control panel VENT FAIL or VENT OVFL annunciator does not go on.

**CORRECTIVE ACTION****1. Check that helicopter connector J376 is connected to vent sensor P376.**

- a. If J376 is connected to P376, go to **2**.
- b. If J376 is not connected to P376, replace auxiliary fuel management control panel (WP 1448 00). Go to **4**.

**2. Check for 360-440 ohms between P856-22 and P856-23.**

- a. If resistance is as specified, replace auxiliary fuel management control panel (WP 1448 00). Go to **4**.
- b. If resistance is not as specified, go to **3**.

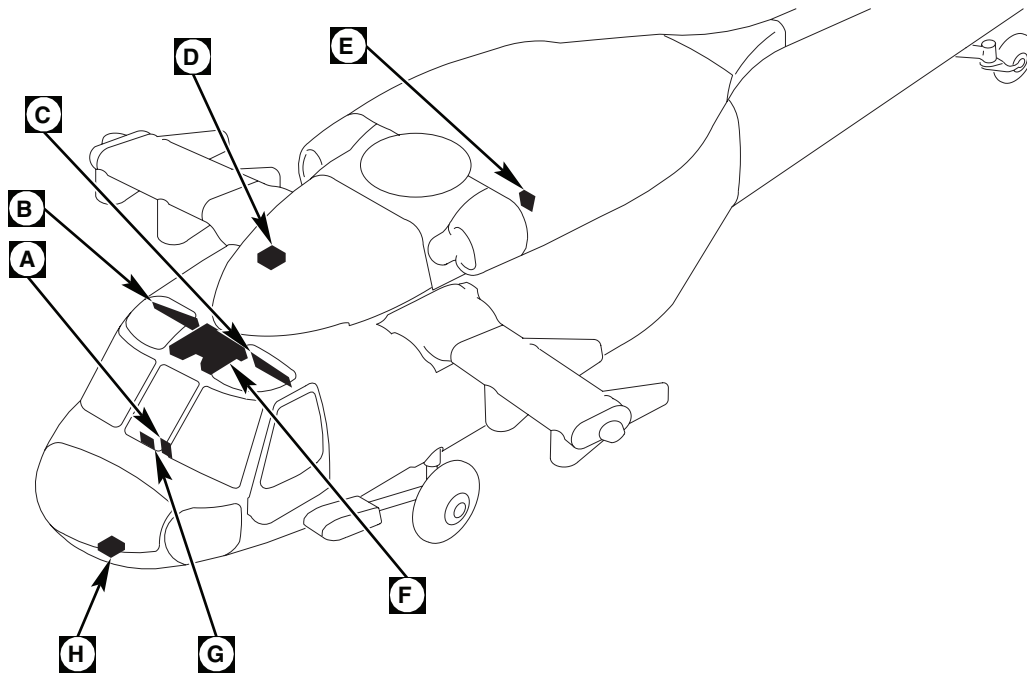
**3. Check overflow sensor resistance as follows:**

- **P376-C and P376-A 360-440 ohms**
- **P376-C and P376-B 360-440 ohms**

- a. If resistance is as specified, repair/replace wiring between (WP 1747 00), then go to **4**:
  - J376-B and P856-22
  - J376-C and P856-23
- b. If resistance is not as specified, replace vent overflow sensor (WP 1457 00). Go to **4**.

**4. Procedure completed.**

## LOCATION



| TERMINAL BOARD /<br>DISCONNECT PLUG /<br>RECEPTACLE | LOCATION /<br>CONNECTION POINT      |
|-----------------------------------------------------|-------------------------------------|
| J299<br>(SEE NOTE 1)                                | CABIN CEILING, BL 10 LH,<br>STA 302 |
| J520                                                | RIGHT EXT FUEL FLOW<br>SENSOR       |
| J521                                                | LEFT EXT FUEL FLOW<br>SENSOR        |
| J522                                                | INT AUX FUEL FLOW<br>SENSOR         |
| J524                                                | FUEL TEMPERATURE<br>TRANSDUCER      |
| P113 / J113                                         | COCKPIT, BL 6 LH, STA 197           |
| P114 / J114                                         | COCKPIT, BL 8 RH, STA 200           |
| P117                                                | CAUTION / ADVISORY<br>PANEL         |
| P137                                                | NO. 2 SIGNAL DATA<br>CONVERTER      |
| P138                                                | NO. 1 SIGNAL DATA<br>CONVERTER      |
| P163 / J163                                         | LEFT DRAG BEAM SWITCH               |

## NOTES

1. **UH-60A**   **UH-60L**
2. **EH-60A**

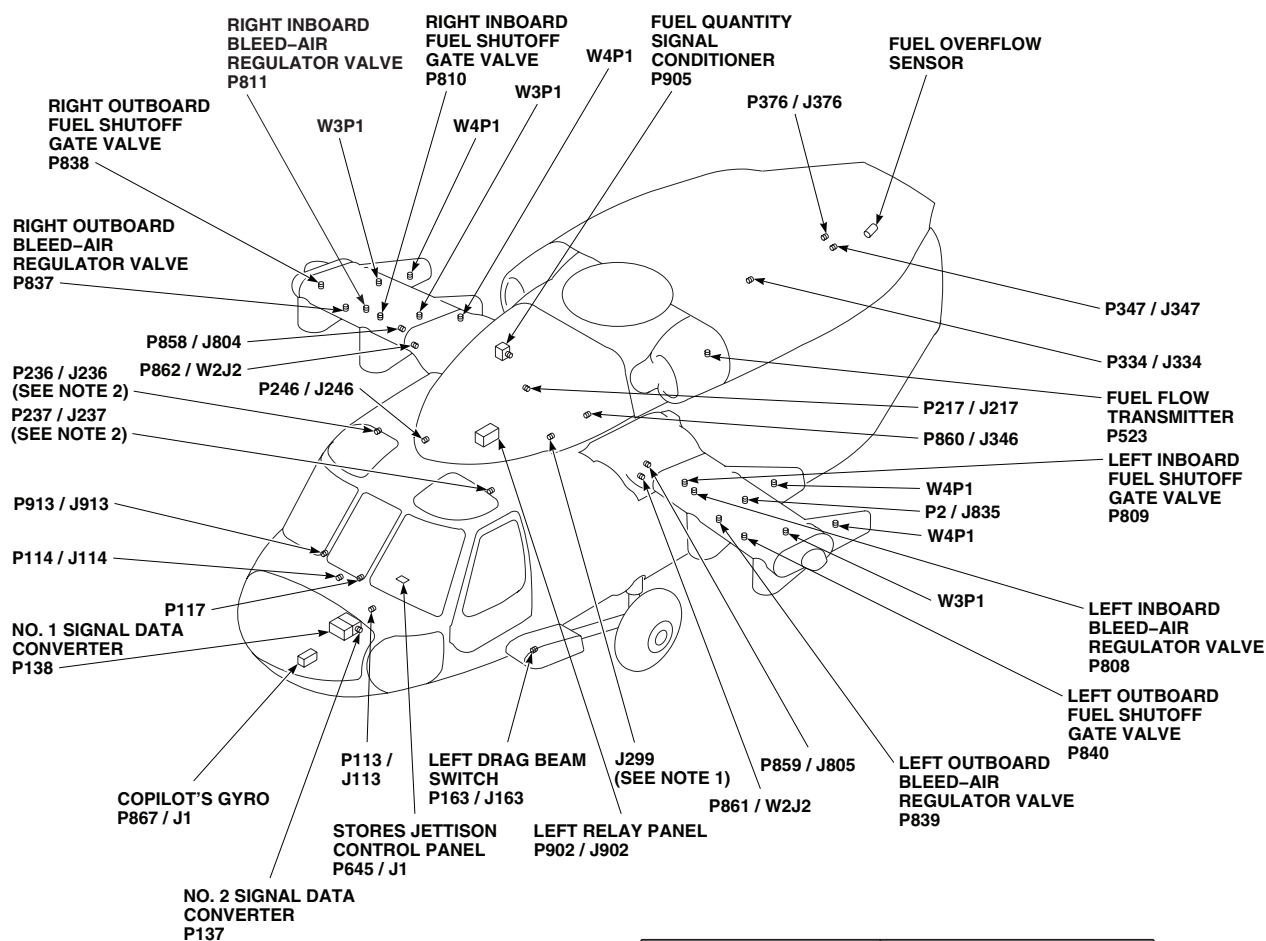
| TERMINAL BOARD /<br>DISCONNECT PLUG /<br>RECEPTACLE | LOCATION /<br>CONNECTION POINT                                  |
|-----------------------------------------------------|-----------------------------------------------------------------|
| P217 / J217                                         | MAIN ROTOR PYLON DECK,<br>BL 0, STA 284                         |
| P236 / J236<br>(SEE NOTE 2)                         | BEHIND PILOT'S CIRCUIT<br>BREAKER PANEL, BL 23 RH               |
| P237 / J237<br>(SEE NOTE 2)                         | BEHIND COPILOT'S<br>CIRCUIT BREAKER PANEL,<br>BL 25 LH, STA 247 |
| P246 / J246                                         | CABIN CEILING, BL 5 RH,<br>STA 247                              |
| P334 / J334                                         | CABIN CEILING, BL 16.5 LH,<br>STA 390                           |
| P347 / J347                                         | TRANSITION SECTION,<br>BL 10 LH, STA 425                        |
| P376 / J376                                         | AFT TRANSITION,<br>BL 18.25 LH, STA 443.5                       |
| P523                                                | FUEL FLOW TRANSMITTER                                           |
| P645 / J1                                           | STORES JETTISON<br>CONTROL PANEL                                |
| P808                                                | LEFT INBOARD BLEED-AIR<br>REGULATOR VALVE                       |
| P809                                                | LEFT INBOARD FUEL<br>SHUTOFF GATE VALVE                         |
| P810                                                | RIGHT INBOARD FUEL<br>SHUTOFF GATE VALVE                        |
| P811                                                | RIGHT INBOARD BLEED-<br>AIR REGULATOR VALVE                     |
| P837                                                | RIGHT OUTBOARD BLEED-<br>AIR REGULATOR VALVE                    |

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Figure 1. ESSS Range Extension System Location Diagram. (Sheet 1 of 5)



## LOCATION - Continued



| TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE | LOCATION / CONNECTION POINT                  |
|-----------------------------------------------|----------------------------------------------|
| P838                                          | RIGHT OUTBOARD FUEL SHUTOFF GATE VALVE       |
| P839                                          | LEFT OUTBOARD BLEED-AIR REGULATOR VALVE      |
| P840                                          | LEFT OUTBOARD FUEL SHUTOFF GATE VALVE        |
| P856 / J1                                     | AUXILIARY FUEL MANAGEMENT CONTROL PANEL      |
| P857 / J2                                     | AUXILIARY FUEL MANAGEMENT CONTROL PANEL      |
| P858 / J804                                   | HORIZONTAL STORES SUPPORT, BL 59 RH, STA 295 |
| P859 / J805                                   | HORIZONTAL STORES SUPPORT, BL 59 LH, STA 295 |
| P860 / J346                                   | CABIN CEILING, BL 20 LH, STA 304             |
| P861 / W2J2                                   | HSS ATTACHMENT FITTING, BL 50 LH, STA 295    |
| P862 / W2J2                                   | HSS ATTACHMENT FITTING, BL 59 LH, STA 295    |

| TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE | LOCATION / CONNECTION POINT        |
|-----------------------------------------------|------------------------------------|
| P867 / J1                                     | COPILOT'S GYRO IN NOSE SECTION     |
| P902 / J902                                   | LEFT RELAY PANEL                   |
| P905                                          | FUEL QUANTITY SIGNAL CONDITIONER   |
| P913 / J913                                   | COCKPIT, BL 24 RH, STA 247         |
| W3P1 / W2J2                                   | LEFT INBOARD FUEL TANK CONNECTOR   |
| W3P1 / W2J2                                   | RIGHT INBOARD FUEL TANK CONNECTOR  |
| W3P1 / W2J3                                   | LEFT OUTBOARD FUEL TANK CONNECTOR  |
| W3P1 / W2J3                                   | RIGHT OUTBOARD FUEL TANK CONNECTOR |
| W4P1 / J1                                     | RIGHT INBOARD FUEL TANK CONNECTOR  |
| W4P1 / J1                                     | LEFT INBOARD FUEL TANK CONNECTOR   |
| W4P1 / J1                                     | LEFT OUTBOARD FUEL TANK CONNECTOR  |
| W4P1 / J1                                     | RIGHT OUTBOARD FUEL TANK CONNECTOR |

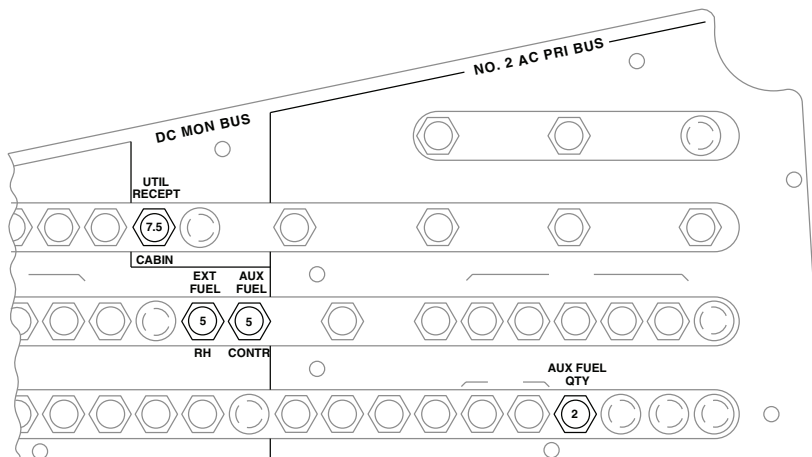
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Figure 1. ESSS Range Extension System Location Diagram. (Sheet 2 of 5)

LOCATION - Continued



CAUTION/ADVISORY PANEL



(SEE NOTE 2)

PILOT'S CIRCUIT BREAKER PANEL

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Figure 1. ESSS Range Extension System Location Diagram. (Sheet 3 of 5)

## LOCATION - Continued

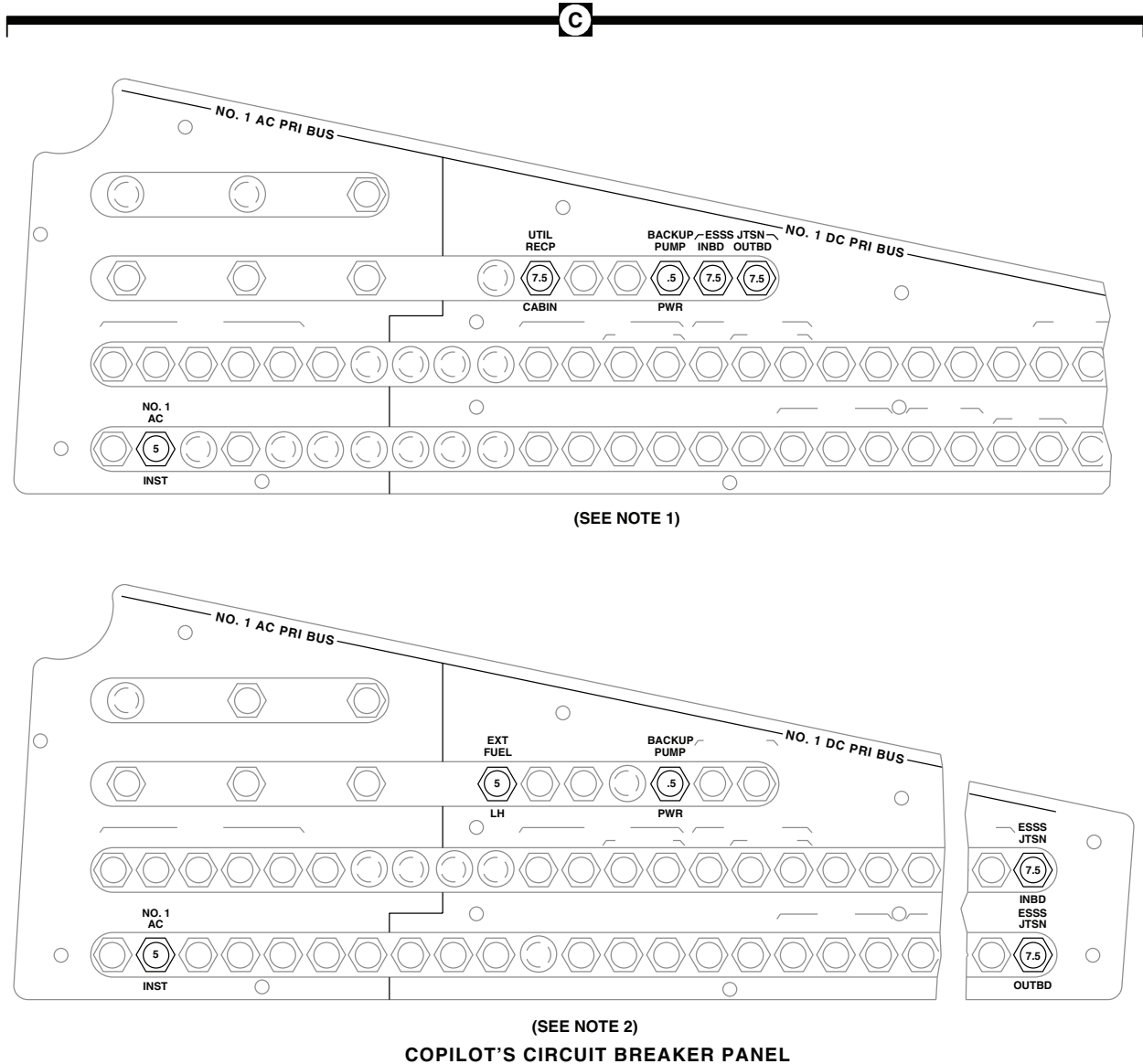
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Figure 1. ESSS Range Extension System Location Diagram. (Sheet 4 of 5)

LOCATION - Continued

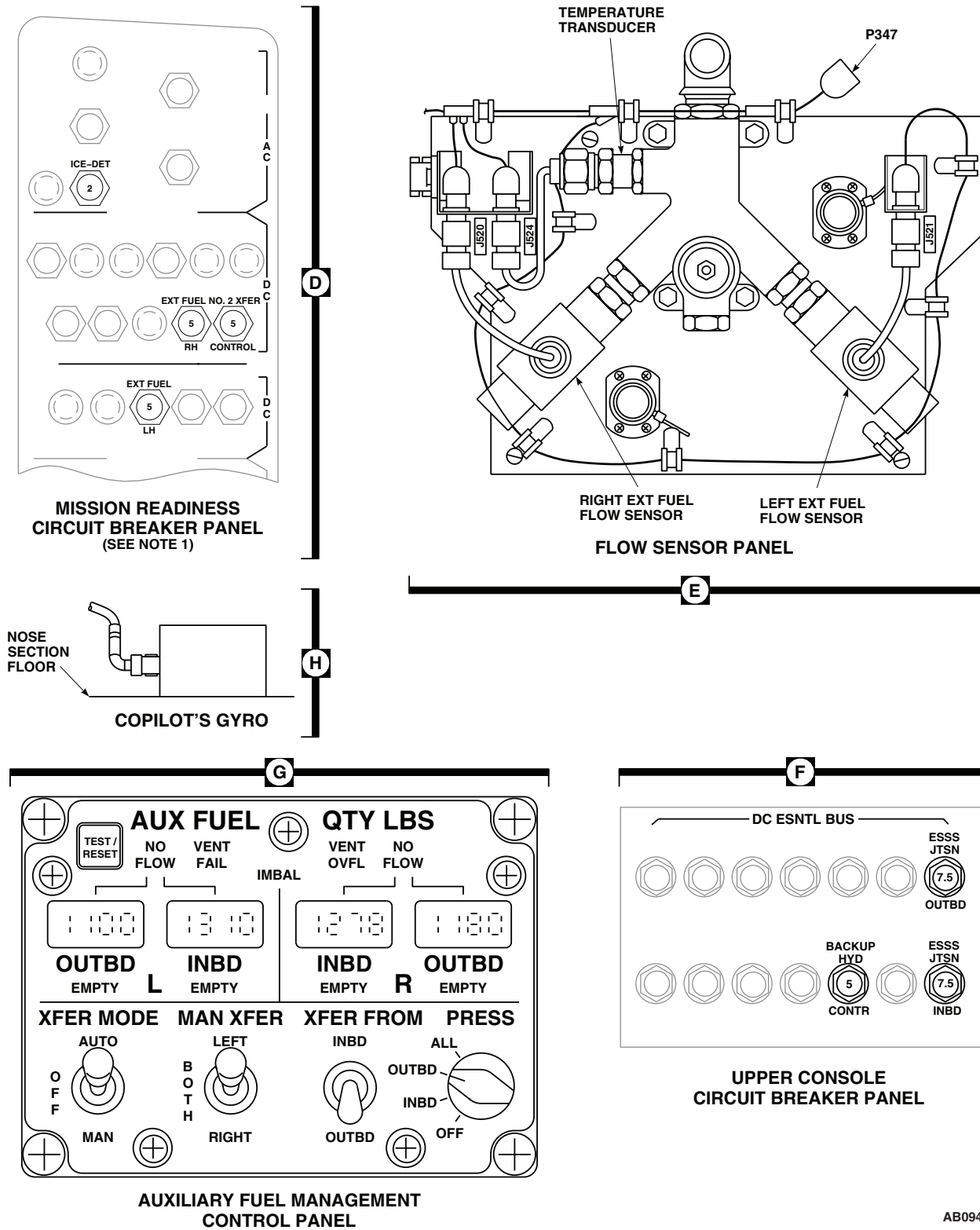


Figure 1. ESSS Range Extension System Location Diagram. (Sheet 5 of 5)

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The diagram illustrates the electrical connections for the Mission Readiness Circuit Breaker Panel and Fuel Quantity Signal Converters. It is divided into three main sections:

- MISSION READINESS CIRCUIT BREAKER PANEL (SEE DETAIL A):** This section shows the internal wiring of the breaker panel. It includes two 28 VDC PRI DC BUS units, each with a 5 AMP circuit breaker (CB14A, CB13A, and CB20A). The panel is connected to external fuel tanks (EXT FUEL RH and EXT FUEL LH) and a NO. 2 XFER CONTROL unit. The wiring is terminated at J346 P860 and J299 B.
- NO. 2 SIGNAL DATA CONVERTER:** This section shows the wiring for the No. 2 Signal Data Converter. It is connected to the 28 VDC PRI DC BUS and the EXT FUEL RH tank. The wiring is terminated at P137 and P138.
- NO. 1 SIGNAL DATA CONVERTER:** This section shows the wiring for the No. 1 Signal Data Converter. It is connected to the 28 VDC PRI DC BUS and the EXT FUEL LH tank. The wiring is terminated at P138 and P114 J114.
- FUEL QUANTITY SIGNAL CONVERTER:** This section shows the wiring for the Fuel Quantity Signal Converter. It is connected to the 28 VDC PRI DC BUS and the EXT FUEL RH and EXT FUEL LH tanks. The wiring is terminated at P905 and P114 J114.

The diagram includes various electrical symbols and labels, such as J346 P860, J299 B, P137, P138, P114 J114, P905, and P217 J217. It also shows the connection to the TO SHEET 7.

1. **UH-60A** **UH-60L**
2. **EH-60A**
3. **W / O EMEP** PIN FILTERED ADAPTER (PFA) IS NOT INSTALLED.

Figure 2. ESSS Range Extension System Schematic Diagram. (Sheet 1 of 8)

SCHEMATICS - Continued

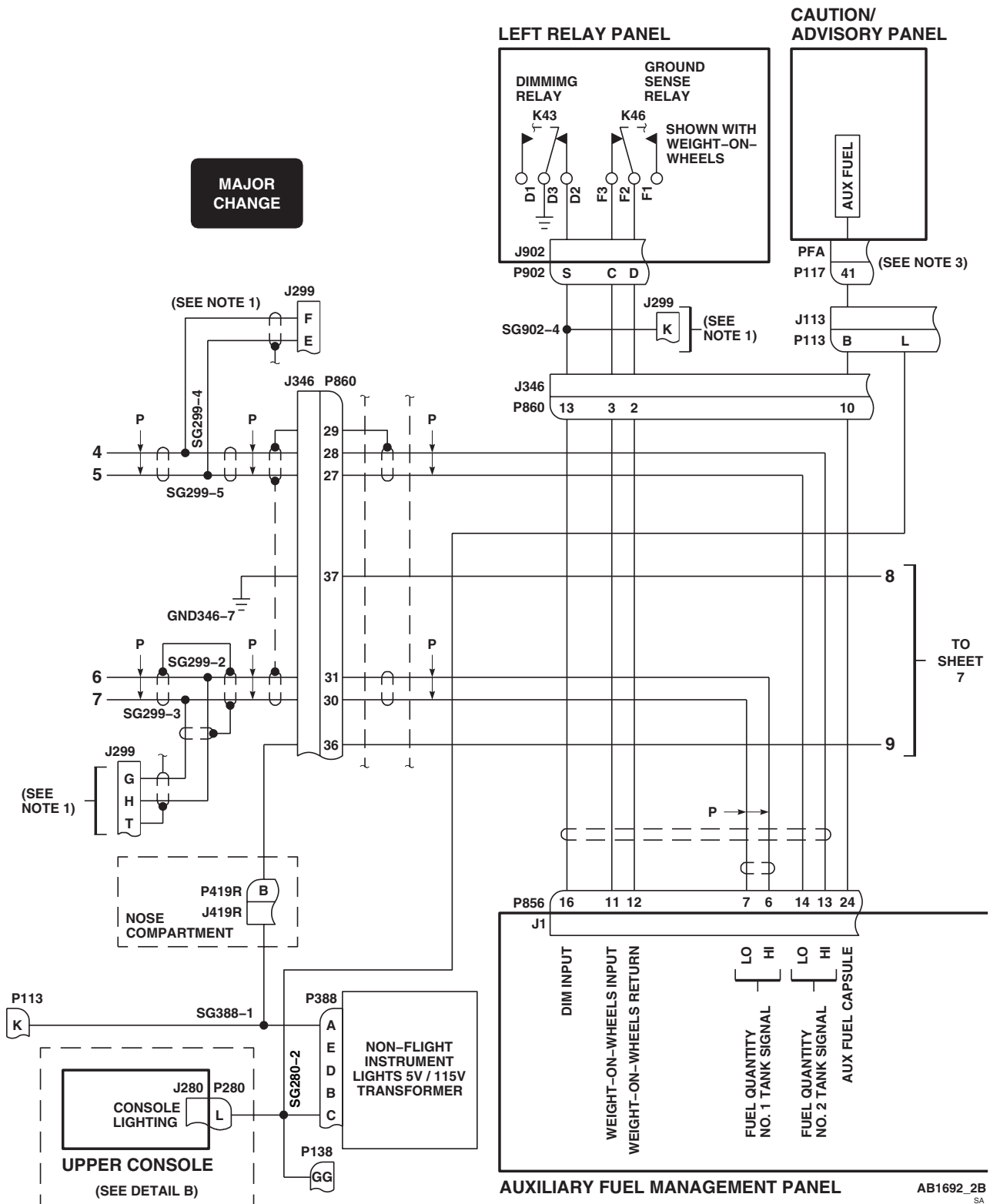
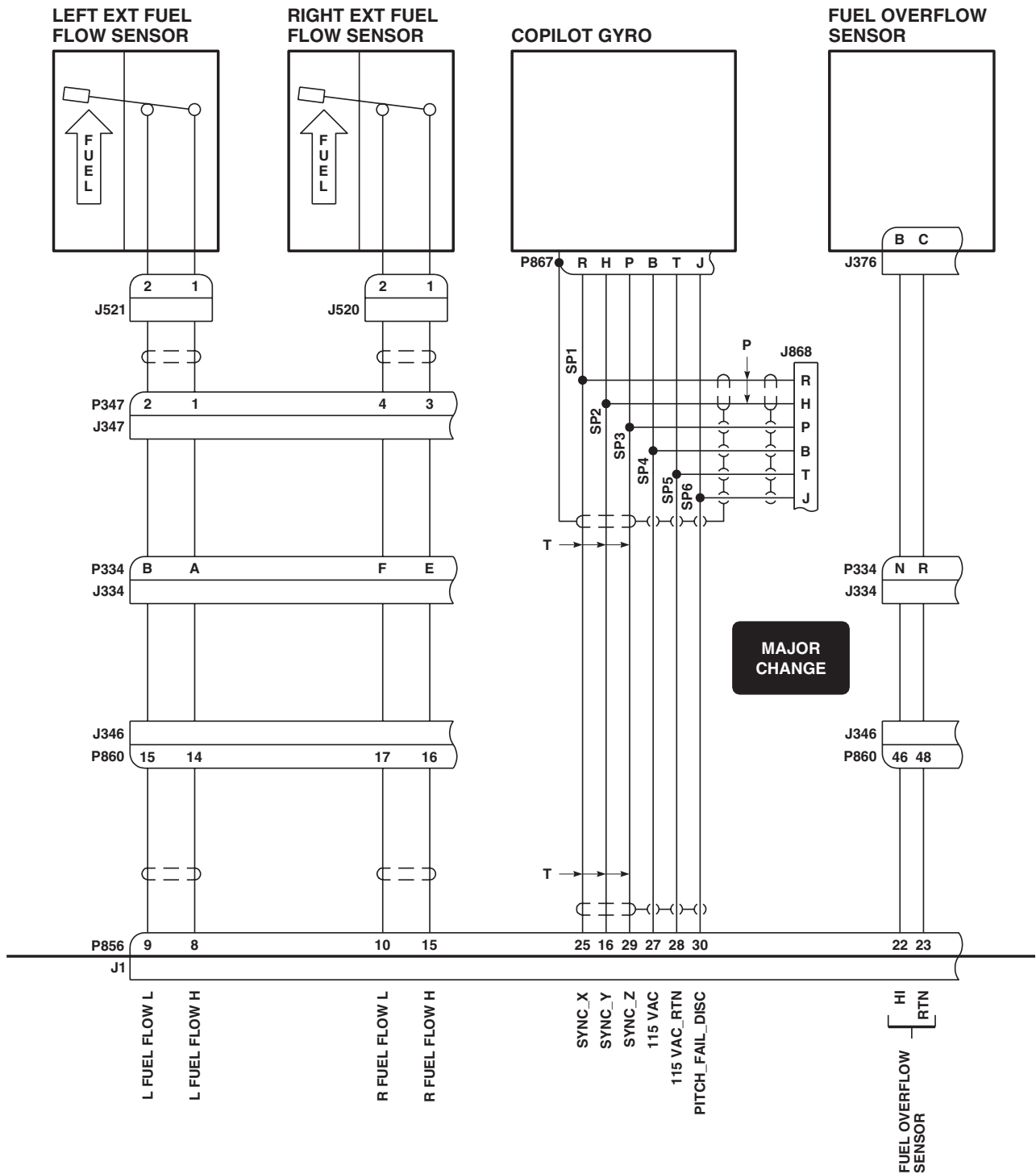


Figure 2. ESSS Range Extension System Schematic Diagram. (Sheet 2 of 8)

SCHEMATICS - Continued

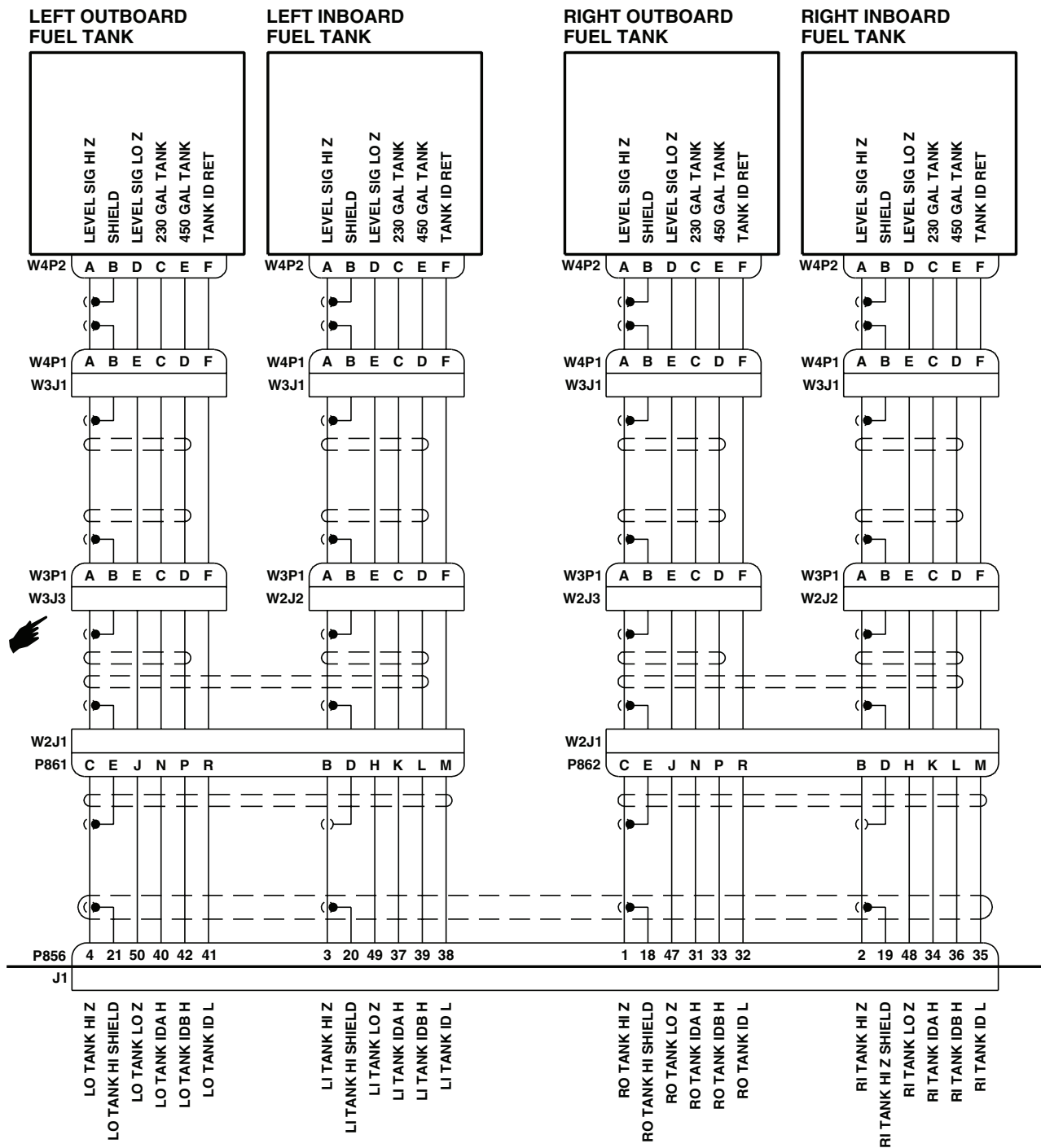


AUXILIARY FUEL MANAGEMENT PANEL

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Figure 2. ESSS Range Extension System Schematic Diagram. (Sheet 3 of 8)

## SCHEMATICS - Continued



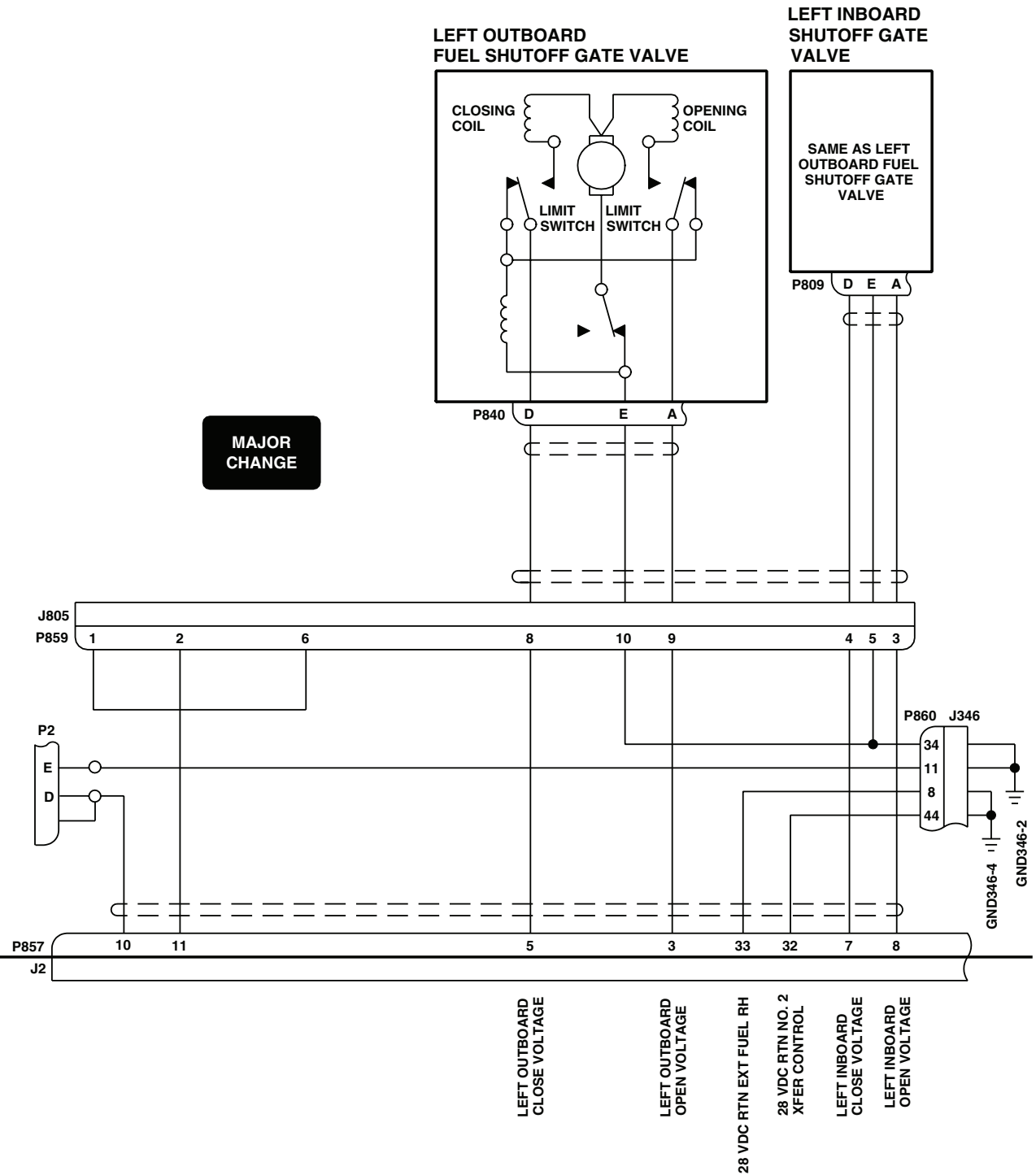
AUXILIARY FUEL MANAGEMENT PANEL

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Figure 2. ESSS Range Extension System Schematic Diagram. (Sheet 4 of 8)



SCHEMATICS - Continued

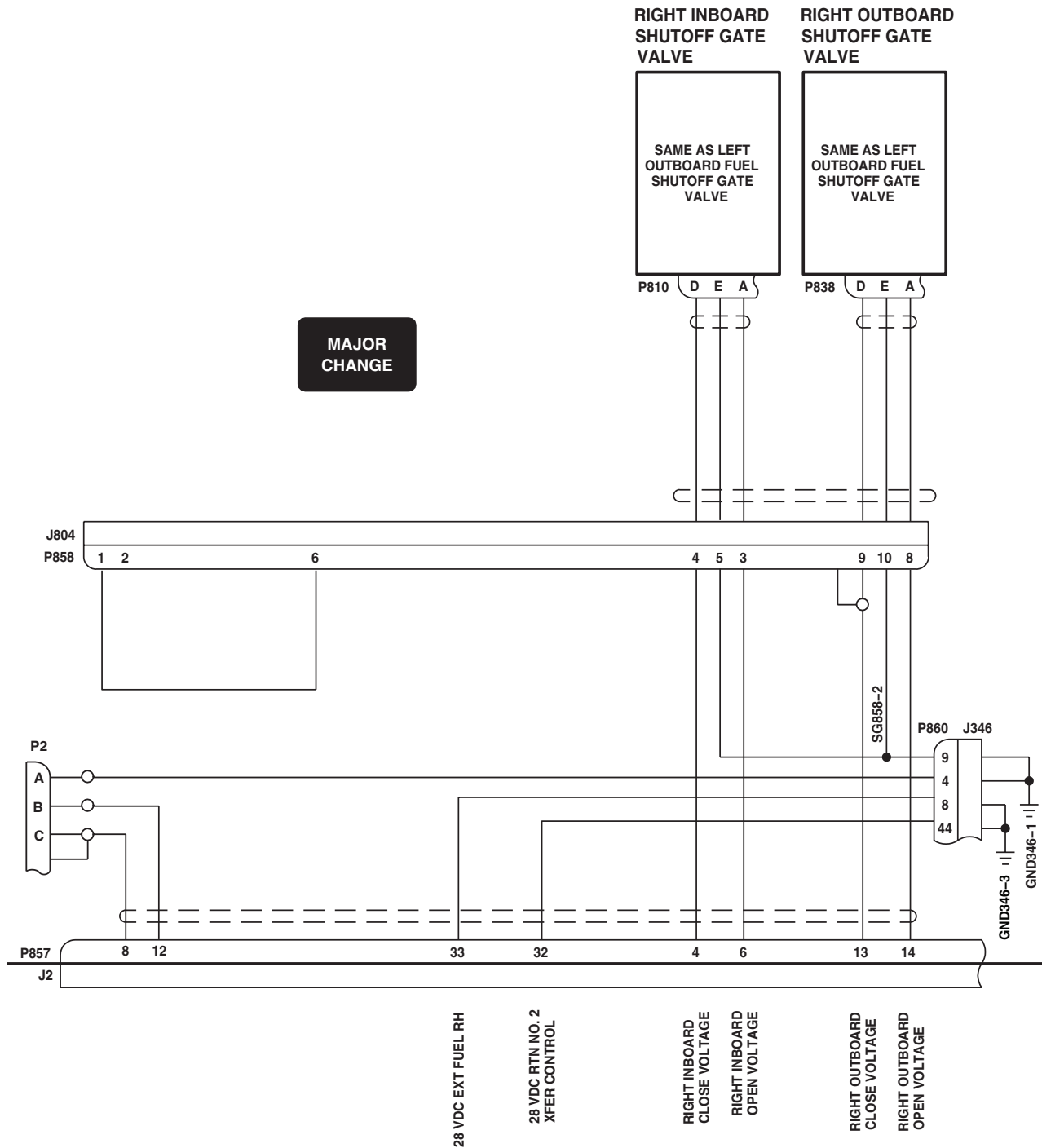


AUXILIARY FUEL MANAGEMENT PANEL

AB1692\_5B  
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Figure 2. ESSS Range Extension System Schematic Diagram. (Sheet 5 of 8)

SCHEMATICS - Continued

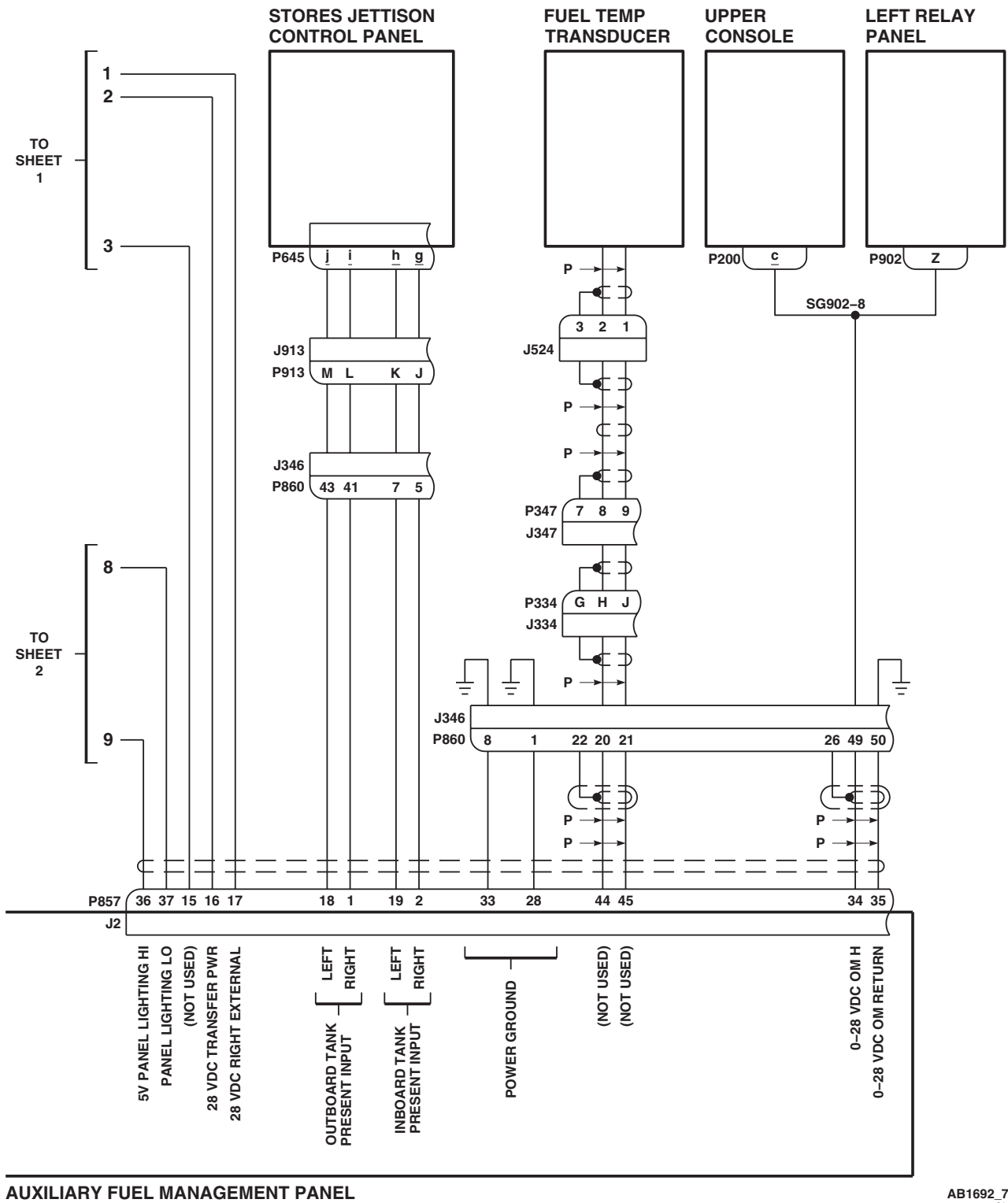


AUXILIARY FUEL MANAGEMENT PANEL

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Figure 2. ESSS Range Extension System Schematic Diagram. (Sheet 6 of 8)

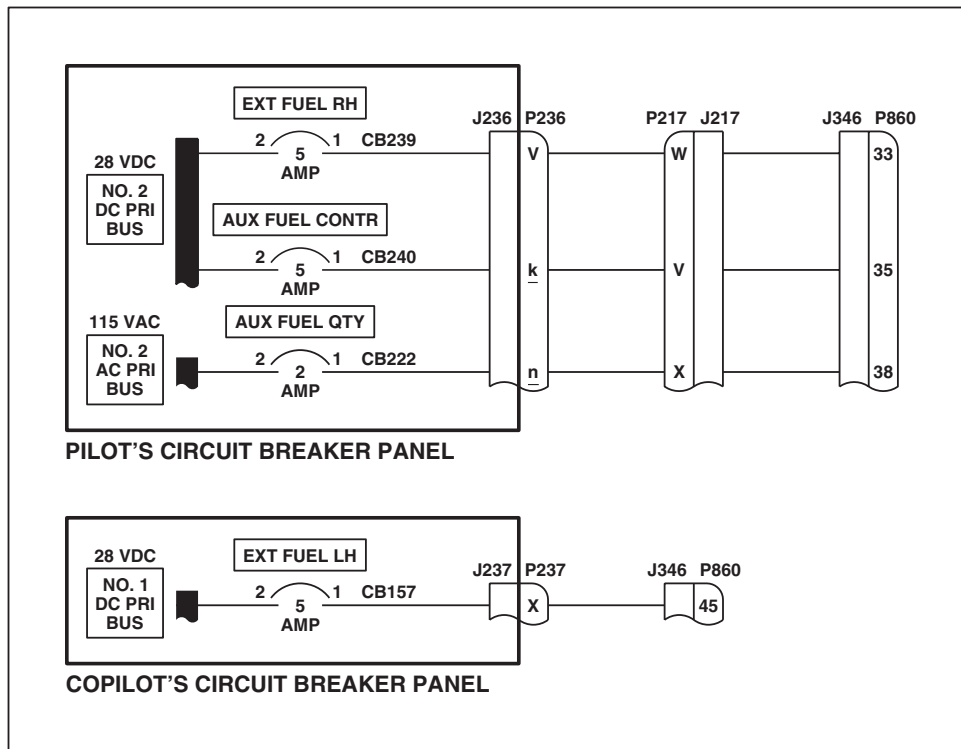
SCHEMATICS - Continued



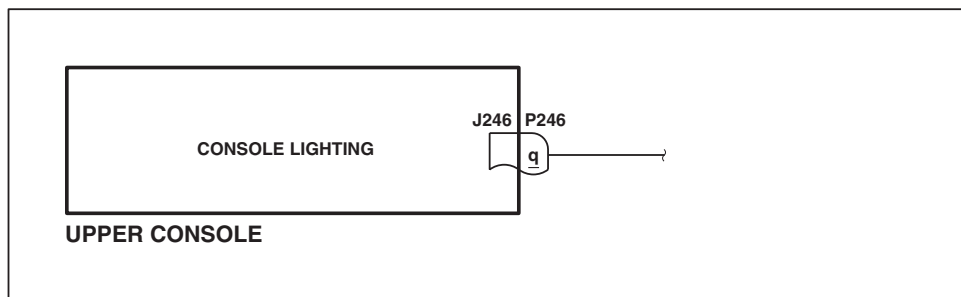
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Figure 2. ESSS Range Extension System Schematic Diagram. (Sheet 7 of 8)

SCHEMATICS - Continued



DETAIL A  
(SEE NOTE 2)



DETAIL B  
(SEE NOTE 2)

AB1692\_8  
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Figure 2. ESSS Range Extension System Schematic Diagram. (Sheet 8 of 8)

END OF WORK PACKAGE

## UNIT LEVEL

## MISSION EQUIPMENT

EXTERNAL STORES SUPPORT SYSTEM (ESSS) EXTERNAL TANK CHECK **MWO 50-78**

## INITIAL SETUP:

## Test Equipment

Aviation Ground Power Unit (AGPU), 83-36A  
 Compressor Unit, MIL-C-13874  
 Locally-Made Fuel Quantity Simulator Test Set, Figure 152, WP 1805 00  
 Multimeter, AN/PSM-45A (2 required)

WP 1445 00  
 WP 1447 00  
 WP 1612 00  
 WP 1613 00  
 WP 1620 00  
 WP 1685 00  
 WP 1805 00

## Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01  
 Container, Suitable  
 Electrical Repairer Toolkit, SC 5180-99-B06  
 Locally-Made External Stores Support System (ESSS)  
 Pneumatic Adapter, Figure 151, WP 1805 00  
 Torque Wrench, GG-W-686

## Personnel Required

Aircraft Electrician MOS 15F (1)  
 UH-60 Helicopter Repairer MOS 15T (1)

## References

TM 1-1500-204-23  
 TM 1-1520-237-10  
 WP 0104 00  
 WP 0160 00  
 WP 0168 00  
 WP 1425 00  
 WP 1442 00

## Equipment Condition

All Explosive Cartridges Removed from Ejector Racks, WP 1442 00  
 or All Explosive Cartridges Removed from Ejector Racks, WP 1445 00  
 External Electrical Power Available, WP 1685 00  
 or APU Operational, TM 1-1520-237-10  
 External Range Extension Tanks **160-190** Pounds of Fuel in Each Outboard, if Installed, WP 1620 00  
 External Range Extension Tanks **220-250** Pounds of Fuel in Each Inboard, if Installed, WP 1620 00  
 Main Fuel Tanks, Not More Than **650** Pounds of Fuel in Each Tank, WP 1612 00  
 or Main Fuel Tanks, Not More Than **650** Pounds of Fuel in Each Tank, WP 1613 00  
 Rear Fairings on Vertical Stores (Pylons), Which Have Tanks Installed, Removed, WP 1447 00

## SPECIAL INSTRUCTIONS

## NOTE

See Figure 1 for component location diagram.

This operational/troubleshooting procedure is subdivided:

- Setup
- Outboard Tanks Installed
- Inboard Tanks Installed
- Shutdown

**SPECIAL INSTRUCTIONS** - Continued**APPLICABLE CONFIGURATION**

This operational/troubleshooting procedure is designed to be utilized when the horizontal (Wings) and vertical (Pylons) stores are installed on a permanent basis, and only the inboard or outboard external tanks are removed and installed.

**SETUP OPERATIONAL/TROUBLESHOOTING PROCEDURE****PROCEDURE****Step****NOTE**

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- As an alternate to using the compressor unit to pressurize the ESSS Range Extension System, the Aviation Ground Power Unit (AGPU) or the APU may be used. If the alternate is selected, steps with **CU** may be skipped.
- The operational/troubleshooting procedure will check the ESSS Range Extension System with any tank configuration, installed on the helicopter, authorized by the Aircraft Operator's Manual (TM 1-1520-237-10).
- If APU is selected for electrical power, it will provide air pressure.
- If AGPU is use for external electrical power, it can provide an external air source when pneumatic line is connected to helicopter.
- Special Icons used in this operational/troubleshooting procedure:

**CU** Steps that are to be done if compressor unit is used.

**INBD** Steps that are to be done if inboard tanks are installed.

**OUTBD** Steps that are to be done if outboard tanks are installed.

**AGPU** Steps that are to be done if AGPU is to be used.

1. Place auxiliary fuel management control panel XFER MODE switch to OFF.
2. **CU** Using locally-made pneumatic adapter (WP 1805 00), tee air compressor into helicopter bleed-air QD and ESSS bleed-air hose. ■
3. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
4. Connect locally-made fuel quantity simulator test set (simulator) (WP 1805 00) power cord to UTIL RCPT +28V (J258) in cabin ceiling.
5. Disconnect helicopter connector P905 from fuel quantity signal conditioner connector J905.
6. Connect simulator connector J905 to helicopter connector P905.
7. Turn simulator TANK 1 and TANK 2 controls fully counterclockwise.
8. Connect No. 1 multimeter to simulator TANK 1 OUTPUT.
9. Connect No. 2 multimeter to simulator TANK 2 OUTPUT.
10. **AGPU** Connect AGPU power cable to helicopter external power receptacle. ■

---

**SETUP OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**

11. **AGPU** Connect AGPU pneumatic hose to helicopter. ■
12. Turn on electrical power.
13. Apply air source, regulated at 35-45 psi, to helicopter's external tanks.
14. Record time.
15. Place auxiliary fuel management control panel PRESS switch to ALL.
16. Inspect fuel and pneumatic hoses and fittings for signs of leaking or damage (TM 1-1500-204-23).
17. Turn simulator TANK 1 control to get 6.0 vdc No. 1 multimeter indication.
18. Turn simulator TANK 2 control to get 6.0 vdc No. 2 multimeter indication.

**Indication/Condition**

6.0 vdc No. 2 multimeter shall be indicated.

**Corrective Action**

None Required

## OUTBOARD TANKS INSTALLED OPERATIONAL/TROUBLESHOOTING PROCEDURE

### PROCEDURE

#### Step

#### NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- As an alternate to using the compressor unit to pressurize the ESSS Range Extension System, the Aviation Ground Power Unit (AGPU) or the APU may be used. If the alternate is selected, steps with **CU** may be skipped.
- The operational/troubleshooting procedure will check the ESSS Range Extension System with any tank configuration, installed on the helicopter, authorized by the Aircraft Operator's Manual (TM 1-1520-237-10).
- If APU is selected for electrical power, it will provide air pressure.
- If AGPU is use for external electrical power, it can provide an external air source when pneumatic line is connected to helicopter.
- Special Icons used in this operational/troubleshooting procedure:
  - CU** Steps that are to be done if compressor unit is used.
  - INBD** Steps that are to be done if inboard tanks are installed.
  - OUTBD** Steps that are to be done if outboard tanks are installed.
  - AGPU** Steps that are to be done if AGPU is to be used.
- If any trouble is encountered during this operational/troubleshooting procedure, go to WP 0168 00.

1. Do SETUP, in this work package, if not already done.
2. Check that auxiliary fuel management control panel L and R OUTBD fuel quantity displays indicate 190 pounds.
3. Verify auxiliary fuel management control panel XFER MODE switch is OFF.
4. Turn simulator TANK 2 control until No. 2 multimeter indicates less than 5.5 vdc.
5. Place auxiliary fuel management control panel XFER FROM switch to OUTBD.
6. Before continuing, make sure that at least 10 minutes have elapsed from time recorded in SETUP, step 14, in this work package.
7. Place auxiliary fuel management control panel XFER MODE switch to AUTO until R OUTBD EMPTY or L OUTBD EMPTY annunciator goes on.
8. Place auxiliary fuel management control panel MAN XFER switch to BOTH.
9. Place auxiliary fuel management control panel XFER MODE switch to MAN.



---

**OUTBOARD TANKS INSTALLED OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Indication/Condition**

- a. Auxiliary fuel management control panel R NO FLOW and L NO FLOW annunciators shall go off, then come back on with 5 minutes.
- b. Auxiliary fuel management control panel fuel quantity displays shall not change when the R NO FLOW and L NO FLOW annunciators are on.

**Corrective Action**

None Required

**Step**

10. Place auxiliary fuel management control panel XFER MODE switch and PRESS switches to OFF.
11. Check for less than 50 mvac and between -50 and 50 mvdc between J705-A and J705-B (left outboard ejector rack).

**Indication/Condition**

Voltage shall be less than 50 mvac and between -50 and 50 mvdc.

**Corrective Action**

If voltage is not as specified, go to WP 0168 00 or WP 0160 00.

**Step**

12. Check for less than 50 mvac and between -50 and 50 mvdc between J710-A and J710-B (right outboard ejector rack).

**Indication/Condition**

Voltage shall be less than 50 mvac and between -50 and 50 mvdc.

**Corrective Action**

If voltage is not as specified, go to WP 0168 00 or WP 0160 00.

**Step**

13. If no other check is to be done, go to SHUTDOWN, in this work package.

**Indication/Condition**

Shutdown shall be performed.

**Corrective Action**

None Required

## INBOARD TANKS INSTALLED OPERATIONAL/TROUBLESHOOTING PROCEDURE

### PROCEDURE

#### Step

#### NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- As an alternate to using the compressor unit to pressurize the ESSS Range Extension System, the Aviation Ground Power Unit (AGPU) or the APU may be used. If the alternate is selected, steps with **CU** may be skipped.
- The operational/troubleshooting procedure will check the ESSS Range Extension System with any tank configuration, installed on the helicopter, authorized by the Aircraft Operator's Manual (TM 1-1520-237-10).
- If APU is selected for electrical power, it will provide air pressure.
- If AGPU is use for external electrical power, it can provide an external air source when pneumatic line is connected to helicopter.
- Special Icons used in this operational/troubleshooting procedure:
  - CU** Steps that are to be done if compressor unit is used.
  - INBD** Steps that are to be done if inboard tanks are installed.
  - OUTBD** Steps that are to be done if outboard tanks are installed.
  - AGPU** Steps that are to be done if AGPU is to be used.
- If any trouble is encountered during this operational/troubleshooting procedure, go to WP 0168 00.

1. Do SETUP, in this work package, if not already done.
2. Check auxiliary fuel management control panel L and R OUTBD fuel quantity displays indicate 250 pounds.
3. Verify auxiliary fuel management control panel XFER MODE switch is OFF.
4. Place auxiliary fuel management control panel XFER FROM switch to INBD.
5. Turn simulator TANK 2 control until No. 2 multimeter indicates less than 5.5 vdc.
6. Before continuing, make sure that at least 10 minutes have elapsed from time recorded in SETUP, step 14.
7. Place auxiliary fuel management control panel XFER MODE switch to AUTO until R INBD EMPTY OR L INBD EMPTY indicator goes on.
8. Place auxiliary fuel management control panel MAN XFER switch to BOTH.
9. Place auxiliary fuel management control panel XFER MODE switch to MAN.

#### Indication/Condition

- a. Auxiliary fuel management control panel R NO FLOW and L NO FLOW annunciators shall go off, then come back on within 5 minutes.

---

**INBOARD TANKS INSTALLED OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued

- b. Auxiliary fuel management control panel fuel quantity displays shall not change when the R NO FLOW and L NO FLOW annunciators are on.

**Corrective Action**

None Required

**Step**

10. Place auxiliary fuel management control panel XFER MODE and PRESS switches to OFF.
11. Check for less than 50 mvac and between -50 and 50 mvdc between J709-B and J709-A (left inboard ejector rack).

**Indication/Condition**

Voltage shall be less than 50 mvac and between -50 and 50 mvdc.

**Corrective Action**

If Indication/Condition is not as specified, go to WP 0168 00 or WP 0160 00.

**Step**

12. Check for less than 50 mvac and between -50 and 50 mvdc between J709-C and J709-N (left inboard ejector rack).

**Indication/Condition**

Voltage shall be less than 50 mvac and between -50 and 50 mvdc.

**Corrective Action**

If Indication/Condition is not as specified, go to WP 0168 00 or WP 0160 00.

**Step**

13. Check for less than 50 mvac and between -50 and 50 mvdc between J706-B and J706-A (right inboard ejector rack).

**Indication/Condition**

Voltage shall be less than 50 mvac and between -50 and 50 mvdc.

**Corrective Action**

If Indication/Condition is not as specified, go to WP 0168 00 or WP 0160 00.

**Step**

14. Check for less than 50 mvac and between -50 and 50 mvdc between J706-C and J706-N (right inboard ejector rack).

**Indication/Condition**

Voltage shall be less than 50 mvac and between -50 and 50 mvdc.

**Corrective Action**

If Indication/Condition is not as specified, go to WP 0168 00 or WP 0160 00.

**Step**

15. If no other check is to be done, go to SHUTDOWN, in this work package.

---

**INBOARD TANKS INSTALLED OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Indication/Condition**

Shutdown shall be performed.

**Corrective Action**

None Required

**SHUTDOWN OPERATIONAL/TROUBLESHOOTING PROCEDURE  
PROCEDURE****Step**

1. Turn off electrical power.
2. Turn off air source.
3. Allow bleed-air regulator valves to vent air to ambient pressure.
4. **CU** Disconnect air compressor, hoses, and locally-made pneumatic adapter. ■
5. **CU** Connect pneumatic hose to outboard side of helicopter bleed-air QD. ■
6. **CU** **TORQUE NUTS TO 135 - 150 INCH-POUNDS.** ■
7. Disconnect test equipment.
8. Connect helicopter electrical connectors.
9. Install all explosive cartridges in ejector racks (WP 1445 00).
10. Install all vertical stores (pylon) rear fairings (WP 1425 00).

**Indication/Condition**

All vertical stores (pylon) rear fairings shall be installed.

**Corrective Action**

None Required

LOCATION

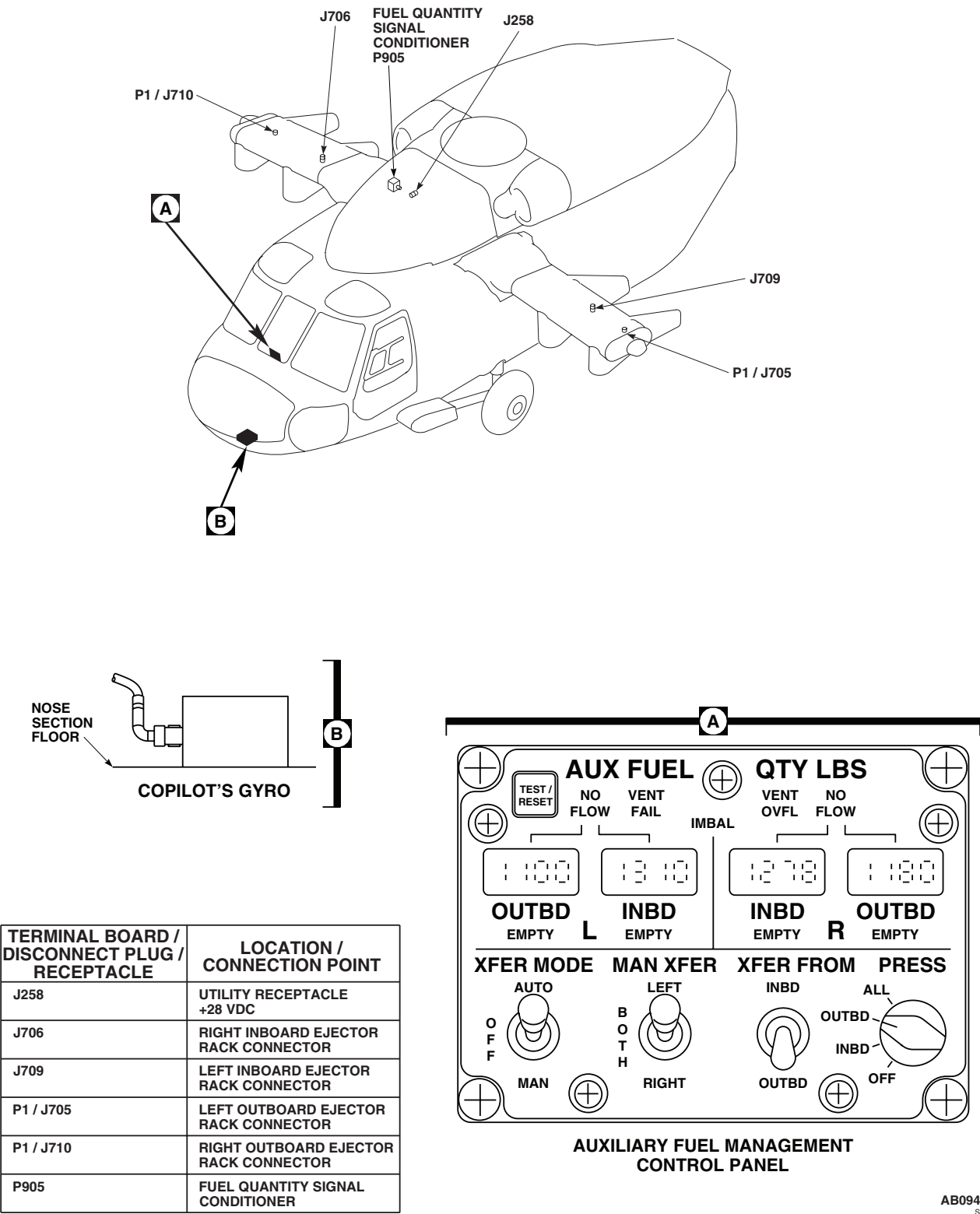
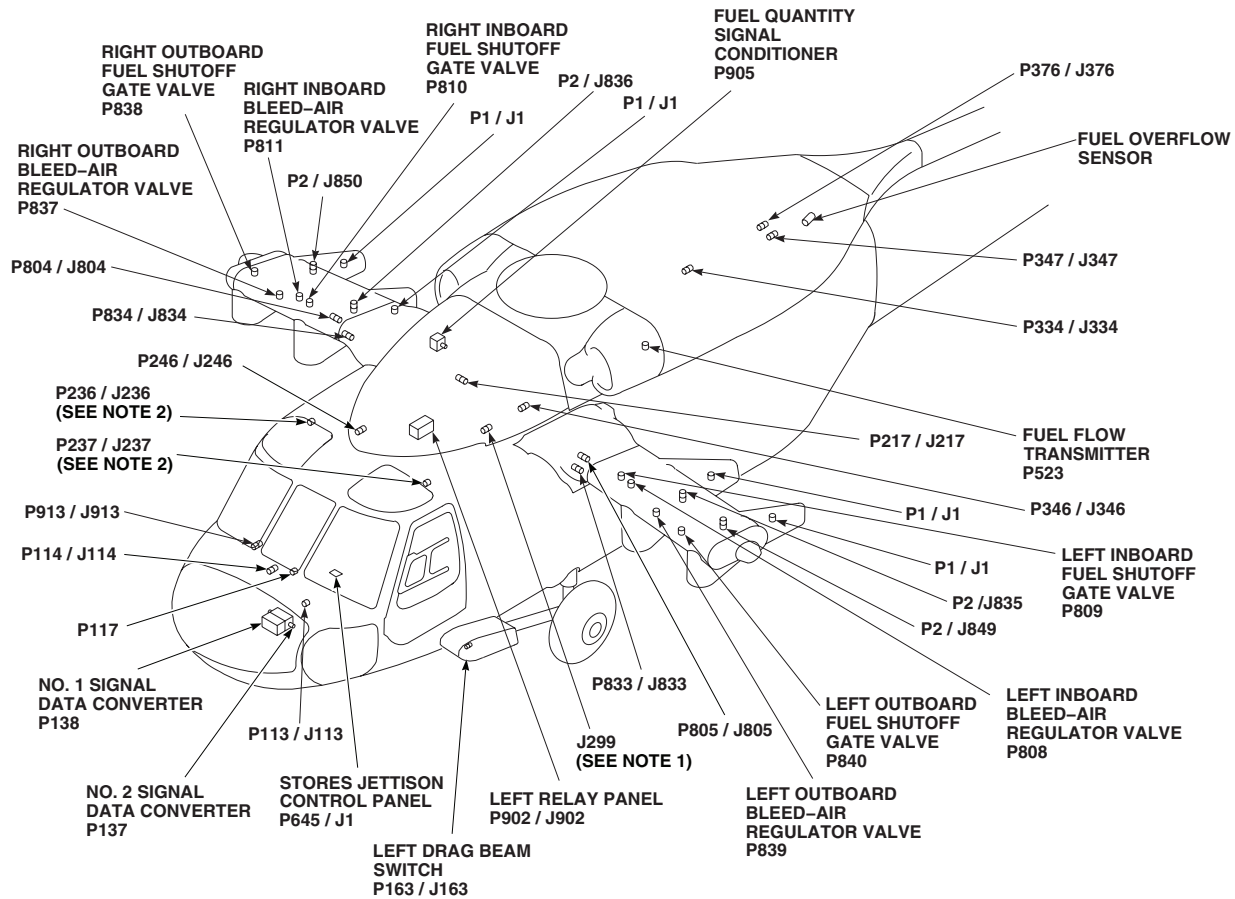


Figure 1. ESSS External Tank Check Location Diagram. (Sheet 1 of 2)

## LOCATION - Continued



| TERMINAL BOARD/<br>DISCONNECT PLUG/<br>RECEPTACLE | LOCATION/<br>CONNECTION POINT                     |
|---------------------------------------------------|---------------------------------------------------|
| P346 / J346                                       | CABIN CEILING<br>BL 20 LH, STA 304                |
| P347 / J347                                       | TRANSITION SECTION<br>BL 10 LH, STA 425           |
| P360 / J360                                       | AUXILIARY FUEL MANAGE-<br>MENT CONTROL PANEL      |
| P361 / J361                                       | AUXILIARY FUEL MANAGE-<br>MENT CONTROL PANEL      |
| P376 / J376                                       | AFT TRANSITION<br>BL 18.25 LH, STA 443.5          |
| P523                                              | FUEL FLOW TRANSMITTER                             |
| P645 / J1                                         | STORES JETTISON<br>CONTROL PANEL                  |
| P804 / J804                                       | HORIZONTAL STORES<br>SUPPORT<br>BL 59 RH, STA 295 |
| P805 / J805                                       | HORIZONTAL STORES<br>SUPPORT<br>BL 59 LH, STA 295 |
| P808                                              | LEFT INBOARD BLEED-AIR<br>REGULATOR VALVE         |

| TERMINAL BOARD/<br>DISCONNECT PLUG/<br>RECEPTACLE | LOCATION/<br>CONNECTION POINT                |
|---------------------------------------------------|----------------------------------------------|
| P809                                              | LEFT INBOARD FUEL SHUT-<br>OFF GATE VALVE    |
| P810                                              | RIGHT INBOARD FUEL<br>SHUTOFF GATE VALVE     |
| P811                                              | RIGHT INBOARD BLEED-AIR<br>REGULATOR VALVE   |
| P833 / J833                                       | HSS ATTACHMENT FITTING<br>BL 59 LH, STA 295  |
| P834 / J834                                       | HSS ATTACHMENT FITTING<br>BL 59 LH, STA 295  |
| P837                                              | RIGHT OUTBOARD BLEED-<br>AIR REGULATOR VALVE |
| P838                                              | RIGHT OUTBOARD FUEL<br>SHUTOFF GATE VALVE    |
| P839                                              | LEFT OUTBOARD BLEED-<br>AIR REGULATOR VALVE  |
| P840                                              | LEFT OUTBOARD FUEL<br>SHUTOFF GATE VALVE     |
| P902 / J902                                       | LEFT RELAY PANEL                             |
| P905                                              | FUEL QUANTITY SIGNAL<br>CONDITIONER          |
| P913 / J913                                       | COCKPIT<br>BL 24 RH, STA 247                 |

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SA

Figure 1. ESSL External Tank Check Location Diagram. (Sheet 2 of 2)

END OF WORK PACKAGE

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**UNIT LEVEL****MISSION EQUIPMENT****FORWARD LOOKING INFRARED (FLIR) HH-60A HH-60L**

**This WP supersedes WP 0170 00, dated 17 April 2006.**

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**INITIAL SETUP:****Test Equipment**

Multimeter, AN/PSM-45A

WP 0104 00

WP 0824 00

WP 1556 00

**Tools and Special Tools**

Electrical Repairer Toolkit, SC 5180-99-B06

WP 1557 00

WP 1558 00

**Personnel Required**

Aircraft Electrician MOS 15F (2)

WP 1685 00

WP 1747 00

**References**

TM 1-1520-237-10

WP 0100 00

**Equipment Condition**

External Electrical Power Available, WP 1685 00  
and/or APU Operational, TM 1-1520-237-10

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**SPECIAL INSTRUCTIONS****NOTE**

See Figure 1 for component location diagram, and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

This operational/troubleshooting procedure is subdivided:

- Setup
- Operational Check
- Drift Compensation
- Shutdown

**SETUP OPERATIONAL/TROUBLESHOOTING PROCEDURE****PROCEDURE****Step****WARNING**

To prevent injury, make sure all personnel are clear of the turret FLIR unit when operational.

**NOTE**

If a circuit breaker pops out during fault isolation procedure, a short circuit is indicated (WP 0104 00).

1. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
2. Turn on electrical power.
3. Turn pilot's and copilot's MFDs switches ON.
4. On CDU press STS key to get to SYSTEM STATUS page.

**Indication/Condition**

The letter G shall appear next to MFD-1 and MFD-2.

**Corrective Action**

If letter F appears next to MFD-1 and/or MFD-2, troubleshoot MFD/caution/advisory warning system (WP 0100 00).

**Step**

5. After caution advisory grid appears, press key B6 (C/A) on both MFDs to clear caution/advisory display.
6. Remove FLIR turret safety pin.
7. On pilot and copilot MFDs, press FLIR key (B5) to select FLIR display.

**Indication/Condition**

FLIR display shall be selected.

**Corrective Action**

None Required



---

## OPERATIONAL CHECK OPERATIONAL/TROUBLESHOOTING PROCEDURE

### PROCEDURE

#### Step

#### WARNING

To prevent injury, make sure all personnel are clear of the turret FLIR unit when operational.

#### NOTE

If a circuit breaker pops out during fault isolation procedure, a short circuit is indicated (WP 0104 00).

1. Do SETUP, in this work package, if not already done.

#### NOTE

Allow five minutes for the FLIR imaging cooler to reach operating temperature.

2. On FLIR control panel, set POWER ON/OFF switch to ON.

#### Indication/Condition

- a. The turret FLIR shall cage (point straight ahead).
- b. Pilot's and copilot's MFD's shall indicate a FLIR image.

#### Corrective Action

1. If Indication/Condition a. is not as specified, and all functions become disabled after 40 seconds, replace turret FLIR unit (WP 1556 00).
2. If Indication/Condition b. is not as specified, go to MFD DOES NOT DISPLAY FLIR IMAGE AND MENU PAGE 1, in this work package.
3. If pilot's MFD does not indicate a FLIR image, go to PILOT'S MFD DOES NOT INDICATE A FLIR IMAGE, in this work package.
4. If copilot's MFD does not indicate a FLIR image, go to COPILOT'S MFD DOES NOT INDICATE A FLIR IMAGE, in this work package.

#### Step

3. On FLIR control panel, press MENU/STOW/LOCK/FREEZE/HOOK (menu control) switch to MENU.

#### Indication/Condition

The pilot and copilot MFDs shall display a FLIR image and MENU page 1.

#### Corrective Action

If Indication/Condition is not as specified, replace central electronic unit (WP 1557 00).

#### Step

4. If necessary, using the joystick on FLIR control panel, scroll up or down to highlight MENU option, and then set menu control switch to HOOK to display MENU page 2.

---

**OPERATIONAL CHECK OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued

5. Using the joystick on FLIR control panel, scroll up or down to highlight FIT option, and then set menu control switch to HOOK to display FAULT ISOLATION TEST MENU.
6. Using the joystick on FLIR control panel, scroll up or down to highlight FIT option, and then set menu control to LOCK to start the fault isolation test.

**Indication/Condition**

On pilot and copilot MFDs, FIT IN PROGRESS followed by FIT OK, shall be displayed at top center of screen.

**Corrective Action**

If an error code is displayed, make a note of error code and go to ERROR CODE IS DISPLAYED IN FAULT ISOLATION TEST SUB-MENU, in this work package.

**Step**

7. If necessary, once the fault isolation test is complete, use the joystick on FLIR control panel to scroll down and highlight EXIT option, then set menu control to HOOK to get back to MENU page 2.
8. If necessary, on FLIR control panel, set menu control to HOOK to exit MENU.

**NOTE**

Have an assistant stand at front of aircraft to observe turret movement.

9. On FLIR control panel, push joystick in any direction for a few seconds to change turret position.
10. On FLIR control panel, toggle FOC OUT/IP HH/FOC IN/POLARITY/CAGE (mode selector) switch to CAGE to move turret FLIR forward.

**Indication/Condition**

Turret FLIR unit shall point straight forward and shall not drift more than 5°/minute.

**Corrective Action**

If a drift of more than 5°/minute is observed, do DRIFT COMPENSATION.

**Step**

11. On FLIR control panel, toggle mode selector to IP/HH (inertial pointing) until INRPT appears on upper status bar on pilot and copilot MFDs.
12. On FLIR control panel, push joystick up for a few seconds.

**Indication/Condition**

Turret FLIR unit shall pivot up.

**Corrective Action**

If Indication/Condition is not as specified, replace FLIR control panel (WP 1558 00).

**Step**

13. On FLIR control panel, push joystick down for a few seconds.

**Indication/Condition**

Turret FLIR unit shall pivot down.

---

**OPERATIONAL CHECK OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued**Corrective Action**

If Indication/Condition is not as specified, replace FLIR control panel (WP 1558 00).

**Step**

14. On FLIR control panel, push joystick left for a few seconds.

**Indication/Condition**

Turret FLIR unit shall rotate counterclockwise.

**Corrective Action**

If Indication/Condition is not as specified, replace FLIR control panel (WP 1558 00).

**Step**

15. On FLIR control panel, push joystick right for a few seconds.

**Indication/Condition**

Turret FLIR unit shall rotate clockwise.

**Corrective Action**

If Indication/Condition is not as specified, replace FLIR control panel (WP 1558 00).

**Step**

16. On FLIR control panel, toggle mode selector to IP/HH (heading hold) until HDHLD is displayed in upper status bar on pilot and copilot MFDs.
17. Press in and release mode selector a few times while observing image.

**Indication/Condition**

Image shall change from white-hot to black-hot and back each time the mode selector is pressed and released.

**Corrective Action**

If Indication/Condition is not as specified, replace turret FLIR unit (WP 1556 00).

**Step**

18. On FLIR control panel, toggle mode selector to FOC OUT a few times while observing image.

**Indication/Condition**

Image sharpness on pilot and copilot MFDs shall decrease.

**Corrective Action**

If Indication/Condition is not as specified, replace turret FLIR unit (WP 1556 00).

**Step**

19. On FLIR control panel, toggle mode selector to FOC IN a few times while observing image.

**Indication/Condition**

Image sharpness on pilot and copilot MFDs shall increase.

---

**OPERATIONAL CHECK OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued**Corrective Action**

If Indication/Condition is not as specified, replace turret FLIR unit (WP 1556 00).

**Step****NOTE**

Toggle menu control between FOC IN and FOC OUT as necessary to get a clear image.

20. On FLIR control panel, set FOV switch to WFOV (wide field of view).

**Indication/Condition**

Pilot and copilot MFDs shall display a wide field of view.

**Corrective Action**

If Indication/Condition is not as specified, replace turret FLIR unit (WP 1556 00).

**Step**

21. On FLIR control panel, set FOV switch to NFOV (narrow field of view).

**Indication/Condition**

Pilot and copilot MFDs shall display a narrow field of view.

**Corrective Action**

If Indication/Condition is not as specified, replace turret FLIR unit (WP 1556 00).

**Step**

22. On FLIR control panel, set FOV switch to 2X ZOOM.

**Indication/Condition**

Pilot and copilot MFDs shall display an enlarged image.

**Corrective Action**

If Indication/Condition is not as specified, replace turret FLIR unit (WP 1556 00).

**Step**

23. On FLIR control panel, set menu control to FREEZE to hold the last image.  
24. On FLIR control panel, press joystick in any direction for a few seconds.

**Indication/Condition**

Turret FLIR unit shall not move and image shall remain fixed.

**Corrective Action**

If Indication/Condition is not as specified, replace turret FLIR unit (WP 1556 00).

**Step**

25. On FLIR control panel, set menu control to FREEZE to release image.  
26. On FLIR control panel, press joystick in any direction for a few seconds.

---

**OPERATIONAL CHECK OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued**Indication/Condition**

Turret FLIR unit shall move and image shall change.

**Corrective Action**

If Indication/Condition is not as specified, replace turret FLIR unit (WP 1556 00).

**Step**

27. On FLIR control panel, press and hold GAIN UP/LVL UP/AUTO GAIN/GAIN ON/LVL DN (gain control) to GAIN DN while observing image and bar at top right of pilot and copilot MFDs.

**Indication/Condition**

The contrast of the image shall decrease and the bar shall widen.

**Corrective Action**

If Indication/Condition is not as specified, replace turret FLIR unit (WP 1556 00).

**Step**

28. On FLIR control panel, press and hold gain control to GAIN UP while observing image and bar at top right of pilot and copilot MFDs.

**Indication/Condition**

The contrast of the image shall increase and the bar shall narrow.

**Corrective Action**

If Indication/Condition is not as specified, replace turret FLIR unit (WP 1556 00).

**Step**

29. On FLIR control panel, press and hold gain control to LVL DN while observing image and bar at top right of pilot and copilot MFDs.

**Indication/Condition**

The image quality shall change and the bar shall move left.

**Corrective Action**

If Indication/Condition is not as specified, replace turret FLIR unit (WP 1556 00).

**Step**

30. On FLIR control panel, press and hold gain control to LVL UP while observing image and bar at top right of pilot and copilot MFDs.

**Indication/Condition**

The image quality shall change and the bar shall move right.

**Corrective Action**

If Indication/Condition is not as specified, replace turret FLIR unit (WP 1556 00).

---

**OPERATIONAL CHECK OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Step**

31. On FLIR control panel, press in and release gain control while observing image and bar at top of pilot and copilot MFDs.

**Indication/Condition**

The image quality shall return to the pre-existing level (improve) and the bar shall disappear.

**Corrective Action**

If Indication/Condition is not as specified, replace turret FLIR unit (WP 1556 00).

**Step**

32. On FLIR control panel, press in and release menu control.

**Indication/Condition**

The turret FLIR UNIT shall rotate forward to center, then roll back and up and previously displayed FLIR image shall disappear.

**Corrective Action**

If Indication/Condition is not as specified, replace turret FLIR unit (WP 1556 00).

**Step**

33. If no other check is to be done, go to SHUTDOWN, in this work package.

**Indication/Condition**

Shutdown shall be performed.

**Corrective Action**

None Required

**DRIFT COMPENSATION OPERATIONAL/TROUBLESHOOTING PROCEDURE  
PROCEDURE****Step****WARNING**

To prevent injury, make sure all personnel are clear of the Turret FLIR Unit when operational.

**NOTE**

If a circuit breaker pops out during fault isolation procedure, a short circuit is indicated (WP 0104 00).

1. Do SETUP, in this work package, unless already done.
2. On FLIR control panel, set POWER switch to ON.
3. On FLIR control panel, set FOV switch to WFOV.
4. On FLIR control panel, set mode selector to CAGE.

---

**DRIFT COMPENSATION OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued

5. Using joystick control on FLIR control panel, position FLIR turret unit to observe a single distant object in the center of pilot and copilot MFDs.
6. On FLIR control panel, set FOV switch to 2X ZOOM and toggle menu control between FOC IN and FOC OUT as necessary to focus centered object.
7. On FLIR control panel, push mode selector to IP/HH until INRPT is displayed on pilot and copilot MFDs upper status bar.
8. On FLIR control panel, push menu control to MENU.
9. If necessary, using the joystick control on FLIR control panel, scroll up or down to highlight MENU option, then set menu control to HOOK to get to MENU page 2.
10. Using the joystick control on FLIR control panel, scroll up or down to highlight CAL option, then set menu control to HOOK to get to CALIBRATION menu.
11. If necessary to reduce lateral drift, using joystick control on FLIR control panel, scroll up or down to highlight DRIFT AZ option and then set menu control to HOOK.

**NOTE**

Decreasing the numerical value reduces drift to the left, while increasing the numerical value reduces drift to the right.

12. While observing a single object in the center of the pilot or copilot MFDs, adjust highlighted value in menu using joystick until minimum or no drift is obtained.
13. On FLIR control panel, set menu control to HOOK to get to left side of menu.
14. If necessary to reduce vertical drift, using joystick control on FLIR control panel, scroll up or down to highlight DRIFT EL option and then set menu control to HOOK.

**NOTE**

Decreasing the numerical value reduces drift downward, while increasing the numerical value reduces drift upward.

15. While observing a single object in the center of the pilot and copilot MFDs, adjust highlighted value in menu using joystick until minimum or no drift is obtained.
16. On FLIR control panel, set menu control to HOOK to get to left side of menu.
17. Using joystick control on FLIR control panel, scroll up or down to highlight EXIT option and then set menu control to HOOK to exit the CALIBRATION menu.
18. On FLIR control panel, press menu control to HOOK to exit MENU.
19. If no other check is to be done, go to SHUTDOWN, in this work package.

**Indication/Condition**

Shutdown shall be performed.

**Corrective Action**

None Required

## SHUTDOWN OPERATIONAL/TROUBLESHOOTING PROCEDURE

### PROCEDURE

#### Step

1. Turn pilot and copilot MFDs switches OFF.
2. On FLIR control panel, select STOW.
3. On FLIR control panel, set POWER switch to OFF.
4. Turn off electrical power.
5. Install FLIR turret safety pin.

#### Indication/Condition

FLIR turret safety pin shall be installed.

#### Corrective Action

None Required

## MFD DOES NOT DISPLAY FLIR IMAGE AND MENU PAGE 1

### SYMPTOM

MFD does not display FLIR image and MENU page 1.

### MALFUNCTION

MFD does not display FLIR image and MENU page 1.

### CORRECTIVE ACTION

1. Check for 28 vdc between:
  - P210R-A and P210R-B
  - P210R-C and P210R-D
  - P210R-E and P210R-F
  - a. If voltage is as specified, go to **3**.
  - b. If voltage is not as specified, go to **2**.
2. Check for 28 vdc between terminal 1 of FLIR circuit breaker and airframe ground.
  - a. If voltage is as specified, repair/replace wiring (WP 1747 00). Go to **7**.
  - b. If voltage is not as specified, replace FLIR circuit breaker (WP 0824 00). Go to **7**.
3. Check for 28 vdc between 1J3-5 and 1J3-6 on central electronic unit.
  - a. If voltage is as specified, go to **4**.
  - b. If voltage is not as specified, replace central electronic unit (WP 1557 00). Go to **7**.
4. Check continuity between P215R-5 and P215R-6.
  - a. If continuity is present, go to **6**.
  - b. If continuity is not present, go to **5**.



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**MFD DOES NOT DISPLAY FLIR IMAGE AND MENU PAGE 1 - Continued**

- 5. Check continuity between 3J1-5 and 3J1-6 on FLIR control panel.**
  - a. If continuity is present, repair/replace wiring (WP 1747 00). Go to **7**.
  - b. If continuity is not present, replace FLIR control panel (WP 1558 00). Go to **7**.
- 6. Check continuity between:**
  - **P1016R-B and P212R-R**
  - **P1015R-B and P213R-R**
  - a. If continuity is present, replace central electronic unit (WP 1557 00). Go to **7**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **7**.
- 7. Procedure completed.**

**PILOT'S MFD DOES NOT INDICATE A FLIR IMAGE****SYMPTOM**

Pilot's MFD does not indicate a FLIR image.

**MALFUNCTION**

Pilot's MFD does not indicate a FLIR image.

**CORRECTIVE ACTION**

- 1. Check continuity between P1016-R and P212R-R.**
  - a. If continuity is present, troubleshoot MFD/caution/advisory warning system (WP 0100 00). Go to **2**.
  - b. If trouble remains, replace central electronic unit (WP 1557 00). Go to **2**.
  - c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **2**.
- 2. Procedure completed.**

**COPILOT'S MFD DOES NOT INDICATE A FLIR IMAGE****SYMPTOM**

Copilot's MFD does not indicate a FLIR image.

**MALFUNCTION**

Copilot's MFD does not indicate a FLIR image.

**CORRECTIVE ACTION**

- 1. Check continuity between P1015-R and P213R-R.**
  - a. If continuity is present, troubleshoot MFD/caution/advisory warning system (WP 0100 00). Go to **2**.
  - b. If trouble remains, replace central electronic unit (WP 1557 00). Go to **2**.
  - c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **2**.
- 2. Procedure completed.**

**ERROR CODE IS DISPLAYED IN FAULT ISOLATION TEST SUB-MENU****SYMPTOM**

Error code is displayed in fault isolation test sub-menu.

**MALFUNCTION**

Error code is displayed in fault isolation test sub-menu.

**CORRECTIVE ACTION****1. Check for error code.**

- a. If error code is 101, 102, 103, 201, 20d, 501, 502, 503, 504, 505, 506, 507, 508, 509, 50a, 50b, 50c, 50d or 50e, replace central electronic unit (WP 1557 00). Go to **8**.
- b. If error code is 601, 602, 603, 604, 605, 700, 701, 705, 706, 708, 709 or 70a, replace turret FLIR unit (WP 1556 00). Go to **8**.
- c. If error code is 202, 206, 207, 208, 209, 20a, 20b, or 20c, go to **2**.
- d. If error code is 203 or 800, go to **3**.
- e. If error code is 204, go to **4**.
- f. If error code is 205 or 801, go to **5**.
- g. If error code is 702, 703, or 704, go to **6**.

**2. Check continuity between:**

- **P215R-1 and P214R-1**
- **P215R-11 and P214R-11**
- **P216R-85 and P217R-85**
- **P216R-93 and P217R-93**

- a. If continuity is present, replace central electronic unit (WP 1557 00). Go to **8**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **8**.

**3. Check continuity between:**

- **P215R-2 and P214R-2**
- **P215R-3 and P214R-3**
- **P215R-4 and P214R-4**

- a. If continuity is present, replace central electronic unit (WP 1557 00). Go to **8**.
- b. If trouble remains, replace FLIR control panel (WP 1558 00). Go to **8**.
- c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **8**.

**4. Check continuity between:**

- **P216R-40 and P217R-40**
- **P216R-41 and P217R-41**
- **P216R-42 and P217R-42**

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**ERROR CODE IS DISPLAYED IN FAULT ISOLATION TEST SUB-MENU - Continued**

- P216R-47 and P217R-47
- P216R-48 and P217R-48
- P216R-49 and P217R-49
- P216R-50 and P217R-50
- P216R-51 and P217R-51
- P216R-52 and P217R-52
- P216R-53 and P217R-53
- P216R-58 and P217R-58
- P216R-59 and P217R-59
- P216R-60 and P217R-60
- P216R-61 and P217R-61
- P216R-62 and P217R-62
- P216R-63 and P217R-63
- P216R-64 and P217R-64
- P216R-69 and P217R-69
- P216R-70 and P217R-70
- P216R-71 and P217R-71
- P216R-72 and P217R-72
- P216R-73 and P217R-73
- P216R-74 and P217R-74
- P216R-79 and P217R-79
- P216R-80 and P217R-80
- P216R-81 and P217R-81
- P216R-87 and P217R-87
- P216R-88 and P217R-88
- P216R-89 and P217R-89
- P216R-94 and P217R-94
- P216R-95 and P217R-95
- P216R-96 and P217R-96

- a. If continuity is present, replace central electronic unit (WP 1557 00). Go to **8**.
- b. If trouble remains, replace FLIR control panel (WP 1558 00). Go to **8**.
- c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **8**.

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**ERROR CODE IS DISPLAYED IN FAULT ISOLATION TEST SUB-MENU - Continued****5. Check continuity between:**

- P216R-8 and P217R-8
- P216R-16 and P217R-16
- P216R-35 and P217R-35
- P216R-25 and P217R-25
- P216R-56 and P217R-56
- P216R-67 and P217R-67
- P216R-78 and P217R-78
- P216R-77 and P217R-77
- P216R-44 and P217R-44
- P216R-43 and P217R-43
- P216R-45 and P217R-45
- P216R-55 and P217R-55
- P216R-65 and P217R-65
- P216R-75 and P217R-75
- P216R-92 and P217R-92
- P216R-100 and P217R-100
- P216R-99 and P217R-99
- P216R-91 and P217R-91
- P216R-98 and P217R-98
- P216R-97 and P217R-97

- a. If continuity is present, replace central electronic unit (WP 1557 00). Go to **8**.
- b. If trouble remains, replace turret FLIR unit (WP 1556 00). Go to **8**.
- c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **8**.

**6. Check for ice in turret FLIR unit.**

- a. If ice is present, remove ice and check heater.
- b. If heater does not go on, go to **7**.
- c. If ice is not present, replace turret FLIR unit (WP 1556 00). Go to **8**.

**7. Check continuity between:**

- P216R-20 and P217R-20
- P216R-21 and P217R-21
- P216R-22 and P217R-22

---

**ERROR CODE IS DISPLAYED IN FAULT ISOLATION TEST SUB-MENU - Continued**

- **P216R-23 and P217R-23**
  - **P216R-31 and P217R-31**
  - **P216R-32 and P217R-32**
- a. If continuity is present, replace central electronic unit (WP 1557 00). Go to **8**.
  - b. If trouble remains, replace turret FLIR unit (WP 1556 00). Go to **8**.
  - c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **8**.
- 8. Procedure completed.**

## LOCATION

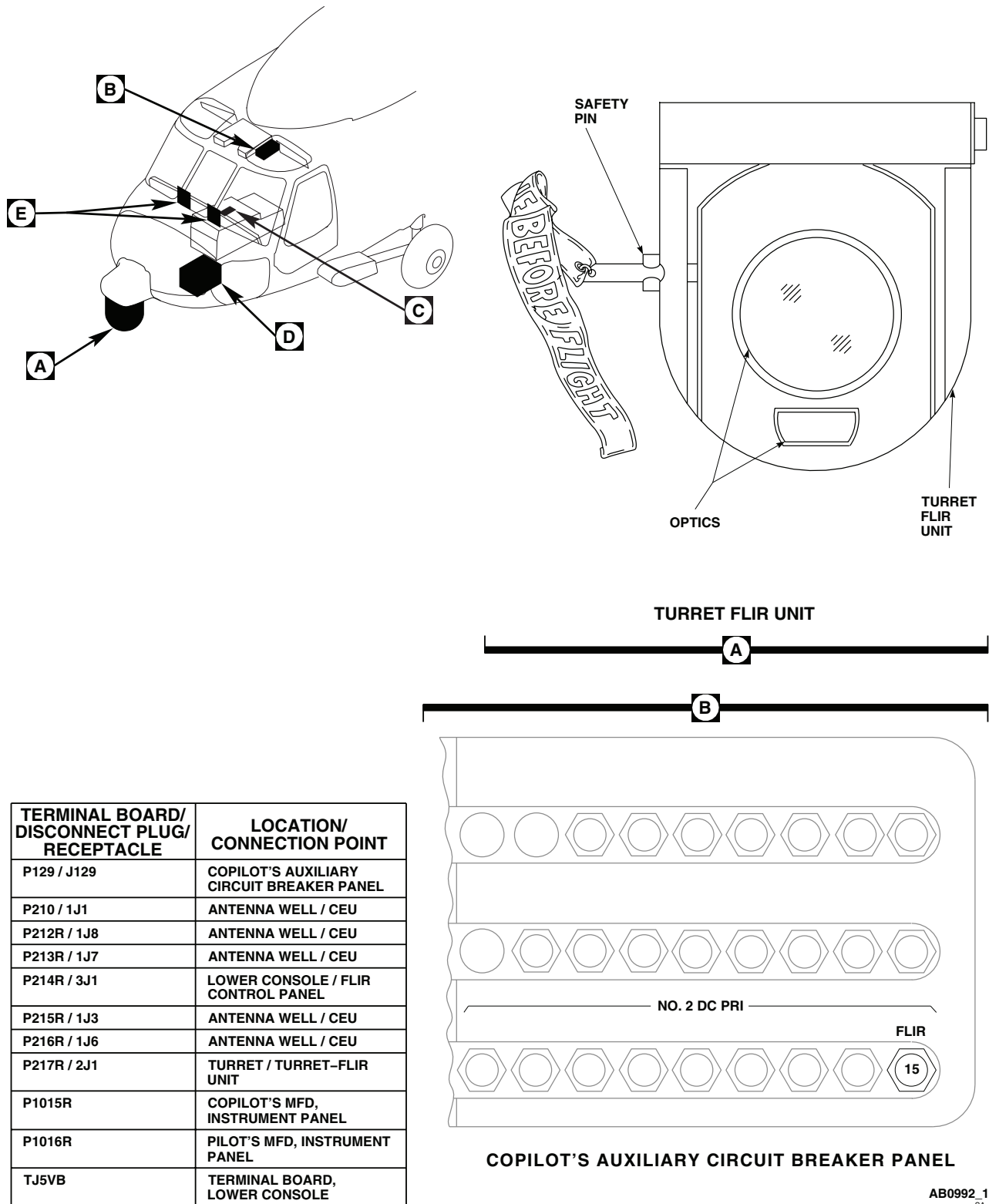


Figure 1. FLIR Location Diagram. (Sheet 1 of 4)

LOCATION - Continued

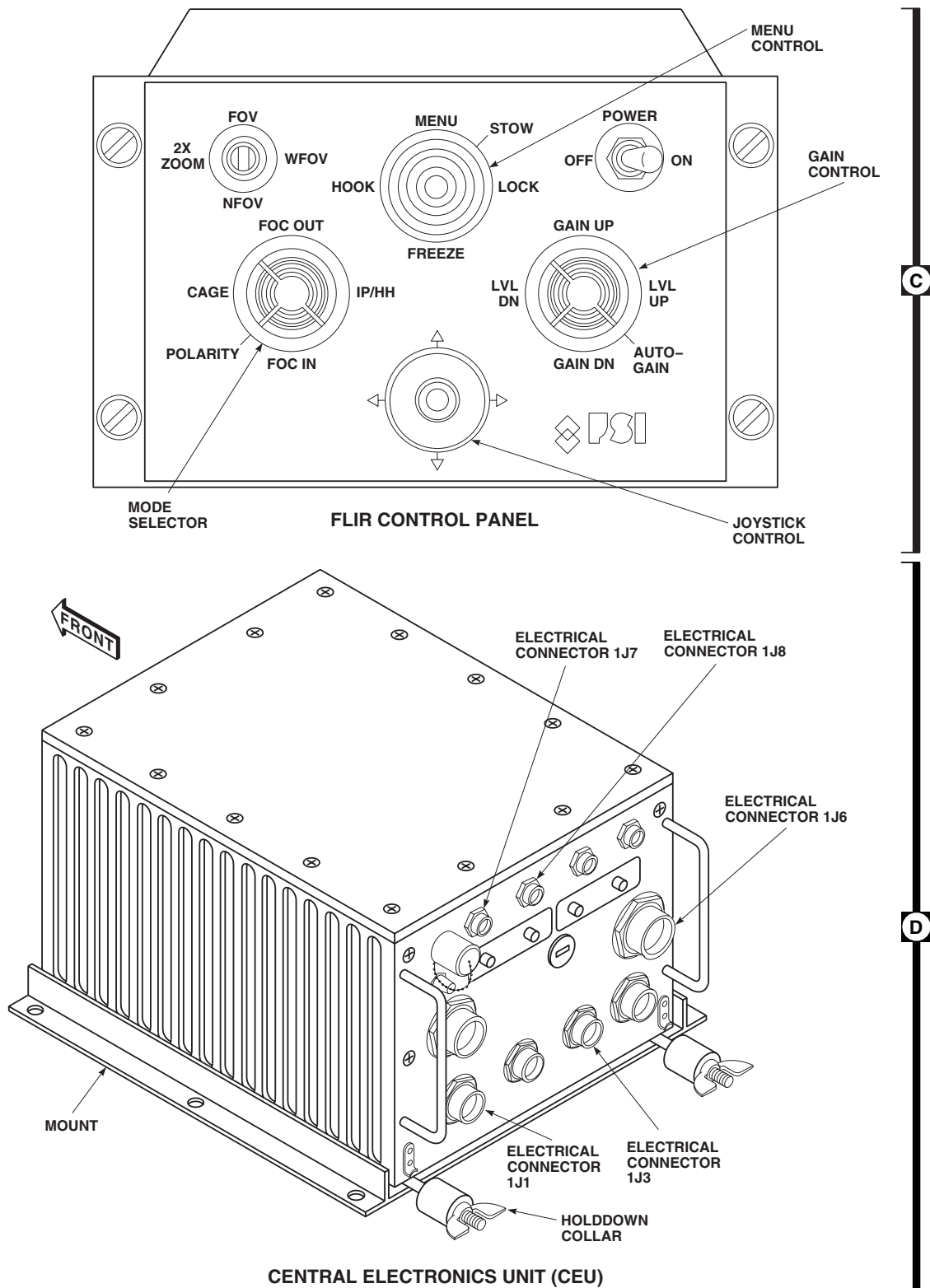


Figure 1. FLIR Location Diagram. (Sheet 2 of 4)

LOCATION - Continued

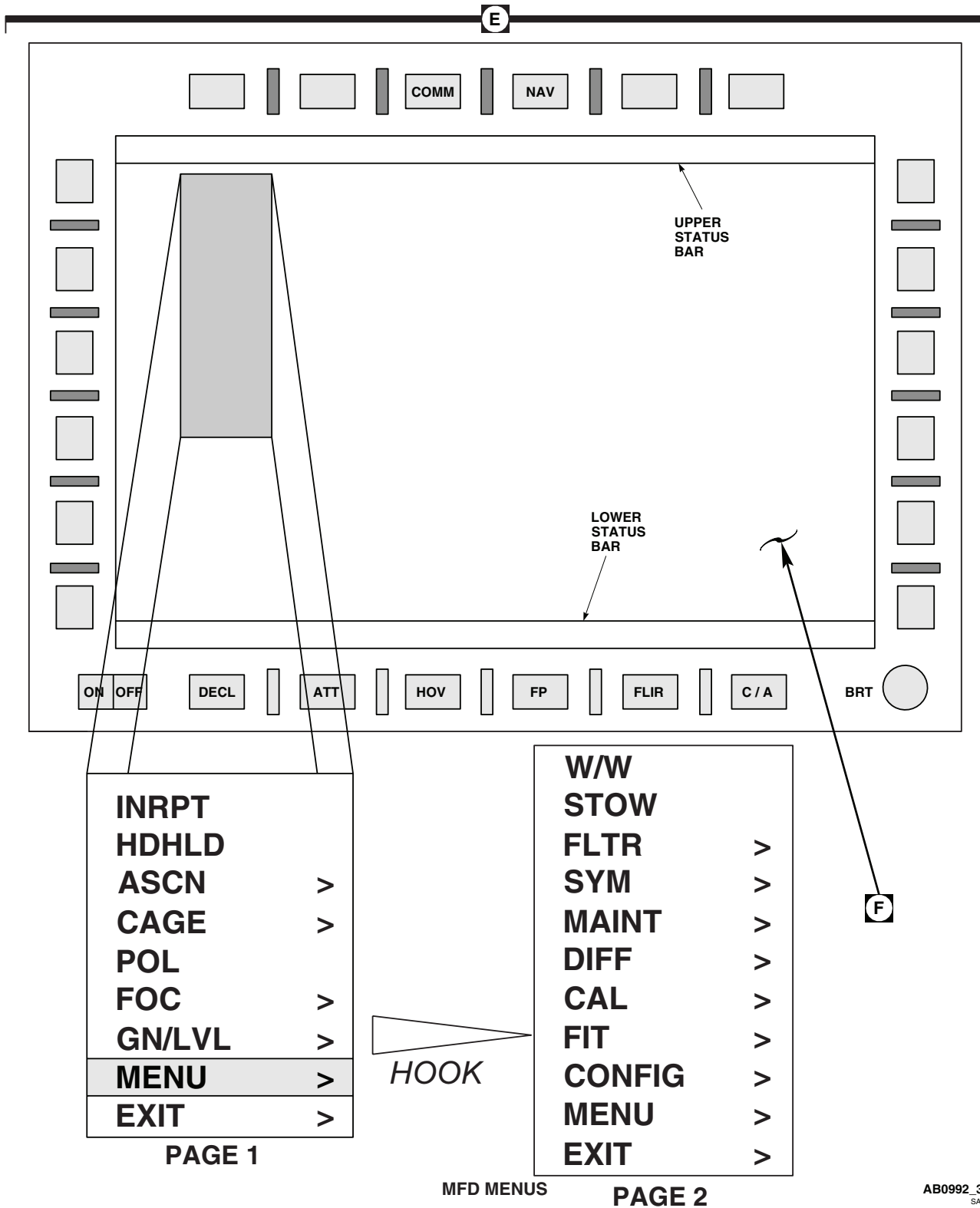
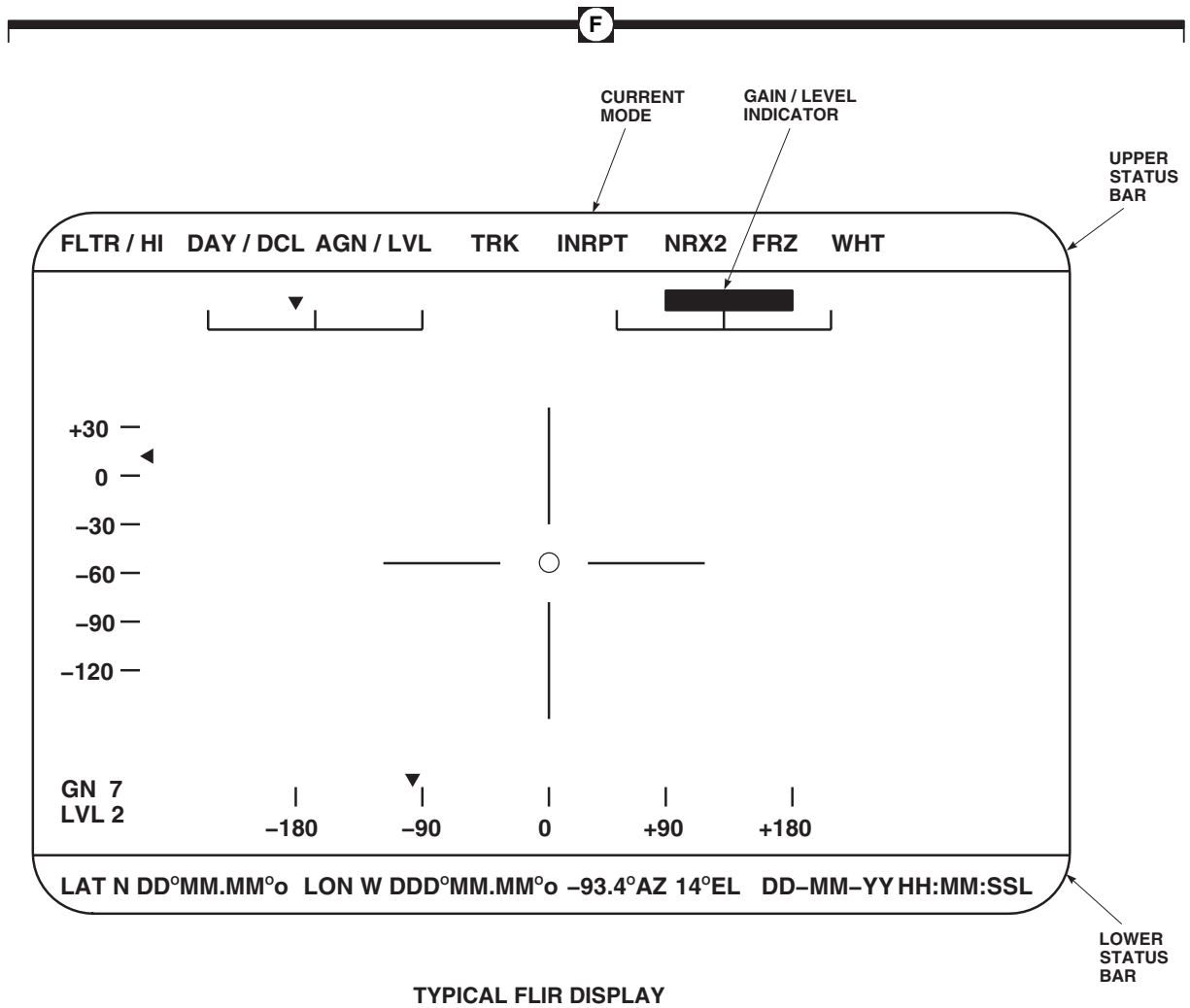


Figure 1. FLIR Location Diagram. (Sheet 3 of 4)



LOCATION - Continued



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Figure 1. FLIR Location Diagram. (Sheet 4 of 4)

SCHEMATICS

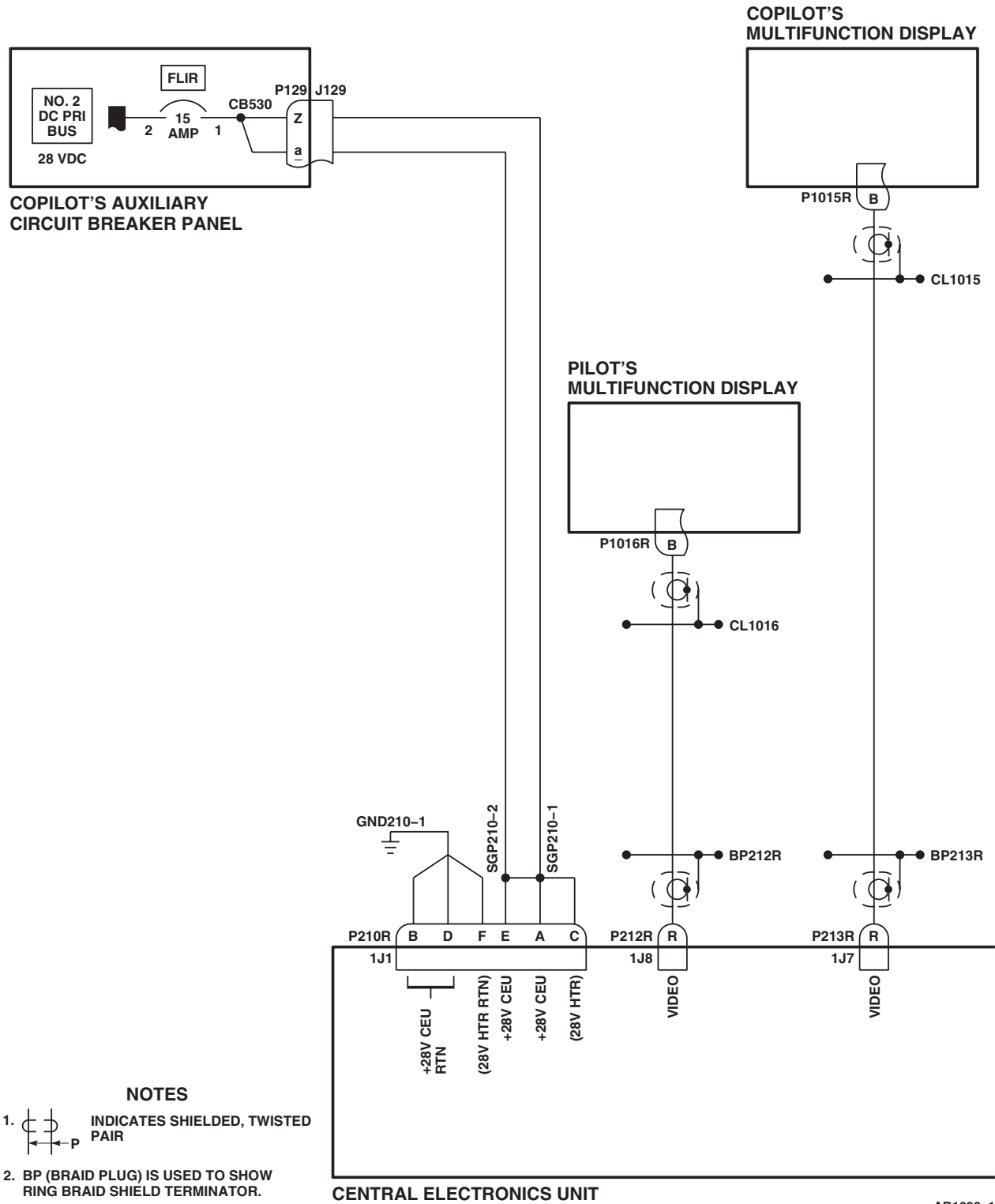


Figure 2. FLIR Schematic Diagram. (Sheet 1 of 5)

## FLIR CONTROL PANEL

The diagram illustrates the rear panel wiring for a device, organized into two main sections: the left side for RS485, power, and video signals, and the right side for video and audio signals.

**Left Section (Connectors 3J1, 1J3):**

- 3J1 (P214R, BP214R):** 10-pin connector. Pins 1, 11, 2, 3, 4, 5, 6, 13, 10, and 9 are shown. Pins 1, 11, 2, 3, 4, 5, 6, and 13 are connected to RS485-, RS485+, +13.8V, 13.8V RETN, +13.8V, FLIR ON, 28V RETURN, and CHASSIS GROUND respectively. Pins 10 and 9 are connected to a transformer labeled TJ5VB, which is also connected to a ground point labeled GND 214-1. A note indicates "LOWER CONSOLE LIGHTS DIMMING" connected to the transformer.
- 1J3 (P215R, BP215R):** 6-pin connector. Pins 1, 11, 2, 3, 4, and 13 are shown. Pins 1, 11, 2, 3, 4, and 13 are connected to RS485-, RS485+, +13.8V, 13.8V RETN, +13.8V, and FLIR ON respectively.

**Right Section (Connectors 2J1, 1J6):**

- 2J1 (P217R, BP217R):** 16-pin connector. Pins 8, 16, 26, 35, 25, 36, 56, 67, 57, 78, 77, 68, 44, 43, and 54 are shown. Pins 8, 16, 26, 35, 25, 36, 56, 67, 57, 78, 77, 68, 44, and 43 are connected to VID1+, VID1-, VID1 SHIELD, VID2+, VID2-, VID2 SHIELD, VID3+, VID3-, VID3 SHIELD, VID4+, VID4-, VID4 SHIELD, AZREF+, and AZREF- respectively. Pin 54 is connected to AZREF SHIELD.
- 1J6 (P216R, BP216R):** 6-pin connector. Pins 8, 16, 26, 35, 25, and 36 are shown. Pins 8, 16, 26, 35, 25, and 36 are connected to VID1+, VID1-, VID1 SHIELD, VID2+, VID2-, and VID2 SHIELD respectively.

**Other Components:**

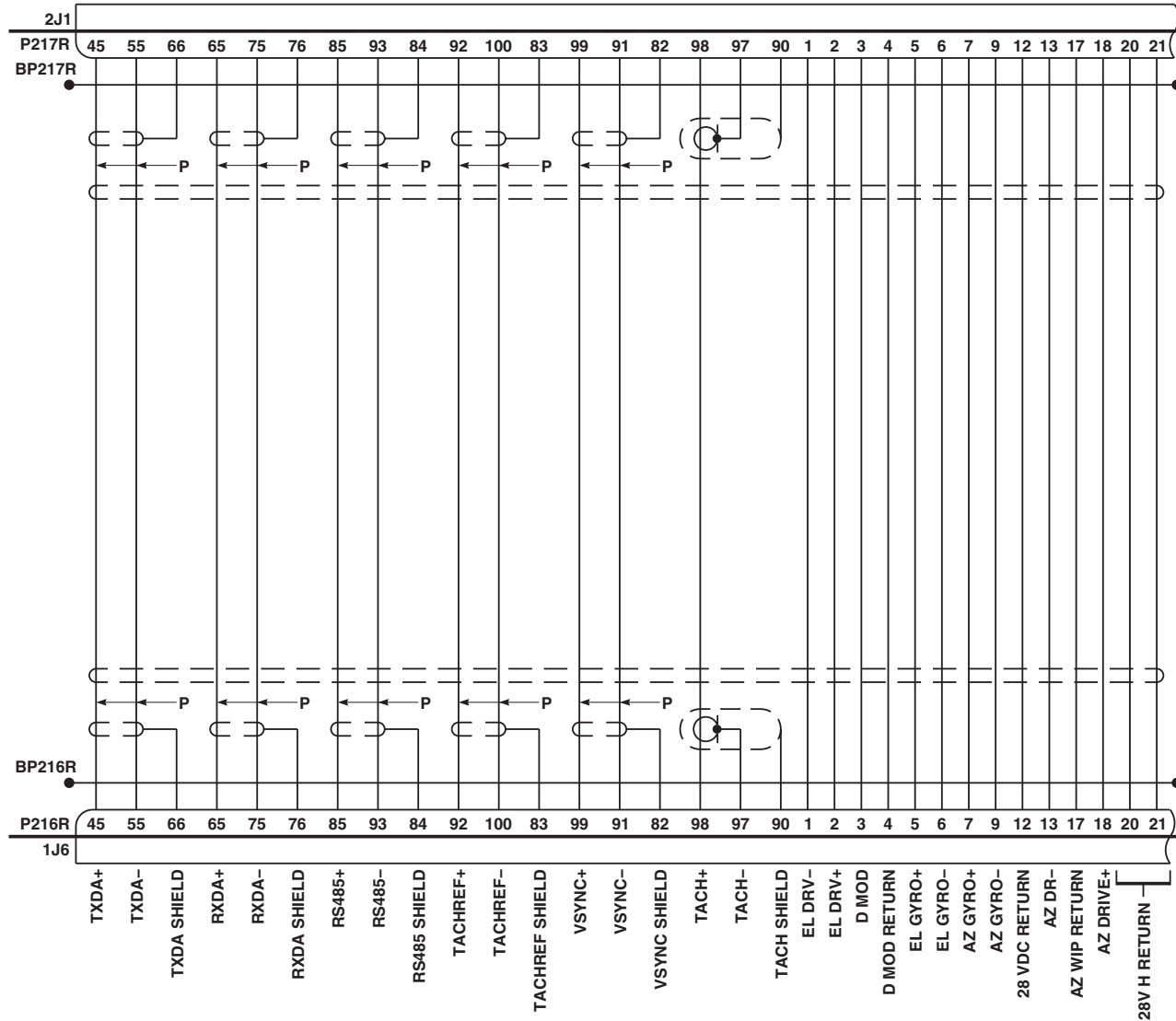
- Resistors:** Various resistors are shown, including a 10k resistor connected to pin 10 of 3J1 and a 10k resistor connected to pin 10 of 1J3.
- Capacitors:** Various capacitors are shown, including a 100nF capacitor connected to pin 10 of 3J1 and a 100nF capacitor connected to pin 10 of 1J3.
- Transformer:** A transformer labeled TJ5VB is connected to pins 10 and 9 of 3J1 and to a ground point labeled GND 214-1.

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### Change 1

SCHEMATICS - Continued

TURRET FLIR UNIT



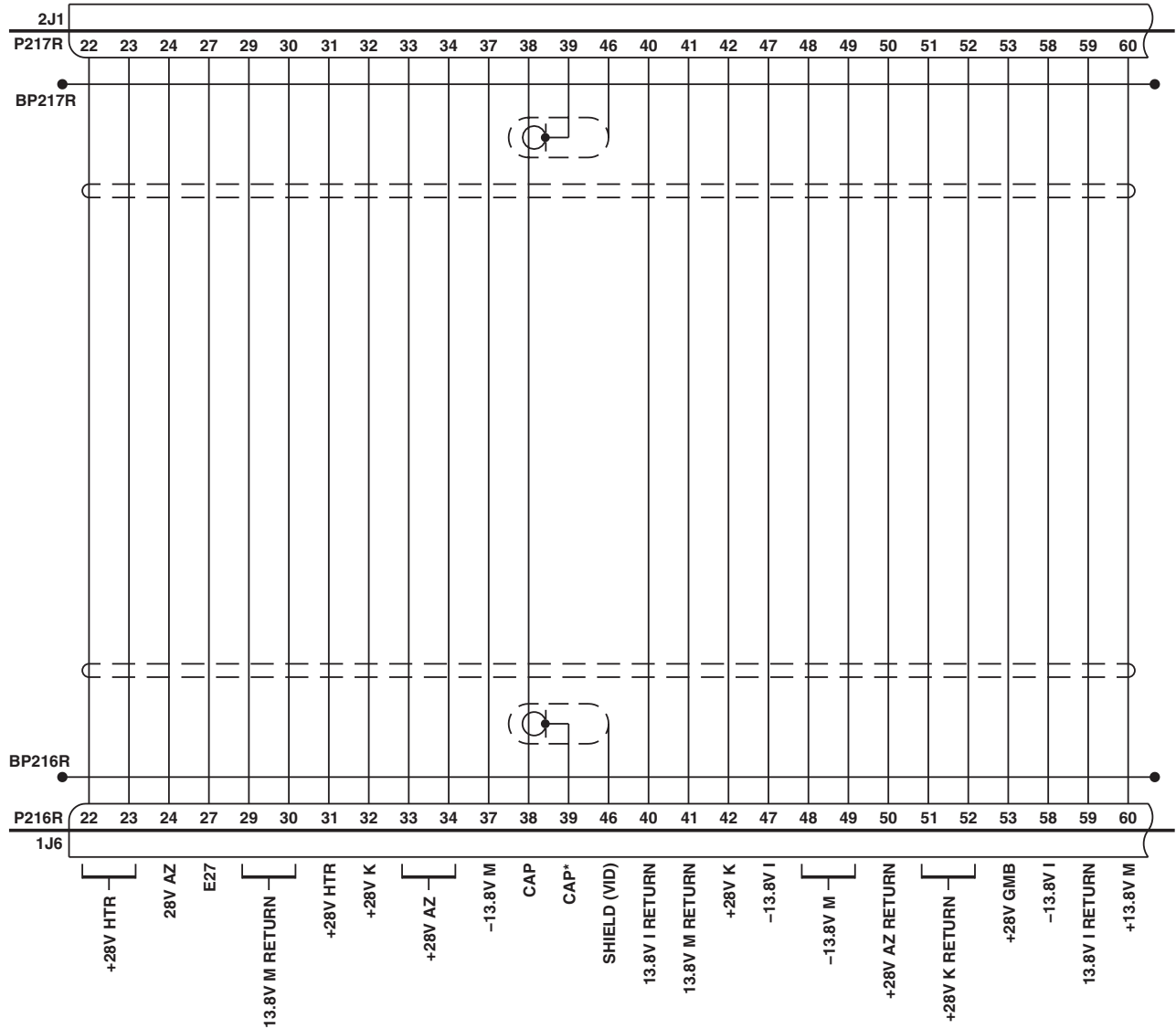
CENTRAL ELECTRONICS UNIT

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SA

Figure 2. FLIR Schematic Diagram. (Sheet 3 of 5)

SCHEMATICS - Continued

TURRET FLIR UNIT



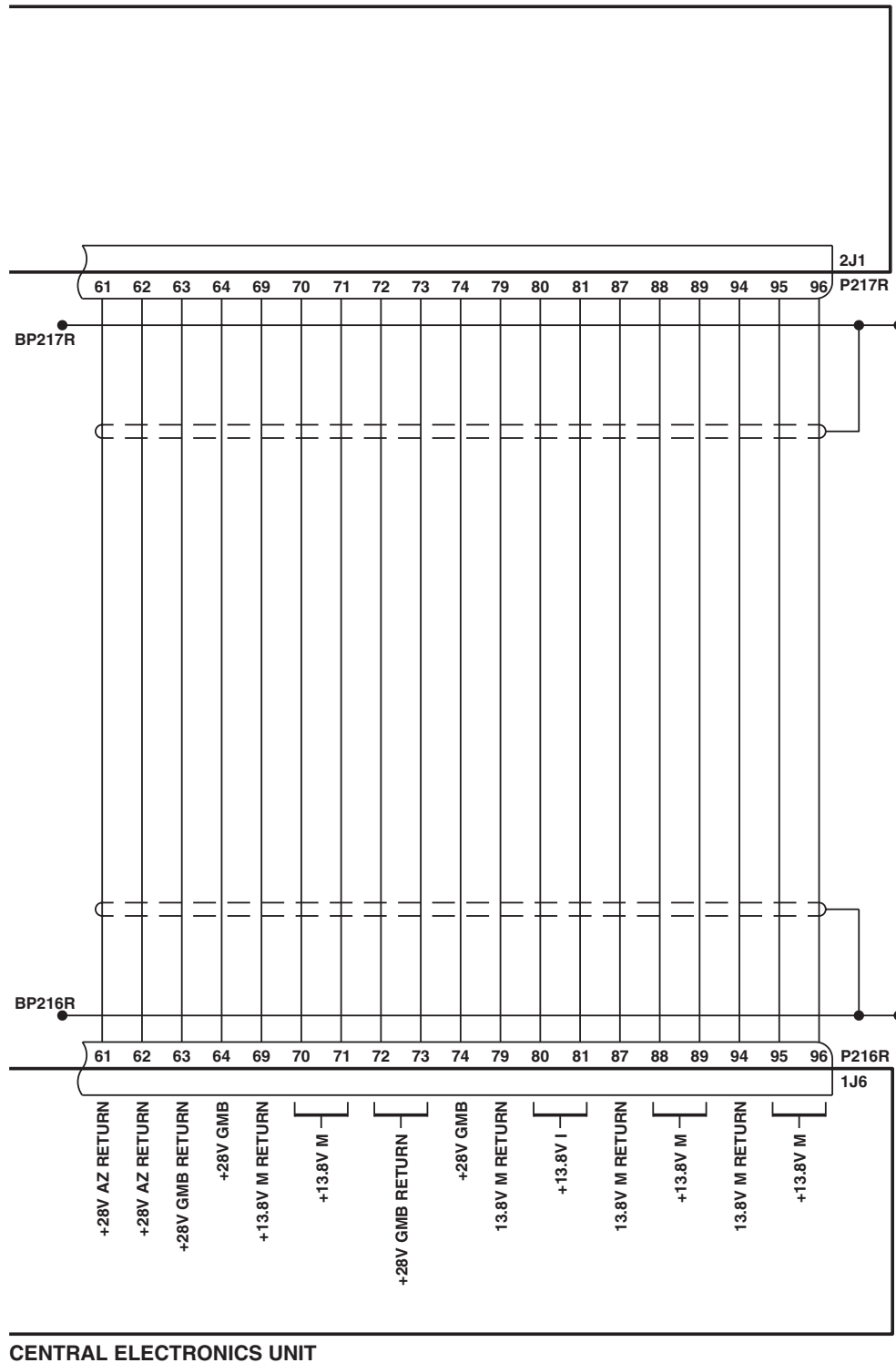
CENTRAL ELECTRONICS UNIT

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Figure 2. FLIR Schematic Diagram. (Sheet 4 of 5)

SCHEMATICS - Continued

TURRET FLIR UNIT



CENTRAL ELECTRONICS UNIT

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Figure 2. FLIR Schematic Diagram. (Sheet 5 of 5)

END OF WORK PACKAGE

**UNIT LEVEL****MISSION EQUIPMENT****FORWARD LOOKING INFRARED (FLIR) HH-60L****INITIAL SETUP:****Test Equipment**

Multimeter, Fluke, Model 37

WP 0824 00

WP 1556 00

WP 1557 00

**Tools and Special Tools**

Electrical Repairer Toolkit, SC 5180-99-B06

WP 1558 00

WP 1685 00

**Personnel Required**

Aircraft Electrician MOS 15F (2)

WP 1747 00

**References**

WP 0100 00

WP 0104 00

**Equipment Condition**External Electrical Power Available, WP 1685 00  
and/or APU Operational, TM 1-1520-237-10**SPECIAL INSTRUCTIONS**

This operational/troubleshooting procedure is subdivided:

- Setup
- Operational Check
- Fault Isolation Test
- FLIR Azimuth Offset Adjustment Procedure
- Shutdown

**SETUP OPERATIONAL/TROUBLESHOOTING PROCEDURE****PROCEDURE****Step****WARNING**

To prevent injury, make sure all personnel are clear of the turret FLIR unit (TFU).

**NOTE**

If a circuit breaker pops out during fault isolation procedure, a short circuit is indicated (WP 0104 00).

1. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.

**SETUP OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****CAUTION**

Ensure the FLIR STOW PIN is removed from the FLIR Turret before applying power to the FLIR.

2. Remove FLIR STOW pin.
3. Turn on electrical power.
4. Turn pilot's and copilot's MFDs power switches to the ON position, and both MFD/NIGHT/DAY switches are in the DAY position. Depress both MFD FLIR switches.

**Indication/Condition**

Both MFDs are set to the FLIR page.

**Corrective Action**

None Required

**Step**

5. Set FLIR hand control unit (HCU) POWER switch to ON.

**Indication/Condition**

The FLIR test screen will be displayed on both MFDs and Built-In-Test (BIT) will automatically be performed. The TFU will enter Cage mode.

**Corrective Action**

None Required

**OPERATIONAL CHECK OPERATIONAL/TROUBLESHOOTING PROCEDURE  
PROCEDURE****Step****WARNING**

To prevent injury, make sure all personnel are clear of the TFU.

**CAUTION**

Place FLIR in STOW position when not in use to prevent window damage.

**NOTE**

If a circuit breaker pops out during fault isolation procedure, a short circuit is indicated (WP 0104 00).

1. Do Setup, if not already done.
2. Allow approximately 3 - 5 minutes for the FLIR to warm-up, complete it's startup BIT test and for the image to stabilize.

**Indication/Condition**

- a. The TFU shall Cage (point straight ahead).
- b. Pilot's and copilot's MFD's shall indicate a FLIR image.



---

**OPERATIONAL CHECK OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued

- c. Pilot's and copilot's MFD's shall not indicate FAULT in the upper status bar of the FLIR display.

**Corrective Action**

1. If result (1) is not as specified, and all functions become disabled after 40 seconds, replace TFU (WP 1556 00).
2. If result (2) is not as specified, go to MFD DOES NOT DISPLAY FLIR IMAGE AND MENU PAGE 1, in this work package..
3. If pilot's MFD does not indicate a FLIR image, go to PILOT'S DISPLAY DOES NOT INDICATE A FLIR IMAGE, in this work package.
4. If copilot's MFD does not indicate a FLIR image, go to COPILOT'S DISPLAY DOES NOT INDICATE A FLIR IMAGE, in this work package.
5. If result (3) is not as specified, perform fault isolation procedure.

**Step**

3. Using the Joystick, toggle MENU/STOW/LOCK/FRZ FRM/HOOK switch to MENU.

**Indication/Condition**

The pilot's and copilot's MFDs shall display a MENU with Exit highlighted.

**Corrective Action**

None Required

**Step**

4. Using the Joystick, highlight Calibration, then toggle the MENU/STOW/LOCK/FRZ FRM/HOOK switch to HOOK.

**Indication/Condition**

Calibration page shall be displayed with Exit highlighted.

**Corrective Action**

None Required

**Step**

5. Using the Joystick, highlight Grayscale and toggle the MENU/STOW/LOCK/FRZ FRM/HOOK switch to HOOK.

**Indication/Condition**

A grayscale bar with many shades of gray will appear across the top of the pilot's and copilot's MFD.

**Corrective Action**

None Required

**Step**

6. Turn the pilot's and copilot's MFD BRT control knob until the far left of the grayscale is dark black and the far right is light white. Toggle the MENU/STOW/LOCK/FRZ FRM/HOOK switch to HOOK.

**Indication/Condition**

The grayscale bar will disappear.

---

**OPERATIONAL CHECK OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued**Corrective Action**

None Required

**Step**

7. Using the Joystick, highlight Set Time/Date and toggle the MENU/STOW/LOCK/FRZ FRM/HOOK switch to HOOK.
8. Using the Joystick, highlight Hour, then toggle the MENU/STOW/LOCK/FRZ FRM/HOOK switch to HOOK.
9. Using the Joystick, change the current hour setting, then toggle the MENU/STOW/LOCK/FRZ FRM/HOOK switch to HOOK.
10. Repeat steps 8. and steps 9. to set the current Minute, Second, Month, Day, and Year.
11. Using the joystick, highlight Update Clock, and then toggle the MENU/STOW/LOCK/FRZ FRM/HOOK switch to HOOK for the new time and date settings to take effect.
12. Using the Joystick, highlight and hook on the Exit label to exit the Set Time/Date sub menu.
13. Using the Joystick, highlight Cage the toggle the MENU/STOW/LOCK/FRZ FRM/HOOK switch to HOOK.

**Indication/Condition**

The Default Cage Positions sub menu appears.

**Corrective Action**

None Required

**Step**

14. Using the Joystick, highlight Cage and toggle the MENU/STOW/LOCK/FRZ FRM/HOOK switch to HOOK to view the Cage sub menu. Adjust if necessary.

**Indication/Condition**

Default Autoscan Cage position shall read 0 AZ and 0 EL. (Default is identified by a ">")

**Corrective Action**

None Required

**Step**

15. Using the Joystick, highlight and hook on the Exit label to exit the Calibration sub-menu.

**OPERATIONAL CHECK OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****CAUTION**

Maintenance menu features are critical to the operation of the FLIR system. In order to prevent damage to the FLIR system, use caution whenever changing the maintenance menu settings.

**NOTE**

The Maintenance menu contains multiple options, access to critical options are limited to access code 374. A restricted maintenance menu can be accessed using access code 000. In order to gain access to the unrestricted maintenance menu, Tumbler Labels (Tumbler 1, Tumbler 2, and Tumbler 3) must be set to 3, 7, 4, respectfully.

16. Using the Joystick, highlight Maintenance then toggle the MENU/STOW/LOCK/FRZ FRM/HOOK switch to HOOK to display Tumbler Labels and settings.
17. Using the Joystick, highlight the desired tumbler label value and change the value to the desired number and select the value by toggling the MENU/STOW/LOCK/FRZ FRM/HOOK switch to HOOK. Set the labels values to 3, 7, 4 (see note above).

**Indication/Condition**

The Tumbler Labels are set to 3, 7, 4.

**Corrective Action**

None Required

**Step**

18. Using the Joystick, highlight OPEN then toggle the MENU/STOW/LOCK/FRZ FRM/HOOK switch to HOOK.
19. Using the Joystick, highlight Gimbal Calibration then toggle the MENU/STOW/LOCK/FRZ FRM/HOOK switch to HOOK.

**Indication/Condition**

The Gimbal Calibration sub-menu screen will be displayed on the MFD.

**Corrective Action**

None Required

**Step**

20. If required, using the Joystick, change the numeric value of each label to value on the right. Toggle the MENU/STOW/LOCK/FRZ FRM/HOOK switch to HOOK to set the numeric value.

**Indication/Condition**

The Gimbal Calibration sub menu shall display the following:

|                    |        |
|--------------------|--------|
| AZIMUTH OFFSET     | 0.0°   |
| 3 AXIS STOW AZ POS | 180.0° |
| 3 AXIS STOW EL POS | 134.0° |

**Corrective Action**

None Required

**OPERATIONAL CHECK OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued**Step**

21. Highlight and hook on the Exit label to exit the Gimbal Calibration sub menu.
22. Using the Joystick, highlight System Configuration then toggle the MENU/STOW/LOCK/FRZ FRM/HOOK switch to HOOK.

**Indication/Condition**

- a. The sub menu shall display the following:

Monitor Type: RS170

Camera: CCD: None

Camera: Spotter Scope: None

Microscan: Installed

Controller Type: xNet HCU

Turret Type: Star II

Laser Type: -

Tracker Type- Octec VME

Exit: -

- b. If required, using the Joystick, change the value of each label to value on the right. Toggle the MENU/STOW/LOCK/FRZ FRM/HOOK switch to HOOK to set the value.

**Corrective Action**

None Required

**Step**

23. Using the Joystick, highlight the Exit label, then toggle the MENU/STOW/LOCK/FRZ FRM/HOOK switch to HOOK. Continue hooking on all Exit labels until menu pages are no longer displayed.
24. Apply a slight amount of pressure on the FLIR Joystick.

**Indication/Condition**

The control breaks out of the Cage mode and enters INRPT (Inertial Point) Mode as shown in the top border of both MFDs.

**Corrective Action**

None Required

**Step**

25. Operate the Joystick to the following positions:

**CONTROL**

UP

**TURRET**

Up Elevation

---

**OPERATIONAL CHECK OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued

---

**CONTROL**

DOWN

LEFT

RIGHT

**TURRET**

Down Elevation

CCW Azimuth (as viewed  
from top of image)CW Azimuth (as viewed  
from top of image)**Indication/Condition**

The FLIR Turret and both MFDs displays the follow the joystick accordingly.

**Corrective Action**

None Required

**Step**

26. Depress the left most FLIR Control Pendant MENU/STOW/LOCK/FRZ FRM/HOOK switch to the CAGE position.

**Indication/Condition**

FLIR proceeds to Cage default position of 0 AZ and 0 EL.

**Corrective Action**

None Required

**Step**

27. Visually inspect the FLIR Turret to verify that the Cage settings position the FLIR Turret to 0° Azimuth and 0° Elevation with respect to aircraft nose.

**Indication/Condition**

- a. The FLIR Turret is positioned to 0° Azimuth and 0° Elevation with respect to aircraft nose.
- b. If the FLIR Turret is not centered on aircraft perform azimuth offset adjustment procedure.

**Corrective Action**

None Required

**Step**

28. Apply some force to the joystick.

**Indication/Condition**

FLIR is in INRPT Mode, shown in the top border of both MFD displays.

**Corrective Action**

None Required

---

**OPERATIONAL CHECK OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued**Step**

29. Depress the right most FLIR Control Pendant MENU/STOW/LOCK/FRZ FRM/HOOK switch up to the GAIN UP position and down to the GAIN DOWN position.

**Indication/Condition**

FLIR Gain changes and the picture quality on the MFDs change. Gain/Level bar shown in the top of both MFDs decreases in width for GAIN UP and increases in width for GAIN DOWN.

**Corrective Action**

None Required

**Step**

30. Initiate the Auto Gain/Level by depressing the right most FLIR Control Pendant MENU/STOW/LOCK/FRZ FRM/HOOK switch.

**Indication/Condition**

FLIR Gain and Level readjust to AUTO settings.

**Corrective Action**

None Required

**Step**

31. Depress in the left most FLIR Control Pendant MENU/STOW/LOCK/FRZ FRM/HOOK switch.

**Indication/Condition**

FLIR Imagery Polarity changes from white hot indication to black hot indication.

**Corrective Action**

None Required

**Step**

32. Depress in the left most FLIR Control Pendant MENU/STOW/LOCK/FRZ FRM/HOOK switch.

**Indication/Condition**

FLIR Imagery Polarity changes from black hot indication to white hot indication.

**Corrective Action**

None Required

**Step**

33. Depress the left most FLIR Control Pendant MENU/STOW/LOCK/FRZ FRM/HOOK switch up to the FOCUS OUT position and down to the FOCUS IN position.

**Indication/Condition**

FLIR Focus changes. Focus bar at top left of both MFD screens moves and picture quality changes.

**Corrective Action**

None Required

---

**OPERATIONAL CHECK OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued**Step**

34. Depress the left most FLIR Control Pendant MENU/STOW/LOCK/FRZ FRM/HOOK switch to the right (IP/HH).

**Indication/Condition**

FLIR Mode changes to Heading Hold. HDHLD is displayed at top of both MFD screens.

**Corrective Action**

None Required

**Step**

35. Depress the left most FLIR Control Pendant MENU/STOW/LOCK/FRZ FRM/HOOK switch to the right (IP/HH).

**Indication/Condition**

FLIR Mode changes to Inertial Point. INRPT is displayed at top of both MFD screens.

**Corrective Action**

None Required

**Step**

36. Depress the FLIR Control Handle FOV Trigger switch to the down position 3 times.

**Indication/Condition**

FLIR FOV changes to ZOOM. IR Nrwx2 is displayed at top of both MFD screens and image changes.

**Corrective Action**

None Required

**Step**

37. Depress the FLIR Control Handle trigger switch to the up position once.

**Indication/Condition**

FLIR FOV changes to Narrow Field of View. IR NRW is displayed at top of both MFD screens and image changes.

**Corrective Action**

None Required

**Step**

38. Depress the FLIR Control Handle trigger switch to the up position once.

**Indication/Condition**

FLIR FOV changes to Medium Field of View. IR MED is displayed at top of both MFD screens and image changes.

**Corrective Action**

None Required

**Step**

39. Depress the FLIR Control Handle trigger switch to the up position once.

---

**OPERATIONAL CHECK OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued**Indication/Condition**

FLIR FOV changes to Wide Field of View. IR WIDE is displayed at top of both MFD screens and image changes.

**Corrective Action**

None Required

**Step**

40. Depress the center most FLIR Control Pendant MENU/STOW/LOCK/FRZ FRM/HOOK switch down to the FRZ FRM position.

**Indication/Condition**

FLIR image on both MFDs is frozen. FRZ is displayed at top of both MFD screens and image changes.

**Corrective Action**

None Required

**Step**

41. Depress the center most FLIR Control Pendant MENU/STOW/LOCK/FRZ FRM/HOOK switch down to the FRZ FRM position.

**Indication/Condition**

FLIR image on both MFDs is unfrozen. FRZ is removed from the top of both MFD screens.

**Corrective Action**

None Required

**Step**

42. Select a distant, contrasting, moving target and center it on the MFD screen. Toggle the Tracker Trigger switch on the underside of the controller.

**Indication/Condition**

A square encases the target and follows it as it moves.

**Corrective Action**

None Required

**Step**

43. Toggle the Tracker Trigger switch again.

**Indication/Condition**

The square is removed from the display and no longer tracks the target.

**Corrective Action**

None Required

**Step**

44. Depress the center most FLIR Control Pendant MENU/FREEZE momentary switch to the MENU position.



---

**OPERATIONAL CHECK OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued**Indication/Condition**

Menu page shall display with the Exit label highlighted.

**Corrective Action**

None Required

**Step**

45. Using the Joystick, highlight Autoscan and toggle the MENU/STOW/LOCK/FRZ FRM/HOOK switch to HOOK to view the Autoscan sub menu.

**Indication/Condition**

Autoscan settings read:

**WIDTH**

> Full

Reduced

**RATE**

High

> Medium

Low

Autolock

Off/On

Autoscan Preferences...

Start Scan

Exit

**Corrective Action**

None Required

**OPERATIONAL CHECK OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Step****NOTE**

If required, using the Joystick, highlight the Full width label and toggle the MENU/STOW/LOCK/FRZ FRM/HOOK switch to HOOK to set as the "default mode" shown with ">" or, if required, using the Joystick, highlight the Medium rate label and toggle the MENU/STOW/LOCK/FRZ FRM/HOOK switch to HOOK to set as the "default mode" shown with ">".

46. Using the Joystick, select the Autoscan Preferences and toggle the MENU/STOW/LOCK/FRZ FRM/HOOK switch to HOOK to view the Autoscan Preferences sub menu.

**Indication/Condition**

The sub menu shall display the following:

## Patterns

Azimuth - Full  $\pm 120$

Azimuth - Reduced  $\pm 30$

Elevation  $\pm 15$

## Rates

High 15°/SEC

Medium 10°/SEC

Low 5°/SEC

Revert to Factory  
Settings

Exit

**Corrective Action**

None Required

**Step****NOTE**

If required, select each label by toggling the MENU/STOW/LOCK/FRZ FRM/HOOK switch to HOOK. Using the Joystick, change the value of each label to value on the right. Toggle the MENU/STOW/LOCK/FRZ FRM/HOOK switch to HOOK to set the value.

47. Using the Joystick, highlight the Exit label then toggle the MENU/STOW/LOCK/FRZ FRM/HOOK switch to HOOK.

**Indication/Condition**

The Autoscan page sub menu screen will be displayed on the MFD.

---

**OPERATIONAL CHECK OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued**Corrective Action**

None Required

**Step**

48. Using the Joystick, highlight the Start Scan label then toggle the MENU/STOW/LOCK/FRZ FRM/HOOK switch to HOOK to initiate an Autoscan pattern.

**Indication/Condition**

Turret FLIR unit shall begin Autoscan pattern for default settings.

**Corrective Action**

None Required

**Step****NOTE**

Exit out all menu's or Depress joystick to stop scan.

49. Initiate STOW by depressing the center MENU/STOW/LOCK/FRZ FRM/HOOK switch.

**Indication/Condition**

Turret points to 180° ( $\pm 1^\circ$ ) azimuth and goes to 134° ( $\pm 1^\circ$ ) elevation off the screen.

**Corrective Action**

None Required

**Step**

50. Using the Joystick, highlight the Exit label then toggle the MENU/STOW/LOCK/FRZ FRM/HOOK switch to HOOK.

**Indication/Condition**

The Main Menu page will be displayed on the MFD.

**Corrective Action**

None Required

**Step**

51. If no other checks are to be done, go to shutdown.

**Indication/Condition**

None Required

**Corrective Action**

None Required

## **FAULT ISOLATION TEST OPERATIONAL/TROUBLESHOOTING PROCEDURE**

### **PROCEDURE**

#### **Step**

1. Do Setup if not already done.

#### **NOTE**

The Fault Isolation Test (FIT) procedure is to be completed only if FAULT is displayed in the upper status bar area upon completion of the Built-In Test (BIT) or if the FLIR System is operating improperly.

2. On the FLIR Control Pendant, toggle the MENU/STOW/LOCK/FRZ FRM/HOOK switch to MENU.

#### **Indication/Condition**

Menu Page shall be displayed on MFD with the EXIT label highlighted.

#### **Corrective Action**

None Required

#### **Step**

3. Using the Joystick, highlight Fault then toggle the MENU/STOW/LOCK/FRZ FRM/HOOK switch to HOOK to enter the Fault Isolation Test sub menu.

#### **Indication/Condition**

The labels Clear, FIT, and Exit will be displayed.

#### **Corrective Action**

None Required

#### **Step**

#### **NOTE**

If a fault code is displayed on the MFD before FIT is initiated, this may be a result of a past FIT. Using the Joystick, highlight the Clear label then toggle the MENU/STOW/LOCK/FRZ FRM/HOOK switch to HOOK. This will clear past FIT results. The Clear option is used to clear any faults that are displayed in the top portion of the menu. Using the Clear option should be the first action taken by an operator, should a fault occur. If the Clear option clears the fault, but the fault is regenerated, the FIT option should be initiated.

4. Using the Joystick, highlight the FIT label. Toggle the MENU/STOW/LOCK/FRZ FRM/HOOK switch to HOOK to initiate FIT.

#### **Indication/Condition**

- a. FIT IN PROGRESS is displayed in the upper status bar area until FIT is completed. If an error is detected, an error code will be displayed on the FIT sub menu screen.
- b. Refer to Table 1: Error Codes and follow the appropriate action to be taken therein.

---

**FAULT ISOLATION TEST OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Corrective Action**

None Required

**Step**

5. Using the Joystick, highlight the Exit label then toggle the MENU/STOW/LOCK/FRZ FRM/HOOK switch to HOOK. Continue hooking on all Exit labels until menu pages are no longer displayed.

**Indication/Condition**

None Required

**Corrective Action**

None Required

**FLIR AZIMUTH OFFSET ADJUSTMENT OPERATIONAL/TROUBLESHOOTING PROCEDURE  
PROCEDURE****Step**

1. Do Setup if not already done.
2. Using the joystick, toggle FOCUS OUT/IP HH/FOCUS IN/POL AF/CAGE switch to CAGE.

**Indication/Condition**

CAGE is displayed in the top border of the pilot's and copilot's MFDs.

**Corrective Action**

None Required

**Step**

3. Using the Joystick, toggle the MENU/STOW/LOCK/FRZ FRM/HOOK switch to MENU.

**Indication/Condition**

The pilot's and copilot's MFDs shall display a menu with Exit highlighted.

**Corrective Action**

None Required

**Step****CAUTION**

Maintenance menu features are critical to the operation of the FLIR system. In order to prevent damage to the FLIR system; use caution whenever changing maintenance menu settings.

4. Using the Joystick, highlight MAINTENANCE then toggle the MENU/STOW/LOCK/FRZ FRM/HOOK switch to HOOK to display tumbler labels and settings.
5. Using the Joystick, highlight the desired tumbler label value and change the value to the desired number and select the value by toggling the MENU/STOW/LOCK/FRZ FRM/HOOK switch to HOOK. Set the label values to 3, 7, and 4.

---

**FLIR AZIMUTH OFFSET ADJUSTMENT OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Indication/Condition**

The tumbler values are set to 3, 7, and 4.

**Corrective Action**

None Required

**Step**

6. Using the joystick, highlight OPEN then toggle the MENU/STOW/LOCK/FRZ FRM/HOOK switch to HOOK.
7. Using the joystick, highlight GIMBAL CALIBRATION then toggle the MENU/STOW/LOCK/FRZ FRM/HOOK switch to HOOK.

**Indication/Condition**

The GIMBAL CALIBRATION sub-menu screen will be displayed on the pilot's and copilot's MFDs.

**Corrective Action**

None Required

**Step**

8. Using the joystick, highlight AZIMUTH OFFSET then toggle the MENU/STOW/LOCK/FRZ FRM/HOOK switch to HOOK.
9. Using the MENU/STOW/LOCK/FRZ FRM/HOOK joystick on the FLIR HCU, reposition the FLIR Line-of-Sight Reticles (crosshairs) with an object centered directly forward of the helicopter.

**Indication/Condition**

AZIMUTH OFFSET value will change with FLIR turret movement.

**Corrective Action**

None Required

**Step**

10. Using the joystick, highlight the Exit label then toggle the MENU/STOW/LOCK/FRZ FRM/HOOK switch to HOOK. Continue hooking on all Exit labels until menu pages are no longer displayed.
11. Verify FLIR is in CAGE.

**Indication/Condition**

CAGE is displayed in the top border of the pilot's and copilot's MFDs.

**Corrective Action**

None Required

**Step**

12. Compare visual reference against FLIR image to verify Caged FLIR turret is pointed directly forward of helicopter.

**Indication/Condition**

- a. FLIR crosshairs are aligned with an object centered directly forward of the helicopter.
- b. Procedure complete.

---

**FLIR AZIMUTH OFFSET ADJUSTMENT OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Corrective Action**

None Required

**SHUTDOWN OPERATIONAL/TROUBLESHOOTING PROCEDURE  
PROCEDURE****Step**

1. Using the Joystick, toggle the MENU/STOW/LOCK/FRZ FRM/HOOK switch to STOW.
2. On HCU, set POWER ON/OFF switch to OFF.
3. Pull out the NO. 2 DC PRI BUS FLIR Circuit Breaker.
4. Turn pilot's and copilot's MFDs switches OFF.
5. Turn off electrical power.
6. Secure pendant and install TFU safety pin.

**Indication/Condition**

None Required

**Corrective Action**

None Required

**MFD DOES NOT DISPLAY FLIR IMAGE AND MENU PAGE 1****SYMPTOM**

MFD Does Not Display FLIR Image and MENU Page 1.

**MALFUNCTION**

MFD Does Not Display FLIR Image and MENU Page 1.

**CORRECTIVE ACTION**

1. **Check for 28 vdc between:**
  - **P210R-A and P210R-E**
  - **P210R-D and P210R-B**
  - a. If voltage is as specified, go to **3**.
  - b. If voltage is not as specified, go to **2**.
2. **Check for 28 vdc between terminal 1 of FLIR circuit breaker CB530 and airframe ground.**
  - a. If voltage is as specified, repair/replace wiring (WP 1747 00). Go to **5**.
  - b. If voltage is not as specified, replace FLIR circuit breaker (WP 0824 00). Go to **5**.
3. **Check for 28 vdc between P214R-5 and P214R-6.**
  - a. If voltage is as specified, replace HCU (WP 1558 00). Go to **5**.
  - b. If voltage is not as specified, go to **4**.
4. **Check continuity between P215R-5 and P215R-6.**

---

**MFD DOES NOT DISPLAY FLIR IMAGE AND MENU PAGE 1 - Continued**

- a. If continuity is present, replace CEU (WP 1557 00). Go to **5**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **5**.

**5. Procedure completed.****PILOT'S MFD DOES NOT INDICATE A FLIR IMAGE****SYMPTOM**

Pilot's MFD Does Not Indicate a FLIR Image.

**MALFUNCTION**

Pilot's MFD Does Not Indicate a FLIR Image.

**CORRECTIVE ACTION****1. Check continuity between P1016R-B and P212R-CC.**

- a. If continuity is present, troubleshoot MFD/caution/advisory warning system (WP 0100 00). Go to **2**.
- b. If trouble remains, replace CEU (WP 1557 00). Go to **2**.
- c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **2**.

**2. Procedure completed.****COPILOT'S MFD DOES NOT INDICATE A FLIR IMAGE****SYMPTOM**

Copilot's MFD Does Not Indicate a FLIR Image.

**MALFUNCTION**

Copilot's MFD Does Not Indicate a FLIR Image.

**CORRECTIVE ACTION****1. Check continuity between P1015R-B and P213R-CC.**

- a. If continuity is present, troubleshoot MFD/caution/advisory warning system (WP 0100 00). Go to **2**.
- b. If trouble remains, replace CEU (WP 1557 00). Go to **2**.
- c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **2**.

**2. Procedure completed.****ERROR CODE IS DISPLAYED IN FAULT ISOLATION TEST SUB-MENU****SYMPTOM**

Error Code is Displayed in Fault Isolation Test Sub-menu.

**MALFUNCTION**

Error Code is Displayed in Fault Isolation Test Sub-menu.

**CORRECTIVE ACTION****1. Check for error code.**



---

**ERROR CODE IS DISPLAYED IN FAULT ISOLATION TEST SUB-MENU - Continued**

- a. If error code is 101, 102, 103, 201, 20d, 501, 502, 503, 504, 505, 506, 507, 508, 509, 50a, 50b, 50c, 50d or 50e, replace CEU (WP 1557 00). Go to **8**.
- b. If error code is 601, 602, 603, 604, 605, 700, 701, 705, 706, 708, 709 or 70a, replace TFU (WP 1556 00). Go to **8**.
- c. If error code is 202, 206, 207, 208, 209, 20a, 20b, or 20c, go to **2**.
- d. If error code is 203 or 800, go to **3**.
- e. If error code is 204, go to **4**.
- f. If error code is 205 or 801, go to **5**.
- g. If error code is 702, 703, or 704, go to **6**.

**2. Check continuity between:**

- P215R-1 and P214R-1
  - P215R-11 and P214R-11
  - P216R-85 and P217R-85
  - P216R-93 and P217R-93
- a. If continuity is present, replace CEU (WP 1557 00). Go to **8**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **8**.

**3. Check continuity between:**

- P215R-2 and P214R-2
  - P215R-3 and P214R-3
  - P215R-4 and P214R-4
- a. If continuity is present, replace CEU (WP 1557 00). Go to **8**.
  - b. If trouble remains, replace HCU (WP 1558 00). Go to **8**.
  - c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **8**.

**4. Check continuity between:**

- **P216R-3 and P217R-3**
- **P216R-4 and P217R-4**
- **P216R-9 and P217R-9**
- **P216R-13 and P217R-13**
- **P216R-17 and P217R-17**
- **P216R-18 and P217R-18**
- **P216R-24 and P217R-24**
- **P216R-29 and P217R-29**
- **P216R-30 and P217R-30**
- **P216R-33 and P217R-33**

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**ERROR CODE IS DISPLAYED IN FAULT ISOLATION TEST SUB-MENU - Continued**

- P216R-34 and P217R-34
- P216R-37 and P217R-37
- P216R-40 and P217R-40
- P216R-41 and P217R-41
- P216R-42 and P217R-42
- P216R-47 and P217R-47
- P216R-48 and P217R-48
- P216R-49 and P217R-49
- P216R-50 and P217R-50
- P216R-51 and P217R-51
- P216R-52 and P217R-52
- P216R-53 and P217R-53
- P216R-58 and P217R-58
- P216R-59 and P217R-59
- P216R-60 and P217R-60
- P216R-61 and P217R-61
- P216R-62 and P217R-62
- P216R-63 and P217R-63
- P216R-64 and P217R-64
- P216R-69 and P217R-69
- P216R-70 and P217R-70
- P216R-71 and P217R-71
- P216R-72 and P217R-72
- P216R-73 and P217R-73
- P216R-74 and P217R-74
- P216R-79 and P217R-79
- P216R-80 and P217R-80
- P216R-81 and P217R-81
- P216R-87 and P217R-87
- P216R-88 and P217R-88
- P216R-89 and P217R-89

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**ERROR CODE IS DISPLAYED IN FAULT ISOLATION TEST SUB-MENU - Continued**

- **P216R-94 and P217R-94**

- **P216R-95 and P217R-95**

- **P216R-96 and P217R-96**

- a. If continuity is present, replace CEU (WP 1557 00). Go to **8**.
- b. If trouble remains, replace TFU (WP 1558 00). Go to **8**.
- c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **8**.

**5. Check continuity between:**

- **P216R-8 and P217R-8**

- **P216R-16 and P217R-16**

- **P216R-35 and P217R-35**

- **P216R-25 and P217R-25**

- **P216R-45 and P217R-45**

- **P216R-55 and P217R-55**

- **P216R-65 and P217R-65**

- **P216R-75 and P217R-75**

- **P216R-92 and P217R-92**

- **P216R-100 and P217R-100**

- **P216R-99 and P217R-99**

- **P216R-91 and P217R-91**

- a. If continuity is present, replace CEU (WP 1557 00). Go to **8**.
- b. If trouble remains, replace TFU (WP 1558 00). Go to **8**.
- c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **8**.

**6. Check for ice in TFU.**

- a. If ice is present, remove ice and check heater.
- b. If heater does not go on, go to **7**.
- c. If ice is not present, replace TFU (WP 1556 00). Go to **8**.

**7. Check continuity between P216R-32 and P217R-32.**

- a. If continuity is present, replace CEU (WP 1557 00). Go to **8**.
- b. If trouble remains, replace TFU (WP 1556 00). Go to **8**.
- c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **8**.

**8. Procedure completed.**

## ERROR CODE IS DISPLAYED IN FAULT ISOLATION TEST SUB-MENU - Continued

Table 1. FIT Error Codes.

| COMPONENT             | DISPLAY    | LONG DISPLAY                        | CORRECTIVE ACTION          |
|-----------------------|------------|-------------------------------------|----------------------------|
| <b>System Control</b> |            |                                     |                            |
| CEU                   | SC Contl   | Controller device not responding    | Check connections          |
| Gimbal                | SC Gimbl   | Gimbal controller not responding    | Check connections          |
| Tracker               | SC Trkr    | Tracker not responding              | Check connections          |
| Payload               | SC Payload | Payload not responding              | Check connections          |
| Laser                 | SC Laser   | Laser device not responding         | Check connections          |
| Imager                | SC Imager  | Imager not responding               | Check connections          |
| CEU                   | SC FitP    | Fit is in progress                  | Wait and try again         |
| <b>Gimbal</b>         |            |                                     |                            |
| Gimbal                | Gimb Fit   | Cannot perform FIT during INIT mode | Wait and try again         |
| Gimbal                | Gyro       | General Servo Failure               | Re-power system            |
| Gimbal                | El Stall   | Failure in the elevation axis       | Re-power system            |
| Gimbal                | AZ Stall   | Failure in the azimuth axis         | Re-power system            |
| Gimbal                | Tracker    | General Tracker Failure             | Re-power system            |
| Gimbal                | Gimb Cal   | Calibration Failure                 | Re-power system            |
| Gimbal                | Gimb EEP   | Gimbal Parameters Lost              | Reset drifts and offsets   |
| Gimbal                | Trkr EEP   | Tracker Parameters Lost             | System using some defaults |
| Gimbal                | GimbCode   | EEPROM store code is missing        | Re-power system            |
| Gimbal                | GyroLink   | Gyro Link Error                     | Re-power system            |
| Gimbal                | GimbFail   | Gyro Failure                        | Re-power system            |

## ERROR CODE IS DISPLAYED IN FAULT ISOLATION TEST SUB-MENU - Continued

Table 1. FIT Error Codes. - Continued

| COMPONENT      | DISPLAY  | LONG DISPLAY                                | CORRECTIVE ACTION |
|----------------|----------|---------------------------------------------|-------------------|
| Gimbal         | GIPNoTlk | Gyro Interface Processor not responding     | Re-power system   |
|                |          | <b>Payload</b>                              |                   |
| Payload        | PEA Gen  | Miscellaneous Failure                       | Re-power system   |
| Payload        | PEA Chks | Payload Parameters Lost                     | Re-power system   |
| Payload        | PEA Humd | Excess Moisture                             | Re-power system   |
| Payload        | PEA Focs | IR Focus Motor Malfunction                  | Re-power system   |
| Payload        | PEA NFOV | IR NFOV MFOV Motor Malfunction              | Re-power system   |
| Payload        | PEA WFOV | IR WFOV/NUC Motor Malfunction               | Re-power system   |
| Payload        | PEA Dith | Dither Mechanism Malfunction                | Re-power system   |
| Payload        | PEA Powr | One or more power supplies out of tolerance | Re-power system   |
| Payload        | PEA ClrP | Cooler power supply out of tolerance        | Re-power system   |
| Payload        | PEA Temp | Detector temperature out of range           | Re-power system   |
| Payload        | PEA Cool | Cooler malfunction                          | Re-power system   |
| Payload        | PEA CCD  | CCD Camera Motor Malfunction                | Re-power system   |
| Payload        | PEA FPGA | FPGA Failure                                | Re-power system   |
| Payload        | PEA 12 C | Intra-board communications failure          | Re-power system   |
| Payload        | PEA IPC  | IPC Communications Failure                  | Re-power system   |
| <b>Tracker</b> |          |                                             |                   |
| Tracker        | Trkr Img | Tracker BIT Failure 0                       | Re-power system   |

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**ERROR CODE IS DISPLAYED IN FAULT ISOLATION TEST SUB-MENU** - Continued

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**Table 1. FIT Error Codes. - Continued**

| <b>COMPONENT</b> | <b>DISPLAY</b> | <b>LONG DISPLAY</b>   | <b>CORRECTIVE ACTION</b> |
|------------------|----------------|-----------------------|--------------------------|
| Tracker          | Trkr TPU       | Tracker BIT Failure 1 | Re-power system          |
| Tracker          | Trkr Fld       | Tracker BIT Failure 2 | Re-power system          |
| Tracker          | Trkr CS        | Tracker BIT Failure 3 | Re-power system          |
| Tracker          | Trkr IVP       | Tracker BIT Failure 4 | Re-power system          |
| Tracker          | Trkr IVP       | Tracker BIT Failure 5 | Re-power system          |

LOCATION

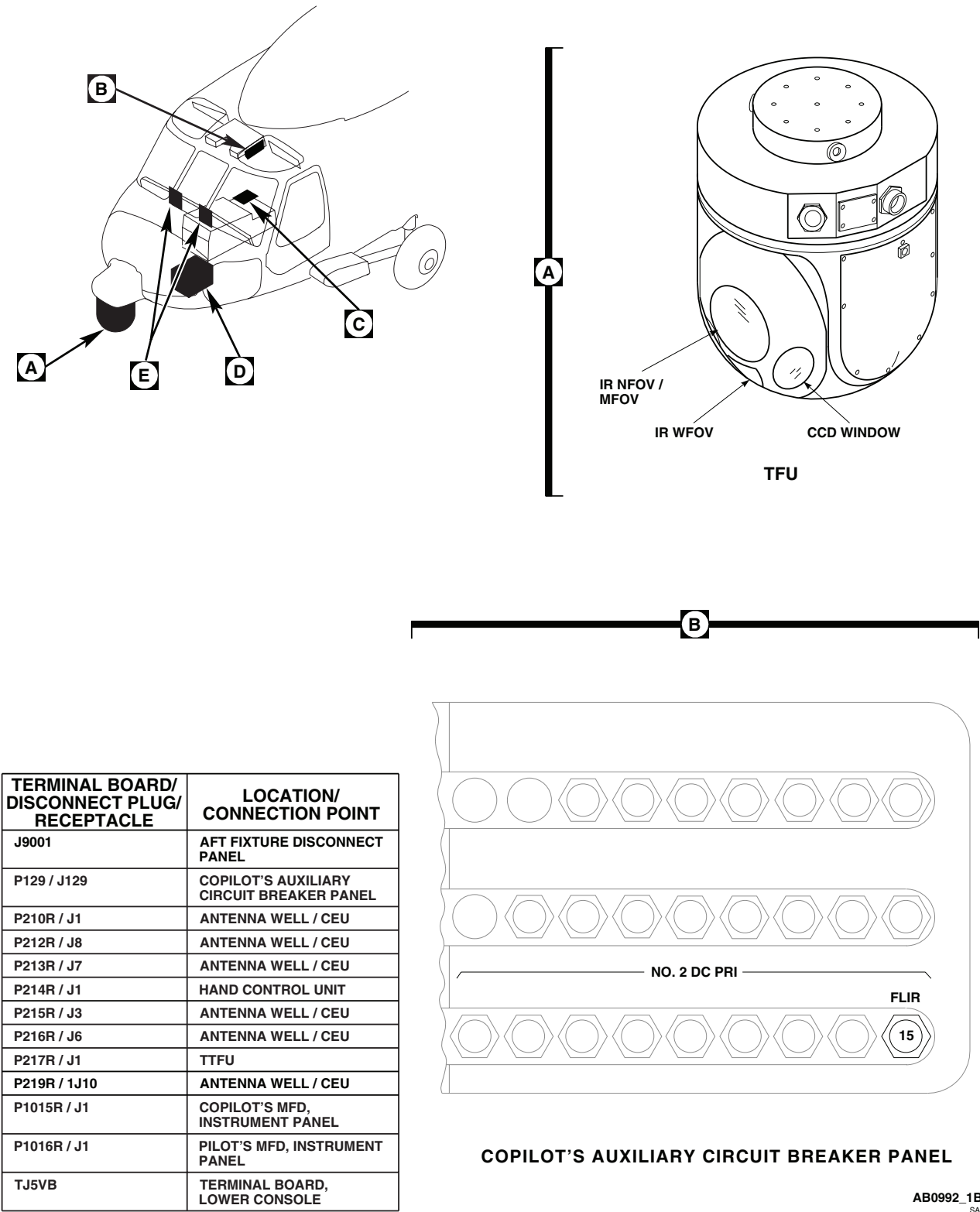


Figure 1. FLIR Location Diagram. (Sheet 1 of 4)

LOCATION - Continued

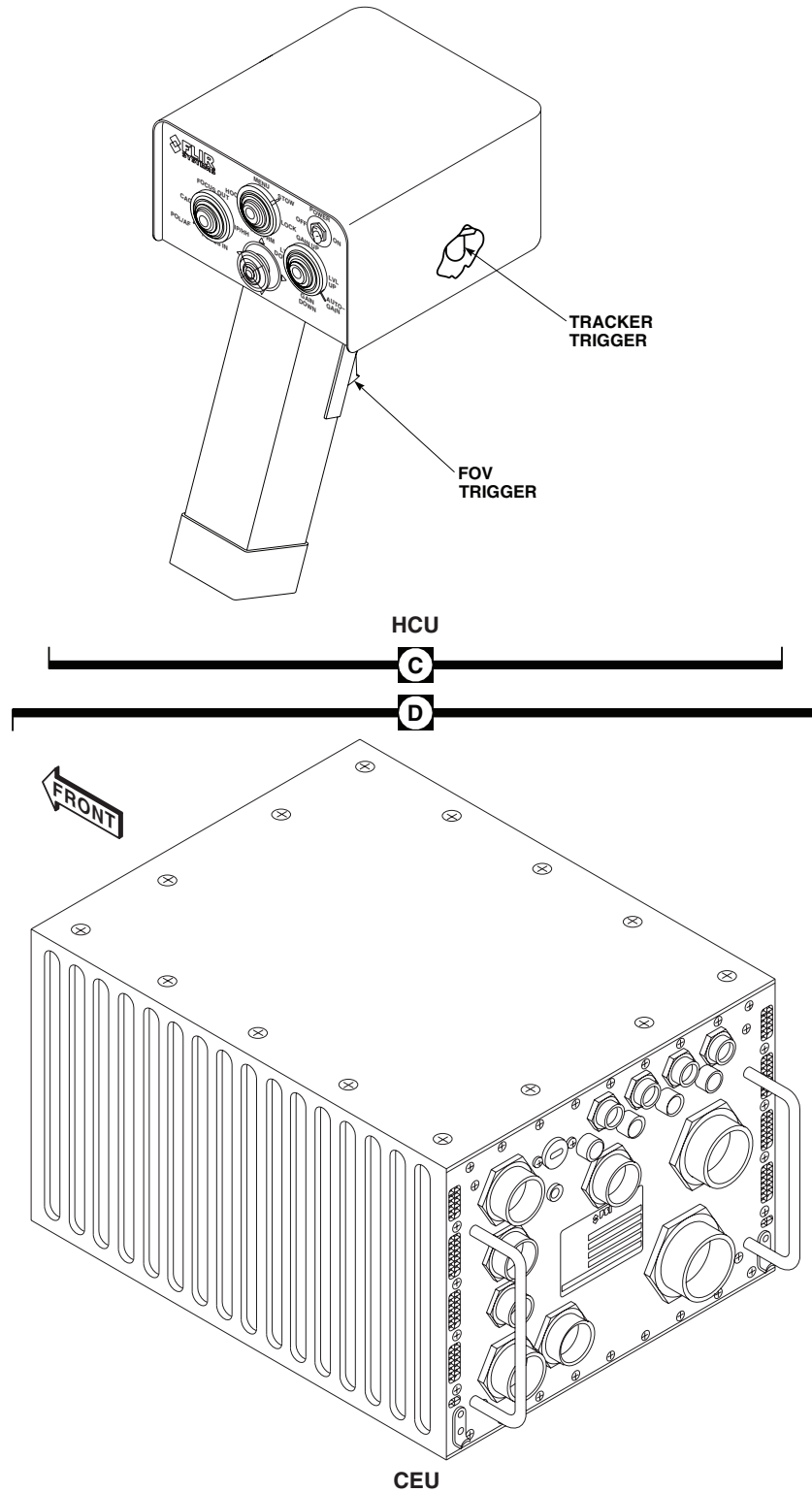


Figure 1. FLIR Location Diagram. (Sheet 2 of 4)

AB0992\_2C  
SA



LOCATION - Continued

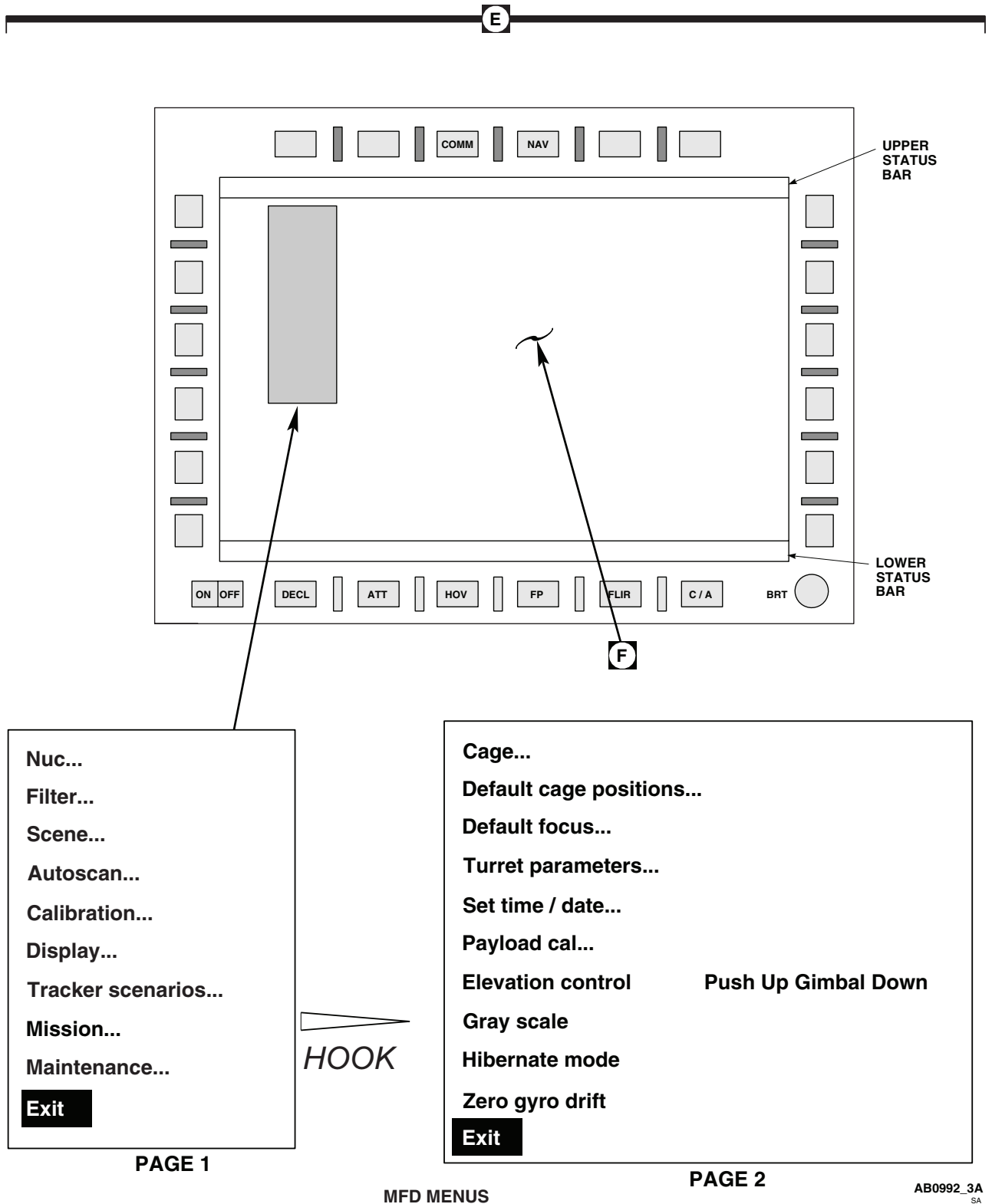
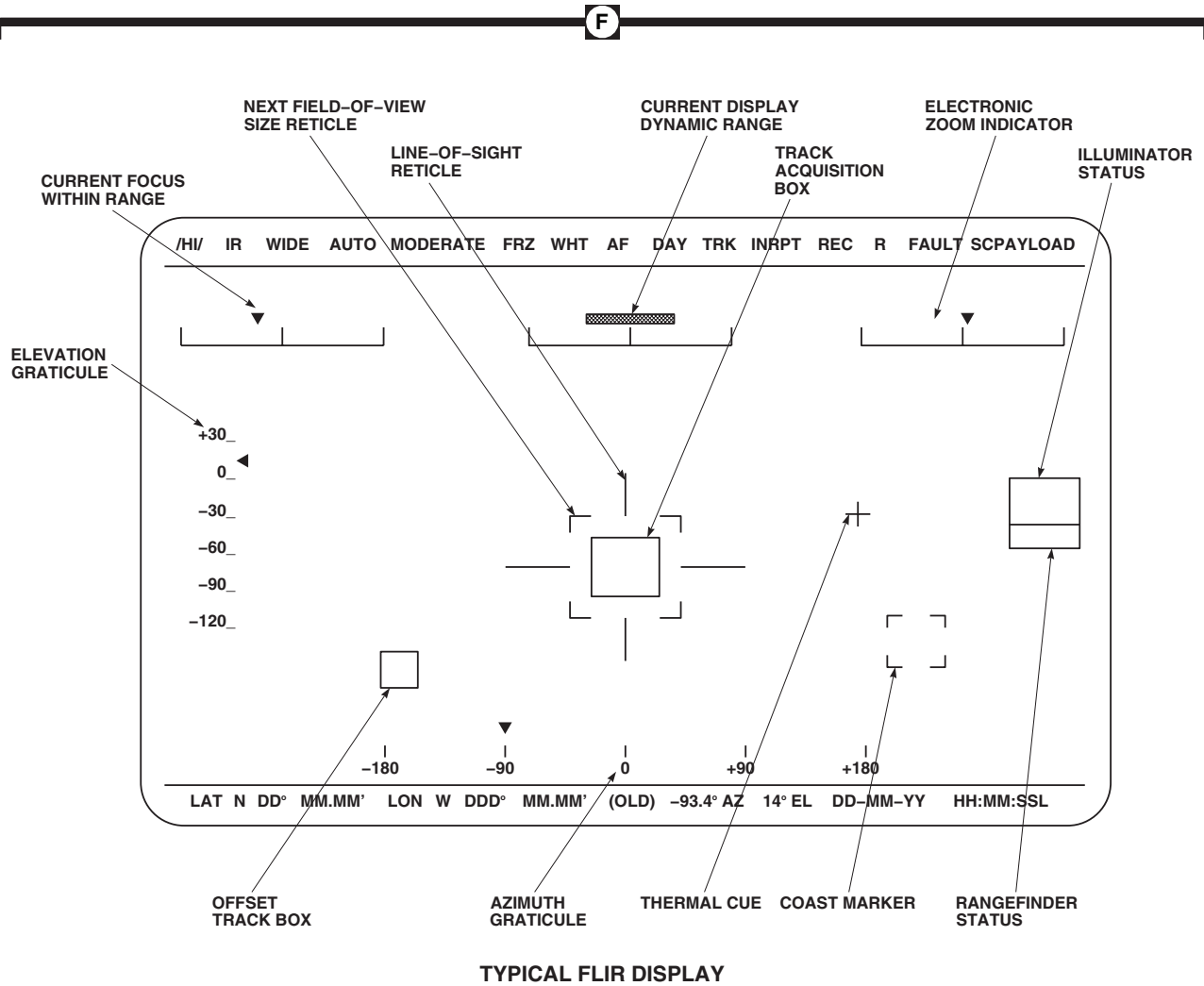


Figure 1. FLIR Location Diagram. (Sheet 3 of 4)

LOCATION - Continued



AB0992\_4A  
SA

Figure 1. FLIR Location Diagram. (Sheet 4 of 4)

**CEU**

**COPILOT'S AUXILIARY CIRCUIT BREAKER PANEL**

NO. 2 DC PRI BUS  
28 VDC

FLIR

15 AMP

CB530

P129

J129

**COPILOT'S MFD**

J1

P1015R

B

CG1015

**PILOT'S MFD**

J1

P1016R

B

CL1016

**AFT FIXTURE DISCONNECT PANEL**

J9001

CC

**CEU**

P224R

J224R

B

E

A

D

GD129-1

P210R

J1

B

E

A

D

28 VDC RTN

28 VDC RTN

28 VDC IN

28 VDC IN

P212R

1J8

CC

VIDEO OUT

P213R

1J7

CC

VIDEO OUT

P219R

1J10

CC

VIDEO OUT

**NOTES**

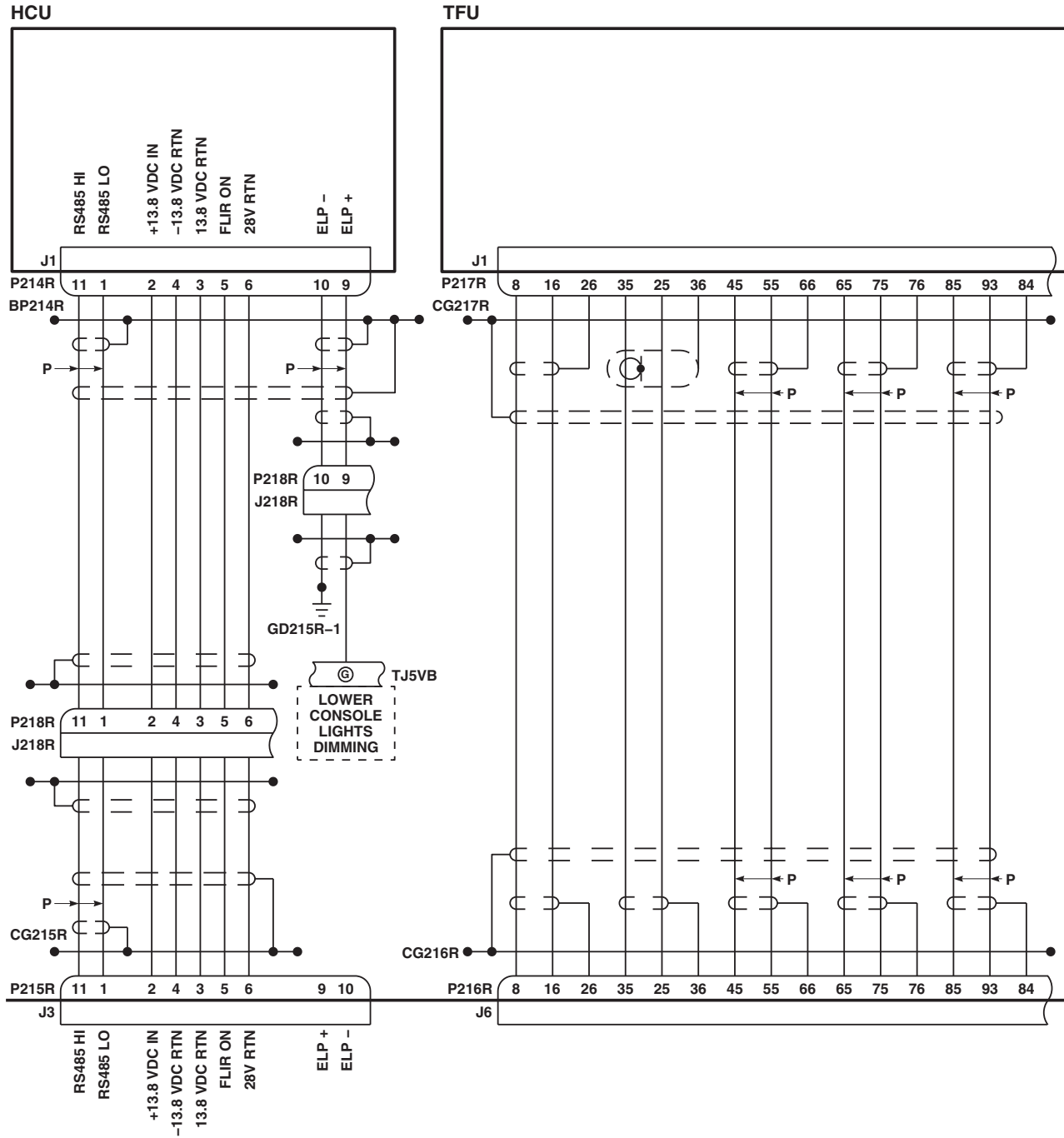
1. INDICATES SHIELDED, TWISTED PAIR

2. BP (BRAID PLUG) IS USED TO SHOW RING BRAID SHIELD TERMINATOR.

**AB1693\_1A**

Figure 2. FLIR Schematic Diagram. (Sheet 1 of 4)

SCHEMATICS - Continued



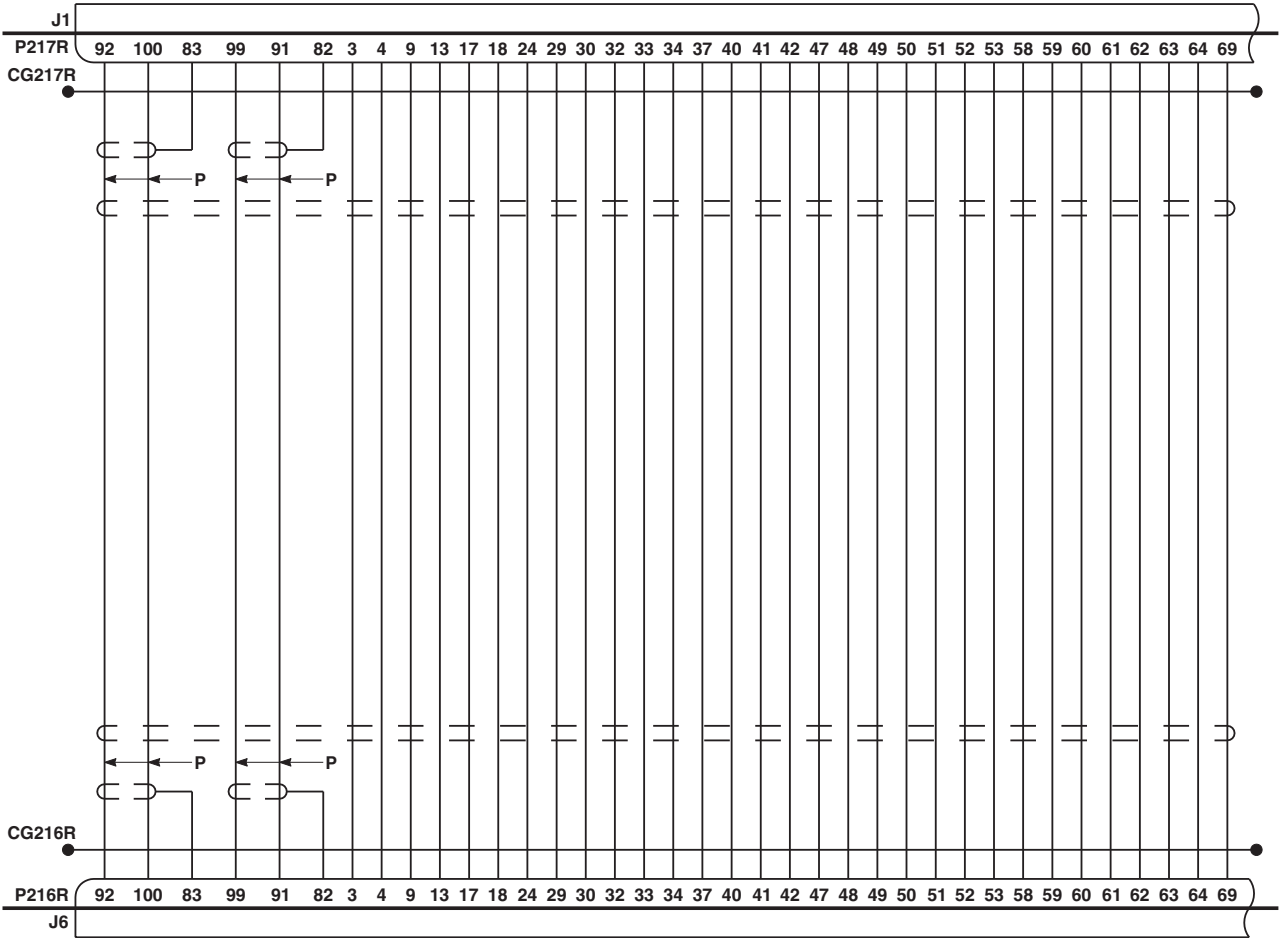
CEU

AB1693\_2A  
SA

Figure 2. FLIR Schematic Diagram. (Sheet 2 of 4)

SCHEMATICS - Continued

TFU

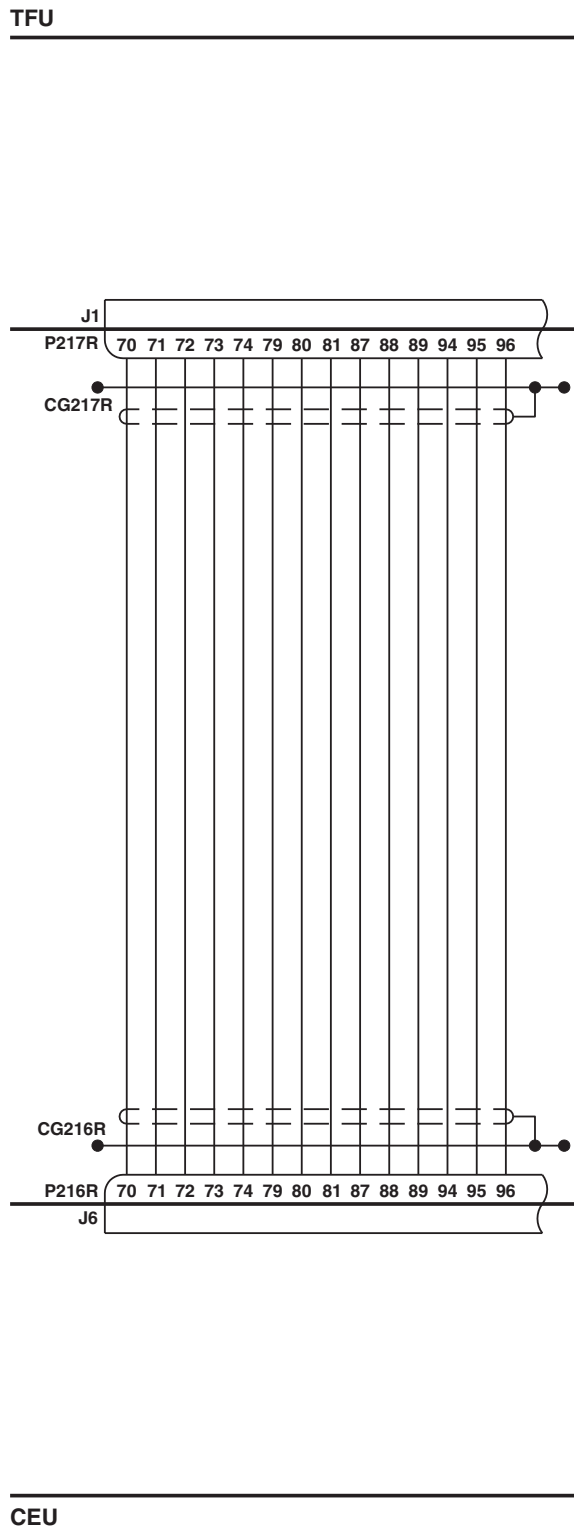


CEU

AB1963\_3A  
SA

Figure 2. FLIR Schematic Diagram. (Sheet 3 of 4)

SCHEMATICS - Continued



AB1963\_4A  
SA

Figure 2. FLIR Schematic Diagram. (Sheet 4 of 4)

END OF WORK PACKAGE

## UNIT LEVEL

## MISSION EQUIPMENT

## OPTION II CABIN HEATER AND VENTILATION SYSTEM

## INITIAL SETUP:

**Test Equipment**

Multimeter, AN/PSM-45A

WP 1269 00

WP 1270 00

WP 1272 00

**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01

WP 1273 00

Electrical Repairer Toolkit, SC 5180-99-B06

WP 1275 00

WP 1276 00

**Personnel Required**

Aircraft Electrician MOS 15F (2)

WP 1279 00

UH-60 Helicopter Repairer 15T (1)

WP 1280 00

WP 1399 00

**References**

TM 1-1520-237-10

WP 1400 00

WP 0079 00

WP 1513 01

WP 0104 00

WP 1513 02

WP 0697 00

WP 1586 01

WP 0698 00

WP 1747 00

WP 0824 00

**Equipment Condition**

WP 0825 00

APU Operational, TM 1-1520-237-10

WP 0828 00

No. 1 and No. 2 Engines Operational,

WP 1259 00

TM 1-1520-237-10

## SETUP OPERATIONAL/TROUBLESHOOTING PROCEDURE

## PROCEDURE

## Step

## NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. If **WINTER** is installed, the ambient air temperature must be very cold (below -25°F (-31.7°C)) to check the winterization mode operation of the cabin heater and ventilation system. If it is not cold enough, omit steps 13. and 14.

## CIRCUIT BREAKER

## CIRCUIT BREAKER PANEL

AIR SOURCE HEAT/START

Copilot's

NO. 1 ENGINE ANTI-ICE

Copilot's

**SETUP OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued

| <b>CIRCUIT BREAKER</b> | <b>CIRCUIT BREAKER PANEL</b> |
|------------------------|------------------------------|
| HEAT & VENT            | Pilot's                      |
| HEAT VENT              | Pilot's                      |
| NO. 2 ENGINE ANTI-ICE  | Pilot's                      |

2. Start APU (TM 1-1520-237-10).

**NOTE**

Open air valves in cockpit and front cabin area before turning on blower.

3. Place upper console VENT BLOWER switch to ON.

**Indication/Condition**

- a. Ventilation blower shall operate.
- b. Air shall come out from each louver.

**Corrective Action**

1. If result a. is not as specified, refer to, VENTILATION BLOWER DOES NOT OPERATE, in this work package.
2. If result b. is not as specified, repair/replace ducting (WP 1279 00).

**Step**

4. Place upper console VENT BLOWER switch to OFF.

**Indication/Condition**

Air shall stop flowing out of louvers.

**Corrective Action**

If result is not as specified, replace right relay panel (WP 0825 00).

**Step****NOTE**

Hot air may cycle on and off during operational/troubleshooting procedure.

5. Place upper console HEATER switch to FWD.

**Indication/Condition**

Mixing valve shall open and warm air shall come out of forward louvers.

**Corrective Action**

If result is not as specified, refer to NO HEATED AIR COMING OUT OF FORWARD LOUVERS, in this work package.



---

**SETUP OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Step**

6. Slowly turn upper console HEATER knob from OFF to HI.

**Indication/Condition**

Temperature of air coming out of forward louvers shall increase.

**Corrective Action**

If result is not as specified, refer to AIR TEMPERATURE CANNOT BE REGULATED, in this work package.

**Step**

7. Turn upper console HEATER knob to OFF.

**Indication/Condition**

Temperature of air coming out of forward louvers shall decrease.

**Corrective Action**

If result is not as specified, refer to AIR TEMPERATURE CANNOT BE REGULATED, in this work package.

**Step**

8. Place upper console HEATER switch to FWD/AFT.

**Indication/Condition**

Mixing valve shall open and warm air shall come out of forward and aft louvers.

**Corrective Action**

If result is not as specified, refer to, NO HEATED AIR COMING OUT OF FORWARD AND AFT LOUVER, in this work package.

**Step**

9. Turn ejector FLOW CONTROL nozzle, in cabin area, from LOW to HIGH.

**Indication/Condition**

Temperature of air coming out of louvers shall increase.

**Corrective Action**

If result is not as specified, replace aft heat lines (WP 1513 01).

**Step**

10. Place upper console HEATER switch to OFF.

**Indication/Condition**

Air shall not be flowing out of forward and aft louvers.

**Corrective Action**

If result is not as specified, replace heater valve (WP 1269 00).

**Step**

11. With qualified personnel at controls, start No. 1 and No. 2 engines (TM 1-1520-237-10).

**SETUP OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued

12. Shut down APU (TM 1-1520-237-10).
13. Place upper console AIR SOURCE HEAT/START switch to ENGINE.

**NOTE**

When HEATER switch is placed to FWD/AFT (engine bleed-air shutoff valves open), a noticeable increase in TGT (T4.5) for both engines should occur.

14. Place upper console HEATER switch to FWD/AFT.

**Indication/Condition**

No. 1 engine and No. 2 engine bleed-air shutoff valves shall open. Mixer valve shall open and warm air shall be coming out of forward and aft louvers.

**Corrective Action**

1. If air is coming out but No. 1 engine TGT does not increase, refer to, NO. 1 ENGINE TGT DOES NOT INCREASE WHEN HEATER SWITCH IS PLACED FWD/AFT, in this work package.
2. If air is coming out but No. 2 engine TGT does not increase, refer to NO. 2 ENGINE TGT DOES NOT INCREASE WHEN HEATER SWITCH IS PLACED FWD/AFT., in this work package.
3. If No. 1 and No. 2 engine TGTs do not increase, refer to, NO. 1 OR NO. 2 ENGINE TGT DOES NOT INCREASE WHEN HEATER SWITCH IS PLACED FWD/AFT, in this work package.
4. **ESSS** If fuel fumes are noticed, clean pneumatic check valve (WP 0698 00). ■

**Step**

15. Turn upper console HEATER knob and cabin area ejector FLOW CONTROL nozzle to HIGH. Note temperature of air coming out of forward and aft louvers.
16. Place upper console ENG ANTI-ICE NO. 1 switch to ON.

**Indication/Condition**

Temperature and flow of air coming out of forward louvers shall decrease and aft louvers shall stop.

**Corrective Action**

If result is not as specified, refer to AIR TEMPERATURE COMING OUT OF LOUVERS DOES NOT CHANGE WHEN ENG ANTI-ICE IS TURNED ON, in this work package.

**Step**

17. Place upper console ENG ANTI-ICE NO. 1 switch to OFF.
18. Place upper console ENG ANTI-ICE NO. 2 switch to ON.

**Indication/Condition**

Temperature and flow of air coming out of forward louvers shall decrease and aft louvers shall stop.

**Corrective Action**

If result is not as specified, replace ENG ANTI-ICE NO. 2 switch (WP 0828 00).

**Step**

19. Place upper console ENG ANTI-ICE NO. 2 switch to OFF.
20. Turn upper console HEATER knob to OFF and ejector FLOW CONTROL nozzle to LOW.

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**SETUP OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**

21. Place upper console HEATER switch to OFF.

**Indication/Condition**

Air shall stop coming out of forward and aft louvers and a noticeable decrease in TGT of both engines shall occur.

**Corrective Action**

If result is not as specified, replace engine bleed-air valve (WP 0697 00).

**Step**

22. Shut down both engines (TM 1-1520-237-10).

**Indication/Condition**

Not Applicable.

**Corrective Action**

Not Applicable.

**VENTILATION BLOWER DOES NOT OPERATE****SYMPTOM**

Ventilation Blower Does Not Operate.

**MALFUNCTION**

Ventilation Blower Does Not Operate.

**CORRECTIVE ACTION**

- 1. Make sure VENT BLOWER switch is placed ON. Go to 2.**
- 2. Check continuity between P437-5 and P437-6.**
  - a. If continuity is present, go to **3**.
  - b. If continuity is not present, replace ventilation blower (WP 1280 00). Go to **14**.
- 3. Check for 28 vdc between J437-6 and ground.**
  - a. If voltage is as specified, go to **4**.
  - b. If voltage is not as specified, go to **11**.
- 4. Check continuity between J437-5 and P241-j.**
  - a. If continuity is present, go to **5**.
  - b. Step 2. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 5. Check continuity between J437-4 and ground.**
  - a. If continuity is present, go to **6**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 6. Install an insulated jumper wire between J437-5 and J437-6 . Go to 7.**
- 7. Check for 115 vac between:**
  - J437-1 and J437-4
  - J437-2 and J437-4

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**VENTILATION BLOWER DOES NOT OPERATE - Continued**

- **J437-3 and J437-4**
  - a. If voltage is as specified, remove jumper wire. Replace ventilation blower (WP 1280 00) then go to **14**.
  - b. If voltage is not as specified, remove jumper wire then go to **8**.
- 8. Check continuity between:**
  - **J437-1 and P241-g**
  - **J437-2 and P241- h**
  - **J437-3 and P241-i**
  - a. If continuity is present, go to **9**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00), as required. Go to **14**.
- 9. Check for 115 vac between HEAT & VENT circuit breaker:**
  - **Terminal A1 and ground**
  - **Terminal B1 and ground**
  - **Terminal C1 and ground**
  - a. If voltage is as specified, go to **10**.
  - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **14**.
- 10. Check continuity between:**
  - **P241-E and terminal A1 (HEAT & VENT circuit breaker)**
  - **P241-n and terminal B1 (HEAT & VENT circuit breaker)**
  - **P241-N and terminal C1 (HEAT & VENT circuit breaker)**
  - a. If continuity is present, replace right relay panel (WP 0825 00). Go to **14**.
  - b. If continuity is not present, repair/replace wiring WP 1747 00), as required. Go to **14**.
- 11. Check for 28 vdc between VENT BLOWER switch terminal 3 and ground.**
  - a. If voltage is as specified, repair/replace wiring between switch terminal 3 and J437-6 (WP 1747 00). Go to **14**.
  - b. If voltage is not as specified, go to **12**.
- 12. Check for 28 vdc between VENT BLOWER switch terminal 2 and ground.**
  - a. If voltage is as specified, replace switch (WP 0828 00). Go to **14**.
  - b. If voltage is not as specified, go to **13**.
- 13. Check for 28 vdc between HEAT & VENT circuit breaker terminal 1 and ground.**
  - a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and VENT BLOWER switch terminal 2 (WP 1747 00). Go to **14**.
  - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **14**.
- 14. Procedure completed.**

**NO HEATED AIR COMING OUT OF FOWARD LOUVERS****SYMPTOM**

No Heated Air Coming Out Of Foward Louvers.

**MALFUNCTION**

No Heated Air Coming Out Of Foward Louvers.

**CORRECTIVE ACTION**

- 1. Check bleed-air line from APU check valve to heater valve for proper connections.**
  - a. If proper connections are visible, go to **2**.
  - b. If proper connections are not visible, repair/replace lines and/or rubber connections as necessary (WP 1272 00). Go to **23**.
- 2. Check for 28 vdc between P436-1 and ground.**
  - a. If voltage is as specified, go to **3**.
  - b. If voltage is not as specified, go to **4**.
- 3. Check continuity between:**
  - **P436-2 and ground**
  - **P436-3 and ground**
  - a. If continuity is present, replace heater valve (WP 1269 00). Go to **23**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **23**.
- 4. Check helicopter configuration.**
  - a. **WINTER** , install an insulated jumper wire between P244B-K and P241-G. Go to **5**.
  - b. **W/O WINTER** , install an insulated jumper wire between P244-K and P241-G. Go to **5**.
- 5. Check for 28 vdc between P436-1 and ground.**
  - a. If voltage is as specified, go to **6**.
  - b. If voltage is not as specified, go to **8**.
- 6. Remove jumper wire. Go to 7.**
- 7. Replace right relay panel (WP 0825 00). Go to 23.**
- 8. Remove jumper wire. Go to 9.**
- 9. Check helicopter configuration.**
  - a. **WINTER** , go to **10**.
  - b. **W/O WINTER** , go to **11**.
- 10. WINTER Check for 28 vdc between P244B-K and ground.**
  - a. If voltage is as specified, go to **12**.
  - b. If voltage is not as specified, replace **WINTER** (WP 1399 00). Go to **23**
  - c. If trouble remains, go to **11**.
- 11. Check for 28 vdc between P244-K and ground.**

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**NO HEATED AIR COMING OUT OF FOWARD LOUVERS** - Continued

- a. If voltage is as specified, repair/replace wiring between P241-G and P436-1 (WP 1747 00). Go to **23**.
  - b. If voltage is not as specified, go to **12**.
- 12. Install an insulated jumper wire across P2902-b and P902-d. Go to 13.**
- 13. Check helicopter configuration.**
- a. **WINTER** , go to **14**.
  - b. **W/O WINTER** , go to **15**.
- 14. Check for 28 vdc between P244B-K and ground.**
- a. If voltage is as specified, go to **16**.
  - b. If voltage is not as specified, replace **WINTER** (WP 1400 00). Go to **23**.
  - c. If trouble remains, go to **15**.
- 15. Check for 28 vdc between P244-K and ground.**
- a. If voltage is as specified, go to **16**.
  - b. If voltage is not as specified, go to **18**.
- 16. Remove jumper wire from P902 . Go to 17.**
- 17. Replace left relay panel (WP 0825 00). Go to 23.**
- 18. Remove jumper wire from P902 . Go to 19.**
- 19. Check for 28 vdc between P902-b and ground.**
- a. If voltage is as specified, repair/replace wiring between P902-d and P244-K (WP 1747 00). Go to **23**.
  - b. If voltage is not as specified, go to **20**.
- 20. Check for 28 vdc between S5-3 **HH-60A HH-60L** S3-3 and ground.**
- a. If voltage is as specified, repair/replace wiring between S5-3 or **HH-60A HH-60L** S3-3 and P902-b (WP 1747 00). Go to **23**.
  - b. If voltage is not as specified, go to **21**.
- 21. Check for 28 vdc between AIR SOURCE HEAT/START circuit breaker terminal 1 and ground.**
- a. If voltage is as specified, go to **22**.
  - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **23**.
- 22. Check continuity between AIR SOURCE HEAT/START circuit breaker terminal 1 and S5-2 or **HH-60A HH-60L** S3-2.**
- a. If continuity is present, replace switch (WP 0828 00). Go to **23**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **23**.
- 23. Procedure completed.**

**AIR TEMPERATURE CANNOT BE REGULATED****SYMPTOM**

Air Temperature Cannot Be Regulated.

**MALFUNCTION**

Air Temperature Cannot Be Regulated.

**CORRECTIVE ACTION**

- 1. Unlock fastener studs securing upper console panel and hinge down panel. Turn HEATER knob and check shaft for proper turning.**
  - a. If shaft turns properly, go to **2**.
  - b. If shaft does not turn properly, secure setscrews attaching shaft to adapter or shaft to HEATER knob. Adjust system as necessary (WP 1259 00). Go to **11**.
- 2. Disconnect control shaft at temperature regulator. Go to 3.**
- 3. Turn HEATER knob, visually check for turning of inner core of control shaft.**
  - a. If inner core of control shaft turns, go to **4**.
  - b. If inner core of control shaft does not turn, go to **9**.
- 4. Check temperature tube for signs of damage or loose connections.**
  - a. If damage or loose connections are visible, secure temperature tube or replace (WP 1270 00). Go to **11**.
  - b. If damage or loose connections are not visible, go to **5**.
- 5. Operate APU (TM 1-1520-237-10). Go to 6.**
- 6. Place HEATER switch to FWD. Go to 7.**
- 7. Turn HEATER knob to HI. Go to 8.**
- 8. Check that temperature tube is warm.**
  - a. If tube is warm, shut down APU (TM 1-1520-237-10), then replace mixture temperature sensor's pressure bellows (WP 1273 00). Go to **11**.
  - b. If trouble remains, shut down APU (TM 1-1520-237-10), then replace mixture temperature sensor (WP 1273 00). Go to **11**.
  - c. If tube is not warm, shut down APU (TM 1-1520-237-10), then replace heater valve (WP 1269 00). Go to **11**.
- 9. Disconnect control shaft at adapter. Go to 10.**
- 10. Turn HEATER knob and visually check for turning of adapter output shaft.**
  - a. If shaft turns, replace control shaft (WP 1275 00). Go to **11**.
  - b. If shaft does not turn, replace adapter (WP 1276 00). Go to **11**.
- 11. Procedure completed.**

**NO HEATED AIR COMING OUT OF FOWARD AND AFT LOUVERS****SYMPTOM**

No Heated Air Coming Out of Foward and Aft Louvers.

**MALFUNCTION**

No Heated Air Coming Out of Foward and Aft Louvers.

**CORRECTIVE ACTION**

- 1. Check for 28 vdc between P456-1 and ground.**
  - a. If voltage is as specified, go to **2**.
  - b. If voltage is not as specified, go to **3**.
- 2. Check continuity between P456-3 and ground.**
  - a. If continuity is present, replace aft solenoid valve (WP 1513 02). Go to **10**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **10**.
- 3. Check continuity between TB26A-G and P456-1.**
  - a. If continuity is present, go to **4**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **10**.
- 4. With No. 1 engine start button not pressed, check for 0 vdc between TB26A-K and ground.**
  - a. If voltage is as specified, go to **5**.
  - b. If voltage is not as specified, troubleshoot engine start and ignition system (WP 0079 00). Go to **10**.
- 5. Check continuity between TB26A-R and TB45A-G.**
  - a. If continuity is present, go to **6**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **10**.
- 6. With No. 2 engine start button not pressed, check for 0 vdc between TB45A-K and ground.**
  - a. If voltage is as specified, go to **7**.
  - b. If voltage is not as specified, troubleshoot engine start and ignition system (WP 0079 00). Go to **10**.
- 7. Check for 28 vdc between TB45A-R and ground.**
  - a. If voltage is as specified, replace heat and vent relay panel (WP 1586 01). Go to **10**.
  - b. If voltage is not as specified, go to **8**.
- 8. Check continuity between TB45A-R and S5-4 HH-60A HH-60L S3-3 .**
  - a. If continuity is present, go to **9**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **10**.
- 9. Check continuity between S5-5 HH-60A HH-60L S3-3 and S23-4.**
  - a. If continuity is present, replace switch (WP 0828 00). Go to **10**.



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**NO HEATED AIR COMING OUT OF FOWARD AND AFT LOUVERS** - Continued

- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **10**.

**10. Procedure completed.**

**NO. 1 ENGINE TGT DOES NOT INCREASE WHEN HEATER SWITCH IS PLACED FWD/AFT****SYMPTOM**

No. 1 Engine TGT Does Not Increase When HEATER Switch is Placed FWD/AFT.

**MALFUNCTION**

No. 1 Engine TGT Does Not Increase When HEATER Switch is Placed FWD/AFT.

**CORRECTIVE ACTION**

- 1. Make sure AIR SOURCE HEAT/START switch is placed to ENGINE. Go to 2.**
- 2. Check for 28 vdc between P434-1 and P434-2.**
  - a. If voltage is as specified, replace No. 1 engine bleed-air shut off valve (WP 0697 00). Go to **10**.
  - b. If voltage is not as specified, go to **3**.
- 3. Check for 28 vdc between P434-1 and ground.**
  - a. If voltage is as specified, repair/replace ground wire on P434-2 (WP 1747 00). Go to **10**.
  - b. If voltage is not as specified, go to **4**.
- 4. Install an insulated jumper wire across P902-a and P902-c. Go to 5.**
- 5. Check for 28 vdc between P434-1 and ground.**
  - a. If voltage is as specified, go to **6**.
  - b. If voltage is not as specified, go to **8**.
- 6. Remove jumper wire from P902 . Go to 7.**
- 7. Replace left relay panel (WP 0825 00). Go to 10.**
- 8. Remove jumper wire from P902. Go to 9.**
- 9. Check for 28 vdc between P902-a and ground.**
  - a. If voltage is as specified, repair/replace wiring between P902-c and P434-1 (WP 1747 00). Go to **10**.
  - b. If voltage is not as specified, repair/replace wiring between P902-a and S18-3 (WP 1747 00). Go to **10**.
- 10. Procedure completed.**

**NO. 2 ENGINE TGT DOES NOT INCREASE WHEN HEATER SWITCH IS PLACED FWD/AFT.****SYMPTOM**

No. 2 Engine TGT Does Not Increase When HEATER Switch is Placed FWD/AFT

**MALFUNCTION**

No. 2 Engine TGT Does Not Increase When HEATER Switch is Placed FWD/AFT

**CORRECTIVE ACTION**

1. Make sure AIR SOURCE HEAT/START switch is placed to ENGINE. Go to 2.
2. Check for 28 vdc between P435-1 and P435-2.
  - a. If voltage is as specified, replace No. 2 engine bleed-air shut off valve (WP 0697 00). Go to 13.
  - b. If voltage is not as specified, go to 3.
3. Check for 28 vdc between P435-1 and ground.
  - a. If voltage is as specified, repair/replace ground wire on P435-2 (WP 1747 00). Go to 13.
  - b. If voltage is not as specified, go to 4.
4. Check helicopter configuration.
  - a. **WINTER** , install an insulated jumper wire between P244B-J and P241-F. Go to 5.
  - b. **W/O WINTER** , install an insulated jumper wire between P244-J and P241-F. Go to 5.
5. Check for 28 vdc between P435-1 and ground.
  - a. If voltage is as specified, go to 6.
  - b. If voltage is not as specified, go to 8.
6. Remove jumper wire. Go to 7.
7. Replace right relay panel (WP 0825 00). Go to 13.
8. Remove jumper wire. Go to 9.
9. Check helicopter configuration.
  - a. **WINTER** , go to 10.
  - b. **W/O WINTER** , go to 12.
10. **WINTER** Check for 28 vdc between P244B-J and ground.
  - a. If voltage is as specified, repair/replace wiring between P241-F and P435-1 (WP 1747 00). Go to 13.
  - b. If voltage is not as specified, go to 11.
11. **WINTER** Check for 28 vdc between P244-J and ground.
  - a. If voltage is as specified, replace **WINTER** (WP 1400 00). Go to 13.
  - b. If voltage is not as specified, repair/replace wiring between P244-J and P231-a (WP 1747 00). Go to 13.
12. Check for 28 vdc between P244-J and ground.

**NO. 2 ENGINE TGT DOES NOT INCREASE WHEN HEATER SWITCH IS PLACED FWD/AFT. - Continued**

- a. If voltage is as specified, repair/replace wiring between P241-F and P435-1 (WP 1747 00). Go to **13**.
- b. If voltage is not as specified, repair/replace wiring between P244-J and P231-a (WP 1747 00). Go to **13**.

**13. Procedure completed.****NO. 1 OR NO. 2 ENGINE TGT DOES NOT INCREASE WHEN HEATER SWITCH IS PLACED FWD/AFT****SYMPTOM**

No. 1 Or No. 2 Engine TGT Does Not Increase When HEATER Switch is Placed FWD/AFT.

**MALFUNCTION**

No. 1 Or No. 2 Engine TGT Does Not Increase When HEATER Switch is Placed FWD/AFT.

**CORRECTIVE ACTION**

- 1. Make sure AIR SOURCE HEAT/START switch is placed to ENGINE. Go to 2.**
- 2. Check for 28 vdc between J231-a and ground.**
  - a. If voltage is as specified, repair/replace wiring between P231-a and P902-a (WP 1747 00). Go to **4**.
  - b. If voltage is not as specified, go to **3**.
- 3. Check wiring between J231-a and switch S18 for proper connections.**
  - a. If wiring connections are good, replace switch (WP 0828 00). Go to **4**.
  - b. If wiring connections are not good, reconnect wiring to switch (WP 1747 00). Go to **4**.
- 4. Procedure completed.**

**AIR TEMPERATURE COMING OUT OF LOUVERS DOES NOT CHANGE WHEN ENG ANTI-ICE IS TURNED ON****SYMPTOM**

Air Temperature Coming Out of Louvers Does Not Change When ENG ANTI-ICE is Turned ON.

**MALFUNCTION**

Air Temperature Coming Out of Louvers Does Not Change When ENG ANTI-ICE is Turned ON.

**CORRECTIVE ACTION**

- 1. With ENG ANTI-ICE switched OFF, check for 28 vdc between J220-C and ground.**
  - a. If voltage is as specified, go to **2**.
  - b. If voltage is not as specified, go to **3**.
- 2. Check for 28 vdc between P436-4 and P436-3.**
  - a. If voltage is as specified, replace heater valve (WP 1269 00). Go to **5**.
  - b. If voltage is not as specified, repair/replace wiring between P436-4 and P220-C (WP 1747 00). Go to **5**.
- 3. Check for 28 vdc between J230-C and ground.**

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**AIR TEMPERATURE COMING OUT OF LOUVERS DOES NOT CHANGE WHEN ENG ANTI-ICE IS TURNED ON** - Continued

- a. If voltage is as specified, repair/replace wiring between J220-C and P230-C (WP 1747 00). Go to **5**.
  - b. If voltage is not as specified, go to **4**.
- 4. Check for broken/disconnected wires between AIR SOURCE HEAT/START switch terminal 2 and J230-C.**
- a. If wiring connections are good, repair/replace wiring between switch S18-2 and J230-C (WP 1747 00). Go to **5**.
  - b. If wiring connections are not good, replace switch NO. 1 ENG ANTI-ICE (WP 0828 00). Go to **5**.
- 5. Procedure completed.**

Diagram illustrating the location of various components in the engine compartment, with labels and part numbers:

- RIGHT RELAY PANEL (SEE NOTE) P241 / J241 P244 / J244
- VENTILATION BLOWER P437 / J437
- MIXING VALVE
- AIR INTAKE
- MIXTURE TEMPERATURE SENSOR P222 / J222
- P220 / J220
- P126 / J126
- P236 / J236
- HEATER CONTROL KNOB
- HEATER CONTROL SHAFT
- P230 / J230
- P127 / J127
- PNEUMATIC LINE
- NO. 2 ENGINE BLEED-AIR SHUTOFF VALVE
- NO. 1 ENGINE BLEED-AIR SHUTOFF VALVE
- EXTERNAL AIR SOURCE CONNECTOR
- P435
- P434
- P436
- LEFT RELAY PANEL P902 / J902
- WINTERIZATION KIT HARNESS ASSEMBLY (SEE NOTE) P244B / J244A
- P221 / J221
- P231 / J231

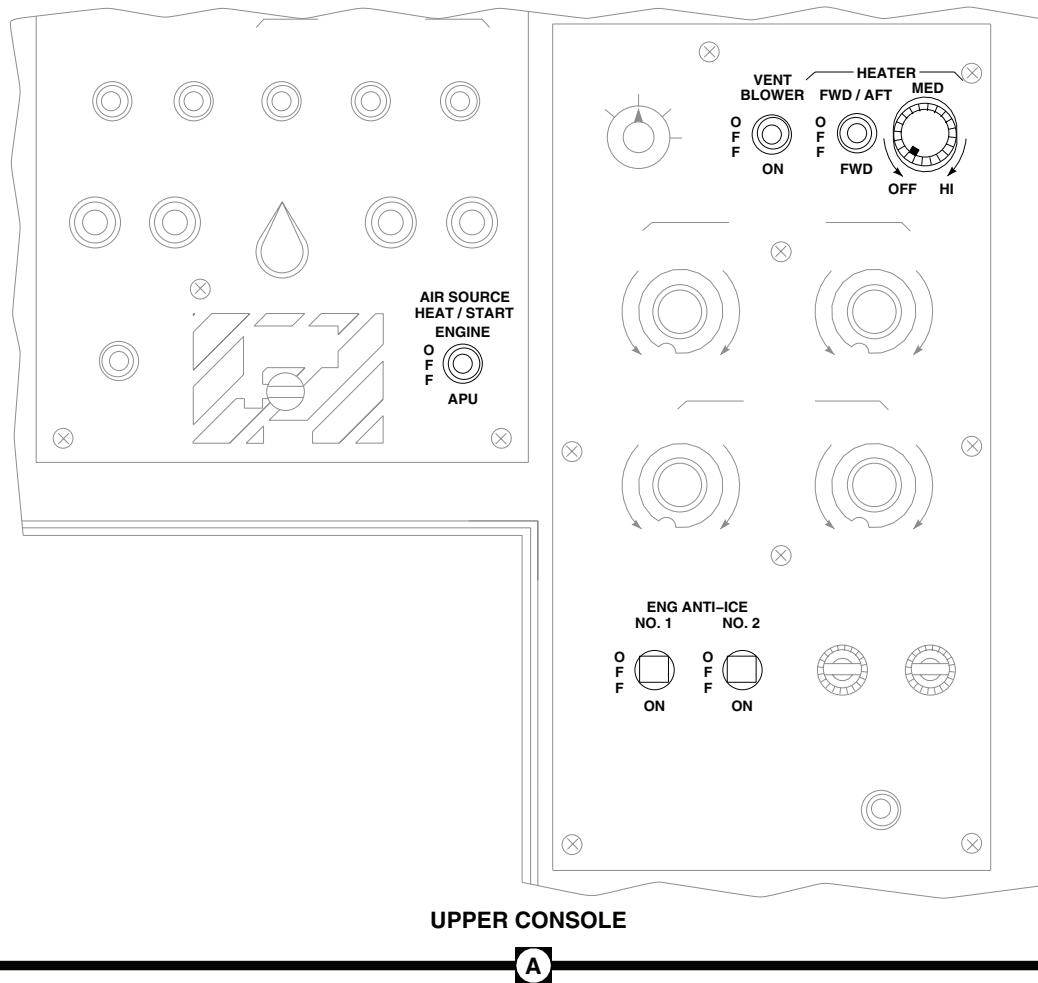
Labels A, B, and C are also present, indicating specific areas of interest.

| <b>TERMINAL BOARD /<br/>DISCONNECT PLUG /<br/>RECEPTACLE</b> | <b>LOCATION /<br/>CONNECTION POINT</b>            |
|--------------------------------------------------------------|---------------------------------------------------|
| P126 / J126                                                  | COCKPIT CEILING, BL 28 RH<br>STA 243              |
| P127 / J127                                                  | COCKPIT CEILING, BL 28 LH,<br>STA 243             |
| P220 / J220                                                  | JUNCTION BOX, MAIN<br>ROTOR PYLON DECK            |
| P221 / J221                                                  | JUNCTION BOX, MAIN<br>ROTOR PYLON DECK            |
| P222 / J222                                                  | UPPER CONSOLE                                     |
| P230 / J230                                                  | UPPER CONSOLE                                     |
| P231 / J231                                                  | CABIN CEILING, BL 9 LH,<br>STA 247                |
| P236 / J236                                                  | BEHIND PILOT'S CIRCUIT<br>BREAKER PANEL, BL 23 RH |
| P241 / J241                                                  | RIGHT RELAY PANEL                                 |
| P242 / J242                                                  | LEFT RELAY PANEL                                  |

WINTER

Figure 1. Option II Cabin Heater and Ventilation System Location Diagram. (Sheet 1 of 3)

LOCATION - Continued

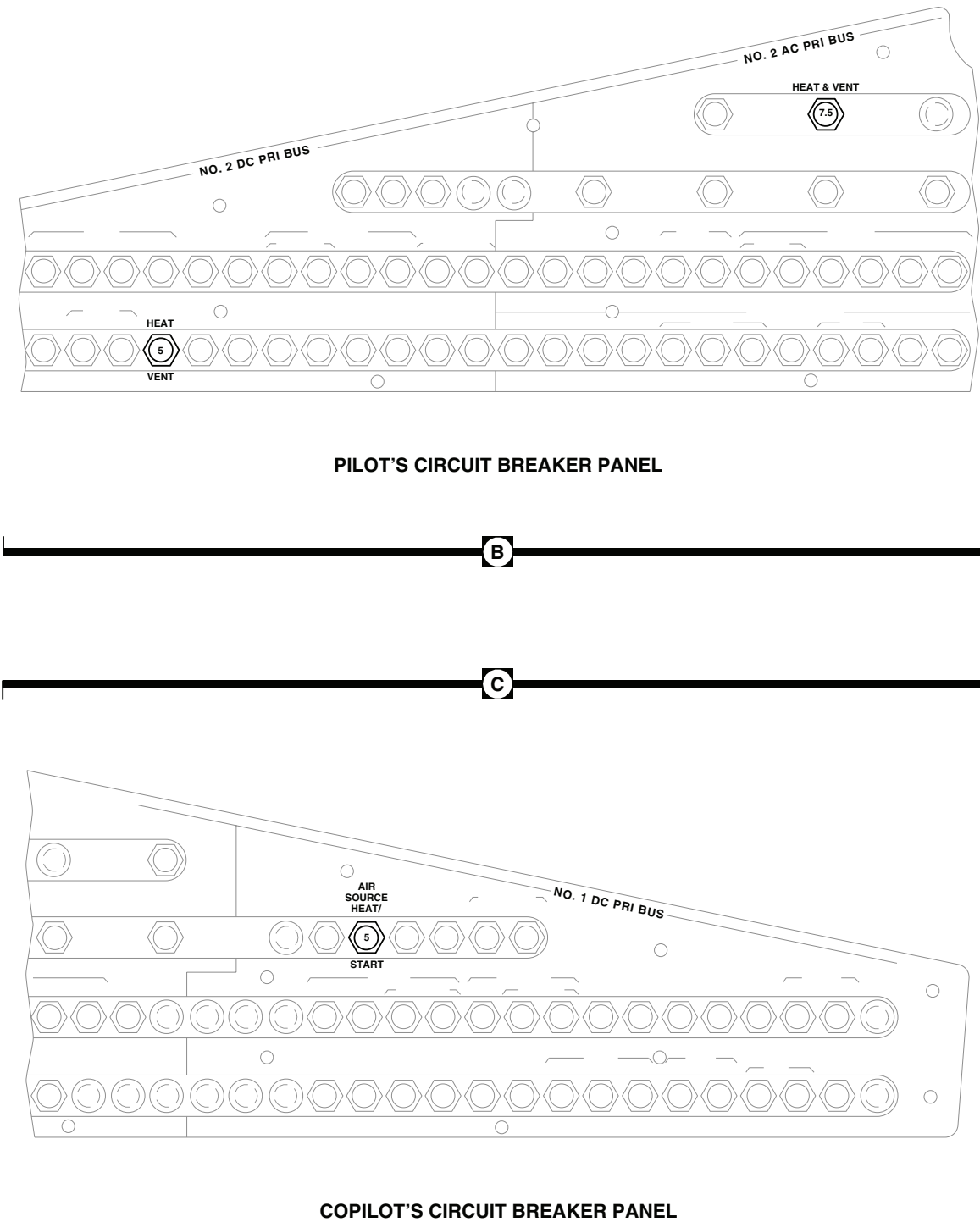


| TERMINAL BOARD /<br>DISCONNECT PLUG /<br>RECEPTACLE | LOCATION /<br>CONNECTION POINT                      |
|-----------------------------------------------------|-----------------------------------------------------|
| P244 / J244                                         | RIGHT RELAY PANEL<br>(SEE NOTE)                     |
| P244B / J244A                                       | WINTERIZATION KIT<br>HARNESS ASSEMBLY (SEE<br>NOTE) |
| P434                                                | NO. 1 ENGINE BLEED-AIR<br>SHUTOFF VALVE             |
| P435                                                | NO. 2 ENGINE BLEED-AIR<br>SHUTOFF VALVE             |
| P436                                                | MIXING VALVE                                        |
| P437 / J437                                         | VENTILATION BLOWER                                  |
| P456                                                | AFT SOLENOID VALVE,<br>BL 12 RH, STA 286, WL 278    |
| P902 / J902                                         | LEFT RELAY PANEL                                    |
| TB26A                                               | HEAT AND VENT RELAY<br>PANEL                        |
| TB45A                                               | HEAT AND VENT RELAY<br>PANEL                        |

AB3315 2  
SA

Figure 1. Option II Cabin Heater and Ventilation System Location Diagram. (Sheet 2 of 3)

LOCATION - Continued



AB3315\_3  
SA

Figure 1. Option II Cabin Heater and Ventilation System Location Diagram. (Sheet 3 of 3)

SCHEMATICS

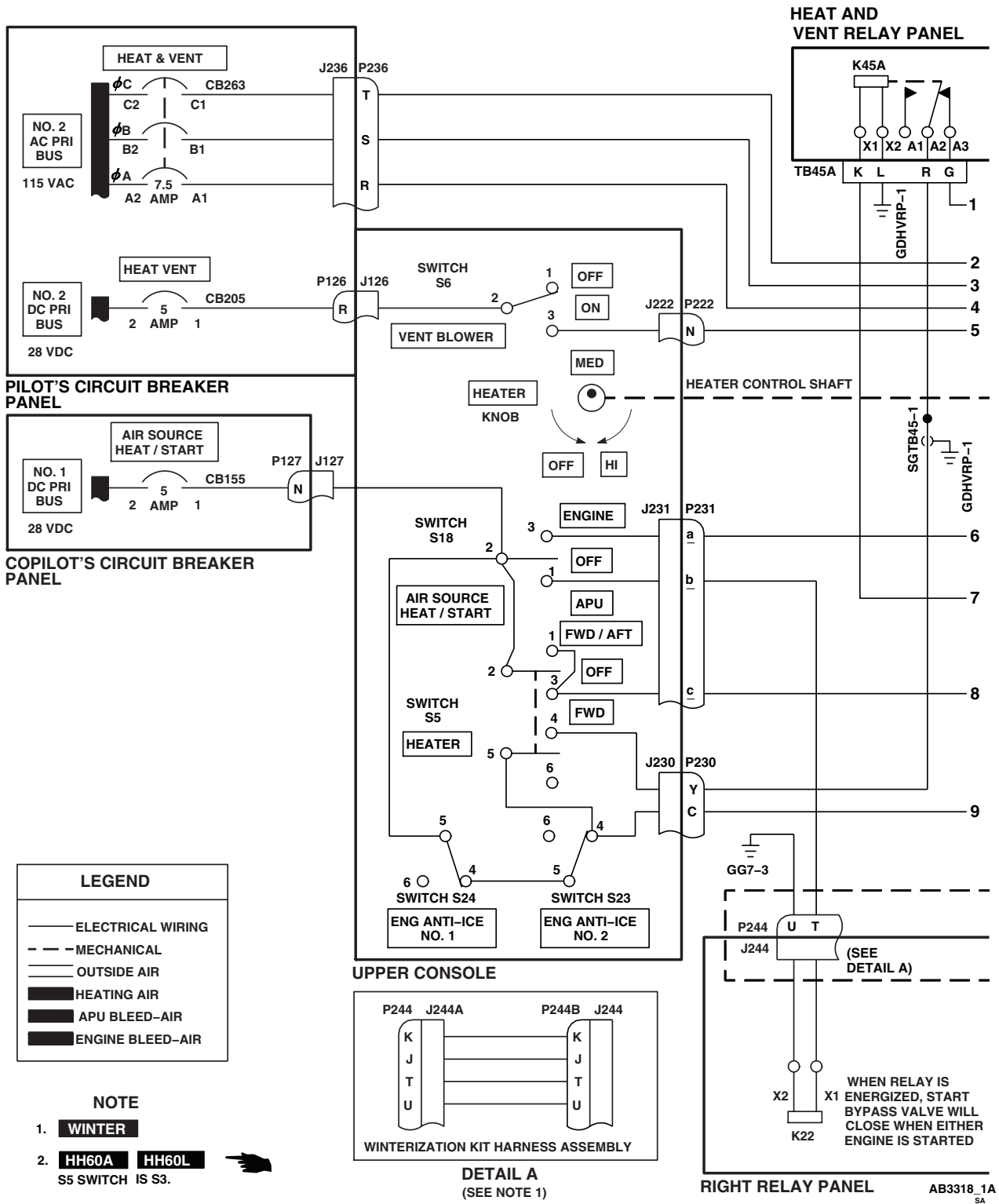


Figure 2. Option II Cabin Heater and Ventilation System Schematic Diagram. (Sheet 1 of 3)



## SCHEMATICS - Continued

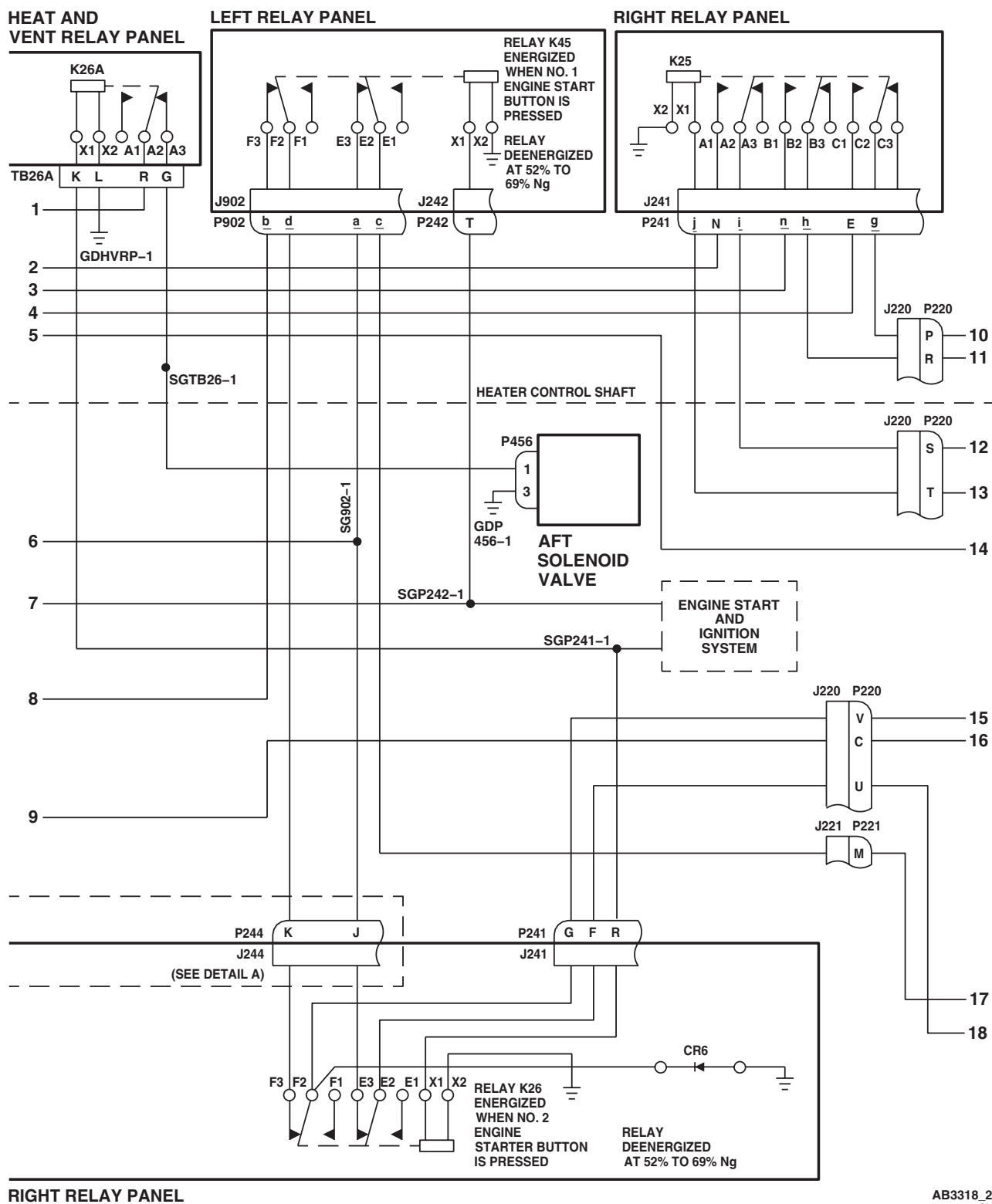
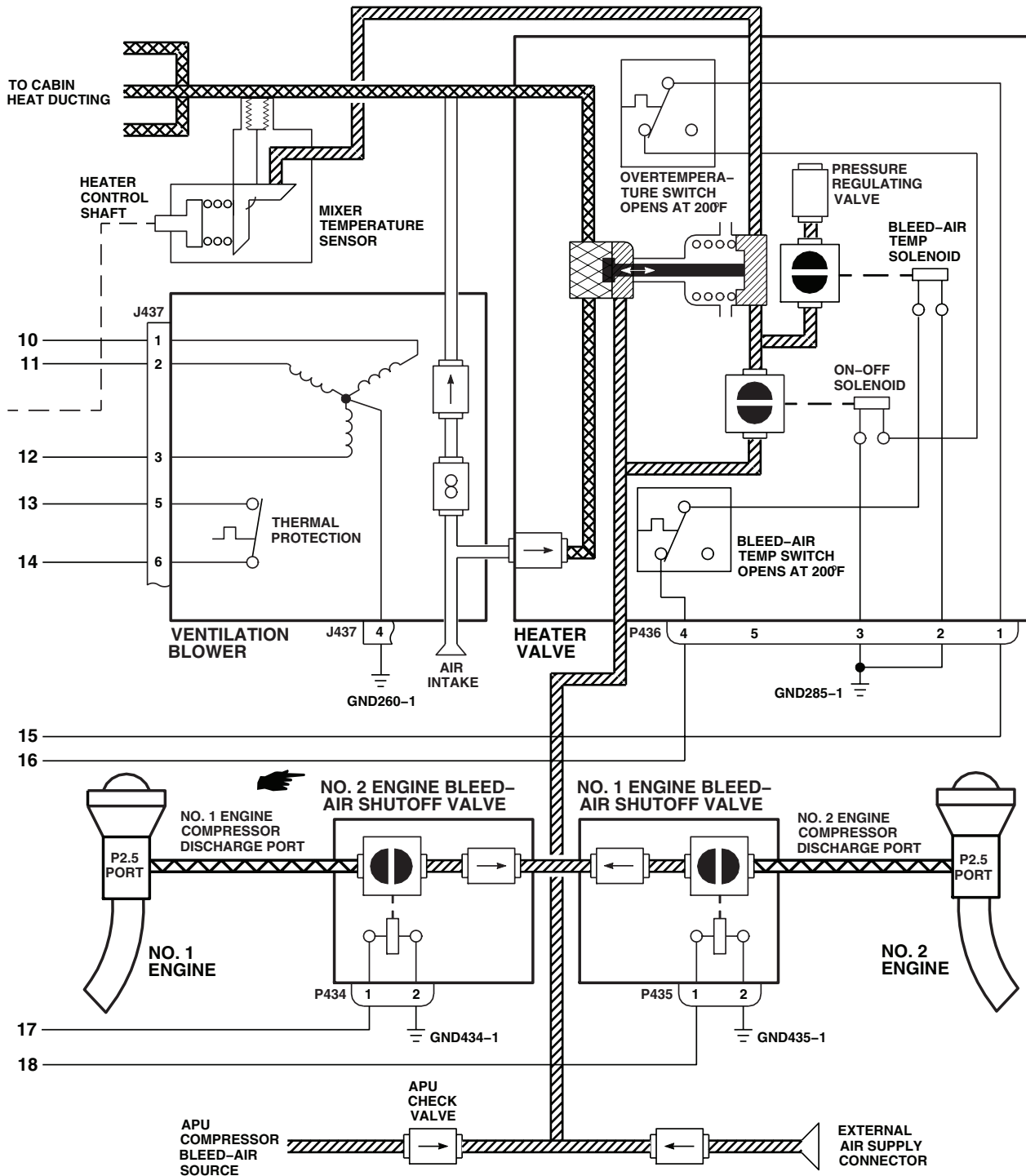


Figure 2. Option II Cabin Heater and Ventilation System Schematic Diagram. (Sheet 2 of 3)

SCHEMATICS - Continued



AB3318\_3A  
SA

Figure 2. Option II Cabin Heater and Ventilation System Schematic Diagram. (Sheet 3 of 3)

END OF WORK PACKAGE

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**UNIT LEVEL****MISSION EQUIPMENT****CRASHWORTHY EXTERNAL FUEL SYSTEM MAINTENANCE OPERATIONAL CHECK**

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**INITIAL SETUP:****Test Equipment**

Multimeter, AN/PSM-45A

WP 1549 00

WP 1574 18

WP 1574 20

**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01

WP 1574 21

WP 1612 01

Electrical Repairer Toolkit, SC 5180-99-B06

WP 1615 01

WP 1685 00

**Personnel Required**

Aircraft Electrician MOS 15F (1)

WP 1747 00

UH-60 Helicopter Repairer MOS 15T (1)

**Equipment Condition****References**

TM 1-1500-204-23-3

TM 1-1520-237-10

TM 1-1560-316-23

WP 0104 00

WP 1440 00

WP 1442 00

WP 1445 00

WP 1447 00

All Explosive Cartridges Removed from Ejector Racks,  
WP 1442 00or All Explosive Cartridges Removed from Ejector  
Racks, WP 1445 00

External Electrical Power Available, WP 1685 00

or APU Operational, TM 1-1520-237-10

Rear Fairings on All Vertical Stores (Pylons) Removed,  
WP 1447 00

## APPLICABLE CONFIGURATION

### NOTE

- The Crashworthy External Fuel System (CEFS) operational/troubleshooting procedure should be performed prior to reinstallation of fairings vapor barrier, and sound proofing to allow visual inspection of fluid connections. Performing this procedure will validate proper installation of the CEFS.
- The CEFS operational/troubleshooting procedure should be performed with the aircraft on level ground. When the aircraft is level, the external fuel tanks are at four degree nose-up attitude. This may cause an inaccurate reading during fuel transfer. When refueling, there may be a delay before the probe indicates rising fuel level; when defueling, there may be a delay before the probe indicates lower fuel level.
- The CEFS operational/troubleshooting procedure assumes that a single tank pair is installed on the outboard pylons. If it is desired to perform the procedure with tanks installed on the inboard pylons, replace outboard with inboard in the following procedure.
- When the wing gate valves and pump package refuel valve are energized, this can be verified by listening for their sound. If there is a problem with this process, it may be necessary to remove the vertical support pylon fairings and/or the soundproofing around the fuel cells to visually check the position of these valves.
- If a circuit breaker pops during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- See Figure 1 for component location diagram and Figures 2, 3, and 4 for schematic diagram as an aid in operational/troubleshooting.

This operational/troubleshooting procedure is subdivided:

- Initial Auxiliary Management Fuel Quantity Check
- Pressure Refueling/Leak Check
- Tanks Full Auxiliary Management Fuel Quantity Check
- Fuel Transfer Preparation
- Fuel Transfer Operation-Manual Mode-Lateral Imbalance Check
- Fuel Transfer Operation-Auto Mode
- Operational/Troubleshooting Procedure Completion
- Right Outboard Wing Gate Valve Not In Commanded Position Troubleshooting
- Right Inboard Wing Gate Valve Not In Commanded Position Troubleshooting
- Left Inboard Wing Gate Valve Not In Commanded Position Troubleshooting
- Left Outboard Wing Gate Valve Not In Commanded Position Troubleshooting
- Refuel Valve Not In Commanded Position Troubleshooting

**APPLICABLE CONFIGURATION** - Continued

- Transfer Pump Does Not Run When Commanded Troubleshooting
- Transfer Pump Cavitation Troubleshooting
- Slow Or No Fuel Transfer Troubleshooting
- Fuel Leakage From Cavity Drain Or Tank Shell Split Line Troubleshooting
- Erroneous Fuel Quantity Display Troubleshooting
- No Empty Light When Fuel Transfer Is Complete Troubleshooting
- Empty Light On When Tank Is Not Empty Troubleshooting

**INITIAL AUXILIARY FUEL MANAGEMENT PANEL FUEL QUANTITY CHECK****PROCEDURE****Step**

1. Set auxiliary fuel management panel (AFMP) switches as follows:

| <b>SWITCH</b> | <b>POSITION</b> |
|---------------|-----------------|
| XFER MODE     | OFF             |
| MAN XFER      | BOTH            |
| XFER FROM     | OUTBD           |
| PRESS         | OFF             |

2. Apply external power to aircraft.
3. Check position of all four wing gate valves.

**Indication/Condition**

All four wing gate valves shall be closed.

**Corrective Action**

If any wing gate valve is not in the closed, refer to, WING GATE VALVE NOT IN COMMANDED POSITION, in this work package.

**Step**

4. Check fuel quantity of outboard tanks and record readings.

**Indication/Condition**

Not Required

**Corrective Action**

Not Required

**PRESSURE REFUELING/LEAK CHECK****PROCEDURE****Step**

1. Set AFMP PRESS switch to OUTBD.
2. Check position of outboard wing gate valves.

**Indication/Condition**

Outboard wing gate valves shall be open.

**Corrective Action**

If either outboard wing gate valve is not open, refer to, WING GATE VALVE NOT IN COMMANDED POSITION, in this work package.

**Step**

3. Check position of pump package refuel valve.

**Indication/Condition**

Refuel valve shall be open.

**Corrective Action**

If refuel valve is not open, refer to, REFUEL VALVE NOT IN COMMANDED POSITION, in this work package.

**Step**

4. Pressure refuel aircraft (WP 1612 01) and visually inspect all fluid couplings for evidence of leakage.

**Indication/Condition**

Leakage shall not be indicated.

**Corrective Action**

If leakage is indicated, repair leaks in accordance with TM 1-1500-204-23-3 prior to continuing with this procedure.

**Step**

5. Check that outboard fuel quantities are rising by comparing readings to those recorded in INITIAL AFMP FUEL CHECK.

**Indication/Condition**

Fuel quantities shall increase.

**Corrective Action**

Not Required

**Step**

6. Continue refueling aircraft until main tanks and CEFS tanks are full (high level shut-off).

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**PRESSURE REFUELING/LEAK CHECK - Continued****Indication/Condition**

Not Required

**Corrective Action**

Not Required

**TANKS FULL AFMP FUEL QUANTITY CHECK****PROCEDURE****Step**

1. Check fuel quantity indication for outboard tanks and record the readings.

**Indication/Condition**

Not Required

**Corrective Action**

Not Required

**Step****NOTE**

Due to residual fuel in external tanks, and/or aircraft attitude, EMPTY lights may not go on or off when expected.

2. Check that tank EMPTY lights are off when tanks are full.

**Indication/Condition**

Left Tank EMPTY light off.

Right Tank EMPTY light off.

**Corrective Action**

If tank EMPTY lights are not off, refer to, EMPTY LIGHT IS ON WHEN TANK IS NOT EMPTY in this work package.

**FUEL TRANSFER PREPARATION****PROCEDURE****Step**

1. Set AFMP PRESS switch to OFF.

**Indication/Condition**

Outboard wing gate valves and pump package refuel valve shall be closed.

**Corrective Action**

Not Required

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**FUEL TRANSFER PREPARATION - Continued****Step**

2. Remove external power from aircraft.

**Indication/Condition**

Not Required

**Corrective Action**

Not Required

**Step**

3. Defuel main fuel tanks until both indicate approximately 900 pounds WP 1615 01.
4. Restore external power to aircraft.

**Indication/Condition**

Not Required

**Corrective Action**

Not Required

**FUEL TRANSFER OPERATION-MANUAL MODE-LATERAL IMBALANCE CHECK****PROCEDURE****Step**

1. Set AFMP XFER MODE switch to MAN.

**Indication/Condition**

Outboard wing gate valves and shall be open and pump package transfer pump shall be running.

**Corrective Action**

If outboard wing gate valves are not open, refer to, WING GATE VALVE NOT IN COMMANDED POSITION, in this work package. If transfer pump runs slow or does not run, refer to, TRANSFER PUMP DOES NOT RUN, in this work package.

**Step**

2. Check fuel quantity indication for outboard and main tanks.

**Indication/Condition**

Fuel quantity indication in outboard tanks shall decrease and main shall tanks increase.

**Corrective Action**

Not Required

**Step**

3. Set AFMP MAN XFER switch to LEFT.



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**FUEL TRANSFER OPERATION-MANUAL MODE-LATERAL IMBALANCE CHECK - Continued****Indication/Condition**

Right outboard wing gate valve shall close and left outboard wing gate valve shall remain open. Fuel quantity indication for left outboard tank shall decrease and main tanks shall increase.

**Corrective Action**

Not Required

**Step**

4. Allow manual fuel transfer from left outboard tank to continue until IMBAL light goes on. Record readings.

**Indication/Condition**

IMBAL light shall go on when the difference between left and right outboard fuel quantity indications is approximately 250 pounds.

**Corrective Action**

Not Required

**Step**

5. Check fuel quantity indication for outboard and main tanks.

**Indication/Condition**

Fuel quantity indication in outboard tanks shall decrease and main tanks shall increase.

**Corrective Action**

Not Required

**Step**

6. Set AFMP MAN XFER switch to RIGHT.

**Indication/Condition**

Left outboard wing gate valve shall close and right outboard wing gate valve shall open. Fuel quantity indication for right outboard tank shall decrease and main tanks shall increase.

**Corrective Action**

Not Required

**Step**

7. Allow manual fuel transfer from right outboard tank to continue until IMBAL light goes off. Record readings.

**Indication/Condition**

IMBAL light shall go off when the difference between left and right outboard fuel quantity indications is approximately 60 pounds.

**Corrective Action**

Not Required

**FUEL TRANSFER OPERATION-AUTO MODE****PROCEDURE****Step**

1. Set AFMP XFER MODE switch to AUTO.

**Indication/Condition**

Outboard wing gate valves and shall be open and pump package transfer pump shall be running.

**Corrective Action**

If outboard wing gate valves are not open, refer to, WING GATE VALVE NOT IN COMMANDED POSITION, in this work package. If transfer pump runs slow or does not run, refer to, TRANSFER PUMP DOES NOT RUN, in this work package.

**Step**

2. Check fuel quantity indication for outboard and main tanks.

**Indication/Condition**

Fuel quantity indication in outboard tanks shall decrease and main tanks shall increase.

**Corrective Action**

Not Required

**Step**

3. Allow fuel transfer to continue to fill main tanks until fuel transfer automatically terminates.

**Indication/Condition**

Fuel transfer shall terminate when the combined main tank fuel quantity rises above 2200 pounds. Pump package transfer pump shall go off and outboard wing gate valves shall close.

**Corrective Action**

Not Required

**CRASHWORTHY EXTERNAL FUEL SYSTEM MAINTENANCE OPERATIONAL CHECK COMPLETION****PROCEDURE****Step**

1. Set AFMP XFER MODE switch to OFF.
2. The aircraft can be refueled (WP 1612 01) or defueled (WP 1615 01).

**NOTE**

If defueling is required, ensure that PRESS REFUEL switch is set to OUTBD prior to removing external power.

3. Remove external power from aircraft.
4. Install all fairings, vapor barriers, and sound proofing as required to restore aircraft to flight status.

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**CRASHWORTHY EXTERNAL FUEL SYSTEM MAINTENANCE OPERATIONAL CHECK COMPLETION -**  
Continued**Indication/Condition**

Not Required

**Corrective Action**

Not Required

**RIGHT OUTBOARD WING GATE VALVE NOT IN COMMANDED POSITION****SYMPTOM**

Right outboard wing gate valve not in commanded position (does not open or close when commanded).

**MALFUNCTION**

Right outboard wing gate valve not in commanded position (does not open or close when commanded).

**CORRECTIVE ACTION**

- 1. With aircraft power on, set auxiliary fuel management control panel (AFMP) XFER MODE switch to OFF and PRESS switch to OUTBD.**
  - a. If right outboard wing gate valve is open, go to **2**.
  - b. If right outboard wing gate valve is not open, go to **6**.
- 2. Set AFMP PRESS switch to OFF.**
  - a. If right outboard wing gate valve is closed, go to **3**.
  - b. If right outboard wing gate valve is not closed, go to **10**.
- 3. Did malfunction occur during fuel transfer mode?**
  - a. If malfunction occurred during fuel transfer mode, go to **4**.
  - b. If fault did not occur during fuel transfer mode, go to **14**.
- 4. Set AFMP XFER MODE switch to MAN, MAN XFR switch to BOTH, and XFER FROM switch to OUTBD.**
  - a. If right outboard wing gate valve is open, go to **5**.
  - b. If right outboard wing gate valve is not open, replace AFMP (WP 1549 00). Go to **14**.
- 5. Set AFMP XFER FROM switch to INBD.**
  - a. If right outboard wing gate valve is not closed, replace AFMP (WP 1549 00). Go to **14**.
- 6. Disconnect P838 from right outboard wing gate valve. Check for 28 vdc between P838 pins A and E.**
  - a. If 28 vdc is not present, go to **7**.
  - b. If 28 vdc is present, replace wing gate valve (WP 1440 00). Go to **14**.
- 7. Turn off aircraft power. Disconnect P857 from AFMP. Check for continuity between P857 pin 14 and P838 pin A.**
  - a. If continuity is present, go to **8**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 8. Disconnect P860 from J346. Check for continuity between P860 pin 9 and P838 pin E.**

**RIGHT OUTBOARD WING GATE VALVE NOT IN COMMANDED POSITION - Continued**

- a. If continuity is present, go to **9**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 9. Check for continuity between J346 pin 9 (mates with P860) to aircraft ground.**
  - a. If continuity is present, replace AFMP (WP 1549 00). Go to **14**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 10. Disconnect P838. Check for 28 vdc between P838 pin D and pin E.**
  - a. If 28 vdc is not present, go to **11**.
  - b. If 28 vdc is present, replace wing gate valve (WP 1440 00). Go to **14**.
- 11. Turn off aircraft power. Disconnect P857 from AFMP. Check for continuity between P857 pin 13 and P838 pin D.**
  - a. If continuity is present, go to **12**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 12. Disconnect P860. Check for continuity between P860 pin 9 and P838 pin E.**
  - a. If continuity is present, go to **13**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 13. Check for continuity between J346 pin 9 to aircraft ground.**
  - a. If continuity is present, replace AFMP (WP 1549 00). Go to **14**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 14. Procedure completed.**

**RIGHT INBOARD WING GATE VALVE NOT IN COMMANDED POSITION****SYMPTOM**

Right inboard wing gate valve not in commanded position (does not open or close when commanded).

**MALFUNCTION**

Right inboard wing gate valve not in commanded position (does not open or close when commanded).

**CORRECTIVE ACTION**

- 1. With aircraft power on, set auxiliary fuel management control panel (AFMP) XFER MODE switch to OFF and PRESS switch to INBD.**
  - a. If right inboard wing gate valve is open, go to **2**.
  - b. If right outboard wing gate valve is not open, go to **6**.
- 2. Set AFMP PRESS switch to OFF.**
  - a. If right inboard wing gate valve is closed, go to **3**.
  - b. If right inboard wing gate valve is not closed, go to **10**.
- 3. Did malfunction occur during fuel transfer mode?**
  - a. If malfunction occurred during fuel transfer mode, go to **4**.
  - b. If fault did not occur during fuel transfer mode, go to **14**.

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**RIGHT INBOARD WING GATE VALVE NOT IN COMMANDED POSITION - Continued**

- 4. Set AFMP XFER MODE switch to MAN, MAN XFR switch to BOTH, and XFER FROM switch to INBD.**
  - a. If right inboard wing gate valve is open, go to **5**.
  - b. If right inboard wing gate valve is not open, replace AFMP (WP 1549 00). Go to **14**.
- 5. Set AFMP XFER FROM switch to OUTBD.**
  - a. If right inboard wing gate valve is not closed, replace AFMP (WP 1549 00). Go to **14**.
- 6. Disconnect P810 from right inboard wing gate valve. Check for 28 vdc between P810 pins A and E.**
  - a. If 28 vdc is not present, go to **7**.
  - b. If 28 vdc is present, replace wing gate valve (WP 1440 00). Go to **14**.
- 7. Turn off aircraft power. Disconnect P857 from AFMP. Check for continuity between P857 pin 6 and P810 pin A.**
  - a. If continuity is present, go to **8**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 8. Disconnect P860 from J346. Check for continuity between P860 pin 9 and P810 pin E.**
  - a. If continuity is present, go to **9**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 9. Check for continuity between J346 pin 9 (mates with P860) to aircraft ground.**
  - a. If continuity is present, replace AFMP (WP 1549 00). Go to **14**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 10. Disconnect P810. Check for 28 vdc between P810 pin D and pin E.**
  - a. If 28 vdc is not present, go to **11**.
  - b. If 28 vdc is present, replace wing gate valve (WP 1440 00). Go to **14**.
- 11. Turn off aircraft power. Disconnect P857 from AFMP. Check for continuity between P857 pin 4 and P810 pin D.**
  - a. If continuity is present, go to **12**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 12. Disconnect P860. Check for continuity between P860 pin 9 and P810 pin E.**
  - a. If continuity is present, go to **13**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 13. Check for continuity between J346 pin 9 to aircraft ground.**
  - a. If continuity is present, replace AFMP (WP 1549 00). Go to **14**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 14. Procedure completed.**

**LEFT INBOARD WING GATE VALVE NOT IN COMMANDED POSITION****SYMPTOM**

Left inboard wing gate valve not in commanded position (does not open or close when commanded).

**MALFUNCTION**

Left inboard wing gate valve not in commanded position (does not open or close when commanded).

**CORRECTIVE ACTION**

- 1. With aircraft power on, set auxiliary fuel management control panel (AFMP) XFER MODE switch to OFF and PRESS switch to INBD.**
  - a. If left inboard wing gate valve is open, go to **2**.
  - b. If left inboard wing gate valve is not open, go to **5**.
- 2. Set AFMP PRESS switch to OFF.**
  - a. If left inboard wing gate valve is closed, go to **3**.
  - b. If left inboard wing gate valve is not closed, go to **9**.
- 3. Did malfunction occur during fuel transfer mode?**
  - a. If malfunction occurred during fuel transfer mode, go to **4**.
  - b. If fault did not occur during fuel transfer mode, go to **14**.
- 4. Set AFMP XFER MODE switch to MAN, MAN XFR switch to BOTH, and XFER FROM switch to INBD.**
  - a. If left inboard wing gate valve is open, go to **5**.
  - b. If left inboard wing gate valve is not open, replace AFMP (WP 1549 00). Go to **14**.
- 5. Set AFMP XFER FROM switch to OUTBD.**
  - a. If left inboard wing gate valve is not closed, replace AFMP (WP 1549 00). Go to **14**.
- 6. Disconnect P809 from left inboard wing gate valve. Check for 28 vdc between P809 pins A and E.**
  - a. If 28 vdc is not present, go to **7**.
  - b. If 28 vdc is present, replace wing gate valve (WP 1440 00). Go to **14**.
- 7. Turn off aircraft power. Disconnect P857 from AFMP. Check for continuity between P857 pin 8 and P809 pin A.**
  - a. If continuity is present, go to **8**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 8. Disconnect P860 from J346. Check for continuity between P860 pin 34 and P809 pin E.**
  - a. If continuity is present, go to **9**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 9. Check for continuity between J346 pin 34 (mates with P860) to aircraft ground.**
  - a. If continuity is present, replace AFMP (WP 1549 00). Go to **14**.

**LEFT INBOARD WING GATE VALVE NOT IN COMMANDED POSITION - Continued**

- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 10. Disconnect P809. Check for 28 vdc between P809 pin D and pin E.**
  - a. If 28 vdc is not present, go to **11**.
  - b. If 28 vdc is present, replace wing gate valve (WP 1440 00). Go to **14**.
- 11. Turn off aircraft power. Disconnect P857 from AFMP. Check for continuity between P857 pin 7 and P809 pin D.**
  - a. If continuity is present, go to **12**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 12. Disconnect P860. Check for continuity between P860 pin 34 and P809 pin E.**
  - a. If continuity is present, go to **13**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 13. Check for continuity between J346 pin 34 to aircraft ground.**
  - a. If continuity is present, replace AFMP (WP 1549 00). Go to **14**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 14. Procedure completed.**

**LEFT OUTBOARD WING GATE VALVE NOT IN COMMANDED POSITION****SYMPTOM**

Left outboard wing gate valve not in commanded position (does not open or close when commanded).

**MALFUNCTION**

Left outboard wing gate valve not in commanded position (does not open or close when commanded).

**CORRECTIVE ACTION**

- 1. With aircraft power on, set auxiliary fuel management control panel (AFMP) XFER MODE switch to OFF and PRESS switch to OUTBD.**
  - a. If left outboard wing gate valve is open, go to **2**.
  - b. If left outboard wing gate valve is not open, go to **5**.
- 2. Set AFMP PRESS switch to OFF.**
  - a. If left outboard wing gate valve is closed, go to **3**.
  - b. If left outboard wing gate valve is not closed, go to **9**.
- 3. Did malfunction occur during fuel transfer mode?**
  - a. If malfunction occurred during fuel transfer mode, go to **4**.
  - b. If fault did not occur during fuel transfer mode, go to **14**.
- 4. Set AFMP XFER MODE switch to MAN, MAN XFR switch to BOTH, and XFER FROM switch to OUTBD.**
  - a. If left outboard wing gate valve is open, go to **5**.
  - b. If left outboard wing gate valve is not open, replace AFMP (WP 1549 00). Go to **14**.

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**LEFT OUTBOARD WING GATE VALVE NOT IN COMMANDED POSITION - Continued**

- 5. Set AFMP XFER FROM switch to INBD.**
  - a. If left outboard wing gate valve is not closed, replace AFMP (WP 1549 00). Go to **14**.
- 6. Disconnect P840 from left outboard wing gate valve. Check for 28 vdc between P840 pins A and E.**
  - a. If 28 vdc is not present, go to **7**.
  - b. If 28 vdc is present, replace wing gate valve (WP 1440 00). Go to **14**.
- 7. Turn off aircraft power. Disconnect P857 from AFMP. Check for continuity between P857 pin 3 and P840 pin A.**
  - a. If continuity is present, go to **8**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 8. Disconnect P860 from J346. Check for continuity between P860 pin 34 and P838 pin E.**
  - a. If continuity is present, go to **9**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 9. Check for continuity between J346 pin 34 (mates with P860) to aircraft ground.**
  - a. If continuity is present, replace AFMP (WP 1549 00). Go to **14**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 10. Disconnect P840. Check for 28 vdc between P840 pin D and pin E.**
  - a. If 28 vdc is not present, go to **11**.
  - b. If 28 vdc is present, replace wing gate valve (WP 1440 00). Go to **14**.
- 11. Turn off aircraft power. Disconnect P857 from AFMP. Check for continuity between P857 pin 5 and P838 pin D.**
  - a. If continuity is present, go to **12**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 12. Disconnect P860. Check for continuity between P860 pin 34 and P840 pin E.**
  - a. If continuity is present, go to **13**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 13. Check for continuity between J346 pin 34 to aircraft ground.**
  - a. If continuity is present, replace AFMP (WP 1549 00). Go to **14**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 14. Procedure completed.**



**REFUEL VALVE NOT IN COMMANDED POSITION****SYMPTOM**

Refuel valve not in commanded position (does not open or close when commanded).

**MALFUNCTION**

Refuel valve not in commanded position (does not open or close when commanded).

**CORRECTIVE ACTION**

- 1. With aircraft power on, set auxiliary fuel management control panel (AFMP) XFER MODE switch to OFF and PRESS switch to OUTBD.**
  - a. If refuel valve is open, go to **2**.
  - b. If refuel valve is not open, go to **5**.
- 2. Set AFMP XFER PRESS switch to OFF.**
  - a. If refuel valve is closed, go to **3**.
  - b. If refuel valve is not closed, go to **9**.
- 3. Set AFMP PRESS switch to INBD.**
  - a. If refuel valve is open, go to **4**.
  - b. If refuel valve is not open, replace AFMP (WP 1549 00). Go to **13**.
- 4. Set AFMP PRESS switch to ALL.**
  - a. If refuel valve is not open, replace AFMP (WP 1549 00). Go to **13**.
- 5. Disconnect connector from refuel valve. Check for 28 vdc between connector pins C and E.**
  - a. If 28 vdc is not present, go to **6**.
  - b. If 28 vdc is present, replace pump package refuel valve (WP 1574 18). Go to **13**.
- 6. Disconnect P2 from pump package relay box J1. Check for 28 vdc between P2 pin C and pin E.**
  - a. If 28 vdc is not present, go to **7**.
  - b. If 28 vdc is present, replace pump package relay box (WP 1574 21). Go to **13**.
- 7. Remove aircraft power and disconnect P857 from AFMP. Check for continuity between P857 pin 9 and P2 pin C.**
  - a. If continuity is present, go to **8**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **13**.
- 8. Disconnect P860 and check for continuity between P860 pin 11 and P2 pin E.**
  - a. If continuity is present, replace AFMP (WP 1549 00). Go to **13**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **13**.
- 9. Disconnect P2 from refuel valve and check for 28 vdc between P2 pin D and pin E.**
  - a. If 28 vdc is not present, go to **10**.
  - b. If 28 vdc is present, replace pump package refuel valve (WP 1574 18). Go to **13**.

**REFUEL VALVE NOT IN COMMANDED POSITION - Continued**

- 10. Disconnect P857 from AFMP. Check for continuity between P857 pin 13 and P838 pin D.**
  - a. If 28 vdc is not present, go to **11**.
  - b. If 28 vdc is present, replace pump package relay box (WP 1574 21). Go to **13**.
- 11. Disconnect P860. Check for continuity between P860 pin 9 and P838 pin E.**
  - a. If continuity is present, go to **12**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **13**.
- 12. Check for continuity between J346 pin 9 and aircraft ground.**
  - a. If continuity is present, replace AFMP (WP 1549 00). Go to **13**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **13**.
- 13. Procedure completed.**

**TRANSFER PUMP DOES NOT RUN WHEN COMMANDED****SYMPTOM**

Transfer pump does not run when commanded.

**MALFUNCTION**

Transfer pump does not run when commanded.

**CORRECTIVE ACTION**

- 1. Apply power to aircraft. Set AFMP XFER MODE switch to MAN.**
  - a. If transfer pump is running, go to **2**.
  - b. If transfer pump is not running, go to **5**.
- 2. Set XFER MODE switch to OFF.**
  - a. If transfer pump shut off, go to **3**.
  - b. If transfer pump is not shut off, go to **11**.
- 3. Did this fault occur during auto mode fuel transfer?**
  - a. If fault occur during auto mode fuel transfer, go to **4**.
  - b. If fault did not occur during auto mode fuel transfer, go to **13**.
- 4. Set AFMP XFER MODE switch to AUTO and XFER FROM switch to select a tank pair for transfer operation (neither tank in selected pair showing an EMPTY light). Check combined main fuel quantity.**
  - a. If combined main fuel quantity is less than 2200 pounds, go to **5**.
  - b. If combined main fuel quantity is more than 2200 pounds, go to **13**.
- 5. Disconnect P1 at the transfer pump. Check for the 115 vac between the following pins:**
  - P1-1 and P1-4 (Phase 1)**
  - P1-2 and P1-4 (Phase 2)**
  - P1-3 and P1-4 (Phase 3)**

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**TRANSFER PUMP DOES NOT RUN WHEN COMMANDED** - Continued

- a. If 115 vac is present, replace transfer pump (WP 1574 20). Go to **13**.
  - b. If 115 vac is not present, go to **7**.
- 6. Disconnect P1 from pump package relay box J1. Check for 115 vac between the following pins:**  
**P1-A and P1-D (Phase 1)**  
**P1-B and P1-D (Phase 2)**  
**P1-C and P1-D (Phase 3)**
  - a. If voltage is present, go to **9**.
  - b. If voltage is not, go to **8**.
- 7. Gain access to mission readiness circuit breaker panel. Check for 115 vac at TB11:**  
**TB11-T2 and aircraft ground**  
**TB11-T3 and aircraft ground**  
**TB11-T4 and aircraft ground**
  - a. If 115 vac is present, go to step **8**.
  - b. If 115 vac is not present, troubleshoot lack of ac power at mission readiness circuit breaker panel TB11.
- 8. Remove aircraft power, check for continuity between the following points:**  
**P1-A and TB11-T2**  
**P1-B and TB11-T3**  
**P1-C and TB11-T4**  
**P1-D and aircraft ground**
  - a. If continuity is present, go to **13**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **13**.
- 9. Disconnect P2, at pump package relay box J2. Check for 28 vdc between pin A and pin B.**
  - a. If 28 vdc is present, remove and replace pump package relay box (WP 1574 21). Go to **13**.
  - b. If 28 vdc is not present, go to **10**.
- 10. Remove aircraft power. Disconnect P857 at AFMP. Check for continuity between P857 pin 12 and P2 pin B.**
  - a. If continuity is present, go to **11**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **12**.
- 11. Remove aircraft power. Check for continuity between P860 pin 4 and P2 pin A.**
  - a. If continuity is present, replace AFMP (WP 1549 00). Go to **13**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **13**.
- 12. Remove connector P2 from pump package relay box connector J2.**

**TRANSFER PUMP DOES NOT RUN WHEN COMMANDED** - Continued

- a. If the transfer pump continues to run, remove and replace pump package relay box (WP 1574 21). Go to **13**.
- b. If not, replace and replace AFMP (WP 1549 00). Go to **13**.

**13. Procedure completed.****TRANSFER PUMP CAVITATION****SYMPTOM**

Transfer pump cavitation.

**MALFUNCTION**

Transfer pump cavitation.

**CORRECTIVE ACTION**

- 1. Is pump cavitation occurring while transferring fuel from a pair of tanks?**
  - a. If pump cavitation occurs while transferring fuel from a pair of tanks, go to **2**.
  - b. If pump cavitation does not occur while transferring fuel from a pair of tanks, go to **4**.
- 2. With auxiliary fuel management panel (AFMP) XFER MODE switch set to MAN, use MAN XFER switch to alternately select individual tanks and note when cavitation occurs.**
  - a. If cavitation occurs regardless of MAN XFER switch position, go to **4**.
  - b. If cavitation does not occur in at least one MAN XFER switch position, go to **5**.
- 3. Pressure refuel aircraft (WP 1612 01) until all tanks are full. Leave truck fueling pressure applied and check transfer pump package and all aircraft fuel plumbing for leaks.**
  - a. If leaks are not present, go to **9**.
  - b. If leaks are present, repair leaks as required (TM 1-1560-316-23). Go to **10**.
- 4. With AFMP XFER MODE switch set to MAN, and MAN XFER switch to the position opposite of the position selected in step 2.**
  - a. If cavitation occurs, go to **3**.
  - b. If cavitation does not occur, go to **5**.
- 5. Check fuel level of tank that causes transfer pump cavitation.**
  - a. If fuel level is adequate for fuel transfer, go to **6**.
  - b. If fuel level is not adequate for fuel transfer, refuel tank. Go to **10**.
- 6. Check for fuel leakage from pylon plumbing between wing gate valve and pull away coupling for tank causing transfer pump cavitation.**
  - a. If fuel leakage is not detected, go to **7**.
  - b. If fuel leakage is detected, repair leaks as required (TM 1-1560-316-23). Go to **10**.
- 7. Has main fuel tank hose been replaced as part of this maintenance activity?**
  - a. If main fuel tank hose been replaced as part of this maintenance activity, go to **8**.

**TRANSFER PUMP CAVITATION - Continued**

- b. If main fuel tank hose has not been replaced as part of this maintenance activity, suction defuel tank (WP 1615 01) and remove tank from aircraft. Replace main in tank fuel hose per TM 1-1560-316-23. Go to **10**.

**8. Has tank internal fuel line packing (union to pull-away socket) been replaced as part this maintenance activity?**

- a. If tank internal fuel line packing been replaced as part this maintenance activity, suction defuel tank (WP 1615 01) and remove tank from aircraft. Replace full pull-away socket per TM 1-1560-316-23. Go to **10**.
- b. If tank internal fuel line packing has not been replaced as part this maintenance activity, suction defuel tank (WP 1615 01) and remove tank from aircraft. Replace packing between pull-away socket and union per TM 1-1560-316-23. Go to **10**.

**9. Have all fuel line fittings been retorqued as part of this maintenance activity?**

- a. If all fuel line fittings have been retorqued as part of this maintenance activity, replace pump package (WP 1574 09). Go to **10**.
- b. If all fuel line fittings have not been retorqued as part of this maintenance activity, retorqued fuel line fittings as required to ensure that all are torqued. Go to **10**.

**10. Procedure completed.****SLOW OR NO FUEL TRANSFER****SYMPTOM**

Slow or no fuel transfer.

**MALFUNCTION**

Slow or no fuel transfer.

**CORRECTIVE ACTION****1. Is auxiliary fuel management panel (AFMP) XFER MODE switch in AUTO?**

- a. If XFER MODE switch is in AUTO, go to **2**.
- b. If XFER MODE switch is not in AUTO, go to **10**.

**2. Is transfer pump running?**

- a. If transfer pump is running, set XFER MODE switch to MAN and confirm that XFER FROM and PRESS switches are set to transfer fuel from tank pair with sufficient fuel (empty lights not on). Go to **17**.
- b. If transfer pump is not running, go to **3**.

**3. Check that total main tank fuel quantity is below 2200 pounds and neither EMPTY light is on for the selected tank pair?**

- a. If total main tank fuel quantity is below 2200 pounds and neither EMPTY light is on for the selected tank pair, go to **4**.
- b. If total main tank fuel quantity is not below 2200 pounds and either EMPTY light is on for the selected tank pair, go to **35**.

**4. Reset CEFS PUMP circuit breaker located on mission readiness circuit breaker panel.**

**SLOW OR NO FUEL TRANSFER** - Continued

- a. If transfer pump runs, go to **35**.
- b. If transfer pump does not run, go to **5**.
- 5. Disconnect connector P1 from fuel transfer pump. Check for 115 vac power between the following pins:**  
**P1-1 and P1-4**  
**P1-2 and P1-4**  
**P1-3 and P1-4**
  - a. If 115 vac is present, replace transfer pump (WP 1574 20). Go to **35**.
  - b. If 115 vac is not present, go to **6**.
- 6. Disconnect connector P1 from pump package relay box. Check for 115 vac power between the following pins:**  
**P1-A and P1-D**  
**P1-B and P1-D**  
**P1-C and P1-D**
  - a. If 115 vac is present, go to **7**.
  - b. If 115 vac is not present, go to **9**.
- 7. Disconnect P2 connector from pump package relay box and check for 28 vdc power between P1-A and P1-B.**
  - a. If 28 vdc is present, replace pump package relay box (WP 1574 21). Go to **35**.
  - b. If 28 vdc is not present, remove aircraft power and go to **8**.
- 8. With aircraft power removed disconnect connector from P857 from AFMP. Check for continuity between P2-B and P857-12.**
  - a. If continuity is present, replace AFMP (WP 1549 00). Go to **35**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **35**.
- 9. Gain access to mission readiness circuit breaker panel. Check for 115 vac at TB11:**  
**TB11-T2 and aircraft ground**  
**TB11-T3 and aircraft ground**  
**TB11-T4 and aircraft ground**
  - a. If 115 vac is present, repair/replace wiring between TB11 and PI (WP 1747 00). Go to **35**.
  - b. If 115 vac is not present, troubleshoot lack of ac power at mission readiness circuit breaker panel TB11. Go to **35**.
- 10. Is transfer pump running?**
  - a. If transfer pump is running, go to **17**.
  - b. If transfer pump is not running, go to **11**.
- 11. Reset CEFS PUMP circuit breaker located in the mission equipment circuit breaker panel.**

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**SLOW OR NO FUEL TRANSFER - Continued**

- a. If pump is running after circuit breaker is reset, go to **35**.
  - b. If pump is not running after circuit breaker is reset, go to **12**.
- 12. Disconnect connector P1 from fuel transfer pump. Check for 115 vac power between the following pins:**
  - P1-1 and P1-4**
  - P1-2 and P1-4**
  - P1-3 and P1-4**
  - a. If 115 vac is present, replace transfer pump (WP 1574 20). Go to **35**.
  - b. If 115 vac is not present, go to **13**.
- 13. Disconnect connector P1 from pump package relay box. Check for 115 vac power between the following pins:**
  - P1-A and P1-D**
  - P1-B and P1-D**
  - P1-C and P1-D**
  - a. If 115 vac is present, go to **14**.
  - b. If 115 vac is not present, go to **16**.
- 14. Disconnect P2 connector from pump package relay box and check for 28 vdc power between P1-A and P1-B.**
  - a. If 28 vdc is present, replace pump package relay box (WP 1574 21). Go to **35**.
  - b. If 28 vdc is not present, remove aircraft power and go to **15**.
- 15. With aircraft power removed disconnect connector from P857 from AFMP. Check for continuity between P2-B and P857-12.**
  - a. If continuity is present, replace AFMP (WP 1549 00). Go to **35**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **35**.
- 16. Gain access to mission readiness circuit breaker panel. Check for 115 vac at TB11:**
  - TB11-T2 and aircraft ground**
  - TB11-T3 and aircraft ground**
  - TB11-T4 and aircraft ground**
  - a. If 115 vac is present, repair/replace wiring between TB11 and P1 (WP 1747 00). Go to **35**.
  - b. If 115 vac is not present, troubleshoot lack of ac power at mission readiness circuit breaker panel TB11. Go to **35**.
- 17. Alternatively select individual tanks and monitor transfer rates. Does the slow or no transfer condition exist regardless of tank selected?**
  - a. If slow or no transfer condition is still present regardless of tank selected, go to **18**.
  - b. If slow or no transfer condition is not present for every tank , go to **26**.
- 18. Check position of refuel valve.**

**SLOW OR NO FUEL TRANSFER - Continued**

- a. If refuel valve is fully closed, go to **24**.
- b. If refuel valve is not fully closed, go to **19**.
- 19. Disconnect connector P2 from pump package refuel valve. Check for 28 vdc between P2-D and P2-E.**
  - a. If 28 vdc is present, replace pump package refuel valve (WP 1574 18). Go to **35**.
  - b. If 28 vdc is not present, remove aircraft power and go to **20**.
- 20. Remove aircraft power. Disconnect connector P2 from pump package relay box and check for continuity between the following relay box connector J2 and relay box harness connector P2 pins.**
  - J2-C and P2-A**
  - J2-D and P2-B**
  - J2-E and P2-E**
  - a. If continuity is present, go to **21**.
  - b. If continuity is not present, replace pump package relay box (WP 1574 21). Go to **35**.
- 21. Disconnect connector P857 from AFMP. Check for continuity between P857-10 and P2-D.**
  - a. If continuity is present, go to **22**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **35**.
- 22. Disconnect P860 from J346 on cabin ceiling. Check for continuity between P860-11 and P2-E.**
  - a. If continuity is present, go to **23**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **35**.
- 23. Check for continuity between P860-11 and aircraft ground.**
  - a. If continuity is present, replace AFMP (WP 1549 00). Go to **35**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **35**.
- 24. Pressure refuel aircraft (WP 1612 01) until all tanks are full. Leave truck fueling pressure applied and check transfer pump package and all aircraft fuel plumbing for leaks.**
  - a. If leaks are not present, go to **25**.
  - b. If leaks are present, repair leaks as required (TM 1-1560-316-23). Go to **35**.
- 25. Have all fuel line fittings been retorqued as part of this maintenance activity?**
  - a. If all fuel line fittings have been retorqued as part of this maintenance activity, replace pump package (WP 1574 09). Go to **35**.
  - b. If all fuel line fittings have not been retorqued as part of this maintenance activity, retorquer fuel line fittings as required to ensure that all are torqued. Go to **35**.
- 26. Check position of wing gate valve for tank selected for fuel transfer. Wing gate valve shall be open.**
  - a. If wing gate valve is open, go to **30**.



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**SLOW OR NO FUEL TRANSFER - Continued**

- b. If wing gate valve is not open, go to **27**.
- 27. Disconnect connector from wing gate valve and check for 28 vdc between connector pins A and E.**
  - a. If 28 vdc is not present, go to **28**.
  - b. If 28 vdc is present, replace wing gate valve (WP 1440 00). Go to **35**.
- 28. Disconnect P857 from AFMP and P860 from J346. Check for continuity between the following points for the effected valve:**
  - Left Outboard
    - P840-A and P857-3**
    - P840-E and P860-34**
  - Left Inboard
    - P809-A and P857-8**
    - P809-E and P860-34**
  - Right Inboard
    - P810-A and P857-6**
    - P810-E and P860-9**
  - Right Outboard
    - P838-A and P857-14**
    - P838-E and P860-9**
  - a. If continuity is present, go to **29**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **35**.
- 29. Check for continuity between P860 pin 34 and aircraft ground for left wing gate valves or P860 pin 9 and aircraft ground for right wing gate valves.**
  - a. If continuity is present, replace AFMP (WP 1549 00). Go to **35**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **35**.
- 30. Check break-away couplings and break-away valves for evidence of breakage on pylon and wing affected by slow or no fuel transfer.**
  - a. If break-away couplings and break-away valves are not broken, go to **31**.
  - b. If any break-away couplings or break-away valves are broken, replace broken couplings and valves (TM 1-1560-316-23). Go to **35**.
- 31. Remove aircraft power and remove fuel tank affected by slow or no fuel transfer. Check to see if tank defuel valve has been replaced as part of this maintenance activity.**
  - a. If tank defuel valve has been replaced as part of this maintenance activity, replace tank high level pilot valve (TM 1-1560-316-23). Go to **32**.

**SLOW OR NO FUEL TRANSFER - Continued**

- b. If if tank defuel valve has not been replaced as part of this maintenance activity, replace tank defuel valve (TM 1-1560-316-23). Go to **35**.
- 32. Check to see if tank high level pilot valve has been replaced as part of this maintenance activity?**
  - a. If tank high level pilot valve has been replaced as part of this maintenance activity, go to **33**.
  - b. If if tank high level pilot valve has not been replaced as part of this maintenance activity, replace tank high level pilot valve (TM 1-1560-316-23). Go to **35**.
- 33. Check to see if effected external tank main in-tank fuel hose been replaced as part of this maintainace action.**
  - a. If effected external tank main in-tank fuel hose has been replaced as part of this maintainace activity, go to **34**.
  - b. If effected external tank main in-tank fuel hose has not been replaced as part of this maintainace activity, replace in-tank fuel hose (TM 1-1560-316-23). Go to **35**.
- 34. Check to see if effected internal tank fuel line packing (union to pull-away socket) has been replaced as part this maintenance activity.**
  - a. If effected internal tank fuel line packing (union to pull-away socket) has been replaced as part this maintenance activity, replace full pull-away socket per TM 1-1560-316-23. Go to **35**.
  - b. If effected internal tank fuel line packing (union to pull-away socket) has not been replaced as part this maintenance activity, replace packing between pull-away socket and union. (TM 1-1560-316-23). Go to **35**.
- 35. Procedure completed.**

**FUEL LEAKAGE FROM CAVITY DRAIN OR TANK SHELL SPLIT LINE****SYMPTOM**

Fuel leakage from cavity drain or tank shell split line.

**MALFUNCTION**

Fuel leakage from cavity drain or tank shell split line.

**CORRECTIVE ACTION**

- 1. Inspect tank surface for evidence of fuel leakage at tank split line or cavity drain.**
  - a. If leakage is present, go to **2**.
  - b. If If leakage is not present, go to **10**.
- 2. Remove forward cover plate per TM 1-1560-316-23. Inspect packing for damage and evidence of fuel leakage.**
  - a. If leakage is present, go to **3**.
  - b. If If leakage is not present, go to **4**.
- 3. Remove forward cover plate per TM 1-1560-316-23. Clean any evidence of fuel from the outside surface of the bladder and inside surface of the forward tank shell. Inspect bladder for evidence of bladder damage or activation due to contact with fuel.**
  - a. If bladder is damaged or activated, replace bladder per TM 1-1560-316-23. Go to **10**.

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**FUEL LEAKAGE FROM CAVITY DRAIN OR TANK SHELL SPLIT LINE - Continued**

- b. If bladder is not damaged or activated, replace damaged packing and reassemble tank per TM 1-1560-316-23. Go to **10**.
- 4. Remove gravity fill spacer per TM 1-1560-316-23. Inspect packing for damage and evidence of fuel leakage.**
  - a. If evidence of fuel leakage is present, go to **5**.
  - b. If evidence of fuel leakage is not present, go to **5**.
- 5. Remove forward tank shell per TM 1-1560-316-23 Clean any evidence of fuel from the outside surface of the bladder and inside surface of the forward tank shell. Inspect bladder for evidence of bladder damage or activation due to contact with fuel.**
  - a. If bladder is damaged or activated, replace bladder per TM 1-1560-316-23 Go to **10**.
  - b. If bladder is not damaged or activated, replace damaged packing and reassemble tank per TM 1-1560-316-23. Go to **10**.
- 6. Remove aft cover plate per TM 1-1560-316-23 Inspect packing for the damage and evidence of fuel leakage.**
  - a. If leakage is present, go to **7**.
  - b. If leakage is not present, go to **8**.
- 7. Remove aft tank shell per TM 1-1560-316-23. Clean any evidence of fuel from the outside surface of the bladder and inside surface of the forward tank shell. Inspect bladder for evidence of bladder damage or activation due to contact with fuel.**
  - a. If bladder is damaged or activated, replace bladder per TM 1-1560-316-23. Go to **10**.
  - b. If bladder is not damaged or activated, replace damaged packing and reassemble tank per TM 1-1560-316-23. Go to **10**.
- 8. Remove aft tank shell per TM 1-1560-316-23. Inspect bladder for evidence of leakage from aft blank cover plate.**
  - a. If leakage present, go to **9**.
  - b. If leakage is not present, reassemble tank per (WP 1747 00). Go to **10**.
- 9. Clean any evidence of fuel from the outside surface of the bladder and inside surface of aft tank shell. Inspect bladder for evidence of bladder damage or activation due to contact with fuel.**
  - a. If bladder is damaged or activated, replace bladder per TM 1-1560-316-23. Go to **10**.
  - b. If bladder is not damaged or activated, remove aft cover plate and replace packing. Reassemble tank per TM 1-1560-316-23. Go to **10**.
- 10. Procedure completed.**

**ERRONEOUS FUEL QUANTITY DISPLAY****SYMPTOM**

Erroneous fuel quantity display.

**MALFUNCTION**

Erroneous fuel quantity display.

**CORRECTIVE ACTION**

- 1. Power up auxiliary fuel management panel (AFMP) and run IBIT by pressing TEST/RESET button. Check error code reading for E04.**
  - a. If error code E04 is not displayed, go to **2**.
  - b. If error code E04 is displayed, replace AFMP (WP 1549 00). Go to **7**.
- 2. Check error code reading for error code E05 and E06.**
  - a. If error code E05 or E06 is not displayed, remove aircraft power and go to **4**.
  - b. If error code E05 or E06 is displayed, remove aircraft power and go to **3**.
- 3. Check continuity as applicable for the tank(s) with erroneous fuel quantity between tank W4P2 and AFMP P856.**

Left Outboard

**W4P2-A and P856-4**

**W4P2-B and Ground**

**W4P2-C and P856-40**

**W4P2-D and P856-50**

**W4P2-E and P856-42**

**W4P2-F and P856-41**

Left Inboard

**W4P2-A and P856-3**

**W4P2-B and Ground**

**W4P2-C and P856-37**

**W4P2-D and P856-49**

**W4P2-E and P856-39**

**W4P2-F and P856-38**

Right Outboard

**W4P2-A and P856-2**

**W4P2-B and Ground**

**W4P2-C and P856-31**

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**ERRONEOUS FUEL QUANTITY DISPLAY - Continued**

**W4P2-D and P856-47**

**W4P2-E and P856-33**

**W4P2-F and P856-32**

Right Inboard

**W4P2-A and P856-2**

**W4P2-B and Ground**

**W4P2-C and P856-34**

**W4P2-D and P856-48**

**W4P2-E and P856-36**

**W4P2-F and P856-35**

- a. If continuity is present, replace external tank fuel quantity probe per TM 1-1560-316-23. Go to **7**.
- b. continuity is not present, repair/replace wiring (WP 1747 00). Go to **7**.

**4. Check operation of copilot's gyro per TM 11-1520-237-23.**

- a. If copilot's gyro operates properly, go to **5**.
- b. If copilot's gyro does not operate properly, troubleshoot CPG pitch gyro per TM 11-1520-237-23. Go to **7**.

**5. Disconnect P867 from CPG pitch gyro and P856 from AFMP. Check for continuity as follows:**

**P856-25 and P857-R**

**P856-26 and P857-H**

**P856-27 and P857-B**

**P856-28 and P857-T**

**P856-29 and P857-P**

**P856-30 and P857-J**

- a. If continuity is present, go to **6**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **7**.

**6. Check to see if AFMP has been replaced as part of this maintenance activity.**

- a. If AFMP has been replaced as part of this maintenance activity, replace external tank fuel probe (TM 1-1560-316-23). Go to **7**.
- b. If AFMP has been replaced as part of this maintenance activity, replace AFMP (WP 1549 00). Go to **7**.

**7. Procedure completed.**

**NO EMPTY LIGHT WHEN FUEL TRANSFER IS COMPLETE****SYMPTOM**

No empty light when fuel transfer is complete.

**MALFUNCTION**

No empty light when fuel transfer is complete.

**CORRECTIVE ACTION**

- 1. Completely defuel affected tank using fuel sample nipple and aircraft hand pump.**
- 2. Power up auxiliary fuel management panel (AFMP) and check for an empty light for affected tank.**
  - a. If empty light is not on, go to **3**.
  - b. If empty light is on, an incomplete fuel transfer occurred during previous use. Go to **5**.
- 3. Disconnect W4P2 from affected tank and measure resistance between probe connector pins C and S.**
  - a. If resistance is between 300 and 500 ohms, go to **4**.
  - b. If resistance is not between 300 and 500 ohms, replace external tank fuel probe (TM 1-1560-316-23). Go to **5**.
- 4. Disconnect P856 from AFMP and check for continuity between the following pins for affected tank:**

Left Outboard

**W4P2-C and P856-40**

**W4P2-F and P856-41**

**W4P2-E and P856-42**

Left Inboard

**W4P2-C and P856-37**

**W4P2-F and P856-38**

**W4P2-E and P856-39**

Right Outboard

**W4P2-C and P856-31**

**W4P2-F and P856-32**

**W4P2-E and P856-33**

Right Inboard

**W4P2-C and P856-34**

**W4P2-F and P856-35**

**W4P2-E and P856-36**

**NO EMPTY LIGHT WHEN FUEL TRANSFER IS COMPLETE** - Continued

- a. If continuity is present, replace AFMP (WP 1549 00). Go to **5**.
- b. continuity is not present, repair/replace wiring (WP 1747 00). Go to **5**.

**5. Procedure completed.****EMPTY LIGHT ON WHEN TANK IS NOT EMPTY****SYMPTOM**

Empty light on when tank is not empty.

**MALFUNCTION**

Empty light on when tank is not empty.

**CORRECTIVE ACTION**

- 1. With aircraft power off, disconnect connector W4P2 from affected tank and measure resistance between probe connector pins C and D.**
  - a. If resistance is not between 300 and 500 ohms, go to **2**.
  - b. If resistance is between 300 and 500 ohms, replace external tank fuel probe (TM 1-1560-316-23). Go to **3**.
- 2. Disconnect P856 from AFMP and check for continuity between the following pins for affected tank:**

Left Outboard

**W4P2-C and P856-40**

**W4P2-F and P856-41**

**W4P2-E and P856-42**

Left Inboard

**W4P2-C and P856-37**

**W4P2-F and P856-38**

**W4P2-E and P856-39**

Right Outboard

**W4P2-C and P856-31**

**W4P2-F and P856-32**

**W4P2-E and P856-33**

Right Inboard

**W4P2-C and P856-34**

**W4P2-F and P856-35**

**W4P2-E and P856-36**

- a. If continuity is present, replace AFMP (WP 1549 00). Go to **3**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **3**.

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**EMPTY LIGHT ON WHEN TANK IS NOT EMPTY - Continued**

**3. Procedure completed.**

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Diagram illustrating the location of various fuel system components and valves on the aircraft fuselage:

- RIGHT OUTBOARD FUEL SHUTOFF GATE VALVE P838**
- RIGHT INBOARD FUEL SHUTOFF GATE VALVE P810**
- W3P1**
- W4P1**
- W4P1**
- W3P1**
- P858 / J804**
- P862 / W2J2**
- COPILOT'S GYRO P867 / J1**
- AUXILIARY FUEL MANAGEMENT CONTROL PANEL P856 / J1 P857 / J2**
- LEFT DRAG BEAM SWITCH P163 / J163**
- LEFT INBOARD FUEL SHUTOFF GATE VALVE P809**
- W3P1**
- W4P1**
- P2 / J835**
- W4P1**
- W3P1**
- LEFT OUTBOARD FUEL SHUTOFF GATE VALVE P840**
- P860 / J346**
- P859 / J805**
- P861 / W2J2**

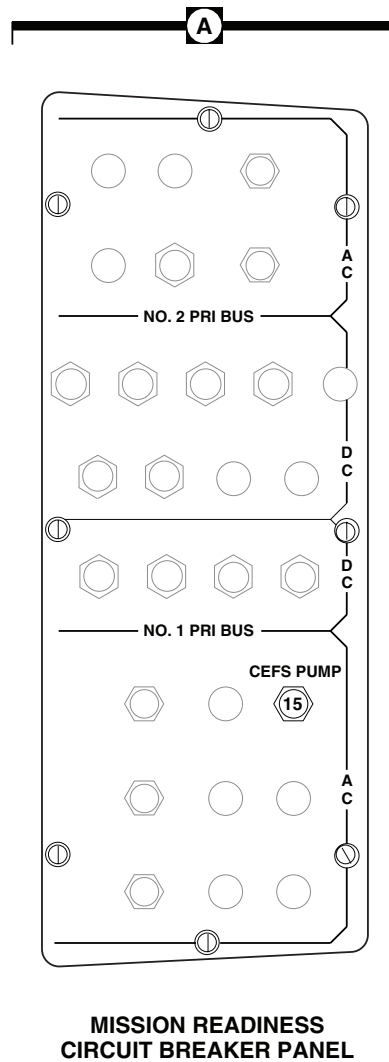
| <b>TERMINAL BOARD /<br/>DISCONNECT PLUG /<br/>RECEPTACLE</b> | <b>LOCATION /<br/>CONNECTION POINT</b>                      |
|--------------------------------------------------------------|-------------------------------------------------------------|
| <b>P809</b>                                                  | <b>LEFT INBOARD FUEL<br/>SHUTOFF GATE VALVE</b>             |
| <b>P810</b>                                                  | <b>RIGHT INBOARD FUEL<br/>SHUTOFF GATE VALVE</b>            |
| <b>P838</b>                                                  | <b>RIGHT OUTBOARD FUEL<br/>SHUTOFF GATE VALVE</b>           |
| <b>P840</b>                                                  | <b>LEFT OUTBOARD FUEL<br/>SHUTOFF GATE VALVE</b>            |
| <b>P856 / J1</b>                                             | <b>AUXILIARY FUEL MANAGE-<br/>MENT CONTROL PANEL</b>        |
| <b>P857 / J2</b>                                             | <b>AUXILIARY FUEL MANAGE-<br/>MENT CONTROL PANEL</b>        |
| <b>P858 / J804</b>                                           | <b>HORIZONTAL STORES<br/>SUPPORT, BL 59 RH,<br/>STA 295</b> |
| <b>P859 / J805</b>                                           | <b>HORIZONTAL STORES<br/>SUPPORT, BL 59 LH,<br/>STA 295</b> |
| <b>P860 / J346</b>                                           | <b>CABIN CEILING, BL 20 LH,<br/>STA 304</b>                 |
| <b>P861 / W2J2</b>                                           | <b>HSS ATTACHMENT FITTING,<br/>BL 50 LH, STA 295</b>        |

| <b>TERMINAL BOARD /<br/>DISCONNECT PLUG /<br/>RECEPTACLE</b> | <b>LOCATION /<br/>CONNECTION POINT</b>               |
|--------------------------------------------------------------|------------------------------------------------------|
| <b>P862 / W2J2</b>                                           | <b>HSS ATTACHMENT FITTING,<br/>BL 59 LH, STA 295</b> |
| <b>P867 / J1</b>                                             | <b>COPILOT'S GYRO IN NOSE<br/>SECTION</b>            |
| <b>W3P1 / W2J2</b>                                           | <b>LEFT INBOARD FUEL TANK<br/>CONNECTOR</b>          |
| <b>W3P1 / W2J2</b>                                           | <b>RIGHT INBOARD FUEL<br/>TANK CONNECTOR</b>         |
| <b>W3P1 / W2J3</b>                                           | <b>LEFT OUTBOARD FUEL<br/>TANK CONNECTOR</b>         |
| <b>W3P1 / W2J3</b>                                           | <b>RIGHT OUTBOARD FUEL<br/>TANK CONNECTOR</b>        |
| <b>W4P1 / J1</b>                                             | <b>RIGHT INBOARD FUEL<br/>TANK CONNECTOR</b>         |
| <b>W4P1 / J1</b>                                             | <b>LEFT INBOARD FUEL<br/>TANK CONNECTOR</b>          |
| <b>W4P1 / J1</b>                                             | <b>LEFT OUTBOARD FUEL<br/>TANK CONNECTOR</b>         |
| <b>W4P1 / J1</b>                                             | <b>RIGHT OUTBOARD FUEL<br/>TANK CONNECTOR</b>        |

AB4343\_1

Figure 1. Crashworthy External Fuel System Location Diagram. (Sheet 1 of 2)

LOCATION - Continued

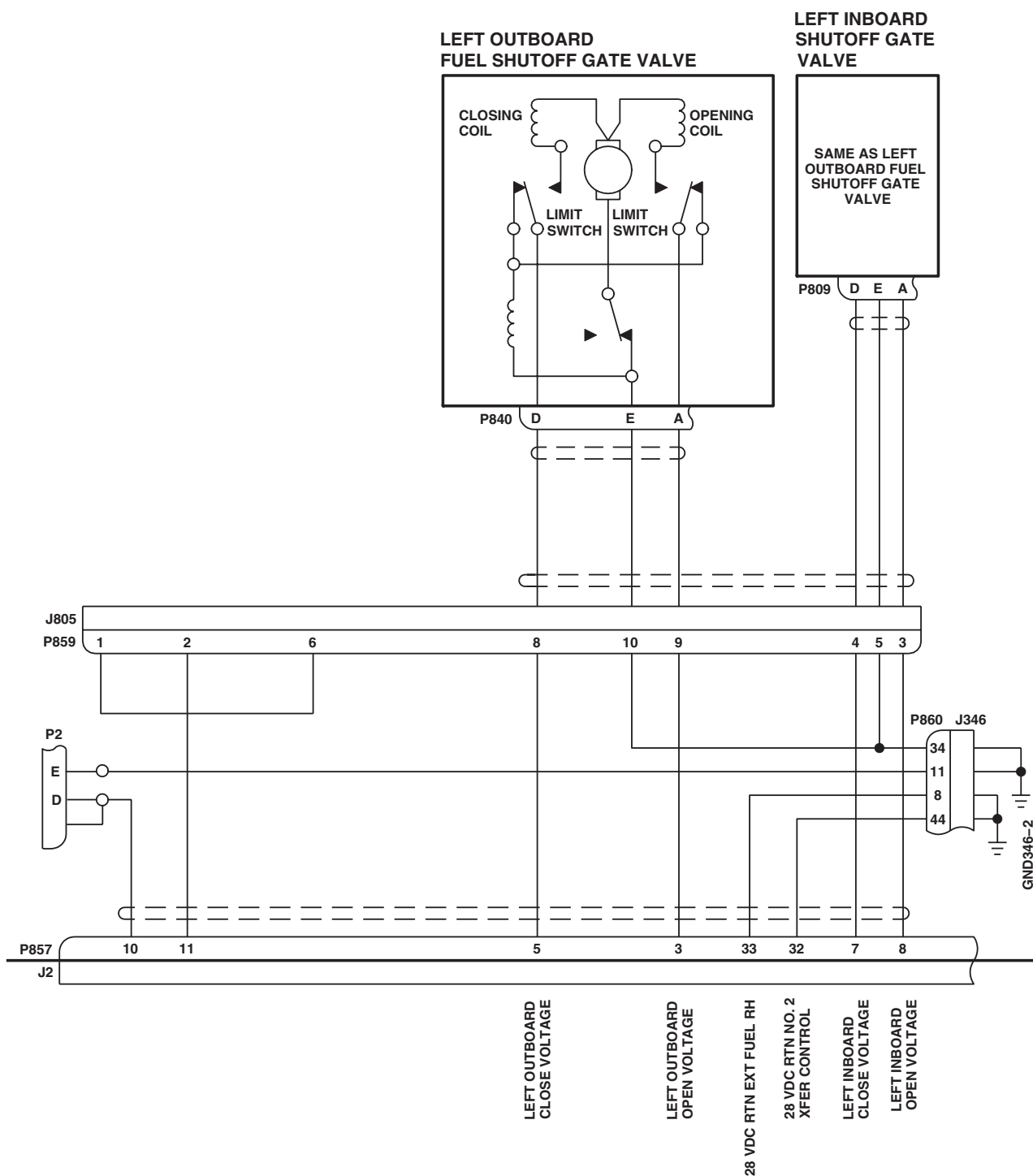


MISSION READINESS  
CIRCUIT BREAKER PANEL

AB4343 2  
SA

Figure 1. Crashworthy External Fuel System Location Diagram. (Sheet 2 of 2)

## SCHEMATICS

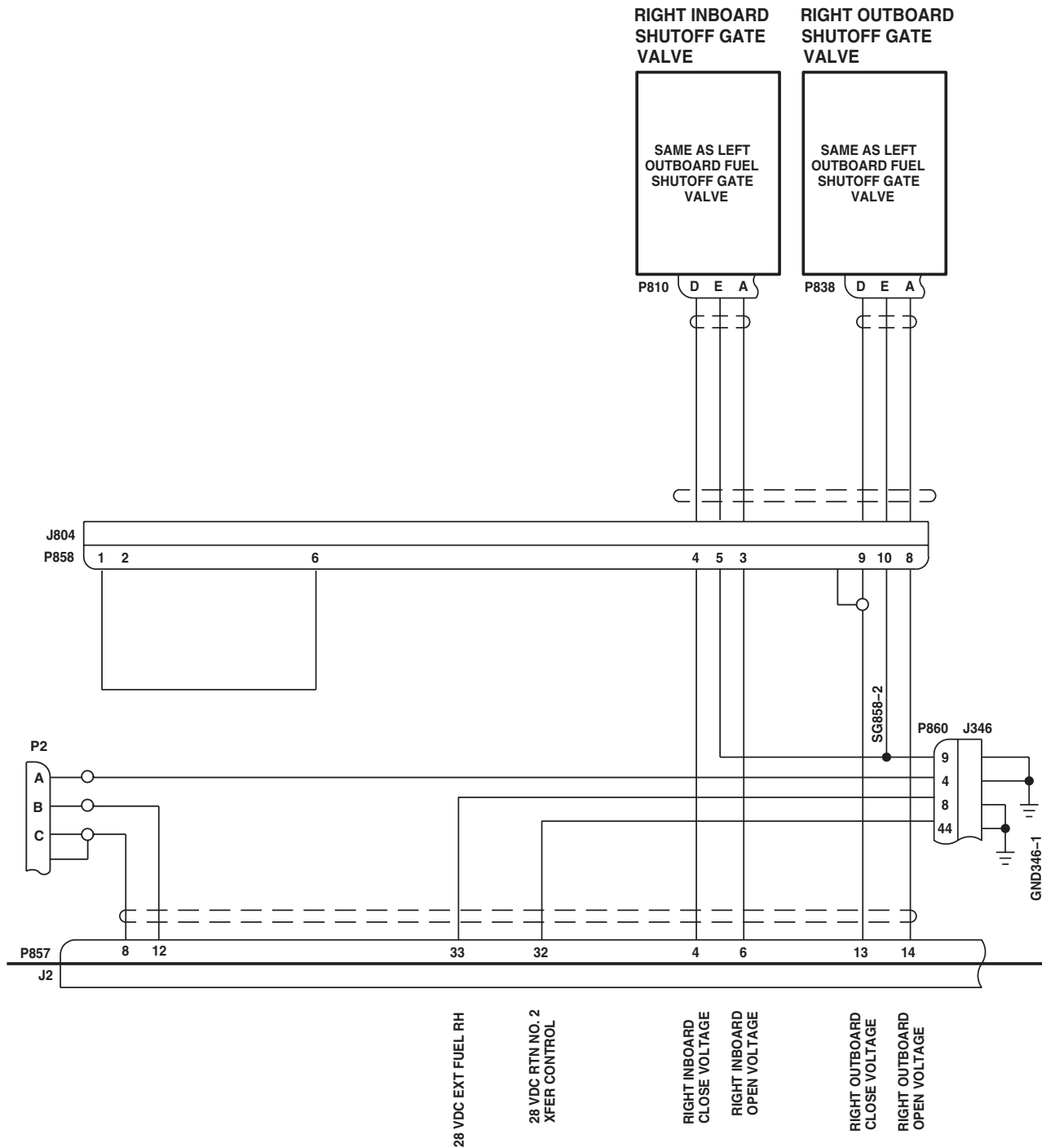


AUXILIARY FUEL MANAGEMENT PANEL

AB4357\_5  
SA

Figure 2. Crashworthy External Fuel System Control Schematic Diagram. (Sheet 1 of 3)

SCHEMATICS - Continued



AUXILIARY FUEL MANAGEMENT PANEL

AB4357 6  
SA

Figure 2. Crashworthy External Fuel System Control Schematic Diagram. (Sheet 2 of 3)

## SCHEMATICS - Continued

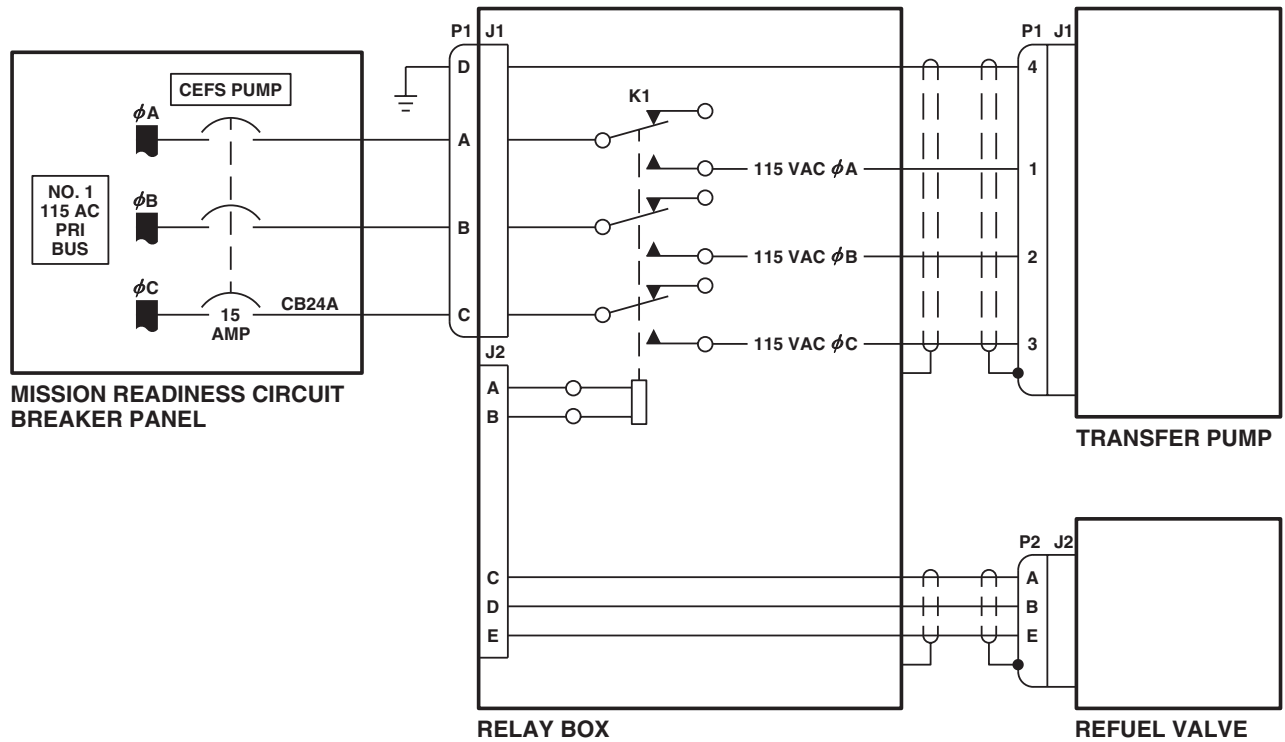
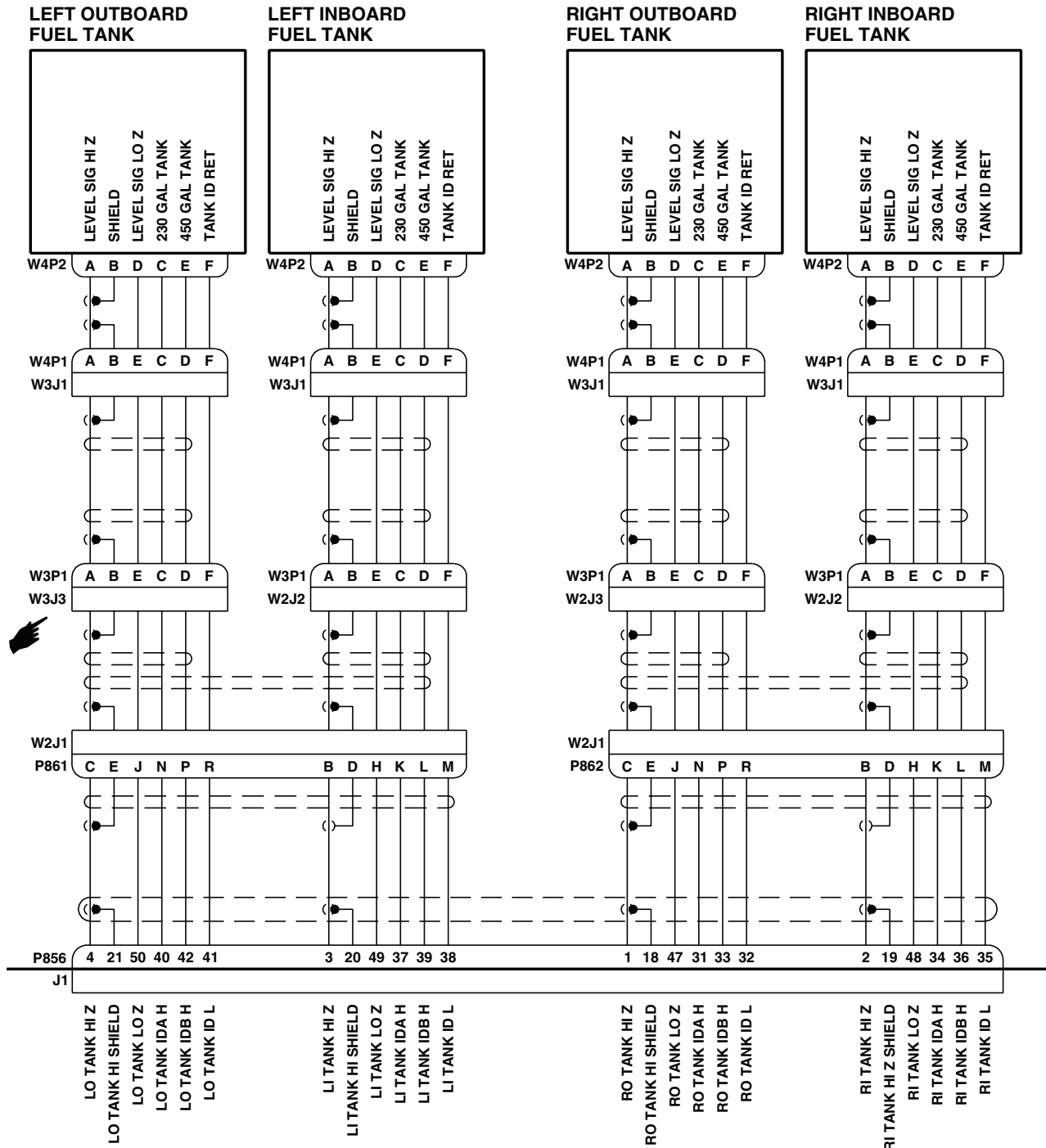
AB4344  
SA

Figure 2. Crashworthy External Fuel System Control Schematic Diagram. (Sheet 3 of 3)

SCHEMATICS - Continued

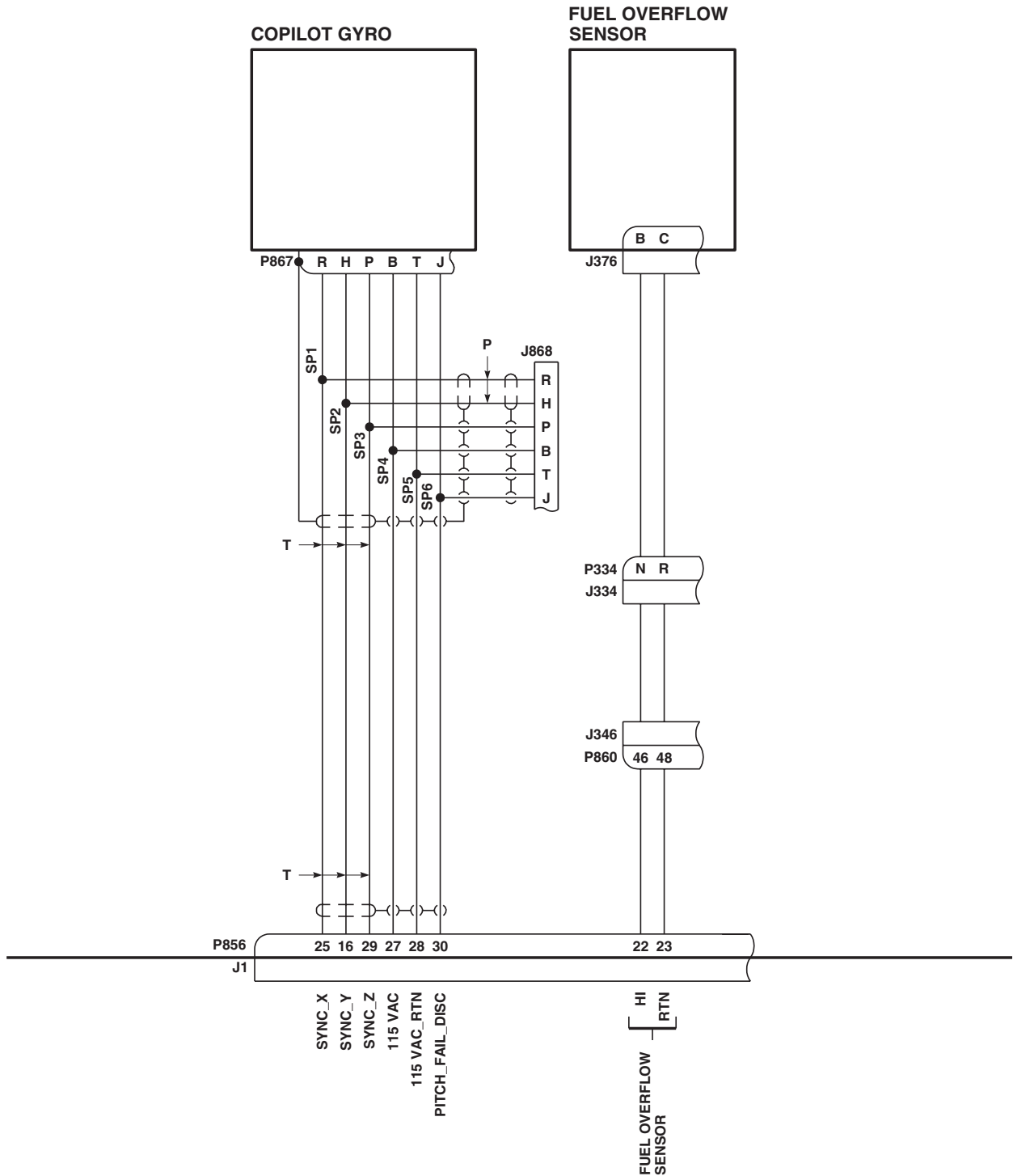


AUXILIARY FUEL MANAGEMENT PANEL

AB1692\_4B  
SA

Figure 3. Crashworthy External Fuel System Fuel Quantity Schematic Diagram.

SCHEMATICS - Continued



AUXILIARY FUEL MANAGEMENT PANEL

AB4357\_3  
SA

Figure 4. Crashworthy External Fuel System Copilot Gyro Schematic Diagram.

END OF WORK PACKAGE





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**UNIT LEVEL****MISSION EQUIPMENT****CRASHWORTHY EXTERNAL FUEL SYSTEM PRESSURE REFUELING MAINTENANCE OPERATIONAL  
CHECK**

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**INITIAL SETUP:****Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01

WP 1549 00

WP 1574 18

WP 1574 21

**Personnel Required**

UH-60 Helicopter Repairer MOS 15T (1)

WP 1612 01

Assistant (1)

WP 1615 01

WP 1685 00

WP 1747 00

**References**

TM 11-1520-237-23

TM 1-1500-204-23-3

TM 1-1560-316-23

WP 1440 00

**Equipment Condition**External Electrical Power Available, WP 1685 00  
or APU Operational, TM 1-1520-237-10

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## CRASHWORTHY EXTERNAL FUEL SYSTEM PRESSURE REFUELING OPERATIONAL/ TROUBLESHOOTING PROCEDURE

### Applicable Configuration

#### NOTE

- The Crashworthy External Fuel System (CEFS) operational /troubleshooting procedure should be performed prior to reinstallation of fairings vapor barrier, and sound proofing to allow visual inspection of fluid connections. Performing this procedure will validate proper installation of the CEFS.
- The CEFS operational /troubleshooting procedure should be performed with the aircraft on level ground. When the aircraft is level, the external fuel tanks are at four degree nose-up attitude. This may cause an inaccurate reading during fuel transfer. When refueling, there may be a delay before the probe indicates rising fuel level; when defueling, there may be a delay before the probe indicates lower fuel level.
- The CEFS operational /troubleshooting procedure assumes that a single tank pair is installed on the outboard pylons. If it is desired to perform the procedure with tanks installed on the inboard pylons, replace outboard with inboard in the following procedure.
- When the wing gate valves and pump package refuel valve are energized, this can be verified by listening for their sound. If there is a problem with this process, it may be necessary to remove the vertical support pylon fairings and/or the soundproofing around the fuel cells to visually check the position of these valves.
- If a circuit breaker pops during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- See Figure 1 for component location diagram and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

This operational/troubleshooting procedure is subdivided:

- Pressure Refueling Operational Check
- Pressure Refueling Operational Check Completion
- Right Outboard Wing Gate Valve Not In Commanded Position Troubleshooting
- Right Inboard Wing Gate Valve Not In Commanded Position Troubleshooting
- Left Inboard Wing Gate Valve Not In Commanded Position Troubleshooting
- Left Outboard Wing Gate Valve Not In Commanded Position Troubleshooting
- Refuel Valve Not In Commanded Position Troubleshooting
- Cannot Pressure Refuel Cefs Tank(s) Troubleshooting
- Tank Fill Level Above Gravity Fill Opening Troubleshooting
- Fuel Leakage From Cavity Drain Or Tank Shell Split Line Troubleshooting

**PRESSURE REFUELING OPERATIONAL CHECK****PROCEDURE****Step**

1. Set auxiliary fuel management panel (AFMP) switches as follows:

| SWITCH    | POSITION |
|-----------|----------|
| XFER MODE | OFF      |
| MAN XFER  | BOTH     |
| XFER FROM | OUTBD    |
| PRESS     | OFF      |

2. Apply external power to aircraft.
3. Check position of all four wing gate valves.

**Indication/Condition**

All four wing gate valves shall be closed.

**Corrective Action**

If any wing gate valve is not in the closed, go to WING GATE VALVE NOT IN COMMANDED POSITION, in this work package.

**Step**

4. Set AFMP PRESS switch to OUTBD.
5. Check position of outboard wing gate valves.

**Indication/Condition**

Outboard wing gate valves shall be open.

**Corrective Action**

If outboard wing gate valves are not open, go to either RIGHT OUTBOARD WING GATE VALVE NOT IN COMMANDED POSITION or, LEFT OUTBOARD WING GATE VALVE NOT IN COMMANDED POSITION, in this work package.

**Step**

6. Check position of pump package refuel valve.

**Indication/Condition**

Refuel valve shall be open.

**Corrective Action**

If refuel valve is not open, go to REFUEL VALVE NOT IN COMMANDED POSITION, in this work package.

---

**PRESSURE REFUELING OPERATIONAL CHECK - Continued****Step**

7. Pressure refuel aircraft (WP 1612 01) and visually inspect all fluid couplings for evidence of leakage.

**Indication/Condition**

Leakage shall not be indicated.

**Corrective Action**

If leakage is indicated, terminate pressure refueling and repair leaks in accordance with TM 1-1500-204-23-3 prior to continuing with this procedure.

**Step**

8. Check that outboard fuel quantities are rising.

**Indication/Condition**

Fuel quantities for outboard tanks shall increase.

**Corrective Action**

None Required

**Step**

9. Continue refueling aircraft until main tanks and CEFS tanks are full (high level shut-off).

**Indication/Condition**

Not Required

**Corrective Action**

Not Required

**Step**

10. Terminate pressure refueling.

**Indication/Condition**

None Required

**Corrective Action**

None Required

**PRESSURE REFUELING OPERATIONAL CHECK COMPLETION****PROCEDURE****Step**

1. Set auxiliary fuel management panel (AFMP) PRESS switch to OFF.
2. The aircraft can be refueled (WP 1612 01) or defueled (WP 1615 01).

---

**PRESSURE REFUELING OPERATIONAL CHECK COMPLETION** - Continued**NOTE**

If defueling is required, ensure that PRESS REFUEL switch is set to OUTBD prior to removing external power.

3. Remove external power from aircraft.
4. Install all fairings, vapor barriers, and sound proofing as required to restore aircraft to flight status.

**Indication/Condition**

Not Required

**Corrective Action**

Not Required

**RIGHT OUTBOARD WING GATE VALVE NOT IN COMMANDED POSITION****SYMPTOM**

Right outboard wing gate valve not in commanded position (does not open or close when commanded).

**MALFUNCTION**

Right outboard wing gate valve not in commanded position (does not open or close when commanded).

**CORRECTIVE ACTION**

- 1. With aircraft power on, set auxiliary fuel management control panel (AFMP) XFER MODE switch to OFF and PRESS switch to OUTBD.**
  - a. If right outboard wing gate valve is open, go to **2**.
  - b. If right outboard wing gate valve is not open, go to **6**.
- 2. Set AFMP PRESS switch to OFF.**
  - a. If right outboard wing gate valve is closed, go to **3**.
  - b. If right outboard wing gate valve is not closed, go to **10**.
- 3. Did malfunction occur during fuel transfer mode?**
  - a. If malfunction occurred during fuel transfer mode, go to **4**.
  - b. If fault did not occur during fuel transfer mode, go to **14**.
- 4. Set AFMP XFER MODE switch to MAN, MAN XFR switch to BOTH, and XFER FROM switch to OUTBD.**
  - a. If right outboard wing gate valve is open, go to **5**.
  - b. If right outboard wing gate valve is not open, replace AFMP (WP 1549 00). Go to **14**.
- 5. Set AFMP XFER FROM switch to INBD.**
  - a. If right outboard wing gate valve is not closed, replace AFMP (WP 1549 00). Go to **14**.
- 6. Disconnect P838 from right outboard wing gate valve. Check for 28 vdc between P838 pins A and E.**
  - a. If 28 vdc is not present, go to **7**.
  - b. If 28 vdc is present, replace wing gate valve (WP 1440 00). Go to **14**.

**RIGHT OUTBOARD WING GATE VALVE NOT IN COMMANDED POSITION - Continued**

- 7. Turn off aircraft power. Disconnect P857 from AFMP. Check for continuity between P857 pin 14 and P838 pin A.**
  - a. If continuity is present, go to **8**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 8. Disconnect P860 from J346. Check for continuity between P860 pin 9 and P838 pin E.**
  - a. If continuity is present, go to **9**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 9. Check for continuity between J346 pin 9 (mates with P860) to aircraft ground.**
  - a. If continuity is present, replace AFMP (WP 1549 00). Go to **14**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 10. Disconnect P838. Check for 28 vdc between P838 pin D and pin E.**
  - a. If 28 vdc is not present, go to **11**.
  - b. If 28 vdc is present, replace wing gate valve (WP 1440 00). Go to **14**.
- 11. Turn off aircraft power. Disconnect P857 from AFMP. Check for continuity between P857 pin 13 and P838 pin D.**
  - a. If continuity is present, go to **12**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 12. Disconnect P860. Check for continuity between P860 pin 9 and P838 pin E.**
  - a. If continuity is present, go to **13**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 13. Check for continuity between J346 pin 9 to aircraft ground.**
  - a. If continuity is present, replace AFMP (WP 1549 00). Go to **14**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 14. Procedure completed.**

**RIGHT INBOARD WING GATE VALVE NOT IN COMMANDED POSITION****SYMPTOM**

Right inboard wing gate valve not in commanded position (does not open or close when commanded).

**MALFUNCTION**

Right inboard wing gate valve not in commanded position (does not open or close when commanded).

**CORRECTIVE ACTION**

- 1. With aircraft power on, set auxiliary fuel management control panel (AFMP) XFER MODE switch to OFF and PRESS switch to INBD.**
  - a. If right inboard wing gate valve is open, go to **2**.
  - b. If right outboard wing gate valve is not open, go to **6**.
- 2. Set AFMP PRESS switch to OFF.**

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**RIGHT INBOARD WING GATE VALVE NOT IN COMMANDED POSITION - Continued**

- a. If right inboard wing gate valve is closed, go to **3**.
- b. If right inboard wing gate valve is not closed, go to **10**.
- 3. Did malfunction occur during fuel transfer mode?**
  - a. If malfunction occurred during fuel transfer mode, go to **4**.
  - b. If fault did not occur during fuel transfer mode, go to **14**.
- 4. Set AFMP XFER MODE switch to MAN, MAN XFR switch to BOTH, and XFER FROM switch to INBD.**
  - a. If right inboard wing gate valve is open, go to **5**.
  - b. If right inboard wing gate valve is not open, replace AFMP (WP 1549 00). Go to **14**.
- 5. Set AFMP XFER FROM switch to OUTBD.**
  - a. If right inboard wing gate valve is not closed, replace AFMP (WP 1549 00). Go to **14**.
- 6. Disconnect P810 from right inboard wing gate valve. Check for 28 vdc between P810 pins A and E.**
  - a. If 28 vdc is not present, go to **7**.
  - b. If 28 vdc is present, replace wing gate valve (WP 1440 00). Go to **14**.
- 7. Turn off aircraft power. Disconnect P857 from AFMP. Check for continuity between P857 pin 6 and P810 pin A.**
  - a. If continuity is present, go to **8**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 8. Disconnect P860 from J346. Check for continuity between P860 pin 9 and P810 pin E.**
  - a. If continuity is present, go to **9**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 9. Check for continuity between J346 pin 9 (mates with P860) to aircraft ground.**
  - a. If continuity is present, replace AFMP (WP 1549 00). Go to **14**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 10. Disconnect P810. Check for 28 vdc between P810 pin D and pin E.**
  - a. If 28 vdc is not present, go to **11**.
  - b. If 28 vdc is present, replace wing gate valve (WP 1440 00). Go to **14**.
- 11. Turn off aircraft power. Disconnect P857 from AFMP. Check for continuity between P857 pin 4 and P810 pin D.**
  - a. If continuity is present, go to **12**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 12. Disconnect P860. Check for continuity between P860 pin 9 and P810 pin E.**
  - a. If continuity is present, go to **13**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.

**RIGHT INBOARD WING GATE VALVE NOT IN COMMANDED POSITION** - Continued**13. Check for continuity between J346 pin 9 to aircraft ground.**

- a. If continuity is present, replace AFMP (WP 1549 00). Go to **14**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.

**14. Procedure completed.****LEFT INBOARD WING GATE VALVE NOT IN COMMANDED POSITION****SYMPTOM**

Left inboard wing gate valve not in commanded position (does not open or close when commanded).

**MALFUNCTION**

Left inboard wing gate valve not in commanded position (does not open or close when commanded).

**CORRECTIVE ACTION**

- 1. With aircraft power on, set auxiliary fuel management control panel (AFMP) XFER MODE switch to OFF and PRESS switch to INBD.**
  - a. If left inboard wing gate valve is open, go to **2**.
  - b. If left inboard wing gate valve is not open, go to **5**.
- 2. Set AFMP PRESS switch to OFF.**
  - a. If left inboard wing gate valve is closed, go to **3**.
  - b. If left inboard wing gate valve is not closed, go to **9**.
- 3. Did malfunction occur during fuel transfer mode?**
  - a. If malfunction occurred during fuel transfer mode, go to **4**.
  - b. If fault did not occur during fuel transfer mode, go to **14**.
- 4. Set AFMP XFER MODE switch to MAN, MAN XFR switch to BOTH, and XFER FROM switch to INBD.**
  - a. If left inboard wing gate valve is open, go to **5**.
  - b. If left inboard wing gate valve is not open, replace AFMP (WP 1549 00). Go to **14**.
- 5. Set AFMP XFER FROM switch to OUTBD.**
  - a. If left inboard wing gate valve is not closed, replace AFMP (WP 1549 00). Go to **14**.
- 6. Disconnect P809 from left inboard wing gate valve. Check for 28 vdc between P809 pins A and E.**
  - a. If 28 vdc is not present, go to **7**.
  - b. If 28 vdc is present, replace wing gate valve (WP 1440 00). Go to **14**.
- 7. Turn off aircraft power. Disconnect P857 from AFMP. Check for continuity between P857 pin 8 and P809 pin A.**
  - a. If continuity is present, go to **8**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 8. Disconnect P860 from J346. Check for continuity between P860 pin 34 and P809 pin E.**



**LEFT INBOARD WING GATE VALVE NOT IN COMMANDED POSITION - Continued**

- a. If continuity is present, go to **9**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 9. Check for continuity between J346 pin 34 (mates with P860) to aircraft ground.**
  - a. If continuity is present, replace AFMP (WP 1549 00). Go to **14**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 10. Disconnect P809. Check for 28 vdc between P809 pin D and pin E.**
  - a. If 28 vdc is not present, go to **11**.
  - b. If 28 vdc is present, replace wing gate valve (WP 1440 00). Go to **14**.
- 11. Turn off aircraft power. Disconnect P857 from AFMP. Check for continuity between P857 pin 7 and P809 pin D.**
  - a. If continuity is present, go to **12**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 12. Disconnect P860. Check for continuity between P860 pin 34 and P809 pin E.**
  - a. If continuity is present, go to **13**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 13. Check for continuity between J346 pin 34 to aircraft ground.**
  - a. If continuity is present, replace AFMP (WP 1549 00). Go to **14**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 14. Procedure completed.**

**LEFT OUTBOARD WING GATE VALVE NOT IN COMMANDED POSITION****SYMPTOM**

Left outboard wing gate valve not in commanded position (does not open or close when commanded).

**MALFUNCTION**

Left outboard wing gate valve not in commanded position (does not open or close when commanded).

**CORRECTIVE ACTION**

- 1. With aircraft power on, set auxiliary fuel management control panel (AFMP) XFER MODE switch to OFF and PRESS switch to OUTBD.**
  - a. If left outboard wing gate valve is open, go to **2**.
  - b. If left outboard wing gate valve is not open, go to **5**.
- 2. Set AFMP PRESS switch to OFF.**
  - a. If left outboard wing gate valve is closed, go to **3**.
  - b. If left outboard wing gate valve is not closed, go to **9**.
- 3. Did malfunction occur during fuel transfer mode?**
  - a. If malfunction occurred during fuel transfer mode, go to **4**.
  - b. If fault did not occur during fuel transfer mode, go to **14**.

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**LEFT OUTBOARD WING GATE VALVE NOT IN COMMANDED POSITION - Continued**

- 4. Set AFMP XFER MODE switch to MAN, MAN XFR switch to BOTH, and XFER FROM switch to OUTBD.**
  - a. If left outboard wing gate valve is open, go to **5**.
  - b. If left outboard wing gate valve is not open, replace AFMP (WP 1549 00). Go to **14**.
- 5. Set AFMP XFER FROM switch to INBD.**
  - a. If left outboard wing gate valve is not closed, replace AFMP (WP 1549 00). Go to **14**.
- 6. Disconnect P840 from left outboard wing gate valve. Check for 28 vdc between P840 pins A and E.**
  - a. If 28 vdc is not present, go to **7**.
  - b. If 28 vdc is present, replace wing gate valve (WP 1440 00). Go to **14**.
- 7. Turn off aircraft power. Disconnect P857 from AFMP. Check for continuity between P857 pin 3 and P840 pin A.**
  - a. If continuity is present, go to **8**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 8. Disconnect P860 from J346. Check for continuity between P860 pin 34 and P838 pin E.**
  - a. If continuity is present, go to **9**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 9. Check for continuity between J346 pin 34 (mates with P860) to aircraft ground.**
  - a. If continuity is present, replace AFMP (WP 1549 00). Go to **14**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 10. Disconnect P840. Check for 28 vdc between P840 pin D and pin E.**
  - a. If 28 vdc is not present, go to **11**.
  - b. If 28 vdc is present, replace wing gate valve (WP 1440 00). Go to **14**.
- 11. Turn off aircraft power. Disconnect P857 from AFMP. Check for continuity between P857 pin 5 and P838 pin D.**
  - a. If continuity is present, go to **12**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 12. Disconnect P860. Check for continuity between P860 pin 34 and P840 pin E.**
  - a. If continuity is present, go to **13**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 13. Check for continuity between J346 pin 34 to aircraft ground.**
  - a. If continuity is present, replace AFMP (WP 1549 00). Go to **14**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 14. Procedure completed.**

**REFUEL VALVE NOT IN COMMANDED POSITION****SYMPTOM**

Refuel valve not in commanded position (does not open or close when commanded).

**MALFUNCTION**

Refuel valve not in commanded position (does not open or close when commanded).

**CORRECTIVE ACTION**

- 1. With aircraft power on, set auxiliary fuel management control panel (AFMP) XFER MODE switch to OFF and PRESS switch to OUTBD.**
  - a. If refuel valve is open, go to **2**.
  - b. If refuel valve is not open, go to **5**.
- 2. Set AFMP XFER PRESS switch to OFF.**
  - a. If refuel valve is closed, go to **3**.
  - b. If refuel valve is not closed, go to **9**.
- 3. Set AFMP PRESS switch to INBD.**
  - a. If refuel valve is open, go to **4**.
  - b. If refuel valve is not open, replace AFMP (WP 1549 00). Go to **13**.
- 4. Set AFMP PRESS switch to ALL.**
  - a. If refuel valve is not open, replace AFMP (WP 1549 00). Go to **13**.
- 5. Disconnect connector from refuel valve. Check for 28 vdc between connector pins C and E.**
  - a. If 28 vdc is not present, go to **6**.
  - b. If 28 vdc is present, replace pump package refuel valve (WP 1574 18). Go to **13**.
- 6. Disconnect P2 from pump package relay box J1. Check for 28 vdc between P2 pin C and pin E.**
  - a. If 28 vdc is not present, go to **7**.
  - b. If 28 vdc is present, replace pump package relay box (WP 1574 21). Go to **13**.
- 7. Remove aircraft power and disconnect P857 from AFMP. Check for continuity between P857 pin 9 and P2 pin C.**
  - a. If continuity is present, go to **8**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **13**.
- 8. Disconnect P860 and check for continuity between P860 pin 11 and P2 pin E.**
  - a. If continuity is present, replace AFMP (WP 1549 00). Go to **13**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **13**.
- 9. Disconnect P2 from refuel valve and check for 28 vdc between P2 pin D and pin E.**
  - a. If 28 vdc is not present, go to **10**.
  - b. If 28 vdc is present, replace pump package refuel valve (WP 1574 18). Go to **13**.

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**REFUEL VALVE NOT IN COMMANDED POSITION - Continued**

- 10. Disconnect P857 from AFMP. Check for continuity between P857 pin 13 and P838 pin D.**
  - a. If 28 vdc is not present, go to **11**.
  - b. If 28 vdc is present, replace pump package relay box (WP 1574 21). Go to **13**.
- 11. Disconnect P860. Check for continuity between P860 pin 9 and P838 pin E.**
  - a. If continuity is present, go to **12**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **13**.
- 12. Check for continuity between J346 pin 9 and aircraft ground.**
  - a. If continuity is present, replace AFMP (WP 1549 00). Go to **13**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **13**.
- 13. Procedure completed.**

**CANNOT PRESSURE REFUEL CEFS TANK(S)****SYMPTOM**

Cannot pressure refuel cefs tank(s).

**MALFUNCTION**

Cannot pressure refuel cefs tank(s).

**CORRECTIVE ACTION**

- 1. With aircraft configured for pressure refueling, check position of refuel valve.**
  - a. If refuel valve is open, go to **7**.
  - b. If refuel valve is not open, go to **2**.
- 2. Disconnect connector from refuel valve. Check for 28 vdc between connector pins C and E.**
  - a. If 28 vdc is not present, remove aircraft power and go to **3**.
  - b. If 28 vdc is present, replace pump package refuel valve (WP 1574 18). Go to **13**.
- 3. With aircraft power removed, disconnect P2 from pump package relay box connector J2. Check for continuity between relay box connector J2 pin C and relay box connector P2 pin A and between relay box connector J2 pin E and relay box connector P2 pin E.**
  - a. If continuity is present, go to **4**.
  - b. If continuity is not present, replace pump package relay box (WP 1574 21). Go to **13**.
- 4. With aircraft power removed, disconnect P857 from AFMP. Check for continuity between P857 pin 9 and P2 pin C.**
  - a. If continuity is present, go to **5**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **13**.
- 5. Disconnect P860 from J346 and check for continuity between P860 pin 11 and P2 pin E.**
  - a. If continuity is present, replace AFMP (WP 1549 00). Go to **13**.

**CANNOT PRESSURE REFUEL CEFS TANK(S) - Continued**

- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **13**.
- 6. Check for continuity between P860 pin 11 and aircraft ground.**
  - a. If continuity is present, replace AFMP (WP 1549 00). Go to **13**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **13**.
- 7. Check position of the wing gate valve corresponding to the selected tank(s).**
  - a. If wing gate valve is open, go to **11**.
  - b. If wing gate valve is not open, to **8**.
- 8. Disconnect connector from selected wing gate valve check for 28 vdc between pins A and E.**
  - a. If 28 vdc is not present, go to **9**.
  - b. If 28 vdc is present, replace wing gate valve (WP 1440 00). Go to **13**.
- 9. Disconnect P857 from AFMP and P860 from J346. Check for continuity between the following points for the effected valve:**
  - Left Outboard
    - P840-A and P857-3**
    - P840-E and P860-34**
  - Left Inboard
    - P809-A and P857-8**
    - P809-E and P860-34**
  - Right Inboard
    - P810-A and P857-6**
    - P810-E and P860-9**
  - Right Outboard
    - P838-A and P857-14**
    - P838-E and P860-9**
  - a. If continuity is present, go to **10**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **13**.
- 10. Check for continuity between P860 pin 34 and aircraft ground for left wing gate valves or P860 pin 9 and aircraft ground for right wing gate valves.**
  - a. If continuity is present, replace AFMP (WP 1549 00). Go to **13**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **13**.
- 11. Check break-away couplings and break-away valves for evidence of breakage.**
  - a. If break-away couplings and break-away valves are not broken, go to **12**.
  - b. If any break-away couplings or break-away valves are broken, replace broken couplings and valves (TM 1-1560-316-23). Go to **13**.

**CANNOT PRESSURE REFUEL CEFS TANK(S) - Continued**

- 12. Remove aircraft power and remove fuel tank that will not pressure refuel. Check to see if tank defuel valve has been replaced as part of this maintenance activity.**
  - a. If tank defuel valve has been replaced as part of this maintenance activity, replace tank high level pilot valve (TM 1-1560-316-23). Go to **13**.
  - b. If tank defuel valve has not been replaced as part of this maintenance activity, replace tank defuel valve (TM 1-1560-316-23). Go to **13**.
- 13. Procedure completed.**

**TANK FILL LEVEL ABOVE GRAVITY FILL OPENING (PRESSURE REFUELED)****SYMPTOM**

Tank fill level above gravity fill opening (pressure refueled).

**MALFUNCTION**

Tank fill level above gravity fill opening (pressure refueled).

**CORRECTIVE ACTION**

- 1. Suction defuel tank (WP 1615 01) and then pressure refuel tank (WP 1612 01). Carefully remove cap from CEFS tank following pressure refueling and check for fuel spillage over gravity fill flange.**
  - a. If fuel spillage is not present, go to **6**.
  - b. If fuel spillage is present, suction defuel tank (WP 1615 01) and then go to **2**.
- 2. Check high level crossover tube and lines for proper attachment and evidence of damage per TM 1-1560-316-23.**
  - a. If high level crossover tube and lines are not damaged, go to **4**.
  - b. If high level crossover tube or lines are damaged, replace lines or fittings and go to **3**.
- 3. Pressure refuel affected tank (WP 1612 01). Carefully remove cap from CEFS tank following pressure refueling and check for fuel spillage over gravity fill flange.**
  - a. If fuel spillage is not present, go to **6**.
  - b. If fuel spillage is present, suction defuel tank (WP 1615 01) and then go to **4**.
- 4. Replace tank high level pilot valve per TM 1-1560-316-23. Pressure refuel affected tank (WP 1612 01). Carefully remove cap from CEFS tank following pressure refueling and check for fuel spillage over gravity fill flange.**
  - a. If fuel spillage is not present, go to **6**.
  - b. If fuel spillage is present, suction defuel tank (WP 1615 01) and then go to **5**.
- 5. Replace tank defuel valve per TM 1-1560-316-23. Pressure refuel affected tank (WP 1612 01). Carefully remove cap from CEFS tank following pressure refueling and check for fuel spillage over gravity fill flange.**
  - a. If fuel spillage is not present, go to **6**.
  - b. If fuel spillage is present, go to **1 through 5**.
- 6. Procedure completed.**

**FUEL LEAKAGE FROM CAVITY DRAIN OR TANK SHELL SPLIT LINE****SYMPTOM**

Fuel leakage from cavity drain or tank shell split line.

**MALFUNCTION**

Fuel leakage from cavity drain or tank shell split line.

**CORRECTIVE ACTION**

- 1. Inspect tank surface for evidence of fuel leakage at tank split line or cavity drain.**
  - a. If leakage is present, go to **2**.
  - b. If If leakage is not present, go to **10**.
- 2. Remove forward cover plate per TM 1-1560-316-23. Inspect packing for damage and evidence of fuel leakage.**
  - a. If leakage is present, go to **3**.
  - b. If If leakage is not present, go to **4**.
- 3. Remove forward cover plate per TM 1-1560-316-23. Clean any evidence of fuel from the outside surface of the bladder and inside surface of the forward tank shell. Inspect bladder for evidence of bladder damage or activation due to contact with fuel.**
  - a. If bladder is damaged or activated, replace bladder per TM 1-1560-316-23. Go to **10**.
  - b. If bladder is not damaged or activated, replace damaged packing and reassemble tank per TM 1-1560-316-23. Go to **10**.
- 4. Remove gravity fill spacer per TM 1-1560-316-23. Inspect packing for damage and evidence of fuel leakage.**
  - a. If evidence of fuel leakage is present, go to **5**.
  - b. If evidence of fuel leakage is not present, go to **5**.
- 5. Remove forward tank shell per TM 1-1560-316-23 Clean any evidence of fuel from the outside surface of the bladder and inside surface of the forward tank shell. Inspect bladder for evidence of bladder damage or activation due to contact with fuel.**
  - a. If bladder is damaged or activated, replace bladder per TM 1-1560-316-23 Go to **10**.
  - b. If bladder is not damaged or activated, replace damaged packing and reassemble tank per TM 1-1560-316-23. Go to **10**.
- 6. Remove aft cover plate per TM 1-1560-316-23 Inspect packing for the damage and evidence of fuel leakage.**
  - a. If leakage is present, go to **7**.
  - b. If leakage is not present, go to **8**.
- 7. Remove aft tank shell per TM 1-1560-316-23. Clean any evidence of fuel from the outside surface of the bladder and inside surface of the forward tank shell. Inspect bladder for evidence of bladder damage or activation due to contact with fuel.**
  - a. If bladder is damaged or activated, replace bladder per TM 1-1560-316-23. Go to **10**.

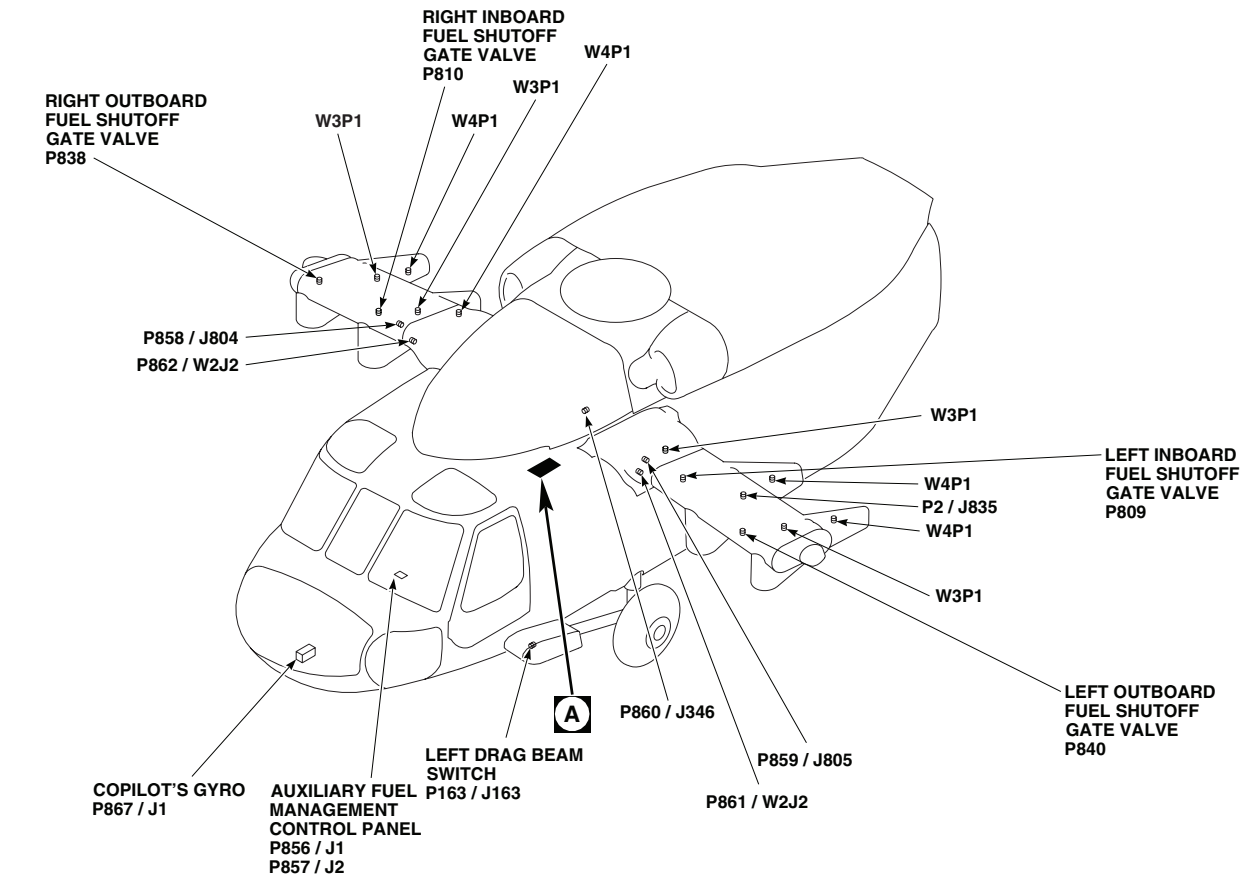
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**FUEL LEAKAGE FROM CAVITY DRAIN OR TANK SHELL SPLIT LINE - Continued**

- b. If bladder is not damaged or activated, replace damaged packing and reassemble tank per TM 1-1560-316-23. Go to **10**.
- 8. Remove aft tank shell per TM 1-1560-316-23. Inspect bladder for evidence of leakage from aft blank cover plate.**
  - a. If leakage present, go to **9**.
  - b. If leakage is not present, reassemble tank per (WP 1747 00). Go to **10**.
- 9. Clean any evidence of fuel from the outside surface of the bladder and inside surface of aft tank shell. Inspect bladder for evidence of bladder damage or activation due to contact with fuel.**
  - a. If bladder is damaged or activated, replace bladder per TM 1-1560-316-23. Go to **10**.
  - b. If bladder is not damaged or activated, remove aft cover plate and replace packing. Reassemble tank per TM 1-1560-316-23. Go to **10**.
- 10. Procedure completed.**



## LOCATION



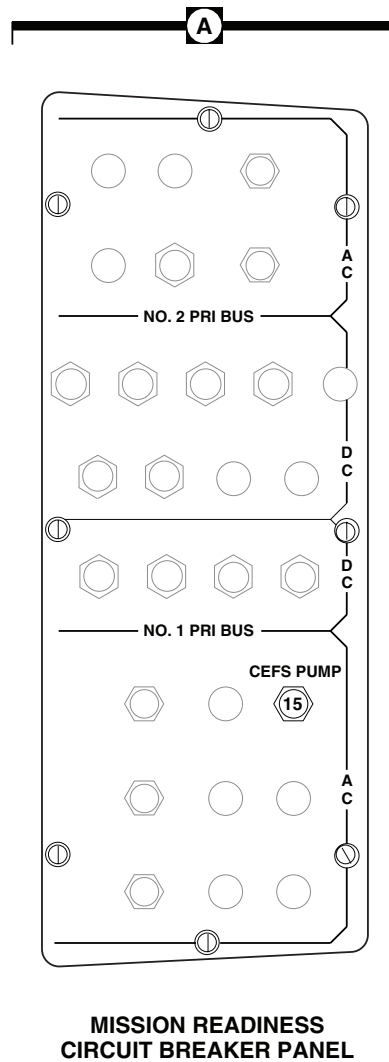
| TERMINAL BOARD /<br>DISCONNECT PLUG /<br>RECEPTACLE | LOCATION /<br>CONNECTION POINT               |
|-----------------------------------------------------|----------------------------------------------|
| P809                                                | LEFT INBOARD FUEL SHUTOFF GATE VALVE         |
| P810                                                | RIGHT INBOARD FUEL SHUTOFF GATE VALVE        |
| P838                                                | RIGHT OUTBOARD FUEL SHUTOFF GATE VALVE       |
| P840                                                | LEFT OUTBOARD FUEL SHUTOFF GATE VALVE        |
| P856 / J1                                           | AUXILIARY FUEL MANAGEMENT CONTROL PANEL      |
| P857 / J2                                           | AUXILIARY FUEL MANAGEMENT CONTROL PANEL      |
| P858 / J804                                         | HORIZONTAL STORES SUPPORT, BL 59 RH, STA 295 |
| P859 / J805                                         | HORIZONTAL STORES SUPPORT, BL 59 LH, STA 295 |
| P860 / J346                                         | CABIN CEILING, BL 20 LH, STA 304             |
| P861 / W2J2                                         | HSS ATTACHMENT FITTING, BL 50 LH, STA 295    |

| TERMINAL BOARD /<br>DISCONNECT PLUG /<br>RECEPTACLE | LOCATION /<br>CONNECTION POINT            |
|-----------------------------------------------------|-------------------------------------------|
| P862 / W2J2                                         | HSS ATTACHMENT FITTING, BL 59 LH, STA 295 |
| P867 / J1                                           | COPILOT'S GYRO IN NOSE SECTION            |
| W3P1 / W2J2                                         | LEFT INBOARD FUEL TANK CONNECTOR          |
| W3P1 / W2J2                                         | RIGHT INBOARD FUEL TANK CONNECTOR         |
| W3P1 / W2J3                                         | LEFT OUTBOARD FUEL TANK CONNECTOR         |
| W3P1 / W2J3                                         | RIGHT OUTBOARD FUEL TANK CONNECTOR        |
| W4P1 / J1                                           | RIGHT INBOARD FUEL TANK CONNECTOR         |
| W4P1 / J1                                           | LEFT INBOARD FUEL TANK CONNECTOR          |
| W4P1 / J1                                           | LEFT OUTBOARD FUEL TANK CONNECTOR         |
| W4P1 / J1                                           | RIGHT OUTBOARD FUEL TANK CONNECTOR        |

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Figure 1. Crashworthy External Fuel System Location Diagram. (Sheet 1 of 2)

LOCATION - Continued

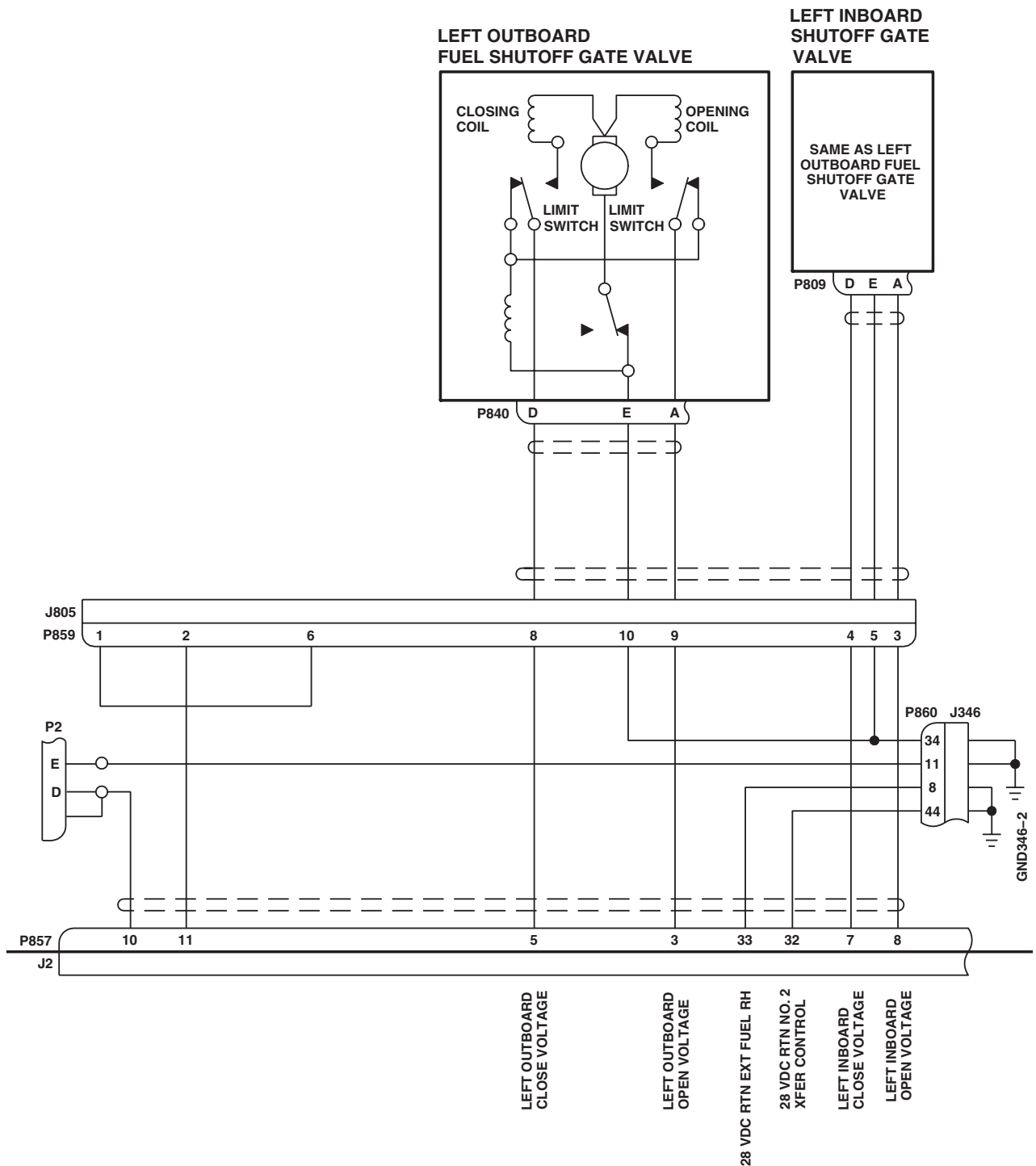


MISSION READINESS  
CIRCUIT BREAKER PANEL

AB4343 2  
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Figure 1. Crashworthy External Fuel System Location Diagram. (Sheet 2 of 2)

## SCHEMATICS

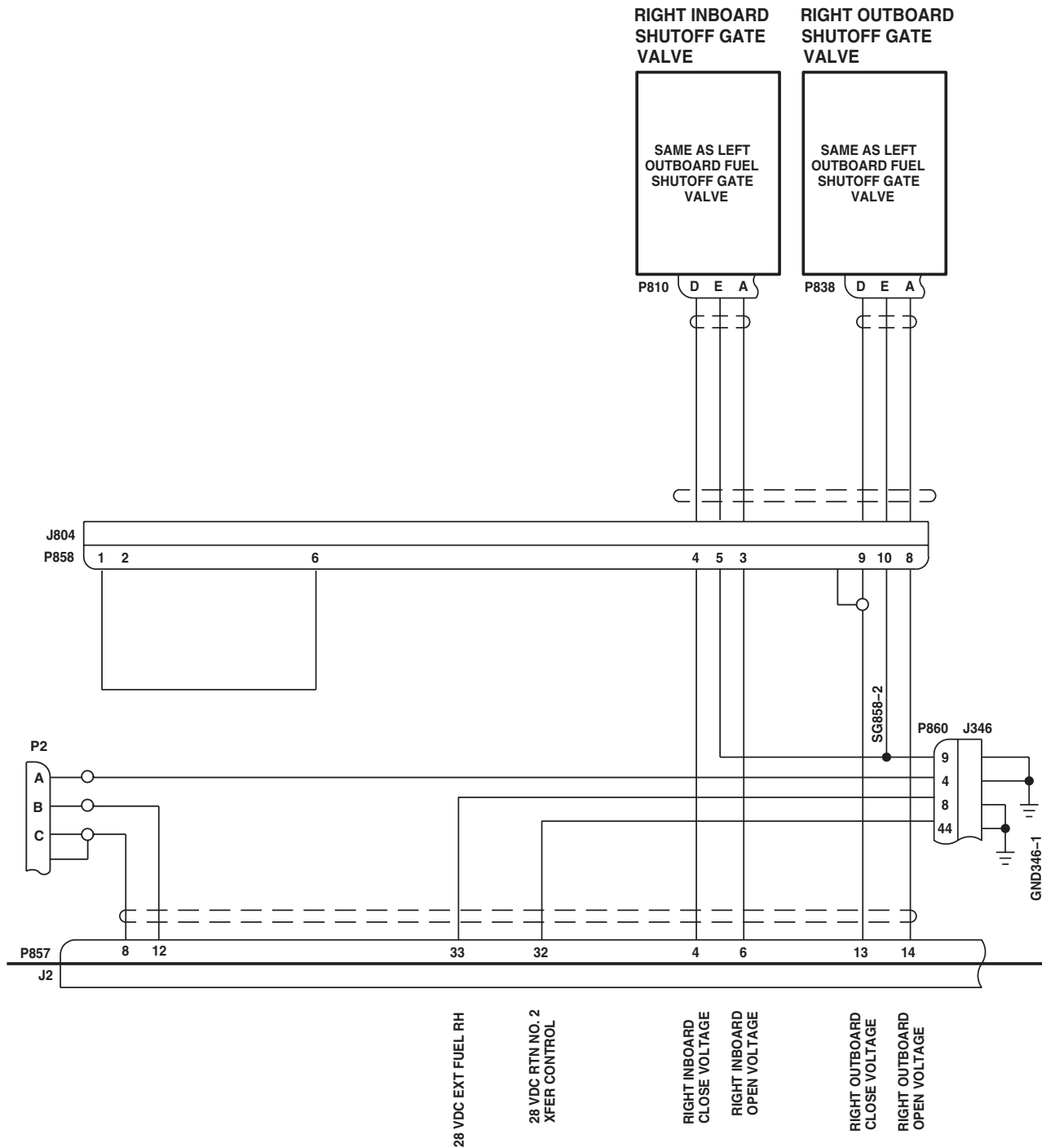


AUXILIARY FUEL MANAGEMENT PANEL

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Figure 2. Crashworthy External Fuel System Control Schematic Diagram. (Sheet 1 of 3)

SCHEMATICS - Continued



AUXILIARY FUEL MANAGEMENT PANEL

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Figure 2. Crashworthy External Fuel System Control Schematic Diagram. (Sheet 2 of 3)

## SCHEMATICS - Continued

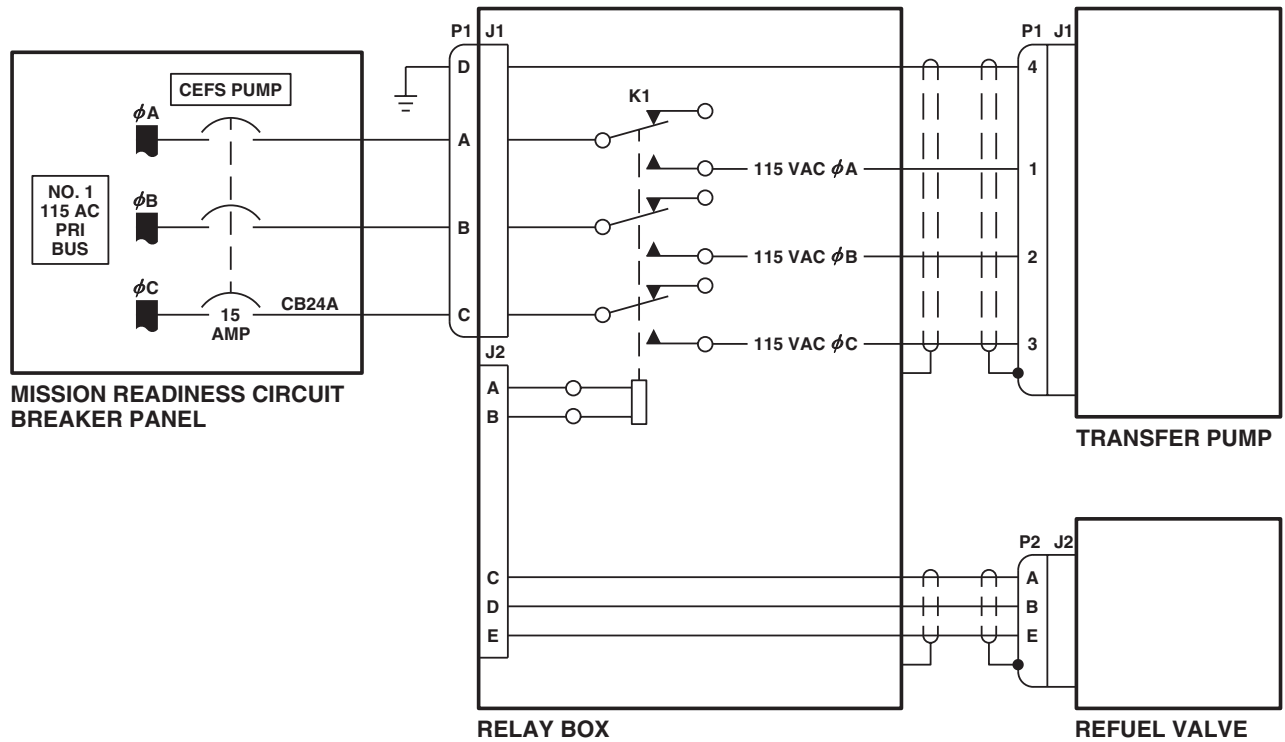
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Figure 2. Crashworthy External Fuel System Control Schematic Diagram. (Sheet 3 of 3)

END OF WORK PACKAGE



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**UNIT LEVEL**  
**EMERGENCY EQUIPMENT**  
**COCKPIT AIR BAG SYSTEM**

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**INITIAL SETUP:****Test Equipment**

Multimeter, AN/PSM-45A

**References**

WP 0104 00

WP 0824 00

**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01

WP 1589 00

Electrical Repairer Toolkit, SC 5180-99-B06

WP 1747 00

**Personnel Required**

Aircraft Electrician MOS 15F (1)

UH-60 Helicopter Repairer MOS 15T (1)

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**COCKPIT AIR BAG SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE**
**PROCEDURE****Step****WARNING**

- The cockpit air bag system (CABS) can cause serious injury or loss of life during maintenance due to inadvertent deployment of the forward or lateral air bags. Pull the CABS circuit breaker to remove power from the electronic crash sensor unit (ECSU) when aircraft is weight-on-wheels.
- Gas generators contain pyrotechnic propellant. The propellant can be ignited by electrostatic discharge and can cause severe injury or loss of life. When working directly on these devices, personal protective equipment and grounding straps must be worn. Equipment, technician and ordnance items must be grounded to the airframe.
- The ECSU contains capacitors that provide backup power in the event system power is lost. To prevent possible electrical shock, remove power and wait five minutes to allow the capacitor to discharge prior to removing CABS components.

**NOTE**

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- See Figure 1 for component location diagram, and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

1. Make sure the CABS circuit breaker on lower console circuit breaker panel is pushed in.
2. Place upper console BATT switch to ON.
3. Allow approximately 60 seconds for ECSU to perform self-test.

---

**COCKPIT AIR BAG SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE** - Continued**Indication/Condition**

SYS FAULT indicator shall be black (indicating a no fault condition).

**Corrective Action**

1. If ECSU does not power up, go to ECSU DOES NOT POWER UP, in this work package.
2. If Indication/Condition is not as specified, go to step 4.

**Step**

4. Press PUSH-TO-DISPLAY button on ECSU.

**Indication/Condition**

- a. All LEDs shall illuminate for approximately three seconds.
- b. After an additional one second, the following LEDs shall go off:
- c. ECSU
- d. A/C ID
- e. No. 1 (pilot's and copilot's)
- f. No. 2 (pilot's and copilot's)

**Corrective Action**

1. If ECSU LED remains on, replace ECSU (WP 1589 00).
2. If A/C LED remains on, go to A/C ID LED REMAINS ON, in this work package.
3. If pilot's or copilot's No. 1 LED remains on, go to PILOT'S OR COPILOT'S NO. 1 LED REMAINS ON, in this work package.
4. If pilot's or copilot's No. 2 LED remains on, go to PILOT'S OR COPILOT'S NO. 2 LED REMAINS ON, in this work package.

**Step**

5. Place upper console BATT switch to OFF.
6. Pull out CABS circuit breaker on the lower console circuit breaker panel.

**Indication/Condition**

Circuit breaker shall be pulled out.

**Corrective Action**

None Required



**ECSU DOES NOT POWER UP****SYMPTOM**

ECSU does not power up.

**MALFUNCTION**

ECSU does not power up.

**CORRECTIVE ACTION**

- 1. Check for 28 vdc between CABS circuit breaker terminal 1 and ground.**
  - a. If voltage is as specified, go to **2**.
  - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **4**.
- 2. Check for 28 vdc between J107-A and ground.**
  - a. If voltage is as specified, go to **3**.
  - b. If voltage is not as specified, repair/replace wiring between J107-A and CABS circuit breaker terminal 1 (WP 1747 00). Go to **4**.
- 3. Check resistance between:**
  - **P4-L and P107-A (Less than 2 ohms)**
  - **P4-J and ground (Less than 2 ohms)**
  - **P4-M and ground (Less than 2 ohms)**
  - a. If resistance is as specified, replace ECSU (WP 1589 00). Go to **4**.
  - b. If resistance is not as specified, repair/replace wiring (WP 1747 00). Go to **4**.
- 4. Procedure completed.**

**A/C ID LED REMAINS ON****SYMPTOM**

A/C ID LED remains on.

**MALFUNCTION**

A/C ID LED remains on.

**CORRECTIVE ACTION**

- 1. Check resistance between:**
  - **P4-A and P4-C (122 ohms - 126 ohms)**
  - **P4-B and P4-C (655 ohms - 659 ohms)**
  - a. If resistance is as specified, replace ECSU (WP 1589 00). Go to **2**.
  - b. If resistance is not as specified, repair/replace wiring (WP 1747 00). Go to **2**.
- 2. Procedure completed.**

**PILOT'S OR COPILOT'S NO. 1 LED REMAINS ON****SYMPTOM**

Pilot's or copilot's No. 1 LED remains on.

**MALFUNCTION**

Pilot's or copilot's No. 1 LED remains on.

**CORRECTIVE ACTION****1. Is pilot's No. 1 LED on?**

- a. If pilot's No. 1 LED is on, go to **2**.
- b. If pilot's No. 1 LED is not on, go to **4**.

**2. Place upper console BATT switch to OFF. Go to 3.****3. Check resistance between:**

- **P2-A and P6-A (Less than 2 ohms)**
- **P2-B and P6-B (Less than 2 ohms)**
- **P2-backshell and P6-backshell (Less than 2 ohms)**

a. If resistance is as specified, replace pilot's gas generator in forward air bag module (WP 1589 00). Go to **6**.

b. If resistance is not as specified, repair/replace wiring (WP 1747 00). Go to **6**.

**4. Place upper console BATT switch to OFF. Go to 5.****5. Check resistance between:**

- **P3-A and P8-A (Less than 2 ohms)**
- **P3-B and P8-B (Less than 2 ohms)**
- **P3-backshell and P8-backshell (Less than 2 ohms)**

a. If resistance is as specified, replace copilot's gas generator in forward air bag module (WP 1589 00). Go to **6**.

b. If resistance is not as specified, repair/replace wiring (WP 1747 00). Go to **6**.

**6. Procedure completed.****PILOT'S OR COPILOT'S NO. 2 LED REMAINS ON****SYMPTOM**

Pilot's or copilot's No. 2 LED remains on.

**MALFUNCTION**

Pilot's or copilot's No. 2 LED remains on.

**CORRECTIVE ACTION****1. Is pilot's No. 2 LED on?**

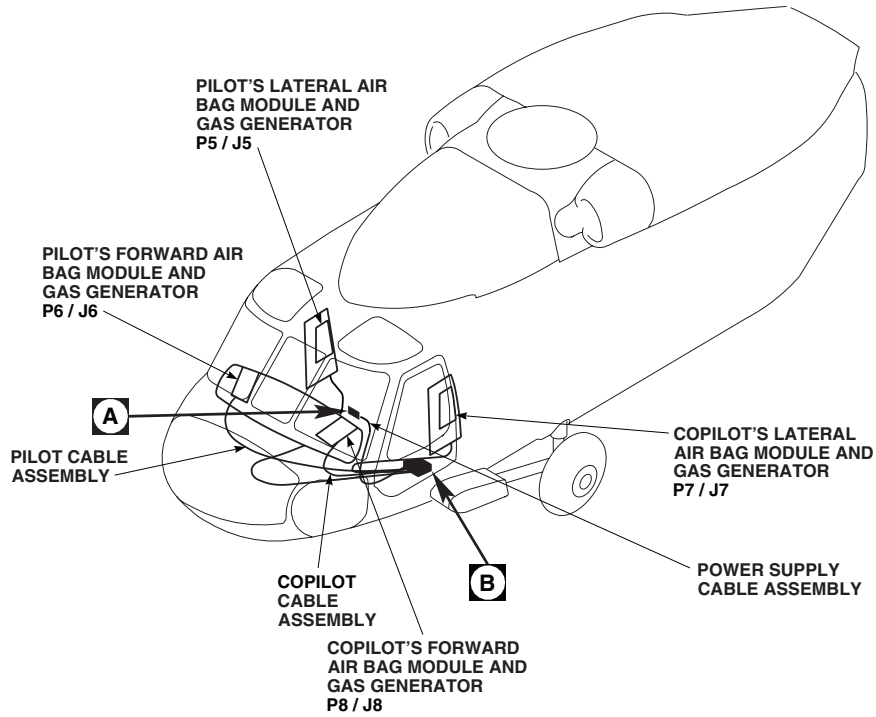
- a. If pilot's No. 2 LED is on, go to **2**.

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**PILOT'S OR COPILOT'S NO. 2 LED REMAINS ON - Continued**

- b. If pilot's No. 2 LED is not on, go to **4**.
- 2. Place upper console BATT switch to OFF. Go to 3.**
- 3. Check resistance between:**
  - **P2-C and P5-A (Less than 2 ohms)**
  - **P2-D and P5-B (Less than 2 ohms)**
  - **P2-backshell and P5-backshell (Less than 2 ohms)**
  - a. If resistance is as specified, replace pilot's gas generator in lateral air bag module (WP 1589 00). Go to **6**.
  - b. If resistance is not as specified, repair/replace wiring (WP 1747 00). Go to **6**.
- 4. Place upper console BATT switch to OFF. Go to 5.**
- 5. Check resistance between:**
  - **P3-C and P7-A (Less than 2 ohms)**
  - **P3-D and P7-B (Less than 2 ohms)**
  - **P3-backshell and P7-backshell (Less than 2 ohms)**
  - a. If resistance is as specified, replace copilot's gas generator in lateral air bag module (WP 1589 00). Go to **6**.
  - b. If resistance is not as specified, repair/replace wiring (WP 1747 00). Go to **6**.
- 6. Procedure completed.**

## LOCATION

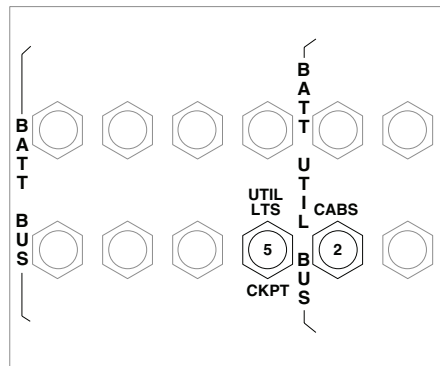


| TERMINAL BOARD /<br>DISCONNECT PLUG /<br>RECEPTACLE | LOCATION /<br>CONNECTION POINT                           |
|-----------------------------------------------------|----------------------------------------------------------|
| P2 / J2<br>P3 / J3<br>P4 / J4                       | ELECTRONIC CRASH<br>SENSOR UNIT                          |
| P5 / J5                                             | PILOT'S LATERAL AIR BAG<br>MODULE AND GAS<br>GENERATOR   |
| P6 / J6                                             | PILOT'S FORWARD AIR<br>BAG MODULE AND GAS<br>GENERATOR   |
| P7 / J7                                             | COPILOT'S LATERAL AIR<br>BAG MODULE AND GAS<br>GENERATOR |
| P8 / J8                                             | COPILOT'S FORWARD AIR<br>BAG MODULE AND GAS<br>GENERATOR |
| P107 / J107                                         | LOWER CONSOLE CIRCUIT<br>BREAKER PANEL<br>DISCONNECT     |

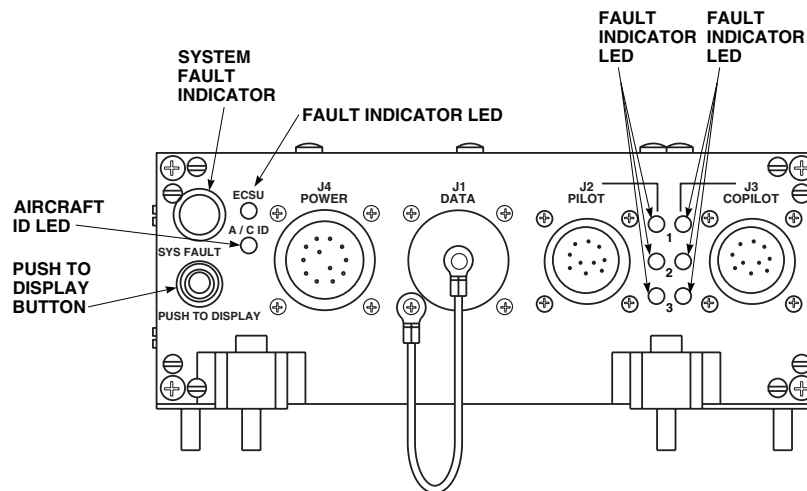
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Figure 1. Cockpit Air Bag System Location Diagram. (Sheet 1 of 2)

LOCATION - Continued



LOWER CONSOLE CIRCUIT BREAKER PANEL



ELECTRONIC CRASH SENSOR UNIT (ECSU)

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Figure 1. Cockpit Air Bag System Location Diagram. (Sheet 2 of 2)

## SCHEMATICS

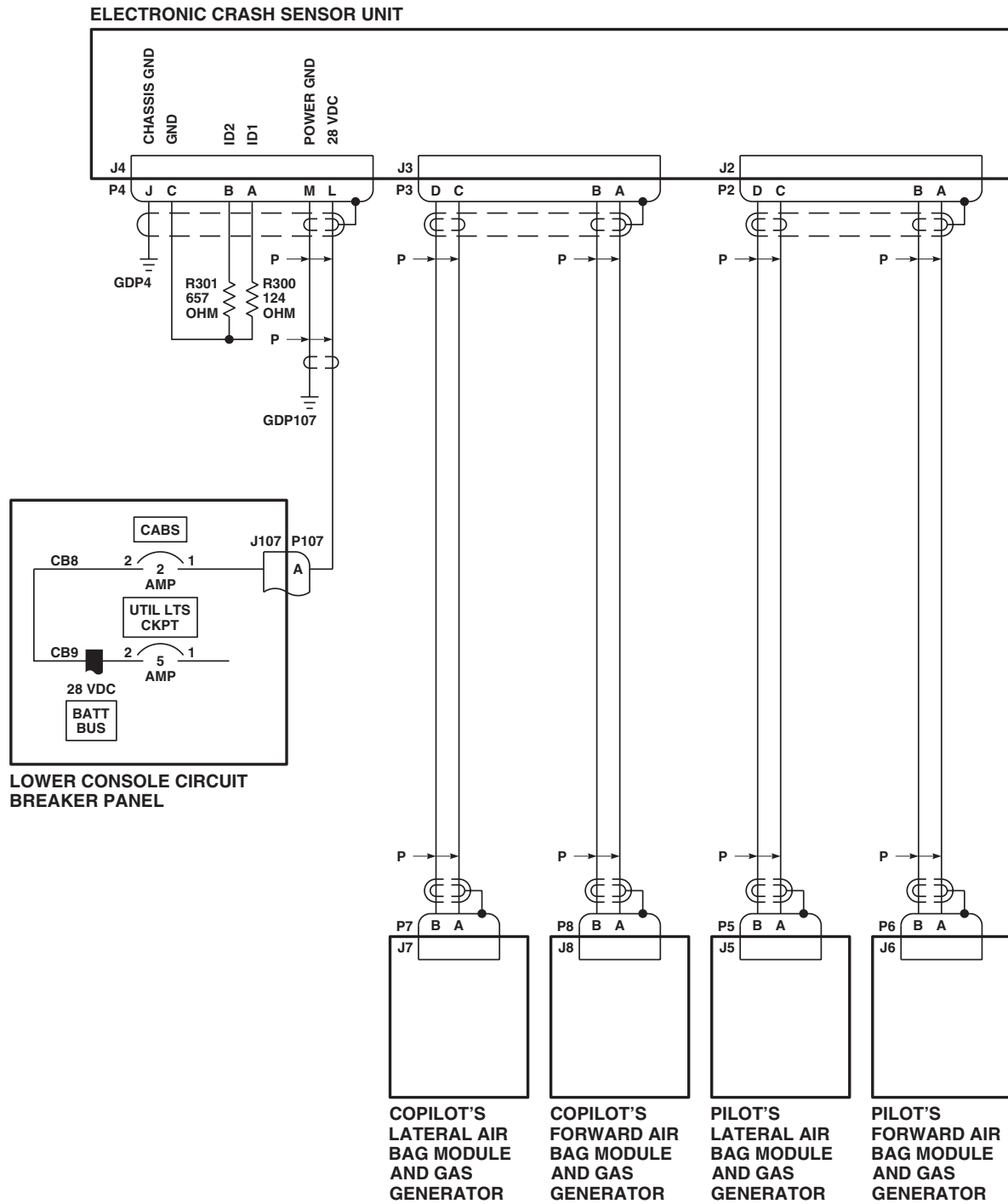


Figure 2. Cockpit Air Bag System Schematic Diagram.

END OF WORK PACKAGE

## UNIT LEVEL

## EMERGENCY EQUIPMENT

## MICROCLIMATE COOLING UNIT AND MASK BLOWER ASSEMBLIES

## INITIAL SETUP:

**Test Equipment**

Multimeter, AN/PSM-45A

WP 1562 00

WP 1563 00

WP 1564 00

**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01

WP 1565 00

Electrical Repairer Toolkit, SC 5180-99-B06

WP 1566 00

WP 1571 00

**Personnel Required**

Aircraft Electrician MOS 15F (1)

WP 1574 00

UH-60 Helicopter Repairer MOS 15T (1)

WP 1685 00

WP 1747 00

**References**

TM 1-1520-237-10

WP 0104 00

WP 1561 00

**Equipment Condition**External Electrical Power Connected, WP 1685 00  
or APU Operational, TM 1-1520-237-10

## SPECIAL INSTRUCTIONS

## NOTE

- Operational/troubleshooting procedure for all microclimate cooling units (MCUs) are the same.
- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- See Figure 1 for component location diagram and Figure 2 for schematic diagrams as an aid in operational/troubleshooting.

The MCU and mask blower assemblies operational/troubleshooting procedure is subdivided as follows:

- Electrical
- Mechanical

## ELECTRICAL OPERATIONAL/TROUBLESHOOTING PROCEDURE

### PROCEDURE

#### Step

1. Make sure the following circuit breakers are pulled out:

| CIRCUIT BREAKER | CIRCUIT BREAKER PANEL |
|-----------------|-----------------------|
| PLT MCU         | Mission Readiness     |
| CPLT MCU        | Mission Readiness     |
| CREW 1 MCU      | Mission Readiness     |
| CREW 2 MCU      | Mission Readiness     |
| MASK BLOWERS    | Pilot's               |

2. Turn on electrical power.
3. Connect microclimate cooling garment (MCG) to pilot's liquid quick disconnect (L-QDC).
4. Push in PLT, CPLT CREW 1, CREW 2 MCU circuit breakers on mission readiness panel.
5. Turn MCU controls on bypass control assembly (BCA) to "FULL".

#### Indication/Condition

MCUs shall run.

#### Corrective Action

1. If #1 MCU is not as specified, go to #1 MCU DOES NOT RUN, in this work package.
2. If #2 MCU is not as specified, go to #2 MCU DOES NOT RUN, in this work package.
3. If #3 MCU is not as specified, go to #3 MCU DOES NOT RUN, in this work package.
4. If #4 MCU is not as specified, go to #4 MCU DOES NOT RUN, in this work package.

#### Step

6. Connect mask blower to pilot's mask blower disconnect J1211.
7. Push in MASK BLOWERS circuit breaker on mission readiness circuit breaker panel.

#### Indication/Condition

Pilot's mask blower shall power up.

#### Corrective Action

If pilot's mask blower is not as specified, go to PILOT'S MASK BLOWER DOES NOT POWER UP, in this work package.



---

**ELECTRICAL OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued****Step**

8. Repeat steps 6. and 7. for copilot's and two crew member's mask blowers.

**Indication/Condition**

Mask blowers shall power up.

**Corrective Action**

1. If copilot's mask blower is not as specified, go to COPILOT'S MASK BLOWER DOES NOT POWER UP, in this work package.
2. If crew mask blower 1 is not as specified, go to CREW MASK BLOWER 1 DOES NOT POWER UP, in this work package.
3. If crew mask blower 2 is not as specified, go to CREW MASK BLOWER 2 DOES NOT POWER UP, in this work package.

**MECHANICAL OPERATIONAL/TROUBLESHOOTING PROCEDURE****PROCEDURE****Step**

1. Adjust pilot's BCA to check cooling effect.

**Indication/Condition**

Coolant flows and MCS cools per BCA range settings.

**Corrective Action**

If Indication/Condition is not as specified, go to COOLANT DOES NOT FLOW AND MCS DOES NOT COOL PER BCA RANGE SETTINGS, in this work package.

**Step**

2. Disconnect pilot's MCG from L-QDC.
3. Disconnect mask blower from connector.
4. Turn off electrical power.

**Indication/Condition**

Electrical power shall be off.

**Corrective Action**

None Required

**#1 MCU DOES NOT RUN****SYMPTOM**

#1 MCU does not run.

**MALFUNCTION**

#1 MCU does not run.

**CORRECTIVE ACTION****1. Check for 28 vdc between P1202-C and ground.**

- a. If voltage is as specified, go to **2**.
- b. If voltage is not as specified, go to **5**.

**2. Check continuity between P1202-A and ground.**

- a. If continuity is present, replace MCU (WP 1565 00), then retest.
- b. If trouble remains, replace MCU cooling hose ((WP 1562 00), (WP 1563 00) or (WP 1564 00). Go to **9**.
- c. If continuity is not present, go to **3**.

**3. Check continuity between J1200-A and ground.**

- a. If continuity is present, repair/replace wiring between P1202-A and P1200-A (WP 1747 00). Go to **9**.
- b. If continuity is not present, go to **4**.

**4. Check continuity between GND BLK 1-F and ground.**

- a. If continuity is present, repair/replace wiring between J1200-A and GND BLK 1-F (WP 1747 00). Go to **9**.
- b. If continuity is not present, replace GND BLK 1 (WP 1574 00). Go to **9**.

**5. Check for 28 vdc between J1200-C and ground.**

- a. If voltage is as specified, repair/replace wiring between P1202-C and P1200-C (WP 1747 00). Go to **9**.
- b. If voltage is not as specified, go to **6**.

**6. Check for 28 vdc between PLT MCU circuit breaker load side and ground.**

- a. If voltage is as specified, repair/replace wiring between PLT MCU circuit breaker terminal and J1200-C (WP 1747 00). Go to **9**.
- b. If voltage is not as specified, go to **7**.

**7. Check for 28 vdc between PLT MCU circuit breaker bus side and ground.**

- a. If voltage is as specified, replace PLT MCU circuit breaker (WP 0824 00). Go to **9**.
- b. If voltage is not as specified, go to **8**.

**8. Check for 28 vdc between TB10-4 and ground.**

- a. If voltage is as specified, repair/replace wiring between TB10-4 and PLT MCU circuit breaker bus side (WP 1747 00). Go to **9**.

---

**#1 MCU DOES NOT RUN - Continued**

- b. If voltage is not as specified, troubleshoot DC electrical system (WP 0102 00 or **HH-60A HH-60L** WP 0123 00 ). Go to 9.

**9. Procedure completed.****#2 MCU DOES NOT RUN****SYMPTOM**

#2 MCU does not run.

**MALFUNCTION**

#2 MCU does not run.

**CORRECTIVE ACTION****1. Check for 28 vdc between P1204-C and ground.**

- a. If voltage is as specified, go to 2.
- b. If voltage is not as specified, go to 5.

**2. Check continuity between P1204-A and ground.**

- a. If continuity is present, replace MCU (WP 1565 00), then retest.
- b. If trouble remains, replace MCU cooling hose (WP 1562 00), (WP 1563 00) or (WP 1564 00). Go to 9.
- c. If continuity is not present, go to 3.

**3. Check continuity between J1201-A and ground.**

- a. If continuity is present, repair/replace wiring between P1204-A and P1201-A (WP 1747 00). Go to 9.
- b. If continuity is not present, go to 4.

**4. Check continuity between GND BLK 1-C and ground.**

- a. If continuity is present, repair/replace wiring between J1201-A and GND BLK 1-C (WP 1747 00). Go to 9.
- b. If continuity is not present, replace GND BLK 1 (WP 1574 00). Go to 9.

**5. Check for 28 vdc between J1201-C and ground.**

- a. If voltage is as specified, repair/replace wiring between P1204-C and P1201-C (WP 1747 00). Go to 9.
- b. If voltage is not as specified, go to 6.

**6. Check for 28 vdc between CPLT MCU circuit breaker load side and ground.**

- a. If voltage is as specified, repair/replace wiring between CPLT MCU circuit breaker terminal and J1201-C (WP 1747 00). Go to 9.
- b. If voltage is not as specified, go to 7.

**7. Check for 28 vdc between CPLT MCU circuit breaker bus side and ground.**

- a. If voltage is as specified, replace CPLT MCU circuit breaker (WP 0824 00). Go to 9.
- b. If voltage is not as specified, go to 8.

---

**#2 MCU DOES NOT RUN** - Continued**8. Check for 28 vdc between TB11-1 and ground.**

- a. If voltage is as specified, repair/replace wiring between TB11-1 and CPLT MCU circuit breaker bus side (WP 1747 00). Go to 9.
- b. If voltage is not as specified, troubleshoot DC electrical system (WP 0102 00 or **HH-60A HH-60L** WP 0123 00 ). Go to 9.

**9. Procedure completed.****#3 MCU DOES NOT RUN****SYMPTOM**

#3 MCU does not run.

**MALFUNCTION**

#3 MCU does not run.

**CORRECTIVE ACTION****1. Check for 28 vdc between P1203-C and ground.**

- a. If voltage is as specified, go to 2.
- b. If voltage is not as specified, go to 5.

**2. Check continuity between P1203-A and ground.**

- a. If continuity is present, replace MCU (WP 1565 00), then retest.
- b. If trouble remains, replace MCU cooling hose (WP 1562 00), (WP 1563 00) or (WP 1564 00). Go to 9.
- c. If continuity is not present, go to 3.

**3. Check continuity between J1200-E and ground.**

- a. If continuity is present, repair/replace wiring between P1203-A and P1200-E (WP 1747 00). Go to 9.
- b. If continuity is not present, go to 4.

**4. Check continuity between GND BLK 1-E and ground.**

- a. If continuity is present, repair/replace wiring between J1200-E and GND BLK 1-E (WP 1747 00). Go to 9.
- b. If continuity is not present, replace GND BLK 1 (WP 1574 00). Go to 9.

**5. Check for 28 vdc between J1200-D and ground.**

- a. If voltage is as specified, repair/replace wiring between P1203-C and P1200-D (WP 1747 00). Go to 9.
- b. If voltage is not as specified, go to 6.

**6. Check for 28 vdc between CREW 1 MCU circuit breaker load side and ground.**

- a. If voltage is as specified, repair/replace wiring between CREW 1 MCU circuit breaker terminal and J1200-D (WP 1747 00). Go to 9.
- b. If voltage is not as specified, go to 7.

**7. Check for 28 vdc between CREW 1 MCU circuit breaker bus side and ground.**

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**#3 MCU DOES NOT RUN - Continued**

- a. If voltage is as specified, replace CREW 1 MCU circuit breaker (WP 0824 00). Go to 9.
- b. If voltage is not as specified, go to 8.

**8. Check for 28 vdc between TB10-4 and ground.**

- a. If voltage is as specified, repair/replace wiring wiring between TB10-4 and CREW 1 MCU circuit breaker bus side (WP 1747 00). Go to 9.
- b. If voltage is not as specified, troubleshoot DC electrical system (WP 0102 00 or **HH-60A HH-60L** WP 0123 00 ). Go to 9.

**9. Procedure completed.****#4 MCU DOES NOT RUN****SYMPTOM**

#4 MCU does not run.

**MALFUNCTION**

#4 MCU does not run.

**CORRECTIVE ACTION****1. Check for 28 vdc between P1205-C and ground.**

- a. If voltage is as specified, go to 2.
- b. If voltage is not as specified, go to 5.

**2. Check continuity between P1205-A and ground.**

- a. If continuity is present, replace MCU (WP 1565 00), then retest.
- b. If trouble remains, replace MCU cooling hose ((WP 1562 00), (WP 1563 00) or (WP 1564 00). Go to 9.
- c. If continuity is not present, go to 3.

**3. Check continuity between J1201-E and ground.**

- a. If continuity is present, repair/replace wiring between P1205-A and P1201-E (WP 1747 00). Go to 9.
- b. If continuity is not present, go to 4.

**4. Check continuity between GND BLK 1-A and ground.**

- a. If continuity is present, repair/replace wiring between J1201-E and GND BLK 1-A (WP 1747 00). Go to 9.
- b. If continuity is not present, replace GND BLK 1 (WP 1574 00). Go to 9.

**5. Check for 28 vdc between J1201-D and ground.**

- a. If voltage is as specified, repair/replace wiring between P1205-C and P1201-D (WP 1747 00). Go to 9.
- b. If voltage is not as specified, go to 6.

**6. Check for 28 vdc between CREW 2 MCU circuit breaker load side and ground.**

- a. If voltage is as specified, repair/replace wiring between CREW 2 MCU circuit breaker terminal and J1201-D (WP 1747 00). Go to 9.

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**#4 MCU DOES NOT RUN - Continued**

- b. If voltage is not as specified, go to 7.
- 7. Check for 28 vdc between CREW 2 MCU circuit breaker bus side and ground.**
  - a. If voltage is as specified, replace CREW 2 MCU circuit breaker (WP 0824 00). Go to 9.
  - b. If voltage is not as specified, go to 8.
- 8. Check for 28 vdc between TB11-1 and ground.**
  - a. If voltage is as specified, repair/replace wiring between TB11-1 and CREW 2 MCU circuit breaker bus side (WP 1747 00). Go to 9.
  - b. If voltage is not as specified, troubleshoot DC electrical system (WP 0102 00 or **HH-60A HH-60L** WP 0123 00 ). Go to 9.
- 9. Procedure completed.**

**PILOT'S MASK BLOWER DOES NOT POWER UP****SYMPTOM**

Pilot's mask blower does not power up.

**MALFUNCTION**

Pilot's mask blower does not power up.

**CORRECTIVE ACTION****NOTE**

The 3.8 vdc output of the power supply can have as much as  $\pm 0.2$  variance.

- 1. Check for 3.8 vdc between J1211-1 and ground.**
  - a. If voltage is as specified, go to 2.
  - b. If voltage is not as specified, go to 4.
- 2. Check continuity between J1211-2 and ground.**
  - a. If continuity is present, replace mask blower (WP 1561 00). Go to 12.
  - b. If continuity is not present, go to 3.
- 3. Check continuity between J1206-B and ground.**
  - a. If continuity is present, repair/replace wiring between P1206-B and J1211-2 (WP 1747 00). Go to 12.
  - b. If continuity is not present, go to 9.
- 4. Check for 3.8 vdc between J1206-A and ground.**
  - a. If voltage is as specified, repair/replace wiring between J1206-A and P1210-C (WP 1747 00). Go to 12.
  - b. If voltage is not as specified, go to 5.
- 5. Check for 28 vdc between P1210-A and ground.**
  - a. If voltage is as specified, go to 9.
  - b. If voltage is not as specified, go to 6.

---

**PILOT'S MASK BLOWER DOES NOT POWER UP - Continued****6. Check for 28 vdc between MASK BLOWER circuit breaker load side and ground.**

- a. If voltage is as specified, repair/replace wiring between P1210-A and MASK BLOWER circuit breaker load side (WP 1747 00). Go to 12.
- b. If voltage is not as specified, go to 7.

**7. Check for 28 vdc between MASK BLOWER circuit breaker bus side and ground.**

- a. If voltage is as specified, replace MASK BLOWER circuit breaker (WP 0083 00). Go to 12.
- b. If voltage is not as specified, go to 8.

**8. Check for 28 vdc between NO. 2 SERVO CONTR circuit breaker bus side and ground.**

- a. If voltage is as specified, repair/replace wiring between NO. 2 SERVO CONTR circuit breaker bus side and MASK BLOWER circuit breaker bus side (WP 1747 00). Go to 12.
- b. If voltage is not as specified, troubleshoot DC electrical system (WP 0102 00 or **HH-60A HH-60L** WP 0123 00 ). Go to 12.

**9. Check for 2.5k ohms resistance between P1210-B and P1210-D.**

- a. If resistance is as specified, go to 10.
- b. If resistance is not as specified, replace R1 (WP 1574 00). Go to 12.
- c. If trouble remains, repair/replace wiring (WP 1747 00) between, as required, then go to 12:
  - P1210-B and ground
  - P1210-D and ground

**10. Check continuity between P1210-B and ground.**

- a. If continuity is present, go to 11.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to 12.

**11. Check continuity between P1210-C and J1211-1.**

- a. If continuity is present, replace power converter (distribution box) (WP 1574 00). Go to 12.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to 12.

**12. Procedure completed.****COPILOT'S MASK BLOWER DOES NOT POWER UP****SYMPTOM**

Copilot's mask blower does not power up.

**MALFUNCTION**

Copilot's mask blower does not power up.

**CORRECTIVE ACTION****NOTE**

The 3.8 vdc output of the power supply can have as much as  $\pm 0.2$  variance.

**1. Check for 3.8 vdc between J1212-1 and ground.**

---

**COPILOT'S MASK BLOWER DOES NOT POWER UP - Continued**

- a. If voltage is as specified, replace mask blower (WP 1561 00). Go to **3**.
- b. If voltage is not as specified, go to **2**.
- 2. Check continuity between J1212-2 and ground.**
  - a. If continuity is present, repair/replace wiring between J1212-1 and P1210-C (WP 1747 00). Go to **3**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **3**.
- 3. Procedure completed.**

**CREW MASK BLOWER 1 DOES NOT POWER UP****SYMPTOM**

Crew mask blower 1 does not power up.

**MALFUNCTION**

Crew mask blower 1 does not power up.

**CORRECTIVE ACTION****NOTE**

The 3.8 vdc output of the power supply can have as much as  $\pm 0.2$  variance.

- 1. Check for 3.8 vdc between J1213-1 and ground.**
  - a. If voltage is as specified, replace mask blower (WP 1561 00). Go to **3**.
  - b. If voltage is not as specified, go to **2**.
- 2. Check continuity between J1213-2 and ground.**
  - a. If continuity is present, repair/replace wiring between J1213-1 and P1210-C (WP 1747 00). Go to **3**.
  - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **3**.
- 3. Procedure completed.**

**CREW MASK BLOWER 2 DOES NOT POWER UP****SYMPTOM**

Crew mask blower 2 does not power up.

**MALFUNCTION**

Crew mask blower 2 does not power up.

**CORRECTIVE ACTION****NOTE**

The 3.8 vdc output of the power supply can have as much as  $\pm 0.2$  variance.

- 1. Check for 3.8 vdc between J1214-1 and ground.**
  - a. If voltage is as specified, replace mask blower (WP 1561 00). Go to **3**.
  - b. If voltage is not as specified, go to **2**.



---

**CREW MASK BLOWER 2 DOES NOT POWER UP - Continued****2. Check continuity between J1214-2 and ground.**

- a. If continuity is present, repair/replace wiring between J1214-1 and P1210-C (WP 1747 00). Go to **3**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **3**.

**3. Procedure completed.****COOLANT DOES NOT FLOW AND MCS DOES NOT COOL PER BCA RANGE SETTINGS****SYMPTOM**

Coolant does not flow and MCS does not cool per BCA range settings.

**MALFUNCTION**

Coolant does not flow and MCS does not cool per BCA range settings.

**CORRECTIVE ACTION****1. Verify all cooling hose connectors are properly mated.**

- a. If unit cools properly and no leaks are visible, system is operational. Go to **12**.
- b. If unit does not cool properly, go to **2**.

**2. Perform MCU charging procedure (WP 1571 00). Go to 3.****3. Check coolant flows freely through MCU, coolant lines, and no leaks are visible.**

- a. If coolant flows freely through MCU, coolant lines, and no leaks are visible, go to **10**.
- b. If coolant does not flow freely through MCU and coolant lines, go to **4**.

**4. Is coolant leaking?**

- a. If no leaks are visible, go to **9**.
- b. If leaks are visible, go to **5**.

**5. Is coolant leaking from L-QDC?**

- a. If coolant is leaking from L-QDC, replace L-QDC (WP 1565 00), then retest.
- b. If coolant is not leaking from L-QDC, go to **6**.

**6. Is coolant leaking from MCU metal fittings?**

- a. If coolant is leaking from MCU metal fittings, replace rubber o-rings on MCU metal fittings (WP 1565 00), then retest.
- b. If coolant is not leaking from MCU metal fittings, go to **7**.

**7. Is coolant leaking from bypass control?**

- a. If coolant is leaking from bypass control, replace entire cooling hose section (WP 1562 00), (WP 1563 00) or (WP 1564 00), then retest.
- b. If coolant is not leaking from bypass control, go to **8**.

**8. Is coolant leaking from damaged tubing?**

- a. If coolant is leaking from damaged tubing, replace cooling hose (WP 1562 00), (WP 1563 00) or (WP 1564 00), then retest.

**9. Check MCU coolant filter is not clogged.**

---

**COOLANT DOES NOT FLOW AND MCS DOES NOT COOL PER BCA RANGE SETTINGS - Continued**

- a. If filter is clogged, replace filter (WP 1566 00). Go to **12**.
- b. If filter is not clogged, replace cooling hose (WP 1562 00), (WP 1563 00) or (WP 1564 00). Go to **12**.
- c. If trouble remains, go to **10**.

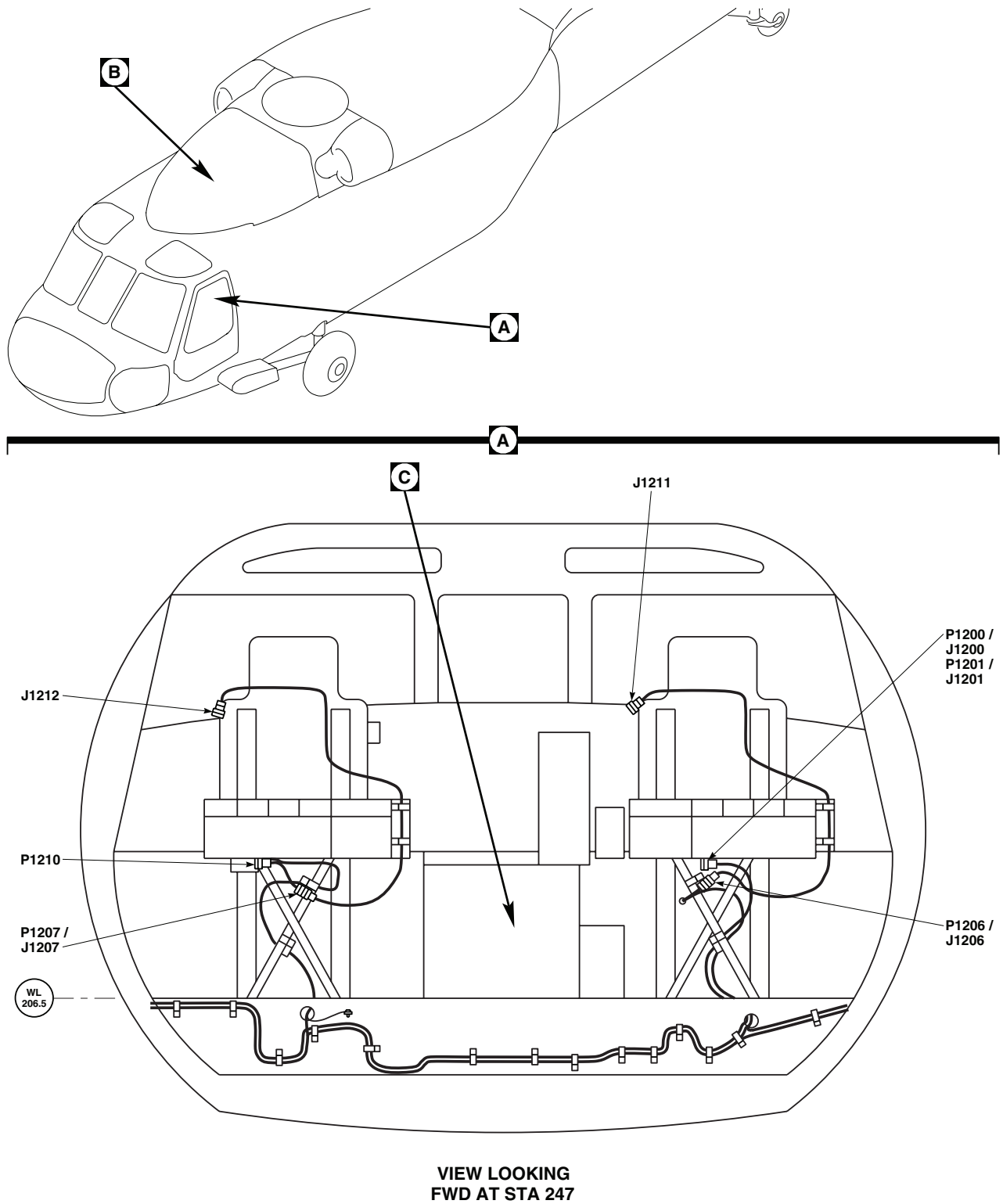
**10. Perform charging procedures on MCG (WP 1571 00). Go to 11.**

**11. Check coolant flows freely through MCG and MCG cools properly.**

- a. If coolant flows freely and MCG cools properly, system is operational. Go to **12**.
- b. If coolant does not flow freely and MCG does not cool properly, repair or replace MCG, then retest.
- c. If trouble remains, replace MCU (WP 1565 00), then retest.

**12. Procedure completed.**

LOCATION



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Figure 1. Microclimate Cooling Unit (MCU) and Mask Blower Assemblies Location Diagram. (Sheet 1 of 4)

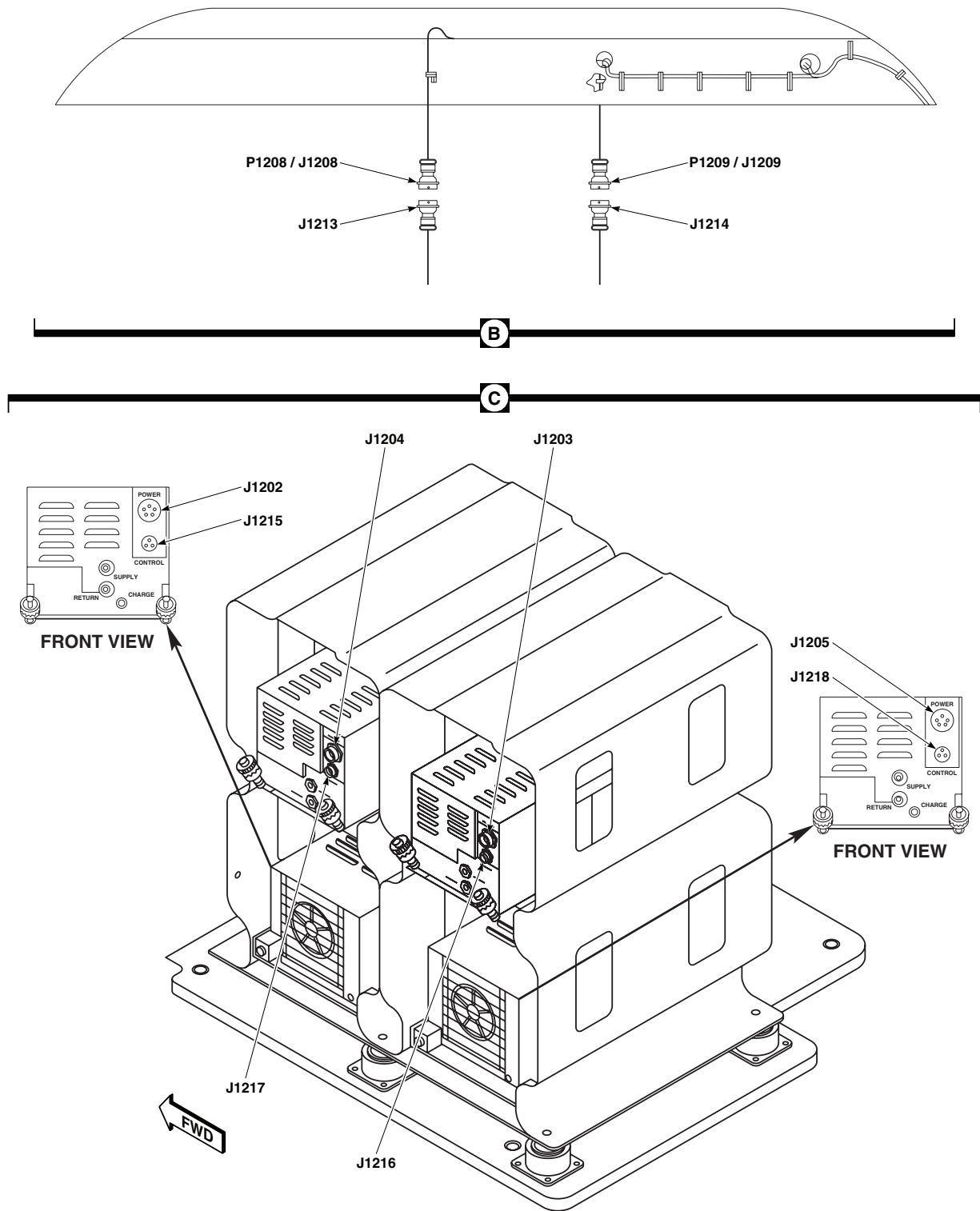
**LOCATION** - Continued

| <b>TERMINAL BOARD /<br/>DISCONNECT PLUG /<br/>RECEPTACLE</b> | <b>LOCATION /<br/>CONNECTION POINT</b>              |
|--------------------------------------------------------------|-----------------------------------------------------|
| <b>MICROCLIMATE<br/>COOLING UNIT (MCU)</b>                   |                                                     |
| P1200 / J1200                                                | MCU 1 & MCU 3<br>DISCONNECT                         |
| P1201 / J1201                                                | MCU 2 & MCU 4<br>DISCONNECT                         |
| J1202                                                        | MCU 1 POWER<br>DISCONNECT                           |
| J1203                                                        | MCU 3 POWER<br>DISCONNECT                           |
| J1204                                                        | MCU 2 POWER<br>DISCONNECT                           |
| J1205                                                        | MCU 4 POWER<br>DISCONNECT                           |
| J1215                                                        | MCU 1 CONTROL<br>DISCONNECT                         |
| J1216                                                        | MCU 3 CONTROL<br>DISCONNECT                         |
| J1217                                                        | MCU 2 CONTROL<br>DISCONNECT                         |
| J1218                                                        | MCU 4 CONTROL<br>DISCONNECT                         |
| <b>MASK BLOWER</b>                                           |                                                     |
| P1206 / J1206                                                | PLT MASK BLOWER<br>POWER DISCONNECT                 |
| P1207 / J1207                                                | CPLT MASK BLOWER<br>POWER DISCONNECT                |
| P1208 / J1208                                                | CREW 1 MASK BLOWER<br>POWER DISCONNECT              |
| P1209 / J1209                                                | CREW 2 MASK BLOWER<br>POWER DISCONNECT              |
| P1210                                                        | MASK BLOWER POWER<br>DISTRIBUTION BOX<br>DISCONNECT |
| J1211                                                        | PLT MASK BLOWER<br>DISCONNECT                       |
| J1212                                                        | CPLT MASK BLOWER<br>DISCONNECT                      |
| J1213                                                        | CREW 1 MASK BLOWER<br>DISCONNECT                    |
| J1214                                                        | CREW 2 MASK BLOWER<br>DISCONNECT                    |

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Figure 1. Microclimate Cooling Unit (MCU) and Mask Blower Assemblies Location Diagram. (Sheet 2 of 4)

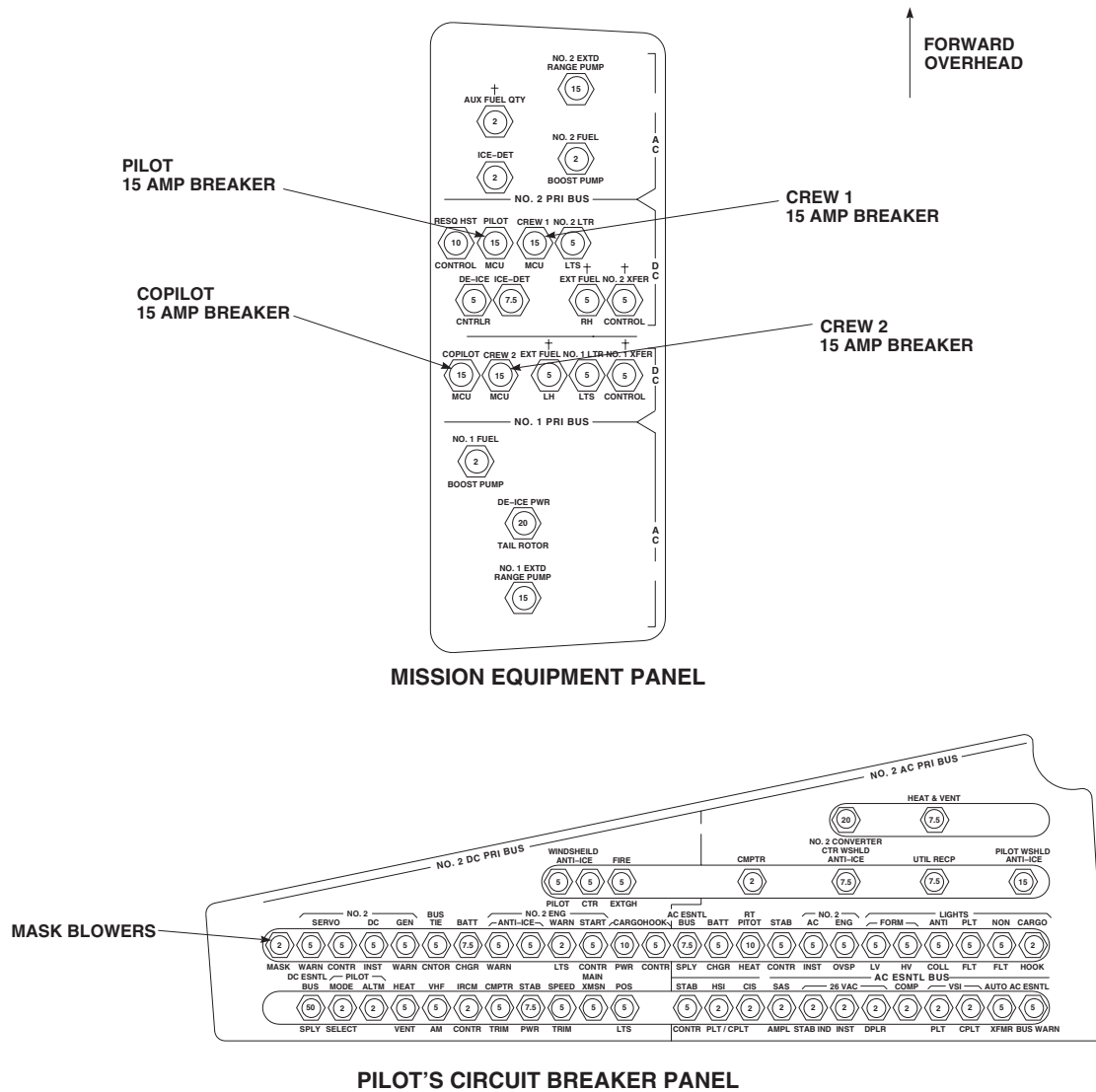
LOCATION - Continued



GF0777  
SA

Figure 1. Microclimate Cooling Unit (MCU) and Mask Blower Assemblies Location Diagram. (Sheet 3 of 4)

LOCATION - Continued



GF0773  
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Figure 1. Microclimate Cooling Unit (MCU) and Mask Blower Assemblies Location Diagram. (Sheet 4 of 4)

## P/O MISSION READINESS CIRCUIT BREAKER PANEL



0172 00-17

## SCHEMATICS - Continued

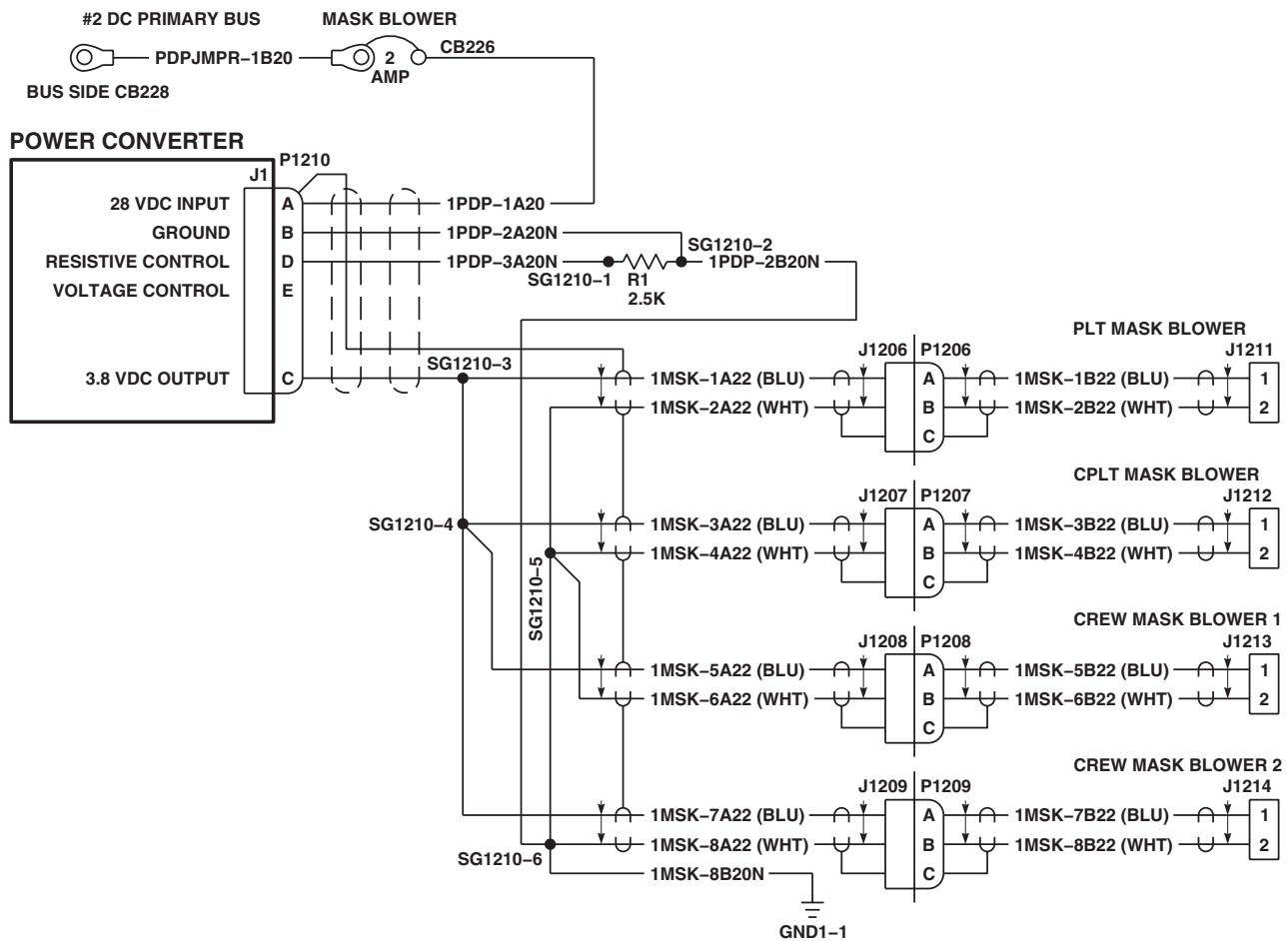
GF0774  
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Figure 2. Microclimate Cooling Unit (MCU) and Mask Blower Assemblies Schematic Diagram. (Sheet 2 of 3)



## SCHEMATICS - Continued

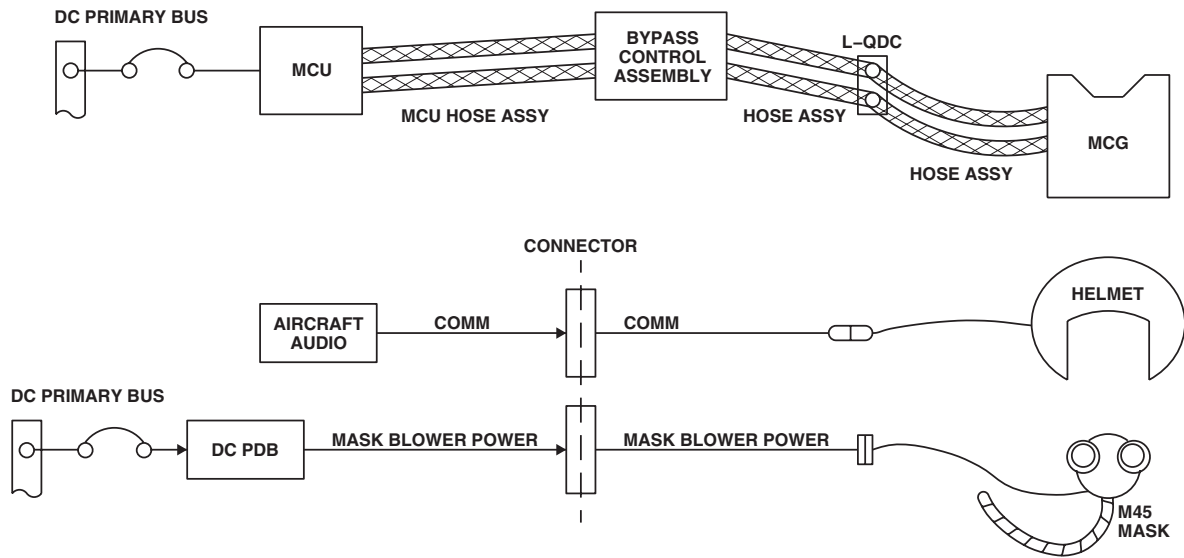
GF0810  
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Figure 2. Microclimate Cooling Unit (MCU) and Mask Blower Assemblies Schematic Diagram. (Sheet 3 of 3)

END OF WORK PACKAGE



By Order of the Secretary of the Army:

Official:

A handwritten signature in black ink, appearing to read "Joyce E. Morrow". The signature is fluid and cursive, with the first name "Joyce" and last name "Morrow" clearly distinguishable.

JOYCE E. MORROW

*Administrative Assistant to the  
Secretary of the Army*

0608615

PETER J. SCHOOMAKER  
*General, United States Army  
Chief of Staff*

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Subject: DA Form 2028

1. From: Joe Smith
2. Unit: home
3. Address: 4300 Park
4. City: Hometown
5. St: MO
6. Zip: 77777
7. Date Sent: 19--OCT--93
8. Pub no: 55--2840--229--23
9. Pub Title: TM
10. Publication Date: 04--JUL--85
11. Change Number: 7
12. Submitter Rank: MSG
13. Submitter FName: Joe
14. Submitter MName: T
15. Submitter LName: Smith
16. Submitter Phone: 123-123-1234
17. Problem: 1
18. Page: 2
19. Paragraph: 3
20. Line: 4
21. NSN: 5
22. Reference: 6
23. Figure: 7
24. Table: 8
25. Item: 9
26. Total: 123
27. Text:

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|                                                                                                                                                                                           |          |          |                       |                                                                                                                   |            |          |                                    |                    |
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|                                                                                                                                                                                           |          |          |                       |                                                                                                                   |            |          |                                    |                    |
| PART III – REMARKS (Any general remarks or recommendations, or suggestions for improvement of publications and blank forms. Additional blank sheets may be used if more space is needed.) |          |          |                       |                                                                                                                   |            |          |                                    |                    |
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|                                                                                                                                                                                        |             |                |               |               |                                                |                                                                                                                  |           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------------|---------------|---------------|------------------------------------------------|------------------------------------------------------------------------------------------------------------------|-----------|
| <b>RECOMMENDED CHANGES TO PUBLICATIONS AND<br/>BLANK FORMS</b><br><br><small>For use of this form, see AR 25-30; the proponent agency is ODISC4.</small>                               |             |                |               |               |                                                | Use Par      reverse) for Repair Parts and Special Tool Lists (RPSTL) and Supply Catalogs/Supply Manuals (SC/SM) | DATE      |
| TO: (Forward to proponent of publication or form)(Include ZIP Code)<br>Commander, U.S. Army Aviation and Missile Command<br>ATTN: AMSAM-MMC-MA-NP<br>Redstone Arsenal, AL 35898 - 5000 |             |                |               |               |                                                | FROM: (Activity and location)(Include ZIP Code)                                                                  |           |
| <b>PART 1 – ALL PUBLICATIONS (EXCEPT RPSTL AND SC/SM) AND BLANK FORMS</b>                                                                                                              |             |                |               |               |                                                |                                                                                                                  |           |
| PUBLICATION/FORM NUMBER                                                                                                                                                                |             |                |               |               |                                                | DATE                                                                                                             | TITLE     |
| ITEM<br>NO.                                                                                                                                                                            | PAGE<br>NO. | PARA-<br>GRAPH | LINE<br>NO. * | FIGURE<br>NO. | TABLE<br>NO.                                   | RECOMMENDED CHANGES AND REASON                                                                                   |           |
|                                                                                                                                                                                        |             |                |               |               |                                                |                                                                                                                  |           |
| <small>* Reference to line numbers within the paragraph or subparagraph.</small>                                                                                                       |             |                |               |               |                                                |                                                                                                                  |           |
| TYPED NAME, GRADE OR TITLE                                                                                                                                                             |             |                |               |               | TELEPHONE EXCHANGE/<br>AUTOVON, PLUS EXTENSION |                                                                                                                  | SIGNATURE |

|                                                                                                                                                                                           |             |             |                          |                                                  |               |             |                                             |                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------|--------------------------|--------------------------------------------------|---------------|-------------|---------------------------------------------|--------------------|
| TO: (Forward direct to addressee listed in publication)<br>Commander, U.S. Army Aviation and Missile Command<br>ATTN: AMSAM-MMC-MA-NP<br>Redstone Arsenal, AL 35898                       |             |             |                          | FROM: (Activity and location) (Include ZIP Code) |               |             | DATE                                        |                    |
| PART II – REPAIR PARTS AND SPECIAL TOOL LISTS AND SUPPLY CATALOGS/SUPPLY MANUALS                                                                                                          |             |             |                          |                                                  |               |             |                                             |                    |
| PUBLICATION NUMBER                                                                                                                                                                        |             |             |                          | DATE                                             |               | TITLE       |                                             |                    |
| PAGE<br>NO.                                                                                                                                                                               | COLM<br>NO. | LINE<br>NO. | NATIONAL STOCK<br>NUMBER | REFERENCE<br>NO.                                 | FIGURE<br>NO. | ITEM<br>NO. | TOTAL NO.<br>OF MAJOR<br>ITEMS<br>SUPPORTED | RECOMMENDED ACTION |
|                                                                                                                                                                                           |             |             |                          |                                                  |               |             |                                             |                    |
| PART III – REMARKS (Any general remarks or recommendations, or suggestions for improvement of publications and blank forms. Additional blank sheets may be used if more space is needed.) |             |             |                          |                                                  |               |             |                                             |                    |
|                                                                                                                                                                                           |             |             |                          |                                                  |               |             |                                             |                    |
| TYPED NAME, GRADE OR TITLE                                                                                                                                                                |             |             |                          | TELEPHONE EXCHANGE/AUTOVON,<br>PLUS EXTENSION    |               |             | SIGNATURE                                   |                    |

## The Metric System and Equivalents

### *Linear Measure*

1 centimeter = 10 millimeters = .39 inch  
 1 decimeter = 10 centimeters = 3.94 inches  
 1 meter = 10 decimeters = 39.37 inches  
 1 dekameter = 10 meters = 32.8 feet  
 1 hectometer = 10 dekameters = 328.08 feet  
 1 kilometer = 10 hectometers = 3,280.8 feet

### *Weights*

1 centigram = 10 milligrams = .15 grain  
 1 decigram = 10 centigrams = 1.54 grains  
 1 gram = 10 decigrams = .035 ounce  
 1 decagram = 10 grams = .35 ounce  
 1 hectogram = 10 decagrams = 3.52 ounces  
 1 kilogram = 10 hectograms = 2.2 pounds  
 1 quintal = 100 kilograms = 220.46 pounds  
 1 metric ton = 10 quintals = 1.1 short tons

### *Liquid Measure*

1 centiliter = 10 milliliters = .34 fl. ounce  
 1 deciliter = 10 centiliters = 3.38 fl. ounces  
 1 liter = 10 deciliters = 33.81 fl. ounces  
 1 dekaliter = 10 liters = 2.64 gallons  
 1 hectoliter = 10 dekaliters = 26.42 gallons  
 1 kiloliter = 10 hectoliters = 264.18 gallons

### *Square Measure*

1 sq. centimeter = 100 sq. millimeters = .155 sq. inch  
 1 sq. decimeter = 100 sq. centimeters = 15.5 sq. inches  
 1 sq. meter (centare) = 100 sq. decimeters = 10.76 sq. feet  
 1 sq. dekameter (are) = 100 sq. meters = 1,076.4 sq. feet  
 1 sq. hectometer (hectare) = 100 sq. dekameters = 2.47 acres  
 1 sq. kilometer = 100 sq. hectometers = .386 sq. mile

### *Cubic Measure*

1 cu. centimeter = 1000 cu. millimeters = .06 cu. inch  
 1 cu. decimeter = 1000 cu. centimeters = 61.02 cu. inches  
 1 cu. meter = 1000 cu. decimeters = 35.31 cu. feet

## Approximate Conversion Factors

| <i>To change</i> | <i>To</i>          | <i>Multiply by</i> | <i>To change</i>   | <i>To</i>     | <i>Multiply by</i> |
|------------------|--------------------|--------------------|--------------------|---------------|--------------------|
| inches           | centimeters        | 2.540              | ounce-inches       | Newton-meters | .007062            |
| feet             | meters             | .305               | centimeters        | inches        | .394               |
| yards            | meters             | .914               | meters             | feet          | 3.280              |
| miles            | kilometers         | 1.609              | meters             | yards         | 1.094              |
| square inches    | square centimeters | 6.451              | kilometers         | miles         | .621               |
| square feet      | square meters      | .093               | square centimeters | square inches | .155               |
| square yards     | square meters      | .836               | square meters      | square feet   | 10.764             |
| square miles     | square kilometers  | 2.590              | square meters      | square yards  | 1.196              |
| acres            | square hectometers | .405               | square kilometers  | square miles  | .386               |
| cubic feet       | cubic meters       | .028               | square hectometers | acres         | 2.471              |
| cubic yards      | cubic meters       | .765               | cubic meters       | cubic feet    | 35.315             |
| fluid ounces     | milliliters        | 29.573             | cubic meters       | cubic yards   | 1.308              |
| pints            | liters             | .473               | milliliters        | fluid ounces  | .034               |
| quarts           | liters             | .946               | liters             | pints         | 2.113              |
| gallons          | liters             | 3.785              | liters             | quarts        | 1.057              |
| ounces           | grams              | 28.349             | liters             | gallons       | .264               |
| pounds           | kilograms          | .454               | grams              | ounces        | .035               |
| short tons       | metric tons        | .907               | kilograms          | pounds        | 2.205              |
| pound-feet       | Newton-meters      | 1.356              | metric tons        | short tons    | 1.102              |
| pound-inches     | Newton-meters      | .11296             |                    |               |                    |

## Temperature (Exact)

|          |                        |                            |                     |           |
|----------|------------------------|----------------------------|---------------------|-----------|
| <b>F</b> | Fahrenheit temperature | 5/9 (after subtracting 32) | Celsius temperature | <b>°C</b> |
|----------|------------------------|----------------------------|---------------------|-----------|

